# PROJECT MANUAL

## STRASBURG POLICE STATION

**115 WEST KING STREET**
**STRASBURG, VA 22657**

**TOWN OF STRASBURG**
**TOWN HALL**
**174 EAST KING STREET**
**STRASBURG, VA 22657**

**DEWBERRY PROJECT NUMBER: 50165756**

## ISSUED FOR BID
## AUGUST 26, 2026

## VOLUME 1 of 1
## DIVISIONS 00 thru 33


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**Structural Engineer of Record:**
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**MEP / FP Engineer of Record:**
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**Technology Designer of Record:**
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Phone: 309.282.8000

# DESIGN PROFESSIONALS OF RECORD: SEALS PAGE

**Architect of Record:**
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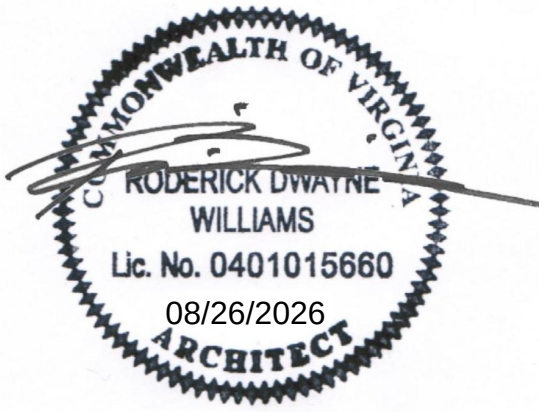
Pages or sheets covered by this seal:
General, Architecture and Interiors
Division 00 - Procurement And Contracting Requirements
Division 01 - General Requirements
Division 03 - Concrete
Division 04 – Masonry
Division 05 – Metals
Division 06 – Wood, Plastics & Composites
Division 07 - Thermal & Moisture Protection
Division 08 - Doors and Windows
Division 09 – Finishes
Division 10 – Specialties
Division 11 – Equipment
Division 12 – Furnishings
Division 14 – Conveying Equipment




**Civil Engineer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Site and Civil
Division 31 – Earthwork
Division 32 – Exterior Improvements
Division 33 – Utilities

**Structural Engineer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Structural
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Division 04 – Masonry
Division 05 – Metals

**Fire Protection and Plumbing Engineer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Mechanical and Plumbing
Division 21 – Fire Suppression
Division 22 – Plumbing

**Mechanical Engineer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Mechanical and Plumbing
Division 23 – Heating Ventilating & Air Conditioning
Division 25 – Integrated Automation

**Electrical Engineer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Electrical
Division 26 – Electrical

**Technology Designer of Record:**
**Dewberry Engineers Inc.**

Pages or sheets covered by this seal:
Technology
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Division 28 – Electronic Safety/Security

08/26/2026

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DAI = DEWBERRY ARCHITECTS INC. (INTERIOR DESIGNER)
DEI = DEWBERRY ENGINEERS INC. (CIVIL ENGINEER)
DEI = DEWBERRY ENGINEERS INC (STRUCTURAL ENGINEER)
DEI = DEWBERRY ENGINEERS INC (FIRE PROT. ENGINEER)
DEI = DEWBERRY ENGINEERS INC. (PLUMBING ENGINEER)
DEI = DEWBERRY ENGINEERS INC (MECHANICAL ENGINEER)
DEI = DEWBERRY ENGINEERS INC. (ELECTRICAL ENGINEER)
DEI = DEWBERRY ENGINEERS INC. (TECHNOLOGY DESIGNER)
F&R = FROEHLING & ROBERTSON INC. (INDUSTRIAL HYGIENIST)


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# DOCUMENT 00 0003 - INVITATION TO BID

## 1.1 PROJECT INFORMATION

A. Notice to Bidders: Bidders are invited to submit bids for Project as described in this Document according to the Instructions to Bidders.

B. Project Identification: Strasburg Police Station, DAI Project #50165756.

1. Project Location: 115 West King Street, Strasburg, VA 22657.

C. Owner: Town of Strasburg, 174 East King Street, Strasburg, VA 22657.

1. Owner's Representatives:

a. Chief Wayne Sager, Strasburg Police Dept., P: 540.465.5230, E: wsager@strasburgva.com

b. Town Manager, Mellanie Shipe, P: 540.465.9197, E: mshipe@strasburgva.com

D. Architect: Roderick Williams, AIA, Dewberry Architects Inc., P: 703.698.9076, E: rwilliams@dewberry.com.

E. Project Description: Project consists of full demolition of existing Architectural, Mechanical, Electrical, plumbing and low voltage elements in the existing building. This will allow the space to be reconfigured to accommodate Administration Offices, Patrol Briefing Room, Records, Detectives Suite with Interview Rooms, and additional support spaces. Additionally, the structural foundations will be required to be upgraded along with the addition of structural cross bracing to provide lateral resistance to the existing structure, allowing the building to comply with the requirements of a risk category 4, essential facility. The two-story building totals approximately 6,440 square feet. The existing site will be reconfigured to provide secure parking for the police station-providing 25 secure and 9 public parking spaces. Site plan revisions include the demolition of the existing teller canopy, canopy over sidewalk, sidewalk and existing parking. The new configuration will involve the installation of a security fence with 2 bi-fold security gates.

F. Construction Contract: Bids will be received for the following Work:

1. General Contract (all trades).

## 1.2 BID SUBMITTAL AND OPENING

A. Owner will receive sealed bids until the bid time and date through eVA at eva.Virginia.gov. Owner will consider bids prepared in compliance with the Instructions to Bidders issued by Owner, and delivered as follows:

1. Bid Date: October 19, 2026.
2. Bid Time: 2:00 p.m. EDT.
3. Location: eva.Virginia.gov

B. Bids will be thereafter publicly opened and read aloud.

1.3 BID SECURITY

A. Bid security shall be submitted with each bid in the amount of 5 percent of the bid amount. No bids may be withdrawn for a period of sixty (60) days after opening of bids. Owner reserves the right to reject any and all bids and to waive informalities and irregularities.

1.4 WITHDRAWAL OF BID

A. After the date and time designated for receipt of Bids, a Bidder who discovers that it made a clerical error in its Bid shall notify the Architect in writing of such error within two (2) business days, accompanied by the original work papers, requesting withdrawal of its Bid per Virginia Code § 2.2-4330. Upon providing evidence of such error to the reasonable satisfaction of the Architect, the Bid shall be withdrawn and not resubmitted.

1.5 PREBID CONFERENCE

A mandatory Prebid conference for all bidders will be held in the Strasburg Town Hall Council Chambers, 174 East King Street, Strasburg, VA 22657 on September 17, 2026 at 10:00 a.m. EDT. Prospective bidders are required to attend. Refer to document 00 0005 “Prebid Meetings” for additional information.

1.6 DOCUMENTS

A. Online Procurement and Contracting Documents: Bidders will receive access on or after September 4, 2026 when notified by the Owner and/or Architect. Online access will be provided to bidders.

1.7 TIME OF COMPLETION

A. Bidders shall begin the Work on receipt of the Notice to Proceed (anticipated issuance, November 10, 2026) and shall complete the Work within the Contract Time.

1.8 BIDDER'S QUALIFICATIONS

A. Bidders must be properly licensed under the laws governing their respective trades and be able to obtain insurance and bonds required for the Work. A Performance Bond, Statutory Bond and Defect, and Insurance in a form acceptable to Owner will be required of the successful Bidder.

END OF DOCUMENT 00 0003

# AIA® Document A701® – 2018

## *Instructions to Bidders*

for the following Project:
*(Name, location, and detailed description)*

Strasburg Police Station
115 West King Street
Strasburg, VA 22657

**THE OWNER:**
*(Name, legal status, address, and other information)*

Town of Strasburg
174 East King Street
Strasburg, VA 22657
540.465.9197

**THE ARCHITECT:**
*(Name, legal status, address, and other information)*

Dewberry Architects Inc.
8401 Arlington Blvd.
Fairfax, VA 22031-4666
703.849.0100

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7 PERFORMANCE BOND AND PAYMENT BOND

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**ADDITIONS AND DELETIONS:**
The author of this document has added information needed for its completion. The author may also have revised the text of the original AIA standard form. An *Additions and Deletions Report* that notes added information as well as revisions to the standard form text is available from the author and should be reviewed. A vertical line in the left margin of this document indicates where the author has added necessary information and where the author has added to or deleted from the original AIA text.

This document has important legal consequences. Consultation with an attorney is encouraged with respect to its completion or modification.

FEDERAL, STATE, AND LOCAL LAWS MAY IMPOSE REQUIREMENTS ON PUBLIC PROCUREMENT CONTRACTS. CONSULT LOCAL AUTHORITIES OR AN ATTORNEY TO VERIFY REQUIREMENTS APPLICABLE TO THIS PROCUREMENT BEFORE COMPLETING THIS FORM.

It is intended that AIA Document G612™–2017, Owner's Instructions to the Architect, Parts A and B will be completed prior to using this document.


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## ARTICLE 1 DEFINITIONS

**§ 1.1** Bidding Documents include the Bidding Requirements and the Proposed Contract Documents. The Bidding Requirements consist of the advertisement or invitation to bid, Instructions to Bidders, supplementary instructions to bidders, the bid form, and any other bidding forms. The Proposed Contract Documents consist of the unexecuted form of Agreement between the Owner and Contractor and that Agreement's Exhibits, Conditions of the Contract (General, Supplementary and other Conditions), Drawings, Specifications, all Addenda, and all other documents enumerated in Article 8 of these Instructions.

**§ 1.2** Definitions set forth in the General Conditions of the Contract for Construction, or in other Proposed Contract Documents apply to the Bidding Documents.

**§ 1.3** Addenda are written or graphic instruments issued by the Architect, which, by additions, deletions, clarifications, or corrections, modify or interpret the Bidding Documents.

**§ 1.4** A Bid is a complete and properly executed proposal to do the Work for the sums stipulated therein, submitted in accordance with the Bidding Documents.

**§ 1.5** The Base Bid is the sum stated in the Bid for which the Bidder offers to perform the Work described in the Bidding Documents, to which Work may be added or deleted by sums stated in Alternate Bids.

**§ 1.6** An Alternate Bid (or Alternate) is an amount stated in the Bid to be added to or deducted from, or that does not change, the Base Bid if the corresponding change in the Work, as described in the Bidding Documents, is accepted.

**§ 1.7** A Unit Price is an amount stated in the Bid as a price per unit of measurement for materials, equipment, or services, or a portion of the Work, as described in the Bidding Documents.

**§ 1.8** A Bidder is a person or entity who submits a Bid and who meets the requirements set forth in the Bidding Documents.

**§ 1.9** A Sub-bidder is a person or entity who submits a bid to a Bidder for materials, equipment, or labor for a portion of the Work.

## ARTICLE 2 BIDDER'S REPRESENTATIONS

**§ 2.1** By submitting a Bid, the Bidder represents that:

**.1** the Bidder has read and understands the Bidding Documents;

**.2** the Bidder understands how the Bidding Documents relate to other portions of the Project, if any, being bid concurrently or presently under construction;

**.3** the Bid complies with the Bidding Documents;

**.4** the Bidder has visited the site, become familiar with local conditions under which the Work is to be performed, and has correlated the Bidder's observations with the requirements of the Proposed Contract Documents;

**.5** the Bid is based upon the materials, equipment, and systems required by the Bidding Documents without exception; and

**.6** the Bidder has read and understands the provisions for liquidated damages, if any, set forth in the form of Agreement between the Owner and Contractor.

**.7** the Bidder understands that the Owner shall withhold five percent (5%) retainage from each progress payment until Substantial Completion, unless permitted by the Contract Documents or modified by Change Order. Upon Substantial Completion, retainage may be reduced or released in accordance with the Contract Documents, provided all conditions for release have been satisfied.

## ARTICLE 3 BIDDING DOCUMENTS

### § 3.1 Distribution

**§ 3.1.1** Bidders shall obtain complete Bidding Documents, as indicated below, from the issuing office designated in the advertisement or invitation to bid, for the deposit sum, if any, stated therein.

*(Indicate how, such as by email, website, host site/platform, paper copy, or other method Bidders shall obtain Bidding Documents.)*

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Beginning September 4, 20206, Bidding Documents will be available for download from the Town website under RFP or on eVA (eva.virginia.gov).

**§ 3.1.2** Any required deposit shall be refunded to Bidders who submit a bona fide Bid and return the paper Bidding Documents in good condition within ten (10) days after receipt of Bids. The cost to replace missing or damaged paper documents will be deducted from the deposit. A Bidder receiving a Contract award may retain the paper Bidding Documents, and the Bidder's deposit will be refunded.

**§ 3.1.3** Bidding Documents will not be issued directly to Sub-bidders unless specifically offered in the advertisement or invitation to bid, or in supplementary instructions to bidders.

**§ 3.1.4** Bidders shall use complete Bidding Documents in preparing Bids. Neither the Owner nor Architect assumes responsibility for errors or misinterpretations resulting from the use of incomplete Bidding Documents.

**§ 3.1.5** The Bidding Documents will be available for the sole purpose of obtaining Bids on the Work. No license or grant of use is conferred by distribution of the Bidding Documents.

**§ 3.2 Modification or Interpretation of Bidding Documents**

**§ 3.2.1** The Bidder shall carefully study the Bidding Documents, shall examine the site and local conditions, and shall notify the Architect of errors, inconsistencies, or ambiguities discovered and request clarification or interpretation pursuant to Section 3.2.2.

**§ 3.2.2** Requests for clarification or interpretation of the Bidding Documents shall be submitted by the Bidder in writing and shall be received by the Architect at least fourteen (14) days prior to the date for receipt of Bids.
*(Indicate how, such as by email, website, host site/platform, paper copy, or other method Bidders shall submit requests for clarification and interpretation.)*

Submit all questions in writing to Roderick Williams, Senior Project Manager, rwilliams@dewberry.com.

**§ 3.2.3** Modifications and interpretations of the Bidding Documents shall be made by Addendum. Modifications and interpretations of the Bidding Documents made in any other manner shall not be binding, and Bidders shall not rely upon them.

**§ 3.3 Substitutions**

**§ 3.3.1** The materials, products, and equipment described in the Bidding Documents establish a standard of required function, dimension, appearance, and quality to be met by any proposed substitution.

**§ 3.3.2 Substitution Process**

**§ 3.3.2.1** Written requests for substitutions shall be received by the Architect no later than Friday, September 25, 2026. Requests shall be submitted in the same manner as that established for submitting clarifications and interpretations in Section 3.2.2.

**§ 3.3.2.2** Bidders shall submit substitution requests on a Substitution Request Form if one is provided in the Bidding Documents.

**§ 3.3.2.3** If a Substitution Request Form is not provided, requests shall include (1) the name of the material or equipment specified in the Bidding Documents; (2) the reason for the requested substitution; (3) a complete description of the proposed substitution including the name of the material or equipment proposed as the substitute, performance and test data, and relevant drawings; and (4) any other information necessary for an evaluation. The request shall include a statement setting forth changes in other materials, equipment, or other portions of the Work, including changes in the work of other contracts or the impact on any Project Certifications (such as LEED), that will result from incorporation of the proposed substitution.

**§ 3.3.3** The burden of proof of the merit of the proposed substitution is upon the proposer. The Architect's decision of approval or disapproval of a proposed substitution shall be final.

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**§ 3.3.4** If the Architect approves a proposed substitution prior to receipt of Bids, such approval shall be set forth in an Addendum. Approvals made in any other manner shall not be binding, and Bidders shall not rely upon them.

**§ 3.3.5** No substitutions will be considered after the Contract award unless specifically provided for in the Contract Documents.

**§ 3.4 Addenda**

**§ 3.4.1** Addenda will be transmitted to Bidders known by the issuing office to have received complete Bidding Documents.
*(Indicate how, such as by email, website, host site/platform, paper copy, or other method Addenda will be transmitted.)*

Bidders that have downloaded bidding documents will be notified by eVA (eva.virginia.gov) when addenda have been issued. Addenda will be available to download from the Town website under RFP.

**§ 3.4.2** Addenda will be available where Bidding Documents are on file.

**§ 3.4.3** Addenda will be issued no later than four (4) days prior to the date for receipt of Bids, except an Addendum withdrawing the request for Bids or one which includes postponement of the date for receipt of Bids.

**§ 3.4.4** Prior to submitting a Bid, each Bidder shall ascertain that the Bidder has received all Addenda issued, and the Bidder shall acknowledge their receipt in the Bid.

## ARTICLE 4 BIDDING PROCEDURES

**§ 4.1 Preparation of Bids**

**§ 4.1.1** Bids shall be submitted on the forms included with or identified in the Bidding Documents.

**§ 4.1.2** All blanks on the bid form shall be legibly executed. Paper bid forms shall be executed in a non-erasable medium.

**§ 4.1.3** Sums shall be expressed in both words and numbers, unless noted otherwise on the bid form. In case of discrepancy, the amount entered in words shall govern.

**§ 4.1.4** Edits to entries made on paper bid forms must be initialed by the signer of the Bid.

**§ 4.1.5** All requested Alternates shall be bid. If no change in the Base Bid is required, enter "No Change" or as required by the bid form.

**§ 4.1.6** Where two or more Bids for designated portions of the Work have been requested, the Bidder may, without forfeiture of the bid security, state the Bidder's refusal to accept award of less than the combination of Bids stipulated by the Bidder. The Bidder shall neither make additional stipulations on the bid form nor qualify the Bid in any other manner.

**§ 4.1.7** Each copy of the Bid shall state the legal name and legal status of the Bidder. As part of the documentation submitted with the Bid, the Bidder shall provide evidence of its legal authority to perform the Work in the jurisdiction where the Project is located. Each copy of the Bid shall be signed by the person or persons legally authorized to bind the Bidder to a contract. A Bid by a corporation shall further name the state of incorporation and have the corporate seal affixed. A Bid submitted by an agent shall have a current power of attorney attached, certifying the agent's authority to bind the Bidder.

**§ 4.1.8** A Bidder shall incur all costs associated with the preparation of its Bid.

**§ 4.2 Bid Security**

**§ 4.2.1** Each Bid shall be accompanied by the following bid security:
*(Insert the form and amount of bid security.)*

Bid Bond in the amount of five percent (5%) of the Bid Proposal submitted.

**§ 4.2.2** The Bidder pledges to enter into a Contract with the Owner on the terms stated in the Bid and shall, if required, furnish bonds covering the faithful performance of the Contract and payment of all obligations arising thereunder. Should

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the Bidder refuse to enter into such Contract or fail to furnish such bonds if required, the amount of the bid security shall be forfeited to the Owner as liquidated damages, not as a penalty. In the event the Owner fails to comply with Section 6.2, the amount of the bid security shall not be forfeited to the Owner.

**§ 4.2.3** If a surety bond is required as bid security, it shall be written on AIA Document A310™, Bid Bond, unless otherwise provided in the Bidding Documents. The attorney-in-fact who executes the bond on behalf of the surety shall affix to the bond a certified and current copy of an acceptable power of attorney. The Bidder shall provide surety bonds from a company or companies lawfully authorized to issue surety bonds in the jurisdiction where the Project is located.

**§ 4.2.4** The Owner will have the right to retain the bid security of Bidders to whom an award is being considered until (a) the Contract has been executed and bonds, if required, have been furnished; (b) the specified time has elapsed so that Bids may be withdrawn; or (c) all Bids have been rejected. However, if no Contract has been awarded or a Bidder has not been notified of the acceptance of its Bid, a Bidder may, beginning sixty (60) days after the opening of Bids, withdraw its Bid and request the return of its bid security.

**§ 4.3 Submission of Bids**

**§ 4.3.1** A Bidder shall submit its Bid as indicated below:
*(Indicate how, such as by website, host site/platform, paper copy, or other method Bidders shall submit their Bid.)*

Bids may be submitted electronically through eVA or delivered as a sealed bid to the Strasburg Town Hall located at 174 E. King Street, Strasburg, VA 22657. All bids, regardless of submission method, must be received not later than 2:00 PM on October 19, 2026. No late bids will be accepted.

**§ 4.3.2** Paper copies of the Bid, the bid security, and any other documents required to be submitted with the Bid shall be enclosed in a sealed opaque envelope. The envelope shall be addressed to the party receiving the Bids and shall be identified with the Project name, the Bidder's name and address, and, if applicable, the designated portion of the Work for which the Bid is submitted. If the Bid is sent by mail, the sealed envelope shall be enclosed in a separate mailing envelope with the notation "SEALED BID ENCLOSED" on the face thereof.

**§ 4.3.3** Bids shall be submitted by the date and time and at the place indicated in the invitation to bid. Bids submitted after the date and time for receipt of Bids, or at an incorrect place, will not be accepted.

**§ 4.3.4** The Bidder shall assume full responsibility for timely delivery at the location designated for receipt of Bids.

**§ 4.3.5** A Bid submitted by any method other than as provided in this Section 4.3 will not be accepted.

**§ 4.4 Modification or Withdrawal of Bid**

**§ 4.4.1** Prior to the date and time designated for receipt of Bids, a Bidder may submit a new Bid to replace a Bid previously submitted, or withdraw its Bid entirely, by notice to the party designated to receive the Bids. Such notice shall be received and duly recorded by the receiving party on or before the date and time set for receipt of Bids. The receiving party shall verify that replaced or withdrawn Bids are removed from the other submitted Bids and not considered. Notice of submission of a replacement Bid or withdrawal of a Bid shall be worded so as not to reveal the amount of the original Bid.

**§ 4.4.2** Withdrawn Bids may be resubmitted up to the date and time designated for the receipt of Bids in the same format as that established in Section 4.3, provided they fully conform with these Instructions to Bidders. Bid security shall be in an amount sufficient for the Bid as resubmitted.

**§ 4.4.3** After the date and time designated for receipt of Bids, a Bidder who discovers that it made a clerical error in its Bid shall notify the Architect in writing of such error within two (2) business days, accompanied by the original work papers, requesting withdrawal of its Bid per Virginia Code § 2.2-4330. Upon providing evidence of such error to the reasonable satisfaction of the Architect, the Bid shall be withdrawn and not resubmitted. If a Bid is withdrawn pursuant to this Section 4.4.3, the bid security will be attended to as follows:
*(State the terms and conditions, such as Bid rank, for returning or retaining the bid security.)*

Bid security for withdrawn bids will be returned within ten (10) business days.

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## ARTICLE 5 CONSIDERATION OF BIDS

### § 5.1 Opening of Bids

If stipulated in an advertisement or invitation to bid, or when otherwise required by law, Bids properly identified and received within the specified time limits will be publicly opened and read aloud. A summary of the Bids may be made available to Bidders.

### § 5.2 Rejection of Bids

Unless otherwise prohibited by law, the Owner shall have the right to reject any or all Bids.

### § 5.3 Acceptance of Bid (Award)

**§ 5.3.1** It is the intent of the Owner to award a Contract to the lowest responsive and responsible Bidder, provided the Bid has been submitted in accordance with the requirements of the Bidding Documents. Unless otherwise prohibited by law, the Owner shall have the right to waive informalities and irregularities in a Bid received and to accept the Bid which, in the Owner's judgment, is in the Owner's best interests. Anticipated date of award is Oct. 5, 2026. Anticipated date of Notice to Proceed is Oct. 20, 2026.

**§ 5.3.2** Unless otherwise prohibited by law, the Owner shall have the right to accept Alternates in any order or combination, unless otherwise specifically provided in the Bidding Documents, and to determine the lowest responsive and responsible Bidder on the basis of the sum of the Base Bid and Alternates accepted.

## ARTICLE 6 POST-BID INFORMATION

### § 6.1 Contractor's Qualification Statement

Bidders to whom award of a Contract is under consideration shall submit to the Architect, upon request and within the timeframe specified by the Architect, a properly executed AIA Document A305™, Contractor's Qualification Statement, unless such a Statement has been previously required and submitted for this Bid.

### § 6.2 Owner's Financial Capability

A Bidder to whom award of a Contract is under consideration may request in writing, fourteen (14) days prior to the expiration of the time for withdrawal of Bids, that the Owner furnish to the Bidder reasonable evidence that financial arrangements have been made to fulfill the Owner's obligations under the Contract. The Owner shall then furnish such reasonable evidence to the Bidder no later than seven (7) days prior to the expiration of the time for withdrawal of Bids. Unless such reasonable evidence is furnished within the allotted time, the Bidder will not be required to execute the Agreement between the Owner and Contractor.

### § 6.3 Submittals

**§ 6.3.1** After notification of selection for the award of the Contract, the Bidder shall, as soon as practicable or as stipulated in the Bidding Documents, submit in writing to the Owner through the Architect:

**.1** a designation of the Work to be performed with the Bidder's own forces;

**.2** names of the principal products and systems proposed for the Work and the manufacturers and suppliers of each; and

**.3** names of persons or entities (including those who are to furnish materials or equipment fabricated to a special design) proposed for the principal portions of the Work.

**§ 6.3.2** The Bidder will be required to establish to the satisfaction of the Architect and Owner the reliability and responsibility of the persons or entities proposed to furnish and perform the Work described in the Bidding Documents.

**§ 6.3.3** Prior to the execution of the Contract, the Architect will notify the Bidder if either the Owner or Architect, after due investigation, has reasonable objection to a person or entity proposed by the Bidder. If the Owner or Architect has reasonable objection to a proposed person or entity, the Bidder may, at the Bidder's option, withdraw the Bid or submit an acceptable substitute person or entity. The Bidder may also submit any required adjustment in the Base Bid or Alternate Bid to account for the difference in cost occasioned by such substitution. The Owner may accept the adjusted bid price or disqualify the Bidder. In the event of either withdrawal or disqualification, bid security will not be forfeited.

**§ 6.3.4** Persons and entities proposed by the Bidder and to whom the Owner and Architect have made no reasonable objection must be used on the Work for which they were proposed and shall not be changed except with the written consent of the Owner and Architect.

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**ARTICLE 7 PERFORMANCE BOND AND PAYMENT BOND**

**§ 7.1 Bond Requirements**

**§ 7.1.1** If stipulated in the Bidding Documents, the Bidder shall furnish bonds covering the faithful performance of the Contract and payment of all obligations arising thereunder.

**§ 7.1.2** If the furnishing of such bonds is stipulated in the Bidding Documents, the cost shall be included in the Bid. If the furnishing of such bonds is required after receipt of bids and before execution of the Contract, the cost of such bonds shall be added to the Bid in determining the Contract Sum.

**§ 7.1.3** The Bidder shall provide surety bonds from a company or companies lawfully authorized to issue surety bonds in the jurisdiction where the Project is located.

**§ 7.1.4** Unless otherwise indicated below, the Penal Sum of the Payment and Performance Bonds shall be the amount of the Contract Sum.
*(If Payment or Performance Bonds are to be in an amount other than 100% of the Contract Sum, indicate the dollar amount or percentage of the Contract Sum.)*

N/A

**§ 7.2 Time of Delivery and Form of Bonds**

**§ 7.2.1** The Bidder shall deliver the required bonds to the Owner not later than ten (10) days following the date of execution of the Contract. If the Work is to commence sooner in response to a letter of intent, the Bidder shall, prior to commencement of the Work, submit evidence satisfactory to the Owner that such bonds will be furnished and delivered in accordance with this Section 7.2.1.

**§ 7.2.2** Unless otherwise provided, the bonds shall be written on AIA Document A312, Performance Bond and Payment Bond.

**§ 7.2.3** The bonds shall be dated on or after the date of the Contract.

**§ 7.2.4** The Bidder shall require the attorney-in-fact who executes the required bonds on behalf of the surety to affix to the bond a certified and current copy of the power of attorney.

**ARTICLE 8 ENUMERATION OF THE PROPOSED CONTRACT DOCUMENTS**

**§ 8.1** Copies of the proposed Contract Documents have been made available to the Bidder and consist of the following documents:

**.1** AIA Document A101™–2017, Standard Form of Agreement Between Owner and Contractor, unless otherwise stated below.
*(Insert the complete AIA Document number, including year, and Document title.)*

**00 0013 Owner Contractor Agreement, A101-2017**

**.2** AIA Document A101™–2017, Exhibit A, Insurance and Bonds, unless otherwise stated below.
*(Insert the complete AIA Document number, including year, and Document title.)*

**.3** AIA Document A201™–2017, General Conditions of the Contract for Construction, unless otherwise stated below.
*(Insert the complete AIA Document number, including year, and Document title.)*

**00 0014 General Conditions, A201-2017**

**.4** Building Information Modeling Exhibit, if completed:

N/A

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**.5** Drawings

| Number | Title | Date |
|---|---|---|
| Sheet G-001 Index | Contract Drawing Index | 8/26/2026 |

**.6** Specifications

| Section | Title | Date | Pages |
|---|---|---|---|
| Project Specification Manual | Table of Contents | 8/26/2026 | 6 |

**.7** Addenda:

| Number | Date | Pages |
|---|---|---|
| To Be Determined | | |

**.8** Other Exhibits:
*(Check all boxes that apply and include appropriate information identifying the exhibit where required.)*

N/A

**.9** Other documents listed below:
*(List here any additional documents that are intended to form part of the Proposed Contract Documents.)*

N/A

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DOCUMENT 00 0005 - PREBID MEETINGS

1.1 PREBID MEETING

A. Owner and Architect will conduct a Prebid meeting as indicated below:

1. Meeting Date: Thursday September 17, 2026.
2. Meeting Time: 10:00 a.m., local time.
3. Location:
   Strasburg Town Hall, Council Chambers
   174 East King Street
   Strasburg, VA 22657

B. Attendance:

1. Prime Bidders: Attendance at Prebid meeting is mandatory.
2. Subcontractors: Attendance at Prebid meeting is recommended.
3. Notice: Bids will only be accepted from prime bidders represented on Prebid Meeting sign-in sheet.

C. Bidder Questions: Submit written questions to be addressed at Prebid meeting minimum of two (2) business days prior to meeting. Deadline to submit written questions prior to bid opening is Friday, September 25, 2026.

D. Agenda: Prebid meeting agenda will include review of topics that may affect proper preparation and submittal of bids, including the following:

1. Procurement and Contracting Requirements:
   a. Advertisement for Bids.
   b. Instructions to Bidders.
   c. Bidder Qualifications.
   d. Bonding.
   e. Insurance.
   f. Bid Security.
   g. Bid Form and Attachments.
   h. Bid Submittal Requirements.
   i. Notice of Award.

2. Communication during Bidding Period:
   a. Obtaining documents.
   b. Access to Project Web site.
   c. Bidder's Requests for Information.
   d. Bidder's Substitution Request/Prior Approval Request.
   e. Addenda.

3. Contracting Requirements:
   a. Agreement.
   b. The General Conditions.
   c. Other Owner requirements.

4. Construction Documents:

    a. Scopes of Work.
    b. Temporary Facilities.
    c. Use of Site.
    d. Work Restrictions.
    e. Alternates, Allowances, and Unit Prices.
    f. Substitutions following award.

5. Separate Contracts:

    a. Work by Owner.
    b. Work of Other Contracts.

6. Schedule:

    a. Project Schedule.
    b. Contract Time.
    c. Other Bidder Questions.

7. Site/facility visit or walkthrough.
8. Post-Meeting Addendum.

E. Minutes: Entity responsible for conducting meeting will record and distribute meeting minutes to attendees. Minutes of meeting are issued as Available Information and do not constitute a modification to the Procurement and Contracting Documents. Modifications to the Procurement and Contracting Documents are issued by written Addendum only.

1. Sign-in Sheet: Minutes will include list of meeting attendees.
2. List of Planholders: Minutes will include list of planholders.

END OF DOCUMENT 00 0005

DOCUMENT 00 0006 - PROCUREMENT SUBSTITUTION PROCEDURES

A. Procurement Substitution Requests: Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Procurement and Contracting Documents, submitted prior to receipt of bids.

B. Substitution Requests: Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Contract Documents, submitted following Contract award. See Section 01 2500 "Substitution Procedures" for conditions under which Substitution requests will be considered following Contract award.

1.2 QUALITY ASSURANCE

A. Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturers.

1.3 PROCUREMENT SUBSTITUTIONS

A. Procurement Substitutions, General: By submitting a bid, the Bidder represents that its bid is based on materials and equipment described in the Procurement and Contracting Documents, including Addenda. Bidders are encouraged to request approval of qualifying substitute materials and equipment when the Specifications Sections list materials and equipment by product or manufacturer name.

B. Procurement Substitution Requests will be received and considered by Owner when the following conditions are satisfied, as determined by Architect; otherwise requests will be returned without action:

1. Extensive revisions to the Contract Documents are not required.
2. Proposed changes are in keeping with the general intent of the Contract Documents, including the level of quality of the Work represented by the requirements therein.
3. The request is fully documented and properly submitted.

1.4 SUBMITTALS

A. Procurement Substitution Request: Submit to Architect. Procurement Substitution Request must be made in writing by a prime contract Bidder only in compliance with the following requirements:

1. Requests for substitution of materials and equipment will be considered if received no later than ten (10) days prior to date of bid opening.
2. Submittal Format: Submit one electronic copies of each written Procurement Substitution Request, using CSI Substitution Request Form 1.5C or approved equivalent.
3. Submittal Format: Submit Procurement Substitution Request, using format provided on Project Web site.

    a. Identify the product or the fabrication or installation method to be replaced in each request. Include related Specifications Sections and drawing numbers.

b. Provide complete documentation on both the product specified and the proposed substitute, including the following information as appropriate:

1) Point-by-point comparison of specified and proposed substitute product data, fabrication drawings, and installation procedures.
2) Copies of current, independent third-party test data of salient product or system characteristics.
3) Samples where applicable or when requested by Architect.
4) Detailed comparison of significant qualities of the proposed substitute with those of the Work specified. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.
5) Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.
6) Research reports, where applicable, evidencing compliance with building code in effect for Project, from ICC-ES.
7) Coordination information, including a list of changes or modifications needed to other parts of the Work and to construction performed by Owner and separate contractors, which will become necessary to accommodate the proposed substitute.

c. Provide certification by manufacturer that the substitute proposed is equal to or superior to that required by the Procurement and Contracting Documents, and that its in-place performance will be equal to or superior to the product or equipment specified in the application indicated.

d. Bidder, in submitting the Procurement Substitution Request, waives the right to additional payment or an extension of Contract Time because of the failure of the substitute to perform as represented in the Procurement Substitution Request.

B. Architect's Action:

1. Architect may request additional information or documentation necessary for evaluation of the Procurement Substitution Request. Architect will notify all bidders of acceptance of the proposed substitute by means of an Addendum to the Procurement and Contracting Documents.

C. Architect's approval of a substitute during bidding does not relieve Contractor of the responsibility to submit required shop drawings and to comply with all other requirements of the Contract Documents.

END OF DOCUMENT 00 0006

# DOCUMENT 00 0007 - BID FORM - STIPULATED SUM (SINGLE-PRIME CONTRACT)

## 1.1 BID INFORMATION

A. Bidder: _____.

B. Project Name: Strasburg Police Station.

C. Project Location: 115 West King Street, Strasburg, VA 22657.

D. Owner: Town of Strasburg, VA

E. Architect: Dewberry Architects Inc.

F. Architect Project Number: 50165756.

## 1.2 CERTIFICATIONS AND BASE BID

A. Base Bid, Single-Prime (All Trades) Contract: The undersigned Bidder, having carefully examined the Procurement and Contracting Requirements, Conditions of the Contract, Drawings, Specifications, and all subsequent Addenda, as prepared by Dewberry Architects Inc. and Architect's consultants, having visited the site, and being familiar with all conditions and requirements of the Work, hereby agrees to furnish all material, labor, equipment and services, including all scheduled allowances, necessary to complete the construction of the above-named project, according to the requirements of the Procurement and Contracting Documents, for the stipulated sum of:

1. _____Dollars ($_____).

## 1.3 BID GUARANTEE

A. The undersigned Bidder agrees to execute a contract for this Work in the above amount and to furnish surety as specified within ten (**10)** days after a written Notice of Award, if offered within sixty (**60)** days after receipt of bids, and on failure to do so agrees to forfeit to Owner the attached cash, cashier's check, certified check, U.S. money order, or bid bond, as liquidated damages for such failure, in the following amount constituting five percent (5%) of the Base Bid amount above:

1. _____Dollars ($_____).

B. In the event Owner does not offer Notice of Award within the time limits stated above, Owner will return to the undersigned the cash, cashier's check, certified check, U.S. money order, or bid bond.

## 1.4 SUBCONTRACTORS AND SUPPLIERS

A. The following companies shall execute subcontracts for the portions of the Work indicated:

1. Concrete Work: _____.

2. Structural Steel Work: _____.
3. Masonry Work: _____.
4. Roofing Work: _____.
5. Plumbing Work: _____.
6. HVAC Work: _____.
7. Electrical Work: _____.
8. Electronic Security Work: _____.

1.5 TIME OF COMPLETION

A. The undersigned Bidder proposes and agrees hereby to commence the Work of the Contract Documents on a date specified in a written Notice to Proceed to be issued by Architect and shall fully complete the Work within the contractually agreed upon calendar days.

1.6 ACKNOWLEDGEMENT OF ADDENDA

A. The undersigned Bidder acknowledges receipt of and use of the following Addenda in the preparation of this Bid:

1. Addendum No. 1, dated _____.
2. Addendum No. 2, dated _____.
3. Addendum No. 3, dated _____.
4. Addendum No. 4, dated _____.
5. Addendum No. 5, dated _____.

1.7 BID SUPPLEMENTS

A. The following supplements are a part of this Bid Form and are attached hereto.

1. Bid Form Supplement – Affidavits.
2. Bid Form Supplement – Bonds and Certificates.

1.8 CONTRACTOR'S LICENSE

A. The undersigned further states that it is a duly licensed contractor, for the type of work proposed, in the State of Virginia, and that all fees, permits, etc., pursuant to submitting this proposal have been paid in full.

1.9 SUBMISSION OF BID

A. Respectfully submitted this ____ day of _____, 2026.

B. Submitted By:_____(Name of bidding firm or corporation).

C. Authorized Signature:_____(Handwritten signature).

D. Signed By:_____(Type or print name).

E. Title:_____(Owner/Partner/President/Vice President).

F. Witnessed By:_____(Handwritten signature).

G. Attest:_____(Handwritten signature).

H. By:_____(Type or print name).

I. Title:_____(Corporate Secretary or Assistant Secretary).

J. Street Address:_____.

K. City, State, Zip:_____.

L. Phone:_____.

M. License No.:_____.

N. Federal ID No.:_____(Affix Corporate Seal Here).

END OF DOCUMENT 00 0007

DOCUMENT 00 0009 — BID FORM SUPPLEMENTS

*(The following bid affidavits are to accompany the bid)*

NON-COLLUSION AFFIDAVIT

STATE OF _____)

) §

COUNTY OF _____)

(1) _____
(Name and Title)

_____
(Company name, City and State)
of lawful age, being first duly sworn, on oath says, that (s)he is the agent authorized by the bidder to submit the attached bid. Affiant further states that the bidder has not been a party to any collusion among bidders in restraint of freedom of competition by agreement to bid at a fixed price or to refrain from bidding; or with any official of the

(2) Town of Strasburg, VA
or any of their employees as to quantity, quality or price in the prospective contract, or any other terms of said prospective contract; or in any discussion between bidders and any official of the

(2) Town of Strasburg, VA
or any of their employees concerning exchange of money or other things of value for special consideration in the letting of the following contract.

(3) Strasburg Police Station

Architect's Job No. 50165756, for the Owner, (4) Town of Strasburg, VA

Firm Name _____

Signature and Title _____

SUBSCRIBED AND SWORN to before me this _____ day of _____, 20___.

_____
Notary Public

My Commission Expires: _____

1. Owner, Partner or Officer of the Firm of, Company Name, City and State.
2. Name of Owner.
3. Name of project as it appears in the Proposal.
4. Name of Owner as it appears in the Proposal.

BUSINESS RELATIONSHIPS AFFIDAVIT

STATE OF _____)

) §

COUNTY OF _____)

(1) _____
(Name and Title)

_____
(Company name, City and State)

of lawful age, being first duly sworn, on oath says that (s)he is the agent authorized by the bidder to submit the attached bid. Affiant further states that the nature of any partnership, joint venture, or other business relationship presently in effect or which existed within one (1) year prior to the date of this statement with the Architect, Engineer, or other party to the project is as follows:

_____

_____

_____

Affiant further states that any such business relationship presently in effect or which existed within one (1) year prior to the date of this statement between any officer or director of the bidding company and any officer or director of the Architectural or Engineering firm or other party to the project is as follows:

_____

_____

_____

Affiant further states that the names of all persons having any such business relationships and the positions they hold with their respective companies or firms are as follows:

_____

_____

(If none of the business relationships hereinabove mentioned exist, Affiant should so state).

Firm Name _____

Signature and Title _____

SUBSCRIBED AND SWORN to before me this_____ day of _____, 20___.

_____
Notary Public

My Commission Expires: _____

END OF DOCUMENT 00 0009

SECTION 00 0010 — BONDS AND CERTIFICATES

## BID BOND

KNOW ALL PERSONS BE THESE PRESENTS:

That _____, as Principal, and _____, as Surety, are hereby held and firmly bound unto the Town of Strasburg, VA, hereinafter called the Owner, in the sum of _____ Dollars ($_____) for the payment of which sum, well and truly to be made, we hereby jointly and severally bind ourselves, our heirs, executors, administrators, successors, and assigns.

WHEREAS, the Bidder has submitted to the Town of Strasburg, VA, a certain bid which is attached hereto and hereby made a part hereof, to enter into agreement, for furnishing labor and materials for the **Strasburg Police Station**.

NOW, THEREFORE, if the Owner shall accept the bid of the Principal and the Principal shall enter into a Contract with the Owner in accordance with the terms of such bid, and give such bond or bonds as may be specified in the bidding of Contract Documents with good and sufficient surety for the faithful performance of such Contract and for the prompt payment of labor and material furnished in the prosecution thereof, or in the event of the failure of the Principal to enter such Contract and give such bond or bonds, if the Principal shall pay to the Owner the difference not to exceed the penalty hereof between the amount specified in the bid and such larger amount for which the Owner may in good faith contract with another party to perform the Work covered by the bid, then this obligation shall be null and void, otherwise to remain in full force and effect.

Signed and sealed this ____ day of _____, 20___.

PRINCIPAL:

(Seal) _____

By _____

ATTEST:

SURETY:

_____

By _____
Title

## PERFORMANCE BOND

KNOW ALL PERSONS BY THESE PRESENTS:

That _____, as Principal, and _____, a corporation organized under the laws of the State of _____ and authorized to transact business in the State of Virginia, hereinafter called Surety, are held and firmly bound unto the Town of Strasburg, VA, as Obligee, hereinafter called the Owner in the amount of _____ Dollars ($_____) in lawful money of the United States of America, for the payment of which, well and truly to be made, we bind ourselves and each of us, our heirs, executors, administrators, trustees, successors, and assigns, jointly and severally, firmly by these presents.

WHEREAS, the Principal has by written agreement dated _____, 20____ entered into a Contract with Owner for construction of the **Strasburg Police Station**, in accordance with Drawings and Specifications prepared by Dewberry; 8401 Arlington Blvd., Fairfax, VA 22031-4666, which contract is by reference made a part hereof and is hereinafter referred to as the Contract.

NOW, THEREFORE, if the Principal shall, in all particulars, well, truly, and faithfully perform and abide by said Contract and each and every covenant, condition, and part thereof and shall fulfill all obligations resting upon the Principal by the terms of said Contract and the specifications; and if the Principal shall promptly pay, or cause to be paid, all labor, materials, and/or repairs and all bills for labor performed on the work, whether by sub-contract or otherwise; and if the Principal shall protect and save harmless the Town of Strasburg, VA, from all loss, damage, and expense to life or property suffered or sustained by any person, firm, or corporation caused by the Principal or his or its agents, servants, or employees in the construction of the work, or by or in consequence of any negligence, carelessness or misconduct in guarding and protecting the same, or form any act or omission of the Principal or his or its agents, servants, or employees, and if the Principal shall protect and save the Town of Strasburg, VA, harmless from all suits and claims of infringement or alleged infringement or patent rights or processes, then this obligation shall be null and void, otherwise to be and remain in full force and effect.

It is further expressly agreed and understood by the parties hereto that no changes or alterations in said Contract and no deviations from the plan or mode of procedure herein fixed shall have the effect of releasing the sureties, or any of them, from the obligations of this Bond.

IN WITNESS WHEREOF, the Principal has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its duly authorized officers, and the Surety has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its attorney-in-fact, duly authorized so to do, the day and year set forth below.

Signed and sealed this ____ day of _____, 20____.

PRINCIPAL:

(Seal) _____

By _____

ATTEST:

SURETY:

_____

By _____

## STATUTORY BOND

KNOW ALL PERSONS BY THESE PRESENTS:

That _____ as Principal, hereinafter called Contractor, and _____, a corporation organized under the laws of the State of _____ and authorized to transact business in the State of Virginia, hereinafter called Surety, are held and firmly bound unto the Town of Strasburg, VA, as Obligee, in the amount of _____ Dollars ($_____) in lawful money of the United States of America, for the payment of which, well and truly to be made, we bind ourselves and each of us, our heirs, executors, administrators, trustees, successors, and assigns, jointly and severally, firmly by these presents.

WHEREAS, Contractor has by written agreement dated _____ , 20___ entered into a Contract with Owner for the **Strasburg Police Station** in accordance with Drawings and Specifications prepared by Dewberry; 8401 Arlington Blvd., Fairfax, VA 22031-4666, which contract is by reference made a part hereof, and is hereinafter referred to as the Contract.

NOW, THEREFORE, if said Contractor shall fail or neglect to pay all indebtedness incurred by said Contractor or subcontractors of said Contractor who perform work in the performance of such contract, for labor and materials and repairs to and parts for equipment used and consumed in the performance of said Contract within thirty (30) days after the same becomes due and payable, the person, firm, or corporation entitled thereto may sue and recover on this bond, the amount so due and unpaid.

It is further expressly agreed and understood by the parties hereto that no changes or alterations in said Contract and no deviations from the plan or mode of procedure herein fixed shall have the effect of releasing the sureties, or any of them, from the obligations of this Bond.

IN WITNESS WHEREOF, the said Contractor has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its duly authorized officers, and the said Surety has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its attorney-in-fact, duly authorized so to do, the day and year first above written.

Signed and sealed this ____ day of _____, 20____.

PRINCIPAL:

_____

By _____

ATTEST:

SURETY:

_____

By _____
Attorney-in-Fact

## DEFECT BOND

KNOW ALL PERSONS BY THESE PRESENTS:

That _____ as Principal, hereinafter called Contractor, and _____, a corporation organized under the laws of the State of _____ and authorized to transact business in the State of Virginia, hereinafter called Surety, are held and firmly bound unto the Town of Strasburg, VA, as Obligee, hereinafter called Owner, in the amount of _____ Dollars ($_____) in lawful money of the United States of America, for the payment of which, well and truly to be made, we bind ourselves and each of us, our heirs, executors, administrators, trustees, successors, and assigns, jointly and severally, firmly by these presents.

WHEREAS, Contractor has by written agreement dated _____ , 20___ entered into a Contract with Owner for construction of the **Strasburg Police Station**, in accordance with Drawings and Specifications prepared by Dewberry; 8401 Arlington Blvd., Fairfax, VA 22031-4666, which contract is by reference made a part hereof, and is hereinafter referred to as the Contract.

NOW, THEREFORE, if said Contractor shall pay or cause to be paid to Owner all damage, loss, and expense which may result by reason of defective materials and/or workmanship in connection with said work, occurring within a period of one (1) year from and after acceptance of said project by Owner then this obligation shall be null and void, otherwise to be and remain in full force and effect.

It is further expressly agreed and understood by the parties hereto that no changes or alterations in said Contract and no deviations from the plan or mode of procedure herein fixed shall have the effect of releasing the sureties, or any of them, from the obligations of this Bond.

IN WITNESS WHEREOF, the said Contractor has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its duly authorized officers, and the said Surety has caused these presents to be executed in its name and its corporate seal to be hereunto affixed by its attorney-in-fact, duly authorized so to do, the day and year set forth below.

Signed and sealed this ____ day of _____, 20____.

PRINCIPAL:

_____

By _____

ATTEST:

SURETY:

_____

By _____
Attorney-in-Fact

# DOCUMENT 00 0012 - LIST OF DRAWINGS AND SPECIFICATIONS

## 1.1 LIST OF DRAWINGS AND SPECIFICATIONS

A. Drawings: Drawings consist of the Contract Drawings and other drawings listed on Drawing G-001 “Index” of the separately bound drawing set titled Strasburg Police Station, Bid Documents, dated August 26, 2026, and as modified by subsequent Addenda and Contract modifications.

B. Specifications: Specifications manuals consist of the Contract documents listed on the “Table of Contents” page(s) of the separately bound project manuals titled Strasburg Police Station, Bid Documents dated August 26, 2026, as modified by subsequent Addenda and Contract modifications.

END OF DOCUMENT 00 0012

# AIA® Document A101® – 2017

## *Standard Form of Agreement Between Owner and Contractor* *where the basis of payment is a Stipulated Sum*

**AGREEMENT** made as of the day of in the year 2026
*(In words, indicate day, month and year.)*

**BETWEEN** the Owner:
*(Name, legal status, address and other information)*

Town of Strasburg
174 East King Street
Strasburg, VA 22657
540.465.9197

and the Contractor:
*(Name, legal status, address and other information)*

for the following Project:
*(Name, location and detailed description)*

Strasburg Police Station
115 West King Street
Strasburg, VA 22657

The Architect:
*(Name, legal status, address and other information)*

Dewberry Architects Inc.
8401 Arlington Blvd.
Fairfax, VA 22031-4666
703.849.0100

The Owner and Contractor agree as follows.

**ADDITIONS AND DELETIONS:**
The author of this document has added information needed for its completion. The author may also have revised the text of the original AIA standard form. An *Additions and Deletions Report* that notes added information as well as revisions to the standard form text is available from the author and should be reviewed. A vertical line in the left margin of this document indicates where the author has added necessary information and where the author has added to or deleted from the original AIA text.

This document has important legal consequences. Consultation with an attorney is encouraged with respect to its completion or modification.

The parties should complete A101®–2017, Exhibit A, Insurance and Bonds, contemporaneously with this Agreement. AIA Document A201®–2017, General Conditions of the Contract for Construction, is adopted in this document by reference. Do not use with other general conditions unless this document is modified.

Init.
/
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**TABLE OF ARTICLES**


1 THE CONTRACT DOCUMENTS

2 THE WORK OF THIS CONTRACT

3 DATE OF COMMENCEMENT AND SUBSTANTIAL COMPLETION

4 CONTRACT SUM

5 PAYMENTS

6 DISPUTE RESOLUTION

7 TERMINATION OR SUSPENSION

8 MISCELLANEOUS PROVISIONS

9 ENUMERATION OF CONTRACT DOCUMENTS

EXHIBIT A INSURANCE AND BONDS


## ARTICLE 1 THE CONTRACT DOCUMENTS

The Contract Documents consist of this Agreement, Conditions of the Contract (General, Supplementary, and other Conditions), Drawings, Specifications, Addenda issued prior to execution of this Agreement, other documents listed in this Agreement, and Modifications issued after execution of this Agreement, all of which form the Contract, and are as fully a part of the Contract as if attached to this Agreement or repeated herein. The Contract represents the entire and integrated agreement between the parties hereto and supersedes prior negotiations, representations, or agreements, either written or oral. An enumeration of the Contract Documents, other than a Modification, appears in Article 9.

## ARTICLE 2 THE WORK OF THIS CONTRACT

The Contractor shall fully execute the Work described in the Contract Documents, except as specifically indicated in the Contract Documents to be the responsibility of others.

## ARTICLE 3 DATE OF COMMENCEMENT AND SUBSTANTIAL COMPLETION

**§ 3.1** The date of commencement of the Work shall be:
*(Check one of the following boxes.)*

[ ] The date of this Agreement.

[ X ] A date set forth in a notice to proceed issued by the Owner.

[ ] Established as follows:
*(Insert a date or a means to determine the date of commencement of the Work.)*

N/A

If a date of commencement of the Work is not selected, then the date of commencement shall be the date of this Agreement.

**§ 3.2** The Contract Time shall be measured from the date of commencement of the Work.

### § 3.3 Substantial Completion

**§ 3.3.1** Subject to adjustments of the Contract Time as provided in the Contract Documents, the Contractor shall achieve Substantial Completion of the entire Work:
*(Check one of the following boxes and complete the necessary information.)*



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[ X ] Not later than Five Hundred Fifty ( 550 ) calendar days from the date of commencement of the Work.

[ ] By the following date:

**§ 3.3.2** Subject to adjustments of the Contract Time as provided in the Contract Documents, if portions of the Work are to be completed prior to Substantial Completion of the entire Work, the Contractor shall achieve Substantial Completion of such portions by the following dates:

| Portion of Work | Substantial Completion Date |
|---|---|

**§ 3.3.3** If the Contractor fails to achieve Substantial Completion as provided in this Section 3.3, liquidated damages, if any, shall be assessed as set forth in Section 4.5.

## ARTICLE 4 CONTRACT SUM

**§ 4.1** The Owner shall pay the Contractor the Contract Sum in current funds for the Contractor's performance of the Contract. The Contract Sum shall be ($ ), subject to additions and deductions as provided in the Contract Documents.

**§ 4.2 Alternates**

**§ 4.2.1** Alternates, if any, included in the Contract Sum: **(N/A)**

**§ 4.2.2 (N/A)**

**§ 4.3** Allowances, if any, included in the Contract Sum:
*(Identify each allowance.)*

| Item | Price |
|---|---|
| **Hazardous Material Remediation** | |
| Abatement: | $65,000 - $85,000 |
| Monitoring: | $16,000 |

**§ 4.4** Unit prices, if any:
*(Identify the item and state the unit price and quantity limitations, if any, to which the unit price will be applicable.)*

| Item | Units and Limitations | Price per Unit ($0.00) |
|---|---|---|
| Removal and offsite disposal: unsuitable soil. | 10 cubic yards | |
| Replacement with suitable soil | 10 cubic yards | |

**§ 4.5** Liquidated damages, if any:
*(Insert terms and conditions for liquidated damages, if any.)*

N/A

**§ 4.6** Other:
*(Insert provisions for bonus or other incentives, if any, that might result in a change to the Contract Sum.)*

N/A


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**User Notes:** (829443192)

## ARTICLE 5 PAYMENTS

### § 5.1 Progress Payments

**§ 5.1.1** Based upon Applications for Payment submitted to the Architect by the Contractor and Certificates for Payment issued by the Architect, the Owner shall make progress payments on account of the Contract Sum to the Contractor as provided below and elsewhere in the Contract Documents.

**§ 5.1.2** The period covered by each Application for Payment shall be one calendar month ending on the last day of the month.

**§ 5.1.3** Provided that a complete and properly documented Application for Payment is received by the Architect in accordance with the Contract Documents, the Architect shall review and certify the Application for Payment as provided in AIA Document A201-2017. Following the Architect's certification, the Owner shall process the certified amount through the Town of Strasburg's normal accounts payable procedures. Payment shall be made within thirty (30) days after the Owner's receipt of the Architect's Certificate for Payment. Payment shall not require separate approval by the Town Council.
*(Federal, state or local laws may require payment within a certain period of time.)*

**§ 5.1.4** Each Application for Payment shall be based on the most recent schedule of values submitted by the Contractor in accordance with the Contract Documents. The schedule of values shall allocate the entire Contract Sum among the various portions of the Work. The schedule of values shall be prepared in such form, and supported by such data to substantiate its accuracy, as the Architect may require. This schedule of values shall be used as a basis for reviewing the Contractor's Applications for Payment.

**§ 5.1.5** Applications for Payment shall show the percentage of completion of each portion of the Work as of the end of the period covered by the Application for Payment.

**§ 5.1.6** In accordance with AIA Document A201™–2017, General Conditions of the Contract for Construction, and subject to other provisions of the Contract Documents, the amount of each progress payment shall be computed as follows:

**§ 5.1.6.1** The amount of each progress payment shall first include:

- **.1** That portion of the Contract Sum properly allocable to completed Work;
- **.2** That portion of the Contract Sum properly allocable to materials and equipment delivered and suitably stored at the site for subsequent incorporation in the completed construction, or, if approved in advance by the Owner, suitably stored off the site at a location agreed upon in writing; and
- **.3** That portion of Construction Change Directives that the Architect determines, in the Architect's professional judgment, to be reasonably justified.

**§ 5.1.6.2** The amount of each progress payment shall then be reduced by:

- **.1** The aggregate of any amounts previously paid by the Owner;
- **.2** The amount, if any, for Work that remains uncorrected and for which the Architect has previously withheld a Certificate for Payment as provided in Article 9 of AIA Document A201–2017;
- **.3** Any amount for which the Contractor does not intend to pay a Subcontractor or material supplier, unless the Work has been performed by others the Contractor intends to pay;
- **.4** For Work performed or defects discovered since the last payment application, any amount for which the Architect may withhold payment, or nullify a Certificate of Payment in whole or in part, as provided in Article 9 of AIA Document A201–2017; and
- **.5** Retainage withheld pursuant to Section 5.1.7.

### § 5.1.7 Retainage

**§ 5.1.7.1** For each progress payment made prior to Substantial Completion of the Work, the Owner may withhold the following amount, as retainage, from the payment otherwise due:
*(Insert a percentage or amount to be withheld as retainage from each Application for Payment. The amount of retainage may be limited by governing law.)*


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Five percent (5%)

**§ 5.1.7.1.1** The following items are not subject to retainage:
*(Insert any items not subject to the withholding of retainage, such as general conditions, insurance, etc.)*

N/A

**§ 5.1.7.2** Reduction or limitation of retainage, if any, shall be as follows:
*(If the retainage established in Section 5.1.7.1 is to be modified prior to Substantial Completion of the entire Work, including modifications for Substantial Completion of portions of the Work as provided in Section 3.3.2, insert provisions for such modifications.)*

Upon substantial completion, retainage shall be released.

**§ 5.1.7.3** Except as set forth in this Section 5.1.7.3, upon Substantial Completion of the Work, the Contractor may submit an Application for Payment that includes the retainage withheld from prior Applications for Payment pursuant to this Section 5.1.7. The Application for Payment submitted at Substantial Completion shall not include retainage as follows:
*(Insert any other conditions for release of retainage upon Substantial Completion.)*

N/A

**§ 5.1.8** If final completion of the Work is materially delayed through no fault of the Contractor, the Owner shall pay the Contractor any additional amounts in accordance with Article 9 of AIA Document A201–2017.

**§ 5.1.9** Except with the Owner's prior approval, the Contractor shall not make advance payments to suppliers for materials or equipment which have not been delivered and stored at the site.

### § 5.2 Final Payment

**§ 5.2.1** Final payment, constituting the entire unpaid balance of the Contract Sum, shall be made by the Owner to the Contractor when

**.1** the Contractor has fully performed the Contract except for the Contractor's responsibility to correct Work as provided in Article 12 of AIA Document A201–2017, and to satisfy other requirements, if any, which extend beyond final payment; and

**.2** a final Certificate for Payment has been issued by the Architect.

**§ 5.2.2** The Owner's final payment to the Contractor shall be made no later than 30 days after the issuance of the Architect's final Certificate for Payment, or as follows:

All punch list items must be complete.

### § 5.3 Interest

Payments due and unpaid under the Contract shall bear interest from the date payment is due at the rate stated below, or in the absence thereof, at the legal rate prevailing from time to time at the place where the Project is located.
*(Insert rate of interest agreed upon, if any.)*

1 % one percent

## ARTICLE 6 DISPUTE RESOLUTION

### § 6.1 Initial Decision Maker

The Architect will serve as the Initial Decision Maker pursuant to Article 15 of AIA Document A201–2017


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**§ 6.2 Binding Dispute Resolution**

For any Claim subject to, but not resolved by, mediation pursuant to Article 15 of AIA Document A201–2017, the method of binding dispute resolution shall be as follows:
*(Check the appropriate box.)*

[ ] Arbitration pursuant to Section 15.4 of AIA Document A201–2017

[ X ] Litigation in a court of competent jurisdiction

[ ] Other *(Specify)*

N/A

If the Owner and Contractor do not select a method of binding dispute resolution, or do not subsequently agree in writing to a binding dispute resolution method other than litigation, Claims will be resolved by litigation in a court of competent jurisdiction.

## ARTICLE 7 TERMINATION OR SUSPENSION

**§ 7.1** The Contract may be terminated by the Owner or the Contractor as provided in Article 14 of AIA Document A201–2017.

**§ 7.1.1** If the Contract is terminated for the Owner's convenience in accordance with Article 14 of AIA Document A201–2017, then the Owner shall pay the Contractor a termination fee as follows:
*(Insert the amount of, or method for determining, the fee, if any, payable to the Contractor following a termination for the Owner's convenience.)*

N/A

**§ 7.2** The Work may be suspended by the Owner as provided in Article 14 of AIA Document A201–2017.

## ARTICLE 8 MISCELLANEOUS PROVISIONS

**§ 8.1** Where reference is made in this Agreement to a provision of AIA Document A201–2017 or another Contract Document, the reference refers to that provision as amended or supplemented by other provisions of the Contract Documents.

**§ 8.2** The Owner's representative:
*(Name, address, email address, and other information)*

Chief Wayne Sager, Strasburg Police Department
174 E King St., Strasburg, VA 22657
P: 540.465.5230

Mellanie Shipe, Town Manager
174 E King St., Strasburg, VA 22657
P: 540.465.9197

**§ 8.3** The Contractor's representative:
*(Name, address, email address, and other information)*


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**§ 8.4** Neither the Owner's nor the Contractor's representative shall be changed without ten days' prior notice to the other party.

**§ 8.5 Insurance and Bonds**

**§ 8.5.1** The Owner and the Contractor shall purchase and maintain insurance as set forth in AIA Document A101™–2017, Standard Form of Agreement Between Owner and Contractor where the basis of payment is a Stipulated Sum, Exhibit A, Insurance and Bonds, and elsewhere in the Contract Documents.

**§ 8.5.2** The Contractor shall provide bonds as set forth in AIA Document A101™–2017 Exhibit A, and elsewhere in the Contract Documents.

**§ 8.6** Notice in electronic format, pursuant to Article 1 of AIA Document A201–2017, may be given in accordance with a building information modeling exhibit, if completed, or as otherwise set forth below:
*(If other than in accordance with a building information modeling exhibit, insert requirements for delivering notice in electronic format such as name, title, and email address of the recipient and whether and how the system will be required to generate a read receipt for the transmission.)*

Roderick Williams, Senior Project Manager, rwilliams@dewberry.com

**§ 8.7** Other provisions:

N/A

**ARTICLE 9 ENUMERATION OF CONTRACT DOCUMENTS**

**§ 9.1** This Agreement is comprised of the following documents:

**.1** AIA Document A101™–2017, Standard Form of Agreement Between Owner and Contractor

**.2** AIA Document A101™–2017, Exhibit A, Insurance and Bonds

**.3** AIA Document A201™–2017, General Conditions of the Contract for Construction

**.4**

*(Paragraphs deleted)*

Drawings

| Number | Title | Date |
|---|---|---|
| Sheet G-001 Index | Contract Drawing Index | 8/26/2026 |

**.6** Specifications

| Section | Title | Date | Pages |
|---|---|---|---|
| Project Specification Manual | Table of Contents | 8/26/2026 | 6 |

**.7** Addenda, if any:

| Number | Date | Pages |
|---|---|---|
| Addendum One | | |

Portions of Addenda relating to bidding or proposal requirements are not part of the Contract Documents unless the bidding or proposal requirements are also enumerated in this Article 9.

**.8** Other Exhibits:
*(Check all boxes that apply and include appropriate information identifying the exhibit where required.)*

[ ] AIA Document E204™–2017, Sustainable Projects Exhibit, dated as indicated below:
*(Insert the date of the E204-2017 incorporated into this Agreement.)*

N/A


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[ ] The Sustainability Plan: N/A

[ ] Supplementary and other Conditions of the Contract: N/A

**.9** Other documents, if any, listed below:
*(List here any additional documents that are intended to form part of the Contract Documents. AIA Document A201™–2017 provides that the advertisement or invitation to bid, Instructions to Bidders, sample forms, the Contractor's bid or proposal, portions of Addenda relating to bidding or proposal requirements, and other information furnished by the Owner in anticipation of receiving bids or proposals, are not part of the Contract Documents unless enumerated in this Agreement. Any such documents should be listed here only if intended to be part of the Contract Documents.)*

N/A

This Agreement entered into as of the day and year first written above.

| | |
|---|---|
| **OWNER** *(Signature)* | **CONTRACTOR** *(Signature)* |
| Mellanie Shipe, Town Manager | |
| *(Printed name and title)* | *(Printed name and title)* |


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# SECTION 01 1000 - SUMMARY

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Project information.
2. Work covered by Contract Documents.
3. Work performed by Owner.
4. Owner-furnished/Contractor-installed (OFCI) products.
5. Owner-furnished/Owner-installed (OFOI) products.
6. Contractor-furnished/Owner-installed (CFOI) products.
7. Contractor's use of site and premises.
8. Work restrictions.
9. Specification and Drawing conventions.

B. Related Requirements:

1. Section 01 5000 "Temporary Facilities and Controls" for limitations and procedures governing temporary use of Owner's facilities.
2. Section 01 7300 "Execution" for coordination of Owner-installed products.

### 1.3 DEFINITIONS

A. Work Package: A group of specifications, drawings, and schedules prepared by the design team to describe a portion of the Project Work for pricing, permitting, and construction.

### 1.4 PROJECT INFORMATION

A. Project Identification: Town of Strasburg, Strasburg Police Station.

1. Project Location: 115 West King Street, Strasburg, VA 22657.

B. Owner: Town of Strasburg, VA

1. Owner's Representatives:

a. Mellanie Shipe, Town Manager
b. Chief Wayne Sager, Police Chief

C. Architect: Dewberry Architects Inc. (DAI).

D. Architect's Consultants: Architect has retained the following design professionals, who have prepared designated portions of the Contract Documents:

1. Dewberry Engineers Inc. (DEI)
2. Froehling & Robertson (F&R)

E. Web-Based Project Software: Project software will be used for purposes of managing communication and documents during the construction stage.

1. See Section 01 3100 "Project Management and Coordination." for requirements for using web-based Project software.

## 1.5 WORK COVERED BY CONTRACT DOCUMENTS

A. The Work of Project is defined by the Contract Documents and includes, but is not limited to, the following:

1. Design and construction for the renovation and repurpose of the 115 W. King Street property located in Strasburg, Virginia. The property is 0.546 acres with an existing 6,440 square foot, two-story building that was built in 1955. The existing building functioned as a bank. The Town has purchased the property with the intention of renovating and repurposing the site for the Strasburg Police Department.

B. Type of Contract:

1. Project will be constructed under a single prime contract.

## 1.6 WORK PERFORMED BY OWNER

A. Cooperate fully with Owner, so work may be carried out smoothly, without interfering with or delaying Work under this Contract or work by Owner. Coordinate the Work of this Contract with work performed by Owner.

B. Subsequent Work: Owner will perform the following additional work at site after Substantial Completion. Completion of that work will depend on successful completion of preparatory Work under this Contract.

## 1.7 OWNER-FURNISHED/CONTRACTOR-INSTALLED (OFCI) PRODUCTS

A. Owner's Responsibilities: Owner will furnish products indicated and perform the following, as applicable:

1. Provide to Contractor Owner-reviewed Product Data, Shop Drawings, and Samples.
2. Provide for delivery of Owner-furnished products to Project site.
3. Upon delivery, inspect, with Contractor present, delivered items.

    a. If Owner-furnished products are damaged, defective, or missing, arrange for replacement.

4. Obtain manufacturer's inspections, service, and warranties.
5. Inform Contractor of earliest available delivery date for Owner-furnished products.

B. Contractor's Responsibilities: The Work includes the following, as applicable:

1. Designate delivery dates of Owner-furnished products in Contractor's construction schedule, utilizing Owner-furnished earliest available delivery dates.
2. Review Owner-reviewed Product Data, Shop Drawings, and Samples, noting discrepancies and other issues in providing for Owner-furnished products in the Work.
3. Receive, unload, handle, store, protect, and install Owner-furnished products.
4. Make building services connections for Owner-furnished products.
5. Protect Owner-furnished products from damage during storage, handling, and installation and prior to Substantial Completion.
6. Repair or replace Owner-furnished products damaged following receipt.

C. Owner-Furnished/Contractor-Installed (OFCI) Products:

1. As indicated in contract drawings.

1.8 OWNER-FURNISHED/OWNER-INSTALLED (OFOI) PRODUCTS

A. The Owner will furnish and install products indicated.

B. Owner-Furnished/Owner-Installed (OFOI) Products:

1. As indicated in contract drawings.

1.9 CONTRACTOR-FURNISHED/OWNER-INSTALLED (CFOI) PRODUCTS

A. Contractor shall furnish products indicated. The Work includes unloading, handling, storing, and protecting Contractor-furnished products as directed and turning them over to Owner at Project closeout.

B. Contractor-Furnished/Owner-Installed (CFOI) Products:

1. As indicated in contract documents.

1.10 CONTRACTOR'S USE OF SITE AND PREMISES

A. Unrestricted Use of Site: Contractor shall have full use of Project site for construction operations during construction period. Contractor's use of Project site is limited only by Owner's right to perform work or to retain other contractors on portions of Project.

B. Limits on Use of Site: Limit use of Project site to areas within the Contract limits indicated. Do not disturb portions of Project site beyond areas in which the Work is indicated.

1. Driveways, Walkways and Entrances: Keep driveways, parking areas, loading areas, and entrances serving premises clear and available to Owner, Owner's employees, and emergency vehicles at all times. Do not use these areas for parking or for storage of materials.

C. Condition of Existing Grounds: Maintain portions of existing grounds, landscaping, and hardscaping affected by construction operations throughout construction period. Repair damage caused by construction operations.

1.11 WORK RESTRICTIONS

A. Comply with restrictions on construction operations.

1. Comply with limitations on use of public streets, work on public streets, rights of way, and other requirements of authorities having jurisdiction.

B. On-Site Work Hours: Limit work to between 7:00 a.m. to 7:00 p.m., Monday through Friday, unless otherwise indicated. Work hours may be modified to meet Project requirements if approved by Owner and authorities having jurisdiction.

1. Weekend Hours: 8:00 a.m. to 5:00 p.m.
2. Early Morning Hours: Coordinate with Owner.
3. Work in Existing Buildings: Coordinate with Owner.
4. Hours for Utility Shutdowns: Coordinate with Owner.

C. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging for temporary utility services according to requirements indicated:

1. Notify Architect, Construction Manager and Owner not less than two (2) days in advance of proposed utility interruptions.
2. Obtain Owner's written permission before proceeding with utility interruptions.

D. Smoking and Controlled Substance Restrictions: Use of tobacco products, alcoholic beverages, and other controlled substances on Owner's property is not permitted.

E. Employee Screening: Comply with Owner's requirements for background screening of Contractor personnel working on Project site as needed.

1. Maintain list of approved screened personnel with Owner's representative.

1.12 SPECIFICATION AND DRAWING CONVENTIONS

A. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:

1. Imperative mood and streamlined language are generally used in the Specifications. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.
2. Specification requirements are to be performed by Contractor unless specifically stated otherwise.

B. Division 00 Contracting Requirements: General provisions of the Contract, including General and Supplementary Conditions, apply to all Sections of the Specifications.

C. Division 01 General Requirements: Requirements of Sections in Division 01 apply to the Work of all Sections in the Specifications.

D. Drawing Coordination: Requirements for materials and products identified on Drawings are described in detail in the Specifications. One or more of the following are used on Drawings to identify materials and products:

1. Terminology: Materials and products are identified by the typical generic terms used in the individual Specifications Sections.
2. Abbreviations: Materials and products are identified by abbreviations scheduled on Drawings and published as part of the U.S. National CAD Standard.
3. Keynoting: Materials and products are identified by reference keynotes referencing Specification Section numbers found in this Project Manual.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 1000

# SECTION 01 2500 - SUBSTITUTION PROCEDURES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes administrative and procedural requirements for post bid substitutions.

B. Related Requirements:

1. Document 00 2600 "Procurement Substitution Procedures" for requirements for substitution requests prior to award of Contract.
2. Section 01 6000 "Product Requirements" for requirements for submitting comparable product submittals for products by listed manufacturers.

### 1.3 DEFINITIONS

A. Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents.

1. Substitutions for Cause: Changes proposed by Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty terms.

### 1.4 ACTION SUBMITTALS

A. Substitution Requests: Submit documentation identifying product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.

1. Substitution Request Form: Use form CSI 13.1A or equivalent form acceptable to Architect.
2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:

a. Statement indicating why specified product or fabrication, or installation method cannot be provided, if applicable.
b. Coordination of information, including a list of changes or revisions needed to other parts of the Work and to construction performed by Owner and separate contractors that will be necessary to accommodate proposed substitution.
c. Detailed comparison of significant qualities of proposed substitutions with those of the Work specified. Include annotated copy of applicable Specification Section. Significant qualities may include attributes, such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific

features and requirements indicated. Indicate deviations, if any, from the Work specified.

d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
e. Samples, where applicable or requested.
f. Certificates and qualification data, where applicable or requested.
g. Material test reports from a qualified testing agency, indicating and interpreting test results for compliance with requirements indicated.
h. Research reports evidencing compliance with building code in effect for Project, from ICC-ES.
i. Detailed comparison of Contractor's construction schedule using proposed substitutions with products specified for the Work, including effect on the overall Contract Time. If specified product or method of construction cannot be provided within the Contract Time, include letter from manufacturer, on manufacturer's letterhead, stating date of receipt of purchase order, lack of availability, or delays in delivery.
j. Cost information, including a proposal of change, if any, in the Contract Sum.
k. Contractor's certification that proposed substitution complies with requirements in the Contract Documents, except as indicated in substitution request, is compatible with related materials and is appropriate for applications indicated.
l. Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of failure of proposed substitution to produce indicated results.

3. Architect's Action: If necessary, Architect will request additional information or documentation for evaluation within seven (7) days of receipt of a request for substitution. Architect will notify Contractor through Construction Manager of acceptance or rejection of proposed substitution within fifteen (15) days of receipt of request, or seven (7) days of receipt of additional information or documentation, whichever is later.

a. Forms of Acceptance: Change Order, Construction Change Directive, or Architect's Supplemental Instructions for minor changes in the Work.
b. Use product specified if Architect does not issue a decision on use of a proposed substitution within time allocated.

## 1.5 QUALITY ASSURANCE

A. Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturers.

## 1.6 PROCEDURES

A. Coordination: Revise or adjust affected work as necessary to integrate work of the approved substitutions.

## 1.7 SUBSTITUTIONS

A. Substitutions for Cause: Submit requests for substitution immediately on discovery of need for change, but not later than fifteen (15) days prior to time required for preparation and review of related submittals.

1. Conditions: Architect will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:

    a. Requested substitution is consistent with the Contract Documents and will produce indicated results.
    b. Substitution request is fully documented and properly submitted.
    c. Requested substitution will not adversely affect Contractor's construction schedule.
    d. Requested substitution has received necessary approvals of authorities having jurisdiction.
    e. Requested substitution is compatible with other portions of the Work.
    f. Requested substitution has been coordinated with other portions of the Work.
    g. Requested substitution provides specified warranty.
    h. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

B. Substitutions for Convenience: Not allowed unless otherwise indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 2500

# SECTION 01 2600 - CONTRACT MODIFICATION PROCEDURES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes administrative and procedural requirements for handling and processing Contract modifications.

B. Related Requirements:

1. Section 01 2500 "Substitution Procedures" for administrative procedures for handling requests for substitutions made after the Contract award.
2. Section 01 3100 "Project Management and Coordination" for requirements for forms for contract modifications provided as part of web-based Project management software.

### 1.3 MINOR CHANGES IN THE WORK

A. Architect will issue through Construction Manager supplemental instructions authorizing minor changes in the Work, not involving adjustment to the Contract Sum or the Contract Time, on AIA Document G710 or equivalent document.

### 1.4 PROPOSAL REQUESTS

A. Owner-Initiated Proposal Requests: Architect will issue a detailed description of proposed changes in the Work that may require adjustment to the Contract Sum or the Contract Time. If necessary, the description will include supplemental or revised Drawings and Specifications.

1. Work Change Proposal Requests issued by Architect are not instructions either to stop work in progress or to execute the proposed change.
2. Within time specified in Proposal Request or twenty (20) days, when not otherwise specified, after receipt of Proposal Request, submit a quotation estimating cost adjustments to the Contract Sum and the Contract Time necessary to execute the change.

    a. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
    b. Indicate applicable taxes, delivery charges, equipment rental, and amounts of trade discounts.
    c. Include costs of labor and supervision directly attributable to the change.
    d. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and

finish times, and activity relationship. Use available total float before requesting an extension of the Contract Time.

e. Quotation Form: Use forms acceptable to Architect.

B. Contractor-Initiated Proposals: If latent or changed conditions require modifications to the Contract, Contractor may initiate a claim by submitting a request for a change to Architect and Construction Manager/Owner.

1. Include a statement outlining reasons for the change and the effect of the change on the Work. Provide a complete description of the proposed change. Indicate the effect of the proposed change on the Contract Sum and the Contract Time.
2. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
3. Indicate applicable taxes, delivery charges, equipment rental, and amounts of trade discounts.
4. Include costs of labor and supervision directly attributable to the change.
5. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and finish times, and activity relationship. Use available total float before requesting an extension of the Contract Time.
6. Comply with requirements in Section 01 2500 "Substitution Procedures" if the proposed change requires substitution of one product or system for product or system specified.
7. Proposal Request Form: Use form acceptable to Architect.

1.5 CHANGE ORDER PROCEDURES

A. On Owner's approval of a Work Change Proposal Request, Architect will issue a Change Order for signatures of Owner and Contractor on AIA Document G701 or equivalent document.

1.6 CONSTRUCTION CHANGE DIRECTIVE

A. Construction Change Directive: Architect may issue a Construction Change Directive on AIA Document G714 or equivalent document. Construction Change Directive instructs Contractor to proceed with a change in the Work, for subsequent inclusion in a Change Order.

1. Construction Change Directive contains a complete description of change in the Work. It also designates method to be followed to determine change in the Contract Sum or the Contract Time.

B. Documentation: Maintain detailed records on a time and material basis of work required by the Construction Change Directive.

1. After completion of change, submit an itemized account and supporting data necessary to substantiate cost and time adjustments to the Contract.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 2600

# SECTION 01 2900 - PAYMENT PROCEDURES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes administrative and procedural requirements necessary to prepare and process Applications for Payment.

B. Related Requirements:

1. Section 01 2600 "Contract Modification Procedures" for administrative procedures for handling changes to the Contract.
2. Section 01 3200 "Construction Progress Documentation" for administrative requirements governing the preparation and submittal of the Contractor's construction schedule.
3. Section 00 0013 Owner Contractor Agreement A101-2017, 5.1.7. Progress payments will be subject to 5% retainage in accordance with the Contract Documents.

### 1.3 SCHEDULE OF VALUES

A. Coordination: Coordinate preparation of the schedule of values with preparation of Contractor's construction schedule.

1. Coordinate line items in the schedule of values with items required to be indicated as separate activities in Contractor's construction schedule.
2. Submit the schedule of values to Architect at earliest possible date, but no later than seven (7) days before the date scheduled for submittal of initial Applications for Payment.
3. Subschedules for Separate Elements of Work: Where the Contractor's construction schedule defines separate elements of the Work, provide subschedules showing values coordinated with each element.

B. Format and Content: Use Project Manual table of contents as a guide to establish line items for the schedule of values. Provide at least one line item for each Specification Section.

1. Identification: Include the following Project identification on the schedule of values:

a. Project name and location.
b. Owner's name.
c. Owner's Project number.
d. Name of Architect.
e. Architect's Project number.
f. Contractor's name and address.
g. Date of submittal.

2. Arrange schedule of values consistent with format of AIA Document G703.

    a. Specification Section in numerical order.
    b. Description of the Work.
    c. Scheduled Value.
    d. Each item is to be broken out by:

        1) Labor.
        2) Materials.

    e. Sections that are definable features of work shall also include line items for:

        1) Submittals.
        2) Shop Drawings.
        3) Fabrication and Delivery.

3. Provide a breakdown of the Contract Sum in enough detail to facilitate continued evaluation of Applications for Payment and progress reports. Provide multiple line items for principal subcontract amounts in excess of five percent of the Contract Sum.
4. Provide a separate line item in the schedule of values for each part of the Work where Applications for Payment may include materials or equipment purchased or fabricated and stored, but not yet installed.

    a. Differentiate between items stored on-site and items stored off-site.

5. Overhead Costs, Separate Line Items: Show cost of temporary facilities and other major cost items that are not direct cost of actual work-in-place as separate line items.
6. Temporary Facilities: Show cost of temporary facilities and other major cost items that are not direct cost of actual work-in-place as separate line items.
7. Closeout Costs. Include separate line items under Contractor and principal subcontracts for Project closeout requirements in an amount totaling five percent of the Contract Sum and subcontract amount.
8. Schedule of Values Revisions: Revise the schedule of values when Change Orders or Construction Change Directives result in a change in the Contract Sum. Include at least one separate line item for each Change Order and Construction Change Directive.

## 1.4 APPLICATIONS FOR PAYMENT

A. Each Application for Payment following the initial Application for Payment shall be consistent with previous applications and payments, as certified by Architect and paid for by Owner.

B. Payment Application Times: The period covered by each Application for Payment shall be as indicated in the Owner/Contractor Agreement. Following review and certification by the Architect, the Owner shall process the certified Application for Payment through the Town of Strasburg's normal accounts payable procedures in accordance with Section 5.1.3 of the Owner/Contractor Agreement.

    1. Submit draft copy of Application for Payment seven (7) days prior to due date for review by Architect.

C. Application for Payment Forms: Use AIA Document G702 and AIA Document G703 as form for Applications for Payment.

D. Application Preparation: Complete every entry on form. Notarize and execute by a person authorized to sign legal documents on behalf of Contractor. Architect will return incomplete applications without action.

1. Entries shall match data on the schedule of values and Contractor's construction schedule. Use updated schedules if revisions were made.
2. Include amounts for work completed following previous Application for Payment, whether or not payment has been received. Include only amounts for work completed at time of Application for Payment.
3. Include amounts of Change Orders and Construction Change Directives issued before last day of construction period covered by application.
4. Indicate separate amounts for work being carried out under Owner-requested project acceleration.

E. Stored Materials: Include in Application for Payment amounts applied for materials or equipment purchased or fabricated and stored, but not yet installed. Differentiate between items stored on-site and items stored off-site.

1. Provide certificate of insurance, evidence of transfer of title to Owner, and consent of surety to payment for stored materials.
2. Provide supporting documentation that verifies amount requested, such as paid invoices. Match amount requested with amounts indicated on documentation; do not include overhead and profit on stored materials.
3. Provide summary documentation for stored materials indicating the following:

    a. Value of materials previously stored and remaining stored as of date of previous Applications for Payment.
    b. Value of previously stored materials put in place after date of previous Application for Payment and on or before date of current Application for Payment.
    c. Value of materials stored since date of previous Application for Payment and remaining stored as of date of current Application for Payment.

F. Transmittal: Electronically submit one (1) signed and notarized Application for Payment to Architect by a method approved by the Architect and Owner. Include required lien waivers and supporting documentation with the submission.

G. Waivers of Mechanic's Lien: With each Application for Payment, submit waivers of mechanic's lien from entities lawfully entitled to file a mechanic's lien arising out of the Contract and related to the Work covered by the payment.

1. Submit partial waivers on each item for amount requested in previous application, after deduction for retainage, on each item.
2. When an application shows completion of an item, submit conditional final or full waivers.
3. Owner reserves the right to designate which entities involved in the Work must submit waivers.
4. Submit final Application for Payment with or preceded by conditional final waivers from every entity involved with performance of the Work covered by the application who is lawfully entitled to a lien.
5. Waiver Forms: Submit executed waivers of lien on forms acceptable to Owner.

H. Initial Application for Payment: Administrative actions and submittals that must precede or coincide with submittal of first Application for Payment include the following:

1. List of subcontractors.

2. Schedule of values.
3. Contractor's construction schedule (preliminary if not final).
4. Products list (preliminary if not final).
5. Sustainable design action plans, including preliminary project materials cost data.
6. Schedule of unit prices.
7. Submittal schedule (preliminary if not final).
8. List of Contractor's staff assignments.
9. List of Contractor's principal consultants.
10. Copies of building permits.
11. Copies of authorizations and licenses from authorities having jurisdiction for performance of the Work.
12. Initial progress report.
13. Report of preconstruction conference.

I. Application for Payment at Substantial Completion: After Architect issues the Certificate of Substantial Completion, submit an Application for Payment showing 100 percent completion for portion of the Work claimed as substantially complete.

1. Include documentation supporting claim that the Work is substantially complete and a statement showing an accounting of changes to the Contract Sum.

a. Complete administrative actions, submittals, and Work preceding this application, as described in Section 01 7700 "Closeout Procedures."

2. This application shall reflect Certificate(s) of Substantial Completion issued previously for Owner occupancy of designated portions of the Work.

J. Final Payment Application: After completing Project closeout requirements, submit final Application for Payment with releases and supporting documentation not previously submitted and accepted, including, but not limited, to the following:

1. Evidence of completion of Project closeout requirements.
2. Certification of completion of final punch list items.
3. Insurance certificates for products and completed operations where required and proof that taxes, fees, and similar obligations were paid.
4. Updated final statement, accounting for final changes to the Contract Sum.
5. AIA Document G706.
6. AIA Document G706A.
7. AIA Document G707.
8. Evidence that claims have been settled.
9. Final meter readings for utilities, a measured record of stored fuel, and similar data as of date of Substantial Completion or when Owner took possession of and assumed responsibility for corresponding elements of the Work.
10. Final liquidated damages settlement statement.
11. Proof that taxes, fees, and similar obligations are paid.
12. Waivers and releases.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 2900

# SECTION 01 3100 - PROJECT MANAGEMENT AND COORDINATION

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes administrative provisions for coordinating construction operations on Project, including, but not limited to, the following:

1. General coordination procedures.
2. Coordination drawings.
3. RFIs.
4. Digital project management procedures.
5. Web-based Project management software package.
6. Project meetings.

B. Related Requirements:

1. Section 01 3200 "Construction Progress Documentation" for preparing and submitting Contractor's construction schedule.
2. Section 01 7300 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
3. Section 01 7700 "Closeout Procedures" for coordinating closeout of the Contract.

### 1.3 DEFINITIONS

A. BIM: Building Information Modeling.

B. RFI: Request for Information. Request from Owner, Construction Manager, Architect, or Contractor seeking information required by or clarifications of the Contract Documents.

### 1.4 INFORMATIONAL SUBMITTALS

A. Subcontract List: Prepare a written summary identifying individuals or firms proposed for each portion of the Work, including those who are to furnish products or equipment fabricated to a special design. Include the following information in tabular form:

1. Name, address, telephone number, and email address of entity performing subcontract or supplying products.
2. Number and title of related Specification Section(s) covered by subcontract.
3. Drawing number and detail references, as appropriate, covered by subcontract.

B. Key Personnel Names: Within fifteen (15) days of starting construction operations, submit a list of key personnel assignments, including superintendent and other personnel in attendance at Project site. Identify individuals and their duties and responsibilities, list addresses, cellular telephone numbers, and e-mail addresses. Provide names, addresses, and telephone numbers of individuals assigned as alternates in the absence of individuals assigned to Project.

1. Post copies of list in Project meeting room, in temporary field office, in web-based Project software directory, and post in prominent location on-site. Keep list current at all times.

## 1.5 GENERAL COORDINATION PROCEDURES

A. Coordination: Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations included in different Sections that depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain the best results, where installation of one part of the Work depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair.
3. Make adequate provisions to accommodate items scheduled for later installation.

B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

## 1.6 COORDINATION DRAWINGS

A. Coordination Drawings, General: Prepare coordination drawings according to requirements in individual Sections, and additionally where installation is not completely indicated on Shop Drawings, where limited space availability necessitates coordination, or if coordination is required to facilitate integration of products and materials fabricated or installed by more than one entity.

1. Content: Project-specific information, drawn accurately to a scale large enough to indicate and resolve conflicts. Do not base coordination drawings on standard printed data. Include the following information, as applicable:

a. Use applicable Drawings as a basis for preparation of coordination drawings. Prepare sections, elevations, and details as needed to describe relationship of various systems and components.
b. Coordinate the addition of trade-specific information to coordination drawings in a sequence that best provides for coordination of the information and resolution of conflicts between installed components before submitting for review.
c. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems.
d. Indicate space requirements for routine maintenance and for anticipated replacement of components during the life of the installation.
e. Show location and size of access doors required for access to concealed dampers, valves, and other controls.
f. Indicate required installation sequences.

g. Indicate dimensions shown on Drawings. Specifically note dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternative sketches to Architect indicating proposed resolution of such conflicts. Minor dimension changes and difficult installations will not be considered changes to the Contract.

B. Coordination Drawing Organization: Organize coordination drawings as follows:

1. Floor Plans and Reflected Ceiling Plans: Show architectural and structural elements, and mechanical, plumbing, fire-protection, fire-alarm, and electrical Work. Show locations of visible ceiling-mounted devices relative to acoustical ceiling grid. Supplement plan drawings with section drawings where required to adequately represent the Work.
2. Plenum Space: Indicate subframing for support of ceiling, raised access floor, and wall systems, mechanical and electrical equipment, and related Work. Locate components within plenums to accommodate layout of light fixtures and other components indicated on Drawings. Indicate areas of conflict between light fixtures and other components.
3. Mechanical Rooms: Provide coordination drawings for mechanical rooms, showing plans and elevations of mechanical, plumbing, fire-protection, fire-alarm, and electrical equipment.
4. Structural Penetrations: Indicate penetrations and openings required for all disciplines.
5. Slab Edge and Embedded Items: Indicate slab edge locations and sizes and locations of embedded items for metal fabrications, sleeves, anchor bolts, bearing plates, angles, door floor closers, slab depressions for floor finishes, curbs and housekeeping pads, and similar items.
6. Mechanical and Plumbing Work: Show the following:

    a. Sizes and bottom elevations of ductwork, piping, and conduit runs, including insulation, bracing, flanges, and support systems.
    b. Dimensions of major components, such as dampers, valves, diffusers, access doors, cleanouts and electrical distribution equipment.
    c. Fire-rated enclosures around ductwork.

7. Electrical Work: Show the following:

    a. Runs of vertical and horizontal conduit 1-1/4 inches (32 mm) in diameter and larger.
    b. Light fixture, exit light, emergency battery pack, smoke detector, and other fire-alarm locations.
    c. Panel board, switchboard, switchgear, transformer, busway, generator, and motor-control center locations.
    d. Location of pull boxes and junction boxes, dimensioned from column center lines.

8. Fire-Protection System: Show the following:

    a. Locations of standpipes, mains piping, branch lines, pipe drops, and sprinkler heads.

9. Review: Architect and Engineer will review coordination drawings to confirm that, in general, the Work is being coordinated, but not for the details of the coordination, which are Contractor's responsibility. If Architect determines that coordination drawings are not being prepared in sufficient scope or detail, or are otherwise deficient, Architect will so inform Contractor, who shall make suitable modifications and resubmit.

C. Coordination Digital Data Files: Prepare coordination digital data files according to the following requirements:

1. File Submittal Format: Submit or post coordination drawing files using PDF format.

2. Architect will furnish Contractor one set of digital data files of Drawings for use in preparing coordination digital data files.

    a. Architect makes no representations as to the accuracy or completeness of digital data files as they relate to Drawings.
    b. Contractor shall execute a data licensing agreement in the form of Agreement form acceptable to Owner and Architect.

## 1.7 REQUEST FOR INFORMATION (RFI)

A. General: Immediately on discovery of the need for additional information, clarification, or interpretation of the Contract Documents, Contractor shall prepare and submit an RFI in the form specified.

1. Architect will return without response those RFIs submitted to Architect by other entities controlled by Contractor.
2. Coordinate and submit RFIs in a prompt manner to avoid delays in Contractor's work or work of subcontractors.

B. Content of the RFI: Include a detailed, legible description of item needing information or interpretation and the following:

1. Project name.
2. Owner name.
3. Owner's Project number.
4. Name of Architect and Construction Manager.
5. Architect's Project number.
6. Date.
7. Name of Contractor.
8. RFI number, numbered sequentially.
9. RFI subject.
10. Specification Section number and title and related paragraphs, as appropriate.
11. Drawing number and detail references, as appropriate.
12. Field dimensions and conditions, as appropriate.
13. Contractor's suggested resolution. If Contractor's suggested resolution impacts the Contract Time or the Contract Sum, Contractor shall state impact in the RFI.
14. Contractor's signature.
15. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.

    a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

C. RFI Forms: Software-generated form with substantially the same content as indicated above, acceptable to Architect.

1. Attachments shall be electronic files in PDF format.

D. Architect's Action: Architect will review each RFI, determine action required, and respond. Allow seven (7) calendar days for Architect's response for each RFI. RFIs received by Architect after 4:00 p.m. will be considered as received the following working day.

1. The following Contractor-generated RFIs will be returned without action:

    a. Requests for approval of submittals.
    b. Requests for approval of substitutions.
    c. Requests for approval of Contractor's means and methods.
    d. Requests for coordination information already indicated in the Contract Documents.
    e. Requests for adjustments in the Contract Time or the Contract Sum.
    f. Requests for interpretation of Architect's actions on submittals.
    g. Incomplete RFIs or inaccurately prepared RFIs.

2. Architect's action may include a request for additional information, in which case Architect's time for response will date from time of receipt by Architect of additional information.
3. Architect's action on RFIs that may result in a change to the Contract Time or the Contract Sum may be eligible for Contractor to submit Change Proposal according to Section 01 2600 "Contract Modification Procedures."

    a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify Architect and Construction Manager in writing within five (5) days of receipt of the RFI response.

E. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log bi-weekly. Use software log that is part of web-based Project management software. Include the following:

1. Project name.
2. Name and address of Contractor.
3. Name and address of Architect and Construction Manager.
4. RFI number, including RFIs that were returned without action or withdrawn.
5. RFI description.
6. Date the RFI was submitted.
7. Date Architect's response was received.
8. Identification of related Minor Change in the Work, Construction Change Directive, and Proposal Request, as appropriate.

F. On receipt of Architect's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify Architect and Construction Manager within three (3) days if Contractor disagrees with response.

## 1.8 DIGITAL PROJECT MANAGEMENT PROCEDURES

A. Architect's Data Files Not Available: Architect will not provide Architect's BIM model digital data files for Contractor's use during construction.

B. Use of Architect's Digital Data Files: Digital data files of Architect's CAD drawings will be provided by Architect for Contractor's use during construction.

1. Digital data files may be used by Contractor in preparing coordination drawings, Shop Drawings, and Project Record Drawings.
2. Architect makes no representations as to the accuracy or completeness of digital data files as they relate to Contract Drawings.
3. Contractor shall execute a data licensing agreement in the form of Agreement form acceptable to Owner and Architect.

    a. Subcontractors and other parties granted access by Contractor to Architect's digital data files shall execute a data licensing agreement in the form of Agreement acceptable to Owner and Architect.

4. The following digital data files will be furnished for each appropriate discipline:

    a. Floor plans.
    b. Reflected ceiling plans.

C. Web-Based Project Management Software Package: Provide, administer, and use web-based Project management software package for purposes of hosting and managing Project communication and documentation until Final Completion.

1. Acceptable Service Providers include the following:

    a. Autodesk Construction Cloud.
    b. Kahua.
    c. Trimble E-Builder.
    d. Procore.

2. Web-based Project management software includes, at a minimum, the following features:

    a. Compilation of Project data, including Contractor, subcontractors, Architect, Architect's consultants, Owner, and other entities involved in Project. Include names of individuals and contact information.
    b. Access control for each entity for each workflow process, to determine entity's digital rights to create, modify, view, and print documents.
    c. Document workflow planning, allowing customization of workflow between project entities.
    d. Creation, logging, tracking, and notification for Project communications required in other Specification Sections, including, but not limited to, RFIs, submittals, Minor Changes in the Work, Construction Change Directives, and Change Orders.
    e. Track status of each Project communication in real time, and log time and date when responses are provided.
    f. Procedures for handling PDFs or similar file formats, allowing markups by each entity. Provide security features to lock markups against changes once submitted.
    g. Processing and tracking of payment applications.
    h. Processing and tracking of contract modifications.
    i. Creating and distributing meeting minutes.
    j. Document management for Drawings, Specifications, and coordination drawings, including revision control.
    k. Management of construction progress photographs.
    l. Mobile device compatibility, including smartphones and tablets.

3. Provide up to six Project management software user licenses as needed for use by Owner and Architect. Provide two hours of software training for web-based Project software users as needed.

4. At completion of Project, provide digital archive in format that is readable by common desktop software applications in format acceptable to Architect. Provide data in locked format to prevent further changes.

D. PDF Document Preparation: Where PDFs are required to be submitted to Architect, prepare as follows:

1. Assemble complete submittal package into a single indexed file, incorporating submittal requirements of a single Specification Section and transmittal form with links enabling navigation to each item.
2. Name file with submittal number or other unique identifier, including revision identifier.
3. Certifications: Where digitally submitted certificates and certifications are required, provide a digital signature with digital certificate on where indicated.

## 1.9 PROJECT MEETINGS

A. General: Architect will schedule and conduct meetings and conferences at Project site unless otherwise indicated.

1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date and time of each meeting. Notify Owner and Architect of scheduled meeting dates and times a minimum of seven (7) days prior to meeting.
2. Agenda: Prepare the meeting agenda. Distribute the agenda to all invited attendees.
3. Minutes: Entity responsible for conducting meeting will record significant discussions and agreements achieved. Distribute the meeting minutes to everyone concerned, including Owner, Construction Manager, and Architect, within seven (7) days of the meeting.

B. Preconstruction Conference: Owner and Architect will schedule and conduct a preconstruction conference before starting construction, at a time convenient to Owner and Architect, but no later than fifteen (15) days after execution of the Agreement.

1. Attendees: Authorized representatives of Owner, Commissioning Authority, Construction Manager, Architect, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Discuss items of significance that could affect progress, including the following:

a. Responsibilities and personnel assignments.
b. Tentative construction schedule.
c. Phasing.
d. Critical work sequencing and long lead items.
e. Designation of key personnel and their duties.
f. Lines of communications.
g. Use of web-based Project software.
h. Procedures for processing field decisions and Change Orders.
i. Procedures for RFIs.
j. Procedures for testing and inspecting.
k. Procedures for processing Applications for Payment.
l. Distribution of the Contract Documents.
m. Submittal procedures.
n. Preparation of Record Documents.
o. Use of the premises and existing buildings.

p. Work restrictions.
q. Working hours.
r. Owner's occupancy requirements.
s. Responsibility for temporary facilities and controls.
t. Procedures for moisture and mold control.
u. Procedures for disruptions and shutdowns.
v. Construction waste management and recycling.
w. Parking availability.
x. Office, work, and storage areas.
y. Equipment deliveries and priorities.
z. First aid.
aa. Security.
bb. Progress cleaning.

3. Minutes: Entity responsible for conducting meeting will record and distribute meeting minutes.

C. Preinstallation Conferences: Contractor to schedule and conduct a preinstallation conference at Project site before each construction activity when required by other Sections and when required for coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise Architect, Construction Manager and Commissioning Authority of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:

a. Contract Documents.
b. Options.
c. Related RFIs.
d. Related Change Orders.
e. Purchases.
f. Deliveries.
g. Submittals.
h. Review of mockups.
i. Possible conflicts.
j. Compatibility requirements.
k. Time schedules.
l. Weather limitations.
m. Manufacturer's written instructions.
n. Warranty requirements.
o. Compatibility of materials.
p. Acceptability of substrates.
q. Temporary facilities and controls.
r. Space and access limitations.
s. Regulations of authorities having jurisdiction.
t. Testing and inspecting requirements.
u. Installation procedures.
v. Coordination with other work.
w. Required performance results.
x. Protection of adjacent work.
y. Protection of construction and personnel.

3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
4. Reporting: Distribute minutes of the meeting to each party present and to other parties requiring information.
5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.

D. Project Closeout Conference: Contractor to schedule and conduct a project closeout conference, at a time convenient to Owner and Architect, but no later than ninety (90) days prior to the scheduled date of Substantial Completion.

1. Conduct the conference to review requirements and responsibilities related to Project closeout.
2. Attendees: Authorized representatives of Owner, Commissioning Authority, Construction Manager, Architect, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the meeting. Participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
3. Agenda: Discuss items of significance that could affect or delay Project closeout, including the following:

    a. Preparation of Record Documents.
    b. Procedures required prior to inspection for Substantial Completion and for final inspection for acceptance.
    c. Procedures for completing and archiving web-based Project software site data files.
    d. Submittal of written warranties.
    e. Requirements for preparing operations and maintenance data.
    f. Requirements for delivery of material samples, attic stock, and spare parts.
    g. Requirements for demonstration and training.
    h. Preparation of Contractor's punch list.
    i. Procedures for processing Applications for Payment at Substantial Completion and for final payment.
    j. Submittal procedures.
    k. Owner's partial occupancy requirements.
    l. Installation of Owner's furniture, fixtures, and equipment.
    m. Responsibility for removing temporary facilities and controls.

4. Minutes: Entity conducting meeting will record and distribute meeting minutes.

E. Progress Meetings: Architect shall conduct progress meetings at biweekly intervals.

1. Coordinate dates of meetings with preparation of payment requests.
2. Attendees: In addition to representatives of Owner, Construction Manager, and Architect, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
3. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.

    a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind

schedule, in relation to Contractor's construction schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.

1) Review schedule for next period. Two-week look behind (progress) and two-week look-ahead.

b. Review present and future needs of each entity present, including the following:

1) Interface requirements.
2) Sequence of operations.
3) Status of submittals.
4) Deliveries.
5) Off-site fabrication.
6) Access.
7) Site use.
8) Temporary facilities and controls.
9) Progress cleaning.
10) Quality and work standards.
11) Status of correction of deficient items.
12) Field observations.
13) Status of RFIs.
14) Status of Proposal Requests.
15) Pending changes.
16) Status of Change Orders.
17) Pending claims and disputes.
18) Documentation of information for payment requests.

4. Minutes: Entity responsible for conducting the meeting will record and distribute the meeting minutes to each party present and to parties requiring information.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 3100

# SECTION 01 3200 - CONSTRUCTION PROGRESS DOCUMENTATION

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:

1. Contractor's Construction Schedule.
2. Construction schedule updating reports.
3. Daily construction reports.
4. Material location reports.
5. Site condition reports.
6. Unusual event reports.

B. Related Requirements:

1. Section 01 2900 "Payment Procedures" for schedule of values and requirements for use of cost-loaded schedule for Applications for Payment.
2. Section 01 4000 "Quality Requirements" for schedule of tests and inspections.

### 1.2 DEFINITIONS

A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction Project. Activities included in a construction schedule consume time and resources.

1. Critical Activity: An activity on the critical path that must start and finish on the planned early start and finish times.
2. Predecessor Activity: An activity that precedes another activity in the network.
3. Successor Activity: An activity that follows another activity in the network.

B. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine the critical path of Project and when activities can be performed.

C. Critical Path: The longest connected chain of interdependent activities through the network schedule that establishes the minimum overall Project duration and contains no float.

D. Event: The starting or ending point of an activity.

E. Float: The measure of leeway in starting and completing an activity.

1. Float time belongs to Owner.
2. Free float is the amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
3. Total float is the measure of leeway in starting or completing an activity without adversely affecting the planned Project completion date.

F. Resource Loading: The allocation of labor and equipment necessary for completing an activity as scheduled.

## 1.3 INFORMATIONAL SUBMITTALS

A. Format for Submittals: Submit required submittals in the following format:

1. Working electronic copy of schedule file.
2. PDF file.

B. Contractor's Construction Schedule: Initial schedule, of size required to display entire schedule for entire construction period.

1. Submit a working digital copy of schedule, using software indicated, and labeled to comply with requirements for submittals.

C. CPM Reports: Concurrent with CPM schedule, submit each of the following reports. Format for each activity in reports to contain activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.

1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
2. Logic Report: List of preceding and succeeding activities for each activity, sorted in ascending order by activity number and then by early start date, or actual start date if known.
3. Total Float Report: List of activities sorted in ascending order of total float.
4. Earnings Report: Compilation of Contractor's total earnings from commencement of the Work until most recent Application for Payment.

D. Construction Schedule Updating Reports: Submit with Applications for Payment.

E. Daily Construction Reports: Submit at weekly intervals.

F. Material Location Reports: Submit at monthly intervals.

G. Site Condition Reports: Submit at time of discovery of differing conditions.

H. Unusual Event Reports: Submit at time of unusual event.

I. Qualification Data: For scheduling consultant.

## 1.4 QUALITY ASSURANCE

A. Scheduling Qualifications: An experienced specialist in CPM scheduling and reporting, with capability of producing CPM reports and diagrams within 48 hours of Architect's request. Scheduler may be employed by General Contractor if qualified.

1.5 COORDINATION

A. Coordinate Contractor's Construction Schedule with the schedule of values, list of subcontracts, submittal schedule, progress reports, payment requests, and other required schedules and reports.

1. Secure time commitments for performing critical elements of the Work from entities involved.
2. Coordinate each construction activity in the network with other activities and schedule them in proper sequence.

1.6 CONTRACTOR'S CONSTRUCTION SCHEDULE

A. Computer Scheduling Software: Prepare schedules using current version of a program that has been developed specifically to manage construction schedules.

1. Use scheduling software agreed to by all parties that is compliant with Project management software package and for current Windows operating system.

B. Scheduling Consultant: Engage a scheduler to provide planning, evaluation, and reporting, using CPM scheduling.

1. In-House Option: Owner may waive requirement to retain a consultant if Contractor employs skilled personnel with experience in CPM scheduling and reporting techniques. Submit qualifications.
2. Meetings: Scheduling consultant to attend all meetings related to Project progress, alleged delays, and time impact.

C. Time Frame: Extend schedule from date established for the Notice of Award to date of Final Completion.

1. Contract completion date to not be changed by submission of a schedule that shows an early completion date, unless specifically authorized by Change Order.

D. Activities: Treat each floor or separate area as a separate numbered activity for each main element of the Work. Comply with the following:

1. Activity Duration: Define activities so no activity is longer than twenty (20) days, unless specifically allowed by Architect.
2. Temporary Facilities: Indicate start and completion dates for the following as applicable:

    a. Securing of approvals and permits required for performance of the Work.
    b. Temporary facilities.
    c. Construction of mock-ups, prototypes, and samples.
    d. Owner interfaces and furnishing of items.
    e. Interfaces with Separate Contracts.
    f. Regulatory agency approvals.
    g. Punch list.

3. Procurement Activities: Include procurement process activities for the following long lead-time items and major items, requiring a cycle of more than sixty (60) days, as separate activities in schedule. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.

a. Structural Steel and Steel Joists.
b. Roofing.
c. Firearms Range Equipment.
d. Packaged Air Handling Units.
e. Direct Expansion Packaged Rooftop Air Handling Units.
f. Building Management System.
g. Switchboards.
h. Electronic Security Systems.
i. Precast Concrete Manhole Structures.

4. Submittal Review Time: Include review and resubmittal times indicated in Section 01 3300 "Submittal Procedures" in schedule. Coordinate submittal review times in Contractor's Construction Schedule with submittal schedule.
5. Startup and Testing Time: Include no fewer than fifteen (15) days for startup and testing.
6. Commissioning Time: Include no fewer than fifteen (15) days for commissioning.
7. Substantial Completion: Indicate completion in advance of date established for Substantial Completion and allow time for Architect's and Construction Manager's administrative procedures necessary for certification of Substantial Completion.
8. Punch List and Final Completion: Include not more than thirty (30) days for completion of punch list items and Final Completion.

E. Constraints: Include constraints and work restrictions indicated in the Contract Documents and as follows in schedule and show how the sequence of the Work is affected.

1. Phasing: Arrange list of activities on schedule by phase.
2. Work under More Than One Contract: Include a separate activity for each contract.
3. Work by Owner: Include a separate activity for each portion of the Work performed by Owner.
4. Products Ordered in Advance: Include a separate activity for each product. Include delivery date indicated in Section 011000 "Summary." Delivery dates indicated stipulate the earliest possible delivery date.
5. Owner-Furnished Products: Include a separate activity for each product. Include delivery date indicated in Section 011000 "Summary." Delivery dates indicated stipulate the earliest possible delivery date.
6. Work Restrictions: Show the effect of the following items on the schedule:

a. Limitations of continued occupancies.
b. Uninterruptible services.
c. Partial occupancy before Substantial Completion.
d. Use-of-premises restrictions.
e. Seasonal variations.
f. Environmental control.

7. Work Stages: Indicate important stages of construction for each major portion of the Work, including, but not limited to, the following:

a. Subcontract awards.
b. Submittals.
c. Mockups.
d. Installation.
e. Tests and inspections.
f. Startup and placement into final use and operation.
g. Commissioning.

F. Milestones: Include milestones indicated in the Contract Documents in schedule, including, but not limited to, the Notice to Proceed, Substantial Completion, and Final Completion, and the following interim milestones:

1. Temporary enclosure and space conditioning.
2. Building dry-in for each location.

G. Upcoming Work Summary: Prepare summary report indicating activities scheduled to occur or commence prior to submittal of next schedule update. Summarize the following issues:

1. Unresolved issues.
2. Unanswered Requests for Information.
3. Rejected or unreturned submittals.
4. Notations on returned submittals.
5. Pending modifications affecting the Work and the Contract Time.

H. Contractor's Construction Schedule Updating: At monthly intervals, update schedule to reflect actual construction progress and activities. Issue schedule with draft application for payment before each regularly scheduled review meeting.

1. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.
2. Include a report with updated schedule that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.
3. As the Work progresses, indicate Final Completion percentage for each activity.

I. Recovery Schedule: When periodic update indicates the Work is fourteen (14) or more calendar days behind the current approved schedule, submit a separate recovery schedule indicating means by which Contractor intends to regain compliance with the schedule. Indicate changes to working hours, working days, crew sizes, equipment required to achieve compliance, and date by which recovery will be accomplished.

J. Distribution: Distribute copies of approved schedule to Architect, Construction Manager, Owner, separate contractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility.

1. Post copies in Project meeting rooms and temporary field offices.
2. When revisions are made, distribute updated schedules to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in performance of construction activities.

## 1.7 CPM SCHEDULE REQUIREMENTS

A. CPM Schedule: Prepare Contractor's Construction Schedule using a time-scaled CPM network analysis diagram for the Work.

1. Develop network diagram in sufficient time to submit CPM schedule, so it can be accepted for use no later than sixty (60) days after date established for the Notice of Award.

a. Failure to include any work item required for performance of this Contract must not excuse Contractor from completing all work within applicable completion dates.

2. Conduct educational workshops to train and inform key Project personnel, including subcontractors' personnel, in proper methods of providing data and using CPM schedule information.
3. Establish procedures for monitoring and updating CPM schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.
4. Use "one workday" as the unit of time for individual activities. Indicate nonworking days and holidays incorporated into the schedule to coordinate with the Contract Time.

B. CPM Schedule Preparation: Prepare a list of all activities required to complete the Work. Using the startup network diagram, prepare a skeleton network to identify probable critical paths.

1. Activities: Indicate the estimated time duration, sequence requirements, and relationship of each activity in relation to other activities. Include estimated time frames for the following activities:

a. Preparation and processing of submittals.
b. Mobilization and demobilization.
c. Purchase of materials.
d. Delivery.
e. Fabrication.
f. Utility interruptions.
g. Installation.
h. Work by Owner that may affect or be affected by Contractor's activities.
i. Testing and inspection.
j. Commissioning.
k. Punch list and Final Completion.
l. Activities occurring following Final Completion.

2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates to be consistent with Contract milestone dates.
3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the CPM schedule within the limitations of the Contract Time.
4. Format: Mark the critical path. Locate the critical path near center of network; locate paths with most float near the edges.

a. Subnetworks on separate sheets are permissible for activities clearly off the critical path.

C. Contract Modifications: For each proposed contract modification and concurrent with its submission, prepare a time-impact analysis using a network fragment to demonstrate the effect of the proposed change on the overall Project schedule.

D. Initial Issue of Schedule: Prepare initial network diagram from a sorted activity list indicating straight "early start-total float." Identify critical activities. Prepare tabulated reports showing the following:

1. Contractor or subcontractor and the Work or activity.
2. Description of activity.
3. Main events of activity.

4. Immediately preceding and succeeding activities.
5. Early and late start dates.
6. Early and late finish dates.
7. Activity duration in workdays.
8. Total float or slack time.
9. Average size of workforce.
10. Dollar value of activity (coordinated with the schedule of values).

E. Schedule Updating: Concurrent with making revisions to schedule, prepare tabulated reports showing the following:

1. Identification of activities that have changed.
2. Changes in early and late start dates.
3. Changes in early and late finish dates.
4. Changes in activity durations in workdays.
5. Changes in the critical path.
6. Changes in total float or slack time.
7. Changes in the Contract Time.

1.8 REPORTS

A. Daily Construction Reports: Prepare a daily construction report recording the following information concerning events at Project site:

1. List of subcontractors at Project site.
2. List of separate contractors at Project site.
3. Approximate count of personnel at Project site.
4. Equipment at Project site.
5. Material deliveries.
6. High and low temperatures and general weather conditions, including presence of rain or snow.
7. Testing and inspection.
8. Accidents.
9. Meetings and significant decisions.
10. Unusual events.
11. Stoppages, delays, shortages, and losses.
12. Meter readings and similar recordings.
13. Emergency procedures.
14. Orders and requests of authorities having jurisdiction.
15. Change Orders received and implemented.
16. Construction Change Directives received and implemented.
17. Services connected and disconnected.
18. Equipment or system tests and startups.
19. Partial completions and occupancies.
20. Substantial Completions authorized.

B. Material Location Reports: At monthly intervals, prepare and submit a comprehensive list of materials delivered to and stored at Project site. List to be cumulative, showing materials previously reported plus items recently delivered. Include with list a statement of progress on and delivery dates for materials or items of equipment fabricated or stored away from Project site. Indicate the following categories for stored materials:

1. Material stored prior to previous report and remaining in storage.
2. Material stored prior to previous report and since removed from storage and installed.

3. Material stored following previous report and remaining in storage.

C. Site Condition Reports: Immediately on discovery of a difference between site conditions and the Contract Documents, prepare and submit a detailed report. Submit with a Request for Information. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.

D. Unusual Event Reports: When an event of an unusual and significant nature occurs at Project site, whether or not related directly to the Work, prepare and submit a special report. List chain of events, persons participating, responses by Contractor's personnel, evaluation of results or effects, and similar pertinent information. Advise Owner in advance when these events are known or predictable.

1. Submit unusual event reports directly to Owner within one day(s) of an occurrence. Distribute copies of report to parties affected by the occurrence.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 3200

# SECTION 01 3233 - PHOTOGRAPHIC DOCUMENTATION

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes administrative and procedural requirements for the following:

1. Preconstruction photographs.
2. Concealed Work photographs.
3. Periodic construction photographs.
4. Preconstruction video recordings.

B. Related Requirements:

1. Section 01 7700 "Closeout Procedures" for submitting photographic documentation as Project Record Documents at Project closeout.
2. Section 01 7900 "Demonstration and Training" for submitting video recordings of demonstration of equipment and training of Owner's personnel.
3. Section 31 1000 "Site Clearing" for photographic documentation before site clearing operations commence.

### 1.2 INFORMATIONAL SUBMITTALS

A. Key Plan: Submit key plan of Project site and building with notation of vantage points marked for location and direction of each photograph and video recording. Indicate elevation or story of construction. Include same information as corresponding photographic documentation.

B. Digital Photographs: Submit image files within three (3) days of taking photographs.

1. Submit photos by uploading to web-based Project management software site. Include copy of key plan indicating each photograph's location and direction.
2. Identification: Provide the following information with each image description in web-based Project management software site:

a. Date photograph was taken.
b. Description of location, vantage point, and direction.
c. Unique sequential identifier keyed to accompanying key plan.

C. Video Recordings: Submit video recordings within seven (7) days of recording.

1. Submit video recordings by uploading to web-based Project management software site. Include copy of key plan indicating each video's location and direction.
2. Identification: With each submittal, provide the following information on web-based Project management software site:

a. Date video recording was recorded.
b. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.

1.3 FORMATS AND MEDIA

A. Digital Photographs: Provide color images in JPG format, produced by a digital camera with minimum sensor size of 12 megapixels, and at an image resolution of not less than 3200 by 2400 pixels, and with vibration-reduction technology. Use flash in low light levels or backlit conditions.

B. Digital Video Recordings: Provide high-resolution, digital video in MPEG format, produced by a digital camera with minimum sensor resolution of 12 megapixels and capable of recording in full high-definition mode with vibration-reduction technology. Provide supplemental lighting in low light levels or backlit conditions.

C. Metadata: Record accurate date and time from camera.

D. File Names: Name media files with date and sequential numbering suffix.

1.4 CONSTRUCTION PHOTOGRAPHS

A. General: Take photographs with maximum depth of field and in focus.

1. Maintain key plan with each set of construction photographs that identifies each photographic location.

B. Preconstruction Photographs: Before commencement of the Work, take photographs of Project site and surrounding properties, including existing items to remain during construction, from different vantage points, as directed by Architect and Engineer.

1. Flag excavation areas and construction limits before taking construction photographs.
2. Take photographs to show existing conditions adjacent to property before starting the Work.
3. Take photographs of existing buildings either on or adjoining property, to accurately record physical conditions at start of construction.
4. Take additional photographs as required to record settlement or cracking of adjacent structures, pavements, and improvements.

C. Concealed Work Photographs: Before proceeding with installing work that will conceal other work, take photographs sufficient in number, with annotated descriptions, to record nature and location of concealed Work, including, but not limited to, the following:

1. Underground utilities.
2. Underslab services.
3. Piping.
4. Electrical conduit.
5. Waterproofing and weather-resistant barriers.
6. Roofing Systems.

D. Periodic Construction Photographs: Take 20 photographs coinciding with the cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last photographs were taken.

1.5 CONSTRUCTION VIDEO RECORDINGS

A. Preconstruction Video Recording: Before starting construction, record video recording of Project site and surrounding properties from different vantage points, as directed by Architect and Engineer.

1. Flag construction limits before recording construction video recordings.
2. Show existing conditions adjacent to Project site before starting the Work.
3. Show existing buildings either on or adjoining Project site to accurately record physical conditions at the start of excavation.
4. Show protection efforts by Contractor.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 3233

# SECTION 01 3300 - SUBMITTAL PROCEDURES

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Submittal schedule requirements.
2. Administrative and procedural requirements for submittals.

B. Related Requirements:

1. Section 01 2900 "Payment Procedures" for submitting Applications for Payment and the schedule of values.
2. Section 01 3100 "Project Management and Coordination" for submitting coordination drawings and subcontract list and for requirements for web-based Project software.
3. Section 01 3200 "Construction Progress Documentation" for submitting schedules and reports, including Contractor's construction schedule.
4. Section 01 3233 "Photographic Documentation" for submitting preconstruction photographs, periodic construction photographs, and Final Completion construction photographs.
5. Section 01 4000 "Quality Requirements" for submitting test and inspection reports, and schedule of tests and inspections.
6. Section 01 7700 "Closeout Procedures" for submitting closeout submittals and maintenance material submittals.
7. Section 01 7823 "Operation and Maintenance Data" for submitting operation and maintenance manuals.
8. Section 01 7839 "Project Record Documents" for submitting record Drawings, record Specifications, and record Product Data.
9. Section 01 7900 "Demonstration and Training" for submitting video recordings of demonstration of equipment and training of Owner's personnel.

### 1.2 DEFINITIONS

A. Action Submittals: Written and graphic information and physical samples that require Architect's action. Action submittals are those submittals indicated in individual Specification Sections as "action submittals."

B. Informational Submittals: Written and graphic information and physical samples that do not require Architect's responsive action. Submittals may be rejected for not complying with requirements. Informational submittals are those submittals indicated in individual Specification Sections as "informational submittals."

### 1.3 SUBMITTAL SCHEDULE

A. Submittal Schedule: Submit, as an action submittal, a list of submittals, arranged in chronological order by dates required by construction schedule. Include time required for review, ordering, manufacturing, fabrication, and delivery when establishing dates. Include

additional time required for making corrections or revisions to submittals noted by Architect and Construction Manager and additional time for handling and reviewing submittals required by those corrections.

1. Coordinate submittal schedule with list of subcontracts, the schedule of values, and Contractor's construction schedule.
2. Initial Submittal Schedule: Submit concurrently with startup construction schedule. Include submittals required during the first sixty (60) days of construction. List those submittals required to maintain orderly progress of the Work and those required early because of long lead time for manufacture or fabrication.
3. Final Submittal Schedule: Submit concurrently with the first complete submittal of Contractor's construction schedule.

    a. Submit revised submittal schedule as required to reflect changes in current status and timing for submittals.

4. Format: Arrange the following information in a tabular format:

    a. Scheduled date for first submittal.
    b. Specification Section number and title.
    c. Submittal Category: Action; informational.
    d. Name of subcontractor.
    e. Description of the Work covered.
    f. Scheduled date for Architect's final release or approval.
    g. Scheduled dates for installation.
    h. Activity or event number.

1.4 SUBMITTAL FORMATS

A. Submittal Information: Include the following information in each submittal:

1. Project name.
2. Date.
3. Name of Architect.
4. Name of Contractor.
5. Name of firm or entity that prepared submittal.
6. Names of subcontractor, manufacturer, and supplier.
7. Unique submittal number, including revision identifier. Include Specification Section number with sequential alphanumeric identifier and alphanumeric suffix for resubmittals.
8. Category and type of submittal.
9. Submittal purpose and description.
10. Number and title of Specification Section, with paragraph number and generic name for each of multiple items.
11. Drawing number and detail references, as appropriate.
12. Indication of full or partial submittal.
13. Location(s) where product is to be installed, as appropriate.
14. Other necessary identification.
15. Remarks.
16. Signature of transmitter.

B. Options: Identify options requiring selection by Architect.

C. Deviations and Additional Information: On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations; include relevant additional information and revisions, other than those requested by Architect on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

D. Electronic Submittals: Prepare submittals as PDF package, incorporating complete information into each PDF file. Name PDF file with submittal number.

E. Submittals Utilizing Web-Based Project Software: Prepare submittals as PDF files or other format indicated by Project management software.

## 1.5 SUBMITTAL PROCEDURES

A. Prepare and submit submittals required by individual Specification Sections. Types of submittals are indicated in individual Specification Sections.

1. Web-Based Project Management Software: Prepare submittals in PDF form, and upload to web-based Project management software website. Enter required data in web-based software site to fully identify submittal.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.

1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
2. Submit all submittal items required for each Specification Section concurrently unless partial submittals for portions of the Work are indicated on approved submittal schedule.
3. Submit action submittals and informational submittals required by the same Specification Section as separate packages under separate transmittals.
4. Coordinate transmittal of submittals for related parts of the Work specified in different Sections, so processing will not be delayed because of need to review submittals concurrently for coordination.

    a. Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

C. Processing Time: Allow time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Architect's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.

1. Initial Review: Allow fifteen (15) days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. Architect will advise Contractor when a submittal being processed must be delayed for coordination.
2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
3. Resubmittal Review: Allow fifteen (15) days for review of each resubmittal.
4. Sequential Review: Where sequential review of submittals by Architect's consultants, Owner, or other parties is indicated, allow twenty-one (21) days for initial review of each submittal.

5. Concurrent Consultant Review: Where the Contract Documents indicate that submittals may be transmitted simultaneously to Architect and to Architect's consultants, allow fifteen (15) days for review of each submittal. Submittal will be returned to Architect before being returned to Contractor.

    a. Submit one copy of submittal to concurrent reviewer in addition to specified number of copies to Architect and Construction Manager.

D. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.

1. Note date and content of previous submittal.
2. Note date and content of revision in label or title block, and clearly indicate extent of revision.
3. Resubmit submittals until they are marked with approval notation from Architect's action stamp.

E. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.

F. Use for Construction: Retain complete copies of submittals on Project site. Use only final action submittals that are marked with approval notation from Architect's action stamp.

## 1.6 SUBMITTAL REQUIREMENTS

A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. All Product Data, Qualification Data, Certifications, Test Reports, Warranties shall be packaged in one submittal regardless of if listed as an action or informational submittal unless otherwise directed by the Architect or Engineer.

2. If information must be specially prepared for submittal because standard published data are unsuitable for use, submit as Shop Drawings, not as Product Data.
3. Mark each copy of each submittal to show which products and options are applicable.
4. Include the following information, as applicable:

    a. Manufacturer's catalog cuts.
    b. Manufacturer's product specifications.
    c. Standard color charts.
    d. Statement of compliance with specified referenced standards.
    e. Testing by recognized testing agency.
    f. Application of testing agency labels and seals.
    g. Notation of coordination requirements.
    h. Availability and delivery time information.

5. For equipment, include the following in addition to the above, as applicable:

    a. Wiring diagrams that show factory-installed wiring.
    b. Printed performance curves.
    c. Operational range diagrams.
    d. Clearances required to other construction, if not indicated on accompanying Shop Drawings.

6. Submit Product Data before Shop Drawings, and before or concurrently with Samples.

B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.

1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:

a. Identification of products.
b. Schedules.
c. Compliance with specified standards.
d. Notation of coordination requirements.
e. Notation of dimensions established by field measurement.
f. Relationship and attachment to adjoining construction clearly indicated.
g. Seal and signature of professional engineer if specified.

2. Paper Sheet Size: Except for templates, patterns, and similar full-size Drawings, submit Shop Drawings on sheets at least 8-1/2 by 11 inches (215 by 280 mm), but no larger than 30 by 42 inches (750 by 1067 mm).

C. Samples: Submit Samples for review of type, color, pattern, and texture for a check of these characteristics with other materials.

1. Transmit Samples that contain multiple, related components, such as accessories together in one submittal package.
2. Identification: Permanently attach label on unexposed side of Samples that includes the following:

a. Project name and submittal number.
b. Generic description of Sample.
c. Product name and name of manufacturer.
d. Sample source.
e. Number and title of applicable Specification Section.
f. Specification paragraph number and generic name of each item.

3. Web-Based Project Management Software: Prepare submittals in PDF form, and upload to web-based Project software website. Enter required data in web-based software site to fully identify submittal.

4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.

a. Samples that may be incorporated into the Work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
b. Samples not incorporated into the Work, or otherwise designated as Owner's property, are the property of Contractor.

5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units, showing the full range of colors, textures, and patterns available.

a. Number of Samples: Submit three full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Architect will return submittal with options selected.

6. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.

   a. Number of Samples: Submit three sets of Samples. Architect will retain one Sample sets; remainder will be returned.

      1) Submit a single Sample where assembly details, workmanship, fabrication techniques, connections, operation, and other similar characteristics are to be demonstrated.
      2) If variation in color, pattern, texture, or other characteristic is inherent in material or product represented by a Sample, submit at least three sets of paired units that show approximate limits of variations.

D. Product Schedule: As required in individual Specification Sections, prepare a written summary indicating types of products required for the Work and their intended location. Include the following information in tabular form:

1. Type of product. Include unique identifier for each product indicated in the Contract Documents or assigned by Contractor if none is indicated.
2. Manufacturer and product name, and model number if applicable.
3. Number and name of room or space.
4. Location within room or space.

E. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, contact information of architects and owners, and other information specified.

F. Design Data: Prepare and submit written and graphic information indicating compliance with indicated performance and design criteria in individual Specification Sections. Include list of assumptions and summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Number each page of submittal.

G. Certificates:

1. Certificates and Certifications Submittals: Submit a statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity. Provide a notarized signature where indicated.
2. Installer Certificates: Submit written statements on manufacturer's letterhead, certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
3. Manufacturer Certificates: Submit written statements on manufacturer's letterhead, certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.

4. Material Certificates: Submit written statements on manufacturer's letterhead, certifying that material complies with requirements in the Contract Documents.
5. Product Certificates: Submit written statements on manufacturer's letterhead, certifying that product complies with requirements in the Contract Documents.
6. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of AWS B2.1/B2.1M on AWS forms. Include names of firms and personnel certified.

H. Test and Research Reports:

1. Compatibility Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for substrate preparation and primers required.
2. Field Test Reports: Submit written reports indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
3. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
4. Preconstruction Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
5. Product Test Reports: Submit written reports indicating that current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
6. Research Reports: Submit written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project. Include the following information:

    a. Name of evaluation organization.
    b. Date of evaluation.
    c. Time period when report is in effect.
    d. Product and manufacturers' names.
    e. Description of product.
    f. Test procedures and results.
    g. Limitations of use.

## 1.7 DELEGATED DESIGN SERVICES

A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.

1. If criteria indicated are insufficient to perform services or certification required, submit a written request for additional information to Architect.

B. Delegated Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally signed PDF file paper copies of certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.

1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

C. BIM Incorporation: Incorporate delegated design drawing and data files into BIM established for Project.

1. Prepare delegated design drawings in the following format: Same digital data software program, version, and operating system as original Drawings.

1.8 CONTRACTOR'S REVIEW

A. Action Submittals and Informational Submittals: Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to Architect and Construction Manager.

B. Contractor's Approval: Indicate Contractor's approval for each submittal with indication in web-based Project management software. Include name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.

1. Architect will not review submittals received from Contractor that do not have Contractor's review and approval.

1.9 ARCHITECT'S REVIEW

A. Action Submittals: Architect will review each submittal, indicate corrections or revisions required, and return.

1. PDF Submittals: Architect will indicate, via markup on each submittal, the appropriate action.

2. Submittals by Web-Based Project Management Software: Architect will indicate, on Project management software website, the appropriate action.

B. Informational Submittals: Architect will review each submittal and will not return it or will return it if it does not comply with requirements. Architect will forward each submittal to appropriate party.

C. Partial submittals prepared for a portion of the Work will be reviewed when use of partial submittals has received prior approval from Architect.

D. Incomplete submittals are unacceptable, will be considered nonresponsive, and will be returned for resubmittal without review.

E. Architect will return without review submittals received from sources other than Contractor.

F. Submittals not required by the Contract Documents will be returned by Architect without action.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 3300

SECTION 01 3523 – PROJECT SITE SAFETY AND SECURITY

PART 1 - GENERAL

1.1 DESCRIPTION

A. Furnish all labor, materials, tools, equipment and services for effective project safety and security operations to meet all regulatory requirements in accordance with applicable codes and contract provisions.

B. The Contractor is to furnish and install any supplementary or miscellaneous items, appurtenances, and devices incidental to or necessary for a safe and secure project operation. The types of security and protection provisions required include, but are not limited to, worker PPE, fire protection, barricades, warning signs/lights, site enclosure fence, sidewalk bridges, building enclosure/lockup, 24/7 site access control, environmental protection, and similar provisions intended to minimize property losses, personal injuries and claims for damages at the project site. No part of such provision(s) shall become the subject of a claim for extension of time or for excess costs or damages by the Contractor.

C. The Contractor is required to submit a project/site specific safety and security plan to the Owner for review/approval no later than the earlier of thirty (30) calendar days after Notice to Proceed or prior to mobilization to site.

1.2 CONSTRUCTION SAFETY

A. It shall be a condition of each construction contract that no contractor, subcontractor, or supplier contracting for any part of the contract work shall require any laborer, mechanic, or other person employed in the performance of the Contract to work in surroundings or under working conditions which are hazardous or dangerous to his/her safety, as determined under construction safety standards promulgated by the U.S. Department of Labor or the Virginia Department of Labor and Industry.

B. No Contractor awarded a construction contract shall knowingly employ or contract with any person, company, or corporation for services pursuant to that contract if such person, company, or corporation could not have been awarded such contract due to the restrictions set forth in the Contract Section, Information for Bidders.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 SAFETY STANDARDS

A. The Contractor shall comply with Federal, State, and local safety and fire codes and regulations and the applicable provisions of the following:

1. Occupational Safety and Health Standards issued by the Secretary of Labor pursuant to the Williams-Steiger Occupational Safety and Health Act of 1970 and as amended.
2. Virginia Occupational Safety and Health Standards for the construction industry.
3. NFPA 70 E – Standards for Electric Safety at Workplace as referenced in OSHA 29 CFR 1926, sub-part K for safety measures, PPE protection from arc flash from electric equipment.

B. Where requirements conflict, the most stringent requirement shall apply.

3.2 ACCIDENT REPORTS

A. The Contractor shall maintain an accurate record of, and shall immediately report orally to the Owner, any accidents resulting in death, traumatic injury, occupational disease, or damage of property, materials, supplies, and equipment incident to Work performed under this Contract. The Contractor shall provide a written preliminary accident report to the Owner within 24 hours of occurrence and a complete written report within seven (7) calendar days for any accident or reportable or recordable injury.

3.3 HARD HATS, SAFETY SHOES, AND EMPLOYEE IDENTIFICATION

A. Hard hats shall be required at the construction site from start to completion of Work. Each Contractor, employee, and visitor at the construction site shall be required to wear a hard hat. The Contractor shall provide hard hats for visitors as necessary and shall enforce the wearing of hard hats by Contractor employees and visitors. Individuals found not wearing hard hats will not be permitted to remain on site. The Contractor shall be responsible for advising all individuals who will be working on the site of this requirement. All Contractor and Subcontractor employees are required to provide employee and company identification on their hard hats or shall carry a separate ID card.

B. The Contractor shall require all of his employees to wear OSHA approved safety shoes and all required PPE on the site at all times.

3.4 HOUSEKEEPING

A. The Contractor shall, at all times, keep the site and surrounding property clean and free of unnecessary and hazardous debris. Any damage to surrounding property shall be repaired upon notification by the Owner at no expense to the Owner.

B. The Contractor shall provide all items required to ensure safety of individuals and property on site.

3.5 ALLAYING DUST, REMOVAL OF CONSTRUCTION DEBRIS, SILTATION CONTROL, AND OTHER HOUSEKEEPING

A. The Contractor will take whatever steps are necessary to ensure that dust, mud and any other construction debris is not tracked onto the surrounding roadways to the satisfaction of and at no additional cost to the Owner.

B. Stormwater Siltation Control Maintenance: Contractor shall maintain all storm water inlets previously installed by others protecting same with silt trap devices in accordance with the overall site plan details. Contractor to maintain and replace silted inlet protection measures at storm water inlets in paved areas used by the Contractor for staging and construction operations. Maintain inlet protection devices as required in response to runoff events and in response to site inspections. Contractor shall maintain the integrity of all silt fencing at all times whether installed or installed by others in the areas of their construction operations.

C. The Contractor shall, at all times, keep the site and surrounding property clean and free of unnecessary and hazardous debris. Any damage to surrounding property shall be repaired upon notification by the Owner at no expense to the Owner.

D. The Contractor shall provide all items required to ensure safety of individuals and property on site.

3.6 COMMUNITY IMPACT

The Contractor shall schedule operations with a minimum of disruption to affected or neighboring properties during construction.

A. Consideration shall be given to adjacent and nearby property access; driveway maintenance including mud-tracking, dust & noise control; staff parking, pedestrian traffic, and overall public safety to the maximum extent possible. Dynamic signage, messaging to community, Maintenance of Traffic (MOT) (to be submitted to Owner's approval prior to execution) as well as designated safety officers from the Contractor will be required and will be provided at no cost to the Owner to maintain public safety.

B. The Contractor shall abide by all applicable noise ordinances, rules, laws and regulations in effect at the job site during the entire project duration. In the case where overlapping constraints exist, the most stringent case shall apply.

C. Staff employed by the Contractor shall recognize that they are working inside the boundaries of an active law enforcement facility and shall behave accordingly.

3.7 FIRE PROTECTION

A. The Contractor is responsible for compliance with OSHA requirements and local fire codes and regulations as enforced by the governing Fire Marshall or his designated representative. Attention should be called, but not limited to, the following:

1. Proper storage and removal of volatile waste
2. Proper storage of bottled gasses
3. On-site storage of fuels
4. Establishment of smoking areas
5. Placement of temporary portable fire extinguishers
6. Employee training
7. Emergency Access

B. Adequate precautions against fire shall be taken throughout all operations. Flammable material shall be kept at an absolute minimum and shall be properly handled and stored. Fires or open salamanders shall not be permitted in any part of the work.

C. Construction practices, including cutting and welding and protection during construction, shall be in accordance with the published standards of the National Fire Protection Association.

D. Store gasoline and other flammable liquids in Underwriters Laboratories-listed safety containers in conformance with the National Board of Fire Underwriters recommendations. Storage, however, shall not be within a building.

E. Make arrangements for periodic inspection of all construction areas by local fire protection authorities and insurance underwriters. Cooperate with these authorities and promptly carry out their recommendations.

3.8 HAZARDOUS MATERIALS

A. Hazard Communication Programs: Contractor shall be responsible for coordinating any exchange of material safety data sheets or other hazard communication information required to be made available to or exchanged between or among employers at the site in accordance with Laws or Regulations.

3.9 FIRE ALARM AUTO DIALER

A. Contractor shall provide, install and make operable a U.L. approved Automatic Telephone Dialer acceptable to authority having jurisdiction and capable of sending out pre-programmed taped message on two separate telephone lines.

B. Contractor shall provide the necessary connections, hardware, and programming.

C. Coordinate with Owner the connection to the Counties central monitoring station. Provide thirty (30) days' notice prior to the need for monitoring services.

3.10 POLLUTION CONTROL

A. The Contractor is responsible for compliance with governing anti-pollution laws and ordinances relative to on-going construction, clean-up, and disposal operations. On-site incineration or burial of items is not allowed.

3.11 SECURITY

A. The Contractor is responsible for overall security of the construction site. This includes but is not necessarily limited to:

1. Preservation of all property adjacent to the work, the removal or destruction of which is not part of the work of the project. The Contractor shall be responsible, at no additional cost, for all damage or injury to property of any character during the prosecution of the work and he shall restore such property to a condition at least similar and equal to that existing before such damage or injury was done by repairing, rebuilding, or restoring, as may be directed.
2. Establishing and maintaining perimeter fencing and access gate(s).
3. Positive site access control including effective use and maintenance of a Visitor Log.
4. Securing of work in progress and materials stored on site from damage by weather, theft, or vandalism.
5. Coordination with local law enforcement when high value construction material/equipment is expected to be stored on site.

END OF SECTION 01 3523

SECTION 01 3543 - HAZARDOUS MATERIALS REMEDIATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:
1. Hazardous materials remediation contractor responsibilities.

2. Limited Regulated Materials Survey Report for the 115 West King Street (Hazardous Materials Report), performed by Froehling & Robertson, dated February 6, 2025, is incorporated into these Contract Documents as Appendix A.

B. Related Requirements:
1. Section 01 1000 "Summary of Work" for use of the premises and phasing requirements.
2. Section 02 8213 "Asbestos Abatement" for asbestos abatement requirements.

1.1 RESPONSIBILITIES

A. The Contractor is responsible for all actions necessary for the safe and effective removal of any/all Hazardous Materials found on the site and as depicted in the Hazardous Materials Report. The Scope of Work shall include providing all labor, material, equipment, administration, and incidentals for performing abatement and monitoring work as required to meet current Federal, State and Local laws, codes and regulations. In the case of conflict in direction from authorities having jurisdiction, the most stringent case shall apply. Personnel performing such action(s) shall be properly trained, insured and licensed in the Commonwealth of Virginia. The Owner, at his sole discretion, may monitor and/or verify any/all activities related to the abatement and monitoring activities performed by the Contractor. Such action by the Owner shall not be construed as replacing the Contractor’s responsibility to perform all required removal, abatement and/or monitoring activities for this Project.

B. Hazardous Materials Awareness Training mandated by authorities having jurisdiction, standard industry work practices and/or per the Contractor’s administrative policies including those of sub-contractors shall be scheduled, conducted, and completed at no additional expense to the Owner prior to crew members performing Work on or near hazardous materials.

PART 2 – PRODUCTS (NOT USED)

PART 3 – EXECUTION

3.1 HAZARDOUS MATERIAL REPORTS

A. Hazardous materials identification and remediation activities are the responsibility of the Contractor as described in paragraph 1.2-A above and in accordance with findings shown in Section 02 8213 “Asbestos Abatement” and Appendix A.

END OF SECTION 01 3543

# SECTION 01 3546 - INDOOR AIR QUALITY PROTECTION DURING CONSTRUCTION

## PART 1 – GENERAL

### 1.1 SUMMARY

A. Protect indoor air, absorbent materials, and mechanical system from contamination. Comply with ASHRAE-62, Chapter 7.

B. Schedule construction and installation to prevent buildup of contaminates that cannot be removed prior to occupancy.

C. Construction and phased occupancy:

1. Comply with SMACNA Guidelines for Occupied Buildings Under Construction.
2. Isolate areas of construction with solid physical barriers. Securely attach and seal at floor and structure above.
3. Schedule adequate time for product installation.
4. Maintain negative air pressure in construction area.
5. Ventilate construction with 100 percent outside air.
6. Do not recirculate air prior to occupancy.
7. Seal return air ducts and use direct exhaust to outside.
8. Factory age sheet goods.
9. Comply with manufacturers' instructions for appropriate dry out times.
10. When installing wet products, such as fluid applied coatings and adhesives, ventilate with outside air. Provide minimum 5 air changes per hour.
11. Protect absorbent materials in existing facilities with polyethylene sheeting.

### 1.2 SCHEDULE OF INSTALLATIONS

A. Install finish materials in the following sequence:

1. Celling Grids.
2. Wall Finishes.
3. Ceiling Tiles.
4. Floor Finishes.

## PART 2 – PRODUCTS (NOT USED)

## PART 3 – EXECUTION (NOT USED)

END OF SECTION 01 3546

# SECTION 01 3547 - INDOOR MOISTURE CONTROL DURING CONSTRUCTION

## PART 1 - GENERAL

### 1.1 QUALITY ASSURANCE

A. Purpose: Prevent the migration of moisture from the exterior to the building interior and prevent the release of moisture from materials, containers or spills that could result in the formation of mold, delamination of adhesive applied materials or other damages caused by water.

B. Process: In addition to requirements specified in individual sections for the Work, Contractor shall be responsible for timely execution of the following actions, independent of the actions of installers, to assure moisture control throughout construction.

1. Inspections of subgrade, and exterior wall and roof construction.
2. Testing of concrete slabs and roof decks and testing of gypsum wallboard applied on the interior side of exterior walls.

C. Corrections: Where deficiencies are found, Work shall be corrected and re-inspected or re-tested until acceptable to the Owner.

1. Corrective procedures may involve use of portable dehumidifiers, mechanical air blowers and other construction drying equipment to provide continuous (24/7) indoor humidity and moisture control.
2. Contractor may also employ use of construction drying equipment concurrent with installations, inspections, and testing.

### 1.2 SUBMITTALS

A. Certification: Letter signed by the Contractor certifying full compliance with the requirements and performance specified herein.

### 1.3 INSPECTIONS

A. Contractor shall verify the following work complies as specified, by visual inspection, before the Work is concealed or covered by subsequent Work.

C. Exterior walls:

1. System to direct moisture to exterior, including but not limited to flashing, weeps, etc.: Complete and functional.
2. Cavity wall air space: Open and clean.
3. Concrete masonry units: Complete and maintained dry in compliance with ASTM-C90 moisture content for Type I units.
4. Damp proofing: Complete and dry.
5. Wall insulation: Complete and dry.
6. Caulking and sealants: Complete.

1.4 TESTING

A. Contractor shall verify the following Work complies as specified, by testing, before the Work is concealed or covered by subsequent Work.

B. Gypsum wallboard applied on the interior side of exterior walls: Test prior to painting or other finish applications.

1. Test: Nondestructive moisture meter.
2. Minimum one test at bottom edge and one test at mid-height of the wall, at maximum 20 FT on center.
3. Acceptable test results: Moisture content of not more than 0.4 percent on gypsum moisture meter or 12 percent on wood meter.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED

END OF SECTION 01 3547

# SECTION 01 4000 - QUALITY REQUIREMENTS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspection services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.

1. Specific quality-assurance and quality-control requirements for individual work results are specified in their respective Specification Sections. Requirements in individual Sections may also cover production of standard products.
2. Specified tests, inspections, and related actions do not limit Contractor's other quality-assurance and quality-control procedures that facilitate compliance with the Contract Document requirements.
3. Requirements for Contractor to provide quality-assurance and quality-control services required by Architect, Owner or authorities having jurisdiction are not limited by provisions of this Section.

### 1.2 DEFINITIONS

A. Experienced: When used with an entity or individual, "experienced," unless otherwise further described, means having successfully completed a minimum of five previous projects similar in nature, size, and extent to this Project; being familiar with special requirements indicated; and having complied with requirements of authorities having jurisdiction.

B. Field Quality-Control Tests and Inspections: Tests and inspections that are performed on-site for installation of the Work and for completed Work.

C. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, subcontractor, or sub-subcontractor, to perform a particular construction operation, including installation, erection, application, assembly, and similar operations.

1. Use of trade-specific terminology in referring to a Work result does not require that certain construction activities specified apply exclusively to specific trade(s).

D. Mockups: Physical assemblies of portions of the Work constructed to establish the standard by which the Work will be judged. Mockups are not Samples.

1. Mockups are used for one or more of the following:

a. Verify selections made under Sample submittals.
b. Demonstrate aesthetic effects.
c. Demonstrate the qualities of products and workmanship.
d. Demonstrate successful installation of interfaces between components and systems.

e. Perform preconstruction testing to determine system performance.

2. Product Mockups: Mockups that may include multiple products, materials, or systems specified in a single Section.
3. In-Place Mockups: Mockups constructed on-site in their actual final location as part of permanent construction.

E. Product Tests: Tests and inspections that are performed by a nationally recognized testing laboratory (NRTL) in accordance with 29 CFR 1910.7, by a testing agency accredited in accordance with NIST's National Voluntary Laboratory Accreditation Program (NVLAP), or by a testing agency qualified to conduct product testing and acceptable to authorities having jurisdiction, to establish product performance and compliance with specified requirements.

F. Source Quality-Control Tests and Inspections: Tests and inspections that are performed at the source (e.g., plant, mill, factory, or shop).

G. Testing Agency: An entity engaged to perform specific tests, inspections, or both. The term "testing laboratory" has the same meaning as the term "testing agency."

H. Quality-Assurance Services: Activities, actions, and procedures performed before and during execution of the Work, to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

I. Quality-Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work, to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Contractor's quality-control services do not include contract administration activities performed by Architect or Construction Manager.

## 1.3 DELEGATED DESIGN SERVICES

A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.

1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to Architect.

B. Delegated Design Services Statement: Submit a statement signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional, indicating that the products and systems are in compliance with performance and design criteria indicated. Include list of codes, loads, and other factors used in performing these services.

## 1.4 CONFLICTING REQUIREMENTS

A. Conflicting Standards and Other Requirements: If compliance with two or more standards or requirements is specified and the standards or requirements establish different or conflicting requirements for minimum quantities or quality levels, inform the Architect regarding the conflict and obtain clarification prior to proceeding with the Work. Refer conflicting requirements that are different, but apparently equal, to Architect for clarification before proceeding.

B. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified is the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Architect for a decision before proceeding.

## 1.5 ACTION SUBMITTALS

A. Contractor's Quality-Control Plan: For quality-assurance and quality-control activities and responsibilities.

B. Qualification Data: For Contractor's quality-control personnel.

C. Mockup Shop Drawings:

1. Include plans, sections, elevations, and details, indicating materials and size of mockup construction.
2. Indicate manufacturer and model number of individual components.
3. Provide axonometric drawings for conditions difficult to illustrate in two dimensions.

## 1.6 INFORMATIONAL SUBMITTALS

A. Testing Agency Qualifications: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in the form of a recent report on the inspection of the testing agency by a recognized authority.

B. Schedule of Tests and Inspections: Prepare in tabular form and include the following:

1. Specification Section number and title.
2. Entity responsible for performing tests and inspections.
3. Description of test and inspection.
4. Identification of applicable standards.
5. Identification of test and inspection methods.
6. Number of tests and inspections required.
7. Time schedule or time span for tests and inspections.
8. Requirements for obtaining samples.
9. Unique characteristics of each quality-control service.

C. Reports: Prepare and submit certified written reports and documents as specified.

D. Permits, Licenses, and Certificates: For Owner's record, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents established for compliance with standards and regulations bearing on performance of the Work.

## 1.7 CONTRACTOR'S QUALITY-CONTROL PLAN

A. Quality-Control Plan, General: Submit quality-control plan within ten (10) days of Notice to Proceed, and not less than five (5) days prior to preconstruction conference. Identify personnel, procedures, controls, instructions, tests, records, and forms to be used to carry out Contractor's

quality-assurance and quality-control responsibilities and to coordinate Owner's quality-assurance and quality-control activities. Coordinate with Contractor's Construction Schedule.

B. Components of Quality Control Plan:

1. Provide, for review and approval, a QC plan submitted with table of contents and pages numbered sequentially that covers both on-site and off-site work and includes the following: (Note: QC plans submitted in any other manner than provided in this section will be rejected).

    a. Contractor’s QC Organization
    b. Names and qualifications
    c. Duties, responsibility, and authority of QC personnel.
    d. Outside organizations
    e. Appointment letters
    f. Submittal procedures and sample register
    g. Testing Laboratory Information
    h. Testing Plan and log including commissioning activities.
    i. Procedures for performing the three phases of control using a quality control checklist.
    j. Procedures to complete rework items
    k. Documentation procedures
    l. List of definable features of work
    m. Personnel Matrix
    n. Material Inspection form
    o. Inspections Report form
    p. Daily report form
    q. RFI form

C. Quality-Control Personnel Qualifications: Engage qualified personnel trained and experienced in managing and executing quality-assurance and quality-control procedures similar in nature and extent to those required for Project.

1. Project quality-control manager may also serve as Project superintendent.
2. Designated individual must meet the requirements established for the QC Manager and shall be Certified.

    a. QC Manager shall have a U.S Army Corps of Engineers certification for Construction Quality Control Management for Contractors or equivalent certification.

D. Submittal Procedure: Describe procedures for ensuring compliance with requirements through review and management of submittal process. Indicate qualifications of personnel responsible for submittal review.

E. Testing and Inspection: In quality-control plan, include a comprehensive schedule of Work requiring testing or inspection, including the following:

1. Contractor-performed tests and inspections, including subcontractor-performed tests and inspections. Include required tests and inspections and Contractor-elected tests and inspections. Distinguish source quality-control tests and inspections from field quality-control tests and inspections.
2. Special inspections required by authorities having jurisdiction and indicated on the Statement of Special Inspections.

3. Any Owner-performed tests and inspections indicated in the Contract Documents, including tests and inspections indicated to be performed by Commissioning Authority.

F. Continuous Inspection of Workmanship: Describe process for continuous inspection during construction to identify and correct deficiencies in workmanship in addition to testing and inspection specified. Indicate types of corrective actions to be required to bring the Work into compliance with standards of workmanship established by Contract requirements and approved mockups.

G. Monitoring and Documentation: Maintain testing and inspection reports, including log of approved and rejected results. Include Work Architect has indicated as nonconforming or defective. Indicate corrective actions taken to bring nonconforming Work into compliance with requirements. Comply with requirements of authorities having jurisdiction.

## 1.8 REPORTS AND DOCUMENTS

A. Test and Inspection Reports: Prepare and submit certified written reports specified in other Sections. Include the following:

1. Date of issue.
2. Project title and number.
3. Name, address, telephone number, and email address of testing agency.
4. Dates and locations of samples and tests or inspections.
5. Names of individuals making tests and inspections.
6. Description of the Work and test and inspection method.
7. Identification of product and Specification Section.
8. Complete test or inspection data.
9. Test and inspection results and an interpretation of test results.
10. Record of temperature and weather conditions at time of sample-taking and testing and inspection.
11. Comments or professional opinion on whether tested or inspected Work complies with the Contract Document requirements.
12. Name and signature of laboratory inspector.
13. Recommendations on retesting and reinspecting.

B. Manufacturer's Technical Representative's Field Reports: Prepare written information documenting manufacturer's technical representative's tests and inspections specified in other Sections. Include the following:

1. Name, address, telephone number, and email address of technical representative making report.
2. Statement on condition of substrates and their acceptability for installation of product.
3. Statement that products at Project site comply with requirements.
4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
6. Statement of whether conditions, products, and installation will affect warranty.
7. Other required items indicated in individual Specification Sections.

C. Factory-Authorized Service Representative's Reports: Prepare written information documenting manufacturer's factory-authorized service representative's tests and inspections specified in other Sections. Include the following:

1. Name, address, telephone number, and email address of factory-authorized service representative making report.
2. Statement that equipment complies with requirements.
3. Results of operational and other tests and a statement of whether observed performance complies with requirements.
4. Statement of whether conditions, products, and installation will affect warranty.
5. Other required items indicated in individual Specification Sections.

1.9 QUALITY ASSURANCE

A. Qualifications paragraphs in this article establish the minimum qualification levels required; individual Specification Sections specify additional requirements.

B. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units. As applicable, procure products from manufacturers able to meet qualification requirements, warranty requirements, and technical or factory-authorized service representative requirements.

C. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Installer Qualifications: A firm or individual experienced in installing, erecting, applying, or assembling work similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.

E. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of the system, assembly, or product that is similar in material, design, and extent to those indicated for this Project.

F. Specialists: Certain Specification Sections require that specific construction activities be performed by entities who are recognized experts in those operations. Specialists will satisfy qualification requirements indicated and engage in the activities indicated.

1. Requirements of authorities having jurisdiction supersede requirements for specialists.

G. Testing and Inspecting Agency Qualifications: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspection indicated, as documented in accordance with ASTM E329, and with additional qualifications specified in individual Sections; and, where required by authorities having jurisdiction, that is acceptable to authorities.

H. Manufacturer's Technical Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to observe and inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

I. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect, demonstrate, repair, and perform service on installations of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

J. Mockups: Before installing portions of the Work requiring mockups, build mockups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:

1. Build mockups of size indicated.
2. Build mockups in location indicated or, if not indicated, as directed by Architect or Construction Manager.
3. Notify Architect and Construction Manager seven (7) days in advance of dates and times when mockups will be constructed.
4. Employ supervisory personnel who will oversee mockup construction. Employ workers who will be employed to perform same tasks during the construction at Project.
5. Demonstrate the proposed range of aesthetic effects and workmanship.
6. Obtain Architect’s and Construction Manager's approval of mockups before starting corresponding Work, fabrication, or construction.

    a. Allow seven (7) days for initial review and each re-review of each mockup.

7. Promptly correct unsatisfactory conditions noted by Architect's preliminary review, to the satisfaction of the Architect, before completion of final mockup.
8. Approval of mockups by the Architect does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
9. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
10. Demolish and remove mockups when directed unless otherwise indicated.

K. Specialty Mockups: See Section 01 4339 "Mockups" for additional construction requirements for integrated exterior mockups and room mockups.

## 1.10 QUALITY CONTROL

A. Owner Responsibilities: Where quality-control services are indicated as Owner's responsibility, Owner will engage a qualified testing agency to perform these services.

1. Owner will furnish Contractor with names, addresses, and telephone numbers of testing agencies engaged and a description of types of testing and inspection they are engaged to perform.

B. Contractor Responsibilities: Tests and inspections not explicitly assigned to Owner are Contractor's responsibility. Perform additional quality-control activities, whether specified or not, to verify and document that the Work complies with requirements.

1. Unless otherwise indicated, provide quality-control services specified and those required by authorities having jurisdiction. Perform quality-control services required of Contractor by authorities having jurisdiction, whether specified or not.
2. Engage a qualified testing agency to perform quality-control services.

a. Contractor will not employ same entity engaged by Owner, unless agreed to in writing by Owner.

3. Notify testing agencies at least 24 hours in advance of time when Work that requires testing or inspection will be performed.
4. Where quality-control services are indicated as Contractor's responsibility, submit a certified written report, in duplicate, of each quality-control service.
5. Testing and inspection requested by Contractor and not required by the Contract Documents are Contractor's responsibility.
6. Submit additional copies of each written report directly to authorities having jurisdiction, when they so direct.

C. Retesting/Reinspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and reinspecting, for construction that replaced Work that failed to comply with the Contract Documents.

D. Testing Agency Responsibilities: Cooperate with Architect, Commissioning Authority, Construction Manager, and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify Architect, Commissioning Authority, Construction Manager, and Contractor promptly of irregularities or deficiencies observed in the Work during performance of its services.
2. Determine the locations from which test samples will be taken and in which in-situ tests are conducted.
3. Conduct and interpret tests and inspections, and state in each report whether tested and inspected Work complies with or deviates from requirements.
4. Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through Contractor.
5. Do not release, revoke, alter, or increase the Contract Document requirements or approve or accept any portion of the Work.
6. Do not perform duties of Contractor.

E. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing as specified in Section 013300 "Submittal Procedures."

F. Manufacturer's Technical Services: Where indicated, engage a manufacturer's technical representative to observe and inspect the Work. Manufacturer's technical representative's services include participation in preinstallation conferences, examination of substrates and conditions, verification of materials, observation of Installer activities, inspection of completed portions of the Work, and submittal of written reports.

G. Contractor's Associated Requirements and Services: Cooperate with agencies and representatives performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide the following:

1. Access to the Work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspection. Assist agency in obtaining samples.
4. Facilities for storage and field curing of test samples.
5. Delivery of samples to testing agencies.

6. Preliminary design mix proposed for use for material mixes that require control by testing agency.
7. Security and protection for samples and for testing and inspection equipment at Project site.

H. Coordination: Coordinate sequence of activities to accommodate required quality-assurance and quality-control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspection.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

I. Schedule of Tests and Inspections: Prepare a schedule of tests, inspections, and similar quality-control services required by the Contract Documents as a component of Contractor's quality-control plan. Coordinate and submit concurrently with Contractor's Construction Schedule. Update and submit with each Application for Payment.

1. Schedule Contents: Include tests, inspections, and quality-control services, including Contractor- and Owner-retained services, commissioning activities, and other Project-required services paid for by other entities.
2. Distribution: Distribute schedule to Owner, Architect, Commissioning Authority, Construction Manager, testing agencies, and each party involved in performance of portions of the Work where tests and inspections are required.

## 1.11 SPECIAL TESTS AND INSPECTIONS

A. Special Tests and Inspections: Engage a qualified testing agency to conduct special tests and inspections required by authorities having jurisdiction as the responsibility of Owner.

1. Notifying Architect. Engineer, Owner and Contractor promptly of irregularities and deficiencies observed in the Work during performance of its services.
2. Submitting a certified written report of each test, inspection, and similar quality-control service to Architect and Commissioning Authority, through Construction Manager, with copy to Contractor and to authorities having jurisdiction.
3. Submitting a final report of special tests and inspections at Substantial Completion, which includes a list of unresolved deficiencies.
4. Interpreting tests and inspections and stating in each report whether tested and inspected Work complies with or deviates from the Contract Documents.
5. Retesting and reinspecting corrected Work.

# PART 2 - PRODUCTS (Not Used)

# PART 3 - EXECUTION

## 3.1 TEST AND INSPECTION LOG

A. Test and Inspection Log: Prepare a record of tests and inspections. Include the following:

1. Date test or inspection was conducted.
2. Description of the Work tested or inspected.
3. Date test or inspection results were transmitted to Architect.

4. Identification of testing agency or special inspector conducting test or inspection.

B. Maintain log at Project site. Post changes and revisions as they occur. Provide access to test and inspection log for Architect's, Commissioning Authority's, Construction Manager's, and authorities' having jurisdiction reference during normal working hours.

1. Submit log at Project closeout as part of Project Record Documents.

## 3.2 REPAIR AND PROTECTION

A. General: On completion of testing, inspection, sample-taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Provide materials and comply with installation requirements specified in other Specification Sections or matching existing substrates and finishes. Restore patched areas and extend restoration into adjoining areas with durable seams that are as invisible as possible. Comply with the Contract Document requirements for cutting and patching in Section 01 7300 "Execution."

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

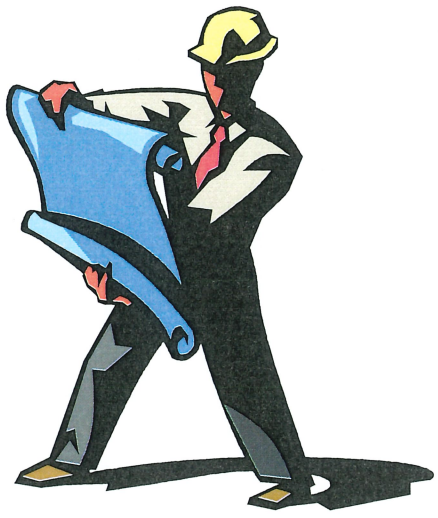
END OF SECTION 01 4000

# County of Shenandoah

## BUILDING AND CODE ENFORCEMENT

SHENANDOAH COUNTY GOVERNMENT CENTER
600 NORTH MAIN STREET SUITE 107
WOODSTOCK, VIRGINIA 22664-1855

# SHENANDOAH COUNTY SPECIAL INSPECTION GUIDELINES AND PROCEDURES



Appendix A

# SHENANDOAH COUNTY STATEMENT OF SPECIAL INSPECTIONS

**PROJECT**

Strasburg Police Station, 115 W King Street

Strasburg, VA 22657

**PERMIT APPLICANT**

**PRIMARY RDP OF RECORD**

**STRUCTURAL ENGINEER OF RECORD**

Joseph Wolhar, PE

Dewberry Engineers Inc.

This Statement of Special Inspections is submitted as a condition for permit issuance in accordance with the International Building Code (IBC) as stated in the Virginia Uniform Statewide Building Code (USBC). It includes a Schedule of Special Inspections applicable to this project as well as the name of the Special Inspector, and the identity of other testing laboratories or agencies intended to be retained for conducting these inspections or tests.

The Special Inspector shall keep records of all inspections, and shall furnish inspection reports to the Building Official, appropriate RDP(s), Owner and Contractor. All discrepancies shall be brought to the immediate attention of the Contractor for correction. If the discrepancies are not corrected, the discrepancies shall be brought to the attention of the Building Official and appropriate registered design professional(s). Interim reports shall be submitted to the Building Official, Owner, Contractor and the appropriate RDP(s) according to the *Shenandoah County Special Inspection Guidelines and Procedures*.

Jobsite safety is solely the responsibility of the contractor. Materials and activities to be Inspected are not to include the contractor's equipment and methods used to erect or install the materials listed. **All fees/costs related to the performance of Special Inspections shall be the responsibility of the Owner. Additionally, the undersigned (RDP or SER) are only acknowledging that the items enumerated on the Schedule of Special Inspections are consistent with the required design elements, the applicable sections of the Uniform Statewide Building Code and their area of expertise.**

## REVIEW, AUTHORIZATION & ACCEPTANCE

**Permit Applicant** *(If not Owner):*

Signature / Date: _____

Printed Name _____

**Owner's Authorization** *(If other than Applicant):*

Signature / Date: _____

Printed Name _____

**Primary RDP of Record:**

*(Review and Acceptance of Schedule)*

Signature / Date: _____

Printed Name _____

**SER of Record:**

*(Review and Acceptance of Schedule)*

Signature / Date: Joseph Wolhar 8/26/2026

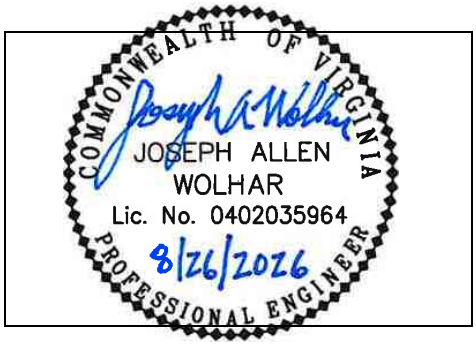
Printed Name Joseph Wolhar, PE - Dewberry Engineers Inc.

**Building Official's Acceptance:**

Signature / Date: _____

Printed Name _____

## SCHEDULE OF SI PREPARED BY:




Virginia RDP Seal of SI Preparer

Joseph Wolhar, PE
Dewberry Engineers Inc.

Printed Name of the Preparer of the Schedule

**Special Inspector:**

Signature/Date:_____

Printed Name:_____

SI Company Name:_____

# SCHEDULE OF SPECIAL INSPECTIONS

| MATERIAL ACTIVITY | TYPE OF INSPECTION | APPLICABLE TO THIS PROJECT | | | |
|---|---|---|---|---|---|
| | | Y/N | EXTENT/REFERENCE | AGENT | COMPLETED |
| ***GENERAL*** | | | | | |
| Pre-Construction conference | Meeting with parties listed in Section 6 of HRRSIGP to discuss Special Inspection procedures | Y | Scheduled by SI with the Contractor prior to commencement of work | | |
| ***EARTHWORK*** | | | | | |
| Site Preparation (building) | Field testing and inspection | N | Field Review, IBC 1704.7 / Chapter 17 | | |
| Fill Placement (building) | Review submittals, field testing and inspection | N | Field Review, IBC 1704.7 / Chapter 17 | | |
| Fill Compaction (building) | In-place density tests | N | Field Review, IBC 1704.7 / Chapter 17 | | |
| Foundation Sub-Grade | Field inspection of foundation subgrade prior to placement of concrete | Y | Field Review, IBC 1704.7 / Chapter 17 | | |
| ***PILE FOUNDATIONS*** | | | | | |
| Test Piles | Monitor driving of test piles | N | IBC 1704.8 / Chapter 17 | | |
| Pile Installation | Monitor driving of piles, including cut off & tip elevation | N | IBC 1704.8 / Chapter 17 | | |
| Pile Load Test | Monitor pile load test | N | IBC 1704.8 / Chapter 17 | | |
| Pile Survey | Review as-driven survey | N | IBC 1704.8 / Chapter 17 | | |
| ***CONCRETE*** | | | | | |
| Materials | Review product supplied versus certificates of compliance and mix design | Y | Chapter 17, ACI 318, Chapter 19 | | |
| Reinforcing Steel | Field inspection of placement | Y | Chapter 17, ACI 318, Chapter 19 | | |
| Formwork Installation | Field inspection | Y | Chapter 17, ACI 318, Chapter 19 | | |
| Concreting Operations & Placement | Field inspection of placement | Y | Chapter 17, ACI 318, Chapter 19 | | |
| Concrete Curing | Field inspection of curing process | Y | Chapter 17, ACI 318, Chapter 19 | | |
| Concrete Strength | Evaluation of concrete strength | Y | Chapter 17, ACI 318, Chapter 19 | | |

# SCHEDULE OF SPECIAL INSPECTIONS

| MATERIAL ACTIVITY | TYPE OF INSPECTION | APPLICABLE TO THIS PROJECT | | | |
|---|---|---|---|---|---|
| | | Y/N | EXTENT/REFERENCE | AGENT | COMPLETED |
| ***PRECAST CONCRETE*** | | | | | |
| Quality Control | In-plant review ** | N | Chapter 17 ACI 318 | | |
| Erection and installation | Field inspection of in-place precast | N | Chapter 17 ACI 318 | | |
| ***MASONRY*** | | | | | |
| Materials | Review of products supplied versus certificate of compliance and material submitted | N | Chapter 17 ACI 530 | | |
| Strength | Testing / review of strength | N | Chapter 17 ACI 530 | | |
| Mortar and Grout | Inspection of proportioning and mixing | N | Chapter 17 ACI 530 | | |
| Reinforcement | Inspect condition, size, location, and spacing | N | Chapter 17 ACI 530 | | |
| Protection | Inspect procedures for protection during cold and hot weather | N | Chapter 17 ACI 530 | | |
| Anchorage | Inspection of anchorages | Y | Chapter 17 ACI 530 | | |
| Masonry operations & installations | Inspection of application and installation | N | Chapter 17 ACI 530 | | |
| ***STRUCTURAL STEEL*** | | | | | |
| Quality Control | In-plant inspection of quality control procedures** | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Bolts, nuts, and washers - materials | Material identification markings. Review of certificate of compliance | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Bolts, nuts, and washers - installation | Inspection of in-place high-strength bolts, bearing type and slip critical connections | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Structural steel - materials | Material identification markings and review of certificate of compliance | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Structural steel - insulation | Inspection of member locations, structural details for bracing, connections, stiffening | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Weld filler materials & welder certification | Review of identification markings, certificate of compliance, and welder certifications | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Welds | Inspection and testing of welds | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| Structural details | Inspection of Structural details | Y | Chapter 17 AISC LRFD ASTM AWS | | |
| ***WOOD*** | | | | | |
| Verify fabrication / quality control procedures | Review submittal and installation | N | Chapter 17 | | |
| High Load Diaphragms - Installation | Inspection of sheathing, framing size, nail and staple diameter and length, number of fastener lines and fastener spacing | N | Chapter 17 | | |

*The qualifications of the Special Inspector and Testing Laboratories are subject to the approval of the Building Official

**Inspection of quality control procedures required only if fabricator is not regularly inspected by and independent inspection agency.

# SCHEDULE OF SPECIAL INSPECTIONS

| MATERIAL ACTIVITY | TYPE OF INSPECTION | APPLICABLE TO THIS PROJECT | | | |
|---|---|---|---|---|---|
| | | Y/N | EXTENT/REFERENCE | AGENT | COMPLETED |
| ***SPRAYED CEMENTITIOUS AND MINERAL FIBER FIRE RESISTIVE MATERIALS*** | | | | | |
| Structural member surface conditions | Field review of surface conditions prior to application | N | Chapter 17 | | |
| Application | Field review of application operations | N | Chapter 17 | | |
| Thickness | Field review of applied thickness | N | Chapter 17 | | |
| ***EXTERIOR INSULATION AND FINISH SYSTEMS*** | | | | | |
| Application | Field review of application / installation | N | Chapter 17 | | |
| ***SPECIAL CASES*** | | | | | |
| Alternative materials and systems | As requested by Building Official, review system and installation | N | Chapter 17 | | |
| ***WIND AND SEISMIC QUALITY ASSURANCE PLANS*** | | | | | |
| Wind and/or seismic requirements | Review of the quality assurance plan and performance of the special inspections required | N | Chapter 17 | | |
| ***SMOKE CONTROL*** | | | | | |
| Special inspection of smoke control systems | Leakage testing and recording of device location. Pressure difference testing, detection, and control verification | N | Chapter 17 | | |

*The qualifications of the Special Inspector and Testing Laboratories are subject to the approval of the Building Official

**Inspection of quality control procedures required only if fabricator is not regularly inspected by and independent inspection agency.

| *INSPECTION AGENTS* | *FIRM* | *ADDRESS* | *TELEPHONE #* |
|---|---|---|---|
| 1. Special Inspector Agent 1 | | | |
| 2. Special Inspector | | | |
| 3. Special Inspector | | | |
| 4. Special Inspector | | | |
| 5. Testing Laboratory | | | |
| 6. Testing Laboratory | | | |

# FINAL REPORT OF SPECIAL INSPECTIONS

**PROJECT**

Strasburg Police Station, 115 W King Street

Strasburg, VA 22657

**PERMIT APPLICANT**

**PRIMARY RDP OF RECORD**

**STRUCTURAL ENGINEER OF RECORD**

Joseph Wolhar, PE

Dewberry Engineers, Inc.

To the best of my information, knowledge, and belief, the special inspections required for this project, and itemized in the statement of special inspections submitted for permit, have been completed.

The following discrepancies that were outstanding since the last interim report dated_____, have been corrected:

_____

_____

_____

_____.

Interim reports submitted prior to this final report, and numbered _____ to _____, form a basis for, and are to be considered an integral part of this final report.

Respectfully submitted,

_____
Signature

_____
Date

_____
Type of Print Name **(Agent 1)**

**Upon completion of all special inspections and testing, the SI shall submit a Final Report of Special Inspections to the Building Official for review and approval. The Building Official review and approval is required to final building inspection approval or issuance of a Certificate of Occupancy.**

SECTION 01 4100 – COORDINATION, FIELD ENGINEERING AND REGULATORY REQUIREMENTS

PART 1 - GENERAL

1.01 SCHEDULES

A. Responsibility for timely installation of all materials rests solely with the Contractor, who shall maintain coordination at all times. Coordination includes but is not limited to the following: installation of services by Shenandoah Valley Electric Cooperative (Electric), Town of Strasburg (Water & Sewer), Shentel (Internet/Cable/Fiber)**.** The responsibility for scheduling all activities including obtaining required permits and sequencing work as required by the Town of Strasburg or other Authorities Having Jurisdiction rests solely with the Contractor. The Contractor shall include such activities, their duration, and the required sequencing in the Construction schedule.

1.02 FIELD DIMENSIONS

A. The drawings accompanying these specifications indicate generally the design and arrangements of all equipment, apparatus, fixtures, accessories, etc., to complete the installation of all systems. The exact location or arrangement of equipment, unless otherwise dimensioned, is subject to minor changes necessitated by field conditions and shall be made as required without additional cost to the Town of Strasburg. Measurements shall be verified by actual observation at construction site and the Contractor shall be responsible for all of the work fitting into place in a satisfactory and workmanlike manner to the approval of the Town of Strasburg and the Architect/Engineer.

1.03 CLEAN-UP DURING CONSTRUCTION

A. During the construction period, the Contractor shall:

1. Clean up all waste materials, rubbish, and debris resulting from his operations weekly or at such frequencies as required by the Town of Strasburg.
2. Place waste materials, rubbish, and debris from ground floor operations outside of building in a designated location.
3. Place waste materials, rubbish, and debris from above ground floor operations in chute.
4. Oversee cleaning and ensure that building and grounds are maintained free from accumulations of debris.
5. Sprinkle dusty debris with water.
6. At reasonable intervals, clean-up site and dispose of debris off site.
7. Be responsible for procuring permits for hauling and dumping rubbish, waste materials and debris from site to area approved by local authorities.
8. Remove grease, dust, dirt, stains, labels, fingerprints, and other foreign materials from interior and exterior surfaces of fixtures, hardware and equipment.
9. Repair, patch, and touch-up marred surfaces to match adjacent finishes.
10. Comply with additional requirements defined in specifications.

1.04 DELIVERY AND STORAGE

A. Storage of materials and equipment in conjunction with this contract is the responsibility of the Contractor. Stored materials and equipment shall be stored on the site where directed by the Town of Strasburg Representative. The Contractor is responsible for security of the site and for protection from the environment. Provide fencing or other security measures. Do not damage pavement or other improvements. Restore to original condition upon completion of work

B. Materials damaged or defective as determined by the Owner or Owner representative, shall be replaced or repaired at no additional cost to the Town of Strasburg.

C. Hazardous material brought to the site by the Contractor shall be accompanied by a Material Safety Data Sheet, Form OSHA-20 or DD Form 1813, pursuant to Federal Standard No. 313A (or latest revision thereof)

D. The Contractor is responsible for receipt and transportation. All shipments to the project shall be addressed to the Contractor and not the Town of Strasburg.

1.05 INTERRUPTION OF EXISTING UTILITIES AND SERVICES

A. The Contractor shall perform the work with a minimum of outage time for all utilities. Interruption shall be at approved sections of the utilities. In some cases, the Contractor may be required to perform the work while the existing utility is in service. The existing utility services may be interrupted only when approved by the Town of Strasburg and utility company involved. The Contractor shall notify the Owner or Owner's Representative in writing at least seven (7) calendar days in advance of the time desired for the existing service to be interrupted.

B. The amount of time requested by the Contractor for interruption of existing utility service shall be as approved by the Town of Strasburg.

1.06 LAYOUT

A. Dimensions and elevations indicated in layout of existing conditions shall be verified by the Contractor. Discrepancies between drawings, specifications and existing conditions shall be referred to the Architect/Engineer for adjustment before work affected is performed. Failure to make such notification shall place responsibility upon the Contractor.

1.07 CONTROLS

A. The Contractor shall establish and preserve all control points and benchmarks, including the property corners.

1.08 STAKES AND GRADES

A. The Contractor shall employ a certified civil engineer or land surveyor to stake out all of the construction. Rough grade stakes shall be set in sufficient number so that the site can be accurately graded to meet the intent of the drawings. Stakes for final grading shall be set for all sanitary, water main and storm sewers and all pavements, curbs and gutters.

B. The Contractor shall perform the following:

1. Provide datum benchmark for the use of all work.
2. Lay out all site work.
3. Lay out interior partitions on concrete slabs.

    a. As soon as possible after completion of floor slabs, and prior to any partitions being done, lay out the entire full-scale floor plan for each floor area.
    b. Layout shall be made by snapping distinct and readable caulk lines or by painting lines, indicating the accurate thicknesses of walls and partitions, locations of swings of doors and similar physical features.
    c. After floor layouts have been made, Contractor and/or appropriate subcontractor shall be responsible for field checking room dimensions, wall thicknesses, dimensions of built-in equipment, and similar items of critical nature.

d. Discrepancies between drawings and field conditions shall be referred to the Architect/Engineer for review and direction before any further work is performed. Failure to make such notification shall place responsibility upon the Contractor.

1.09 CODE COMPLIANCE

A. The Contractor shall comply with all applicable Town of Strasburg, Commonwealth of Virginia and Federal regulations, ordinances and laws. All violations shall be corrected within 24 hours from notice of violation.

B. The Contractor is responsible for ensuring that necessary inspections are made, regarding the Town of Strasburg and all other authorities having jurisdiction. It is the Contractor's responsibility to ensure that inspections are requested in a timely manner, and that the inspections are made, and approvals granted.

PART 2 - MATERIALS - NOT USED

PART 3 - EXECUTION – NOT USED

END OF SECTION 01 4100

# SECTION 01 4339 - MOCKUPS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. This section includes general information for the application, construction, use, inspection, and acceptance of mockups. Refer to technical specification sections and drawings for additional detail regarding specific mockup requirements and inspection processes.

B. In the event that this general specification conflicts with technical specifications, local laws, rules or code requirements, the Contractor shall immediately bring such conditions to the Owner's attention for resolution before commencing with the related work. It is noted that the intent in the case of such conflicts is that the more stringent case / detailed process shall govern.

C. Section Includes:

1. Integrated exterior mockups.
2. Room mockups.

D. Related Requirements:

1. Section 01 4000 "Quality Requirements" for quality assurance requirements for aesthetic and workmanship mockups specified in other Sections.

### 1.2 DEFINITIONS

A. Integrated Exterior Mockups: Mockups of the exterior envelope constructed on-site as freestanding temporary built elements, consisting of multiple products, assemblies, and subassemblies.

B. Room Mockups: Mockups of typical interior spaces complete with wall, floor, and ceiling finishes; doors; windows; millwork; casework; specialties; furnishings and equipment; and lighting as indicated.

### 1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site and/or virtually.

1. Meet with Owner, Construction Manager, Architect, testing and inspecting agency representative, and installers of major systems whose Work is included in integrated exterior and room mockups.
2. Review coordination of equipment and furnishings provided by the Owner for room mockups.
3. Review locations and extent of mockups.
4. Review testing procedures to be performed on mockups.
5. Review and finalize schedule for mockups, and verify availability of materials, personnel, equipment, and facilities needed to complete mockups and maintain schedule for the Work.

1.4 ACTION SUBMITTALS

A. The Contractor shall submit a listing of mockups to be executed throughout Contract duration under this specification section. Failure to provide a complete listing does not relieve the Contractor of any Contract obligations.

B. Shop Drawings: For integrated exterior mockups.

1. Include plans, elevations, sections, and mounting, attachment and support details.
2. Indicate manufacturer and model number of individual components, subassemblies, and assemblies.
3. Include site location drawing indicating orientation of mockup.
4. Revise and resubmit Shop Drawings to reflect approved modifications in details and component interfaces resulting from changes made during testing procedures.

C. Delegated Design Submittal: For temporary structural supports for mockups not attached to building structure, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.5 INFORMATIONAL SUBMITTALS

A. Room Mockup Coordination Drawings: drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved include but are not limited to the following:

1. Room finishes.

1.6 QUALITY ASSURANCE

A. Testing Agency Qualifications: Qualified in accordance with ASTM E699 for testing indicated and acceptable to Owner and Architect.

B. Build mockups to do the following:

1. Verify selections made under Sample submittals.
2. Demonstrate aesthetic effects.
3. Demonstrate the qualities of products and workmanship.
4. Demonstrate acceptable coordination between components and systems.
5. Perform preconstruction testing, such as window air- and water-leakage testing.

C. Fabrication: Before fabricating or installing portions of the Work requiring mockups, build mockups for each form of construction and finish required. Use materials and installation methods as required for the Work.

1. Build mockups of size indicated.
2. Build mockups in location indicated or, if not indicated, as directed by Architect.
3. Employ supervisory personnel who will oversee mockup construction. Employ workers who will be employed to perform same tasks during the construction at Project.
4. Demonstrate the proposed range of aesthetic effects and workmanship.
5. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
6. Demolish and remove mockups when directed unless otherwise indicated.

D. Notifications:

1. Notify Architect and Construction Manager seven (7) days in advance of the dates and times when mockups will be constructed.
2. Allow seven (7) days for initial review and each re-review of each mockup.

E. Approval: Obtain Architect's approval of mockups before starting fabrication or construction of corresponding Work.

1. Unless otherwise indicated, approved mockups establish the standard by which the Work will be judged.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, prior approved in-place and room mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

## 1.7 COORDINATION

A. Coordinate schedule for construction of mockups, so construction, testing, and review of mockups do not impact Project schedule.

# PART 2 - PRODUCTS

## 2.1 INTEGRATED EXTERIOR MOCKUPS

A. Construct integrated exterior mockups according to approved mockup Shop Drawings. Construct mockups to demonstrate constructability, coordination of trades, and sequencing of Work; and to ensure materials, components, subassemblies, assemblies, and interfaces integrate into a system complying with indicated performance and aesthetic requirements.

B. Design and construct foundation and superstructure to support free-standing integrated exterior mockups.

C. Build integrated exterior mockups using installers and construction methods that will be used in completed construction.

D. Use specified products that have been approved by Architect. Coordinate installation of materials and products specified in individual Specification Sections that include Work included in integrated exterior mockups.

E. The Work of integrated exterior mockups includes mockup panels, including but not limited to, the following:

1. Air and weather barriers.
2. Thermal insulation.
3. Through-wall flashing.
4. Flashing and sheet metal trim.
5. Joint sealants.
6. Masonry.
7. Stone.

8. Window Openings.
9. Glazing.

2.2 ROOM MOCKUPS

A. Build room mockups according to approved mockup Shop Drawings to evaluate constructability, demonstrate the coordination of trades and sequencing of Work, and to demonstrate aesthetic requirements. Include each visible finish, component, and equipment item within room mockups; include operable lighting.

B. The Work of room mockups for both new buildings, as proposed by contractor and accepted by Architect(s), includes but is not limited to, the following:

1. Millwork and casework.
2. Doors and frames.
3. Access doors and frames.
4. Glazing.
5. Metal framing.
6. Gypsum board.
7. Ceramic tiling.
8. Acoustical ceilings.
9. Resinous flooring.
10. Carpet.
11. Painting.
12. Registers and grilles.
13. Wiring devices.
14. Security devices.
15. Lighting.

PART 3 - NOT USED

END OF SECTION 01 4339

# SECTION 01 5000 - TEMPORARY FACILITIES AND CONTROLS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

B. Related Requirements:

1. Section 01 1000 "Summary" for work restrictions and limitations on utility interruptions.

### 1.2 USE CHARGES

A. Installation, removal, and use charges for temporary facilities to be included in the Contract Sum unless otherwise indicated. Allow other entities engaged in the Project to use temporary services and facilities without cost, including, but not limited to, Owner and Owner Representatives, Architect, testing agencies, and authorities having jurisdiction.

B. Sewer Service: Pay sewer-service use charges for sewer usage by all entities for construction operations. Provide connections and extensions of services and metering as required for construction operations.

C. Water Service: Pay water-service use charges for water used by all entities for construction operations. Provide connections and extensions of services and metering as required for construction operations.

D. Electric Power Service: Pay electric-power-service use charges for electricity used by all entities for construction operations. Provide connections and extensions of services and metering as required for construction operations.

### 1.3 INFORMATIONAL SUBMITTALS

A. Site Utilization Plan: Show temporary facilities, temporary utility lines and connections, staging areas, construction site entrances, vehicle circulation, and parking areas for construction personnel.

B. Project Identification and Temporary Signs: Show fabrication and installation details, including plans, elevations, details, layouts, typestyles, graphic elements, and message content.

C. Fire-Safety Program: Show compliance with requirements of NFPA 241 and authorities having jurisdiction. Indicate Contractor personnel responsible for management of fire-prevention program.

1.4 QUALITY ASSURANCE

A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.

B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.5 PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Engage Installer of each permanent service to assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Chain-Link Fencing: Minimum 2-inch (50-mm), 0.148-inch- (3.8-mm-) thick, galvanized-steel, chain-link fabric fencing; minimum 6 feet (1.8 m) high with galvanized-steel pipe posts; minimum 2-3/8-inch- (60-mm-) OD line posts and 2-7/8-inch- (73-mm-) OD corner and pull posts, with 1-5/8-inch- (42-mm-) OD top rails.

B. Portable Chain-Link Fencing: Minimum 2-inch (50-mm), 0.148-inch- (3.8-mm-) thick, galvanized-steel, chain-link fabric fencing; minimum 6 feet (1.8 m) high with galvanized-steel pipe posts; minimum 2-3/8-inch- (60-mm-) OD line posts and 2-7/8-inch- (73-mm-) OD corner and pull posts, with 1-5/8-inch- (42-mm-) OD top and bottom rails. Provide concrete bases for supporting posts.

C. Fencing Windscreen Privacy Screen: Polyester fabric scrim with grommets for attachment to chain-link fence, sized to height of fence, in color selected by Architect from manufacturer's standard colors.

D. Polyethylene Sheet: Reinforced, fire-resistive sheet, 10-mil (0.25-mm) minimum thickness, with flame-spread rating of 15 or less in accordance with ASTM E84 and passing NFPA 701 Test Method 2.

E. Dust-Control Adhesive-Surface Walk-Off Mats: Provide mats, minimum 36 by 60 inches (914 by 1524 mm).

2.2 TEMPORARY FACILITIES

A. Field Offices:

1. Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

B. Common-Use Field Office: Of sufficient size to accommodate needs of Owner, Architect, Construction Manager, and construction personnel office activities and to

accommodate Project meetings specified in other Division 01 Sections. Keep office clean and orderly. Furnish and equip offices as follows:

1. Furniture required for Project-site documents, including file cabinets, plan tables, plan racks, and bookcases.
2. Conference room of sufficient size to accommodate meetings of 10 or more individuals. Provide electrical power service and 120-V ac duplex receptacles, with no fewer than one receptacle on each wall. Furnish room with conference table, chairs, and 4-foot- (1.2-m-) square tack and marker boards.
3. Drinking water and private toilet.
4. Heating and cooling equipment necessary to maintain a uniform indoor temperature of 68 to 72 deg F (20 to 22 deg C).
5. Lighting fixtures capable of maintaining average illumination of 20 fc (215 lx) at desk height.

C. Storage and Fabrication Sheds: Provide sheds sized, furnished, and equipped to accommodate materials and equipment for construction operations.

1. Store combustible materials apart from building.

## 2.3 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

B. HVAC Equipment: Unless Owner authorizes use of permanent HVAC system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.

1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.
2. Heating, Cooling, and Dehumidifying Units: Listed and labeled for type of fuel being consumed, by a qualified testing agency acceptable to authorities having jurisdiction and marked for intended location and application.
3. Permanent HVAC System: If Owner authorizes use of permanent HVAC system for temporary use during construction, provide filter with MERV of 8 at each return-air grille in system and remove at end of construction and clean HVAC system as required in Section 017700 "Closeout Procedures."

# PART 3 - EXECUTION

## 3.1 TEMPORARY FACILITIES, GENERAL

A. Conservation: Coordinate construction and use of temporary facilities with consideration given to conservation of energy, water, and materials. Coordinate use of temporary utilities to minimize waste.

3.2 INSTALLATION, GENERAL

A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

1. Locate facilities to limit site disturbance as specified in Section 011000 "Summary."

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.3 TEMPORARY UTILITY INSTALLATION

A. General: Install temporary service or connect to existing service.

1. Arrange with utility company, Owner, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.

B. Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.

C. Water Service:

1. Connect to Owner's existing water service facilities. Clean and maintain water service facilities in a condition acceptable to Owner. At Substantial Completion, restore these facilities to condition existing before initial use.

D. Sanitary Facilities: Provide temporary toilets, wash facilities, safety shower and eyewash facilities, and drinking water for use of construction personnel. Comply with requirements of authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.

1. Use of Permanent Toilets: Use of Owner's new toilet facilities is not permitted.

E. Temporary Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.

1. Provide temporary dehumidification systems when required to reduce ambient and substrate moisture levels to level required to allow installation or application of finishes and their proper curing or drying.

F. Electric Power Service:

1. Connect to Owner's existing electric power service. Maintain equipment in a condition acceptable to Owner.
2. Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.

a. Connect temporary service to Owner's existing power source, as directed by Owner.

G. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.

H. Telephone Service: Provide temporary telephone service in common-use facilities for use by all construction personnel. Install WiFi cell phone access equipment and land-based telephone line(s) for each field office.

1. At each location, post a list of important telephone numbers.

a. Police and fire departments.
b. Contractor's home office.
c. Contractor's emergency after-hours telephone number.
d. Owner's office.
e. Principal subcontractors' field and home offices.

I. Electronic Communication Service: Provide secure WiFi wireless connection to internet with provisions for access by Architect and Owner.

## 3.4 SUPPORT FACILITIES INSTALLATION

A. Comply with the following:

1. Provide construction for temporary field offices, shops, and sheds located within construction area or within 30 feet (9 m) of building lines that is noncombustible in accordance with ASTM E136. Comply with NFPA 241.
2. Maintain support facilities until Architect schedules Substantial Completion inspection. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Owner.

B. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations. Locate temporary roads and paved areas as indicated on Drawings.

1. Provide dust-control treatment that is nonpolluting and non-tracking. Reapply treatment as required to minimize dust.

C. Temporary Use of Planned Permanent Roads and Paved Areas: Locate temporary roads and paved areas in same location as permanent roads and paved areas. Construct and maintain temporary roads and paved areas adequate for construction operations. Extend temporary roads and paved areas, within construction limits indicated, as necessary for construction operations.

1. Coordinate elevations of temporary roads and paved areas with permanent roads and paved areas.
2. Prepare subgrade and install subbase and base for temporary roads and paved areas in accordance with Section 312000 "Earth Moving."
3. Recondition base after temporary use, including removing contaminated material, regrading, proofrolling, compacting, and testing.

D. Traffic Controls: Comply with requirements of authorities having jurisdiction.

1. Protect existing site improvements to remain, including curbs, pavement, and utilities.
2. Maintain access for fire-fighting equipment and access to fire hydrants.

E. Parking: Provide parking areas for construction personnel.

F. Storage and Staging: Provide areas for storage and staging needs.

G. Dewatering Facilities and Drains: Comply with requirements of authorities having jurisdiction. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in a lawful manner that will not result in flooding Project or adjoining properties or endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

H. Project Signs: Provide Project signs as indicated. Unauthorized signs are not permitted.

1. Identification Signs: Provide Project identification signs as indicated on Drawings or at location approved by Owner and Architect.
2. Temporary Signs: Provide other signs as indicated and as required to inform public and individuals seeking entrance to Project.

a. Provide temporary, directional signs for construction personnel and visitors.

3. Maintain and touch up signs, so they are legible at all times.

I. Waste Disposal Facilities:

1. Comply with requirements specified in Section 01 7419 "Construction Waste Management and Disposal."

J. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.

K. Temporary Stairs: Until permanent stairs are available, provide temporary stairs, use of wood or temporary ladders are not acceptable.

L. Temporary Use of Permanent Stairs: Use of new stairs for construction traffic will be permitted, provided stairs are protected and finishes restored to new condition at time of Substantial Completion.

## 3.5 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Protection of Existing Facilities: Protect existing vegetation, equipment, structures, utilities, and other improvements at Project site and on adjacent properties, except those indicated to be removed or altered. Repair damage to existing facilities.

1. Where access to adjacent properties is required in order to affect protection of existing facilities, obtain written permission from adjacent property owner to access property for that purpose.

B. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction as required to comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.

1. Comply with work restrictions specified in Section 01 1000 "Summary."

C. Temporary Erosion and Sedimentation Control:

1. Comply with requirements of EPA Construction General Permit or authorities having jurisdiction, whichever is more stringent, and requirements specified in Section 31 1000 "Site Clearing."

D. Stormwater Control: Comply with requirements of authorities having jurisdiction. Provide barriers in and around excavations and subgrade construction to prevent flooding by runoff of stormwater from heavy rains.

E. Tree and Plant Protection:

1. Install temporary fencing located as indicated or outside the drip line of trees to protect vegetation from damage from construction operations. Protect tree root systems from damage, flooding, and erosion.

F. Site Enclosure Fence: Before construction operations begin, furnish and install site enclosure fence in a manner that will prevent people from easily entering site except by entrance gates.

1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
2. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Furnish one set of keys to Owner.

G. Security Enclosure and Lockup: Install temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security. Lock entrances at end of each workday.

H. Barricades, Warning Signs, and Lights: Comply with requirements of authorities having jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.

I. Temporary Egress: Provide temporary egress from existing occupied facilities as indicated and as required by authorities having jurisdiction. Provide signage directing occupants to temporary egress.

J. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is incomplete, insulate temporary enclosures.

K. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241; manage fire-prevention program.

1. Prohibit smoking in construction areas. Comply with additional limits on smoking specified in other Sections.

2. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition in accordance with requirements of authorities having jurisdiction.
3. Develop and supervise an overall fire-prevention and -protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign, stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

## 3.6 MOISTURE AND MOLD CONTROL

A. Moisture and Mold Protection: Protect stored materials and installed Work in accordance with Moisture and Mold Protection Plan.

B. Exposed Construction Period: Before installation of weather barriers, when materials are subject to wetting and exposure and to airborne mold spores, protect as follows:

1. Protect porous materials from water damage.
2. Protect stored and installed material from flowing or standing water.
3. Keep porous and organic materials from coming into prolonged contact with concrete.
4. Remove standing water from decks.
5. Keep deck openings covered or dammed.

C. Partially Enclosed Construction Period: After installation of weather barriers but before full enclosure and conditioning of building, when installed materials are still subject to infiltration of moisture and ambient mold spores, protect as follows:

1. Do not load or install drywall or other porous materials or components, or items with high organic content, into partially enclosed building.
2. Keep interior spaces reasonably clean and protected from water damage.
3. Periodically collect and remove waste containing cellulose or other organic matter.
4. Discard or replace water-damaged material.
5. Do not install material that is wet.
6. Discard and replace stored or installed material that begins to grow mold.
7. Perform work in a sequence that allows wet materials adequate time to dry before enclosing the material in gypsum board or other interior finishes.

D. Controlled Construction Period: After completing and sealing of the building enclosure but prior to the full operation of permanent HVAC systems, maintain as follows:

1. Control moisture and humidity inside building by maintaining effective dry-in conditions.
2. Use temporary or permanent HVAC system to control humidity within ranges specified for installed and stored materials.
3. Comply with manufacturer's written instructions for temperature, relative humidity, and exposure to water limits.

    a. Hygroscopic materials that may support mold growth, including wood and gypsum-based products, that become wet during the course of construction and remain wet for 48 hours are considered defective and require replacing.
    b. Measure moisture content of materials that have been exposed to moisture during construction operations or after installation. Record readings beginning at time of exposure and continuing daily for 48 hours. Identify materials containing moisture levels higher than allowed. Report findings in writing to Architect.

c. Remove and replace materials that cannot be completely restored to their manufactured moisture level within 48 hours.

## 3.7 OPERATION, TERMINATION, AND REMOVAL

A. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.

B. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.

C. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. Owner reserves right to take possession of Project identification signs.
2. Remove temporary roads and paved areas not intended for or acceptable for integration into permanent construction. Where area is intended for landscape development, remove soil and aggregate fill that do not comply with requirements for fill or subsoil. Remove materials contaminated with road oil, asphalt and other petrochemical compounds, and other substances that might impair growth of plant materials or lawns. Repair or replace street paving, curbs, and sidewalks at temporary entrances, as required by authorities having jurisdiction.
3. At Substantial Completion, repair, renovate, and clean permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 01 7700 "Closeout Procedures."

END OF SECTION 01 5000

# SECTION 01 6000 - PRODUCT REQUIREMENTS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. The Work of This Section Includes: Administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and comparable products.

B. Related Requirements:

1. Section 01 2500 "Substitution Procedures" for requests for substitutions after bidding.
2. Section 01 7700 "Closeout Procedures" for submitting warranties.

### 1.2 DEFINITIONS

A. Products: Items obtained for incorporating into the Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.

1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature that is current as of date of the Contract Documents.
2. New Products: Items that have not previously been incorporated into another project or facility. Salvaged items or items reused from other projects are not considered new products. Items that are manufactured or fabricated to include recycled content materials are considered new products unless otherwise indicated.
3. Comparable Product: Product by named manufacturer that is demonstrated and approved through the comparable product submittal process described in "Comparable Products" Article, to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.

B. Basis-of-Design Product Specification: A specification in which a single manufacturer's product is named and accompanied by the words "basis-of-design product," including make or model number or other designation. Published attributes and characteristics of basis-of-design product establish salient characteristics of products.

1. Evaluating Comparable Products: In addition to the basis-of-design product description, product attributes and characteristics may be listed to establish the significant qualities related to type, function, in-service performance and physical properties, weight, dimension, durability, visual characteristics, and other special features and requirements for purposes of evaluating comparable products of additional manufacturers named in the specification. Manufacturer's published attributes and characteristics of basis-of-design product also establish salient characteristics of products for purposes of evaluating comparable products.

C. Subject to Compliance with Requirements: Where the phrase "Subject to compliance with requirements" introduces a product selection procedure in an individual Specification Section,

provide products qualified under the specified product procedure. In the event that a named product or product by a named manufacturer does not meet the other requirements of the specifications, select another named product or product from another named manufacturer that does meet the requirements of the specifications; submit a comparable product request or substitution request, if applicable.

D. Comparable Product Request Submittal: An action submittal requesting consideration of a comparable product, including the following information:

1. Identification of basis-of-design product or fabrication or installation method to be replaced, including Specification Section number and title and Drawing numbers and titles.
2. Data indicating compliance with the requirements specified in "Comparable Products" Article.

E. Basis-of-Design Product Specification Submittal: An action submittal complying with requirements in Section 013300 "Submittal Procedures."

F. Substitution: Refer to Section 012500 "Substitution Procedures" for definition and limitations on substitutions.

## 1.3 QUALITY ASSURANCE

A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, select product compatible with products previously selected, even if previously selected products were also options.

B. Identification of Products: Except for required labels and operating data, do not attach or imprint manufacturer or product names or trademarks on exposed surfaces of products or equipment that will be exposed to view in occupied spaces or on the exterior.

1. Labels: Locate required product labels and stamps on a concealed surface, or, where required for observation following installation, on a visually accessible surface that is inconspicuous.
2. Equipment Nameplates: Provide a permanent nameplate on each item of service- or power-operated equipment. Locate on a visually accessible but inconspicuous surface. Include information essential for operation, including the following:

    a. Name of product and manufacturer.
    b. Model and serial number.
    c. Capacity.
    d. Speed.
    e. Ratings.

3. See individual identification Sections in Divisions 21, 22, 23, and 26 for additional equipment identification requirements.

## 1.4 COORDINATION

A. Modify or adjust affected work as necessary to integrate work of approved comparable products and approved substitutions.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle products, using means and methods that will prevent damage, deterioration, and loss, including theft and vandalism. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to determine compliance with the Contract Documents and that products are undamaged and properly protected.

C. Storage:

1. Provide a secure location and enclosure at Project site for storage of materials and equipment.
2. Store products to allow for inspection and measurement of quantity or counting of units.
3. Store materials in a manner that will not endanger Project structure.
4. Store products that are subject to damage by the elements under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation and with adequate protection from wind.
5. Protect foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
7. Protect stored products from damage and liquids from freezing.

1.6 PRODUCT WARRANTIES

A. Warranties specified in other Sections are to be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of the Contract Documents.

1. Manufacturer's Warranty: Written standard warranty form furnished by individual manufacturer for a particular product and issued in the name of Owner or endorsed by manufacturer to Owner.
2. Special Warranty: Written warranty required by the Contract Documents to provide specific rights for Owner and issued in the name of Owner or endorsed by manufacturer to Owner.

B. Special Warranties: Prepare a written document that contains appropriate terms and identification, ready for execution.

1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.

2. Specified Form: When specified forms are included in the Project Manual, prepare a written document, using indicated form properly executed.
3. See other Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 017700 "Closeout Procedures."

## PART 2 - PRODUCTS

### 2.1 PRODUCT SELECTION PROCEDURES

A. General Product Requirements: Provide products that comply with the Contract Documents, are undamaged and, unless otherwise indicated, are new at time of installation.

1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
3. Owner reserves the right to limit selection to products with warranties meeting requirements of the Contract Documents.
4. Where products are accompanied by the term "as selected," Architect will make selection.
5. Descriptive, performance, and reference standard requirements in Specifications establish salient characteristics of products.

B. Product Selection Procedures:

1. Sole Product: Where Specifications name a single manufacturer and product, provide the named product that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.

    a. Sole product may be indicated by the phrase "Subject to compliance with requirements, provide the following."

2. Sole Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.

    a. Sole manufacturer/source may be indicated by the phrase "Subject to compliance with requirements, provide products by the following."

3. Limited List of Products: Where Specifications include a list of names of both manufacturers and products, provide one of the products listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless otherwise indicated.

    a. Limited list of products may be indicated by the phrase "Subject to compliance with requirements, provide one of the following."

4. Limited List of Manufacturers: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless otherwise indicated.

    a. Limited list of manufacturers is indicated by the phrase "Subject to compliance with requirements, provide products by one of the following."

5. Basis-of-Design Product: Where Specifications name a product, or refer to a product indicated on Drawings, and include a list of manufacturers, provide the specified or indicated product or a comparable product by one of the other named manufacturers. Drawings and Specifications may additionally indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product by one of the other named manufacturers.

    a. For approval of products by unnamed manufacturers, comply with requirements in Section 012500 "Substitution Procedures" for substitutions for convenience.

C. Visual Matching Specification: Where Specifications require the phrase "match Architect's sample," provide a product that complies with requirements and matches Architect's sample. Architect's decision will be final on whether a proposed product matches.

1. If no product available within specified category matches and complies with other specified requirements, comply with requirements in Section 012500 "Substitution Procedures" for proposal of product.

D. Visual Selection Specification: Where Specifications include the phrase "as selected by Architect from manufacturer's full range" or a similar phrase, select a product that complies with requirements. Architect will select color, gloss, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

## 2.2 COMPARABLE PRODUCTS

A. Conditions for Consideration of Comparable Products: Architect will consider Contractor's request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Architect may return requests without action, except to record noncompliance with the following requirements:

1. Evidence that proposed product does not require revisions to the Contract Documents, is consistent with the Contract Documents, will produce the indicated results, and is compatible with other portions of the Work.
2. Detailed comparison of significant qualities of proposed product with those of the named basis-of-design product. Significant product qualities include attributes such as type, function, in-service performance and physical properties, weight, dimension, durability, visual characteristics, and other specific features and requirements.
3. Evidence that proposed product provides specified warranty.
4. List of similar installations for completed projects, with project names and addresses and names and addresses of architects and owners, if requested.
5. Samples, if requested.

B. Architect's Action on Comparable Products Submittal: If necessary, Architect will request additional information or documentation for evaluation within seven (7) days of receipt of a

request for a comparable product. Architect will notify Contractor of approval or rejection of proposed comparable product within fifteen (15) days of receipt of request, or seven (7) days of receipt of additional information or documentation, whichever is later.

1. Architect's Approval of Submittal: Marked with approval notation from Architect's action stamp and Indication of approval in web-based Project management software. See Section 013300 "Submittal Procedures."
2. Use product specified if Architect does not issue a decision on use of a comparable product request within time allocated.

C. Submittal Requirements, Two-Step Process: Approval by Architect of Contractor's request for use of comparable product is not intended to satisfy other submittal requirements. Comply with specified submittal requirements.

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 6000

# SECTION 01 7300 - EXECUTION

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes general administrative and procedural requirements governing execution of the Work, including, but not limited to, the following:

1. Construction layout.
2. Field engineering.
3. Installation.
4. Cutting and patching.
5. Progress cleaning.
6. Starting and adjusting.
7. Protection of installed construction.
8. Correction of the Work.

B. Related Requirements:

1. Section 01 3300 "Submittal Procedures" for submitting surveys.
2. Section 01 7700 "Closeout Procedures" for submitting final property survey with Project Record Documents, recording of Owner-accepted deviations from indicated lines and levels, replacing defective work, and final cleaning.
3. Section 07 8413 "Penetration Firestopping" for patching penetrations in fire-rated construction.

### 1.2 DEFINITIONS

A. Cutting: Removal of in-place construction necessary to permit installation or performance of subsequent work.

B. Patching: Fitting and repair work required to restore construction to original conditions after installation of subsequent work.

### 1.3 PREINSTALLATION MEETINGS

A. Layout Conference: Conduct conference at Project site.

1. Prior to establishing layout of new and existing perimeter and structural column grid(s), review building location requirements. Review benchmark, control point, and layout and dimension requirements. Inform Architect and Construction Manager of scheduled meeting. Require representatives of each entity directly concerned with Project layout to attend, including the following:

a. Contractor's superintendent.
b. Professional licensed surveyor responsible for performing Project surveying and layout.

2. Review meanings and intent of dimensions, notes, terms, graphic symbols, and other layout information indicated on the Drawings.
3. Review requirements for including layouts on Shop Drawings and other submittals.
4. Review areas of potential interference and conflict. Coordinate procedures and resolve potential conflicts before proceeding.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For land surveyor.

B. Certified Surveys: Submit two copies signed by professional engineer.

C. Certificates: Submit certificate signed by land surveyor, certifying that location and elevation of improvements comply with requirements.

1.5 CLOSEOUT SUBMITTALS

A. Final Property Survey: Submit 10 copies showing the Work performed and record survey data.

1.6 QUALITY ASSURANCE

A. Land Surveyor Qualifications: A professional land surveyor who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing land-surveying services of the kind indicated.

B. Professional Engineer Qualifications: Refer to Section 01 4000 "Quality Requirements."

C. Cutting and Patching: Comply with requirements for and limitations on cutting and patching of construction elements.

1. Structural Elements: When cutting and patching structural elements, or when encountering the need for cutting and patching of elements whose structural function is not known, notify Architect of locations and details of cutting and await directions from Architect before proceeding. Shore, brace, and support structural elements during cutting and patching. Do not cut and patch structural elements in a manner that could change their load-carrying capacity or increase deflection.

2. Visual Elements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch exposed construction in a manner that would, in Architect's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

D. Manufacturer's Installation Instructions: Obtain and maintain on-site manufacturer's written recommendations and instructions for installation of specified products and equipment.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Comply with requirements specified in other Sections.

B. In-Place Materials: Use materials for patching identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.

1. If identical materials are unavailable or cannot be used, use materials that, when installed, will provide a match acceptable to Architect for the visual and functional performance of in-place materials. Use materials that are not considered hazardous.

C. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

1. Use cleaning products that comply with Green Seal's GS-37, or if GS-37 is not applicable, use products that comply with the California Code of Regulations maximum allowable VOC levels.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Existing Conditions: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify the existence and location of underground utilities, mechanical and electrical systems, and other construction affecting the Work.

1. Before construction, verify the location and invert elevation at points of connection of sanitary sewer, storm sewer, gas service piping, and water-service piping; underground electrical services; and other utilities.
2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.

B. Examination and Acceptance of Conditions: Before proceeding with each component of the Work, examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.

1. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
2. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
3. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.

C. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 PREPARATION

A. Existing Utility Information: Furnish information to local utility and owner that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate with authorities having jurisdiction.

B. Field Measurements: Take field measurements as required to fit the Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

C. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

D. Review of Contract Documents and Field Conditions: Immediately on discovery of the need for clarification of the Contract Documents, submit a request for information to Architect in accordance with requirements in Section 01 3100 "Project Management and Coordination."

3.3 CONSTRUCTION LAYOUT

A. Verification: Before proceeding to lay out the Work, verify layout information shown on Drawings, in relation to the property survey and existing benchmarks and existing conditions. If discrepancies are discovered, notify Architect promptly.

B. Engage a land surveyor experienced in laying out the Work, using the following accepted surveying practices:

1. Establish benchmarks and control points to set lines and levels at each story of construction and elsewhere as needed to locate each element of Project.
2. Establish limits on use of Project site.
3. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
4. Inform installers of lines and levels to which they must comply.
5. Check the location, level and plumb, of every major element as the Work progresses.
6. Notify Architect and Construction Manager when deviations from required lines and levels exceed allowable tolerances.
7. Close site surveys with an error of closure equal to or less than the standard established by authorities having jurisdiction.

C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and rim and invert elevations.

D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

E. Record Log: Maintain a log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make the log available for reference by Architect and Construction Manager.

3.4 FIELD ENGINEERING

A. Identification: Owner will provide existing benchmarks, control points, and property corners.

B. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning the Work. Preserve and protect permanent benchmarks and control points during construction operations.

1. Do not change or relocate existing benchmarks or control points without prior written approval of Architect or Construction Manager. Report lost or destroyed permanent benchmarks or control points promptly. Report the need to relocate permanent benchmarks or control points to Architect and Construction Manager before proceeding.
2. Replace lost or destroyed permanent benchmarks and control points promptly. Base replacements on the original survey control points.

C. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with authorities having jurisdiction for type and size of benchmark.

1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.
2. Where the actual location or elevation of layout points cannot be marked, provide temporary reference points sufficient to locate the Work.
3. Remove temporary reference points when no longer needed. Restore marked construction to its original condition.

D. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

E. Final Property Survey: Engage a land surveyor to prepare a final property survey showing significant features (real property) for Project. Include on the survey a certification, signed by professional engineer, that principal metes, bounds, lines, and levels of Project are accurately positioned as shown on the survey.

1. Show boundary lines, monuments, streets, site improvements and utilities, existing improvements and significant vegetation, adjoining properties, acreage, grade contours, and the distance and bearing from a site corner to a legal point.

3.5 INSTALLATION

A. Locate the Work and components of the Work accurately, in correct alignment and elevation, as indicated.

1. Make vertical work plumb and make horizontal work level.
2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
3. Conceal pipes, ducts, and wiring in finished areas unless otherwise indicated.
4. Maintain minimum headroom clearance of 96 inches (2440 mm) in occupied spaces and 90 inches (2300 mm) in unoccupied spaces, unless otherwise indicated on Drawings.

B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions that will ensure satisfactory results as judged by Architect. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations, so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy of type expected for Project.

E. Sequence the Work and allow adequate clearances to accommodate movement of construction items on-site and placement in permanent locations.

F. Tools and Equipment: Select tools or equipment that minimize production of excessive noise levels.

G. Templates: Obtain and distribute to the parties involved templates for Work specified to be factory prepared and field installed. Check Shop Drawings of other portions of the Work to confirm that adequate provisions are made for locating and installing products to comply with indicated requirements.

H. Attachment: Provide blocking and attachment plates and anchors and fasteners of adequate size and number to securely anchor each component in place, accurately located and aligned with other portions of the Work. Where size and type of attachments are not indicated, verify size and type required for load conditions with manufacturer.

1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Architect.
2. Allow for building movement, including thermal expansion and contraction.
3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

I. Joints: Make joints of uniform width. Where joint locations in exposed Work are not indicated, arrange joints for the best visual effect, as judged by Architect. Fit exposed connections together to form hairline joints.

## 3.6 CUTTING AND PATCHING

A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.

1. Cut in-place construction to provide for installation of other components or performance of other construction, and subsequently patch as required to restore surfaces to their original condition.

B. Temporary Support: Provide temporary support of Work to be cut.

C. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

D. Adjacent Occupied Areas: Where interference with use of adjoining areas or interruption of free passage to adjoining areas is unavoidable, coordinate cutting and patching in accordance with requirements in Section 01 1000 "Summary."

E. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to minimize interruption to occupied areas.

F. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer; comply with original Installer's written recommendations.

1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots neatly to minimum size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
2. Finished Surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
3. Concrete and Masonry: Cut using a cutting machine, such as an abrasive saw or a diamond-core drill.
4. Excavating and Backfilling: Comply with requirements in applicable Sections where required by cutting and patching operations.
5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. Cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
6. Proceed with patching after construction operations requiring cutting are complete.

G. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams that are as invisible as practicable, as judged by Architect. Provide materials and comply with installation requirements specified in other Sections, where applicable.

1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate physical integrity of installation.
2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.

    a. Clean piping, conduit, and similar features before applying paint or other finishing materials.
    b. Restore damaged pipe covering to its original condition.

H. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials from adjacent finished surfaces.

## 3.7 PROGRESS CLEANING

A. Clean Project site and work areas daily, including common areas. Enforce requirements strictly. Dispose of materials lawfully.

1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.
2. Do not hold waste materials more than seven (7) days during normal weather or three (3) days if the temperature is expected to rise above 80 deg F (27 deg C).
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, in accordance with regulations.

    a. Use containers intended for holding waste materials of type to be stored.

B. Site: Maintain Project site free of waste materials and debris.

C. Work Areas: Clean areas where Work is in progress to the level of cleanliness necessary for proper execution of the Work.

1. Remove liquid spills promptly.
2. Where dust would impair proper execution of the Work, broom-clean or vacuum the entire work area, as appropriate.

D. Installed Work: Keep installed work clean. Clean installed surfaces in accordance with written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and that will not damage exposed surfaces.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at time of Substantial Completion.

G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways. Comply with waste disposal requirements in Section 017419 "Construction Waste Management and Disposal."

H. During handling and installation, clean and protect construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

I. Clean and provide maintenance on completed construction as frequently as necessary through the remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

J. Limiting Exposures: Supervise construction operations to ensure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period.

## 3.8 STARTING AND ADJUSTING

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Adjust equipment for proper operation. Adjust operating components for proper operation without binding.

C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

D. Manufacturer's Field Service: Comply with qualification requirements in Section 01 4000 "Quality Requirements."

3.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Protection of Existing Items: Provide protection and ensure that existing items to remain undisturbed by construction are maintained in condition that existed at commencement of the Work.

C. Comply with manufacturer's written instructions for temperature and relative humidity.

3.10 CORRECTION OF THE WORK

A. Repair or remove and replace damaged, defective, or nonconforming Work. Restore damaged substrates and finishes.

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Repair Work previously completed and subsequently damaged during construction period. Repair to like-new condition.

C. Restore permanent facilities used during construction to their specified condition.

D. Remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence of repair.

E. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

F. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01 7300

# SECTION 01 7419 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

## 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

## 1.2 SUMMARY

A. Section includes administrative and procedural requirements for the following:

1. Recycling nonhazardous demolition and construction waste.
2. Disposing of nonhazardous demolition and construction waste.

B. Related Requirements:

1. Section 04 2200 "Unit Masonry" for disposal requirements for masonry waste.
2. Section 31 1000 "Site Clearing" for disposition of waste resulting from site clearing and removal of above- and below-grade improvements.

## 1.3 DEFINITIONS

A. Construction Waste: Building, structure, and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.

B. Demolition Waste: Site improvement materials resulting from demolition operations.

C. Disposal: Removal of demolition or construction waste and subsequent salvage, sale, recycling, or deposit in landfill, incinerator acceptable to authorities having jurisdiction, or designated spoil areas on Owner’s property.

D. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

## 1.4 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition and construction waste become property of Contractor.

## 1.5 QUALITY ASSURANCE

A. Waste Management Coordinator Qualifications: Experienced firm, or individual employed and assigned by General Contractor, with a record of successful waste management coordination of projects with similar requirements. Superintendent may serve as Waste Management Coordinator.

B. Waste Management Conference(s): Conduct conference(s) at Project site to comply with requirements in Section 01 3100 "Project Management and Coordination." Review methods and procedures related to waste management including, but not limited to, the following:

1. Review and discuss waste management plan including responsibilities of each contractor and waste management coordinator.
2. Review requirements for documenting quantities of each type of waste and its disposition.
3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
5. Review waste management requirements for each trade.

## 1.6 WASTE MANAGEMENT PLAN

A. General: Develop a waste management plan according to requirements in this Section. Plan shall consist of waste identification, waste reduction work plan. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.

B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing and construction waste generated by the Work. Use approved form for construction waste and for demolition waste.

C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Use approved form for construction waste and for demolition waste. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.

1. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
2. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
3. Handling and Transportation Procedures: Include method that will be used for separating recyclable waste including sizes of containers, container labeling, and designated location where materials separation will be performed.

# PART 2 - PRODUCTS

## 2.1 PERFORMANCE REQUIREMENTS

A. General: Achieve end-of-Project rates for salvage/recycling of minimum 75 percent by weight of total nonhazardous solid waste generated by the Work. Practice efficient waste management in the use of materials in the course of the Work. Use all reasonable means to divert construction and demolition waste from landfills and incinerators. Facilitate recycling and salvage of materials. Including but not limited to the following:

1. Demolition Waste:

a. Asphalt paving.

b. Soil and rocks.
c. Concrete.
d. Concrete reinforcing steel.
e. Brick.
f. Concrete masonry units.

2. Construction Waste:

a. Masonry and CMU.
b. Lumber.
c. Wood sheet materials.
d. Wood trim.
e. Metals.
f. Roofing.
g. Insulation.
h. Carpet.
i. Gypsum board.
j. Piping.
k. Electrical conduit.
l. Packaging: Regardless of salvage/recycle goal indicated in "General" Paragraph above, salvage or recycle 100 percent of the following uncontaminated packaging materials:

1) Paper.
2) Cardboard.
3) Boxes.
4) Plastic sheet and film.
5) Polystyrene packaging.
6) Wood crates.
7) Wood pallets.
8) Plastic pails.

m. Construction Office Waste: Regardless of salvage/recycle goal indicated in "General" Paragraph above, salvage or recycle 100 percent of the following construction office waste materials:

1) Paper.
2) Aluminum and plastic.
3) Glass containers.

## PART 3 - EXECUTION

### 3.1 PLAN IMPLEMENTATION

A. General: Implement approved waste management plan. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during the entire duration of the Contract.

1. Comply with operation, termination, and removal requirements in Section 01 5000 "Temporary Facilities and Controls."

B. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work.

1. Distribute waste management plan to everyone concerned within three (3) days of submittal return.
2. Distribute waste management plan to entities when they first begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.

C. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

1. Designate and label specific areas on Project site necessary for separating materials that are to be salvaged and recycled.
2. Comply with Section 01 5000 "Temporary Facilities and Controls" for controlling dust and dirt, environmental protection, and noise control.

## 3.2 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

A. General: Recycle paper and beverage containers used by on-site workers.

B. Preparation of Waste: Prepare and maintain recyclable waste materials according to recycling or reuse facility requirements. Maintain materials free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process.

C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to the maximum extent practical according to approved construction waste management plan.

1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.

    a. Inspect containers and bins for contamination and remove contaminated materials if found.

2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
4. Store components off the ground and protect from the weather.
5. Remove recyclable waste from Owner's property and transport to recycling receiver or processor as often as required to prevent overfilling bins.

## 3.3 RECYCLING DEMOLITION WASTE

A. Asphalt Paving: Grind asphalt to maximum 1-1/2-inch (38-mm) size.

1. Crush asphaltic concrete paving and screen to comply with requirements in Section 31 1000 "Site Clearing" for use as general fill.

B. Asphalt Paving: Break up and transport paving to asphalt-recycling facility.

C. Concrete: Remove reinforcement and other metals from concrete and sort with other metals.

1. Pulverize concrete to maximum 1-1/2-inch (38-mm) size.
2. Crush concrete and screen to comply with requirements in Section 31 2000 "Site Clearing" for use as satisfactory soil for fill or subbase.

D. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.

1. Pulverize masonry to maximum 3/4-inch (19-mm) size.

a. Crush masonry and screen to comply with requirements in Section 31 1000 "Site Clearing" for use as general fill.
b. Crush masonry and screen to comply with requirements in Section 32 9300 "Plant Material and Operations" for use as mineral mulch.

2. Clean and stack undamaged, whole masonry units on wood pallets.

E. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.

F. Metals: Separate metals by type.

1. Structural Steel: Stack members according to size, type of member, and length.
2. Remove and dispose of bolts, nuts, washers, and other rough hardware.

## 3.4 RECYCLING CONSTRUCTION WASTE

A. Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in a dry location.
2. Polystyrene Packaging: Separate and bag materials.
3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

B. Wood Materials:

1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
2. Clean Sawdust: Bag sawdust that does not contain painted or treated wood.

C. Gypsum Board: Stack large clean pieces on wood pallets or in container and store in a dry location.

1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.

D. Paint: Seal containers and store by type.

3.5 DISPOSAL OF WASTE

A. General: Except for items or materials to be salvaged or recycled, remove waste materials from Project site and legally dispose of them in a landfill or incinerator acceptable to authorities having jurisdiction.

1. Except as otherwise specified, do not allow waste materials that are to be disposed of accumulate on-site.
2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.

B. General: Except for items or materials to be salvaged or recycled, remove waste materials and legally dispose of at designated spoil areas on Owner's property.

C. Burning: Do not burn waste materials.

END OF SECTION 01 7419

# SECTION 01 7700 - CLOSEOUT PROCEDURES

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes administrative and procedural requirements for Contract closeout, including, but not limited to, the following:

1. Substantial Completion procedures.
2. Final Completion procedures.
3. List of incomplete items.
4. Submittal of Project warranties.
5. Final cleaning.

B. Related Requirements:

1. Section 01 2900 "Payment Procedures" for requirements for Applications for Payment for Substantial Completion and Final Completion.
2. Section 01 3233 "Photographic Documentation" for submitting Final Completion construction photographic documentation.
3. Section 01 7823 "Operation and Maintenance Data" for additional operation and maintenance manual requirements.
4. Section 01 7839 "Project Record Documents" for submitting Record Drawings, Record Specifications, and Record Product Data.
5. Section 01 7900 "Demonstration and Training" for requirements to train Owner's maintenance personnel to adjust, operate, and maintain products, equipment, and systems.

### 1.2 DEFINITIONS

A. List of Incomplete Items: Contractor-prepared list of items to be completed or corrected, prepared for the Architect's use prior to Architect's inspection, to determine if the Work is substantially complete.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of cleaning agent.

B. Contractor's List of Incomplete Items: Initial submittal at Substantial Completion.

C. Certified List of Incomplete Items: Final submittal at Final Completion.

### 1.4 CLOSEOUT SUBMITTALS

A. Certificates of Release: From authorities having jurisdiction.

B. Certificate of Insurance: For continuing coverage.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Schedule of Maintenance Material Items: For maintenance material submittal items required by other Sections.

1.6 SUBSTANTIAL COMPLETION PROCEDURES

A. Contractor's List of Incomplete Items: Prepare and submit a list of items to be completed and corrected (Contractor's "punch list"), indicating the value of each item on the list and reasons why the Work is incomplete.

B. Submittals Prior to Substantial Completion: Complete the following a minimum of ten (10) days prior to requesting inspection for determining date of Substantial Completion. List items below that are incomplete at time of request.

1. Certificates of Release: Obtain and submit releases from authorities having jurisdiction, permitting Owner unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
2. Submit closeout submittals specified in other Division 01 Sections, including Project Record Documents, operation and maintenance manuals, damage or settlement surveys, property surveys, and similar final record information.
3. Submit closeout submittals specified in individual Sections, including specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
4. Submit maintenance material submittals specified in individual Sections, including tools, spare parts, extra materials, and similar items, and deliver to location designated by Construction Manager. Label with manufacturer's name and model number.
5. Submit testing, adjusting, and balancing records.
6. Submit sustainable design submittals not previously submitted.
7. Submit changeover information related to Owner's occupancy, use, operation, and maintenance.

C. Procedures Prior to Substantial Completion: Complete the following a minimum of ten (10) days prior to requesting inspection for determining date of Substantial Completion. List items below that are incomplete at time of request.

1. Advise Owner of pending insurance changeover requirements.
2. Make final changeover of permanent locks and deliver keys to Owner. Advise Owner's personnel of changeover in security provisions.
3. Complete startup and testing of systems and equipment.
4. Perform preventive maintenance on equipment used prior to Substantial Completion.
5. Instruct Owner's personnel in operation, adjustment, and maintenance of products, equipment, and systems. Submit demonstration and training video recordings specified in Section 01 7900 "Demonstration and Training."
6. Advise Owner of changeover in utility services.
7. Participate with Owner in conducting inspection and walkthrough with local emergency responders.
8. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.
9. Complete final cleaning requirements.
10. Touch up paint and otherwise repair and restore marred exposed finishes to eliminate visual defects.

D. Inspection: Submit a written request for inspection to determine Substantial Completion a minimum of ten (10) days prior to date the Work will be completed and ready for final inspection and tests. On receipt of request, Architect and Construction Manager will either proceed with inspection or notify Contractor of unfulfilled requirements. Architect will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by Architect, that must be completed or corrected before certificate will be issued.

1. Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.
2. Results of completed inspection will form the basis of requirements for Final Completion.

## 1.7 FINAL COMPLETION PROCEDURES

A. Submittals Prior to Final Completion: Before requesting final inspection for determining Final Completion, complete the following:

1. Submit a final Application for Payment and supporting documentation in accordance with Section 01 29001.4.J "Final Payment Application."
2. Certified List of Incomplete Items: Submit certified copy of Architect's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by Architect. Certified copy of the list will state that each item has been completed or otherwise resolved for acceptance.
3. Certificate of Insurance: Submit evidence of final, continuing insurance coverage complying with insurance requirements.
4. Submit pest-control final inspection report.
5. Submit Final Completion photographic documentation.

B. Inspection: Submit a written request for final inspection to determine acceptance a minimum of ten (10) days prior to date the Work will be completed and ready for final inspection and tests. On receipt of request, Architect and Construction Manager will either proceed with inspection or notify Contractor of unfulfilled requirements. Architect will prepare a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.

1. Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.

## 1.8 LIST OF INCOMPLETE ITEMS

A. Organization of List: Include name and identification of each space and area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by Contractor that are outside the limits of construction.

1. Organize list of spaces in sequential order, starting with exterior areas first and proceeding from lowest floor to highest floor, listed by room or space number.
2. Organize items applying to each space by major element, including categories for ceilings, individual walls, floors, equipment, and building systems.
3. Include the following information at the top of each page:

    a. Project name.
    b. Date.

c. Name of Architect and Construction Manager.
d. Name of Contractor.
e. Page number.

4. Submit list of incomplete items one of the following formats:

a. MS Excel Electronic File: Architect will include items in punchlist.
b. PDF Electronic File: Architect will include items in punchlist.

5. Web-Based Project Software: Contractor shall utilize software feature for creating and updating list of incomplete items (punch list).

## 1.9 SUBMITTAL OF PROJECT WARRANTIES

A. Time of Submittal: Submit written warranties on request of Architect for designated portions of the Work where warranties are indicated to commence on dates other than date of Substantial Completion, or when delay in submittal of warranties might limit Owner's rights under warranty.

B. Organize warranty documents into an orderly sequence based on the table of contents of Project Manual.

C. Warranty Electronic File: Provide warranties and bonds in PDF format. Assemble complete warranty and bond submittal package into a single electronic PDF file with bookmarks enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

1. Submit by uploading to web-based project software site.

D. Provide additional copies of each warranty to include in operation and maintenance manuals.

# PART 2 - PRODUCTS

## 2.1 MATERIALS

A. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

# PART 3 - EXECUTION

## 3.1 FINAL CLEANING

A. General: Perform final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a designated portion of Project:

    a. Clean Project site of rubbish, waste material, litter, and other foreign substances.
    b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
    c. Rake grounds that are not planted, mulched, or paved to a smooth, even-textured surface.
    d. Remove tools, construction equipment, machinery, and surplus material from Project site.
    e. Remove snow and ice to provide safe access to building.
    f. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
    g. Remove debris and surface dust from limited-access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
    h. Clean flooring, removing debris, dirt, and staining; clean in accordance with manufacturer's instructions.
    i. Vacuum and mop concrete.
    j. Vacuum carpet and similar soft surfaces, removing debris and excess nap; clean in accordance with manufacturer's instructions if visible soil or stains remain.
    k. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision-obscuring materials. Polish mirrors and glass, taking care not to scratch surfaces.
    l. Remove labels that are not permanent.
    m. Wipe surfaces of mechanical and electrical equipment and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
    n. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.
    o. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
    p. Clean ducts, blowers, and coils if units were operated without filters during construction or that display contamination with particulate matter on inspection.
    q. Clean luminaires, lamps, globes, and reflectors to function with full efficiency.
    r. Clean strainers.
    s. Leave Project clean and ready for occupancy.

C. Pest Control: Comply with pest control requirements in Section 01 5000 "Temporary Facilities and Controls." Prepare written report.

D. Construction Waste Disposal: Comply with waste-disposal requirements in Section 01 5000 "Temporary Facilities and Controls." and Section 01 7419 "Construction Waste Management and Disposal."

## 3.2 CORRECTION OF THE WORK

A. Complete repair and restoration operations required by "Correction of the Work" Article in Section 01 7300 "Execution" before requesting inspection for determination of Substantial Completion.

END OF SECTION 01 7700

# SECTION 01 7823 - OPERATION AND MAINTENANCE DATA

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes: Administrative and procedural requirements for preparing operation and maintenance manuals, including the following:

1. Operation and maintenance documentation directory manuals.
2. Systems and equipment operation manuals.
3. Systems and equipment maintenance manuals.
4. Product maintenance manuals.

B. Related Requirements:

1. Section 01 3300 "Submittal Procedures" for submitting copies of submittals for operation and maintenance manuals.
2. Section 01 9113 "General Commissioning Requirements" for verification and compilation of data into operation and maintenance manuals.

### 1.2 DEFINITIONS

A. System: An organized collection of parts, equipment, or subsystems united by regular interaction.

B. Subsystem: A portion of a system with characteristics similar to a system.

### 1.3 CLOSEOUT SUBMITTALS

A. Submit operation and maintenance manuals indicated. Provide content for each manual as specified in individual Specification Sections, and as reviewed and approved at the time of Section submittals. Submit reviewed manual content formatted and organized as required by this Section.

1. Architect and Engineer will comment on whether content of operation and maintenance submittals is acceptable.
2. Where applicable, clarify and update reviewed manual content to correspond to revisions and field conditions.

B. Format: Submit operation and maintenance manuals in the following format:

1. Submit on digital media acceptable to Architect by uploading to web-based project software site. Enable reviewer comments on draft submittals.

C. Initial Manual Submittal: Submit draft copy of each manual at least thirty (30) days before commencing demonstration and training. Architect and Engineer will comment on whether general scope and content of manual are acceptable.

D. Final Manual Submittal: Submit each manual in final form prior to requesting inspection for Substantial Completion and at least fifteen (15) days before commencing demonstration and training. Architect and Commissioning Authority will return copy with comments.

1. Correct or revise each manual to comply with Architect's and Engineer's comments. Submit copies of each corrected manual within fifteen (15) days of receipt of Architect's and Engineer's comments and prior to commencing demonstration and training.

E. Comply with Section 01 7700 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

## 1.4 FORMAT OF OPERATION AND MAINTENANCE MANUALS

A. Manuals, Electronic Files: Submit manuals in the form of a multiple file composite electronic PDF file for each manual type required.

1. Electronic Files: Use electronic files prepared by manufacturer where available. Where scanning of paper documents is required, configure scanned file for minimum readable file size.
2. File Names and Bookmarks: Bookmark individual documents based on file names. Name document files to correspond to system, subsystem, and equipment names used in manual directory and table of contents. Group documents for each system and subsystem into individual composite bookmarked files, then create composite manual, so that resulting bookmarks reflect the system, subsystem, and equipment names in a readily navigated file tree. Configure electronic manual to display bookmark panel on opening file.

## 1.5 REQUIREMENTS FOR OPERATION AND MAINTENANCE MANUALS

A. Organization of Manuals: Unless otherwise indicated, organize each manual into a separate section for each system and subsystem, and a separate section for each piece of equipment not part of a system. Each manual to contain the following materials, in the order listed:

1. Title page.
2. Table of contents.
3. Manual contents.

B. Title Page: Include the following information:

1. Subject matter included in manual.
2. Name and address of Project.
3. Name and address of Owner.
4. Date of submittal.
5. Name and contact information for Contractor.
6. Name and contact information for Construction Manager.
7. Name and contact information for Architect.
8. Name and contact information for Commissioning Authority.
9. Names and contact information for major consultants to the Architect that designed the systems contained in the manuals.
10. Cross-reference to related systems in other operation and maintenance manuals.

C. Table of Contents: List each product included in manual, identified by product name, indexed to the content of the volume, and cross-referenced to Specification Section number in Project Manual.

## 1.6 SYSTEMS AND EQUIPMENT OPERATION MANUALS

A. Systems and Equipment Operation Manual: Assemble a complete set of data indicating operation of each system, subsystem, and piece of equipment not part of a system. Include information required for daily operation and management, operating standards, and routine and special operating procedures.

1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by Owner's operating personnel.

B. Content: In addition to requirements in this Section, include operation data required in individual Specification Sections and the following information:

1. System, subsystem, and equipment descriptions. Use designations for systems and equipment indicated on Contract Documents.
2. Performance and design criteria if Contractor has delegated design responsibility.
3. Operating standards.
4. Operating procedures.
5. Operating logs.
6. Wiring diagrams.
7. Control diagrams.
8. Piped system diagrams.
9. Precautions against improper use.
10. License requirements including inspection and renewal dates.

C. Descriptions: Include the following:

1. Product name and model number. Use designations for products indicated on Contract Documents.
2. Manufacturer's name.
3. Equipment identification with serial number of each component.
4. Equipment function.
5. Operating characteristics.
6. Limiting conditions.
7. Performance curves.
8. Engineering data and tests.
9. Complete nomenclature and number of replacement parts.

D. Operating Procedures: Include the following, as applicable:

1. Startup procedures.
2. Equipment or system break-in procedures.
3. Routine and normal operating instructions.
4. Regulation and control procedures.
5. Instructions on stopping.
6. Normal shutdown instructions.
7. Seasonal and weekend operating instructions.

8. Required sequences for electric or electronic systems.
9. Special operating instructions and procedures.

E. Systems and Equipment Controls: Describe the sequence of operation, and diagram controls as installed.

F. Piped Systems: Diagram piping as installed and identify color coding where required for identification.

1.7 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

A. Systems and Equipment Maintenance Manuals: Assemble a complete set of data indicating maintenance of each system, subsystem, and piece of equipment not part of a system. Include manufacturers' maintenance documentation, preventive maintenance procedures and frequency, repair procedures, wiring and systems diagrams, lists of spare parts, and warranty information.

1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by Owner's operating personnel.

B. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranties and bonds as described below.

C. Manufacturers' Maintenance Documentation: Include the following information for each component part or piece of equipment:

1. Standard maintenance instructions and bulletins; include only sheets pertinent to product or component installed. Mark each sheet to identify each product or component incorporated into the Work. If data include more than one item in a tabular format, identify each item using appropriate references from the Contract Documents. Identify data applicable to the Work and delete references to information not applicable.

    a. Prepare supplementary text if manufacturers' standard printed data are not available and where the information is necessary for proper operation and maintenance of equipment or systems.

2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
3. Identification and nomenclature of parts and components.
4. List of items recommended to be stocked as spare parts.

D. Maintenance Procedures: Include the following information and items that detail essential maintenance procedures:

1. Test and inspection instructions.
2. Troubleshooting guide.
3. Precautions against improper maintenance.
4. Disassembly; component removal, repair, and replacement; and reassembly instructions.
5. Aligning, adjusting, and checking instructions.

6. Demonstration and training video recording, if available.

E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.

1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.

F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.

G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.

H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.

1. Include procedures to follow and required notifications for warranty claims.

I. Drawings: Prepare drawings supplementing manufacturers' printed data to illustrate the relationship of component parts of equipment and systems and to illustrate control sequence and flow diagrams. Coordinate these drawings with information contained in record Drawings to ensure correct illustration of completed installation.

1. Do not use original project record documents as part of maintenance manuals.

## 1.8 PRODUCT MAINTENANCE MANUALS

A. Product Maintenance Manual: Assemble a complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into the Work.

B. Content: Organize manual into a separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

C. Source Information: List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

D. Product Information: Include the following, as applicable:

1. Product name and model number.
2. Manufacturer's name.
3. Color, pattern, and texture.
4. Material and chemical composition.
5. Reordering information for specially manufactured products.

E. Maintenance Procedures: Include manufacturer's written recommendations and the following:

1. Inspection procedures.
2. Types of cleaning agents to be used and methods of cleaning.
3. List of cleaning agents and methods of cleaning detrimental to product.
4. Schedule for routine cleaning and maintenance.
5. Repair instructions.

F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.

G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.

1. Include procedures to follow and required notifications for warranty claims.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 7823

# SECTION 01 7839 - PROJECT RECORD DOCUMENTS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes administrative and procedural requirements for Project Record Documents, including the following:

1. Record Drawings.
2. Record specifications.
3. Record Product Data.
4. Miscellaneous record submittals.

B. Related Requirements:

1. Section 01 7300 "Execution" for final property survey.
2. Section 01 7700 "Closeout Procedures" for general closeout procedures.
3. Section 01 7823 "Operation and Maintenance Data" for operation and maintenance manual requirements.

### 1.3 CLOSEOUT SUBMITTALS

A. Record Drawings: Comply with the following:

1. Number of Copies: Submit one set(s) of marked-up record prints.
2. Number of Copies: Submit copies of Record Drawings as follows:

    a. Initial Submittal:
        1) Submit PDF electronic files of scanned record prints and one set(s) of file prints.
        2) Architect will indicate whether general scope of changes, additional information recorded, and quality of drafting are acceptable.

    b. Final Submittal:
        1) Submit PDF electronic files of scanned Record Prints.
        2) Print each drawing, whether or not changes and additional information were recorded.

B. Record Specifications: Submit annotated PDF electronic files of Project's Specifications, including addenda and Contract modifications.

C. Reports: Submit written report monthly, indicating items incorporated into Project Record Documents concurrent with progress of the Work, including revisions, concealed conditions, field changes, product selections, and other notations incorporated.

1.4 RECORD DRAWINGS

A. Record Prints: Maintain one set of marked-up paper copies of the Contract Drawings and Shop Drawings, incorporating new and revised drawings as modifications are issued.

1. Preparation: Mark record prints to show the actual installation, where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to provide information for preparation of corresponding marked-up record prints.

a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
b. Accurately record information in an acceptable drawing technique.
c. Record data as soon as possible after obtaining it.
d. Record and check the markup before enclosing concealed installations.
e. Cross-reference record prints to corresponding photographic documentation.

2. Content: Types of items requiring marking include, but are not limited to, the following:

a. Dimensional changes to Drawings.
b. Revisions to details shown on Drawings.
c. Depths of foundations.
d. Locations and depths of underground utilities.
e. Revisions to routing of piping and conduits.
f. Revisions to electrical circuitry.
g. Actual equipment locations.
h. Duct size and routing.
i. Locations of concealed internal utilities.
j. Changes made by Change Order or Construction Change Directive.
k. Changes made following Architect's written orders.
l. Details not on the original Contract Drawings.
m. Field records for variable and concealed conditions.
n. Record information on the Work that is shown only schematically.

3. Mark the Contract Drawings and Shop Drawings completely and accurately. Use personnel proficient at recording graphic information in production of marked-up record prints.
4. Mark record prints with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at same location.
5. Mark important additional information that was either shown schematically or omitted from original Drawings.
6. Note Construction Change Directive numbers, alternate numbers, Change Order numbers, and similar identification, where applicable.

B. Record Digital Data Files: Immediately before inspection for Certificate of Substantial Completion, review marked-up record prints with Architect and Construction Manager. When authorized, prepare a full set of corrected digital data files of the Contract Drawings, as follows:

1. Format: Annotated PDF electronic file with comment function enabled.

2. Incorporate changes and additional information previously marked on record prints. Delete, redraw, and add details and notations where applicable.
3. Refer instances of uncertainty to Architect through Construction Manager for resolution.
4. Architect will furnish Contractor with one set of digital data files of the Contract Drawings for use in recording information.

    a. See Section 013100 "Project Management and Coordination" for requirements related to use of Architect's digital data files.
    b. Architect will provide data file layer information. Record markups in separate layers.

C. Format: Identify and date each Record Drawing; include the designation "RECORD DRAWINGS" in a prominent location.

1. Record Prints: Organize record prints into manageable sets. Bind each set with durable paper cover sheets. Include identification on cover sheets.
2. Format: Annotated PDF electronic file.
3. Record Digital Data Files: Organize digital data information into separate electronic files that correspond to each sheet of the Contract Drawings. Name each file with the sheet identification. Include identification in each digital data file.
4. Identification: As follows:

    a. Project name.
    b. Date.
    c. Designation "RECORD DRAWINGS."
    d. Name of Architect.
    e. Name of Contractor.

## 1.5 RECORD SPECIFICATIONS

A. Preparation: Mark Specifications to indicate the actual product installation, where installation varies from that indicated in Specifications, addenda, and Contract modifications.

1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
3. Record the name of manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
4. Note related Change Orders and Record Drawings where applicable.

B. Format: Submit record specifications as annotated PDF electronic file.

## 1.6 RECORD PRODUCT DATA

A. Recording: Maintain one copy of each submittal during the construction period for Project Record Document purposes. Post changes and revisions to Project Record Documents as they occur; do not wait until end of Project.

B. Preparation: Mark Product Data to indicate the actual product installation where installation varies substantially from that indicated in Product Data submittal.

1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
3. Note related Change Orders, Record Specifications, and Record Drawings where applicable.

C. Format: Submit Record Product Data as annotated PDF electronic file.

1. Include Record Product Data directory organized by Specification Section number and title, electronically linked to each item of Record Product Data.

1.7 MISCELLANEOUS RECORD SUBMITTALS

A. Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

B. Format: Submit miscellaneous record submittals as PDF electronic file.

1. Include miscellaneous record submittals directory organized by Specification Section number and title, electronically linked to each item of miscellaneous record submittals.

1.8 MAINTENANCE OF RECORD DOCUMENTS

A. Maintenance of Record Documents: Store Record Documents in the field office apart from the Contract Documents used for construction. Do not use Project Record Documents for construction purposes. Maintain Record Documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Provide access to Project Record Documents for Architect's and Construction Manager's reference during normal working hours.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 7839

# SECTION 01 7900 - DEMONSTRATION AND TRAINING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes administrative and procedural requirements for instructing Owner's personnel, including the following:

1. Instruction in operation and maintenance of systems, subsystems, and equipment.
2. Demonstration and training video recordings.

### 1.2 INFORMATIONAL SUBMITTALS

A. Instruction Program: Submit outline of instructional program for demonstration and training, including a list of training modules and a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1. Indicate proposed training modules using manufacturer-produced demonstration and training video recordings for systems, equipment, and products in lieu of video recording of live instructional module.

B. Qualification Data: For instructor.

C. Attendance Record: For each training module, submit list of participants and length of instruction time.

D. Evaluations: For each participant and for each training module, submit results and documentation of performance-based test.

### 1.3 CLOSEOUT SUBMITTALS

A. Demonstration and Training Video Recordings: Submit two copies within seven (7) days of end of each training module.

1. Identification: On each copy, provide an applied label with the following information:

a. Name of Project.
b. Name of Owner.
c. Name and address of Contractor.
d. Date of video recording.

2. Transcript:

a. Prepared in PDF electronic format. Include a cover sheet with same label information as the corresponding video recording and a table of contents with links to corresponding training components. Include name of Project and date of video recording on each page.

3. At completion of training, submit complete training manual(s) for Owner's use prepared in same PDF file format required for operation and maintenance manuals specified in Section 01 7823 "Operation and Maintenance Data."

1.4 QUALITY ASSURANCE

A. Facilitator Qualifications: A firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for this Project, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: A factory-authorized service representative, complying with requirements in Section 014000 "Quality Requirements," experienced in operation and maintenance procedures and training.

C. Videographer Qualifications: A professional videographer who is experienced photographing demonstration and training events similar to those required.

D. Pre-instruction Conference: Conduct conference at Project site to comply with requirements in Section 01 3100 "Project Management and Coordination." Review methods and procedures related to demonstration and training including, but not limited to, the following:

1. Inspect and discuss locations and other facilities required for instruction.
2. Review and finalize instruction schedule and verify availability of educational materials, instructors' personnel, audiovisual equipment, and facilities needed to avoid delays.
3. Review required content of instruction.
4. For instruction that must occur outside, review weather and forecasted weather conditions and procedures to follow if conditions are unfavorable.

1.5 COORDINATION

A. Coordinate instruction schedule with Owner's operations. Adjust schedule as required to minimize disrupting Owner's operations and to ensure availability of Owner's personnel.

B. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data have been reviewed and approved by Architect.

1.6 INSTRUCTION PROGRAM

A. Program Structure: Develop an instruction program that includes individual training modules for each system and for equipment not part of a system, as required by individual Specification Sections.

B. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instruction for the following as applicable to the system, equipment, or component:

1. Basis of System Design, Operational Requirements, and Criteria: Include the following:

a. System, subsystem, and equipment descriptions.
b. Performance and design criteria if Contractor is delegated design responsibility.
c. Operating standards.
d. Regulatory requirements.
e. Equipment function.
f. Operating characteristics.
g. Limiting conditions.
h. Performance curves.

2. Documentation: Review the following items in detail:

a. Emergency manuals.
b. Systems and equipment operation manuals.
c. Systems and equipment maintenance manuals.
d. Product maintenance manuals.
e. Project Record Documents.
f. Identification systems.
g. Warranties and bonds.
h. Maintenance service agreements and similar continuing commitments.

3. Emergencies: Include the following, as applicable:

a. Instructions on meaning of warnings, trouble indications, and error messages.
b. Instructions on stopping.
c. Shutdown instructions for each type of emergency.
d. Operating instructions for conditions outside of normal operating limits.
e. Sequences for electric or electronic systems.
f. Special operating instructions and procedures.

4. Operations: Include the following, as applicable:

a. Startup procedures.
b. Equipment or system break-in procedures.
c. Routine and normal operating instructions.
d. Regulation and control procedures.
e. Control sequences.
f. Safety procedures.
g. Instructions on stopping.
h. Normal shutdown instructions.
i. Operating procedures for emergencies.
j. Operating procedures for system, subsystem, or equipment failure.
k. Seasonal and weekend operating instructions.
l. Required sequences for electric or electronic systems.
m. Special operating instructions and procedures.

5. Adjustments: Include the following:

a. Alignments.
b. Checking adjustments.
c. Noise and vibration adjustments.
d. Economy and efficiency adjustments.

6. Troubleshooting: Include the following:

a. Diagnostic instructions.
b. Test and inspection procedures.

7. Maintenance: Include the following:

a. Inspection procedures.
b. Types of cleaning agents to be used and methods of cleaning.
c. List of cleaning agents and methods of cleaning detrimental to product.
d. Procedures for routine cleaning.
e. Procedures for preventive maintenance.
f. Procedures for routine maintenance.
g. Instruction on use of special tools.

8. Repairs: Include the following:

a. Diagnosis instructions.
b. Repair instructions.
c. Disassembly; component removal, repair, and replacement; and reassembly.
d. Instructions for identifying parts and components.
e. Review of spare parts needed for operation and maintenance.

## 1.7 PREPARATION

A. Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a training manual organized in coordination with requirements in Section 01 7823 "Operation and Maintenance Data."

B. Set up instructional equipment at instruction location.

## 1.8 INSTRUCTION

A. Facilitator: Engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Owner for number of participants, instruction times, and location.

B. Engage qualified instructors to instruct Owner's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

1. Owner will furnish Contractor with names and positions of participants.

C. Scheduling: Provide instruction at mutually agreed-on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.

1. Schedule training with Owner, through Construction Manager, with at least seven (7) days' advance notice.

D. Training Location and Reference Material: Conduct training on-site in the completed and fully operational facility using the actual equipment in-place. Conduct training using final operation and maintenance data submittals.

E. Cleanup: Collect used and leftover educational materials and give to Owner. Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.

1.9 DEMONSTRATION AND TRAINING VIDEO RECORDINGS

A. Digital Video Recordings: Provide high-resolution, digital video in MPEG format, produced by a digital camera with minimum sensor resolution of 12 megapixels and capable of recording in full HD mode with vibration reduction technology.

1. Submit video recordings on CD-ROM or thumb drive and by uploading to web-based Project software site.
2. File Hierarchy: Organize folder structure and file locations in accordance with Project Manual table of contents. Provide complete screen-based menu.
3. File Names: Utilize file names based on name of equipment generally described in video segment, as identified in Project specifications.
4. Contractor and Installer Contact File: Using appropriate software, create a file for inclusion on the equipment demonstration and training recording that describes the following for each Contractor involved on the Project, arranged in accordance with Project Manual table of contents:

a. Name of Contractor/Installer.
b. Business address.
c. Business phone number.
d. Point of contact.
e. Email address.

B. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to adequately cover area of demonstration and training. Display continuous running time.

1. Film training session(s) in segments not to exceed 15 minutes.

a. Produce segments to present a single significant piece of equipment per segment.
b. Organize segments with multiple pieces of equipment to follow order of Project Manual table of contents.
c. Where a training session on a particular piece of equipment exceeds 15 minutes, stop filming and pause training session. Begin training session again upon commencement of new filming segment.

C. Light Levels: Verify light levels are adequate to properly light equipment. Verify equipment markings are clearly visible prior to recording.

1. Furnish additional portable lighting as required.

D. Preproduced Video Recordings: Provide video recordings used as a component of training modules in same format as recordings of live training.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 7900

SECTION 01 8800 – ADA ACCESSIBILITY

PART 1 - GENERAL

1.1 The Contract Documents have been designed to comply with ICC/ANSI A117.1 as well as with the Department of Justice's 2010 Americans with Disabilities Act (ADA) Standards for Accessible Design, with the more stringent of the two standards governing.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 ADA COMPONENT PROJECT SUBMITTALS AND CONTRACTOR SCHEDULED PROJECT SITE MEETINGS:

Within thirty (30) days after issuance of the Notice-to-Proceed (NTP), the Contractor shall submit for Owner review an ADA Compliance Submittal that outlines the Contractor's project specific ADA related work and execution plan. Subsequent to the Contractor making his ADA Compliance submittal, he shall schedule on- site review meetings to assure all work shall comply with all ADA provisions as reflected in the Contract Drawings and Specifications. At a minimum, the referenced plan shall reflect the following activities by the Contractor:

a. Contractor to schedule and host multiple ADA pre-construction meetings with the Owner to review all ADA aspects of the project prior to the start of various ADA sensitive installation activities. Activities include, but are not limited to, partition framing, MEP rough-in, ceramic tile installation, millwork installation, toilet room plumbing fixture placement and clearances, toilet accessory component placement and installation, water cooler placement and clearances, handrail and guardrail installation, construction of interior and exterior ramps, sidewalks, stairs reviewing slope & cross pitch, limited degree of pitch, at all path of travel surfaces including landings adjacent to ramps, stairs and entries, etc.

b. Review Contract Documents to identify items designed for conformance with accessibility criteria.

c. Review installation/construction procedures and goals as they relate to compliance with applicable accessibility criteria and measurement criteria.

d. Review all applicable accessibility criteria and methods of measurement of compliance.

e. Review subcontractor shop drawings for adherence to all applicable accessibility criteria as shown in the Contract Drawings and Specifications.

f. Develop a comprehensive checklist of all applicable accessibility criteria as shown on the Contract Drawings and Specifications for the purpose of periodic Quality Control inspections during construction and at Substantial Completion.

g. Contractor to schedule and host a construction-phase field meeting with the Owner on site prior to close-in to review all ADA aspects of the Project and to review compliance at rough-in phase.

h. Contractor to schedule and host a site hardscape review phase construction meeting, with the Owner present to review rough-ins of ADA aspects of the Project. This meeting is to be scheduled by the Contractor prior to concrete placement along the site "accessible path of travel" pavement. Railing and hardscape features shall be included. Concurrence regarding the ADA "path of travel" criteria including, but not limited to, accessible route(s) from

handicapped parking stalls on site to the building public entrance(s) shall be addressed and confirmed.

i. Field measurements and measurement criteria shall be applied to the accessible path(s) of travel slopes and cross slopes prior to placing concrete and shall be included as an agenda item of this meeting.

j. Provide ADA component updates and field installation schedules for Owner review during regular Progress Meetings.

END OF SECTION 01 8800

# SECTION 02 4119 - SELECTIVE DEMOLITION

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Demolition and removal of selected portions of building or structure.
2. Demolition and removal of selected site elements.
3. Salvage of existing items to be reused or recycled.

B. Related Requirements:

1. Section 01 1100 "Summary of Work" for restrictions on use of the premises, Owner-occupancy requirements, and phasing requirements.
2. Section 01 3453 “Hazardous Materials Remediation” for Hazardous Materials, responsibility and removal.
3. Section 31 1000 "Site Clearing" for site clearing and removal of above- and below-grade improvements not part of selective demolition.

### 1.2 DEFINITIONS

A. Remove: Detach items from existing construction and dispose of them off-site unless indicated to be salvaged or reinstalled.

B. Remove and Salvage: Detach items from existing construction, in a manner to prevent damage, and deliver to Owner ready for reuse or as indicated.

C. Remove and Reinstall: Detach items from existing construction, in a manner to prevent damage, prepare for reuse, and reinstall where indicated.

D. Existing to Remain: Leave existing items that are not to be removed and that are not otherwise indicated to be salvaged or reinstalled.

E. Dismantle: To remove by disassembling or detaching an item from a surface, using gentle methods and equipment to prevent damage to the item and surfaces; disposing of items unless indicated to be salvaged or reinstalled.

### 1.3 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition waste becomes property of Contractor.

B. Historic items, relics, antiques, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, and other items of interest or value to Owner that may be uncovered during demolition remain the property of Owner.

1. Carefully salvage in a manner to prevent damage and promptly return to Owner.

1.4 PREINSTALLATION MEETINGS

A. Pre-demolition Conference: Conduct conference at Project site.

1. Inspect and discuss condition of construction to be selectively demolished.
2. Review structural load limitations of existing structure.
3. Review and finalize selective demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
4. Review requirements of work performed by other trades that rely on substrates exposed by selective demolition operations.
5. Review areas where existing construction is to remain and requires protection.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For refrigerant recovery technician.

B. Proposed Protection Measures: Submit report, including Drawings, that indicates the measures proposed for protecting individuals and property, for environmental protection, for dust control and, for noise control. Indicate proposed locations and construction of barriers.

C. Schedule of Selective Demolition Activities: Indicate the following:

1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity. Ensure Owner's building manager's and other tenants' on-site operations are uninterrupted.
2. Interruption of utility services. Indicate how long utility services will be interrupted.
3. Coordination for shutoff, capping, and continuation of utility services.
4. Use of elevator and stairs.
5. Coordination of Owner's continuing occupancy of portions of existing building and of Owner's partial occupancy of completed Work.

D. Pre-demolition Photographs or Video: Show existing conditions of adjoining construction, including finish surfaces, that might be misconstrued as damage caused by salvage and demolition operations. Comply with Section 01 3233 "Photographic Documentation." Submit before Work begins.

E. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating that all refrigerant that was present was recovered and that recovery was performed according to EPA regulations. Include name and address of technician and date refrigerant was recovered.

F. Warranties: Documentation indicating that existing warranties are still in effect after completion of selective demolition.

1.6 CLOSEOUT SUBMITTALS

A. Inventory: Submit a list of items that have been removed and salvaged.

1.7 QUALITY ASSURANCE

A. Refrigerant Recovery Technician Qualifications: Certified by an EPA-approved certification program.

1.8 FIELD CONDITIONS

A. Owner will occupy portions of building immediately adjacent to selective demolition area. Conduct selective demolition so Owner's operations will not be disrupted.

B. Conditions existing at time of inspection for bidding purpose will be maintained by Owner as far as practical.

C. Notify Architect of discrepancies between existing conditions and Drawings before proceeding with selective demolition.

D. Hazardous Materials: Present in buildings and structures to be selectively demolished. A report on the presence of hazardous materials is on file for review and use. Examine report to become aware of locations where hazardous materials are present.

1. Hazardous material remediation is specified elsewhere in the Contract Documents.
2. Do not disturb hazardous materials or items suspected of containing hazardous materials except under procedures specified elsewhere in the Contract Documents.
3. Owner will provide material safety data sheets for suspected hazardous materials that are known to be present in buildings and structures to be selectively demolished because of building operations or processes performed there.

E. Storage or sale of removed items or materials on-site is not permitted.

F. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.

1. Maintain fire-protection facilities in service during selective demolition operations.

1.9 WARRANTY

A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during selective demolition, by methods and with materials and using approved contractors so as not to void existing warranties. Notify warrantor before proceeding.

B. Notify warrantor on completion of selective demolition and obtain documentation verifying that existing system has been inspected and warranty remains in effect. Submit documentation at Project closeout.

1.10 COORDINATION

A. Arrange selective demolition schedule so as not to interfere with Owner's operations.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

B. Standards: Comply with ANSI/ASSP A10.6 and NFPA 241.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that utilities have been disconnected and capped before starting selective demolition operations.

B. Verify that hazardous materials have been remediated before proceeding with building demolition operations.

C. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs or video and templates.

1. Inventory and record the condition of items to be removed and salvaged.

3.2 PREPARATION

A. Refrigerant: Before starting demolition, remove refrigerant from mechanical equipment according to 40 CFR 82 and regulations of authorities having jurisdiction.

3.3 UTILITY SERVICES AND MECHANICAL/ELECTRICAL SYSTEMS

A. Existing Services/Systems to Remain: Maintain services/systems indicated to remain and protect them against damage.

B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off utility services and mechanical/electrical systems serving areas to be selectively demolished.

1. Arrange to shut off utilities with utility companies.
2. If services/systems are required to be removed, relocated, or abandoned, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to other parts of building.
3. Disconnect, demolish, and remove fire-suppression systems, plumbing, and HVAC systems, equipment, and components indicated on Drawings to be removed.

a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.

b. Piping to Be Abandoned in Place: Drain piping and cap or plug piping with same or compatible piping material and leave in place.
c. Equipment to Be Removed: Disconnect and cap services and remove equipment.
d. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
e. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and deliver to Owner.
f. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
g. Ducts to Be Abandoned in Place: Cap or plug ducts with same or compatible ductwork material and leave in place.

## 3.4 PROTECTION

A. Temporary Protection: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.

1. Provide protection to ensure safe passage of people around selective demolition area and to and from occupied portions of building.
2. Provide temporary weather protection, during interval between selective demolition of existing construction on exterior surfaces and new construction, to prevent water leakage and damage to structure and interior areas.
3. Protect walls, ceilings, floors, and other existing finish work that are to remain or that are exposed during selective demolition operations.
4. Cover and protect furniture, furnishings, and equipment that have not been removed.
5. Comply with requirements for temporary enclosures, dust control, heating, and cooling specified in Section 01 5000 "Temporary Facilities and Controls."

B. Temporary Shoring: Design, provide, and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.

1. Strengthen or add new supports when required during progress of selective demolition.

C. Remove temporary barricades and protections where hazards no longer exist.

## 3.5 SELECTIVE DEMOLITION, GENERAL

A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:

1. Proceed with selective demolition systematically, from higher to lower level. Complete selective demolition operations above each floor or tier before disturbing supporting members on the next lower level.
2. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping. Temporarily cover openings to remain.

3. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
4. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations.
5. Maintain fire watch during and for at least <**Insert number**> hours after flame-cutting operations.
6. Maintain adequate ventilation when using cutting torches.
7. Remove decayed, vermin-infested, or otherwise dangerous or unsuitable materials and promptly dispose of off-site.
8. Remove structural framing members and lower to ground by method suitable to avoid free fall and to prevent ground impact or dust generation.
9. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.
10. Dispose of demolished items and materials promptly. Comply with requirements in Section 015000 "Temporary Construction Facilities, Utilities and Controls".

B. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

C. Removed and Salvaged Items:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in a secure area until delivery to Owner.
4. Transport items to Owner's storage area designated by Owner.
5. Protect items from damage during transport and storage.

D. Removed and Reinstalled Items:

1. Clean and repair items to functional condition adequate for intended reuse.
2. Pack or crate items after cleaning and repairing. Identify contents of containers.
3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.

E. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by Architect, items may be removed to a suitable, protected storage location during selective demolition **and cleaned** and reinstalled in their original locations after selective demolition operations are complete.

## 3.6 SELECTIVE DEMOLITION PROCEDURES FOR SPECIFIC MATERIALS

A. Concrete: Demolish in sections. Cut concrete full depth at junctures with construction to remain and at regular intervals using power-driven saw, and then remove concrete between saw cuts.

B. Masonry: Demolish in small sections. Cut masonry at junctures with construction to remain, using power-driven saw, and then remove masonry between saw cuts.

C. Concrete Slabs-on-Grade: Saw-cut perimeter of area to be demolished, and then break up and remove.

D. Resilient Floor Coverings: Remove floor coverings and adhesive according to recommendations in RFCI's "Recommended Work Practices for the Removal of Resilient Floor Coverings. Do not use methods requiring solvent-based adhesive strippers.

E. Roofing: Remove no more existing roofing than what can be covered in one day by re-roofing and so that building interior remains watertight and weathertight. See Section 07 0150 Preparation for reroofing requirements.

1. Remove existing roof membrane, flashings, copings, and roof accessories as indicated in contract documents.

## 3.7 DISPOSAL OF DEMOLISHED MATERIALS

A. Remove demolition waste materials from Project site and dispose of them in an EPA-approved construction and demolition waste landfill acceptable to authorities having jurisdiction. and recycle or dispose of them in accordance with Section 01 7419 "Construction Waste Management and Disposal."

1. Do not allow demolished materials to accumulate on-site.
2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
3. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.

B. Burning: Do not burn demolished materials.

## 3.8 CLEANING

A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

END OF SECTION 02 4119

## Part 1 General

### 1.1 Background

The former BB&T Bank and associated breezeway and semi-detached drive-through located at 115 W. King Street in Strasburg, VA is the subject of planned renovation; abatement of certain accessible Asbestos Containing Materials (ACMs) that will be impacted by the renovation work is planned in conjunction with the renovation. Note that wholesale abatement of all known ACMs is not currently anticipated.

An existing Limited Hazardous Materials Survey Reports dated February 6, 2025 prepared by Froehling and Robertson (F&R) for the Town of Strasburg (the Owner) as part of F&R project 59C-0339 was used in preparation of this section and, based on the report, the following building materials were identified as ACM (or with trace, <1% concentrations of asbestos) in the areas of potential impact, based on the scope of the renovation as understood at the time of the survey; the Contractor shall be familiar with the report and the homogeneous areas (HA) where ACM were identified and where presumed ACM (PACM) are possible. Note that the Sample Material Location(s) included in the table simply reflects the location(s) where representative sample collection was performed and does not describe all locations of the material at the facility. Reference attached drawings for locations of ACMs associated with vinyl and terrazzo flooring and ceiling skim coat.

Note that the Material Location and attached drawings presents known or strongly suspected locations of materials but may not recount all possible locations, particularly those concealed within walls, behind finishes, beneath newer flooring, et cetera. The original sample locations are recounted in the referenced February 2025 report. Additionally, note that the HA designations are based, in part, on assigned identity from the referenced report.

**ASBESTOS CONTAINING MATERIALS INVENTORY**

| HA | Material Description | Sample Material Location(s) | Asbestos Content | EPA/NESHAP Category | Estimated Quantity[1] |
|---|---|---|---|---|---|
| 2 | Vinyl Composite Tile (VCT): White w/Green Streaks 9”x 9” and Black Mastic | See Note 2 | 4%-6% Chrysotile | Non-friable Category I | 881 SF |
| 3 | VCT: Tan w/Brown and Yellow Streaks 9”x 9” and Black Mastic | See Note 3 | 5%-6% Chrysotile | Non-friable Category I | 486 SF |
| 4 | VCT: Tan w/Green Streaks 9”x 9” and Black Mastic | See Note 4 | 4%-5% Chrysotile | Non-friable Category I | 191 SF |
| 7 | VCT: Dark Brown w/Brown and White Streaks 9”x 9” and Black Mastic | Entrance: Vault 3; Entrance: Elect. Cabinet | 3% Chrysotile | Non-friable Category I | 614 SF |
| 15 & 16 | VCT: White w/Green Streaks 9”x 9” (remnant tile, beneath two carpet types with yellow mastic) | Entrance Door: Teller Booth; Rear Counter Behind Teller Station #2; Office 3: Near Entrance Door (Beneath carpet) | 3% Chrysotile | Non-friable Category I | Included in HA 2 Total |
| 23[5] | Skim Coat (Insulation) | Mech. Room: Ceiling Near AH Units | 3% Chrysotile 2% Amosite 2% Crocidolite | Friable RACM | 324 SF |

| HA | Material Description | Sample Material Location(s) | Asbestos Content | EPA/NESHAP Category | Estimated Quantity[1] |
|---|---|---|---|---|---|
| 24[6] | Terrazzo Tile: Speckled Pattern | Main Vestibule; Side Vestibule | Trace <1% Chrysotile (confirmed 0.50% Chrysotile via 400 Point Count) | Non-ACM (would be rendered friable through disturbance) | 1108 SF |
| NA[7] | Black Floor Mastic associated with asbestos containing Vinyl Floor Tile associated with HAs 2, 3, 4, 7, 15, and 16 | Throughout Facility | <1-6% Chrysotile | Non-friable Category II | 2172 SF |
| NA[8] | Fire Door interiors | Presumed[7] | | | Presumed[7] |
| NA[8] | Gaskets or packing materials | | | | Presumed[7] |
| NA[8] | Electrical panels/arc deflectors/spark arrestors | | | | Presumed[7] |
| NA[8] | Woven or paper-like wiring insulation or other live electrical system components | | | | Presumed[7] |
| NA[8] | HVAC components such as vibration dampers (if observed) or concealed components | | | | Presumed[7] |

[1]Estimated Quantity: SF = Square Feet, LF = Linear Feet, Units = Number of Units

[2]HA2 Sample Locations: Side Vestibule, Corridor near Office 5; Corridor, Behind Vault 2 Door; Entrance, Offices 4,5,6; Breakroom Entrance; Corridor Entrance near Vault 1; Corridor 109 near Rear Vestibule Entrance; Rear Counter behind Teller Station #2; Behind Vault 2 Door

[3]HA3 Sample Locations: Vault 2: Near C/D Corner; Vault 3: Near C/D Corner; Near Entrance Door Vault 2 (Side C); Near Entrance Door Vault 3 (Side C)

[4]HA4 Sample Locations: Corner near Entrance Door-Room 115 (Staff Bathroom); Plumbing Access, Space under Stairs; Breakroom Closet Entrance; Corridor to Rear Vestibule

[5]Lab analysis of Skim Coat and Insulation material (Sample 23B) removed from the Second Floor Mechanical Room ceiling indicated an asbestos content of 3% Chrysotile, 2% Amosite and 2% Crocidolite. As detailed in the referenced report, three supplemental samples of the Skim Coat and Insulation material were collected on January 24, 2025 from within the same area with lab analysis results of "No Asbestos Detected (NAD)" for each of the three samples. Based upon this analysis, F&R concludes the one sample is an anomaly pending any new information or further investigation. F&R therefore recommends vigilance during renovation work in order to determine whether the source of the anomalous ACM is revealed; such materials could constitute residual thermal systems insulation (TSI) or other materials which had been present but subjected to previous abatement efforts. Note that the ceiling in the mechanical room is not anticipated to be impacted by the project and that future occupants or owners should be informed of this condition.

[6]Trace (<1%) asbestos was identified in the terrazzo flooring; as detailed in the referenced report, this material was quantitatively confirmed to contain asbestos at 0.50%. It is therefore not an ACM; however, certain elements

of the OSHA standard are applicable to the material. As a best practice, F&R recommends that the material be treated as ACM for disturbance & handling purposes. Debris resulting from this material can be treated as non-ACM; details are presented in section 2.1.5. For the purposes of this specification, F&R has assumed that the terrazzo, where impacted, will be treated as ACM. The Contractor shall coordinate with the Architect and Owner to determine where such materials may be impacted to determine scope of potential abatement.

[7]Trace concentrations of asbestos were detected in several HAs. As noted in the referenced report, where materials have been documented to be ACMs and where those materials are similar to other materials which have not been found to be positive, those similar materials will be considered to be ACMs (i.e. where one material was analyzed and found to be positive, it is prudent to consider other similar materials positive, despite potential analytical data to the contrary). Specifically, for this site, F&R encountered a disparate results between multiple instances of black mastic collected in multiple locations across and throughout the facility. Results included concentrations ranging from NAD to 6% with instances of trace (<1%) and 5%; the data also indicated that both positive and negative mastics coexist. F&R intentionally reviewed the site in order to establish legitimately discernable trends such that certain black mastics could be treated as negative in order to minimize future abatement efforts. F&R attempted to identify characteristics such as associated floor tiles, office locations, apparent renovation vintage, and other aspects that sometimes permit distinguishing positive from negative black mastic; however, the efforts were not successful. Consequently, F&R recommends that all instances of black mastic encountered at the facility be treated as ACM, despite data to the contrary (including, for example, black mastic associated with HAs 15 & 16). Note that yellow mastic was found to be non-ACM in every instance where it was tested; consequently, instances of yellow mastic may be treated as non-ACM provided it is not concurrent with or contaminated by black mastic.

[8]Other ACMs may be present in the building which are not documented here but which, if present, could be impacted as part of the demolition. If work is performed that will impact presumed or other newly discovered materials that have not been sampled, F&R recommends that work that would impact that material would cease or that the material be treated as asbestos in accordance with this document. Samples of suspect materials should be collected by a Virginia licensed asbestos inspector and submitted to a NVLAP accredited laboratory for analysis and handled accordingly prior to disturbance.

### 1.1.1 Document Intent

The intent of this document is to provide guidance with regard to the abatement of known ACMs which will be impacted by the renovation activities within the former bank building and associated structures for conversion into a Police Station for the Town of Strasburg. While some of the renovation work may not impact the ACMs in the project area, historically a proactive approach to have **all ACMs removed in the project area** in order to preclude potential future impact has been recommended.

### 1.1.2 Consultant Contact and Signatures

This section was prepared under F&R project number 59B-0348 by:

**FROEHLING & ROBERTSON, INC.**
1734 Seibel Drive, N.E.
Roanoke, Virginia 24012
(540) 344-7939

Digitally signed by Jesse D. Phillips, CIH
Reason: J Phillips signed on behalf of S Bowman

Steven M. Bowman
Industrial Hygienist

Jesse D. Phillips, CIH
Senior Environmental Professional
VA Asbestos Designer # 3305 001167

### 1.1.3 Work Included

A. The work includes the furnishing of all labor, materials, equipment, insurance and services necessary for the completion of asbestos removal and related work.

B. Comply with all governing regulations which this Plan supplements.

C. Comply with applicable contract documents and any additional Town of Strasburg Requirements.

#### 1.1.3.1 Scope of Work

The Abatement Contractor has the responsibility for determining actual quantities of materials to be removed and reviewing the scope of work and developing an understanding of the existing report and specification documents; the values described below are for budgetary purposes and may not accurately reflect the values encountered in the field or the actual scope of abatement warranted by the project. The contractor shall provide lump sum costs for abatement of the ACMs at the quantities identified. Unit rates shall also be provided for each of the noted materials to be abated and for each of the materials identified in other areas of the building in the event that values greater than those anticipated are encountered during site work. Unit rates should also be provided for the noted presumed materials in the event that they are encountered during work.

The Owner may elect to reduce the scope of abatement work if it is determined that specific identified PACMs will not require removal in order to accomplish the proposed renovations, or if the materials are not present within the areas of renovation. The Contractor's unit rates may be utilized to adjust the costs associated with the abatement project. As an example, this is specifically the case for the terrazzo flooring.

The Contractor should allow under their base bid for the removal of all materials as described in the survey report and referenced in this specification. All mobilizations and permit notifications shall be the Abatement Contractor's responsibility. The base lump sum value will be reduced or expanded by the respective unit rates based on the actual quantity of materials encountered during the abatement work.

This scope of work includes, but is not limited to, the abatement of the materials identified in the table presented above in section 1.1. As has been stated repeatedly, the eventual abatement scope shall be confirmed in order to prevent impact to materials that may be designed to remain, such as terrazzo flooring.

We note that this document is specific to the asbestos abatement and that the asbestos abatement contractor and other site personnel are responsible for developing and implementing protective measures for their respective personnel and compliance with all federal and state requirements for asbestos abatement activities, including exposure monitoring.

A. The Scope of Work includes, but is not necessarily limited, to the following:

1) The Abatement Contractor shall be responsible for removal of all asbestos containing materials (currently identified as PACMs, and referred to interchangeably as either ACM or PACM throughout this document) noted above as required by the Building Owner or Architect for completion of the project.

2) All mobilizations permits, and notifications shall be the Abatement Contractor's responsibility.

3) This section includes all work necessary to prevent air concentrations of asbestos above the specified level and maintain the specified asbestos control limits during the life of the contract. It also contains procedures for removal, containment, and disposal of ACMs. The work specified in this document consists of the provision of services for the removal and disposal of ACMs present in materials identified at the site.

4) All ACM abatement and cleanup work will be performed by competent, licensed (by DPOR) asbestos workers and/or supervisors trained, qualified, and knowledgeable in the techniques of abatement, handling, and disposal of ACMs and materials contaminated by asbestos, in accordance with pertinent local, state, and federal regulations.

5) The Contractor is responsible for verifying quantities of ACM for abatement and providing unit rates, as applicable. Alternate cost estimates as acceptable to the Building Owner and/or his representative(s) may be arranged.

Requirements regarding the execution of the abatement are specified below, in Part 4.

#### 1.1.3.2 Bidding

The Abatement Contractor shall submit unit costs and base costs for abatement of the following items, based on the quantity depicted and the specified execution methods for each item, in accordance with Section 4.1 and other access area requirements as noted below. For presumed materials, unit rates shall be furnished.

| HA | Material Description | Estimated Quantity | Abatement Technique Options[1] |
|---|---|---|---|
| 2 | Vinyl Composite Tile (VCT): White w/Green Streaks 9”x 9” and Black Mastic | 881 SF | 4.1.1.1<br>4.1.1.2, or<br>4.1.1.5<br><br>**NOTICE: If abatement techniques cause non-friable materials to become friable (e.g. using a buffer with an abrasive pad and solvent to remove floor mastics) the work will be performed using wet methods within a full negative pressure containment consisting of critical barriers on openings, ceiling, floors, and walls (as applicable) with a minimum pressure differential of 0.02" water column as detailed below in Sections 4.1.1.2 and 4.1.1.5 (full and mini containment areas, respectively). All removed material shall be containerized and areas gross cleaned on a daily basis. Removed material will not remain on the floor overnight or during any non-work period.** |
| 3 | VCT: Tan w/Brown and Yellow Streaks 9”x 9” and Black Mastic | 486 SF | |
| 4 | VCT: Tan w/Green Streaks 9”x 9” and Black Mastic | 191 SF | |
| 7 | VCT: Dark Brown w/Brown and White Streaks 9”x 9” and Black Mastic | 614 SF | |
| 15 & 16 | VCT: White w/Green Streaks 9”x 9” (remnant tile, beneath two carpet types with yellow mastic) | Included in HA 2 Total | |
| 23 | Skim Coat (Insulation) | 334 SF | |

| HA | Material Description | Estimated Quantity | Abatement Technique Options[1] |
|---|---|---|---|
| 24 | Terrazzo Tile: Speckled Pattern | 1108 SF | |
| Various | Black Mastic | Included in values above; see note beneath table in section 1.1. | |

[1]The abatement technique options presented here accord with Sections 4, with access area details outlined in Section 4.1.1 eq. seq. Reference these sections.

**A. <u>Presumed Materials</u>**

a. If fire doors are detected and if the door is to be removed and replaced as part of the renovation, the following abatement procedures will be followed. The abatement contractor may remove the fire door in a Regulated Area via whole component removal. The door shall be wrapped in two layers of 6-mil thick polyethylene sheeting, sealed with duct tape, and labeled as asbestos waste. If it is necessary to remove the door frame, that work would take place in a negative pressure mini-containment with an appropriately sized HEPA filtered vacuum providing the air flow to create the negative pressure. After the mini-containment is constructed and negative pressure is created, and approved by the QP, the abatement crew with the appropriate PPE to include air filtering respirators can conduct destructive investigative measures to determine if suspect insulation is present in the door frame components. If insulation or other suspect ACM is not noted within the interior of the door frame the components can be removed and disposed of as regular construction debris. If suspect ACM insulation is noted the door frame components will be removed in as intact a state as possible, double wrapped in 6-mil polyethylene sheeting, wet-wiped and then passed out to the Regulated Area for transport to an appropriate container. After the door frame components have been removed, the interior of the mini-containment will be cleaned via HEPA filtered vacuum and wet wiping. At the discretion of the QP an air sample may be collected in the mini-containment prior to releasing for decommissioning.

b. Asbestos-containing gaskets and packing materials were not detected during the previous survey documentation. If, during the renovation, it is determined that gaskets and/or packing materials are ACM and it is necessary to remove the components to which the gaskets and/or packing materials are installed within, the following abatement procedures will be followed. Prior to any work that involves "breaking" a line the abatement contractor will first conduct all necessary lock out tag out (LOTO) procedures to include de-energizing the line and applying locks at appropriate locations. The abatement contractor will conduct the LOTO procedures with either the general contractor or an appropriate representative of the Town of Strasburg Department of Public Works & Utilities or the Shenandoah Valley Electric Cooperative (SVEC) dependent upon the line involved.. The abatement contractor may remove the gasket/packing material in a Regulated Area via whole component removal to the extent possible. If the whole component can be removed within which the gasket/packing materials are installed the component will be wrapped in two layers of 6-mil polyethylene sheeting, sealed with duct tape, and labeled as asbestos waste. If whole component removal is not a viable option and the gasket/packing materials must be physically removed this work will be performed in a Regulated Area using techniques that do not cause the material to become friable. If it is not possible to abate the material in a manner that does not cause it to become friable the abatement contractor shall either perform the removal via glove bag technology or within a mini-containment. If a mini-containment is constructed the guidance for the fire door frame procedure will be followed.

c. Electrical panel backing/arc deflectors/spark arresters were not tested during the previous survey documentation. If it is necessary to remove any of these electrical components and if appropriate testing has confirmed the presence of ACM (or based on the discretion of the QP

the material is designated as a PACM) the following abatement procedures will be followed. Prior to any work beginning the appropriate LOTO procedures will be conducted as per the direction for gaskets/packing materials. When de-energizing is confirmed, the abatement contractor will designate and demarcate an adequately sized Regulated Area within which to do the abatement. Abatement can be effected via whole component if possible by removing the component, wrapping in two layers of 6-mil thick polyethylene sheeting, sealing with duct tape, and labeling the parcel as asbestos waste. If whole component removal is not a viable option, the ACM will be removed in as intact a state as possible using hand-held tools and wet removal with a drop cloth under the immediate work area. If the ACM cannot be removed without rendering it friable the work will take place in a negative pressure mini-containment following the guidance as developed for the fire door frames.

d. CMU block walls were not investigated for the presence of vermiculite during the previous survey documentation. Previous penetrations of interior block walls within the Second Floor Mechanical Room were observed during the survey. If it is necessary to penetrate the CMU block walls during the renovation and testing has indicated that asbestos is present in the vermiculite, the following abatement procedures will be followed. The abatement contractor will coordinate with the general contractor or their designated representative to determine the exact area, if possible, or the vicinity of the area where the CMU block wall will be penetrated. The abatement contractor will construct a negative pressure mini-containment of sufficient size to allow two crew persons to work inside and have space to temporarily store bags of removed vermiculite. Workers will don the appropriate PPE to include air filtering respirators. After the QP has inspected and approved the mini-containment and after negative pressure has been created via a HEPA filtered vacuum, a drill fitted with a masonry bit will be used to create an opening in the CMU block with a diameter of at least 2.0 inches. The crew should anticipate that the possibility exists for a considerable amount of material to exit the interior of the wall and should be prepared to staunch the flow with a temporary seal if necessary. Vermiculite that initially flows from the wall should be wetted and carefully placed into a 6-mil thick polyethylene container labeled for asbestos. When the flow of vermiculite ceases and/or did not flow from the wall the abatement crew will utilize a HEPA filtered vacuum to remove as much vermiculite from the wall cavity as possible. The abatement will be deemed complete when vermiculite is no longer flowing from the opening and/or the interior of the wall cavity has been vacuumed to the extent that is practicable. After vacuuming of the interior of the wall is complete, the abatement contractor will spray an encapsulant into the wall cavity. The hole will be temporarily patched with a material that will offer more durability than duct tape. All vermiculite will be containerized, double-bagged, and handed out of the mini-containment. The interior of the mini-containment will be cleaned via HEPA filtered vacuum and wet wiping. At the discretion of the QP, air sampling may be performed prior to releasing the mini-containment for decommissioning. This procedure should be repeated at each area where the CMU wall will be penetrated. See 4.2 B for additional guidance.

e. Woven or paper-like wiring insulation and other live electrical system components were not tested during the previous survey documentation. If it is necessary to remove any wiring or any of these electrical components and if appropriate testing has confirmed the presence of ACM (or based on the discretion of the QP the material is designated as a PACM) the following abatement procedures will be followed. Prior to any work beginning the appropriate LOTO procedures will be conducted as per the direction for gaskets/packing materials. When de-energizing is confirmed, the abatement contractor will designate and demarcate adequately sized Regulated Areas within which to perform abatement; Regulated Areas shall be established at the point where wiring is accessed for removal and where the wiring will be cut (opposite end).

Abatement can be effected via whole component if possible by removing the component, wrapping in two layers of 6-mil thick polyethylene sheeting, sealing with duct tape, and labeling the parcel as asbestos waste.

If whole component removal is not a viable option, the ACM will be removed in as intact a state as possible using hand-held tools and wet removal with a drop cloth under the immediate work area. If the ACM cannot be removed without rendering it friable, the work will take place in a negative pressure mini-containment following the guidance as developed for the fire door frames.

Removal shall be effected via manual extraction with a Regulated Area at the point of removal and at the opposite termination point of the wire; HEPA filtered local exhaust ventilation shall be used at the point of extraction and the wire will be immediately containerized at the point of generation.

If conduit from which the material is being extracted is to be reused or is to remain in place, a light nylon or similar cord shall be connected to the terminus such that, during abatement, the string is accessible at the extraction point following abatement. Following abatement, negative pressure shall be applied to the conduit a HEPA filtered vacuum in order to remove errant fibers remaining in the space but preventing the cord from being pulled through; the cord will be then used to connect a brush or similar material roughly the inner diameter of the conduit, treated with amended water, at the opposite end of the conduit and pulled through the conduit to remove fibers. The brush shall also be connected to a cord or material to be used as electrical pull wire for subsequent wiring installation, if necessary.

f. Plumbing or electrical equipment located within or behind the ceiling/wall interiors were not tested during the previous survey documentation. If, during the renovation, it is determined that one or more of these components are ACM or if asbestos-containing TSI is present and it is necessary to remove the components the following abatement procedures will be followed. Prior to any work that involves “breaking” a line or handling electrical equipment the abatement contractor will first conduct all necessary lock out tag out (LOTO) procedures to include de-energizing the line and applying locks at appropriate locations. The abatement contractor will conduct the LOTO procedures with either the general contractor or an appropriate Owner representative. The abatement contractor may remove pipe TSI in a Regulated Area via whole component removal and/or glove bag. If the pipe TSI can be abated via whole component removal, it will still be necessary to remove a swath of the TSI via glove bag to create areas where the pipe can be cut. Pipe lengths should not exceed eight feet in length. For electrical equipment abatement can be effected via whole component if possible by removing the component, wrapping in two layers of 6-mil thick polyethylene sheeting, sealing with duct tape, and labeling the parcel as asbestos waste. If whole component removal is not a viable option, the ACM will be removed in as intact a state as possible using hand-held tools and wet removal with a drop cloth under the immediate work area. If the ACM cannot be removed without rendering it friable, the work will take place in a negative pressure mini-containment following the guidance as developed for the fire door frames.

## 1.2 Regulations

A. All work shall conform to the requirements of the U.S. Environmental Protection Agency (EPA), U.S. Department of Labor - Occupational Safety and Health Administration (OSHA), National Institute of Occupational Safety and Health (NIOSH) and applicable Commonwealth of Virginia regulations relating to asbestos and the specification contained herein. Where a conflict exists between these regulations and the project specifications, the most stringent shall apply. It is the intent of these

specifications to apply requirements over and above those minimum requirements of the listed regulations.

B. The contractor shall comply with all EPA and OSHA Regulations, and shall follow EPA workplace guidelines unless they are shown not to be applicable. Pertinent OSHA standards are the applicable sections of 29 CFR 1910 and 29 CFR 1926 including, but not limited to, 1910.1001, 1910.134 and 1926.1101. EPA workplace guidelines include those pertinent sections of Part I and II, EPA guideline document 560/5-85- 024,"Guidance for Controlling Asbestos Containing Materials in Buildings", and include but are not limited to EPA regulations 40 CFR Part 61 Subparts A and B. Also include all documents incorporated by reference herein.

C. The contractor shall remove, transport and dispose of the asbestos from the job site in compliance with 40 CFR Appendix D to Subpart E of Part 763 of the EPA regulation, the Virginia Department of Environmental Quality Regulations and these specifications. The contractor shall be responsible for obtaining the approval document for a waste disposal site in compliance with 40 CFR Appendix D to Subpart E of Part 763 of the EPA regulations and other governing regulations.

D. The contractor shall be licensed in accordance with the provision of Chapter 5 (54.1-503 et. seq.) of Title 54. The contractor shall provide a copy of their current, valid license with submittals.

E. The EPA and OSHA regulations shall be posted or shall be readily available digitally at the job site for the duration of the work; posting shall be in a location clearly visible or accessible to employees and others in the area.

**1.3 Definitions**

Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Any drawings must be recognized as diagrammatic in nature and not completely descriptive of the requirements indicated therein.

A. Abatement: Procedures to control fiber release from ACMs. Includes removal, encapsulation, enclosure, demolition, and renovation activities related to ACM.

B. Accredited/Accreditation: When referring to a person, Contractor or laboratory, means that such person is accredited in accordance with Section 206 of Title II of the Toxic Substances Control Act (AHERA Regulations).

C. Adequately wet: Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the ACM, then that material has not been adequately wetted.

D. Aerosol: A system consisting of particles, solid or liquid, suspended in air.

E. Aggressive Sampling: High-activity level air sampling which results in settled asbestos remaining airborne and uniformly disturbed through the use of special entrainment and mixing techniques. This makes any settled asbestos fibers accessible to the sampling filters for subsequent detection. The technique is described in 40 CFR 763.90, Appendix A to Subpart E; and Guidance for Controlling ACM in Buildings, Appendix M.

F. Air Filtration Device (AFD): Air filtration device (AFD) is part of the pressure differential system in which the air is filtered. The AFD is to be equipped with HEPA filters.

G. Airlock: shall mean a system for permitting entrance and exit while restricting air movement between a contaminated area and an uncontaminated area.

H. Air monitoring: The process of measuring the fiber content of a known volume of air collected over a specified period of time. The NIOSH 7400 Method, Issue 2 is used to determine the fiber levels in air. For personal samples and clearance air testing using Phase Contrast Microscopy (PCM) analysis. NIOSH Method 7402 can be used when it is necessary to confirm fibers counted by PCM as being asbestos. The AHERA TEM analysis may be used for background, area samples and clearance samples when required by this specification, or at the discretion of the IH as appropriate.

I. Air sample filter: The filter used to collect fibers which are then counted. The filter is made of mixed cellulose ester membrane with a 0.8 micrometer (µm) pore size for PCM (Phase Contrast Microscopy) analysis and a 0.45 µm pore size for TEM (Transmission Electron Microscopy) analysis; polycarbonate may also be used for sampling for TEM analysis.

J. Amended Water: Water to which a surfactant has been added to increase the penetrating ability of the liquid.

K. Approve: Where used in conjunction with the QP's response to submittals, requests, applications, inquiries, reports, and claims by the Contractor, "approved" will be held to limitations of QP's responsibilities and duties and does not release the Contractor from responsibilities to fulfill requirements of the Contract Documents. Approved shall also mean consent by U.S. EPA of training programs and the like.

L. Architect: Dewberry

M. Asbestos: The asbestiform varieties of serpentinite (chrysotile), riebeckite (crocidolite), cummingtonite-grunerite (amosite), anthophyllite, actinolite, and/or tremolite minerals. Both the asbestiform and non-asbestiform varieties of the above minerals and any of these materials that have been chemically treated and/or altered shall be considered to be asbestos.

N. Asbestos-Containing Material (ACM): Any material containing more than 1% by weight of asbestos of any type or mixture of types. In this document, ACM includes PACM or materials with trace (<1%) concentrations of asbestos, unless otherwise.

O. Asbestos-Containing Waste Material: Any material which is or is suspected of being or any material contaminated with an asbestos-containing material which is to be removed from a Work Area for disposal.

P. Air Lock: A system for permitting ingress or egress without permitting air movement from a contaminated area into an uncontaminated area, typically consisting of two curtained doorways at least 3 feet apart.

Q. Authorized Visitor: Personnel authorized by the Project Officer, testing lab personnel, or a representative of any Federal, State or local regulatory agency having authority over the project are considered authorized visitors.

R. Barrier: Any surface that seals off the Work Area to inhibit the movement of fibers.

S. Breathing Zone: A hemisphere forward of the shoulders with a radius of approximately 6 to 9 inches.

T. Ceiling Concentration: The concentration of an airborne substance that shall not be exceeded.

U. Certified Industrial Hygienist (CIH): An industrial hygienist certified in Comprehensive Practice by the American Board of Industrial Hygiene (ABIH) or Board for Global Credentialing (BGC).

V. Critical Barrier: Two layers of 6 mil polyethylene sheeting on walls or three layers on floors; spray foam, or duct tape used to completely seal off the Work Area to prevent spread of fibers to surrounding areas.

W. Containment: A Negative Pressure Enclosure with a pressure differential such that the air pressure within interior of the enclosure is less than that outside. This is achieved by application of Critical Barriers at windows, doors, ducts, or other similar openings in order to reduce communication between the interior and the exterior of the space. Pressure differential shall be at least 0.02 with the negative pressure within the work area.

X. Decontamination (Decon) Area: An enclosed area adjacent and connected to the regulated area and consisting of an equipment room, air lock, shower room, air lock, and a clean room which is used for the decontamination of workers, materials and certain equipment contaminated with asbestos. This shall serve as the only entrance or exit to the Work Area.

Y. Demolition: The wrecking or taking out of any building component, system, finish or assembly of a facility together with any related handling operations.

Z. Disposal Bag: A 6 mil thick, leak-proof polyethylene bag used for transporting asbestos waste from the work area to the disposal site. Each is labeled in compliance with OSHA 1926.1101 as follows:

**DANGER**
**CONTAINS ASBESTOS FIBERS**
**MAY CAUSE CANCER**
**CAUSES DAMAGE TO LUNGS**
**DO NOT BREATH DUST**
**AVOID CREATING DUST**

and U.S. DOT ORM-E label for Asbestos-Hazardous Material (including Asbestos Waste Manifest) and statements as required.

AA. Disturbance: means activities that disrupt the matrix of ACM or PACM, crumble or pulverize ACM or PACM, or generate visible debris from ACM or PACM, including cutting away small amounts of ACM and PACM.

BB. Encapsulant: A material that surrounds or embeds asbestos fibers in an adhesive matrix to prevent release of fibers.

a. Bridging Encapsulant: An encapsulant that forms a discrete layer on the surface of an in situ asbestos matrix.

b. Penetrating Encapsulant: An encapsulant that is absorbed by the in situ asbestos matrix without leaving a discrete surface layer.

c. Removal Encapsulant: A penetrating encapsulant specifically designed for removal of ACMs rather than for in situ encapsulation.

d. Lockdown Encapsulant: An encapsulant that is applied following removal which is applied in an aerosol form in order to capture and "lockdown" the microscopic airborne asbestos fibers to substrates (including poly sheeting) by the formation of a flexible resinous barrier.

CC. Encapsulation: Treatment of ACM with an encapsulant.

DD. Enclosure: The construction of an air-tight, impermeable, permanent barrier around asbestos-containing material to control the release of asbestos fibers into the air.

EE. Filter: A media component used in respirators to remove solid or liquid particles from the respired air.

FF. Friable Asbestos Containing Material: Material that contains more than 1.0% asbestos by Polarized Light Microscopy (PLM), and that can be crumbled, pulverized, or reduced to powder by hand pressure when dry. This includes previously non-friable material which becomes damaged to the extent that, when dry, may be crumbled, pulverized or reduced to powder by hand pressure.

GG. Furnish: Except as otherwise defined in greater detail, the term "furnish" is used to mean supply and deliver to project site, ready for unloading, unpacking, assembly, installation, etc., as applicable in each instance.

HH. General Supervisor: Site Superintendent, Foreman: is the Contractor's representative at the work site. This person can be the Competent Person required by OSHA, 29 CFR 1926.1101.

II. Glovebag: A sack (typically constructed of 6 mil transparent polyethylene) with two inward projecting long sleeve gloves, which are designed to enclose an object from which ACM is to be removed.

JJ. HEPA Filter: A high efficiency particulate air (HEPA) filter that removes from air 99.97% or more of monodispersed dioctylphthalate (DOP) or dioctylsebacate (DOS) particles having a mean particle diameter of 0.3 microns.

KK. HEPA Filter Vacuum Collection Equipment (or vacuum cleaner): HEPA filtered vacuum collection equipment with a filter system capable of collecting and retaining asbestos fibers. Filters shall be 99.97% efficiency for retaining fibers with an aerodynamic diameter of 0.3 microns.

LL. Indicated: The term "Indicated" is a cross-reference for Notes or Schedules on Drawings, to other paragraphs or Schedules in the Specifications, and to similar means of recording requirements in Contract Documents.

MM. Install: Unless defined in greater detail, "install" is used to describe operations at the project site including unloading, unpacking, assembly, erection, placing, anchoring, applying, working on dimension, finishing, curing, protecting, cleaning and similar operations, as applicable in each instance.

NN. Installer: The "installer" is defined as the entity (person or firm) engaged by the Contractor or Sub-Contractor to perform a particular trade at the work site, including installation, erection, application and similar required operations. Such entities (installers) shall be experts in operations they perform.

OO. Landfill Receipt: Document signed by a landfill operator acknowledging the receipt of ACM waste.

PP. Manifest: A document detailing chain of custody for ACM waste hauled.

QQ. Negative Pressure Glovebag: A glovebag which is composed of flexible plastic that can be subjected to negative pressure without collapsing.

RR. Negative Pressure Respirator: A respirator in which the air pressure inside the respiratory-inlet covering is positive during exhalation in relation to the air pressure of the outside atmosphere and negative during inhalation in relation to the air pressure of the outside atmosphere.

SS. Owner: Town of Strasburg Virginia

TT. Permissible Exposure Limit (PEL): The Contractor shall ensure that no employee is exposed to an airborne fiber concentration of asbestos in excess of the PEL expressed as an 8-hour TWA as determined by the OSHA Reference Method of 29 CFR 1926.1101 (Current PEL for asbestos is 0.1 fiber/cc.).

UU. Personal Sampling Monitoring: Air samples taken in the breathing zone of workers as required by OSHA 29 CFR 1926.1101.

VV. Pressure Differential: Air pressure lower than surrounding areas, caused by exhausting air from a sealed space (Work Area).

WW. Pressure Differential System: A local exhaust system, utilizing HEPA filtration, capable of maintaining a negative pressure differential inside the Work Area and a constant airflow from adjacent areas into the Work Area and exhausting that filtered air outside the Work Area.

XX. Project Manager (Contractor): The asbestos Contractor's employee responsible for the total oversight of the project.

YY. Plasticize: Means to cover floors and walls with polyethylene sheeting as herein specified and in accordance with the temporary Enclosure Section.

ZZ. Protection Factor: The ratio of the ambient concentration of an airborne substance to the concentration of the substance inside the respirator at the breathing zone of the wearer. The protection factor is a measure of the degree of protection provided by a respirator to the wearer.

AAA. Provide: Except as otherwise defined in greater detail, the term "provide" means furnish and install, complete and ready for intended use, as applicable in each instance.

BBB. Qualified Person (QP): A Registered Architect, Professional Engineer, or Industrial Hygienist who has successfully completed training and is therefore accredited under a legitimate State Model Accreditation Plan as described in 40 CFR 763 as a Building Inspector, Management Planner, Project Monitor, or Asbestos Project Designer. The QP must be qualified to perform visual inspections as indicated in ASTM E 1368. The QP shall be appropriately licensed in the State of Virginia as a Project Designer and/or Project Monitor. For this project, Froehling & Robertson, Inc. (F&R) personnel may serve as the QP. Individuals completing work in service of Section 3.2 shall be licensed as a Project Monitor.

CCC. Regulated ACM: Means friable ACM, non-friable ACM that has become friable, non-friable ACM that will be or has been subjected to sanding, grinding, cutting or abrading or non-friable ACM that has a high probability of becoming or has become crumbled, pulverized or reduced to powder by the forces expected to act on the ACM during renovation or demolition.

DDD. Regulated Area: An area where asbestos removal operations are performed which is isolated by physical boundaries to prevent entry of unauthorized persons or the spread of asbestos dust, fibers or debris. Within this area, the airborne concentration of asbestos could reasonably be expected to exceed the PEL. Critical barriers shall be applied within each Regulated Area.

EEE. Removal: The taking out or stripping of all ACM from a damaged area or associated area or space.

FFF. Respirator: A device designed to protect the wearer from the inhalation of harmful atmospheres.

GGG. Short-Term Exposure Limit (STEL): A "ceiling" concentration, identified in OSHA regulations, of an airborne substance that shall not be exceeded for a duration of any 30-minute period (Current STEL for asbestos is 1.0 fiber/cc and is typically referred to as the excursion limit.).

HHH. Submittal: Items which are required to be presented to the Project Officer and/or the QP for review, consideration, or decision.

III. Surfacing Material: Material in a building that is sprayed-on, troweled-on or otherwise applied to surfaces or structural members for acoustical, fireproofing or other purposes.

JJJ. Surfactant: A chemical wetting agent added to water to improve penetration, thus reducing the quantity of water required for a given operation or area.

KKK. Testing Laboratory: The "testing laboratory" is an independent entity to perform specific air sampling and analysis at the work site and associated areas, to report and (if required) interpret results. Analysis shall be performed by a laboratory accredited by the American Industrial Hygiene Association (AIHA) and having demonstrated a proficient rating in AIHA's Proficiency Analytical Testing (PAT) Program. The laboratory shall be licensed by DPOR as an Asbestos Analytical Laboratory. The laboratory shall also be accredited by the National Institute of Standards and Technology (NIST) through the National Voluntary Laboratory Accreditation Program (NVLAP) for bulk sample analysis and air sample analysis by TEM (TEM Method of 40 CFR 763, Subpart E, Appendix A).

LLL. Time Weighted Average (TWA): The average concentration of a contaminant in air during a specific time period.

MMM. Visible Emissions: Any emissions containing particulate asbestos material that are visually detectable without the aid of instruments. This does not include condensed water vapor.

NNN. Waste Shipment Record: Means the original shipping document, originated and signed by the waste generator (Abatement Contractor) used to track and substantiate the disposal of ACM waste as described in 40 CFR Part 61.

OOO. Waste Generator: Means the licensed Asbestos Abatement Contractor removing ACM waste from the property.

PPP. Wet Cleaning: The process of eliminating asbestos contamination from building surfaces and objects by using cloths, mops, or other cleaning utensils that have been dampened with amended water or diluted removal encapsulant and afterwards thoroughly decontaminated or disposed of as asbestos-containing waste.

QQQ. Work Area: The area where asbestos-related work or removal operations are performed; the Work Area is defined and/or isolated to prevent the spread of asbestos dust, fibers, or debris, and entry by unauthorized personnel. The Work Area is a Regulated Area as defined by 29 CFR 1926.1101.

RRR. Work Site: The term "work site" is defined as the space available to the Contractor for performance of the work either exclusively or in conjunction with others performing other work as part of the project. The extent of the project site is shown on the Drawings, and may or may not be identical with the description of land upon which the project is to be built.

**1.4 Abbreviations and names**

The following acronyms or abbreviations referenced in Contract Documents are defined to mean the associated names. Both names and addresses are subject to change and are believed to be, but are not assured to be, accurate and up-to-date as of the date of the Contract Documents:

| | |
|---|---|
| ACM | Asbestos Containing Material |
| AIA | American Institute of Architects<br>1735 New York Avenue, N.W.<br>Washington, D.C. 20006<br>(800) 242-3837 |
| ANSI | American National Standards Institute<br>25 West 43rd Street<br>4th Floor<br>New York, NY 10036<br>(212) 642-4900 |
| ASTM | American Society for Testing and Materials<br>100 Barr Harbor Drive<br>West Conshohocken, PA 19428<br>(610) 832-9500 |
| CFR | Code of Federal Regulations<br>Available from Government Printing Office<br>Washington, D.C 20402 (Usually first published in Federal Register) |
| EPA | Environmental Protection Agency<br>1650 Arch Street<br>Philadelphia, PA 19103<br>(800) 438-2474 |
| f/cc | fibers per cubic centimeter |
| NIOSH | National Institute for Occupational Safety and Health<br>Patriots Plaza 1<br>395 E Street, SW, Suite 9200<br>Washington, DC 20201<br>(800) 232-4636 |
| NIST | National Institute of Standards and Technology<br>(U.S. Department of Commerce)<br>100 Bureau Drive |

| | |
|---|---|
| | Gaithersburg, MD 20899<br>(301) 975-2000 |
| OSHA | Occupational Safety and Health Administration<br>(U.S. Department of Labor)<br>200 Constitution Avenue, NW<br>Washington, DC 20210<br>(800) 321-6742 |
| TEM | Transmission Electron Microscopy |
| TSI | Thermal Systems Insulation |
| VA DOLI | Virginia Department of Labor and Industry<br>Main Street Center<br>600 East Main Street, Suite 207<br>Richmond, VA 23219<br>(804) 371-2327 |
| VA DPOR | Virginia Department of Professional and Occupational Regulation<br>Perimeter Center<br>Suite 400<br>9960 Mayland Drive<br>Richmond, VA 23233<br>(804) 367-8500 |
| UL | Underwriters Laboratories<br>333 Pfingsten Road<br>Northbrook, IL 60062<br>(847) 272-8800 |

**1.5 Quality Controls**

A. The asbestos removal Contractor's Supervisor shall be on the job each day during removal and he shall be knowledgeable, experienced and competent in this type of work and shall enforce the use of all safety procedures and equipment. He shall be knowledgeable of relevant EPA, OSHA and NIOSH requirements and guidelines.

B. The Supervisor shall have a current, valid asbestos supervisor's license issued by the Virginia DPOR in accordance with the provisions of Chapter 5 (54.1-503 et. seq.) of Title 54.

C. The asbestos removal Contractor shall be responsible for any damage to the building and its contents resulting from leakage or spillage of water.

D. Authorities of the Commonwealth of Virginia shall be notified of the starting date of the asbestos removal project by the asbestos removal Contractor, as required. Notifications shall be maintained and all work will be conducted within the specific notification times and dates specified on the notification or amendments. All amendments will be the responsibility of the Abatement Contractor and said contractor shall furnish copies of amendments and related documentation illustrating submission to DPOR to the QP/Project Monitor.

E. The Owner reserves the right to halt the project work until hazardous or potentially hazardous conditions are corrected.

F. The Owner reserves the right to independently perform such analysis and tests at any time as he deems necessary to ensure and protect safety of the project.

## 1.6 Worker Protection

A. Comply with all EPA and OSHA Regulations, and follow EPA workplace guidelines.

B. Provide and maintain negative air systems for all work areas when necessary, for the duration of asbestos removal work.

### 1.6.1 Respirators

A. Workers shall wear properly fitted respirators in the work area. Respirator selection shall be based on personal air monitoring as required by 29 CFR 1926.1101 and OSHA 1910.134, as suitable for the asbestos exposure in the work area. Long sideburns, beards etc., which interfere with proper fit are unacceptable. The Contractor shall provide sufficient filters for replacement of disposable type filters.

B. In accordance with 29 CFR 1926.1101(h)(3)(iv) et. seq., all work involving the removal of surfacing or TSI constituting ACMs requires the use of Type "C' supplied air respirators unless respirators of lower protection factor are authorized by the Owner or his representative AND it can be demonstrated that conditions are anticipated to warrant a lower protection factor. Supplied air quality shall be C.G.A. grade D breathing air or better from a certified air source. Copies of these certificates will be supplied to the Owner or its representative. Consequently, half face negative pressure respirators are assumed acceptable provided they offer an adequate protection factor. However, OSHA compliance is the responsibility of the Contractor; initial exposure assessments shall be conducted in order to document that the respiratory protection used is adequate for potential exposures or to document a negative exposure assessment in accordance with Section 1.6.6, below.

C. Supplied air respirator systems must include a back-up provision approved for maintaining air flow long enough for escape and decontamination from a contaminate atmosphere in the event of loss of the primary source of breathing air.

D. All workers shall be instructed in and be knowledgeable of the following:

1) The hazards of asbestos exposure.
2) Use of respirators and protective clothing.
3) Use of personal air monitoring equipment.
4) Use of decontamination facilities and designated showers.

### 1.6.2 Respiratory Protection Program

Provide a copy of a Written Respiratory Protection Program on the job site at all times. The Contractor shall develop and implement a written Respiratory Protection Program (RPP) which is in compliance with the OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.134. ANSI Standard Z88.2 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH approved for asbestos abatement activities. The written RPP shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c)(1)(i - ix) - Respiratory Protection Program.

### 1.6.3 Respirator Fit Check

The Competent Person shall assure that the positive/negative pressure user seal check is performed each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that

prevents an effective face piece to face seal as evidenced by failure of a user seal check shall preclude that person from wearing a respirator inside the regulated area until resolution of the problem.

### 1.6.4 Training of Abatement Personnel

Prior to beginning any abatement activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.1101 (k)(9) and any additional State/Local requirements. Training must include, at a minimum, the elements listed at 29 CFR 1926.1101 (k)(9)(viii). Training shall have been conducted by a third party, EPA/State approved trainer meeting the requirements of EPA 40 CFR 763 Appendix C (AHERA MAP). Initial training certificates and/or current refresher and accreditation proof must be submitted for each person working at the site PRIOR TO the trained individual performing work.

### 1.6.5 Medical Examinations

Medical examinations meeting the requirements of 29 CFR 1926.1101 (m) shall be provided for all personnel working in the regulated area, regardless of exposure levels. A current (dated within the preceding 12 months) physician's written opinion as required by 29 CFR 1926.1101 (m)(4) shall be provided for each person and shall include in the medical opinion the person has been evaluated for working in a heat and cold stress environment while wearing personal protective equipment (PPE) and is able to perform the work without risk of material health impairment. Documentation of current medical evaluation allowing the use of PPE onsite must be submitted for each person working at the site PRIOR TO the trained individual performing work.

### 1.6.6 Personnel Air Monitoring

The abatement contractor shall provide air monitoring in the work areas throughout all asbestos stripping, removal and cleaning operations to ensure that the workers are adequately protected at all times. All personal air monitoring for OSHA compliance shall be the responsibility of the Contractor. In lieu of air monitoring, a negative exposure assessment (NEA) developed in conjunction with an initial exposure assessment (IEA) that is not older than 12 months old will be accepted. The IEA must represent the same conditions as will be encountered during this abatement activity. If an IEA is not available, the contractor can utilize the results of personal air monitoring on the first full day of abatement to develop the NEA, for each similar exposure group present (SEG) onsite; this would require testing for each material type and removal method combination used.

Samples for air monitoring shall be collected by a competent person in accordance with methods prescribed in Chapter 1 of Section II of the Federal OSHA Technical Manual or by equivalent procedures. Air monitoring shall be in compliance with 1910.1001 (g) of the OSHA standards. Air samples must be analyzed by NIOSH method 7400 by a laboratory that participates in the AIHA administered PAT Round program and is currently rated as proficient.

Air monitoring (protection of the Contractor's employees) shall be provided throughout the removal and cleaning operations. Air monitoring shall be conducted and evaluated by a testing laboratory employed by the asbestos removal Contractor to ensure that the Contractor is complying with applicable EPA and OSHA regulations.

#### 1.6.6.1 Personal Monitoring Results

When necessary, all personal monitoring (Personnel breathing zone samples) results shall be made by the asbestos removal Contractor on a daily basis for determination of both 8-hour time weighted average (TWA) and ceiling concentrations of employee exposures and are to be received within 24 hours and retained at the work site where they may be reviewed by the Owner's representative. Submit copies of these results at the completion of the project.

The sampling schedule may be posted outside of the containment area showing sample frequency, duration of the sample, and pump flow rates. Posted results should include a synopsis of work activities of which the results are representative.

### 1.6.7 Ambient Air Monitoring

1. Environmental samples outside of containment/regulated area and clearance sampling shall be performed by the QP.

2. Area samples shall be collected outside the containment/regulated area in areas of highest risk of contamination or in areas of concern based on the professional experience and judgment of the QP.

3. When glove bag activities are occurring, air samples shall be collected on a daily basis within the area where the glove bag(s) are located.

4. All analytical results shall be presented as signed "Certificates of Analysis". The form shall state:

    i. A specific sample ID for each sample
    ii. Date sampling and analysis occurred.
    iii. Volume of air collected for each sample.
    iv. The number of fibers and fields counted during each analysis.
    v. Concentration of fibers in fibers per square millimeter and cubic centimeters of air.
    vi. Specific area/individual sampled.
    vii. Signature of Analyst and reviewer.

5. Analytical results shall be available at the job site within 48 hours of sample analysis (excluding non-working days).

6. The Owner and Abatement Contractor shall be informed immediately of any area samples outside the containment or regulated work area results in excess of 0.1 fibers/cc. The Contractor shall be informed immediately of any area samples outside the containment or regulated work area with concentrations in excess of 0.01 f/cc. Work may be halted at the discretion of the QP.

7. Copies of the results of all samples made in areas where Owner's employees are or may be exposed shall be given to the Owner to assure maintenance of records in compliance with OSHA standard 1926.1101 (n).

8. Operations shall be discontinued immediately at any time visible emissions are observed emanating from the containment/regulated area or from any glovebag.

9. For this project, full-time daily project monitoring may be performed at the discretion of the Owner. As a best practice, full time daily monitoring should be performed.

## Part 2 Submittals

Immediately upon award of the Contract and before any work is commenced, Contractor shall submit the following to the Owner and QP in quantity to allow the Owner to retain two copies and the QP to retain one copy (or digitally). The provided information shall show compliance with the requirements of the Contract Documents and governing regulations. **Failure to comply with this request will cause delays and complications associated with the DOLI permit. The Contractor will not be allowed to begin any abatement-associated activity until such time that ALL requested submittals have been received, reviewed, and any discrepancies resolved. Requested submittals are summarized as follows and greater detail is provided in the following sections:**

- A Written Asbestos Abatement Plan
- A copy of the permit notification to DOLI, if required
- Name and other relevant information of the laboratory used for personal air sample analysis
- Copies of Safety Data Sheets (SDS) for chemicals and other materials that will be on site
- The name and address of the landfill where containerized asbestos materials and waste will be transported to for disposal. Additional information for the landfill where contaminated materials with trace (<1%) asbestos or where soil will be disposed of if different than the asbestos landfill
- ANSI Z9.2 certificates of compliance for equipment used during the project
- Qualifications of supervisor and workers to include:
    - Current DPOR license
    - Proof of initial or most recent training course not older than 12 months
    - Proof of fit test not older than 12 months
    - Proof of training for procedures in the event of an emergency
- Copy of a medical surveillance program to include physician's statement that employees that will don respirators have been medically cleared
- A copy of the company Written Respiratory Protection Program
- An example of the daily log and visitor log that the abatement supervisor will maintain during the project
- Previous sampling data representative of planned work and supporting documentation (if relaxed respiratory protection requirements will be practiced, as noted in section 1.6.1 item B.).

## 2.1 Pre-Start Contractor Submittals

The following items shall be submitted and approved by the Building Owner with the exception being the landfill manifest receipt, logs, and air monitoring results for review prior to mobilization. This requirement may be waived by the Building Owner; however, these items must be provided and maintained at the job site prior to beginning work and maintained onsite during all work with other materials or documentation, as noted elsewhere herein.

### 2.1.1 Asbestos Plan

Submit a detailed abatement plan of the work procedures to be used in the removal of the materials containing asbestos. Such plans shall include the location of the asbestos work areas, change rooms, layout of change rooms, interface with other trades involved in this project if applicable, sequencing of asbestos related work (method and means of removal and encapsulation of ACMs and containment and shrouding procedures, including unusual conditions), disposal plan, type of wetting agent, asbestos sealer, air monitoring (sampling plan), and detailed description of the methods to be employed in order to control air pollution.

Should the contractor subsequently desire to vary from this plan, he shall submit a written request to do so. Work shall not commence until the abatement plan is approved by the Owner and the QP.

### 2.1.2 Notification

Submit a copy of the required notification which will be submitted to the Department of Labor and Industry, as warranted. The following is excerpted from the Virginia Department of Labor & Industry FACT Sheet. DOLI's Asbestos Notification and Permit Program regulations require written notification by licensed asbestos contractors for any asbestos abatement project that is at least ten linear or ten square feet. A notification is not required for non-friable asbestos containing roofing, flooring, or siding materials which when installed, encapsulated, or removed do not become friable. Notifications must be submitted 20 calendar days prior to starting the asbestos project. Notifications should be sent to the address below by certified mail or hand delivery. Notifications may also be sent by facsimile (FAX) transmission for projects that do not require a fee or projects with fees paid by credit card. A notification form can be obtained online from DOLI at:

https://www.doli.virginia.gov/asbestos-and-lead-programs/asbestos-and-lead-forms/ or by contacting the Department in writing at the address below:

Asbestos Program Department of Labor and Industry
600 East Main Street, Suite 207
Richmond, Virginia 23219
Phone (804) 371-2327
Fax (804) 371-7634.

In accordance with a December 22, 2021 EPA Memorandum, notifications are no longer required by the Region 3 of the EPA. As noted in the document, "Complete and timely notifications must be sent to the state or local authority where the work is occurring, but they do not need to be sent to EPA. This applies to any demolition or renovation work occurring in Pennsylvania, including Allegheny County and Philadelphia; Delaware; Maryland; West Virginia: and Virginia. For any demolition or renovation work occurring in the District of Columbia, asbestos notifications are still required to be sent to both DC and EPA Region 3."

Based on the anticipated quantities of ACM, notification to DOLI is required. Additionally, if buffer or other equipment or methods are used which will render category I or II materials into friable RACM, the notification would have to be amended. **NOTICE: If abatement techniques cause non-friable materials (including floor tiles, mastic, or otherwise) to become friable (e.g. using a buffer with an abrasive pad and solvent) the work will be performed using wet methods within a full negative pressure containment consisting of critical barriers on openings, ceiling, floors, and walls (as applicable) with a minimum pressure differential of 0.02" water column as detailed below in Sections 4.1.1.2 and 4.1.1.5 (full and mini containment areas, respectively).**

### 2.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected for the monitoring of airborne concentrations of asbestos fibers along with a copy of the Commonwealth of Virginia Asbestos Analytical Laboratory License. Also include evidence that the laboratory is accredited to analyze airborne asbestos fibers counts. Provide a copy of the American Industrial Hygiene Association (AIHA) accreditation.

### 2.1.4 SDS

Submit the manufacturer's technical Safety Data Sheets (SDS) on proposed surfactant, encapsulant, mastic removers, water amendments, etc.

### 2.1.5 Landfill

When friable materials are abated, submit written evidence (copy of permit) that the landfill for disposal is approved for disposal of friable asbestos materials by the US EPA and state or local regulatory agencies and where the landfill site is located. Manifest documentation of disposal at landfills accepting non-friable ACM shall also be provided and shall clearly document the designation of the material as ACM. Within thirty-five days of the deposit of a load of waste from this project at the designated landfill, the contractor shall submit a copy of the waste system record to the Owner. The Owner shall have received all acceptable waste manifests prior to making any payments to the contractor.

For disposal of materials quantitatively confirmed to contain detectable but less than one percent (<1%) asbestos, the contractor shall coordinate with a landfill and document that the receiving landfill is aware of the potential for asbestos to be in the waste material prior to transfer. Alternately, such materials may be treated as RACM for disposal purposes.

### 2.1.6 Certificate of Compliance

A copy of the manufacturer's certificate of compliance with ANSI Z9.2 for each brand and model of vacuum, ventilation and other equipment used by the contractor to contain or remove asbestos fibers.

### 2.1.7 Qualifications

Prior approval by the owner is required of all proposed asbestos removal personnel. Approval shall be based on review and acceptance of contractor's written submittals that all of its personnel working in this asbestos project:

1. Have a current, valid asbestos Worker's or Supervisor's license issued by DPOR in accordance with the provisions of Chapter 5 (54.1-503 et. seq.) of Title 54.

2. Have taken an initial training course for the particular discipline and license, annual refresher training on an on-going basis as required to maintain accreditation, and completed a refresher course in the last twelve months.

3. Have been provided with a respirator fit test in accordance with 29 CFR 1926.134 at the time of the initial fitting, when facial conditions change, and at least every twelve months thereafter for each employee wearing a negative-pressure respirator.

4. Documentation of compliance with part 1.6.5, above, requiring annual medical evaluation in order to demonstrate ability to perform the scope of work safely using the required PPE.

5. Have been trained in the proper procedures to follow in the event of an emergency.

### 2.1.8 Medical Requirements

Submit certification that contractor has an established medical surveillance program in compliance with OSHA regulations 29 CFR 1926.1101. This submittal shall include copies of the physician's statement that each employee (working on this project) is able to perform his duties while wearing a respirator.

### 2.1.9 Respirator Program

Submit a copy of a Written Respiratory Protection Program that complies with OSHA regulation 29 CFR 1910.134.

### 2.1.10 Logs

Copies of daily progress log and visitor's log.

## Part 3 Products

### 3.1 Products and Equipment

Prior to the start of work, the contractor shall provide and maintain a sufficient quantity of materials and equipment to assure continuous and efficient work throughout the duration of the project.

A. Protective plastic (polyethylene) sheeting of minimum 6 mil thickness and size to provide protection to all equipment, floors, walls, ceilings, and all other exposed areas (as necessary), with minimum frequency of joints, where required. Plastic shall not be moved and reused between work areas but shall be containerized for disposal as ACM waste in accordance with Section 4.2. However, for areas where

a Modified Containment will be established, plastic may not be required on ceilings or full wall height, at the discretion of the QP.

B. Seal tape shall be glass fiber reinforced or other type capable of sealing joints of adequate sheets of plastic for the attachment of plastic sheeting to finished or unfinished surfaces of dissimilar materials under either dry or wet conditions, including use of amended water.

C. Disposal Containers: Bags and drums to be used for disposal of asbestos waste shall be suitable to receive and retain any asbestos-containing or contaminated materials until disposal at an EPA approved and certified waste disposal site. Bags shall be 6 mil thicknesses. If using drums, the material must be double bagged or lined before being placed in the drum. Drums entering the work area must be covered with poly and upon exiting the work area to the load out area the poly is to be removed and the drum is to be wet-wiped.

    a. Disposal bags – 2 layers of 6 mil poly for asbestos waste shall be pre-printed with labels, markings, and the Facility's address as required by OSHA, EPA and DOT regulations.

    Note that the Contractor may use a burlap bag on the interior to reduce tears or breakage of outer bags but this does NOT preclude the need to double bag as noted above.

D. Installation and plumbing hardware, showers, hoses, drain pans, sump pumps, and waste water filtration system shall be provided by the Contractor.

E. An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, disposable mops, rags, sponges, staple guns, shovels, ladders and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the abatement project shall be provided by the contractor. All electrically operated hand tools, equipment, electric cords shall be connected to ground fault circuit interrupter (GFCI) protection.

F. Special protection for objects in the regulated area shall be detailed (e.g., plywood over electrical panels or intact non-ACM pipe TSI to prevent damage from ladders, water and falling material).

G. The owner and QP shall be provided an advance copy of the SDS as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication in the pre-start submittal. Chlorinated compounds shall not be used with any spray adhesive, mastic remover, or other product. Appropriate encapsulant(s) shall be provided.

H. OSHA DANGER demarcation signs, as many and as required by OSHA 29 CFR 1926.1101(k)(7) shall be provided and placed by the Competent Person. All other posters and notices required by Federal and State regulations shall be posted or shall be readily available digitally in the Clean Room.

I. Adequate and appropriate PPE for the project and number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d). The Contractor shall furnish disposable coveralls to visitors or QP personnel sufficient to inspect work areas as needed.

J. Surfactant (wetting agent for amended water): Acceptable surfactant manufactured for that purpose.

K. Encapsulant: Acceptable encapsulant manufactured for that purpose. When remaining ACM is present and it is acceptable by the Owner, the bridging encapsulant shall be appropriately tinted with pigment to readily indicate adequate coverage.

L. Solvent: The contractor shall confirm with the Owner that solvents, if used, are compatible with future finishes and associated adhesives. Failure to comply with this requirement would result in the Contractor being responsible for resolution due to incompatibility. Alternate treatment schemes to denature solvents, if required, shall be approved by the Owner prior to performance.

### 3.1.1 Equipment

A. All protective equipment requires the approval of the Owner. Workers shall wear disposable, full body coveralls (including foot coverings), disposable head covers, and disposable footwear in the work area. Required protective equipment also includes goggles or safety glasses if half-faced masks are used, and disposable gloves taped to the sleeves of the suit. If heavy reusable gloves are necessary, they shall be of a type approved by the Owner. Such gloves shall be rinsed at the end of each day. Prior to disposal, contaminated water shall be filtered to retain particles 5.0 microns or larger or disposed of as asbestos waste. Removal of coverall foot coverings shall NOT be allowed unless non-porous boots are donned prior to entering the work area and the legs of coveralls are taped to boots.

B. Make available to the owner's representative complete sets of personal protective equipment as required herein for entry to the asbestos control area at any time for inspection of the asbestos control area.

C. Air Handling Equipment: Equipment shall be high-efficiency particulate air (HEPA) absolute filtration system, equipped with filtration equipment in compliance with ANSI Z9.2. No air movement system or air equipment shall discharge asbestos fibers outside the work area.

### 3.1.2 Warning Signs and Labels

A. Warning Signs: As required by OSHA Regulation 29 CFR 1926.1101 (k)(7). Provide warning signs at all approaches to the asbestos control areas containing concentrations of airborne asbestos fibers. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scraps, waste, debris and other products contaminated with asbestos. Signage shall be in English and the predominant language of the asbestos workers are required.

a. Warning sign: The sign shall bear the following information:

DANGER
ASBESTOS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
AUTHORIZED PERSONNEL ONLY

In addition, where the use of respirators and protective clothing is required in the regulated area the warning sign shall include the following:

WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA

b. Container Label: provide labels of sufficient size to be clearly legible conforming to requirements of 1926.1101.

### 3.1.3 Decontamination Areas

A. The contractor shall set up a decontamination area(s) including an equipment room, clean room, and a shower/decon chamber outside the work area(s). These areas shall include a clean room connected by a double barrier to a walk through shower which is also connected by a double barrier to an equipment room. Adequate storage containers shall be provided for each worker's street clothes. The equipment, clean room and shower areas shall be cleaned daily.

   a. In lieu of a shower, and assuming that all asbestos abatement is performed using glovebag techniques or removal of non-friable Category II ACM in a regulated area (not within a negative pressure containment), the contractor, upon approval by the QP, can construct a decontamination (decon) chamber within which will be placed a CLEAN HEPA filtered vacuum, disposable cloths and a source of water for worker decontamination. These areas shall include a clean room connected by a double barrier to a walk through shower/decon chamber which is also connected by a double barrier to an equipment room. Adequate storage containers shall be provided for each worker's street clothes. The equipment, clean room and shower/decon chamber areas shall be cleaned daily. At the discretion of the QP and based upon the nature of the material to be abated, the decontamination area may exclude double barrier corridors; the equipment room may constitute an equipment staging area if maintained adequately clean.

   b. In lieu of a shower, and assuming that all asbestos abatement of non-friable Category II ACM is performed in a regulated area (not within a negative pressure containment), the contractor can construct a decontamination (decon) area within which will be placed a CLEAN HEPA filtered vacuum, disposable cloths and a source of water for worker decontamination.

   c. In lieu of a shower, and assuming that all asbestos abatement of non-friable Category II ACM is performed within a negative pressure mini-containment, the contractor can, upon approval by the QP, construct a decontamination (decon) chamber within which will be placed a CLEAN HEPA filtered vacuum, disposable cloths and a source of water for worker decontamination.

B. The clean room, shower/decon chamber, and equipment room must be sealed completely to ensure that the sole source of airflow through these areas originate from uncontaminated areas outside the asbestos removal, demolition, or renovation enclosure. The shower, if used for decontamination, must be drained and filtered to retain particles 5.0 microns or larger after each use to ensure that contaminated water is not released to uncontaminated areas; alternatively, the water can be collected for disposal. If waste water is inadvertently released, it shall be cleaned up as soon as possible to prevent any asbestos in the water from drying and becoming airborne in areas outside the work area. The contractor will be responsible for the required water source.

C. In all cases, except removal of Category I or II ACM removed in a regulated area where rooms are not present (not within a negative pressure containment), access between contaminated and uncontaminated rooms or area shall be through an airlock.

## 3.2 Monitoring, Inspection, and Testing

### 3.2.1 General

A. The QP will perform monitoring, inspection and testing inside and around the regulated area in accordance with the OSHA requirements and these specifications throughout abatement work to include visual inspections and final air clearance testing. The QP shall monitor conditions to document compliance with these specifications, as warranted.

B. If fiber counts are detected outside the regulated area during abatement work utilizing the NIOSH 7400 air monitoring method that exceed the specified respective limits, the Contractor shall stop work. The Contractor may request confirmation of the results by analysis of the samples by TEM. The request for TEM analysis must be in writing and submitted to the Client's representative. The cost for the confirmation of results will be borne by the Contractor for both the collection and analysis of samples and for the time delay that may/does result for this confirmation. Work may resume upon correction of cause of elevations upon authorization by the QP.

### 3.2.2 Scope of Services of the QP Consultant

The purpose of the work of the QP is to: document adherence to the specification; facilitate resolution of problems; check for the spread of contamination beyond the regulated area and, when noted recommend and document corrective measures; collect supplemental samples of newly discovered or previously unrepresented suspect ACM and submit to a testing laboratory for analysis; and document clearance at the end of the project, as warranted. In addition, their work includes performing the final inspection and testing to determine whether the regulated area(s) have been adequately decontaminated. All air monitoring sample analysis will be performed via PCM. The QP will perform the following tasks:

a. Task 1: Perform as needed air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect faults in the regulated area isolation and check for adverse impact on the surroundings from regulated area activities.

b. Task 2: Perform visits to spot check overall compliance of work with the contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area and all aspects of the operation except personnel monitoring.

c. Task 3: Perform final inspection and testing of a decontaminated regulated area at the conclusion of the abatement to document general compliance with all applicable regulations. If all asbestos abatement is performed using glovebag technology, area air samples will be collected continuously during abatement. At the end of the shift and/or completion of abatement in a regulated area all area air samples will be analyzed. If the area air samples are at or less than 0.01 f/cc and the area is cleared with a visual inspection, then the regulated area will be considered clear.

d. Task 4: Provide documentation of abatement and "clearance," as needed.

e. Task 5: Document quantities of asbestos removed based on the applicable units. Documentation assumes quantities will be recorded by the abatement contractor and presented daily to QP for corroboration.

f. Task 6: Field modify requirements included in this specification, as warranted and reasonable based on the determination and professional opinion of the QP.

## Part 4 Execution

This section includes all work necessary to reduce air concentrations of asbestos to the specified level and maintain the specified asbestos control limits during the life of the contract. The work specified in this document consists of the provision of services for the containment, removal and disposal of ACMs identified at the site.

All asbestos abatement work will be performed using wet methods, within established Regulated Area(s) by competent, licensed (by DPOR) persons trained, qualified, and knowledgeable in the techniques of abatement, handling, and disposal of ACMs and materials contaminated by asbestos, in accordance with pertinent local,

state, and federal regulations. Regulated work shall NOT be conducted by unlicensed individuals even if the removal does not constitute a “Project” as defined by DPOR.

The Abatement Contractor shall coordinate with the Mechanical, Electrical, Plumbing, and General Contractors and/or the site representative to ensure that all appropriate systems that will be impacted by the abatement have been properly decommissioned prior to the start of any work.

The Abatement Contractor shall coordinate with the site representative for this project to verify that the structures will support the planned activities and comply with local building codes and OSHA requirements.

This scope of work includes, but is not limited to, the abatement of those materials presented in Section 1.1 and 1.1.3.2. However, the scope of the renovation may reduce the abatement of known ACMs (including materials with trace (<1%) asbestos); the Contractor shall coordinate with the Architect and/or Owner to confirm scope of abatement.

During all containment work, air filtration advices (AFD) shall be employed and should be in place during work area preparation and maintained in use until clearance is issued, if warranted. At no point during abatement (following approval of containment(s) until clearance is issued or the space is visually approved as cleared, as warranted) shall the pressure differential fall below 0.02" water column (wc) where containments are in place. In addition (as noted below), filtration units shall be sized to provide a minimum of four air changes per hour in the containment area. **Provide continuous, direct-read instrumentation to document pressure differential (manometer). The instrument must be calibrated daily with documentation provided to the IH.**

**Fiber concentrations immediately outside any regulated area are not to exceed 0.01 fibers per cubic centimeter of air (f/cc). If this limit is exceeded, further removal shall be suspended until appropriate actions are taken to reduce the fiber concentration.** If other trades or work is being completed in the work area which may result in elevations of fibers that are not likely to constituted asbestos AND if the ambient areas remain unoccupied by casual visitors, owner employees, or other unrelated persons, the acceptable threshold may be modified to 0.1 f/cc, at the discretion of the QP.

Before start of work, the Contractor shall submit the design and layout of the regulated areas and the negative air machines. The submittal shall indicate the number of, location of and size of negative air machines. The point(s) of exhaust, air flow within the regulated area, anticipated negative pressure differential, and supporting calculations for sizing shall be provided.

### 4.1 Preparation of Work Area: Installation

A. Establish each work area as a regulated area and place warning signs as discussed in OSHA Regulation 29 CFR 1926.1101 (k) (7) at the regulated area perimeters to prevent entrance by non-authorized personnel. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator. Place barricade tape at the regulated area perimeter as a physical barrier to prevent access by unauthorized persons.

B. Clean the proposed work areas using HEPA filtered vacuum equipment or wet cleaning methods as appropriate. Methods that raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters, shall not be used. Preliminary cleaning may be omitted at the discretion of the Contractor with the permission of the Owner; however, elevations in airborne fibers resulting from failure to pre-clean is may result in later requirements for additional dust mitigation measures.

   Any areas of deteriorated ACM which would be disturbed during establishment of engineering controls or access areas shall, at the discretion of the contractor and with approval of the QP, be cleaned immediately following establishment of regulated area and, if warranted, placement of poly sheeting drop cloths.

C. If abatement will take place in a containment of the type detailed in 4.1.1.2, the following actions will be taken. Seal off all openings before covering walls and ceilings, including, but not limited to: corridors, access ways, skylights, ducts, grilles, diffusers, and any other penetrations of the work area, with one layer of 6 mil plastic sheeting sealed with tape. Access ways and corridors which will not be used for passage during work must be also sealed. For modified containments, only critical barriers and drop cloths are required, with splash guards were solvent based removal of floor mastic is performed; covering walls and ceilings are not necessary for work performed in accordance with 4.1.1.1, 4.1.1.3, or 4.1.1.4. Use of buffers or creation of RACM will trigger full containment construction including plasticization of full wall height and ceilings.

    a. For work areas that require complete enclosure:

        i. Cover floor, wall and ceiling surfaces that are to remain with plastic sheeting sealed with tape such that a continuous seal is maintained. Use a minimum of two layers (three in carpeted areas) of 6 mil plastic on floor and walls. Cover floors first so plastic extends at least 12 inches up the wall, then cover the walls with minimum of 6 mil plastic sheeting to the floor level, thus overlapping the floor materials by a minimum of 12 inches, then continuously tape edges to provide a complete seal.

        ii. Cover entire ceiling surface that are to remain with one layer (overlapping 12" at any seams) of 6 mil plastic sheeting extending 12" down walls and sealed with tape, thus creating a second layer over penetrations previously sealed.

        iii. Cover all wall surfaces that are to remain with one layer (overlapping 12" at any seams) of 6 mil plastic and sealed with tape at ceiling and floor or baseboard levels, thus creating a second layer over penetrations previously sealed.

        iv. Maintain emergency and fire exits from the work area or establish alternative exits satisfactory to the applicable fire officials.

    b. For enclosed areas where only asbestos glovebagging, overwrapping (whole component removal), or non-friable exterior abatement utilizing wet methods will be conducted:

        i. Cover all surfaces in the immediate area with one layer of 6 mil plastic and sealed with tape at overlapped areas (minimum of 3 feet overlap) and at walls, and ensure creation of a second layer over any penetrations. Place barrier tape and signage to delineate Regulated Area. Regulated area shall be a minimum of 10 feet from work area in all directions, unless wall barrier is present.

    c. For mini-containments where asbestos removal associated with non-friable category I or II materials is to be completed:

        i. Construct a containment structure which completely encloses the work area using a minimum of two layers of 6 mil plastic on floor and walls.

        ii. Place an access/decon chamber at one side with a negative air machine on the opposite side such that air is pulled across the work area from the exterior of the mini-containment.

    d. As noted above, for containments where asbestos removal associated with non-friable floor tile or mastic materials is to be completed, placement of poly sheeting on walls and ceilings is not required; however, splash guards, critical barriers, decontamination chambers, and negative pressure shall be implemented.

e. Splash guards shall be placed on walls prior to removal of flooring mastic in order to provide clean, uncontaminated wall surfaces at completion of project.

f. In all cases where abatement does not include flooring removal, placement of drop clothes under work is required.

D. Isolate the work areas for the duration of the work by completely sealing off all openings and fixtures, including HVAC and other ducts or air movement systems in the work area with plastic sheeting taped and glued securely. Ensure HVAC system is nonfunctional prior to covering system openings. Cover all critical barriers with two layers of 6 mil plastic sheeting. As abatement requires removal of materials which decreases continuity of the containment, barrier installation must be performed in order to maintain containment system.

E. Maintain enclosures and regulated work areas in tidy conditions. Ensure that barriers and plastic linings are effectively sealed and taped and that the regulated area is effectively demarcated such that physical boundaries are maintained intact for the duration of the project. Repair damaged barriers and remedy defects immediately upon discovery. Visually inspect enclosures at the beginning and end of each work period. Use smoke methods, as necessary, to test effectiveness of barriers, where applicable.

F. If buffer equipment is used during the mastic removal process where the mastic consists of ACM, the removal shall be conducted within a negative pressure containment including polyethylene sheeting on walls and ceilings in accordance with 4.1.1.1. A reduction in buffer RPM or manufacturer designation of non-abrasive for the buffer pad shall not negate this requirement.

G. Each unit of asbestos filtration systems shall consist of a blower filter system, equipped with HEPA inline filtration which, as a minimum, continuously captures particles with an aerodynamic diameter of 0.3 microns at 99.97% efficiency. Each unit shall be equipped with the following minimum controls:

1) A warning light and audible alarm to indicate reduced air flow due to dirty filters.
2) Automatic shutdown, with warning light, to ensure against continued operation of units in event of clogged or damaged filters.

H. The asbestos filtration system shall be in operation during all removal operations, until final clearance air samples are received or approved by the QP. An adequate number of filtration units shall be used to assure maintenance of pressure differential of 0.02" water column (wc). In addition, filtration units shall be sized to provide a minimum of four air changes per hour in the containment area. Provide continuous, direct-read instrumentation to document pressure differential.

I. Maintain the EPA and OSHA regulations or any applicable state and local government regulations at the job site in locations accessible to employees and others. Attention is directed to all requirements of the Contract Documents concerning precautionary procedures mandated thereby and by OSHA and EPA for the protection of personnel, the public, and the environment from exposure to or possible contamination by asbestos fibers.

J. In addition to requirements for asbestos protection, comply with all other applicable requirements of 29 CFR 1910 and 1926 or others, as applicable.

K. Provide hard hats, eye protection, and foot protection in those areas where such protective measures are required by OSHA regulations.

L. Workers shall always wear a respirator properly fitted on the face while in the work area unless a Negative Exposure Assessment in accordance with Section 1.6.6 has been completed and is applicable to this abatement effort. Instruct and train workers to use respirators properly in accordance with the

requirements of the American National Standards Practices for Respiratory Protection (ANSI Z88.2). Ensure that workers wear the appropriate respirator at all times while in the work area. Each employee shall be tested for respirator fit in accordance with the cited ANSI standard.

M. Workers shall wear disposable full body coveralls and disposable head and foot coverings in the work area. If non-disposable footwear such as protective shoes are required and disposal foot coverings are not suitable, the non-disposable protective footwear shall be left in the work area at all times until disposal at job completion, then disposed of as asbestos contaminated waste. Employees shall not modify suits to allow ambient air to the interior of the suit by cutting off feet unless legs are taped to non-porous boots which are decontaminated prior to exiting the regulated area. This requirement may be modified upon approval by the QP for certain non-friable Cat I or Cat II materials.

N. All removed material shall be containerized and areas gross cleaned on a daily basis. Removed material will not remain on the floor overnight or during any non-work period

### 4.1.1 Access Areas (Decontamination)

#### 4.1.1.1 Modified Containment Areas

For the removal of Category I and II non-friable materials which will NOT be rendered friable during the abatement effort. If work WILL result in liberation of fibers above the OSHA PEL, work shall be performed in accordance with 4.1.1.2, below.

A. Contractor shall provide the Outside Clean Room, Decontamination Room, and Equipment Room prior to start of work within building work areas. Personnel lockers or equivalent in the Clean Room and facilities for disposal of contaminated clothing in the Equipment or Decontamination Rooms shall be provided. Egress openings shall consist of a minimum of two sheets of plastic taped across the opening head and down opposite jambs, one leaf shall be taped on one side of the jamb, the other on the opposite jamb.

    a. Containment partitions separating a contaminated area from a clean area shall be constructed of wood studs (or similar) and two sheets of minimum 6 mil polyethylene plastic. The inner plastic barrier shall face the contaminated area, the outer barrier, the clean area.

B. The Contractor shall establish decontamination procedures for each work area. All persons without exception shall pass through these decontamination areas for any purpose. Procedures shall, as a minimum, consist of the following:

    1) Outside Clean room Area: In this room, the worker or individual may remove normal street clothing and replace with clean work clothing, including disposable coveralls, respiratory protective equipment, and all other protective gear. No asbestos contaminated items shall enter this room with the exception of reusable respirators which are to be placed in a bin or other suitable receptacle approved by the Contractor's technical representative. Provide suitable lockers or other secure storage areas for the employee's clothing. The outside clean room may consist of the ambient space outside the work area and does not require walled enclosure. At this facility, the covered walkway may be used as an open-air clean room provided adequate space for storage and activities is available.

    2) Equipment Room: Provide an area in which work equipment, footwear and contaminated work clothing can be placed in suitable receptacles for reuse or disposal prior to entry into the shower room and thence to the outside clean room. The equipment room may comprise a single stage pop up on the clean room side of the decontamination area. If all disposable materials are discarded in the decontamination room, then an equipment room is not required.

3) Decontamination Room: In this room, which may comprise a single stage pop-up, decontamination procedures detailed below shall be implemented such that contaminated clothing or materials are safely discarded as asbestos-contaminated or adequately cleaned prior to removal or doffing. A HEPA vacuum as well as materials for wet wiping exposed skin and equipment shall be available in this space.

4) Decontamination Procedures: Maintain a protection program to ensure that workers and others follow an established decontamination sequence utilizing the aforementioned facilities. They shall ensure that gross contamination and debris is removed from protective clothing and equipment prior to egress from the work area. Respiratory protective equipment shall be removed last, during shower, to prevent inhalation of fibers during removal of contaminated clothing. The Contractor shall provide a plan for receipt, inspection, cleaning and storage of respiratory protective equipment in such a manner as to avoid contamination of clean areas. Suggested decontamination sequencing is as follows:

    a. Still wearing respirators, proceed to decontamination room. Doff disposable coveralls and place in asbestos disposal bag.
    b. With respirator still in place thoroughly vacuum exposed skin, respirator face piece, all exterior parts of the respirator, and all equipment.
    c. Wet wipe equipment or skin exhibiting contamination and dispose of wipe materials in asbestos disposal bag.
    d. Take a deep breath, hold it and/or exhale slowly. While still holding breath, move out of decontamination room, remove respirator and hold it away from face before starting to breathe.
    e. Carefully clean face piece of respirator inside and out.
    f. Vacuum debris on room walls and floor prior to exit.
    g. Proceed to changing (clean) room and change into street clothes or new disposable work items.

After decontamination, each employee shall inspect, clean and repair his respirator as needed. The respirator shall be dried, placed in a suitable storage bag and properly stored.

C. Bag Out Area: A dedicated structure (including an airlock) for removal of bagged and labelled bags from the containment may be utilized for transferring abated asbestos materials from the work area to an appropriate disposal container. Alternate methods may be reviewed and approved by the QP.

#### 4.1.1.2 Full Containment Areas

If Class I work is performed or if Class II work is performed which will render non-friable materials into friable RACM, full containment is required.

A. Contractor shall provide the Outside Clean Room, Shower Room, and Equipment Room prior to start of work within building work areas. Personnel lockers or equivalent in the Clean Room and facilities for disposal of contaminated clothing in the Equipment Room shall be provided. Egress openings shall consist of a minimum of two sheets of plastic taped across the opening head and down opposite jambs, one leaf shall be taped on one side of the jamb, the other on the opposite jamb.

    a. Containment partitions separating a contaminated area from a clean area shall be constructed of wood studs (or similar) and two sheets of minimum 6 mil polyethylene plastic. The inner plastic barrier shall face the contaminated area, the outer barrier, the clean area.

B. The Contractor shall establish decontamination procedures for each work area. All persons without exception shall pass through these decontamination areas for any purpose. Procedures shall, as a minimum, consist of the following:

1) Outside Clean room Area: In this room, the worker or individual may remove normal street clothing and replace with clean work clothing, including disposable coveralls, respiratory protective equipment, and all other protective gear. No asbestos contaminated items shall enter this room with the exception of reusable respirators which are to be placed in a bin or other suitable receptacle approved by the Contractor's technical representative. Provide suitable lockers or other secure storage areas for the employee's clothing.

2) Showers: A shower room or similar facility shall be provided for transit by cleanly dressed workers entering the work area from the outside clean room, or by workers headed for the showers after undressing in the contaminated equipment room or area. Except in cases of emergency, no person shall leave the work area without first having taken a shower. Water from the showers shall be passed through 5 micron water filters and then piped into the building floor drain or collected and disposed of by the Contractor. Provide water for the showers. Cold water supply from existing system may be tapped by use of garden hoses, clamps and control valves. The taps and extensions shall be provided by the Contractor. The change facility shall be equipped with adequate water heating capacity to provide for hot water showers. The decontamination facility shall be equipped with thermostatically controlled heating system for the clean room and equipment room.

3) Equipment Room: Provide an area in which work equipment, footwear and contaminated work clothing can be placed in suitable receptacles for reuse or disposal prior to entry into the shower room and thence to the outside clean room.
4) Decontamination Procedures: Maintain a protection program to ensure that workers and others follow an established decontamination sequence utilizing the aforementioned facilities. They shall ensure that gross contamination and debris is removed from protective clothing and equipment prior to egress from the work area. Respiratory protective equipment shall be removed last, during shower, to prevent inhalation of fibers during removal of contaminated clothing. The Contractor shall provide a plan for receipt, inspection, cleaning and storage of respiratory protective equipment in such a manner as to avoid contamination of clean areas. Suggested decontamination sequencing is as follows:

   Still wearing respirators, proceed to showers. Showering is mandatory. Care must be taken to follow reasonable procedures in removing the respirator to avoid asbestos fibers while showering. The following procedure is required as a minimum:

   h. Thoroughly wet body including hair and face.
   i. With respirator still in place thoroughly wash body, hair, respirator face piece, and all exterior parts of the respirator.
   j. Take a deep breath, hold it and/or exhale slowly, completely wet hair, face and respirator. While still holding breath, remove respirator and hold it away from face before starting to breathe.
   k. Carefully wash face piece of respirator inside and out.
   l. Shower completely with soap and water; rinse thoroughly.
   m. Rinse shower room walls and floor prior to exit.
   n. Proceed from shower to changing (clean) room and change into street clothes or new disposable work items.

   After showering, each employee shall inspect, clean and repair his respirator as needed. The respirator shall be dried, placed in a suitable storage bag and properly stored.

C. Bag Out Area: A dedicated structure (including an airlock) for removal of bagged and labelled bags from the containment may be utilized for transferring abated asbestos materials from the work area to an appropriate disposal container. Alternate methods may be reviewed and approved by the QP.

#### 4.1.1.3 Glovebag Abatement or Component Removal Areas

A. Contractor shall provide a Single Stage Decontamination Room or a Decontamination Corridor prior to start of work within building work areas where glovebag abatement or whole component removal is to be conducted. Egress openings in a Decontamination Room shall consist of a minimum of two sheets of plastic taped across the opening head and down opposite jambs, one leaf shall be taped on one side of the jamb, the other on the opposite jamb. The Decontamination Corridor shall comprise a minimum of 6 mil poly sheeting on the floor, partitioned from the work area and the outside of the regulated area and equipped with a HEPA vacuum, and disposable cloths and a source of water provided for worker decontamination. Disposal containers shall be maintained for disposal of contaminated PPE in a manner which prevents release of contamination.

    a. Containment partitions separating a contaminated area from a clean area shall be constructed of wood studs (or similar) and two sheets of minimum 6 mil polyethylene plastic. The inner plastic barrier shall face the contaminated area, the outer barrier, the clean area.

B. Decontamination Procedures: Work in this area shall be conducted utilizing double suit decontamination procedures; double suit use may be avoided by using a shower. Maintain a protection program to ensure that workers and others follow an established decontamination sequence utilizing the aforementioned facilities. They shall ensure that gross contamination and debris is removed from the outer layer of protective clothing and equipment first and then remove the underlying layer of protective clothing prior to egress from the work area. Respiratory protective equipment shall be removed last, to prevent inhalation of fibers during removal of contaminated clothing. The Contractor shall provide a plan for receipt, inspection, cleaning and storage of respiratory protective equipment in such a manner as to avoid contamination of clean areas. Protective clothing shall be left within the regulated area and disposed of as asbestos contaminated.

#### 4.1.1.4 Category I and II Non-Friable ACM Abatement Areas

A. Contractor shall provide a Decontamination Room or a Decontamination Corridor prior to start of work in areas on building exteriors only wet methods and HEPA Vacuums will be conducted. The Decontamination Corridor shall comprise a minimum of 6 mil poly sheeting on the floor, partitioned from the work area and the outside of the regulated area and equipped with a HEPA vacuum, and disposable cloths and a source of water provided for worker decontamination. Disposal containers shall be maintained for disposal of contaminated PPE in a manner which prevents release of contamination. All removal shall be conducted in a way which prevents release of fibers or generation of Regulated (friable) ACM. Materials should be sliced or snipped, rather than cut using abrasive methods.

B. Decontamination Procedures: Work in this area shall be conducted utilizing double suit decontamination procedures. Maintain a protection program to ensure that workers and others follow an established decontamination sequence utilizing the aforementioned facilities. They shall ensure that gross contamination and debris is removed from the outer layer of protective clothing and equipment first and then remove the underlying layer of protective clothing prior to egress from the work area. Respiratory protective equipment shall be removed last, to prevent inhalation of fibers during removal of contaminated clothing. The Contractor shall provide a plan for receipt, inspection, cleaning and storage of respiratory protective equipment in such a manner as to avoid contamination of clean areas. Protective clothing shall be left within the regulated area and disposed of as asbestos contaminated. If boots or gloves are used within the regulated area, they must be left in the decontamination area/corridor unless and until they are appropriately decontaminated or disposed of as asbestos waste. Porous materials which cannot be decontaminated shall be disposed of as asbestos at the end of the job and may not be allowed outside the regulated area.

#### 4.1.1.5 Mini-Containment Abatement Areas

A. Contractor shall provide a Single Stage Decontamination Room or a Decontamination Corridor prior to start of work within building work areas where mini-containments are to be used. For RACM removal within a mini-containment, the decontamination area shall be enclosed, contiguous with, and sealed against the containment space. Egress openings in a Decontamination Room shall consist of a minimum of two sheets of plastic taped across the opening head and down opposite jambs, one leaf shall be taped on one side of the jamb, the other on the opposite jamb. The Decontamination Corridor shall comprise a minimum of 6 mil poly sheeting on the floor, partitioned from the work area and the outside of the regulated area and equipped with a HEPA vacuum, and disposable cloths and a source of water provided for worker decontamination. Disposal containers shall be maintained for disposal of contaminated PPE in a manner which prevents release of contamination.

    b. Containment partitions separating a contaminated area from a clean area shall be constructed of wood studs (or similar) and two sheets of minimum 6 mil polyethylene plastic. The inner plastic barrier shall face the contaminated area, the outer barrier, the clean area.

B. Decontamination Procedures: Work in these areas shall be conducted utilizing double suit decontamination procedures. Maintain a protection program to ensure that workers and others follow an established decontamination sequence utilizing the aforementioned facilities. They shall ensure that gross contamination and debris is removed from the outer layer of protective clothing and equipment first and then remove the underlying layer of protective clothing prior to egress from the work area. Respiratory protective equipment shall be removed last, to prevent inhalation of fibers during removal of contaminated clothing. The Contractor shall provide a plan for receipt, inspection, cleaning and storage of respiratory protective equipment in such a manner as to avoid contamination of clean areas. Protective clothing shall be left within the regulated area and disposed of as asbestos contaminated. If boots or gloves are used within the regulated area, they must be left in the decontamination area/corridor unless and until they are appropriately decontaminated or disposed of as asbestos waste. Porous materials which cannot be decontaminated shall be disposed of as asbestos at the end of the job and may not be allowed outside the regulated area.

**4.2 Method of Removal**

A. A low pressure fine spray of amended water shall be applied to reduce fiber release preceding removal. The asbestos shall be saturated sufficiently to retard emission of airborne fibers. If the asbestos containing material is thick and detaches in chunks having dry bottoms, amended water shall be sprayed over the material as it is loosened and removed.

B. For vermiculite (if encountered), once the negative pressure containment is approved by the QP, an opening may be created in a shell at the bottom of the CMU wall with a HEPA vacuum placed immediately and the inlet hose sealed against the wall. An opening may then be created at the top of the wall, vertically in line with the bottom to allow air movement through the CMU cores from the top to the bottom and vacuuming of the filler material. The face shells may be removed from top to bottom and residual vermiculite gently brushed from top to bottom to encourage capture with the HEPA vacuum. In no case shall vermiculite be permitted to flow from the concealed space without immediate capture.

    This process shall be continued horizontally to allow each core to be evacuated and the face of the CMU wall removed to expose the cores. CMU elements shall be placed in accordance with section 4.2.3. Materials to remain shall be subjected to application of a penetrating encapsulant to the exposed elements where vermiculite was removed. Visual approval of the removal shall be granted by the QP, after which a tinted bridging encapsulant shall be applied to readily indicate proper removal and sealing has occurred.

C. Remove material and immediately place in plastic disposable bags, 6-mil minimum. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. All disposed ACMs shall be in two discrete bags, the first one sealed closed and placed in a second bag, and then the second bag sealed. Where unusual circumstances prohibit the use of 6 mil plastic bags, alternative methods may be used. For example, in the case where large sections of carpet are removed, the carpet can be placed on two layers of 6 mil polyethylene sheeting, sealed individually, such that each layer comprises a watertight barrier, with the ends and seems sealed with tape. Asbestos containing material shall be containerized while wet and containerized within two discrete 6 mil plastic layers, individually applied, taped, and sealed. At no time shall asbestos material be allowed to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in 40 CFR 61-SUBPART M. Submit any alternate proposal for containment of asbestos fibers to the Owner for approval. Pre-printed labels, markings and address as required by OSHA, EPA and DOT regulation shall be affixed to all containers prior to disposal. Use of trash chutes shall not be used.

a. For whole component removal: containerize component in accordance with 4.2 B. (above) and 4.2.2 (below). If piping is removed as a component where non-friable Category II ACM sealant is present, remove non-ACM TSI on either side of the ACM sealant followed by wrapping the ACM with poly and “candy stripe” with duct tape; then cut pipe on either side of ACM material and containerize as noted.

b. Removal Procedures - Removal of TSI or non-friable Category II sealant on TSI:

Glovebag Removal Procedures

All applicable OSHA requirements and glovebag manufacturer’s recommendations shall be met during glove bagging operations. In cases where live steam lines are present, the lines must be shut down prior to any work being performed on the system. No abatement work shall be conducted on live, pressurized steam lines. The Contractor may choose to use a High Temperature Glovebag in which a temperature rating ranges from 300˚F to 700˚F on steam lines that have recently been shut down and remain at high temperature for some time. In the case where a glovebag is not feasible, the Contractor will need to build a full negative pressure containment of sufficient size and follow all regulations as it pertains to removal.

i. Mix the surfactant with water in the garden sprayer, following the manufacturer's directions.

ii. Have each employee don a HEPA filtered respirator approved for asbestos and check the fit using the positive/negative fit check.

iii. Have each employee put on a disposable full-body suit. The hood shall be worn over the respirator straps.

iv. Check closely the integrity of the glove bag to be used. Check all seams, gloves, sleeves, and glove openings. OSHA requires the bottom of the bag to be seamless.

v. Check the pipe where the work will be performed. If it is damaged (broken lagging, hanging, etc.), wrap the entire length of the pipe in poly sheeting and "candy stripe" it with duct tape.

vi. Attach glovebag with required tools per manufacturer’s instructions.

vii. Using the smoke tube and aspirator bulb, test 10% of glovebags by placing the tube into the water porthole (two-inch opening to glove bag), and fill the bag with smoke and squeeze it. If leaks are found, they should be taped closed using duct tape and the bag should be retested with smoke. QP personnel will conduct smoke testing at their discretion prior to any abatement and will be notified by the contractor prior to abatement in order to provide opportunity to evaluate the glovebag system.

viii. Insert the wand from the water sprayer through the water porthole.

ix. Insert the hose end from a HEPA vacuum into the upper portion of the glove bag.

x. Wet and remove the pipe insulation. Where exposed TSI will remain apply a lockdown encapsulant.

xi. If the section of pipe is covered with an aluminum jacket, remove it first using the wire cutters to cut any bands and the tin snips to remove the aluminum. It is important to fold the sharp edges in to prevent cutting the bag when placing it in the bottom.

xii. When the work is complete, spray the upper portion of the bag and clean-push all residue into the bottom of the bag with the other waste material. Be very thorough. Use adequate water.

xiii. Put all tools, after washing them off in the bag, in one of the sleeves of the glove bag and turn it inside out, drawing it outside of the bag. Twist the sleeve tightly several times to seal it and tape it several tight turns with duct tape. Cut through the middle of the duct tape and remove the sleeve. Put the sleeve in the next glove bag or put it in a bucket of water to decontaminate the tools after cutting the sleeve open.

xiv. Turn on the HEPA vacuum and collapse the bag completely. Remove the vacuum nozzle, seal the hole with duct tape, twist the bag tightly several times in the middle, and tape it to keep the material in the bottom during removal of the glove bag from the pipe.

xv. Slip a disposal bag over the glove bag (still attached to the pipe). Remove the tape securing the ends, and slit open the top of the glove bag and carefully fold it down into the disposal bag. Double bag and gooseneck waste materials.

For Negative Pressure Glovebag Procedure, in addition to the above requirements, the HEPA vacuum shall be run continuously during the glovebag procedure until completion at which time the glovebag will be collapsed by the HEPA vacuum prior to removal from the pipe/component. Additionally, the HEPA vacuum shall be attached and operated as needed to prevent collapse of the glovebag during the removal process.

D. After completion of stripping work, all surfaces from which asbestos has been removed or contaminated shall be wire brushed and/or wet sponged or cleaned by an equivalent method to remove all visible material. During this work, the surfaces being cleaned shall be kept wet.

E. All containers shall be cleaned and thoroughly decontaminated before leaving the work area by being passed through the shower or bag out area, or through the airlock and container cleaning assembly, as follows:

1) Containers shall first be gross-cleaned by vacuuming and then damp-wiped, before being placed into shower container or cleaning airlock and removed from the work area. .

2) If a container being transferred from the work area via a shower has dried, it shall be wet-wiped again before being transferred past the shower.

F. Transport the sealed container or skips to an EPA approved and certified waste disposal site. The Contractor shall provide the Owner with a signed certificate listing the quantity of materials delivered to the disposal site, a description of the location of the site, and a statement attesting to the fact that the site is an EPA and State approved disposal location. The signatures of the asbestos removal Contractor, transporter, and site operator must appear on the certificate. The Contractor shall ensure that the operator leaves damaged bags in the delivery containers and that the entire contaminated container is buried; however, sealed plastic bags may be dumped from the containers into the burial site and uncontaminated containers may be reused. The Contractor shall certify that any reused containers have not contained damaged or broken bags of asbestos or other asbestos-contaminated material.

G. Disposal of all asbestos waste shall be at a prearranged disposal site in accordance with regulations of the Virginia Department of Environmental Quality-Waste Division (or the regulations governing the eventual location of disposal) and OSHA Regulation 29 CFR 1910.1001 as well as section 2.1.5 of this Section. Documentation of disposal shall be provided to the Owner.

H. Following removal of asbestos-containing material, all plastic sheeting, tape, cleaning material, clothing and all other disposal materials or items used in the work area shall be packed into sealable plastic bags (6 mil minimum), sealed and placed into a second sealed plastic bag or into metal or fiber containers or skips for transport. The containers or skips shall be labeled as prescribed by OSHA Specifications 29 CFR 1910.1001 (g). Plastic sheeting shall not be reused in other containment(s).

I. If present, seal edges of ACM exposed by removal work which is inaccessible, such as a sleeve, wall penetration, etc., with bridging encapsulant. Prior to sealing, permit the exposed edges to dry completely to permit penetration of the bridging encapsulant.

### 4.2.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as full-height partitions, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. The asbestos lock-down encapsulant shall be tinted a contrasting color. It shall be spray-applied by airless method. Thoroughness of the sealing operation shall be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants shall comply with the performance requirements specified herein.

### 4.2.2 Packaging Waste For Disposal

The abatement contractor will place the asbestos waste into appropriately labeled 6-mil thick impermeable polyethylene plastic disposal bags. The bags should not be filled more than 40 percent by volume to avoid tearing. The ACM waste must be wetted with sufficient water to insure that it is adequately wet. Prior to gooseneck sealing, the bag should be collapsed to remove any trapped air that may later contribute to bursting. After the exterior of the sealed bag is wet cleaned to remove any gross material, it is then double bagged and sealed by goose necking and taping. The bagged and labeled ACM waste will then be stored in a secured container to await transport to the disposal site. Asbestos waste cannot be mixed with any other type of construction waste from the site. Personnel handling ACM waste outside of the containment must wear, as minimum, protective clothing and a HEPA equipped half-mask, air-purifying respirator.

### 4.2.3 OSHA Warning Labels

In compliance with the OSHA Asbestos Standard, 29 CFR 1926.1101, the OSHA Hazard Communication Standard, 29 CFR 1926.59, and the EPA NESHAP Standard, all of which require each employer to ensure that all containers of hazardous chemicals in the workplace are labeled, tagged, or marked with the identity of the hazardous chemical contained therein, and an appropriate warning of the hazards of the chemical, all asbestos-containing waste disposal bags utilized by the abatement contractor will be appropriately marked as follows:

**DANGER**
**CONTAINS ASBESTOS FIBERS**
**MAY CAUSE CANCER**
**CAUSES DAMAGE TO LUNGS**
**DO NOT BREATHE DUST**
**AVOID CREATING DUST**

These labels must be at least three (3) inches high by five (5) inches wide and conform to the requirements specified in 29 CFR 1910.145 for danger labels or signs.

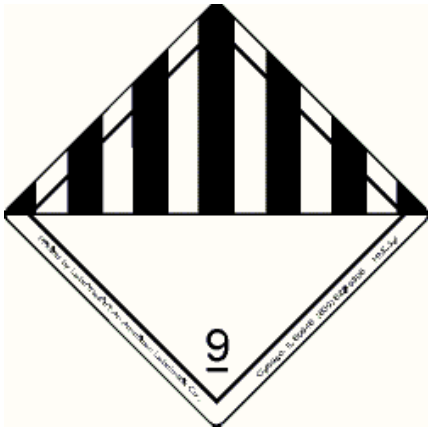
F&R acknowledges that section 1926.1101(k)(8)(vi) and 1926.1101(k)(8)(vi)(B) state that labeling is NOT required for concentrations of <1% asbestos. This exemption can be applied to waste stream comprising ONLY materials with concentrations of <1% asbestos. However, best practices would be to label the materials to inhibit unintentional exposure.

### 4.2.4 Generator Labels

NESHAP requires that each bag of asbestos-containing material that is to be transported off of the facility site, contain a visible label identifying the name of the waste generator (owner) and the location at which the waste was generated. This generator label must either be printed on the outside of the bag or it must be a separate label attached to the bag, or it can be a separate label that is inserted between the two poly bags if the outside bag is a clear bag that would allow the label to be read without opening the bag. Labels must be applied prior to transport from the site and shall be of sufficient quality to prevent detachment during handling.

### 4.2.5 DOT Transportation Labels

In addition to the warning labels required by both OSHA and the EPA, the United States Department of Transportation (DOT) requires a “CLASS 9” warning label on each bag of ACMs that is transported away from the removal site. The label should be as follows:



The "CLASS 9" (Miscellaneous Hazardous Materials) label should be approximately 100 mm by 100 mm in size and in addition to complying with 49 CFR 172.407; the background on the CLASS 9 label must be white with seven black vertical stripes on the top half. The black vertical stripes must be spaced, so that, visually, they appear equal in width to the six white spaces between them. The lower half of the label must be white with the class number "9"' underlined and centered at the bottom.

Collect asbestos waste, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 6 mils thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with 40 CFR 61-SUBPART M. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 6 mils thick bags with the approved warnings and DOT labeling preprinted on the bag. The name of the waste generator and the location at which the waste was generated shall be clearly indicated on the outside of each container. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete. Dispose of waste asbestos material at an Environmental Protection Agency (EPA) or State-approved asbestos landfill per waste disposal contract requirement. For temporary storage, store sealed impermeable bags in asbestos waste drums or skids.

**4.3 Bag Out Procedures**

A. All containers shall be cleaned and thoroughly decontaminated before leaving the work area by being passed through the airlock and container cleaning assembly (or decon area), as follows:

   1) Containers shall first be gross-cleaned by vacuuming and then damp-wiped, before being placed in the single stage decontamination unit and removed from the work area.

   2) If a container being transferred from the work area via a shower has dried, it shall be wet-wiped again before being transferred past the decontamination unit.

B. Transport the sealed container or skips to an EPA approved and certified waste disposal site. The Contractor shall provide the Owner with a signed certificate listing the quantity of materials delivered to the disposal site, a description of the location of the site, and a statement attesting to the fact that the site is an EPA and State approved disposal location, as applicable. The signatures of the asbestos removal Contractor, transporter, and site operator must appear on the certificate. The Contractor shall ensure that the operator leaves damaged bags in the delivery containers and that the entire contaminated container is buried; however, sealed plastic bags may be dumped from the containers into the burial site and uncontaminated containers may be reused. The Contractor shall certify that any reused containers have not contained damaged or broken bags of asbestos or other asbestos-contaminated material.

C. Disposal of all asbestos waste shall be at a prearranged disposal site in accordance with regulations of the Virginia Department of Environmental Quality-Waste Division (or the applicable regulations of the state where eventual disposal is completed) and OSHA Regulation 29 CFR 1910.1001. Documentation of disposal shall be provided to the Owner.

D. Following removal of ACM, all plastic sheeting, tape, cleaning material, clothing and all other disposal materials or items used in the work area shall be packed into sealable plastic bags (6 mil minimum), sealed and placed into a second sealed plastic bag or into metal or fiber containers or skips for transport. The containers or skips shall be labeled as prescribed by OSHA Specifications 29 CFR 1910.1001 (f). Plastic sheeting shall not be reused in other containment(s) or work areas.

E. Seal edges of ACM exposed by removal work which is inaccessible, such as TSI within wall cavities at the perimeter of the work area, etc., with penetrating and/or bridging encapsulant. Prior to sealing, permit the exposed edges to dry completely to permit penetration of the bridging encapsulant. Bridging encapsulant shall be tinted to provide visible confirmation of completion.

### 4.4 Decontamination of Work Area

All removed ACM shall be containerized and each regulated area shall be gross cleaned on a daily basis. Removed material will not remain on the floor overnight or during any non-work period.

#### 4.4.1 General

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Project Monitor and Contractor will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order.

A. If applicable, replace pre-filter and the intermediate filter in the Air Filtration Device. Clean all surfaces of the Work Area, including the outside surface of critical barrier sheeting, tools, scaffolding and/or staging, by HEPA-filtered vacuuming, then damp cleaning and mopping. Do not dry-dust or dry-sweep. Continue cleaning until there is no visible dust, debris or residue on polyethylene sheeting and other surfaces.

B. Perform a complete visual inspection of all Work Area surfaces and contents. If any debris or residue is found, repeat the first cleaning and continue decontamination procedure from that point. For glovebag operations a visual inspection will be performed subsequent to the removal of each glovebag.

C. Prior to removal, decontaminate all equipment and materials which are no longer required in the space. Enclosure within goose necked and taped waste bags is permitted provided they are cleaned prior to bagging and prior to removal from the work area.

D. If applicable, allow sufficient time for the Work Area to completely dry while operating HEPA filtered fan units. Maintain operation of negative pressure differential system in operation during the drying period.

E. If applicable, the QP shall conduct a visual inspection of the Work Area when the abatement and decontamination is complete and when the Contractor's supervisor requests such inspection.

F. If applicable, after the visual inspection, an approved lock down encapsulant shall be applied to all the surfaces in the Work Area. The encapsulant used shall not impede installation of new materials. After sufficient drying time, determined by the QP, the final clearance can take place. If an opaque bridging encapsulant is used, the QP shall be afforded the opportunity to review the removal prior to and following application.

G. Additional cleaning required after the first final cleaning will be performed at the expense of the contractor. Additional hours required by the QP will also be an expense paid for by the Contractor, as well as necessary repeat final air clearance analyses.

H. Area air samples will be continuously collected within the regulated area in the vicinity of the glovebag operations, whole component removal, or uncontained outdoor abatement, if conducted. The air samples will be analyzed at the end of each shift and/or upon completion of abatement in a regulated area. If area air samples are found to meet clearance criteria, the contractor will remove and dispose of critical barriers and completely dismantle and remove Decontamination Area. If area air samples exceed the clearance criteria a visual inspection will be performed and additional cleaning will occur as necessary. An additional set of air samples will be collected and analyzed on site using PCM.

I. While they are operating, seal HEPA filtered AFDs with 6 mil polyethylene sheeting and duct tape to form a tight seal at intake and before unit is wet wiped and moved from the Work Area.

**4.5 Final Inspection and Testing**

A. If abatement is performed within a containment or a mini-containment, after cleaning and decontamination of the workspace has been conducted, and if a high degree of cleanliness has been achieved, notify the QP that the workspace is ready for inspection and final testing. The QP will visually inspect each Work Area where such activity was conducted to determine whether the clean-up has been properly completed and to detect any visible asbestos dust or contamination. The QP shall conduct a visual inspection of the Work Area when the abatement and decontamination is complete and when the Contractor's supervisor requests such inspection. The visual inspection will be conducted in general compliance with ASTM E 1368, Standard Practice for Visual Inspection of Asbestos Abatement Projects.

B. If the visual inspection does not reveal any dust or other visible signs of contamination, the final air monitoring will take place.

C. Final air clearance testing shall be conducted by the IH using aggressive air sampling techniques or modified aggressive air sampling techniques in the Work Area in accordance with the procedures detailed in Title 1 of the Virginia Administrative Code Agency 20 Chapter 110 (1VAC30-110) and EPA 40 CFR Part 763.90(i), (2, i) and Appendix A. F&R will collect a minimum of two clearance samples per work area; if the contained space is less than approximately 50 SF, fewer samples may be collected at the discretion of the QP. For larger containment areas, additional samples may be conducted.

D. For abatement areas which are not conducted within negative pressure enclosures, area monitoring samples collected during abatement shall serve as proxy provided they are below 0.01 fibers/cubic centimeter (f/cc). If area sampling in the regulated area or along the regulated area perimeter exceeds 0.01 fibers/cubic centimeter (f/cc), clearance sampling shall be conducted.

E. Phase Contrast Microscopy (PCM) analysis will be performed in accordance with NIOSH Method 7400. Final test results shall show contamination levels not to exceed 0.01 fibers/cubic centimeter (f/cc). Air samples shall have a minimum volume of 1,200 liters per sample but may vary depending on size of Work Area and other variables. If PCM sample results exceed the 0.01 f/cc threshold, the Contractor shall re-clean the containment and re-testing will be performed. The cost of re-testing failed clearance shall be the responsibility of the Contractor. Alternatively, at the request and expense of the Contractor, the Consultant may submit the samples to a qualified laboratory for TEM analysis.

    a. Transmission electron microscopy (TEM) analysis will be performed in accordance with NIOSH Method 7402 at the Owner’s request. Final test results shall show contamination levels not to exceed 70 structures per square mm. Air samples shall have a minimum volume of 1,200 liters per sample but may vary depending on size of Work Area and other variables.

F. The number of air samples required for clearances is as follows, generally:

| Square feet of Enclosure | Number of Samples |
|---|---|
| Less than 100 | 2 |
| 100-500 | 3 |
| 500-1,000 | 4 |
| 1,000-10,000* | 5 |

*For areas greater than 10,000 square feet, the number of inside air samples to be obtained will increase by one sample for each 5,000 square feet over 10,000 square feet.

Within mini containments, a single clearance sample may be collected at the discretion of the QP.

G. For areas where abatement is performed using glovebag technology, where whole component removal is effected, or where work is conducted outside in a regulated area using wet methods and HEPA Vacuum(s), air samples will be continuously collected within the regulated area established for the work. Air samples will be analyzed at the end of the shift, and/or when all abatement is complete in a regulated area. The area air samples will be analyzed by PCM in general conformity to the NIOSH Method 7400. If analysis indicates fiber concentrations at or less than 0.01 f/cc air testing will be considered completed. If all pipe surfaces and surfaces beneath and adjacent to the glovebag work are clean and free of any visible TSI then the area will be considered clear for re-occupancy. If, during periods where other trades are performing work which generated significant amounts of dust and debris, concentrations are above 0.01 f/cc but below the OSHA 0.1 f/cc PEL, the space will be considered completed, at the discretion of the QP. If warranted, following abatement work, a limited clearance sampling event may be completed to demonstrate that airborne concentrations are less than 0.01 f/cc.

**4.6 Abatement Completion**

After thorough decontamination, seal negative air machines with 2 layers of 6 mil poly and duct tape to form a tight seal at the intake/outlet ends before removal from the regulated area in accordance with section 4.4.1. I. Complete asbestos abatement work upon meeting the regulated area visual and air clearance criteria and fulfilling the following:

A. Remove remaining decontaminated equipment and materials from the project area.

B. Dispose of all packaged ACM waste as required, to include poly sheeting from the containment. All ACM and containment poly sheeting shall be disposed of as ACM at a permitted facility.

C. Repair or replace all interior finishes damaged during the abatement work, as required.

D. Fulfill other project closeout requirements as required in this specification including furnishing waste manifest, logs, and other documentation outlined in section 2.1.

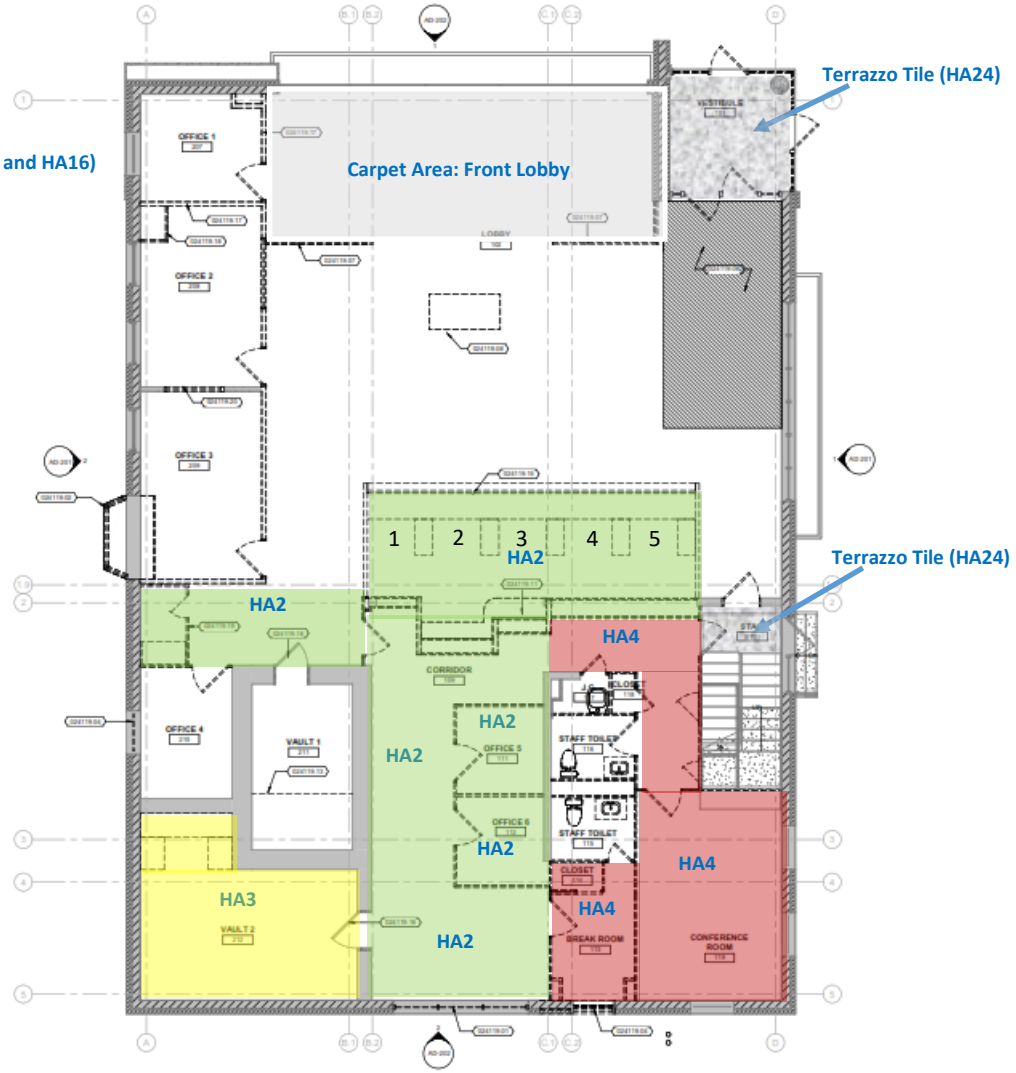
E. Place warning labels on ACM to remain which will be concealed by to-be-installed finish materials

-End of Section 028213-

## Key:

### Homogeneous Area:

- HA2 (Note: HA2 includes HA15 and HA16)
- HA3
- HA4
- HA24




1 FIRST FLOOR DEMOLITION PLAN

## Former BB&T Bank-First Floor


**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701


| | |
|---|---|
| Client: | Town of Strasburg |
| Project: | IH Services: Planned Strasburg Police Station |
| Location: | Strasburg, VA 22657 |
| Project Number: | 59C-0339 |
| Data Source: | Client Provided Drawings |
| Date: | 2025 |

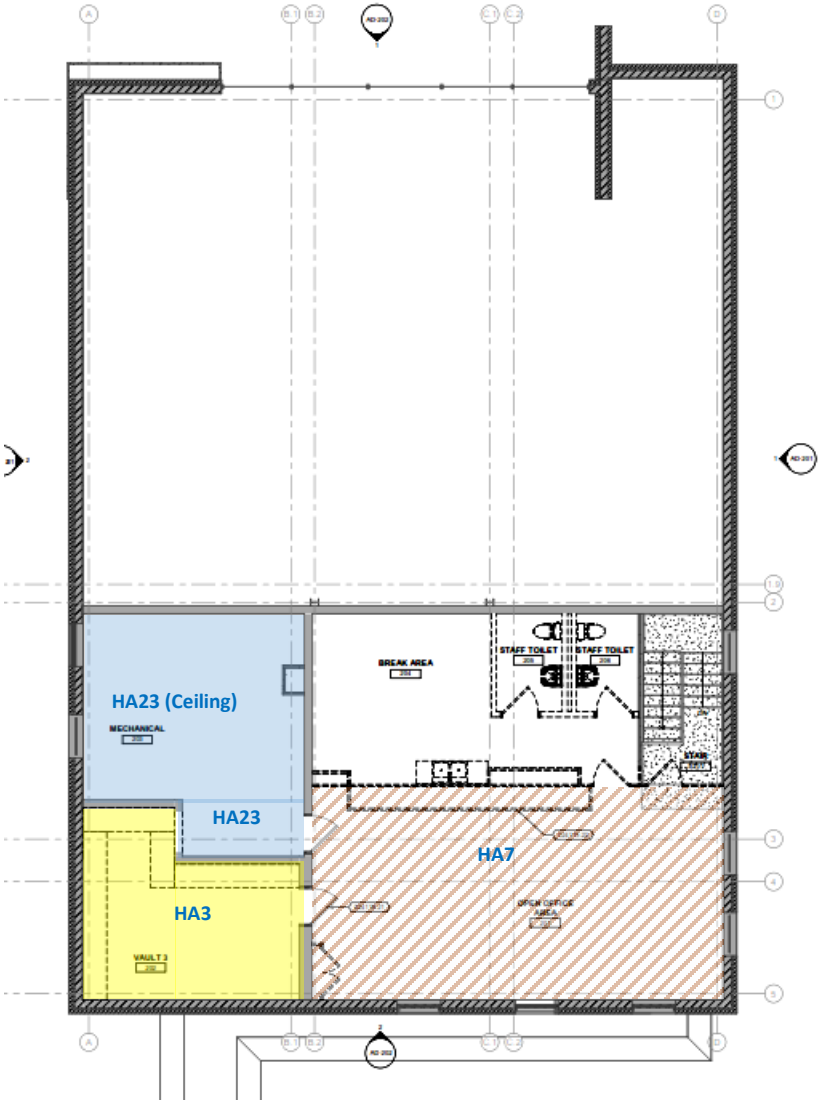
Scale: Not to Scale

FIGURE No.: 1

## Key:

### Homogeneous Area:

- HA3
- HA7
- HA23

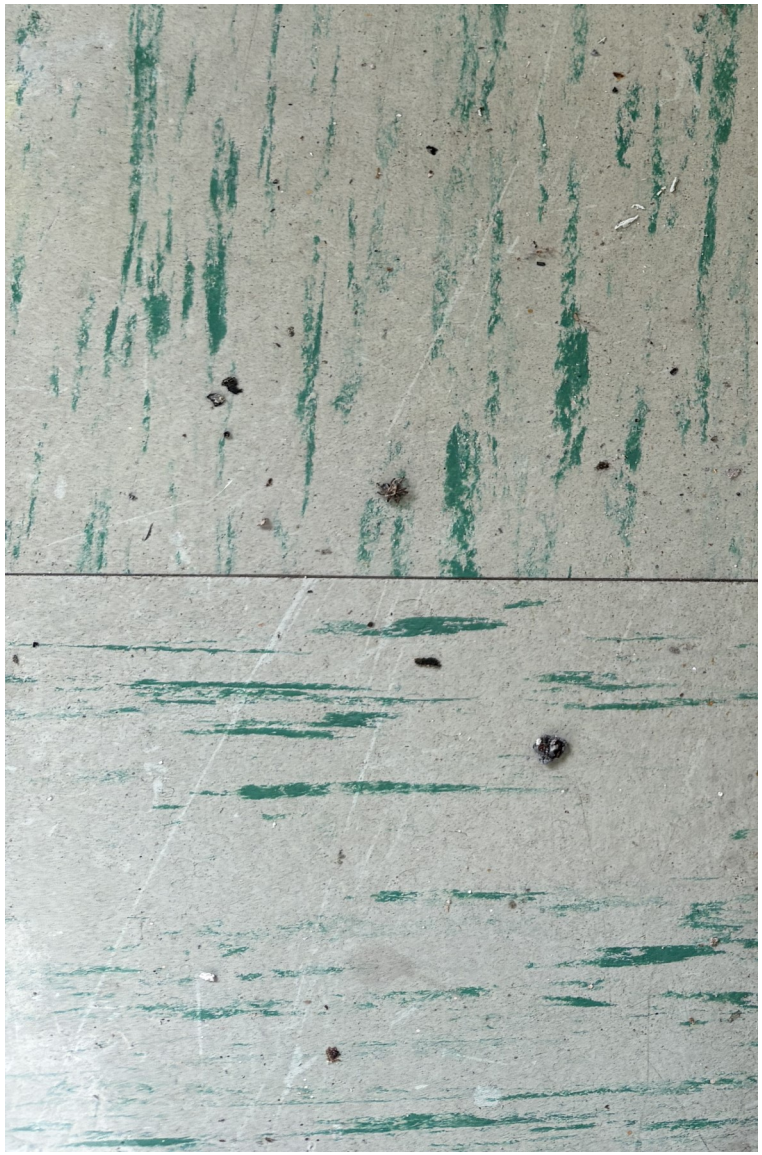



**Former BB&T Bank-Second Floor**

**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701

| | | | |
|---|---|---|---|
| Client: | Town of Strasburg | | |
| Project: | IH Services: Planned Strasburg Police Station | | |
| Location: | Strasburg, VA 22657 | | |
| Project Number: | 59C-0339 | | FIGURE No.: 2 |
| Data Source: | Client provided Drawings | | |
| Date: | 2025 | Scale: Not to Scale | |

# HA2



# HA3

**Former BB&T Bank**

**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701

| | | | |
|---|---|---|---|
| Client: | Town of Strasburg | | |
| Project: | IH Services: Planned Strasburg Police Station | | |
| Location: | Strasburg, VA 22657 | | |
| Project Number: | 59C-0339 | | FIGURE No.: 3 |
| Data Source: | | | |
| Date: | 2025 | Scale: Not to Scale | |

# HA4

# HA7

**Former BB&T Bank**

**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701

| | | | |
|---|---|---|---|
| Client: | Town of Strasburg | | |
| Project: | IH Services: Planned Strasburg Police Station | | |
| Location: | | | |
| Project Number: | 59C-0339 | | FIGURE No.: 4 |
| Data Source: | | | |
| Date: | 2025 | Scale: Not to Scale | |

# Mechanical Room Ceiling (HA23)

**Former BB&T Bank**

**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701

| | | | |
|---|---|---|---|
| Client: | Town of Strasburg | | |
| Project: | IH Services: Planned Strasburg Police Station | | |
| Location: | Strasburg, VA 22657 | | |
| Project Number: | 59C-0339 | | FIGURE No.: 5 |
| Data Source: | | | |
| Date: | 2025 | Scale: Not to Scale | |

# Terrazzo Tile (HA24)

**Former BB&T Bank**

**Froehling & Robertson, Inc.**
3015 Dumbarton Road
Richmond, Virginia 23228
T 804.264.2701

| | | | |
|---|---|---|---|
| Client: | Town of Strasburg | | |
| Project: | IH Services: Planned Strasburg Police Station | | |
| Location: | Strasburg, VA 22657 | | |
| Project Number: | 59C-0339 | | FIGURE No.: 6 |
| Data Source: | | | |
| Date: | 2025 | Scale: Not to Scale | |

SECTION 033000 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section specifies cast-in place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes, for the following:

1. Footings.
2. Slabs-on-grade.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Design Mixtures: For each concrete mixture. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

1. Indicate amounts of mixing water to be withheld for later addition at Project site.

C. Steel Reinforcement Shop Drawings: Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, splices and laps, mechanical connections, tie spacing, hoop spacing, and supports for concrete reinforcement.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For installer and manufacturer.

B. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
2. Admixtures.
3. Form materials and form-release agents.
4. Steel reinforcement and accessories.
5. Curing compounds.
6. Floor and slab treatments.
7. Bonding agents.
8. Adhesives.
9. Vapor barriers.
10. Repair materials.

C. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:

1. Aggregates. Include service record data indicating absence of deleterious expansion of concrete due to alkali aggregate reactivity.

D. Floor surface flatness and levelness measurements indicating compliance with specified tolerances.

E. Minutes of preinstallation conference.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and a supervisor who is an ACI-certified Concrete Flatwork Technician.

B. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94 requirements for production facilities and equipment. Manufacturer shall be certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

C. Testing Agency Qualifications: An independent agency, qualified according to ASTM C 1077 and ASTM E 329 for testing indicated, as documented according to ASTM E 548.

1. Personnel conducting field tests shall be qualified as ACI Concrete Field Testing Technician, Grade 1, according to ACI CP-01 or an equivalent certification program.
2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician - Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician - Grade II.

D. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from one source, and obtain admixtures through one source from a single manufacturer.

E. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301, "Specification for Structural Concrete,"
2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

F. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.

G. Preinstallation Conference: Conduct conference at Project site.

1. Before submitting design mixtures, review concrete design mixture and examine procedures for ensuring quality of concrete materials. Require representatives of each entity directly concerned with cast-in-place concrete to attend, including the following:

a. Contractor's superintendent.
b. Independent testing agency responsible for concrete design mixtures.
c. Ready-mix concrete manufacturer.
d. Concrete subcontractor.
e. Special Inspections Representative.
f. Structural Engineer of Record (optional).

2. Review special inspection and testing and inspecting agency procedures for field quality control, concrete finishes and finishing, cold- and hot-weather concreting procedures,

curing procedures, construction contraction and isolation joints, and joint-filler strips, forms and form removal limitations, shoring and reshoring procedures, vapor-barrier installation, anchor rod and anchorage device installation tolerances, steel reinforcement installation, floor and slab flatness and levelness measurement, concrete repair procedures, and concrete protection.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Steel Reinforcement: Deliver, store, and handle steel reinforcement to prevent bending and damage.

B. Waterstops: Store waterstops under cover to protect from moisture, sunlight, dirt, oil, and other contaminants.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. In other Part 2 articles where titles below introduce lists, the following requirements apply to product selection:

1. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, products specified.
2. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, manufacturers specified.

2.2 FORM-FACING MATERIALS

A. Smooth-Formed Finished Concrete: Form-facing panels that will provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

1. Exterior-grade plywood panels, suitable for concrete forms, complying with DOC PS 1, and as follows:

a. Medium-density overlay, Class 1 or better; mill-release agent treated and edge sealed.

B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

C. Forms for Cylindrical Columns, Pedestals, and Supports: Metal, glass-fiber-reinforced plastic, paper, or fiber tubes that will produce surfaces with gradual or abrupt irregularities not exceeding specified formwork surface class. Provide units with sufficient wall thickness to resist plastic concrete loads without detrimental deformation.

D. Void Forms: Biodegradable paper surface, treated for moisture resistance, structurally sufficient to support weight of plastic concrete and other superimposed loads.

E. Chamfer Strips: Wood, metal, PVC, or rubber strips, 3/4 by 3/4 inch, minimum.

F. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.

1. Formulate form-release agent with rust inhibitor for steel form-facing materials.

G. Form Ties: Factory-fabricated, removable or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.

1. Furnish units that will leave no corrodible metal closer than 1 inch to the plane of exposed concrete surface.
2. Furnish ties that, when removed, will leave holes no larger than 1 inch in diameter in concrete surface.

2.3 STEEL REINFORCEMENT

A. Reinforcing Bars: ASTM A 615, Grade 60, deformed.

B. Low-Alloy-Steel Reinforcing Bars (where weldable rebar is specified): ASTM A 706, deformed.

C. Plain-Steel Wire: ASTM A 82.

D. Plain-Steel Welded Wire Reinforcement: ASTM A 1064, plain, fabricated from as-drawn steel wire into flat sheets.

2.4 REINFORCEMENT ACCESSORIES

A. Joint Dowel Bars: ASTM A 615, Grade 60, plain-steel bars, cut bars true to length with ends square and free of burrs.

B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice," of greater compressive strength than concrete. For concrete surfaces exposed to view where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected steel wire or CRSI Class 2 stainless-steel bar supports.

2.5 CONCRETE MATERIALS

A. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:

1. Portland Cement: ASTM C 150, Type I / II or ASTM C 595, Type IL.

a. Fly Ash: ASTM C 618, Class C or F.

B. Normal-Weight Aggregates: ASTM C 33, Class 3S coarse aggregate or better, graded. Provide aggregates from a single source.

1. Maximum Coarse-Aggregate Size: 1 inch nominal.
2. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.

C. Water: ASTM C 94 and potable.

2.6 ADMIXTURES

A. Air-Entraining Admixture: ASTM C 260.

B. Chemical Admixtures: Provide admixtures certified by manufacturer to be compatible with other admixtures and that will not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride. Anti-freeze admixtures are not permitted.

1. Water-Reducing Admixture: ASTM C 494, Type A.
2. Retarding Admixture: ASTM C 494, Type B.
3. Water-Reducing and Retarding Admixture: ASTM C 494, Type D.
4. High-Range, Water-Reducing Admixture: ASTM C 494, Type F.
5. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494, Type G.
6. Plasticizing and Retarding Admixture: ASTM C 1017, Type II.

## 2.7 VAPOR BARRIERS

A. Plastic Vapor Barrier: ASTM E 1745, Class A with a Perm Rating less than or equal to 0.012 perms (grains / ($ft^2$ *hr *in. Hg)) as tested by ASTM E 96.

1. Available Products:

a. Stego Wrap (15 mil) Vapor Barrier by Stego Industries, LLC.
b. Perminator (15 mil) by W.R. Meadows.
c. Zero-Perm (15 mil) by Alumiseal Industries, Inc.

B. Vapor Barrier Accessories:

1. Seal Tape: Water vapor transmission rate per ASTM E 96 of 0.3 perms or less.
2. Vapor Proofing Mastic: Water vapor transmission rate per ASTM E 96 of 0.3 perms or less.
3. Pipe Boots: Construct pipe boots from vapor barrier material, pressure sensitive tape and/or mastic per manufacturer’s instructions.

## 2.8 FLOOR AND SLAB TREATMENTS

A. Penetrating Liquid Floor Treatment: Use where interior concrete floor slabs will be permanently exposed to view. Clear, chemically reactive, waterborne solution of inorganic silicate or siliconate materials and proprietary components; odorless; colorless; that penetrates, hardens, and densifies concrete surfaces.

1. Available Products:

a. Burke by Edoco; Titan Hard.
b. ChemMasters; Chemisil Plus.
c. Curecrete Distribution Inc.; Ashford Formula.
d. Dayton Superior Corporation; Day-Chem Sure Hard.
e. Euclid Chemical Company (The); Euco Diamond Hard.
f. Meadows, W. R., Inc.; Liqui-Hard.
g. Metalcrete Industries; Floorsaver.
h. Symons Corporation, a Dayton Superior Company; Buff Hard.
i. Vexcon Chemicals, Inc.; Vexcon StarSeal PS.

## 2.9 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

1. Available Products:

    a. Axim Concrete Technologies; Cimfilm.
    b. Burke by Edoco; BurkeFilm.
    c. ChemMasters; Spray-Film.
    d. Dayton Superior Corporation; Sure Film.
    e. Euclid Chemical Company (The); Eucobar.
    f. Kaufman Products, Inc.; Vapor Aid.
    g. Lambert Corporation; Lambco Skin.
    h. L&M Construction Chemicals, Inc.; E-Con.
    i. Meadows, W. R., Inc.; Sealtight Evapre.
    j. Metalcrete Industries; Waterhold.
    k. Nox-Crete Products Group, Kinsman Corporation; Monofilm.
    l. Sika Corporation, Inc.; SikaFilm.
    m. Symons Corporation, a Dayton Superior Company; Finishing Aid.
    n. Vexcon Chemicals, Inc.; Certi-Vex EnvioAssist.

B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

C. Water: Potable.

D. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating.

1. Available Products:

    a. Anti-Hydro International, Inc.; AH Curing Compound #2 DR WB.
    b. Burke by Edoco; Aqua Resin Cure.
    c. ChemMasters; Safe-Cure Clear.
    d. Dayton Superior Corporation; Day Chem Rez Cure (J-11-W).
    e. Euclid Chemical Company (The); Kurez DR VOX.
    f. Kaufman Products, Inc.; Thinfilm 420.
    g. Lambert Corporation; Aqua Kure-Clear.
    h. Meadows, W. R., Inc.; 1100 Clear.
    i. Nox-Crete Products Group, Kinsman Corporation; Resin Cure E.
    j. Symons Corporation, a Dayton Superior Company; Resi-Chem Clear Cure.
    k. Tamms Industries, Inc.; Horncure WB 30.
    l. Vexcon Chemicals, Inc.; Certi-Vex Enviocure 100.

2.10 RELATED MATERIALS

A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber or ASTM D 1752, cork or self-expanding cork.

B. Semirigid Joint Filler: Two-component, semirigid, 100 percent solids, aromatic polyurea with a Type A shore durometer hardness range of 90 to 95 per ASTM D 2240.

C. Bonding Agent: ASTM C 1059, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

D. Epoxy Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade to suit requirements, and as follows:

1. Types IV and V, load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

E. High-Strength Epoxy Doweling System: Epoxy doweling system shall consist of an injectable two-part epoxy complying with the requirements of ASTM C881-90, Type IV, Grade 3, Class B and C except gel times. Epoxy doweling system shall be tested in accordance with ICC Acceptance Criteria 308 demonstrating compliance with the performance features of ACI 355.2. Epoxy doweling system shall have an ICC-ES Evaluation Report (ESR) indicating it is intended for use in both cracked and uncracked normal weight concrete in Seismic Design Categories A through F. The epoxy doweling system shall be installed according to manufacturer's instructions.

1. Available Products:

a. Hilti HIT-HY 200 Epoxy Adhesive Anchoring System (ICC ESR-3187)
b. ITW Red Head EPCON G5 Adhesive Anchoring System (ICC ESR-1137)
c. Simpson Strong-Tie SET-XP Epoxy Adhesive Anchors (ICC ESR-2508)

F. Expansion Wedge Anchors: Expansion wedge anchors shall consist of threaded stud bolt body and integral wedge expander, nut, and washer complying with the requirements of Federal Specification A-A 1923A, Type 4. Expansion wedge anchors shall be tested in accordance with ICC Acceptance Criteria 193 demonstrating compliance with the performance features of ACI 355.2. Expansion wedge anchors shall have an ICC-ES Evaluation Report (ESR) indicating it is intended for use in both cracked and uncracked normal weight concrete in Seismic Design Categories A through F. Anchors shall be made of zinc-plated carbon steel meeting ASTM B633 or Type 304 stainless steel.

1. Available Products:

a. Hilti Kwik Bolt TZ Carbon and Stainless Steel Anchors (ICC ESR-1917)
b. ITW Red Head TRUBOLT+ Wedge Anchors [3/8"-5/8" diameters only] (ICC ESR-2427)
c. Simpson STRONG-BOLT/STRONG-BOLT 2 Wedge Anchor (ICC ESR-1771/ESR-3037)

2.11 REPAIR MATERIALS

A. Repair Underlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch and that can be feathered at edges to match adjacent floor elevations.

1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
2. Primer: Product of underlayment manufacturer recommended for substrate, conditions, and application.
3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch or coarse sand as recommended by underlayment manufacturer.
4. Compressive Strength: Not less than 4000 psi at 28 days when tested according to ASTM C 109.

B. Repair Overlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch and that can be feathered at edges to match adjacent floor elevations.

1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
2. Primer: Product of topping manufacturer recommended for substrate, conditions, and application.
3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch or coarse sand as recommended by topping manufacturer.
4. Compressive Strength: Not less than 5000 psi at 28 days when tested according to ASTM C 109.

2.12 CONCRETE MIXTURES, GENERAL

A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.

B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland cement in concrete as follows:

1. Fly Ash: 25 percent.

C. Limit water-soluble, chloride-ion content in hardened concrete to 0.30 percent by weight of cement.

D. Admixtures: Use admixtures according to manufacturer's written instructions.

1. Use water-reducing, high-range water-reducing, or plasticizing admixture in concrete, as required, for placement and workability.
2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
3. Use water-reducing admixture in pumped concrete and concrete with a water-cementitious materials ratio below 0.50.

2.13 CONCRETE MIXTURES FOR BUILDING ELEMENTS

A. Footings: Proportion normal weight concrete mixture as follows:

1. Minimum Compressive Strength: 3,000 psi at 28 days.
2. Slump Limit: 4 inches, plus or minus 1 inch. 8 inches for concrete with verified slump of 2 to 4 inches before adding high-range water-reducing admixture or plasticizing admixture.
3. Air Content: 5 percent, plus or minus 1 percent at point of delivery for 1-inch nominal maximum aggregate size.

B. Slabs-on-Grade: Proportion normal weight concrete mixture as follows:

1. Minimum Compressive Strength: 4,000 psi at 28 days.
2. Slump Limit: 4 inches, plus or minus 1 inch. 8 inches for concrete with verified slump of 2 to 4 inches before adding high-range water-reducing admixture or plasticizing admixture.
3. Air Content: Do not allow air content of trowel-finished interior floors to exceed 3 percent.

2.14 FABRICATING REINFORCEMENT

A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.15 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94 and ASTM C 1116, and furnish batch ticket information.

1. When air temperature is between 85 and 90 deg F, reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F, reduce mixing and delivery time to 60 minutes.

B. Project-Site Mixing: Not Allowed.

PART 3 - EXECUTION

3.1 FORMWORK

A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.

B. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.

C. Limit concrete surface irregularities, designated by ACI 347R as abrupt or gradual, as follows:

1. Class A, 1/8 inch for smooth-formed finished surfaces.
2. Class B, 1/4 inch for rough-formed finished surfaces.

D. Construct forms tight enough to prevent loss of concrete mortar.

E. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.

1. Install keyways, reglets, recesses, and the like, for easy removal.
2. Do not use rust-stained steel form-facing material.

F. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.

G. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.

H. Chamfer exterior corners and edges of permanently exposed concrete.

I. Form openings, chases, offsets, sinkages, keyways, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.

J. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

K. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

L. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

3.2 EMBEDDED ITEMS

A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

1. Install anchor rods, accurately located, to elevations required and complying with tolerances in Section 7.5 of AISC's "Code of Standard Practice for Steel Buildings and Bridges."
2. Install dovetail anchor slots in concrete structures as indicated.

3.3 REMOVING AND REUSING FORMS

A. General: Formwork for sides of walls, piers, and similar parts of the Work that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F for 24 hours after placing concrete, if concrete is hard enough to not be damaged by form-removal operations and curing and protection operations are maintained.

1. Leave formwork for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved at least 70 percent of its 28-day design compressive strength.
2. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.

C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Architect.

3.4 VAPOR BARRIERS

A. Plastic Vapor Barriers: Place, protect, and repair vapor retarders according to ASTM E 1643 and manufacturer's written instructions. Lap joints 6 inches and seal with manufacturer's recommended tape.

3.5 STEEL REINFORCEMENT

A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.

1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.

C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars. Weld reinforcing bars according to AWS D1.4, where indicated.

1. Weld reinforcing bars according to AWS D1.4 where indicated.

D. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

E. Install welded wire reinforcement in longest practicable lengths on bar supports spaced to minimize sagging. Lap edges and ends of adjoining sheets at least one mesh spacing. Offset laps of adjoining sheet widths to prevent continuous laps in either direction. Lace overlaps with wire.

3.6 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints in walls, beams, columns, piers, and footings (not applicable to slab-on-grade): Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Architect.

1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints, unless otherwise indicated.
2. Form keyed joints as indicated. Embed keys at least 1-1/2 inches into concrete.
3. Locate horizontal joints in walls and columns at underside of floors, slabs, beams, and girders and at the top of footings or floor slabs.
4. Use epoxy-bonding adhesive at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.

C. Doweled Construction Joints in Slab-on-Grade: Install dowel bars and support assemblies at joints where indicated. Lubricate or asphalt coat one-half of dowel length to prevent concrete bonding to one side of joint. Do not continue reinforcement through sides of strip placements of slabs.

D. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness as follows:

1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint to a radius of 1/8 inch. Repeat grooving of contraction joints after applying surface finishes. Eliminate groover tool marks on concrete surfaces.
2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch-wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.

E. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, and other locations, as indicated.

1. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface, unless otherwise indicated.

3.7 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.

1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.

C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

D. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.

1. Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
2. Maintain reinforcement in position on chairs during concrete placement.
3. Screed slab surfaces with a straightedge and strike off to correct elevations.
4. Slope surfaces uniformly to drains where required.
5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.

E. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.

1. When average high and low temperature is expected to fall below 40 deg F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

F. Hot-Weather Placement: Comply with ACI 301 and as follows:

1. Maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.8 FINISHING FORMED SURFACES

A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defects repaired and patched. Remove fins and other projections that exceed specified limits on formed-surface irregularities.

1. Apply to concrete surfaces not exposed to public view.

B. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.

1. Apply to concrete surfaces exposed to public view and for concrete surfaces indicated to receive a rubbed finish.

C. Rubbed Finish: Apply the following to smooth-formed finished as-cast concrete where indicated:

1. Smooth-Rubbed Finish: Not later than one day after form removal, moisten concrete surfaces and rub with carborundum brick or another abrasive until producing a uniform color and texture. Do not apply cement grout other than that created by the rubbing process.

D. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces, unless otherwise indicated.

3.9 FINISHING FLOORS AND SLABS

A. General: Comply with ACI 302.1R recommendations for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Float Finish: Consolidate surface with power-driven floats or by hand floating if area is small or inaccessible to power driven floats. Restraighten, cut down high spots, and fill low spots. Repeat float passes and restraightening until surface is left with a uniform, smooth, granular texture.

1. Apply float finish to surfaces to receive trowel finish.

C. Trowel Finish: After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel. Continue troweling passes and restraighten until surface is free of trowel marks and uniform in texture and appearance. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.

1. Apply a trowel finish to surfaces exposed to view or to be covered with resilient flooring, carpet, ceramic or quarry tile set over a cleavage membrane, paint, or another thin-film-finish coating system.
2. Finish surfaces to the following tolerances, according to ASTM E 1155, for a randomly trafficked floor surface:

    a. Carpeted Slabs: Specified overall values of flatness, F(F) 25; and of levelness, F(L) 20; with minimum local values of flatness, F(F) 17; and of levelness, F(L) 15.
    b. Thin Floor Coverings on Slabs-on-Grade: Specified overall values of flatness, F(F) 35; and of levelness, F(L) 25; with minimum local values of flatness, F(F) 24; and of levelness, F(L) 17.
    c. Thin Floor Coverings on Suspended Slabs: Specified overall values of flatness, F(F) 30; and of levelness, F(L) 20; with minimum local values of flatness, F(F) 24; and of levelness, F(L) 15.

D. Trowel and Fine-Broom Finish: Apply a first trowel finish to surfaces where ceramic or quarry tile is to be installed by either thickset or thin-set method. While concrete is still plastic, slightly scarify surface with a fine broom.

1. Comply with flatness and levelness tolerances for trowel finished floor surfaces.

E. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, and ramps, and elsewhere as indicated.

1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route. Coordinate required final finish with Architect before application.

## 3.10 MISCELLANEOUS CONCRETE ITEMS

A. Filling In: Fill in holes and openings left in concrete structures, unless otherwise indicated, after work of other trades is in place. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete the Work.

## 3.11 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.

E. Cure concrete according to ACI 308.1, by one or a combination of the following methods:

1. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

    a. Use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
    b. Use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
    c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of floor covering used on Project.

2. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.

    a. After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound will not interfere with bonding of floor covering used on Project.

3. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

## 3.12 LIQUID FLOOR TREATMENTS

A. Penetrating Liquid Floor Treatment: Prepare, apply, and finish penetrating liquid floor treatment according to manufacturer's written instructions. Apply penetrating liquid floor treatment to interior slab-on-grade floors which will remain exposed to view.

1. Remove curing compounds, sealers, oil, dirt, laitance, and other contaminants and complete surface repairs.
2. Do not apply to concrete that is less than 14 days' old.
3. Apply liquid until surface is saturated, scrubbing into surface until a gel forms; rewet; and repeat brooming or scrubbing. Rinse with water; remove excess material until surface is dry. Apply a second coat in a similar manner if surface is rough or porous.

B. Sealing Coat: Uniformly apply a continuous sealing coat of curing and sealing compound to hardened concrete by power spray or roller according to manufacturer's written instructions.

## 3.13 CONCRETE SURFACE REPAIRS

A. Defective Concrete: Repair and patch defective areas when approved by Architect. Remove and replace concrete that cannot be repaired and patched to Architect's approval.

B. Patching Mortar: Mix dry-pack patching mortar, consisting of one part portland cement to two and one-half parts fine aggregate passing a No. 16 sieve, using only enough water for handling and placing.

C. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and stains and other discolorations that cannot be removed by cleaning.

1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 inch in any dimension in solid concrete, but not less than 1 inch in depth. Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent. Fill and compact with patching mortar before bonding agent has dried. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
2. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.
3. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by Architect.

D. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template.

1. Repair finished surfaces containing defects. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.01 inch wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
2. After concrete has cured at least 14 days, correct high areas by grinding.
3. Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
4. Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
5. Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 1/4 inch to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
6. Repair defective areas, except random cracks and single holes 1 inch or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least a 3/4-inch clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mixture as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
7. Repair random cracks and single holes 1 inch or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hours.

E. Perform structural repairs of concrete, subject to Architect's approval, using epoxy adhesive and patching mortar.

F. Repair materials and installation not specified above may be used, subject to Architect's approval.

3.14 FIELD QUALITY CONTROL

A. Testing and Inspecting: Owner will engage a Special Inspector and qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

B. Inspections: See Statement of Special Inspections.

C. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:

1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd., but less than 25 cu. yd., plus one set for each additional 50 cu. yd. or fraction thereof.

    a. When frequency of testing will provide fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.

2. Slump: ASTM C 143; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
3. Air Content: ASTM C 231, pressure method, for normal-weight concrete; ASTM C 173, volumetric method, for structural lightweight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
4. Concrete Temperature: ASTM C 1064; one test hourly when air temperature is 40 deg F and below and when 80 deg F and above, and one test for each composite sample.
5. Compression Test Specimens: ASTM C 31. Cast and laboratory cure one set of four standard cylinder specimens for each composite sample.
6. Compressive-Strength Tests: ASTM C 39; test one laboratory-cured specimen at 7 days and two specimens at 28 days, and retain one specimen for later testing at 56 days if 28 day strength falls below the required specified strength.

    a. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.

7. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength and no compressive-strength test value falls below specified compressive strength by more than 500 psi.
8. Test results shall be reported in writing to Architect, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.
9. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by Architect but will not be used as sole basis for approval or rejection of concrete.

10. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Architect. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C 42 or by other methods as directed by Architect.
11. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
12. Correct deficiencies in the Work that test reports and inspections indicate dos not comply with the Contract Documents.

D. Measure floor and slab flatness and levelness according to ASTM E 1155 within 24 hours of finishing.

3.15 PROTECTION OF LIQUID FLOOR TREATMENTS

A. Protect liquid floor treatment from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by liquid floor treatments installer.

END OF SECTION 033000

# SECTION 04 0110 - MASONRY CLEANING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes cleaning the following:

1. Masonry surfaces.

### 1.3 DEFINITIONS

A. Very Low-Pressure Spray: Under 100 psi (690 kPa).

B. Low-Pressure Spray: 100 to 400 psi (690 to 2750 kPa); 4 to 6 gpm (0.25 to 0.4 L/s).

### 1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to cleaning masonry including, but not limited to, the following:

a. Verify masonry-cleaning equipment and facilities needed to make progress and avoid delays.
b. Materials, material application, and sequencing.
c. Cleaning program.
d. Coordination with building occupants.

### 1.5 SEQUENCING AND SCHEDULING

A. Work Sequence: Perform masonry-cleaning work in the following sequence:

1. Remove plant growth.
2. Inspect for open mortar joints. Where repairs are required, delay further cleaning work until after repairs are completed, cured, and dried to prevent the intrusion of water and other cleaning materials into the wall.
3. Remove paint.
4. Clean masonry surfaces.

B. As scaffolding is removed, patch anchor holes used to attach scaffolding. Patch holes in masonry units according to masonry repair Sections. Patch holes in mortar joints according to masonry repointing Sections.

## 1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include material descriptions and application instructions.
2. Include test data substantiating that products comply with requirements.

## 1.7 INFORMATIONAL SUBMITTALS

A. Qualification Data: For paint-remover manufacturer and chemical-cleaner manufacturer.

B. Cleaning program.

## 1.8 QUALITY ASSURANCE

A. Chemical-Cleaner Manufacturer Qualifications: A firm regularly engaged in producing masonry cleaners that have been used for similar applications with successful results, and with factory-authorized service representatives who are available for consultation and Project-site inspection] and on-site assistance.

B. Cleaning Program: Prepare a written cleaning program that describes cleaning process in detail, including materials, methods, and equipment to be used; protection of surrounding materials; and control of runoff during operations. Include provisions for supervising worker performance and preventing damage.

1. If materials and methods other than those indicated are proposed for any phase of cleaning work, add a written description of such materials and methods, including evidence of successful use on comparable projects and demonstrations to show their effectiveness for this Project.

## 1.9 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit masonry-cleaning work to be performed according to product manufacturers' written instructions and specified requirements.

B. Clean masonry surfaces only when air temperature is 40 deg F (4 deg C) and above and is predicted to remain so for at least seven days after completion of cleaning.

PART 2 - PRODUCTS

2.1 PAINT REMOVERS

A. Alkaline Paste Paint Remover: Manufacturer's standard alkaline paste or gel formulation, for removing paint from masonry; containing no methylene chloride.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering equivalent products that may be incorporated into the Work include, but are not limited to the following:

a. Cathedral Stone Products, Inc.
b. Diedrich Technologies, Inc.; a Hohmann & Barnard company.
c. PROSOCO, Inc.

2.2 CLEANING MATERIALS

A. Water: Potable.

B. Hot Water: Water heated to a temperature of 140 to 160 deg F (60 to 71 deg C).

C. Detergent Solution, Job Mixed: Solution prepared by mixing 2 cups (0.5 L) of tetrasodium pyrophosphate (TSPP), 1/2 cup (125 mL) of laundry detergent, and 20 quarts (20 L) of hot water for every 5 gal. (20 L) of solution required.

D. Mold, Mildew, and Algae Remover, Job Mixed: Solution prepared by mixing 2 cups (0.5 L) of tetrasodium pyrophosphate (TSPP), 5 quarts (5 L) of 5 percent sodium hypochlorite (bleach), and 15 quarts (15 L) of hot water for every 5 gal. (20 L) of solution required.

E. Nonacidic Liquid Cleaner: Manufacturer's standard mildly alkaline liquid cleaner formulated for removing mold, mildew, and other organic soiling from ordinary building materials, including polished stone, brick, aluminum, plastics, and wood.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering equivalent products that may be incorporated into the Work include, but are not limited to the following:

a. Cathedral Stone Products, Inc.
b. Diedrich Technologies, Inc.; a Hohmann & Barnard company.
c. PROSOCO, Inc.

2.3 ACCESSORY MATERIALS

A. Liquid Strippable Masking Agent: Manufacturer's standard liquid, film-forming, strippable masking material for protecting glass, metal, glazed masonry, and polished stone surfaces from damaging effects of acidic and alkaline masonry cleaners.

2.4 CHEMICAL CLEANING SOLUTIONS

A. Dilute chemical cleaners with water to produce solutions not exceeding concentration recommended in writing by chemical-cleaner manufacturer.

B. Acidic Cleaner Solution for Nonglazed Masonry and Unpolished Stone: Dilute acidic cleaner with water to produce hydrofluoric acid content of 3 percent or less, but not greater than that recommended in writing by chemical-cleaner manufacturer.

## PART 3 - EXECUTION

### 3.1 PROTECTION

A. Comply with each manufacturer's written instructions for protecting building and other surfaces against damage from exposure to its products. Prevent paint removers and chemical cleaning solutions from coming into contact with people, motor vehicles, landscaping, buildings, and other surfaces that could be harmed by such contact.

1. Cover adjacent surfaces with materials that are proven to resist paint removers and chemical cleaners used unless products being used will not damage adjacent surfaces. Use protective materials that are waterproof and UV resistant. Apply masking agents according to manufacturer's written instructions. Do not apply liquid strippable masking agent to painted or porous surfaces. When no longer needed, promptly remove masking to prevent adhesive staining.
2. Do not apply chemical solutions during winds of enough force to spread them to unprotected surfaces.
3. Neutralize alkaline and acid wastes before disposal.
4. Dispose of runoff from operations by legal means and in a manner that prevents soil erosion, undermining of paving and foundations, damage to landscaping, and water penetration into building interiors.

### 3.2 CLEANING MASONRY, GENERAL

A. Cleaning Appearance Standard: Cleaned surfaces are to have a uniform appearance as viewed from 20 feet (6 m) away by Architect.

B. Proceed with cleaning in an orderly manner; work from top to bottom of each scaffold width and from one end of each elevation to the other. Ensure that dirty residues and rinse water do not wash over dry, cleaned surfaces.

C. Use only those cleaning methods indicated for each masonry material and location.

1. Brushes: Do not use wire brushes or brushes that are not resistant to chemical cleaner being used.
2. Spray Equipment: Use spray equipment that provides controlled application at volume and pressure indicated, measured at nozzle. Adjust pressure and volume to ensure that cleaning methods do not damage surfaces, including joints.

    a. Equip units with pressure gages.
    b. For chemical-cleaner spray application, use low-pressure tank or chemical pump suitable for chemical cleaner indicated, equipped with nozzle having a cone-shaped spray.
    c. For water-spray application, use fan-shaped spray that disperses water at an angle of 25 to 50 degrees.
    d. For heated water-spray application, use equipment capable of maintaining temperature between 140 and 160 deg F (60 and 71 deg C) at flow rates indicated.

e. For steam application, use steam generator capable of delivering live steam at nozzle.

D. Perform each cleaning method indicated in a manner that results in uniform coverage of all surfaces, including corners, moldings, and interstices, and that produces an even effect without streaking or damaging masonry surfaces. Keep wall wet below area being cleaned to prevent streaking from runoff.

E. Perform additional general cleaning, paint and stain removal, and spot cleaning of small areas that are noticeably different when viewed according to the "Cleaning Appearance Standard" Paragraph, so that cleaned surfaces blend smoothly into surrounding areas.

F. Water Application Methods:

1. Water-Soak Application: Soak masonry surfaces by applying water continuously and uniformly to limited area for time indicated. Apply water at low pressures and low volumes in multiple fine sprays using perforated hoses or multiple spray nozzles. Erect a protective enclosure constructed of polyethylene sheeting to cover area being sprayed.
2. Water-Spray Applications: Unless otherwise indicated, hold spray nozzle at least 6 inches (150 mm) from masonry surface and apply water in horizontal back-and-forth sweeping motion, overlapping previous strokes to produce uniform coverage.

G. Chemical-Cleaner Application Methods: Apply chemical cleaners to masonry surfaces according to chemical-cleaner manufacturer's written instructions; use brush or spray application. Do not allow chemicals to remain on surface for periods longer than those indicated or recommended in writing by manufacturer.

H. Rinse off chemical residue and soil by working upward from bottom to top of each treated area at each stage or scaffold setting. Periodically during each rinse, test pH of rinse water running off of cleaned area to determine that chemical cleaner is completely removed.

1. Apply neutralizing agent and repeat rinse if necessary to produce tested pH of between 6.7 and 7.5.

I. After cleaning is complete, remove protection no longer required. Remove tape and adhesive marks.

## 3.3 PRELIMINARY CLEANING

A. Removing Plant Growth: Completely remove visible plant, moss, and shrub growth from masonry surfaces. Carefully remove plants, creepers, and vegetation by cutting at roots and allowing remaining growth to dry as long as possible before removal. Remove loose soil and plant debris from open joints to whatever depth they occur.

B. Preliminary Cleaning: Before beginning general cleaning, remove extraneous substances that are resistant to planned cleaning methods. Extraneous substances include paint, calking, asphalt, and tar.

1. Carefully remove heavy accumulations of rigid materials from masonry surface with sharp chisel. Do not scratch or chip masonry surface.
2. Remove paint and calking with alkaline paint remover.

3.4 CLEANING MASONRY

A. Water Wash: Use water applied by low-pressure spray.

B. Detergent Cleaning:

1. Wet surface with water applied by low-pressure spray.
2. Scrub surface with detergent solution using medium-soft brushes until soil is thoroughly dislodged and can be removed by rinsing. Use small brushes to remove soil from mortar joints and crevices. Dip brush in solution often to ensure that adequate fresh detergent is used and that surface remains wet.
3. Rinse with water applied by low-pressure spray to remove detergent solution and soil.
4. Repeat cleaning procedure above where required to produce cleaning effect established by mockup.

C. Mold, Mildew, and Algae Removal:

1. Wet surface with water applied by low-pressure spray.
2. Apply mold, mildew, and algae remover by brush or low-pressure spray.
3. Scrub surface with medium-soft brushes until mold, mildew, and algae are thoroughly dislodged and can be removed by rinsing. Use small brushes for mortar joints and crevices. Dip brush in mold, mildew, and algae remover often to ensure that adequate fresh cleaner is used and that surface remains wet.
4. Rinse with water applied by low-pressure spray to remove mold, mildew, and algae remover and soil.
5. Repeat cleaning procedure above where required to produce cleaning effect established by mockup.

D. Nonacidic Liquid Chemical Cleaning:

1. Wet surface with water applied by low-pressure spray.
2. Apply cleaner to surface in two applications by brush or low-pressure spray.
3. Let cleaner remain on surface for period recommended in writing by chemical-cleaner manufacturer.
4. Rinse with water applied by low-pressure spray to remove chemicals and soil.
5. Repeat cleaning procedure above where required to produce cleaning effect established by mockup. Do not repeat more than once. If additional cleaning is required, use steam cleaning.

3.5 FINAL CLEANING

A. Clean adjacent non-masonry surfaces of spillage and debris. Use detergent and soft brushes or cloths.

B. Remove debris from gutters and downspouts. Rinse off roof and flush gutters and downspouts.

C. Remove masking materials, leaving no residues that could trap dirt.

END OF SECTION 04 0110

# SECTION 040120.64 - BRICK MASONRY REPOINTING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Repointing joints with mortar.

### 1.3 DEFINITIONS

A. Low-Pressure Spray: 100 to 400 psi (690 to 2750 kPa); 4 to 6 gpm (0.25 to 0.4 L/s).

### 1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to repointing brick masonry including, but not limited to, the following:

    a. Verify brick masonry repointing specialist's personnel, equipment, and facilities needed to make progress and avoid delays.
    b. Materials, material application, sequencing, tolerances, and required clearances.
    c. Quality-control program.
    d. Coordination with building occupants.

### 1.5 SEQUENCING AND SCHEDULING

A. Order materials after approval of samples and mockups. Take delivery of and store at Project site enough quantity to complete Project.

B. Work Sequence: Perform brick masonry repointing work in the following sequence, which includes work specified in this and other Sections:

1. Remove plant growth.
2. Inspect masonry for open mortar joints and permanently or temporarily point them before cleaning to prevent the intrusion of water and other cleaning materials into the wall.
3. Remove paint.
4. Clean masonry.
5. Rake out mortar from joints surrounding masonry to be replaced and from joints adjacent to masonry repairs along joints.

6. Repair masonry, including replacing existing masonry with new masonry materials.
7. Rake out mortar from joints to be repointed.
8. Point mortar and sealant joints.
9. After repairs and repointing have been completed and cured, perform a final cleaning to remove residues from this work.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Include recommendations for product application and use.
3. Include test data substantiating that products comply with requirements.

B. Shop Drawings:

1. Include plans, elevations, sections, and locations of repointing work on the structure.
2. Show provisions for expansion joints or other sealant joints.
3. Show locations of scaffolding and points of scaffolding in contact with masonry. Include details of each point of contact or anchorage.

C. Samples for Initial Selection: For the following:

1. Pointing Mortar: Submit sets of mortar for pointing in the form of sample mortar strips, 6 inches (150 mm) long by 1/4 inch (6 mm) wide, set in aluminum or plastic channels.

    a. Have each set contain a close color range of at least three Samples of different mixes of colored sands and cements that produce a mortar matching existing, cleaned mortar when cured and dry.
    b. Submit with precise measurements on ingredients, proportions, gradations, and source of colored sands from which each Sample was made.

2. Sealant materials.
3. Include similar Samples of accessories involving color selection.

1.7 INFORMATIONAL SUBMITTALS

A. Qualification Data: For brick masonry repointing specialist including field supervisors and workers.

B. Preconstruction Test Reports: For existing bricks and mortar.

C. Quality-control program.

1.8 QUALITY ASSURANCE

A. Brick Masonry Repointing Specialist Qualifications: Engage an experienced brick masonry repointing firm to perform work of this Section. Firm shall have completed work similar in material, design, and extent to that indicated for this Project with a record of successful in-

service performance. Experience in only installing masonry is insufficient experience for masonry repointing work.

1. Field Supervision: Brick masonry repointing specialist firms shall maintain experienced full-time supervisors on Project site during times that brick masonry repointing work is in progress.

B. Quality-Control Program: Prepare a written quality-control program for this Project to systematically demonstrate the ability of personnel to properly follow methods and use materials and tools without damaging masonry. Include provisions for supervising performance and preventing damage.

C. Mockups: Prepare mockups of brick masonry repointing to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Repointing: Rake out joints in two separate areas, each approximately 36 inches (900 mm) high by 48 inches (1200 mm) wide for each type of repointing required, and repoint one of the areas.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

## 1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver packaged materials to Project site in manufacturer's original and unopened containers, labeled with manufacturer's name and type of products.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store hydrated lime in manufacturer's original and unopened containers. Discard lime if containers have been damaged or have been opened for more than two days.

D. Store sand where grading and other required characteristics can be maintained and contamination avoided.

## 1.10 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit repointing work to be performed according to product manufacturers' written instructions and specified requirements.

B. Temperature Limits: Repoint mortar joints only when air temperature is between 40 and 90 deg F (4 and 32 deg C) and is predicted to remain so for at least seven days after completion of the Work unless otherwise indicated.

C. Cold-Weather Requirements: Comply with the following procedures for mortar-joint pointing unless otherwise indicated:

1. When air temperature is below 40 deg F (4 deg C), heat mortar ingredients and existing masonry walls to produce temperatures between 40 and 120 deg F (4 and 49 deg C).
2. When mean daily air temperature is below 40 deg F (4 deg C), provide enclosure and heat to maintain temperatures above 32 deg F (0 deg C) within the enclosure for seven days after pointing.

D. Hot-Weather Requirements: Protect mortar-joint pointing when temperature and humidity conditions produce excessive evaporation of water from mortar materials. Provide artificial shade and wind breaks, and use cooled materials as required to minimize evaporation. Do not apply mortar to substrates with temperatures of 90 deg F (32 deg C) and above unless otherwise indicated.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Source Limitations: Obtain each type of material for repointing brick masonry (cement, sand, etc.) from single source with resources to provide materials of consistent quality in appearance and physical properties.

### 2.2 MORTAR MATERIALS

A. Portland Cement: ASTM C150/C150M, Type I or Type II, except Type III may be used for cold-weather construction; white[or gray, or both where required for color matching of mortar.

1. Provide cement containing not more than 0.60 percent total alkali when tested according to ASTM C114.

B. Hydrated Lime: ASTM C207, Type S.

C. Masonry Cement: ASTM C91/C91M.

D. Mortar Sand: ASTM C144.

1. Match size, texture, and gradation of existing mortar sand as closely as possible. Blend several sands if necessary to achieve suitable match.
2. Color: Natural sand or ground marble, granite, or other sound stone of color necessary to produce required mortar color.

E. Mortar Pigments: ASTM C979/C979M, compounded for use in mortar mixes, and having a record of satisfactory performance in masonry mortars.

F. Water: Potable.

### 2.3 ACCESSORY MATERIALS

A. Sealant Materials:

1. Sealant manufacturer's standard elastomeric sealant(s) of base polymer and characteristics indicated below and according to applicable requirements in Section 079200 "Joint Sealants."

    a. Type: Single-component, nonsag urethane sealant.

2. Colors: Provide colors of exposed sealants to match colors of mortar adjoining installed sealant unless otherwise indicated.

B. Joint-Sealant Backing:

1. Cylindrical Sealant Backings: ASTM C1330, Type C (closed-cell material with a surface skin) or Type B (bicellular material with a surface skin), and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.
2. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended in writing by sealant manufacturer for preventing sealant from adhering to rigid, inflexible, joint-filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable.

C. Masking Tape: Nonstaining, nonabsorbent material; compatible with mortar, joint primers, sealants, and surfaces adjacent to joints; and that easily comes off entirely, including adhesive.

D. Other Products: Select materials and methods of use based on the following, subject to approval of a mockup:

1. Previous effectiveness in performing the work involved.
2. Minimal possibility of damaging exposed surfaces.
3. Consistency of each application.
4. Uniformity of the resulting overall appearance.
5. Do not use products or tools that could leave residue on surfaces.

## 2.4 MORTAR MIXES

A. Measurement and Mixing: Measure cementitious materials and sand in a dry condition by volume or equivalent weight. Do not measure by shovel; use known measure. Mix materials in a clean, mechanical batch mixer.

1. Mixing Pointing Mortar: Thoroughly mix cementitious materials and sand together before adding any water. Then mix again, adding only enough water to produce a damp, unworkable mix that retains its form when pressed into a ball. Maintain mortar in this dampened condition for 15 to 30 minutes. Add remaining water in small portions until mortar reaches desired consistency. Use mortar within one hour of final mixing; do not retemper or use partially hardened material.

B. Colored Mortar: Produce mortar of color required by using specified ingredients. Do not alter specified proportions without Architect's approval.

1. Mortar Pigments: Where mortar pigments are indicated, do not add pigment exceeding 10 percent by weight of the cementitious or binder materials, except for carbon black which is limited to 2 percent, unless otherwise demonstrated by a satisfactory history of performance.

C. Do not use admixtures in mortar unless otherwise indicated.

D. Mixes: Mix mortar materials in the following proportions:

1. Pointing Mortar by Type: ASTM C270, Proportion Specification, Type N> unless otherwise indicated; with cementitious material limited to portland cement and lime masonry cement. Add mortar pigments to produce mortar colors required.

## PART 3 - EXECUTION

### 3.1 PROTECTION

A. Prevent mortar from staining face of surrounding masonry and other surfaces.

1. Cover sills, ledges, and other projecting items to protect them from mortar droppings.
2. Keep wall area wet below pointing work to discourage mortar from adhering.
3. Immediately remove mortar splatters in contact with exposed masonry and other surfaces.

B. Remove gutters and downspouts and associated hardware adjacent to masonry and store during masonry repointing. Reinstall when repointing is complete.

1. Provide temporary rain drainage during work to direct water away from building.

### 3.2 MASONRY REPOINTING, GENERAL

A. Appearance Standard: Repointed surfaces are to have a uniform appearance as viewed from 20 feet (6 m) away by Architect.

### 3.3 REPOINTING

1. Joints at locations of the following defects:

a. Holes and missing mortar.
b. Cracks that can be penetrated 1/4 inch (6 mm) or more by a knife blade 0.027 inch (0.7 mm) thick.
c. Cracks 1/16 inch (1.6 mm) or more in width and of any depth.
d. Hollow-sounding joints when tapped by metal object.
e. Eroded surfaces 1/4 inch (6 mm) or more deep.
f. Deterioration to point that mortar can be easily removed by hand, without tools.
g. Joints filled with substances other than mortar.

B. Do not rake out and repoint joints where not required.

C. Rake out joints as follows, according to procedures demonstrated in approved mockup:

1. Remove mortar from joints to depth of joint width plus 1/8 inch (3 mm). Do not remove unsound mortar more than 2 inches (50 mm) deep; consult Architect for direction.
2. Remove mortar from brick and other masonry surfaces within raked-out joints to provide reveals with square backs and to expose masonry for contact with pointing mortar. Brush, vacuum, or flush joints to remove dirt and loose debris.

3. Do not spall edges of brick or other masonry units or widen joints. Replace or patch damaged brick or other masonry units as directed by Architect.

D. Notify Architect of unforeseen detrimental conditions including voids in mortar joints, cracks, loose masonry units, rotted wood, rusted metal, and other deteriorated items.

E. Pointing with Mortar:

1. Rinse joint surfaces with water to remove dust and mortar particles. Time rinsing application so, at time of pointing, joint surfaces are damp but free of standing water. If rinse water dries, dampen joint surfaces before pointing.
2. Apply pointing mortar first to areas where existing mortar was removed to depths greater than surrounding areas. Apply in layers not greater than 3/8 inch (9 mm) until a uniform depth is formed. Fully compact each layer and allow it to become thumbprint hard before applying next layer.
3. After deep areas have been filled to same depth as remaining joints, point joints by placing mortar in layers not greater than 3/8 inch (9 mm). Fully compact each layer and allow to become thumbprint hard before applying next layer. Where existing masonry units have worn or rounded edges, slightly recess finished mortar surface below face of masonry to avoid widened joint faces. Take care not to spread mortar beyond joint edges onto exposed masonry surfaces or to featheredge the mortar.
4. When mortar is thumbprint hard, tool joints to match original appearance of joints as demonstrated in approved mockup. Remove excess mortar from edge of joint by brushing.
5. Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours, including weekends and holidays.
6. Hairline cracking within mortar or mortar separation at edge of a joint is unacceptable. Completely remove such mortar and repoint.

## 3.4 FINAL CLEANING

A. After mortar has fully hardened, thoroughly clean exposed masonry surfaces of excess mortar and foreign matter; use wood scrapers, stiff-nylon or -fiber brushes, and clean water applied by low-pressure spray.

1. Do not use metal scrapers or brushes.
2. Do not use acidic or alkaline cleaners.

B. Clean adjacent nonmasonry surfaces. Use detergent and soft brushes or cloths.

C. Clean mortar and debris from roof; remove debris from gutters and downspouts. Rinse off roof and flush gutters and downspouts.

D. Remove masking materials, leaving no residues that could trap dirt.

## 3.5 FIELD QUALITY CONTROL

A. Testing Agency: Owner will engage qualified testing agency to perform tests and inspections. Allow inspectors use of lift devices and scaffolding, as needed, to perform inspections.

B. Architect's Project Representatives: Architect will assign Project representatives to help carry out Architect's responsibilities at the site, including observing progress and quality of portion of

the Work completed. Allow Architect's Project representatives use of lift devices and scaffolding, as needed, to observe progress and quality of portion of the Work completed.

C. Notify inspectors and Architect's Project representatives in advance of times when lift devices and scaffolding will be relocated. Do not relocate lift devices and scaffolding until inspectors have had reasonable opportunity to make inspections and observations of work areas at lift device or scaffold location.

END OF SECTION 04 0120

# SECTION 04 2000 - UNIT MASONRY

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Concrete masonry units.
2. Lintels.
3. Brick.
4. Mortar and grout materials.
5. Steel reinforcing bars.
6. Masonry-joint reinforcement.
7. Ties and anchors.
8. Embedded flashing.
9. Miscellaneous masonry accessories.

B. Products Installed but not Furnished under This Section:

1. Steel shelf angles for supporting unit masonry.
2. Cavity wall insulation.

C. Related Requirements:

1. Section 01 4339 "Mockup Requirements" for integrated exterior mockup requirements.
2. Section 07 2100 "Thermal Insulation" for cavity wall insulation.
3. Section 07 6200 "Sheet Metal Flashing and Trim" for exposed sheet metal flashing and for furnishing manufactured reglets installed in masonry joints.

### 1.3 DEFINITIONS

A. CMU(s): Concrete masonry unit(s).

B. Reinforced Masonry: Masonry containing reinforcing steel in grouted cells.

### 1.4 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For the following:

1. Masonry Units: Show sizes, profiles, coursing, and locations of special shapes.
2. Reinforcing Steel: Detail bending, lap lengths, and placement of unit masonry reinforcing bars. Comply with ACI 315. Show elevations of reinforced walls.
3. Fabricated Flashing: Detail corner units, end-dam units, and other special applications.
4. Axonometric details within the shop drawing submittals to illustrate complicated interfaces and expansion provisions.

C. Samples for Verification: For each type and color of the following:

1. Hollow and molded brick.
2. Special brick shapes.
3. Pigmented and colored-aggregate mortar. Make Samples using same sand and mortar ingredients to be used on Project.
4. Weep holes and cavity vents.
5. Accessories embedded in masonry.

1.6 INFORMATIONAL SUBMITTALS

A. List of Materials Used in Constructing Mockups: List generic product names together with manufacturers, manufacturers' product names, model numbers, lot numbers, batch numbers, source of supply, and other information as required to identify materials used. Include mix proportions for mortar and grout and source of aggregates.

1. Submittal is for information only. Receipt of list does not constitute approval of deviations from the Contract Documents unless such deviations are specifically brought to the attention of Architect and approved in writing.

B. Qualification Data: For testing agency.

C. Material Certificates: For each type and size of the following:

1. Masonry units.

a. Include material test reports substantiating compliance with requirements.
b. For brick, include size-variation data verifying that actual range of sizes falls within specified tolerances.
c. For exposed brick, include test report for efflorescence according to ASTM C 67.
d. For surface-coated brick, include test report for durability of surface appearance after 50 cycles of freezing and thawing according to ASTM C 67.
e. For masonry units used in structural masonry, include data and calculations establishing average net-area compressive strength of units.

2. Integral water repellent used in CMUs.
3. Cementitious materials. Include name of manufacturer, brand name, and type.
4. Mortar admixtures.
5. Pre-blended, dry mortar mixes. Include description of type and proportions of ingredients.
6. Grout mixes. Include description of type and proportions of ingredients.
7. Reinforcing bars.
8. Joint reinforcement.
9. Anchors, ties, and metal accessories.

D. Mix Designs: For each type of mortar and grout Include description of type and proportions of ingredients.

1. Include test reports for mortar mixes required to comply with property specification. Test according to ASTM C 109/C 109M for compressive strength, ASTM C 1506 for water retention, and ASTM C 91/C 91M for air content.
2. Include test reports, according to ASTM C 1019, for grout mixes required to comply with compressive strength requirement.

E. Statement of Compressive Strength of Masonry: For each combination of masonry unit type and mortar type, provide statement of average net-area compressive strength of masonry units, mortar type, and resulting net-area compressive strength of masonry determined according to TMS 602/ACI 530.1/ASCE 6.

F. Cold-Weather and Hot-Weather Procedures: Detailed description of methods, materials, and equipment to be used to comply with requirements.

1.7 QUALITY ASSURANCE

A. Testing Agency Qualifications: Qualified according to ASTM C 1093 for testing indicated.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Build mockups for each type of exposed unit masonry construction in sizes approximately 60 inches (1500 mm) long by 60 inches (1500 mm) high by full thickness, including face and backup wythes and accessories.

a. Include a sealant-filled joint at least 16 inches (400 mm) long in each mockup.
b. Include lower corner of window opening, framed with stone trim, at upper corner of exterior wall mockup. Make opening approximately 12 inches (300 mm) wide by 16 inches (400 mm) high.
c. Include through-wall flashing installed for a 24-inch (600-mm) length in corner of exterior wall mockup approximately 16 inches (400 mm) down from top of mockup, with a 12-inch (300-mm) length of flashing left exposed to view (omit masonry above half of flashing).
d. Include metal studs, sheathing, sheathing joint-and-penetration treatment, veneer anchors, flashing, cavity drainage material, and weep holes in exterior masonry-veneer wall mockup.

2. Where masonry is to match existing, erect mockups adjacent and parallel to existing surface.
3. Clean one-half of exposed faces of mockups with masonry cleaner as indicated.
4. Protect accepted mockups from the elements with weather-resistant membrane.
5. Approval of mockups is for color, texture, and blending of masonry units; relationship of mortar and sealant colors to masonry unit colors; tooling of joints; and aesthetic qualities of workmanship.

a. Approval of mockups is also for other material and construction qualities specifically approved by Architect in writing.
b. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.

6. Contractor is not authorized to proceed with manufacture, procurement or delivery of all masonry elements until all shop drawings, product data, mix designs, color samples and required mock-ups are approved by the Architect and Owner.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store aggregates where grading and other required characteristics can be maintained and contamination avoided.

D. Deliver pre-blended, dry mortar mix in moisture-resistant containers. Store pre-blended, dry mortar mix in delivery containers on elevated platforms in a dry location or in covered weatherproof dispensing silos.

E. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.9 FIELD CONDITIONS

A. Protection of Masonry: During construction, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.

1. Extend cover a minimum of 24 inches (600 mm) down both sides of walls, and hold cover securely in place.
2. Where one wythe of multiwythe masonry walls is completed in advance of other wythes, secure cover a minimum of 24 inches (600 mm) down face next to unconstructed wythe, and hold cover in place.

B. Do not apply uniform floor or roof loads for at least 12 hours and concentrated loads for at least three days after building masonry walls or columns.

C. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil that come in contact with such masonry.

1. Protect base of walls from rain-splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
2. Protect sills, ledges, and projections from mortar droppings.
3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.

D. Cold-Weather Requirements: Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost

or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

1. Cold-Weather Cleaning: Use liquid cleaning methods only when air temperature is 40 deg F (4 deg C) and higher and will remain so until masonry has dried, but not less than seven days after completing cleaning.

E. Hot-Weather Requirements: Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

A. Source Limitations for Masonry Units: Obtain exposed masonry units of a uniform texture and color, or a uniform blend within the ranges accepted for these characteristics, from single source from single manufacturer for each product required.

B. Source Limitations for Mortar Materials: Obtain mortar ingredients of a uniform quality, including color for exposed masonry, from single manufacturer for each cementitious component and from single source or producer for each aggregate.

### 2.2 PERFORMANCE REQUIREMENTS

A. Provide unit masonry that develops indicated net-area compressive strengths at 28 days.

1. Determine net-area compressive strength of masonry from average net-area compressive strengths of masonry units and mortar types (unit-strength method) according to TMS 602/ACI 530.1/ASCE 6.

### 2.3 UNIT MASONRY, GENERAL

A. Masonry Standard: Comply with TMS 602/ACI 530.1/ASCE 6, except as modified by requirements in the Contract Documents.

B. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other defects exceeding limits stated. Do not use units where such defects are exposed in the completed Work and will be within 20 feet (6 m) vertically and horizontally of a walking surface.

C. Fire-Resistance Ratings: Comply with requirements for fire-resistance-rated assembly designs indicated.

1. Where fire-resistance-rated construction is indicated, units shall be listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction.

### 2.4 CONCRETE MASONRY UNITS

A. Regional Materials: CMUs shall be manufactured within 500 miles (800 km) of Project site.

B. Shapes: Provide shapes indicated and as follows, with exposed surfaces matching exposed faces of adjacent units unless otherwise indicated.

1. Provide special shapes for lintels, corners, jambs, sashes, movement joints, headers, bonding, and other special conditions.
2. Provide square-edged units for outside corners and bullnose for all exposed interior corners unless otherwise indicated.

C. Insulated CMUs: Where indicated, units shall contain rigid, specially shaped, cellular thermal insulation units complying with ASTM C 578, Type I, designed for installing in cores of masonry units.

D. CMUs: ASTM C 90.

1. Unit Compressive Strength: Provide units with minimum average net-area compressive strength of 2150 psi (14.8 MPa).
2. Density Classification: Normal weight unless otherwise indicated.
3. Size (Width): Manufactured to dimensions 3/8 inch (10 mm) less than nominal dimensions.
4. Exposed Faces: Provide color and texture matching the range represented by Architect's sample.
5. Faces to Receive Plaster: Where units are indicated to receive a direct application of plaster, provide textured-face units made with gap-graded aggregates.

## 2.5 CONCRETE AND MASONRY LINTELS

A. General: Provide one of the following:

B. Masonry Lintels: Prefabricated or built-in-place masonry lintels made from bond beam CMUs matching adjacent CMUs in color, texture, and density classification, with reinforcing bars placed as indicated and filled with coarse grout. Cure precast lintels before handling and installing. Temporarily support built-in-place lintels until cured.

## 2.6 BRICK

A. Regional Materials: Brick shall be manufactured within 500 miles (800 km) of Project site from materials that have been extracted, harvested, or recovered, as well as manufactured, within 500 miles (800 km) of Project site.

B. General: Provide shapes indicated and as follows, with exposed surfaces matching finish and color of exposed faces of adjacent units:

1. For ends of sills and caps and for similar applications that would otherwise expose unfinished brick surfaces, provide units without cores or frogs and with exposed surfaces finished.
2. Provide special shapes for applications as indicated on drawings where stretcher units cannot accommodate special conditions, including those at corners, movement joints, bond beams, sashes, and lintels.

C. Molded Brick: Complying with ASTM C216, Grade SW, Type FBS.

1. Basis-of-Design Product: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

a. Acme Brick Company.

b. Belden Brick Company (The).
c. Bowerston Shale Company.
d. General Shale Inc.
e. Glen-Gery Corporation.

2. Brick Schedule:

a. Brick-1: Match Existing Molded Brick or approved equivalent.

3. Unit Compressive Strength: Provide units with minimum average net-area compressive strength of 3000 psi (20.68 MPa).
4. Size (Actual Dimensions): 3-5/8 inches (92 mm) wide by 2-1/4 inches (57 mm) high by 7-5/8 inches (194 mm) long.
5. Provide Type I (single-faced units) where only one finished face is exposed when units are installed, and Type II (double-faced units) where two opposite finished faces are exposed when units are installed.
6. Application: Use where indicated on drawings.

2.7 MORTAR AND GROUT MATERIALS

A. Regional Materials: Aggregate for mortar and grout, cement, and lime shall be manufactured within 500 miles (800 km) of Project site.

B. Portland Cement: ASTM C 150/C 150M, Type I or II, except Type III may be used for cold-weather construction. Provide natural color or white cement as required to produce mortar color indicated.

1. Alkali content shall not be more than 0.1 percent when tested according to ASTM C 114.

C. Hydrated Lime: ASTM C 207, Type S.

D. Portland Cement-Lime Mix: Packaged blend of portland cement and hydrated lime containing no other ingredients.

E. Masonry Cement: ASTM C 91/C 91M.

F. Colored Cement Products: Packaged blend made from masonry cement and mortar pigments, all complying with specified requirements, and containing no other ingredients.

1. Colored Masonry Cement:

a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1) Argos USA LLC.
2) Holcim (US) Inc.
3) Lafarge North America Inc.
4) Lehigh Hanson; Heidelberg Cement Group.
5) Workrite Cements; A Division of York Building Products

2. Colors: As selected by Architect from manufacturer's full range.
3. Pigments shall not exceed 5 percent of masonry cement by weight.

G. Aggregate for Mortar: ASTM C 144.

1. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.
2. For joints less than 1/4 inch (6 mm) thick, use aggregate graded with 100 percent passing the No. 16 (1.18-mm) sieve.
3. White-Mortar Aggregates: Natural white sand or crushed white stone.
4. Colored-Mortar Aggregates: Natural sand or crushed stone of color necessary to produce required mortar color.

H. Aggregate for Grout: ASTM C 404.

I. Cold-Weather Admixture: Nonchloride, noncorrosive, accelerating admixture complying with ASTM C 494/C 494M, Type C, and recommended by manufacturer for use in masonry mortar of composition indicated.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. BASF Corporation; Construction Systems.
b. Euclid Chemical Company (The); an RPM company.
c. GCP Applied Technologies Inc. (formerly Grace Construction Products).

J. Water: Potable.

2.8 REINFORCEMENT

A. Uncoated-Steel Reinforcing Bars: ASTM A 615/A 615M or ASTM A 996/A 996M, Grade 60 (Grade 420).

B. Reinforcing Bar Positioners: Wire units designed to fit into mortar bed joints spanning masonry unit cells and to hold reinforcing bars in center of cells. Units are formed from 0.148-inch (3.77-mm) steel wire, hot-dip galvanized after fabrication. Provide units designed for number of bars indicated.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Dur-O-Wal; a Hohmann & Barnard company.
b. Heckmann Building Products, Inc.
c. Hohmann & Barnard, Inc.
d. Lock Rite.
e. Wire-Bond.

C. Masonry-Joint Reinforcement, General: ASTM A 951/A 951M.

1. Interior Walls: Hot-dip galvanized carbon steel.
2. Exterior Walls: Hot-dip galvanized carbon steel.
3. Wire Size for Side Rods: 0.148-inch (3.77-mm) diameter.
4. Wire Size for Cross Rods: 0.148-inch (3.77-mm) diameter.
5. Wire Size for Veneer Ties: 0.148-inch (3.77-mm) diameter.
6. Spacing of Cross Rods, Tabs, and Cross Ties: Not more than 16 inches (407 mm) o.c.
7. Provide in lengths of not less than 10 feet (3 m), with prefabricated corner and tee units.

D. Masonry-Joint Reinforcement for Single-Wythe Masonry: Ladder or truss type with single pair of side rods.

E. Masonry-Joint Reinforcement for Multiwythe Masonry:

1. Adjustable (two-piece) type, either ladder or truss design, with one side rod at each face shell of backing wythe and with separate adjustable ties with pintle-and-eye connections having a maximum horizontal play of 1/16 inch (1.5 mm) and maximum vertical adjustment of 1-1/4 inches (32 mm). Size ties to extend at least halfway through facing wythe but with at least 5/8-inch (16-mm) cover on outside face. Ties have hooks or clips to engage a continuous horizontal wire in the facing wythe.

F. Masonry-Joint Reinforcement for Veneers Anchored with Seismic Masonry-Veneer Anchors: Single 0.187-inch- (4.76-mm-) diameter, hot-dip galvanized carbon-steel continuous wire.

2.9 TIES AND ANCHORS

A. General: Ties and anchors shall extend at least 1-1/2 inches (38 mm) into veneer but with at least a 5/8-inch (16-mm) cover on outside face.

B. Materials: Provide ties and anchors specified in this article that are made from materials that comply with the following unless otherwise indicated:

1. Mill-Galvanized, Carbon-Steel Wire: ASTM A 82/A 82M, with ASTM A 641/A 641M, Class 1 coating.
2. Hot-Dip Galvanized, Carbon-Steel Wire: ASTM A 82/A 82M, with ASTM A 153/A 153M, Class B-2 coating.
3. Stainless-Steel Wire: ASTM A 580/A 580M, Type 304.
4. Galvanized-Steel Sheet: ASTM A 653/A 653M, Commercial Steel, G60 (Z180) zinc coating.
5. Steel Sheet, Galvanized after Fabrication: ASTM A 1008/A 1008M, Commercial Steel, with ASTM A 153/A 153M, Class B coating.
6. Stainless-Steel Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304.
7. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
8. Stainless-Steel Bars: ASTM A 276 or ASTM A 666, Type 304.

C. Individual Wire Ties: Rectangular units with closed ends and not less than 4 inches (100 mm) wide.

1. Z-shaped ties with ends bent 90 degrees to provide hooks not less than 2 inches (50 mm) long may be used for masonry constructed from solid units.
2. Where wythes do not align or are of different materials, use adjustable ties with pintle-and-eye connections having a maximum adjustment of 1-1/4 inches (32 mm).
3. Wire: Fabricate from 3/16-inch- (4.76-mm-) diameter, hot-dip galvanized steel

D. Adjustable Anchors for Connecting to Structural Steel Framing: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.

1. Anchor Section for Welding to Steel Frame: Crimped 1/4-inch- (6.35-mm-) diameter, hot-dip galvanized steel wire. Mill-galvanized wire may be used at interior walls unless otherwise indicated.

2. Tie Section: Triangular-shaped wire tie made from 0.187-inch- (4.76-mm-) diameter, hot-dip galvanized steel wire. Mill-galvanized wire may be used at interior walls unless otherwise indicated.

E. Adjustable Anchors for Connecting to Concrete: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.

1. Corrugated-Metal Ties: Metal strips not less than 7/8 inch (22 mm) wide with corrugations having a wavelength of 0.3 to 0.5 inch (7.6 to 12.7 mm) and an amplitude of 0.06 to 0.10 inch (1.5 to 2.5 mm) made from 0.060-inch- (1.52-mm-) thick steel sheet, galvanized after fabrication with dovetail tabs for inserting into dovetail slots in concrete.

a. 0.064-inch- (1.63-mm-) thick galvanized sheet may be used at interior walls unless otherwise indicated.

F. Partition Top Anchors: 0.105-inch- (2.66-mm-) thick metal plate with a 3/8-inch- (9.5-mm-) diameter metal rod 6 inches (152 mm) long welded to plate and with closed-end plastic tube fitted over rod that allows rod to move in and out of tube. Fabricate from steel, hot-dip galvanized after fabrication.

G. Adjustable Masonry-Veneer Anchors:

1. General: Provide anchors that allow vertical adjustment but resist a 100-lbf (445-N) load in both tension and compression perpendicular to plane of wall without deforming or developing play in excess of 1/16 inch (1.5 mm).

2. Contractor's Option: Unless otherwise indicated, provide any of the adjustable masonry-veneer anchors specified.

3. Screw-Attached, Masonry-Veneer Anchors: Wire tie and a rib-stiffened, sheet metal anchor section with screw holes top and bottom, with a projecting vertical tab having a slotted hole for inserting wire tie.

4. Seismic Masonry-Veneer Anchors: Connector section and rib-stiffened, sheet metal anchor section with screw holes top and bottom, with projecting tabs having slotted holes for inserting vertical leg of connector section. Connector section consists of a rib-stiffened, sheet metal bent plate with down-turned leg designed to fit in anchor section slot and with integral tabs designed to engage continuous wire.

5. Polymer-Coated, Steel Drill Screws for Steel Studs: ASTM C 954 except manufactured with hex washer head and neoprene or EPDM washer, No. 10 (4.83-mm) diameter by length required to penetrate steel stud flange with not less than three exposed threads, and with organic polymer coating with salt-spray resistance to red rust of more than 800 hours according to ASTM B 117.

2.10 EMBEDDED FLASHING MATERIALS

A. Metal Flashing: Provide metal flashing complying with SMACNA's "Architectural Sheet Metal Manual" and as follows:

1. Stainless Steel: ASTM A 240/A 240M or ASTM A 666, Type 304, 0.016 inch (0.40 mm) thick.

2. Fabricate continuous flashings in sections 96 inches (2400 mm) long minimum, but not exceeding 12 feet (3.7 m). Provide splice plates at joints of formed, smooth metal flashing.

3. Fabricate through-wall metal flashing embedded in masonry from stainless steel, with ribs at 3-inch (76-mm) intervals along length of flashing to provide an integral mortar bond.

    a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

        1) Cheney Flashing Company.
        2) Hohmann & Barnard, Inc.
        3) Keystone Flashing Company, Inc.
        4) Sandell Manufacturing Co., Inc.

4. Fabricate through-wall flashing with snaplock receiver on exterior face where indicated to receive counterflashing.
5. Fabricate through-wall flashing with drip edge where indicated. Fabricate by extending flashing 1/2 inch (13 mm) out from wall, with outer edge bent down 30 degrees and hemmed.
6. Fabricate through-wall flashing with sealant stop unless otherwise indicated. Fabricate by bending metal back on itself 3/4 inch (19 mm) at exterior face of wall and down into joint 1/4 inch (6 mm) to form a stop for retaining sealant backer rod.
7. Fabricate metal drip edges from stainless steel. Extend at least 3 inches (76 mm) into wall and 1/2 inch (13 mm) out from wall, with outer edge bent down 30 degrees and hemmed.
8. Fabricate metal sealant stops from stainless steel. Extend at least 3 inches (76 mm) into wall and out to exterior face of wall. At exterior face of wall, bend metal back on itself for 3/4 inch (19 mm) and down into joint 1/4 inch (6 mm) to form a stop for retaining sealant backer rod.
9. Fabricate metal expansion-joint strips from stainless steel to shapes indicated.
10. Solder metal items at corners.

B. Flexible Flashing: Use the following unless otherwise indicated:

1. Self-Adhering, Stainless Steel Fabric Flashing: Composite, flashing product consisting of 2 mil (0.05 mm) of Type 304 stainless steel sheet, bonded to a layer of polymeric fabric with a butyl adhesive, to produce an overall thickness of 10 mil (0.25 mm).

    a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

        1) Hohmann & Barnard, Inc. – Mighty-Flash SA
        2) VaproShield LLC. – Vapro-SS.
        3) Wire-Bond – Bond-N-Flash SA
        4) York Flashings (York Manufacturing) – 304 SA.

    b. Accessories: Provide preformed corners, end dams, other special shapes, and seaming materials produced by flashing manufacturer.

C. Solder and Sealants for Sheet Metal Flashings:

1. Solder for Stainless Steel: ASTM B 32, Grade Sn60, with acid flux of type recommended by stainless-steel sheet manufacturer.

D. Adhesives, Primers, and Seam Tapes for Flashings: Flashing manufacturer's standard products or products recommended by flashing manufacturer for bonding flashing sheets to each other and to substrates.

E. Termination Bars for Flexible Flashing: Stainless steel bars 0.075 inch by 1 inch (1.90 mm by 25 mm)

2.11 MISCELLANEOUS MASONRY ACCESSORIES

A. Compressible Filler: Premolded filler strips complying with ASTM D 1056, Grade 2A1; compressible up to 35 percent; of width and thickness indicated; formulated from neoprene, urethane or PVC.

B. Preformed Control-Joint Gaskets: Made from styrene-butadiene-rubber compound, complying with ASTM D 2000, Designation M2AA-805 or PVC, complying with ASTM D 2287, Type PVC-65406 and designed to fit standard sash block and to maintain lateral stability in masonry wall; size and configuration as indicated.

C. Bond-Breaker Strips: Asphalt-saturated felt complying with ASTM D 226/D 226M, Type I (No. 15 asphalt felt).

D. Weep/Cavity Vent Products: Use one of the following unless otherwise indicated:

1. Cellular Plastic Weep/Vent: One-piece, flexible extrusion made from UV-resistant polypropylene copolymer, full height and width of head joint and depth 1/8 inch (3 mm) less than depth of outer wythe, in color selected from manufacturer's standard.

a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1) Advanced Building Products Inc.
2) Hohmann & Barnard, Inc.
3) Wire-Bond.

E. Cavity Drainage Material: Free-draining mesh, made from polymer strands that will not degrade within the wall cavity.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Advanced Building Products Inc.
b. Hohmann & Barnard, Inc.
c. Mortar Net Solutions.
d. Wire-Bond.

2. Configuration: Provide one of the following:

a. Strips, full depth of cavity and 10 inches (250 mm) high, with dovetail-shaped notches 7 inches (175 mm) deep that prevent clogging with mortar droppings.

b. Strips, not less than 3/4 inch (19 mm)]thick and 10 inches (250 mm) high, with dimpled surface designed to catch mortar droppings and prevent weep holes from clogging with mortar.

2.12 MASONRY CLEANERS

A. Proprietary Acidic Cleaner: Manufacturer's standard-strength cleaner designed for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cleaner manufacturer and manufacturer of masonry units being cleaned.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Diedrich Technologies, Inc.; a division of Sandell Construction Solutions.
b. EaCo Chem, Inc.
c. PROSOCO, Inc.

2.13 MORTAR AND GROUT MIXES

A. General: Do not use admixtures, including pigments, air-entraining agents, accelerators, retarders, water-repellent agents, antifreeze compounds, or other admixtures unless otherwise indicated.

1. Do not use calcium chloride in mortar or grout.
2. Use portland cement-lime mortar unless otherwise indicated.
3. For exterior masonry, use portland cement-lime mortar.
4. For reinforced masonry, use portland cement-lime mortar.
5. Add cold-weather admixture (if used) at same rate for all mortar that will be exposed to view, regardless of weather conditions, to ensure that mortar color is consistent.

B. Preblended, Dry Mortar Mix: Furnish dry mortar ingredients in form of a preblended mix. Measure quantities by weight to ensure accurate proportions, and thoroughly blend ingredients before delivering to Project site.

C. Mortar for Unit Masonry: Comply with ASTM C 270, Property Specification. Provide the following types of mortar for applications stated unless another type is indicated or needed to provide required compressive strength of masonry.

1. For masonry below grade or in contact with earth, use Type M.
2. For reinforced masonry, use Type M.
3. For mortar parge coats, use Type S or Type N.
4. For exterior, above-grade, load-bearing and nonload-bearing walls and parapet walls; for interior load-bearing walls; for interior nonload-bearing partitions; and for other applications where another type is not indicated, use Type N.
5. For interior nonload-bearing partitions, Type O may be used instead of Type N.

D. Pigmented Mortar: Use colored cement product or select and proportion pigments with other ingredients to produce color required. Do not add pigments to colored cement products.

1. Pigments shall not exceed 10 percent of portland cement by weight.
2. Mix to match Architect's sample.

E. Colored-Aggregate Mortar: Produce required mortar color by using colored aggregates and natural color or white cement as necessary to produce required mortar color.

1. Mix to match Architect's sample.

F. Grout for Unit Masonry: Comply with ASTM C 476.

1. Use grout of type indicated or, if not otherwise indicated, of type (fine or coarse) that will comply with TMS 602/ACI 530.1/ASCE 6 for dimensions of grout spaces and pour height.
2. Proportion grout in accordance with ASTM C 476, Table 1 or paragraph 4.2.2 for specified 28-day compressive strength indicated, but not less than 2000 psi (14 MPa).
3. Provide grout with a slump of 8 to 11 inches (200 to 280 mm) as measured according to ASTM C 143/C 143M.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

1. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
2. Verify that foundations are within tolerances specified.
3. Verify that reinforcing dowels are properly placed.
4. Verify that substrates are free of substances that impair mortar bond.

B. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of piping connections.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Thickness: Build cavity and composite walls and other masonry construction to full thickness shown. Build single-wythe walls to actual widths of masonry units, using units of widths indicated.

B. Build chases and recesses to accommodate items specified in this and other Sections.

C. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match construction immediately adjacent to opening.

D. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.

E. Select and arrange units for exposed unit masonry to produce a uniform blend of colors and textures. Mix units from several pallets or cubes as they are placed.

F. Wetting of Brick: Wet brick before laying if initial rate of absorption exceeds 30 g/30 sq. in. (30 g/194 sq. cm) per minute when tested according to ASTM C 67. Allow units to absorb water so they are damp but not wet at time of laying.

3.3 TOLERANCES

A. Dimensions and Locations of Elements:

1. For dimensions in cross section or elevation, do not vary by more than plus 1/2 inch (12 mm) or minus 1/4 inch (6 mm).
2. For location of elements in plan, do not vary from that indicated by more than plus or minus 1/2 inch (12 mm).
3. For location of elements in elevation, do not vary from that indicated by more than plus or minus 1/4 inch (6 mm) in a story height or 1/2 inch (12 mm) total.

B. Lines and Levels:

1. For bed joints and top surfaces of bearing walls, do not vary from level by more than 1/4 inch in 10 feet (6 mm in 3 m), or 1/2-inch (12-mm) maximum.
2. For conspicuous horizontal lines, such as lintels, sills, parapets, and reveals, do not vary from level by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2-inch (12-mm) maximum.
3. For vertical lines and surfaces, do not vary from plumb by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2-inch (12-mm) maximum.
4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2-inch (12-mm) maximum.
5. For lines and surfaces, do not vary from straight by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2-inch (12-mm) maximum.
6. For vertical alignment of exposed head joints, do not vary from plumb by more than 1/4 inch in 10 feet (6 mm in 3 m), or 1/2-inch (12-mm) maximum.
7. For faces of adjacent exposed masonry units, do not vary from flush alignment by more than 1/16 inch (1.5 mm) except due to warpage of masonry units within tolerances specified for warpage of units.

C. Joints:

1. For bed joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm), with a maximum thickness limited to 1/2 inch (12 mm).
2. For exposed bed joints, do not vary from bed-joint thickness of adjacent courses by more than 1/8 inch (3 mm).
3. For head and collar joints, do not vary from thickness indicated by more than plus 3/8 inch (9 mm) or minus 1/4 inch (6 mm).
4. For exposed head joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm). Do not vary from adjacent bed-joint and head-joint thicknesses by more than 1/8 inch (3 mm).
5. For exposed bed joints and head joints of stacked bond, do not vary from a straight line by more than 1/16 inch (1.5 mm) from one masonry unit to the next.

3.4 LAYING MASONRY WALLS

A. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thicknesses and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less-than-half-size units, particularly at corners, jambs, and, where possible, at other locations.

B. Bond Pattern for Exposed Masonry: Unless otherwise indicated, lay exposed masonry in bond patterns indicated on Drawings; do not use units with less-than-nominal 4-inch (100-mm) horizontal face dimensions at corners or jambs.

C. Stopping and Resuming Work: Stop work by stepping back units in each course from those in course below; do not tooth. When resuming work, clean masonry surfaces that are to receive mortar, remove loose masonry units and mortar, and wet brick if required before laying fresh masonry.

D. Built-in Work: As construction progresses, build in items specified in this and other Sections. Fill in solidly with masonry around built-in items.

E. Fill space between steel frames and masonry solidly with mortar unless otherwise indicated.

F. Where built-in items are to be embedded in cores of hollow masonry units, place a layer of metal lath, wire mesh, or plastic mesh in the joint below, and rod mortar or grout into core.

G. Fill cores in hollow CMUs with grout 24 inches (600 mm) under bearing plates, beams, lintels, posts, and similar items unless otherwise indicated.

H. Build nonload-bearing interior partitions full height of story to underside of solid floor or roof structure above unless otherwise indicated.

1. Install compressible filler in joint between top of partition and underside of structure above.
2. Fasten partition top anchors to structure above and build into top of partition. Grout cells of CMUs solidly around plastic tubes of anchors and push tubes down into grout to provide 1/2-inch (13-mm) clearance between end of anchor rod and end of tube. Space anchors 48 inches (1200 mm) o.c. unless otherwise indicated.
3. Wedge nonload-bearing partitions against structure above with small pieces of tile, slate, or metal. Fill joint with mortar after dead-load deflection of structure above approaches final position.
4. At fire-rated partitions, treat joint between top of partition and underside of structure above to comply with Section 078443 "Joint Firestopping."

3.5 MORTAR BEDDING AND JOINTING

A. Lay hollow brick and CMUs as follows:

1. Bed face shells in mortar and make head joints of depth equal to bed joints.
2. Bed webs in mortar in all courses of piers, columns, and pilasters.
3. Bed webs in mortar in grouted masonry, including starting course on footings.
4. Fully bed entire units, including areas under cells, at starting course on footings where cells are not grouted.
5. Fully bed units and fill cells with mortar at anchors and ties as needed to fully embed anchors and ties in mortar.

B. Lay solid masonry units and hollow brick with completely filled bed and head joints; butter ends with sufficient mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.

C. Set cast-stone trim units in full bed of mortar with full vertical joints. Fill dowel, anchor, and similar holes.

1. Clean soiled surfaces with fiber brush and soap powder and rinse thoroughly with clear water.
2. Allow cleaned surfaces to dry before setting.
3. Wet joint surfaces thoroughly before applying mortar.
4. Rake out mortar joints for pointing with sealant.

D. Rake out mortar joints at pre-faced CMUs and glazed brick to a uniform depth of 1/4 inch (6 mm) and point with epoxy mortar to comply with epoxy-mortar manufacturer's written instructions.

E. Tool exposed joints slightly concave when thumbprint hard, using a jointer larger than joint thickness unless otherwise indicated.

1. For glazed masonry units, use a nonmetallic jointer 3/4 inch (19 mm) or more in width.

F. Cut joints flush for masonry walls to receive plaster or other direct-applied finishes (other than paint) unless otherwise indicated.

G. Cut joints flush where indicated to receive waterproofing, cavity wall insulation and air barriers unless otherwise indicated.

3.6 CAVITY WALLS

A. Bond wythes of cavity walls together using one of the following methods:

1. Individual Metal Ties: Provide ties as shown installed in horizontal joints, but not less than one metal tie for 4.5 sq. ft. (0.42 sq. m) of wall area spaced not to exceed 36 inches (914 mm) o.c. horizontally and 16 inches (406 mm) o.c. vertically. Stagger ties in alternate courses. Provide additional ties within 12 inches (305 mm) of openings and space not more than 36 inches (915 mm) apart around perimeter of openings. At intersecting and abutting walls, provide ties at no more than 24 inches (610 mm) o.c. vertically.

a. Where bed joints of wythes do not align, use adjustable-type (two-piece-type) ties.
b. Where one wythe is of clay masonry and the other of concrete masonry, use adjustable-type (two-piece-type) ties to allow for differential movement regardless of whether bed joints align.

2. Masonry-Joint Reinforcement: Installed in horizontal mortar joints.

a. Where bed joints of both wythes align, use ladder-type reinforcement extending across both wythes.
b. Where bed joints of wythes do not align, use adjustable-type (two-piece-type) reinforcement with continuous horizontal wire in facing wythe attached to ties.
c. Where one wythe is of clay masonry and the other of concrete masonry, use adjustable-type (two-piece-type) reinforcement with continuous horizontal wire in facing wythe attached to ties to allow for differential movement regardless of whether bed joints align.

3. Header Bonding: Provide masonry unit headers extending not less than 3 inches (76 mm) into each wythe. Space headers not more than 8 inches (203 mm) clear horizontally and 16 inches (406 mm) clear vertically.
4. Masonry-Veneer Anchors: Comply with requirements for anchoring masonry veneers.

B. Bond wythes of cavity walls together using bonding system indicated on Drawings.

C. Keep cavities clean of mortar droppings and other materials during construction. Bevel beds away from cavity, to minimize mortar protrusions into cavity. Do not attempt to trowel or remove mortar fins protruding into cavity.

D. Parge cavity face of backup wythe in a single coat approximately 3/8 inch (10 mm) thick. Trowel face of parge coat smooth.

E. Installing Cavity Wall Insulation: Place small dabs of adhesive, spaced approximately 12 inches (300 mm) o.c. both ways, on inside face of insulation boards, or attach with plastic fasteners designed for this purpose. Fit courses of insulation between wall ties and other confining obstructions in cavity, with edges butted tightly both ways. Press units firmly against inside wythe of masonry or other construction as shown.

1. Fill cracks and open gaps in insulation with crack sealer compatible with insulation and masonry.

3.7 ANCHORED MASONRY VENEERS

A. Anchor masonry veneers to wall framing and concrete and masonry backup with masonry-veneer anchors to comply with the following requirements:

1. Fasten screw-attached and seismic anchors through sheathing to wall framing and] to concrete and masonry backup with metal fasteners of type indicated. Use two fasteners unless anchor design only uses one fastener.
2. Embed connector sections and continuous wire in masonry joints.
3. Locate anchor sections to allow maximum vertical differential movement of ties up and down.
4. Space anchors as indicated, but not more than 18 inches (458 mm) o.c. vertically and 24 inches (610 mm) o.c. horizontally, with not less than one anchor for each 2 sq. ft. (0.2 sq. m) of wall area. Install additional anchors within 12 inches (305 mm) of openings and at intervals, not exceeding 8 inches (203 mm), around perimeter.

B. Provide not less than 2 inches (50 mm) of airspace between back of masonry veneer and face of sheathing.

1. Keep airspace clean of mortar droppings and other materials during construction. Bevel beds away from airspace, to minimize mortar protrusions into airspace. Do not attempt to trowel or remove mortar fins protruding into airspace.

3.8 MASONRY-JOINT REINFORCEMENT

A. General: Install entire length of longitudinal side rods in mortar with a minimum cover of 5/8 inch (16 mm) on exterior side of walls, 1/2 inch (13 mm) elsewhere. Lap reinforcement a minimum of 6 inches (150 mm).

1. Space reinforcement not more than 16 inches (406 mm) o.c.
2. Space reinforcement not more than 8 inches (203 mm) o.c. in foundation walls and parapet walls.
3. Provide reinforcement not more than 8 inches (203 mm) above and below wall openings and extending 12 inches (305 mm) beyond openings.

B. Interrupt joint reinforcement at control and expansion joints unless otherwise indicated.

C. Provide continuity at wall intersections by using prefabricated T-shaped units.

D. Provide continuity at corners by using prefabricated L-shaped units.

E. Cut and bend reinforcing units as directed by manufacturer for continuity at corners, returns, offsets, column fireproofing, pipe enclosures, and other special conditions.

3.9 ANCHORING MASONRY TO STRUCTURAL STEEL AND CONCRETE

A. Anchor masonry to structural steel and concrete, where masonry abuts or faces structural steel or concrete, to comply with the following:

1. Provide an open space not less than 1 inch (25 mm) wide between masonry and structural steel or concrete unless otherwise indicated. Keep open space free of mortar and other rigid materials.
2. Anchor masonry with anchors embedded in masonry joints and attached to structure.
3. Space anchors as indicated, but not more than 24 inches (610 mm) o.c. vertically and 36 inches (915 mm) o.c. horizontally.

3.10 CONTROL AND EXPANSION JOINTS

A. General: Install control- and expansion-joint materials in unit masonry as masonry progresses. Do not allow materials to span control and expansion joints without provision to allow for in-plane wall or partition movement.

B. Form control joints in concrete masonry using one of the following methods:

1. Fit bond-breaker strips into hollow contour in ends of CMUs on one side of control joint. Fill resultant core with grout, and rake out joints in exposed faces for application of sealant.
2. Install preformed control-joint gaskets designed to fit standard sash block.
3. Install interlocking units designed for control joints. Install bond-breaker strips at joint. Keep head joints free and clear of mortar, or rake out joint for application of sealant.
4. Install temporary foam-plastic filler in head joints, and remove filler when unit masonry is complete for application of sealant.

C. Form expansion joints in brick as follows:

1. Build flanges of metal expansion strips into masonry. Lap each joint 4 inches (100 mm) in direction of water flow. Seal joints below grade and at junctures with horizontal expansion joints if any.
2. Build flanges of factory-fabricated, expansion-joint units into masonry.
3. Build in compressible joint fillers where indicated.
4. Form open joint full depth of brick wythe and of width indicated, but not less than 3/8 inch (10 mm) for installation of sealant and backer rod specified in Section 079200 "Joint Sealants."

D. Provide horizontal, pressure-relieving joints by either leaving an airspace or inserting a compressible filler of width required for installing sealant and backer rod specified in Section 079200 "Joint Sealants," but not less than 3/8 inch (10 mm).

1. Locate horizontal, pressure-relieving joints beneath shelf angles supporting masonry.

3.11 LINTELS

A. Install steel lintels where indicated.

B. Provide concrete or masonry lintels where shown and where openings of more than 12 inches (305 mm) for brick-size units and 24 inches (610 mm) for block-size units are shown without structural steel or other supporting lintels.

C. Provide minimum bearing of 8 inches (200 mm) at each jamb unless otherwise indicated.

3.12 FLASHING, WEEP HOLES, AND CAVITY VENTS

A. General: Install embedded flashing and weep holes in masonry at shelf angles, lintels, ledges, other obstructions to downward flow of water in wall, and where indicated.

B. Install flashing as follows unless otherwise indicated:

1. Prepare masonry surfaces so they are smooth and free from projections that could puncture flashing. Where flashing is within mortar joint, place through-wall flashing on sloping bed of mortar and cover with mortar. Before covering with mortar, seal penetrations in flashing with adhesive, sealant, or tape as recommended by flashing manufacturer.
2. At multiwythe masonry walls, including cavity walls, extend flashing through outer wythe, turned up a minimum of 4 inches (100 mm), and through inner wythe to within 1/2 inch (13 mm) of the interior face of wall in exposed masonry. Where interior face of wall is to receive furring or framing, carry flashing completely through inner wythe and turn flashing up approximately 2 inches (50 mm) on interior face.
3. At masonry-veneer walls, extend flashing through veneer, across airspace behind veneer, and up face of sheathing at least 8 inches (200 mm); with upper edge tucked under water-resistive barrier, lapping at least 4 inches (100 mm). Fasten upper edge of flexible flashing to sheathing through termination bar.
4. At lintels and shelf angles, extend flashing a minimum of 6 inches (150 mm) into masonry at each end. At heads and sills, extend flashing 6 inches (150 mm) at ends and turn up not less than 2 inches (50 mm) to form end dams.
5. Install metal drip edges beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall, and adhere flexible flashing to top of metal drip edge.
6. Install metal flashing termination beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall, and adhere flexible flashing to top of metal flashing termination.
7. Cut flexible flashing off flush with face of wall after masonry wall construction is completed.

C. Install single-wythe CMU flashing system in bed joints of CMU walls where indicated to comply with manufacturer's written instructions. Install CMU cell pans with upturned edges located below face shells and webs of CMUs above and with weep spouts aligned with face of wall. Install CMU web covers so that they cover upturned edges of CMU cell pans at CMU webs and extend from face shell to face shell.

D. Install reglets and nailers for flashing and other related construction where they are shown to be built into masonry.

E. Install weep holes in exterior wythes and veneers in head joints of first course of masonry immediately above embedded flashing.

1. Use specified weep/cavity vent products or open-head joints to form weep holes.
2. Space weep holes 24 inches (600 mm) o.c. unless otherwise indicated.

3. Cover cavity side of weep holes with plastic insect screening at cavities insulated with loose-fill insulation.

F. Place cavity drainage material in cavities to comply with configuration requirements for cavity drainage material in "Miscellaneous Masonry Accessories" Article.

G. Install cavity vents in head joints in exterior wythes at spacing indicated. Use specified weep/cavity vent products or open-head joints to form cavity vents.

1. Close cavities off vertically and horizontally with blocking in manner indicated. Install through-wall flashing and weep holes above horizontal blocking.

3.13 REINFORCED UNIT MASONRY INSTALLATION

A. Temporary Formwork and Shores: Construct formwork and shores as needed to support reinforced masonry elements during construction.

1. Construct formwork to provide shape, line, and dimensions of completed masonry as indicated. Make forms sufficiently tight to prevent leakage of mortar and grout. Brace, tie, and support forms to maintain position and shape during construction and curing of reinforced masonry.
2. Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and that of other loads that may be placed on them during construction.

B. Placing Reinforcement: Comply with requirements in TMS 602/ACI 530.1/ASCE 6.

C. Grouting: Do not place grout until entire height of masonry to be grouted has attained enough strength to resist grout pressure.

1. Comply with requirements in TMS 602/ACI 530.1/ASCE 6 for cleanouts and for grout placement, including minimum grout space and maximum pour height.
2. Limit height of vertical grout pours to not more than 60 inches (1520 mm).

3.14 FIELD QUALITY CONTROL

A. Testing and Inspecting: Engage special inspectors to perform tests and inspections and prepare reports. Allow inspectors access to scaffolding and work areas as needed to perform tests and inspections. Retesting of materials that fail to comply with specified requirements shall be done at Contractor's expense.

B. Inspections: Special inspections according to Level B in TMS 402/ACI 530/ASCE 5.

1. Begin masonry construction only after inspectors have verified proportions of site-prepared mortar.
2. Place grout only after inspectors have verified compliance of grout spaces and of grades, sizes, and locations of reinforcement.
3. Place grout only after inspectors have verified proportions of site-prepared grout.

C. Testing Prior to Construction: One set of tests.

D. Concrete Masonry Unit Test: For each type of unit provided, according to ASTM C 140 for compressive strength.

E. Mortar Aggregate Ratio Test (Proportion Specification): For each mix provided, according to ASTM C 780.

F. Mortar Test (Property Specification): For each mix provided, according to ASTM C 780. Test mortar for mortar air content and compressive strength.

G. Grout Test (Compressive Strength): For each mix provided, according to ASTM C 1019.

H. Prism Test: For each type of construction provided, according to ASTM C 1314 at 7 days and at 28 days.

## 3.15 REPAIRING, POINTING, AND CLEANING

A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.

B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.

C. In-Progress Cleaning: Clean unit masonry as work progresses by dry brushing to remove mortar fins and smears before tooling joints.

D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:

1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes. Obtain Architect's approval of sample cleaning before proceeding with cleaning of masonry.
3. Protect adjacent stone and nonmasonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.
4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
5. Clean brick by bucket-and-brush hand-cleaning method described in BIA Technical Notes 20.
6. Clean concrete masonry by applicable cleaning methods indicated in NCMA TEK 8-4A.
7. Clean masonry with a proprietary acidic cleaner applied according to manufacturer's written instructions.
8. Clean stone trim to comply with stone supplier's written instructions.
9. Clean limestone units to comply with recommendations in ILI's "Indiana Limestone Handbook."

## 3.16 MASONRY WASTE DISPOSAL

A. Salvageable Materials: Unless otherwise indicated, excess masonry materials are Contractor's property. At completion of unit masonry work, remove from Project site.

B. Waste Disposal as Fill Material: Dispose of clean masonry waste, including excess or soil-contaminated sand, waste mortar, and broken masonry units, by crushing and mixing with fill material as fill is placed.

1. Crush masonry waste to less than 4 inches (100 mm) in each dimension.
2. Mix masonry waste with at least two parts of specified fill material for each part of masonry waste. Fill material is specified in Section 312000 "Earth Moving."
3. Do not dispose of masonry waste as fill within 18 inches (450 mm) of finished grade.

C. Masonry Waste Recycling: Return broken CMUs not used as fill to manufacturer for recycling.

D. Excess Masonry Waste: Remove excess clean masonry waste that cannot be used as fill, as described above or recycled, and other masonry waste, and legally dispose of off Owner's property.

END OF SECTION 04 2000

# SECTION 05 0553 – TAMPER PROOF METAL FASTENERS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. This Section includes the following:

1. Tamper proof metal fasteners.
2. Accessories.

B. Related Sections include the following:

1. Division 05 for Metals
2. Division 08 for Openings
3. Division 10 for Specialties
4. Division 11 for Equipment
5. Division 21 for Fire Suppression
6. Division 22 for Plumbing
7. Division 23 for Heating, Ventilating, and Air-Conditioning (HVAC)
8. Division 26 for Electrical
9. Division 27 for Communications
10. Division 28 for Electronic Safety and Security

### 1.3 SUBMITTALS

A. Product Data: For each type of product.

### 1.4 DELIVERY, STORAGE AND HANDLING

A. Deliver, store and handle per manufacturer's requirements.

## PART 2 - PRODUCTS

### 2.1 TAMPER PROOF METAL FASTENERS

A. Exposed Security Fasteners:

1. Torx-head (star design with center pin) security fasteners.
2. Finish shall match that specified of the item anchored.

B. Fabrication:

1. Fabricate removable tamper proof fasteners to allow removal only by tools produced by fastener manufacturer or other licensed fabricator specifically for individual tamper proof fastener design.
2. Plating: Cadmium, zinc, nickel, phosphate and chrome to match adjacent materials.
3. Limit size and shape variations such that no more than six (6) different tools are required for each type of tamper proof fastener used on project.

2.2 ACCESSORIES

A. Screw-thread Adhesive Sealant: Loctite No. 242 or acceptable substitute.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install work using proper sized tamper proof fastener, matched to configuration, structural loading, and size.

B. Install fasteners with the proper amount of torque as recommended by the manufacturer.

C. Set tamper proof fasteners with screw thread adhesive sealant in accordance with manufacturer's instructions.

D. Store and maintain inventory control for each installing tool used for installation of security fasteners. After use, installers shall return tools for inventory control. At completion of the project, installing tools shall be turned over to the facility.

3.2 SPARE PARTS

A. Provide five (5) security fastener kits to the Owner. Kits shall contain all fasteners found on project and a tool with bits.

END OF SECTION 05 0553

SECTION 051200 – STRUCTURAL STEEL FRAMING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Structural steel.
2. Grout.

1.3 DEFINITIONS

A. Structural Steel: Elements of structural-steel frame, as classified by AISC 303, "Code of Standard Practice for Steel Buildings and Bridges."

1.4 PERFORMANCE REQUIREMENTS

A. Connections: Provide details of simple shear connections required by the Contract Documents to be selected or completed by structural-steel fabricator to withstand loads indicated and comply with other information and restrictions indicated.

1. Select and complete connections using schematic details indicated and AISC 360.
2. Use Load and Resistance Factor Design (LRFD); data are given at factored load level.
3. Engineering Responsibility: Fabricator's responsibilities include using a qualified professional engineer to prepare structural details and design calculations for braced frame connections.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Shop Drawings: Show fabrication of structural-steel components.

1. Include details of cuts, connections, splices, camber, holes, and other pertinent data.
2. Include embedment drawings.
3. Indicate welds by standard AWS symbols, distinguishing between shop and field welds, and show size, length, and type of each weld. Show backing bars that are to be removed and supplemental fillet welds where backing bars are to remain.
4. Indicate type, size, and length of bolts, distinguishing between shop and field bolts. Identify pretensioned and slip-critical high-strength bolted connections.
5. For braced frame connections indicated to comply with design loads, include structural analysis details and calculations signed and sealed by the qualified professional engineer responsible for their preparation. Professional engineer shall be currently registered in the Commonwealth of Virginia. Structural engineer of record will review braced frame connection details and calculations apply shop drawing stamp indicating approval.

1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified fabricator and professional engineer.

B. Welding certificates.

C. Mill test reports for structural steel, including chemical and physical properties.

D. Product Test Reports: For the following:

1. Bolts, nuts, and washers including mechanical properties and chemical analysis.
2. Tension-control, high-strength bolt-nut-washer assemblies.
3. Nonshrink grout.

E. Source quality-control reports.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: Engage an experienced Installer who has completed structural steel work similar in material, design, and extent to that indicated for this Project and with a record of successful in-service performance.

B. Fabricator Qualifications: Engage a firm experienced in fabricating structural steel similar to that indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to fabricate structural steel without delaying the Work.

C. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code - Steel."

1. Welders and welding operators performing work on bottom-flange, demand-critical welds shall pass the supplemental welder qualification testing, as required by AWS D1.8. FCAW-S and FCAW-G shall be considered separate processes for welding personnel qualification.

D. Comply with applicable provisions of the following specifications and documents:

1. AISC 303.
2. AISC 360.
3. RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1.8 DELIVERY, STORAGE, AND HANDLING

A. Store materials to permit easy access for inspection and identification. Keep steel members off ground and spaced by using pallets, dunnage, or other supports and spacers. Protect steel members and packaged materials from corrosion and deterioration.

1. Do not store materials on structure in a manner that might cause distortion, damage, or overload to members or supporting structures. Repair or replace damaged materials or structures as directed.

B. Store fasteners in a protected place in sealed containers with manufacturer's labels intact.

1. Fasteners may be repackaged provided Design-Builder's testing and inspecting agency observes repackaging and seals containers.

2. Clean and relubricate bolts and nuts that become dry or rusty before use.
3. Comply with manufacturers' written recommendations for cleaning and lubricating ASTM F 1852 fasteners and for retesting fasteners after lubrication.

1.9 COORDINATION

A. Coordinate installation of anchorage items to be embedded in or attached to other construction without delaying the Work. Provide setting diagrams, sheet metal templates, instructions, and directions for installation.

PART 2 - PRODUCTS

2.1 STRUCTURAL STEEL MATERIALS

A. W-Shapes: ASTM A 992, Grade 50.

B. Channels, Angles: ASTM A 36.

C. Plate and Bar: ASTM A 36.

D. Cold-Formed Hollow Structural Sections: ASTM A 500, Grade C, structural tubing.

E. Steel Pipe: ASTM A 53, Type E or S, Grade B.

F. Steel Castings: ASTM A 216, Grade WCB with supplementary requirement S11.

G. Steel Forgings: ASTM A 668.

H. Welding Electrodes: Comply with AWS requirements.

2.2 BOLTS, CONNECTORS, AND ANCHORS

A. High-Strength Bolts, Nuts, and Washers: ASTM A 325, Type 1, heavy-hex steel structural bolts; ASTM A 563, Grade C, heavy-hex carbon-steel nuts; and ASTM F 436, Type 1, hardened carbon-steel washers; all with plain finish.

B. Tension-Control, High-Strength Bolt-Nut-Washer Assemblies: ASTM F 1852, Type 1, heavy-hex head assemblies consisting of steel structural bolts with splined ends, heavy-hex carbon-steel nuts, and hardened carbon-steel washers.

1. Finish: Plain

C. Unheaded Anchor Rods: ASTM F 1554, Grade 36 and grade 55 as indicated in drawings.

1. Configuration: Straight.
2. Nuts: ASTM A 563 heavy-hex carbon steel.
3. Plate Washers: ASTM A 36 carbon steel.
4. Washers: ASTM F 436, Type 1, hardened carbon steel.
5. Finish: Plain.

D. Headed Anchor Rods: ASTM F 1554, Grade 36 and Grade 55 as indicated in drawings, straight.

1. Nuts: ASTM A 563 heavy-hex carbon steel.

2. Plate Washers: ASTM A 36 carbon steel.
3. Washers: ASTM F 436, Type 1, hardened carbon steel.
4. Finish: Plain.

E. Threaded Rods: ASTM A 36.

1. Nuts: ASTM A 563 heavy-hex carbon steel.
2. Washers: ASTM F 436, Type 1, hardened carbon steel.
3. Finish: Plain.

2.3 PRIMER

A. Primer: Fabricator's standard lead- and chromate-free, nonasphaltic, rust-inhibiting primer complying with MPI#79 and compatible with topcoat. Color: Grey.

B. Galvanizing Repair Paint: ASTM A 780.

2.4 GROUT

A. Nonmetallic, Shrinkage-Resistant Grout: ASTM C 1107, factory-packaged, nonmetallic aggregate grout, noncorrosive and nonstaining, mixed with water to consistency suitable for application and a 30-minute working time.

2.5 FABRICATION

A. Structural Steel: Fabricate and assemble in shop to greatest extent possible. Fabricate according to AISC's "Code of Standard Practice for Steel Buildings and Bridges" and AISC 360.

1. Camber structural-steel members where indicated.
2. Fabricate beams with rolling camber up.
3. Identify high-strength structural steel according to ASTM A 6 and maintain markings until structural steel has been erected.
4. Mark and match-mark materials for field assembly.
5. Complete structural-steel assemblies, including welding of units, before starting shop-priming operations.

B. Thermal Cutting: Perform thermal cutting by machine to greatest extent possible.

1. Plane thermally cut edges to be welded to comply with requirements in AWS D1.1.

C. Bolt Holes: Cut, drill, or punch standard bolt holes perpendicular to metal surfaces.

D. Finishing: Accurately finish ends of columns and other members transmitting bearing loads.

E. Cleaning: Clean and prepare steel surfaces that are to remain unpainted according to SSPC-SP 1, "Solvent Cleaning or SSPC-SP 2, "Hand Tool Cleaning."

F. Holes: Provide holes required for securing other work to structural steel and for other work to pass through steel framing members.

1. Cut, drill, or punch holes perpendicular to steel surfaces. Do not thermally cut bolt holes or enlarge holes by burning.
2. Baseplate Holes: Cut, drill, mechanically thermal cut, or punch holes perpendicular to steel surfaces.

3. Weld threaded nuts to framing and other specialty items indicated to receive other work.

2.6 SHOP CONNECTIONS

A. High-Strength Bolts: Shop install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: Snug tightened unless indicated as Pretensioned or Slip Critical on drawings.

B. Weld Connections: Comply with AWS D1.1 for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

1. Assemble and weld built-up sections by methods that will maintain true alignment of axes without exceeding tolerances in AISC 303 for mill material.

2.7 SHOP PRIMING

A. Shop prime steel surfaces except the following:

1. Surfaces embedded in concrete or mortar. Extend priming of partially embedded members to a depth of 2 inches.
2. Surfaces to be field welded.
3. Surfaces of high-strength bolted, slip-critical connections.
4. Surfaces to receive sprayed fire-resistive materials (applied fireproofing).
5. Galvanized surfaces.

B. Surface Preparation: Clean surfaces to be painted. Remove loose rust and mill scale and spatter, slag, or flux deposits. Prepare surfaces according to the following specifications and standards:

1. SSPC-SP 3, "Power Tool Cleaning."

C. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils. Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

1. Stripe paint corners, crevices, bolts, welds, and sharp edges.
2. Apply two coats of shop paint to surfaces that are inaccessible after assembly or erection. Change color of second coat to distinguish it from first.

2.8 GALVANIZING

A. Hot-Dip Galvanized Finish: Apply zinc coating by the hot-dip process to structural steel according to ASTM A 123.

1. Fill vent and drain holes that will be exposed in the finished Work unless they will function as weep holes, by plugging with zinc solder and filing off smooth.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify, with steel Erector present, elevations of concrete- and masonry-bearing surfaces and locations of anchor rods, bearing plates, and other embedments for compliance with requirements.

1. Prepare a certified survey of bearing surfaces, anchor rods, bearing plates, and other embedments showing dimensions, locations, angles, and elevations.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Provide temporary shores, guys, braces, and other supports during erection to keep structural steel secure, plumb, and in alignment against temporary construction loads and loads equal in intensity to design loads. Remove temporary supports when permanent structural steel, connections, and bracing are in place unless otherwise indicated.

3.3 ERECTION

A. Set structural steel accurately in locations and to elevations indicated and according to AISC 303 and AISC 360.

B. Base Bearing and Leveling Plates: Clean concrete-bearing surfaces of bond-reducing materials, and roughen surfaces prior to setting plates. Clean bottom surface of plates.

1. Set plates for structural members on wedges, shims, or setting nuts as required.
2. Weld plate washers to top of baseplate.
3. Snug-tighten anchor rods after supported members have been positioned and plumbed. Do not remove wedges or shims but, if protruding, cut off flush with edge of plate before packing with grout.
4. Promptly pack grout solidly between bearing surfaces and plates so no voids remain. Neatly finish exposed surfaces; protect grout and allow to cure. Comply with manufacturer's written installation instructions for shrinkage-resistant grouts.

C. Maintain erection tolerances of structural steel within AISC's "Code of Standard Practice for Steel Buildings and Bridges."

D. Align and adjust various members that form part of complete frame or structure before permanently fastening. Before assembly, clean bearing surfaces and other surfaces that will be in permanent contact with members. Perform necessary adjustments to compensate for discrepancies in elevations and alignment.

1. Level and plumb individual members of structure.
2. Make allowances for difference between temperature at time of erection and mean temperature when structure is completed and in service.

E. Splice members only where indicated.

F. Do not use thermal cutting during erection unless approved by Architect. Finish thermally cut sections within smoothness limits in AWS D1.1.

G. Do not enlarge unfair holes in members by burning or using drift pins. Ream holes that must be enlarged to admit bolts.

H. Shear Connectors: Prepare steel surfaces as recommended by manufacturer of shear connectors. Use automatic end welding of headed-stud shear connectors according to AWS D1.1 and manufacturer's written instructions.

3.4 FIELD CONNECTIONS

A. High-Strength Bolts: Install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: Snug tightened unless indicated as Pretensioned or Slip Critical on drawings.

B. Weld Connections: Comply with AWS D1.1 for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

1. Comply with AISC 303 and AISC 360 for bearing, alignment, adequacy of temporary connections, and removal of paint on surfaces adjacent to field welds.
2. Remove backing bars or runoff tabs where indicated, back gouge, and grind steel smooth.
3. Assemble and weld built-up sections by methods that will maintain true alignment of axes without exceeding tolerances in AISC's "Code of Standard Practice for Steel Buildings and Bridges" for mill material.

3.5 FIELD QUALITY CONTROL

A. Testing Agency: Owner will engage a Special Inspector and qualified independent testing and inspecting agency to inspect field welds and high-strength bolted connections and prepare test reports.

B. Inspections: See Statement of Special Inspections.

C. Bolted Connections: Bolted connections will be tested and inspected according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

D. Welded Connections: Field welds will be visually inspected according to AWS D1.1.

1. In addition to visual inspection, field welds will be tested and inspected according to AWS D1.1 and the following inspection procedures, at testing agency's option:

a. Liquid Penetrant Inspection: ASTM E 165.
b. Magnetic Particle Inspection: ASTM E 709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration will not be accepted.
c. Ultrasonic Inspection: ASTM E 164.
d. Radiographic Inspection: ASTM E 94.

E. Correct deficiencies in Work that test reports and inspections indicate does not comply with the Contract Documents.

3.6 REPAIRS AND PROTECTION

A. Touchup Painting: Immediately after erection, clean exposed areas where primer is damaged or missing and paint with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

1. Clean and prepare surfaces by SSPC-SP 2 hand-tool cleaning or SSPC-SP 3 power-tool cleaning.

B. Galvanized Surfaces: Clean areas where galvanizing is damaged or missing and repair galvanizing to comply with ASTM A 780.

END OF SECTION 051200

SECTION 052100 - STEEL JOIST FRAMING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. K-series steel joists.
2. Joist accessories.

1.3 DEFINITIONS

A. SJI "Specifications": Steel Joist Institute's "Standard Specifications, Load Tables and Weight Tables for Steel Joists and Joist Girders".

1.4 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide special joists and connections capable of withstanding design loads indicated.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of joist, accessory, and product indicated.

B. Shop Drawings: Show layout, mark, number, type, location, and spacings of joists. Include joining and anchorage details, bracing, bridging, accessories; splice and connection locations and details; and attachments to other construction.

1. Indicate locations and details of anchorage devices and bearing plates to be embedded in other construction.

C. Welding Certificates: Copies of certificates for welding procedures and personnel.

D. Manufacturer certificates.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: A firm experienced in manufacturing joists similar to those indicated for this Project and with a record of successful in-service performance.

1. Manufacturer must be certified by SJI to manufacture joists complying with SJI standard specifications and load tables.
2. Assumes responsibility for engineering special joists to comply with performance requirements. This responsibility includes preparation of Shop Drawings and comprehensive engineering analysis by a qualified professional engineer.

B. SJI Specifications: Comply with SJI's "Standard Specifications Load Tables and Weight Tables for Steel Joists and Joist Girders" (hereafter, "Specifications"), applicable to types of joists indicated.

C. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel".

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle joists as recommended in SJI's "Specifications".

B. Protect joists from corrosion, deformation, and other damage during delivery, storage, and handling.

1.8 SEQUENCING

A. Deliver, store, and handle joists as recommended in SJI's "Specifications".

B. Protect joists from corrosion, deformation, and other damage during delivery, storage, and handling.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Steel: Comply with SJI's "Specifications" for web and steel-angle chord members.

B. High-Strength Bolts and Nuts: ASTM A 325, Type 1, heavy hex steel structural bolts, heavy hex carbon-steel nuts, and hardened carbon-steel washers.

C. Welding Electrodes: Comply with AWS standards.

2.2 PRIMERS

A. Primer: SSPC-Paint 15, Type I, red oxide; FS TT-P-636, red oxide; or manufacturer's standard shop primer complying with performance requirements of either of these red-oxide primers. Color: Grey.

2.3 K-SERIES STEEL JOISTS

A. Manufacture steel joists of type indicated according to "Standard Specifications for Open Web Steel Joists, K-Series" in SJI's "Specifications", with steel-angle top- and bottom-chord members, underslung ends, and parallel top chord.

1. Joist Type: K-series steel joists.

B. Comply with AWS requirements and procedures for shop welding, appearance, quality of welds, and methods used in correcting welding work.

C. Extended Ends: Extend bearing ends of joists with SJI's Type R extended ends where indicated, complying with SJI's "Specifications".

D. Camber joists according to SJI’s “Specifications”.

E. Equip bearing ends of joists with manufacturer's standard beveled ends or sloped shoes if joist slope exceeds 1/4 inch per 12 inches.

2.4 JOIST ACCESSORIES

A. Bridging: Fabricate as indicated and according to SJI's "Specifications". Furnish additional erection bridging if required for stability.

B. Fabricate steel bearing plates with integral anchorages of sizes and thicknesses indicated. Shop prime paint.

C. Supply miscellaneous accessories, including splice plates and bolts required by joist manufacturer to complete joist installation.

2.5 CLEANING AND SHOP PAINTING

A. Clean and remove loose scale, heavy rust, and other foreign materials from fabricated joists and accessories to be primed by hand-tool cleaning, SSPC-SP 2 or power-tool cleaning, SSPC-SP 3.

B. Apply one shop coat of primer to joists and joist accessories to be primed to provide a continuous, dry paint film not less than 1 mil thick.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine supporting substrates, embedded bearing plates, and abutting structural framing for compliance with requirements for installation tolerances and other conditions affecting performance.

1. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Do not install joists until supporting construction is in place and secured.

B. Install joists and accessories plumb, square, and true to line; securely fasten to supporting construction according to SJI's "Specifications", joist manufacturer's written recommendations, and requirements in this Section.

1. Before installation, splice joists delivered to Project site in more than one piece.
2. Space, adjust, and align joists accurately in location before permanently fastening.
3. Install temporary bracing and erection bridging, connections, and anchors to ensure that joists are stabilized during construction.

C. Field weld joists to supporting steel bearing plates and framework. Coordinate welding sequence and procedure with placement of joists. Comply with AWS requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.

D. Install and connect bridging concurrently with joist erection, before construction loads are applied. Anchor ends of bridging lines at top and bottom chords if terminating at walls or beams.

3.3 FIELD QUALITY CONTROL

A. Testing and Inspecting Agency: Contractor will engage a Special Inspector and qualified independent testing and inspecting agency to perform field tests and inspections and prepare test reports per the requirements in the Statement of Special Inspections.

B. Visually inspect field welds according to AWS D1.1.

C. In addition to visual inspection, field welds will be tested according to AWS D1.1 and the following procedures, as applicable, at testing agency's option:

1. Radiographic Testing: ASTM E 94.
2. Magnetic Particle Inspection: ASTM E 709.
3. Liquid Penetrant Inspection: ASTM E 165.

D. High-strength, field-bolted connections will be tested and verified according to procedures in RCSC's "Specification for Structural Joints Using ASTM A 325 or ASTM A 490 Bolts".

E. Correct deficiencies in Work that test and inspection reports have indicated are not in compliance with specified requirements.

F. Additional testing will be performed to determine compliance of corrected Work with specified requirements.

3.4 REPAIRS AND PROTECTION

A. Touchup Painting: After installation, promptly clean, prepare, and prime or reprime field connections, rust spots, and abraded surfaces of prime-painted joists and accessories, bearing plates and abutting structural steel.

1. Clean and prepare surfaces by hand-tool cleaning, SSPC-SP 2, or power-tool cleaning, SSPC-SP 3.
2. Apply a compatible primer of the same type as the shop primer used on adjacent surfaces.

B. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and installer, that ensure joists and accessories are without damage or deterioration at time of Substantial Completion.

END OF SECTION 052100

# SECTION 05 5000 - METAL FABRICATIONS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Steel framing and supports for countertops.
2. Steel framing and supports for mechanical and electrical equipment.
3. Steel framing and supports for applications where framing and supports are not specified in other Sections.
4. Shelf angles.
5. Metal ladders.
6. Miscellaneous steel trim.
7. Metal Bollards.
8. Loose bearing and leveling plates for applications where they are not specified in other Sections.

B. Products furnished, but not installed, under this Section include the following:

1. Loose steel lintels.
2. Anchor bolts, steel pipe sleeves, slotted-channel inserts, and wedge-type inserts indicated to be cast into concrete or built into unit masonry.

C. Related Requirements:

1. Section 04 2000 "Unit Masonry" for installing loose lintels, anchor bolts, and other items built into unit masonry.

### 1.3 COORDINATION

A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written instructions to ensure that shop primers and topcoats are compatible with one another.

B. Coordinate installation of metal fabrications that are anchored to or that receive other work. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.4 ACTION SUBMITTALS

A. Product Data: For the following:

1. Nonslip aggregates and nonslip-aggregate surface finishes.
2. Fasteners.
3. Shop primers.
4. Metal ladders.

B. Shop Drawings: Show fabrication and installation details. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items. Provide Shop Drawings for the following:

1. Steel framing and supports for countertops.
2. Steel tube reinforcement for low partitions.
3. Steel framing and supports for mechanical and electrical equipment.
4. Steel framing and supports for applications where framing and supports are not specified in other Sections.
5. Shelf angles.
6. Metal ladders.
7. Metal bollards.
8. Loose steel lintels.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For professional engineer's experience with providing delegated-design engineering services of the kind indicated, including documentation that engineer is licensed in the jurisdiction in which Project is located.

B. Mill Certificates: Signed by stainless steel manufacturers, certifying that products furnished comply with requirements.

C. Welding certificates.

D. Paint Compatibility Certificates: From manufacturers of topcoats applied over shop primers, certifying that shop primers are compatible with topcoats.

E. Research Reports: For post-installed anchors.

1.6 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel in accordance with the following:

1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
2. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.7 FIELD CONDITIONS

A. Field Measurements: Verify actual locations of walls, floor slabs, decks, and other construction contiguous with metal fabrications by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes acting on exterior metal fabrications by preventing buckling, opening of joints, overstressing of components, failure of connections, and other detrimental effects.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 METALS

A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.

B. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

C. Steel Plates, Shapes, and Bars: ASTM A36/A36M.

D. Stainless Steel Sheet, Strip, and Plate: ASTM A240/A240M or ASTM A666, Type 304.

E. Stainless Steel Bars and Shapes: ASTM A276/A276M, Type 304.

F. Rolled-Steel Floor Plate: ASTM A786/A786M, rolled from plate complying with ASTM A36/A36M or ASTM A283/A283M, Grade C or D.

G. Rolled-Stainless Steel Floor Plate: ASTM A793.

H. Steel Tubing: ASTM A500/A500M, cold-formed steel tubing.

I. Steel Pipe: ASTM A53/A53M, Standard Weight (Schedule 40) unless otherwise indicated.

J. Stainless Steel Wire Rope: Wire rope manufactured from stainless steel wire complying with ASTM A492, Type 316.

1. Wire Rope Fittings: Stainless steel connectors, Type 316, with capability to sustain, without failure, a load equal to minimum breaking strength of wire rope with which they are used.

K. Steel Prestressing Strand: ASTM A416/A416M, Grade 270 (Grade 1860), low-relaxation, seven-wire, with 0.9-lb/sq. ft. (4.39-kg/sq. m)zinc coating.

1. Steel Prestressing Strand Fittings: Hot-dip galvanized-steel anchors and connectors with capability to sustain, without failure, a load equal to minimum breaking strength of steel prestressing strand with which they are used.

L. Cast Iron: Either gray iron, ASTM A48/A48M, or malleable iron, ASTM A47/A47M, unless otherwise indicated.

M. Aluminum Plate and Sheet: ASTM B209 (ASTM B209M), Alloy 6061-T6.

N. Aluminum Extrusions: ASTM B221 (ASTM B221M), Alloy 6063-T6.

O. Aluminum-Alloy Rolled Tread Plate: ASTM B632/B632M, Alloy 6061-T6.

P. Aluminum Castings: ASTM B26/B26M, Alloy 443.0-F.

2.3 FASTENERS

A. General: Unless otherwise indicated, provide Type 304 stainless steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, at exterior walls. Select fasteners for type, grade, and class required.

1. Provide stainless steel fasteners for fastening aluminum, stainless steel or nickel silver.
2. Provide bronze fasteners for fastening bronze.

B. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A307, Grade A (ASTM F568M, Property Class 4.6); with hex nuts, ASTM A563 (ASTM A563M); and, where indicated, flat washers.

C. High-Strength Bolts, Nuts, and Washers: ASTM F3125/F3125M, Grade A325 (Grade A325M), Type 3, heavy-hex steel structural bolts; ASTM A563, Grade DH3, (ASTM A563M, Class 10S3) heavy-hex carbon-steel nuts; and where indicated, flat washers.

D. Stainless Steel Bolts and Nuts: Regular hexagon-head annealed stainless steel bolts, ASTM F593 (ASTM F738M); with hex nuts, ASTM F594 (ASTM F836M); and, where indicated, flat washers; Alloy Group 1 (A1).

E. Anchor Bolts: ASTM F1554, Grade 36, of dimensions indicated; with nuts, ASTM A563 (ASTM A563M); and, where indicated, flat washers.

1. Hot-dip galvanize or provide mechanically deposited, zinc coating where item being fastened is indicated to be galvanized.

F. Anchors, General: Capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing in accordance with ASTM E488/E488M, conducted by a qualified independent testing agency.

G. Cast-in-Place Anchors in Concrete: Either threaded or wedge type unless otherwise indicated; galvanized ferrous castings, either ASTM A47/A47M malleable iron or ASTM A27/A27M cast steel. Provide bolts, washers, and shims as needed, all hot-dip galvanized per ASTM F2329/F2329M.

H. Post-Installed Anchors: Torque-controlled expansion anchors or chemical anchors.

1. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, unless otherwise indicated.
2. Material for Exterior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 (A1) stainless steel bolts, ASTM F593 (ASTM F738M), and nuts, ASTM F594 (ASTM F836M).

2.4 MISCELLANEOUS MATERIALS

A. Shop Primers: Provide primers that comply with Section 099113 "Exterior Painting," Section 099123 "Interior Painting," and Section 099600 "High-Performance Coatings."

B. Universal Shop Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with MPI#79 and compatible with topcoat.

1. Use primer containing pigments that make it easily distinguishable from zinc-rich primer.

C. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.

D. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

E. Shrinkage-Resistant Grout: Factory-packaged, nonmetallic, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.

F. Concrete: Comply with requirements in Section 033000 "Cast-in-Place Concrete" for normal-weight, air-entrained concrete with a minimum 28-day compressive strength of 3000 psi (20 MPa).

2.5 FABRICATION, GENERAL

A. Shop Assembly: Preassemble items in the shop to greatest extent possible. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.

B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.

C. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.

D. Form exposed work with accurate angles and surfaces and straight edges.

E. Weld corners and seams continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

F. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) fasteners unless otherwise indicated. Locate joints where least conspicuous.

G. Fabricate seams and other connections that are exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.

H. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.

I. Provide for anchorage of type indicated; coordinate with supporting structure. Space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.

J. Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors, 1/8 by 1-1/2 inches (3.2 by 38 mm), with a minimum 6-inch (150-mm) embedment and 2-inch (50-mm) hook, not less than 8 inches (200 mm) from ends and corners of units and 24 inches (600 mm) o.c., unless otherwise indicated.

## 2.6 MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.

B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.

1. Fabricate units from slotted channel framing where indicated.
2. Furnish inserts for units installed after concrete is placed.

C. Galvanize miscellaneous framing and supports where indicated.

D. Prime miscellaneous framing and supports with primer specified in Section 099600 "High-Performance Coatings" where indicated.

## 2.7 SHELF ANGLES

A. Fabricate shelf angles from steel angles of sizes indicated and for attachment to concrete framing. Provide horizontally slotted holes to receive 3/4-inch (19-mm) bolts, spaced not more than 6 inches (150 mm) from ends and 24 inches (600 mm) o.c., unless otherwise indicated.

1. Provide mitered and welded units at corners.
2. Provide open joints in shelf angles at expansion and control joints. Make open joint approximately 2 inches (50 mm) larger than expansion or control joint.

B. Galvanize shelf angles located in exterior walls.

C. Furnish wedge-type concrete inserts, complete with fasteners, to attach shelf angles to cast-in-place concrete.

## 2.8 METAL LADDERS

A. General:

1. Comply with ANSI A14.3, except for elevator pit ladders.

B. Aluminum Ladders:

1. Basis-of-Design Product: Subject to compliance with requirements, Provide O'Keefe's #533A Ladder with Roofover Platform, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

    a. Kattsafe.
    b. O'Keeffe's Inc.
    c. Precision Ladders, LLC.
    d. Royalite Manufacturing, Inc.

2. Source Limitations: Obtain aluminum ladders from single source from single manufacturer.
3. Space siderails 24 inches (609.6 mm) apart unless otherwise indicated.
4. Siderails: Continuous extruded-aluminum channels or tubes, not less than 2-1/2 inches (64 mm) deep, 3/4 inch (19 mm) wide, and 1/8 inch (3.2 mm) thick.
5. Rungs: Extruded-aluminum tubes, not less than 3/4 inch (19 mm) deep and not less than 1/8 inch (3.2 mm) thick, with ribbed tread surfaces.
6. Fit rungs in centerline of siderails; fasten by welding or with stainless steel fasteners or brackets and aluminum rivets.
7. Support each ladder at top and bottom and not more than 60 inches (1500 mm) o.c. with welded or bolted aluminum brackets.
8. Safety cages: Required on ladders that exceed 20 feet (6096 mm) in height.

    a. Fabricate and assemble by welding. Spacing of primary hoops, secondary hoops and vertical bars shall not exceed that required by code.
    b. Hoops and Bars: 3/16 inch (5 mm) by 2 inches (51 mm) aluminum bar.

9. Provide minimum 72-inch- (1830-mm-) high, hinged security door with padlock hasp at foot of ladder to prevent unauthorized ladder use.

## 2.9 MISCELLANEOUS STEEL TRIM

A. Unless otherwise indicated, fabricate units from steel shapes, plates, and bars of profiles shown with continuously welded joints and smooth exposed edges. Miter corners and use concealed field splices where possible.

B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work.

1. Provide with integrally welded steel strap anchors for embedding in concrete or masonry construction.

C. Galvanize and prime exterior miscellaneous steel trim.

## 2.10 METAL BOLLARDS

A. Fabricate metal bollards from 40 steel pipe.

1. Cap bollards with 1/4-inch- (6.4-mm-) thick, steel with sloped top.
2. Where bollards are indicated to receive controls for door operators, provide cutouts for controls and holes for wire.

B. Fabricate bollards with 3/8-inch- (9.5-mm-) thick, steel baseplates for bolting to concrete slab. Drill baseplates at all four corners for 3/4-inch (19-mm) anchor bolts.

1. Where bollards are to be anchored to sloping concrete slabs, angle baseplates for plumb alignment of bollards.

C. Fabricate sleeves for bollard anchorage from steel or stainless-steel pipe with 1/4-inch- (6.4-mm-) thick, steel or stainless steel plate welded to bottom of sleeve. Make sleeves not less than 8 inches (200 mm) deep and 3/4 inch (19 mm) larger than OD of bollard.

D. Prime steel bollards with zinc-rich primer.

## 2.11 LOOSE BEARING AND LEVELING PLATES

A. Provide loose bearing and leveling plates for steel items bearing on masonry or concrete construction. Drill plates to receive anchor bolts and for grouting.

B. Galvanize bearing and leveling plates.

## 2.12 LOOSE STEEL LINTELS

A. Fabricate loose steel lintels from steel angles and shapes of size indicated for openings and recesses in masonry walls and partitions at locations indicated. Fabricate in single lengths for each opening unless otherwise indicated. Weld adjoining members together to form a single unit where indicated.

B. Size loose lintels to provide bearing length at each side of openings equal to 1/12 of clear span, but not less than 8 inches (200 mm) unless otherwise indicated.

C. Galvanize loose steel lintels located in exterior walls.

## 2.13 STEEL WELD PLATES AND ANGLES

A. Provide steel weld plates and angles not specified in other Sections, for items supported from concrete construction as needed to complete the Work. Provide each unit with no fewer than two integrally welded steel strap anchors for embedding in concrete.

## 2.14 GENERAL FINISH REQUIREMENTS

A. Finish metal fabrications after assembly.

B. Finish exposed surfaces to remove tool and die marks and stretch lines, and to blend into surrounding surface.

## 2.15 STEEL AND IRON FINISHES

A. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A153/A153M for steel and iron hardware and with ASTM A123/A123M for other steel and iron products.

1. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.

B. Preparation for Shop Priming Galvanized Items: After galvanizing, thoroughly clean galvanized surfaces of grease, dirt, oil, flux, and other foreign matter, and treat with metallic phosphate process.

C. Shop prime iron and steel items not indicated to be galvanized unless they are to be embedded in concrete, sprayed-on fireproofing, or masonry, or unless otherwise indicated.

1. Shop prime with primers specified in Section 099600 "High-Performance Coatings" are indicated.

D. Preparation for Shop Priming: Prepare surfaces to comply with requirements indicated below:

1. Exterior Items: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
2. Items Indicated to Receive Zinc-Rich Primer: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
3. Items Indicated to Receive Primers Specified in Section 099600 "High-Performance Coatings": SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
4. Other Steel Items: SSPC-SP 3, "Power Tool Cleaning."
5. Galvanized-Steel Items: SSPC-SP 16, "Brush-off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals."

E. Shop Priming: Apply shop primer to comply with SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.

1. Stripe paint corners, crevices, bolts, welds, and sharp edges.

2.16 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, Class I, AA-M12C22A41.

## PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.

B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.

C. Field Welding: Comply with the following requirements:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.

4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.

E. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.

F. Corrosion Protection: Coat concealed surfaces of aluminum that come into contact with grout, concrete, masonry, wood, or dissimilar metals with the following:

1. Cast Aluminum: Heavy coat of bituminous paint.
2. Extruded Aluminum: Two coats of clear lacquer.

## 3.2 INSTALLATION OF MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings.

B. Anchor supports for ceiling hung toilet partitions, overhead doors and overhead grilles securely to, and rigidly brace from, building structure.

C. Anchor shelf angles securely to existing construction with expansion anchors.

D. Support steel girders on solid grouted masonry, concrete, or steel pipe columns. Secure girders with anchor bolts embedded in grouted masonry or concrete or with bolts through top plates of pipe columns.

1. Where grout space under bearing plates is indicated for girders supported on concrete or masonry, install as specified in "Installing Bearing and Leveling Plates" Article.

E. Install pipe columns on concrete footings with grouted baseplates. Position and grout column baseplates as specified in "Installation of Bearing and Leveling Plates" Article.

1. Grout baseplates of columns supporting steel girders after girders are installed and leveled.

## 3.3 INSTALLATION OF BEARING AND LEVELING PLATES

A. Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen to improve bond to surfaces. Clean bottom surface of plates.

B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with shrinkage-resistant grout. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.4 REPAIRS

A. Touchup Painting:

1. Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

a. Apply by brush or spray to provide a minimum 2.0-mil (0.05-mm) dry film thickness.

2. Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint are specified in Section 09 9113 "Exterior Painting."

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

END OF SECTION 05 5000

# SECTION 055213 - PIPE AND TUBE RAILINGS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:
   1. Aluminum railings.

### 1.2 COORDINATION

A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written recommendations to ensure that shop primers and topcoats are compatible with one another.

B. Coordinate installation of anchorages for railings. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

### 1.3 ACTION SUBMITTALS

A. Product Data:

   1. Manufacturer's product lines of mechanically connected railings.
   2. Fasteners.
   3. Post-installed anchors.
   4. Handrail brackets.
   5. Shop primer.
   6. Intermediate coats and topcoats.
   7. Bituminous paint.
   8. Nonshrink, nonmetallic grout.
   9. Anchoring cement.
   10. Metal finishes.
   11. Paint products.

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.

C. Samples for Initial Selection: For products involving selection of color, texture, or design.

D. Samples for Verification: For each type of exposed finish required.

   1. Sections of each distinctly different linear railing member, including handrails, top rails, posts, and balusters, including finish.
   2. Fittings and brackets.

E. Delegated Design Submittal: For railings, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For delegated design professional engineer.

B. Welding certificates.

C. Paint Compatibility Certificates: From manufacturers of topcoats applied over shop primers, certifying that shop primers are compatible with topcoats.

D. Product Test Reports: For tests on railings performed by a qualified testing agency, in accordance with ASTM E894 and ASTM E935.

E. Research Reports: For post-installed anchors, from ICC-ES or other qualified testing agency acceptable to authorities having jurisdiction.

1.5 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel in accordance with the following:
1. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.6 DELIVERY, STORAGE, AND HANDLING

A. Protect mechanical finishes on exposed surfaces of railings from damage by applying a strippable, temporary protective covering before shipping.

1.7 FIELD CONDITIONS

A. Field Measurements: Verify actual locations of walls and other construction contiguous with railings by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design railings, including attachment to building construction.

B. Structural Performance: Railings, including attachment to building construction, withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:

1. Handrails and Top Rails of Guards:

a. Uniform load of 50 lbf/ ft. (0.73 kN/m) applied in any direction.
b. Concentrated load of 200 lbf (0.89 kN) applied in any direction.
c. Uniform and concentrated loads need not be assumed to act concurrently.

2. Infill of Guards:

a. Concentrated load of 50 lbf (0.22 kN) applied horizontally on an area of 1 sq. ft. (0.093 sq. m).
b. Infill load and other loads need not be assumed to act concurrently.

C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

## 2.2 METALS, GENERAL

A. Metal Surfaces, General: Provide materials with smooth surfaces, without seam marks, roller marks, rolled trade names, stains, discolorations, or blemishes.

B. Brackets, Flanges, and Anchors: Cast or formed metal of same type of material and finish as supported rails unless otherwise indicated.

1. Provide type of bracket with predrilled hole for exposed bolt anchorage and that provides 1-1/2-inch (38-mm) clearance from inside face of handrail to finished wall surface.

## 2.3 ALUMINUM RAILINGS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the work include, but are not limited to the following:
1. Hallmark Iron Works, Inc.
2. American Iron Works, Inc.
3. Architectural Metal Works
4. Wagner, R & B Inc.

B. Source Limitations: Obtain each type of railing from single source from single manufacturer.

C. Aluminum, General: Provide alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than the strength and durability properties of alloy and temper designated below for each aluminum form required.

D. Extruded Structural Pipe and Round Tubing: ASTM B429/B429M, Alloy 6063-T6.

1. Provide Standard Weight (Schedule 40) pipe unless otherwise indicated.

E. Drawn Seamless Tubing: ASTM B210/B210M, Alloy 6063-T832.

## 2.4 FASTENERS

A. Fastener Materials:
1. Aluminum Railing Components: Type 304 stainless steel fasteners.
2. Finish exposed fasteners to match appearance, including color and texture, of railings.

B. Fasteners for Anchoring Railings to Other Construction: Select fasteners of type, grade, and class required to produce connections suitable for anchoring railings to other types of construction and capable of withstanding design loads.

C. Fasteners for Interconnecting Railing Components:
   1. Provide concealed fasteners for interconnecting railing components and for attaching them to other work, unless exposed fasteners are unavoidable or are the standard fastening method for railings indicated.

D. Post-Installed Anchors: Fastener systems with working capacity greater than or equal to the design load, according to an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC193.

## 2.5 MISCELLANEOUS MATERIALS

A. Handrail Brackets: Cast aluminum, center of handrail 2-1/2 inches (63.5 mm) from wall.

B. Welding Rods and Bare Electrodes: Select in accordance with AWS specifications for metal alloy welded.

   1. For aluminum railings, provide type and alloy as recommended by producer of metal to be welded and as required for color match, strength, and compatibility in fabricated items.

C. Shop Primers: Provide primers that comply with Section 099600 "High-Performance Coatings."

D. Intermediate Coats and Topcoats: Provide products that comply with Section 099600 "High-Performance Coatings."

E. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout, complying with ASTM C1107/C1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.

F. Anchoring Cement: Factory-packaged, nonshrink, nonstaining, hydraulic-controlled expansion cement formulation for mixing with water at Project site to create pourable anchoring, patching, and grouting compound.

   1. Water-Resistant Product: At exterior locations, provide formulation that is resistant to erosion from water exposure without needing protection by a sealer or waterproof coating and that is recommended by manufacturer for exterior use.

## 2.6 FABRICATION

A. General: Fabricate railings to comply with requirements indicated for design, dimensions, member sizes and spacing, details, finish, and anchorage, but not less than that required to support structural loads.

B. Shop assemble railings to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations.

   1. Clearly mark units for reassembly and coordinated installation.
   2. Use connections that maintain structural value of joined pieces.

C. Cut, drill, and punch metals cleanly and accurately.

1. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated.
2. Remove sharp or rough areas on exposed surfaces.

D. Form work true to line and level with accurate angles and surfaces.

E. Fabricate connections that are exposed to weather in a manner that excludes water.

1. Provide weep holes where water may accumulate.
2. Locate weep holes in inconspicuous locations.

F. Cut, reinforce, drill, and tap as indicated to receive finish hardware, screws, and similar items.

G. Connections: Fabricate railings with welded or nonwelded connections unless otherwise indicated.

H. Gates: Form gates from steel tube of same size and shape as top rails, with infill to match guards. Provide with cam-type, self-closing hinges for fastening to wall and overlapping stop with rubber bumper to prevent gate from opening in direction opposite egress.

I. Welded Connections for Aluminum Pipe: Fabricate railings to interconnect members with concealed internal welds that eliminate surface grinding, using manufacturer's standard system of sleeve and socket fittings.

J. Nonwelded Connections: Connect members with concealed mechanical fasteners and fittings. Fabricate members and fittings to produce flush, smooth, rigid, hairline joints.

K. Form changes in direction as follows:

1. As detailed.
2. by inserting prefabricated flush-elbow fittings.

L. Close exposed ends of hollow railing members with prefabricated cap and end fittings of same metal and finish as railings.

M. Provide wall returns at ends of wall-mounted handrails unless otherwise indicated. Close ends of returns unless clearance between end of rail and wall is 1/4 inch (6 mm) or less.

N. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors to interconnect railing members to other work unless otherwise indicated.

O. Provide inserts and other anchorage devices for connecting railings to concrete or masonry work.

1. Fabricate anchorage devices capable of withstanding loads imposed by railings.
2. Coordinate anchorage devices with supporting structure.

P. For railing posts set in concrete, provide stainless steel sleeves not less than 6 inches (150 mm) long with inside dimensions not less than 1/2 inch (13 mm) greater than outside dimensions of post, with metal plate forming bottom closure.

2.7 ALUMINUM FINISHES

A. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in the same piece are unacceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

B. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

1. Color and Gloss: As selected by Architect from manufacturer's full range.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Perform cutting, drilling, and fitting required for installing railings.

1. Fit exposed connections together to form tight, hairline joints.
2. Install railings level, plumb, square, true to line; without distortion, warp, or rack.
3. Set railings accurately in location, alignment, and elevation; measured from established lines and levels.
4. Do not weld, cut, or abrade surfaces of railing components that are coated or finished after fabrication and that are intended for field connection by mechanical or other means without further cutting or fitting.
5. Set posts plumb within a tolerance of 1/16 inch in 3 feet (2 mm in 1 m).
6. Align rails so variations from level for horizontal members and variations from parallel with rake of steps and ramps for sloping members do not exceed 1/4 inch in 12 feet (6 mm in 3.5 m).

B. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

1. Coat concealed surfaces of aluminum that will be in contact with grout, concrete, masonry, wood, or dissimilar metals, with a heavy coat of bituminous paint.

C. Adjust railings before anchoring to ensure matching alignment at abutting joints.

D. Fastening to In-Place Construction: Use anchorage devices and fasteners where necessary for securing railings and for properly transferring loads to in-place construction.

3.2 RAILING CONNECTIONS

A. Nonwelded Connections: Use mechanical or adhesive joints for permanently connecting railing components. Use wood blocks and padding to prevent damage to railing members and fittings. Seal recessed holes of exposed locking screws, using plastic cement filler colored to match finish of railings.

B. Welded Connections: Use fully welded joints for permanently connecting railing components. Comply with requirements for welded connections in "Fabrication" Article, whether welding is performed in the shop or in the field.

## 3.3 ANCHORING POSTS

A. Use stainless steel pipe sleeves preset and anchored into concrete for installing posts. After posts are inserted into sleeves, fill annular space between post and sleeve with nonshrink, nonmetallic grout or anchoring cement, mixed and placed to comply with anchoring material manufacturer's written instructions.

B. Cover anchorage joint with flange of same metal as post, attached to post with setscrews.

C. Anchor posts to metal surfaces with flanges, angle type, or floor type, as required by conditions, connected to posts and to metal supporting members as follows:
1. For aluminum railings, attach posts as indicated, using fittings designed and engineered for this purpose.
2.

## 3.4 ATTACHING RAILINGS

A. Anchor railing ends to concrete and masonry with brackets on underside of rails connected to railing ends and anchored to wall construction with anchors and bolts.

B. Attach handrails to walls with wall brackets, except where end flanges are used. Provide brackets with 1-1/2-inch (38-mm) clearance from inside face of handrail and finished wall surface.

C. Install railing gates level, plumb, and secure for full opening without interference.

1. Attach hardware using tamper-resistant or concealed means.
2. Adjust hardware for smooth operation.

## 3.5 REPAIR

A. Touchup Painting:

1. Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with the same material used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

    a. Apply by brush or spray to provide a minimum 2.0-mil (0.05-mm) dry film thickness.

2. Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint are specified in Section 099600 "High-Performance Coatings."

## 3.6 CLEANING

A. Clean aluminum by washing thoroughly with clean water and soap and rinsing with clean water.

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas, and repair galvanizing to comply with ASTM A780/A780M.

3.7 PROTECTION

A. Protect finishes of railings from damage during construction period with temporary protective coverings approved by railing manufacturer. Remove protective coverings at time of Substantial Completion.

B. Restore finishes damaged during installation and construction period, so no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit, or provide new units.

END OF SECTION 055213

# SECTION 06 1000 - ROUGH CARPENTRY

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Wood products.
2. Wood-preservative-treated lumber.
3. Fire-retardant-treated lumber.
4. Dimension lumber framing.
5. Miscellaneous lumber.
6. Plywood backing panels.

### 1.3 DEFINITIONS

A. Boards or Strips: Lumber of less than 2 inches nominal (38 mm actual) size in least dimension.

B. Dimension Lumber: Lumber of 2 inches nominal (38 mm actual) or greater size but less than 5 inches nominal (114 mm actual) size in least dimension.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of process and factory-fabricated product. Indicate component materials and dimensions and include construction and application details.

1. Include data for wood-preservative treatment from chemical treatment manufacturer and certification by treating plant that treated materials comply with requirements. Indicate type of preservative used and net amount of preservative retained.
2. Include data for fire-retardant treatment from chemical treatment manufacturer and certification by treating plant that treated materials comply with requirements. Include physical properties of treated materials based on testing by a qualified independent testing agency.
3. For fire-retardant treatments, include physical properties of treated lumber both before and after exposure to elevated temperatures, based on testing by a qualified independent testing agency according to ASTM D 5664.
4. For products receiving a waterborne treatment, include statement that moisture content of treated materials was reduced to levels specified before shipment to Project site.

1.5 INFORMATIONAL SUBMITTALS

A. Material Certificates:

1. For dimension lumber specified to comply with minimum allowable unit stresses. Indicate species and grade selected for each use and design values approved by the ALSC Board of Review.
2. For preservative-treated wood products. Indicate type of preservative used and net amount of preservative retained.

1.6 QUALITY ASSURANCE

A. Testing Agency Qualifications: For testing agency providing classification marking for fire-retardant-treated material, an inspection agency acceptable to authorities having jurisdiction that periodically performs inspections to verify that the material bearing the classification marking is representative of the material tested.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Stack lumber flat with spacers beneath and between each bundle to provide air circulation. Protect lumber from weather by covering with waterproof sheeting, securely anchored. Provide for air circulation around stacks and under coverings.

PART 2 - PRODUCTS

2.1 WOOD PRODUCTS, GENERAL

A. Regional Materials: Dimension lumber, except treated materials, shall be manufactured within 100 miles (160 km) of Project site from materials that have been extracted, harvested, or recovered, as well as manufactured, within 100 miles (160 km) of Project site.

B. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, provide lumber that complies with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Provide lumber graded by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.

1. Factory mark each piece of lumber with grade stamp of grading agency.
2. For exposed lumber indicated to receive a stained or natural finish, mark grade stamp on end or back of each piece or omit grade stamp and provide certificates of grade compliance issued by grading agency.
3. Dress lumber, S4S, unless otherwise indicated.

C. Maximum Moisture Content of Lumber: 15 percent for 2-inch nominal (38-mm actual) thickness or less, 19 percent for more than 2-inch nominal (38-mm actual) thickness unless otherwise indicated.

2.2 WOOD-PRESERVATIVE-TREATED MATERIALS

A. Preservative Treatment by Pressure Process: AWPA U1; Use Category UC2 for interior construction not in contact with ground, Use Category UC3b for exterior construction not in contact with ground, and Use Category UC4a for items in contact with ground.

1. Preservative Chemicals: Acceptable to authorities having jurisdiction and containing no arsenic or chromium. Do not use inorganic boron (SBX) for sill plates.
2. For exposed items indicated to receive a stained or natural finish, chemical formulations shall not require incising, contain colorants, bleed through, or otherwise adversely affect finishes.

B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or does not comply with requirements for untreated material.

C. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.

1. For exposed lumber indicated to receive a stained or natural finish, mark end or back of each piece or omit marking and provide certificates of treatment compliance issued by inspection agency.

D. Application: Treat items indicated on Drawings, and the following:

1. Wood cants, nailers, curbs, equipment support bases, blocking, stripping, and similar members in connection with roofing, flashing, vapor barriers, and waterproofing.
2. Wood sills, sleepers, blocking, furring, stripping, and similar concealed members in contact with masonry or concrete.
3. Wood framing and furring attached directly to the interior of below-grade exterior masonry or concrete walls.
4. Wood framing members that are less than 18 inches (460 mm) above the ground in crawlspaces or unexcavated areas.
5. Wood floor plates that are installed over concrete slabs-on-grade.

2.3 FIRE-RETARDANT-TREATED MATERIALS

A. General: Where fire-retardant-treated materials are indicated, materials shall comply with requirements in this article, that are acceptable to authorities having jurisdiction, and with fire-test-response characteristics specified as determined by testing identical products per test method indicated by a qualified testing agency.

B. Fire-Retardant-Treated Lumber and Plywood by Pressure Process: Products with a flame-spread index of 25 or less when tested according to ASTM E 84, and with no evidence of significant progressive combustion when the test is extended an additional 20 minutes, and with the flame front not extending more than 10.5 feet (3.2 m) beyond the centerline of the burners at any time during the test.

1. Treatment shall not promote corrosion of metal fasteners.
2. Exterior Type: Treated materials shall comply with requirements specified above for fire-retardant-treated lumber and plywood by pressure process after being subjected to accelerated weathering according to ASTM D 2898. Use for exterior locations and where indicated.

3. Interior Type A: Treated materials shall have a moisture content of 28 percent or less when tested according to ASTM D 3201 at 92 percent relative humidity. Use where exterior type is not indicated.

C. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent.

D. Identify fire-retardant-treated wood with appropriate classification marking of qualified testing agency.

1. For exposed lumber indicated to receive a stained or natural finish, mark end or back of each piece or omit marking and provide certificates of treatment compliance issued by inspection agency.

E. For exposed items indicated to receive a stained or natural finish, chemical formulations shall not bleed through, contain colorants, or otherwise adversely affect finishes.

F. Application: Treat items indicated on Drawings, and the following:

1. Framing for raised platforms.
2. Concealed blocking.
3. Roof framing and blocking.
4. Wood cants, nailers, curbs, equipment support bases, blocking, and similar members in connection with roofing.
5. Plywood backing panels.

## 2.4 DIMENSION LUMBER FRAMING

A. Non-Load-Bearing Interior Partitions: Construction or No. 2 any of the following species:

1. Hem-fir (north); NLGA.
2. Mixed southern pine or southern pine; SPIB.
3. Spruce-pine-fir; NLGA.
4. Hem-fir; WCLIB or WWPA.
5. Spruce-pine-fir (south); NeLMA, WCLIB, or WWPA.
6. Northern species; NLGA.
7. Eastern softwoods; NeLMA.
8. Western woods; WCLIB or WWPA.

B. Other Framing: Construction or No. 2 grade of any of the following species:

1. Hem-fir (north); NLGA.
2. Southern pine; SPIB.
3. Douglas fir-larch; WCLIB or WWPA.
4. Southern pine or mixed southern pine; SPIB.
5. Spruce-pine-fir; NLGA.
6. Douglas fir-south; WWPA.
7. Hem-fir; WCLIB or WWPA.
8. Douglas fir-larch (north); NLGA.
9. Spruce-pine-fir (south); NeLMA, WCLIB, or WWPA.

2.5 MISCELLANEOUS LUMBER

A. General: Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:

1. Blocking.
2. Nailers.
3. Rooftop equipment bases and support curbs.
4. Cants.
5. Furring.
6. Grounds.
7. Utility shelving.

B. Dimension Lumber Items: Construction or No. 2 any of the following species:

1. Hem-fir (north); NLGA.
2. Mixed southern pine or southern pine; SPIB.
3. Spruce-pine-fir; NLGA.
4. Hem-fir; WCLIB or WWPA.
5. Spruce-pine-fir (south); NeLMA, WCLIB, or WWPA.
6. Western woods; WCLIB or WWPA.
7. Northern species; NLGA.
8. Eastern softwoods; NeLMA.

C. Utility Shelving: Lumber with 19 percent maximum moisture content of any of the following species and grades:

1. Eastern white pine, Idaho white, lodgepole, ponderosa, or sugar pine; Premium or No. 2 Common (Sterling) grade; NeLMA, NLGA, WCLIB, or WWPA.
2. Mixed southern pine or southern pine No. 1 grade; SPIB.
3. Hem-fir or hem-fir (north), Construction or No. 2 Common grade; NLGA, WCLIB, or WWPA.
4. Spruce-pine-fir (south) or spruce-pine-fir, Construction or No. 2 Common grade; NeLMA, NLGA, WCLIB, or WWPA.

D. Concealed Boards: 19 percent maximum moisture content of any of the following species and grades:

1. Mixed southern pine or southern pine, No. 2 grade; SPIB.
2. Hem-fir or hem-fir (north), Construction or No. 2 Common grade; NLGA, WCLIB, or WWPA.
3. Spruce-pine-fir (south) or spruce-pine-fir, Construction or No. 2 Common grade; NeLMA, NLGA, WCLIB, or WWPA.
4. Eastern softwoods, No. 2 Common grade; NELMA.
5. Northern species, No. 2 Common grade; NLGA.
6. Western woods, Construction or No. 2 Common grade; WCLIB or WWPA.

E. For blocking not used for attachment of other construction, Utility, Stud, or No. 3 grade lumber of any species may be used provided that it is cut and selected to eliminate defects that will interfere with its attachment and purpose.

F. For blocking and nailers used for attachment of other construction, select and cut lumber to eliminate knots and other defects that will interfere with attachment of other work.

G. For furring strips for installing plywood or hardboard paneling, select boards with no knots capable of producing bent-over nails and damage to paneling.

## 2.6 PLYWOOD BACKING PANELS

A. Equipment Backing Panels: Plywood, DOC PS 1, Exposure 1, C-D Plugged in thickness indicated or, if not indicated, not less than 3/4-inch (19-mm) nominal thickness.

## 2.7 FASTENERS

A. General: Provide fasteners of size and type indicated that comply with requirements specified in this article for material and manufacture.

1. Where carpentry is exposed to weather, in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners with hot-dip zinc coating complying with ASTM A 153/A 153M.

B. Nails, Brads, and Staples: ASTM F 1667.

C. Screws for Fastening to Metal Framing: ASTM C 1002, length as recommended by screw manufacturer for material being fastened.

D. Power-Driven Fasteners: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

E. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 or ICC-ES AC308 as appropriate for the substrate.

1. Material: Carbon-steel components, zinc plated to comply with ASTM B 633, Class Fe/Zn 5.

## 2.8 METAL FRAMING ANCHORS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Cleveland Steel Specialty Co.
2. KC Metals Products, Inc.
3. Simpson Strong-Tie Co., Inc.

B. Galvanized-Steel Sheet: Hot-dip, zinc-coated steel sheet complying with ASTM A 653/A 653M, G60 (Z180) coating designation.

1. Use for interior locations unless otherwise indicated.

C. Hot-Dip, Heavy-Galvanized Steel Sheet: ASTM A 653/A 653M; Structural Steel (SS), high-strength low-alloy steel Type A (HSLAS Type A), or high-strength low-alloy steel Type B (HSLAS Type B); G185 (Z550) coating designation; and not less than 0.036 inch (0.9 mm) thick.

1. Use for wood-preservative-treated lumber and where indicated.

D. Stainless-Steel Sheet: ASTM A 666, Type 304.

1. Use for exterior locations and where indicated.

2.9 MISCELLANEOUS MATERIALS

A. Adhesives for Gluing Furring and Sleepers to Concrete or Masonry: Formulation complying with ASTM D 3498 that is approved for use indicated by adhesive manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.

B. Flexible Flashing: Composite, self-adhesive, flashing product consisting of a pliable, butyl rubber or rubberized-asphalt compound, bonded to a high-density polyethylene film, aluminum foil, or spunbonded polyolefin to produce an overall thickness of not less than 0.025 inch (0.6 mm).

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Framing Standard: Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.

B. Set carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Fit carpentry accurately to other construction. Locate furring, nailers, blocking, grounds, and similar supports to comply with requirements for attaching other construction.

C. Install plywood backing panels by fastening to studs; coordinate locations with utilities requiring backing panels. Install fire-retardant-treated plywood backing panels with classification marking of testing agency exposed to view.

D. Install metal framing anchors to comply with manufacturer's written instructions. Install fasteners through each fastener hole.

E. Do not splice structural members between supports unless otherwise indicated.

F. Provide blocking and framing as indicated and as required to support facing materials, fixtures, specialty items, and trim.

1. Provide metal clips for fastening gypsum board or lath at corners and intersections where framing or blocking does not provide a surface for fastening edges of panels. Space clips not more than 16 inches (406 mm) o.c.

G. Provide fire blocking in furred spaces, stud spaces, and other concealed cavities as indicated and as follows:

1. Fire block furred spaces of walls, at each floor level, at ceiling, and at not more than 96 inches (2438 mm) o.c. with solid wood blocking or noncombustible materials accurately fitted to close furred spaces.

2. Fire block concealed spaces of wood-framed walls and partitions at each floor level, at ceiling line of top story, and at not more than 96 inches (2438 mm) o.c. Where fire blocking is not inherent in framing system used, provide closely fitted solid wood blocks of same width as framing members and 2-inch nominal (38-mm actual) thickness.
3. Fire block concealed spaces between floor sleepers with same material as sleepers to limit concealed spaces to not more than 100 sq. ft. (9.3 sq. m) and to solidly fill space below partitions.
4. Fire block concealed spaces behind combustible cornices and exterior trim at not more than 20 feet (6 m) o.c.

H. Sort and select lumber so that natural characteristics do not interfere with installation or with fastening other materials to lumber. Do not use materials with defects that interfere with function of member or pieces that are too small to use with minimum number of joints or optimum joint arrangement.

I. Comply with AWPA M4 for applying field treatment to cut surfaces of preservative-treated lumber.

1. Use inorganic boron for items that are continuously protected from liquid water.
2. Use copper naphthenate for items not continuously protected from liquid water.

J. Where wood-preservative-treated lumber is installed adjacent to metal decking, install continuous flexible flashing separator between wood and metal decking.

K. Securely attach carpentry work to substrate by anchoring and fastening as indicated, complying with the following:

1. Table 2304.9.1, "Fastening Schedule," in ICC's International Building Code.
2. Table R602.3(1), "Fastener Schedule for Structural Members," and Table R602.3(2), "Alternate Attachments," in ICC's International Residential Code for One- and Two-Family Dwellings.
3. ICC-ES evaluation report for fastener.

L. Use steel common nails unless otherwise indicated. Select fasteners of size that will not fully penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections between members. Install fasteners without splitting wood. Drive nails snug but do not countersink nail heads unless otherwise indicated.

## 3.2 WOOD BLOCKING AND NAILER INSTALLATION

A. Install where indicated and where required for screeding or attaching other work. Form to shapes indicated and cut as required for true line and level of attached work. Coordinate locations with other work involved.

B. Attach items to substrates to support applied loading. Recess bolts and nuts flush with surfaces unless otherwise indicated.

C. Provide permanent grounds of dressed, pressure-preservative-treated, key-beveled lumber not less than 1-1/2 inches (38 mm) wide and of thickness required to bring face of ground to exact thickness of finish material. Remove temporary grounds when no longer required.

3.3 WOOD FURRING INSTALLATION

A. Install level and plumb with closure strips at edges and openings. Shim with wood as required for tolerance of finish work.

B. Furring to Receive Plywood or Hardboard Paneling: Install 1-by-3-inch nominal- (19-by-63-mm actual-) size furring horizontally and vertically at 24 inches (610 mm) o.c.

C. Furring to Receive Gypsum Board: Install 1-by-2-inch nominal- (19-by-38-mm actual-) size furring vertically at 16 inches (406 mm) o.c.

3.4 PROTECTION

A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

B. Protect miscellaneous rough carpentry from weather. If, despite protection, miscellaneous rough carpentry becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

END OF SECTION 06 1000

# SECTION 06 4116 - PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Plastic-laminate-clad architectural cabinets.
2. Cabinet hardware and accessories.
3. Miscellaneous materials.
4. Wood furring, blocking, shims, and hanging strips for installing plastic-laminate-faced architectural cabinets that are not concealed within other construction.

B. Related Requirements:

1. Section 06 1000 “Rough Carpentry" for wood furring, blocking, shims, and hanging strips required for installing cabinets that are concealed within other construction before cabinet installation.
2. Section 12 3661.16 "Solid Surfacing Countertops."

### 1.3 COORDINATION

A. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections to support loads imposed by installed and fully loaded cabinets.

B. Hardware Coordination: Distribute copies of approved hardware schedule specified in Section 087100 "Door Hardware" to manufacturer of architectural cabinets; coordinate Shop Drawings and fabrication with hardware requirements.

### 1.4 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Plastic-laminate-clad architectural cabinets.
2. Cabinet hardware and accessories.
3. Miscellaneous materials.
4. Include data for fire-retardant treatment from chemical-treatment manufacturer and certification by treating plant that treated materials comply with requirements.

B. Shop Drawings: For plastic-laminate-faced and sheet-wall-protection faced architectural cabinets.

1. Include plans, elevations, sections, and attachment details.
2. Show large-scale details.
3. Show locations and sizes of furring, blocking, and hanging strips, including concealed blocking and reinforcement specified in other Sections.
4. Show locations and sizes of cutouts and holes for items installed in plastic-laminate and sheet-wall-protection faced architectural cabinets.
5. Include product data for all associated cabinet hardware.

C. Samples: For each exposed product and for each color and texture specified, in manufacturer's or fabricator's standard size.

D. Samples for Verification: For the following:

1. Plastic Laminates: 8 by 10 inches, for each type, color, pattern, and surface finish required.
2. PVC Edge Banding: 6” long strip by 1 ½” wide x thickness as specified for each type, color, pattern, and surface finish required.

1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Product Certificates: For each type of product, including but not limited to adhesives and sealants.

C. Evaluation Reports: For fire-retardant-treated materials, from ICC-ES.

1.7 QUALITY ASSURANCE

A. Fabricator Qualifications: Shop that employs skilled workers who custom fabricate products similar to those required for this Project and whose products have a record of successful in-service performance.

B. Installer Qualifications: Fabricator of products.

C. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Build mockups of typical architectural cabinets as shown on Drawings.
2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Do not deliver cabinets until painting and similar finish operations that might damage architectural cabinets have been completed in installation areas. Store cabinets in installation areas or in areas where environmental conditions comply with requirements specified in "Field Conditions" Article.

1.9 FIELD CONDITIONS

A. Environmental Limitations: Do not deliver or install cabinets until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature and relative humidity at levels planned for building occupants during the remainder of the construction period.

B. Field Measurements: Where cabinets are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1. Locate concealed framing, blocking, and reinforcements that support cabinets by field measurements before being enclosed/concealed by construction and indicate measurements on Shop Drawings.

C. Established Dimensions: Where cabinets are indicated to fit to other construction, establish dimensions for areas where cabinets are to fit. Provide allowance for trimming at site, and coordinate construction to ensure that actual dimensions correspond to established dimensions.

## PART 2 - PRODUCTS

### 2.1 PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS

A. Accessibility Standard: Comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design" and ICC A117.1.

B. Quality Standard: Unless otherwise indicated, comply with the "Architectural Woodwork Standards" for grades of cabinets indicated for construction, finishes, installation, and other requirements.

C. Grade: Premium.

D. Type of Construction: Frameless.

E. Door and Drawer-Front Style: Flush overlay.

F. High-Pressure Decorative Laminate: ISO 4586-3, grades as indicated or if not indicated, as required by quality standard.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

a. ABET Inc.
b. Formica Corporation.
c. Nevamar LLC.
d. Pionite; a Panolam Industries International, Inc. brand.
e. Wilsonart LLC.

2. PLAM-1: Subject to compliance with requirements, Provide Wilsonart, Brushed Walnut, #Y0643K-12, Soft Grain Finish (58) or approved equivalent product.

3. PLAM-2: Subject to compliance with requirements, Provide Wilsonart, Slate Velvet, #15507, Ultra Matte Finish or approved equivalent product.

G. Exposed Surfaces:

1. Plastic-Laminate Grade: HGS.
2. Edges: PVC edge banding, 0.12-inch thick, matching face laminate in color, pattern, and finish.

3. Pattern Direction: Vertically for drawer fronts, doors, and fixed panels unless noted otherwise.

H. Semi-Exposed Surfaces:

1. Surfaces Other Than Drawer Bodies: High-pressure decorative laminate, ISO 4586-3.

   a. Edges of Plastic-Laminate Shelves: PVC tape, 0.018-inch (0.460-mm) minimum thickness, matching laminate in color, pattern, and finish.
   b. Edges of Thermally Fused Laminate Panel Shelves: PVC or polyester edge banding.
   c. For semi-exposed backs of panels with exposed plastic-laminate surfaces, provide surface of high-pressure decorative laminate, ISO 4586-3, grade to match exposed surface.

2. Drawer Sides and Backs: Thermally fused laminate panels with PVC or polyester edge banding.
3. Drawer Bottoms: Thermally fused laminate panels.

I. Dust Panels: 1/4-inch plywood or tempered hardboard above compartments and drawers unless located directly under tops.

J. Concealed Backs of Panels with Exposed Plastic-Laminate Surfaces: High-pressure decorative laminate, ISO 4583-3, grade to match exposed surface.

K. Drawer Construction: Fabricate with exposed fronts fastened to subfront with mounting screws from interior of body.

1. Join subfronts, backs, and sides with glued rabbeted joints supplemented by mechanical fasteners or glued dovetail joints only, dowel method is not acceptable.

L. Colors, Patterns, and Finishes: Provide materials and products that result in colors and textures of exposed laminate surfaces complying with the following requirements:

1. As indicated by laminate manufacturer's designations.
2. Match Architect’s sample, refer to Room Finish List and Notes as indicated on the drawings.

2.2 WOOD MATERIALS

A. Wood Products: Provide materials that comply with requirements of referenced quality standard for each type of architectural cabinet and quality grade specified unless otherwise indicated.

1. Wood Moisture Content: 5 to 10 percent.

B. Composite Wood Products: Provide materials that comply with requirements of referenced quality standard for each type of architectural cabinet and quality grade specified unless otherwise indicated.

1. Medium-Density Fiberboard (MDF): ANSI A208.2, Grade 130.
2. Softwood Plywood: DOC PS 1, medium-density overlay.
3. Thermally Fused Laminate (TFL) Panels: Particleboard or MDF finished with thermally fused, melamine-impregnated decorative paper and complying with requirements of ISO 4586. White melamine interiors are acceptable for all interior casework within

cabinets and drawers. Any exposed panels, faces, and surrounds shall be matching PLAM finish as scheduled.

C. Composite Wood Products: Products shall be made without urea formaldehyde.

2.3 FIRE-RETARDANT-TREATED MATERIALS

A. Fire-Retardant-Treated Materials, General: Where fire-retardant-treated materials are indicated, use materials that are acceptable to authorities having jurisdiction and with fire-test-response characteristics specified as determined by testing identical products per test method indicated by a qualified testing agency.

1. Use treated materials that comply with requirements of referenced quality standard. Do not use materials that are warped, discolored, or otherwise defective.
2. Use fire-retardant-treatment formulations that do not bleed through or otherwise adversely affect finishes. Do not use colorants to distinguish treated materials from untreated materials.
3. Identify fire-retardant-treated materials with appropriate classification marking of qualified testing agency in the form of removable paper label or imprint on surfaces that will be concealed from view after installation.

B. Fire-Retardant-Treated Lumber and Plywood: Products with a flame-spread index of 25 or less when tested according to ASTM E 84, with no evidence of significant progressive combustion when the test is extended an additional 20 minutes, and with the flame front not extending more than 10.5 feet beyond the centerline of the burners at any time during the test.

1. Kiln-dry lumber and plywood after treatment to a maximum moisture content of 19 and 15 percent, respectively.
2. Mill lumber after treatment within limits set for wood removal that do not affect listed fire-test-response characteristics, using a woodworking shop certified by testing and inspecting agency.

C. Fire-Retardant Fiberboard: MDF panels complying with ANSI A208.2, made from softwood fibers, synthetic resins, and fire-retardant chemicals mixed together at time of panel manufacture to achieve flame-spread index of 25 or less and smoke-developed index of 200 or less per ASTM E 84.

2.4 CABINET HARDWARE AND ACCESSORIES

A. General: Provide cabinet hardware and accessory materials associated with architectural cabinets except for items specified in Section 08 7100 "Door Hardware."

B. Source Limitations: Obtain each type of cabinet hardware and accessory from single source and from single manufacturer unless other indicated or approved by Architect.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following unless otherwise noted or approved:

a. Accuride International Inc.
b. Häfele America Co
c. Julius Blum & Co., Inc.
d. Knape & Vogt Manufacturing Company.
e. Richelieu Hardware Inc.

C. Frameless Concealed Hinges (European Type): BHMA A156.9, B01602, 170 degrees of opening, soft closing. Basis of Design: Blum BluMotion or approved equivalent.

D. Back and edge mounted pulls, 6 inches (152 mm) long, 2-1/4 inches (58 mm) deep.

1. Subject to compliance with requirements, provide Richelieu 5632 in Matte Black finish, or approved equivalent.

E. Catches: Magnetic catches, ANSI/BHMA A156.9, B03141.

F. Adjustable Shelf Standards and Supports: ANSI/BHMA A156.9, B04071; with shelf rests, B04081.

G. Shelf Rests: BHMA A156.9, B04013; metal, two-pin type with shelf hold-down clip.

H. Drawer Slides: ANSI/BHMA A156.9.

1. Standard Duty (Grade 1 and Grade 2): Side mount.
2. Heavy-Duty (Grade 1HD-100 and Grade 1HD-200): Side mount.

a. Type: Full extension.
b. Material: Zinc-plated ball bearing slides.
c. Motion Feature: Self-closing mechanism.
d. Basis of Design: Blum Tandem or approved equivalent.

3. Pencil drawers not more than 3 inches (75 mm) high and not more than 24 inches (600 mm) wide, provide 50 lb (22.7 kg) load capacity.
4. General-purpose drawers more than 3 inches (75 mm) high, but not more than 6 inches (150 mm) high and not more than 24 inches (600 mm) wide, provide 75 lb (34 kg) load capacity.
5. File drawers more than 6 inches (150 mm) high or more than 24 inches (600 mm) wide, provide 100 lb (45 kg) load capacity.
6. Lateral file drawers more than 6 inches (150 mm) high and more than 24 inches (600 mm) but not more than 30 inches (762 mm) wide, provide 150 lb (68 kg) load capacity.
7. For trash bins not more than 20 inches high and 16 inches wide, provide Grade 1HD-100.

I. Door Locks: ANSI/BHMA A156.11, E07121.

J. Drawer Locks: ANSI/BHMA A156.11, E07041.

K. Door and Drawer Silencers: BHMA A156.16, L03011.

L. Pullout Waste Container Systems: Subject to compliance with requirements, provide Rev-a-Shelf soft close series 5349, or an approved equivalent product.

1. Waste Containers: Subject to compliance with requirements, provide waste bin sizes that fit into the pullout container systems as indicated on the drawing, or an approved equivalent product.

M. ADA Toe Kick: Subject to compliance with requirements, provide an ADA toe kick bracket as indicated on the drawings similar to one at www.adatoekick.com, or an approved equivalent product.

N. Closet Rod: Subject to compliance with requirements, Provide Knape & Vogt 660 Series Heavy Duty Round Closet Rod with Flanges, or equivalent product.

O. Grommets for Cable Passage: Refer to section 12 3661 “Solid Surfacing Materials”.

1. Grommet locations to be identified and located in shop drawings.

P. Exposed Hardware Finishes: For exposed hardware, provide finish that complies with BHMA A156.18 for BHMA finish number indicated.

1. Satin Stainless Steel: BHMA 630.

Q. For concealed hardware, provide manufacturer's standard finish that complies with product class requirements in BHMA A156.9.

## 2.5 MISCELLANEOUS MATERIALS

A. Furring, Blocking, Shims, and Hanging Strips: Softwood or hardwood lumber, kiln-dried to less than 15 percent moisture content.

B. Anchors: Select material, type, size, and finish required for each substrate for secure anchorage. Provide metal expansion sleeves or expansion bolts for post-installed anchors. Use nonferrous-metal or hot-dip galvanized anchors and inserts at inside face of exterior walls and at floors.

C. Adhesives: Do not use adhesives that contain urea formaldehyde.

D. Adhesive for Bonding Plastic Laminate: Un-pigmented contact cement.

1. Adhesive for Bonding Edges: Hot-melt adhesive or adhesive specified above for faces.

## 2.6 FABRICATION

A. Fabricate architectural cabinets to dimensions, profiles, and details indicated.

B. Complete fabrication, including assembly and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.

1. Trial fit assemblies at fabrication shop that cannot be shipped completely assembled. Install dowels, screws, bolted connectors, and other fastening devices that can be removed after trial fitting. Verify that various parts fit as intended and check measurements of assemblies against field measurements before disassembling for shipment.

C. Shop-cut openings to maximum extent possible to receive hardware, appliances, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.

PART 3 - EXECUTION

3.1 PREPARATION

A. Before installation, condition cabinets to humidity conditions in installation areas for not less than 72 hours.

3.2 INSTALLATION

A. Grade: Install cabinets to comply with quality standard grade of item to be installed.

B. Assemble cabinets and complete fabrication at Project site to extent that it was not completed in the shop.

C. Anchor cabinets to anchors or blocking built in or directly attached to substrates. Secure with wafer-head cabinet installation screws.

D. Install cabinets level, plumb, and true in line to a tolerance of 1/8 inch in 96 inches using concealed shims.

1. Scribe and cut cabinets to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.
2. Install cabinets without distortion so doors and drawers fit openings and are accurately aligned. Adjust hardware to center doors and drawers in openings and to provide unencumbered operation. Complete installation of hardware and accessory items as indicated.
3. Fasten wall cabinets through back, near top and bottom, and at ends not more than 16 inches o.c. with No. 10 wafer-head screws sized for not less than 1-1/2-inch penetration into wood framing, blocking, or hanging strips.

3.3 ADJUSTING AND CLEANING

A. Repair damaged and defective cabinets, where possible, to eliminate functional and visual defects. Where not possible to repair, replace architectural cabinets. Adjust joinery for uniform appearance.

B. Clean, lubricate, and adjust hardware.

C. Clean cabinets on exposed and semi-exposed surfaces.

END OF SECTION 06 4116

# SECTION 07 0150 - PREPARATION FOR REROOFING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. The Work of This Section Includes:

1. Partial roof tear-off.
2. Temporary roofing.
3. Roof re-cover preparation.
4. Disposal.

B. Related Requirements:

1. Section 01 1000 "Summary" for use of premises and for phasing requirements.
2. Section 01 5000 "Temporary Facilities and Controls" for temporary construction and environmental-protection measures for reroofing preparation.
3. Section 07 5000 “Roofing Coating” for re-roofing coating requirements.

### 1.2 DEFINITIONS

A. EPS: Molded (expanded) polystyrene.

B. Partial Roof Tear-off: Removal of selected components and accessories from existing roofing system.

C. OSB: Oriented strand board.

D. Roofing Terminology: Definitions in ASTM D1079 and glossary of NRCA's "The NRCA Roofing Manual: Membrane Roof Systems" apply to work of this Section.

E. Roof Re-Cover Preparation: Existing roofing system is to remain and be prepared for new roof installed over it.

### 1.3 PREINSTALLATION MEETINGS

A. Preliminary Roofing Conference: Before starting removal Work, conduct conference at Project site.

1. Meet with Owner, Architect, Construction Manager, Owner's insurer if applicable, testing and inspecting agency representative, roofing Installer, and installers whose work interfaces with or affects roofing, including installers of roof accessories and roof-mounted equipment.
2. Review methods and procedures related to roofing tear-off, including, but not limited to, the following:

a. Reroofing preparation, including roofing system manufacturer's written instructions.

b. Temporary protection requirements for existing roofing system components that are to remain.
c. Existing roof drains and roof drainage during each stage of reroofing, and roof-drain plugging and plug removal.
d. Construction schedule and availability of materials, Installer's personnel, equipment, and facilities needed to avoid delays.
e. Existing roof deck conditions requiring Architect notification.
f. Existing roof deck removal procedures and Owner notifications.
g. Condition and acceptance of existing roof deck and base flashing substrate for reuse.
h. Structural loading limitations of roof deck during reroofing.
i. Base flashings, special roofing details, drainage, penetrations, equipment curbs, and condition of other construction that affect reroofing.
j. HVAC shutdown and sealing of air intakes.
k. Shutdown of fire-suppression, -protection, and -alarm and -detection systems.
l. Existing conditions that may require Architect notification before proceeding.

## 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Temporary Roofing Submittal: Product data and description of temporary roofing system.

1. If temporary roof remains in place, include surface preparation requirements needed to receive permanent roof, and submit a letter from roofing manufacturer stating acceptance of the temporary roof and that its inclusion does not adversely affect the new roofing system's resistance to fire and wind or specified special warranty.

## 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

1. Include certificate that Installer is approved by warrantor of existing roofing system.

B. Photographs or Video: Show existing conditions of adjoining construction and site improvements, including exterior and interior finish surfaces, that might be misconstrued as having been damaged by reroofing operations.

1. Submit before Work begins.

## 1.6 QUALITY ASSURANCE

A. Regulatory Requirements:

1. Comply with governing EPA notification regulations before beginning roofing removal.
2. Comply with hauling and disposal regulations of authorities having jurisdiction.

## 1.7 FIELD CONDITIONS

A. Existing Roofing System: Membrane roofing.

B. Owner will not occupy portions of building immediately below reroofing area.

C. Protect building to be reroofed, adjacent buildings, walkways, site improvements, exterior plantings, and landscaping from damage or soiling from reroofing operations.

D. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities.

E. Conditions existing at time of inspection for bidding will be maintained by Owner as far as practical.

F. Weather Limitations: Proceed with reroofing preparation only when existing and forecasted weather conditions permit Work to proceed without water entering existing roofing system or building.

1. Remove only as much roofing in one day as can be made watertight in the same day.

G. Hazardous Materials:

1. It is not expected that hazardous materials, such as asbestos-containing materials, will be encountered in the Work.

## PART 2 - PRODUCTS

### 2.1 TEMPORARY PROTECTION MATERIALS

A. EPS Insulation: ASTM C578.

B. Plywood: DOC PS 1, Grade CD, Exposure 1.

C. OSB: DOC PS 2, Exposure 1.

### 2.2 TEMPORARY ROOFING MATERIALS

A. Design and selection of materials for temporary roofing are Contractor's responsibilities.

### 2.3 INFILL AND REPLACEMENT MATERIALS

A. Use infill materials matching existing roofing system materials unless otherwise indicated.

B. Wood blocking, curbs, and nailers are specified in Section 06 1000 "Rough Carpentry."

C. Fasteners: Factory-coated steel fasteners with metal or plastic plates listed in FM Approvals' RoofNav, and acceptable to new roofing system manufacturer.

### 2.4 AUXILIARY REROOFING MATERIALS

A. General: Use auxiliary reroofing preparation materials recommended by roofing system manufacturer for intended use and compatible with components of new roofing system.

PART 3 - EXECUTION

3.1 PREPARATION

A. Protection of In-Place Conditions:

1. Protect existing roofing system that is not to be reroofed.

B. Seal or isolate windows that may be exposed to airborne substances created in removal of existing materials.

C. Shut off rooftop utilities and service piping before beginning the Work.

D. Test existing roof drains to verify that they are not blocked or restricted.

1. Immediately notify Architect of any blockages or restrictions.

E. During removal operations, have sufficient and suitable materials on-site to facilitate rapid installation of temporary protection in the event of unexpected rain.

F. Maintain roof drains in functioning condition to ensure roof drainage at end of each workday.

1. Prevent debris from entering or blocking roof drains and conductors.

a. Use roof-drain plugs specifically designed for this purpose.
b. Remove roof-drain plugs at end of each workday, when no work is taking place, or when rain is forecast.

2. If roof drains are temporarily blocked or unserviceable due to roofing system removal or partial installation of new roofing system, provide alternative drainage method to remove water and eliminate ponding.

a. Do not permit water to enter in or under existing roofing system components that are to remain.

3.2 ROOF TEAR-OFF

A. Lower removed roofing materials to ground and onto lower roof levels, using dust-tight chutes or other acceptable means of removing materials from roof areas.

B. Remove pavers and accessories from roofing.

1. Store and protect pavers and accessories for reuse in manner not to exceed structural loading limitations of roof deck.
2. Discard cracked pavers.

C. Partial Roof Tear-off: Where indicated on Drawings, remove existing roofing down to existing roof deck and immediately check for presence of moisture.

1. Engage a qualified testing agency to perform the following test:

2. Survey exposed substrate that is to remain using infrared color thermography according to ASTM C1153.

    a. Prepare survey report of initial scan indicating locations of entrapped moisture, if any, and area calculations of locations of entrapped moisture.

3. Remove wet or damp materials below existing roofing and above deck as directed by Architect.

    a. Removal is paid for by adjusting the Contract Sum included in the Contract Documents.

4. Inspect wood blocking, curbs, and nailers for deterioration and damage.

    a. If wood blocking, curbs, or nailers have deteriorated, immediately notify Architect.
    b. Removal is paid for by adjusting the Contract Sum included in the Contract Documents.

## 3.3 TEMPORARY ROOFING

A. Install approved temporary roofing over area to be reroofed.

B. Remove temporary roofing before installing new roofing.

## 3.4 ROOF RE-COVER PREPARATION

A. Remove blisters, ridges, buckles and other substrate irregularities from existing roofing that inhibit new recover boards from conforming to substrate.

1. Broom clean existing substrate.
2. Coordinate with inspector to schedule times for tests and inspections.
3. Verify that existing substrate is dry.

    a. Spot check substrates with an electrical capacitance moisture-detection meter.

4. Remove materials that are wet or damp.

    a. Removal will be paid for by adjusting the Contract Sum included in the Contract Documents.

## 3.5 BASE FLASHING REMOVAL

A. Remove existing base flashings.

1. Clean substrates of contaminants, such as asphalt, sheet materials, dirt, and debris.

B. Do not damage metal counter flashings that are to remain.

1. Replace metal counter flashings damaged during removal with counter flashings specified in Section 07 7100 "Roof Specialties."

C. Inspect sheathing, wood blocking, curbs, and nailers for deterioration and damage.

1. If sheathing, wood blocking, curbs, or nailers have deteriorated, immediately notify Architect.

D. When directed by Architect, replace parapet framing, wood blocking, curbs, and nailers to comply with 06 1000 "Rough Carpentry."

3.6 DISPOSAL

A. Collect demolished materials and place in containers.

1. Promptly dispose of demolished materials.
2. Do not allow demolished materials to accumulate on-site.
3. Storage or sale of demolished items or materials on-site is not permitted.

B. Transport and legally dispose of demolished materials off Owner's property.

END OF SECTION 07 0150

SECTION 07 2100 - THERMAL INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Extruded polystyrene foam-plastic board.
2. Glass-fiber blanket.
3. Mineral-wool blanket insulation.
4. Loose-fill insulation.
5. Spray-applied cellulosic insulation.
6. Spray-applied thermal/acoustical insulation.

B. Related Requirements:

1. Section 04 2000 "Unit Masonry" for insulation installed in masonry cells.
2. Section 09 2900 "Gypsum Board" for sound attenuation blanket used as acoustic insulation.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Include Sustainable Design Data:

1. Product Data: For recycled content, indicating postconsumer and pre-consumer recycled content and cost.
2. Product Data: For adhesives, indicating VOC content.

1.4 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For each product, for tests performed by a qualified testing agency.

B. Evaluation Reports: For foam-plastic insulation, from ICC-ES.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Protect insulation materials from physical damage and from deterioration due to moisture, soiling, and other sources. Store inside and in a dry location. Comply with manufacturer's written instructions for handling, storing, and protecting during installation.

B. Protect foam-plastic board insulation as follows:

1. Do not expose to sunlight except to necessary extent for period of installation and concealment.
2. Protect against ignition at all times. Do not deliver foam-plastic board materials to Project site until just before installation time.
3. Quickly complete installation and concealment of foam-plastic board insulation in each area of construction.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Surface-Burning Characteristics: Maximum flame-spread and smoke-developed indexes less than Class A, 25 and 450 when tested in accordance with ASTM E84.

B. Fire-Resistance Ratings: Comply with ASTM E119 or UL 263; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Indicate design designations from UL's "Fire Resistance Directory" or from listings of another qualified testing agency.

C. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.

D. Labeling: Provide identification of mark indicating R-value of each piece of insulation 12 inches (305 mm) and wider in width.

E. Thermal-Resistance Value (R-Value): R-value as indicated on Drawings accordance with ASTM C518.

### 2.2 EXTRUDED POLYSTYRENE FOAM-PLASTIC BOARD INSULATION (XPS)

A. Extruded Polystyrene Board Insulation, Type IV: ASTM C578, Type IV, 25-psi (173-kPa) minimum compressive strength; unfaced.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Owens Corning Foamular 250 Insulation, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. DiversiFoam Products.
b. Dow Chemical Company (The).
c. Owens Corning.

2. Board Insulation for vertical installation as indicated on drawings.

### 2.3 GLASS-FIBER BLANKET

A. Recycled Content: Postconsumer recycled content plus one-half of pre-consumer recycled content not less than 40 percent.

B. Glass-Fiber Blanket, Semi-Rigid, Unfaced ASTM C 665, Type I; with maximum flame-spread and smoke-developed indexes of 0 and 0, respectively, per ASTM E 84; passing ASTM E 136 for non-combustion characteristics.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. CertainTeed Corporation.
b. Johns Manville; a Berkshire Hathaway company.
c. Knauf Insulation.
d. Owens Corning.
e. Roxul Inc.

C. Glass-Fiber Blanket, Unfaced ASTM C 665, Type I; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively, per ASTM E 84; passing ASTM E 136 for combustion characteristics.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. CertainTeed Corporation.
b. Johns Manville; a Berkshire Hathaway company.
c. Knauf Insulation.
d. Owens Corning.
e. Roxul Inc.

## 2.4 MINERAL-WOOL BLANKET INSULATION

A. Mineral-Wool Blanket Insulation, Unfaced: ASTM C665, Type I (blankets without membrane facing); consisting of fibers; passing ASTM E136 for combustion characteristics.

## 2.5 LOOSE-FILL INSULATION

A. Recycled Content: Postconsumer recycled content plus one-half of pre-consumer recycled content not less than 25 percent.

B. Glass-Fiber Loose-Fill Insulation ASTM C 764, Type I for pneumatic application or Type II for poured application; with maximum flame-spread and smoke-developed indexes of 5, per ASTM E 84.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. CertainTeed Corporation.
b. Johns Manville; a Berkshire Hathaway company.
c. Knauf Insulation.

2.6 SPRAY-APPLIED CELLULOSIC INSULATION

A. Recycled Content: Postconsumer recycled content plus one-half of pre-consumer recycled content not less than 25 percent.

B. Self-Supported, Spray-Applied Cellulosic Insulation: ASTM C 1149, Type I (materials applied with liquid adhesive; suitable for either exposed or enclosed applications), chemically treated for flame-resistance, processing, and handling characteristics.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Central Fiber LLC.
b. GreenFiber.
c. Hamilton Manufacturing Inc.
d. International Cellulose Corp.
e. Nu-Wool Co., Inc.

2.7 SPRAY-APPLIED THERMAL/ACOUSTICAL INSULATION

A. Recycled Content: Postconsumer recycled content plus one-half of pre-consumer recycled content not less than 25 percent.

B. Self-Supported, Spray-Applied Insulation: Spray-applied materials shall be Spray-On White Fiber conforming to ASTM E-136 using Liquid Bonding Adhesive. Fibers shall consist of Bio Soluble fiberglass

1. Basis-of-Design Product: Subject to compliance with requirements, provide Monoglass Spray-On Insulation, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. International Cellulose Corp.
b. Monoglass Incorporated.
c. Urecoat Inc.

C. Insulation shall not contain asbestos, free crystalline silica or combustible fibers.

2.8 INSULATION FASTENERS

A. Adhesively Attached, Spindle-Type Anchors: Plate welded to projecting spindle; capable of holding insulation of specified thickness securely in position with self-locking washer in place.

1. Plate: Perforated, galvanized carbon-steel sheet, 0.030-inch (0.762 mm) thick by 2 inches (50 mm) square.
2. Spindle: Copper-coated, low-carbon steel; fully annealed; 0.105 inch (2.67 mm) in diameter; length to suit depth of insulation.

B. Adhesively Attached, Angle-Shaped, Spindle-Type Anchors: Angle welded to projecting spindle; capable of holding insulation of specified thickness securely in position with self-locking washer in place.

1. Angle: Formed from 0.030-inch- (0.762-mm-) thick, perforated, galvanized carbon-steel sheet with each leg 2 inches (50 mm) square.
2. Spindle: Copper-coated, low-carbon steel; fully annealed; 0.105 inch (2.67 mm) in diameter; length to suit depth of insulation.

2.9 ACCESSORIES

A. Insulation for Miscellaneous Voids:

1. Glass-Fiber Insulation: ASTM C 764, Type II, loose fill; with maximum flame-spread and smoke-developed indexes of 5, per ASTM E 84.
2. Spray Polyurethane Foam Insulation: ASTM C 1029, Type II, closed cell, with maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, per ASTM E 84.

B. Adhesive for Bonding Insulation: Product compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.

1. Adhesives shall have a VOC content of 70 g/L or less.

C. Asphalt Coating for Cellular-Glass Block Insulation: Cutback asphalt or asphalt emulsion of type recommended by manufacturer of cellular-glass block insulation.

D. Eave Ventilation Troughs: Preformed, rigid fiberboard or plastic sheets designed and sized to fit between roof framing members and to provide ventilation between insulated attic spaces and vented eaves.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean substrates of substances that are harmful to insulation, including removing projections capable of puncturing insulation or vapor retarders, or that interfere with insulation attachment.

3.2 INSTALLATION, GENERAL

A. Comply with insulation manufacturer's written instructions applicable to products and applications.

B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.

C. Extend insulation to envelop entire area to be insulated. Fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with placement.

D. Provide sizes to fit applications and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units unless multiple layers are otherwise shown or required to make up total thickness or to achieve R-value.

3.3 INSTALLATION OF SLAB INSULATION

A. On vertical slab edge and foundation surfaces, set insulation units using manufacturer's recommended adhesive according to manufacturer's written instructions.

1. If not otherwise indicated, extend insulation a minimum of 24 inches (610 mm) below exterior grade line.

B. On horizontal surfaces, loosely lay insulation units according to manufacturer's written instructions. Stagger end joints and tightly abut insulation units.

1. If not otherwise indicated, extend insulation a minimum of 24 inches (610 mm) in from exterior walls.

3.4 INSTALLATION OF FOUNDATION WALL INSULATION

A. Butt panels together for tight fit.

B. Anchor Installation: Install board insulation on concrete substrates by adhesively attached, spindle-type insulation anchors as follows:

1. Fasten insulation anchors to concrete substrates with insulation anchor adhesive according to anchor manufacturer's written instructions. Space anchors according to insulation manufacturer's written instructions for insulation type, thickness, and application.
2. Apply insulation standoffs to each spindle to create cavity width indicated on Drawings between concrete substrate and insulation.
3. After adhesive has dried, install board insulation by pressing insulation into position over spindles and securing it tightly in place with insulation-retaining washers, taking care not to compress insulation.
4. Where insulation will not be covered by other building materials, apply capped washers to tips of spindles.

C. Adhesive Installation: Install with adhesive or press into tacky waterproofing or dampproofing according to manufacturer's written instructions.

3.5 INSTALLATION OF CAVITY-WALL INSULATION

A. Installing Cavity Wall Insulation: Place small dabs of adhesive, spaced approximately 12 inches (300 mm) o.c. both ways, on inside face of insulation boards, or attach with plastic fasteners designed for this purpose. Fit courses of insulation between wall ties and other confining obstructions in cavity, with edges butted tightly both ways. Press units firmly against inside wythe of masonry or other construction as shown.

1. Fill cracks and open gaps in insulation with crack sealer compatible with insulation and masonry.

2. Supplement adhesive attachment of insulation by securing boards with two-piece wall ties designed for this purpose and specified in Section 042000 "Unit Masonry."

3.6 INSTALLATION OF INSULATION IN FRAMED CONSTRUCTION

A. Blanket Insulation: Install in cavities formed by framing members according to the following requirements:

1. Use insulation widths and lengths that fill the cavities formed by framing members. If more than one length is required to fill the cavities, provide lengths that will produce a snug fit between ends.
2. Place insulation in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
3. Maintain 3-inch (76-mm) clearance of insulation around recessed lighting fixtures not rated for or protected from contact with insulation.
4. Attics: Install eave ventilation troughs between roof framing members in insulated attic spaces at vented eaves.
5. For metal-framed wall cavities where cavity heights exceed 96 inches (2438 mm), support unfaced blankets mechanically and support faced blankets by taping flanges of insulation to flanges of metal studs.

B. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation using the following materials:

1. Glass-Fiber Insulation: Compact to approximately 40 percent of normal maximum volume equaling a density of approximately 2.5 lb/cu. ft. (40 kg/cu. m).
2. Spray Polyurethane Insulation: Apply according to manufacturer's written instructions.

C. Loose-Fill Insulation: Apply according to ASTM C 1015 and manufacturer's written instructions. Level horizontal applications to uniform thickness as indicated, lightly settle to uniform density, but do not compact excessively.

1. For cellulosic-fiber loose-fill insulation, comply with CIMA's Bulletin #2, "Standard Practice for Installing Cellulose Insulation."

D. Spray-Applied Cellulosic Insulation: Apply spray-applied insulation according to manufacturer's written instructions. Do not apply insulation until installation of pipes, ducts, conduits, wiring, and electrical outlets in walls is completed and windows, electrical boxes, and other items not indicated to receive insulation are masked. After insulation is applied, make flush with face of studs by using method recommended by insulation manufacturer.

3.7 PROTECTION

A. Protect installed insulation from damage due to harmful weather exposures, physical abuse, and other causes. Provide temporary coverings or enclosures where insulation is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

END OF SECTION 07 2100

# SECTION 07 2726 - FLUID-APPLIED MEMBRANE AIR BARRIERS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. High-build air barriers, vapor permeable, for use on CMU walls or approved locations.

### 1.2 DEFINITIONS

A. Air-Barrier Accessory: A transitional component of the air barrier that provides continuity.

B. Air-Barrier Assembly: The collection of air-barrier materials and accessories applied to an opaque wall, including joints and junctions to abutting construction, to control air movement through the wall.

C. Air-Barrier Material: A primary element that provides a continuous barrier to the movement of air.

### 1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review air-barrier requirements and installation, special details, mockups, air-leakage and bond testing, air-barrier protection, and work scheduling that covers air barriers.

### 1.4 ACTION SUBMITTALS

A. Product Data: Include manufacturer's written instructions for evaluating, preparing, and treating each substrate; technical data; dry film thickness; and tested physical and performance properties of products.

1. High-build air barriers, vapor permeable.

B. Shop Drawings: For air-barrier assemblies.

1. Show locations and extent of air-barrier materials, accessories, and assemblies specific to Project conditions.
2. Include details for substrate joints and cracks, counterflashing strips, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.
3. Include details of interfaces with other materials that form part of air barrier.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer. Include list of ABAA-certified installers and supervisors employed by Installer, who work on Project.

B. Product Certificates: From air-barrier manufacturer, certifying compatibility of air barriers and accessory materials with Project materials that connect to or that come in contact with the barrier.

C. Product Test Reports: For each air-barrier assembly, for tests performed by a qualified testing agency.

D. Field quality-control reports.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1. Installer to be licensed by ABAA in accordance with ABAA's Quality Assurance Program and to employ ABAA-certified installers and supervisors on Project.

B. Mockups: Build mockups to set quality standards for materials and execution.

1. Build integrated mockups of exterior wall assembly 150 sq. ft. (14 sq. m), incorporating backup wall construction, external cladding, window, storefront, door frame and sill, insulation, ties and other penetrations, and flashing to demonstrate surface preparation, crack and joint treatment, application of air barriers, and sealing of gaps, terminations, and penetrations of air-barrier assembly.

a. Coordinate construction of mockups to permit inspection and testing of air barrier before external insulation and cladding are installed.
b. Include junction with roofing membrane, building corner condition, and foundation wall intersection.
c. If Architect determines mockups do not comply with requirements, reconstruct mockups and apply air barrier until mockups are approved.

2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Remove and replace liquid materials that cannot be applied within their stated shelf life.

B. Protect stored materials from direct sunlight.

1.8 FIELD CONDITIONS

A. Environmental Limitations: Apply air barrier within the range of ambient and substrate temperatures recommended in writing by air-barrier manufacturer.

1. Protect substrates from environmental conditions that affect air-barrier performance.
2. Do not apply air barrier to a damp or wet substrate or during snow, rain, fog, or mist.

## PART 2 - PRODUCTS

### 2.1 SOURCE LIMITATIONS

A. Obtain primary air-barrier materials and air-barrier accessories from single manufacturer.

### 2.2 PERFORMANCE REQUIREMENTS

A. Air-Barrier Performance: Air-barrier assembly and seals with adjacent construction to be capable of performing as a continuous air barrier and as a liquid-water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Air-barrier assemblies to be capable of accommodating substrate movement and of sealing substrate expansion and control joints, construction material changes, penetrations, tie-ins to installed waterproofing, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.

B. Air-Barrier Assembly Air Leakage: Maximum 0.04 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft. (0.2 L/s x sq. m of surface area at 75 Pa), when tested in accordance with ASTM E2357.

C. Ultimate Elongation: Minimum 500> percent; ASTM D412, Die C.

D. Adhesion to Substrate: Minimum 16 lbf/sq. in. (110 kPa) when tested in accordance with ASTM D4541.

E. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.

F. UV Resistance: Can be exposed to sunlight for 60 days in accordance with manufacturer's written instructions.

### 2.3 HIGH-BUILD AIR BARRIERS, VAPOR PERMEABLE

A. High-Build, Vapor-Permeable Air Barrier, Synthetic Polymer Type: Synthetic polymer membrane with an installed dry film thickness, according to manufacturer's written instructions, of 35 mils (0.9 mm) or thicker over smooth, void-free substrates.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Carlisle Coatings & waterproofing Inc. - Barriseal
    b. GCP Applied Technologies Inc. – Perm-A-Barrier NPL 10
    c. Henry Company; a Carlisle company. – Air-Bloc
    d. W.R. Meadows, Inc. – Air Shield

B. Vapor Permeance: Minimum 1 perms (290 ng/Pa x s x sq. m) Procedure B, Water Method.

2.4 ACCESSORY MATERIALS

A. Provide primers, transition strips, termination strips, joint reinforcing fabric and strips, joint sealants, counterflashing strips, flashing sheets and metal termination bars, termination mastic, substrate patching materials, adhesives, tapes, foam sealants, lap sealants, and other accessory materials that are recommended in writing by air-barrier manufacturer to produce a complete air-barrier assembly and that are compatible with primary air-barrier material and adjacent construction to which they may seal.

B. Primer: Liquid waterborne primer recommended for substrate by air-barrier material manufacturer.

C. Stainless Steel Sheet: ASTM A240/A240M, Type 304, 0.0187 inch (0.5 mm) thick, and Series 300 stainless steel fasteners.

D. Preformed Silicone Extrusion: Manufacturer's standard system consisting of cured low-modulus silicone extrusion, sized to fit opening widths, with a single-component, neutral-curing, Class 100/50 (low-modulus) silicone sealant for bonding extrusions to substrates.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that substrates are sound and free of oil, grease, dirt, excess mortar, or other contaminants.
2. Verify that substrates have cured and aged for minimum time recommended in writing by air-barrier manufacturer.
3. Verify that substrates are visibly dry and free of moisture. Test concrete substrates for capillary moisture by plastic sheet method in accordance with ASTM D4263.
4. Verify that masonry joints are flush and completely filled with mortar.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SURFACE PREPARATION

A. Clean, prepare, treat, fill, and seal substrate and joints and cracks in substrate in accordance with manufacturer's written instructions and details. Provide clean, dust-free, and dry substrate for air-barrier application.

B. Mask off adjoining surfaces not covered by air barrier to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, bitumen, form-release agents, paints, curing compounds, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, mortar, and other projections and fill honeycomb, aggregate pockets, holes, and other voids in concrete with substrate-patching material.

E. Remove excess mortar from masonry ties, shelf angles, and other obstructions.

F. At changes in substrate plane, apply sealant or termination mastic beads at sharp corners and edges to form a smooth transition from one plane to another.

G. Cover gaps in substrate plane and form a smooth transition from one substrate plane to another with stainless steel sheet mechanically fastened to structural framing to provide continuous support for air barrier.

3.3 INSTALLATION OF ACCESSORIES

A. Install accessory materials in accordance with air-barrier manufacturer's written instructions and details to form a seal with adjacent construction and ensure continuity of air and water barrier.

1. Coordinate the installation of air barrier with installation of roofing membrane and base flashing to ensure continuity of air barrier with roofing membrane.
2. Install transition strip on roofing membrane or base flashing so that a minimum of 3 inches (75 mm) of coverage is achieved over each substrate.
3. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
4. Apply primer to substrates at required rate and allow it to dry. Limit priming to areas that will be covered by air-barrier material on same day. Reprime areas exposed for more than 24 hours.

B. Connect and seal exterior wall air-barrier material continuously to roofing-membrane air barrier, concrete below-grade structures, floor-to-floor construction, exterior glazing and window systems, glazed curtain-wall systems, storefront systems, exterior louvers, exterior door framing, and other construction used in exterior wall openings, using accessory materials.

C. At end of each working day, seal top edge of strips and transition strips to substrate with termination mastic.

D. Apply joint sealants forming part of air-barrier assembly within manufacturer's recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.

E. Wall Openings: Prime concealed, perimeter frame surfaces of windows, curtain walls, storefronts, and doors. Apply preformed silicone extrusion so that a minimum of 3 inches (75 mm) of coverage is achieved over each substrate. Maintain 3 inches (75 mm) of full contact over firm bearing to perimeter frames, with not less than 1 inch (25 mm) of full contact.

1. Preformed Silicone Extrusion: Set in full bed of silicone sealant applied to walls, frame, and air-barrier material.

F. Fill gaps in perimeter frame surfaces of windows, curtain walls, storefronts, and doors, and miscellaneous penetrations of air-barrier material with foam sealant.

G. Seal strips and transition strips around masonry reinforcing or ties and penetrations with termination mastic.

H. Seal top of through-wall flashings to air barrier with an additional 6-inch- (150-mm-) wide, transition strip.

I. Seal exposed edges of strips at seams, cuts, penetrations, and terminations not concealed by metal counterflashings or ending in reglets with termination mastic.

J. Repair punctures, voids, and deficient lapped seams in strips and transition strips. Slit and flatten fishmouths and blisters. Patch with transition strips extending 6 inches (150 mm) beyond repaired areas in strip direction.

## 3.4 INSTALLATION OF PRIMARY AIR-BARRIER MATERIAL

A. Apply air-barrier material to form a seal with strips and transition strips and to achieve a continuous air barrier in accordance with air-barrier manufacturer's written instructions and details. Apply air-barrier material within manufacturer's recommended application temperature ranges.

1. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
2. Limit priming to areas that will be covered by air-barrier material on same day. Reprime areas exposed for more than 24 hours.
3. Where multiple prime coats are needed to achieve required bond, allow adequate drying time between coats.

B. High-Build Air Barriers: Apply continuous unbroken air-barrier material to substrates according to the following thickness. Apply air-barrier material in full contact around protrusions such as masonry ties.

1. Vapor-Permeable, High-Build Air Barrier: Total dry film thickness as recommended in writing by manufacturer to comply with performance requirements, but not less than 35 mils (0.9 mm), applied in one or more equal coats.

C. Do not cover air barrier until it has been tested and inspected by testing agency.

D. Correct deficiencies in or remove air barrier that does not comply with requirements; repair substrates and reapply air-barrier components.

## 3.5 FIELD QUALITY CONTROL

A. ABAA Quality Assurance Program: Perform examinations, preparation, installation, testing, and inspections under ABAA's Quality Assurance Program.

B. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

C. Inspections: Air-barrier materials, accessories, and installation are subject to inspection for compliance with requirements. Inspections may include the following:

1. Continuity of air-barrier system has been achieved throughout the building envelope with no gaps or holes.
2. Air-barrier dry film thickness.
3. Continuous structural support of air-barrier system has been provided.
4. Masonry and concrete surfaces are smooth, clean, and free of cavities, protrusions, and mortar droppings.
5. Site conditions for application temperature and dryness of substrates have been maintained.
6. Maximum exposure time of materials to UV deterioration has not been exceeded.
7. Surfaces have been primed, if applicable.

8. Laps in strips and transition strips have complied with minimum requirements and have been shingled in the correct direction (or mastic has been applied on exposed edges), with no fishmouths.
9. Termination mastic has been applied on cut edges.
10. Strips and transition strips have been firmly adhered to substrate.
11. Compatible materials have been used.
12. Transitions at changes in direction and structural support at gaps have been provided.
13. Connections between assemblies (air-barrier and sealants) have complied with requirements for cleanliness, surface preparation and priming, structural support, integrity, and continuity of seal.
14. All penetrations have been sealed.

D. Tests: As determined by testing agency from among the following tests:

1. Air-Leakage-Volume Testing: Air-barrier assemblies will be tested for air-leakage rate in accordance with ASTM E783 or ASTM E2357.
2. Adhesion Testing: Air-barrier assemblies will be tested for required adhesion to substrate in accordance with ASTM D4541 for each 600 sq. ft. (56 sq. m) of installed air barrier or part thereof.

E. Air barriers will be considered defective if they do not pass tests and inspections.

1. Apply additional air-barrier material, in accordance with manufacturer's written instructions, where inspection results indicate insufficient thickness.
2. Remove and replace deficient air-barrier components for retesting as specified above.

F. Repair damage to air barriers caused by testing; follow manufacturer's written instructions.

G. Prepare test and inspection reports.

3.6 CLEANING AND PROTECTION

A. Protect air-barrier system from damage during application and remainder of construction period, in accordance with manufacturer's written instructions.

1. Protect air barrier from exposure to UV light and harmful weather exposure as recommended in writing by manufacturer. If exposed to these conditions for longer than recommended, remove and replace air barrier or install additional, full-thickness, air-barrier application after repairing and preparing the overexposed materials in accordance with air-barrier manufacturer's written instructions.
2. Protect air barrier from contact with incompatible materials and sealants not approved by air-barrier manufacturer.

B. Clean spills, stains, and soiling from construction that would be exposed in the completed work using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

C. Remove masking materials after installation.

END OF SECTION 07 2726

# SECTION 07 5700 – ROOFING COATING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes: Spray or roller-applied, roof coating.

1. Silicone coatings.
2. Accessory materials.

B. Related Requirements:

1. Section 06 1000 "Rough Carpentry" for wood blocking, curbs, cants, and nailers.
2. Section 07 9200 "Joint Sealants" for joint sealants, fillers, and preparation.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to roofing coating, including, but not limited to, the following:

a. Load limitations on in-place roofing.
b. Construction schedule. Verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
c. Surface preparation specified in other Sections.
d. Minimum curing period.
e. Forecasted weather conditions.
f. Special details and sheet flashings.
g. Repairs.

### 1.3 ACTION SUBMITTALS

A. Product Data: Include manufacturer's written instructions for evaluating, preparing, and treating substrate; technical data; and tested physical and performance properties.

1. Silicone coatings.
2. Acrylic coatings.
3. Accessory materials.

B. Samples for Initial Selection: For each type of exposed product, finish, and color.

1. Include Samples of accessory materials and accessories involving color and finish selection.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For SPFA-qualified Installer and applicators.

B. Product Certificates: For each type of roofing coating.

C. Evaluation Reports: For coated roofing, from ICC-ES.

D. Field quality-control reports.

E. Sample Warranty: For special warranty.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For roofing coating to include in maintenance manuals.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: A qualified coated roofing installer who is approved, authorized, or licensed by coating manufacturer for installation of coating manufacturer's product.

B. Comply with recommendations in SPFA AY-104.

C. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for fabrication and installation.

1. Build mockup of typical roof area as shown on Drawings.

a. Size: 10 feet (3 m) by 10 feet (3 m).

2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials to Project site with manufacturer's name, product brand name and type, date of manufacture, shelf life, and directions for storing and mixing with other components.

B. Store materials in their original undamaged containers in a clean, dry, protected location and within the temperature range required by manufacturer.

C. Remove and replace material that cannot be applied within its stated shelf life.

1.8 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit roofing work to be performed in accordance with manufacturer's written instructions and warranty requirements.

1. Apply materials within the range of ambient and substrate temperatures recommended in writing by material manufacturers, but not below 50 deg F (10 deg C).
2. Apply materials within range of relative humidity recommended in writing by manufacturer of each component, but not when relative humidity exceeds 85 percent, or when temperatures are less than 5 deg F (3 deg C) above dew point.
3. Do not apply materials to damp or wet surfaces.
4. Do not apply primers or coatings in snow, rain, fog, or mist, or when such weather conditions are imminent during the application and curing period.
5. Do not apply coatings when wind conditions prevent uniform coating application.

1.9 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace coated roofing that does not comply with requirements or that does not remain watertight within specified warranty period.

1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain coating system from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. General Performance: Coated roofing to withstand exposure to weather without failure due to defective manufacture, installation, or other defects in construction. Membrane roofing to remain watertight.

1. Material Compatibility: Provide coatings and accessory materials that are compatible with one another and with substrate under conditions of service and application, as demonstrated by manufacturer based on testing and field experience.

2.3 SILICONE COATINGS

A. Silicone Coating: Liquid silicone elastomeric coating system complying with ASTM D6694/D6694M and specifically formulated for coating roofing.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Carlisle Syntec Systems.

b. GAF.
c. Gaco; Holcim Building Envelope.
d. The Garland Company, Inc.
e. General Coatings Manufacturing Corp.

2. Composition: One- or two-component silicone.
3. Base-Coat Color: Contrasting with topcoat.
4. Topcoat Color: Gray.
5. Topcoat Color at Walkways: White.
6. Vapor Permeance: Minimum 5.0 perms (286 ng/Pa x s x sq. m) at 20 mils (0.5 mm) thick in accordance with ASTM E96/E96M, Desiccant Method, Procedure A.

## 2.4 ACCESSORY MATERIALS

A. General: Accessory materials recommended in writing by roofing manufacturer for intended use.

1. Liquid-type accessory materials must comply with VOC limits of authorities having jurisdiction.

B. Vapor Retarder: Fluid applied or as recommended by manufacturer if required.

C. Reinforcement: Flexible polyester or fiberglass mat of weight, type, and composition recommended in writing by coating manufacturer for embedment in liquid coating.

D. Sealant: ASTM C920, Class 25, Use NT, Grade NS, Type M, multicomponent urethane or Type S, one-component, neutral- or acid-curing silicone, and as recommended in writing by coated roofing manufacturer for substrate and joint conditions and for compatibility with roofing materials.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Verify that related work is complete. Do not install coated roofing until roof openings, curbs, and parapets, if any, are complete and roof drains, vents, and other roof penetrations are in place.

B. Examine substrates, areas, and conditions under which coated roofing will be applied, with Installer present, for compliance with requirements.

C. Proceed with installation only after unsatisfactory conditions have been corrected and substrates are dry.

## 3.2 SURFACE PREPARATION

A. General: Clean and prepare substrate in accordance with coated roofing manufacturer's written instructions. Provide clean, dust-free, dew-free, and dry substrate for coated roofing application.

B. Remove grease, oil, form-release agents, curing compounds, and other contaminants from substrate.

C. Prepare substrate for re-covering in accordance with Section 07 0150 "Preparation for Re-Roofing" and to coated roofing manufacturer's written instructions.

D. Cover and mask adjoining surfaces not receiving coated roofing to prevent overspray or spillage affecting other construction. Temporarily close off roof drains, removing roof-drain plugs when not doing coated roofing work or when rain is forecast.

E. Prime substrate as recommended in writing by coated roofing manufacturer.

F. Fill, cover, or tape joints and cracks in substrate that exceed a width of 1/4 inch (6 mm). Remove dust and dirt from narrower joints and cracks before applying coating.

G. Install vapor retarder in accordance with coated roofing manufacturer's written instructions.

## 3.3 APPLICATION OF COATING

A. Apply coating system by spray, roller, or other suitable application method in accordance with coating manufacturer's written instructions.

B. Apply base coat and one or more topcoats to obtain a uniform, seamless membrane free of blisters and pinholes. Apply each coat at right angles to preceding coat, using contrasting color tints for successive coats.

1. Apply topcoat(s) after removing dust, dirt, water, and other contaminants from base coat.
2. Silicone Coating: Apply coating system to a minimum dry film thickness recommended in writing by coated roofing manufacturer.

C. Height at Terminations: Apply coating system at wall terminations and other vertical surfaces to extend vertically beyond by a minimum of 4 inches (100 mm).

D. Mineral Granules: Apply mineral granules over wet topcoat, using pressure equipment at the rate of 0.5 lb/sq. ft. (2.45 kg/sq. m). Remove excess granules after topcoat has cured.

E. Sealant: Apply sealant to perimeter and other terminations where indicated on Drawings or required by coated roofing manufacturer.

F. Spread with care to prevent puncturing coating and to minimize damage to substrate.

## 3.4 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

1. Testing agency will identify, seal, and certify samples of materials taken from Project site, with Contractor present.
2. Testing agency will perform tests for product characteristics specified or cited in manufacturer's product data.

    a. Six slit-test samples will be required for each 10,000 sq. ft. (900 sq. m) of roof area to determine, as a minimum, the number of coats applied and dry film thickness of coating.

3. Testing agency will verify that surfaces slope to drain.

B. Coated roofing will be considered defective if it does not pass tests and inspections.

C. Refill cores, repair slits, and re-coat test areas.

D. Prepare test and inspection reports.

## 3.5 REPAIR AND RE-COATING

A. Correct deficiencies in, or remove coatings that do not comply with requirements; fill and repair substrates and reapply materials.

B. Repair and re-coat coated roofing in accordance with ASTM D6705/D6705M and manufacturer's written instructions.

## 3.6 CURING, PROTECTING, AND CLEANING

A. Cure coatings in accordance with manufacturer's written instructions, taking care to prevent contamination and damage during application stages and curing. Do not permit traffic on uncured coatings.

B. Protect coated roofing from damage and wear during remainder of construction period.

C. Clean overspray and spillage from adjacent construction using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

END OF SECTION 07 5700

# SECTION 07 6200 - SHEET METAL FLASHING AND TRIM

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes: Custom flashing and trim fabrications, made from the following:

1. Sheet metal materials.
2. Miscellaneous materials.

B. Related Requirements:

1. Section 06 1000 "Rough Carpentry" for wood nailers, curbs, and blocking.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review construction schedule. Verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
2. Review special roof details, roof drainage, roof-penetration flashing, equipment curbs, and condition of other construction that affect sheet metal flashing and trim.
3. Review requirements for insurance and certificates if applicable.
4. Review sheet metal flashing observation and repair procedures after flashing installation.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For sheet metal flashing and trim.

1. Plans, elevations, sections, and attachment details.
2. Fabrication and installation layouts, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled Work.
3. Identification of material, thickness, weight, and finish for each item and location in Project.
4. Details for forming, including profiles, shapes, seams, and dimensions.
5. Details for joining, supporting, and securing, including layout and spacing of fasteners, cleats, clips, and other attachments. Include pattern of seams.
6. Details of termination points and assemblies.
7. Details of roof-penetration flashing.
8. Details of edge conditions, including eaves, ridges, valleys, rakes, crickets, flashings, and counter flashings.
9. Details of special conditions.
10. Details of connections to adjoining work.

C. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type of exposed finish.

1.4 INFORMATIONAL SUBMITTALS

A. Certificates: For each type of coping and roof edge flashing that is ANSI/SPRI/FM 4435/ES-1 tested.

B. Product Test Reports: For each product, for tests performed by a qualified testing agency.

C. Qualification Statements: For fabricator.

D. Sample warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For sheet metal flashing and trim, and its accessories.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: Entity that employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage.

1. Store sheet metal flashing and trim materials away from uncured concrete and masonry.
2. Protect stored sheet metal flashing and trim from contact with water.

B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to extent necessary for period of sheet metal flashing and trim installation.

1.8 COORDINATION

A. Coordinate sheet metal flashing and trim layout and seams with sizes and locations of penetrations to be flashed, and joints and seams in adjacent materials.

B. Coordinate sheet metal flashing and trim installation with adjoining roofing and wall materials, joints, and seams to provide leakproof, secure, and noncorrosive installation.

1.9 WARRANTY

A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: 20 years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Sheet metal flashing and trim assemblies, including cleats, anchors, and fasteners, are to withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim are not to rattle, leak, or loosen, and are to remain watertight.

B. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's "The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control, and Reroofing" and SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.

C. SPRI Wind Design Standard: Manufacture and install roof edge flashings and copings tested in accordance with ANSI/SPRI/FM 4435/ES-1 and capable of resisting the following design pressure:

1. Design Pressure: As indicated on Drawings.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

### 2.2 SHEET METAL MATERIALS

A. Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.

B. Aluminum Sheet: Coil-coated sheet, ASTM B209/B209M, alloy as standard with manufacturer, with temper as required to suit forming operations and structural performance required.

1. Thickness: 0.032 inch (0.81 mm).
2. Surface: Smooth, flat.
3. Exposed Coil-Coated Finish:

a. Three-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

4. Color: As selected by Architect from manufacturer's full range.
5. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with minimum total dry film thickness of 0.5 mil (0.013 mm).

C. Stainless Steel Sheet: ASTM A240/A240M, Type 304, dead soft, fully annealed.

1. Nominal Thickness: 0.0156 inch (0.396 mm).
2. Surface: Smooth, flat.
3. Exterior Finish: ASTM A480/A480M, No. 2D (dull, cold rolled).

a. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.
b. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.

1) Run grain of directional finishes with long dimension of each piece.
2) When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

## 2.3 MISCELLANEOUS MATERIALS

A. Provide materials and types of fasteners, solder, protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.

B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item.

1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.

a. Exposed Fasteners: Heads matching color of sheet metal using plastic caps or factory-applied coating. Provide metal-backed EPDM or PVC sealing washers under heads of exposed fasteners bearing on weather side of metal.
b. Blind Fasteners: High-strength aluminum or stainless steel rivets suitable for metal being fastened.
c. Spikes and Ferrules: Same material as gutter; with spike with ferrule matching internal gutter width.

2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.
3. Fasteners for Stainless Steel Sheet: Series 300 stainless steel.

C. Solder:

1. For Zinc-Coated (Galvanized) Steel: ASTM B32, Grade Sn50, 50 percent tin and 50 percent lead or Grade Sn60, 60 percent tin and 40 percent lead.
2. For Stainless Steel: ASTM B32, Grade Sn60, with acid flux of type recommended by stainless steel sheet manufacturer.

D. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch (13 mm) wide and 1/8 inch (3 mm) thick.

E. Elastomeric Sealant: ASTM C920, elastomeric polyurethane polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.

F. Butyl Sealant: ASTM C1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for hooked-type expansion joints with limited movement.

G. Epoxy Seam Sealer: Two-part, noncorrosive, aluminum seam-cementing compound, recommended by aluminum manufacturer for exterior nonmoving joints, including riveted joints.

H. Bituminous Coating: Cold-applied asphalt emulsion in accordance with ASTM D1187/D1187M.

I. Asphalt Roofing Cement: ASTM D4586/D4586M, asbestos free, of consistency required for application.

## 2.4 FABRICATION, GENERAL

A. Custom fabricate sheet metal flashing and trim to comply with details indicated and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required.

1. Fabricate sheet metal flashing and trim in shop to greatest extent possible.
2. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
3. Verify shapes and dimensions of surfaces to be covered and obtain field measurements for accurate fit before shop fabrication.
4. Form sheet metal flashing and trim to fit substrates without excessive oil-canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
5. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.

B. Fabrication Tolerances:

1. Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 ft. (6 mm in 6 m) on slope and location lines indicated on Drawings and within 1/8-inch (3-mm) offset of adjoining faces and of alignment of matching profiles.

C. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.

1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with butyl sealant concealed within joints.
2. Use lapped expansion joints only where indicated on Drawings.

D. Sealant Joints: Where movable, nonexpansion-type joints are required, form metal in accordance with cited sheet metal standard to provide for proper installation of elastomeric sealant.

E. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.

F. Seams:

1. Fabricate nonmoving seams with flat-lock seams. Tin edges to be seamed, form seams, and solder.
2. Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use.
3. Seams for Aluminum: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer.

G. Do not use graphite pencils to mark metal surfaces.

## 2.5 WALL SHEET METAL FABRICATIONS

A. Through-Wall Flashing: Fabricate continuous flashings in minimum 96-inch- (2400-mm-) long, but not exceeding 12 ft. (3.6 m) long, sections, under copings, and at shelf angles. Fabricate discontinuous lintel, sill, and similar flashings to extend 6 inches (150 mm) beyond each side of wall openings; and form with 2-inch- (50-mm-) high, end dams. Fabricate from the following materials:

1. Stainless Steel: 0.0156 inch (0.396 mm) thick.

B. Opening Flashings in Frame Construction: Fabricate head, sill, jamb, and similar flashings to extend 4 inches (100 mm) beyond wall openings. Form head and sill flashing with 2-inch- (50-mm-) high, end dams. Fabricate from the following materials:

1. Galvanized Steel: 0.022 inch (0.56 mm) thick.
2. Aluminum: 0.032 inch (0.81 mm) thick.
3. Stainless Steel: 0.0156 inch (0.396 mm) thick.

## 2.6 MISCELLANEOUS SHEET METAL FABRICATIONS

A. Equipment Support Flashing: Fabricate from the following materials:

1. Galvanized Steel: 0.028 inch (0.71 mm) thick.
2. Stainless Steel: 0.0188 inch (0.477 mm) thick.

B. Overhead-Piping Safety Pans: Fabricate from the following materials:

1. Galvanized Steel: 0.040 inch (1.02 mm) thick.
2. Stainless Steel: 0.0250 inch (0.635 mm) thick.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrates, and other conditions affecting performance of the Work.

1. Verify compliance with requirements for installation tolerances of substrates.
2. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
3. Verify that air- or water-resistant barriers have been installed over substrate to prevent air infiltration or water penetration.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF SHEET METAL FLASHING AND TRIM, GENERAL

A. Install sheet metal flashing and trim to comply with details indicated and recommendations of cited sheet metal standard that apply to installation characteristics required unless otherwise indicated on Drawings.

1. Install fasteners, solder, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
2. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of solder, welds, and sealant.
3. Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement.
4. Install sheet metal flashing and trim to fit substrates and to result in watertight performance.
5. Install continuous cleats with fasteners spaced not more than 12 inches (300 mm) o.c.
6. Install exposed sheet metal flashing and trim with limited oil-canning, and free of buckling and tool marks.
7. Do not field cut sheet metal flashing and trim by torch.
8. Do not use graphite pencils to mark metal surfaces.

B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.

1. Coat concealed side of uncoated-aluminum and stainless steel sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
2. Underlayment: Where installing sheet metal flashing and trim directly on cementitious or wood substrates, install underlayment and cover with slip sheet.

C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim.

1. Space movement joints at maximum of 10 ft. (3 m) with no joints within 24 inches (600 mm) of corner or intersection.

2. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with sealant concealed within joints.
3. Use lapped expansion joints only where indicated on Drawings.

D. Fasteners: Use fastener sizes that penetrate substrate not less than recommended by fastener manufacturer to achieve maximum pull-out resistance.

E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.

F. Seal joints as required for watertight construction.

1. Use sealant-filled joints unless otherwise indicated.

a. Form joints to completely conceal sealant.
b. When ambient temperature at time of installation is between 40 and 70 deg F (4 and 21 deg C), set joint members for 50 percent movement each way.
c. Adjust setting proportionately for installation at higher ambient temperatures.

1) Do not install sealant-type joints at temperatures below 40 deg F (4 deg C).

2. Prepare joints and apply sealants to comply with requirements in Section 079200 "Joint Sealants."

G. Soldered Joints: Clean surfaces to be soldered, removing oils and foreign matter.

1. Pretin edges of sheets with solder to width of 1-1/2 inches (38 mm); however, reduce pretinning where pretinned surface would show in completed Work.
2. Do not use torches for soldering.
3. Heat surfaces to receive solder, and flow solder into joint.

a. Fill joint completely.
b. Completely remove flux and spatter from exposed surfaces.

4. Stainless Steel Soldering:

a. Tin edges of uncoated sheets, using solder for stainless steel and acid flux.
b. Promptly remove acid-flux residue from metal after tinning and soldering.
c. Comply with solder manufacturer's recommended methods for cleaning and neutralization.

5. Copper Soldering: Tin edges of uncoated sheets, using solder for copper.
6. Copper-Clad Stainless Steel Soldering: Tin edges of uncoated sheets, using solder for copper-clad stainless steel.

H. Rivets: Concealed rivet joints in uncoated aluminum where necessary for strength.

## 3.3 INSTALLATION OF WALL SHEET METAL FABRICATIONS

A. Install sheet metal wall flashing to intercept and exclude penetrating moisture in accordance with cited sheet metal standard unless otherwise indicated. Coordinate installation of wall flashing with installation of wall-opening components such as windows, doors, and louvers.

B. Opening Flashings in Frame Construction: Install continuous head, sill, jamb, and similar flashings to extend 4 inches (100 mm) beyond wall openings.

C. Reglets: Installation of reglets is specified in Section 033000 "Cast-in-Place Concrete." and Section 042000 "Unit Masonry."

## 3.4 INSTALLATION OF MISCELLANEOUS SHEET METAL FABRICATIONS

A. Equipment Support Flashing:

1. Coordinate installation of equipment support flashing with installation of roofing and equipment.
2. Weld or seal flashing with elastomeric sealant to equipment support member.

B. Overhead-Piping Safety Pans:

1. Suspend pans from structure above, independent of other overhead items such as equipment, piping, and conduit, unless otherwise indicated on Drawings.
2. Pipe and install drain line to plumbing waste or drainage system.

## 3.5 INSTALLATION TOLERANCES

A. Shim and align sheet metal flashing and trim within installed tolerance of 1/4 inch in 20 ft. (6 mm in 6 m) on slope and location lines indicated on Drawings and within 1/8-inch (3-mm) offset of adjoining faces and of alignment of matching profiles.

## 3.6 CLEANING

A. Clean and neutralize flux materials. Clean off excess solder.

B. Clean off excess sealants.

## 3.7 PROTECTION

A. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.

B. On completion of sheet metal flashing and trim installation, remove unused materials and clean finished surfaces as recommended in writing by sheet metal flashing and trim manufacturer.

C. Maintain sheet metal flashing and trim in clean condition during construction.

D. Replace sheet metal flashing and trim that have been damaged or that have deteriorated beyond successful repair by finish touchup or similar minor repair procedures, as determined by Architect.

END OF SECTION 07 6200

# SECTION 07 8413 - PENETRATION FIRESTOPPING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Penetrations in fire-resistance-rated walls.
2. Penetrations in horizontal assemblies.
3. Penetrations in smoke barriers.

B. Related Requirements:

1. Section 07 8443 "Joint Firestopping" for joints in or between fire-resistance-rated construction, at exterior curtainwall / floor intersections, and in smoke barriers.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.
2. Laboratory Test Reports: For sealants, indicating compliance with requirements for low-emitting materials.

C. Product Schedule: For each penetration firestopping system. Include location, illustration of firestopping system, and design designation of qualified testing and inspecting agency.

1. Engineering Judgments: Where Project conditions require modification to a qualified testing and inspecting agency's illustration for a particular penetration firestopping system, submit illustration, with modifications marked, approved by penetration firestopping system manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly. Obtain approval of authorities having jurisdiction prior to submittal.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Product Test Reports: For each penetration firestopping system, for tests performed by a qualified testing agency.

1.6 CLOSEOUT SUBMITTALS

A. Installer Certificates: From Installer indicating that penetration firestopping systems have been installed in compliance with requirements and manufacturer's written instructions.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: A firm that has been approved by UL and found to comply with its "Qualified Firestop Contractor Program Requirements."

1.8 PROJECT CONDITIONS

A. Environmental Limitations: Do not install penetration firestopping system when ambient or substrate temperatures are outside limits permitted by penetration firestopping system manufacturers or when substrates are wet because of rain, frost, condensation, or other causes.

B. Install and cure penetration firestopping materials per manufacturer's written instructions using natural means of ventilations or, where this is inadequate, forced-air circulation.

1.9 COORDINATION

A. Coordinate construction of openings and penetrating items to ensure that penetration firestopping systems can be installed according to specified firestopping system design.

B. Coordinate sizing of sleeves, openings, core-drilled holes, or cut openings to accommodate penetration firestopping systems.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics:

1. Perform penetration firestopping system tests by a qualified testing agency acceptable to authorities having jurisdiction.
2. Test per testing standards referenced in "Penetration Firestopping Systems" Article. Provide rated systems complying with the following requirements:

    a. Penetration firestopping systems shall bear classification marking of a qualified testing agency.

1) UL in its "Fire Resistance Directory."
2) Intertek Group in its "Directory of Listed Building Products."

2.2 PENETRATION FIRESTOPPING SYSTEMS

A. Penetration Firestopping Systems: Systems that resist spread of fire, passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated. Penetration firestopping systems shall be compatible with one another, with the substrates forming openings, and with penetrating items if any.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. 3M Fire Protection Products.
b. Hilti, Inc.
c. RectorSeal.
d. Specified Technologies, Inc.
e. Tremco, Inc.

B. Penetrations in Fire-Resistance-Rated Walls: Penetration firestopping systems with ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg (2.49 Pa).

1. F-Rating: Not less than the fire-resistance rating of constructions penetrated.

C. Penetrations in Horizontal Assemblies: Penetration firestopping systems with ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg (2.49 Pa).

1. F-Rating: At least one hour, but not less than the fire-resistance rating of constructions penetrated.
2. T-Rating: At least one hour, but not less than the fire-resistance rating of constructions penetrated except for floor penetrations within the cavity of a wall.
3. W-Rating: Provide penetration firestopping systems showing no evidence of water leakage when tested according to UL 1479.

D. Penetrations in Smoke Barriers: Penetration firestopping systems with ratings determined per UL 1479, based on testing at a positive pressure differential of 0.30-inch wg (74.7 Pa).

1. L-Rating: Not exceeding 5.0 cfm/sq. ft. (0.025 cu. m/s per sq. m) of penetration opening at and no more than 50-cfm (0.024-cu. m/s) cumulative total for any 100 sq. ft. (9.3 sq. m) at both ambient and elevated temperatures.

E. Exposed Penetration Firestopping Systems: Flame-spread and smoke-developed indexes of less than 25 and 450, respectively, per ASTM E 84.

1. Sealant shall have a VOC content of 250 g/L or less.

F. Accessories: Provide components for each penetration firestopping system that are needed to install fill materials and to maintain ratings required. Use only those components specified by penetration firestopping system manufacturer and approved by qualified testing and inspecting agency for conditions indicated.

1. Permanent forming/damming/backing materials.
2. Substrate primers.
3. Collars.
4. Steel sleeves.

## 2.3 FILL MATERIALS

A. Cast-in-Place Firestop Devices: Factory-assembled devices for use in cast-in-place concrete floors and consisting of an outer sleeve lined with an intumescent strip, a flange attached to one end of the sleeve for fastening to concrete formwork, and a neoprene gasket.

B. Latex Sealants: Single-component latex formulations that do not re-emulsify after cure during exposure to moisture.

C. Firestop Devices: Factory-assembled collars formed from galvanized steel and lined with intumescent material sized to fit specific diameter of penetrant.

D. Intumescent Composite Sheets: Rigid panels consisting of aluminum-foil-faced intumescent elastomeric sheet bonded to galvanized-steel sheet.

E. Intumescent Putties: Nonhardening, water-resistant, intumescent putties containing no solvents or inorganic fibers.

F. Intumescent Wrap Strips: Single-component intumescent elastomeric sheets with aluminum foil on one side.

G. Mortars: Prepackaged dry mixes consisting of a blend of inorganic binders, hydraulic cement, fillers and lightweight aggregate formulated for mixing with water at Project site to form a nonshrinking, homogeneous mortar.

H. Pillows/Bags: Reusable heat-expanding pillows/bags consisting of glass-fiber cloth cases filled with a combination of mineral-fiber, water-insoluble expansion agents, and fire-retardant additives. Where exposed, cover openings with steel-reinforcing wire mesh to protect pillows/bags from being easily removed.

I. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

J. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants.

## 2.4 MIXING

A. Penetration Firestopping Materials: For those products requiring mixing before application, comply with penetration firestopping system manufacturer's written instructions for accurate proportioning of materials, water (if required), type of mixing equipment, selection of mixer speeds, mixing containers, mixing time, and other items or procedures needed to produce products of uniform quality with optimum performance characteristics for application indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning: Before installing penetration firestopping systems, clean out openings immediately to comply with manufacturer's written instructions and with the following requirements:

1. Remove from surfaces of opening substrates and from penetrating items foreign materials that could interfere with adhesion of penetration firestopping materials.
2. Clean opening substrates and penetrating items to produce clean, sound surfaces capable of developing optimum bond with penetration firestopping materials. Remove loose particles remaining from cleaning operation.
3. Remove laitance and form-release agents from concrete.

B. Prime substrates where recommended in writing by manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

3.3 INSTALLATION

A. General: Install penetration firestopping systems to comply with manufacturer's written installation instructions and published drawings for products and applications.

B. Install forming materials and other accessories of types required to support fill materials during their application and in the position needed to produce cross-sectional shapes and depths required to achieve fire ratings.

1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not forming permanent components of firestopping.

C. Install fill materials by proven techniques to produce the following results:

1. Fill voids and cavities formed by openings, forming materials, accessories and penetrating items to achieve required fire-resistance ratings.
2. Apply materials so they contact and adhere to substrates formed by openings and penetrating items.
3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.4 IDENTIFICATION

A. Wall Identification: Permanently label walls containing penetration firestopping systems with the words "FIRE AND/OR SMOKE BARRIER - PROTECT ALL OPENINGS," using lettering not less than 3 inches (76 mm) high and with minimum 0.375-inch (9.5-mm) strokes.

1. Locate in accessible concealed floor, floor-ceiling, or attic space at 15 feet (4.57 m) from end of wall and at intervals not exceeding 30 feet (9.14 m).

B. Penetration Identification: Identify each penetration firestopping system with legible metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches (150 mm) of penetration firestopping system edge so labels are visible to anyone seeking to remove penetrating items or firestopping systems. Use mechanical fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:

1. The words "Warning - Penetration Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
2. Contractor's name, address, and phone number.
3. Designation of applicable testing and inspecting agency.
4. Date of installation.
5. Manufacturer's name.
6. Installer's name.

3.5 FIELD QUALITY CONTROL

A. Owner will engage a qualified testing agency to perform tests and inspections according to ASTM E 2174.

B. Where deficiencies are found or penetration firestopping system is damaged or removed because of testing, repair or replace penetration firestopping system to comply with requirements.

C. Proceed with enclosing penetration firestopping systems with other construction only after inspection reports are issued and installations comply with requirements.

3.6 CLEANING AND PROTECTION

A. Clean off excess fill materials adjacent to openings as the Work progresses by methods and with cleaning materials that are approved in writing by penetration firestopping system manufacturers and that do not damage materials in which openings occur.

B. Provide final protection and maintain conditions during and after installation that ensure that penetration firestopping systems are without damage or deterioration at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, immediately cut out and remove damaged or deteriorated penetration firestopping material and install new materials to produce systems complying with specified requirements.

END OF SECTION 07 8413

# SECTION 07 8443 - JOINT FIRESTOPPING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Joints in or between fire-resistance-rated constructions.
2. Joints at exterior wall/floor intersections.

B. Related Requirements:

1. Section 07 8413 "Penetration Firestopping" for penetrations in fire-resistance-rated walls, horizontal assemblies, and smoke barriers.
2. Section 09 2216 "Non-Structural Metal Framing" for firestop tracks for metal-framed partition heads.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.
2. Laboratory Test Reports: For sealants, indicating compliance with requirements for low-emitting materials.

C. Product Schedule: For each joint firestopping system. Include location, illustration of firestopping system, and design designation of qualified testing agency.

1. Engineering Judgments: Where Project conditions require modification to a qualified testing agency's illustration for a particular joint firestopping system condition, submit illustration, with modifications marked, approved by joint firestopping system manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Product Test Reports: For each joint firestopping system, for tests performed by a qualified testing agency.

1.6 CLOSEOUT SUBMITTALS

A. Installer Certificates: From Installer indicating that joint firestopping systems have been installed in compliance with requirements and manufacturer's written instructions.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: A firm that has been approved by UL and found to comply with UL's "Qualified Firestop Contractor Program Requirements."

1.8 PROJECT CONDITIONS

A. Environmental Limitations: Do not install joint firestopping systems when ambient or substrate temperatures are outside limits permitted by joint firestopping system manufacturers or when substrates are wet due to rain, frost, condensation, or other causes.

B. Install and cure joint firestopping systems per manufacturer's written instructions using natural means of ventilation or, where this is inadequate, forced-air circulation.

1.9 COORDINATION

A. Coordinate construction of joints to ensure that joint firestopping systems can be installed according to specified firestopping system design.

B. Coordinate sizing of joints to accommodate joint firestopping systems.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics:

1. Perform joint firestopping system tests by a qualified testing agency acceptable to authorities having jurisdiction.
2. Test per testing standards referenced in "Joint Firestopping Systems" Article. Provide rated systems complying with the following requirements:

a. Joint firestopping systems shall bear classification marking of a qualified testing agency.

1) UL in its "Fire Resistance Directory."
2) Intertek Group in its "Directory of Listed Building Products."

2.2 JOINT FIRESTOPPING SYSTEMS

A. Joint Firestopping Systems: Systems that resist spread of fire, passage of smoke and other gases, and maintain original fire-resistance rating of assemblies in or between which joint firestopping systems are installed. Joint firestopping systems shall accommodate building movements without impairing their ability to resist the passage of fire and hot gases.

B. Joints in or between Fire-Resistance-Rated Construction: Provide joint firestopping systems with ratings determined per ASTM E 1966 or UL 2079.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. 3M Fire Protection Products.
b. Hilti, Inc.
c. RectorSeal.
d. Roxul Inc.

2. Fire-Resistance Rating: Equal to or exceeding the fire-resistance rating of the wall, floor, or roof in or between which it is installed.

C. Joints at Exterior Wall/Floor Intersections: Provide joint firestopping systems with rating determined per ASTM E 2307.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. 3M Fire Protection Products.
b. Hilti, Inc.
c. RectorSeal.
d. Roxul Inc.

2. F-Rating: Equal to or exceeding the fire-resistance rating of the floor assembly.

D. Exposed Joint Firestopping Systems: Flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E 84.

1. Sealant shall have a VOC content of 250 g/L or less.

E. Accessories: Provide components of fire-resistive joint systems, including primers and forming materials, that are needed to install elastomeric fill materials and to maintain ratings required. Use only components specified by joint firestopping system manufacturer and approved by the qualified testing agency for conditions indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for joint configurations, substrates, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning: Before installing fire-resistive joint systems, clean joints immediately to comply with fire-resistive joint system manufacturer's written instructions and the following requirements:

1. Remove from surfaces of joint substrates foreign materials that could interfere with adhesion of elastomeric fill materials or compromise fire-resistive rating.
2. Clean joint substrates to produce clean, sound surfaces capable of developing optimum bond with elastomeric fill materials. Remove loose particles remaining from cleaning operation.
3. Remove laitance and form-release agents from concrete.

B. Prime substrates where recommended in writing by joint firestopping system manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

3.3 INSTALLATION

A. General: Install fire-resistive joint systems to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.

B. Install forming materials and other accessories of types required to support elastomeric fill materials during their application and in position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.

1. After installing elastomeric fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of fire-resistive joint system.

C. Install elastomeric fill materials for fire-resistive joint systems by proven techniques to produce the following results:

1. Elastomeric fill voids and cavities formed by joints and forming materials as required to achieve fire-resistance ratings indicated.
2. Apply elastomeric fill materials so they contact and adhere to substrates formed by joints.
3. For elastomeric fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.4 IDENTIFICATION

A. Joint Identification: Identify joint firestopping systems with legible metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches (150 mm) of joint edge so labels are visible to anyone seeking to remove or joint firestopping system. Use mechanical fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:

1. The words "Warning - Joint Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
2. Contractor's name, address, and phone number.
3. Designation of applicable testing agency.
4. Date of installation.
5. Manufacturer's name.
6. Installer's name.

3.5 FIELD QUALITY CONTROL

A. Inspecting Agency: Owner will engage a qualified testing agency to perform tests and inspections according to ASTM E 2393.

B. Where deficiencies are found or joint firestopping systems are damaged or removed due to testing, repair or replace joint firestopping systems so they comply with requirements.

C. Proceed with enclosing joint firestopping systems with other construction only after inspection reports are issued and installations comply with requirements.

3.6 CLEANING AND PROTECTION

A. Clean off excess elastomeric fill materials adjacent to joints as the Work progresses by methods and with cleaning materials that are approved in writing by joint firestopping system manufacturers and that do not damage materials in which joints occur.

B. Provide final protection and maintain conditions during and after installation that ensure joint firestopping systems are without damage or deterioration at time of Substantial Completion. If damage or deterioration occurs despite such protection, cut out and remove damaged or deteriorated fire-resistive joint systems immediately and install new materials to produce fire-resistive joint systems complying with specified requirements.

3.7 JOINT FIRESTOPPING SYSTEM SCHEDULE

A. Where UL-classified systems are indicated, they refer to system numbers in UL's "Fire Resistance Directory" under product Category XHBN or Category XHDG.

B. Perimeter Joint Firestopping Systems: Refer to “G” or “LS” series drawings.

END OF SECTION 07 8443

# SECTION 07 9200 - JOINT SEALANTS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Non-staining silicone joint sealants.
2. Urethane joint sealants.
3. Mildew-resistant joint sealants.
4. Butyl joint sealants.
5. Latex joint sealants.

B. Related Requirements:

1. Section 07 9219 "Acoustical Joint Sealants" for sealing joints in sound-rated construction.
2. Section 32 1373 "Concrete Paving Joint Sealants" for sealing joints in paved roads, parking lots, walkways, and curbing.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each joint-sealant product.

B. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.
2. Laboratory Test Reports: For sealants, indicating compliance with requirements for low-emitting materials.

C. Samples for Initial Selection: Manufacturer's color charts consisting of strips of cured sealants showing the full range of colors available for each product exposed to view.

D. Samples for Verification: For each kind and color of joint sealant required, provide Samples with joint sealants in 1/2-inch-wide joints formed between two 6-inch-long strips of material matching the appearance of exposed surfaces adjacent to joint sealants.

E. Joint-Sealant Schedule: Include the following information:

1. Joint-sealant application, joint location, and designation.
2. Joint-sealant manufacturer and product name.
3. Joint-sealant formulation.
4. Joint-sealant color.

## 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified testing agency.

B. Product Test Reports: For each kind of joint sealant, for tests performed by manufacturer and witnessed by a qualified testing agency.

C. Preconstruction Laboratory Test Schedule: Include the following information for each joint sealant and substrate material to be tested:

1. Joint-sealant location and designation.
2. Manufacturer and product name.
3. Type of substrate material.
4. Proposed test.
5. Number of samples required.

D. Preconstruction Laboratory Test Reports: From sealant manufacturer, indicating the following:

1. Materials forming joint substrates and joint-sealant backings have been tested for compatibility and adhesion with joint sealants.
2. Interpretation of test results and written recommendations for primers and substrate preparation are needed for adhesion.

E. Preconstruction Field-Adhesion-Test Reports: Indicate which sealants and joint preparation methods resulted in optimum adhesion to joint substrates based on testing specified in "Preconstruction Testing" Article.

F. Field-Adhesion-Test Reports: For each sealant application tested.

G. Sample Warranties: For special warranties.

## 1.6 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.

B. Product Testing: Test joint sealants using a qualified testing agency.

1. Testing Agency Qualifications: Qualified according to ASTM C 1021 to conduct the testing indicated.

## 1.7 PRECONSTRUCTION TESTING

A. Preconstruction Field-Adhesion Testing: Before installing sealants, field test their adhesion to Project joint substrates as follows:

1. Locate test joints where indicated on Project or, if not indicated, as directed by Architect.
2. Conduct field tests for each kind of sealant and joint substrate.
3. Notify Architect seven days in advance of dates and times when test joints will be erected.
4. Arrange for tests to take place with joint-sealant manufacturer's technical representative present.

    a. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1.1 in ASTM C 1193 or Method A, Tail Procedure, in ASTM C 1521.

        1) For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

5. Report whether sealant failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each kind of product and joint substrate. For sealants that fail adhesively, retest until satisfactory adhesion is obtained.
6. Evaluation of Preconstruction Field-Adhesion-Test Results: Sealants not evidencing adhesive failure from testing, in absence of other indications of noncompliance with requirements, will be considered satisfactory. Do not use sealants that fail to adhere to joint substrates during testing.

## 1.8 FIELD CONDITIONS

A. Do not proceed with installation of joint sealants under the following conditions:

1. When ambient and substrate temperature conditions are outside limits permitted by joint-sealant manufacturer or are below 40 deg F.
2. When joint substrates are wet.
3. Where joint widths are less than those allowed by joint-sealant manufacturer for applications indicated.
4. Where contaminants capable of interfering with adhesion have not yet been removed from joint substrates.

## 1.9 WARRANTY

A. Special Installer's Warranty: Installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

B. Special Manufacturer's Warranty: Manufacturer agrees to furnish joint sealants to repair or replace those joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Five years from date of Substantial Completion.

C. Special warranties specified in this article exclude deterioration or failure of joint sealants from the following:

1. Movement of the structure caused by stresses on the sealant exceeding sealant manufacturer's written specifications for sealant elongation and compression.
2. Disintegration of joint substrates from causes exceeding design specifications.
3. Mechanical damage caused by individuals, tools, or other outside agents.
4. Changes in sealant appearance caused by accumulation of dirt or other atmospheric contaminants.

## PART 2 - PRODUCTS

### 2.1 JOINT SEALANTS, GENERAL

A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer, based on testing and field experience.

B. VOC Content: Sealants and sealant primers shall comply with the following:

1. Architectural sealants shall have a VOC content of 250 g/L or less.
2. Sealants and sealant primers for nonporous substrates shall have a VOC content of 250 g/L or less.
3. Sealants and sealant primers for porous substrates shall have a VOC content of 775 g/L or less.

C. Colors of Exposed Joint Sealants: As selected by Architect from manufacturer's full range.

### 2.2 NONSTAINING SILICONE JOINT SEALANTS

A. Non-staining Joint Sealants: No staining of substrates when tested according to ASTM C 1248.

B. Silicone, Non-staining, S, NS, 100/50, NT: Non-staining, single-component, nonsag, plus 100 percent and minus 50 percent movement capability, nontraffic-use, neutral-curing silicone joint sealant; ASTM C 920, Type S, Grade NS, Class 100/50, Use NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Pecora Corporation.
b. Sika Corporation; Joint Sealants.
c. Tremco Incorporated.

### 2.3 URETHANE JOINT SEALANTS

A. Urethane, S, NS, 25, NT: Single-component, nonsag, nontraffic-use, plus 25 percent and minus 25 percent movement capability, urethane joint sealant; ASTM C 920, Type S, Grade NS, Class 25, Use NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. BASF Corporation; Construction Systems.
b. Pecora Corporation.
c. Sika Corporation; Joint Sealants.
d. Tremco Incorporated.

B. Urethane, S, P, 25, T, NT: Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C 920, Type S, Grade P, Class 25, Uses T and NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. BASF Corporation; Construction Systems.
b. Pecora Corporation.
c. Sika Corporation; Joint Sealants.
d. Tremco Incorporated.

C. Urethane, M, P, 50, T, NT: Multicomponent, pourable, plus 50 percent and minus 50 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C 920, Type M, Grade P, Class 50, Uses T and NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. BASF Corporation; Construction Systems.
b. Pecora Corporation.
c. Sika Corporation; Joint Sealants.
d. Tremco Incorporated.

## 2.4 MILDEW-RESISTANT JOINT SEALANTS

A. Mildew-Resistant Joint Sealants: Formulated for prolonged exposure to humidity with fungicide to prevent mold and mildew growth.

B. Silicone, Mildew Resistant, Acid Curing, S, NS, 25, NT: Mildew-resistant, single-component, nonsag, plus 25 percent and minus 25 percent movement capability, nontraffic-use, acid-curing silicone joint sealant; ASTM C 920, Type S, Grade NS, Class 25, Use NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Dow Corning Corporation.
b. GE Construction Sealants; Momentive Performance Materials Inc.
c. May National Associates, Inc.; a subsidiary of Sika Corporation.
d. Pecora Corporation.
e. Tremco Incorporated.

2.5 BUTYL JOINT SEALANTS

A. Butyl-Rubber-Based Joint Sealants: ASTM C 1311.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Bostik, Inc.
b. Pecora Corporation.
c. Tremco Incorporated.

2.6 LATEX JOINT SEALANTS

A. Acrylic Latex: Acrylic latex or siliconized acrylic latex, ASTM C 834, Type OP, Grade NF.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Franklin International.
b. May National Associates, Inc.; a subsidiary of Sika Corporation.
c. Pecora Corporation.
d. Sherwin-Williams Company (The).
e. Tremco Incorporated.

2.7 JOINT-SEALANT BACKING

A. Sealant Backing Material, General: Nonstaining; compatible with joint substrates, sealants, primers, and other joint fillers; and approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Alcot Plastics Ltd.
b. BASF Corporation; Construction Systems.
c. Construction Foam Products; a division of Nomaco, Inc.

B. Cylindrical Sealant Backings: ASTM C 1330, Type C (closed-cell material with a surface skin), and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.

C. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint. Provide self-adhesive tape where applicable.

2.8 MISCELLANEOUS MATERIALS

A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.

B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants to joint substrates.

C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions and the following requirements:

1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
2. Clean, porous joint substrate surfaces by brushing, grinding, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining after cleaning operations above by vacuuming or blowing out joints with oil-free compressed air. Porous joint substrates include the following:

    a. Concrete.
    b. Masonry.
    c. Unglazed surfaces of ceramic tile.

3. Remove laitance and form-release agents from concrete.
4. Clean nonporous joint substrate surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants. Nonporous joint substrates include the following:

    a. Metal.
    b. Glass.
    c. Porcelain enamel.

d. Glazed surfaces of ceramic tile.

B. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.

C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

## 3.3 INSTALLATION OF JOINT SEALANTS

A. General: Comply with joint-sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.

B. Sealant Installation Standard: Comply with recommendations in ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.

C. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.

1. Do not leave gaps between ends of sealant backings.
2. Do not stretch, twist, puncture, or tear sealant backings.
3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.

D. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and backs of joints.

E. Install sealants using proven techniques that comply with the following and at the same time backings are installed:

1. Place sealants so they directly contact and fully wet joint substrates.
2. Completely fill recesses in each joint configuration.
3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.

F. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified in subparagraphs below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.

1. Remove excess sealant from surfaces adjacent to joints.
2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
3. Provide concave joint profile per Figure 8A in ASTM C 1193 unless otherwise indicated.
4. Provide flush joint profile at locations indicated on Drawing according to Figure 8B in ASTM C 1193.
5. Provide recessed joint configuration of recess depth and at locations indicated on Drawings according to Figure 8C in ASTM C 1193.

    a. Use masking tape to protect surfaces adjacent to recessed tooled joints.

3.4 FIELD QUALITY CONTROL

A. Field-Adhesion Testing: Field test joint-sealant adhesion to joint substrates as follows:

1. Extent of Testing: Test completed, and cured sealant joints as follows:

    a. Perform 10 tests for the first 1000 feet of joint length for each kind of sealant and joint substrate.
    b. Perform one test for each 1000 feet of joint length thereafter or one test per each floor per elevation.

2. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C 1193 or Method A, Tail Procedure, in ASTM C 1521.

    a. For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

3. Inspect tested joints and report on the following:

    a. Whether sealants filled joint cavities and are free of voids.
    b. Whether sealant dimensions and configurations comply with specified requirements.
    c. Whether sealants in joints connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each kind of product and joint substrate. Compare these results to determine if adhesion complies with sealant manufacturer's field-adhesion hand-pull test criteria.

4. Record test results in a field-adhesion-test log. Include dates when sealants were installed, names of persons who installed sealants, test dates, test locations, whether joints were primed, adhesion results and percent elongations, sealant material, sealant configuration, and sealant dimensions.
5. Repair sealants pulled from test area by applying new sealants following same procedures used originally to seal joints. Ensure that original sealant surfaces are clean and that new sealant contacts original sealant.

B. Evaluation of Field-Adhesion-Test Results: Sealants not evidencing adhesive failure from testing or noncompliance with other indicated requirements will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements.

3.5 CLEANING

A. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

3.6 PROTECTION

A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out, remove, and repair damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from original work.

3.7 JOINT-SEALANT SCHEDULE

A. Joint-Sealant Application: Exterior joints in horizontal traffic surfaces.

1. Joint Locations:

a. Control and expansion joints in brick pavers.
b. Isolation and contraction joints in cast-in-place concrete slabs.
c. Joints between plant-precast architectural concrete paving units.
d. Joints in stone paving units, including steps.
e. Tile control and expansion joints.
f. Joints between different materials listed above.
g. Other joints as indicated on Drawings.

2. Joint Sealant: Urethane, M, P, 50, T, NT.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

B. Joint-Sealant Application: Exterior joints in vertical surfaces and horizontal nontraffic surfaces.

1. Joint Locations:

a. Construction joints in cast-in-place concrete.
b. Joints between plant-precast architectural concrete units.
c. Control and expansion joints in unit masonry.
d. Joints in dimension stone cladding.
e. Joints in glass unit masonry assemblies.
f. Joints in exterior insulation and finish systems.
g. Joints between metal panels.
h. Joints between different materials listed above.
i. Perimeter joints between materials listed above and frames of doors, windows and louvers.
j. Control and expansion joints in ceilings and other overhead surfaces.
k. Other joints as indicated on Drawings.

2. Joint Sealant: Silicone, nonstaining, S, NS, 100/50, NT.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

C. Joint-Sealant Application: Interior joints in horizontal traffic surfaces.

1. Joint Locations:

   a. Isolation joints in cast-in-place concrete slabs.
   b. Control and expansion joints in tile flooring.
   c. Other joints as indicated on Drawings.

2. Joint Sealant: Urethane, S, P, 25, T, NT.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

D. Joint-Sealant Application: Interior joints in vertical surfaces and horizontal nontraffic surfaces.

1. Joint Locations:

   a. Control and expansion joints on exposed interior surfaces of exterior walls.
   b. Tile control and expansion joints.
   c. Vertical joints on exposed surfaces of unit masonry, concrete, walls and partitions.
   d. Joints on underside of plant-precast structural concrete beams and planks.
   e. Other joints as indicated on Drawings.

2. Joint Sealant: Urethane, S, NS, 25, NT.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

E. Joint-Sealant Application: Interior joints in vertical surfaces and horizontal nontraffic surfaces not subject to significant movement.

1. Joint Locations:

   a. Control joints on exposed interior surfaces of exterior walls.
   b. Perimeter joints between interior wall surfaces and frames of interior doors, windows and elevator entrances.
   c. Other joints as indicated on Drawings.

2. Joint Sealant: Acrylic latex.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

F. Joint-Sealant Application: Mildew-resistant interior joints in vertical surfaces and horizontal nontraffic surfaces.

1. Joint Locations:

   a. Joints between plumbing fixtures and adjoining walls, floors, and counters.
   b. Tile control and expansion joints where indicated.
   c. Other joints as indicated on Drawings.

2. Joint Sealant: Silicone, mildew resistant, acid curing, S, NS, 25, NT.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

G. Joint-Sealant Application: Concealed mastics.

1. Joint Locations:

   a. Aluminum thresholds.
   b. Sill plates.
   c. Other joints as indicated on Drawings.

2. Joint Sealant: Butyl-rubber based.
3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.

END OF SECTION 07 9200

# SECTION 07 9219 - ACOUSTICAL JOINT SEALANTS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes: Acoustical joint sealants.

B. Related Requirements:

1. Section 07 9200 "Joint Sealants" for elastomeric, latex, and butyl-rubber-based joint sealants for non-acoustical applications.
2. Section 09 2900 “Gypsum Board” for acoustical sealants at gypsum boards assemblies.

### 1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples for Initial Selection: Manufacturer's color charts consisting of strips of cured sealants, showing full range of available colors for each product exposed to view.

C. Samples for Verification: For each type and color of acoustical joint sealant required.

1. Size: 1/2-inch- (13-mm-) wide sealant joints formed between two 6-inch- (150-mm-) long strips of material matching the appearance of exposed surfaces adjacent to joint sealants.

D. Acoustical Joint-Sealant Schedule: Include the following information:

1. Joint-sealant application, joint location, and designation.
2. Joint-sealant manufacturer and product name.
3. Joint-sealant formulation.
4. Joint-sealant color.

### 1.3 INFORMATIONAL SUBMITTALS

A. Test and Evaluation Reports:

1. Product Test Reports: For each type of acoustical joint sealant, for tests performed by qualified testing agency.

B. Sample warranties.

### 1.4 CLOSEOUT SUBMITTALS

A. Warranty Documentation:

1. Manufacturers' special warranties.

2. Installer's special warranties.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained between 40 and 95 deg F (4 and 35 deg C).

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.6 FIELD CONDITIONS

A. Apply coatings only when temperature of surfaces to be coated and ambient air temperatures are between 60 and 90 deg F (16 and 32 deg C).

1.7 WARRANTY

A. Installer's Special Warranty: Installer agrees to repair or replace acoustical joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

B. Manufacturer's Special Warranty: Manufacturer agrees to furnish acoustical joint sealants to repair or replace those joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 ACOUSTICAL JOINT SEALANTS

A. Acoustical joint-sealant products that effectively reduce airborne sound transmission through perimeter joints and openings in building construction, as demonstrated by testing representative assemblies in accordance with ASTM E90.

B. Acoustical Sealant for Exposed and Concealed Joints: Manufacturer's standard nonsag, paintable, nonstaining latex acoustical sealant complying with ASTM C834.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Grabber Construction Products, Inc.
b. OSI Sealants; Henkel Corporation.
c. Pecora Corporation.
d. Specified Technologies Inc.

2. Colors of Exposed Acoustical Joint Sealants: As selected by Architect from manufacturer's full range of colors.

2.2 MISCELLANEOUS MATERIALS

A. Primer: Material recommended by acoustical joint-sealant manufacturer where required for adhesion of sealant to joint substrates.

B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants to joint substrates.

C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine joints indicated to receive acoustical joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning of Joints: Clean out joints immediately before installing acoustical joint sealants to comply with joint-sealant manufacturer's written instructions.

B. Joint Priming: Prime joint substrates where recommended by acoustical joint-sealant manufacturer. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.

C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 INSTALLATION OF ACOUSTICAL JOINT SEALANTS

A. Comply with acoustical joint-sealant manufacturer's written installation instructions unless more stringent requirements apply.

B. STC-Rated Assemblies: Seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical joint sealant. Install acoustical joint sealants at both faces of partitions, at perimeters, and through penetrations. Comply with ASTM C919, ASTM C1193, and manufacturer's written instructions for closing off sound-

flanking paths around or through assemblies, including sealing partitions to underside of floor slabs above acoustical ceilings.

C. Acoustical Ceiling Areas: Apply acoustical joint sealant at perimeter edge moldings of acoustical ceiling areas in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.

3.4 CLEANING

A. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of acoustical joint sealants and of products in which joints occur.

3.5 PROTECTION

A. Protect acoustical joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out, remove, and repair damaged or deteriorated acoustical joint sealants immediately so installations with repaired areas are indistinguishable from original work.

END OF SECTION 07 9219

# SECTION 07 9222 – SECURITY JOINT SEALANTS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. This Section includes the following:

1. Security Joint Sealant.
2. Security Gap Filler.

B. Related Sections: The following Sections contain requirements that relate to this Section:

1. Division 3 Section “Cast-in-place Concrete”
2. Division 4 Section “Unit Masonry Assemblies”
3. Division 8 Section “Steel Doors and Frames”
4. Division 22 for Plumbing
5. Division 23 for Mechanical
6. Division 26 for Electrical
7. Division 28 for Security Electronics

### 1.3 SUBMITTALS

A. Product Data: For each joint sealant product indicated.

1. Certification by joint sealant manufacturer that sealants plus the primers and cleaners required for sealant installation comply with local regulations controlling use of volatile organic compounds.

B. Certificates by Manufacturer: That products supplied complies with performance requirements specified and are suitable for the use indicated.

### 1.4 QUALITY ASSURANCE

A. Installer Qualifications: Engage an experienced Installer who has completed joint sealant applications similar in material, design, and extent to that indicated for Project that have resulted in construction with a record of successful in-service performance.

B. Mockups: Install sealant in mockups of assemblies specified in other Sections that are indicated to receive joint sealants specified in this Section. Use materials and installation methods specified in this Section.

1. Provide mockup of security joint sealant at top and bottom of wall base for review and

approval by Architect and Owner.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials to Project site in original unopened containers or bundles with labels indicating manufacturer, product name and designation, color, expiration period for use, pot life, curing time, and mixing instructions for multicomponent materials.

B. Store and handle materials in compliance with manufacturer's recommendations to prevent their deterioration or damage due to moisture, high or low temperatures, contaminants, or other causes.

1.6 PROJECT CONDITIONS

A. Environmental Conditions: Do not proceed with installation of joint sealants under the following conditions:

1. When ambient and substrate temperature conditions are outside the limits permitted by joint sealant manufacturer.
2. When ambient and substrate temperature conditions are outside the limits permitted by joint sealant manufacturer or below 40 deg F (4.4 deg C).
3. When joint substrates are wet.

B. Joint Width Conditions: Do not proceed with installation of joint sealants where joint widths are less than allowed by joint sealant manufacturer for application indicated.

C. Joint Substrate Conditions: Do not proceed with installation of joint sealants until contaminants capable of interfering with their adhesion are removed from joint substrates.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Compatibility: Provide joint sealants, joint fillers, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.

B. Colors: Provide color of exposed joint sealants to comply with the following and as indicated on Finish Plans:

1. Provide selections made by Architect from manufacturer's full range of standard colors for products of type indicated (PRC-1 thru PRC-5).
2. Provide custom color PRC-6 to match Sherwin Williams “Privilege Green” or approved equivalent.

2.2 SECURITY JOINT SEALANT (PRC)

A. Product/Manufacturer:

1. SikaFlex –2c NS TG / Sika Corp.
2. Dyna Flex / Pecora Corp.
3. Sonolastic Ultra / BASF

2.3 SECURITY EPOXY RESIN GAP FILLER (LOW-MOD GEL)

A. Product/Manufacturer:

1. Sika-Dur 23 / Sika Corp.
2. DynaPoxy / Pecora Corp.
3. Concresive Paste LPL / BASF

B. Materials

1. Epoxy resin adhesive binder:

a. Component “A” shall be a modified epoxy resin of the epichlorohydrin bisphenol A type containing suitable viscosity control agents. It shall not contain butyl glycidyl ether.
b. Component “B” shall be a blend of aliphatic amines containing suitable viscosity control agents, pigments and accelerators.
c. The ratio of Component “A”: Component “B” shall be 1:1 by volume.
d. The material shall not contain asbestos.

C. Performance Criteria

1. Properties of the mixed epoxy resin adhesive:

a. Pot Life: 45-65 minutes
b. Tack-Free Time to Touch (20 mil thickness): 1.5-2.5 hours
c. Consistency (½ in. thick) : non-sag
d. Color: gray

2. Properties of the mixed epoxy resin adhesive:

a. Compressive Properties (ASTM D-695) at 28 days
b. Compressive Strength: 4400 psi min
c. Modulus of Elasticity: 1.5 x 105 psi min
d. Tensile Properties (ASTM D-638) at 14 days

1) Tensile Strength: 1725 psi min
2) Elongation at Break: 5.5% min
3) Modulus of Elasticity: 2.7 x 105 psi min

e. Flexural Properties (ASTM D-790) at 14 days

1) Flexural Strength (Modulus of Rupture): 4100 psi min.
2) Tangent Modulus of Elasticity in Bending: 4.0 x 105 psi min

f. Shear Strength (ASTM D-732) at 14 days: 2600 psi min
g. Total Water Absorption (ASTM D-570) at 7 days: .5% max. (2 hour boil)
h. Bond Strength (ASTM C-882) Hardened Concrete to Hardened Concrete

1) 2 day (Dry cure): 2200 psi min
2) 14 day (moist cure): 1500 psi min

i. Deflection Temperature (ASTM D-648) at 14 days: 87F min (fiber stress loading = 66 psi)
j. The epoxy resin adhesive shall conform to ASTM C-881 and AASHTO M 235-90.
k. The epoxy resin adhesive binder shall be approved by the United States Department of Agriculture.

3. Properties of the epoxy resin mortar (epoxy resin/aggregate* =1/1 by loose volume):

a. Compressive Properties (ASTM D-695) at 28 days

1) Compressive Strength: 6100 psi min.
2) Modulus of Elasticity: L 3.4 x 105 psi min.

b. Tensile Properties (ASTM D-638) at 14 days

1) Tensile Strength: 2050 psi min.
2) Elongation at Break: 0.85% min.
3) Modulus of Elasticity: 5.2 x 105 psi min.

c. Flexural Properties (ASTM D-790) at 14 days

1) Flexural Strength (Modulus of Rupture): 3300 psi min.
2) Tangent Modulus of Elasticity in Bending: 5.8 x 105 psi min.

d. Shear Strength (ASTM D-732) at days: 2800 psi min.
e. Aggregate used shall conform to ASTM C-190.

## 2.4 JOINT SEALANT BACKING

A. General: Provide sealant backings of material that are nonstaining; are compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.

B. Cylindrical Sealant Backings: ASTM C 1330, Type C (closed-cell material with a surface skin) as approved in writing by joint-sealant manufacturer for joint application indicated, and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.

## 2.5 MISCELLANEOUS MATERIALS

A. Primer: Material recommended by joint sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint sealant-substrate tests and field tests.

B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming in any way joint substrates and adjacent nonporous surfaces, and formulated to promote optimum adhesion of sealants with joint substrates.

C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

## PART 3 - EXECUTION

### 3.1 LOCATIONS

A. Provide security joint sealants and security gap filler work as required under the General, Mechanical, and Electrical Sections. See security matrix at the end of this Section.

### 3.2 EXAMINATION

A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting joint sealant performance. Do not proceed with installation of joint sealants until unsatisfactory conditions have been corrected.

### 3.3 PREPARATION

A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with recommendations of joint sealant manufacturer and the following requirements:

1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
2. Clean concrete, masonry, unglazed surfaces of ceramic tile, and similar porous joint substrate surfaces by brushing, grinding, blast cleaning, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining from above cleaning operations by vacuuming or blowing out joints with oil-free compressed air.
3. Remove laitance and form release agents from concrete.
4. Clean metal, glass, porcelain enamel, glazed surfaces of ceramic tile, and other nonporous surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants.

B. Joint Priming: Prime joint substrates where indicated or where recommended by joint sealant manufacturer based on preconstruction joint sealant-substrate tests or prior experience. Apply primer to comply with joint sealant manufacturer's recommendations. Confine primers to areas of joint sealant bond; do not allow spillage or migration onto adjoining surfaces.

C. Masking Tape: Use masking tape where required to prevent contact of sealant with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

### 3.4 INSTALLATION OF JOINT SEALANTS

A. Comply with joint sealant manufacturer's printed installation instructions applicable to products

and applications indicated, except where more stringent requirements apply.

3.5 CLEANING

A. Clean off excess sealants or sealant smears adjacent to joints as work progresses by methods and with cleaning materials approved by manufacturers of joint sealants and of products in which joints occur.

3.6 PROTECTION

A. Protect joint sealants during and after curing period from contact with contaminating substances or from damage resulting from construction operations or other causes so that they are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated joint sealants immediately so that and installations with repaired areas are indistinguishable from original work.

3.7 SEALANT MATRIX

A. In the Detention/Secure Areas provide Security Gap Filler in the following locations:

1. Security Metal Ceiling Assemblies
2. Detention Furnishings and Equipment
3. Mechanical Grilles/ Diffusers
4. Security Light Fixtures
5. Exposed Conduit /Raceways
6. Security Plumbing Fixtures

B. In the Detention/Secure Areas provide Security Joint Sealant as indicated and per Finish Plans:

1. Architectural Door Frames
2. Switch/Outlet
3. Duress Plates/buttons
4. Wall, Flooring and Ceiling Joints
5. Plumbing Fixtures
6. Intercom Call Stations
7. Toilet Accessories (Mirrors, Grab Bars, Towel Hooks).

END OF SECTION 07 9222

# SECTION 08 1113 - HOLLOW METAL DOORS AND FRAMES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Standard and custom hollow metal doors and frames.
2. Steel sidelight, borrowed lite and transom frames.
3. Light frames and glazing installed in hollow metal doors.

B. Related Sections:

1. Division 04 Section "Unit Masonry" for embedding anchors for hollow metal work into masonry construction.
2. Division 08 Section "Glazing" for glass view panels in hollow metal doors.
3. Division 08 Section "Door Hardware".
4. Division 09 Sections "Exterior Painting" and "Interior Painting" for field painting hollow metal doors and frames.
5. Division 28 Section "Access Control" for access controls at hollow metal doors & frames.

C. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.

1. ANSI/SDI A250.8 - Recommended Specifications for Standard Steel Doors and Frames.
2. ANSI/SDI A250.4 - Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames, Frames Anchors and Hardware Reinforcing.
3. ANSI/SDI A250.6 - Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames.
4. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.
5. ANSI/SDI A250.11 - Recommended Erection Instructions for Steel Frames.
6. ASTM A1008 - Standard Specification for Steel Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability.
7. ASTM A653 - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
8. ASTM A924 - Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process.
9. ASTM C 1363 - Standard Test Method for Thermal Performance of Building Assemblies by Means of a Hot Box Apparatus.
10. ANSI/BHMA A156.115 - Hardware Preparation in Steel Doors and Frames.
11. ANSI/SDI 122 - Installation and Troubleshooting Guide for Standard Steel Doors and Frames.
12. ANSI/NFPA 80 - Standard for Fire Doors and Fire Windows; National Fire Protection Association.

13. ANSI/NFPA 105: Standard for the Installation of Smoke Door Assemblies.
14. NFPA 252 - Standard Methods of Fire Tests of Door Assemblies; National Fire Protection Association.
15. UL 10C - Positive Pressure Fire Tests of Door Assemblies.
16. UL 1784 - Standard for Air Leakage Tests of Door Assemblies.

1.3 SUBMITTALS

A. Product Data: For each type of product indicated. Include construction details, material descriptions, core descriptions, hardware reinforcements, profiles, anchors, fire-resistance rating, and finishes.

B. Door hardware supplier is to furnish templates, template reference number and/or physical hardware to the steel door and frame supplier in order to prepare the doors and frames to receive the finish hardware items.

C. Shop Drawings: Include the following:

1. Details of doors, including vertical and horizontal edge details and metal thicknesses.
2. Frame details for each frame type, including dimensioned profiles and metal thicknesses.
3. Locations of reinforcement and preparations for hardware.
4. Details of anchorages, joints, field splices, and connections.
5. Details of accessories.
6. Details of conduit and preparations for power, signal, and control systems.

D. Samples for Verification:

1. Samples are only required by request of the architect and for manufacturers that are not current members of the Steel Door Institute.

1.4 QUALITY ASSURANCE

A. Source Limitations: Obtain hollow metal doors and frames through one source from a single manufacturer wherever possible.

B. Quality Standard: In addition to requirements specified, furnish SDI-Certified manufacturer products that comply with ANSI/SDI A250.8, latest edition, "Recommended Specifications for Standard Steel Doors and Frames".

C. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to UL10C (neutral pressure at 40” above sill) or UL 10C.

1. Temperature-Rise Limit: Where indicated and at vertical exit enclosures (stairwell openings) and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F (250 deg C) above ambient after 30 minutes of standard fire-test exposure.

2. Smoke Control Door Assemblies: Comply with NFPA 105.

1. Smoke "S" Label: Doors to bear “S” label and include smoke and draft control gasketing applied to frame and on meeting stiles of pair doors.

D. Fire-Rated, Borrowed-Light Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled, by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on testing according to NFPA 257. Provide labeled glazing material.

E. Pre-Submittal Conference: Conduct conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representatives of Supplier, Installer, and Contractor to review proper methods and procedures for installing hollow metal doors and frames and to verify installation of electrical knockout boxes and conduit at frames with electrified or access control hardware.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver hollow metal work palletized, wrapped, or crated to provide protection during transit and Project site storage. Do not use non-vented plastic.

B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.

C. Store hollow metal work under cover at Project site. Place in stacks of five units maximum in a vertical position with heads up, spaced by blocking, on minimum 4-inch high wood blocking. Do not store in a manner that traps excess humidity.

1. Provide minimum 1/4-inch space between each stacked door to permit air circulation. Door and frames to be stacked in a vertical upright position.

1.6 PROJECT CONDITIONS

A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

1.7 COORDINATION

A. Coordinate installation of anchorages for hollow metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

1.8 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.

B. Warranty includes installation and finishing that may be required due to repair or replacement of defective doors.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, provide steel doors and frames from a SDI Certified manufacturer:

1. CECO Door Products (C).
2. Curries Company (CU).

2.2 MATERIALS

A. Cold-Rolled Steel Sheet: ASTM A 1008/A 1008M, Commercial Steel (CS), Type B; suitable for exposed applications.

B. Metallic-Coated Steel Sheet: ASTM A 653/A 653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

C. Frame Anchors: ASTM A 653/A 653M, Commercial Steel (CS), Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

2.3 HOLLOW METAL DOORS

A. General: Provide 1-3/4 inch doors of design indicated, not less than thickness indicated; fabricated with smooth surfaces, without visible joints or seams on exposed faces unless otherwise indicated. Comply with ANSI/SDI A250.8 and ANSI/NAAMM HMMA 867.

B. Exterior Doors (Energy Efficient): Face sheets fabricated of commercial quality hot-dipped zinc coated steel that complies with ASTM A924 A60. Provide doors complying with requirements indicated below by referencing ANSI/SDI A250.8 for level and model, ANSI/SDI A250.4 for physical performance level, and HMMA 867 for door construction.

1. Design: Flush panel.
2. Core Construction: Foamed in place polyurethane and steel stiffened laminated core with no stiffener face welds, in compliance with HMMA 867 “Laminated Core”.

    1. Provide 18-gauge steel stiffeners at 6 inches on-center internally welded at 5" on-center to integral core assembly, foamed in place polyurethane core chemically bonded to all interior surfaces. No stiffener face welding is permitted.
    2. Thermal properties to rate at a fully operable minimum U-Factor 0.37 and R-Value 2.7, including insulated door, thermal-break frame and threshold.
    3. Kerf Type Frames: Thermal properties to rate at a fully operable minimum U-Factor 0.38 and R-Value 2.6, including insulated door, kerf type frame, and threshold.

3. Level/Model: Level 3 and Physical Performance Level A (Extra Heavy Duty), Minimum 16 gauge thick steel, Model 2, unless otherwise noted.

    1. Maximum-Duty Doors and Frames: Level 4 and Physical Performance Level A, minimum 14-gauge thick steel, Model 2, where indicated.

4. Vertical Edges: Vertical edges to be mechanically interlocked with hairline seam. Beveled Lock Edge, 1/8 inch in 2 inches.
5. Top and Bottom Edges: Reinforce tops and bottoms of doors with a continuous steel channel not less than 16 gauge, extending the full width of the door and welded to the face sheet. Doors with an inverted top channel to include a steel closure channel, screw attached, with the web of the channel flush with the face sheets of the door. Plastic or composite channel fillers are not acceptable.
6. Hinge Reinforcement: Minimum 7 gauge (3/16") plate 1-1/4" x 9".

7. Hardware Reinforcements: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.

C. Interior Doors (Energy Efficient): Face sheets fabricated of commercial quality cold rolled steel that complies with ASTM A366 or 620. Provide doors complying with requirements indicated below by referencing ANSI/SDI A250.8 for level and model and ANSI/SDI A250.4 for physical performance level:

1. Design: Flush panel.
2. Core Construction: Steel stiffened laminated core with fiberglass filler with no stiffener face welds, in compliance with HMMA 867 “Laminated Core”.
    1. Provide 22-gauge steel-stiffeners at 6 inches on-center internally welded at 5" on-center to integral core assembly, no stiffener face welding is permitted.
    2. Acoustical sound transmission rating shall be no less than STC 38 complying with ASTM E 90 and must be visible on factory applied labels.
3. Level/Model: Level 2 and Physical Performance Level B (Heavy Duty), Minimum 18 gauge thick steel, Model 2.
4. Vertical Edges: Vertical edges-to be mechanically interlocked with hairline seam. Beveled Lock Edge, 1/8 inch in 2 inches.
5. Top and Bottom Edges: Reinforce tops and bottoms of doors with a continuous steel channel not less than 16 gauge, extending the full width of the door and welded to the face sheet. Doors with an inverted top channel to include a steel closure channel, screw attached, with the web of the channel flush with the face sheets of the door. Plastic or composite channel fillers are not acceptable.
6. Hinge Reinforcement: Minimum 7 gauge (3/16") plate 1-1/4" x 9".
7. Hardware Reinforcements: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.

D. Manufacturers Basis of Design:

1. CECO Door Products (C) Energy Efficient - Trio-E Series.
2. Curries Company (CU) - Energy Efficient - 777 Trio-E Series.

## 2.4 HOLLOW METAL FRAMES

A. General: Comply with ANSI/SDI A250.8 and with details indicated for type and profile.

B. Exterior Frames: Fabricated of hot-dipped zinc coated steel that complies with ASTM A 653/A 653M, Coating Designation A60.

1. Fabricate frames with mitered or coped corners. Profile as indicated on drawings.
2. Frames: Minimum 14 gauge (0.067-inch -1.7-mm) thick steel sheet.
3. Provide thermally broken frames for exterior door assemblies.

C. Manufacturers Basis of Design:

1. CECO Door Products (C) – SU SR Series.
2. Curries Company (CU) – M Series.

D. Interior Frames: Fabricated from cold-rolled steel sheet that complies with ASTM A 1008/A 1008M.

1. Fabricate frames with mitered or coped corners. Profile as indicated on drawings.
2. Frames: Minimum 16 gauge thick steel sheet.

E. Manufacturers Basis of Design:

1. CECO Door Products (C) - SU Series.
2. Curries Company (CU) - M Series.

F. Fire rated frames: Fabricate frames in accordance with NFPA 80, listed and labeled by a qualified testing agency, for fire-protection ratings indicated.

G. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 Table 4 with reinforcement plates from same material as frames.

2.5 FRAME ANCHORS

A. Jamb Anchors:

1. Masonry Type: Adjustable strap-and-stirrup or T-shaped anchors to suit frame size, formed from A60 metallic coated material, not less than 0.042 inch thick, with corrugated or perforated straps not less than 2 inches wide by 10 inches long; or wire anchors not less than 0.177 inch thick.
2. Stud Wall Type: Designed to engage stud and not less than 0.042 inch thick.
3. Compression Type for Drywall Slip-on (Knock-Down) Frames: Adjustable compression anchors.

B. Floor Anchors: Floor anchors to be provided at each jamb, formed from A60 metallic coated material, not less than 0.042 inches thick.

C. Mortar Guards: Formed from same material as frames, not less than 0.016 inches thick.

2.6 LOUVERS

A. Metal Louvers: Unless otherwise indicated provide louvers to meet the following requirements.

1. Blade Type: Vision proof inverted V or inverted Y.
2. Metal and Finish: Galvanized steel, 0.040 inch thick, factory primed for paint finish with baked enamel or powder coated finish. Match pre-finished door paint color where applicable.

2.7 LIGHT OPENINGS AND GLAZING

A. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with butted or mitered hairline joints at fabricator’s shop. Fixed and removable stops to allow multiple glazed lites each to be removed independently. Coordinate frame rabbet widths between fixed and removable stops with the type of glazing and installation indicated.

B. Moldings for Glazed Lites in Doors and Loose Stops for Glazed Lites in Frames: Minimum 20 gauge thick, fabricated from same material as door face sheet in which they are installed.

C. Fixed Frame Moldings: Formed integral with hollow metal frames, a minimum of 5/8 inch high unless otherwise indicated. Provide fixed frame moldings and stops on outside of exterior and on secure side of interior doors and frames.

D. Preformed Metal Frames for Light Openings: Manufacturer's standard frame formed of 0.048-inch-thick, cold rolled steel sheet; with baked enamel or powder coated finish; and approved for use in doors of fire protection rating indicated. Match pre-finished door paint color where applicable.

2.8 ACCESSORIES

A. Mullions and Transom Bars: Join to adjacent members by welding or rigid mechanical anchors.

B. Grout Guards: Formed from same material as frames, not less than 0.016 inches thick.

2.9 FABRICATION

A. Fabricate hollow metal work to be rigid and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. When shipping limitations so dictate, frames for large openings are to be fabricated in sections for splicing or splining in the field by others.

B. Tolerances: Fabricate hollow metal work to tolerances indicated in ANSI/SDI A250.8.

C. Hollow Metal Doors:

1. Exterior Doors: Provide optional weep-hole openings in bottom of exterior doors to permit moisture to escape where specified.
2. Glazed Lites: Factory cut openings in doors with applied trim or kits to fit. Factory install glazing where indicated.
3. Astragals: Provide overlapping astragals as noted in door hardware sets in Division 08 Section "Door Hardware" on one leaf of pairs of doors where required by NFPA 80 for fire-performance rating or where indicated. Extend minimum 3/4 inch beyond edge of door on which astragal is mounted.
4. Continuous Hinge Reinforcement: Provide welded continuous 12 gauge strap for continuous hinges specified in hardware sets in Division 08 Section "Door Hardware".
5. Electrical Raceways: Provide hollow metal doors to receive electrified hardware with concealed wiring harness and standardized Molex™ plug connectors on both ends to accommodate up to twelve wires. Coordinate connectors on end of the wiring harness to plug directly into the electrified hardware and the through-wire transfer hardware or wiring harness specified in hardware sets in Division 08 Section "Door Hardware" and Division 28 "Access Controls". Wire nut connections are not acceptable.

D. Hollow Metal Frames:

1. Shipping Limitations: Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.
2. Welded Frames: Weld flush face joints continuously; grind, fill, dress, and make smooth, flush, and invisible.

1. Welded frames are to be provided with two steel spreaders temporarily attached to the bottom of both jambs to serve as a brace during shipping and handling. Spreader bars are for bracing only and are not to be used to size the frame opening.

3. Sidelight and Transom Bar Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material as door frame. Fasten members at crossings and to jambs by butt welding.
4. High Frequency Hinge Reinforcement: Provide high frequency hinge reinforcements at door openings 48-inches and wider with mortise butt type hinges at top hinge locations.
5. Continuous Hinge Reinforcement: Provide welded continuous 12 gauge straps for continuous hinges specified in hardware sets in Division 08 Section "Door Hardware".
6. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated for removable stops, provide security screws at exterior locations.
7. Mortar Guards: Provide guard boxes at back of hardware mortises in frames at all hinges and strike preps regardless of grouting requirements.
8. Floor Anchors: Weld anchors to bottom of jambs and mullions with at least four spot welds per anchor.
9. Jamb Anchors: Provide number and spacing of anchors as follows:

    1. Masonry Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:

        1) Two anchors per jamb up to 60 inches high.
        2) Three anchors per jamb from 60 to 90 inches high.
        3) Four anchors per jamb from 90 to 120 inches high.
        4) Four anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 120 inches high.

    2. Stud Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:

        1) Three anchors per jamb up to 60 inches high.
        2) Four anchors per jamb from 60 to 90 inches high.
        3) Five anchors per jamb from 90 to 96 inches high.
        4) Five anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 96 inches high.
        5) Two anchors per head for frames above 42 inches wide and mounted in metal stud partitions.

10. Door Silencers: Except on weather-stripped or gasketed doors drill stops to receive door silencers. Silencers to be supplied by frame manufacturer regardless if specified in Division 08 Section "Door Hardware".
11. Bituminous Coating: Where frames are fully grouted with an approved Portland Cement based grout or mortar, coat inside of frame throat with a water based bituminous or asphaltic emulsion coating to a minimum thickness of 3 mils DFT, tested in accordance with UL 10C and applied to the frame under a 3rd party independent follow-up service procedure.

E. Hardware Preparation: Factory prepare hollow metal work to receive template mortised hardware; include cutouts, reinforcement, mortising, drilling, and tapping according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware."

1. Locate hardware as indicated, or if not indicated, according to ANSI/SDI A250.8.
2. Reinforce doors and frames to receive non-template, mortised and surface mounted door hardware.
3. Comply with applicable requirements in ANSI/SDI A250.6 and ANSI/DHI A115 Series specifications for preparation of hollow metal work for hardware.
4. Coordinate locations of conduit and wiring boxes for electrical connections with Division 26 Sections.

2.10 STEEL FINISHES

A. Prime Finishes: Doors and frames to be cleaned, and chemically treated to insure maximum finish paint adhesion. Surfaces of the door and frame exposed to view to receive a factory applied coat of rust inhibiting shop primer.

1. Shop Primer: Manufacturer's standard, fast-curing, lead and chromate free primer complying with ANSI/SDI A250.10 acceptance criteria; recommended by primer manufacturer for substrate; and compatible with substrate and field-applied coatings.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. General Contractor to verify the accuracy of dimensions given to the steel door and frame manufacturer for existing openings or existing frames (strike height, hinge spacing, hinge back set, etc.).

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Remove welded in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces.

B. Prior to installation, adjust and securely brace welded hollow metal frames for square, level, twist, and plumb condition.

C. Tolerances shall comply with SDI-117 "Manufacturing Tolerances Standard Steel Doors and Frames."

D. Drill and tap doors and frames to receive non-template, mortised, and surface-mounted door hardware.

E. Verify tolerances against manufacturers installations instructions for tornado and hurricane storm shelter openings.

3.3 INSTALLATION

A. General: Install hollow metal work plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions.

B. Hollow Metal Frames: Install hollow metal frames of size and profile indicated. Comply with ANSI/SDI A250.11 and NFPA 80 at fire rated openings.

1. Set frames accurately in position, plumbed, leveled, aligned, and braced securely until permanent anchors are set. After wall construction is complete and frames properly set and secured, remove temporary braces, leaving surfaces smooth and undamaged. Shim as necessary to comply with installation tolerances.
2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with post-installed expansion anchors.
3. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with mortar.
4. Grout Requirements: Do not grout head of frames unless reinforcing has been installed in head of frame. Do not grout vertical or horizontal closed mullion members.

C. Hollow Metal Doors: Fit hollow metal doors accurately in frames, within clearances specified below. Shim as necessary.

1. Non-Fire-Rated Standard Steel Doors:

1. Jambs and Head: 1/8 inch plus or minus 1/16 inch.
2. Between Edges of Pairs of Doors: 1/8 inch plus or minus 1/16 inch.
3. Between Bottom of Door and Top of Threshold: Maximum 3/8 inch.
4. Between Bottom of Door and Top of Finish Floor (No Threshold): Maximum 3/4 inch.

2. Fire-Rated Doors: Install doors with clearances according to NFPA 80.

D. Field Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.

3.4 FIELD QUALITY CONTROL

A. Inspection Agency: Engage a qualified inspector to perform inspections and to furnish reports to Architect.

B. Inspections:

1. Fire-Rated Door Inspections: Inspect each fire-rated door in accordance with NFPA 80, Section 5.2.

C. Repair or remove and replace installations where inspections indicate that they do not comply with specified requirements.

D. Reinspect repaired or replaced installations to determine if replaced or repaired door assembly installations comply with specified requirements.

E. Prepare and submit separate inspection report for each fire-rated door assembly indicating compliance with each item listed in NFPA 80 and NFPA 101.

3.5 ADJUSTING AND CLEANING

A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or otherwise unacceptable.

B. Remove grout and other bonding material from hollow metal work immediately after installation.

C. Prime-Coat and Painted Finish Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat, or painted finishes, and apply touchup of compatible air drying, rust-inhibitive primer, zinc rich primer (exterior and galvanized openings) or finish paint.

END OF SECTION 08 1113

# SECTION 08 1416 - FLUSH WOOD DOORS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Solid-core five-ply flush wood veneer-faced doors for transparent finish.
2. Light frames and louvers.

B. Related Requirements:

1. Section 08 8000 "Glazing" for glass view panels in flush wood doors.

### 1.2 ACTION SUBMITTALS

A. Product Data:

1. Solid-core five-ply flush wood veneer-faced doors for transparent finish.
2. Light frames and louvers.

B. Product Data Submittals: For each product, including the following:

1. Door core materials and construction.
2. Door edge construction
3. Door face type and characteristics.
4. Door louvers.
5. Door trim for openings.
6. Factory-machining criteria.
7. Factory-finishing specifications.

C. Shop Drawings: Indicate location, size, and hand of each door; elevation of each type of door; construction details not covered in Product Data; and the following:

1. Door schedule indicating door location, type, size, fire protection rating, and swing.
2. Door elevations, dimension and locations of hardware, lite and louver cutouts, and glazing thicknesses.
3. Details of frame for each frame type, including dimensions and profile.
4. Details of electrical raceway and preparation for electrified hardware, access control systems, and security systems.
5. Dimensions and locations of blocking for hardware attachment.
6. Dimensions and locations of mortises and holes for hardware.
7. Clearances and undercuts.
8. Requirements for veneer matching.
9. Doors to be factory finished and application requirements.

D. Samples for Initial Selection: For factory-finished doors.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Data: For door inspector.

1. Submit copy of DHI's Fire and Egress Door Assembly Inspector (FDAI) certificate.

B. Field quality-control reports.

C. Sample Warranty: For special warranty.

1.4 CLOSEOUT SUBMITTALS

A. Special warranties.

B. Record Documents: For fire-rated doors, list of door numbers and applicable room name and number to which door accesses.

1.5 QUALITY ASSURANCE

A. Fire-Rated Door Inspector Qualifications: Inspector for field quality-control inspections of fire-rated door assemblies complies with qualifications set forth in NFPA 80, Section 5.2.3.1 and the following:

1. DHI's Fire and Egress Door Assembly Inspector (FDAI) certification.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Comply with requirements of referenced standard and manufacturer's written instructions.

B. Package doors individually in cardboard cartons and wrap bundles of doors in plastic sheeting.

C. Mark each door on top and bottom rail with opening number used on Shop Drawings.

1.7 FIELD CONDITIONS

A. Environmental Limitations:

1. Do not deliver or install doors until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, and HVAC system is operating and maintaining temperature and relative humidity at levels designed for building occupants for the remainder of construction period.
2. Do not deliver or install doors until building is enclosed and weathertight, wet work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F (16 and 32 deg C) and relative humidity between 43 and 70 percent during remainder of construction period.

1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Delamination of veneer.
b. Warping (bow, cup, or twist) more than 1/4 inch (6.4 mm) in a 42-by-84-inch (1067-by-2134-mm) section.
c. Telegraphing of core construction in face veneers exceeding 0.01 inch in a 3-inch (0.25 mm in a 76.2-mm) span.

2. Warranty also includes installation and finishing that may be required due to repair or replacement of defective doors.

3. Warranty Period for Solid-Core Interior Doors: Life of installation.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain flush wood doors from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Fire-Rated Wood Door and Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction, for fire-protection ratings on Drawings, based on testing at positive pressure in accordance with UL 10C or NFPA 252.

2.3 FLUSH WOOD DOORS AND FRAMES, GENERAL

A. Quality Standard: In addition to requirements specified, comply with AWI/AWMAC/WI's "Architectural Woodwork Standards."

2.4 FLUSH WOOD VENEER-FACED DOORS FOR TRANSPARENT FINISH

A. Interior Doors, Solid-Core Five-Ply Veneer-Faced (WD):

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Masonite Architectural.
b. Oregon Door.
c. Oshkosh Door Company.
d. VT Industries, Inc.

2. Performance Grade: ANSI/WDMA I.S. 1A Extra Heavy Duty.

3. ANSI/WDMA I.S. 1A Quality Grade: Premium.
4. Architectural Woodwork Standards Quality Grade: Premium.
5. Faces: Single-plywood veneer not less than 1/50 inch (0.508 mm) thick.

    a. Species: Select white maple.
    b. Cut: Plain sliced (flat sliced).
    c. Match between Veneer Leaves: Book match.
    d. Assembly of Veneer Leaves on Door Faces: Center-balance match.
    e. Pair and Set Match: Provide for doors hung in same opening or separated only by mullions.

6. Exposed Vertical and Top Edges: Same species as faces - Architectural Woodwork Standards edge Type A.

    a. Fire-Rated Single Doors: Provide edge construction with intumescent seals concealed by outer stile. Comply with specified requirements for exposed vertical edges.
    b. Fire-Rated Pairs of Doors:

        1) Provide formed-steel edges and astragals with intumescent seals.

            a) Finish steel edges and astragals with baked enamel same color as doors.

    c. Mineral-Core Doors: At hinge stiles, provide laminated-edge construction with improved screw-holding capability and split resistance. Comply with specified requirements for exposed edges.

        1) Screw-Holding Capability: Minimum of 400 lbf (1780 N) in accordance with WDMA T.M. 10.

7. Core for Non-Fire-Rated Doors:

    a. ANSI A208.1, Grade LD-1 particleboard.

        1) Blocking: Provide wood blocking in particleboard-core doors as needed to eliminate through-bolting hardware.

        2) Provide doors with glued-wood-stave or WDMA I.S. 10 structural-composite-lumber cores instead of particleboard cores for doors scheduled to receive exit devices in Section 087100 "Door Hardware."

    b. Glued wood stave.
    c. WDMA I.S. 10 structural composite lumber.

        1) Screw Withdrawal, Door Face: Minimum 400 lbf (1780 N).

    d. Either glued wood stave or WDMA I.S. 10 structural composite lumber.

8. Core for Fire-Rated Doors: As required to achieve fire-protection rating indicated on Drawings.

a. Blocking for Mineral-Core Doors: Provide composite blocking with improved screw-holding capability approved for use in doors of fire-protection ratings indicated on Drawings as needed to eliminate through-bolting hardware.

9. Construction: Five plies, hot-pressed bonded (vertical and horizontal edging is bonded to core), with entire unit abrasive planed before veneering.

## 2.5 LIGHT FRAMES AND LOUVERS

A. Wood Beads for Light Openings in Wood Doors: Provide manufacturer's standard wood beads unless otherwise indicated.

1. Wood Species: Same species as door faces.
2. Profile: Flush rectangular beads.
3. At wood-core doors with 20-minute fire-protection ratings, provide wood beads and metal glazing clips approved for such use.

B. Wood-Veneered Beads for Light Openings in Fire-Rated Doors: Manufacturer's standard wood-veneered noncombustible beads matching veneer species of door faces and approved for use in doors of fire-protection rating indicated on Drawings. Include concealed metal glazing clips where required for opening size and fire-protection rating indicated.

## 2.6 FABRICATION

A. Factory fit doors to suit frame-opening sizes indicated.

1. Comply with clearance requirements of referenced quality standard for fitting unless otherwise indicated.
2. Comply with NFPA 80 requirements for fire-rated doors.

B. Factory machine doors for hardware that is not surface applied.

1. Locate hardware to comply with DHI-WDHS-3.
2. Comply with final hardware schedules, door frame Shop Drawings, ANSI/BHMA-156.115-W, and hardware templates.
3. Coordinate with hardware mortises in metal frames, to verify dimensions and alignment before factory machining.
4. For doors scheduled to receive electrified locksets, provide factory-installed raceway and wiring to accommodate specified hardware.
5. Metal Astragals: Factory machine astragals and formed-steel edges for hardware for pairs of fire-rated doors.

C. Openings: Factory cut and trim openings through doors.

1. Light Openings: Trim openings with moldings of material and profile indicated.
2. Glazing: Factory install glazing in doors indicated to be factory finished. Comply with applicable requirements in Section 088000 "Glazing."

## 2.7 FACTORY FINISHING

A. Comply with referenced quality standard for factory finishing.

1. Complete fabrication, including fitting doors for openings and machining for hardware that is not surface applied, before finishing.
2. Finish faces, all four edges, edges of cutouts, and mortises.
3. Stains and fillers may be omitted on top and bottom edges, edges of cutouts, and mortises.

B. Factory finish doors.

C. Transparent Finish:

1. Architectural Woodwork Standards Grade: Premium.

a. System-9, UV Curable, Acrylated Epoxy, Polyester or Urethane.

2. Finish: As selected by Architect from manufacturer's full range.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine doors and installed door frames, with Installer present, before hanging doors.

1. Verify that installed frames comply with indicated requirements for type, size, location, and swing characteristics and have been installed with level heads and plumb jambs.
2. Reject doors with defects.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

A. Hardware: For installation, see Section 08 7100 "Door Hardware."

B. Install doors and frames to comply with manufacturer's written instructions and referenced quality standard, and as indicated.

C. Install frames level, plumb, true, and straight.

1. Shim as required with concealed shims. Install level and plumb to a tolerance of 1/8 inch in 96 inches (3.2 mm in 2400 mm).
2. Anchor frames to anchors or blocking built in or directly attached to substrates.

a. Secure with countersunk, concealed fasteners and blind nailing.
b. Use fine finishing nails or finishing screws for exposed fastening, countersunk and filled flush with woodwork.

1) For factory-finished items, use filler matching finish of items being installed.

3. Install fire-rated doors and frames in accordance with NFPA 80.

D. Factory-Fitted Doors: Align in frames for uniform clearance at each edge.

E. Factory-Finished Doors: Restore finish before installation if fitting or machining is required at Project site.

3.3 ADJUSTING

A. Operation: Rehang or replace doors that do not swing or operate freely.

B. Finished Doors: Replace doors that are damaged or that do not comply with requirements. Doors may be repaired or refinished if Work complies with requirements and shows no evidence of repair or refinishing.

END OF SECTION 08 1416

# SECTION 08 3113 - ACCESS DOORS AND FRAMES

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Access doors and frames.

B. Related Requirements:

1. Section 09 5113 “Acoustical Panels Ceilings” for Coordination Drawings.
2. Division 23 "Heating, Ventilating and Air Conditioning" for heating and air-conditioning duct access doors.

### 1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, fire ratings, material descriptions, dimensions of individual components and profiles, and finishes.

B. Product Schedule: For access doors and frames.

C. Coordination Drawings: Wall and ceiling plans, drawn to scale, on which the access doors are indicated and coordinated with other trades and with input from installers of all items involved requiring wall or ceiling access.

### 1.3 CLOSEOUT SUBMITTALS

A. Record Documents: For fire-rated doors, list of applicable room name and number in which access door is located.

## PART 2 - PRODUCTS

### 2.1 ACCESS DOORS AND FRAMES

A. Interior Flush Access Doors with Concealed Flanges:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ACUDOR Products, Inc.
b. Babcock-Davis.
c. JL Industries; Activar Construction Products Group, Inc.
d. Nystrom, Inc.

2. Description: Face of door flush with frame; with concealed flange for gypsum board installation and concealed hinge.
3. Locations: Walls.
4. Door Size: Minimum 15”x15”
5. Metallic-Coated Steel Sheet for Door: Nominal 0.064 inch (1.63 mm), 16 gage factory primed.
6. Frame Material: Same material and thickness as door.
7. Latch and Lock: Cylinder, key operated with interior release.

B. Interior Flush GFRG Access Doors with Concealed Flanges:

1. Manufacturers: Subject to compliance with requirements, provide products by the following or approved equivalent:

   a. Armstrong World Industries – CASTWORKS Access panels.

2. Description: Face of concealed-hinge door flush with frame, with concealed flange for gypsum board installation.
3. Locations: Ceilings.
4. Door Size: Minimum 24”x24”.
5. Door Type Concealed-hinge, radius corner.
6. Door and Frame Material: Unpainted glass-fiber-reinforced gypsum, with frames reinforced for hardware and fastenings.
7. Latch and Lock: Cam latch, key operated.

C. Exterior Flush Access Doors:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

   a. Babcock-Davis.
   b. JL Industries, Inc.; a division of the Activar Construction Products Group.
   c. Nystrom, Inc.

2. Description: Weatherproof assembly, with face of door fit flush with frame and with exposed frame. Include extruded door gaskets and minimum 2-inch-thick (50-mm-thick) fiberglass insulation.
3. Locations: Walls.
4. Door Size: Minimum 15”x15”.
5. Stainless-Steel Sheet for Door: Nominal 0.062 inch (1.59 mm), 16 gage, No. 4 finish.
6. Frame Material: Same material, thickness, and finish as door.
7. Latch and Lock: Cam latch operated by handle, with keyed lock in handle.

## 2.2 MATERIALS

A. Steel Plates, Shapes, and Bars: ASTM A36/A36M.

B. Steel Sheet: Uncoated or electrolytic zinc coated, ASTM A879/A879M, with cold-rolled steel sheet substrate complying with ASTM A1008/A1008M, Commercial Steel (CS), exposed.

C. Metallic-Coated Steel Sheet: ASTM A653/A653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

D. Stainless Steel Plate, Sheet, and Strip: ASTM A240/A240M or ASTM A666, Type 304. Remove tool and die marks and stretch lines, or blend into finish.

E. Aluminum Extrusions: ASTM B221 (ASTM B221M), Alloy 6063.

F. Aluminum Sheet: ASTM B209 (ASTM B209M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

G. Frame Anchors: Same material as door face.

H. Inserts, Bolts, and Anchor Fasteners: Hot-dip galvanized steel according to ASTM A153/A153M or ASTM F2329.

2.3 FABRICATION

A. General: Provide access door and frame assemblies manufactured as integral units ready for installation.

B. Metal Surfaces: For metal surfaces exposed to view in the completed Work, provide materials with smooth, flat surfaces without blemishes. Do not use materials with exposed pitting, seam marks, roller marks, rolled trade names, or roughness.

C. Doors and Frames: Grind exposed welds smooth and flush with adjacent surfaces. Furnish mounting holes, attachment devices and fasteners of type required to secure access doors to types of supports indicated.

1. For concealed flanges with drywall bead, provide edge trim for gypsum panels securely attached to perimeter of frames.
2. For concealed flanges with plaster bead for full-bed plaster applications, provide zinc-coated expanded-metal lath and exposed casing bead welded to perimeter of frames.

D. Recessed Access Doors: Form face of panel to provide recess for application of applied finish. Reinforce panel as required to prevent buckling. Provide access sleeves for each latch operator and install in holes cut through finish.

1. For recessed doors with plaster infill, provide self-furring expanded-metal lath attached to door panel.

E. Latch and Lock Hardware:

1. Quantity: Furnish number of latches and locks required to hold doors tightly closed.
2. Keys: Furnish two keys per lock and key all locks alike.
3. Mortise Cylinder Preparation: Where indicated, prepare door panel to accept cylinder specified in Section 08 7100 "Door Hardware."

F. Aluminum: After fabrication, apply manufacturer's standard protective coating on aluminum that will come in contact with concrete.

2.4 FINISHES

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

D. Painted Finishes: Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

1. Factory Primed: Apply manufacturer's standard, lead- and chromate-free, universal primer immediately after surface preparation and pretreatment.
2. Factory Finished: Apply manufacturer's standard baked-enamel or powder-coat finish immediately after cleaning and pretreating, with minimum dry-film thickness of 1 mil (0.025 mm) for topcoat.

a. Color: As selected by Architect from full range of industry colors.

E. Stainless Steel Finishes:

1. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.
2. Polished Finish: ASTM A480/A480M No. 4 finish. Grind and polish surfaces to produce uniform finish, free of cross scratches.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

A. Comply with manufacturer's written instructions for installing access doors and frames.

### 3.3 FIELD QUALITY CONTROL

A. Repair or remove and replace installations where inspections indicate that they do not comply with specified requirements.

B. Reinspect repaired or replaced installations to determine if replaced or repaired door assembly installations comply with specified requirements.

### 3.4 ADJUSTING

A. Adjust doors and hardware, after installation, for proper operation.

END OF SECTION 08 3113

# SECTION 08 4113 - ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Aluminum-framed storefront systems.
2. Aluminum-framed entrance door systems.
3. Aluminum break-metal to match aluminum storefront framing.

B. Related Requirements:

1. Section 088000 "Glazing" for storefront glazing and glazing sealants.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

B. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.
2. Product Data: For recycled content, indicating postconsumer and preconsumer recycled content and cost.
3. Product Certificates: For regional materials, indicating location of material manufacturer and point of extraction, harvest, or recovery for each raw material. Include distance to Project and cost for each regional material.

C. Shop Drawings: For aluminum-framed entrances and storefronts. Include plans, elevations, sections, full-size details, and attachments to other work.

1. Include details of provisions for assembly expansion and contraction and for draining moisture occurring within the assembly to the exterior.
2. Include full-size isometric details of each type of vertical-to-horizontal intersection of aluminum-framed entrances and storefronts, showing the following:

a. Joinery, including concealed welds.
b. Anchorage.
c. Expansion provisions.
d. Glazing.
e. Flashing and drainage.

3. Show connection to and continuity with adjacent thermal, weather, air, and vapor barriers.
4. Include point-to-point wiring diagrams showing the following:

a. Power requirements for each electrically operated door hardware.
b. Location and types of switches, signal device, conduit sizes, and number and size of wires.

D. Samples for Initial Selection: For units with factory-applied color finishes.

E. Entrance Door Hardware Schedule: Prepared by or under supervision of supplier, detailing fabrication and assembly of entrance door hardware, as well as procedures and diagrams. Coordinate final entrance door hardware schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of entrance door hardware.

F. Delegated Design Submittals: For aluminum-framed entrances and storefront systems, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

## 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer**.**

B. Energy Performance Certificates: For aluminum-framed entrances and storefronts, accessories, and components, from manufacturer.

1. Basis for Certification: NFRC-certified energy performance values for each aluminum-framed entrance and storefront.

C. Product Test Reports: For aluminum-framed entrances and storefronts, for tests performed by manufacturer and witnessed by a qualified testing agency.

D. Source quality-control reports.

E. Field quality-control reports.

F. Sample Warranties: For special warranties.

## 1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For aluminum-framed entrances and storefronts to include in maintenance manuals.

B. Maintenance Data for Structural Sealant: For structural-sealant-glazed storefront to include in maintenance manuals. Include ASTM C1401 recommendations for post-installation-phase quality-control program.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

B. Delegated Design Engineer Qualifications: A professional engineer who is legally qualified to practice in Commonwealth of Virginia where project is located and who is experienced in providing engineering services of the type indicated.

C. Product Options: Information on Drawings and in Specifications establishes requirements for aesthetic effects and performance characteristics of assemblies. Aesthetic effects are indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sightlines, to one another, and to adjoining construction.

1. Do not change intended aesthetic effects, as judged solely by Architect, except with Architect's approval. If changes are proposed, submit comprehensive explanatory data to Architect for review.

D. Structural-Sealant Glazing: Comply with ASTM C1401 for design and installation of storefront systems.

1.8 MOCKUPS

A. Mockups: Build mockups to verify selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for fabrication and installation.

1. Build mockup of typical wall area as shown on Drawings.
2. Testing shall be performed on mockups according to requirements in "Field Quality Control" Article.
3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 WARRANTY

A. Special Warranty: Installer agrees to repair or replace components of aluminum-framed entrances and storefronts that do not comply with requirements or that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures, including, but not limited to, excessive deflection.
b. Noise or vibration created by wind and thermal and structural movements.
c. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
d. Water penetration through fixed glazing and framing areas.
e. Failure of operating components.

2. Warranty Period: Two years from date of Substantial Completion.

B. Special Finish Warranty: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Hunter units when tested according to ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested according to ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Warranty Period: 20 years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

A. Source Limitations: Obtain all components of aluminum-framed entrance and storefront system, including framing spandrel panels and accessories, from single manufacturer.

### 2.2 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 4000 "Quality Requirements," to design aluminum-framed entrance and storefront systems.

B. General Performance: Comply with performance requirements specified, as determined by testing of aluminum-framed entrances and storefronts representing those indicated for this Project without failure due to defective manufacture, fabrication, installation, or other defects in construction.

1. Aluminum-framed entrances and storefronts shall withstand movements of supporting structure, including, but not limited to, twist, column shortening, long-term creep, and deflection from uniformly distributed and concentrated live loads.
2. Failure also includes the following:

a. Thermal stresses transferring to building structure.
b. Glass breakage.
c. Noise or vibration created by wind and thermal and structural movements.
d. Loosening or weakening of fasteners, attachments, and other components.
e. Failure of operating units.

C. Structural Loads:

1. Wind Loads: As indicated on Structural Drawings.
2. Other Design Loads: As indicated on Structural Drawings.

D. Deflection of Framing Members: At design wind pressure, as follows:

1. Deflection Normal to Wall Plane: Limited to 1/175 of clear span for spans of up to 13 feet 6 inches (4.1 m) and to 1/240 of clear span plus 1/4 inch (6.35 mm) for spans greater than 13 feet 6 inches (4.1 m) or an amount that restricts edge deflection of individual glazing lites to 3/4 inch (19.1 mm), whichever is less.

a. Operable Units: Provide a minimum 1/16-inch (1.6-mm) clearance between framing members and operable units.

2. Cantilever Deflection: Where framing members overhang an anchor point, as follows:

a. Perpendicular to Plane of Wall: No greater than 1/240 of clear span plus 1/4 inch (6.35 mm) for spans greater than 11 feet 8-1/4 inches (3.6 m) or 1/175 times span, for spans of less than 11 feet 8-1/4 inches (3.6 m).

E. Structural: Test according to ASTM E330/E330M as follows:

1. When tested at 150 percent of positive and negative wind-load design pressures, storefront assemblies, including entrance doors and anchorage, do not evidence material failures, structural distress, or permanent deformation of main framing members exceed 0.2 percent of span.
2. Test Durations: As required by design wind velocity, but not less than 10 seconds.

F. Air Infiltration:

1. Fixed Glazing and Framing Areas: Air leakage for the system of not more than 0.06 cfm/sq. ft. (0.30 L/s per sq. m) at a static-air-pressure differential of 6.24 lbf/sq. ft. (300 Pa) when tested in accordance with ASTM E283.

2. Entrance Doors: Air leakage of not more than 1.0 cfm/sq. ft. (5.08 L/s per sq. m) at a static-air-pressure differential of 1.57 lbf/sq. ft. (75 Pa).

G. Water Penetration under Static Pressure: Test according to ASTM E331 as follows:

1. No evidence of water penetration through fixed glazing and framing areas, including entrance doors, when tested according to a minimum static-air-pressure differential of 20 percent of positive wind-load design pressure, but not less than 6.24 lbf/sq. ft. (300 Pa).

H. Seismic Performance: Aluminum-framed entrances and storefronts shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. Seismic Drift Causing Glass Fallout: Complying with criteria for passing based on building occupancy type when tested according to AAMA 501.6 at design displacement and 1.5 times the design displacement.

I. Energy Performance: Certified and labeled by manufacturer for energy performance as follows:

1. Thermal Transmittance (U-factor):

a. Fixed Glazing and Framing Areas: U-factor for the system of not more than 0.41 Btu/sq. ft. x h x deg F (2.33 W/sq. m x K) as determined in accordance with NFRC 100.
b. Entrance Doors: U-factor of not more than 0.68 Btu/sq. ft. x h x deg F (3.86 W/sq. m x K) as determined in accordance with NFRC 100.

2. Solar Heat-Gain Coefficient (SHGC):

a. Fixed Glazing and Framing Areas: SHGC for the system of not more than 0.26 as determined in accordance with NFRC 200.
b. Entrance Doors: SHGC of not more than 0.22 as determined in accordance with NFRC 200.

J. Thermal Movements: Allow for thermal movements resulting from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

K. Structural-Sealant Joints:

1. Designed to carry gravity loads of glazing.

L. Structural Sealant: ASTM C1184. Capable of withstanding tensile and shear stresses imposed by structural-sealant-glazed, aluminum-framed entrances and storefronts without failing adhesively or cohesively. When tested for preconstruction adhesion and compatibility, cohesive failure of sealant shall occur before adhesive failure.

1. Adhesive failure occurs when sealant pulls away from substrate cleanly, leaving no sealant material behind.
2. Cohesive failure occurs when sealant breaks or tears within itself but does not separate from each substrate, because sealant-to-substrate bond strength exceeds sealant's internal strength.

## 2.3 ALUMINUM-FRAMED STOREFRONT SYSTEMS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Kawneer Versa Glaze 451/451T Framing System, or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. EFCO Corporation.
2. Kawneer Company, Inc.; Arconic Corporation.
3. OldCastle BuildingEnvelope (OBE).
4. Tubelite Inc.
5. YKK AP America Inc.

B. Framing Members: Manufacturer's extruded- or formed-aluminum framing members of thickness required and reinforced as required to support imposed loads.

1. Exterior Framing Construction: Thermally broken.
2. Interior Vestibule Framing Construction: Nonthermal.
3. Glazing System: Retained mechanically with gaskets on four sides.
4. Glazing Plane: Center.
5. Finish: Clear anodic finish.
6. Fabrication Method: Field-fabricated stick system.
7. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish indicated.
8. Steel Reinforcement: As required by manufacturer.
9. Sill Flashing: Manufacturers integrated flashing.

C. Backer Plates: Manufacturer's standard, continuous backer plates for framing members, if not integral, where framing abuts adjacent construction.

D. Brackets and Reinforcements: Manufacturer's standard high-strength aluminum with nonstaining, nonferrous shims for aligning system components.

2.4 ENTRANCE DOOR SYSTEMS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Kawneer 350T Insulpour, or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. EFCO Corporation.
2. Kawneer North America, an Arconic company.
3. Oldcastle BuildingEnvelope™.
4. YKK AP America Inc.
5. Tubelite Inc.

B. Entrance Doors: Manufacturer's standard glazed entrance doors for manual-swing or automatic operation.

1. Door Construction: 2-inch (50.8-mm) overall thickness, with minimum 0.188-inch- (4.8-mm-) thick, extruded-aluminum tubular rail and stile members. Mechanically fasten corners with reinforcing brackets that are deeply penetrated and fillet welded or that incorporate concealed tie rods.
2. Door Design: Medium stile; 3-1/2-inch (88.9-mm) nominal width.
3. Glazing Stops and Gaskets: Beveled, snap-on, extruded-aluminum stops and preformed gaskets.

    a. Provide nonremovable glazing stops on outside of door.

4. Finish: Match adjacent storefront framing finish.

2.5 ENTRANCE DOOR HARDWARE

A. Entrance Door Hardware: If hardware specified in this Section is also specified in Section 08 7100 "Door Hardware." then Section 087100 takes precedence over this section for all door Hardware and shall be coordinated by the Contractor. For hardware not specified in Section 08 7100 “Door Hardware” comply with the following requirements.

1. Sequence of Operation: Provide electrified door hardware function, sequence of operation, and interface with other building control systems indicated.

2. Opening-Force Requirements:

    a. Egress Doors: Not more than 15 lbf (67 N) to release the latch and not more than 30 lbf (133 N) to set the door in motion and not more than 15 lbf (67 N) to open the door to its minimum required width.
    b. Accessible Interior Doors: Not more than 5 lbf (22.2 N) to fully open door.

B. Cylinders: As specified in Section 087100 "Door Hardware."

C. Pivot Hinges: BHMA A156.4, Grade 1.

1. Offset-Pivot Hinges: Provide top, bottom, and intermediate offset pivots at each door leaf.

D. Continuous-Gear Hinges: BHMA A156.26.

E. Mortise Auxiliary Locks: BHMA A156.5, Grade 1.

F. Manual Flush Bolts: BHMA A156.16, Grade 1.

G. Automatic and Self-Latching Flush Bolts: BHMA A156.3, Grade 1.

H. Panic Exit Devices: BHMA A156.3, Grade 1, listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for panic protection, based on testing according to UL 305.

I. Strikes: Provide strike with black-plastic dust box for each latch or lock bolt; fabricated for aluminum framing.

J. Operating Trim: BHMA A156.6.

K. Removable Mullions: BHMA A156.3 extruded aluminum.

1. When used with panic exit devices, provide keyed removable mullions listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for panic protection, based on testing according to UL 305. Use only mullions that have been tested with exit devices to be used.

L. Closers: BHMA A156.4, Grade 1, with accessories required for a complete installation, sized as required by door size, exposure to weather, and anticipated frequency of use; adjustable to comply with field conditions and requirements for opening force.

M. Concealed Overhead Holders and Stops: BHMA A156.8, Grade 1.

N. Doorstops: BHMA A156.16, Grade 1, floor or wall mounted, as appropriate for door location indicated, with integral rubber bumper.

O. Weather Stripping: Manufacturer's standard replaceable components.

1. Compression Type: Made of ASTM D2000 molded neoprene or ASTM D2287 molded PVC.
2. Sliding Type: AAMA 701/702, made of wool, polypropylene, or nylon woven pile with nylon-fabric or aluminum-strip backing.

P. Weather Sweeps: Manufacturer's standard exterior-door bottom sweep with concealed fasteners on mounting strip.

Q. Thresholds: BHMA A156.21 raised thresholds beveled with a slope of not more than 1:2, with maximum height of 1/2 inch (12.7 mm).

R. Finger Guards: Manufacturer's standard collapsible neoprene or PVC gasket anchored to frame hinge-jamb at center-pivoted doors.

## 2.6 GLAZING

A. Glazing: Comply with Section 08 8000 "Glazing."

B. Glazing Gaskets: Manufacturer's standard sealed-corner pressure-glazing system of black, resilient elastomeric glazing gaskets, setting blocks, and shims or spacers.

C. Glazing Sealants: Comply with Section 088000 "Glazing."

1. Sealant shall have a VOC content of 250 g/L or less.

D. Structural Glazing Sealants: ASTM C1184 chemically curing silicone formulation that is compatible with system components with which it comes in contact; specifically formulated and tested for use as structural sealant and approved by structural-sealant manufacturer for use in storefront system indicated.

1. Color: As selected by Architect from manufacturer's full range of colors.

E. Weatherseal Sealants: ASTM C920 for Type S; Grade NS; Class 25; Uses NT, G, A, and O; chemically curing silicone formulation that is compatible with structural sealant and other system components with which it comes in contact; recommended by structural-sealant, weatherseal-sealant, and structural-sealant-glazed storefront manufacturers for this use.

1. Color: Match structural sealant.

## 2.7 MATERIALS

A. Sheet and Plate: ASTM B209 (ASTM B209M).

B. Extruded Bars, Rods, Profiles, and Tubes: ASTM B221 (ASTM B221M).

C. Extruded Structural Pipe and Tubes: ASTM B429/B429M.

D. Structural Profiles: ASTM B308/B308M.

E. Steel Reinforcement:

1. Structural Shapes, Plates, and Bars: ASTM A36/A36M.
2. Cold-Rolled Sheet and Strip: ASTM A1008/A1008M.
3. Hot-Rolled Sheet and Strip: ASTM A1011/A1011M.
4. Primer: Manufacturer's standard zinc-rich, corrosion-resistant primer complying with SSPC-PS Guide No. 12.00; applied immediately after surface preparation and pretreatment. Select surface preparation methods according to recommendations in SSPC-SP COM and prepare surfaces according to applicable SSPC standard.

F. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

G. Recycled Content of Aluminum Components: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

## 2.8 ACCESSORIES

A. Fasteners and Accessories: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding fasteners and accessories compatible with adjacent materials.

1. Use self-locking devices where fasteners are subject to loosening or turning out from thermal and structural movements, wind loads, or vibration.
2. Reinforce members as required to receive fastener threads.
3. Use exposed fasteners with countersunk Phillips screw heads, finished to match framing system.

B. Anchors: Three-way adjustable anchors with minimum adjustment of 1 inch (25.4 mm) that accommodate fabrication and installation tolerances in material and finish compatible with adjoining materials and recommended by manufacturer.

1. Concrete and Masonry Inserts: Hot-dip galvanized cast-iron, malleable-iron, or steel inserts complying with ASTM A123/A123M or ASTM A153/A153M requirements.

C. Concealed Flashing: Manufacturer's integrated standard corrosion-resistant, nonstaining, nonbleeding flashing compatible with adjacent materials.

D. Bituminous Paint: Cold-applied asphalt-mastic paint containing no asbestos, formulated for 30-mil (0.762-mm) thickness per coat.

E. Rigid PVC Filler.

## 2.9 FABRICATION

A. Form or extrude aluminum shapes before finishing.

B. Weld in concealed locations to greatest extent possible to minimize distortion or discoloration of finish. Remove weld spatter and welding oxides from exposed surfaces by descaling or grinding.

C. Fabricate components that, when assembled, have the following characteristics:

1. Profiles that are sharp, straight, and free of defects or deformations.
2. Accurately fitted joints with ends coped or mitered.
3. Physical and thermal isolation of glazing from framing members.
4. Accommodations for thermal and mechanical movements of glazing and framing to maintain required glazing edge clearances.
5. Provisions for field replacement of glazing from exterior.
6. Fasteners, anchors, and connection devices that are concealed from view to greatest extent possible.

D. Mechanically Glazed Framing Members: Fabricate for flush glazing without projecting stops.

E. Structural-Sealant-Glazed Framing Members: Include accommodations for using temporary support device to retain glazing in place while structural sealant cures.

F. Storefront Framing: Fabricate components for assembly using screw-spline system.

G. Entrance Door Frames: Reinforce as required to support loads imposed by door operation and for installing entrance door hardware.

1. At interior and exterior doors, provide compression weather stripping at fixed stops.

H. Entrance Doors: Reinforce doors as required for installing entrance door hardware.

1. At pairs of exterior doors, provide sliding-type weather stripping retained in adjustable strip and mortised into door edge.
2. At all exterior doors, provide weather sweeps applied to door bottoms.

I. Entrance Door Hardware Installation: Factory install entrance door hardware to the greatest extent possible. Cut, drill, and tap for factory-installed entrance door hardware before applying finishes.

J. After fabrication, clearly mark components to identify their locations in Project according to Shop Drawings.

2.10 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, AA-M12C22A41, Class I, 0.018 mm or thicker.

2.11 SOURCE QUALITY CONTROL

A. Structural Sealant: Perform quality-control procedures complying with ASTM C1401 recommendations, including, but not limited to, assembly material qualification procedures, sealant testing, and assembly fabrication reviews and checks.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Prepare surfaces that are in contact with structural sealant according to sealant manufacturer's written instructions, to ensure compatibility and adhesion. Preparation includes, but is not limited to, cleaning and priming surfaces.

3.3 INSTALLATION

A. General:

1. Comply with manufacturer's written instructions.
2. Do not install damaged components.
3. Fit joints to produce hairline joints free of burrs and distortion.
4. Rigidly secure non-movement joints.
5. Install anchors with separators and isolators to prevent metal corrosion and electrolytic deterioration and to prevent impeding movement of moving joints.
6. Seal perimeter and other joints watertight unless otherwise indicated.

B. Metal Protection:

1. Where aluminum is in contact with dissimilar metals, protect against galvanic action by painting contact surfaces with materials recommended by manufacturer for this purpose or by installing nonconductive spacers.

2. Where aluminum is in contact with concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

C. Set continuous sill members and flashing in full sealant bed, as specified in Section 079200 "Joint Sealants," to produce weathertight installation.

D. Install components plumb and true in alignment with established lines and grades.

E. Install operable units level and plumb, securely anchored, and without distortion. Adjust weather-stripping contact and hardware movement to produce proper operation.

F. Install glazing as specified in Section 088000 "Glazing."

G. Install weatherseal sealant according to Section 079200 "Joint Sealants" and according to sealant manufacturer's written instructions to produce weatherproof joints. Install joint filler behind sealant as recommended by sealant manufacturer.

H. Entrance Doors: Install doors to produce smooth operation and tight fit at contact points.

1. Exterior Doors: Install to produce weathertight enclosure and tight fit at weather stripping.
2. Field-Installed Entrance Door Hardware: Install surface-mounted entrance door hardware according to entrance door hardware manufacturers' written instructions using concealed fasteners to greatest extent possible.

## 3.4 ERECTION TOLERANCES

A. Erection Tolerances: Install aluminum-framed entrances and storefronts to comply with the following maximum tolerances:

1. Plumb: 1/8 inch in 10 feet (3.2 mm in 3 m); 1/4 inch in 40 feet (6.35 mm in 12.2 m).
2. Level: 1/8 inch in 20 feet (3.2 mm in 6 m); 1/4 inch in 40 feet (6.35 mm in 12.2 m).
3. Alignment:

    a. Where surfaces abut in line or are separated by reveal or protruding element up to 1/2 inch (12.7 mm) wide, limit offset from true alignment to 1/16 inch (1.6 mm).
    b. Where surfaces are separated by reveal or protruding element from 1/2 to 1 inch (12.7 to 25.4 mm) wide, limit offset from true alignment to 1/8 inch (3.2 mm).
    c. Where surfaces are separated by reveal or protruding element of 1 inch (25.4 mm) wide or more, limit offset from true alignment to 1/4 inch (6 mm).

4. Location: Limit variation from plane to 1/8 inch in 12 feet (3.2 mm in 3.6 m); 1/2 inch (12.7 mm) over total length.

## 3.5 FIELD QUALITY CONTROL

A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.

B. Field Quality-Control Testing: Perform the following test on representative areas of aluminum-framed entrances and storefronts.

1. Water-Spray Test: Before installation of interior finishes has begun, areas designated by Architect shall be tested according to AAMA 501.2 and shall not evidence water penetration.

a. Perform a minimum of two tests in areas as directed by Architect.

2. Air Infiltration: ASTM E783 at 1.5 times the rate specified for laboratory testing in "Performance Requirements" Article but not more than 0.09 cfm/sq. ft. (0.45 L/s per sq. m) at a static-air-pressure differential of 1.57 lbf/sq. ft. (75 Pa).

a. Perform a minimum of two tests in areas as directed by Architect.

3. Water Penetration: ASTM E1105 at a minimum uniform and cyclic static-air-pressure differential of 0.67 times the static-air-pressure differential specified for laboratory testing in "Performance Requirements" Article, but not less than 6.24 lbf/sq. ft. (300 Pa), and shall not evidence water penetration.

C. Structural-Sealant Adhesion: Test structural sealant according to recommendations in ASTM C1401, Destructive Test Method A, "Hand Pull Tab (Destructive)," Appendix X2.

1. Test a minimum of two areas on each building facade.
2. Repair installation areas damaged by testing.

D. Aluminum-framed entrances and storefronts will be considered defective if they do not pass tests and inspections.

E. Prepare test and inspection reports.

3.6 MAINTENANCE SERVICE

A. Entrance Door Hardware:

1. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of entrance door hardware.

END OF SECTION 08 4113

# SECTION 08 5113 - ALUMINUM WINDOWS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Aluminum windows.

B. Related Requirements:

1. Section 08 4113 “Aluminum Framed Entrances and Storefronts” for coordinating finish among aluminum fenestration units.

### 1.3 DEFINITIONS

A. Combination Assemblies: An assembly formed by a combination of two or more separate fenestration products whose frames are mulled together utilizing a combination mullion or reinforcing mullion.

B. Combination Mullions: A horizontal or vertical member formed by joining two or more individual fenestration units together without a mullion stiffener.

C. Reinforcing Mullions: A horizontal or vertical member with an added continuous mullion stiffener and joining two or more individual fenestration units along the sides of the mullion stiffener.

### 1.4 COORDINATION

A. Finish Matching: Coordinate all exposed exterior aluminum components and trim to ensure uniform and consistent color and appearance. Use products specified in section 08 4113 “Aluminum Framed Entrances and Storefronts” as a benchmark. Architect's decision will be final as to whether a proposed product matches.

### 1.5 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
2. Review and discuss finishing of aluminum windows that is required to be coordinated with finishing of other aluminum work for color and finish matching.

3. Review, discuss, and coordinate interrelationship of aluminum windows with other exterior wall components. Include provisions for anchoring, flashing, weeping, sealing perimeters, and protecting finishes.
4. Review and discuss sequence of work required to construct a watertight and weathertight exterior building envelope.
5. Inspect and discuss condition of substrate and other preparatory work performed by other trades.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, glazing and fabrication methods, dimensions of individual components and profiles, hardware, and finishes.

B. Shop Drawings:

1. Plans, elevations, sections, hardware, accessories, insect screens, operational clearances, and details of installation, including anchor, flashing, and sealant installation.

C. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type of exposed finish.

D. Samples for Verification: Actual sample of finished products for each type of exposed finish:

1. Exposed Aluminum Finishes: Manufacturers' standard size.

E. Product Schedule: For aluminum windows. Use same designations indicated on Drawings.

F. Sustainable Design Submittals:

1. Shall have a Type III Product-Specific EPD created from a Product Category Rule specific to North America.

1.7 INFORMATIONAL SUBMITTALS

A. Test and Evaluation Reports:

1. Product Test Reports: For each aluminum window, for tests performed by qualified testing agency.

B. Field Quality-Control Reports: For aluminum windows.

C. Qualification Statements: For manufacturer and Installer.

1.8 CLOSEOUT SUBMITTALS

A. Warranty Documentation:

1. Manufacturers' special warranties.

1.9 QUALITY ASSURANCE

A. Manufacturer Qualifications: A manufacturer capable of fabricating aluminum windows that meet or exceed performance requirements indicated and of documenting this performance by test reports and calculations.

B. Installer Qualifications: Authorized representative who is trained and approved by aluminum window manufacturer.

1.10 MOCKUPS

A. Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution. See Section 014339 "Mockups" for additional construction requirements for integrated exterior mockups.

1. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Owner specifically approves such deviations by Change Order.

1.11 DELIVERY, STORAGE, AND HANDLING

A. Deliver aluminum windows to Project site in original, unopened packages and store them in accordance with manufacturer's written instructions. Protect aluminum windows against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

B. Handle aluminum windows in a manner that prevents damage before, during, and after installation.

1.12 FIELD CONDITIONS

A. Environmental Limitations: Do not install aluminum windows outside of limits recommended in writing by manufacturer.

1.13 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace aluminum windows that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Failure to meet performance requirements.
b. Structural failures, including excessive deflection, water leakage, condensation, and air infiltration.
c. Faulty operation of movable sash and hardware.
d. Deterioration of materials and finishes beyond normal weathering.
e. Failure of insulating glass.

2. Warranty Period:

a. Window: Two years from date of Substantial Completion.
b. Aluminum Finish: 10 years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 SOURCE LIMITATIONS

A. Obtain aluminum windows from single source from single manufacturer.

### 2.2 PERFORMANCE REQUIREMENTS

A. Product Standard: Comply with AAMA/WDMA/CSA 101/I.S.2/A440 for definitions and minimum standards of performance, materials, components, accessories, and fabrication unless more stringent requirements are indicated.

B. Performance Class and Grade: AAMA/WDMA/CSA 101/I.S.2/A440 as follows:

1. Minimum Performance Class: AW.
2. Minimum Performance Grade: 100.
3. Mulled Window Systems: Evaluate and rate combination assemblies as single systems as determined by AAMA 450 in accordance with AAMA/WDMA/CSA 101/I.S.2/A440 requirements.

C. Energy Performance: Certified and labeled by manufacturer for energy performance as follows:

1. Thermal Transmittance (U-factor): As determined in accordance with NFRC 100:

a. Fixed Windows: Not more than 0.36 Btu/sq. ft. x h x deg F (2.04 W/sq. m x K).

D. Thermal Movements: Provide aluminum windows, including anchorage, which allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change: 120 deg F (67 deg C) ambient; 180 deg F (100 deg C) material surfaces.

### 2.3 ALUMINUM WINDOWS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Kawneer 8410TL Thermal Windows or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. Boyd Aluminum Mfg. Co.
2. EFCO Corporation.
3. Kawneer Company, Inc.; Arconic Corporation.
4. OldCastle BuildingEnvelope (OBE).
5. YKK AP America Inc.

B. Provide manufacturer's standard aluminum window assemblies consisting of frames, sashes, glass, hardware, fasteners, and all components and accessories as required for a complete installation.

C. Operating Types: Provide the following operating types in locations indicated on Drawings:

1. Fixed.

D. Frames and Sashes: Aluminum extrusions of alloy, temper, and strength complying with AAMA/WDMA/CSA 101/I.S.2/A440.

1. Thermally Broken Construction: Fabricate frames, sashes, and muntins with an integral, concealed, low-conductance thermal barrier located between exterior materials and window members exposed on interior side in a manner that eliminates direct metal-to-metal contact.

E. Weather Stripping: Provide manufacturer's standard full-perimeter weather stripping for each operable sash unless otherwise indicated.

F. Fasteners: Noncorrosive and compatible with window members, trim, hardware, anchors, and other components.

1. Exposed Fasteners: Avoid exposed fasteners to greatest extent possible. For application of hardware, use fasteners that match finish hardware being fastened.

G. Mullions: Provide manufacturer's standard combination and reinforcing mullions and cover plates matching window units, complete with anchors for support to structure. Allow for erection tolerances and provide for movement of window units due to thermal expansion and building deflections. Provide reinforcing mullions and cover plates capable of withstanding design wind loads of window units.

## 2.4 ACCESSORIES

A. Dividers (False Muntin’s): Provide manufacturer's standard extruded-aluminum divider grilles in designs indicated for each sash lite.

1. Type: Permanently located at exterior lite.
2. Pattern: As indicated on Drawings.
3. Profile: As selected by Architect from manufacturer's full range.

B. Subsills: Thermally broken, extruded-aluminum subsills in configurations indicated on Drawings.

C. Interior Trim: Extruded-aluminum profiles in sizes and configurations indicated on Drawings.

D. Panning Trim: Profiles in sizes and configurations indicated on Drawings.

E. Nail Fins: Manufacturer's standard mounting flanges with holes pre-punched for mechanical fasteners.

2.5 FABRICATION

A. Fabricate aluminum windows in sizes indicated. Include a complete system for assembling components and anchoring windows.

B. Glaze aluminum windows in the factory.

C. Weather strip each operable sash to provide weathertight installation.

D. Weep Holes: Provide weep holes and internal passages to conduct infiltrating water to exterior.

E. Provide water-shed members above side-hinged sashes and similar lines of natural water penetration.

F. Window Assemblies: Provide window units in configuration indicated on Drawings. Provide window frames, sashes, hardware, and other trim and components necessary for a complete, secure, and weathertight installation, including the following:

1. Combination and reinforcing mullions with interior and exterior trim.
2. Interior and exterior extension and trim.
3. Exterior head and sill casings and trim.

G. Complete fabrication, assembly, finishing, hardware application, and other work in the factory to greatest extent possible. Disassemble components only as necessary for shipment and installation.

2.6 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM/NOMMA AMP 500 "Metal Finishes Manual for Architectural and Metal Products," for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

2.7 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, AA-M12C22A41, Class I, 0.018 mm or thicker.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine openings, substrates, structural support, anchorage, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Verify rough opening dimensions, levelness of sill plate, and operational clearances.

C. Examine wall flashings, vapor retarders, air and weather barriers, and other built-in components to ensure weathertight window installation.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 INSTALLATION

A. Comply with manufacturer's written instructions for installing windows, hardware, accessories, and other components. For installation procedures and requirements not addressed in manufacturer's written instructions, comply with installation requirements in ASTM E2112.

B. Install windows level, plumb, square, true to line, without distortion or impeding thermal movement, anchored securely in place to structural support, and in proper relation to wall flashing and other adjacent construction to produce weathertight construction.

C. Mullions: Install combination and reinforcing mullions for combination assemblies in accordance with manufacturer's written instructions.

D. Install windows and components to drain water passing joints and condensation to the exterior.

E. Separate aluminum from sources of corrosion or electrolytic action at points of contact with other materials.

## 3.3 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

1. Testing and inspecting agency will interpret tests and state in each report whether tested work complies with or deviates from requirements.

B. Testing Services: Testing and inspection of installed windows as follows:

1. Testing Methodology: Testing of windows for air infiltration and water resistance in accordance with AAMA 502.
2. Air-Infiltration Testing:

    a. Test Pressure: As required to determine compliance in accordance with AAMA/WDMA/CSA 101/I.S.2/A440 for performance class indicated.
    b. Allowable Air-Leakage Rate: 1.5 times the applicable AMA/WDMA/CSA 101/I.S.2/A440 rate for product type and performance class rounded down to one decimal place.

3. Water-Resistance Testing:

    a. Test Pressure: Two-thirds times test pressure required to determine compliance with AAMA/WDMA/CSA 101/I.S.2/A440 performance grade indicated.
    b. Allowable Water Infiltration: No water penetration.

4. Testing Extent: One window(s) of each type as selected by Architect and a qualified independent testing and inspecting agency. Windows will be tested after installation.

5. Test Reports: Prepared in accordance with AAMA 502.

C. Tests and Inspections:

1. Windows will be considered defective if they do not pass tests and inspections.

D. Prepare test and inspection reports.

## 3.4 ADJUSTING

A. Adjust operating sashes and hardware for a tight fit at contact points and weather stripping for smooth operation and weathertight closure.

## 3.5 CLEANING AND PROTECTION

A. Clean exposed surfaces immediately after installing windows using manufacturer's written instructions. Avoid damaging finishes. Remove excess sealants, glazing materials, dirt, and other substances.

B. Remove and replace glass that has been broken, chipped, cracked, abraded, or damaged during construction period.

C. Protect window surfaces from contact with contaminating substances resulting from construction operations. If contaminating substances do contact window surfaces, remove contaminants immediately in accordance with manufacturer's written instructions.

END OF SECTION 08 5113

# SECTION 08 5653 - SECURITY WINDOWS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Fixed, transaction security windows.

### 1.2 COORDINATION

A. Coordinate installation of anchorages for security windows. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in adjacent construction.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, weights and finishes for window units.

B. Shop Drawings: For security windows.

1. Include plans, elevations, sections, and attachment details.
2. Full-size section details of framing members, including internal armoring, reinforcement, and stiffeners.
3. Location of weep holes.
4. Hardware for sliding window units.
5. Glazing details.
6. Details of deal tray, transaction counter and speaking aperture.

C. Samples for Initial Selection: For frame members with factory-applied color finishes.

### 1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Welding certificates.

C. Product Test Reports: For each type of security window and accessory indicated as ballistics or forced-entry resistant, for tests performed by a qualified testing agency.

D. Configuration Disclosure Drawing: For each type of forced-entry-resistant security window, complying with ASTM F1233.

E. Examination reports documenting inspections of substrates, areas, and conditions.

F. Anchor inspection reports documenting inspections of built-in and cast-in anchors.

G. Field quality-control reports documenting inspections of installed products.

1. Field quality-control certification signed by Contractor.

H. Sample Warranty: For special warranty.

## 1.5 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer for installation and maintenance of units required for this Project.

B. Welding Qualifications: Qualify procedures and personnel in accordance with the following:

1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
2. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."
3. AWS D1.3/D1.3M, "Structural Welding Code - Sheet Steel."
4. AWS D1.6, "Structural Welding Code - Stainless Steel."

## 1.6 DELIVERY, STORAGE, AND HANDLING

A. Pack security windows in wood crates for shipment. Crate glazing separate from frames unless factory glazed.

B. Label security window packaging with drawing designation.

C. Store crated security windows on raised blocks to prevent moisture damage.

## 1.7 FIELD CONDITIONS

A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

## 1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace security windows that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures including deflections exceeding 1/4 inch (6 mm).
b. Failure of welds.
c. Excessive air leakage.
d. Faulty operation of sliding window hardware.
e. Faulty operation of transaction drawers.
f. Deterioration of metals, metal finishes, and other materials beyond normal weathering and use.

2. Warranty Period: Three years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Attack Resistance: Provide units identical to those tested for compliance with requirements indicated, and as follows:

1. Ballistics Resistance, UL 752: Level 3 in accordance with UL 752.
2. Forced-Entry Resistance, SD-STD-01.01: 30-minute protection level in accordance with SD-STD-01.01.

2.2 FIXED, TRANSACTION SECURITY WINDOWS

A. Provide fixed, transaction security windows with operable sash or ventilator capable of allowing transfer of currency and documents.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

a. C.R. Laurence Co., Inc.; CRH Americas, Inc.
b. Chicago Bullet Proof Systems.
c. Krieger Specialty Products Company.
d. National Bullet Proof, Inc.
e. Ross Technology Company.

B. Configuration: As indicated on Drawings.

C. Framing: Fabricate perimeter framing, mullions, and glazing stops from aluminum as follows:

1. Profile: Manufacturer's standard with minimum face dimension indicated.

a. Minimum Face Dimension: 2 inches (50 mm).

2. Depth: Manufacturer's standard.

D. Head and Jamb Framing: Designed for sealant glazing.

E. Channel-Frame Sill: Formed from stainless steel and designed for sealant glazing.

1. Transaction Counter Tray: Baffle type window as indicated on drawings.

F. Voice-Communication-Type: Formed from stainless steel and designed to allow passage of speech at normal speaking volume without distortion.

G. Glazing and Glazing Materials: Comply with requirements in Section 08 8853 "Security Glazing."

H. Glazing Meeting Edges: Polished glazing.

I. Materials:

1. Mild Steel Plates, Shapes, and Bars: ASTM A36/A36M.

2. Cold-Rolled Steel Sheet: ASTM A1008/A1008M, CS (Commercial Steel), Type B; suitable for exposed applications.
3. Metallic-Coated Steel Sheet: ASTM A653/A653M, CS (Commercial Steel), Type B; with G60 (Z180) zinc (galvanized) or A60 (ZF180) zinc-iron-alloy (galvannealed) coating designation.
4. Hot-Rolled Steel Sheet: ASTM A1011/A1011M, CS (Commercial Steel), Type B; free of scale, pitting, or surface defects; pickled and oiled.
5. Stainless Steel Sheet, Strip, Plate, and Flat Bars: ASTM A666 or ASTM A240/A240M, austenitic stainless steel, Type 304.
6. Aluminum Extrusions: ASTM B221 (ASTM B221M). Provide alloy and temper recommended by manufacturer for strength, corrosion resistance, and application of required finish, but not less than 22,000-psi (150-MPa) ultimate tensile strength.
7. Aluminum Sheet and Plate: ASTM B209 (ASTM B209M).

## 2.3 FABRICATION

A. General: Fabricate security windows to provide a complete system for assembly of components and anchorage of window units.

1. Provide units that are reglazable from the secure side without dismantling the attack side of framing.
2. Prepare security windows for field glazing unless preglazing at the factory is indicated.

B. Provide weep holes and internal water passages for exterior security windows to conduct infiltrating water to the exterior.

C. Thermally Improved or Thermally Broken Construction: Fabricate framing with an integral, concealed, low-conductance thermal barrier, located between exterior materials and members exposed on interior in a manner that eliminates direct metal-to-metal contact.

D. Framing: Miter or cope corners the full depth of framing; weld and dress smooth.

1. Fabricate framing with manufacturer's standard, internal opaque armoring in thicknesses required for security windows to comply with ballistics-resistance performance indicated.

E. Glazing Stops: Finish glazing stops to match security window framing.

1. Attack-Side (Exterior) Glazing Stops: Welded or integral to framing.
2. Secure-Side (Interior) Glazing Stops: Removable, coordinated with glazing indicated.

F. Welding: Weld components to comply with referenced AWS standard. To greatest extent possible, weld before finishing and in concealed locations to minimize distortion or discoloration of finish. Remove weld spatter and welding oxides from exposed surfaces by descaling or grinding.

G. Metal Protection: Separate dissimilar metals to protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose.

H. Factory-cut openings in glazing for speaking apertures.

I. Preglazed Fabrication: Preglaze window units at factory, where required for applications indicated. Installation orientation of glazing to meet performance requirements. Comply with requirements in Section 088853 "Security Glazing."

## 2.4 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM/NOMMA 500 for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

## 2.5 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, AA-M12C22A41, Class I, 0.018 mm or thicker.

B. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

1. Color and Gloss: selected by Architect from manufacturer's full range.

## 2.6 STAINLESS STEEL FINISHES

A. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.

B. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.

1. Run grain of directional finishes with long dimension of each piece.
2. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

C. Stainless Steel Sheet and Plate Finishes:

1. Directional Satin Finish: ASTM A480/A480M, No. 4.

## 2.7 ACCESSORIES

A. Recessed, Non-ricochet Deal Trays: Formed from stainless steel; fabricated with recessed bullet trap to ricochet bullets away from secure side, with exposed flanges for recessed installation into horizontal surface, and with sliding stainless steel cover.

1. Clear Opening Size: 10 inches wide by 7 inches deep by 1-1/2 inches high (254 mm wide by 178 mm deep by 38 mm high).
2. Bullet Trap Location: Secure side.
3. Ballistics Resistance: Same as security window.
4. Listed and labeled as bullet resisting in accordance with UL 752.

B. Transaction Drawers: Formed from stainless steel; with ball-bearing, telescoping sliding mechanism; with cover on secure side of top of drawer that automatically closes when drawer is extended to attack side.

1. Inside Dimensions: 15-3/8 inches wide by 8-1/2 inches deep by 4-3/8 inches high (390 mm wide by 216 mm deep by 111 mm high).
2. Operation: Manual.
3. Ballistics Resistance: Same as security window.
4. Listed and labeled as bullet resisting in accordance with UL 752.

C. Speaking Apertures: Fabricate from stainless steel, designed to allow passage of speech at normal speaking volume without distortion and ADA compliant.

1. Shape: Circular.
2. Ballistics Resistance: Same as security window.
3. Listed and labeled as bullet resisting in accordance with UL 752.

D. Concealed Bolts: ASTM A307, Grade A unless otherwise indicated.

E. Cast-in-Place Anchors in Concrete: Fabricated from corrosion-resistant materials capable of sustaining, without failure, a load equal to four times the load imposed, as determined by testing in accordance with ASTM E488/E488M, conducted by a qualified testing agency; of type indicated below.

1. Threaded or wedge type; galvanized ferrous castings, either ASTM A27/A27M cast steel or ASTM A47/A47M malleable iron. Provide bolts, washers, and shims as required; hot-dip galvanized in accordance with ASTM A153/A153M or ASTM F2329/F2329M.

F. Embedded Plate Anchors: Fabricated from mild steel shapes and plates, minimum 3/16 inch (4.8 mm) thick; with minimum 1/2-inch- (12.7-mm-) diameter, headed studs welded to back of plate.

G. Welding Rods and Bare Electrodes: Select in accordance with AWS specifications for metal alloy welded.

H. Glazing Strips and Weather Stripping: Manufacturer's standard replaceable components.

1. Compression Type: Molded EPDM or neoprene gaskets complying with ASTM D2000, Designations 2BC415 to 3BC620; molded PVC gaskets complying with ASTM D2287; or molded, expanded EPDM or neoprene gaskets complying with ASTM C509, Grade 4.
2. Sliding Type: AAMA 701/702, made of wool, polypropylene, or nylon woven pile with nylon-fabric backing.

I. Miscellaneous Glazing Materials: Provide material, size, and shape complying with requirements of glass manufacturers and with a proven record of compatibility with surfaces contacted in installation.

1. Cleaners, Primers, and Sealers: Type recommended by sealant or gasket manufacturer.
2. Setting Blocks: Elastomeric material with a Shore A durometer hardness of 85, plus or minus 5.
3. Spacers: Elastomeric blocks or continuous extrusions with a Shore A durometer hardness required by glass manufacturer to maintain glass lites in place for installation indicated.

4. Edge Blocks: Elastomeric material of hardness needed to limit glass lateral movement (side walking).

J. Anchors, Clips, and Window Accessories: Stainless steel; hot-dip, zinc-coated steel or iron, complying with ASTM B633; provide sufficient strength to withstand design pressures indicated.

K. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

L. Sealants: For sealants required within fabricated security windows, provide type recommended by manufacturer for joint size and movement. Sealant remains permanently elastic, nonshrinking, and nonmigrating.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of security windows.

B. Examine roughing-in for embedded and built-in anchors to verify actual locations of security window connections before security window installation.

C. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of security windows.

D. Inspect built-in and cast-in anchor installations, before installing security windows, to verify that anchor installations comply with requirements. Prepare inspection reports.

1. Remove and replace anchors where inspections indicate that they do not comply with specified requirements. Reinspect after repairs or replacements are made.
2. Perform additional inspections to determine compliance of replaced or additional work. Prepare anchor inspection reports.

E. For factory-installed glazing materials whose orientation (secure or attack side) is critical for performance, verify installation orientation.

F. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

A. Coordination: Furnish layouts for cast-in-place anchors, clips, and other security window anchors whose installation is specified in other Sections.

1. Furnish cast-in-place anchors and similar devices to other trades for installation well in advance of time needed for coordinating other work.

3.3 INSTALLATION

A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing security windows to in-place construction. Include threaded fasteners for inserts, security fasteners, and other connectors.

1. Install an attached or integral flange to secure side of security windows extending over rough-in opening gap so that gap has same forced-entry-resistance and ballistics-resistance performance as security window.

B. Voice-Communication-Type Framing: Attach removable glass spacers to jambs and head of glazing, located not more than 6 inches (152 mm) from each corner and spaced not more than 12 inches (305 mm) o.c.

C. Glazed Framing: Provide gasket-glazed framing. Comply with installation requirements in Section 088853 "Security Glazing."

D. Removable Glazing Stops and Trim: Fasten components with security fasteners.

E. Fasteners: Install security windows using fasteners recommended by manufacturer with head style appropriate for installation requirements, strength, and finish of adjacent materials. Provide stainless steel fasteners in stainless steel materials.

F. Sealants: Comply with requirements in Section 079200 "Joint Sealants" for installing sealants, fillers, and gaskets.

1. Set continuous sill members and flashing in a full sealant bed to provide weathertight construction unless otherwise indicated.
2. Seal frame perimeter with sealant to provide weathertight construction unless otherwise indicated.

G. Metal Protection: Where dissimilar metals will contact each other, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended in writing by manufacturer for this purpose. Where aluminum will contact concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

3.4 FIELD QUALITY CONTROL

A. Inspect installed products to verify compliance with requirements. Prepare inspection reports and indicate compliance with and deviations from the Contract Documents.

B. Perform additional inspections to determine compliance of replaced or additional work. Prepare inspection reports.

C. Prepare field quality-control certification that states installed products and their installation comply with requirements in the Contract Documents.

3.5 ADJUSTING

A. Adjust horizontal-sliding, transaction security windows to provide a tight fit at contact points for smooth operation and a secure enclosure.

B. Adjust transaction drawers to provide a tight fit at contact points for smooth operation and secure enclosure.

C. Remove and replace defective work, including security windows that are warped, bowed, or otherwise unacceptable.

## 3.6 CLEANING AND PROTECTION

A. Clean surfaces promptly after installation of security windows. Take care to avoid damaging the finish. Remove excess glazing and sealant compounds, dirt, and other substances.

1. Lubricate transaction drawer hardware.

B. Clean glass of preglazed security windows promptly after installation. Comply with requirements in Section 08 8853 "Security Glazing" for cleaning and maintenance.

C. Provide temporary protection to ensure that security windows are without damage at time of Substantial Completion.

## 3.7 DEMONSTRATION

A. Train Owner's maintenance personnel to adjust, operate, and maintain windows with transaction drawers.

END OF SECTION 08 5653

# SECTION 08 7100 - DOOR HARDWARE

## PART 1 - GENERAL

### 1.01 SUMMARY

A. Section includes:

1. Mechanical and electrified door hardware
2. Electronic access control system components

B. Section excludes:

1. Windows
2. Cabinets (casework), including locks in cabinets
3. Signage
4. Toilet accessories
5. Overhead doors

C. Related Sections:

1. Division 01 "General Requirements" sections for Allowances, Alternates, Owner Furnished Contractor Installed, Project Management and Coordination.
2. Division 06 Section "Rough Carpentry"
3. Division 07 Section "Joint Sealants" for sealant requirements applicable to threshold installation specified in this section.
4. Division 08 Sections:
   a. "Metal Doors and Frames"
   b. "Flush Wood Doors"
   c. "Aluminum-Framed Entrances and Storefronts"
5. Division 26 "Electrical" sections for connections to electrical power system and for low-voltage wiring.
6. Division 28 "Electronic Safety and Security" sections for coordination with other components of electronic access control system and fire alarm system.

### 1.02 REFERENCES

A. UL LLC

1. UL 10B - Fire Test of Door Assemblies
2. UL 10C - Positive Pressure Test of Fire Door Assemblies
3. UL 1784 - Air Leakage Tests of Door Assemblies
4. UL 305 - Panic Hardware

B. DHI - Door and Hardware Institute

1. Sequence and Format for the Hardware Schedule
2. Recommended Locations for Builders Hardware
3. Keying Systems and Nomenclature
4. Installation Guide for Doors and Hardware

C. NFPA – National Fire Protection Association

1. NFPA 70 – National Electric Code
2. NFPA 80 – 2016 Edition – Standard for Fire Doors and Other Opening Protectives
3. NFPA 101 – Life Safety Code
4. NFPA 105 – Smoke and Draft Control Door Assemblies
5. NFPA 252 – Fire Tests of Door Assemblies

D. ANSI - American National Standards Institute

1. ANSI A117.1 – 2017 Edition – Accessible and Usable Buildings and Facilities
2. ANSI/BHMA A156.1 - A156.29, and ANSI/BHMA A156.31 - Standards for Hardware and Specialties
3. ANSI/BHMA A156.28 - Recommended Practices for Keying Systems
4. ANSI/WDMA I.S. 1A - Interior Architectural Wood Flush Doors
5. ANSI/SDI A250.8 - Standard Steel Doors and Frames

1.03 SUBMITTALS

A. General:

1. Submit in accordance with Conditions of Contract and Division 01 Submittal Procedures.
2. Prior to forwarding submittal:
   a. Review drawings and Sections from related trades to verify compatibility with specified hardware.
   b. Highlight, encircle, or otherwise specifically identify on submittals: deviations from Contract Documents, issues of incompatibility or other issues which may detrimentally affect the Work.

B. Action Submittals:

1. Product Data: Submit technical product data for each item of door hardware, installation instructions, maintenance of operating parts and finish, and other information necessary to show compliance with requirements.
2. Riser and Wiring Diagrams: After final approval of hardware schedule, submit details of electrified door hardware, indicating:
   a. Wiring Diagrams: For power, signal, and control wiring and including:
      1) Details of interface of electrified door hardware and building safety and security systems.
      2) Schematic diagram of systems that interface with electrified door hardware.
      3) Point-to-point wiring.
      4) Risers.

3. Samples for Verification: If requested by Architect, submit production sample of requested door hardware unit in finish indicated and tagged with full description for coordination with schedule.
   a. Samples will be returned to supplier. Units that are acceptable to Architect may, after final check of operations, be incorporated into Work, within limitations of key coordination requirements.

4. Door Hardware Schedule:

a. Submit concurrent with submissions of Product Data, Samples, and Shop Drawings. Coordinate submission of door hardware schedule with scheduling requirements of other work to facilitate fabrication of other work critical in Project construction schedule.
b. Submit under direct supervision of a Door Hardware Institute (DHI) certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) with hardware sets in vertical format as illustrated by Sequence of Format for the Hardware Schedule published by DHI.
c. Indicate complete designations of each item required for each opening, include:
    1) Door Index: door number, heading number, and Architect's hardware set number.
    2) Quantity, type, style, function, size, and finish of each hardware item.
    3) Name and manufacturer of each item.
    4) Fastenings and other pertinent information.
    5) Location of each hardware set cross-referenced to indications on Drawings.
    6) Explanation of all abbreviations, symbols, and codes contained in schedule.
    7) Mounting locations for hardware.
    8) Door and frame sizes and materials.
    9) Degree of door swing and handing.
    10) Operational Description of openings with electrified hardware covering egress, ingress (access), and fire/smoke alarm connections.

5. Key Schedule:
    a. After Keying Conference, provide keying schedule that includes levels of keying, explanations of key system's function, key symbols used, and door numbers controlled.
    b. Use ANSI/BHMA A156.28 "Recommended Practices for Keying Systems" as guideline for nomenclature, definitions, and approach for selecting optimal keying system.
    c. Provide 3 copies of keying schedule for review prepared and detailed in accordance with referenced DHI publication. Include schematic keying diagram and index each key to unique door designations.
    d. Index keying schedule by door number, keyset, hardware heading number, cross keying instructions, and special key stamping instructions.
    e. Provide one complete bitting list of key cuts and one key system schematic illustrating system usage and expansion. Forward bitting list, key cuts and key system schematic directly to Owner, by means as directed by Owner.
    f. Prepare key schedule by or under supervision of supplier, detailing Owner's final keying instructions for locks.

C. Informational Submittals:

1. Provide Qualification Data for Supplier, Installer and Architectural Hardware Consultant.
2. Provide Product Data:
    a. Certify that door hardware approved for use on types and sizes of labeled fire-rated doors complies with listed fire-rated door assemblies.
    b. Include warranties for specified door hardware.

D. Closeout Submittals:

1. Operations and Maintenance Data: Provide in accordance with Division 01 and include:
    a. Complete information on care, maintenance, and adjustment; data on repair and replacement parts, and information on preservation of finishes.
    b. Catalog pages for each product.
    c. Final approved hardware schedule edited to reflect conditions as installed.
    d. Final keying schedule

e. Copy of warranties including appropriate reference numbers for manufacturers to identify project.
f. As-installed wiring diagrams for each opening connected to power, both low voltage and 110 volts.

E. Inspection and Testing:

1. Submit written reports to the Owner and Authority Having Jurisdiction (AHJ) of the results of functional testing and inspection for:
   a. Fire door assemblies, in compliance with NFPA 80.
   b. Required egress door assemblies, in compliance with NFPA 101.

1.04 QUALITY ASSURANCE

A. Qualifications and Responsibilities:

1. Supplier: Recognized architectural hardware supplier with a minimum of 5 years documented experience supplying both mechanical and electromechanical door hardware similar in quantity, type, and quality to that indicated for this Project. Supplier to be recognized as a factory direct distributor by the manufacturer of the primary materials with a warehousing facility in the Project's vicinity. Supplier to have on staff, a certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) available to Owner, Architect, and Contractor, at reasonable times during the Work for consultation.
2. Installer: Qualified tradesperson skilled in the application of commercial grade hardware with experience installing door hardware similar in quantity, type, and quality as indicated for this Project.
3. Architectural Hardware Consultant: Person who is experienced in providing consulting services for door hardware installations that are comparable in material, design, and extent to that indicated for this Project and meets these requirements:
   a. For door hardware: DHI certified AHC or DHC.
   b. Can provide installation and technical data to Architect and other related subcontractors.
   c. Can inspect and verify components are in working order upon completion of installation.
   d. Capable of producing wiring diagram and coordinating installation of electrified hardware with Architect and electrical engineers.

4. Single Source Responsibility: Obtain each type of door hardware from single manufacturer.

B. Certifications:

1. Fire-Rated Door Openings:
   a. Provide door hardware for fire-rated openings that complies with NFPA 80 and requirements of authorities having jurisdiction.
   b. Provide only items of door hardware that are listed products tested by UL LLC, Intertek Testing Services, or other testing and inspecting organizations acceptable to authorities having jurisdiction for use on types and sizes of doors indicated, based on testing at positive pressure and according to NFPA 252 or UL 10C and in compliance with requirements of fire-rated door and door frame labels.

2. Smoke and Draft Control Door Assemblies:
   a. Provide door hardware that meets requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105

b. Comply with the maximum air leakage of 0.3 cfm/sq. ft. (3 cu. m per minute/sq. m) at tested pressure differential of 0.3-inch wg (75 Pa) of water.

3. Electrified Door Hardware
   a. Listed and labeled as defined in NFPA 70, Article 100, by testing agency acceptable to authorities having jurisdiction.

4. Accessibility Requirements:
   a. Comply with governing accessibility regulations cited in "REFERENCES" article 087100, 1.02.D3 herein for door hardware on doors in an accessible route. This project must comply with all Federal Americans with Disability Act regulations and all Local Accessibility Regulations.

C. Pre-Installation Meetings

1. Keying Conference
   a. Incorporate keying conference decisions into final keying schedule after reviewing door hardware keying system including:
      1) Function of building, flow of traffic, purpose of each area, degree of security required, and plans for future expansion.
      2) Preliminary key system schematic diagram.
      3) Requirements for key control system.
      4) Requirements for access control.
      5) Address for delivery of keys.

2. Pre-installation Conference
   a. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
   b. Inspect and discuss preparatory work performed by other trades.
   c. Inspect and discuss electrical roughing-in for electrified door hardware.
   d. Review sequence of operation for each type of electrified door hardware.
   e. Review required testing, inspecting, and certifying procedures.
   f. Review questions or concerns related to proper installation and adjustment of door hardware.

3. Electrified Hardware Coordination Conference:
   a. Prior to ordering electrified hardware, schedule and hold meeting to coordinate door hardware with security, electrical, doors and frames, and other related suppliers.

## 1.05 DELIVERY, STORAGE, AND HANDLING

A. Inventory door hardware on receipt and provide secure lock-up for hardware delivered to Project site. Promptly replace products damaged during shipping

B. Tag each item or package separately with identification coordinated with final door hardware schedule, and include installation instructions, templates, and necessary fasteners with each item or package. Deliver each article of hardware in manufacturer's original packaging.

C. Maintain manufacturer-recommended environmental conditions throughout storage and installation periods.

D. Provide secure lock-up for door hardware delivered to Project. Control handling and installation of hardware items so that completion of Work will not be delayed by hardware losses both before and after installation.

E. Handle hardware in manner to avoid damage, marring, or scratching. Correct, replace or repair products damaged during Work. Protect products against malfunction due to paint, solvent, cleanser, or any chemical agent.

F. Deliver keys to manufacturer of key control system for subsequent delivery to Owner.

1.06 COORDINATION

A. Coordinate layout and installation of floor-recessed door hardware with floor construction. Cast anchoring inserts into concrete.

B. Installation Templates: Distribute for doors, frames, and other work specified to be factory or shop prepared. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing door hardware to comply with indicated requirements.

C. Security: Coordinate installation of door hardware, keying, and access control with Owner's security consultant.

D. Electrical System Roughing-In: Coordinate layout and installation of electrified door hardware with connections to power supplies and building safety and security systems.

1.07 WARRANTY

A. Manufacturer's standard form in which manufacturer agrees to repair or replace components of door hardware that fail in materials or workmanship within published warranty period.

1. Warranty does not cover damage or faulty operation due to improper installation, improper use or abuse.
2. Warranty Period: Beginning from date of Substantial Completion, for durations indicated in manufacturer's published listings.
    a. Mechanical Warranty
        1) Locks
            a) Schlage L Series: 10 years
        2) Exit Devices
            a) Von Duprin: 10 years
        3) Closers
            a) LCN 4000 Series: 30 years
        4) Automatic Operators
            a) LCN: 2 years
    b. Electrical Warranty
        1) Locks
            a) Schlage: 3 years
        2) Exit Devices
            a) Von Duprin: 3 years

1.08 MAINTENANCE

A. Furnish complete set of special tools required for maintenance and adjustment of hardware, including changing of cylinders.

B. Turn over unused materials to Owner for maintenance purposes.

## PART 2 - PRODUCTS

### 2.01 MANUFACTURERS

A. The Owner requires use of certain products for their unique characteristics and project suitability to ensure continuity of existing and future performance and maintenance standards. After investigating available product offerings, the Awarding Authority has elected to prepare proprietary specifications. These products are specified with the notation: "No Substitute."

1. Where "No Substitute" is noted, submittals and substitution requests for other products will not be considered.

B. Approval of alternate manufacturers and/or products other than those listed as "Scheduled Manufacturer" or "Acceptable Manufacturers" in the individual article for the product category are only to be considered by official substitution request in accordance with section 01 25 00.

C. Approval of products from manufacturers indicated in "Acceptable Manufacturers" is contingent upon those products providing all functions and features and meeting all requirements of scheduled manufacturer's product.

D. Where specified hardware is not adaptable to finished shape or size of members requiring hardware, furnish suitable types having same operation and quality as type specified, subject to Architect's approval.

### 2.02 MATERIALS

A. Fabrication

1. Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. provide screws according to manufacturer’s recognized installation standards for application intended.
2. Finish exposed screws to match hardware finish, or, if exposed in surfaces of other work, to match finish of this other work including prepared for paint surfaces to receive painted finish.
3. Provide concealed fasteners wherever possible for hardware units exposed when door is closed. Coordinate with “Metal Doors and Frames”, “Flush Wood Doors”, “Stile and Rail Wood Doors” to ensure proper reinforcements. Advise the Architect where visible fasteners, such as thru bolts, are required.

B. Provide screws, bolts, expansion shields, drop plates and other devices necessary for hardware installation.

1. Where fasteners are exposed to view: Finish to match adjacent door hardware material.

C. Cable and Connectors:

1. Where scheduled in the hardware sets, provide each item of electrified hardware and wire harnesses with number and gage of wires enough to accommodate electric function of specified hardware.
2. Provide Molex connectors that plug directly into connectors from harnesses, electric locking and power transfer devices.
3. Provide through-door wire harness for each electrified locking device installed in a door and wire harness for each electrified hinge, electrified continuous hinge, electrified pivot, and electric power transfer for connection to power supplies.

## 2.03 HINGES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Ives 5BB series

2. Acceptable Manufacturers and Products:
   a. Best FBB series
   b. McKinney TB series

B. Requirements:

1. Provide hinges conforming to ANSI/BHMA A156.1.
2. Provide five knuckle, ball bearing hinges.
3. 1-3/4 inch (44 mm) thick doors, up to and including 36 inches (914 mm) wide:
   a. Exterior: Standard weight, bronze or stainless steel, 4-1/2 inches (114 mm) high
   b. Interior: Standard weight, steel, 4-1/2 inches (114 mm) high

4. 1-3/4 inch (44 mm) thick doors over 36 inches (914 mm) wide:
   a. Exterior: Heavy weight, bronze/stainless steel, 5 inches (127 mm) high
   b. Interior: Heavy weight, steel, 5 inches (127 mm) high

5. 2 inches or thicker doors:
   a. Exterior: Heavy weight, bronze or stainless steel, 5 inches (127 mm) high
   b. Interior: Heavy weight, steel, 5 inches (127 mm) high

6. Adjust hinge width for door, frame, and wall conditions to allow proper degree of opening.
7. Provide three hinges per door leaf for doors 90 inches (2286 mm) or less in height, and one additional hinge for each 30 inches (762 mm) of additional door height.
8. Hinge Pins: Except as otherwise indicated, provide hinge pins as follows:
   a. Steel Hinges: Steel pins
   b. Non-Ferrous Hinges: Stainless steel pins
   c. Out-Swinging Exterior Doors: Non-removable pins
   d. Out-Swinging Interior Lockable Doors: Non-removable pins
   e. Interior Non-lockable Doors: Non-rising pins

9. Provide hinges with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware. Locate electric hinge at second hinge from bottom or nearest to electrified locking component. Provide mortar guard for each electrified hinge specified.

## 2.04 CONTINUOUS HINGES

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. ABH
   b. Select

B. Requirements:

1. Provide aluminum geared continuous hinges conforming to ANSI/BHMA A156.26, Grade 1.
2. Provide aluminum geared continuous hinges, where specified in the hardware sets, fabricated from 6063-T6 aluminum.
3. Provide split nylon bearings at each hinge knuckle for quiet, smooth, self-lubricating operation.
4. Provide hinges capable of supporting door weights up to 450 pounds, and successfully tested for 1,500,000 cycles.
5. On fire-rated doors, provide aluminum geared continuous hinges classified for use on rated doors by testing agency acceptable to authority having jurisdiction.
6. Provide aluminum geared continuous hinges with electrified option scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
7. Provide hinges 1 inch (25 mm) shorter in length than nominal height of door, unless otherwise noted or door details require shorter length and with symmetrical hole pattern.

## 2.05 ELECTRIC POWER TRANSFER

A. Manufacturers:

1. Scheduled Manufacturer and Product:
   a. Von Duprin EPT-10

2. Acceptable Manufacturers and Products:
   a. ABH PT1000
   b. Security Door Controls PTM

B. Requirements:

1. Provide power transfer with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
2. Locate electric power transfer per manufacturer's template and UL requirements, unless interference with operation of door or other hardware items.

## 2.06 DOOR CORDS

A. Manufacturers:

1. Scheduled Manufacturer and Product:
   a. Schlage 788/798 Series

2. Acceptable Manufacturers and Products:
   a. Alarm Controls DL Series
   b. Locknetics DC Series

B. Requirements:

1. Provide door cords with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
2. Locate electric power transfer per manufacturer's template and UL requirements, unless interference with operation of door or other hardware items.

## 2.07 MORTISE LOCKS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage L9000 series

2. Acceptable Manufacturers and Products:
   a. Sargent 8200 series
   b. Best 45H series

B. Requirements:

1. Provide mortise locks conforming to ANSI/BHMA A156.13 Series 1000, Grade 1, and UL Listed for 3-hour fire doors.
2. Indicators: Where specified, provide indicator window measuring a minimum 2-inch x 1/2 inch with 180-degree visibility. Provide messages color-coded with full text and/or symbols, as scheduled, for easy visibility.
3. Provide locks manufactured from heavy gauge steel, containing components of steel with a zinc dichromate plating for corrosion resistance.
4. Provide lock case that is multi-function and field reversible for handing without opening case. Cylinders: Refer to “KEYING” article, herein.
5. Provide locks with standard 2-3/4 inches (70 mm) backset with full 3/4 inch (19 mm) throw stainless steel mechanical anti-friction latchbolt. Provide deadbolt with full 1-inch (25 mm) throw, constructed of stainless steel.
6. Provide standard ASA strikes unless extended lip strikes are necessary to protect trim.
7. Provide electrified options as scheduled in the hardware sets. Where scheduled, provide switches and sensors integrated into the locks and latches. Provide motor based electrified and motor based latch retraction locksets that comply with the following requirements:
   a. Universal input voltage – single chassis accepts 12 or 24VDC to allow for changes in the field without changing lock chassis.
   b. Fail Safe/Fail Secure – changing mode between electrically locked (fail safe) and electrically unlocked (fail secure) is field selectable without opening the lock case
   c. Low maximum current draw – maximum 0.4 amps (Lever control) and maximum 2.0 amps (Latch retraction) to allow for multiple locks on a single power supply.
   d. Low holding current (Lever control or latch retraction) – maximum 0.01 amps to produce minimal heat, eliminate “hot levers” in electrically locked applications and motorized latch retraction applications, and to provide reliable operation in wood doors that provide minimal ventilation and air flow.
   e. Connections – provide quick-connect Molex system standard.

8. (OPTION Key Override) Provide locks with a key override feature built into the chassis that allows the outside key to retract the deadbolt and/or latchbolt, overriding the inside thumbturn when it is being held in the locked position - where the XL13-439 option is specified in the hardware sets.
9. Lever Trim: Solid brass, bronze, or stainless steel, cast or forged in design specified, with wrought roses and external lever spring cages. Provide thru-bolted levers with 2-piece spindles.
    a. (OPTION Vandlgard) Provide levers with vandal resistant technology for use at heavy traffic or abusive applications.
    b. Lever Design: 07A.

## 2.08 EXIT DEVICES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
    a. Von Duprin 98/35A series

2. Acceptable Manufacturers and Products:
    a. Detex Advantex series
    b. Precision APEX 2000 series

B. Requirements:

1. Provide exit devices tested to ANSI/BHMA A156.3 Grade 1 and UL listed for Panic Exit or Fire Exit Hardware.
2. Cylinders: Refer to "KEYING" article, herein.
3. Provide smooth touchpad type exit devices, fabricated of brass, bronze, stainless steel, or aluminum, plated to standard architectural finishes to match balance of door hardware.
4. Touchpad must extend a minimum of one half of door width. No plastic inserts are allowed in touchpads.
5. Provide exit devices with deadlatching feature for security and for future addition of alarm kits and/or other electrified requirements.
6. Provide exit devices with weather resistant components that can withstand harsh conditions of various climates and corrosive cleaners used in outdoor pool environments.
7. Provide flush end caps for exit devices.
8. Provide exit devices with manufacturer's approved strikes.
9. Provide exit devices cut to door width and height. Install exit devices at height recommended by exit device manufacturer, allowable by governing building codes, and approved by Architect.
10. Mount mechanism case flush on face of doors or provide spacers to fill gaps behind devices. Where glass trim or molding projects off face of door, provide glass bead kits.
11. Provide cylinder or hex-key dogging as specified at non fire-rated openings.
12. Removable Mullions: 2 inches (51 mm) x 3 inches (76 mm) steel tube. Where scheduled as keyed removable mullion, provide type that can be removed by use of a keyed cylinder, which is self-locking when re-installed.
13. Provide factory drilled weep holes for exit devices used in full exterior application, highly corrosive areas, and where noted in hardware sets.
14. Provide electrified options as scheduled.
15. Top latch mounting: double- or single-tab mount for steel doors, face mount for aluminum doors eliminating requirement of tabs, and double tab mount for wood doors.
16. Provide exit devices with optional trim designs to match other lever and pull designs used on the project.

## 2.09 ELECTRIC STRIKES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Von Duprin 6000 Series

2. Acceptable Manufacturers and Products:
   a. Folger Adam 300 Series
   b. HES 1006 Series

B. Requirements:

1. Provide electric strikes designed for use with type of locks shown at each opening.
2. Provide electric strikes UL Listed as burglary resistant that are tested to a minimum endurance test of 1,000,000 cycles.
3. Where required, provide electric strikes UL Listed for fire doors and frames.
4. Provide transformers and rectifiers for each strike as required. Verify voltage with electrical contractor.

## 2.10 PASSIVE INFRARED MOTION SENSORS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage SCAN II Series

2. Acceptable Manufacturers and Products:
   a. RCI 915 Series
   b. Security Door Controls MD-31D Series

B. Requirements:

1. Provide motion sensors as specified in hardware groups.

## 2.11 POWER SUPPLIES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage/Von Duprin PS900 Series

2. Acceptable Manufacturers and Products:
   a. Dynalock 5000 series
   b. Security Door Controls 600 series

B. Requirements:

1. Provide power supplies approved by manufacturer of supplied electrified hardware.

2. Provide appropriate quantity of power supplies necessary for proper operation of electrified locking components as recommended by manufacturer of electrified locking components with consideration for each electrified component using power supply, location of power supply, and approved wiring diagrams. Locate power supplies as directed by Architect.
3. Provide regulated and filtered 24 VDC power supply, and UL class 2 listed.
4. Provide power supplies with the following features:
   a. 12/24 VDC Output, field selectable.
   b. Class 2 Rated power limited output.
   c. Universal 120-240 VAC input.
   d. Low voltage DC, regulated and filtered.
   e. Polarized connector for distribution boards.
   f. Fused primary input.
   g. AC input and DC output monitoring circuit w/LED indicators.
   h. Cover mounted AC Input indication.
   i. Tested and certified to meet UL294.
   j. NEMA 1 enclosure.
   k. Hinged cover w/lock down screws.
   l. High voltage protective cover.

## 2.12 CYLINDERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage Everest 29 T

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide cylinders/cores compliant with ANSI/BHMA A156.5; latest revision; cylinder face finished to match lockset; manufacturer’s series as indicated. Refer to “KEYING” article, herein.
2. Provide cylinders in the below-listed configuration(s), distributed throughout the Project as indicated.
   a. Patented Restricted: cylinder with interchangeable core with patented, restricted keyway.

3. Patent Protection: Cylinders/cores requiring use of restricted, patented keys, patent protected.
4. Nickel silver bottom pins.

## 2.13 KEYING

A. Scheduled System:

1. New factory registered system:
   a. Provide a factory registered keying system, complying with guidelines in ANSI/BHMA A156.28, incorporating decisions made at keying conference.

B. Requirements:

1. Construction Keying:
   a. Replaceable Construction Cores.
      1) Provide temporary construction cores replaceable by permanent cores, furnished in accordance with the following requirements.
         a) 3 construction control keys
         b) 12 construction change (day) keys.
      2) Owner or Owner's Representative will replace temporary construction cores with permanent cores.

2. Permanent Keying:
   a. Provide permanent cylinders/cores keyed by the manufacturer according to the following key system.
      1) Master Keying system as directed by the Owner.
   b. Forward bitting list and keys separately from cylinders, by means as directed by Owner. Failure to comply with forwarding requirements will be cause for replacement of cylinders/cores involved at no additional cost to Owner.
   c. Provide keys with the following features:
      1) Material: Nickel silver; minimum thickness of .107-inch (2.3mm)
      2) Patent Protection: Keys and blanks protected by one or more utility patent(s).
   d. Identification:
      1) Mark permanent cylinders/cores and keys with applicable blind code for identification. Do not provide blind code marks with actual key cuts.
      2) Identification stamping provisions must be approved by the Architect and Owner.
      3) Stamp cylinders/cores and keys with Owner's unique key system facility code as established by the manufacturer; key symbol and embossed or stamped with "DO NOT DUPLICATE" along with the "PATENTED" or patent number to enforce the patent protection.
      4) Failure to comply with stamping requirements will be cause for replacement of keys involved at no additional cost to Owner.
      5) Forward permanent cylinders/cores to Owner, separately from keys, by means as directed by Owner.
   e. Quantity: Furnish in the following quantities.
      1) Permanent Control Keys: 3.
      2) Master Keys: 6.
      3) Change (Day) Keys: 3 per cylinder/core that is keyed differently
      4) Key Blanks: Quantity as determined in the keying meeting.

## 2.14 KEY CONTROL SYSTEM

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Telkee

2. Acceptable Manufacturers:
   a. HPC
   b. Lund

B. Requirements:

1. Provide key control system, including envelopes, labels, tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet, all as recommended by system manufacturer, with capacity for 150% of number of locks required for Project.

a. Provide complete cross index system set up by hardware supplier, and place keys on markers and hooks in cabinet as determined by final key schedule.
b. Provide hinged-panel type cabinet for wall mounting.

## 2.15 DOOR CLOSERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. LCN 4010/4110/4020 series

2. Acceptable Manufacturers and Products:
   a. Corbin-Russwin DC8000 series
   b. Sargent 281 series

B. Requirements:

1. Provide door closers conforming to ANSI/BHMA A156.4 Grade 1 requirements by BHMA certified independent testing laboratory. Certify surface mounted mechanical closers to meet fifteen million (15,000,000) full load cycles. ISO 9000 certify closers. Stamp units with date of manufacture code.
2. Provide door closers with fully hydraulic, full rack and pinion action with high strength cast iron cylinder, and full complement bearings at shaft.
3. Cylinder Body: 1-1/2-inch (38 mm) diameter with 11/16-inch (17 mm) diameter double heat-treated pinion journal.
4. Hydraulic Fluid: Fireproof, passing requirements of UL10C, and requiring no seasonal closer adjustment for temperatures ranging from 120 degrees F to -30 degrees F.
5. Spring Power: Continuously adjustable over full range of closer sizes, and providing reduced opening force as required by accessibility codes and standards.
6. Hydraulic Regulation: By tamper-proof, non-critical valves, with separate adjustment for latch speed, general speed, and backcheck.
7. Provide closers with solid forged steel main arms and factory assembled heavy-duty forged forearms for parallel arm closers. When closers are parallel arm mounted, provide closers which mount within 6-inch (152 mm) top rail without use of mounting plate so that closer is not visible through vision panel from pull side.
8. Pressure Relief Valve (PRV) Technology: Not permitted.
9. Finish for Closer Cylinders, Arms, Adapter Plates, and Metal Covers: Powder coating finish which has been certified to exceed 100 hours salt spray testing as described in ANSI/BHMA Standard A156.4 and ASTM B117, or has special rust inhibitor (SRI).
10. Provide special templates, drop plates, mounting brackets, or adapters for arms as required for details, overhead stops, and other door hardware items interfering with closer mounting.

## 2.16 ELECTRO-HYDRAULIC AUTOMATIC OPERATORS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. LCN 4600 series

2. Acceptable Manufacturers and Products:
   a. Precision D4990 series
   b. Besam Power Swing

B. Requirements:

1. Provide low energy automatic operator units with hydraulic closer complying with ANSI/BHMA A156.19.
2. Hydraulic Fluid: Fireproof, passing requirements of UL10C, and requiring no seasonal closer adjustment for temperatures ranging from 120 degrees F to -30 degrees F.
3. Provide units with conventional door closer opening and closing forces unless power operator motor is activated. Provide door closer assembly with adjustable spring size, back-check, and opening and closing speed adjustment valves to control door
4. Provide units with on/off switch for manual operation, motor start up delay, vestibule interface delay, electric lock delay, and door hold open delay.
5. Provide drop plates, brackets, and adapters for arms as required for details.
6. Provide actuator switches and receivers for operation as specified.
7. Provide weather-resistant actuators at exterior applications.
8. Provide key switches with LED's, recommended and approved by manufacturer of automatic operator as required for function described in operation description of hardware group below. Cylinders: Refer to "KEYING" article, herein.
9. Provide complete assemblies of controls, switches, power supplies, relays, and parts/material recommended and approved by manufacturer of automatic operator for each individual leaf. Actuators control both doors simultaneously at pairs. Sequence operation of exterior and vestibule doors with automatic operators to allow ingress or egress through both sets of openings as directed by Architect. Locate actuators, key switches, and other controls as directed by Architect.
10. Provide units with vestibule inputs that allow sequencing operation of two units, and SPDT relay for interfacing with latching or locking devices.

## 2.17 DOOR TRIM

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Accurate Lock & Hardware

2. Acceptable Manufacturers:
   a. Burns
   b. Trimco

B. Requirements:

1. Provide push plates, push bars, pull plates, pulls, and hands-free reversible door pulls with diameter and length as scheduled.

## 2.18 PROTECTION PLATES

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. Burns
   b. Trimco

B. Requirements:

1. Provide protection plates with a minimum of 0.050 inch (1 mm) thick, beveled four edges as scheduled. Furnish with sheet metal or wood screws, finished to match plates.
2. Sizes plates 2 inches (51 mm) less width of door on single doors, pairs of doors with a mullion, and doors with edge guards. Size plates 1 inch (25 mm) less width of door on pairs without a mullion or edge guards.
3. At fire rated doors, provide protection plates over 16 inches high with UL label.

## 2.19 OVERHEAD STOPS AND OVERHEAD STOP/HOLDERS

A. Manufacturers:

1. Scheduled Manufacturers:
   a. Glynn-Johnson

2. Acceptable Manufacturers:
   a. ABH
   b. Rixson

B. Requirements:

1. Provide overhead stop at any door where conditions do not allow for a wall stop or floor stop presents tripping hazard.

## 2.20 DOOR STOPS AND HOLDERS

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. Burns
   b. Trimco

B. Provide door stops at each door leaf:

1. Provide wall stops wherever possible. Provide concave type where lockset has a push button of thumbturn.
2. Where a wall stop cannot be used, provide universal floor stops.
3. Where wall or floor stop cannot be used, provide overhead stop.
4. Provide roller bumper where doors open into each other and overhead stop cannot be used.

## 2.21 THRESHOLDS, SEALS, DOOR SWEEPS, AUTOMATIC DOOR BOTTOMS, AND GASKETING

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Zero International

2. Acceptable Manufacturers:
   a. National Guard
   b. Reese

B. Requirements:

1. Provide thresholds, weather-stripping, and gasketing systems as specified and per architectural details. Match finish of other items.
2. Smoke- and Draft-Control Door Assemblies: Where smoke- and draft-control door assemblies are required, provide door hardware that meets requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105.
3. Provide door sweeps, seals, astragals, and auto door bottoms only of type where resilient or flexible seal strip is easily replaceable and readily available.
4. Size thresholds 1/2 inch (13 mm) high by 5 inches (127 mm) wide by door width unless otherwise specified in the hardware sets or detailed in the drawings.

## 2.22 SILENCERS

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. Steelcraft
   b. Republic

B. Requirements:

1. Provide "push-in" type silencers for hollow metal or wood frames.
2. Provide one silencer per 30 inches (762 mm) of height on each single frame, and two for each pair frame.
3. Omit where gasketing is specified.

## 2.23 DOOR POSITION SWITCHES

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Schlage

2. Acceptable Manufacturers:
   a. GE-Interlogix
   b. Sargent

B. Requirements:

1. Provide recessed or surface mounted type door position switches as specified.
2. Coordinate door and frame preparations with door and frame suppliers. If switches are being used with magnetic locking device, provide minimum of 4 inches (102 mm) between switch and magnetic locking device.

2.24 FINISHES

A. FINISH: BHMA 626/652 (US26D); EXCEPT:

1. Hinges at Exterior Doors: BHMA 630 (US32D)
2. Aluminum Geared Continuous Hinges: BHMA 628 (US28)
3. Push Plates, Pulls, and Push Bars: BHMA 630 (US32D)
4. Protection Plates: BHMA 630 (US32D)
5. Overhead Stops and Holders: BHMA 630 (US32D)
6. Door Closers: Powder Coat to Match
7. Wall Stops: BHMA 630 (US32D)
8. Latch Protectors: BHMA 630 (US32D)
9. Weatherstripping: Clear Anodized Aluminum
10. Thresholds: Mill Finish Aluminum

PART 3 - EXECUTION

3.01 EXAMINATION

A. Prior to installation of hardware, examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, labeled fire-rated door assembly construction, wall and floor construction, and other conditions affecting performance. Verify doors, frames, and walls have been properly reinforced for hardware installation.

B. Examine roughing-in for electrical power systems to verify actual locations of wiring connections before electrified door hardware installation.

C. Submit a list of deficiencies in writing and proceed with installation only after unsatisfactory conditions have been corrected.

3.02 INSTALLATION

A. Mount door hardware units at heights to comply with the following, unless otherwise indicated or required to comply with governing regulations.

1. Standard Steel Doors and Frames: ANSI/SDI A250.8.
2. Custom Steel Doors and Frames: HMMA 831.
3. Interior Architectural Wood Flush Doors: ANSI/WDMA I.S. 1A
4. Installation Guide for Doors and Hardware: DHI TDH-007-20

B. Install door hardware in accordance with NFPA 80, NFPA 101 and provide post-install inspection, testing as specified in section 1.03.E unless otherwise required to comply with governing regulations.

C. Install each hardware item in compliance with manufacturer's instructions and recommendations, using only fasteners provided by manufacturer.

D. Do not install surface mounted items until finishes have been completed on substrate. Protect all installed hardware during painting.

E. Set units level, plumb and true to line and location. Adjust and reinforce attachment substrate as necessary for proper installation and operation.

F. Drill and countersink units that are not factory prepared for anchorage fasteners. Space fasteners and anchors according to industry standards.

G. Install operating parts so they move freely and smoothly without binding, sticking, or excessive clearance.

H. Hinges: Install types and in quantities indicated in door hardware schedule but not fewer than quantity recommended by manufacturer for application indicated.

I. Lock Cylinders:

1. Install construction cores to secure building and areas during construction period.
2. Replace construction cores with permanent cores as indicated in keying section.
3. Furnish permanent cores to Owner for installation.

J. Wiring: Coordinate with Division 26, ELECTRICAL and Division 28 ELECTRONIC SAFETY AND SECURITY sections for:

1. Conduit, junction boxes and wire pulls.
2. Connections to and from power supplies to electrified hardware.
3. Connections to fire/smoke alarm system and smoke evacuation system.
4. Connection of wire to door position switches and wire runs to central room or area, as directed by Architect.
5. Connections to panel interface modules, controllers, and gateways.
6. Testing and labeling wires with Architect's opening number.

K. Key Control System: Tag keys and place them on markers and hooks in key control system cabinet, as determined by final keying schedule.

L. Continuous Hinges: Re-locate the door and frame fire rating labels where they will remain visible so that the hinge does not cover the label once installed.

M. Door Closers & Auto Operators: Mount closers/operators on room side of corridor doors, inside of exterior doors, and stair side of stairway doors from corridors. Mount closers/operators so they are not visible in corridors, lobbies and other public spaces unless approved by Architect.

N. Overhead Stops/Holders: Mount overhead stops/holders on room side of corridor doors, inside of exterior doors, and stair side of stairway doors.

O. Power Supplies: Locate power supplies as indicated or, if not indicated, above accessible ceilings or in equipment room, or alternate location as directed by Architect.

P. Thresholds: Set thresholds in full bed of sealant complying with requirements specified in Division 07 Section "Joint Sealants."

Q. Stops: Provide floor stops for doors unless wall or other type stops are indicated in door hardware schedule. Do not mount floor stops where they may impede traffic or present tripping hazard.

R. Perimeter Gasketing: Apply to head and jamb, forming seal between door and frame.

S. Meeting Stile Gasketing: Fasten to meeting stiles, forming seal when doors are closed.

T. Door Bottoms and Sweeps: Apply to bottom of door, forming seal with threshold when door is closed.

## 3.03 ADJUSTING

A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

1. Spring Hinges: Adjust to achieve positive latching when door can close freely from an open position of 30 degrees.
2. Electric Strikes: Adjust horizontal and vertical alignment of keeper to properly engage lock bolt.
3. Door Closers: Adjust sweep period to comply with accessibility requirements and requirements of authorities having jurisdiction.

B. Occupancy Adjustment: Approximately three to six months after date of Substantial Completion, examine and readjust each item of door hardware, including adjusting operating forces, as necessary to ensure function of doors and door hardware.

## 3.04 CLEANING AND PROTECTION

A. Clean adjacent surfaces soiled by door hardware installation.

B. Clean operating items per manufacturer's instructions to restore proper function and finish.

C. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of Substantial Completion.

## 3.05 DOOR HARDWARE SCHEDULE

A. The intent of the hardware specification is to specify the hardware for interior and exterior doors, and to establish a type, continuity, and standard of quality. However, it is the door hardware supplier's responsibility to thoroughly review existing conditions, schedules, specifications, drawings, and other Contract Documents to verify the suitability of the hardware specified.

B. Discrepancies, conflicting hardware, and missing items are to be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application.

C. Hardware items are referenced in the following hardware schedule. Refer to the above specifications for special features, options, cylinders/keying, and other requirements.

D. Hardware Sets:

Legend:
Link to catalog cut sheet
Electrified Opening

Hardware Group No. 00

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | POWER SUPPLY | PS906 900-4RL 900-4RL KL900 120/240 VAC | | LGR | SCE |
| 2 | EA | POWER SUPPLY | PS906 900-4RL-FA 900-4RL-FA KL900 120/240 VAC | | LGR | SCE |

OPENINGS 102, 104, 106, 107, 109, 003A & 003B TO SHARE A PS906 900-4RL, 900-4RL POWER SUPPLY.
OPENINGS 110, 111, 113, 114, 116, 120 & 119 TO SHARE A PS906 900-4RL-FA, 900-4RL-FA POWER SUPPLY.
OPENINGS 201A, 201B, 203, 206, 207, 209 & 210 TO SHARE A PS906 900-4RL-FA, 900-4RL-FA POWER SUPPLY.
OPENINGS 001A, 001B & 121 TO SHARE A PS906 900-4RL, 900-4RL POWER SUPPLY.
OPENINGS 002, E130, 130 & 131 TO SHARE A PS906 900-4RL, 900, 4RL POWER SUPPLY.

Hardware Group No. 01

Provide each PR door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 2 | EA | CONT. HINGE | 112XY EPT | | | 628 | IVE |
| 2 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-EO-CON 24 VDC | | ✓ | 626 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-NL-OP-110MD-CON 24 VDC | | ✓ | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 2 | EA | 90 DEG OFFSET PULL | 8190HD 10" O | | | 630 | IVE |
| 1 | EA | OH STOP | 100SE<br>MOUNT ON RHR DOOR LEAF. | | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH<br>MOUNT ON LHR DOOR LEAF. | | | 689 | LCN |
| 1 | EA | SURF. AUTO OPERATOR | 4642 TBWMS 120 VAC<br>MOUNT ON RHR DOOR LEAF. | | ✓ | 689 | LCN |
| 1 | EA | WEATHER RING | 8310-801 | | | | LCN |
| 2 | EA | ACTUATOR, TOUCH | 8310-853T | | ✓ | | LCN |
| 1 | EA | MOUNT BOX | 8310-867S | | | | LCN |
| 1 | EA | BOLLARD POST | B-4X6-FT-PC-SM-RT9 | | ✓ | 630 | WIK |
| 2 | EA | KICK PLATE | 8400 10" X 1" LDW B-CS | | | 630 | IVE |
| 1 | EA | GASKETING/SEALS | PROVIDED BY DOOR & FRAME MANUFACTURER | | | | |
| 2 | EA | DOOR SWEEP | 8198AA | | | AA | ZER |
| 1 | EA | THRESHOLD | 655A | | | A | ZER |
| 2 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 2 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | PUSH-TO-LOCK COVER W. RELAY SWITCH | STI-13230CR | | ✓ | | |
| 2 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | PUSH-TO-LOCK PUSH BUTTON | SS2-009-LD-EN | | ✓ | | |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DURING NORMAL BUSINESS HOURS:
2. DOORS CAN BE UNLOCKED BY ACCESS CONTROL SYSTEM AND ACTUATORS ARE ACTIVE.
3. ACTIVATION OF ACTUATORS WILL ALLOW AUTOMATIC OPERATOR TO OPEN ONE DOOR LEAF.

1. DURING NON-BUSINESS HOURS:
2. DOORS NORMALLY CLOSED AND UNLOCKED.
3. EMERGENCY PUSH BUTTON IN VESTIBULE TO LOCK EXTERIOR DOORS AND TURN OFF ACTUATORS.

1. AT ALL TIMES:
2. PRESENTING VALID CREDENTIAL TO EXTERIOR READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLTS, ALLOWING ENTRY.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PADS WILL BYPASS DOOR CONTACTS SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 02

Provide each PR door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 2 | EA | CONT. HINGE | 112XY EPT | | 628 | IVE |
| 2 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-EO-CON 24 VDC | | 626 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-NL-OP-110MD-CON 24 VDC | | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 2 | EA | 90 DEG OFFSET PULL | 8190HD 10" O | | 630 | IVE |
| 1 | EA | OH STOP | 100SE<br>MOUNT ON RHR DOOR LEAF. | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH<br>MOUNT ON LHR DOOR LEAF. | | 689 | LCN |
| 1 | EA | SURF. AUTO OPERATOR | 4642 TBWMS 120 VAC<br>MOUNT ON RHR DOOR LEAF. | | 689 | LCN |
| 2 | EA | ACTUATOR, TOUCH | 8310-853T | | | LCN |
| 2 | EA | MOUNT BOX | 8310-867S | | | LCN |
| 2 | EA | KICK PLATE | 8400 10" X 1" LDW B-CS | | 630 | IVE |
| 2 | EA | SILENCER | SR64 | | GRY | IVE |
| 2 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 2 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DESK MOUNT BUTTON | 660-T4 | | 628 | SCE |
| 2 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DURING NORMAL BUSINESS HOURS:
2. DOORS CAN BE UNLOCKED BY ACCESS CONTROL SYSTEM AND ACTUATORS ARE ACTIVE.
3. ACTIVATION OF ACTUATORS WILL ALLOW AUTOMATIC OPERATOR TO OPEN ONE DOOR LEAF.
4. REMOTE PUSH BUTTON TO LOCK DOORS AND TURN OFF ACTUATORS IN THE EVENT OF AN EMERGENCY.

1. DURING NON-BUSINESS HOURS:
2. DOORS NORMALLY CLOSED AND LOCKED.
3. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLTS AND ENABLES VESTIBULE ACTUATOR. KEY OVERRIDE AVAILABLE.

1. AT ALL TIMES.
2. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PADS WILL BYPASS DOOR CONTACTS SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
3. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 03

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 EDA | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DESK MOUNT BUTTON | 660-PB | ✓ | 628 | SCE |
| 1 | EA | DOOR CONTACT | 679-05HM | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY REMOTE RELEASE OR PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 04

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | OH STOP | 90S | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4011 | | 689 | LCN |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DESK MOUNT BUTTON | 660-PB | ✓ | 628 | SCE |
| 1 | EA | DOOR CONTACT | 679-05HM | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY REMOTE RELEASE OR PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 05

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | | 689 | LCN |
| 1 | EA | WALL STOP | WS406/407CCV | | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 06

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 07

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | STOREROOM LOCK | L9080T 07A | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 08

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 EDA | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | FLOOR STOP | FS436 | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DESK MOUNT BUTTON | 660-PB | ✓ | 628 | SCE |
| 1 | EA | DOOR CONTACT | 679-05HM | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY REMOTE RELEASE OR PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 09

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | STOREROOM W/DEADBOLT | L9480T 07A 09-544 RX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | ELECTRIC STRIKE | 6216 FS 12/16/24/28 VAC/VDC | | 630 | VON |
| 1 | EA | SURFACE CLOSER | 4111 EDA | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | MOTION SENSOR | SCANII 12/24 VDC | | WHT | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RELEASES ELECTRIC STRIKE, ALLOWING ENTRY. KEY OVERRIDE AVAILABLE.
3. OUTSIDE CARD READER WILL NOT UNLOCK DOOR WHEN INSIDE DEADBOLT IS THROWN.
4. FROM EGRESS SIDE, MOTION SENSOR (SCAN II) TO SERVE AS REQUEST TO EXIT (RX) AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED EGRESS.
5. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
6. FREE EGRESS AT ALL TIMES.
7. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 10

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX LX CON 12/24 VDC | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | OH STOP | 90S | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4011 | | 689 | LCN |
| 1 | EA | KICK PLATE | 8402 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | GASKETING | 488SBK PSA | | BK | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, OR ACTIVATION OF FIRE ALARM SYSTEM, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 11

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | PASSAGE SET | L9010 07A | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 12

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | | 689 | LCN |
| 1 | EA | WALL STOP | WS406/407CCV | | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 13

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 14

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX LX CON 12/24 VDC | | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 1 | EA | OH STOP | 90S | | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4011 | | | 689 | LCN |
| 1 | EA | KICK PLATE | 8402 10" X 2" LDW B-CS | | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 15

Provide each PR door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 2 | EA | CONT. HINGE | 112XY EPT | | 628 | IVE |
| 2 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-EO-CON 24 VDC | | 626 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-9847-NL-OP-110MD-CON 24 VDC | | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 2 | EA | 90 DEG OFFSET PULL | 8190HD 10" O | | 630 | IVE |
| 1 | EA | OH STOP | 100SE<br>MOUNT ON RHR DOOR LEAF. | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH<br>MOUNT ON LHR DOOR LEAF. | | 689 | LCN |
| 1 | EA | SURF. AUTO OPERATOR | 4642 TBWMS 120 VAC<br>MOUNT ON RHR DOOR LEAF. | | 689 | LCN |
| 1 | EA | WEATHER RING | 8310-801 | | | LCN |
| 2 | EA | ACTUATOR, TOUCH | 8310-853T | | | LCN |
| 1 | EA | MOUNT BOX | 8310-867S | | | LCN |
| 1 | EA | BOLLARD POST | B-4X6-FT-PC-SM-RT9 | | 630 | WIK |
| 2 | EA | KICK PLATE | 8400 10" X 1" LDW B-CS | | 630 | IVE |
| 1 | EA | GASKETING/SEALS | PROVIDED BY DOOR & FRAME MANUFACTURER | | | |
| 2 | EA | DOOR SWEEP | 8198AA | | AA | ZER |
| 1 | EA | THRESHOLD | 655A | | A | ZER |
| 2 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 2 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 2 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOORS NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLTS AND ENABLES EXTERIOR ACTUATOR. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PADS WILL BYPASS DOOR CONTACTS SIGNALING AUTHORIZED EGRESS. INTERIOR ACTUATOR ENABLED. FREE EGRESS AT ALL TIMES.
4. AUTOMATIC OPERATOR ALLOWS FOR ONE DOOR TO OPEN UPON ACTIVATION OF ACTUATORS.
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 16

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | PRIVACY LOCK | L9040 07A 09-544 | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | MOP PLATE | 8400 4" X 1" LDW B-CS | | 630 | IVE |
| 1 | EA | FLOOR STOP | FS436 | | 626 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 17

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | STOREROOM LOCK | L9080T 07A | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | OH STOP | 90S | | 689 | GLY |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 18

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 630 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-98-NL-CON 24 VDC | | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH | | 689 | LCN |
| 1 | EA | RAIN DRIP | 142AA | | AA | ZER |
| 1 | SET | GASKETING | 429AA-S | | AA | ZER |
| 1 | EA | DOOR SWEEP | 8198AA | | AA | ZER |
| 1 | EA | THRESHOLD | 655A | | A | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLT ALLOWING ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PAD WILL BYPASS DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 19

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | ELEC PANIC HARDWARE | RX-QEL-98-NL-OP-110MD-CON 24 VDC | | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | 90 DEG OFFSET PULL | 8190HD 10" O | | 630 | IVE |
| 1 | EA | SURFACE CLOSER | 4111 EDA | | 689 | LCN |
| 1 | EA | KICK PLATE | 8402 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 1 | EA | GASKETING | 488SBK PSA | | BK | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLT ALLOWING ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PAD WILL BYPASS DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 20

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 2 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | STOREROOM LOCK | L9080T 07A | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 2 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 21

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | | FINISH | MFR |
|---|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | ✓ | 689 | VON |
| 1 | EA | EL MORTISE LOCK | L9095TEL 07A RX LX CON 12/24 VDC | | ✓ | 626 | SCH |
| 2 | EA | PRIMUS CORE | 20-740-XP | | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 EDA | | | 689 | LCN |
| 1 | EA | KICK PLATE | 8402 10" X 2" LDW B-CS | | | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | | | 626 | IVE |
| 1 | EA | GASKETING | 488SBK PSA | | | BK | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | | VON |
| 2 | EA | CARD READER | PROVIDED BY DIV 28 | | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | | |

1. DOOR NORMALLY CLOSED AND LOCKED ON BOTH SIDES.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. EGRESS BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS INSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED EGRESS. KEY OVERRIDE AVAILABLE.
4. UPON LOSS OF POWER, OR ACTIVATION OF FIRE ALARM SYSTEM, DOOR TO REMAIN CLOSED AND UNLOCKED (FAIL SAFE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 22

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 23

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | FINISH | MFR |
|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | 652 | IVE |
| 1 | EA | PRIVACY LOCK | L9040 07A 09-544 | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | 630 | IVE |
| 1 | EA | MOP PLATE | 8400 4" X 1" LDW B-CS | 630 | IVE |
| 1 | EA | WALL STOP | WS406/407CCV | 626 | IVE |
| 3 | EA | SILENCER | SR64 | GRY | IVE |

Hardware Group No. 24

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | PRIVACY LOCK | L9040 07A 09-544 | | 626 | SCH |
| 1 | EA | OH STOP | 100S | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4011 ST-1544 | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 1 | EA | MOP PLATE | 8400 4" X 1" LDW B-CS | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 25

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9095TEU 07A RX LX CON 12/24 VDC | ✓ | 626 | SCH |
| 2 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | OH STOP | 100S | | 689 | GLY |
| 1 | EA | SURFACE CLOSER | 4011 ST-1544 | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 2 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED ON BOTH SIDES.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. EGRESS BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS INSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED EGRESS. KEY OVERRIDE AVAILABLE.
4. UPON LOSS OF POWER, OR ACTIVATION OF FIRE ALARM SYSTEM, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 26

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 NRP | | 652 | IVE |
| 1 | EA | STOREROOM LOCK | L9080T 07A | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH | | 689 | LCN |
| 1 | EA | KICK PLATE | 8400 10" X 2" LDW B-CS | | 630 | IVE |
| 3 | EA | SILENCER | SR64 | | GRY | IVE |

Hardware Group No. 27

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 3 | EA | HINGE | 5BB1 4.5 X 4.5 | | 652 | IVE |
| 1 | EA | POWER TRANSFER | EPT10 CON | ✓ | 689 | VON |
| 1 | EA | EU MORTISE LOCK | L9092TEU 07A RX CON 12/24 VDC | ✓ | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | SURFACE CLOSER | 4011 | | 689 | LCN |
| 1 | EA | WALL STOP | WS406/407CCV | | 626 | IVE |
| 1 | EA | GASKETING | 488SBK PSA | | BK | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | ✓ | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY UNLOCKS OUTSIDE LEVER HANDLE AND BYPASSES DOOR CONTACT, SIGNALING AUTHORIZED ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN THE LEVER HANDLE BYPASSES DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. UPON LOSS OF POWER, OR ACTIVATION OF FIRE ALARM SYSTEM, DOOR TO REMAIN CLOSED AND LOCKED (FAIL SECURE).
5. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

Hardware Group No. 28

Provide each SGL door(s) with the following:

| QTY | | DESCRIPTION | CATALOG NUMBER | | FINISH | MFR |
|---|---|---|---|---|---|---|
| 1 | EA | CONT. HINGE | 157XY | | 628 | IVE |
| 1 | EA | DOOR CORD | 798C-18 WITH 20 GAUGE WIRES | | 626 | SCE |
| 1 | EA | ELEC PANIC HARDWARE | LX-RX-QEL-98-NL-OP-110MD-CON 24 VDC | | 626 | VON |
| 1 | EA | RIM CYLINDER | 20-057 ICX | | 626 | SCH |
| 1 | EA | PRIMUS CORE | 20-740-XP | | 626 | SCH |
| 1 | EA | 90 DEG OFFSET PULL | 8190HD 10" O | | 630 | IVE |
| 1 | EA | SURFACE CLOSER | 4111 SCUSH | | 689 | LCN |
| 1 | EA | CUSH SHOE SUPPORT | 4110-30 SRT | | 689 | LCN |
| 1 | EA | BLADE STOP SPACER | 4110-61 SRT | | 689 | LCN |
| 1 | EA | GASKETING/SEALS | PROVIDED BY DOOR & FRAME MANUFACTURER | | | |
| 1 | EA | DOOR SWEEP | 8198AA | | AA | ZER |
| 1 | EA | THRESHOLD | 655A | | A | ZER |
| 1 | EA | WIRE HARNESS | CON-XXX (LOCK/EXIT TO HINGE FRAME) | | | VON |
| 1 | EA | WIRE HARNESS | CON-XXP (FRAME TO POWER SUPPLY) | | | VON |
| 1 | EA | CARD READER | PROVIDED BY DIV 28 | | | |
| 1 | EA | DOOR CONTACT | 679-05HM | | BLK | SCE |
| 1 | EA | POWER SUPPLY | REFER TO HARDWARE SET '00' | | | |

GENERAL CONTRACTOR TO FIELD VERIFY EXISTING CONDITIONS COMPATIBILITY WITH HARDWARE SPECIFIED. PATCH, FILL AND PAINT ALL HOLES AS NEEDED.

1. DOOR NORMALLY CLOSED AND LOCKED.
2. ENTRY BY PRESENTING VALID CREDENTIAL TO READER MOMENTARILY RETRACTS EXIT DEVICE LATCH BOLT ALLOWING ENTRY. KEY OVERRIDE AVAILABLE.
3. FROM EGRESS SIDE, REQUEST TO EXIT (RX) IN PANIC DEVICE PUSH PAD WILL BYPASS DOOR CONTACT SIGNALING AUTHORIZED EGRESS. FREE EGRESS AT ALL TIMES.
4. CONDUIT, WIRING, POWER, AND ACCESS CONTROL SYSTEM INTERFACE AND REQUIREMENTS BY DIV 26/28.

END OF SECTION 08 7100

# SECTION 08 8000 - GLAZING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes:

1. Glass products.
2. Laminated glass.
3. Insulating glass.
4. Glazing sealants.
5. Glazing tapes.
6. Miscellaneous glazing materials.

B. Related Requirements:

1. Section 08 4113 "Aluminum-Framed Entrances and Storefronts".
2. Section 08 5113 “Aluminum Windows”.

### 1.3 DEFINITIONS

A. Glass Manufacturers: Firms that produce primary glass, fabricated glass, or both, as defined in referenced glazing publications.

B. Glass Thicknesses: Indicated by thickness designations in millimeters according to ASTM C 1036.

C. IBC: International Building Code.

D. Interspace: Space between lites of an insulating-glass unit.

### 1.4 COORDINATION

A. Coordinate glazing channel dimensions to provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances to achieve proper safety margins for glazing retention under each design load case, load case combination, and service condition.

### 1.5 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

1. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.

2. Review temporary protection requirements for glazing during and after installation.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Glass Samples: For each type of glass product other than clear monolithic vision glass; 12 inches square.

C. Glazing Schedule: List glass types and thicknesses for each size opening and location. Use same designations indicated on Drawings.

1.7 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer, manufacturers of insulating-glass units with sputter-coated, low-E coatings, glass testing agency and sealant testing agency.

B. Product Certificates: For glass.

C. Product Test Reports: For coated glass, insulating glass and glazing sealants, for tests performed by a qualified testing agency.

1. For glazing sealants, provide test reports based on testing current sealant formulations within previous 36-month period.

D. Preconstruction adhesion and compatibility test report.

E. Sample Warranties: For special warranties.

1.8 QUALITY ASSURANCE

A. Manufacturer Qualifications for Insulating-Glass Units with Sputter-Coated, Low-E Coatings: A qualified insulating-glass manufacturer who is approved and certified by coated-glass manufacturer.

B. Installer Qualifications: A qualified installer who employs glass installers for this Project who are certified under the National Glass Association's Certified Glass Installer Program.

C. Glass Testing Agency Qualifications: A qualified independent testing agency accredited according to the NFRC CAP 1 Certification Agency Program.

D. Sealant Testing Agency Qualifications: An independent testing agency qualified according to ASTM C 1021 to conduct the testing indicated.

E. Mockups: Build mockups to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Install glazing in mockups specified in Section 084113 "Aluminum-framed entrances and storefronts" to match glazing systems required for Project, including glazing methods.
2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Protect glazing materials according to manufacturer's written instructions. Prevent damage to glass and glazing materials from condensation, temperature changes, direct exposure to sun, or other causes.

B. Comply with insulating-glass manufacturer's written instructions for venting and sealing units to avoid hermetic seal ruptures due to altitude change.

1.10 FIELD CONDITIONS

A. Environmental Limitations: Do not proceed with glazing when ambient and substrate temperature conditions are outside limits permitted by glazing material manufacturers and when glazing channel substrates are wet from rain, frost, condensation, or other causes.

1. Do not install glazing sealants when ambient and substrate temperature conditions are outside limits permitted by sealant manufacturer or are below 40 deg F.

1.11 WARRANTY

A. Manufacturer's Special Warranty for Laminated Glass: Manufacturer agrees to replace laminated-glass units that deteriorate within specified warranty period. Deterioration of laminated glass is defined as defects developed from normal use that are not attributed to glass breakage or to maintaining and cleaning laminated glass contrary to manufacturer's written instructions. Defects include edge separation, delamination materially obstructing vision through glass, and blemishes exceeding those allowed by referenced laminated-glass standard.

1. Warranty Period: 10 years from date of Substantial Completion.

B. Manufacturer's Special Warranty for Insulating Glass: Manufacturer agrees to replace insulating-glass units that deteriorate within specified warranty period. Deterioration of insulating glass is defined as failure of hermetic seal under normal use that is not attributed to glass breakage or to maintaining and cleaning insulating glass contrary to manufacturer's written instructions. Evidence of failure is the obstruction of vision by dust, moisture, or film on interior surfaces of glass.

1. Warranty Period: 10 years from date of Substantial Completion.

C. Manufacturer’s Special Warranty, Monolithic Polycarbonate Security Glazing: Manufacturer agrees to replace monolithic polycarbonate security glazing that fails in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Defects developed from normal use that are not attributed to maintaining and cleaning monolithic polycarbonate security glazing contrary to manufacturer's written instructions. Defects include yellowing and loss of light transmission.

2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations for Glass: Obtain tinted and coated glass from single source from single manufacturer.

B. Source Limitations for Glazing Accessories: For each product and installation method, obtain from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. General: Installed glazing systems shall withstand normal thermal movement and wind and impact loads (where applicable) without failure, including loss or glass breakage attributable to the following: defective manufacture, fabrication, or installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; or other defects in construction.

B. Structural Performance: Glazing shall withstand the following design loads within limits and under conditions indicated determined according to the IBC and ASTM E 1300.

1. Design Wind Pressures: Determine design wind pressures applicable to Project according to ASCE/SEI 7, based on heights above grade indicated on Drawings.

    a. Wind Design Data: As indicated on Drawings.
    b. Basic Wind Speed: 111 mph.
    c. Exposure Category: C.

2. Design Snow Loads: As indicated on Drawings.
3. Probability of Breakage for Sloped Glazing: For glass surfaces sloped more than 15 degrees from vertical, design glass for a probability of breakage not greater than 0.001.
4. Maximum Lateral Deflection: For glass supported on all four edges, limit center-of-glass deflection at design wind pressure to not more than 1/50 times the short-side length or 1 inch, whichever is less.
5. Thermal Loads: Design glazing to resist thermal stress breakage induced by differential temperature conditions and limited air circulation within individual glass lites and insulated glazing units.

C. Safety Glazing: Where safety glazing is indicated, provide glazing that complies with 16 CFR 1201, Category II.

D. Thermal and Optical Performance Properties: Provide glass with performance properties specified, as indicated in manufacturer's published test data, based on procedures indicated below:

1. For monolithic glass lites, properties are based on units with lites 6 mm thick.
2. For insulating-glass units, properties are based on units of thickness indicated for overall unit and for each lite.
3. U-Factors: Center-of-glazing values, according to NFRC 100 and based on LBL's WINDOW 5.2 computer program, expressed as Btu/sq. ft. x h x deg F.
4. Solar Heat-Gain Coefficient and Visible Transmittance: Center-of-glazing values, according to NFRC 200 and based on LBL's WINDOW 5.2 computer program.
5. Visible Reflectance: Center-of-glazing values, according to NFRC 300.

2.3 GLASS PRODUCTS, GENERAL

A. Glazing Publications: Comply with published recommendations of glass product manufacturers and organizations below unless more stringent requirements are indicated. See these publications for glazing terms not otherwise defined in this Section or in referenced standards.

1. GANA Publications: Glazing Manual."
2. AAMA Publications: AAMA GDSG-1, "Glass Design for Sloped Glazing," and AAMA TIR A7, "Sloped Glazing Guidelines."
3. IGMA Publication for Sloped Glazing: IGMA TB-3001, "Guidelines for Sloped Glazing."
4. IGMA Publication for Insulating Glass: SIGMA TM-3000, "North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial and Residential Use."

B. Safety Glazing Labeling: Where safety glazing is indicated, permanently mark glazing with certification label of the SGCC or another certification agency acceptable to authorities having jurisdiction or manufacturer. Label shall indicate manufacturer's name, type of glass, thickness, and safety glazing standard with which glass complies.

C. Insulating-Glass Certification Program: Permanently marked either on spacers or on at least one component lite of units with appropriate certification label of IGCC.

D. Thickness: Where glass thickness is indicated, it is a minimum.

1. Minimum Glass Thickness for Exterior Lites: 6 mm.
2. Thickness of Tinted Glass: Provide same thickness for each tint color indicated throughout Project.

E. Strength: Where annealed float glass is indicated, provide annealed float glass, heat-strengthened float glass, or fully tempered float glass. Where heat-strengthened float glass is indicated, provide heat-strengthened float glass or fully tempered float glass as needed to comply with "Performance Requirements" Article. Where fully tempered float glass is indicated, provide fully tempered float glass.

2.4 GLASS PRODUCTS

A. Ultraclear Float Glass: ASTM C 1036, Type I, Class I (clear), Quality-Q3; and with visible light transmission of not less than 91 percent.

B. Fully Tempered Float Glass: ASTM C 1048, Kind FT (fully tempered), Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.

1. Fabrication Process: By horizontal (roller-hearth) process with roll-wave distortion parallel to bottom edge of glass as installed unless otherwise indicated.

C. Heat-Strengthened Float Glass: ASTM C 1048, Kind HS (heat strengthened), Type I, Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.

1. Fabrication Process: By horizontal (roller-hearth) process with roll-wave distortion parallel to bottom edge of glass as installed unless otherwise indicated.

D. Silicone-Coated Spandrel Glass: ASTM C 1048, Type I, Condition C, Quality-Q3

E. Reflective- and Low-E-Coated Spandrel Glass: ASTM C1376, Kind CS.

F. Transparent Mirror (One Way Vision Glass):

1. ASTM C1036, Type I, Class 1, Quality q2 or Class 3, Quality q3; Grey Glass.
2. Thickness: 1/4 inch.
3. Coated one face with a hard adherent reflective film of chromium or other coating of proven equivalent durability.
4. Visible light transmittance; eight percent, plus or minus two percent.
5. Visible reflectance; sixty percent, plus or minus five percent.
6. Light ratio; mirror side 10 or more; observer side one or less.
7. Assemble with coating covered and protected with a layer of clear glass not less than 3 mm (1.8 inch) thick.
8. Clean interface glass prior to assembly.
9. Tape edge to seal interface and hold panes together.

## 2.5 LAMINATED GLASS

A. Laminated Glass: ASTM C 1172. Use materials that have a proven record of no tendency to bubble, discolor, or lose physical and mechanical properties after fabrication and installation.

1. Construction: Laminate glass with polyvinyl butyral interlayer to comply with interlayer manufacturer's written instructions.
2. Interlayer Thickness: Provide thickness not less than that indicated and as needed to comply with requirements.
3. Interlayer Color: Clear unless otherwise indicated.

## 2.6 INSULATING GLASS

A. Insulating-Glass Units: Factory-assembled units consisting of sealed lites of glass separated by a dehydrated interspace, qualified in accordance with ASTM E2190.

1. Sealing System: Dual seal, with manufacturer's standard primary and secondary sealants.
2. Perimeter Spacer: Manufacturer's standard spacer material and construction.

a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include:

1) AGC Glass Company North America, Inc.
2) Guardian Industries Corp.; SunGuard.
3) Pilkington North America.
4) Saint-Gobain Glass Corp.
5) Viracon, Inc.
6) Vitro Architectural Glass

3. Desiccant: Molecular sieve or silica gel, or a blend of both.

## 2.7 GLAZING SEALANTS

A. General:

1. Compatibility: Compatible with one another and with other materials they contact, including glass products, seals of insulating-glass units, and glazing channel substrates,

under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.

2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
3. Sealant shall have a VOC content of 250 g/L or less.
4. Colors of Exposed Glazing Sealants: As selected by Architect from manufacturer's full range.

B. Glazing Sealant: Neutral-curing silicone glazing sealant complying with ASTM C 920, Type S, Grade NS, Class 100/50, Use NT.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

   a. Dow Corning Corporation.
   b. GE Construction Sealants; Momentive Performance Materials Inc.
   c. Pecora Corporation.
   d. Sika Corporation.
   e. Tremco Incorporated.

2. Applications: Perimeter joints between materials and frames of doors, windows and louvers.

## 2.8 GLAZING TAPES

A. Back-Bedding Mastic Glazing Tapes: Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C 1281 and AAMA 800 for products indicated below:

1. AAMA 804.3 tape, where indicated.
2. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
3. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

B. Expanded Cellular Glazing Tapes: Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

1. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.
2. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

## 2.9 MISCELLANEOUS GLAZING MATERIALS

A. General: Provide products of material, size, and shape complying with referenced glazing standard, with requirements of manufacturers of glass and other glazing materials for application indicated, and with a proven record of compatibility with surfaces contacted in installation.

B. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

C. Setting Blocks: Elastomeric material with a Shore, Type A durometer hardness of 85, plus or minus 5.

D. Spacers: Elastomeric blocks or continuous extrusions of hardness required by glass manufacturer to maintain glass lites in place for installation indicated.

E. Edge Blocks: Elastomeric material of hardness needed to limit glass lateral movement (side walking).

F. Cylindrical Glazing Sealant Backing: ASTM C 1330, Type O (open-cell material), of size and density to control glazing sealant depth and otherwise produce optimum glazing sealant performance.

2.10 FABRICATION OF GLAZING UNITS

A. Fabricate glazing units in sizes required to fit openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with written instructions of product manufacturer and referenced glazing publications, to comply with system performance requirements.

1. Allow for thermal movements from ambient and surface temperature changes acting on glass framing members and glazing components.

a. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

B. Clean-cut or flat-grind vertical edges of butt-glazed monolithic lites to produce square edges with slight chamfers at junctions of edges and faces.

C. Grind smooth and polish exposed glass edges and corners.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine framing, glazing channels, and stops, with Installer present, for compliance with the following:

1. Manufacturing and installation tolerances, including those for size, squareness, and offsets at corners.
2. Presence and functioning of weep systems.
3. Minimum required face and edge clearances.
4. Effective sealing between joints of glass-framing members.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean glazing channels and other framing members receiving glass immediately before glazing. Remove coatings not firmly bonded to substrates.

B. Examine glazing units to locate exterior and interior surfaces. Label or mark units as needed so that exterior and interior surfaces are readily identifiable. Do not use materials that leave visible marks in the completed Work.

## 3.3 GLAZING, GENERAL

A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.

B. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site and legally dispose of off Project site. Damaged glass includes glass with edge damage or other imperfections that, when installed, could weaken glass, impair performance, or impair appearance.

C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.

D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.

E. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.

F. Provide spacers for glass lites where length plus width is larger than 50 inches.

1. Locate spacers directly opposite each other on both inside and outside faces of glass. Install correct size and spacing to preserve required face clearances, unless gaskets and glazing tapes are used that have demonstrated ability to maintain required face clearances and to comply with system performance requirements.
2. Provide 1/8-inch (3-mm) minimum bite of spacers on glass and use thickness equal to sealant width. With glazing tape, use thickness slightly less than final compressed thickness of tape.

G. Provide edge blocking where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and according to requirements in referenced glazing publications.

H. Set glass lites in each series with uniform pattern, draw, bow, and similar characteristics.

I. Set glass lites with proper orientation so that coatings face exterior, or interior as specified.

## 3.4 TAPE GLAZING

A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.

B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.

C. Cover vertical framing joints by applying tapes to heads and sills first, then to jambs. Cover horizontal framing joints by applying tapes to jambs, then to heads and sills.

D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.

E. Do not remove release paper from tape until right before each glazing unit is installed.

F. Apply heel bead of elastomeric sealant.

G. Center glass lites in openings on setting blocks and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.

H. Apply cap bead of elastomeric sealant over exposed edge of tape.

## 3.5 GASKET GLAZING (DRY)

A. Cut compression gaskets to lengths recommended by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.

B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.

C. Installation with Pressure-Glazing Stops: Center glass lites in openings on setting blocks and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.

D. Install gaskets so they protrude past face of glazing stops.

## 3.6 SEALANT GLAZING (WET)

A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.

B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.

C. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

## 3.7 CLEANING AND PROTECTION

A. Immediately after installation remove nonpermanent labels and clean surfaces.

B. Protect glass from contact with contaminating substances resulting from construction operations. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.

1. If, despite such protection, contaminating substances do come into contact with glass, remove substances immediately as recommended in writing by glass manufacturer. Remove and replace glass that cannot be cleaned without damage to coatings.

C. Remove and replace glass that is damaged during construction period.

D. Wash glass on both exposed surfaces not more than four days before date scheduled for inspections that establish date of Substantial Completion. Wash glass as recommended in writing by glass manufacturer.

## 3.8 MONOLITHIC GLASS SCHEDULE

A. Glass Type (**GL-2**): Clear fully tempered float glass.

1. Minimum Thickness: 0.25 inch (6 mm).
2. Safety glazing required.

## 3.9 INSULATING GLASS SCHEDULE

A. Low-E-Coated, Clear Insulating Glass Type (**GL-1**)

1. Basis-of-Design Product: Subject to compliance with requirements, Provide Vitro Architectural Glass, Solarban 60 (2) Clear + Clear, or equivalent product from available manufacturers offering products that may be incorporated into the Work.

2. Overall Unit Thickness: 1 inch (25 mm).
3. Minimum Thickness of Each Glass Lite: 6 mm.
4. Outdoor Lite: Heat-strengthened float glass.
5. Interspace Content: Air.
6. Indoor Lite: Heat-strengthened float glass.
7. Low-E Coating: Sputtered on second surface.
8. Winter Nighttime U-Factor: 0.29 maximum.
9. Summer Daytime U-Factor: 0.27 maximum.
10. Visible Light Transmittance: 70 percent minimum.
11. SGHC: 0.39 maximum.
12. Safety glazing required.

END OF SECTION 08 8000

# SECTION 08 8853 - SECURITY GLAZING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes:

1. Polycarbonate security glazing.
2. Security window film.

B. Related Requirements:

1. Section 08 5653 "Security Windows."

### 1.2 DEFINITIONS

A. Glazing Manufacturers: Firms that produce primary glass, monolithic plastic glazing, or fabricated security glazing, as defined in referenced glazing publications.

B. Interspace: Space between lites of air-gap security glazing or insulating security glazing.

### 1.3 COORDINATION

A. Coordinate glazing channel dimensions to provide necessary bite on security glazing, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples for Verification:

1. Glazing: Actual sample of finished products for each type of security glazing.

a. Size: Manufacturers' standard size.

C. Security Glazing Schedule: List security glazing types and thicknesses for each size opening and location. Use same designations indicated on Drawings. Indicate coordinated dimensions of security glazing and construction that receives security glazing, including clearances and glazing channel dimensions.

### 1.5 INFORMATIONAL SUBMITTALS

A. Test and Evaluation Reports:

1. Product Test Reports:

a. For each type of security glazing, for tests performed by manufacturer and witnessed by a qualified testing agency.
b. For each type of glazing sealant, for tests performed by a qualified testing agency.

1) Provide test reports based on testing current sealant formulations within previous 36-month period.

B. Qualification Statements: For installers.

C. Sample warranties.

1.6 QUALITY ASSURANCE

A. Qualifications:

1. Installers: Entity that employs installers and supervisors who are trained and approved by manufacturer or Authorized representative who is trained and approved by manufacturer.
2. Security Glazing Testing Agency: Subject to compliance with requirements, testing agency is one of the following:

a. Intertek.
b. Underwriters Laboratories, Inc.
c. Wiss, Janney, Elstner Associates, Inc.

3. Sealant Testing Agency: An independent testing agency qualified in accordance with ASTM C1021 to conduct the testing indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Protect security glazing and glazing materials according to manufacturer's written instructions. Prevent damage from condensation, temperature changes, direct exposure to sun, or other causes.

1.8 FIELD CONDITIONS

A. Environmental Limitations: Do not proceed with glazing when ambient and substrate temperature conditions are outside limits permitted by glazing material manufacturers and when glazing channel substrates are wet from rain, frost, condensation, or other causes.

1. Do not install glazing sealants when ambient and substrate temperature conditions are outside limits permitted by sealant manufacturer or below 40 deg F (4.4 deg C).

1.9 WARRANTY

A. Special Warranty, Glass-Clad and Polycarbonate Security Glazing: Manufacturer agrees to replace glass-clad polycarbonate security glazing that fails in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Defects developed from normal use that are not attributed to glass breakage or to maintaining and cleaning glass-clad polycarbonate security glazing contrary to manufacturer's written instructions. Defects include edge separation, delamination materially obstructing vision through glazing, blemishes exceeding those allowed by referenced polycarbonate standard, yellowing, and loss of light transmission.

2. Warranty Period: Five years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 SOURCE LIMITATIONS

A. Obtain each type of security glazing from single source from single manufacturer.

B. Obtain glazing gaskets from single source from single manufacturer for each product and installation method.

### 2.2 PERFORMANCE REQUIREMENTS

A. General:

1. Installed security glazing will withstand normal thermal movement and wind and impact loads (where applicable) without failure, including loss or breakage attributable to defective manufacture, fabrication, or installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; or other defects in construction.
2. Installed security glazing will withstand security-related loads and forces without damage to the glazing beyond that allowed by referenced standards.

B. Structural Performance: Glazing will withstand the following design loads within limits and under conditions indicated.

1. Design Procedure for Glass: ASTM E1300 and the IBC.

C. Safety Glazing: Where safety glazing is indicated, provide glazing that complies with 16 CFR 1201, Category II.

### 2.3 SECURITY GLAZING, GENERAL

A. Glazing Publications: Comply with published recommendations of security glazing and glazing material manufacturers and organizations below unless more stringent requirements are indicated. Refer to these publications for glazing terms not otherwise defined in this Section or in referenced standards.

1. AAMA Publications: AAMA GDSG-1 and AAMA TIR-A7.
2. NGA Publications: "Laminated Glazing Reference Manual" and "GANA Glazing Manual."

B. Plastic Glazing Labeling: Identify plastic sheets with appropriate markings of applicable testing and inspecting agency, indicating compliance with required fire-test-response characteristics.

C. Safety Glazing Labeling: Where safety glazing is indicated, permanently mark glazing with certification label of the Safety Glazing Certification Council or manufacturer. Label will indicate manufacturer's name, type of glazing, glass thickness, and safety glazing standard with which glazing complies.

D. Fire-Test-Response Characteristics of Polycarbonate Sheets: As determined by testing polycarbonate sheets identical to those used in security glazing products by a qualified testing agency acceptable to authorities having jurisdiction.

1. Self-ignition temperature of 650 deg F (343 deg C) or more when tested in accordance with ASTM D1929 on plastic sheets in thicknesses indicated for the Work.
2. Smoke-Developed Index of 450 or less when tested in accordance with ASTM E84 or UL 723, or smoke density of 75 or less when tested in accordance with ASTM D2843 on plastic sheets in thicknesses indicated for the Work.
3. Burning extent of 1 inch (25 mm) or less when tested in accordance with ASTM D635 at a nominal thickness of 0.060 inch (1.52 mm) or thickness indicated for the Work.

E. Optical Performance Properties: Provide security glazing with performance properties specified, as indicated in manufacturer's published test data, based on construction products indicated and on procedures indicated below:

1. SHGC and Visible Transmittance: Center-of-glazing values, in accordance with NFRC 200 and based on most current non-beta version of LBL's WINDOW 7.7 computer program.
2. Visible Reflectance: Center-of-glazing values, in accordance with NFRC 300.

## 2.4 POLYCARBONATE SECURITY GLAZING

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

1. Dlubak Specialty Glass Corporation; Consolidated Glass Holdings, Inc.
2. Global Security Glazing; Consolidated Glass Holdings, Inc.
3. OldCastle BuildingEnvelope (OBE).
4. Total Security Solutions.

B. Glass-Clad Polycarbonate Security Glazing: ASTM C1349. One or more cores of polycarbonate sheet, clad with glass, bonded with clear urethane interlayer. Use materials that have a proven record of no tendency to bubble, discolor, or lose physical and mechanical properties after fabrication and installation.

## 2.5 SPALL-RESISTANT FILM

A. Composite of clear polyvinyl butyral film and clear abrasion-resistant polyester film.

B. Laminating Process: Factory laminate spall-resistant film to glazing assemblies to produce laminated lites free of foreign substances, air, and glass pockets.

## 2.6 SECURITY WINDOW FILM

A. Composite of clear polyvinyl butyral film and clear abrasion-resistant polyester film.

1. Manufacturers: Subject to compliance with requirements, provide products by the following or approved equivalent:

    a. Riotlite by Impact Security.
    b. 3M

B. Process: Adhere security window film to glazing assemblies to produce laminated lites free of foreign substances, air, and glass pockets.

## 2.7 GLAZING SEALANTS

A. General:

1. Compatibility: Provide glazing sealants that are compatible with one another and with other materials they contact, including security glazing, seals of insulating security glazing and air-gap security glazing, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
2. Suitability: Comply with sealant and security glazing manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
3. Colors of Exposed Glazing Sealants: As indicated by manufacturer's designations.

B. Glazing Sealant:

C. Security Sealant: Manufacturer's standard, nonsag, tamper-resistant sealant for joints with low movement complying with ASTM C920, Grade NS, Class 25, Use NT, and with a Shore A hardness of at least 45 when tested in accordance with ASTM C661.

## 2.8 GLAZING TAPES

A. Back-Bedding Mastic Glazing Tapes: Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and security glazing manufacturers for application indicated; and complying with ASTM C1281 and AAMA 800 for products indicated below:

1. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

## 2.9 MISCELLANEOUS GLAZING MATERIALS

A. General: Provide products of material, size, and shape complying with referenced glazing standard, recommended in writing by manufacturers of security glazing and other glazing materials for application indicated, and with a proven record of compatibility with surfaces contacted in installation.

B. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

C. Setting Blocks:

1. EPDM with Shore A durometer hardness of 85, plus or minus 5.

2. Type recommended in writing by sealant or glass manufacturer.

D. Spacers:

1. Neoprene blocks or continuous extrusions of hardness required by security glazing manufacturer to maintain security glazing lites in place for installation indicated.
2. Type recommended in writing by sealant or security glazing manufacturer.

E. Edge Blocks:

1. EPDM with Shore A durometer hardness in accordance with manufacturer's written instructions.
2. Type recommended in writing by sealant or security glazing manufacturer.

## 2.10 FABRICATION OF SECURITY GLAZING

A. Fabricate security glazing in sizes required to fit openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with written instructions of product manufacturer and referenced glazing publications, to comply with system performance requirements.

B. Grind smooth and polish exposed security glazing edges and corners.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine framing for security glazing, with Installer present, for compliance with the following:

1. Manufacturing and installation tolerances, including those for size, squareness, and offsets at corners.
2. Presence and functioning of weep system.
3. Minimum required face or edge clearances.
4. Minimum required bite.
5. Effective sealing between joints of framing members.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 PREPARATION

A. Clean glazing channels and other framing members receiving security glazing immediately before glazing. Remove coatings not firmly bonded to substrates.

## 3.3 GLAZING, GENERAL

A. Comply with combined written instructions of manufacturers of security glazing, sealants, gaskets, and other glazing materials unless more stringent requirements are indicated, including those in referenced glazing publications.

B. Protect edges of security glazing from damage during handling and installation. Remove damaged security glazing from Project site and legally dispose of it off Project site. Damaged security glazing includes units with edge or face damage or other imperfections that, when installed, could weaken security glazing and impair performance and appearance.

C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.

D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications unless otherwise required by glazing unit manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.

E. Do not exceed edge pressures stipulated by security glazing manufacturers for installing lites.

F. Provide spacers for security glazing lites where the length plus width is larger than 50 inches (1270 mm).

1. Locate spacers directly opposite each other on both inside and outside faces of security glazing. Install correct size and spacing to preserve required face clearances unless gaskets and glazing tapes are used that have demonstrated ability to maintain required face clearances and to comply with performance requirements.
2. Provide 1/8-inch (3-mm) minimum bite of spacers on glazing lites and use thickness equal to sealant width. With glazing tape, use thickness of slightly less than final compressed thickness of tape.

G. Provide edge blocking where indicated or needed to prevent security glazing from moving sideways in glazing channel, as recommended in writing by security glazing manufacturer and in accordance with requirements in referenced glazing publications.

H. Set security glazing in each series with uniform pattern, draw, bow, and similar characteristics.

I. Set coated security glazing with proper orientation so that coatings and films face exterior or interior as specified.

## 3.4 TAPE GLAZING

A. Position tapes on fixed stops so that, when compressed by security glazing, their exposed edges are flush with or protrude slightly above sightline of stops.

B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.

C. Cover vertical framing joints by applying tapes to heads and sills first and then to jambs. Cover horizontal framing joints by applying tapes to jambs and then to heads and sills.

D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.

E. Do not remove release paper from tape until just before each glazing unit is installed.

F. Apply heel bead of elastomeric sealant.

G. Center security glazing in openings on setting blocks and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.

3.5 GASKET GLAZING (DRY)

A. Cut compression gaskets to lengths recommended in writing by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.

B. Insert soft compression gasket between glazing unit and frame or fixed stop, so it is securely in place with joints miter cut and bonded together at corners.

C. Installation with Drive-in Wedge Gaskets: Center security glazing in openings on setting blocks and press firmly against soft compression gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weathertight seal without developing bending stresses in security glazing. Seal gasket joints with sealant recommended in writing by gasket manufacturer.

D. Installation with Pressure-Glazing Stops: Center security glazing in openings on setting blocks and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending stresses in security glazing. Seal gasket joints with sealant recommended in writing by gasket manufacturer.

E. Install gaskets so they protrude past face of glazing stops.

3.6 SEALANT GLAZING (WET)

A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between security glazing and glazing stops to maintain face clearances and to prevent sealant from extruding into glazing channel and blocking weep systems. Secure spacers, or spacers and backings, in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.

B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to security glazing and channel surfaces.

C. Tool exposed surfaces of sealants to provide a substantial wash away from security glazing.

3.7 CLEANING AND PROTECTION

A. Immediately after installation remove nonpermanent labels and clean surfaces.

B. Protect security glazing from contact with contaminating substances resulting from construction operations, including weld splatter. Examine security glazing surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.

1. If, despite such protection, contaminating substances do contact with security glazing, remove substances immediately as recommended in writing by security glazing manufacturer. Remove and replace security glazing that cannot be cleaned without damage.

3.8 GLASS-CLAD POLYCARBONATE SECURITY GLAZING SCHEDULE

A. Security Glazing (GL-3): Clear symmetrical glass-clad polycarbonate.

1. Forced-Entry Resistance, ASTM F1233: Class 2.0 in accordance with ASTM F1233.
2. Forced-Entry Resistance, HPW-TP-0500.03: Level III in accordance with HPW-TP-0500.03.
3. Ballistic Resistance, UL 752: Minimum Level 3 in accordance with UL 752.
4. Maximum Overall Unit Thickness: 1-1/4-inch (32 mm).
5. Multiple Core:
   a. Inner Core Plies: Polycarbonate.
6. Outer Ply: 6 mm heat-strengthened float glass.
7. Interlayer Material: Polyurethane.
8. Provide safety glazing labeling.

3.9 SECURITY WINDOW FILM

A. Security Window Film:

1. Forced-Entry Resistance, ASTM F3561: Class 3.0 in accordance with ASTM F3561.
2. Forced-Entry Resistance, ASTM F588: Type D, Grade 40.

END OF SECTION 08 8853

# SECTION 09 2216 - NON-STRUCTURAL METAL FRAMING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Non-load-bearing steel framing systems for interior partitions.
2. Suspension systems for interior ceilings and soffits.
3. Grid suspension systems for gypsum board ceilings.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For recycled content, indicating postconsumer and preconsumer recycled content and cost.

### 1.4 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of code-compliance certification for studs and tracks.

B. Evaluation Reports: For embossed steel studs and tracks, firestop tracks, post-installed anchors and power-actuated fasteners, from ICC-ES or other qualified testing agency acceptable to authorities having jurisdiction.

### 1.5 QUALITY ASSURANCE

A. Code-Compliance Certification of Studs and Tracks: Provide documentation that framing members are certified according to the product-certification program of the Certified Steel Stud Association, the Steel Framing Industry Association or the Steel Stud Manufacturers Association.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics: For fire-resistance-rated assemblies that incorporate non-load-bearing steel framing, provide materials and construction identical to those tested in assembly indicated, according to ASTM E 119 by an independent testing agency.

B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated on Drawings, according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.

2.2 FRAMING SYSTEMS

A. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

B. Framing Members, General: Comply with ASTM C 754 for conditions indicated.

1. Steel Sheet Components: Comply with ASTM C 645 requirements for metal unless otherwise indicated.
2. Protective Coating: ASTM A 653/A 653M, G40 (Z120), hot-dip galvanized unless otherwise indicated.

C. Studs and Tracks: ASTM C 645. Use either steel studs and tracks or embossed steel studs and tracks.

1. Steel Studs and Tracks:

a. Minimum Base-Metal Thickness: 0.0329 inch (0.836 mm).
b. Depth: As indicated on Drawings.

D. Slip-Type Head Joints: Where indicated, provide one of the following:

1. Single Long-Leg Track System: ASTM C 645 top track with 2-inch- (51-mm-) deep flanges in thickness not less than indicated for studs, installed with studs friction fit into top track and with continuous bridging located within 12 inches (305 mm) of the top of studs to provide lateral bracing.
2. Double-Track System: ASTM C 645 top outer tracks, inside track with 2-inch- (51-mm-) deep flanges in thickness not less than indicated for studs and fastened to studs, and outer track sized to friction-fit over inner track.
3. Deflection Track: Steel sheet top track manufactured to prevent cracking of finishes applied to interior partition framing resulting from deflection of structure above; in thickness not less than indicated for studs and in width to accommodate depth of studs.

a. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1) ClarkDietrich Building Systems.
2) MarinoWARE.
3) MBA Building Supplies.
4) SCAFCO Steel Stud Company.
5) Steel Network, Inc. (The).

E. Firestop Tracks: Top track manufactured to allow partition heads to expand and contract with movement of structure while maintaining continuity of fire-resistance-rated assembly indicated; in thickness not less than indicated for studs and in width to accommodate depth of studs.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ClarkDietrich Building Systems.
b. MarinoWARE.
c. SCAFCO Steel Stud Company.
d. Steel Network, Inc. (The).

F. Flat Strap and Backing Plate: Steel sheet for blocking and bracing in length and width indicated.

1. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm)

G. Cold-Rolled Channel Bridging: Steel, 0.0538-inch (1.367-mm) minimum base-metal thickness, with minimum 1/2-inch- (13-mm-) wide flanges.

1. Depth: 1-1/2 inches (38 mm).

2. Clip Angle: Not less than 1-1/2 by 1-1/2 inches (38 by 38 mm), 0.068-inch- (1.72-mm-) thick, galvanized steel.

H. Hat-Shaped, Rigid Furring Channels: ASTM C 645.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ClarkDietrich Building Systems.
b. MarinoWARE.
c. MRI Steel Framing, LLC.
d. SCAFCO Steel Stud Company.

2. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm).

3. Depth: As indicated on Drawings.

I. Resilient Furring Channels: 1/2-inch- (13-mm-) deep, steel sheet members designed to reduce sound transmission.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. ClarkDietrich Building Systems.
b. MarinoWARE.
c. MRI Steel Framing, LLC.
d. SCAFCO Steel Stud Company.

2. Configuration: Asymmetrical or hat shaped.

J. Cold-Rolled Furring Channels: 0.053-inch (1.34-mm) uncoated-steel thickness, with minimum 1/2-inch- (13-mm-) wide flanges.

1. Depth: As indicated on Drawings.
2. Furring Brackets: Adjustable, corrugated-edge-type steel sheet with minimum uncoated-steel thickness of 0.0329 inch (0.8 mm).
3. Tie Wire: ASTM A 641/A 641M, Class 1 zinc coating, soft temper, 0.062-inch- (1.59-mm-) diameter wire, or double strand of 0.048-inch- (1.21-mm-) diameter wire.

K. Z-Shaped Furring: With slotted or nonslotted web, face flange of 1-1/4 inches (32 mm), wall attachment flange of 7/8 inch (22 mm), minimum uncoated-metal thickness of 0.0179 inch (0.455 mm), and depth required to fit insulation thickness indicated.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ClarkDietrich Building Systems.
b. MarinoWARE.
c. MRI Steel Framing, LLC.
d. SCAFCO Steel Stud Company.

## 2.3 SUSPENSION SYSTEMS

A. Tie Wire: ASTM A 641/A 641M, Class 1 zinc coating, soft temper, 0.062-inch- (1.59-mm-) diameter wire, or double strand of 0.048-inch- (1.21-mm-) diameter wire.

B. Hanger Attachments to Concrete:

1. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 or AC30 as appropriate for the substrate.

a. Uses: Securing hangers to structure.
b. Type: Torque-controlled, expansion anchor or adhesive anchor.
c. Material for Interior Locations: Carbon-steel components zinc-plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.
d. Material for Exterior or Interior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 (A1) stainless-steel bolts, ASTM F 593 (ASTM F 738M), and nuts, ASTM F 594 (ASTM F 836M).

2. Power-Actuated Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

C. Wire Hangers: ASTM A 641/A 641M, Class 1 zinc coating, soft temper, 0.16 inch (4.12 mm) in diameter.

D. Flat Hangers: Steel sheet, in size indicated on Drawings by length indicated.

E. Carrying Channels (Main Runners): Cold-rolled, commercial-steel sheet with a base-metal thickness of 0.0538 inch (1.367 mm) and minimum 1/2-inch- (13-mm-) wide flanges.

1. Depth: As indicated on Drawings.

F. Furring Channels (Furring Members):

1. Cold-Rolled Channels: 0.0538-inch (1.367-mm) uncoated-steel thickness, with minimum 1/2-inch- (13-mm-) wide flanges, 3/4 inch (19 mm) deep.
2. Steel Studs and Tracks: ASTM C 645.

a. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm).
b. Depth: As indicated on Drawings.

3. Hat-Shaped, Rigid Furring Channels: ASTM C 645, 7/8 inch (22 mm) deep.

a. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm).

G. Grid Suspension System for Gypsum Board Ceilings: ASTM C 645, direct-hung system composed of main beams and cross-furring members that interlock.

1. Manufacturers: Subject to compliance with requirements, Provide Armstrong Frame All Drywall Grid Systems, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Armstrong World Industries, Inc.
b. CertainTeed Corporation.
c. Chicago Metallic Corporation.
d. United States Gypsum Company.

H. Grid Suspension System for Gypsum Board Ceilings (14’-0” wide or less span): ASTM C 645, direct-hung system composed of main beams and cross-furring members that interlock.

1. Manufacturers: Subject to compliance with requirements, Provide Armstrong Shortspan Drywall Grid System, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Armstrong World Industries, Inc.
b. CertainTeed Corporation.
c. Chicago Metallic Corporation.
d. United States Gypsum Company.

I. Grid Suspension System for Transitions with Elevation Changes and Flush: ASTM C 645, direct-hung system composed of main beams and cross-furring members that interlock.

1. Manufacturers: Subject to compliance with requirements, Provide Armstrong Axiom Transitions, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Armstrong World Industries, Inc.
b. CertainTeed Corporation.

c. Chicago Metallic Corporation.
d. United States Gypsum Company.

2.4 AUXILIARY MATERIALS

A. General: Provide auxiliary materials that comply with referenced installation standards.

1. Fasteners for Steel Framing: Of type, material, size, corrosion resistance, holding power, and other properties required to fasten steel members to substrates.

B. Isolation Strip at Exterior Walls: Provide one of the following:

1. Asphalt-Saturated Organic Felt: ASTM D 226/D 226M, Type I (No. 15 asphalt felt), nonperforated.
2. Foam Gasket: Adhesive-backed, closed-cell vinyl foam strips that allow fastener penetration without foam displacement, 1/8 inch (3.2 mm) thick, in width to suit steel stud size.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and substrates, with Installer present, and including welded hollow-metal frames, cast-in anchors, and structural framing, for compliance with requirements and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Suspended Assemblies: Coordinate installation of suspension systems with installation of overhead structure to ensure that inserts and other provisions for anchorages to building structure have been installed to receive hangers at spacing required to support the Work and that hangers will develop their full strength.

1. Furnish concrete inserts and other devices indicated to other trades for installation in advance of time needed for coordination and construction.

B. Coordination with Sprayed Fire-Resistive Materials:

1. Before sprayed fire-resistive materials are applied, attach offset anchor plates or ceiling tracks to surfaces indicated to receive sprayed fire-resistive materials. Where offset anchor plates are required, provide continuous plates fastened to building structure not more than 24 inches (610 mm) o.c.
2. After sprayed fire-resistive materials are applied, remove them only to extent necessary for installation of non-load-bearing steel framing. Do not reduce thickness of fire-resistive materials below that are required for fire-resistance ratings indicated. Protect adjacent fire-resistive materials from damage.

3.3 INSTALLATION, GENERAL

A. Installation Standard: ASTM C 754.

1. Gypsum Plaster Assemblies: Also comply with requirements in ASTM C 841 that apply to framing installation.
2. Portland Cement Plaster Assemblies: Also comply with requirements in ASTM C 1063 that apply to framing installation.
3. Gypsum Veneer Plaster Assemblies: Also comply with requirements in ASTM C 844 that apply to framing installation.
4. Gypsum Board Assemblies: Also comply with requirements in ASTM C 840 that apply to framing installation.

B. Install framing and accessories plumb, square, and true to line, with connections securely fastened.

C. Install supplementary framing, and blocking to support fixtures, equipment services, heavy trim, grab bars, toilet accessories, furnishings, or similar construction.

D. Install bracing at terminations in assemblies.

E. Do not bridge building control and expansion joints with non-load-bearing steel framing members. Frame both sides of joints independently.

3.4 INSTALLING FRAMED ASSEMBLIES

A. Install framing system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.

1. Single-Layer Application: 16 inches (406 mm) o.c. unless otherwise indicated.
2. Multilayer Application: 16 inches (406 mm) o.c. unless otherwise indicated.
3. Tile Backing Panels: 16 inches (406 mm) o.c. unless otherwise indicated.

B. Where studs are installed directly against exterior masonry walls or dissimilar metals at exterior walls, install isolation strip between studs and exterior wall.

C. Install studs so flanges within framing system point in same direction.

D. Install tracks at floors and overhead supports. Extend framing full height to structural supports or substrates above suspended ceilings except where partitions are indicated to terminate at suspended ceilings. Continue framing around ducts that penetrate partitions above ceiling.

1. Slip-Type Head Joints: Where framing extends to overhead structural supports, install to produce joints at tops of framing systems that prevent axial loading of finished assemblies.
2. Door Openings: Screw vertical studs at jambs to jamb anchor clips on door frames; install track section (for cripple studs) at head and secure to jamb studs.

    a. Install two studs at each jamb unless otherwise indicated.

b. Install cripple studs at head adjacent to each jamb stud, with a minimum 1/2-inch (13-mm) clearance from jamb stud to allow for installation of control joint in finished assembly.
c. Extend jamb studs through suspended ceilings and attach to underside of overhead structure.

3. Other Framed Openings: Frame openings other than door openings the same as required for door openings unless otherwise indicated. Install framing below sills of openings to match framing required above door heads.
4. Fire-Resistance-Rated Partitions: Install framing to comply with fire-resistance-rated assembly indicated and support closures and to make partitions continuous from floor to underside of solid structure.

a. Firestop Track: Where indicated, install to maintain continuity of fire-resistance-rated assembly indicated.

5. Sound-Rated Partitions: Install framing to comply with sound-rated assembly indicated.
6. Curved Partitions:

a. Bend track to uniform curve and locate straight lengths so they are tangent to arcs.
b. Begin and end each arc with a stud, and space intermediate studs equally along arcs. On straight lengths of no fewer than two studs at ends of arcs, place studs 6 inches (150 mm) o.c.

E. Direct Furring:

1. Screw to wood framing.
2. Attach to concrete or masonry with stub nails, screws designed for masonry attachment, or powder-driven fasteners spaced 24 inches (610 mm) o.c.

F. Z-Shaped Furring Members:

1. Erect insulation, specified in Section 072100 "Thermal Insulation," vertically and hold in place with Z-shaped furring members spaced 24 inches (610 mm) o.c.
2. Except at exterior corners, securely attach narrow flanges of furring members to wall with concrete stub nails, screws designed for masonry attachment, or powder-driven fasteners spaced 24 inches (610 mm) o.c.
3. At exterior corners, attach wide flange of furring members to wall with short flange extending beyond corner; on adjacent wall surface, screw-attach short flange of furring channel to web of attached channel. At interior corners, space second member no more than 12 inches (305 mm) from corner and cut insulation to fit.

G. Installation Tolerance: Install each framing member so fastening surfaces vary not more than 1/8 inch (3 mm) from the plane formed by faces of adjacent framing.

## 3.5 INSTALLING CEILING SUSPENSION SYSTEMS

A. Install suspension system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.

1. Hangers: 48 inches (1219 mm) o.c.
2. Carrying Channels (Main Runners): 48 inches (1219 mm) o.c.
3. Furring Channels (Furring Members): 16 inches (406 mm) o.c.

B. Isolate suspension systems from building structure where they abut or are penetrated by building structure to prevent transfer of loading imposed by structural movement.

C. Suspend hangers from building structure as follows:

1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structural or suspension system.

    a. Splay hangers only where required to miss obstructions and offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.

2. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with locations of hangers required to support standard suspension system members, install supplemental suspension members and hangers in the form of trapezes or equivalent devices.

    a. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced installation standards.

3. Wire Hangers: Secure by looping and wire tying, either directly to structures or to inserts, eye screws, or other devices and fasteners that are secure and appropriate for substrate, and in a manner that will not cause hangers to deteriorate or otherwise fail.
4. Flat Hangers: Secure to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices and fasteners that are secure and appropriate for structure and hanger, and in a manner that will not cause hangers to deteriorate or otherwise fail.
5. Do not attach hangers to steel roof deck.
6. Do not attach hangers to permanent metal forms. Furnish cast-in-place hanger inserts that extend through forms.
7. Do not attach hangers to rolled-in hanger tabs of composite steel floor deck.
8. Do not connect or suspend steel framing from ducts, pipes, or conduit.

D. Fire-Resistance-Rated Assemblies: Wire tie furring channels to supports.

E. Seismic Bracing: Sway-brace suspension systems with hangers used for support.

F. Grid Suspension Systems: Attach perimeter wall track or angle where grid suspension systems meet vertical surfaces. Mechanically join main beam and cross-furring members to each other and butt-cut to fit into wall track.

G. Installation Tolerances: Install suspension systems that are level to within 1/8 inch in 12 feet (3 mm in 3.6 m) measured lengthwise on each member that will receive finishes and transversely between parallel members that will receive finishes.

END OF SECTION 09 2216

SECTION 09 2900 - GYPSUM BOARD

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Interior gypsum board.
2. Exterior gypsum board for ceilings and soffits.

B. Related Requirements:

1. Section 06 1600 "Sheathing" for gypsum sheathing for exterior walls.
2. Section 09 2216 "Non-Structural Metal Framing" for non-structural steel framing and suspension systems that support gypsum board panels.
3. Section 09 3013 "Ceramic Tiling" for cementitious backer units installed as substrates for ceramic tile.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: Show locations and installation of control and expansion joints, including plans, elevations, sections, details of components, and attachments to other work.

1.4 QUALITY ASSURANCE

A. Fire-Test-Response Characteristics: For gypsum board assemblies with fire-resistance ratings, provide materials and construction identical to those tested in assembly indicated according to ASTM E 119, and shall carry a classification label from an independent testing and inspecting agency acceptable to authorities having jurisdiction.

1. Fire-Resistance-Rated Assemblies: Indicated by design designations from UL's "Fire Resistance Directory," or International Accreditation Service certified independent testing agency.

B. Sound Transmission Characteristics: For gypsum board assemblies with STC ratings, provide materials and construction identical to those tested in assembly indicated according to ASTM E90 and classified according to ASTM E413 by a qualified independent testing agency.

C. STC-Rated Assemblies: Indicated by design designations from GA-600, "Fire Resistance Design Manual."

1.5 DELIVERY, STORAGE AND HANDLING

A. Deliver materials in original packages, containers, or bundles bearing brand name and identification of manufacturer or supplier.

B. Store materials inside under cover and keep them dry and protected against weather, condensation, direct sunlight, construction traffic, and other potential causes of damage. Stack panels flat and supported on risers on a flat platform to prevent sagging.

1.6 FIELD CONDITIONS

A. Environmental Limitations: Comply with ASTM C 840 requirements or gypsum board manufacturer's written instructions, whichever are more stringent.

B. Do not install paper-faced gypsum panels until installation areas are enclosed and conditioned.

C. Do not install panels that are wet, moisture damaged, and mold damaged.

1. Indications that panels are wet, or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Resistance-Rated Assemblies: For fire-resistance-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 119 by an independent testing agency.

B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.

2.2 GYPSUM BOARD, GENERAL

A. Size: Provide maximum lengths and widths available that will minimize joints in each area and that correspond with support system indicated.

2.3 INTERIOR GYPSUM BOARD

A. Gypsum Board, Type X: ASTM C 1396/C 1396M.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. American Gypsum.

b. Georgia-Pacific Building Products.
c. National Gypsum Company.
d. United States Gypsum Company.

2. Thickness: 5/8 inch.
3. Long Edges: Tapered and featured (rounded or beveled) for prefilling.
4. Location: Vertical surfaces, unless otherwise indicated.

B. Abuse-Resistant Gypsum Board: ASTM C 1396/C 1396M gypsum board, tested according to ASTM C 1629/C 1629M.

1. Manufacturers: Basis of Design: Gold-Bond Hi-Impact XP Gypsum Board. Subject to compliance with requirements, provide products by one of the following:

a. American Gypsum.
b. Georgia-Pacific Building Products.
c. National Gypsum Company.
d. United States Gypsum Company.

2. Core: Mold, mildew, moisture, and fire-resistant Type X gypsum core with fiberglass mesh embedded.

3. Surface Abrasion: ASTM C 1629/C 1629M, meets or exceeds Level 3 requirements.
4. Indentation: ASTM C 1629/C 1629M, meets or exceeds Level 3 requirements.
5. Soft-Body Impact: ASTM C 1629/C 1629M, meets or exceeds Level 3 requirements.
6. Long Edges: Tapered.
7. Mold Resistance: ASTM D 3273, score of 10 as rated according to ASTM D 3274.

C. Mold-Resistant Gypsum Board: ASTM C 1396/C 1396M. With moisture- and mold-resistant core and paper surfaces.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. American Gypsum.
b. Georgia-Pacific Building Products.
c. National Gypsum Company.
d. United States Gypsum Company.

2. Core: As indicated.
3. Long Edges: Tapered.
4. Mold Resistance: ASTM D 3273, score of 10 as rated according to ASTM D 3274.

## 2.4 SPECIALTY GYPSUM BOARD

A. Acoustically Enhanced Gypsum Board: ASTM C1766. Multilayer products constructed of two layers of gypsum boards sandwiching a viscoelastic sound-absorbing polymer core.

1. Basis-of-Design Product: Subject to compliance with requirements, provide PABCO Gypsum; QuietRock 510 or comparable product by one of the following:

a. CertainTeed; SAINT-GOBAIN.
b. Gold Bond Building Products, LLC provided by National Gypsum Company.
c. PABCO Gypsum.

2. Core: As indicated on drawings.
3. Long Edges: Tapered.

## 2.5 EXTERIOR GYPSUM BOARD FOR CEILINGS AND SOFFITS

A. Exterior Gypsum Soffit Board: ASTM C 1396/C 1396M, with manufacturer's standard edges.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

    a. CertainTeed Corporation.
    b. Georgia-Pacific Building Products.
    c. National Gypsum Company.
    d. United States Gypsum Company.

2. Core: 5/8-inch, Type X.

## 2.6 TRIM ACCESSORIES

A. Interior Trim: ASTM C 1047.

1. Material: Galvanized or aluminum-coated steel sheet, rolled zinc, plastic, or paper-faced galvanized-steel sheet.
2. Shapes:

    a. Cornerbead.
    b. Bullnose bead.
    c. LC-Bead: J-shaped; exposed long flange receives joint compound.
    d. L-Bead: L-shaped; exposed long flange receives joint compound.
    e. U-Bead: J-shaped; exposed short flange does not receive joint compound.
    f. Expansion (control) joint.
    g. Curved-Edge Cornerbead: With notched or flexible flanges.

B. Exterior Trim: ASTM C 1047.

1. Material: Hot-dip galvanized-steel sheet, plastic, or rolled zinc.
2. Shapes:

    a. Cornerbead.
    b. LC-Bead: J-shaped; exposed long flange receives joint compound.
    c. Expansion (Control) Joint: One-piece, rolled zinc with V-shaped slot and removable strip covering slot opening.

## 2.7 JOINT TREATMENT MATERIALS

A. General: Comply with ASTM C 475/C 475M.

B. Joint Tape:

1. Interior Gypsum Board: Paper unless noted otherwise.

2. Interior High-Impact Abuse-Resistant gypsum board – Fiberglass mesh tape on joints, typ.
3. Exterior Gypsum Soffit Board: Paper.
4. Glass-Mat Gypsum Sheathing Board: 10-by-10 glass mesh.
5. Tile Backing Panels: As recommended by panel manufacturer.

C. Joint Compound for Interior Gypsum Board: For each coat, use formulation that is compatible with other compounds applied on previous or for successive coats.

1. Prefilling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting-type taping compound.
2. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use setting-type taping compound.

    a. Use setting-type compound for installing paper-faced metal trim accessories.

3. Fill Coat: For second coat, use setting-type, sandable topping compound.
4. Finish Coat: For third coat, use setting-type, sandable topping compound.
5. Skim Coat: For final coat of Level 5 finish, use setting-type, sandable topping compound.

D. Joint Compound for Exterior Applications:

1. Exterior Gypsum Soffit Board: Use setting-type taping compound and setting-type, sandable topping compound.
2. Glass-Mat Gypsum Sheathing Board: As recommended by sheathing board manufacturer.

E. Joint Compound for Tile Backing Panels:

1. Glass-Mat, Water-Resistant Backing Panel: As recommended by backing panel manufacturer.
2. Cementitious Backer Units: As recommended by backer unit manufacturer.
3. Water-Resistant Gypsum Backing Board: Use setting-type taping compound and setting-type, sandable topping compound.

## 2.8 AUXILIARY MATERIALS

A. General: Provide auxiliary materials that comply with referenced installation standards and manufacturer's written instructions.

B. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.

1. Adhesives shall have a VOC content of 50 g/L or less.

C. Steel Drill Screws: ASTM C 1002 unless otherwise indicated.

1. Use screws complying with ASTM C 954 for fastening panels to steel members from 0.033 to 0.112 inch thick.
2. For fastening cementitious backer units, use screws of type and size recommended by panel manufacturer.

D. Thermal Insulation: As specified in Section 07 2100 "Thermal Insulation."

E. Acoustical Sealant: Manufacturer's standard nonsag, paintable, nonstaining latex sealant complying with ASTM C834. Product effectively reduces airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies in accordance with ASTM E90.

1. Basis-of-Design Product: Subject to compliance with requirements, provide PABCO Gypsum; QuietSeal Pro or comparable product by one of the following:

a. CertainTeed Gypsum.
b. PABCO Building Products, LLC.
c. USG Corporation.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine areas and substrates including welded hollow-metal frames and support framing, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

B. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 APPLYING AND FINISHING PANELS, GENERAL

A. Comply with ASTM C 840.

B. Install ceiling panels across framing to minimize the number of abutting end joints and to avoid abutting end joints in central area of each ceiling. Stagger abutting end joints of adjacent panels not less than one framing member.

C. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch of open space between panels. Do not force into place.

D. Locate edge and end joints over supports, except in ceiling applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Do not place tapered edges against cut edges or ends. Stagger vertical joints on opposite sides of partitions. Do not make joints other than control joints at corners of framed openings.

E. Form control and expansion joints with space between edges of adjoining gypsum panels.

F. Cover both faces of support framing with gypsum panels in concealed spaces (above ceilings, etc.), except in chases braced internally.

1. Unless concealed application is indicated or required for sound, fire, air, or smoke ratings, coverage may be accomplished with scraps of not less than 8 sq. ft. in area.
2. Fit gypsum panels around ducts, pipes, and conduits.
3. Where partitions intersect structural members projecting below underside of floor/roof slabs and decks, cut gypsum panels to fit profile formed by structural members; allow 1/4- to 3/8-inch-wide joints to install sealant.

G. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments. Provide 1/4- to 1/2-inch-wide spaces at these locations and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.

H. Attachment to Steel Framing: Attach panels so leading edge or end of each panel is attached to open (unsupported) edges of stud flanges first.

I. STC-Rated Assemblies: Seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Comply with ASTM C 919 and with manufacturer's written instructions for locating edge trim and closing off sound-flanking paths around or through assemblies, including sealing partitions above acoustical ceilings.

J. Install sound attenuation blankets before installing gypsum panels unless blankets are readily installed after panels have been installed on one side.

## 3.3 APPLYING INTERIOR GYPSUM BOARD

A. Install interior gypsum board in the following locations:

1. Wallboard Type: As indicated on Drawings.
2. Type X: Where required for fire-resistance-rated assembly.
3. Ceiling Type: As indicated on Drawings.
4. Abuse-Resistant Type: As indicated on Drawings.
5. Impact-Resistant Type: As indicated on Drawings.
6. Mold-Resistant Type: As indicated on Drawings.
7. Glass-Mat Interior Type: As indicated on Drawings.
8. Acoustically Enhanced Type: As indicated on Drawings.

B. Single-Layer Application:

1. On ceilings, apply gypsum panels before wall/partition board application to greatest extent possible and at right angles to framing unless otherwise indicated.
2. On partitions/walls, apply gypsum panels horizontally (perpendicular to framing) unless otherwise indicated or required by fire-resistance-rated assembly, and minimize end joints.

    a. Stagger abutting end joints not less than one framing member in alternate courses of panels.
    b. At stairwells and other high walls, install panels horizontally unless otherwise indicated or required by fire-resistance-rated assembly.

3. On Z-shaped furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.
4. Fastening Methods: Apply gypsum panels to supports with steel drill screws.

C. Multilayer Application:

1. On ceilings, apply gypsum board indicated for base layers before applying base layers on walls/partitions; apply face layers in same sequence. Apply base layers at right angles to framing members and offset face-layer joints one framing member, 16 inches (400 mm) minimum, from parallel base-layer joints, unless otherwise indicated or required by fire-resistance-rated assembly.

2. On partitions/walls, apply gypsum board indicated for base layers and face layers vertically (parallel to framing) with joints of base layers located over stud or furring member and face-layer joints offset at least one stud or furring member with base-layer joints unless otherwise indicated or required by fire-resistance-rated assembly. Stagger joints on opposite sides of partitions.
3. On Z-shaped furring members, apply base layer vertically (parallel to framing) and face layer either vertically (parallel to framing) or horizontally (perpendicular to framing) with vertical joints offset at least one furring member. Locate edge joints of base layer over furring members.
4. Fastening Methods: Fasten base layers with screws; fasten face layers with adhesive and supplementary fasteners.

D. Laminating to Substrate: Where gypsum panels are indicated as directly adhered to a substrate (other than studs, joists, furring members, or base layer of gypsum board), comply with gypsum board manufacturer's written instructions and temporarily brace or fasten gypsum panels until fastening adhesive has set.

## 3.4 APPLYING EXTERIOR GYPSUM PANELS FOR CEILINGS AND SOFFITS

A. Apply panels perpendicular to supports, with end joints staggered and located over supports.

1. Install with 1/4-inch open space where panels abut other construction or structural penetrations.
2. Fasten with corrosion-resistant screws.

## 3.5 INSTALLING TRIM ACCESSORIES

A. General: For trim with back flanges intended for fasteners, attach to framing with same fasteners used for panels. Otherwise, attach trim according to manufacturer's written instructions.

B. Control Joints: Install control joints according to ASTM C 840 and in specific locations approved by Architect for visual effect.

C. Interior Trim: Install in the following locations:

1. Cornerbead: Use at outside corners unless otherwise indicated.
2. Bullnose Bead: Use at outside corners.
3. LC-Bead: Use at exposed panel edges.
4. L-Bead: Use where indicated.
5. U-Bead: Use at exposed panel edges.
6. Curved-Edge Cornerbead: Use at curved openings.

D. Exterior Trim: Install in the following locations:

1. Cornerbead: Use at outside corners.
2. LC-Bead: Use at exposed panel edges.

E. Aluminum Trim: Install in locations indicated on Drawings.

3.6 FINISHING GYPSUM BOARD

A. General: Treat gypsum board joints, interior angles, edge trim, control joints, penetrations, fastener heads, surface defects, and elsewhere as required to prepare gypsum board surfaces for decoration. Promptly remove residual joint compound from adjacent surfaces.

B. Prefill open joints, rounded or beveled edges, and damaged surface areas.

C. Apply joint tape over gypsum board joints, except for trim products specifically indicated as not intended to receive tape.

D. Gypsum Board Finish Levels: Finish panels to levels indicated below and according to ASTM C 840:

1. Level 1: Ceiling plenum areas, concealed areas, and where indicated.
2. Level 2: Panels that are substrate for tile.
3. Level 3: Where indicated on Drawings.
4. Level 4: At panel surfaces that will be exposed to view unless otherwise indicated.

    a. Primer and its application to surfaces are specified in Section 099123 "Interior Painting."

5. Level 5: Where indicated on Drawings.

    a. Primer and its application to surfaces are specified in Section 099123 "Interior Painting."

E. Glass-Mat Gypsum Sheathing Board: Finish according to manufacturer's written instructions for use as exposed soffit board.

F. Glass-Mat Faced Panels: Finish according to manufacturer's written instructions.

G. Clean-Up After Installation: Clean top surface of gypsum board ceilings during installation to remove all debris including but not limited to fasteners, connectors, grid components, wire, and al other loose items that are not part of the completed work.

3.7 PROTECTION

A. Protect adjacent surfaces from drywall compound and promptly remove from floors and other non-drywall surfaces. Repair surfaces stained, marred, or otherwise damaged during drywall application.

B. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.

C. Remove and replace panels that are wet, moisture damaged, and mold damaged.

1. Indications that panels are wet, or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION 09 2900

SECTION 09 3013 - CERAMIC TILING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Porcelain tile.
2. Waterproof membrane for thinset applications.
3. Crack isolation membrane.
4. Metal wall access.
5. Metal edge protection.

B. Related Requirements:

1. Section 07 9200 "Joint Sealants" for sealing of expansion, contraction, control, and isolation joints in tile surfaces.

1.3 DEFINITIONS

A. General: Definitions in the ANSI A108 series of tile installation standards and in ANSI A137.1 apply to Work of this Section unless otherwise specified.

B. Module Size: Actual tile size plus joint width indicated.

C. Face Size: Actual tile size, excluding spacer lugs.

D. Large Format Tile: Tile with at least one edge 15 inches (381 mm) or longer.

1.4 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

1. Review requirements in ANSI A108.01 for substrates and for preparation by other trades.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: Show locations of each type of tile and tile pattern. Show widths, details, and locations of expansion, contraction, control, and isolation joints in tile substrates and finished tile surfaces.

C. Samples for Initial Selection: For grout and accessories involving color selection.

D. Samples for Verification:

1. Full-size units of each type and composition of tile and for each color and finish required. For ceramic mosaic tile in color blend patterns, provide full sheets of each color blend.
2. Full-size units of each type of trim and accessory.
3. Metal edge strips in 6-inch (150-mm) lengths.

1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Master Grade Certificates: For each shipment, type, and composition of tile, signed by tile manufacturer and Installer.

C. Product Certificates: For each type of product.

D. Product Test Reports: For tile-setting and -grouting products.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match and are from same production runs as products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Tile and Trim Units: Furnish quantity of full-size units equal to 3 percent of amount installed for each type, composition, color, pattern, and size indicated.
2. Grout: Furnish quantity of grout equal to 3 percent of amount installed for each type, composition, and color indicated.

1.8 QUALITY ASSURANCE

A. Installer Qualifications:

1. Installer is a five-star member of the National Tile Contractors Association or a Trowel of Excellence member of the Tile Contractors' Association of America.
2. The above qualification can be waived as long as the installer provides at least one installer for Project that has completed the Advanced Certification for Tile Installers (ACT) certification or is Ceramic Tile Education Foundation Certified.
3. Installer's supervisor for Project holds the International Masonry Institute's Foreman Certification.

B. Mockups: Build mockups to verify selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Build mockup of wall tile installation.

    a. Mockup floor and wall tile layout of staff restroom and do not proceed with grout installation until reviewed and approved by Architect and Owner.

2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver and store packaged materials in original containers with seals unbroken and labels intact until time of use. Comply with requirements in ANSI A137.1 for labeling tile packages.

B. Store tile and cementitious materials on elevated platforms, under cover, and in a dry location.

C. Store aggregates where grading and other required characteristics can be maintained and contamination can be avoided.

D. Store liquid materials in unopened containers and protected from freezing.

E. Handle tile that has temporary protective coating on exposed surfaces to prevent coated surfaces from contacting backs or edges of other units. If coating does contact bonding surfaces of tile, remove coating from bonding surfaces before setting tile.

1.10 FIELD CONDITIONS

A. Environmental Limitations: Do not install tile until construction in spaces is complete and ambient temperature and humidity conditions are maintained at the levels indicated in referenced standards and manufacturer's written instructions.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Tile: Obtain tile of each type and color or finish from single source or producer.

1. Obtain tile of each type and color or finish from same production run and of consistent quality in appearance and physical properties for each contiguous area.

B. Accessory Products: Obtain each of the following products specified in this Section from a single manufacturer:

1. Stone thresholds.
2. Backer units.

2.2 MANUFACTURERS

A. Source Limitations for Tile: Obtain tile of each type, manufacturer and color or finish from single source or producer.

1. Obtain tile of each type and color or finish from same production run and of consistent quality in appearance and physical properties for each contiguous area.

B. Source Limitations for Setting and Grouting Materials: Obtain ingredients of a uniform quality for each mortar, adhesive, and grout component from single manufacturer and each aggregate from single source or producer.

1. Obtain setting and grouting materials, except for unmodified Portland cement and aggregate, from single manufacturer.

2. Obtain waterproof membrane and crack isolation membrane, except for sheet products, from manufacturer of setting and grouting materials.

2.3 PRODUCTS, GENERAL

A. ANSI Ceramic Tile Standard: Provide tile that complies with ANSI A137.1 for types, compositions, and other characteristics indicated.

1. Provide tile complying with Standard grade requirements unless otherwise indicated.

B. ANSI Standards for Tile Installation Materials: Provide materials complying with ANSI A108.02, ANSI standards referenced in other Part 2 articles, ANSI standards referenced by TCNA installation methods specified in tile installation schedules, and other requirements specified.

C. Factory Blending: For tile exhibiting color variations within ranges, blend tile in factory and package so tile units taken from one package show same range in colors as those taken from other packages and match approved Samples.

D. Mounting: For factory-mounted tile, provide back- or edge-mounted tile assemblies as standard with manufacturer unless otherwise indicated.

1. Where tile is indicated for installation on exteriors or in wet areas, do not use back- or edge-mounted tile assemblies unless tile manufacturer specifies in writing that this type of mounting is suitable for installation indicated and has a record of successful in-service performance.

2.4 TILE PRODUCTS

A. Porcelain and Mosaic Tile Type: Unglazed and glazed.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. American Olean; a brand of Dal-Tile Corporation.
b. Crossville, Inc.
c. Daltile; a brand of Dal-Tile Corporation.
d. Portobello America.
e. Porcelanosa USA.

B. Basis of Design Products: Subject to compliance with requirements; provide products indicated by manufacturer's designations or approved equivalent.

a. PT-1: Daltile, Volume 1.0, Electric Moss VL79, Matte Finish.
b. PT-2: Daltile, Volume 1.0, Vapor VL63, Matte Finish.
c. PT-3: Pocelanosa USA, Xtone, Liem Grey Nature (Full Slab for Countertop).
d. PT-4: Portobello America, Color Market, Aubergine 25893E.
e. PT-5: Portobello America, Color Market, Industrial Garage 25839E.
f. PT-6: Portobello America, Color Market, Mirtilo 25892E.
g. PTB-1: Daltile, Volume 1.0, Electric Moss VL79, Matte Finish.

2. Certification: Tile certified by the Porcelain Tile Certification Agency.
3. Face Size: As indicated on drawings.
4. Product Use Classification: Interior, Wet (IW).

5. Physical Properties: Chemical resistant when tested with indicated chemicals in accordance with ASTM C650.
6. Tile Color, Glaze, and Pattern: Match Architect's designation on drawing A-114 and A-603 or approved equivalent.
7. Grout Color: As selected by Architect from manufacturer's full range.
8. Precoat with temporary protective coating.

C. Trim Units: Coordinated with sizes and coursing of adjoining flat tile where applicable and matching characteristics of adjoining flat tile. Unless noted otherwise, provide shapes as follows, selected from manufacturer's standard shapes:

1. Base for Portland Cement Mortar Installations: Coved, module size to match adjoining flat tile.
2. External Corners for Thin-Set Mortar Installations: Surface bullnose, same size as adjoining flat tile.
3. Internal Corners: Field-butted square corners. For coved base and cap use angle pieces designed to fit with stretcher shapes.

2.5 WATERPROOF MEMBRANE

A. General: Manufacturer's standard product, selected from the following that complies with ANSI A118.10 and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

B. Polyethylene Sheet: Polyethylene faced on both sides with fleece webbing; 0.008-inch (0.203-mm) nominal thickness.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Schluter Systems L.P. – Kerdi System

C. Fluid-Applied Membrane: Liquid-latex rubber or elastomeric polymer.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ARDEX Americas.
b. H.B. Fuller Construction Products Inc. / TEC.
c. MAPEI Corporation – Mapelastic Aqua Defense.

D. Waterproofing and Tile-Setting Adhesive: One-part, fluid-applied product intended for use as both waterproofing and tile-setting adhesive in a two-step process.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Boiardi Products Corporation; a QEP company.
b. Bostik, Inc.

2. Adhesives shall have a VOC content of 65 g/L or less.

2.6 CRACK ISOLATION MEMBRANE

A. General: Manufacturer's standard product, selected from the following that complies with ANSI A118.12 for standard performance and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

B. Polyethylene Sheet: Polyethylene faced on both sides with fleece webbing; 0.008-inch nominal thickness.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Schluter Systems L.P. – Kerdi System

C. Fluid-Applied Membrane: Liquid-latex rubber or elastomeric polymer.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ARDEX Americas.
b. H.B. Fuller Construction Products Inc. / TEC.
c. MAPEI Corporation – Mapelastic Aqua Defense.

2.7 SETTING MATERIALS

A. Improved Modified Dry-Set Mortar (Thinset): ANSI A118.4, specific for large format tile (Floors).

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ARDEX Americas.
b. Bostik, Inc.
c. Custom Building Products.
d. LATICRETE SUPERCAP, LLC.
e. MAPEI Corporation.
f. Merkrete; a Parex USA, Inc.

2. Provide prepackaged, dry-mortar mix combined with acrylic resin or styrene-butadiene-rubber liquid-latex additive at Project site.
3. For wall applications, provide mortar that complies with requirements for nonsagging mortar in addition to the other requirements in ANSI A118.4.

B. Medium-Bed, Modified Dry-Set Mortar (Walls): Comply with requirements in ANSI A118.4. Provide product that is approved by manufacturer for application thickness of 1/4 inch.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ARDEX Americas.
b. Bostik, Inc.
c. Custom Building Products.
d. LATICRETE SUPERCAP, LLC;
e. MAPEI Corporation.
f. Merkrete; a Parex USA, Inc. brand

2. Provide prepackaged, dry-mortar mix combined with acrylic resin or styrene-butadiene-rubber liquid-latex additive at Project site.

2.8 GROUT MATERIALS

A. Pre-Mixed, Ready-to-Use Grout: Professional-grade, ready-to-use color consistent, non-staining grout, water cleanable, for use with grout joints 1/16” to 1/2”.

1. Products: Subject to compliance with requirements, provide one of the following, or approved equivalent:

a. Bostik, Inc; Quartzlock™ Rapidcure™.
b. Laticrete International, Inc.; Plasma™ Translucent Grout.
c. MAPEI Corporation; Flexcolor™ CQ.

B. Polymer-Modified Glass Tile Grout: ANSI A118.3, premixed high-performance grout for use with glass tile.

1. Products: Subject to compliance with requirements, provide one of the following, or approved equivalent:

a. Bostik, Inc; Dimension® RapidCure™ Grout.
b. Laticrete International, Inc; Plasma™ Translucent Grout.
c. MAPEI Corporation; Flexlucent™.

C. Urethane-Modified Tile Grout: ANSI A118.3, premixed high-performance grout

1. Products: Subject to compliance with requirements, provide one of the following, or approved equivalent:

a. Bostik, Inc; 1900™ Grout.
b. Laticrete International, Inc; Latapoxy™ 210 Grout.
c. MAPEI Corporation; Kerapoxy™.

D. Grout Colors: As selected by Architect from manufacturer's full range.

2.9 MISCELLANEOUS MATERIALS

A. Trowelable Underlayments and Patching Compounds: Latex-modified, portland cement-based formulation provided or approved by manufacturer of tile-setting materials for installations indicated.

B. Vapor-Retarder Membrane: Polyethylene sheeting, ASTM D 4397, 4.0 mils (0.1 mm) thick.

C. Metal Wall Access System: Wall access panel system featuring aluminum brackets with molded casings containing magnets that are clamped to lateral, movable guide shoes, and ferromagnetic metal counter plates.

1. Manufacturers: Subject to compliance with requirements, Provide Schluter-REMA, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

a. Babcock-Davis.
b. Nystrom

c. Schluter Systems L.P.

D. Metal Edge Strips: As specified on the drawings, angle or L-shaped, height to match tile and setting-bed thickness, metallic or combination of metal and PVC or neoprene base, designed specifically for flooring applications; color, material and finish as specified, exposed-edge material.

1. Manufacturers: Subject to compliance with requirements, Provide Schluter-DILEX-AHK, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

a. Blanke Corporation.
b. Ceramic Tool Company, Inc.
c. Schluter Systems L.P.

E. Tile Cleaner: A neutral cleaner capable of removing soil and residue without harming tile and grout surfaces, specifically approved for materials and installations indicated by tile and grout manufacturers.

F. Floor Sealer: Manufacturer's standard product for sealing grout joints and that does not change color or appearance of grout.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Bonsal American, an Oldcastle company.
b. Custom Building Products.
c. Southern Grouts & Mortars, Inc.
d. Summitville Tiles, Inc.

2.10 MIXING MORTARS AND GROUT

A. Mix mortars and grouts to comply with referenced standards and mortar and grout manufacturers' written instructions.

B. Add materials, water, and additives in accurate proportions.

C. Obtain and use type of mixing equipment, mixer speeds, mixing containers, mixing time, and other procedures to produce mortars and grouts of uniform quality with optimum performance characteristics for installations indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions where tile will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

1. Verify that substrates for setting tile are firm; dry; clean; free of coatings that are incompatible with tile-setting materials, including curing compounds and other substances that contain soap, wax, oil, or silicone; and comply with flatness tolerances required by ANSI A108.01 for installations indicated.

2. Verify that installation of grounds, anchors, recessed frames, electrical and mechanical units of work, and similar items located in or behind tile has been completed.
3. Verify that joints and cracks in tile substrates are coordinated with tile joint locations; if not coordinated, adjust joint locations in consultation with Architect.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Fill cracks, holes, and depressions in concrete substrates for tile floors installed with adhesives or thinset mortar with trowelable leveling and patching compound specifically recommended by tile-setting material manufacturer.

B. Where indicated, prepare substrates to receive waterproofing by applying a reinforced mortar bed that complies with ANSI A108.1A and is sloped 1/4 inch per foot (1:50) toward drains.

C. Blending: For tile exhibiting color variations, verify that tile has been factory blended and packaged so tile units taken from one package show same range of colors as those taken from other packages and match approved Samples. If not, factory blended, either return to manufacturer or blend tiles at Project site before installing.

D. Porcelain Slab PT-3:

1. The supplied pieces are not rectified, so they must be rectified by cutting at least 2 cm (6/8”) on each side of the piece, which will release the tension in the slab.
2. To prevent breakage when the blade enters or comes out of the piece, the speed recommended by the manufacturer shall be reduced to half on the initial and final sections in 20/25 cm (77/8”/97/8”) width.

3.3 CERAMIC TILE INSTALLATION

A. Comply with TCNA's "Handbook for Ceramic, Glass, and Stone Tile Installation" for TCNA installation methods specified in tile installation schedules. Comply with parts of the ANSI A108 series "Specifications for Installation of Ceramic Tile" that are referenced in TCNA installation methods, specified in tile installation schedules, and apply to types of setting and grouting materials used.

1. For the following installations, follow procedures in the ANSI A108 series of tile installation standards for providing 95 percent mortar coverage:

a. Tile floors in wet areas.
b. Tile floors consisting of tiles 8 by 8 inches or larger.
c. Tile floors consisting of rib-backed tiles.

B. Extend tile work into recesses and under or behind equipment and fixtures to form complete covering without interruptions unless otherwise indicated. Terminate work neatly at obstructions, edges, and corners without disrupting pattern or joint alignments.

C. Accurately form intersections and returns. Perform cutting and drilling of tile without marring visible surfaces. Carefully grind cut edges of tile abutting trim, finish, or built-in items for straight aligned joints. Fit tile closely to electrical outlets, piping, fixtures, and other penetrations so plates, collars, or covers overlap tile.

D. Provide manufacturer's standard trim shapes where necessary to eliminate exposed tile edges.

E. Where accent tile differs in thickness from field tile, vary setting-bed thickness so that tiles are flush.

F. Jointing Pattern: Lay tile in grid pattern unless otherwise indicated. Lay out tile work and center tile fields in both directions in each space or on each wall area. Lay out tile work to minimize the use of pieces that are less than half of a tile. Provide uniform joint widths unless otherwise indicated.

1. For tile mounted in sheets, make joints between tile sheets same width as joints within tile sheets so joints between sheets are not apparent in finished work.
2. Where adjoining tiles on floor, base, walls, or trim are specified or indicated to be same size, align joints.
3. Where tiles are specified or indicated to be whole integer multiples of adjoining tiles on floor, base, walls, or trim, align joints unless otherwise indicated.

G. Joint Widths: Unless otherwise indicated, install tile with the following joint widths:

1. Ceramic Mosaic Tile: 1/16 inch.
2. Glazed Wall Tile: 1/16 inch.
3. Porcelain Tile: 1/8 inch.

H. Lay out tile wainscots to dimensions indicated or to next full tile beyond dimensions indicated.

I. Expansion Joints: Provide expansion joints and other sealant-filled joints, including control, contraction, and isolation joints, where indicated. Form joints during installation of setting materials, mortar beds, and tile. Do not saw-cut joints after installing tiles.

1. Where joints occur in concrete substrates, locate joints in tile surfaces directly above them.
2. Do not extend waterproofing or crack isolation membrane under thresholds set in standard dry-set or modified dry-set mortar. Fill joints between such thresholds and adjoining tile set on waterproofing or crack isolation membrane with elastomeric sealant.

J. Metal Edge Strips: Install at locations indicated and unless otherwise noted, where exposed edge of tile flooring meets carpet, wood, or other flooring that finishes flush with or below top of tile and no threshold is indicated.

K. Floor Sealer: Apply floor sealer to grout joints in tile floors according to floor-sealer manufacturer's written instructions. As soon as floor sealer has penetrated grout joints, remove excess sealer and sealer from tile faces by wiping with soft cloth.

## 3.4 TILE BACKING PANEL INSTALLATION

A. Install panels and treat joints according to ANSI A108.11 and manufacturer's written instructions for type of application indicated. Use modified dry-set mortar for bonding material unless otherwise directed in manufacturer's written instructions.

## 3.5 WATERPROOFING INSTALLATION

A. Install waterproofing to comply with ANSI A108.13 and manufacturer's written instructions to produce waterproof membrane of uniform thickness that is bonded securely to substrate.

B. Allow waterproofing to cure and verify by testing that it is watertight before installing tile or setting materials over it.

3.6 CRACK ISOLATION MEMBRANE INSTALLATION

A. Install crack isolation membrane to comply with ANSI A108.17 and manufacturer's written instructions to produce membrane of uniform thickness that is bonded securely to substrate.

B. Allow crack isolation membrane to cure before installing tile or setting materials over it.

3.7 ADJUSTING AND CLEANING

A. Remove and replace tile that is damaged or that does not match adjoining tile. Provide new matching units, installed as specified and in a manner to eliminate evidence of replacement.

B. Cleaning: On completion of placement and grouting, clean all ceramic tile surfaces so they are free of foreign matter.

1. Remove grout residue from tile as soon as possible.
2. Clean grout smears and haze from tile according to tile and grout manufacturer's written instructions. Use only cleaners recommended by tile and grout manufacturers and only after determining that cleaners are safe to use by testing on samples of tile and other surfaces to be cleaned. Protect metal surfaces and plumbing fixtures from effects of cleaning. Flush surfaces with clean water before and after cleaning.

3.8 PROTECTION

A. Protect installed tile work with kraft paper or other heavy covering during construction period to prevent staining, damage, and wear. If recommended by tile manufacturer, apply coat of neutral protective cleaner to completed tile walls and floors.

B. Prohibit foot and wheel traffic from tiled floors for at least seven days after grouting is completed.

C. Before final inspection, remove protective coverings and rinse neutral protective cleaner from tile surfaces.

3.9 TILE INSTALLATION SCHEDULE

A. Interior Floor Installations, Concrete Subfloor:

1. Tile Installation TCNA F111: Thin-set mortar.

a. Location: Areas indicated to receive tile, unless otherwise indicated.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout.

2. Tile Installation TCNA F125: Thin-set mortar on bonded waterproofing membrane.

a. Location: Toilet room floors.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout.

B. Interior Wall Installations, Metal Studs or Furring:

1. Tile Installation TCNA W247: Thin-set mortar on coated glass-mat, water-resistant gypsum backer board.

a. Location: Walls indicated to receive tile, unless otherwise indicated.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout.

2. Tile Installation TCNA W245: Thin-set mortar on abuse and impact resistant, coated glass-mat, water-resistant gypsum backer board.

a. Location: Toilet rooms and other wet walls, including those behind plumbing fixtures.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout, unless otherwise indicated

C. Interior Wall Installations, Masonry or Concrete:

1. TCNA W202I: Thinset mortar over waterproof membrane.

a. Location: Areas indicated to receive tile, unless otherwise indicated.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout.

D. Shower Wall Installations:

1. TCNA B419: Thinset mortar over waterproof membrane.

a. Location: Areas indicated to receive tile, unless otherwise indicated.
b. Thin-Set Mortar: Latex-portland cement mortar.
c. Grout: Pre-mixed, urethane grout.

END OF SECTION 09 3013

# SECTION 09 5113 - ACOUSTICAL PANEL CEILINGS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Acoustical panels.
2. Metal suspension system.
3. Metal edge moldings and trim.

B. Section includes acoustical panels and exposed suspension systems for interior ceilings and direct applied acoustical panels for security mesh ceilings

C. Products furnished, but not installed under this Section, include anchors, clips, and other ceiling attachment devices to be cast in concrete.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.3 ACTION SUBMITTALS

A. Product Data:

1. Acoustical panels.
2. Metal edge moldings and trim.

B. Samples for Verification: For each component indicated and for each exposed finish required, prepared on Samples of sizes indicated below:

1. Acoustical Panels: Set of 6-inch- (150-mm-) square samples of each type, color, pattern, and texture.

### 1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the

B. following items are shown and coordinated with each other, using input from installers of the items involved:

1. Ceiling suspension-system members.
2. Structural members to which suspension systems will be attached.
3. Method of attaching hangers to building structure.
4. Carrying channels or other supplemental support for hanger-wire attachment where conditions do not permit installation of hanger wires at required spacing.
5. Size and location of initial access modules for acoustical panels.

6. Items penetrating finished ceiling and ceiling-mounted items including the following:

    a. Lighting fixtures.
    b. Diffusers.
    c. Grilles.
    d. Speakers.
    e. Sprinklers.
    f. Access panels.
    g. Perimeter moldings.

7. Show operation of hinged and sliding components covered by or adjacent to acoustical panels.
8. Minimum Drawing Scale: 1/4 inch = 1 foot (1:48).

C. Qualification Data: For testing agency.

D. Product Test Reports: For each acoustical panel ceiling, for tests performed by manufacturer and witnessed by a qualified testing agency.

E. Evaluation Reports: For each acoustical panel ceiling suspension systemand anchor and fastener type, from ICC-ES.

F. Field quality-control reports.

## 1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For finishes to include in maintenance manuals.

## 1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Acoustical Ceiling Units: Full-size panels equal to 2 percent of quantity installed.
2. Suspension-System Components: Quantity of each exposed component equal to 2 percent of quantity installed.
3. Hold-Down Clips: Equal to 2percent of quantity installed.

## 1.7 QUALITY ASSURANCE

## 1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver acoustical panels, suspension-system components, and accessories to Project site and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

B. Before installing acoustical panels, permit them to reach room temperature and a stabilized moisture content.

1.9 FIELD CONDITIONS

A. Environmental Limitations: Do

B. not install acoustical panel ceilings until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

1. Pressurized Plenums: Operate ventilation system for not less than 48 hours before beginning acoustical panel ceiling installation.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Source Limitations for Ceiling System: Obtain each type of acoustical ceiling panel and its supporting suspension system from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Surface-Burning Characteristics: Comply with ASTM E84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Flame-Spread Index: Class Ain accordance with ASTM E1264.
2. Smoke-Developed Index: 50 or less.

2.3 ACOUSTICAL PANELS (APC-1)

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

B. Basis-of-Design Product: Subject to compliance with requirements, provide Armstrong Ultima High NRC #2080. or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. Armstrong World Industries
2. CertainTeed; SAINT-GOBAIN.
3. USG Corporation.

C. Acoustical Panel Standard: Provide manufacturer's standard panels in accordance with ASTM E1264 and designated by type, form, pattern, acoustical rating, and light reflectance unless otherwise indicated.

D. Color: White.
E. Light Reflectance (LR): Not less than 0.85
F. Ceiling Attenuation Class (CAC): Not less than 35
G. Noise Reduction Coefficient (NRC): Not less than 0.85
H. Edge/Joint Detail: Square Lay-In.
I. Modular Size: 24 by 24 inches (610 by 610 mm)

J. Antimicrobial Treatment: Manufacturer's standard broad spectrum, antimicrobial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested in accordance with ASTM D3273, ASTM D3274, or ASTM G21 and evaluated in accordance with ASTM D3274 or ASTM G21.

2.4 ACOUSTICAL PANELS (APC-2)

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

B. Basis-of-Design Product: Subject to compliance with requirements, provide Armstrong CALL High CAC #8806. or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. Armstrong World Industries
2. CertainTeed; SAINT-GOBAIN.
3. USG Corporation.

C. Acoustical Panel Standard: Provide manufacturer's standard panels in accordance with ASTM E1264 and designated by type, form, pattern, acoustical rating, and light reflectance unless otherwise indicated.

D. Color: White.
E. Light Reflectance (LR): Not less than 0.85
F. Ceiling Attenuation Class (CAC): Not less than 40
G. Noise Reduction Coefficient (NRC): Not less than 0.80
H. Edge/Joint Detail: Square tegular.
I. Modular Size: 24 by 24 inches (610 by 610 mm)

J. Antimicrobial Treatment: Manufacturer's standard broad spectrum, antimicrobial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested in accordance with ASTM D3273, ASTM D3274, or ASTM G21 and evaluated in accordance with ASTM D3274 or ASTM G21.

2.5 DIRECT APPLIED ACOUSTICAL PANEL (SEC-1)

A. Manufacturers: Subject to compliance with requirements, Provide Armstrong Fine Fissured #746, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Armstrong World Industries.
2. CertainTeed; SAINT-GOBAIN.
3. USG Corporation.

B. Acoustical Panel Standard: Provide manufacturer's standard panels in accordance with ASTM E1264 and designated by type, form, pattern, acoustical rating, and light reflectance unless otherwise indicated.

C. Color: White.
D. Light Reflectance (LR): Not less than 0.82.

E. Noise Reduction Coefficient (NRC): Not less than 0.55.
F. Edge/Joint Detail: Beveled, concealed.
G. Modular Size: 12 by 12 inches (304.8 by 304.8 mm).

H. Antimicrobial Treatment: Manufacturer's standard broad spectrum, antimicrobial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested according to ASTM D 3273, ASTM D 3274, or ASTM G 21 and evaluated according to ASTM D 3274 or ASTM G 21.

## 2.6 METAL SUSPENSION SYSTEM (APC-1 and APC-2)

A. Narrow-Face, Capped, Double-Web, Steel Suspension System: Main and cross runners roll formed from cold-rolled steel sheet; prepainted, electrolytically zinc coated, or hot-dip galvanized, G30 (Z90) coating designation; with prefinished 9/16-inch- (15-mm-) wide metal caps on flanges.

B. Basis-of-Design Product: Subject to compliance with requirements, provide Armstrong Prelude XL 15*16”. or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. Armstrong World Industries.
2. CertainTeed; SAINT-GOBAIN.
3. USG Corporation.

C. Metal Suspension-System Standard: Provide manufacturer's standard, direct-hung, metal suspension system and accessories in accordance with ASTM C635/C635M and designated by type, structural classification, and finish indicated.

D. Cap Finish: White.

## 2.7 ACCESSORIES

A. Wire Hangers, Braces, and Ties: Provide wires as follows:

1. Zinc-Coated, Carbon-Steel Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper.
2. Stainless Steel Wire: ASTM A580/A580M, Type 304, nonmagnetic.
3. Nickel-Copper-Alloy Wire: ASTM B164, nickel-copper-alloy UNS No. N04400.
4. Size: Wire diameter sufficient for its stress at three times hanger design load (ASTM C635/C635M, Table 1, "Direct Hung") will be less than yield stress of wire, but not less than 0.106-inch- (2.69-mm-) diameter wire.

B. Hanger Rods: Mild steel, zinc coated or protected with rust-inhibitive paint.

C. Flat Hangers: Mild steel, zinc coated or protected with rust-inhibitive paint.

D. Angle Hangers: Angles with legs not less than 7/8 inch (22 mm) wide; formed with 0.04-inch- (1-mm-) thick, galvanized-steel sheet complying with ASTM A653/A653M, G90 (Z275) coating designation; with bolted connections and 5/16-inch- (8-mm-) diameter bolts.

E. Hold-Down Clips: Manufacturer's standard hold-down.

2.8 METAL EDGE MOLDINGS AND TRIM

A. Roll-Formed, Sheet-Metal Edge Moldings and Trim: Type and profile indicated or, if not indicated, manufacturer's standard moldings for edges and penetrations that comply with seismic design requirements; formed from sheet metal of same material, finish, and color as that used for exposed flanges of suspension-system.

1. Edge moldings to fit acoustical panel edge details and suspension systems indicated and match width and configuration of exposed runners unless otherwise indicated.
2. For lay-in panels with reveal edge details, provide stepped edge molding that forms reveal of same depth and width as that formed between edge of panel and flange at exposed suspension member.
3.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, including structural framing to which acoustical panel ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and with requirements for installation tolerances and other conditions affecting performance of acoustical panel ceilings.

B. Examine acoustical panels before installation. Reject acoustical panels that are wet, moisture damaged, or mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Measure each ceiling area and establish layout of acoustical panels to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width panels at borders unless otherwise indicated, and comply with layout shown on reflected ceiling plans.

B. Layout openings for penetrations centered on the penetrating items.

3.3 INSTALLATION OF ACOUSTICAL PANEL CEILINGS

A. Install acoustical panel ceilings in accordance with ASTM C636/C636Mand manufacturer's written instructions.

B. Suspend ceiling hangers from building's structural members and as follows:

1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
2. Splay hangers only where required and, if permitted with fire-resistance-rated ceilings, to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.

3. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
4. Secure wire hangers to ceiling-suspension members and to supports above with a minimum of three tight turns. Connect hangers directly to structure or to inserts, eye screws, or other devices that are secure and appropriate for substrate and that will not deteriorate or otherwise fail due to age, corrosion, or elevated temperatures.
5. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both the structure to which hangers are attached and the type of hanger involved. Install hangers in a manner that will not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
6. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to cast-in-place hanger inserts, postinstalled mechanical or adhesive anchors, or power-actuated fasteners that extend through forms into concrete.
7. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
8. Do not attach hangers to steel deck tabs.
9. Do not attach hangers to steel roof deck. Attach hangers to structural members.
10. Space hangers not more than 48 inches (1200 mm) o.c. along each member supported directly from hangers unless otherwise indicated; provide hangers not more than 8 inches (200 mm) from ends of each member.
11. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards.

C. Secure bracing wires to ceiling suspension members and to supports with a minimum of four tight turns. Suspend bracing from building's structural members as required for hangers, without attaching to permanent metal

D. forms, steel deck, or steel deck tabs. Fasten bracing wires into concrete with cast-in-place or postinstalled anchors.

E. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical panels.

1. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
2. Screw attach moldings to substrate at intervals not more than 16 inches (400 mm) o.c. and not more than 3 inches (75 mm) from ends. Miter corners accurately and connect securely.
3. Do not use exposed fasteners, including pop rivets, on moldings and trim.

F. Install suspension-system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.

G. Install acoustical panels with undamaged edges and fit accurately into suspension-system runners and edge moldings. Scribe and cut panels at borders and penetrations to provide precise fit.

a. As indicated on reflected ceiling plans.

2. For square-edged panels, install panels with edges fully hidden from view by flanges of suspension-system runners and moldings.

3. For reveal-edged panels on suspension-system runners, install panels with bottom of reveal in firm contact with top surface of runner flanges.
4. For reveal-edged panels on suspension-system members with box-shaped flanges, install panels with reveal surfaces in firm contact with suspension-system surfaces and panel faces flush with bottom face of runners.
5. Paint cut edges of panel remaining exposed after installation; match color of exposed panel surfaces using coating recommended in writing for this purpose by acoustical panel manufacturer.
6. Install hold-downclips in areas indicated; space in accordance with panel manufacturer's written instructions unless otherwise indicated.

    a. Hold-Down Clips: Space 24 inches (610 mm) o.c. on all cross runners.

## 3.4 ERECTION TOLERANCES

A. Suspended Ceilings: Install main and cross runners level to a tolerance of 1/8 inch in 12 feet (3 mm in 3.6 m) , non-cumulative.

B. Moldings and Trim: Install moldings and trim to substrate and level with ceiling suspension system to a tolerance of 1/8 inch in 12 feet (3 mm in 3.6 m) , non-cumulative.

## 3.5 FIELD QUALITY CONTROL

A. Special Inspections: Engage a qualified special inspector to perform the following special inspections:

1. Periodic inspection during the installation of suspended ceiling grids in accordance with ASCE/SEI 7.

B. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

C. Perform the following tests and inspections of completed installations of acoustical panel ceiling hangers and anchors and fasteners in successive stages

D. and when installation of ceiling suspension systems on each floor has reached 20 percent completion, but no panels have been installed. Do not proceed with installations of acoustical panel ceiling hangers for the next area until test results for previously completed installations of acoustical panel ceiling hangers show compliance with requirements.

E. Acoustical panel ceiling hangers, anchors, and fasteners will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports.

## 3.6 CLEANING

A. Clean exposed surfaces of acoustical panel ceilings, including trim, edge moldings, and suspension-system members. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage.

B. Remove and replace ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

END OF SECTION 09 5113

# SECTION 09 5423 - LINEAR METAL CEILINGS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Linear metal ceilings.

### 1.2 COORDINATION

A. Coordinate layout and installation of linear metal pans and suspension system with other construction that penetrates ceilings or is supported by them, including light fixtures, HVAC equipment, fire-suppression system, and partition assemblies.

### 1.3 ACTION SUBMITTALS

A. Product Data: For linear metal ceilings.

B. Shop Drawings: For linear metal ceilings.

1. Include reflected ceiling plans, sections, and details, drawn to scale, showing the following:

a. Linear ceiling patterns and joints.
b. Ceiling suspension members.
c. Method of attaching hangers to building structure and locations of cast-in-place anchors, clips, and other ceiling attachment devices whose installation is specified in other Sections.
d. Ceiling-mounted items including, but not limited to, light fixtures, diffusers, grilles, speakers, sprinklers, and access panels.
e. Ceiling perimeter and penetrations through ceiling; trim and moldings.

C. Samples for Initial Selection: For units with factory-applied colors and finishes.

1. Include Samples of accessories involving color and finish selections.

### 1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For testing agency.

B. Product Test Reports: For each linear metal ceiling, for tests performed by a qualified testing agency.

C. Evaluation Reports: For linear-metal-ceiling framing systems.

D. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For finishes to include in maintenance manuals.

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Linear-Metal-Ceiling Components: Quantity of each pan, carrier, accessory, and exposed molding and trim equal to 2 percent of quantity installed.

1.7 QUALITY ASSURANCE

A. Testing Agency Qualifications: Accredited by National Voluntary Laboratory Accreditation Program for testing indicated.

B. Mockups: Build mockups to verify selections made under Sample submittals and to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Build mockup of each type of linear metal ceiling as shown on Drawings.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver ceiling components and accessories to Project site in original, unopened packages and store them in a fully enclosed, conditioned space where they are protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

B. Handle ceiling components and accessories in a manner that prevents damage.

1.9 PROJECT CONDITIONS

A. Environmental Limitations: Do not install interior ceilings until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, work above ceilings is complete, and HVAC system is operating and maintaining temperature and relative humidity at levels planned for building occupants during the remainder of the construction period.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Structural Performance: Exterior linear metal ceilings to withstand exterior exposure, the effects of gravity loads, and the following loads and stresses without showing permanent deformation of ceiling system components, including pans and suspension system; noise or metal fatigue caused by vibration, deflection, and displacement of ceiling pans; or permanent damage to fasteners and anchors:

B. Seismic Criteria: Provide linear metal ceilings designed and installed to withstand the effects of earthquake motions in accordance with ASCE/SEI 7 and requirements of authorities having jurisdiction.

2.2 LINEAR METAL CEILINGS

A. Pans and Suspension System (VWD-1):

1. Basis-of-Design Product: Subject to compliance with requirements, provide Armstrong Metal Works Linear Classics 5492 Series Metal Ceiling System, or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

   a. Armstrong Ceiling & Wall Solutions.
   b. Rockfon; ROCKWOOL International.
   c. USG Corporation.

B. Metal Pans: Complying with ASTM E1264 for Type XX and formed to snap on to carriers securely, without separate fasteners.

1. Surface-Burning Characteristics: For metal-pan assemblies, including backings, determined by testing in accordance with ASTM E84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

   a. Flame-Spread Index: 25 or less.
   b. Smoke-Developed Index: 50 or less.

2. Metal: Aluminum sheet, ASTM B209 (ASTM B209M), alloy and temper recommended by producer and finisher for type of use and finish indicated.
3. Form: Perforated.

   a. Perforation Pattern: As indicated by manufacturers designation.

4. Noise Reduction Coefficient (NRC) Rating: Not less than 0.70 when tested in accordance with ASTM C423.
5. Backing: Manufacturer's standard to provide NRC rating indicated for perforation pattern indicated.
6. Pan Thickness: Not less than 0.020 inch (0.51 mm).
7. Pan Edge Detail: Manufacturer's standard.
8. Pan Width: 6-inch (152-mm) module width and 5-1/4-inch (133-mm) face width.
9. Pan Depth: As indicated on Drawings.

10. Metal-Pan Finish: Protected on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping and as follows:

    a. Laminated-Film Finish: Provide manufacturer's standard, PVC-free film permanently bonded to metal pan with adhesive.

        1) Color and Pattern: As selected by Architect from manufacturer's full range.

C. Pan Splices: Formed for snap fit into butt-cut pans, 4 inches (102 mm) long.

1. Finish: Matte black.

D. End Caps: Manufacturer's standard material fabricated to fit and conceal exposed ends of pans.

1. Finish: Matching pan.

E. Filler Strips: Manufacturer's standard, fabricated to close voids between pans.

1. Finish: Matching pan.

F. Moldings and Trim: Manufacturer's standard for exposed members, to conceal edges of penetrations through ceiling, to conceal ends of pans and carriers, for fixture trim and adapters, for fasciae at changes in ceiling height, and for other conditions; of metal and finish matching linear metal pans or extruded plastic unless otherwise indicated.

1. For Circular Penetrations of Ceiling: Fabricate edge moldings to diameter required to fit penetration exactly.

G. Carrier Suspension System: Manufacturer's standard complying with requirements in ASTM C635/C635M for applications indicated; complete with carriers, splice sections, stabilizing components, connector clips, alignment clips, leveling clips, hangers, molding, trim, retention clips, load-resisting struts, fixture adapters, and other suspension components required to support ceiling units and other ceiling-supported construction.

1. Material: ASTM A653/A653M, hot-dip galvanized, cold-rolled sheet steel, G60 (Z180) coating designation.
2. Adaptable Carriers: Manufacturer's standard carriers for direct attachment to existing suspended tees.
3. Flexible Radial Carriers: Manufacturer's standard radial carriers.
4. Expansion Carriers: Manufacturer's standard carriers allowing for irregularities or other unusual space conditions.
5. Stabilizer Channels, Tees, and Bars: Manufacturer's standard components for stabilizing main carriers.
6. Carrier Splices: Same metal, profile, and finish as for carriers.
7. Hold-Down Clips: Manufacturer's standard hold-down clips spaced as standard with manufacturer.
8. Carrier Finish: Flat black.

## 2.3 CARRIER-SYSTEM HANGERS, BRACES, AND TIES

A. Attachment Devices: Size for 5 times the design load indicated in ASTM C635/C635M, Table 1, Direct Hung, unless otherwise indicated.

1. Power-Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hangers of type indicated, and with capability to sustain, without failure, a load equal to [10] <Insert safety factor> times that imposed by ceiling construction as determined by testing in accordance with ASTM E1190 conducted by a qualified testing and inspecting agency.

B. Wire Hangers, Braces, and Ties: Provide wire complying with the following requirements:

1. Zinc-Coated, Carbon-Steel Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper.
2. Stainless Steel Wire: ASTM A580/A580M, Type 304, nonmagnetic.
3. Nickel-Copper-Alloy Wire: ASTM B164, nickel-copper-alloy UNS No. N04400.
4. Size: Select wire diameter so its stress at 3 times the hanger design load indicated in ASTM C635/C635M, Table 1, Direct Hung is less than yield stress of wire, but provides not less than 0.106-inch- (2.69-mm-) diameter wire.

C. Rods and Flat Hangers: Mild steel, zinc coated or protected with rust-inhibitive paint.

D. Angle Hangers: Angles with legs not less than 7/8 inch (22 mm) wide; formed from 0.04-inch- (1.0-mm-) thick, galvanized-steel sheet complying with ASTM A653/A653M, G90 (Z275) coating designation; with bolted connections and 5/16-inch- (8-mm-) diameter bolts.

E. Seismic Struts: Suspension-system manufacturer's standard compression struts designed to accommodate seismic forces.

## 2.4 ACCESSORIES

A. Access Panels: For access at locations indicated, provide door hinge assembly, retainer clip, and retainer bar, assembled with ceiling panels and carrier sections into access doors permitting upward or downward opening.

1. Size: As indicated on Drawings, 24 inches (610 mm) square.

B. Air-Distribution Devices: Where indicated on Drawings, provide independently suspended air-distribution devices that are relocatable and adjustable from below finished ceiling, that do not interrupt ceiling components, and that are fully concealed by and integrated with ceiling system.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, including structural framing and substrates to which linear metal ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and with requirements for installation tolerances and other conditions affecting performance of linear metal ceilings.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Measure each ceiling area and establish layout of linear metal pans.

1. Balance border widths at opposite edges of each ceiling.
2. Avoid using less-than-half-width pans at borders.

3.3 INSTALLATION OF LINEAR METAL CEILINGS

A. Comply with ASTM C636/C636M and seismic requirement indicated, in accordance with manufacturer's written instructions and CISCA's "Ceiling Systems Handbook."

B. Suspend ceiling hangers from building's structural members and as follows:

1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
2. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
3. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
4. Secure wire hangers to ceiling suspension members and to supports above with a minimum of three tight turns in 3 inches (76 mm). Connect hangers directly either to structures or to inserts, eye screws, or other devices that are secure and appropriate for substrate to which hangers are attached and for type of hanger involved.
5. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both structure to which hangers are attached and type of hanger involved. Install hangers in a manner that does not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
6. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to cast-in-place hanger inserts, power-actuated fasteners, or postinstalled mechanical or adhesive anchors that extend through forms into concrete.
7. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
8. Do not attach hangers to steel deck tabs.
9. Do not attach hangers to steel roof deck. Attach hangers to structural members.
10. Space hangers not more than 48 inches (1219 mm) o.c. along each member supported directly from hangers unless otherwise indicated; provide hangers not more than 8 inches (203 mm) from ends of each member.
11. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards and publications.

C. Install edge moldings and trim at perimeter of linear metal ceiling area and where necessary to conceal edges and ends of linear metal pans.

1. Screw attach moldings to substrate at intervals of not more than 16 inches (406 mm) o.c. and not more than 3 inches (76 mm) from ends, leveling with ceiling suspension system to a tolerance of 1/8 inch in 12 feet (3 mm in 3.7 m). Miter corners accurately and connect securely.
2. Do not use exposed fasteners, including pop rivets, on moldings and trim.

D. Install suspension-system carriers so they are aligned and securely interlocked with one another.

1. Install stabilizer channels, tees, and bars at regular intervals to stabilize carriers and at light fixtures, air-distribution equipment, access doors, and other equipment; spaced as standard with manufacturer for use indicated.
2. Remove and replace dented, bent, or kinked members.

E. Cut linear metal pans for accurate fit at borders and at interruptions and penetrations by other work through ceilings. Stiffen edges of cut units as required to eliminate evidence of buckling or variations in flatness.

F. Install linear metal pans in coordination with suspension system and exposed moldings and trim.

1. Align joints in adjacent courses to form uniform, straight joints parallel to room axis in both directions unless otherwise indicated on Drawings.
2. Fit adjoining units to form flush, tight joints. Scribe and cut units for accurate fit at borders and around construction penetrating ceiling.

a. Install pans with butt joints aligned as indicated on Drawings using internal pan splices.

3. Install directionally textured or patterned metal pans in directions indicated.
4. Where metal pan ends are visible, install end caps unless trim is indicated.
5. Install filler strips where indicated on Drawings.
6. Install sound-absorbent pads at right angle to perforated metal pans so pads do not hang unsupported.

G. Install hold-down clips where indicated.

## 3.4 CLEANING

A. Clean exposed surfaces of linear metal ceilings, including trim and edge moldings, after removing strippable, temporary protective covering if any. Comply with manufacturer's written instructions for stripping of temporary protective covering, cleaning, and touchup of minor finish damage. Remove and replace ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage, including dented and bent units.

END OF SECTION 09 5423

# SECTION 096513 - RESILIENT BASE AND ACCESSORIES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Thermoplastic-rubber base.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples for Verification: For each type of product indicated and for each color, texture, and pattern required in manufacturer's standard-size Samples, but not less than 12 inches (300 mm) long.

C. Product Schedule: For resilient base and accessory products. Use same designations indicated on Drawings.

### 1.4 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Furnish not less than 50 linear feet (15 linear m) or one unopened box of each type, color, pattern, and size of resilient product installed.

### 1.5 DELIVERY, STORAGE, AND HANDLING

A. Store resilient products and installation materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than 50 deg F (10 deg C) or more than 90 deg F (32 deg C).

### 1.6 FIELD CONDITIONS

A. Maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F (13 deg C) or more than 95 deg F (35 deg C), in spaces to receive resilient products during the following periods:

1. 48 hours before installation.
2. During installation.
3. 48 hours after installation.

B. After installation and until Substantial Completion, maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F (13 deg C) or more than 95 deg F (35 deg C).

C. Install resilient products after other finishing operations, including painting, have been completed.

## PART 2 - PRODUCTS

### 2.1 THERMOPLASTIC-RUBBER BASE (RB-1)

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following or approved equivalent:

1. Armstrong World Industries, Inc.
2. Johnsonite; a Tarkett company.
3. Roppe Corporation; Roppe Holding Company.

B. Product Standard: ASTM F1861, Type TP (rubber, thermoplastic).

1. Group: I (solid, homogeneous).
2. Style and Location:

    a. Style A, straight: Provide in all areas scheduled.

C. Thickness: 0.125 inch (3.2 mm).

D. Height: 4 inches (102 mm) unless otherwise indicated.

E. Lengths: Cut lengths 48 inches (1219 mm) long or coils in manufacturer's standard length.

F. Outside Corners: Job formed or preformed.

G. Inside Corners: Job formed or preformed.

H. Colors: Match Architect's sample: Tarkett TG6 Mink WG.

### 2.2 INSTALLATION MATERIALS

A. Trowelable Leveling and Patching Compounds: Latex-modified, portland-cement-based or blended hydraulic-cement-based formulation provided or approved by resilient-product manufacturer for applications indicated.

B. Adhesives: Water-resistant type recommended by resilient-product manufacturer for resilient products and substrate conditions indicated.

1. Adhesives shall have a VOC content of 50 g/L or less.

C. Metal Edge Strips: Extruded aluminum with mill finish, nominal 2 inches (50.8 mm) wide, of height required to protect exposed edges of flooring, and in maximum available lengths to minimize running joints.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resilient products.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

1. Installation of resilient products indicates acceptance of surfaces and conditions.

### 3.2 PREPARATION

A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of resilient products.

B. Do not install resilient products until materials are the same temperature as space where they are to be installed.

1. At least 48 hours in advance of installation, move resilient products and installation materials into spaces where they will be installed.

### 3.3 RESILIENT BASE INSTALLATION

A. Comply with manufacturer's written instructions for installing resilient base.

B. Apply resilient base to walls, columns, pilasters, casework and cabinets in toe spaces, and other permanent fixtures in rooms and areas where base is required.

C. Install resilient base in lengths as long as practical without gaps at seams and with tops of adjacent pieces aligned.

D. Tightly adhere resilient base to substrate throughout length of each piece, with base in continuous contact with horizontal and vertical substrates.

E. Do not stretch resilient base during installation.

F. On masonry surfaces or other similar irregular substrates, fill voids along top edge of resilient base with manufacturer's recommended adhesive filler material.

G. Preformed Corners: Install preformed corners before installing straight pieces.

H. Job-Formed Corners:

1. Outside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 4 inches (101 mm) in length.

    a. Form without producing discoloration (whitening) at bends.

2. Inside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 4 inches (101 mm) in length.

    a. Miter or cope corners to minimize open joints.

## 3.4 CLEANING AND PROTECTION

A. Comply with manufacturer's written instructions for cleaning and protecting resilient products.

B. Perform the following operations immediately after completing resilient-product installation:

1. Remove adhesive and other blemishes from surfaces.

C. Protect resilient products from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.

END OF SECTION 09 6513

# SECTION 09 6519 - RESILIENT FLOORING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Solid vinyl floor tile (LVT)
2. Resilient trims and transition strips (TS)

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For each type of resilient floor material and product as specified in drawings.

1. Include floor tile layouts, edges, columns, doorways, enclosing partitions, built-in furniture, cabinets, and cutouts.
2. Show details of special patterns.

C. Samples for Verification: Full-size units of each color and pattern of floor tile required.

1. In manufacturer’s standard size, but not less than 12-by-12-inch sections of each different color and pattern of resilient flooring specified, showing the full range of variations expected in these characteristics.
2. Resilient Wall Base and Accessories: Manufacturer’s standard-size samples, but not less than 12 inches long, of each resilient product, color and pattern required.
3. For heat-welding bead, manufacturer's standard-size Samples, but not less than 9 inches long, of each color required.

D. Product Schedule: For all products. Use same designations indicated on Drawings.

### 1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer; signed by manufacturer certifying that installers comply with specified requirements.

### 1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of flooring, base and trim accessory to include in maintenance manuals.

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials as described below from the same product run that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Resilient Sheet Flooring: Furnish not less than 10 linear feet for every 500 linear feet or fraction thereof, in roll form and in full roll width for each type, color, and pattern of flooring installed.
2. Floor Tile: Furnish one box for every 50 boxes or fraction thereof, of each type, color, and pattern of floor tile installed.
3. Resilient Wall Base and Accessories: Furnish not less than 10 linear feet for every 500 linear feet or fraction thereof, of each type, color, pattern, and size of resilient product installed.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are competent in techniques required by manufacturer for floor tile installation and seaming method indicated.

1. Engage an installer who employs workers for this Project who are trained or certified by floor tile manufacturer for installation techniques required.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Coordinate mockups in this Section with mockups specified in other Sections.
2. Refer to drawings for locations, patterns and materials to be used in mock-ups.
3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Store flooring, base, trim accessories and installation materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than 50 deg F or more than 90 deg F. Store floor tiles on flat surfaces.

1.9 FIELD CONDITIONS

A. Maintain ambient temperatures within range recommended by manufacturer, but not less than 70 deg F or more than 95 deg F, in spaces to receive floor tile during the following periods:

1. 48 hours before installation.
2. During installation.
3. 48 hours after installation.

B. After installation and until Substantial Completion, maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F or more than 95 deg F.

C. Close spaces to traffic during resilient flooring installation and for time period after installation recommended in writing by manufacturer.

D. Install floor tile after other finishing operations, including painting, have been completed.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics: For resilient floor tile, as determined by testing identical products according to ASTM E 648 or NFPA 253 by a qualified testing agency.

1. Critical Radiant Flux Classification: Class I, not less than 0.45 W/sq. cm.

2.2 LUXURY VINYL FLOOR TILE (**LVT-1)**

A. Manufacturers: Subject to compliance with requirements, Provide Tarkett Event Wood Collection, or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

1. Armstrong World Industries, Inc.
2. Johnsonite; a Tarkett company.
3. Mohawk Group.
4. J&J Flooring Group

B. Tile Standard: ASTM F 1700.

1. Class: Class III, Printed Film Vinyl Tile.
2. Type: B, Embossed Surface.

C. Thickness: 0.12 inch (3 mm).
D. Size: 6 by 48 inches.
E. Slip Resistance: ADA Compliant.
F. Seamless-Installation Method: Unidirectional.
G. Colors and Patterns: 11220 Heirloom.
H. Added antimicrobial: ZPT, Zinc Protective Technology.
I. Backing Class: Commercial Grade.
J. Commercial Traffic: Heavy Commercial.

2.3 RESILIENT MOLDING ACCESSORY

**A.** Products: Subject to compliance with requirements, provide products as scheduled by Architect from manufacturer’s full range of colors, and as indicated on the drawings.

B. INSTALLATION MATERIALS

C. Trowelable Leveling and Patching Compounds: Polymer-modified, portland-cement-based or blended hydraulic-cement-based formulation provided or approved by floor tile manufacturer for applications indicated.

D. Adhesives: Water-resistant type recommended by floor tile and adhesive manufacturers to suit floor tile and substrate conditions indicated.

1. Adhesives shall have a VOC content of 50 g/L or less.

E. Seamless-Installation Accessories:

1. Chemical-Bonding Compound: Manufacturer's product for chemically bonding seams.
2. Chemical-Bonding Compound shall have a VOC content of 510 g/L or less.
3. Heat-Welding Bead: Solid-strand product of floor covering manufacturer for heat-welding seams. Match color and pattern of resilient sheet flooring, as selected by Architect from manufacturer's full range of colors.
4. Floor Polish / Sealer: Provide protective, liquid floor-polish products recommended by resilient sheet flooring manufacturer.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of floor tile.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

A. Prepare substrates according to floor tile manufacturer's written instructions to ensure adhesion of resilient products.

B. Concrete Substrates: Prepare according to ASTM F 710.

1. Verify that substrates are dry and free of curing compounds, sealers, and hardeners.
2. Remove substrate coatings and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, using mechanical methods recommended by floor tile manufacturer. Do not use solvents.
3. Alkalinity and Adhesion Testing: Perform tests recommended by floor tile manufacturer. Proceed with installation only after substrate alkalinity falls within range on pH scale recommended by manufacturer in writing, but not less than 5 or more than 10 pH.

C. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound; remove bumps and ridges to produce a uniform and smooth substrate.

D. Moisture Testing: Proceed with installation only after substrates pass testing according to floor tile manufacturer's written recommendations, but not less stringent than the following:

1. Perform anhydrous calcium chloride test according to ASTM F 1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate not to exceed manufacturer's recommendations for each type of product installed.
2. Perform relative humidity test using in situ probes according to ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level.

E. Do not install floor tiles until materials are the same temperature as space where they are to be installed.

1. At least 48 hours in advance of installation, move resilient floor tile and installation materials into spaces where they will be installed.

F. Immediately before installation, sweep and vacuum clean substrates to be covered by resilient floor tile.

## 3.3 FLOOR TILE INSTALLATION

A. Comply with manufacturer's written instructions for installing floor tile.

B. Lay out floor tiles from center marks established with principal walls, discounting minor offsets, so tiles at opposite edges of room are of equal width. Adjust as necessary to avoid using cut widths that equal less than one-half tile at perimeter.

1. Lay tiles in pattern indicated.

C. Match floor tiles for color and pattern by selecting tiles from cartons in the same sequence as manufactured and packaged, if so numbered. Discard broken, cracked, chipped, or deformed tiles.

1. Lay tiles in pattern of colors and sizes indicated.

D. Scribe, cut, and fit resilient flooring to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, door frames, thresholds, and nosings.

E. Extend resilient flooring into toe spaces, door reveals, closets, and similar openings. Extend resilient flooring to center of door openings unless noted otherwise.

F. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on floor tiles as marked on substrates. Use chalk or other nonpermanent marking device.

G. Install resilient flooring on covers for telephone and electrical ducts, building expansion-joint covers, and similar items in installation areas. Maintain overall continuity of color and pattern between pieces of tile installed on covers and adjoining tiles. Tightly adhere tile edges to substrates that abut covers and to cover perimeters.

H. Adhere resilient flooring to substrates to comply with floor covering manufacturer's written instructions, including those for trowel notching, adhesive mixing, and adhesive open and working times. Produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.

3.4 SEAMLESS FLOOR INSTALLATION

A. Unroll resilient sheet flooring and allow it to stabilize before cutting and fitting, if recommended in writing by manufacturer.

B. Scribe, cut, and fit resilient flooring to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, door frames, thresholds, and nosings.

C. Extend resilient flooring into toe spaces, door reveals, closets, and similar openings. Extend resilient flooring to center of door openings unless noted otherwise.

D. Install resilient flooring on covers for telephone and electrical ducts, building expansion-joint covers, and similar items in installation areas. Maintain overall continuity of color and pattern between pieces of tile installed on covers and adjoining tiles. Tightly adhere tile edges to substrates that abut covers and to cover perimeters.

E. Adhere resilient flooring to substrates to comply with floor covering manufacturer’s written instructions, including those for trowel notching, adhesive mixing, and adhesive open and working times. Produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections

F. Chemically Bonded Seams: Bond seams with chemical-bonding compound to fuse sections permanently into a seamless flooring installation. Prepare seams and apply compound to produce tightly fitted seams without gaps, overlays, or excess bonding compound on flooring surfaces:

G. Heat-Welded Seams: Rout joints and heat weld with welding bead, permanently fusing sections into a seamless floor covering. Prepare, weld, and finish seams according to manufacturer’s written instructions and ASTM F 1516 to product surfaces flush with adjoining floor covering surfaces.

3.5 RESILIENT ACCESSORY INSTALLATION

A. Resilient Molding Accessories:

1. Butt to adjacent materials and tightly adhere to substrates throughout length of each piece. Install reducer strips at edges of floor coverings that would otherwise be exposed.

3.6 CLEANING AND PROTECTION

A. Comply with manufacturer's written instructions for cleaning and protecting floor tile.

B. Perform the following operations immediately after completing floor tile installation:

1. Remove adhesive and other blemishes from surfaces using cleaner recommended by floor covering manufacturer.
2. Sweep and vacuum surfaces thoroughly.
3. Do not wash floor covering until after time period recommended by floor covering manufacturer.
4. Damp-mop surfaces to remove marks and soil.

C. Protect flooring, base and trim from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period. Use protection methods indicated or recommended in writing by floor covering manufacturer.

D. Floor Polish/Sealer: Apply protective floor polish/sealer to resilient sheet flooring surfaces that are free from soil, visible adhesive, and surface blemishes, as recommended in writing by resilient flooring manufacturer.

1. Use commercially available product acceptable to floor covering manufacturer.
2. Coordinate selection of flor polish/sealer with Owner's maintenance service.
3. Cover resilient flooring with undyed, untreated building paper until inspection for Substantial Completion.
4. Do not move heavy and sharp objects directly over resilient flooring. Place plywood or hardboard panels over floor coverings and under objects while they are being moved. Slide or roll objects over panels without moving panels.

E. Joint Sealant: Apply sealant to joints according to manufacturer's recommendations.

END OF SECTION 09 6519

# SECTION 09 6723 - RESINOUS FLOORING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Resinous flooring.
2. Integral cove base accessories.
3. Vapor reduction membrane.

### 1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review manufacturer's written instructions for substrate preparation and environmental conditions affecting resinous flooring installation.
2. Review details of integral cove bases.
3. Review manufacturer's written instructions for installing resinous flooring systems.
4. Review protection measures for adjacent construction and installed flooring, floor drainage requirements, curbs, base details, and so forth.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include manufacturer's technical data, installation instructions, and recommendations for each resinous flooring component required.

B. Samples for Initial Selection: For each type of exposed finish required.

### 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Material Certificates: For each resinous flooring component.

### 1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For resinous flooring to include in maintenance manuals.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.

1. Engage an installer who is certified in writing by resinous flooring manufacturer as qualified to apply resinous flooring systems indicated.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Simulate finished lighting conditions for Architect's review of mockups.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating brand name and directions for storage and mixing with other components.

1.9 FIELD CONDITIONS

A. Environmental Limitations: Comply with resinous flooring manufacturer's written instructions for substrate temperature, ambient temperature, moisture, ventilation, and other conditions affecting resinous flooring installation.

B. Lighting: Provide permanent lighting or, if permanent lighting is not in place, simulate permanent lighting conditions during resinous flooring installation.

C. Close spaces to traffic during resinous flooring installation and for 24 hours after installation unless manufacturer recommends a longer period.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Flammability: Self-extinguishing in accordance with ASTM D635.

2.2 RESINOUS FLOORING (**EUF-1**)

A. Resinous Flooring System: Abrasion-, impact-, and chemical-resistant, aggregate-filled, resin-based monolithic floor surfacing designed to produce a seamless floor and integral cove base.

1. Basis-of-Design Product: Subject to compliance with requirements, Provide Tnemec Co., Stratashield, or approved equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

    a. Sherwin-Williams High Performance Flooring.
    b. Sika Corporation; Flooring.
    c. Stonhard, Inc.
    d. Tnemec Co. Inc,

B. Source Limitations: Obtain primary resinous flooring materials, including primers, resins, hardening agents, grouting coats, and topcoats, from single source from single manufacturer. Obtain secondary materials, including patching and fill material, joint sealant, and repair materials, of type and from manufacturer recommended in writing by manufacturer of primary materials.

C. Primer: Type recommended in writing by resinous flooring manufacturer for substrate and resinous flooring system indicated.

1. Products:

    a. Tnemec Series N241 Ultra Tread MVT.

        1) Thickness of coats: As recommended by manufacturer.

D. Body Coats

1. Products:

    a. Tnemec Series N241 Ultra Tread MVT

        1) Thickness of coats: As recommended by manufacturer.

    b. Tnemec Series 222 Deco-Tread.

        1) Thickness of coats: As recommended by manufacturer.

    c. Broadcast aggregate to refusal
    d. Lightly sand/screed after cure to knock down peaks.

E. Topcoat: Sealing or finish coats (urethane or epoxy sealer).

1. Products:

    a. Tnemec Series 248 Everthane.

        1) Thickness of coats: 3 mils (0.1 mm).

    b. Finish Color: As selected by Architect from manufacturer’s full range.

F. Waterproofing Membrane: Type recommended in writing by resinous flooring manufacturer for substrate and resinous flooring system indicated.

G. Reinforcing Membrane: Flexible resin formulation that is recommended in writing by resinous flooring manufacturer for substrate and resinous flooring system indicated and that inhibits substrate cracks from reflecting through resinous flooring.

H. Patching and Fill Material: Resinous product of or approved by resinous flooring manufacturer and recommended in writing by manufacturer for installation indicated.

A. System Physical Properties: Provide resinous flooring system with the following minimum physical property requirements when tested according to test methods indicated:

1. Compressive Strength: Minimum 10,000 psi per ASTM C 579.
2. Tensile Strength: Minimum 1900 psi per ASTM C 307.
3. Water Absorption: 0.1% max per ASTM C-413.
4. Impact Resistance: No chipping, cracking, or delamination and not more than 1/16-inch (1.6-mm) permanent indentation per MIL-D-3134.
5. Flammability: Self-extinguishing per ASTM D 635.
6. Hardness: Minimum 70, Shore D per ASTM D 2240.
7. Bond Strength: Minimum 400 psi, 100 percent concrete failure per ACI 503R.

## 2.3 INTEGRAL COVE BASE ACCESSORIES

A. Precast, Integral Cove Base: Impact-resistant, polymer-resin, cove base moldings with a grit profile to promote adhesion of resinous flooring and recommended in writing by resinous flooring manufacturer.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following or approved equivalent:

   a. SpeedCove, Inc.

2. Radius Cove Base: 6-inch- (152-mm-) high base molding that provides approximately 1-inch (25-mm) radius cove at floor-to-wall joint; for adhesive installation as substrate for resinous flooring system to form an integral cove base.

   a. Preformed Inside and Outside Corners: Provide manufacturer's standard square inside and square outside corners.

B. Installation Adhesive: As recommended in writing by accessory manufacturer.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resinous flooring systems.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 PREPARATION

A. Prepare and clean substrates in accordance with resinous flooring manufacturer's written instructions for substrate indicated to ensure adhesion.

B. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.

1. Roughen concrete substrates as follows:

a. Shot-blast surfaces with an apparatus that abrades the concrete surface, contains the dispensed shot within the apparatus, and recirculates the shot by vacuum pickup.
b. Comply with requirements in SSPC-SP 13/NACE No. 6, with a Concrete Surface Profile of 3 or greater in accordance with ICRI Technical Guideline No. 310.2R, unless manufacturer's written instructions are more stringent.

2. Repair damaged and deteriorated concrete in accordance with resinous flooring manufacturer's written instructions.

3. Moisture Testing: Perform tests so that each test area does not exceed 200 sq. ft. (18.6 sq. m) and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.

a. Relative Humidity Test: Using in-situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement or Conduct quantitative moisture testing in accordance with ASTM-F2659-10 utilizing Tramex type impedance moisture meter.

4. Alkalinity and Adhesion Testing: Perform tests recommended in writing by resinous flooring manufacturer. Proceed with installation only after substrate alkalinity is not less than 6 or more than 8 pH unless otherwise recommended in writing by flooring manufacturer,

C. Patching and Filling: Use patching and fill material to fill holes and depressions in substrates in accordance with manufacturer's written instructions.

1. Control Joint Treatment: Treat control joints and other nonmoving substrate cracks to prevent cracks from reflecting through resinous flooring in accordance with manufacturer's written instructions.

D. Resinous Materials: Mix components and prepare materials in accordance with resinous flooring manufacturer's written instructions.

E. Provide vapor reduction membrane over all concrete floor substrates that do not meet the maximum allowable moisture level, prior to installation of resinous flooring materials.

F. Provide kerfs or sawcuts at all floor drains, and where product abuts differing flooring materials to maintain full depth product thickness

## 3.3 INSTALLATION

A. Apply components of resinous flooring system in accordance with manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness specified.

1. Coordinate installation of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.

2. Cure resinous flooring components in accordance with manufacturer's written instructions. Prevent contamination during installation and curing processes.
3. Expansion and Isolation Joint Treatment: At substrate expansion and isolation joints, comply with resinous flooring manufacturer's written instructions.

B. Primer: Apply primer over prepared substrate at spreading rate recommended in writing by manufacturer.

C. Waterproofing Membrane: Apply waterproofing membrane where indicated on Drawings, in thickness recommended in writing by manufacturer.

1. Apply waterproofing membrane to integral cove base substrates.

D. Reinforcing Membrane: Apply reinforcing membrane to substrate cracks.

E. Integral Cove Base Accessories: Adhesively install precast accessories before applying flooring coats and in accordance with manufacturer's written instructions.

F. Field-Formed Integral Cove Base: Apply cove base mix to wall surfaces before applying flooring coats. Apply in accordance with manufacturer's written instructions and details, including those for taping, mixing, priming, troweling, sanding, and topcoating of cove base. Round internal and external corners.

1. Integral Cove Base: As indicated on Drawings.

G. Troweled or Screeded Body Coats: Apply troweled or screeded body coats in thickness specified for flooring system. Hand or power trowel and grout to fill voids. When body coats are cured, remove trowel marks and roughness using method recommended in writing by manufacturer.

H. Grout Coat: Apply grout coat to fill voids in surface of final body coat.

I. Topcoats: Apply topcoats in number indicated for flooring system specified, at spreading rates recommended in writing by manufacturer, and to produce wearing surface specified.

## 3.4 CLEANING, PROTECTING, AND CURING

A. Protect resinous flooring materials from damage and wear during construction operation. Where temporary covering is required for this purpose, comply with manufacturer's recommendations for protective materials and method of application. General Contractor is responsible for protection and cleaning of surfaces after final coats.

B. Cleaning: Remove temporary covering and clean resinous flooring just prior to final inspection. Use cleaning materials and procedures recommended by resinous flooring manufacturer.

END OF SECTION 09 6723

# SECTION 09 6813 - TILE CARPETING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Carpet tile (CPT).
2. Walk-off carpet tile (MAT).

B. Related Requirements:

1. Section 09 6513 "Resilient Base and Accessories" for resilient wall base and accessories installed with carpet tile.

### 1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include manufacturer's written data on physical characteristics, durability, and fade resistance.
2. Include manufacturer's written installation recommendations for each type of substrate.

B. Shop Drawings: For carpet tile installation, showing the following:

1. Columns, doorways, enclosing walls or partitions, built-in cabinets, and locations where cutouts are required in carpet tiles.
2. Carpet tile type, color, and dye lot.
3. Type of installation.
4. Pattern of installation.
5. Pattern type, location, and direction.
6. Type, color, and location of edge, transition, and other accessory strips.
7. Transition details to other flooring materials.

C. Samples for Verification: Actual sample of finished products for each of the following products and for each color and texture required. Label each Sample with manufacturer's name, material description, color, pattern, and designation indicated on Drawings and in schedules.

1. Carpet Tile: Full-size Sample.
2. Exposed Edge, Transition, and Other Accessory Stripping: 12-inch- (300-mm-) long Samples.

D. Product Schedule: For carpet tile. Use same designations indicated on Drawings.

### 1.3 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For carpet tile, for tests performed by a qualified testing agency.

B. Qualification Statements: For Installer.

C. Sample Warranties: For carpet tile.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance Data: For carpet tiles. Include the following:

1. Methods for maintaining carpet tile, including cleaning and stain-removal products and procedures and manufacturer's recommended maintenance schedule.
2. Precautions for cleaning materials and methods that could be detrimental to carpet tile.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials, from the same production run, to Owner that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Carpet Tile: Full-size units equal to [5] <Insert number> percent of amount installed for each type indicated, but no fewer than 10 full-size units.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is certified by the International Certified Floorcovering Installers Association at the Commercial II certification level.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Comply with CRI 104.

1.8 FIELD CONDITIONS

A. Comply with CRI 104 for temperature, humidity, and ventilation limitations.

B. Environmental Limitations: Do not deliver or install carpet tiles until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, and ambient temperature and humidity conditions are maintained at levels planned for building occupants during the remainder of the construction period.

C. Do not install carpet tiles over concrete slabs until slabs have cured and are sufficiently dry to bond with adhesive and concrete slabs have pH range recommended in writing by carpet tile manufacturer.

D. Where demountable partitions or other items are indicated for installation on top of carpet tiles, install carpet tiles before installing these items.

1.9 WARRANTY

A. Special Warranty for Carpet Tiles: Manufacturer agrees to repair or replace components of carpet tile installation that fail in materials or workmanship within specified warranty period.

1. Warranty does not include deterioration or failure of carpet tile due to unusual traffic, failure of substrate, vandalism, or abuse.
2. Failures include, but are not limited to, the following:

a. More than 10 percent loss of face fiber, edge raveling, snags, and runs.
b. Loss of tuft-bind strength.
c. Excess static discharge.
d. Delamination.
e. Dimensional instability.

3. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 CARPET TILE PRODUCTS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following or approved equivalent:

1. Interface, Inc.
2. J&J Flooring Group LLC.
3. Mannington Commercial; a business unit of Mannington Mills, Inc.
4. Mohawk Group.
5. Patcraft & Shaw Industries Group, Inc.; A Berkshire Hathaway Company.

B. Applied Treatments:

1. Soil-Resistance Treatment: Manufacturer's standard treatment.
2. Antimicrobial Treatment: Manufacturer's standard treatment that protects carpet tiles as follows:

a. Antimicrobial Activity: Not less than 2 mm halo of inhibition for gram-positive bacteria, not less than 1 mm halo of inhibition for gram-negative bacteria, and no fungal growth, in accordance with AATCC 174.

C. Performance Characteristics:

1. Texture Appearance Retention Rating (TARR): Heavy traffic, 3.0 minimum in accordance with ASTM D7330.
2. Critical Radiant Flux Classification: Not less than 0.45 W/sq. cm in accordance with NFPA 253.
3. Dry Breaking Strength: Not less than 100 lbf (445 N) in accordance with ASTM D2646.
4. Colorfastness to Crocking: Not less than 4, wet and dry, in accordance with AATCC 165.
5. Electrostatic Propensity: Less than 3.5 kV in accordance with AATCC 134.

2.2 CARPET TILE (**CPT-1**)

A. Basis-of-Design Product: J&J Flooring.
B. Style/Color: Match Architect's samples, Thrive Collection, Plum Path #3276.
C. Installation Pattern: Parquet.
D. Fiber Content: 100 percent nylon 6,6.
E. Pile Characteristic: Patterned Loop pile.
F. Density: 6775 oz./cu. yd.
G. Pile Thickness: 0.25 inches (6.35 mm) for finished carpet tile in accordance with ASTM D6859.
H. Stitches: 10 stitches per inch (mm).
I. Gage: 1/12 per inch (mm).
J. Surface Pile Weight: 21 oz./sq. yd.
K. Primary Backing/Backcoating: Nexus Modular.
L. Secondary Backing: Manufacturer's standard material.
M. Size: 18 by 36 inches (457 by 914 mm).

2.1 WALKOFF CARPET TILE (**MAT-1**)

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

1. Interface, Inc.
2. Milliken & Company
3. Mohawk Group
4. Tarkett USA.

B. Basis-of-Design Product: Tarkett USA.
C. Style/Color: Match Architect's samples, Abrasive Action II, Charcoal 19100.
D. Fiber Content: 100 percent Solution Dyed.
E. Pile Characteristic: Patterned loop.
F. Pile Thickness: 0.187 inches (4.8 mm) for finished carpet tile in accordance with ASTM D6859.
G. Stitches: 8.0 stitches per inch (mm).
H. Gage: 1/12 per inch (mm).
I. Surface Pile Weight: 24 oz./sq. yd. (949 g/sq. m).
J. Primary Backing/Backcoating: Manufacturer's standard composite materials.
K. Secondary Backing: Manufacturer's standard material.
L. Size: 24 by 24 inches (609 by 609 mm).

2.2 INSTALLATION ACCESSORIES

A. Trowelable Leveling and Patching Compounds: Latex-modified, hydraulic-cement-based formulation provided or recommended in writing by carpet tile manufacturer.

B. Adhesives: Water-resistant, mildew-resistant, nonstaining, pressure-sensitive types to suit products and subfloor conditions indicated, that comply with flammability requirements for installed carpet tile, and that are recommended in writing by carpet tile manufacturer for releasable installation.

C. Metal Edge/Transition Strips: Extruded aluminum with mill finish of profile and width shown, of height required to protect exposed edge of carpet, and of maximum lengths to minimize running joints.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for maximum moisture content, alkalinity range, installation tolerances, and other conditions affecting carpet tile performance.

B. Examine carpet tile for type, color, pattern, and potential defects.

C. Concrete Slabs: Verify that finishes comply with requirements specified in Section 033000 "Cast-in-Place Concrete" and that surfaces are free of cracks, ridges, depressions, scale, and foreign deposits.

1. Moisture Testing: Perform tests so that each test area does not exceed 200 sq. ft. (20 sq. m) and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.

a. Relative Humidity Test: Using in situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.
b. Perform additional moisture tests recommended in writing by adhesive and carpet tile manufacturers. Proceed with installation only after substrates pass testing.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. General: Comply with CRI 104 and with carpet tile manufacturer's written installation instructions for preparing substrates.

B. Use trowelable leveling and patching compounds, in accordance with manufacturer's written instructions, to fill cracks, holes, depressions, and protrusions in substrates. Fill or level cracks, holes and depressions 1/8 inch (3 mm) wide or wider, and protrusions more than 1/32 inch (0.8 mm) unless more stringent requirements are required by manufacturer's written instructions.

C. Concrete Substrates: Remove coatings, including curing compounds, and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, without using solvents. Use mechanical methods recommended in writing by adhesive and carpet tile manufacturers.

D. Metal Substrates: Clean grease, oil, soil and rust, and prime if recommended in writing by adhesive manufacturer. Rough sand painted metal surfaces and remove loose paint. Sand aluminum surfaces, to remove metal oxides, immediately before applying adhesive.

E. Broom and vacuum clean substrates to be covered immediately before installing carpet tile.

3.3 INSTALLATION

A. General: Comply with CRI 104, Section 10, "Carpet Tile," and with carpet tile manufacturer's written installation instructions.

B. Installation Method: As recommended in writing by carpet tile manufacturer.

1. Installation pattern: Ashlar.

C. Maintain dye-lot integrity. Do not mix dye lots in same area.

D. Maintain pile-direction patterns indicated on Shop Drawings.

E. Cut and fit carpet tile to butt tightly to vertical surfaces, permanent fixtures, and built-in furniture including cabinets, pipes, outlets, edgings, thresholds, and nosings. Bind or seal cut edges as recommended in writing by carpet tile manufacturer.

F. Extend carpet tile into toe spaces, door reveals, closets, open-bottomed obstructions, removable flanges, alcoves, and similar openings.

G. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on carpet tile as marked on subfloor. Use nonpermanent, nonstaining marking device.

H. Install pattern parallel to walls and borders.

3.4 CLEANING AND PROTECTION

A. Perform the following operations immediately after installing carpet tile:

1. Remove excess adhesive and other surface blemishes using cleaner recommended in writing by carpet tile manufacturer.
2. Remove yarns that protrude from carpet tile surface.
3. Vacuum carpet tile using commercial machine with face-beater element.

B. Protect carpet tile against damage from construction operations and placement of equipment and fixtures during the remainder of construction period. Use protection methods indicated or recommended in writing by carpet tile manufacturer.

END OF SECTION 09 6813

# SECTION 098433 - SOUND-ABSORBING WALL UNITS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes shop-fabricated, acoustical panel units tested for acoustical performance, including the following:

1. Sound-absorbing wall panels.

### 1.3 DEFINITIONS

A. NRC: Noise Reduction Coefficient.

B. SAA: Sound Absorption Average.

### 1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include fabric facing, panel edge, core material, and mounting indicated.

B. Shop Drawings: For unit assembly and installation.

1. Include plans, elevations, sections, and mounting devices and details.
2. Include details at panel head, base, joints, and corners; and details at ceiling, floor base, and wall intersections. Indicate panel edge profile and core materials.
3. Include details at cutouts and penetrations for other work.
4. Include direction of fabric weave and pattern matching.

C. Samples for Initial Selection: For each type of fabric facing.

1. Include Samples of hardware and accessories involving color or finish selection.

1.6 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Elevations and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Electrical outlets, switches, and thermostats.
2. Items penetrating or covered by units including but not limited to the following:

a. Alarms.
b. Access panels.
c. Electronic security devices.

B. Product Certificates: For each type of unit.

C. Sample Warranty: For manufacturer's special warranty.

1.7 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of unit to include in maintenance manuals. Include fabric manufacturers' written cleaning and stain-removal instructions.

1.8 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials from same production run that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Fabric: For each fabric, color, and pattern installed, provide length equal to 10 percent of amount installed, but no fewer than 10 sq. yd. (9 sq. m), full width of bolt.
2. Mounting Devices: Full-size units equal to 5 percent of amount installed, but no fewer than five devices, including unopened adhesives.

1.9 QUALITY ASSURANCE

A. Certified Wood: Provide an invoice including vendor's chain-of-custody number, product cost, and entity being invoiced.

B. Vendor Qualifications: A vendor that is certified for chain of custody by an FSC-accredited certification body.

C. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials, fabrication, and installation.

1. Build mockup of typical wall area 48 inches (1200 mm) wide by full height, Include intersection of wall and ceiling, corners, and perimeters.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.10 DELIVERY, STORAGE, AND HANDLING

A. Comply with fabric and unit manufacturers' written instructions for minimum and maximum temperature and humidity requirements for shipment, storage, and handling.

B. Deliver materials and units in unopened bundles and store in a temperature-controlled dry place with adequate air circulation.

1.11 FIELD CONDITIONS

A. Environmental Limitations: Do not install units until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, work at and above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

B. Lighting: Do not install units until a permanent level of lighting is provided on surfaces to receive the units.

C. Air-Quality Limitations: Protect units from exposure to airborne odors, such as tobacco smoke, and install units under conditions free from odor contamination of ambient air.

D. Field Measurements: Verify unit locations and actual dimensions of openings and penetrations by field measurements before fabrication and indicate them on Shop Drawings.

1.12 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace units and components that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to the following:

a. Acoustical performance.
b. Fabric sagging, distorting, or releasing from panel edge.
c. Warping of core.

2. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain wall units specified in this Section from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics: Units shall comply with "Surface-Burning Characteristics" or "Fire Growth Contribution" Subparagraph below, or both, as determined by testing identical

products by UL or another testing and inspecting agency acceptable to authorities having jurisdiction:

1. Surface-Burning Characteristics: Comply with ASTM E84 or UL 723; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

    a. Flame-Spread Index: 25 or less.
    b. Smoke-Developed Index: 450 <Insert value> or less.

2. Fire Growth Contribution: Comply with acceptance criteria of local code and authorities having jurisdiction when tested according to NFPA 265 Method B Protocol or NFPA 286.

## 2.3 SOUND-ABSORBING WALL UNITS

A. Sound-Absorbing Wall Panel (AWP-1): Manufacturer's standard panel construction consisting of facing material laminated to front face, edges, and back edge border of core.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Armstrong World Industries, Inc.
2. Conwed Designscape; an Owens Corning company.
3. Decoustics Limited; a Saint Gobain company.
4. Golterman & Sabo.
5. Lamvin, Inc.

    a. Finish Color at Exposed Edges: Match color of facing material.

6. Mounting: Back mounted with manufacturer's standard metal clips or bar hangers, secured to substrate.
7. Core: Glass-fiber boar.

    a. Core-Face Layer: Manufacturer's standard impact resistant, acoustically transparent, copolymer sheet.

8. Edge Construction: Manufacturer's standard chemically hardened core with no frame.
9. Edge Profile: Square.
10. Corner Detail in Elevation: Square with continuous edge profile indicated.
11. Reveals between Panels: Flush reveals.
12. Facing Material: As indicated on Drawings.
13. Acoustical Performance: Sound absorption NRC or SAA of not less than 1.00 according to ASTM C423 for Type A mounting according to ASTM E795.
14. Nominal Overall Panel Thickness: 2 inch (50.8 mm).
15. Panel Width: As indicated on Drawings.
16. Panel Height: As indicated on Drawings.

C. Acoustical Wall Covering (AWC-1) Fabric from same dye lot; color and pattern as selected by Architect from manufacturer's full range.

1. Manufacturer: Koroseal, or approved equivalent.
2. Product Line/Pattern: Chords.
3. Color: Ghost CHD-210.
4. Fiber Content: 100 percent woven polyester

5. Width: 54 inches (1371 mm).
6. Applied Treatments: Stain resistance and flame retardant.

## 2.4 MATERIALS

A. Composite Wood Products: Verify products are made using ultra-low-emitting formaldehyde resins, as defined in the California Air Resources Board's "Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products," or are made with no added formaldehyde.

B. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

C. Core Materials: Manufacturer's standard.

1. Glass-Fiber Board: ASTM C612; of type standard with manufacturer; nominal density of 6 to 7 lb/cu. ft. (96 to 112 kg/cu. m), unfaced, and dimensionally stable, molded rigid board; and with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.

D. Fabric from same dye lot; color and pattern as indicated on Finish Schedule.

E. Mounting Devices: Concealed on back of unit, recommended by manufacturer to support weight of unit, and as follows:

1. Verify adhesives have a VOC content of 70> g/L or less.
2. Metal Clips or Bar Hangers: Manufacturer's standard two-part metal "Z" clips, with one part of each clip mechanically attached to back of unit and the other part to substrate, designed to permit unit removal.

## 2.5 FABRICATION

A. Standard Construction: Use manufacturer's standard construction unless otherwise indicated; with facing material applied to face, edges, and back border of dimensionally stable core; and with rigid edges to reinforce panel perimeter against warpage and damage.

B. Edge Hardening: For glass-fiber board cores, chemically harden core edges and areas of core where mounting devices are attached.

C. Core-Face Layer: Evenly stretched over core face and edges and securely attached to core; free from puckers, ripples, wrinkles, or sags.

D. Facing Material: Apply fabric facing fully covering visible surfaces of unit; with material stretched straight, on the grain, tight, square, and free from puckers, ripples, wrinkles, sags, blisters, seams, adhesive, or other visible distortions or foreign matter.

1. Square Corners: Tailor corners. Heat-seal vinyl fabric seams at corners.
2. Radius and Other Nonsquare Corners: Attach facing material so there are no seams or gathering of material.
3. Fabrics with Directional or Repeating Patterns or Directional Weave: Mark fabric top and attach fabric in same direction so pattern or weave matches in adjacent units.

E. Dimensional Tolerances of Finished Units: Plus or minus 1/16 inch (1.6 mm) for the following:

1. Thickness.
2. Edge straightness.
3. Overall length and width.
4. Squareness from corner to corner.
5. Chords, radii, and diameters.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine fabric, fabricated units, substrates, areas, and conditions for compliance with requirements, installation tolerances, and other conditions affecting unit performance.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

A. Install units in locations indicated. Unless otherwise indicated, install units with vertical surfaces and edges plumb, top edges level and in alignment with other units, faces flush, and scribed to fit adjoining work accurately at borders and at penetrations.

B. Comply with manufacturer's written instructions for installation of units using type of mounting devices indicated. Mount units securely to supporting substrate.

C. Align fabric pattern and grain with adjacent units.

### 3.3 INSTALLATION TOLERANCES

A. Variation from Plumb and Level: Plus or minus 1/16 inch (1.6 mm) in 48 inches (1200 mm), noncumulative.

B. Variation of Joint Width: Not more than 1/16-inch (1.6-mm) variation from hairline in 48 inches (1200 mm), noncumulative.

### 3.4 CLEANING

A. Clip loose threads; remove pills and extraneous materials.

B. Clean panels on completion of installation to remove dust and other foreign materials according to manufacturer's written instructions.

END OF SECTION 098433

# SECTION 09 9113 - EXTERIOR PAINTING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes:

1. Primers.
2. Finish coatings.

B. Related Requirements:

1. Section 05 1200 "Structural Steel Framing" for shop priming of metal substrates.
2. Section 05 5000 "Metal Fabrications" for shop priming metal fabrications.
3. Section 09 9600 “High-Performance Coatings” for epoxy coatings.

### 1.3 DEFINITIONS

A. MPI Gloss Level 1: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.

B. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

C. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.

D. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.

E. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.

F. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product. Include preparation requirements and application instructions.

1. Include preparation requirements and application instructions.
2. Indicate VOC content.

B. Samples for Verification: For each type of paint system and each color and gloss of topcoat.

1. Submit Samples on rigid backing, 8 inches (200 mm) square.
2. Apply coats on Samples in steps to show each coat required for system.
3. Label each coat of each Sample.
4. Label each Sample for location and application area.

C. Product Schedule: Use same designations indicated on Drawings and in the Interior Painting Schedule to cross-reference paint systems specified in this Section. Include color designations.

## 1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Paint: 5 percent, but not less than 1 gal. (3.8 L) of each material and color applied with paint type indicated on each unopened container. No partial containers accepted.

## 1.6 QUALITY ASSURANCE

A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Architect will select one surface to represent surfaces and conditions for application of each paint system.

    a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).
    b. Other Items: Architect will designate items or areas required.

2. Final approval of color selections will be based on mockups.

    a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

## 1.7 DELIVERY, STORAGE, AND HANDLING

A. Delivery and Handling: Deliver products to Project site in an undamaged condition in manufacturer's original sealed containers, complete with labels and instructions for handling, storing, unpacking, protecting, and installing. Packaging shall bear the manufacture’s label with the following information:

1. Product name and type (description).
2. Batch date.
3. Color number.
4. VOC content.
5. Environmental handling requirements.
6. Surface preparation requirements.

7. Application instructions

B. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.8 FIELD CONDITIONS

A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).

B. Do not apply paints in snow, rain, fog, or mist; when relative humidity exceeds 85 percent; at temperatures less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include:

1. Behr Paint Company (Behr Process LLC).
2. Benjamin Moore & Co.
3. PPG Paints; PPG Industries, Inc.
4. Sherwin-Williams Company (The).

2.2 PAINT, GENERAL

A. Material Compatibility:

1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.

B. VOC Content: For field applications, paints and coatings shall comply with VOC content limits of authorities having jurisdiction and the following VOC content limits:

1. Flat Paints and Coatings: 50 g/L.
2. Nonflat Paints and Coatings: 50 g/L.
3. Dry-Fog Coatings: 150 g/L.
4. Primers, Sealers, and Undercoaters: 100 g/L.
5. Rust-Preventive Coatings: 100 g/L.
6. Zinc-Rich Industrial Maintenance Primers: 100 g/L.
7. Pretreatment Wash Primers: 420 g/L.
8. Shellacs, Clear: 730 g/L.
9. Shellacs, Pigmented: 550 g/L.

C. Colors As indicated in a finish schedules (Match Architect's samples).

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:

1. Concrete: 12 percent.
2. Fiber-Cement Board: 12 percent.
3. Masonry (Clay and CMUs): 12 percent.
4. Wood: 15 percent.
5. Portland Cement Plaster: 12 percent.
6. Gypsum Board: 12 percent.

C. Portland Cement Plaster Substrates: Verify that plaster is fully cured.

D. Exterior Gypsum Board Substrates: Verify that finishing compound is sanded smooth.

E. Verify suitability of substrates, including surface conditions and compatibility, with existing finishes and primers.

F. Proceed with coating application only after unsatisfactory conditions have been corrected.

1. Application of coating indicates acceptance of surfaces and conditions.

### 3.2 PREPARATION

A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and paint systems indicated.

B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection.

C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems indicated.

D. Concrete Substrates: Remove release agents, curing compounds, efflorescence, and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces to be painted exceeds that permitted in manufacturer's written instructions.

E. Masonry Substrates: Remove efflorescence and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces or mortar joints exceeds that permitted in manufacturer's written instructions.

F. Steel Substrates: Remove rust, loose mill scale, and shop primer if any. Clean using methods recommended in writing by paint manufacturer, but not less than the following:

1. SSPC-SP 2.
2. SSPC-SP 3.
3. SSPC-SP 7/NACE No. 4.
4. SSPC-SP 11.

G. Shop-Primed Steel Substrates: Clean field welds, bolted connections, and areas where shop paint is abraded. Paint exposed areas with the same material as used for shop priming to comply with SSPC-PA 1 for touching up shop-primed surfaces.

H. Galvanized-Metal Substrates: Remove grease and oil residue from galvanized sheet metal by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied paints.

I. Aluminum Substrates: Remove loose surface oxidation.

J. Wood Substrates:

1. Scrape and clean knots. Before applying primer, apply coat of knot sealer recommended in writing by topcoat manufacturer for exterior use in paint system indicated.
2. Sand surfaces that will be exposed to view and dust off.
3. Prime edges, ends, faces, undersides, and backsides of wood.
4. After priming, fill holes and imperfections in the finish surfaces with putty or plastic wood filler. Sand smooth when dried.

K. Plastic Trim Fabrication Substrates: Remove dust, dirt, and other foreign material that might impair bond of paints to substrates.

## 3.3 APPLICATION

A. Apply paints according to manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual."

1. Use applicators and techniques suited for paint and substrate indicated.
2. Paint surfaces behind movable items same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed items with prime coat only.
3. Paint both sides and edges of exterior doors and entire exposed surface of exterior door frames.
4. Paint entire exposed surface of window frames and sashes.
5. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.
6. Primers specified in painting schedules may be omitted on items that are factory primed or factory finished if acceptable to topcoat manufacturers.

B. Tint undercoats same color as topcoat but tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Provide sufficient difference in shade of undercoats to distinguish each separate coat.

C. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.

D. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

E. Painting Fire Suppression, Plumbing, HVAC, Electrical, Communication, and Electronic Safety and Security Work:

1. Paint the following work where exposed to view:

a. Uninsulated metal piping.
b. Uninsulated plastic piping.
c. Pipe hangers and supports.
d. Metal conduit.
e. Plastic conduit.
f. Tanks that do not have factory-applied final finishes.
g. Items as directed by Architect

## 3.4 FIELD QUALITY CONTROL

A. Dry Film Thickness Testing: Owner may engage the services of a qualified testing and inspecting agency to inspect and test paint for dry film thickness.

1. Contractor shall touch up and restore painted surfaces damaged by testing.
2. If test results show that dry film thickness of applied paint does not comply with paint manufacturer's written recommendations, Contractor shall pay for testing and apply additional coats as needed to provide dry film thickness that complies with paint manufacturer's written recommendations.

## 3.5 CLEANING AND PROTECTION

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

B. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

## 3.6 EXTERIOR PAINTING SCHEDULE

A. Concrete Substrates, Nontraffic Surfaces:

1. Latex System MPI EXT 3.1A:

a. Prime Coat: Primer, alkali resistant, water based, MPI #3.

1) S-W Loxon Concrete & Masonry Primer Sealer, A24W8300, at 8.0 mils (0.203 mm) wet, 3.2 mils (0.081 mm) dry.

b. Intermediate Coat: Latex, exterior, matching topcoat.

c. Topcoat: Latex, exterior, MPI #11.

1) S-W Solo Acrylic, A76 Series, at 4.0 mils (0.102 mm) wet, 1.5 mils (0.038 mm) dry, per coat.

d. Topcoat: Latex, exterior, MPI #119.

1) S-W A-100 Exterior Latex Gloss, A8 Series, at 4.0 mils (0.102 mm) wet, 1.3 mils (0.033 mm) dry, per coat.

B. CMU Substrates:

1. Latex System:

a. Prime Coat: Block filler, latex, interior/exterior, MPI #4.

1) S-W PrepRite Block Filler, B25W25, at 75 to 125 sq. ft. per gal. (1.84 to 3.07 sq. m per liter).

b. Intermediate Coat: Latex, exterior, matching topcoat.

c. Topcoat: Latex, exterior, MPI #11.

1) S-W Solo Acrylic, A76 Series, at 4.0 mils (0.102 mm) wet, 1.5 mils (0.038 mm) dry, per coat.

C. Steel and Iron Substrates:

1. Water-Based Light Industrial Coating System:

a. Prime Coat: Primer, rust inhibitive, water based MPI #107.

1) S-W Pro Industrial Pro-Cryl Universal Primer, B66-310 Series, 5.0 to 10.0 mils (0.127 to 0.254 mm) wet, 2.0 to 4.0 mils (0.051 to 0.102 mm) dry.

b. Intermediate Coat: Light industrial coating, exterior, water based, matching topcoat.

c. Topcoat: Light industrial coating, exterior, water based (MPI Gloss Level 3), MPI #161.

1) S-W Pro Industrial Eg-Shel Acrylic B66-660 Series, at 2.5 to 4.0 mils (0.064 to 0.102 mm) dry, per coat.

D. Galvanized-Metal Substrates:

1. Latex System:

a. Prime Coat: Primer, galvanized, cementitiou[, MPI #26.

b. Prime Coat: Primer, galvanized, water based, MPI #134.
   1) S-W Pro Industrial Pro-Cryl Universal Primer, B66-310 Series.

c. Intermediate Coat: Latex, exterior, matching topcoat.

d. Topcoat: Latex, exterior, low sheen (MPI Gloss Level 3-4), MPI #15.

   1) S-W Resilience Exterior Acrylic Latex, K43W00051 Series.

END OF SECTION 09 9113

# SECTION 09 9123 - INTERIOR PAINTING

## 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

## 1.2 SUMMARY

A. Section includes surface preparation and the application of paint systems on interior substrates.

1. Primers.
2. Water-based finish coatings.
3. Solvent-based finish coatings.
4. Floor sealers and paints.
5. Dry fall coatings.

B. Related Requirements:

1. Section 05 1200 "Structural Steel Framing" for shop priming structural steel.
2. Section 05 5000 "Metal Fabrications" for shop priming metal fabrications.
3. Section 09 9600 “High-Performance Coatings” for epoxy coatings.

## 1.3 DEFINITIONS

A. MPI Gloss Level 1: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.

B. MPI Gloss Level 2: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

C. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

D. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.

E. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.

F. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.

G. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

## 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product. Include preparation requirements and application instructions.

1. Include Printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
2. Indicate VOC content.

B. Sustainable Design Submittals:

1. Product Data: For paints and coatings, indicating VOC content.

C. Samples for Verification: For each type of paint system and in each color and gloss of topcoat.

1. Submit Samples on rigid backing, 8 inches (200 mm) square.
2. Apply coats on Samples in steps to show each coat required for system.
3. Label each coat of each Sample.
4. Label each Sample for location and application area.

D. Product List: Cross-reference to paint system and locations of application areas. Use same designations indicated on Drawings and in schedules. Include color designations.

## 1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Paint: 5 percent, but not less than 1 gal. (3.8 L) of each material and color applied with paint type indicated on each unopened container. No partial containers accepted.

## 1.6 QUALITY ASSURANCE

A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Architect will select one surface to represent surfaces and conditions for application of each paint system.

a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).
b. Other Items: Architect will designate items or areas required.

2. Final approval of color selections will be based on mockups.

a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.8 FIELD CONDITIONS

A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).

B. Do not apply paints when relative humidity exceeds 85 percent; at temperatures less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include:

1. Behr Paint Company (Behr Process LLC).
2. Benjamin Moore & Co.
3. PPG Paints; PPG Industries, Inc.
4. Sherwin-Williams Company (The).

2.2 PAINT, GENERAL

A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."

B. Material Compatibility:

1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.

C. VOC Content: For field applications that are inside the weatherproofing system, paints and coatings shall comply with VOC content limits of authorities having jurisdiction and the following VOC content limits:

1. Flat Paints and Coatings: 50 g/L.
2. Nonflat Paints and Coatings: 150 g/L.
3. Dry-Fog Coatings: 400 g/L.
4. Primers, Sealers, and Undercoaters: 200 g/L.
5. Anticorrosive and Antirust Paints Applied to Ferrous Metals: 250 g/L.

6. Zinc-Rich Industrial Maintenance Primers: 340 g/L.
7. Pretreatment Wash Primers: 420 g/L.
8. Shellacs, Clear: 730 g/L.
9. Shellacs, Pigmented: 550 g/L.

D. Colors: As indicated in finish schedules (Match Architect's samples).

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:

1. Concrete: 12 percent.
2. Fiber-Cement Board: 12 percent.
3. Masonry (Clay and CMUs): 12 percent.
4. Wood: 15 percent.
5. Gypsum Board: 12 percent.
6. Plaster: 12 percent.

C. Gypsum Board Substrates: Verify that finishing compound is sanded smooth.

D. Plaster Substrates: Verify that plaster is fully cured.

E. Spray-Textured Ceiling Substrates: Verify that surfaces are dry.

F. Verify suitability of substrates, including surface conditions and compatibility, with existing finishes and primers.

G. Proceed with coating application only after unsatisfactory conditions have been corrected.

1. Application of coating indicates acceptance of surfaces and conditions.

### 3.2 PREPARATION

A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and paint systems indicated.

B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.

C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems indicated.

D. Concrete Substrates: Remove release agents, curing compounds, efflorescence, and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces to be painted exceeds that permitted in manufacturer's written instructions.

E. Masonry Substrates: Remove efflorescence and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces or mortar joints exceeds that permitted in manufacturer's written instructions.

F. Steel Substrates: Remove rust, loose mill scale, and shop primer, if any. Clean using methods recommended in writing by paint manufacturer, but not less than the following:

1. SSPC-SP 2.
2. SSPC-SP 3.
3. SSPC-SP 7/NACE No. 4.
4. SSPC-SP 11.

G. Shop-Primed Steel Substrates: Clean field welds, bolted connections, and areas where shop paint is abraded. Paint exposed areas with the same material as used for shop priming to comply with SSPC-PA 1 for touching up shop-primed surfaces.

H. Galvanized-Metal Substrates: Remove grease and oil residue from galvanized sheet metal by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied paints.

I. Aluminum Substrates: Remove loose surface oxidation.

J. Wood Substrates:

1. Scrape and clean knots, and apply coat of knot sealer before applying primer.
2. Sand surfaces that will be exposed to view, and dust off.
3. Prime edges, ends, faces, undersides, and backsides of wood.
4. After priming, fill holes and imperfections in the finish surfaces with putty or plastic wood filler. Sand smooth when dried.

K. Cotton or Canvas Insulation Covering Substrates: Remove dust, dirt, and other foreign material that might impair bond of paints to substrates.

## 3.3 APPLICATION

A. Apply paints according to manufacturer's written instructions and to recommendations in "MPI Manual."

1. Use applicators and techniques suited for paint and substrate indicated.
2. Paint surfaces behind movable equipment and furniture same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed equipment or furniture with prime coat only.
3. Paint front and backsides of access panels, removable or hinged covers, and similar hinged items to match exposed surfaces.
4. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.

5. Primers specified in painting schedules may be omitted on items that are factory primed or factory finished if acceptable to topcoat manufacturers.

B. Tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Tint undercoats to match color of topcoat, but provide sufficient difference in shade of undercoats to distinguish each separate coat.

C. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.

D. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

E. Painting Fire Suppression, Plumbing, HVAC, Electrical, Communication, and Electronic Safety and Security Work:

1. Paint the following work where exposed in equipment rooms:

a. Uninsulated metal piping.
b. Metal conduit.
c. Tanks that do not have factory-applied final finishes.
d. Duct, equipment, and pipe insulation having cotton or canvas insulation covering or other paintable jacket material.
e. Other items as directed by Architect

2. Paint the following work where exposed in occupied spaces:

a. Uninsulated metal piping.
b. Pipe hangers and supports.
c. Metal conduit.
d. Duct, equipment, and pipe insulation having cotton or canvas insulation covering or other paintable jacket material.
e. Steel columns, beams and miscellaneous steel items.
f. Other items as directed by Architect.

3. Paint portions of internal surfaces of metal ducts, without liner, behind air inlets and outlets that are visible from occupied spaces.

4. Prevent paint overspray on IT/Telecom cables

## 3.4 FIELD QUALITY CONTROL

A. Dry Film Thickness Testing: Owner may engage the services of a qualified testing and inspecting agency to inspect and test paint for dry film thickness.

1. Contractor shall touch up and restore painted surfaces damaged by testing.
2. If test results show that dry film thickness of applied paint does not comply with paint manufacturer's written recommendations, Contractor shall pay for testing and apply additional coats as needed to provide dry film thickness that complies with paint manufacturer's written recommendations.

3.5 CLEANING AND PROTECTION

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

B. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.6 INTERIOR PAINTING SCHEDULE

A. Refer to finish schedules for color and finish of each paint type:

B. Concrete Substrates, Nontraffic Surfaces:

1. Latex System:

a. Prime Coat: Primer sealer, latex, interior, MPI #3:

1) S-W Loxon Concrete & Masonry Primer Sealer, A24W8300, at 8.0 mils wet, 3.2 mils dry.

b. Intermediate Coat: Latex, interior, matching topcoat.

c. Topcoat: Latex, interior, MPI #43 X-Green:

1) S-W ProMar 200 Zero VOC Latex B31-2600 Series, at 4.0 mils wet, 1.6 mils dry, per coat.

C. CMU Substrates:

1. Latex System:

a. Block Filler: Block filler, latex, interior/exterior, MPI #4 X-Green]: S-W PrepRite Block Filler, B25W25, at 100 to 200 sq. ft. per gal (2.4 to 4.9 sq. m per l).

b. Intermediate Coat: Latex, interior, matching topcoat.

c. Topcoat: Latex, interior, MPI #54:

1) S-W ProMar 200 Latex B11-2200 Series, at 4.0 mils wet, 1.5 mils dry, per coat.

D. Steel Substrates:

1. Water-Based Dry-Fall System:

a. Prime Coat: matching topcoat.

b. Topcoat: Dry-fall latex, MPI #226: S-W Pro Industrial Waterborne Acrylic Dry Fall B42-80 Series, at 5.8 mils wet, 2.3 mils dry.

E. Gypsum Board Substrates:

1. Latex System:

a. Prime Coat: Primer, latex, interior, MPI #149 X-Green]: S-W ProMar 200 Zero VOC Latex Primer, B28W2600, at 4.0 mils wet, 1.5 mils dry.

b. Intermediate Coat: Latex, interior, matching topcoat.

c. Topcoat: Latex, interior, MPI #43 X-Green:

1) S-W ProMar 200 Zero VOC Latex, B31-2600 Series, at 4.0 mils wet, 1.6 mils dry, per coat.

END OF SECTION 09 9123

# SECTION 09 9600 - HIGH-PERFORMANCE COATINGS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes surface preparation and the application of high-performance coating systems on the following substrates:

1. Interior Substrates:

    a. Concrete, vertical and horizontal surfaces.
    b. Concrete masonry units (CMUs).
    c. Steel and metal.
    d. Gypsum board.

B. Related Requirements:

1. Section 09 9113 "Exterior Painting" for general field painting.
2. Section 09 9123 "Interior Painting" for general field painting.

### 1.3 DEFINITIONS

A. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D523.
B. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D523.
C. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D523.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product. Include preparation requirements and application instructions.

1. Include printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
2. Indicate VOC content.

B. Sustainable Design Submittals:

1. Product Data: For paints and coatings, indicating VOC content.
2. Laboratory Test Reports: For paints and coatings, indicating compliance with requirements for low-emitting materials.

C. Product List: Cross-reference to coating system and locations of application areas. Use same designations indicated on Drawings and in schedules. Include color designations.

D. Samples for Verification: For each type of coating system and each color and gloss of topcoat indicated.

1. Submit Samples on rigid backing, 8 inches (200 mm) square.
2. Apply coats on Samples in steps to show each coat required for system.
3. Label each coat of each Sample.
4. Label each Sample for location and application area.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Coatings: 5 percent, but not less than 1 gal. (3.8 L) each material and color applied.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated dry areas with ambient temperatures continuously maintained at not less than 65 deg F.

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.7 FIELD CONDITIONS

A. Apply coatings only when temperature of surfaces to be coated and ambient air temperatures are between 60 and 85 deg F.

B. Do not apply coatings when relative humidity exceeds 85 percent; at temperatures less than 5 deg F above the dew point; or to damp or wet surfaces.

C. Do not apply exterior coatings in snow, rain, fog, or mist.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. PPG Paints; PPG Industries Inc.
2. Sherwin-Williams Company (The).
3. Tnemec Company, Inc.

B. Products: Subject to compliance with requirements, provide one of the products listed in the Exterior High-Performance Coating Schedule or Interior High-Performance Coating Schedule for the coating category indicated.

## 2.2 HIGH-PERFORMANCE COATINGS, GENERAL

A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."

B. Material Compatibility:

1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
3. Products shall be of same manufacturer for each coat in a coating system.

C. VOC Content: For field applications, verify paints and coatings comply with VOC content limits of authorities having jurisdiction and following VOC content limits:

1. Flat Paints and Coatings: 50 g/L.
2. Nonflat Paints and Coatings: 50 g/L.
3. Primers, Sealers, and Undercoaters: 100 g/L.
4. Rust-Preventive Coatings: 100 g/L.
5. Zinc-Rich Industrial Maintenance Primers: 100 g/L.
6. Pretreatment Wash Primers: 420 g/L.
7. Floor Coatings: 50 g/L.
8. Shellacs, Clear: 730 g/L.
9. Shellacs, Pigmented: 550 g/L.

D. Colors: As indicated in finish schedule, refer to sheet A-114.

## 2.3 SOURCE QUALITY CONTROL

A. Testing of Coating Materials: Owner reserves the right to invoke the following procedure:

1. Owner will engage the services of a qualified testing agency to sample coating materials. Contractor will be notified in advance and may be present when samples are taken. If coating materials have already been delivered to Project site, samples may be taken at Project site. Samples will be identified, sealed, and certified by testing agency.
2. Testing agency will perform tests for compliance with product requirements.
3. Owner may direct Contractor to stop applying coatings if test results show materials being used do not comply with product requirements. Contractor shall remove noncomplying coating materials from Project site, pay for testing, and recoat surfaces coated with rejected materials. Contractor will be required to remove rejected materials from previously coated surfaces if, on recoating with complying materials, the two coatings are incompatible.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:

1. Concrete: 12 percent.
2. Masonry (Clay and CMUs): 12 percent.
3. Gypsum Board: 12 percent.

C. Gypsum Board Substrates: Verify that finishing compound is sanded smooth.

D. Verify suitability of substrates, including surface conditions and compatibility, with existing finishes and primers.

E. Proceed with coating application only after unsatisfactory conditions have been corrected.

1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and coating systems indicated.

B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.

C. Clean substrates of substances that could impair bond of coatings, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce coating systems indicated.

D. Concrete Substrates: Remove release agents, curing compounds, efflorescence, and chalk. Do not coat surfaces if moisture content or alkalinity of surfaces to be coated exceeds that permitted in manufacturer's written instructions.

E. Masonry Substrates: Remove efflorescence and chalk. Do not coat surfaces if moisture content, alkalinity of surfaces, or alkalinity of mortar joints exceeds that permitted in manufacturer's written instructions.

F. Galvanized-Metal Substrates: Remove grease and oil residue from galvanized sheet metal by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied coatings.

3.3 APPLICATION

A. Apply high-performance coatings according to manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual."

1. Use applicators and techniques suited for coating and substrate indicated.
2. Coat surfaces behind movable equipment and furniture same as similar exposed surfaces. Before final installation, coat surfaces behind permanently fixed equipment or furniture with prime coat only.
3. Coat backsides of access panels, removable or hinged covers, and similar hinged items to match exposed surfaces.
4. Do not apply coatings over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.

B. Tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of the same material are to be applied. Tint undercoats to match color of finish coat but provide sufficient difference in shade of undercoats to distinguish each separate coat.

C. If undercoats or other conditions show through final coat, apply additional coats until cured film has a uniform coating finish, color, and appearance.

D. Apply coatings to produce surface films without cloudiness, spotting, holidays, laps, brush marks, runs, sags, ropiness, or other surface imperfections. Produce sharp glass lines and color breaks.

3.4 CLEANING, PROTECTION AND CURING

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

B. After completing coating application, clean spattered surfaces. Remove spattered coatings by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from coating operation. Correct damage to work of other trades by cleaning, repairing, replacing, and recoating, as approved by Architect, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced coated surfaces.

E. Cure resinous flooring materials in compliance with manufacturer’s directions, taking care to prevent contamination during stages of application and prior to completion of curing process. Close area of application for a minimum of 24 hours.

3.5 INTERIOR HIGH-PERFORMANCE COATING SCHEDULE

A. CMU Substrates: (**EP)**

1. Epoxy System:

a. Block Filler: Block filler, epoxy.

1) Sherwin Williams Conflex Block Filler, CF01W0050 White
2) Coats: Two.

b. Topcoat: Epoxy, Eggshell.

1) Sherwin Williams Pro Industrial Pre-Catalyzed Waterbased Epoxy K45-1150
2) Coats: (2) two

B. Gypsum Board Substrates: (**EP)**

1. Epoxy System:

a. Primer:

1) Sherwin Williams ProMar 200 Zero VOC Primer
2) Coats: (1) One.

b. Topcoat: Epoxy, Eggshell.

1) Sherwin Williams Pro Industrial Pre-Catalyzed Waterbased Epoxy K45-1150
2) Coats: (2) Two.

C. Steel and Metal Substrates: **EP (Interior and Exterior Metal)**

1. Epoxy System:

a. Prime Coat: Pro-Cryl Universal primer Primer, epoxy, anti-corrosive, for metal.

b. Topcoat: Pro Industrial water-based catalyzed epoxy, Gloss

1) Sherwin Williams Pro Industrial Water Based Catalyzed Epoxy, B73-300 Series
2) Coats: (2) Two.

END OF SECTION 09 9600

# SECTION 10 1419 - DIMENSIONAL LETTER SIGNAGE

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Dimensional characters.

a. Cast dimensional characters.
b. Fabricated channel dimensional characters.

### 1.3 DEFINITIONS

A. Illuminated: Illuminated by lighting source integrally constructed as part of the sign unit.

### 1.4 COORDINATION

A. Furnish templates for placement of electrical service embedded in permanent construction by other installers.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.

C. Shop Drawings: For signs.

1. Include fabrication and installation details and attachments to other work.
2. Show sign mounting heights, locations of supplementary supports to be provided by other installers, and accessories.
3. Show message list, typestyles, graphic elements, and layout for each sign at least half size.

D. Samples for Initial Selection: For each type of sign assembly, exposed component, and exposed finish.

1. Include representative Samples of available typestyles and graphic symbols.

E. Samples for Verification: For each type of sign assembly showing all components and with the required finish(es), in manufacturer's standard size unless otherwise indicated and as follows:

1. Dimensional Characters: Half-size Sample of each type of dimensional character.
2. Exposed Accessories: Half-size Sample of each accessory type.

F. Product Schedule: For dimensional letter signs. Use same designations indicated on Drawings or specified.

## 1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer and manufacturer.

B. Sample Warranty: For special warranty.

## 1.7 CLOSEOUT SUBMITTALS

A. Maintenance Data: For signs to include in maintenance manuals.

## 1.8 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

## 1.9 FIELD CONDITIONS

A. Field Measurements: Verify locations of electrical service embedded in permanent construction by other installers by field measurements before fabrication and indicate measurements on Shop Drawings.

## 1.10 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of signs that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Deterioration of finishes beyond normal weathering.
b. Separation or delamination of sheet materials and components.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Structural Performance: Signs and supporting elements shall withstand the effects of gravity and other loads within limits and under conditions indicated.

B. Thermal Movements: For exterior fabricated channel dimensional characters, allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 DIMENSIONAL CHARACTERS

A. Cast Characters: Characters with uniform faces, sharp corners, and precisely formed lines and profiles, and as follows:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. ACE Sign Systems, Inc.
b. ASI Sign Systems, Inc.
c. Matthews International Corporation; Bronze Division.
d. Metallic Arts.
e. Southwell Company (The).

2. Character Material: Sheet or plate aluminum
3. Material Thickness: Manufacturer's standard for size and design of character.
4. Character Height: As indicated on Drawings.
5. Character Depth: As indicated on Drawings.
6. Finishes:

a. Integral Aluminum Finish: Clear anodized

7. Mounting: Manufacturer's standard for size and design of character.

a. Hold characters at manufacturer's recommended distance from wall surface.

2.3 DIMENSIONAL CHARACTER MATERIALS

A. Aluminum Castings: ASTM B 26/B 26M, alloy and temper recommended by sign manufacturer for casting process used and for type of use and finish indicated.

B. Aluminum Sheet and Plate: ASTM B 209 (ASTM B 209M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

C. Aluminum Extrusions: ASTM B 221 (ASTM B 221M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

D. Stainless-Steel Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304, stretcher-leveled standard of flatness.

## 2.4 ACCESSORIES

A. Fasteners and Anchors: Manufacturer's standard as required for secure anchorage of signs, noncorrosive and compatible with each material joined, and complying with the following:

1. Use concealed fasteners and anchors unless indicated to be exposed.
2. For exterior exposure, furnish stainless-steel or hot-dip galvanized devices unless otherwise indicated.

3. Sign Mounting Fasteners:

    a. Concealed Studs: Concealed (blind), threaded studs welded or brazed to back of sign material, screwed into back of sign assembly, or screwed into tapped lugs cast integrally into back of cast sign material, unless otherwise indicated.
    b. Projecting Studs: Threaded studs with sleeve spacer, welded or brazed to back of sign material, screwed into back of sign assembly, or screwed into tapped lugs cast integrally into back of cast sign material, unless otherwise indicated.
    c. Through Fasteners: Exposed metal fasteners matching sign finish, with type of head indicated, installed in predrilled holes.

B. Adhesive: As recommended by sign manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.

C. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187/D 1187M.

## 2.5 FABRICATION

A. General: Provide manufacturer's standard sign assemblies according to requirements indicated.

1. Preassemble signs and assemblies in the shop to greatest extent possible. Disassemble signs and assemblies only as necessary for shipping and handling limitations. Clearly mark units for reassembly and installation; apply markings in locations concealed from view after final assembly.
2. Mill joints to a tight, hairline fit. Form assemblies and joints exposed to weather to resist water penetration and retention.
3. Comply with AWS for recommended practices in welding and brazing. Provide welds and brazes behind finished surfaces without distorting or discoloring exposed side. Clean exposed welded and brazed connections of flux, and dress exposed and contact surfaces.
4. Conceal connections if possible; otherwise, locate connections where they are inconspicuous.
5. Internally brace dimensional characters for stability, to meet structural performance loading without oil-canning or other surface deformation, and for securing fasteners.
6. Provide rabbets, lugs, and tabs necessary to assemble components and to attach to existing work. Drill and tap for required fasteners. Use concealed fasteners where possible; use exposed fasteners that match sign finish.
7. Castings: Fabricate castings free of warp, cracks, blowholes, pits, scale, sand holes, and other defects that impair appearance or strength. Grind, wire brush, sandblast, and buff

castings to remove seams, gate marks, casting flash, and other casting marks before finishing.

B. Brackets: Fabricate brackets, fittings, and hardware for bracket-mounted signs to suit sign construction and mounting conditions indicated. Modify manufacturer's standard brackets as required.

1. Aluminum Brackets: Factory finish brackets with baked-enamel or powder-coat finish to match sign-background color unless otherwise indicated.
2. Stainless-Steel Brackets: Factory finish brackets to match sign background finish unless otherwise indicated.

## 2.6 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

C. Directional Finishes: Run grain with long dimension of each piece and perpendicular to long dimension of finished trim or border surface unless otherwise indicated.

D. Organic, Anodic, and Chemically Produced Finishes: Apply to formed metal after fabrication but before applying contrasting polished finishes on raised features unless otherwise indicated.

## 2.7 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, Class I, 0.018 mm or thicker.

B. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

## 2.8 STAINLESS-STEEL FINISHES

A. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.

B. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.

1. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.
2. Directional Satin Finish: No. 4.
3. Dull Satin Finish: No. 6.
4. Reflective, Directional Polish: No. 7.
5. Mirror like Reflective, Non-directional Polish: No. 8.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance.

B. Verify that sign-support surfaces are within tolerances to accommodate signs without gaps or irregularities between backs of signs and support surfaces unless otherwise indicated.

C. Verify that electrical service is correctly sized and located to accommodate signs.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. General: Install signs using mounting methods indicated and according to manufacturer's written instructions.

1. Install signs level, plumb, true to line, and at locations and heights indicated, with sign surfaces free of distortion and other defects in appearance.
2. Before installation, verify that sign surfaces are clean and free of materials or debris that would impair installation.
3. Corrosion Protection: Coat concealed surfaces of exterior aluminum in contact with grout, concrete, masonry, wood, or dissimilar metals, with a heavy coat of bituminous paint.

B. Mounting Methods:

1. Concealed Studs: Using a template, drill holes in substrate aligning with studs on back of sign. Remove loose debris from hole and substrate surface.

    a. Masonry Substrates: Fill holes with adhesive. Leave recess space in hole for displaced adhesive. Place sign in position and push until flush to surface, embedding studs in holes. Temporarily support sign in position until adhesive fully sets.
    b. Thin or Hollow Surfaces: Place sign in position and flush to surface, install washers and nuts on studs projecting through opposite side of surface, and tighten.

2. Projecting Studs: Using a template, drill holes in substrate aligning with studs on back of sign. Remove loose debris from hole and substrate surface.

    a. Masonry Substrates: Fill holes with adhesive. Leave recess space in hole for displaced adhesive. Place spacers on studs, place sign in position, and push until spacers are pinched between sign and substrate, embedding the stud ends in holes. Temporarily support sign in position until adhesive fully sets.
    b. Thin or Hollow Surfaces: Place spacers on studs, place sign in position with spacers pinched between sign and substrate and install washers and nuts on stud ends projecting through opposite side of surface, and tighten.

3. Through Fasteners: Drill holes in substrate using predrilled holes in sign as template. Countersink holes in sign if required. Place sign in position and flush to surface. Install through fasteners and tighten.

4. Back Bar and Brackets: Remove loose debris from substrate surface and install backbar or bracket supports in position, so that signage is correctly located and aligned.
5. Adhesive: Clean bond-breaking materials from substrate surface and remove loose debris. Apply linear beads or spots of adhesive symmetrically to back of sign and of suitable quantity to support weight of sign after cure without slippage. Keep adhesive away from edges to prevent adhesive extrusion as sign is applied and to prevent visibility of cured adhesive at sign edges. Place sign in position and push to engage adhesive. Temporarily support sign in position until adhesive fully sets.
6. Two-Face Tape: Clean bond-breaking materials from substrate surface and remove loose debris. Apply tape strips symmetrically to back of sign and of suitable quantity to support weight of sign without slippage. Keep strips away from edges to prevent visibility at sign edges. Place sign in position and push to engage tape adhesive.

## 3.3 ADJUSTING AND CLEANING

A. Remove and replace damaged or deformed characters and signs that do not comply with specified requirements. Replace characters with damaged or deteriorated finishes or components that cannot be successfully repaired by finish touchup or similar minor repair procedures.

B. Remove temporary protective coverings and strippable films as signs are installed.

C. On completion of installation, clean exposed surfaces of signs according to manufacturer's written instructions and touch up minor nicks and abrasions in finish. Maintain signs in a clean condition during construction and protect from damage until acceptance by Owner.

END OF SECTION 10 1419

# SECTION 10 1423 - PANEL SIGNAGE

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Panel signs.
2. Field-applied, vinyl-character signs.

B. Related Requirements:

1. Section 10 1423.16 "Room-Identification Panel Signage" for room-identification signs that are directly attached to the building.
2. Section 22 0553 "Identification for Plumbing Piping and Equipment" for labels, tags, and nameplates for plumbing systems and equipment.
3. Section 2 30553 "Identification for HVAC Piping and Equipment" for labels, tags, and nameplates for HVAC systems and equipment.
4. Section 26 0553 "Identification for Electrical Systems" for labels, tags, and nameplates for electrical equipment.

### 1.3 DEFINITIONS

A. Accessible: In accordance with the accessibility standard.

B. Illuminated: Illuminated by lighting source integrally constructed as part of the sign unit.

### 1.4 COORDINATION

A. Furnish templates for placement of sign-anchorage devices embedded in permanent construction by other installers.

B. Furnish templates for placement of electrical service embedded in permanent construction by other installers.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.
2. Laboratory Test Reports: For adhesives, indicating compliance with requirements for low-emitting materials.

C. Shop Drawings: For panel signs.

1. Include fabrication and installation details and attachments to other work.
2. Show sign mounting heights, locations of supplementary supports to be provided by other installers, and accessories.
3. Show message list, typestyles, graphic elements, including raised characters and Braille, and layout for each sign at least half size.

D. Samples for Initial Selection: For each type of sign assembly, exposed component, and exposed finish.

1. Include representative Samples of available typestyles and graphic symbols.

E. Samples for Verification: For each type of sign assembly showing all components and with the required finish(es), in manufacturer's standard size unless otherwise indicated and as follows:

1. Panel Signs: Not less than 12 inches (300 mm) square, including corner.
2. Field-Applied, Vinyl-Character Signs: Half-size Sample of characters on glass.
3. Variable Component Materials: 8-inch (200-mm) Sample of each base material, character (letter, number, and graphic element) in each exposed color and finish not included in Samples above.
4. Exposed Accessories: Of each accessory type.

F. Product Schedule: For panel signs. Use same designations indicated on Drawings or specified.

## 1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer and manufacturer.

B. Evaluation Reports: For post-installed anchors and power-actuated fasteners, from ICC-ES or other qualified testing agency acceptable to authorities having jurisdiction.

C. Sample Warranty: For special warranty.

## 1.7 CLOSEOUT SUBMITTALS

A. Maintenance Data: For signs to include in maintenance manuals.

## 1.8 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Variable Component Materials: 12 replaceable text inserts and interchangeable characters (letters, numbers, and graphic elements) of each type.
2. Tools: One set(s) of specialty tools for assembling signs and replacing variable sign components.

1.9 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.10 FIELD CONDITIONS

A. Field Measurements: Verify locations of **anchorage devices** embedded in permanent construction by other installers by field measurements before fabrication and indicate measurements on Shop Drawings.

1.11 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of signs that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Deterioration of finishes beyond normal weathering.
b. Deterioration of embedded graphic image.
c. Separation or delamination of sheet materials and components.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: Comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design", the ABA standards of the Federal agency having jurisdiction and ICC A117.1.

2.2 PANEL SIGNS

A. Police Badge Panel Signage: Sign with smooth, uniform surfaces; with message and characters having uniform faces, sharp corners, and precisely formed lines and profiles; and as follows:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. ASI Sign Systems, Inc.
b. Best Sign Systems, Inc.
c. Bunting Graphics, Inc.
d. Inpro Corporation.
e. Seton Identification Products; a Brady Corporation company.

2. Solid-Sheet Sign: Back Frosted Acrylic Sign Panel with flame polished edges

a. Thickness: As indicated on drawings.
b. Applied graphics – Architect/Owner to provide exact drawing file with signage details for shop drawings.

3. Sign-Panel Perimeter: Finish edges smooth.

a. Edge Condition at Vertical and Horizontal Edges: Flame polished edges
b. Corner Condition in Elevation: Flame polished edges

4. Frame: Entire perimeter.

a. Material: Acrylic Sign Panel
b. Frame Depth: As indicated on Drawings.
c. Profile: refer to drawings
d. Corner Condition in Elevation: Rounded to radius indicated.
e. Finish and Color: Refer to drawings

5. Mounting: 1” metal standoff
6. Surface Finish and Graphics:

a. Text is grooved, clear etched against a frosted background
b. Badge is etched or applied onto acrylic panel

7. Text and Typeface: As indicated.
8. Flatness Tolerance: Sign shall remain flat or uniformly curved under installed conditions as indicated on Drawings and within a tolerance of plus or minus 1/16 inch (1.5 mm) measured diagonally from corner to corner.

B. No Smoking Panel Sign: Sign with smooth, uniform surfaces; with message and characters having uniform faces, sharp corners, and precisely formed lines and profiles; and as follows:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. ASI Sign Systems, Inc.
b. Best Sign Systems, Inc.
c. Bunting Graphics, Inc.
d. Inpro Corporation.
e. Seton Identification Products; a Brady Corporation company.

2. Solid-Sheet Sign: Aluminum sheet:

a. Thickness: 0.25 inch (6.35 mm).
b. Surface-Applied, Flat Graphics: Applied vinyl film.
c. Surface-Applied, Raised Graphics: Applied polymer characters and Braille

## 2.3 FIELD-APPLIED, VINYL-CHARACTER SIGNS

A. Field-Applied, Vinyl-Character Sign: Prespaced characters die cut from 3- to 3.5-mil (0.076- to 0.089-mm) thick, weather-resistant vinyl film with release liner on the back and carrier film on the front for on-site alignment and application.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

    a. Allen Markings.
    b. APCO Graphics, Inc.
    c. Inpro Corporation.
    d. Seton Identification Products; a Brady Corporation company.
    e. Mowhawk Sign Systems.

2. Size: As indicated on Drawings.
3. Substrate: As indicated on Drawings
4. Text and Font: As indicated on Drawings.

2.4 PANEL-SIGN MATERIALS

A. Aluminum Sheet and Plate: ASTM B209 (ASTM B209M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

B. Aluminum Extrusions: ASTM B221 (ASTM B221M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

C. Steel Materials:

1. Metallic-Coated Steel Sheet: ASTM A653/A653M, G90 (Z275)] coating, either commercial or forming steel.
2. Steel Sheet: Uncoated, cold-rolled, ASTM A1008/A1008M, commercial steel, Type B, exposed or electrolytic zinc-coated, ASTM A879/A879M, Coating Designation 08Z (24G), with steel-sheet substrate according to ASTM A1008/A1008M, commercial steel, exposed.
3. Steel Members Fabricated from Plate or Bar Stock: ASTM A529/A529M or ASTM A572/A572M, 42,000-psi (290-MPa) minimum yield strength.
4. For steel exposed to view on completion, provide materials having flat, smooth surfaces without blemishes. Do not use materials whose surfaces exhibit pitting, seam marks, roller marks, rolled trade names, or roughness.

D. Stainless-Steel Sheet: ASTM A240/A240M or ASTM A666, Type 304, stretcher-leveled standard of flatness.

E. Acrylic Sheet: ASTM D4802, category as standard with manufacturer for each sign, Type UVF (UV filtering).

F. Fiberglass Sheet: Multiple laminations of glass-fiber-reinforced polyester resin with UV-light stable, colorfast, nonfading, weather- and stain-resistant, colored polyester gel coat, and with manufacturer's standard finish.

G. Polycarbonate Sheet: ASTM C1349, Appendix X1, Type II (coated, mar-resistant, UV-stabilized polycarbonate), with coating on both sides.

H. PVC Sheet: Manufacturer's standard, UV-light stable, PVC plastic.

I. Plastic-Laminate Sheet: NEMA LD 3, general-purpose HGS grade, 0.048-inch (1.2-mm) nominal thickness.

J. Vinyl Film: UV-resistant vinyl film of nominal thickness indicated, with pressure-sensitive, permanent adhesive on back; die cut to form characters or images as indicated on Drawings and suitable for exterior applications.

K. Paints and Coatings for Sheet Materials: Inks, dyes, and paints that are recommended by manufacturer for optimum adherence to surface and are UV and water resistant for colors and exposure indicated.

2.5 ACCESSORIES

A. Fasteners and Anchors: Manufacturer's standard as required for secure anchorage of signs, noncorrosive and compatible with each material joined, and complying with the following unless otherwise indicated:

1. Use concealed fasteners and anchors unless indicated to be exposed.
2. For exterior exposure, furnish stainless-steel or hot-dip galvanized devices unless otherwise indicated.
3. Exposed Metal-Fastener Components, General:

    a. Fabricated from same basic metal and finish of fastened metal unless otherwise indicated.
    b. Fastener Heads: For nonstructural connections, use flathead or oval countersunk screws and bolts with tamper-resistant Allen-head or one-way-head slots unless otherwise indicated.

4. Sign Mounting Fasteners:

    a. Concealed Studs: Concealed (blind), threaded studs welded or brazed to back of sign material or screwed into back of sign assembly unless otherwise indicated.
    b. Projecting Studs: Threaded studs with sleeve spacer, welded or brazed to back of sign material or screwed into back of sign assembly, unless otherwise indicated.
    c. Through Fasteners: Exposed metal fasteners matching sign finish, with type of head indicated, and installed in predrilled holes.

5. Inserts: Furnish inserts to be set by other installers into concrete or masonry work.

B. Post-Installed Anchors: Fastener systems with bolts of same basic metal as fastened metal, if visible, unless otherwise indicated; with working capacity greater than or equal to the design load, according to an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 or ICC-ES AC308 as appropriate for the substrate.

1. Uses: Securing signs with imposed loads to structure.
2. Type: Torque-controlled, expansion anchor or adhesive anchor.
3. Material for Exterior or Interior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 (A1) stainless-steel bolts, ASTM F593 (ASTM F738M), and nuts, ASTM F594 (ASTM F836M).

C. Power-Actuated Anchors: Fastener systems with working capacity greater than or equal to the design load, according to an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

D. Adhesive: As recommended by sign manufacturer.

1. Adhesives shall have a VOC content of **70** g/L or less.

E. Two-Face Tape: Manufacturer's standard high-bond, foam-core tape, 0.045 inch (1.14 mm) thick, with adhesive on both sides.

F. Hook-and-Loop Tape: Manufacturer's standard two-part tape consisting of hooked part on sign back and looped side on mounting surface.

G. Magnetic Tape: Manufacturer's standard magnetic tape with adhesive on one side.

H. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

## 2.6 FABRICATION

A. General: Provide manufacturer's standard sign assemblies according to requirements indicated.

1. Preassemble signs in the shop to greatest extent possible. Disassemble signs and assemblies only as necessary for shipping and handling limitations. Clearly mark units for reassembly and installation; apply markings in locations concealed from view after final assembly.
2. Mill joints to a tight, hairline fit. Form assemblies and joints exposed to weather to resist water penetration and retention.
3. Comply with AWS for recommended practices in welding and brazing. Provide welds and brazes behind finished surfaces without distorting or discoloring exposed side. Clean exposed welded and brazed connections of flux, and dress exposed and contact surfaces.
4. Conceal connections if possible; otherwise, locate connections where they are inconspicuous.
5. Internally brace signs for stability, to meet structural performance loading without oil-canning or other surface deformation, and for securing fasteners.
6. Provide rabbets, lugs, and tabs necessary to assemble components and to attach to existing work. Drill and tap for required fasteners. Use concealed fasteners where possible; use exposed fasteners that match sign finish.

B. Subsurface-Applied Graphics: Apply graphics to back face of clear face-sheet material to produce precisely formed image. Image shall be free of rough edges.

C. Shop- and Subsurface-Applied Vinyl: Align vinyl film in final position and apply to surface. Firmly press film from the middle outward to obtain good bond without blisters or fishmouths.

D. Brackets: Fabricate brackets, fittings, and hardware for bracket-mounted signs to suit sign construction and mounting conditions indicated. Modify manufacturer's standard brackets as required.

1. Aluminum Brackets: Factory finish brackets with baked-enamel or powder-coat finish to match sign-background color unless otherwise indicated.
2. Stainless-Steel Brackets: Factory finish brackets to match sign background finish unless otherwise indicated.

2.7 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

C. Directional Finishes: Run grain with long dimension of each piece and perpendicular to long dimension of finished trim or border surface unless otherwise indicated.

D. Organic, Anodic, and Chemically Produced Finishes: Apply to formed metal after fabrication but before applying contrasting polished finishes on raised features unless otherwise indicated.

2.8 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, Class I, 0.018 mm or thicker.

B. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

2.9 METALLIC-COATED STEEL FINISHES

A. Surface Preparation: Clean surfaces of oil and other contaminants. Use cleaning methods that do not leave residue. After cleaning, apply a conversion coating compatible with the organic coating to be applied over it. Clean welds, mechanical connections, and abraded areas and apply galvanizing repair paint, complying with SSPC-Paint 20, to comply with ASTM A780/A780M.

B. Factory Prime Finish: After cleaning and pretreating, apply an air-dried primer compatible with the organic coating to be applied over it.

C. Baked-Enamel or Powder-Coat Finish: After cleaning and pretreating, apply manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat to a minimum dry film thickness of 2 mils (0.05 mm).

2.10 STEEL FINISHES

A. Surface Preparation: Remove mill scale and rust, if present, from uncoated steel, and prepare for coating according to coating manufacturer's written instructions.

1. For Baked-Enamel or Powder-Coat Finish: After cleaning, apply a conversion coating compatible with the organic coating to be applied over it.

B. Factory Prime Finish: After surface preparation and pretreatment, apply manufacturer's standard, fast-curing, lead- and chromate-free, universal primer.

C. Baked-Enamel or Powder-Coat Finish: After cleaning and pretreating, apply manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat to a minimum dry film thickness of 2 mils (0.05 mm).

## 2.11 STAINLESS-STEEL FINISHES

A. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.

B. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.

1. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.
2. Directional Satin Finish: No. 4.
3. Dull Satin Finish: No. 6.
4. Reflective, Directional Polish: No. 7.
5. Mirrorlike Reflective, Nondirectional Polish: No. 8.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Verify that sign-support surfaces are within tolerances to accommodate signs without gaps or irregularities between backs of signs and support surfaces unless otherwise indicated.

C. Verify that anchorage devices embedded in permanent construction are correctly sized and located to accommodate signs.

D. Verify that electrical service is correctly sized and located to accommodate signs.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 INSTALLATION

A. General: Install signs using mounting methods indicated and according to manufacturer's written instructions.

1. Install signs level, plumb, true to line, and at locations and heights indicated, with sign surfaces free of distortion and other defects in appearance.
2. Install signs so they do not protrude or obstruct according to the accessibility standard.
3. Before installation, verify that sign surfaces are clean and free of materials or debris that would impair installation.
4. Corrosion Protection: Coat concealed surfaces of exterior aluminum in contact with grout, concrete, masonry, wood, or dissimilar metals, with a heavy coat of bituminous paint.

B. Accessible Signage: Install in locations on walls **as indicated on Drawings and according to the accessibility standard**.

C. Mounting Methods:

1. Concealed Studs: Using a template, drill holes in substrate aligning with studs on back of sign. Remove loose debris from hole and substrate surface.

    a. Masonry Substrates: Fill holes with adhesive. Leave recess space in hole for displaced adhesive. Place sign in position and push until flush to surface, embedding studs in holes. Temporarily support sign in position until adhesive fully sets.
    b. Thin or Hollow Surfaces: Place sign in position and flush to surface, install washers and nuts on studs projecting through opposite side of surface, and tighten.

2. Projecting Studs: Using a template, drill holes in substrate aligning with studs on back of sign. Remove loose debris from hole and substrate surface.

    a. Masonry Substrates: Fill holes with adhesive. Leave recess space in hole for displaced adhesive. Place spacers on studs, place sign in position, and push until spacers are pinched between sign and substrate, embedding the stud ends in holes. Temporarily support sign in position until adhesive fully sets.
    b. Thin or Hollow Surfaces: Place spacers on studs, place sign in position with spacers pinched between sign and substrate, and install washers and nuts on stud ends projecting through opposite side of surface, and tighten.

3. Through Fasteners: Drill holes in substrate using predrilled holes in sign as template. Countersink holes in sign if required. Place sign in position and flush to surface. Install through fasteners and tighten.
4. Brackets: Remove loose debris from substrate surface and install backbar or bracket supports in position so that signage is correctly located and aligned.
5. Adhesive: Clean bond-breaking materials from substrate surface and remove loose debris. Apply linear beads or spots of adhesive symmetrically to back of sign and of suitable quantity to support weight of sign after cure without slippage. Keep adhesive away from edges to prevent adhesive extrusion as sign is applied and to prevent visibility of cured adhesive at sign edges. Place sign in position, and push to engage adhesive. Temporarily support sign in position until adhesive fully sets.
6. Shim-Plate Mounting: Provide 1/8-inch- (3-mm-) thick, concealed aluminum shim plates with predrilled and countersunk holes, at locations indicated, and where other direct mounting methods are impractical. Attach plate with fasteners and anchors suitable for secure attachment to substrate. Attach signs to plate using method specified above.

D. Field-Applied, Vinyl-Character Signs: Clean and dry substrate. Align sign characters in final position before removing release liner. Remove release liner in stages, and apply and firmly press characters into final position. Press from the middle outward to obtain good bond without blisters or fishmouths. Remove carrier film without disturbing applied vinyl film.

E. Signs Mounted on Glass: Provide opaque sheet matching sign material and finish onto opposite side of glass to conceal back of sign.

## 3.3 ADJUSTING AND CLEANING

A. Remove and replace damaged or deformed signs and signs that do not comply with specified requirements. Replace signs with damaged or deteriorated finishes or components that cannot be successfully repaired by finish touchup or similar minor repair procedures.

B. Remove temporary protective coverings and strippable films as signs are installed.

C. On completion of installation, clean exposed surfaces of signs according to manufacturer's written instructions, and touch up minor nicks and abrasions in finish. Maintain signs in a clean condition during construction and protect from damage until acceptance by Owner.

END OF SECTION 10 1423

# SECTION 10 1423.16 - ROOM-IDENTIFICATION SIGNAGE

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes room-identification signs that are directly attached to the building.

B. Related Requirements:

1. Section 10 1419 "Dimensional Letter Signage" for letter signage.
2. Section 10 1423 "Panel Signage" for panel and applied signage.

### 1.3 DEFINITIONS

A. Accessible: In accordance with the accessibility standard.

### 1.4 COORDINATION

A. Furnish templates for placement of sign-anchorage devices embedded in permanent construction by other installers.

B. Furnish templates for placement of electrical service embedded in permanent construction by other installers.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.

C. Shop Drawings: For room-identification signs.

1. Include fabrication and installation details and attachments to other work.
2. Show sign mounting heights, locations of supplementary supports to be provided by other installers, and accessories.
3. Show message list, typestyles, graphic elements, including raised characters and Braille, and layout for each sign at least half size.

D. Samples for Initial Selection: For each type of sign assembly, exposed component, and exposed finish.

1. Include representative Samples of available typestyles and graphic symbols.

E. Samples for Verification: For each type of sign assembly showing all components and with the required finish(es), in manufacturer's standard size unless otherwise indicated and as follows:

1. Room-Identification Signs: Full-size Sample.
2. Variable Component Materials: Full-size Sample of each base material, character (letter, number, and graphic element) in each exposed color and finish not included in Samples above.
3. Exposed Accessories: Full-size Sample of each accessory type.

F. Product Schedule: For room-identification signs. Use same designations indicated on Drawings or specified.

## 1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer and manufacturer.

B. Sample Warranty: For special warranty.

## 1.7 CLOSEOUT SUBMITTALS

A. Maintenance Data: For signs to include in maintenance manuals.

## 1.8 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Variable Component Materials: 24 replaceable text inserts and interchangeable characters (letters, numbers, and graphic elements) of each type.
2. Tools: Two set(s) of specialty tools for assembling signs and replacing variable sign components.

## 1.9 QUALITY ASSURANCE

A. Installer Qualifications: Manufacturer of products.

## 1.10 FIELD CONDITIONS

A. Field Measurements: Verify locations of anchorage devices and electrical service bedded in permanent construction by other installers by field measurements before fabrication and indicate measurements on Shop Drawings.

1.11 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of signs that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Deterioration of finishes beyond normal weathering.
b. Deterioration of embedded graphic image.
c. Separation or delamination of sheet materials and components.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: Comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design" and ICC A117.1.

2.2 ROOM-IDENTIFICATION SIGNS

A. Room-Identification Sign: Sign system with smooth, uniform surfaces; with message and characters having uniform faces, sharp corners, and precisely formed lines and profiles; and as follows:

B. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

1. ASI Sign Systems, Inc.
2. Best Sign Systems, Inc.
3. Mohawk Sign Systems.
4. Seton Identification Products.
5. Signature Signs, Inc.
6. Systems 2/90, Inc.

7. Acrylic Sheet Sign: Frosted acrylic sheet with raised graphics and painted subsurface.

a. Sheet Thickness: ¼”.
b. Surface-Applied Graphics: 5/8” Applied raised text.
c. Subsurface Graphics: Slide-in changeable insert as indicated on drawings
d. Subsurface Graphics: Magnetic Name Plate as indicated on drawings
e. Color(s): As selected by Architect from manufacturer's full range.
f. Trim: Aluminum Trim Accent
g. Font: Arial unless otherwise noted.

8. Sign-Panel Perimeter: Finish Flame Polished Edges Smooth.

a. Edge Condition at Vertical Edges: Square.
b. Corner Condition in Elevation: Square.

9. Mounting: Surface mounted to wall with two-faced tape.
10. Text and Typeface: Accessible raised characters and Braille as selected by Architect from manufacturer's full range and variable content as scheduled. Finish raised characters to contrast with background color, and finish Braille to match background color.

## 2.3 SIGN MATERIALS

A. Aluminum Sheet and Plate: ASTM B 209 (ASTM B 209M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

B. Aluminum Extrusions: ASTM B 221 (ASTM B 221M), alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

C. Acrylic Sheet: ASTM D 4802, category as standard with manufacturer for each sign, Type UVF (UV filtering).

D. Vinyl Film: UV-resistant vinyl film with pressure-sensitive, permanent adhesive; die cut to form characters or images as indicated on Drawings.

E. Paints and Coatings for Sheet Materials: Inks, dyes, and paints that are recommended by manufacturer for optimum adherence to surface and are UV and water resistant for colors and exposure indicated.

## 2.4 ACCESSORIES

A. Fasteners and Anchors: Manufacturer's standard as required for secure anchorage of signs, noncorrosive and compatible with each material joined, and complying with the following:

1. Use concealed fasteners and anchors unless indicated to be exposed.
2. For exterior exposure, furnish stainless-steel or hot-dip galvanized devices unless otherwise indicated.
3. Exposed Metal-Fastener Components, General:

    a. Fabricated from same basic metal and finish of fastened sign unless otherwise indicated.
    b. Fastener Heads: Use flathead or oval countersunk screws and bolts with tamper-resistant spanner-head or one-way-head slots unless otherwise indicated.

4. Sign Mounting Fasteners:

    a. Concealed Studs: Concealed (blind), threaded studs welded or brazed to back of sign material or screwed into back of sign assembly unless otherwise indicated.
    b. Through Fasteners: Exposed metal fasteners matching sign finish, with type of head indicated, and installed in predrilled holes.

B. Adhesive: As recommended by sign manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.

C. Two-Face Tape: Manufacturer's standard high-bond, foam-core tape, 0.045 inch (1.14 mm) thick, with adhesive on both sides.

D. Hook-and-Loop Tape: Manufacturer's standard two-part tape consisting of hooked part on sign back and looped side on mounting surface.

E. Magnetic Tape: Manufacturer's standard magnetic tape with adhesive on one side.

## 2.5 FABRICATION

A. General: Provide manufacturer's standard sign assemblies according to requirements indicated.

1. Preassemble signs and assemblies in the shop to greatest extent possible. Disassemble signs and assemblies only as necessary for shipping and handling limitations. Clearly mark units for reassembly and installation; apply markings in locations concealed from view after final assembly.
2. Mill joints to a tight, hairline fit. Form assemblies and joints exposed to weather to resist water penetration and retention.
3. Conceal connections if possible; otherwise, locate connections where they are inconspicuous.
4. Provide rabbets, lugs, and tabs necessary to assemble components and to attach to existing work. Drill and tap for required fasteners. Use concealed fasteners where possible; use exposed fasteners that match sign finish.

B. Subsurface-Applied Graphics: Apply graphics to back face of clear face-sheet material to produce precisely formed image. Image shall be free of rough edges.

C. Subsurface-Etched Graphics: Reverse etch back face of clear face-sheet material. Fill resulting copy with manufacturer's standard enamel. Apply opaque manufacturer's standard background color coating over enamel-filled copy.

D. Signs with Changeable Message Capability: Fabricate signs to allow insertion of changeable messages as follows:

1. For snap-in changeable inserts beneath removable face sheet, furnish one suction or other device to assist in removing face sheet. Furnish initial changeable insert. Furnish two blank inserts for each sign for Owner's use.
2. For slide-in changeable inserts, fabricate slot without burrs or constrictions that inhibit function. Furnish initial changeable insert. Furnish two blank inserts for each sign for Owner's use.

## 2.6 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

## 2.7 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, Class I, 0.018 mm or thicker.

B. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils (0.04 mm). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

## PART 3 - EXECUTION

### 3.1 INSTALLATION

A. General: Install signs using mounting methods indicated and according to manufacturer's written instructions.

1. Install signs level, plumb, true to line, and at locations and heights indicated, with sign surfaces free of distortion and other defects in appearance.
2. Install signs so they do not protrude or obstruct according to the accessibility standard.
3. Before installation, verify that sign surfaces are clean and free of materials or debris that would impair installation.

B. Accessibility: Install signs in locations on walls as indicated on Drawings and according to the accessibility standard.

C. Mounting Methods:

1. Concealed Studs: Using a template, drill holes in substrate aligning with studs on back of sign. Remove loose debris from hole and substrate surface.

    a. Masonry Substrates: Fill holes with adhesive. Leave recess space in hole for displaced adhesive. Place sign in position and push until flush to surface, embedding studs in holes. Temporarily support sign in position until adhesive fully sets.
    b. Thin or Hollow Surfaces: Place sign in position and flush to surface, install washers and nuts on studs projecting through opposite side of surface, and tighten.

2. Through Fasteners: Drill holes in substrate using predrilled holes in sign as template. Countersink holes in sign if required. Place sign in position and flush to surface. Install through fasteners and tighten.
3. Adhesive: Clean bond-breaking materials from substrate surface and remove loose debris. Apply linear beads or spots of adhesive symmetrically to back of sign and of suitable quantity to support weight of sign after cure without slippage. Keep adhesive away from edges to prevent adhesive extrusion as sign is applied and to prevent visibility of cured adhesive at sign edges. Place sign in position, and push to engage adhesive. Temporarily support sign in position until adhesive fully sets.
4. Two-Face Tape: Clean bond-breaking materials from substrate surface and remove loose debris. Apply tape strips symmetrically to back of sign and of suitable quantity to support weight of sign without slippage. Keep strips away from edges to prevent visibility at sign edges. Place sign in position, and push to engage tape adhesive.
5. Hook-and-Loop Tape: Clean bond-breaking materials from substrate surface and remove loose debris. Apply sign component of two-part tape strips symmetrically to back of sign and of suitable quantity to support weight of sign without slippage; push to engage tape adhesive. Keep tape strips 0.250 inch (6.35 mm) away from edges to prevent visibility at sign edges when sign is initially installed or reinstalled. Apply substrate component of tape to substrate in locations aligning with tape on back of sign; push and rub well to fully engage tape adhesive to substrate.

6. Magnetic Tape: Clean bond-breaking materials from substrate surface and remove loose debris. Apply tape strips symmetrically to back of sign and of suitable quantity to support weight of sign without slippage. Keep strips away from edges to prevent visibility at sign edges. Place sign in position.

3.2 ADJUSTING AND CLEANING

A. Remove and replace damaged or deformed signs and signs that do not comply with specified requirements. Replace signs with damaged or deteriorated finishes or components that cannot be successfully repaired by finish touchup or similar minor repair procedures.

B. Remove temporary protective coverings and strippable films as signs are installed.

C. On completion of installation, clean exposed surfaces of signs according to manufacturer's written instructions and touch up minor nicks and abrasions in finish. Maintain signs in a clean condition during construction and protect from damage until acceptance by Owner.

END OF SECTION 10 1423.16

# SECTION 10 2213 - WIRE MESH CEILINGS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Wire mesh ceilings.

B. Related Requirements:

1. Section 09 5113 “Acoustical Panels Ceilings” for ceiling tiles adhered to mesh ceilings.

### 1.2 DEFINITIONS

A. Intermediate Crimp: Wires pass over one and under the next adjacent wire in both directions, with wires crimped before weaving and with extra crimps between the intersections.

B. Lock Crimp: Deep crimps at points of the intersection that lock wires securely in place.

### 1.3 ACTION SUBMITTALS

A. Product Data:

1. Wire mesh ceilings.

B. Shop Drawings:

1. Include plans, elevations, sections, and attachment details.

### 1.4 INFORMATIONAL SUBMITTALS

A. Certificates:

1. Welding certificates.

B. Qualification Statements: For Installer.

### 1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For wire mesh partition hardware.

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials to Owner that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1.7 QUALITY ASSURANCE

A. Qualifications:

1. Installers: Authorized representative who is trained and approved by manufacturer.
2. Welding Qualifications: Qualify procedures and personnel in accordance with the following welding codes:

a. AWS D1.3/D1.3M.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Inventory wire mesh partition material on receipt and provide secure lockup for wire mesh partition door hardware delivered to Project site.

1. Tag each item or package separately with identification and include basic installation instructions with each item or package.

1.9 FIELD CONDITIONS

A. Field Measurements: Verify actual dimensions of construction contiguous with wire mesh units by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Acorn Wire & Iron Works.
2. Folding Guard, Incorporated.
3. Jesco Industries, Inc.
4. SpaceGuard Products.
5. WireCrafters, LLC.

2.2 SOURCE LIMITATIONS

A. For wire mesh products, obtain each color, grade, finish, type, and variety from single source with resources to provide products of consistent quality in appearance and physical properties.

2.3 PERFORMANCE REQUIREMENTS

A. Structural Performance: Wire mesh units to withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated.

1. Concentrated load of 50 lbf (0.22 kN) applied horizontally on an area of 1 sq. ft. (0.093 sq. m) at any location on a panel.
2. Total load of 200 lbf (0.89 kN) applied uniformly over each panel.
3. Concentrated load and total load need not be assumed to act concurrently.

2.4 WIRE MESH CEILINGS

A. Mesh, Framing, and Stiffeners: Fabricated from same mesh and framing as wire mesh partition panels.

B. Perimeter Partition Supports: 1-1/2-by-1-1/2-by-1/8-inch (38-by-38-by-3.2-mm)] steel angle, with holes for 1/4-inch- (6-mm-) diameter bolts aligned for bolting to top of wire mesh partitions and to sides of wire mesh ceiling panels.

C. Wall Supports: 1-1/2-by-1-1/2-by-1/8-inch (38-by-38-by-3.2-mm) steel angle punched for attachment to wall and wire mesh ceiling panels.

D. Intermediate Supports: Steel I-beams or rectangular tubes, as recommended by manufacturer.

E. Finishes: Match adjacent wire mesh partitions.

2.5 MATERIALS

A. Steel Wire: ASTM A510/A510M.

B. Steel Plates, Channels, Angles, and Bars: ASTM A36/A36M.

C. Steel Sheet: Cold-rolled steel sheet, ASTM A1008/A1008M, Commercial Steel (CS), Type B.

D. Steel Tubing: ASTM A500/A500M, cold-formed structural-steel tubing or ASTM A513/A513M, Type 5, mandrel-drawn mechanical tubing.

E. Metallic-Coated Steel Sheet: ASTM A653/A653M, Commercial Steel (CS), Type B; with G60 (Z180) zinc (galvanized) or A60 (ZF180) zinc-iron-alloy (galvannealed) coating designation.

F. Panel-to-Panel Fasteners: Manufacturer's standard steel bolts, nuts, and washers.

G. Post-Installed Anchors: Capable of sustaining, without failure, a load equal to 6 times the load imposed when installed in unit masonry and 4 times the load imposed when installed in concrete, as determined by testing in accordance with ASTM E488/E488M, conducted by a qualified independent testing agency.

1. Material for Interior Locations: Carbon-steel components are zinc plated to comply with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, unless otherwise indicated.

H. Power-Driven Fasteners: ICC-ES AC70.

I. Universal Shop Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer, complying with MPI#79.

1. Use primer containing pigments that make it easily distinguishable from zinc-rich primer.

J. Zinc-Rich Primer: Compatible with topcoat, complying with SSPC-Paint 20 or SSPC-Paint 29.

K. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds in steel, complying with SSPC-Paint 20.

## 2.6 FABRICATION

A. General: Fabricate wire mesh items from components of sizes not less than those indicated. Use larger-sized components as recommended by wire mesh item manufacturer. Furnish bolts, hardware, and accessories required for complete installation with manufacturer's standard finishes.

1. Fabricate wire mesh items to be readily disassembled.
2. Welding: Weld corner joints of framing and grind smooth, leaving no evidence of joint.

B. Wire Mesh Ceilings: Fabricate wire mesh partitions with cutouts for pipes, ducts, beams, and other items indicated. Finish edges of cutouts to provide a neat, protective edge.

1. Mesh: Securely clinch mesh to framing.
2. Framing: Fabricate framing with mortise-and-tenon corner construction.

a. Provide stiffeners as indicated or, if not indicated, as required by panel span and as recommended by wire mesh ceiling manufacturer. Weld stiffeners to framing.

## 2.7 STEEL AND IRON FINISHES

A. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A153/A153M for steel and iron hardware and with ASTM A123/A123M for other steel and iron products.

B. Preparation for Shop Priming Galvanized Items: After galvanizing, thoroughly clean items of grease, dirt, oil, flux, and other foreign matter, and treat with metallic phosphate process.

C. Preparation for Shop Priming: Prepare surfaces to comply with SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."

D. Shop Priming: Apply shop primer to uncoated surfaces of wire mesh units unless otherwise indicated. Comply with SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine floors for suitable conditions where wire mesh items will be installed.

C. Examine walls to which wire mesh items will be attached for properly located blocking, grounds, and other solid backing for attachment of support fasteners.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 INSTALLATION OF WIRE MESH CEILINGS

A. Anchor wall support angle to walls at 12 inches (300 mm) o.c. and as follows:

1. For concrete and solid masonry anchorage, use expansion anchors.
2. For hollow masonry anchorage, use toggle bolts.
3. For steel-framed gypsum board assemblies, fasten brackets directly to steel framing or concealed reinforcements using self-tapping screws of size and type required to support structural loads.

B. Attach wire mesh ceiling panels to wall support angles with bolts at 12 inches (305 mm) o.c.

C. Attach wire mesh ceiling panels to intermediate supports as recommended by manufacturer.

## 3.3 REPAIR

A. Repair Painting:

1. Wire brush and clean rust spots, welds, and abraded areas immediately after installation, and apply repair paint with same material as used for shop painting to comply with SSPC-PA 1 requirements for touching up shop-painted surfaces.

    a. Apply by brush or spray to provide a minimum 2.0-mil (0.05-mm) dry film thickness.

B. Repair of Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

## 3.4 PROTECTION

A. Remove and replace defective work, including doors and framing that are warped, bowed, or otherwise unacceptable.

END OF SECTION 10 2213

# SECTION 10 2600 - WALL PROTECTION

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Corner guards (CG).
2. Prefinished fiberglass reinforced wall protection (FRP).

B. Related Requirements:

1. Section 08 7100 "Door Hardware" for metal and plastic protective trim units, according to BHMA A156.6, used for armor, kick, mop, and push plates.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, impact strength, dimensions of individual components and profiles, and finishes.
2. Include fire ratings of units recessed in fire-rated walls and listings for door-protection items attached to fire-rated doors.

B. Shop Drawings: For each type of wall and door protection showing locations and extent.

1. Include plans, elevations, sections, and attachment details.

C. Samples for Initial Selection: For each type of impact-resistant wall-protection unit indicated, in each color and texture specified.

1. Include Samples of accent strips and accessories to verify color selection.

D. Samples for Verification: For each type of exposed finish on the following products, prepared on Samples of size indicated below:

1. Corner Guards: 12 inches (300 mm) long. Include example top caps.
2. Abuse-Resistant Wall Covering: 6 by 6 inches (150 by 150 mm) square.

### 1.4 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of handrail.

B. Material Certificates: For each type of exposed plastic material.

C. Sample Warranty: For special warranty.

## 1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of wall and door protection product to include in maintenance manuals.

1. Include recommended methods and frequency of maintenance for maintaining best condition of plastic covers under anticipated traffic and use conditions. Include precautions against using cleaning materials and methods that may be detrimental to finishes and performance.

## 1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Wall-Guard Covers: Full-size plastic covers of maximum length equal to 2 percent of each type, color, and texture of cover installed, but no fewer than two, 96-inch- (2400-mm-) long units.
2. Corner-Guard Covers: Full-size plastic covers of maximum length equal to 2 percent of each type, color, and texture of cover installed, but no fewer than two, 48-inch- (1200-mm-) long units.
3. Mounting and Accessory Components: Amounts proportional to the quantities of extra materials. Package mounting and accessory components with each extra material.

## 1.7 DELIVERY, STORAGE, AND HANDLING

A. Store wall and door protection in original undamaged packages and containers inside well-ventilated area protected from weather, moisture, soiling, extreme temperatures, and humidity.

1. Maintain room temperature within storage area at not less than 70 deg F (21 deg C) during the period plastic materials are stored.
2. Keep plastic materials out of direct sunlight.
3. Store plastic wall- and door-protection components for a minimum of 72 hours, or until plastic material attains a minimum room temperature of 70 deg F (21 deg C).

a. Store corner-guard covers in a vertical position.
b. Store wall-guard covers in a horizontal position.

## 1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of wall- and door-protection units that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures including detachment of components from each other or from the substrates, delamination, and permanent deformation beyond normal use.

b. Deterioration of metals, metal finishes, plastics, and other materials beyond normal use.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Surface Burning Characteristics: Comply with ASTM E 84 or UL 723; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Flame-Spread Index: 25 or less.
2. Smoke-Developed Index: 450 or less.

B. Regulatory Requirements: Comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines for Buildings and Facilities and ICC A117.1.

2.2 CORNER GUARDS (CG)

A. Surface-Mounted, flexible Corner Guards: Fabricated as one piece polymer flexible PVC with formed edges; fabricated with 90- or 135-degree turn to match wall condition.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Babcock-Davis.
b. Koroseal Interior Products, LLC.
c. inpro Corporation.

2. Wing Size: Nominal 1-1/2 by 1-1/2 inches (38 by 38 mm).
3. Height: 4 feet (1.2 m).
4. Thickness: Minimum 0.110 inch (2.8 mm).
5. Mounting: Manufacturer’s recommended mounting method.
6. Color and Texture: As selected by Architect from manufacturer's full range.

2.3 FIBERGLASS REINFORCED WALL PROTECTION (FRP)

A. Abuse-Resistant Sheet Wall Covering: Fabricated from semi-rigid, fiberglass reinforced sheet wall-covering material.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include:

a. Construction Specialties, Inc.
b. InPro Corporation (IPC).
c. Korogard Wall Protection Systems; a division of RJF International Corporation.
d. Marlite, Inc.

2. Size: 48 by 96 inches for sheet.
3. Sheet Thickness: 0.090 inch (2.2 mm).
4. Color and Texture: Marlite, Standard FRP, Pebbled, P100 White or approved equivalent.
5. Height: As indicated on drawings
6. Trim and Joint Moldings: Extruded rigid plastic that matches wall-covering color.
7. Mounting: Adhesive.

## 2.4 MATERIALS

A. Plastic Materials: Chemical- and stain-resistant, high-impact-resistant plastic with integral color throughout; extruded and sheet material as required, thickness as indicated.

B. Polycarbonate Plastic Sheet: ASTM D 6098, S-PC01, Class 1 or Class 2, abrasion resistant; with a minimum impact-resistance rating of 15 ft.-lbf/in. (800 J/m) of notch when tested according to ASTM D 256, Test Method A.

C. Fasteners: Aluminum, nonmagnetic stainless-steel, or other noncorrosive metal screws, bolts, and other fasteners compatible with items being fastened. Use security-type fasteners where exposed to view.

D. Adhesive: As recommended by protection product manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.

## 2.5 FABRICATION

A. Fabricate wall and door protection according to requirements indicated for design, performance, dimensions, and member sizes, including thicknesses of components.

B. Factory Assembly: Assemble components in factory to greatest extent possible to minimize field assembly. Disassemble only as necessary for shipping and handling.

C. Quality: Fabricate components with uniformly tight seams and joints and with exposed edges rolled. Provide surfaces free of wrinkles, chips, dents, uneven coloration, and other imperfections. Fabricate members and fittings to produce flush, smooth, and rigid hairline joints.

## 2.6 FINISHES

A. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and wall areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine walls to which wall and door protection will be attached for blocking, grounds, and other solid backing that have been installed in the locations required for secure attachment of support fasteners.

1. For wall and door protection attached with adhesive, verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Complete finishing operations, including painting, before installing wall and door protection.

B. Before installation, clean substrate to remove dust, debris, and loose particles.

3.3 INSTALLATION

A. Installation Quality: Install wall and door protection according to manufacturer's written instructions, level, plumb, and true to line without distortions. Do not use materials with chips, cracks, voids, stains, or other defects that might be visible in the finished Work.

B. Mounting Heights: Install wall and door protection in locations and at mounting heights indicated on Drawings.

C. Accessories: Provide splices, mounting hardware, anchors, trim, joint moldings, and other accessories required for a complete installation.

1. Provide anchoring devices and suitable locations to withstand imposed loads.
2. Adjust end and top caps as required to ensure tight seams.

D. Abuse-Resistant Wall Covering: Install top and edge moldings, corners, and divider bars as required for a complete installation.

3.4 CLEANING

A. Immediately after completion of installation, clean plastic covers and accessories using a standard ammonia-based household cleaning agent.

B. Remove excess adhesive using methods and materials recommended in writing by manufacturer.

END OF SECTION 10 2600

# SECTION 10 2641 – BULLET RESISTANT PANELS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Bullet resistant fiberglass panels.

B. Related Requirements:

1. Section 09 2216 "Non-Structural Metal Framing" for metal framing to receive bullet-resistant panels.

### 1.3 REFERENCES

A. ASTM International ASTM E119 - Standard Test Methods for Fire Tests of Building Construction and Materials.

B. Underwriters Laboratories UL 752 - Bullet Resisting Equipment.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Product Data: Include product description for bullet-resistant sheet including bullet-resistant ratings.
2. Certificates: Submit printed data to indicate compliance with following requirements.

    a. UL Listing Verification and UL752 Current Test Results as provided by Underwriters Laboratories.
    b. ASTM E119-98 One-Hour Fire Rating of Building Construction and Materials.

B. Sustainable Design Submittals:

1. Product Data: For recycled content, indicating postconsumer and pre-consumer recycled content and cost.
2. Product Certificates: For materials manufactured within 500 miles (804 km) of Project, indicating location of material manufacturer and point of extraction, harvest, or recovery for each raw material. Include distance to Project and cost for each raw material.

C. Shop Drawings: For each type of wall showing locations and extent.

1. Include plans, elevations, sections, and attachment details.

1.5 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of panel.

B. Material Certificates: For each type of exposed plastic material.

C. Sample Warranty: For special warranty.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of product to include in maintenance manuals.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials to project with manufacturer's UL listed Labels intact and legible.

B. Handle material with care to prevent damage. Store materials inside under cover, stack flat and off the floor.

C. Store materials in original undamaged packages and containers inside well-ventilated area protected from weather, moisture, soiling, extreme temperatures, and humidity.

1. Maintain room temperature within storage area at not less than 70 deg F (21 deg C) during the period plastic materials are stored.
2. Keep plastic materials out of direct sunlight.

1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of bullet resistant fiberglass panel that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Failures including detachment of components from each other or from the substrates, delamination, and permanent deformation beyond normal use.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain bullet resistant fiberglass panels from single source and from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Bullet Resistant Fiberglass Panels shall be non-ricochet type to permit the encapture and retention of an attacking projectile lessening the potential of a random injury or lateral penetration.

B. Panel Rating: UL752 Level 3.

C. Bullet resistance of joints: equal to that of the panel.

2.3 BULLET RESISTANT FIBERGLASS PANELS

A. Bullet Resistant Fiberglass Sheet: Manufactured from multiple layers of woven roving ballistic grade fiberglass cloth impregnated with thermoset polyester resin, compressed into flat rigid sheets.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. AmorCore (Waco Composites Ltd.)
b. Armortex.
c. CR Laurence Co., Inc.
d. US Bullet Proofing
e. TSS Bullet Proofing

2. Ballistic Level 3 tested to UL 752.
3. Fire rating: 1 hour, tested to ASTM E119.

2.4 MATERIALS

A. Panels fabricated of multiple layers of woven roving ballistic grade fiberglass cloth impregnated with a thermoset polyester resin and compressed into flat rigid sheets.

B. Thickness: Level 3: 7/16” nominal thickness.

C. Nominal Weight: Level 3: 5 pounds per sq. ft.

2.5 ACCESSORIES

A. Adhesives and Fasteners: Type recommended by bullet resistant panel manufacturer for specific application.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and wall areas, with Installer present, for compliance with requirements for installation tolerances, fire rating, and other conditions affecting performance of the Work.

B. Prior to starting installation, verify work of related trades required in contract documents is complete to the point where work of this Section may properly commence.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 JOINTS

A. Reinforce joints with a back-up layer of bullet resistive material. Minimum width of reinforcing layer at joint shall be 4-inches, centered on panel joints.

3.3 INSTALLATION

A. Installation Quality: Install bullet resistant panels according to manufacturer's written instructions, level, plumb, and true to line without distortions. Do not use materials with chips, cracks, voids, stains, or other defects.

B. Secure armor panels using screws, bolts, or an industrial adhesive.

1. Method of application shall install panels minimizing vulnerabilities by fitting tightly to adjacent surfaces including concrete floor slab, concrete roof slab, bullet resistive door frames, bullet resistive window frames, and the like.

C. Fasten or adhere panels to supports in manner to maintain bullet-resistive rating at perimeter, junctures with other materials, and penetrations.

3.4 CLEANING

A. Remove excess adhesive using methods and materials recommended in writing by manufacturer.

END OF SECTION 10 2641

# SECTION 10 2800 - TOILET, BATH, AND LAUNDRY ACCESSORIES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Public-use washroom accessories.
2. Private-use shower room accessories.
3. Warm-air dryers.
4. Underlavatory guards.
5. Custodial accessories.
6. Automatic external defibrillator (AED).

B. Related Requirements:

1. Section 09 3013 "Ceramic Tiling" for ceramic toilet and bath accessories.

### 1.3 COORDINATION

A. Coordinate accessory locations with other work to prevent interference with clearances required for access by people with disabilities, and for proper installation, adjustment, operation, cleaning, and servicing of accessories.

B. Deliver inserts and anchoring devices set into concrete or masonry as required to prevent delaying the Work.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Include anchoring and mounting requirements, including requirements for cutouts in other work and substrate preparation.
3. Include electrical characteristics.

B. Product Schedule: Indicating types, quantities, sizes, and installation locations by room of each accessory required.

1. Identify locations using room designations indicated.
2. Identify accessories using designations indicated.

1.5 INFORMATIONAL SUBMITTALS

A. Sample Warranty: For manufacturer's special warranty.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For accessories to include in maintenance manuals.

1.7 WARRANTY

A. Manufacturer's Special Warranty for Mirrors: Manufacturer agrees to repair or replace mirrors that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, visible silver spoilage defects.
2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Structural Performance: Design accessories and fasteners to comply with the following requirements:

1. Grab Bars: Installed units are able to resist 250 lbf (1112 N) concentrated load applied in any direction and at any point.
2. Shower Seats: Installed units are able to resist 250 lbf (1112 N) applied in any direction and at any point.

C. Regulatory Requirements: Comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines for Buildings and Facilities and ICC A117.1 for toilet accessories designated as accessible.

2.2 PUBLIC-USE WASHROOM ACCESSORIES

A. Source Limitations: Obtain public-use washroom accessories from single source from single manufacturer unless otherwise indicated.

B. Toilet Tissue (Roll) Dispenser (**TPD-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-4288 Multi-Roll Toilet Tissue Dispenser, or an equivalent product by one of the following manufacturers:

a. AJW Architectural Products.
b. American Specialties, Inc.
c. Bobrick Washroom Equipment, Inc.
d. Bradley Corporation.
e. GAMCO Specialty Accessories; a division of Bobrick.

2. Description: Double-roll dispenser.
3. Mounting: Surface mounted.
4. Operation: Spindleless with tension-spring controlled delivery.
5. Capacity: Designed for 5-inch- (127-mm-) diameter tissue rolls.
6. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (satin).

C. Grab Bar (**GB-1,2,3)**:

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-9806 Stainless Steel Grab Bar, or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Mounting: Flanges with concealed fasteners.
3. Material: Stainless steel, 0.05 inch (1.3 mm) thick.

    a. Finish: Smooth, No. 4 finish (satin) on ends and slip-resistant texture in grip area.

4. Outside Diameter: 1-1/2 inches (38 mm).
5. Configuration and Length: As indicated on Drawings.

D. Sanitary-Napkin Disposal Unit **(SND-1:**

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-35139, or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Mounting: Surface mounted.
3. Door or Cover: Self-closing, disposal-opening cover and hinged face panel with tumbler lockset.
4. Receptacle: Removable.
5. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (satin).

E. Mirror Unit (**MR-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-290 Mirror or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Frame: Stainless steel, fixed tilt.

3. Hangers: Manufacturer's standard rigid, tamper and theft resistant
    a. Wall bracket of galvanized steel, equipped with concealed locking devices requiring a special tool to remove.
4. Size: As indicated on Drawings.

F. High-Speed Air Hand Dryer **(HD-1):**

1. Basis-of Design Product: Subject to compliance with requirements, provide Excel Thin Air handryer TA-SB, or an equivalent product by one of the following manufacturers:
    a. Bobrick Washroom Equipment, Inc.
    b. Dyson Inc.
    c. Excel Dryer Inc.
2. Description: High-speed, warm-air hand dryer for rapid hand drying.
3. Mounting: Surface mounted.
    a. Protrusion Limit: Installed unit protrudes maximum 4 inches (102 mm) from wall surface.
4. Operation: Infrared sensor activated with timed power cutoff switch.
    a. Average Dry Time: 12 seconds.
    b. Automatic Shutoff: At 60 seconds.
5. Maximum Sound Level: 69 dB
6. Cover Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (satin).
7. Electrical Requirements: 208 to 240 V, 9 to 10 A, 1900 to 2300 W.

2.3 PRIVATE-USE SHOWER ROOM ACCESSORIES

A. Source Limitations: Obtain public-use shower room accessories from single source from single manufacturer.

B. Robe Hook (**HK-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-6777 Stainless Steel Robe Hook, or an equivalent product by one of the following manufacturers:
    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.
2. Description: Single-prong unit.
3. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (satin).

C. Shower Curtain Rod (**SCR-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-6107 Heavy-Duty Shower Curtain Rod or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Description: 1-1/4-inch (32-mm) OD; fabricated from nominal 0.0375-inch- (0.95-mm-) thick stainless steel
3. Mounting Flanges: Stainless-steel flanges designed for concealed fasteners.
4. Finish: Stainless steel, No. 4 finish (satin).

D. Shower Curtain (**SCA-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-204-3 Vinyl Shower Curtain, or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Size: Minimum 6 inches (152 mm) wider than opening by 72 inches (1828 mm) high.
3. Material: Nylon-reinforced vinyl, minimum 10 oz. (284 g) or 0.008-inch- (0.2-mm-) thick vinyl, with integral antibacterial agent.
4. Color: White.
5. Grommets: Corrosion resistant at minimum 6 inches (152 mm) o.c. through top hem.
6. Shower Curtain Hooks: Chrome-plated or stainless-steel, spring wire curtain hooks with snap fasteners, sized to accommodate specified curtain rod. Provide one hook per curtain grommet.

E. Folding Shower Seat (**SB-1**):

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-51815 Folding Shower Seat, or an equivalent product by one of the following manufacturers:

    a. AJW Architectural Products.
    b. American Specialties, Inc.
    c. Bobrick Washroom Equipment, Inc.
    d. Bradley Corporation.
    e. GAMCO Specialty Accessories; a division of Bobrick.

2. Configuration: L-shaped seat, designed for wheelchair access.
3. Seat: Phenolic or polymeric composite of slat-type or one-piece construction in color as selected by Architect.
4. Mounting Mechanism: Stainless steel, No. 4 finish (satin).

2.4 UNDERLAVATORY GUARDS

A. Underlavatory Guard:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Buckaroos, Inc.
b. Plumberex Specialty Products, Inc.
c. Truebro by IPS Corporation.

2. Description: Insulating pipe covering for supply and drain piping assemblies that prevents direct contact with and burns from piping; allow service access without removing coverings.

3. Material and Finish: Antimicrobial, molded plastic, white.

2.5 CUSTODIAL ACCESSORIES

A. Source Limitations: Obtain custodial accessories from single source from single manufacturer.

B. Mop and Broom Holder:

1. Basis-of Design Product: Subject to compliance with requirements, provide Bobrick B-223 Utility Shelf with Mop/Broom Holders, or an equivalent product by one of the following manufacturers

a. AJW Architectural Products.
b. American Specialties, Inc.
c. Bobrick Washroom Equipment, Inc.
d. Bradley Corporation.
e. GAMCO Specialty Accessories; a division of Bobrick.

2. Description: Unit with hooks, holders, and rod suspended beneath shelf.

3. Length: 24 inches (610 mm).
4. Hooks: Three.
5. Mop/Broom Holders: Three, spring-loaded, rubber hat, cam type.
6. Material and Finish: Stainless steel, No. 4 finish (satin).

a. Shelf: Not less than nominal 0.05-inch- (1.3-mm-) thick stainless steel.
b. Rod: Approximately 1/4-inch- (6-mm-) diameter stainless steel.

2.6 AUTOMATIC EXTERNAL DEFIBRILLATOR (AED)

A. Basis-of-Design Product: Subject to compliance with requirements, provide Physio Control; Lifepak 1000 AED Graphical Display, or approved equivalent.

1. Provide recessed wall cabinet, wall sign, and check tag.

2.7 MATERIALS

A. Stainless Steel: ASTM A 666, Type 304, 0.031-inch (0.8-mm) minimum nominal thickness unless otherwise indicated.

B. Steel Sheet: ASTM A 1008/A 1008M, Designation CS (cold rolled, commercial steel), 0.036-inch (0.9-mm) minimum nominal thickness.

C. Fasteners: Screws, bolts, and other devices of same material as accessory unit and tamper-and-theft resistant where exposed, and of galvanized steel where concealed.

D. Mirrors: ASTM C 1503, Mirror Glazing Quality, clear-glass mirrors, nominal 6.0 mm thick.

2.8 FABRICATION

A. General: Fabricate units with tight seams and joints, and exposed edges rolled. Hang doors and access panels with full-length, continuous hinges. Equip units for concealed anchorage and with corrosion-resistant backing plates.

B. Keys: Provide universal keys for internal access to accessories for servicing and resupplying. Provide minimum of six keys to Owner's representative.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install accessories according to manufacturers' written instructions, using fasteners appropriate to substrate indicated and recommended by unit manufacturer. Install units level, plumb, and firmly anchored in locations and at heights indicated.

1. Remove temporary labels and protective coatings.

B. Grab Bars: Install to comply with specified structural-performance requirements.

C. Shower Seats: Install to comply with specified structural-performance requirements.

3.2 ADJUSTING AND CLEANING

A. Adjust accessories for unencumbered, smooth operation. Replace damaged or defective items.

B. Remove temporary labels and protective coatings.

C. Clean and polish exposed surfaces according to manufacturer's written instructions.

END OF SECTION 10 2800

# SECTION 10 4413 - FIRE PROTECTION CABINETS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Fire-protection cabinets for the following:

a. Portable fire extinguisher.

B. Related Requirements:

1. Section 10 4416 "Fire Extinguishers" for portable, hand-carried fire extinguishers accommodated by fire-protection cabinets.
2. Division 21 "Fire Suppression" for fire-hose connections.

### 1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Show door hardware, cabinet type, trim style, and panel style. Include roughing-in dimensions and details showing recessed-, semirecessed-, or surface-mounting method and relationships of box and trim to surrounding construction.

B. Shop Drawings: For fire-protection cabinets.

1. Include plans, elevations, sections, details, and attachments to other work.

C. Product Schedule: For fire-protection cabinets. Indicate whether recessed, semirecessed, or surface mounted. Coordinate final fire-protection cabinet schedule with fire-extinguisher schedule to ensure proper fit and function. Use same designations indicated on Drawings.

### 1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For fire-protection cabinets to include in maintenance manuals.

### 1.4 COORDINATION

A. Coordinate size of fire-protection cabinets to ensure that type and capacity of fire extinguishers indicated are accommodated.

B. Coordinate sizes and locations of fire-protection cabinets with wall depths.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain fire-protection cabinets, accessories, and fire extinguishers from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Fire-Rated Fire-Protection Cabinets: Listed and labeled to comply with requirements in ASTM E814 for fire-resistance rating of walls where they are installed.

2.3 FIRE-PROTECTION CABINET

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Babcock-Davis.
2. Guardian Fire Equipment, Inc.
3. JL Industries; Activar Construction Products Group, Inc.
4. Larsen's Manufacturing Company.
5. Nystrom, Inc.
6. Potter Roemer LLC; a Division of Morris Group International.

B. Fire-Protection Cabinet Type: Suitable for fire extinguisher.

C. Cabinet Construction: Nonrated, One-hour fire rated, and Two-hour fire rated.

1. Fire-Rated Cabinets: Construct fire-rated cabinets with double walls fabricated from 0.043-inch- (1.09-mm-) thick cold-rolled steel sheet lined with minimum 5/8-inch- (16-mm-) thick fire-barrier material. Provide factory-drilled mounting holes.

D. Cabinet Material: Cold-rolled steel sheet.

1. Shelf: Same metal and finish as cabinet.

E. Recessed Cabinet:

1. Exposed Flat Trim: One-piece combination trim and perimeter door frame overlapping surrounding wall surface, with exposed trim face and wall return at outer edge (backbend).

F. Semi recessed Cabinet: One-piece combination trim and perimeter door frame overlapping surrounding wall surface, with exposed trim face and wall return at outer edge (backbend).

1. Square-Edge Trim: 1-1/4- to 1-1/2-inch (32- to 38-mm) backbend depth.

G. Surface-Mounted Cabinet: Cabinet box fully exposed and mounted directly on wall with no trim.

H. Cabinet Trim Material: Same material and finish as door.

I. Door Material: Steel sheet.

J. Door Style: Fully glazed panel with frame.

K. Door Glazing: Acrylic sheet.

1. Acrylic Sheet Color: White.

L. Door Hardware: Manufacturer's standard door-operating hardware of proper type for cabinet type, trim style, and door material and style indicated.

1. Provide projecting lever handle with cam-action latch.
2. Provide continuous hinge, of same material and finish as trim, permitting door to open 180 degrees.

M. Accessories:

1. Mounting Bracket: Manufacturer's standard steel, designed to secure fire extinguisher to fire-protection cabinet, of sizes required for types and capacities of fire extinguishers indicated, with plated or baked-enamel finish.
2. Break-Glass Strike: Manufacturer's standard metal strike, complete with chain and mounting clip, secured to cabinet.
3. Lettered Door Handle: One-piece, cast-iron door handle with the word "FIRE" embossed into face.
4. Door Lock: Cylinder lock, keyed alike to other cabinets.
5. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated.

    a. Identify fire extinguisher in fire-protection cabinet with the words "FIRE EXTINGUISHER."

        1) Location: Applied to cabinet door.
        2) Application Process: Silk-screened.
        3) Lettering Color: Black.
        4) Orientation: Horizontal.

6. Alarm: Manufacturer's standard alarm that actuates when fire-protection cabinet door is opened and that is powered by batteries.

N. Materials:

1. Cold-Rolled Steel: ASTM A1008/A1008M, Commercial Steel (CS), Type B.

    a. Finish: Baked enamel, TGIC polyester powder coat, HAA polyester powder coat, epoxy powder coat, or polyester/epoxy hybrid powder coat, complying with AAMA 2603.
    b. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
    c. Color: As selected by Architect from manufacturer's full range.

2. Tempered Float Glass: ASTM C1048, Kind FT, Condition A, Type I, Quality q3, 3 mm thick, Class 1 (clear).

2.4 FABRICATION

A. Fire-Protection Cabinets: Provide manufacturer's standard box (tub) with trim, frame, door, and hardware to suit cabinet type, trim style, and door style indicated.

1. Weld joints and grind smooth.
2. Miter corners and grind smooth.
3. Provide factory-drilled mounting holes.
4. Prepare doors and frames to receive locks.
5. Install door locks at factory.

B. Cabinet Doors: Fabricate doors according to manufacturer's standards, from materials indicated and coordinated with cabinet types and trim styles.

1. Fabricate door frames with tubular stiles and rails and hollow-metal design, minimum 1/2 inch (13 mm) thick.
2. Fabricate door frames of one-piece construction with edges flanged.
3. Miter and weld perimeter door frames and grind smooth.

C. Cabinet Trim: Fabricate cabinet trim in one piece with corners mitered, welded, and ground smooth.

2.5 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM's AMP 500, "Metal Finishes Manual for Architectural and Metal Products," for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces of fire-protection cabinets from damage by applying a strippable, temporary protective covering before shipping.

C. Finish fire-protection cabinets after assembly.

D. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine walls and partitions for suitable framing depth and blocking where recessed and semi recessed cabinets will be installed.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Prepare recesses for recessed and semi recessed fire-protection cabinets as required by type and size of cabinet and trim style.

3.3 INSTALLATION OF FIRE-PROTECTION CABINETS

A. General: Install fire-protection cabinets in locations and at mounting heights indicated or, if not indicated, at height indicated below:

1. Fire-Protection Cabinet Mounting Height: 42 inches (1067 mm) above finished floor to top of fire extinguisher.

B. Fire-Protection Cabinets: Fasten cabinets to structure, square and plumb.

1. Unless otherwise indicated, provide recessed fire-protection cabinets. If wall thickness is inadequate for recessed cabinets, provide semi recessed fire-protection cabinets.
2. Provide inside latch and lock for break-glass panels.
3. Fasten mounting brackets to inside surface of fire-protection cabinets, square and plumb.
4. Fire-Rated Cabinets:

a. Install cabinet with not more than 1/16-inch (1.6-mm) tolerance between pipe OD and knockout OD. Center pipe within knockout.
b. Seal through penetrations with firestopping sealant as specified in Section 078413 "Penetration Firestopping."

C. Identification:

1. Apply vinyl lettering at locations indicated.

3.4 ADJUSTING AND CLEANING

A. Remove temporary protective coverings and strippable films, if any, as fire-protection cabinets are installed unless otherwise indicated in manufacturer's written installation instructions.

B. Adjust fire-protection cabinet doors to operate easily without binding. Verify that integral locking devices operate properly.

C. On completion of fire-protection cabinet installation, clean interior and exterior surfaces as recommended by manufacturer.

D. Touch up marred finishes or replace fire-protection cabinets that cannot be restored to factory-finished appearance. Use only materials and procedures recommended or furnished by fire-protection cabinet and mounting bracket manufacturers.

E. Replace fire-protection cabinets that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 10 4413

# SECTION 10 4416 - FIRE EXTINGUISHERS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes portable, hand-carried fire extinguishers and mounting brackets for fire extinguishers.

B. Related Requirements:

1. Section 10 4413 "Fire Protection Cabinets."

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product. Include rating and classification, material descriptions, dimensions of individual components and profiles, and finishes for fire extinguisher and mounting brackets.

B. Product Schedule: For fire extinguishers. Coordinate final fire-extinguisher schedule with fire-protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

### 1.4 INFORMATIONAL SUBMITTALS

A. Warranty: Sample of special warranty.

### 1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For fire extinguishers to include in maintenance manuals.

### 1.6 COORDINATION

A. Coordinate type and capacity of fire extinguishers with fire-protection cabinets to ensure fit and function.

### 1.7 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace fire extinguishers that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Failure of hydrostatic test according to NFPA 10.
b. Faulty operation of valves or release levers.

2. Warranty Period: Six years from date of Substantial Completion.

a. Initial certification date tagged for all fire extinguishers shall be the date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. NFPA Compliance: Fabricate and label fire extinguishers to comply with NFPA 10, "Portable Fire Extinguishers."

B. Fire Extinguishers: Listed and labeled for type, rating, and classification by an independent testing agency acceptable to authorities having jurisdiction.

1. Provide fire extinguishers approved, listed, and labeled by FM Global.

### 2.2 PORTABLE, HAND-CARRIED FIRE EXTINGUISHERS

A. Fire Extinguishers: Type, size, and capacity for each fire-protection cabinet and mounting bracket indicated.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Amerex Corporation.
b. Badger Fire Protection.
c. JL Industries, Inc.; a division of the Activar Construction Products Group.
d. Kidde Residential and Commercial Division.

2. Valves: Manufacturer's standard.
3. Handles and Levers: Manufacturer's standard.
4. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B and bar coding for documenting fire-extinguisher location, inspections, maintenance, and recharging.

B. Multipurpose Dry-Chemical Type in Aluminum Container: UL-rated 3-A:40-B:C, 5-lb (2.3-kg) nominal capacity, with monoammonium phosphate-based dry chemical in enameled-aluminum container.

### 2.3 MOUNTING BRACKETS

A. Mounting Brackets: Manufacturer's standard galvanized steel, designed to secure fire extinguisher to wall or structure, of sizes required for types and capacities of fire extinguishers indicated, with plated or red baked-enamel finish.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

    a. Amerex Corporation.
    b. Badger Fire Protection.
    c. JL Industries, Inc.; a division of the Activar Construction Products Group.
    d. Kidde Residential and Commercial Division.

B. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by Architect.

1. Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

    a. Orientation: Vertical.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine fire extinguishers for proper charging and tagging.

1. Remove and replace damaged, defective, or undercharged fire extinguishers.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

A. General: Install fire extinguishers and mounting brackets in locations indicated and in compliance with requirements of authorities having jurisdiction.

1. Mounting Brackets: 48 inches (1219 mm) above finished floor to top of fire extinguisher.

B. Mounting Brackets: Fasten mounting brackets to surfaces, square and plumb, at locations indicated.

END OF SECTION 10 4416

# SECTION 10 5113 - METAL LOCKERS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Evidence Lockers
2. Gear Lockers

B. Related Requirements:

1. Section 23 3001 “Ductwork and Sheet Metal” for attachment to building HVAC system.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles and finishes for each type of metal locker and bench.

B. Shop Drawings: For each type of metal locker.

1. Include plans, elevations, sections, and attachment details.
2. Show locker trim and accessories.
3. Include locker identification system and numbering sequence.

C. Samples for Initial Selection: Manufacturer's color charts showing the full range of colors available.

D. Product Schedule: For lockers.

### 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Sample Warranty: For special warranty.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For adjusting, repairing, and replacing locker doors and latching mechanisms to include in maintenance manuals.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. The following metal locker hardware items equal to 10 percent of amount installed for each type and finish installed, but no fewer than five units:

a. Locks.
b. Blank identification plates.
c. Hooks.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Do not deliver metal lockers until spaces to receive them are clean, dry, and ready for their installation.

1.9 FIELD CONDITIONS

A. Field Measurements: Verify actual dimensions of recessed openings by field measurements before fabrication.

1.10 COORDINATION

A. Coordinate sizes and locations of concrete bases for metal lockers.

B. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of work specified in other Sections to ensure that metal lockers can be supported and installed as indicated.

1.11 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of metal lockers that fail in materials or workmanship, excluding finish, within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures.
b. Faulty operation of latches and other door hardware.

2. Damage from deliberate destruction and vandalism is excluded.
3. Warranty Period for Welded Metal Lockers: Lifetime from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain metal lockers, and accessories from single source from single locker manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: For lockers indicated to be accessible, comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design" and ICC A117.1.

2.3 EVIDENCE LOCKERS (PASS-THRU)

A. Basis-of-Design Product: Subject to compliance with requirements, provide Spacesaver Pass-Thru Evidence Lockers, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

1. DSM Law Enforcement Products.
2. Fasco Security Products.
3. Spacesaver Corporation
4. Tiffin Metal Products.

B. Locker Arrangement: As indicated on drawings and selected by Architect from manufacturer’s range of product options.

C. Size: As indicated on drawings.

D. Material: Cold-rolled steel sheet.

E. Body: Assembled by welding body components together. Fabricate from unperforated steel sheet with thicknesses as follows:

1. Tops, Bottoms, and Sides: 14-gauge thickness.
2. Backs: 14-gauge thickness.
3. Shelves: 14-gauge, with double bend at front and single bend at sides and back.

F. Frames: Channel formed; fabricated from 14-gauge steel sheet; lapped and factory welded at corners; with top and bottom main frames factory welded into vertical main frames. Form continuous, integral door strike full height on vertical main frames.

1. Cross Frames between Tiers: Channel formed and fabricated from same material as main frames; welded to vertical main frames.

G. Doors: One piece; fabricated from 14-gauge nominal-thickness steel sheet; formed into channel shape with double bend at vertical edges and with right-angle single bend at horizontal edges.

1. Reinforcement: Manufacturer's standard reinforcing angles, channels, or stiffeners for doors more than 15 inches (381 mm) wide; welded to inner face of doors.

H. Hinges: Welded to door and attached to door frame with no fewer than two factory-installed rivets per hinge that are completely concealed and tamper resistant when door is closed; fabricated to swing 180 degrees; self-closing.

1. Continuous Hinges: Manufacturer's standard, steel, full height.

I. Recessed Door Handle and Latch: Stainless-steel cup with integral door pull, recessed so locking device does not protrude beyond face of door; pry and vandal resistant.

1. Multipoint Latching: Finger-lift latch control designed for use with built-in combination locks or padlocks; positive automatic latching and prelocking.

a. Latching Mechanism: Manufacturer's standard, rattle-free latching mechanism and moving components isolated to prevent metal-to-metal contact, and incorporating a prelocking device that allows locker door to be locked while door is open and then closed without unlocking or damaging lock or latching mechanism.

2. Locks:

a. Keyless locking system on deposit side, push button system to secure door.
b. Lock resets from rear (secure side) when rear door is in open position only.

J. Equipment: Equip each compartment with an identification plate.

K. Accessories:

1. Legs: 6 inches (152 mm) high; formed by extending vertical frame members or fabricated from 0.075-inch (1.90-mm) nominal-thickness steel sheet; welded to bottom of locker.

a. Closed Front and End Bases: Fabricated from 0.036-inch (0.91-mm) nominal-thickness steel sheet.

2. Recess Trim: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.
3. Filler Panels: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.
4. Boxed End Panels: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.

L. Finish: Powder Coat.

1. Color: As selected by Architect from manufacturer's full range.

## 2.4 GEAR LOCKERS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Spacesaver Gear Lockers, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

1. DSM Law Enforcement Products.
2. Fasco Security Products.
3. Spacesaver Corporation
4. Tiffin Metal Products.

B. Locker Arrangement: Refer to drawings for overall arrangement.

C. Size: Refer to drawings for overall dimensions.

D. Material: Cold-rolled steel sheet.

E. Body: Assembled by welding body components together. Fabricate from unperforated steel sheet with thicknesses as follows:

1. Tops, Bottoms, and Sides: 14-gauge thickness.
2. Backs: 14-gauge thickness.
3. Shelves: 14-gauge, with double bend at front and single bend at sides and back.

F. Frames: Channel formed; fabricated from 14-gauge steel sheet; lapped and factory welded at corners; with top and bottom main frames factory welded into vertical main frames. Form continuous, integral door strike full height on vertical main frames.

1. Cross Frames between Tiers: Channel formed and fabricated from same material as main frames; welded to vertical main frames.

G. Doors: No doors, units will be open.

H. Equipment: Equip each metal locker compartment with an identification plate.

I. Accessories:

1. Legs: 6 inches (152 mm) high; formed by extending vertical frame members or fabricated from 0.075-inch (1.90-mm) nominal-thickness steel sheet; welded to bottom of locker.

a. Closed Front and End Bases: Fabricated from 0.036-inch (0.91-mm) nominal-thickness steel sheet.

2. Recess Trim: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.
3. Filler Panels: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.
4. Boxed End Panels: Fabricated from 0.048-inch (1.21-mm) nominal-thickness steel sheet.

J. Finish: Powder Coat.

1. Color: As selected by Architect from manufacturer's full range.

## 2.5 FABRICATION

A. Fabricate metal lockers square, rigid, without warp, and with metal faces flat and free of dents or distortion. Make exposed metal edges safe to touch and free of sharp edges and burrs.

1. Form body panels, doors, shelves, and accessories from one-piece steel sheet unless otherwise indicated.
2. Provide fasteners, filler plates, supports, clips, and closures as required for complete installation.

B. Fabricate each metal locker with an individual door and frame; individual top, bottom, and back; and common intermediate uprights separating compartments.

C. Equipment: Provide each locker with an identification plate and the following equipment:

1. Single-Tier Units: Shelf, one double-prong ceiling hook, and two single-prong wall hooks.
2. Double-Tier Units: One double-prong ceiling hook and two single-prong wall hooks.
3. Triple-Tier Units: One double-prong ceiling hook.

D. Welded Construction: Factory preassemble metal lockers by welding all joints, seams, and connections; with no bolts, nuts, screws, or rivets used in assembly of main locker groups. Factory weld main locker groups into one-piece structures. Grind exposed welds smooth and flush.

E. Accessible Lockers: Fabricate as follows:

1. Locate bottom shelf no lower than 15 inches (381 mm) above the floor.
2. Where hooks, coat rods, or additional shelves are provided, locate no higher than 48 inches (1219 mm) above the floor.

F. Continuous Zee Base: Fabricated in lengths as long as practical to enclose base and base ends; finished to match lockers.

G. Continuous Sloping Tops: Fabricated in lengths as long as practical, without visible fasteners at splice locations; finished to match lockers.

1. Sloping-top corner fillers, mitered.

H. Recess Trim: Fabricated with minimum 2-1/2-inch (64-mm) face width and in lengths as long as practical; finished to match lockers.

I. Filler Panels: Fabricated in an unequal leg angle shape; finished to match lockers. Provide slip-joint filler angle formed to receive filler panel.

J. Boxed End Panels: Fabricated with 1-inch- (25-mm-) wide edge dimension, and designed for concealing fasteners and holes at exposed ends of nonrecessed metal lockers; finished to match lockers.

1. Provide one-piece panels for double-row (back-to-back) locker ends.

K. Finished End Panels: Fabricated to conceal unused penetrations and fasteners, except for perimeter fasteners, at exposed ends of nonrecessed metal lockers; finished to match lockers.

1. Provide one-piece panels for double-row (back-to-back) locker ends.

L. Center Dividers: Full-depth, vertical partitions between bottom and shelf; finished to match lockers.

## 2.6 ACCESSORIES

A. Fasteners: Zinc- or nickel-plated steel, slotless-type, exposed bolt heads; with self-locking nuts or lock washers for nuts on moving parts.

B. Anchors: Material, type, and size required for secure anchorage to each substrate.

1. Provide nonferrous-metal or hot-dip galvanized anchors and inserts on inside face of exterior walls, and elsewhere as indicated, for corrosion resistance.
2. Provide toothed-steel or lead expansion sleeves for drilled-in-place anchors.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine walls and floors or support bases, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install lockers level, plumb, and true; shim as required, using concealed shims.
1. Anchor locker runs at ends and at intervals recommended by manufacturer, but not more than 36 inches (910 mm) o.c. Using concealed fasteners, install anchors through backup reinforcing plates, channels, or blocking as required to prevent metal distortion.
2. Anchor single rows of metal lockers to walls near top and bottom of lockers.
3. Anchor back-to-back metal lockers to floor.

B. Welded Lockers: Connect groups together with manufacturer's standard fasteners, with no exposed fasteners on face frames.

C. Equipment:
1. Attach hooks with at least two fasteners.
2. Attach door locks on doors using security-type fasteners.
3. Identification Plates: Identify metal lockers with identification indicated on Drawings.
    a. Attach plates to each locker door, near top, centered, with at least two aluminum rivets.
    b. Attach plates to upper shelf of each open-front metal locker, centered, with a least two aluminum rivets.

D. Trim: Fit exposed connections of trim, fillers, and closures accurately together to form tight, hairline joints, with concealed fasteners and splice plates.
1. Attach recess trim to recessed metal lockers with concealed clips.
2. Attach filler panels with concealed fasteners. Locate filler panels where indicated on Drawings.
3. Attach sloping-top units to metal lockers, with closures at exposed ends.
4. Attach boxed end panels using concealed fasteners to conceal exposed ends of nonrecessed metal lockers.
5. Attach finished end panels using fasteners only at perimeter to conceal exposed ends of nonrecessed metal lockers.

E. Fixed Benches: Provide no fewer than two pedestals for each bench, uniformly spaced not more than 72 inches (1830 mm) apart. Securely fasten tops of pedestals to undersides of bench tops, and anchor bases to floor.

F. Electrical System:

1. UL Listed electrical wiring system with plug in play component design.
2. Receptacles: Standard 20A duplex receptacles and 20A GFCI duplex receptacles.

3. Manufacturer and/or manufacturer installer shall furnish and install plug and play wiring system and receptacles. Connect to building electrical system provided by others.

3.3 ADJUSTING

A. Clean, lubricate, and adjust hardware. Adjust doors and latches to operate easily without binding. Verify that integral locking devices operate properly.

3.4 PROTECTION

A. Protect metal lockers from damage, abuse, dust, dirt, stain, or paint. Do not permit use during construction.

B. Touch up marred finishes or replace metal lockers that cannot be restored to factory-finished appearance. Use only materials and procedures recommended or furnished by locker manufacturer.

END OF SECTION 10 5113

# SECTION 10 5626 - MOBILE STORAGE SHELVING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Mechanically assisted systems.

B. Products Furnished but Not Installed Under This Section:

1. Inserts in cast-in-place concrete.

### 1.3 COORDINATION

A. Recessed Tracks: Coordinate size and location of recesses and inserts in concrete with installation of recessed tracks.

### 1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include mobile operation, construction details, material descriptions, dimensions of individual components and profiles, and finishes for mobile storage shelving systems and accessories.

B. Shop Drawings:

1. Include plans, elevations, sections, and details.
2. Show shelving layout.
3. Show location and extent of rail system including depth required if recessed.
4. Show clear-aisle widths from face of carriages.
5. Detail fabrication and installation of mobile shelving systems including methods of anchoring shelves to carriages and rails to building structure as required for seismic restraint.
6. Include diagrams for power and control wiring.
7. Include methods of connection for motors and controls and location of electrical rough-in.

C. Samples: For each exposed product and for each color and texture specified.

1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer and manufacturer.

B. Sample Warranty: For manufacturer's special warranty.

1.7 CLOSEOUT SUBMITTALS

A. Maintenance Data: For mobile shelving systems and operating manuals to include in maintenance manuals.

1.8 QUALITY ASSURANCE

A. Manufacturer Qualifications: A manufacturer capable of fabricating mobile storage shelving that meets or exceeds performance requirements indicated and of documenting this performance by test reports, and calculations.

B. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.9 PROJECT CONDITIONS

A. Field Measurements: Verify actual locations of support rail anchors, depressed slab, embedded conduit, and other construction contiguous with mobile storage shelving by field measurements before fabrication.

1.10 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of mobile shelving systems that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Deterioration of metals, metal finishes, and other materials beyond normal wear.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain mobile storage systems including shelving from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Mobile shelving systems shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

B. Structural Performance: Provide mobile shelving systems capable of supporting the following:

1. Load per Linear Foot of Carriage: 1000 lb/ft. (1488 kg/m).

C. Operating Force: For manually operated systems, maximum 1 lbf (4.45 N) required to move 1000 lb (453 kg).

2.3 SYSTEMS AND COMPONENTS

A. General: Provide manufacturer's standard mobile storage shelving systems and components. Where components are not otherwise indicated, provide manufacturer's standard components as required for a complete system.

B. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

C. Inserts: Furnish required concrete inserts and similar anchorage devices for installing track system and furnish other components of work where installation of devices is specified in another Section.

D. Flooring: Underlayment thickness required to bring aisle floor finish flush with rail tops.

1. Plywood Underlayment: DOC PS 1, Interior, Underlayment.

a. Fire-Retardant-Treated Plywood by Pressure Process: Products with a flame-spread index of 25 or less when tested according to ASTM E84.

2. Ramps: Manufacturer's standard metallic-coated, cold-rolled steel ramp not steeper than 1:12, with non-slip finish.
3. Floor Finish: As indicated on Drawings.

E. Tracks: Steel rails with tops machined to mate with guide wheels and with ends designed to provide smooth, secure continuity between sections without field welding. Provide mounting brackets, anchorage devices, adjustable leveling devices, and stops at terminations of rails to prevent carriages from running off track ends.

1. Mounting: Surface.

F. Carriages: Rigid frames consisting of C-shaped cold-formed steel beams and cross beams, designed to allow secure anchorage of shelving units.

1. Carriage Width: Refer to drawings.
2. Carriage Length: Refeer to drawings.
3. Wheels: Manufacturer's standard number of bearing-mounted, steel wheels, precision ground to mate with tracks.
4. Bumpers: Provide two rubber bumpers with minimum depth of 1/2 inch (13 mm) each side.

G. Anti-Tip Brackets: Mount on carriage for engagement with track system to secure units against tipping.

H. Carriage End Panels: Full depth and height of shelving units. Provide at both ends of each range.

1. Material: manufacturer's standard.

2.4 MECHANICALLY ASSISTED SYSTEMS

A. Basis-of-Design Product: Subject to compliance with requirements, provide products by the following or approved equivalent:

1. Spacesaver Corporation – Spacesaver Interiors Mechanical Assist Synchro System.

B. Drive Systems: Geared transmission and chain systems with tensioning device to provide mechanical assistance and uniform movement along entire length of each carriage. Permanently shielded and lubricated.

C. Drive Shaft: Continuous tubular or solid steel shaft, capable of transmitting torque from drive system without distortion.

D. Locking Pins: Located on range end panels to allow locking of individual range carriage when depressed.

2.5 MATERIALS

A. High-Pressure Decorative Laminate: NEMA LD 3, Grade VGS.

1. Colors, Textures, and Patterns: As selected by Architect from manufacturer's full range.

2.6 STEEL FINISHES

A. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

B. Baked-Enamel or Powder-Coat Finish: Manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat to achieve a minimum dry film thickness of 2 mils (0.05 mm).

1. Color and Gloss: As selected by Architect from manufacturer's full range.

2.7 SYSTEM ACCESSORIES

A. Floor Lock: Key-operated floor lock capable of securing entire system. Provide two keys.

B. Aisle Lights: Manufacturer's standard aisle lighting system that automatically turns on when aisle is open and shuts off when aisle is closed. Provide one light unit per aisle.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances, location of framing and reinforcements and other conditions affecting performance of mobile shelving systems.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Level and plumb tracks to a tolerance of 0.09 inch in 120 inches (2.4 mm in 3.048 m) with no more than 0.06-inch (1.5-mm) variation between adjacent rails. Use permanent shims or non-shrink grout as indicated by manufacturer.

B. Surface-Mounted Track Systems: Install underlayment, ramps, and finish flooring according to manufacturer's written instructions and flush with track surfaces. Do not extend ramps beyond ends of carriages.

C. Carriage Installation: Mount mobile carriages on track system with anti-tip brackets engaged by rails and adjust for smooth operation. Provide non-moving carriages securely fixed to rails where indicated.

3.3 SHELVING INSTALLATION

A. Attach shelving units to carriages according to manufacturer's written instructions and as required to prevent vibration during movement.

1. Level and plumb shelving units to a tolerance of 1/8 inch in 96 inches (3 mm in 2438 mm).

B. Starter/Adder Units: Connect groups together with standard fasteners according to manufacturer's written instructions, using concealed fasteners where possible.

C. Install shelves in shelving units at locations indicated on Drawings and according to manufacturer's written instructions.

D. Shelving Enclosure Panels: Install end panels and canopy tops with concealed fasteners.

3.4 CLEANING AND PROTECTING

A. Repair or remove and replace defective work as directed on completion of installation.

B. Clean finished surfaces, touch up as required, and remove or refinish damaged or soiled areas to match original factory finish, as approved by Architect.

C. Protect installed products from damage during remainder of the construction period.

3.5 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain mobile storage shelving.

END OF SECTION 10 5626

SECTION 10 7316 – METAL CANOPIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Section includes design. furnishing and installation of extruded aluminum overhead cantilever supported canopies.

B. Related Requirements:

1. Section 05 4000 Cold-Formed Metal Framing.
2. Section 05 5000 Metal Fabrications.
3. Section 07 6200 Sheet Metal Flashing and Trim.

1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: manufacturer’s product data sheets describing materials, accessories, material color options, and installation details.

B. Shop Drawings:

1. Showing structural component locations and positions.
2. Material dimensions
3. Details of construction and assembly to demonstrate compliance with project design intent.

C. Samples:

1. For Selection: Manufacturer’s standard color charts or color chips.
2. For Verification: 6-inch by 6-inch metal sample with selected metal finish and color.

D. Delegated-Design Submittal: For metal canopies indicated to comply with performance requirements and design criteria, including structural calculations signed and sealed by the qualified professional engineer responsible for their preparation.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For professional engineer responsible for canopy engineering.

B. Qualification Data: For testing agency.

C. Welding certificates

D. Product Test Reports: For each listed product, for tests performed by manufacturer and witnessed by a qualified testing agency.

E. Evaluation Reports: For nonstandard steel framing post-installed anchors and power-actuated fasteners, from ICC-ES or other qualified testing agency acceptable to authorities having jurisdiction.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

B. Engineer Qualifications: Qualified professional engineer, registered in Commonwealth of Virginia, and responsible for delegated design.

C. Field Measurement: Confirm dimensions prior to preparation of shop drawings, when possible.

1.7 WARRANTY

A. Special Warranty: Manufacturer agrees to repair finish or replace shelters that fail in materials or workmanship within specified warranty period.

1. Warranty Period: One year from date of Substantial Completion.

PART 2 – PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 013300 "Submittal Procedures," to design cold-formed steel framing.

B. Structural Performance, Canopy Design: Conform to local building codes and local site design loads, including seismic performance criteria.

C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change (Range): 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

D. Manufacturer and/or Delegated Designer to provide the design loads to the light gauge framing Delegated Designer for inclusion as part of the “054000 Cold-Formed Metal Framing” design.

2.2 MANUFACTURERS

A. Source Limitations: Obtain metal canopies from single source from single manufacturer.

B. Basis-of-Design Product: Subject to compliance with requirements, Provide Mapes Architectural Canopies, Super Lumishade Cantilver Flat Soffit or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Mapes Architectural Canopies
b. MASA Architectural Canopies
c. Mitchell Metals, LLC.
d. Skyscape Architectural Canopies

2.3 MATERIALS

A. Metal Decking: Interlocking roll-formed, 2-1/2-inch (63.5-mm) deep “W” shaped pan; minimum 0.032-inch (0.81-mm) thick aluminum.

B. Intermediate Framing Members: Extruded aluminum, alloy 6063-T6, in profile and thickness per manufacturer’s standard materials and details.

C. Hanger Rods and Attachment Hardware: Manufacturer’s standard components.

D. Fascia: Manufacturer’s Standard 8-inch (203-mm) deep extruded “J”-shaped fascia; minimum 0.125-inch (3.2 -mm) aluminum.

2.4 FABRICATION

A. Fabricate canopies in preassembled sections for ease of transport and installation.

B. Connections: Mechanically assemble utilizing 3/16-inch (4.75-mm) fasteners with a minimum shear stress of 350 pounds.

C. Decking: Designed and manufactured with interlocking roll-formed aluminum members.

D. Concealed Drainage: Fabricate canopies so that water drains from covered surfaces into intermediate trough and then, directed to either the front for front drainage or to the rear for ground level discharge via one or more designated downspouts.

2.5 FINISHES

A. Framing, Decking, Fascia and Accessories:

B. High-Performance Organic Finish, Two-Coat PVDF: Fluoropolymer finish complying with AAMA 2605 and containing not less than 70 percent PVDF resin by weight in color coat.

1. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
2. Color and Gloss: Match Kawneer “Charcoal”.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify surrounding substrates are ready for canopy installation.

B. Confirm dimensions and elevations agree with reviewed shop drawings.

C. Perform installation using a manufacturer’s approved installer and schedule installation after adjacent concrete, masonry and roofing is completed.

3.2 INSTALLATION

A. Install under supervision of the manufacturer's authorized representative to comply with performance requirements.

B. Installation shall be in strict accordance with reviewed shop drawings. Give attention to protecting the canopy finish during handling and installation.

3.3 CLEANING AND ADJUSTING

A. Verify canopy assembly is installed level, plumb, and in accordance with reviewed shop drawings. Adjust installation as necessary to achieve accurate, complete installation.

B. After installation, clean entire canopy assembly.

C. Repair and clean adjacent building surfaces that may have been damaged or soiled during canopy installation.

END OF SECTION 10 7316

# SECTION 11 1916 - DETENTION GUN LOCKERS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Gun Lockers.

### 1.3 COORDINATION

A. Coordinate installation of anchorages for detention gun lockers. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors that are to be embedded in adjacent construction. Deliver such items to Project site in time for installation.

B. Coordinate size and location of recesses in wall construction to receive recessed detention gun lockers.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for all detention gun lockers.

B. Shop Drawings: For detention gun lockers.

1. Include plans, elevations, sections, and attachment details.
2. Indicate locations, dimensions, and profiles of wall and floor reinforcements.
3. Indicate locations and installation details of built-in anchors.
4. Show elevations and indicate dimensions of detention gun lockers, preparations for receiving anchors, and locations of anchorage.
5. Show details of attachment of detention gun lockers to built-in anchors.

C. Samples for Initial Selection: For detention gun lockers with factory-applied color finishes.

### 1.5 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to the following:

1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
2. AWS D1.3, "Structural Welding Code - Sheet Steel."

3. AWS D1.6, "Structural Welding Code - Stainless Steel."

1.6 FIELD CONDITIONS

A. Field Measurements: Verify openings for recessed detention gun lockers by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 GUN LOCKERS

A. Manufacturers: Subject to compliance with requirements, provide products from available manufacturers offering products that may be incorporated into the Work include the following or approved equivalent:

1. Fasco Security Products.
2. Patterson Pope
3. Southwest Solution Group.
4. Spacesaver Corporation, a Division of KI.
5. Tiffin Metal Products.

B. Cabinet: Formed from 0.134-inch nominal-thickness steel sheet.

1. Compartments: Refer to drawings for arrangement, locations and details.

C. Compartments: Doors formed from same material as cabinet, supported by heavy-duty continuous bottom hinge, with attached tilt-out compartment with formed metal sides. Line each compartment with mothproofed felt or non-absorbing, closed-cell padding.

D. Locks: Cylinder type, keyed differently and master keyed; provide one lock for each compartment.

E. Mounting: Floor mounted with mounting flange formed from same material as body.

F. Materials:

1. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
2. Cold-Rolled Steel Sheet: ASTM A 1008/A 1008M, CS (Commercial Steel), Type B; suitable for exposed applications.
3. Hot-Rolled Steel Sheet: ASTM A 1011/A 1011M, CS (Commercial Steel), Type B; free of scale, pitting, or surface defects; pickled and oiled.
4. Stainless-Steel Sheet, Strip, Plate and Flat Bars: ASTM A 666 or ASTM A 240/A 240M, austenitic stainless steel, Type 304.

G. Finishes:

1. Steel Factory Finish: Clean, pretreat, and apply manufacturer's standard two-coat, baked- on finish consisting of prime coat and thermosetting topcoat, with a minimum dry film thickness of 1 mil for topcoat.

a. Color and Gloss: As selected by Architect from manufacturer's full range.

2.1 FABRICATION

A. Shop Assembly: Preassemble items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.

B. Coordinate dimensions and attachment methods of detention gun lockers with those of adjoining construction to produce integrated assemblies with closely fitting joints and with edges and surfaces aligned unless otherwise indicated.

C. Shear and punch metals cleanly and accurately. Remove burrs.

D. Form and grind edges and corners to be free of sharp edges or rough areas.

E. Form metal in maximum lengths to minimize joints. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing the Work.

F. Weld corners and seams continuously to comply with referenced AWS standard and the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. Finish exposed welds and surfaces smooth and blended at exposed connections so that no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
5. Weld before finishing components to greatest extent possible. Remove weld spatter and welding oxides from exposed surfaces by descaling or grinding.

G. Provide for anchorage of type indicated; coordinate with supporting structure. Fabricate and space anchoring devices to secure detention gun lockers rigidly in place and to support expected

H. loads. Build in straps, plates, and brackets as needed to support and anchor fabricated items to adjoining construction. Reinforce formed-metal units as needed to attach and support other construction.

I. Cut, reinforce, drill, and tap detention gun lockers as indicated to receive hardware, fasteners, and similar items.

J. Form exposed work true to line and level with accurate angles, surfaces, and straight sharp edges.

K. Form exposed connections with hairline joints, flush and smooth using concealed fasteners where possible. Locate joints where least conspicuous.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of detention gun lockers.

B. Examine roughing-in for embedded and built-in anchors to verify actual locations of detention gun lockers before detention gun locker installation.

C. Inspect built-in and cast-in anchor installations, before installing detention gun lockers, to verify that anchor installations comply with requirements. Prepare inspection reports.

1. Remove and replace anchors where inspections indicate that they do not comply with specified requirements. Re-inspect after repairs or replacements are made.

2. Perform additional inspections to determine compliance of replaced or additional work. Prepare inspection reports.

D. Verify locations of detention gun lockers with those indicated on Shop Drawings.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing detention gun lockers to in-place construction.

B. Cutting, Fitting, and Placement: Obtain manufacturer's written approval for cutting, drilling, and fitting required for installing detention gun lockers. Set detention gun lockers accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.

C. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations.

D. Adjust doors and latches of detention gun lockers to operate easily without binding. Verify that integral locking devices operate properly.

E. Assemble detention gun lockers requiring field assembly with security fasteners with no exposed fasteners on exposed faces and frames.

3.3 FIELD QUALITY CONTROL

A. Inspect installed products to verify compliance with requirements. Prepare inspection reports and indicate compliance with and deviations from the Contract Documents.

B. Remove and replace detention work if inspections indicate that work does not comply with specified requirements. Remove malfunctioning units; replace with new units.

C. Perform additional inspections to determine compliance of replaced or additional work. Prepare inspection reports.

END OF SECTION 11 1916

# SECTION 11 1963 – DETENTION FURNISHINGS AND EQUIPMENT

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. This Section includes the following:

1. Detention Mirror with Embedded Mounting
2. Detention Grab Bars
3. Detention Toilet Paper Holder
4. Detention Access Panel (DAP)

B. Related Sections include the following:

1. Division 04 Section “Unit Masonry Assemblies”
2. Division 05 Section “Metal Fabrications”
3. Division 05 Section “Tamper Proof Metal Fasteners”
4. Division 07 Section “Security Joint Sealants”
5. Division 09 Section “Painting”

### 1.3 SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each type of detention furnishing indicated.

B. Shop Drawings: For each type of detention furnishing.

1. Include plans, elevations, sections, details, and attachments to other Work.

C. Coordination Drawings: Drawings of each built-in anchor supporting detention furnishings, including those to be installed as work of other Sections, drawn to scale and coordinating anchorage with detention furnishings. Show the following:

1. Locations, dimensions, and profiles of wall and floor reinforcements.
2. Locations and installation details of built-in anchors.
3. Elevations of each detention furnishing showing dimensions of furnishing, preparations for receiving anchors, and locations of anchorage.
4. Details of attachment of each detention furnishing to built-in anchors.

D. Samples: For each type of detention furnishing with factory-applied color finishes.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: Engage an experienced installer who has completed detention furnishings and equipment applications similar in material, design, and extent to that indicated for Project that have resulted in construction with a record of successful in-service performance.

B. Provide detention furnishings and equipment manufactured by a single firm specializing in the production of this type of work.

1.5 DELIVERY, STORAGE AND HANDLING

A. Deliver, store and handle detention furnishings and equipment per manufacturer's requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURER OF DETENTION FURNISHINGS AND EQUIPMENT

A. Acceptable Manufacturers:

1. American Jail Products; Troy, NY
2. CortechUSA; Willowbrook, IL
3. Jail Correctional Products; Minster, OH
4. Maximum Security Products Corp.; Albany, NY
5. Norix Group, Inc.; West Chicago, IL
6. PSI LLC.; Minneapolis, MN
7. PDI; Orange, CA
8. Slate Security System, Inc.; Hartselle, AL
9. Southern Steel Company; San Antonio, TX
10. Sweeper Metal Fabricators Corp.; Drumright, OK
11. Viking Products; Orange, CA
12. Willo Products; Decatur, AL

2.2 DETENTION MIRROR WITH EMBEDDED MOUNTING PLATE:

A. Mirror:

1. Mirror frame dimensions shall be 12-1/2 inches by 16-1/2 inches (317.5 mm by 419 mm), fabricated from 0.0598 inch (1.5 mm) mild steel. 5/16 inch by 1 inch (8 mm by 25.4mm) mirror frame to be chromium plated.
2. Mirror opening shall be 10-1/2 inches by 14-1/2 inches (266.7 mm by 368.3 mm). Mirror shall be made of 0.035 inch stainless steel polished for high reflectivity.
3. Include chromium plated security fasteners.

B. Embedded Mounting Plate:

1. Embedded mounting plate to be constructed of 1/4 inch plate steel with two (2) 0.1196 inch by 3 inch wide bent steel anchors with minimum 1inch (25.4-mm) bend.
2. Embed plate shall be drilled and tapped for security fasteners.
3. Embedded mounting plate shall be provided with one (1) shop coat primer.

C. Mount top of mirror at 6 feet 2 inches above finish floor typically, in handicap areas mount bottom of mirror at 3 feet 4 inches above finish floor.

D. Provide low-mod gel, equal to Sikadur 23 by Sika, around perimeter of frame and embed.

## 2.3 DETENTION GRAB BARS

A. Construction:

1. Grab Bars: 1-1/2 inches (38.1 mm) in diameter; formed from 0.038 inch (0.95 mm) thick, stainless-steel tubing, with 3 inch (76.2 mm) diameter flanges formed from 0.125 inch (3.18 mm) thick, stainless steel. Closure plates formed from 0.125 inch (3.18 mm) thick, stainless steel. All-welded construction. Provide No. 4 finish.
2. Length: As indicated on Drawings.
3. Mounting: Front mounting with security fasteners.

## 2.4 DETENTION TOILET PAPER HOLDER – FRONT MOUNT

A. Construction:

1. One-piece drawn 16 gauge type 304 stainless steel seamless well, welded to 14 gauge stainless flange. Bead blasted satin finish.
2. Face: 7 inch (178 mm) square face flange.

## 2.5 DETENTION ACCESS PANEL (DAP)

A. Flush Detention Access Doors and Frames with Exposed Trim:

1. Frame: 3 inches by 2 inches by 3/16 inch steel angles, with 1 inch by 1 inch by 1/8 inch (25.4 mm by 25.4 mm by 3 mm) angle stops on 3 sides.
2. Door: 3/16 inch steel with 1-1/4 inch flange on all four sides.
3. Latch: Equip the panel with one (1) No. 1010-1/Southern Steel Co. deadlock.

B. Provide mounting holes in frames to attach masonry anchors in masonry construction.

C. Provide low-mod gel, equal to Sikadur 23 by Sika, around perimeter of angle frame.

D. Assembly shall be provided with one (1) shop coat of primer. Field finish paint by Section 09 9123, color as selected by Architect. Verify primer compatibility with finish paint as specified in Section 09 9123.

## 2.6 DETENTION EQUIPMENT ACCESSORIES

A. Provide accessories, anchorage inserts and security fasteners for a complete, tamperproof installation.

B. Exposed Security Fasteners:

1. Provide torx-head (star design with center pin) security fasteners for anchoring work in exposed detention areas. Comply with specification section 05 0553.
2. Finish shall match that specified of the item anchored.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine roughing-in for embedded and built-in anchors to verify actual locations before installation.

C. Notify the General Contractor in writing of conditions detrimental to the proper and timely completion of the work.

D. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Expenses carried by the Architect/Engineer, Project Manager or Owner in troubleshooting equipment problems caused by inadequate workmanship or other form of poor performance on the part of the Contractor, shall be borne by the Contractor.

B. Comply with manufacturer's printed installation instructions.

C. Touch-up painting of factory finished or factory primed items is the Installer's responsibility.

D. Fill voids between materials of the detention equipment and embeds or other physical construction with low-mod gel, equal to Sikadur 23, by Sika.

3.3 CLEANING

A. Clean equipment thoroughly prior to Substantial Completion.

3.4 PROTECTION

A. Protect equipment and finishes until Substantial Completion.

B. Replace damaged equipment as directed by the Architect.

END OF SECTION 11 1963

# SECTION 11 3013 - RESIDENTIAL APPLIANCES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Cooking appliances.
2. Refrigeration appliances.

B. Related Requirements:

1. Section 22 4000 " Plumbing Fixtures" for kitchen sinks, dishwasher air-gap fittings, waste (garbage) disposers, and instant hot-water dispensers.

### 1.3 PREINSTALLATION MEETINGS

A. Pre-installation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include installation details, material descriptions, dimensions of individual components, and finishes for each appliance.
2. Include rated capacities, operating characteristics, electrical characteristics, and furnished accessories.

B. Samples: For each exposed product and for each color and texture specified, in manufacturer's standard form or size.

C. Product Schedule: For appliances. Use same designations indicated on Drawings.

### 1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For manufacturer.

B. Product Certificates: For each type of appliance.

C. Field quality-control reports.

D. Sample Warranties: For manufacturers' special warranties.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For each residential appliance to include in operation and maintenance manuals.

1.7 QUALITY ASSURANCE

A. Manufacturer Qualifications: Maintains, within 50 miles (80.4 km) of Project site, a service center capable of providing training, parts, and emergency maintenance repairs.

1.8 WARRANTY

A. Special Warranties: Manufacturer agrees to repair or replace residential appliances or components that fail in materials or workmanship within specified warranty period except as qualified below:

B. Electric Cooktop and Range: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on surface-burner elements.

1. Warranty Period: Two years from date of Substantial Completion.

C. Microwave Oven: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on the magnetron tube.

1. Warranty Period: Two years from date of Substantial Completion.

D. Refrigerator/Freezer, Sealed System: Full warranty, including parts and labor, for on-site service on the product.

1. Warranty Period for Sealed Refrigeration System: Five years from date of Substantial Completion.
2. Warranty Period for Other Components: Two years from date of Substantial Completion.

E. Medical Refrigerator, Sealed System: Full warranty, including parts and labor

1. Warranty Period for Parts and Labor: Two years from date of Substantial Completion.
2. Warranty Period for Compressor: Five years from date of Substantial Completion.

F. Icemaker: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on the product.

1. Warranty Period for Sealed System: Five years from date of Substantial Completion.
2. Warranty Period for Other Components: Two years from date of Substantial Completion.

G. Dishwasher: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on the product.

1. Warranty Period for Deterioration of Tub and Metal Door Liner: Three years from date of Substantial Completion.

2. Warranty Period for Other Components: Two years from date of Substantial Completion.

H. Clothes Washer: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on the product.

1. Warranty Period: Two years from date of Substantial Completion.

I. Clothes Dryer: Limited warranty, including parts and labor for first year and parts thereafter, for on-site service on the product.

1. Warranty Period: Two years from date of Substantial Completion.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

A. Source Limitations: Obtain residential appliances from single source.

### 2.2 PERFORMANCE REQUIREMENTS

A. Electrical Appliances: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Accessibility: Where residential appliances are indicated to comply with accessibility requirements, comply with applicable provisions in the DOJ's 2010 ADA Standards for Accessible Design and ICC A117.1.

### 2.3 COOKING APPLIANCES

A. Electric Range (**OV-1**): Freestanding, Slide-in range with one oven(s) and complying with AHAM ER-1.

1. Manufacturers: Subject to compliance with requirements, Provide the following or approved equivalent:

a. General Electric. – Drop-In Electric Range JD630ST/DT.

2. Width: 30 inches (762 mm).
3. Electric Burner Elements: Four.

a. Induction Type: Manufacturer's standard.
b. Controls: Digital panel controls, located on front.

4. Oven Features:

a. Capacity: 4.4 cu. ft. (0.15 cu. m).
b. Operation: Baking, convection and pyrolytic self-cleaning or catalytic continuous cleaning.
c. Broiler: Located in top of oven.

d. Oven Door(s): Counterbalanced, removable, with observation window and full-width handle.
e. Electric Power Rating:

1) Oven(s): 3400 W.
2) Broiler: 3400 W.

f. Controls: Digital panel controls and timer display, located on front.

5. Anti-Tip Device: Manufacturer's standard.
6. Electric Power Supply: 240 V, 60 Hz, 1 phase, 40 A.
7. Material: Stainless steel with manufacturer's standard cooktop.

B. Microwave Oven (**MW-1)**:

1. Manufacturers: Subject to compliance with requirements, Provide GE Appliances PCWK15C1WSS, or equivalent product from available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. General Electric Company (GE Appliances).
b. KitchenAid; a division of Whirlpool Corporation.
c. LG Electronics.
d. Samsung.
e. Sharp Electronics Corp.

2. Mounting: Countertop.
3. Type: Conventional.
4. Dimensions:

a. Width: 22 inches (559 mm).
b. Depth: 16 inches (406 mm).
c. Height: 13 inches (330 mm).

5. Capacity: 1.5 cu. ft. (0.04 cu. m).
6. Oven Door: Door with observation window and pushbutton latch release.
7. Microwave Power Rating: 1000 W.
8. Electric Power Supply: 120 V, 60 Hz, 1 phase, 15 A.
9. Controls: Digital panel controls and timer display.
10. Other Features: Turntable.
11. Material: Stainless steel.

## 2.4 REFRIGERATION APPLIANCES

A. Refrigerator/Freezer (**REF-1**): Two-door refrigerator/freezer with freezer on bottom and complying with AHAM HRF-1.

1. Manufacturers: Subject to compliance with requirements, Provide GE Appliances Counter Depth French Door Refrigerator GYE21JYLFS, or equivalent product from available manufacturers offering products that may be incorporated into the Work include the following:

a. Electrolux.
b. GE Appliances; Haier Group.

2. Type: Freestanding, Built in.
3. Dimensions:

    a. Width: 35 inches (889 mm).
    b. Depth: 31 inches (787 mm).
    c. Height: 70 inches (1778 mm).

4. Storage Capacity:

    a. Refrigeration Compartment Volume: 12.5 cu. ft. (0.35 cu. m).
    b. Freezer Volume: 5.01 cu. ft. (0.14 cu. m).
    c. Shelf Area: Four adjustable glass shelves.

5. General Features:

    a. Door Configuration: Framed.
    b. Dispenser in door for ice and cold water.
    c. Built-in water-filtration system.
    d. Dual refrigeration systems.
    e. Separate touch-pad temperature controls for each compartment.

6. Refrigerator Features:

    a. Interior light in refrigeration compartment.
    b. Compartment Storage: Vegetable crisper and meat compartment.
    c. Door Storage: Modular compartments and 1 gal. (3.8 L) milk-container storage.
    d. Temperature-controlled meat/deli bin.

7. Freezer Features: One freezer compartment(s) configured as pull-out drawer(s).

    a. Automatic defrost.
    b. Interior light in freezer compartment.
    c. Automatic icemaker and storage bin.

8. Energy Star: Provide appliances that qualify for the EPA/DOE Energy Star product-labeling program.
9. Front Panel(s): Stainless steel.
10. Appliance Color/Finish: Stainless steel.

B. Refrigerator/Freezer **(REF-2)**: One-door refrigerator with freezer compartment inside and complying with AHAM HRF-1.

1. Manufacturers: Subject to compliance with requirements, Provide the following or approved equivalent:

    a. Summit Appliance Division – AL54PLHD.

2. Type: Undercounter.
3. Dimensions:

    a. Width: 24 inches (610 mm).
    b. Depth: 24 inches (610 mm).
    c. Height: 32 inches (813 mm).

4. Storage Capacity:

    a. Refrigeration Compartment Volume: 4.8 cu. ft. (0.14 cu. m).
    b. Shelf Area: Three adjustable glass shelves,

5. General Features:

    a. Door Configuration: Overlay.

6. Refrigerator Features:

    a. Interior light in refrigeration compartment.
    b. Door Storage: Modular compartments.
    c. Temperature-controlled

7. Freezer Features: One freezer compartment(s).

    a. Manual defrost.

8. Energy Star: Provide appliances that qualify for the EPA/DOE Energy Star product-labeling program.
9. Front Panel(s): Manufacturer's standard Lavender Purple.

10. Appliance Color/Finish: Stainless steel.

## 2.5 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, power connections, and other conditions affecting installation and performance of residential appliances.

B. Examine roughing-in for piping systems to verify actual locations of piping connections before appliance installation.

C. Examine walls, ceilings, and roofs for suitable conditions where overhead exhaust hoods will be installed.

D. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 INSTALLATION

A. Install appliances according to manufacturer's written instructions.

B. Built-in Equipment: Securely anchor units to supporting cabinets or countertops with concealed fasteners. Verify that clearances are adequate for proper functioning and that rough openings are completely concealed.

C. Freestanding Equipment: Place units in final locations after finishes have been completed in each area. Verify that clearances are adequate to properly operate equipment.

D. Range Anti-Tip Device: Install at each range according to manufacturer's written instructions.

## 3.3 FIELD QUALITY CONTROL

A. Perform the following tests and inspections with the assistance of a factory-authorized service representative:

1. Perform visual, mechanical, and electrical inspection and testing for each appliance according to manufacturers' written recommendations. Certify compliance with each manufacturer's appliance-performance parameters.
2. Leak Test: After installation, test for leaks. Repair leaks and retest until no leaks exist.
3. Operational Test: After installation, start units to confirm proper operation.
4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and components.

B. An appliance will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

## 3.4 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain residential appliances.

END OF SECTION 11 3013

# SECTION 12 2113 - HORIZONTAL LOUVER BLINDS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Horizontal louver blinds, aluminum slats.

B. Related Requirements:

1. Section 06 1000 "Rough Carpentry" for wood blocking and grounds for mounting horizontal louver blinds and accessories.

### 1.2 ACTION SUBMITTALS

A. Product Data:

1. Horizontal louver blinds, aluminum slats.

B. Product Data Submittals: For each product.

1. Construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Rated capacities, operating characteristics and furnished accessories.

C. Shop Drawings: For horizontal louver blinds.

1. Fabrication and installation details.

D. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type and color of horizontal louver blind.

1. Include Samples of accessories involving color selection.

E. Product Schedule: For horizontal louver blinds. Use same designations indicated on Drawings.

### 1.3 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For horizontal louver blinds with polymer slats that have been tested for compliance with NFPA 701, for tests performed by a qualified testing agency or manufacturer and witnessed by a qualified testing agency.

### 1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For horizontal louver blinds.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials to Owner that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Horizontal Louver Blinds: Full-size units equal to 5 percent of quantity installed for each size, color, texture, pattern, and gloss indicated, but no fewer than two units. Include brackets.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver horizontal louver blinds in factory packages, marked with manufacturer, product name, and location of installation, using same designations indicated on Drawings.

1.7 FIELD CONDITIONS

A. Environmental Limitations: Do not install horizontal louver blinds until construction and wet-work and finish work in spaces, including painting, is complete and dry and ambient temperature and humidity conditions are maintained at levels indicated for Project when occupied for its intended use.

B. Field Measurements: Where horizontal louver blinds are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Allow clearances for operating hardware of operable glazed units through entire operating range. Notify Architect of installation conditions that vary from Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of motorized products that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Faulty operation of motorized operating system components.

2. Warranty Period: 5 year(s) from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain horizontal louver blinds from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Window Covering Safety Standard: Provide horizontal louver blinds that comply with WCMA A100.1.

2.3 HORIZONTAL LOUVER BLINDS, ALUMINUM SLATS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. CACO, Inc.
2. Hunter Douglas Architectural.
3. Springs Window Fashions; SWF contract.

B. Slats: Aluminum; alloy and temper recommended by producer for type of use and finish indicated; with crowned profile and radius corners.

1. Width: 1 inch (25 mm).
2. Thickness: Manufacturer's standard.
3. Spacing: Manufacturer's standard.
4. Finish: Ionized antistatic, dust-repellent, baked polyester finish.
5. Features:

a. Privacy Slats: Manufacturer's standard privacy slats with no inner lift-cord route holes or with minimum size inner lift-cord route holes located near back (outside) edge of slat to maximize slat overlap and minimize light gaps between slats.
b. Inverted slats.

C. Ladders: Evenly spaced across headrail at spacing that prevents long-term slat sag.

1. Type: Braided cord.

D. Headrail: Formed steel or extruded aluminum; long edges returned or rolled. Headrails fully enclose control mechanisms on three sides.

1. Capacity: One blind(s) per headrail unless otherwise indicated.
2. Ends: Capped or plugged.

E. Manual Corded Operation:

1. Lift Mechanism:

a. Lift-Cord Lock: Variable; stops lift cord at user-selected position within blind full operating range.

2. Lift Operator: Extension of inner lift cord(s) through lift-cord lock mechanism to form lift cord.
3. Lift Cord: Manufacturer's standard braided cord.
4. Lift-Cord Length: Short cord, not greater than 8 inches (203 mm) when blind is fully closed.
5. Tilt Mechanism: Enclosed worm-gear mechanism and linkage rod that adjusts ladders.
6. Tilt Position:

a. Full.

7. Tilt Operator: Clear-plastic wand.
8. Tilt Over-Rotation Protection: Manufacturer's detachable operator or slip clutch to prevent over-rotation of gear.
9. Tilt-Operator Length, Wand or Rod: 36 inches (914 mm).
10. Lift-Cord and Tilt-Operator Locations: Left side of headrail unless otherwise indicated.

F. Bottom Rail: Formed-steel or extruded-aluminum tube that secures and protects ends of ladders and inner lift cords and has plastic- or metal-capped ends.

1. Type: Bottom contoured to minimize light gaps.

G. Mounting Brackets: With spacers and shims required for blind placement and alignment indicated.

1. Type: As indicated.
2. Intermediate Support: Provide intermediate support brackets to produce support spacing recommended by horizontal louver blind manufacturer for weight and size of blind.

H. Side Channels and Perimeter Light Gap Seals: Manufacturer's standard.

I. Colors, Textures, Patterns, and Gloss:

1. Slats: As selected by Architect from manufacturer's full range.
2. Components: Provide rails, cords, ladders, and materials exposed to view matching or coordinating with slat color unless otherwise indicated.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, operational clearances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION OF HORIZONTAL LOUVER BLINDS

A. Install horizontal louver blinds level and plumb, aligned and centered on openings, and aligned with adjacent units in accordance with manufacturer's written instructions.

1. Locate so exterior slat edges are not closer than 1 inch (25 mm) from interior faces of glass and not closer than 1/2 inch (13 mm) from interior faces of glazing frames through full operating ranges of blinds.
2. Install mounting brackets to prevent deflection of headrails.
3. Install with clearances that prevent interference with adjacent blinds, adjacent construction, and operating hardware of glazed openings, other window treatments, and similar building components and furnishings.

3.3 ADJUSTING

A. Adjust horizontal louver blinds to operate free of binding or malfunction through full operating ranges.

3.4 CLEANING AND PROTECTION

A. Clean horizontal louver blind surfaces after installation in accordance with manufacturer's written instructions.

B. Provide final protection and maintain conditions in a manner acceptable to manufacturer and Installer that ensures that horizontal louver blinds are without damage or deterioration at time of Substantial Completion.

C. Replace damaged horizontal louver blinds that cannot be repaired in a manner approved by Architect before time of Substantial Completion.

END OF SECTION 12 2113

# SECTION 12 3661 – SOLID SURFACING MATERIALS

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Solid surface material countertops.
2. Solid surface material back, side and end splashes.
3. Solid surface material apron fronts.
4. Solid surface material sinks.

B. Related Requirements:

1. Section 06 4116 “Plastic Laminate Faced Architectural Cabinets”
2. Section 22 4000 “Plumbing Fixtures” for sinks and plumbing fittings.

### 1.3 ACTION SUBMITTALS

A. Product Data: For countertop materials and sink.

B. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.
2. Laboratory Test Reports: For adhesives, indicating compliance with requirements for low-emitting materials.

C. Shop Drawings: For countertops. Show materials, finishes, edge and backsplash profiles, methods of joining, and cutouts for plumbing fixtures.

1. Show locations and details of joints.
2. Show direction of directional pattern, if any.
3. Dimensions
4. Component sizes
5. Fabrication details
6. Attachment provisions
7. Coordination requirement with adjacent work

D. Samples for Verification: For each type of material exposed to view.

A. 6” x 6” square min. size samples in each specified color

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For fabricator.

B. Warranty: Submit specimen copy of specified warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For solid surface material and quartz agglomerate countertops to include in maintenance manuals. Include Product Data for care products used or recommended by Installer and names, addresses, and telephone numbers of local sources for products.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: Shop that employs skilled workers who custom-fabricate countertops similar to that required for this Project, and whose products have a record of successful in-service performance.

B. Installer Qualifications: Fabricator of countertops.

C. Mockups: Build mockups to demonstrate aesthetic effects and to set quality standards for fabrication and execution. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1. Build mockup of typical solid surface countertop integrated with millwork as part of the room mockups requirement.

1.7 DELIVERY, STORAGE AND HANDLING

A. Do not deliver countertops and solid surfacing materials until painting and similar operations that could damage solid surfacing have been completed in installation areas. If solid surfacing must be stored in other than installation areas, store only in areas where environmental conditions comply with requirements specified in “Project Conditions” article.

1.8 PROJECT CONDITIONS

A. Environmental Limitations: Do not deliver or install solid surfacing materials until building is enclosed, wet work is complete, and HVAC system is operating and maintaining temperature and relative humidity at occupancy levels during the remainder of the construction period.

B. Field Measurements: Where countertops are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work. Locate concealed framing, blocking, and reinforcements that support woodwork by field measurements before being enclosed and indicate measurements on Shop Drawings. after base cabinets are installed but before countertop and fascia fabrication is complete

C. Adhesives: Acclimate adhesives to occupancy room temperatures with maximum temperature not to exceed 75 degrees F.

1.9 COORDINATION

A. Coordinate field measurements and fabrication schedule with construction progress to avoid construction delays.

B. Coordinate locations of utilities that will penetrate countertops or backsplashes.

C. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections to ensure that interior architectural woodwork can be supported and installed as indicated.

1.10 WARRANTY

A. Manufacturer's Limited Warranty: Standard 10 Year Commercial and Residential Limited Warranty against defects in solid surfacing materials.

PART 2 - PRODUCTS

2.1 SOLID SURFACE MATERIALS (SSM-1, SSM-2, SSM-3)

A. Solid Surface Material: Homogeneous-filled acrylic resin complying with ISFA 2-01.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include the following:

    1. DuPont; DuPont de Nemours, Inc.
    2. Durasein Solid Surface; a brand of Relang International, LLC.
    3. Formica Corporation.
    4. Wilsonart LLC.

2. Type: Provide Standard type or Veneer type made from material complying with requirements for Standard type, as indicated unless Special Purpose type is indicated.

3. Integral Sink Bowls: Comply with CSA B45.5/IAPMO Z124.

4. Thickness: 1/2" unless noted otherwise.

5. Colors and Patterns, Match Architect's samples:

    1. SSM-1: Subject to compliance with requirements, Provide Corian, Limestone Prima, or approved equivalent product.

    2. SSM-2: Subject to compliance with requirements, Provide Corian, Silver Linear, or approved equivalent product.

    3. SSM-3: Subject to compliance with requirements, Provide Corian, Venaro White, or approved equivalent product.

2.2 FABRICATION

A. Fabricate countertops and fascia according to solid surface material manufacturer's written instructions and to the AWI/AWMAC/WI's "Architectural Woodwork Standards."

1. Grade: Premium.

B. Configuration:

1. Front: 1-1/2”-inch square.
2. Backsplash: Straight, slightly eased at corner.
3. End Splash: Where indicated, matching backsplash.

C. Countertops: 1/2-inch-thick, solid surface material with front edge built up with same material.

1. Fabricate in accordance with manufacturer’s instructions and approved shop drawings.
2. Fabricate components in shop to greatest extent possible, to sizes and shapes indicated.
3. Provide components in largest practical pieces.
4. Form joints between pieces using joint adhesive, to be inconspicuous and without voids. Use reinforcement strips on seams
5. Fabricate splashes, skirts and trim from solid surfacing in color to match countertops.

D. Backsplashes: 1/2-inch-thick, solid surface material.

E. Fabricate tops with shop-applied edges and backsplashes unless otherwise indicated. Comply with solid surface material manufacturer's written instructions for adhesives, sealers, fabrication, and finishing.

1. Fabricate with integral backsplashes for field assembly.
2. Install integral sink bowls in countertops in the shop.
3. Countertop Adhesive: Type: Silicone or flexible neoprene
4. Maximum volatile organic compound (VOC) content: 70 grams per liter

F. Joints: Fabricate countertops without joints wherever possible.

G. Joints: Fabricate countertops in sections for joining in field.

1. Joint Locations: Not within 18 inches of a sink or cooktop and not where a countertop section less than 36 inches long would result, unless unavoidable.
2. Splined Joints: Accurately cut kerfs in edges at joints for insertion of metal splines to maintain alignment of surfaces at joints. Make width of cuts slightly more than thickness of splines to provide snug fit. Provide at least three splines in each joint.
3. Joint Adhesive: Two component, type as recommended by solid surfacing manufacture
4. Maximum volatile organic compound (VOC) content: 70 grams per liter

H. Cutouts and Holes:

1. Undercounter Plumbing Fixtures: Make cutouts for fixtures in shop using template or pattern furnished by fixture manufacturer. Form cutouts to smooth, even curves.

    1. Provide vertical edges, slightly eased at juncture of cutout edges with top and bottom surfaces of countertop and projecting 3/16 inch into fixture opening.
    2. Provide 3/4-inch full bullnose edges projecting 3/8 inch into fixture opening.

2. Counter-Mounted Plumbing Fixtures: Prepare countertops in shop for field cutting openings for counter-mounted fixtures. Mark tops for cutouts and drill holes at corners of cutout locations. Make corner holes of largest radius practical.
3. Fittings: Drill countertops in shop for grommets, plumbing fittings, undercounter soap dispensers, and similar items. Exposed edges to be polished unless otherwise indicated.
4. Ease exposed edges 1/16”
5. Finish exposed edges to smooth with uniform profile.

## 2.3 INSTALLATION MATERIALS

A. Sealants: Comply with applicable requirements in Section 079200 "Joint Sealants."

B. Construction Adhesive: Countertop manufacturer’s recommended silicone-based construction adhesive for backsplashes, endsplashes, and other applications according to manufacturer’s published fabrication instructions.

C. Shower Wall Adhesive: Product recommended by material manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.

## 2.4 BRACKETS

A. Subject to compliance with requirements, provide 3/8-inch steel brackets, by A&M Hardware, or an approved equivalent.

1. Bracket types:

1. Concealed Flat Brackets (Heavy-Duty)
2. Workstation Brackets (Heavy-Duty)

2. Finish: Powder-coated
3. Color: White
4. Weight Capacity: Minimum 1,000 lbs.

# PART 3 - EXECUTION

## 3.1 EXAMINATION

A. Examine substrates to receive solid surface and quartz agglomerate material countertops and conditions under which countertops will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of countertops.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

## 3.2 INSTALLATION

A. Install countertops level to a tolerance of 1/8 inch in 8 feet, 1/4 inch maximum. Do not exceed 1/64-inch difference between planes of adjacent units.

B. Fasten countertops by screwing through corner blocks of base units into underside of countertop. Predrill holes for screws as recommended by manufacturer. Align adjacent surfaces and, using adhesive in color to match countertop, form seams to comply with manufacturer's written instructions. Carefully dress joints smooth, remove surface scratches, and clean entire surface.

C. Fasten subtops to cabinets by screwing through subtops into cornerblocks of base cabinets. Shim as needed to align subtops in a level plane.

D. Secure countertops to subtops with adhesive according to solid surface material manufacturer's written instructions. Align adjacent surfaces and, using adhesive in color to match countertop, form seams to comply with manufacturer's written instructions. Carefully dress joints smooth, remove surface scratches, and clean entire surface.

E. Bond joints with adhesive and draw tight as countertops are set. Mask areas of countertops adjacent to joints to prevent adhesive smears.

1. Install metal splines in kerfs in countertop edges at joints. Fill kerfs with adhesive before inserting splines and remove excess immediately after adjoining units are drawn into position.
2. Clamp units to temporary bracing, supports, or each other to ensure that countertops are properly aligned, and joints are of specified width.

F. Install backsplashes and end splashes by adhering to wall and countertops with adhesive. Mask areas of countertops and splashes adjacent to joints to prevent adhesive smears.

G. Install aprons to backing and countertops with adhesive. Mask areas of countertops and splashes adjacent to joints to prevent adhesive smears. Fasten by screwing through backing. Predrill holes for screws as recommended by manufacturer.

H. Complete cutouts not finished in shop. Mask areas of countertops adjacent to cutouts to prevent damage while cutting. Make cutouts to accurately fit items to be installed, and at right angles to finished surfaces unless beveling is required for clearance. Ease edges slightly to prevent snipping.

1. Seal edges of cutouts in particleboard subtops by saturating with varnish.

I. Apply sealant to gaps at walls; comply with Section 07 9200 "Joint Sealants."

J. Clean surface components according to manufacturer’s published maintenance instructions.

K. Completely remove excess adhesives and sealants from finished surfaces.

L. Protect surfaces from damage with non-staining coverings.

END OF SECTION 12 3661

SECTION 22 00 00 - PLUMBING

PART 1 - GENERAL

1.1 DESCRIPTION

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

B. It is the intent of these Specifications for the Contractor to provide plumbing systems complete, fully operational, fully adjusted, and ready for use.

C. All contract requirements indicated in the Division 21, 22, 23, 25, 26, and 28 documents, including drawings and specifications, shall be considered supplemental to and in addition to the contract requirements indicated in the documents of other trades and the Division 01 requirements.

1.2 REQUIRED WORK DESCRIBED IN OTHER SECTIONS

A. Section 22 00 10 - Division 22 General Requirements
B. Section 22 00 15 - Division 22 Design Conditions
C. Section 22 00 70 - Division 22 Schedule of Submittal Data

D. Section 22 05 13 - Common Motor Requirements for Plumbing Equipment
E. Section 22 05 30 - Division 22 Demolition
F. Section 22 05 53 - Identification for Plumbing Piping and Equipment

G. Section 22 05 76 - Facility Drainage Piping Cleanouts
H. Section 22 05 93 - Testing, Adjusting, and Balancing for Plumbing
I. Section 22 07 00 - Plumbing Insulation
J. Section 22 11 16 - Domestic Water Piping
K. Section 22 11 19 - Domestic Water Piping Specialties
L. Section 22 11 23 - Domestic Water Pumps
M. Section 22 13 16 - Sanitary Waste and Vent Piping
N. Section 22 13 19 - Sanitary Waste Piping Specialties

O. Section 22 33 00 - Electric Domestic Water Heaters
P. Section 22 40 00 - Plumbing Fixtures

1.3 PARTIAL LIST OF WORK NOT INCLUDED IN DIVISION 22

A. Electrical connections for motors:
B. Individual motor controllers except motor controllers furnished as integral parts of pieces of equipment:
C. Painting (except as otherwise specified herein):

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. General: All equipment shall be of the capacity and type as specified herein, as shown in the Schedule of Capacities and as indicated on the Drawings. Equipment shall be provided by a

specified manufacturer and in accordance with the space limitations of the Project.

B. Single Source: To maximize ease of maintenance and part replacement, equipment of a similar nature shall be provided by a single manufacturer.

C. Approved Equal: Equipment and materials selected by the Contractor within the context of "equal as approved by the Engineer", "approved equal", "equivalent as determined by the Engineer" and similar terminology shall be submitted to the Engineer for review, approval and inclusion into the Contract Documents prior to the finalization of the contract between the Owner and the Contractor, and prior to the shop drawing submittal phase of the Project. All equipment and materials submitted to the Engineer under the terms of "approved equal" during the shop drawing phase of the Project without prior review and approval shall be returned to the Contractor without review under the status of "No Action".

2.2 MATERIAL

A. All material required for a complete and proper installation shall be as specified and as selected by the Contractor subject to the approval of the Architect.

2.3 CONTRACTOR APPROVALS

A. The contractor shall submit in writing and obtain written approval from the Owner, Architect, and/or Engineer for all equipment substitutions, installation deviations from that shown on the Contract Documents, and all other miscellaneous approvals required from the Owner, Architect, and/or Engineer as referenced throughout these specifications.

B. A request for a substitution or deviation from the Contract Document constitutes that the Contractor:

1. Has investigated the proposed product and determined that it is equal to or superior in all respects to that specified.

2. Will provide the same warranties for the substitution as for the product specified.

3. Will coordinate the installation of the substitution into the work, and make such other changes as may be required to make the work complete in all respects.

4. Waives all claims for additional costs, under his responsibility, which may subsequently become apparent.

5. Will absorb all costs incurred by the substitution when affecting other trades including but not limited to electrical, structural, architectural, etc.

6. Will absorb any cost incurred by the Owner's Representative in review of the substituted product if the acceptance of the substituted item creates the need for system modification and/or redesign, or if the substituting Contractor exhibits negligence in his substituting procedure thus submitting inferior, mis-applied or mis-sized equipment.

PART 3 - EXECUTION

3.1 CONDITIONS

A. Inspection: Prior to proceeding with the work of this Division, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation

may properly commence. Verify that the work of this Division may be completed in strict accordance with all pertinent codes and regulations, the reviewed shop drawings, and the manufacturers' recommendations.

B. Discrepancies: In the event of discrepancy, immediately notify the Owner's Representative. Do not proceed in areas of discrepancy until all such discrepancies have been resolved. If there is a discrepancy between the Drawings and the Specifications, the Specifications shall typically govern. However, any discrepancy of this type shall be immediately brought to the attention of the Owner's Representative for formal interpretation prior to proceeding with the work.

C. Interpretation of Documents: Any and all contractual requirements may be indicated solely on the Drawings, solely in the Specifications, in both the Specifications and on the Drawings, in reference standards indicated in the Specifications and/or in the Owner's and Contractor's Contract. If Contract requirements are indicated in both the Specifications and the Drawings, the Contractor shall comply with both requirements unless the requirements are mutually exclusive of each other. If Contract requirements are indicated in both a reference standard and the Specifications, the more stringent requirement shall apply. Any and all contractual requirements shall be interpreted within the overall context of the complete scope of work. All materials, equipment, systems and installation methods shall be suitable for the intended service, coordinated with other trades and be complete, fully operational, adjusted, tested and ready for use by the Owner.

## 3.2 INSTALLATION OF EQUIPMENT

A. Locations: Install all equipment in the locations shown on the reviewed shop drawings, except where specifically otherwise approved on the job by the Architect and/or Owner's Representative.

B. Interferences: Avoid interference with structure and with work of other trades, preserving adequate headroom and clearing all doors and passageways to the approval of the Architect and/or Owner's Representative.

C. Inspection: Check each piece of equipment in the system for defects. Verify that all parts are properly furnished and installed, function properly, and that all adjustments have been made.

## 3.3 CONNECTIONS TO EQUIPMENT

A. The Contractor shall make final connections to all equipment for a complete and operational system.

## 3.4 CLOSING-IN OF UNINSPECTED WORK

A. General: Do not allow or cause any of the work of this Division to be covered up or enclosed until it has been inspected, tested, and approved by the Architect, Owner's Representative, and by all other authorities having jurisdiction.

B. Uncovering: Should any of the work of this Division be covered up or enclosed before it has been completely inspected, tested and approved, the Contractor shall provide all services, labor, materials and equipment necessary to uncover such work without added expense to the Owner. After the work has been completely inspected, tested, and approved, the Contractor shall provide all services, labor, materials and equipment to make all repairs necessary to restore the work to its original and proper condition at no additional cost to the Owner.

## 3.5 COOPERATION WITH OTHER TRADES

A. Coordinate and cooperate with other trades in order that all systems in the scope of the Contract may be installed in the best arrangement. Coordinate and cooperate as required with all other trades which share space in common areas and maximize the access to each system and their respective components.

## 3.6 CLEANING

A. It is the intent of these Specifications that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment. Level of cleanliness shall be defined as "broom" clean unless noted otherwise. All exterior surfaces of Division 22 equipment shall be wiped down and cleaned of all dust and dirt. All interior surfaces of equipment shall be wiped down and vacuum cleaned so as to be delivered to the Owner in factory new condition. Surfaces to be painted shall be cleaned and prepared in accordance with architectural division of the contract and as noted in other sections herein.

## 3.7 COMPLETENESS

A. It is the intent of these Specifications to provide complete systems. Completeness shall mean that all materials, equipment, and systems as installed and operating on this project have been installed properly with the best practices of the trade; are suitable for the intended purpose, location, and environment; properly fit within the physical space limitations for the project; are in conformance with applicable codes and reference standards; have been started-up, tested, adjusted, and commissioned for the intended use; have maintained applicable UL Listings; are in compliance with manufacturer's recommendations and warranty requirements; ready for the Owner’s use, and in the opinion of the Architect, performing as designed.

## 3.8 ADJUSTMENT OF CONTROLS

A. The Contractor shall provide the personnel and equipment to completely adjust the controls to the satisfaction of the Architect. At the completion of the project, the Architect will arrange a meeting at the job site to allow the Contractor to demonstrate the proper operation of the electrical controls.

## 3.9 NOISE

A. It is the intent of these Specifications to provide a system free from objectionable audible noise and vibration. Any equipment that is generating objectionable noise or vibration, in the opinion of the Architect, shall be corrected and dampened as required to eliminate the objectionable level.

END OF SECTION 22 00 00

SECTION 22 00 10 - DIVISION 22 GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 DESCRIPTION

A. General

1. The work under this Section shall conform to the requirements of the "General Conditions" and "Supplementary Conditions" of these Specifications.

B. Work Included

1. The work included in this Section consists of the general requirements for the work more specifically described in Division 21, 22, 23, 25, 26 and 28 where referenced.

C. Definitions

1. The Contractor shall provide all supervision, labor, material, equipment, machinery, and any and all other items necessary to complete the mechanical and electrical systems. All items of equipment are specified in the singular; however, the Contractor shall provide the number of items of equipment as indicated on the Drawings, and as required for complete systems.

D. Intent

1. It is the intention of these Specifications and Drawings to call for finished work that has been completed, tested, adjusted, commissioned, fully operational, approved by the local code authorities, and ready for the Owner’s use. Wherever the word "provide" is used, it shall mean "furnish and install complete and ready to use."

2. It is the intention of these specifications and drawings that wherever the word “weather” is used throughout the various Division 21, 22, 23, 25, 26 and 28 specification sections it refers to “locations exposed to weather”. “Locations exposed to weather” shall mean areas exposed to outside environmental temperature, humidity, or moisture conditions. These locations include but are not limited to areas on the exterior of the building, on the roof, within architectural exhaust and intake plenums, within loading docks, and other areas as indicated on the drawings or in the specifications.

3. All labor, materials, apparatus, and appliances essential to the complete functioning of the systems described and indicated herein, or which may be reasonably implied as essential whether mentioned in the Drawings and Specifications or not, shall be furnished and installed by the Contractor.

4. In addition to the requirements indicated throughout the Drawings and Specifications, as completely installed and operating on this project; equipment and systems provided for this project shall meet the following minimum requirements without additional cost to the Project: a. Be suitable for the intended service, location, and environment; b. Properly fit within the physical space limitations for the project; c. Meet the minimum requirements of applicable codes and reference standards as determined and approved by the local code authorities; d. Maintain applicable UL listings; e. Be coordinated between Divisions 21, 22, 23, 25, 26 and 28 in a timely manner; f. Be coordinated with the work of other trades in a timely manner; g. Not be purchased or installed prior to the resolution of any conflicts or coordination that could result in additional cost to the Owner; h. Be in

accordance with the manufacturer's recommendations and warranty requirements; and I. Be ready for use by the Owner.

5. In cases of doubt as to the Work intended, or in the event of need for explanation thereof, the Contractor shall request supplementary instructions from the Architect.

E. Codes, Rules, Regulations, Permits and Fees

1. Refer to the architectural specification Divisions for additional requirements. The architectural specification Divisions shall govern these requirements. The following information shall be in addition to the architectural specification Divisions.

2. The Contractor shall give all necessary notices, obtain all permits and pay all government sales taxes, fees, and other costs, including utility connections or extensions, in connection with the Contractor's work; file all necessary plans, prepare all documents and obtain all necessary approvals of all governmental departments having jurisdiction; obtain all required certificates of inspection of the Contractor's work and deliver same to the Architect before request for acceptance and final payment for work.

3. The Contractor shall comply with the most recently revised versions of all applicable laws, rules, regulations, and ordinances of Federal, State, and Local Authorities. Modifications required by the above said Authorities shall be made without additional charge to the Owner.

4. The Contractor shall comply with applicable building codes as adopted and modified by the Local Code Authorities. Modifications required by the above said building codes shall be made without additional charge to the Owner.

5. All equipment shall comply with all applicable requirements of laws, codes, ordinances, legislation, etc., of all Federal, State, and Local Authorities, whether indicated on the Contract Documents or not.

6. Where Contract Drawings and Specifications requirements are in excess of Code requirements and are permitted under the Code, the Contract Drawings and Specifications shall govern.

7. All rules and regulations of the State Fire Insurance Regulatory Body, Underwriters Laboratories, Americans with Disabilities Act (ADA), and National Electrical Code, shall be complied with whether indicated in the Contract Drawings and Specifications or not.

8. All materials, equipment, and systems provided on this project shall be in compliance with the applicable energy codes and the energy efficiency ASHRAE Standard 90.1 as adopted by the applicable building codes.

9. Project shall meet or exceed the requirements of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality Guideline for Occupied Buildings Under Construction. In addition, all stored on-site or installed materials shall be protected at all times from moisture damage. Provide photographs to the owner, architect, or engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

10. All products and systems including but not limited to piping, pumps, valves, fittings, meters, filters, faucets, drinking fountains, water heaters, sealants, gaskets, adhesives,

coatings, lubricants, etc… which are installed directly in contact with potable water shall be certified and labeled in accordance with NSF 61 standards.

11. Where alterations to or deviations from the Contract Drawings and Specifications are required by the above Authorities report the requirements to the Architect and secure Architect's approval before starting the alterations.

12. Third Party Certification: All packaged equipment shall be independently Third Party labeled as a system for it's intended use by a Nationally Recognized Testing Laboratory (NRTL), such as UL or ETL, in accordance with OSHA Federal Regulations 29 CFR 1910.303 & 1910.399, as well as NFPA 70, National Electric Code (NEC), Article 90-7.

1.2 QUALITY ASSURANCE

A. Surveys and Measurements

1. The Contractor shall base all measurements, both horizontal and vertical, from established benchmarks. All work shall agree with these established lines and levels. Verify all measurements at site and check the correctness of same as related to the work.

2. Should the Contractor discover any discrepancy between actual measurements and those indicated, which prevents following good practice or the intent of the Drawings and Specifications, the Contractor shall notify the Architect, and shall not proceed with the work until instruction has been received from the Architect.

B. Drawings

1. Drawings are diagrammatic and indicate the general arrangement of systems and work included in the contract. Drawings are not to be scaled. The Architectural Drawings and Details shall be examined for exact location of fixtures and equipment. Where they are not definitely located, this information shall be obtained from the Owner's Representative.

2. The Contractor shall follow Drawings in laying out work and check Drawings of other trades to verify spaces in which work will be installed. Maintain maximum headroom and space conditions at all points. Where headroom or space conditions appear inadequate, the Owner's Representative shall be notified before proceeding with installation.

3. If directed by the Owner's Representative, the Contractor shall, without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of the work, if such directions are made prior to the performance of the affected work.

C. Cooperation with Other Trades

1. The Subcontractors shall give full cooperation to other trades and shall furnish in writing to the General Contractor any information necessary to permit the work of all trades to be installed satisfactorily and with the least possible interference or delay.

2. Where Division 21, 22, 23, 25, 26 and 28 work will be installed in close proximity to, or will interfere with, work of other trades, composite shop drawings will be prepared to accomplish coordination among trades in working out space conditions. All necessary adjustments to work out space conditions shall be reflected on the composite shop drawings. The Contractor shall prepare composite shop drawings and sections at suitable scale not less than 1/4" = 1'-0", clearly showing how work of all affected trades is to be

installed. If work is installed before coordinating with other trades, or so as to cause any interference with work of other trades, then the necessary changes in the work shall be made to correct the condition without additional expense to the Owner.

3. The Contractor shall obtain and transmit between affected trades, all necessary templates, patterns, setting plans, and shop details for the proper installation of work and for the purpose of coordinating adjacent work.

D. Protection

1. The Contractor shall protect all work and material from damage by the Contractor's work or workmen, subcontractors' work or workmen, and shall be liable for all damage thus caused.

2. The Contractor shall be responsible for work and equipment until "Final Review"; the Contractor shall protect work against theft, injury or damage; and shall carefully store material and equipment received on site which are not immediately installed. The Contractor shall close open ends of work with temporary covers or plugs during storage and construction to prevent entry of obstructing material.

3. All stored on-site or installed materials shall be protected at all times from moisture damage.

E. Material and Workmanship

1. Work shall be executed in strict accordance with the best practice of the trades in a thorough, substantial, workmanlike manner by competent workmen. The Contractor shall furnish the services of a full-time, experienced superintendent, who shall be constantly in charge of the installation of the work, together with all skilled workmen, fitters, metal workers, welders, helpers, and laborers required to unload, transfer, erect, connect, adjust, start, operate, and test each system.

F. Manufacturers' Recommendations

1. In addition to the requirements indicated in the drawings and specifications; manufactured products, articles, materials, and equipment shall be applied, installed, connected, erected, used, cleaned, conditioned, inspected, started-up, tested, operated, and commissioned in accordance with the manufacturer's current printed recommendations. Keep copies of such printed recommendations at job site and make them available as required. When conditions on this project are not covered by the manufacturer's printed recommendations, at the discretion of and as requested by the owner's representative, the manufacturer's authorized representative shall review the conditions and provide written supplemental recommendations to address the special situation. If the manufacturer’s recommendations are in conflict with the requirements of the drawings and specifications, the Contractor shall advise the Owner’s representative of the conflict prior to purchase and installation.

G. Space Limitations

1. MEP systems and equipment shall be provided and installed to accommodate the structural systems, architectural systems, MEP systems, other trades, and the physical space limitations of the project.

2. Equipment shall be provided which will fit into the physical spaces provided and indicated, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearance in accordance with the Code requirements, the requirements of the local inspection department, and the equipment manufacturers' recommendations.

3. In the preparation of Drawings, a reasonable effort to accommodate equipment manufacturers' space requirements has been made. However, since space requirements, equipment arrangement, access and service clearances, conduit, piping, ductwork and other connections, manufacturers' recommendations, and other variables vary according to each manufacturer; due to the multiple manufacturers listed for each product and the possible number of different equipment combinations; the responsibility for access, proper fit, coordination, and accommodating the space limitations of the project rests with the Contractor. Refer to shop drawing submittal requirements herein for additional requirements.

4. MEP systems and equipment shall be installed to maximize headroom clearance below the bottom of material. MEP systems and equipment shall be installed to maximize vertical clearances between the bottom of material and top of specified ceiling construction. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, lighting, and other MEP systems shall be prepared to indicate compliance with the Space Limitations and Maximum Headroom Clearance requirements. Conflicts with vertical and horizontal clearances, ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data, clouded, and clearly marked so as to bring the conflict to the attention of the Architect in a timely manner prior to installation. For the purpose of definition, the term MEP refers to all systems, material, equipment, services, means and methods described the Division 21, 22, 23, 25, 26 and 28 documents including the Division 21, 22, 23, 25, 26 and 28 specifications and drawings. Refer to the specifications for additional requirements.

5. Refer to the drawings for additional requirements.

6. Physical dimensions and arrangements of equipment to be installed shall be subject to the Architects' review.

H. Coordination with Installation Conditions

1. The requirements of the Division 21, 22, 23, 25, 26 and 28 systems shall be coordinated with the installation conditions. The Contractor shall visit the site, survey the installation conditions, and verify exact measurements. Means and methods of building access, rigging, and modifications to installation conditions to accommodate the installation of new Division 21, 22, 23, 25, 26 and 28 work shall be the responsibility of the Contractor. Disassembly, reassembly, and certification testing of equipment as required to accommodate the installation conditions shall be included in the base bid. Prior to the ordering, purchasing, and fabricating of Division 21, 22, 23, 25, 26 and 28 systems, coordination with installation conditions shall be completed by the Contractor.

I. Coordination Between Divisions 21, 22, 23, 25, 26 and 28:

1. The Division 22 Contractors shall cooperate with the General Contractor and the Division 21, 23, 25, 26 and 28 Contractors to provide coordination between the Division 21, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 22, it shall be the responsibility of the Division 22 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service

characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

J. Commissioning:

1. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience, and expertise at the general contractor and sub-contractor levels to properly install, check, verify, supervise, start-up, trouble-shoot, de-bug, pre-test, commission, and make the necessary corrections to all MEP systems in accordance with the Contract Documents. The Contractor shall be responsible for the operational demonstration of the completed MEP systems to the Owner’s Representative, Architect and Engineer. The Contractor shall complete the MEP systems in accordance with the contract documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems prior to requesting that the Owner’s Representative, Architect, and Engineer attend the operational demonstrations of the MEP systems. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner. Refer to the various sections of the specifications for additional commissioning requirements.

K. Additional Information

1. Information provided by the Engineer to the Contractors during construction must be reviewed and approved by the Architect and Owner's Representative. The Contractor shall not proceed without prior review and authorization from the Architect and Owner's Representative. The Contractor shall not proceed with work that will increase the cost of the project or impact the project schedule without prior review and authorization from the Architect and Owner's Representative. If information provided by the Engineer to the Contractor will increase the cost of the project or impact the project schedule, the Contractor shall advise the Architect and Owner's Representative in a reasonably timely manner so as to minimize the cost and impact to the project. Refer to the General Conditions of the Contract and the Architectural documents for additional information.

1.3 SHOP DRAWINGS, PRODUCT DATA AND SAMPLES

A. After the Contract is awarded but prior to proceeding with the Work requiring final reviewed shop drawings, product data and samples, the Contractor shall provide the Architect and Engineer with a submittal log schedule indicating when each submittal is to be sent to the Architect and Engineer for review in order for the Architect and Engineer to schedule personnel and plan accordingly. Unless otherwise indicated, a minimum period of two (2) weeks, exclusive of transmittal time, will be required in the Engineer's office each time a shop drawing, product data and/or samples are submitted or resubmitted for review. This time period will commence once the shop drawing, product data and/or samples are received in the Engineer’s office contingent upon the shop drawing, product data and/or samples being received in the Engineer’s office before 3:00 PM. If the shop drawing, product data and/or samples are received in the Engineer’s office after 3:00 PM, the time period will not start until the next business day. This time period shall be considered by the Contractor when scheduling the Work. The submittal log schedule should be updated and reissued by the Contractor on a regular basis to avoid submittal review delays. If a submittal arrives before or after the scheduled time indicated within the Contractor’s submittal log schedule, or if a number of submittals arrive at the same time without scheduled notification, the Architect and Engineer may not be able to review all of the submittals within the minimum two (2) week time period. The Contractor shall obtain complete shop drawings, product data and samples from the

manufacturers, suppliers, vendors, subcontractors, and sub-subcontractors for all materials, systems and equipment specified herein; and submit data and details of such materials, systems and equipment for review by the Architect and Engineer. Prior to submission of the shop drawings, product data and samples to the Architect and Engineer, the Contractor shall review and certify that the shop drawings, product data and samples are in compliance with the Contract Documents. The Contractor shall check all materials and equipment after their arrival on the job site and verify their compliance with the Contract Documents. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience and expertise at the general contractor and sub-contractor levels to properly check, verify, and correct if necessary shop drawing submittal data to ensure compliance with the Contract Documents prior to delivery to the Architect and Engineer. The Division 22 shop drawing submittal requirements, including but not limited to the Division 22 certification requirements, shall be in addition to the requirements indicated in the Architectural Divisions of the Contract Documents.

B. The Architect's approval and Engineer's review of shop drawings, product data, and samples shall not relieve the Contractor of the responsibility for dimensions or errors that may be contained therein, or for deviations from requirements in the Contract Documents. It shall be clearly understood that the Engineer's noting some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents shall govern the Work and are neither waived nor superseded in any way by the review of shop drawings, product data and samples.

C. Before submitting shop drawings, product data and samples the Contractor shall certify that data and details set forth on each shop drawing, product data, or samples for each item of equipment and materials complies with the Contract Documents for this Project. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the Contractor, or by a person duly authorized to sign binding agreements for the Contractor.

Unless certified, shop drawings, product data and samples will not be reviewed, and will be returned unchecked to the Contractor. Certifications shall be in the form of rubber stamp impressions or a digitally certified stamp which state:

> I hereby certify that this shop drawing, product data, and/or sample has been checked prior to submittal and that it has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents and physical space limitations for the Project.
>
> (Name of the Plumbing Subcontractor)
>
> Signed:_____
>
> Position:_____
>
> Date:_____

D. The Contractor shall submit shop drawings, product data and/or samples, equipment and other items listed in Section 22 00 70 and for any other equipment for which such submittals may be required in various Sections of the Specifications. Prior to submitting shop drawings, product data and samples for all items of equipment listed in Section 22 00 70 and shown with an asterisk (*), each submittal package including subsequent submittal packages shall be certified by the manufacturer as to performance and compliance with Contract Documents. Such

certification shall be made by the Owner, a Partner, a Corporate Officer of the manufacturer, or be a person duly authorized to sign for the Owner, Partner, or corporation.

Unless certified, shop drawings, product data and/or samples will not be reviewed and will be returned unchecked to the Contractor. The certification shall be as follows:

> I hereby certify that the equipment shown on this shop drawing, product data and/or sample, has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents for this Project. I further certify that all data shown herein as to performance dimensions, construction, materials and other pertinent items is true and correct.
>
> (Name of the Manufacturer)
>
> Signed:_____
>
> Position:_____
>
> Date:_____

E. Inadequate or incomplete shop drawings, product data and/or samples will not be reviewed by the Architect or the Engineer and shall be returned to the Contractor for resubmittal.

F. The Contractor shall observe the following rules when submitting shop drawings, product data and samples:

1. Identification: Each shop drawing shall indicate in the lower right hand corner, and each product data brochure shall indicate on the front cover the following: title of the sheet or brochure; name and location of the Project, names of the Architect and Engineer, Contractor, Subcontractor, manufacturer, supplier, vendor; the date of submittal; and the date of each correction and revision. So far as is practical, each shop drawing, product data and/or sample shall bear a cross-reference note to the page number or numbers of sheets of the Drawings and/or Specifications showing the Work. Unless the above information is included, the submittal will be returned for resubmittal.

2. Quality Assurance: Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification paragraph. A hand written marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

3. Shop drawings shall be done in an easily legible scale and shall contain sufficient plans, elevations, sections and schematics to clearly describe the apparatus. Drawings shall be

prepared by an engineer-draftsman skilled in this type of work. Unless otherwise indicated, all sheet metal, piping and similar shop drawings shall be drawn to at least 1/4" = 1'-0" scale. Typical floor sheet metal ductwork shop drawings may be drawn at 1/8"=1'-0" scale. The Contractor shall submit shop drawings as described below. Shop drawings which do not comply with these requirements will be returned for resubmittal. Refer to "Cooperation with Other Trades", this section, for additional composite shop drawing requirements. Unless otherwise indicated in the Contract Documents, the submittal shall consist of three prints of the shop drawings. The Architect and Engineer will each retain one print and the remaining copies will be returned to the Contractor.

4. Product data to be submitted shall be published by the manufacturers and shall contain complete and detailed engineering and dimensional information. The Contractor shall submit product data as described below. Product data which does not comply with these requirements will be returned for resubmittal. Unless otherwise indicated in the Contract Documents, the Contractor shall submit eight (8) copies of each product data brochure. The Architect and Engineer will each retain one copy of the submittal and the remaining copies will be returned to the Contractor. Product data submitted shall contain only information relevant to the particular equipment or material to be furnished. The Contractor shall not submit catalogs which describe several different items in addition to those items to be used, unless all irrelevant information is marked out, or unless relevant information is clearly marked. The submittal shall consist of product data from each manufacturer. Product data from each manufacturer shall be identified and submitted separately.

5. Coordination between Division 21, 22, 23, 25, 26 and 28: Each Division 22 shop drawing and product data submitted for review, which has Division 22 equipment requiring electrical service, shall contain a statement indicating that the Division 22 Contractor has provided the Division 26 Contractor with the necessary electrical characteristics of the actual Division 22 equipment to be provided.

6. Shop Drawing and Product Data - Submittal Review Status:

    a. Shop drawing and product data submittals will be returned marked with one of the following status:

        1) "REVIEWED NO EXCEPTIONS"
        2) "REVIEWED EXCEPTIONS NOTED"
        3) "INFORMATION ONLY"
        4) "REVISE AND RESUBMIT"
        5) "REJECTED"

    b. If stamped “REVISE AND RESUBMIT" or “REJECTED”, the shop drawing or product data shall be revised in accordance with the Contract Documents. If marked "REVIEWED, EXCEPTIONS NOTED" documentation that addresses only the noted exceptions in the form of a letter, supplemental information, or complete resubmittal shall be forwarded to the Owner, Architect and Engineer for record purposes only. If marked "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” no additional submittal will be required.

    c. If the copy stamped "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” is altered for any reason after it has been stamped, the "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” shall automatically be voided and the submittal status revised to "REVIEWED EXCEPTIONS NOTED", "REVISE AND RESUBMIT", or “REJECTED” depending on the alterations.

d. All work shall be done in accordance with shop drawings stamped "REVIEWED NO EXCEPTIONS", "REVIEWED EXCEPTIONS NOTED", or “INFORMATION ONLY” insofar as these are in agreement with the Contract Documents. Wherever differences occur between the shop drawings and the Contract Documents, the Contract Documents shall govern the Work.

G. Record Shop Drawings

1. After receiving the Engineer’s and Architect’s final approval for each shop drawing, the Contractor shall provide the Engineer with AutoCAD files of each final approved shop drawing. All of the previous review comments by the Engineer and Architect associated with each shop drawing shall be incorporated into the AutoCAD files of each shop drawing prior to sending the AutoCAD files to the Engineer. The AutoCAD files shall be in AutoCAD 2015 version software or later version as approved by the Engineer. The Contractor shall be responsible for contacting the Engineer to determine the approved AutoCAD software to use prior to preparing the AutoCAD files.

H. Maximum Ceiling Clearance and Headroom Requirements

1. All material for systems and equipment shall be installed to accommodate the structural systems, architectural ceiling construction, base building and future lighting systems, and other physical space limitations of the project. All systems and equipment shall be installed to maximize clearances between the bottom of material and top of ceiling construction. All systems and equipment shall be installed to maximize headroom clearance below the bottom of material. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, and other systems that will be located where ceiling plenum clearances and headroom clearances are limited; shall be prepared to address these issues. Conflicts with ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data and clearly marked so as to bring the situation to the attention of the Engineer.

I. Equipment and Layout Deviations

1. Layout, connections, supports, access & maintenance clearances, related architectural enclosures, and structural supports have been based on selected products and manufacturers. In some cases, the manufacturer’s model number has been provided. Other products by specified manufacturers may be used if they comply with requirements of the contract documents including but not limited to capacity ratings. However, where the Contractor proposes to use an item of equipment that has alternative but equivalent minimum capacity ratings or that requires any redesign of the structure, partitions, foundations, piping, wiring or any other part of the mechanical, electrical, structural, civil, utility, or architectural systems; all such redesign, and all new drawings and detailing required therefore, shall be prepared by the Contractor at the Contractor's own expense as approved by the Owner’s Representative. Where such approved deviation requires a different quantity, capacity, and arrangement of ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, and starters from that specified or indicated on the drawings, the Contractor shall furnish and install any such ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, starters and any other additional equipment required by the system, at no additional cost to the Owner.

1.4 RESPONSE TO SHOP DRAWING, SUBMITTAL, AND FIELD REPORT REVIEW COMMENTS

A. Contractor shall respond in written form to each shop drawing, submittal, or field report review comment from the Engineer within 10 business days from the date indicated on the Engineer's review form. Contractor's written response to each review comment should indicate the date and specifically what corrective action was or will be taken and include the name of the person and associated company responding to each review comment. Refer to the Architectural documents for additional requirements.

1.5 MOCK-UP REQUIREMENTS

A. Contractor shall provide the following mock-up requirements specified herein. Mock-up requirements shall be completed after the applicable shop drawing review but prior to the fabrication and installation of the multiple systems, equipment, and accessories throughout the facility which are to be included as part of the mock-up requirements. The Contractor shall advise the Engineer and Owner's Representative after the mock-up is complete and ready for review and inspection. The Contractor shall arrange a time mutually agreeable to these parties so they can meet at the project site, review the mock-up installation, and determine any changes that need to be made for the installation to be acceptable to the Owner's Representative and Engineer. The Contractor shall make adjustments to the mock-up arrangement as requested by the Owner's Representative and Engineer prior to the fabrication and installation of the remaining systems, equipment, and accessories throughout the facility which are included as part of the mock-up arrangement. The Contractor shall account for this requirement in the schedule of construction so this procedure does not delay the construction progress. If multiple systems, equipment, and accessories which are included as part of the mock-up arrangement are installed in multiple locations throughout the facility prior to the mock-up approval, the Contractor shall be responsible for the remedial work required to comply with the approved mock-up condition at no additional cost to the Owner. Mock-up arrangements shall be located on site within the facility at an actual useable location. The Contractor shall provide advance notice to the appropriate parties of the various mock-up inspections a minimum of seven working days prior to the meeting. If additional or follow-up field inspections of the mock-up modifications are required, the Contractor shall provide these modifications and additional follow-up field inspections as required without additional cost to the Owner.

1. Not Used.

1.6 WARRANTY, SERVICE, AND REPLACEMENT PARTS

A. All equipment supplied for this Contract shall be free from defects in material, workmanship, and title, and shall be of the kind and quality described herein. If it appears within one (1) year from the date of the building final acceptance by the Owner that the equipment does not meet the warranties specified above or the equipment is recalled by the equipment manufacturer, the Contractor shall correct any defect or replace any recalled equipment, including nonconformance with these Specifications, at no additional cost to the Owner.

B. The manufacturers for each of the various selected Division 22 pieces of equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner's acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

1.7 CONFERENCE PRIOR TO START OF WORK

A. Immediately upon the award of this Contract, but prior to commencing any Work, the Contractor together with designated major Subcontractors shall confer with the Owner's Representative, the Architect and Engineer concerning the Work under this Contract.

B. Contractors and Subcontractors attending this meeting shall include but not be limited to the following: General; Mechanical; Plumbing; Fire Protection; Automatic Temperature Controls and Building Management; Sheet Metal; Insulation; Electrical; Fire Alarm; and Testing, Balancing and Adjusting. Contractor and Subcontractor representatives shall be familiar with the Drawings, Specifications and shop drawing submittal requirements and be prepared to intelligently discuss the project requirements.

C. The conference will be at a mutually agreed place and acceptable time.

1.8 ELECTRONIC FILE TRANSFER AGREEMENT

A. Release of the Dewberry CAD, BIM, and other electronic files will require the User of the files to sign the Dewberry Electronic File Transfer Agreement.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials and equipment shall be new and in good condition. The commercially standard items of equipment and the specific names mentioned herein are intended to identify standards of quality and performance necessary for the proper functioning of the Work.

B. Since manufacturing methods vary, reasonable minor variations are expected; however, performance and material requirements specified are the minimum standards acceptable. The Engineer retains the right to judge the quality of equipment that deviate from the Contract Documents.

2.2 PRODUCTS, COMPONENTS AND MATERIALS IN PLENUMS

A. All products, components, materials and composite materials within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723 as required by applicable mechanical code.

2.3 EQUIPMENT GUARDS

A. Provide equipment guards for the safe operation of equipment as recommended by the equipment manufacturer and as required by applicable codes. Provide easily removable expanded metal guards for all belts, couplings, exposed fan inlets and outlets, and other moving parts of machinery. Provide tachometer openings in the guards at least 2" in diameter, for all belt-driven or variable speed machinery.

2.4 EQUIPMENT MOUNTING/SUPPORT DEVICES

A. Unless otherwise required to accommodate the particular requirements and unique conditions of this project, all floor, wall or ceiling mounting/support devices shall be factory fabricated devices as provided by the equipment manufacturer for the intended application. In addition to

the factory-fabricated devices, the Contractor shall provide additional support members, framework, mounting hardware, brackets, steel angles, anchors, restraints, support stands, attachment devices, and other miscellaneous mounting and support devices as required for adequate equipment support.

B. The Engineer retains the right to judge the quality of the design, application, and installation of all field and shop fabricated mounting/support devices using steel angle, steel channels, Uni-Strut, brackets, all-thread rods, etc. Any mounting or support devices that will rust, corrode, or in the opinion of the Engineer become unsightly or a maintenance problem will not be acceptable.

C. Support devices not factory finished shall be hot dipped galvanized or properly primed and painted with a minimum of two coats of rust inhibiting paint. Refer to the Architectural Painting Systems for additional requirements.

D. All-thread rods shall be galvanized or cadmium plated.

E. All plywood used as mounting boards in association with Division 22, unless specified otherwise by the Architect, shall be APA C-D plugged interior 5/8" to 3/4". All plywood used shall be painted (front, back and edges) with a minimum of one prime coat and one finish coat prior to equipment installation. Refer to the Architectural Specifications for additional requirements. Color shall be determined by the Architect.

## 2.5 ACCESS DOORS

A. *Wherever access is required through walls, ceilings or fire-rated enclosures to valves, fire dampers, heating coils, damper actuators, drain line cleanouts, or other concealed equipment installed under this Division, unless otherwise noted in the architectural documents, the Contractor shall furnish a hinged access door and frame as follows:

1. Drywall construction--Milcor Style DW.

2. Finished acoustical tile ceiling--Milcor Style AT in applications where there is not adequate clearance space to properly push an acoustical ceiling tile up for removal and replacement.

3. Finished plaster ceiling--Milcor Style AP.

4. Finished plaster walls or ceramic tile--Milcor Style M.

5. Plaster or masonry walls and ceilings outside offices and in other finished areas exposed to view--Milcor Style K or M.

6. Access doors shall be "B" label fire construction where required.

7. Access doors shall be installed under another Division.

8. No access door shall be installed until location and type have been approved by the Architect.

## 2.6 ESCUTCHEONS

A. Furnish and install heavy chrome-plated or nickel-plated steel plates of approved pattern on all pipes passing through walls and ceiling in finished areas. Escutcheons shall be B & C No. 10 or

approved equal with concealed hinges. Pattern shall be approved by the Architect.

2.7 OVERCURRENT PROTECTION

A. Fused and circuit breaker type overcurrent protection devices required for any equipment furnished under Division 22 shall be in accordance with Division 26 Specification Section 26 28 00.

B. Unless specifically noted otherwise on the Drawings or in these Specifications, all Division 22 equipment shall be rated for use with both fused and circuit breaker overcurrent protection. This requirement shall be clearly indicated on the manufacturers' nameplate.

2.8 SEALANTS AND ADHESIVES

A. All sealants and adhesives shall comply with the current Volatile Organic Compound (VOC) content limits of the latest edition of the South Coast Air Quality Management District (SCAQM) Rule 1168 entitled “Adhesive and Sealant Applications”. Piping sealants shall have no more than 250 grams/liter VOC content for all sealants and adhesives to be used within the building interior locations.

B. The Contractor shall submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each adhesive and sealant used within the building interior locations.

PART 3 - EXECUTION

3.1 SCAFFOLDING, RIGGING, HOISTING

A. The Contractor shall furnish all scaffolding, rigging, hoisting, and services necessary for erection and delivery of any equipment and apparatus furnished to the job site. Remove same from job site when no longer required.

3.2 SLEEVES, CUTTING AND PATCHING

A. The Contractor shall be responsible for the timely placing of openings and sleeves for all piping passing through walls, partitions, beams, floors, louvers, and roofs while the same are under construction. A pipe sleeve shall be round and one size larger than the size of pipe it serves except where "Thunderline/Link Seal" casing seals are used in penetrations through walls below grade. If the pipe is insulated, its pipe sleeve shall be one size larger than the outside diameter of the insulation around the pipe. All sleeves in floors shall extend two inches above the finished floor. All pipes passing through concrete or masonry walls above grade shall have 18 gauge galvanized steel sleeves or plastic "hole-outs". All sleeves set in concrete floor construction shall be standard weight galvanized steel. Sleeves set in concrete floor construction directly supporting riser piping 4" and larger shall be standard weight galvanized steel with three 6" long, 3/8" diameter steel reinforcing rods welded at 120o spacing to the sleeve and shall be installed with the rods embedded in the concrete slab. Where the pipe passes through a sleeve, no point of the pipe or its insulation shall touch the sleeve. Caulk around such pipe with sufficient quantity of fire safing insulation to equal fire rating of construction penetrated by the sleeve. In addition, seal the fire safing material with a non-hardening mastic.

B. Seal all pipe penetrations and unused sleeves located in fire rated construction with factory built devices or with manufactured fill, void, or cavity materials "Classified" by Underwriters Laboratories, Inc. for use as a Through-Penetration Firestop. All firestop devices and systems shall be approved for such use by the authority having jurisdiction. The fire safing system used

shall maintain the fire resistance rating of the building component that is penetrated. Fire safing systems and devices shall comply with ASTM E 814 (UL 1479). Submittal data for fire safing systems shall include the U.L. System Numbers listed in the U.L. Building Materials Directory under which the material was tested in accordance with ASTM E 814 (UL 1479) for use in a Through-Penetration Firestop System. Excessive shrinkage of the fire stopping materials which would permit the transmission of smoke or water prior to exposure to a Fire condition is unacceptable. Where a mastic coating is used to seal the surface of the firestop, the mastic shall be non-hardening. The fire safing system used shall accommodate expansion and contraction of the floating mechanical piping systems without damaging the firestop or reducing its effectiveness as a smoke barrier or water seal. See paragraph titled "Vibration Isolation for Piping" in Section 22 05 48.

C. Sleeves penetrating walls below grade shall be “WS Wall Sleeves” by Pipeline Seal and Insulator, Inc. Houston, Texas, with 2” water seal stop. The water seal stop shall be located in the middle of the wall. Seal off annular opening between pipe and sleeve with "Thunderline/Link-Seal" casing seal as manufactured by Pipeline Seal & Insulator, Inc., Houston, Texas. The pipe sleeve shall be sized to accommodate the “Thunderline/Link-Seal” casing seal. Casing seals shall be Model “C” and shall be sized per the manufacturer’s recommendations.

D. If holes and/or sleeves are not properly installed and cutting and patching becomes necessary, it shall be done at no expense to the Owner. The Contractor shall undertake no cutting or patching without first securing the Architect's written approval.

E. Sleeves in floor slabs with water proof membranes shall be modified as required to accept a Zurn Z-197 riser sleeve and also meet all other sleeve requirements as specified herein before. The Z-197 riser sleeve shall be constructed of dura-coated cast iron body, double threaded pipe connector with combination membrane flashing flange.

F. The location and size of all penetrations and openings through existing walls, slabs, beams, roofs and other existing building construction shall be provided by the contractor to accommodate the new work. Penetrations and openings through existing building construction shall be reviewed by the structural engineer and architect prior to cutting the penetration. Unless otherwise required by the architectural and structural divisions, metal sleeve inserts will not be required in pipe penetrations except as required to meet UL fire rating requirements. The contractor shall perform a structural survey, including x-rays or graphs of all floor slabs and supporting walls to locate existing structural bars, cables, in slab conduits and any other interferences that may be in conflict with proposed sleeve/penetration locations. The x-ray or graph shall be performed prior to any cutting or chipping of structural supporting walls. In the event of a conflict, the contractor shall notify the architect prior to proceeding with the work.

G. Provide sleeves for future use as indicated on the drawings.

## 3.3 EXCAVATION AND BACKFILL

A. The Contractor shall make all necessary excavations, cutting of paving, concrete, etc., and do all backfilling and paving repairs necessary for the proper execution of the Work. Remove all dirt and debris out of and away from the building as directed. Backfill shall be mechanically compacted to a density of 95% of the maximum dry density at optimum moisture content as determined by the Standard Proctor Compaction Test.

B. Backfill shall be compacted and repairs to paving or concrete shall be accomplished to the satisfaction of the Architect and the Local Authorities.

C. See the various Sections of the Division for additional excavation and backfill requirements.

3.4 PAINTING

A. Painting, except as specified or indicated otherwise, shall be done under another Division. This Division shall cooperate with the other Division to determine the size of equipment, sizes and lengths of pipes, etc., to be painted.

B. Equipment furnished under this Division shall be factory-finished. If the factory finish is damaged during shipment, installation, etc., it shall be repainted subject to the Architect's approval.

C. Refer to Section 22 05 53 for additional requirements.

3.5 DRAYAGE AND HAULING

A. Include all drayage, hauling, hoisting, and placement in the building of equipment specified herein. The Contractor shall be responsible for the timely introduction of equipment to the Project. If any item of equipment is received prior to the time it is required, the Contractor shall be responsible for its proper storage and protection until such time as it may be required. The Contractor shall pay for all costs of demurrage or storage.

3.6 IDENTIFICATION OF EQUIPMENT

A. Equipment and piping shall be identified in accordance with the requirements indicated in specification Section 22 05 53.

B. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

3.7 EQUIPMENT NOISE AND VIBRATION

A. It is the intention to specify and for the Contractor to provide equipment and systems that, as defined herein, shall be quiet and free of apparent vibration in operation. To this end, both the balancing of rotating machinery and the installation of vibration isolation at various locations are required.

B. It shall be the responsibility of the Contractor to obtain equipment that is quiet in operation as compared to other available equipment of its size, capacity, and type; to install equipment so that a minimum amount of noise and/or vibration is transmitted to the building; and to fabricate the duct system so that air noises generated in the system are held to a minimum.

C. Any additional precautions deemed necessary to provide a quiet installation shall be done as part of the Work of the Contractor, subject to review by the Engineer, without additional cost to the Owner. After the system is in operation, it shall be the responsibility of the Contractor to make any changes to equipment or Work installed that may be required to provide a system which is quiet in operation as defined herein.

D. The system noise level shall be equal to or less than those indicated in Section 23 00 15. Sound levels within the occupied spaces must meet the criteria described above and with all building, wall partition, floor and ceiling construction in place as they exist for the individual

spaces. The attenuation through boundary construction of equipment rooms must be considered in selecting equipment for acceptable noise level.

## 3.8 AUTOMATIC TEMPERATURE CONTROL AND EMCS COORDINATION

A. It shall be the responsibility of any equipment manufacturer who furnishes equipment for this Project to coordinate and cooperate with the requirements of Section 25 00 90. Equipment automatically and remotely controlled and/or monitored via the EMCS and/or interlocked with some other piece of equipment as specified in Section 25 00 90 shall be complete with all necessary contact points and interface devices to accomplish the control strategies and sequences described in Section 25 00 90. It shall be the responsibility of all applicable equipment manufacturers to fully participate in the coordination procedure described in Section 25 00 90 under "Composite Wiring Diagram". By accepting this Contract; the Contractor, all Subcontractors, and all applicable equipment manufacturers expressly agree to cooperate with the EMCS Contractor to fully comply with the "Composite Wiring Diagram" requirements and expedite the obtaining of all necessary information.

## 3.9 LUBRICATION

A. Provide means for lubricating all bearings and other machine parts. If a part requiring lubrication is concealed or inaccessible, extend a lubrication tube with suitable fitting to an accessible location and suitably identify it.

B. After installation, properly lubricate all parts requiring lubrication and keep them adequately lubricated until final acceptance by the Owner.

## 3.10 SPARE PARTS

A. General: The Contractor shall provide all necessary spare parts for the equipment supplied in this Project as indicated hereafter and in various other Sections of the Specifications. Spare parts shall be obtained from the original equipment manufacturer.

## 3.11 DOCUMENTATION

A. For all testing, training and review procedures the Contractor shall maintain adequate records in written form that clearly documents the event. The written documentation for each testing, training and review procedure shall include but not be limited to the following: Name and company of attending personnel, general description of procedure(s), date(s), time of day, deficiencies found, corrective action taken, and nature of final results. Documentation records shall be kept in an orderly, legible and up to date format. Contractor's documentation records shall be accessible for review by the Owner Representative upon request. Each testing, training and review procedure document shall bear the name and signature of the authorized representative of the General Contractor, Mechanical Contractor, and manufacturer or Subcontractor if applicable.

## 3.12 COMPLETION, COMMISSIONING AND FINAL REVIEW OF THE DIVISION 22 SYSTEMS

A. Start-up: After the Contractor and equipment manufacturer representative determine that the equipment and related systems have been properly installed and are ready for start-up; the equipment manufacturer representative shall perform the manufacturer’s start-up and check-out procedures to ensure proper operation of the equipment in accordance with the contract documents and the manufacturer’s recommendations. Start-up and check-out of each piece of equipment shall be in accordance with the equipment manufacturer’s recommendations and warranty requirements. Start-up services shall be performed by factory-trained personnel of the

equipment manufacturer or factory-trained personnel authorized by the manufacturer and their official representatives. Start-up personnel shall be certified by the equipment manufacturer as properly trained in the manufacturer's start-up and check-out procedures. Provide documentation of the start-up and check-out procedures in accordance with the contract documents and the manufacturer's recommendations. Documentation of the start-up and check-out procedures shall be included within and submitted with the Operational and Maintenance Manuals.

B. Trouble-shooting, Debugging, and Pre-testing: After Start-Up but prior to the Operational Demonstrations; the contractor shall perform operational pre-testing, trouble-shooting, and de-bugging of all equipment and systems. At the completion of the Trouble-shooting, Debugging, and Pre-testing process, all equipment and systems shall be complete and operating in accordance with the requirements of the contract documents. Trouble-shooting, Debugging, and Pre-testing of equipment and systems shall be performed to the extent necessary for the Contractor to determine that all systems are complete and operating in accordance with the Contract Documents. Prior to final Commissioning and Operational Demonstrations all equipment and systems shall be certified by the Contractor as complete and operating in accordance with the Contract Documents.

C. Operational Demonstrations and Final Commissioning: Prior to Substantial Completion and within a schedule mutually agreed to by the contractor and the Owner's Representative, the Contractor shall demonstrate the operation of all equipment and systems as necessary to demonstrate compliance with the Contract Documents. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner.

D. Final Review and Owner Acceptance:

1. After completion of the Operational Demonstrations, Final Commissioning, and Owner's Close-out Package requirements; at a time designated and mutually agreed by the Contractor and Owner's Representative; a final walk-through review of the project will be performed by the Contractor and Owner's Representative. The review will be of a general nature with the intent of addressing outstanding issues prior to the Owner's final acceptance of the project.

2. The entire system shall be clean and operating properly with all systems balanced and all controls adjusted. All labels shall be removed from the plumbing fixtures, and the fixtures shall be clean and in operating condition.

3. All floor drains and traps shall be clean and tested in the presence of the Building Owner's Representative allowing a minimum of five (5) gallons of clean water to rapidly pass through the drain and trap.

4. Certificates, Documents, and Close-out Package required herein shall be complete, in order and presented to the Architect at least two weeks prior to the review.

5. After the review, any changes or corrections noted as necessary for the Work to comply with these Specifications and the Drawings shall be accomplished without delay in order to secure final acceptance of the Work.

E. Schedule: The Contractor shall complete the MEP systems in accordance with the Contract Documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems a minimum of three (3) weeks prior to the date of "Substantial Completion".

The three (3) weeks prior to the date of "Substantial Completion" shall be reserved by the Contractor for the Owner's Representative, Architect, and Engineer to witness the operational demonstrations of the completed fully commissioned MEP systems. If more than three (3) weeks is deemed necessary by the Contractor for the operational demonstrations, the Contractor shall notify the Owner's representative in written form and determine an operational demonstration schedule mutually agreeable to the Owner's Representative and Contractor that completes the operational demonstrations prior to "Substantial Completion". Additional costs incurred by the Owner related to reviewing, witnessing, testing, inspecting, and commissioning of the mechanical systems by the Architect and Engineer after the date of "Substantial Completion" shall be reimbursed by the Contractor to the Owner. As part of the shop drawing submittal process, the Contractor shall provide a detailed schedule of completion indicating when each MEP system will be complete and ready for owner acceptance. The schedule shall indicate time periods and milestone dates of Start-up; Trouble-shooting, Debugging, and Pre-testing; Operational Demonstrations and Final Commissioning; Final Review and Owner Acceptance.

F. Additional Requirements:
1. Commissioning Requirements: Refer to specification Section 22 08 00 for additional building commissioning general requirements.

2. Owner's Close-out Package: Refer to the Owner's Close-out Package requirements within Section 22 00 10 for additional requirements.

3. Refer to the various sections of the specifications for additional requirements.

## 3.13 OPERATING INSTRUCTION

A. The Contractor shall provide the services of a factory trained specialist to supervise the operation of all equipment specified herein and to instruct the Owner's operators for a five-day, eight hours per day, operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. See individual Sections for additional instructions by the manufacturer's trained specialists.

B. Training schedule, duration, and subject matter shall also be in accordance with a training syllabus provided by owner's operating staff. This syllabus will be provided prior to the final review.

C. The Owner shall be notified in writing at least five days before each operating instruction period begins. The Contractor shall not commence instruction period until the Owner has issued a written acceptance of the starting time.

## 3.14 OPERATING AND MAINTENANCE BOOKS

A. The Contractor shall provide operating instruction and maintenance data books for all equipment and materials furnished under this Division.

B. Submit three (3) hard copies and three (3) electronic copies of the operating and maintenance data books for review at least eight weeks before final review of the Project. Assemble all data in a completely indexed volume or volumes and identify the size, model, and features indicated for each item. The Engineer's review period for the operating and maintenance data books shall be a minimum of 4 weeks, exclusive of transmittal time. The Contractor's schedule shall plan to accommodate this review period requirement.

C. Maintenance instruction manuals shall include complete oiling, cleaning, and servicing data compiled in clearly and easily understandable form. Data shall show all model numbers of each piece of equipment, complete lists of replacement parts, motor ratings, and actual loads.

D. Include the following information where applicable:

1. Identifying name and mark number as indicated on the Drawings and in the Specification.

2. Locations (where several similar items are used, provide a list).

3. Complete nameplate data.

4. Certified record drawings and "Final Reviewed" or completely corrected shop drawings and/or submittals.

5. Parts lists.

6. Performance curves and data including air and water design flow rate set points.

7. Wiring diagrams.

8. Lubrication charts.

9. Start-up and Check-out procedures and check-lists.

10. Manufacturers' recommended operating and maintenance instructions with all non-applicable information deleted. This shall include but not be limited to the manufacturer's service and repair manual.

11. Additional information as recommended by the equipment manufacturer or system provider for the proper operation, maintenance, and warranty of the equipment and systems.

## 3.15 RECORD DRAWINGS

A. The Contractor shall maintain on a daily basis at the project site a complete set of "Record Drawings". The “Record Drawings” shall consist of a set of black line prints and AutoCAD files of the Contract Drawings for this Division. The black line prints shall be marked and the AutoCAD files shall be electronically updated regularly by the Contractor during the construction period to reflect an accurate dimensional record of all buried or concealed work. In addition, the set of black line prints shall be marked and the set of AutoCAD files shall be electronically updated regularly to show the precise location of concealed work and equipment including buried, concealed or embedded piping, conduit and valves and all changes and deviations in the mechanical work from that shown on the Contract Documents. This requirement shall not be construed as authorization for the Contractor to make changes in the layout or work without definite instructions from the Architect. The set of black line prints and AutoCAD files of the Contract Drawings for this Division shall remove or blacken out the Engineer's seal and Engineer's firm name.

B. Prior to commencing work, the Contractor shall need to purchase from the Architect or Engineer a set of black line prints to be used for the "Record Drawings" and obtain from the Engineer a set of AutoCAD files of the Contract Drawings for this Division in preparing “Records Drawings” for the project based on the Contractor’s work. The AutoCAD files shall be in AutoCad 2015 version software or later version as provided by the Engineer. In order to obtain a set of

AutoCAD files of the Contract Drawings for this Division from the Engineer, the Contractor shall need to contact the Engineer, Architect or Client and obtain a written release and indemnification statement from the Engineer. This release and indemnification statement shall be signed by an owner, partner, corporate officer or other person duly authorized to agree to and sign legally binding documents for the Contractor's respective company. Upon the Engineer receiving the signed unedited release and indemnification statement back from the Contractor, the Engineer will forward the AutoCAD files of the Contract Drawings for this Division to the Architect or Client who in turn will forward the AutoCAD files of the Contract Drawings for this Division to the Contractor for his use in preparing "Records Drawings" for the project based on the Contractor's work. If the Contractor desires to have architectural and structural information in preparing "Records Drawings" for the project, the Contractor shall need to contact the Architect and Structural Engineer to obtain the pertinent architectural and structural information directly from the Architect and the Structural Engineer.

C. Record dimensions shall clearly and accurately delineate the Work as installed; locations shall be suitably identified by at least two dimensions to permanent structures.

D. The owner's representative shall retain the right to periodically review the "Record Drawings" as the project progresses. Final acceptance of as-built "Record Drawings" shall be conditional upon the Contractor clearly "red-marking" all changes from the contract documents, including "request for information" changes and other supplemental instruction changes, directly on the black line set of prints and clouding all of the changes in the electronic AutoCAD files. All of the clouded changes on each drawing in the electronic AutoCAD files shall be on a separate drawing layer which can be turned off without affecting the appearance of each drawing.

E. The Contractor shall mark all "Record Drawings" on the front lower right hand corner with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWINGS (3/8" high letters)
To be used for recording Field
Deviations and Dimensional Data
Only". (5/16" high letters)

F. Upon completion of work, the Contractor shall certify all "Record Drawings" on the front lower right hand corner adjacent to the above marking with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWING CERTIFIED CORRECT (3/8" high letters)

(Name of Plumbing Subcontractor)

By: _____

Date:_____

For all black line prints to be used for the "Record Drawings", the Contractor must sign the above certification. For all AutoCAD files to be used for the "Record Drawings", the Contractor must fill in the name of the person responsible for certifying the above certification. Both the black line prints and the AutoCAD files shall be certified by the same person.

G. Prior to final acceptance of the Work of this Division, the Contractor shall submit properly certified "Record Drawings" to the Architect for review and shall make changes, corrections, or additions as the Architect may require to the "Record Drawings". After all changes have been made to the "Record Drawings", the Contractor shall deliver to the Owner the final set of

"Record Drawings" consisting of the black line prints and the AutoCAD files. The AutoCAD files to be delivered by the Contractor to the Owner shall be in AutoCAD 2015 version software or later version as originally provided to the Contractor by the Engineer.

3.16 OWNER'S CLOSE-OUT PACKAGE

A. Contractor shall provide a master project close-out log and corresponding schedule for submission to the Owner and Engineer for review and approval.

B. Contractor shall provide the Owner with two (2) copies of all warranties bound in a notebook with a corresponding warranty log. Warranty log shall clearly indicate the subcontractor, address, telephone number, and contact person.

C. Contractor shall provide the Owner with one (1) complete set of approved submittals and a copy of the final submittal log.

D. Contractor shall provide the Owner with a complete list of all subcontractors and major suppliers with address, telephone number, and contact persons.

E. Contractor shall provide the Owner with a complete inventory of attic stock materials and spare parts as applicable and a copy of the transmittal signed by the Owner's representative acknowledging receipt of the items.

F. Contractor shall provide the Owner and Engineer with documentation of compliance for all specified requirements associated with equipment start-up, commissioning, demonstrations, training and warranties. Documentation shall include, but not be limited to, equipment start-up sheets as completed by the manufacturer's representative.

G. Contractor shall provide the Owner with approved water balancing reports including three (3) copies of water balance reports and two (2) sets of water balance prints.

H. Contractor shall provide the Owner with a comprehensive valve tag chart.

I. Contractor shall provide the Owner with documentation that all storm and roof drains have been thoroughly cleaned immediately prior to Substantial Completion.

J. Contractor shall provide the Owner and Engineer with certification that the domestic water system has been properly cleaned, flushed and sterilized. The Contractor shall advise the Owner of the time and date of the pipe cleaning, flushing, and sterilization procedures a minimum of two weeks prior to the procedures. The Owner shall have the option of witnessing the complete procedures. Any cleaning, flushing, and sterilization procedures performed prior to Owner notification, at the Owner's discretion, shall be performed again in the presence of the Owner.

K. Contractor shall provide the Owner and Engineer with documentation of acceptable water lead levels. The Owner reserves the right to contract with an independent consultant to perform additional water lead testing. If it is determined that unacceptable water lead levels exist, all costs associated with additional tests and time required to correct and uncover the deficiencies by the independent consultant will be the responsibility of the Contractor unless the deficiencies are not design related.

L. When the Contractor considers the work to be ready for final acceptance, the Contractor shall provide the Owner and Engineer with written certification stating the following:

1. All work has been completed in accordance with the Contract Documents.

2. All field review items and other deficiencies identified by the Certificate of Substantial Completion have been properly corrected.

3. All work has been inspected for compliance with the Contract Documents.

4. All mechanical systems have been tested in the presence of the Owner's representative and are fully operational in compliance with the Contract Documents.

5. As-built documents have been reviewed, are complete and are accurate.

M. The Contractor shall provide the Owner with a letter from the Engineer of Record stating that the equipment installed fully complies with the Contract Documents.

3.17 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 22 08 00 and the alternate pricing requirements indicated in Section 22 00 00 for additional building commissioning general requirements.

END OF SECTION 22 00 10

SECTION 22 00 15 - DIVISION 22 DESIGN CONDITIONS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

PART 2 - REQUIREMENTS

2.1 DESIGN CONDITIONS

A. Working Pressures:

| Description | Working Pressure | Normal Operating Temperature Range | Pressure Zones |
|---|---|---|---|
| Domestic Cold Water System | 150 psig | 45°F to 80°F | None |
| Domestic Hot Water System | 150 psig | 105°F to 140°F | None |

1. Refer to the drawings and Schedules for Plumbing for additional information regarding working pressures.

2. Refer to the drawings for additional information regarding pressure zones.

B. Refer to the drawings, specification section Automatic Temperature Control System, and specification section Schedules for Plumbing for additional design conditions.

C. All piping systems downstream of domestic water pressure reducing stations shall have an operating pressure of less than 80 psig.

END OF SECTION 22 00 15

SECTION 22 00 70 - DIVISION 22 SCHEDULE OF SUBMITTAL DATA

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Furnish the shop drawing submittal data for the Architect's and Engineer's review as indicated herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Division 21, 22, 23, 25, 26 and 28: All Sections

PART 2 - PART 2 - SHOP DRAWING SUBMITTAL DATA

2.1 GENERAL

A. The submittal data to be furnished for this Project shall comply with the Specifications hereinbefore. The referenced Sections herein scheduled are for general information purposes only and shall not be construed to be limiting to the submittal data scheduled and/or required within the individual Sections of these Specifications.

2.2 CONTRACT REQUIREMENTS

A. Shop drawing submittal data shall clearly and concisely address all Contract Document requirements. Standard manufacturer data that does not indicate these requirements shall be supplemented with additional information provided by the Contractor or equipment manufacturer. Refer to sections 22 00 10 for additional requirements, including submittal log schedule requirements.

2.3 DEVIATIONS FROM THE CONTRACT DOCUMENTS AND PERTINENT DATA

A. All deviations from the contract documents shall be noted in the form of a cover letter included with the shop drawing submittal package. All pertinent data such as material options, equipment arrangements, control options, diagrams, etc. that applies to this Project shall be clearly marked. All data and diagrams that do not apply to this Project shall be omitted from the package or clearly marked out.

2.4 QUALITY ASSURANCE

A. Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification

paragraph. A hand written marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

2.5 SPECIAL REQUIREMENTS

A. The shop drawing submittals for the Operation and Maintenance (O&M) Manuals shall require an Owner/Consultant review period of four (4) weeks. The Contractor shall plan for and arrange the shop drawing submittal review schedule to accommodate this requirement.

2.6 SPARE PARTS

A. The submittal data for each piece of equipment shall include a detailed itemized list of recommended spare parts. Spare parts shall be clearly identified by name and part number.

2.7 ORGANIZATION OF PACKAGES

A. Unless otherwise noted in the specifications and in the following schedule of shop drawing, the Contractor shall provide a single shop drawing submittal package for each of the listed specification sections. Unless specifically noted otherwise, multiple submittal packages per section will not be acceptable under the terms of the Contract.

2.8 AUTOMATIC TEMPERATURE CONTROL AND EMCS COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall be responsible for full compliance with the automatic temperature control and EMCS coordination requirements of Sections 22 00 10 and 25 00 90. Particular attention is called to the "Composite Wiring Diagram" requirements of Section 25 00 90.

2.9 COORDINATION BETWEEN DIVISIONS 21, 22, 23, 25, 26 AND 28

A. The Division 22 Contractors shall cooperate with the General Contractor and the Division 21, 23, 25, 26 and 28 Contractors to provide coordination between the Division 21, 22, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 22, it shall be the responsibility of the Division 22 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

2.10 MANUFACTURER'S CERTIFICATION

A. All items or equipment listed herein with asterisks (*) shall be certified by the manufacturer. Refer to Section 22 00 10 for certification requirements.

2.11 SHOP DRAWING SUBMITTAL SCHEDULE

A. Provide a comprehensive list of Divisions 21, 22, 23, 25, 26 and 28 shop drawing submittal

packages and a schedule indicating the date when each package will be submitted to the contractor and the date when each package will be submitted to the design team. The intent of this list and schedule is to allow the contractor and design team to plan ahead and schedule the necessary review and coordination resources in advance of the arrival of shop drawing submittal packages. Shop drawing submittal schedule shall be submitted to the Architect not less than four weeks prior to the submission of the first Division 21, 22, 23, 25, 26 and 28 shop drawing submittal packages. Shop drawing submittal schedule shall be periodically updated by the contractor and issued to the Architect to reflect current status of the project. Shop drawing submittal packages shall be listed on the schedule not less than four weeks prior to submission to the contractor and not less than six prior to the submission to the design team. Packages not listed on this schedule will required up to two additional weeks of review time.

2.12 SHOP DRAWING SUBMITTAL DATA

A. The submittal data to be furnished for this Project shall include but not be limited to the following:

1. Section 22 00 10 - Overcurrent Protection: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

2. Section 22 05 13 - AC Electric Motors: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

3. Section 22 05 13 - Motor Controllers: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

4. Section 22 05 48 - Vibration and Seismic Controls for Plumbing Piping and Equipment *

5. Section 22 05 76 - Facility Drainage Piping Cleanouts:

6. Section 22 05 93 - Water Balancing Procedures and Recording Forms.

7. Section 22 05 93 - Testing, Adjusting, and Balancing Plumbing Report for entire project.

8. Section 22 07 00 - Plumbing Insulation

9. Section 22 11 16 - Domestic Water Piping

10. Section 22 11 19 - Domestic Water Piping Specialties

11. Section 22 11 23 - Domestic Water Pumps

12. Section 22 13 16 - Sanitary Waste and Vent Piping

13. Section 22 13 19 - Sanitary Waste Piping Specialties

14. Section 22 33 00 - Electric Domestic Water Heaters

15. Section 22 40 00 - Plumbing Fixtures

16. Section 22 00 70 - Shop drawing submittal schedule.

17. Drawing Packages:
    a. Plumbing drawings shall be limited to a maximum of two (2) packages.

2.13 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 22 08 00 and the alternate pricing requirements indicated in Section 22 00 00 for additional building commissioning general requirements.

END OF SECTION 22 00 70

# SECTION 22 05 13 - COMMON MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

### 1.2 WORK INCLUDED

B. Furnish and install all electric motors required for equipment furnished under this Division. Electric motors shall be factory-mounted on equipment wherever possible and shall be constructed as specified in this Section. In general, all motor controllers will be furnished and installed under Division 26 unless the motor controller is an integral part of a piece of equipment and/or specifically required to be furnished under this Division.

### 1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Sections pertaining to equipment with electric motors and controllers.

### 1.4 REFERENCE STANDARDS

A. DOE Code of Federal Regulations (CFR)
1. 10 CFR 431 Subpart B - Electric Motors

B. Institute of Electrical and Electronics Engineers (IEEE)
1. 112 Standard Test Procedure for Polyphase Induction Motors and Generators

C. National Electrical Manufacturers Association (NEMA)
1. MG 1 Motors and Generators

### 1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawings for all equipment requiring motors and controllers shall be submitted for review. Drawings shall state the motor manufacturer, motor horsepower, voltage, speed, efficiency, insulation class, motor winding material, special torque requirements, and other special requirements.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

A. If they comply with these Specifications, the following manufacturers will be acceptable.

1. Motors
   a. Baldor-Reliance
   b. Siemens
   c. Century
   d. General Electric
   e. Louis Allis
   f. Toshiba

2. Controllers
   a. ITE/Gould
   b. General Electric
   c. Allen-Bradley
   d. Cutler-Hammer
   e. Federal
   f. Danfoss Graham
   g. ABB Industrial Systems

## 2.2 MOTORS

A. Voltage and Motor Horsepower: In general, motor voltages shall be as follows, unless specified or indicated otherwise.

1. 3/4 hp and larger: 208 volt, 3 phase, 60 hertz

2. Smaller than 3/4 hp: 120 volt, 1 phase, 60 hertz

B. Motor Efficiency

1. All motors shall meet or exceed the NEMA "Premium Efficiency" standards and nominal full load efficiency values as required by applicable codes, currently adopted Table 12-12 of NEMA MG 1, and as defined in accordance with the currently adopted NEMA standards. Motor efficiency shall be determined and rated in accordance with IEEE 112 Test Method B.

2. Applicable motors shall be in compliance with the currently adopted DOE Energy Policy Act (EPACT), the Energy Independence and Security Act (EISA) and Code of Federal Regulations (CFR).

C. General Construction for All Motors: All motors will be started across-the-line, unless specified otherwise. Motors shall be selected with low starting current and shall be designed for continuous duty to provide the running torque and pull-in torque required to suit the load. Unless otherwise specified all motors shall be single speed, 1750 rpm. All motors shall have standard open drip proof (ODP) type enclosure unless otherwise specified. All motors exposed to the weather shall have totally enclosed fan cooled (TEFC), totally enclosed air over (TEAO) or totally enclosed non-ventilated (TENV) type enclosures. All motors, except TEFC, TEAO and TENV types, shall have a service factor of at least 1.15. Motors shall be selected to operate at design conditions without exceeding nameplate ratings of a 1.0 service factor is assumed. Motors shall have sealed or field-lubricated type ball bearings unless noted otherwise. Field-lubricated motors shall be provided with grease fittings. A field-lubricated motor shall have the bearing cage drilled and tapped with a plug to accommodate a grease fitting.

1. All motors shall have copper windings.

2. Motors shall be selected for quiet operation.

3. All motors shall have a shaft grounding attachment on the rotor to mitigate shaft voltage and bearing currents.

D. ODP Motors: ODP three phase motors 10 horsepower and smaller shall have cast aluminum end bells with steel frames. Three phase motors 15 horsepower and larger shall have cast iron housings. ODP single phase motors shall have cast aluminum end bells with steel frames.

E. TEFC, TEAO and TENV Motors: TEFC, TEAO and TENV single and three phase motors shall have cast iron housings; Class F insulation; epoxy varnish coated for extra protection; cadmium

plated hardware; stator and rotor to be epoxy enameled for extra protection; corrosion-resistant fan (TEFC only).

F. Single Phase Motors

1. Unless otherwise noted, single phase motors shall be capacitor start type with starting, pull-in, and running characteristics to suit the load.

2. 1/8 horsepower and smaller motors on direct drive fan applications may be of the shaded pole type.

3. Motors larger than 1/8 horsepower up to and including 1/3 horsepower on direct drive fan applications may be of the permanent split capacitor type.

G. Variable Speed Motors

1. Motors used for variable speed operation shall be specifically designed and UL listed for variable speed applications.

2. Motors shall be inverter duty type and comply with the requirements of NEMA Standard MG 1 Part 31.

3. Motors shall be selected for compatibility with specified variable speed drive.

4. Motors shall be designed and selected for the intended load and application.

5. Motors shall be selected for quiet operation throughout the entire operational range of the equipment. Quiet operation shall be defined as follows:

   a. Noise from a motor when operating at any point in the variable speed mode shall not increase more than 3 dB in any noise criteria (NC) curve frequency band when compared to the same motor operating in the full bypass non-variable speed mode.

   b. Noise from a motor, as installed as required by the Contract Documents, shall not cause the sound in areas outside of the air handling unit room to exceed the established NC curve range. This requirement applies to variable speed, non-variable speed, start-up and shut-down modes of operation.

6. Motors shall be designed and selected to operate at any continuous duty point from 10% to 100% of rated motor speed without motor damage and motor overheating.

7. Motors used in conjunction with variable speed drives shall be provided with a shaft grounding kit/bearing protection ring to prevent variable speed drive induced electrical current damage to the motor.

8. It shall be the joint responsibility of the motor manufacturer, equipment manufacturer, and variable speed drive manufacturer to coordinate and cooperate in meeting these requirements. The equipment manufacturer and drive manufacturer shall provide the motor manufacturer with all information necessary, such as fan and pump curves, to insure proper motor selection.

H. Nameplate: A motor nameplate shall be securely affixed to each motor and shall clearly indicate the class of insulation, winding material and the service factor in addition to the usual electrical data.

I. Special Requirements: Refer to various Sections of this Division for special requirements for specific items of equipment requiring motors.

2.3 MOTOR CONTROLLERS

A. General: Motor controllers for all motors will be furnished and installed under Division 26 unless indicated or specified otherwise. Any motor controller that is an integral part of a piece of equipment shall be furnished under this Division and shall be in accordance with the following Specifications. All motor controllers shall be coordinated with the Division 26 contractor's fault current and short circuit coordination study.

B. Special Requirements: The motor controllers shall be steel mounted. Controllers shall be front wired, and all terminals shall be accessible for wiring directly from the front. No slate or ebony asbestos shall be permitted on any size controller from Size 0 through Size 8. All contacts shall be solid-silver cadmium oxide alloy which will not require any filling, dressing, or cleaning throughout the life of the control equipment. Bare copper or silver flashed contacts shall not be permitted. Operating coils shall be pressure molded. When a coil fails under overvoltage condition, the motor controller shall drop out.

C. Overload, Low Voltage and Phase Loss Protection: All three phase full voltage magnetic motor controllers shall have overload protection and electronic low voltage and phase loss protection in all three phases. All single phase full voltage magnetic motor controllers shall have overload protection in both legs. All two-speed full voltage magnetic motor controllers shall have overload protection in all six legs of the controller. Overload relays shall be of the hand-reset, trip-free variety so that blocking the reset mechanism in the reset position will not prevent the starter from dropping out if the motor is overloaded. This specifically excludes those overload relays which change to automatic reset from hand reset when the reset mechanism is blocked unless the automatic reset feature can be removed or voided. Accidentally depressing the reset button or mechanism shall not shut off the motor. Overload relays shall not be field convertible from hand to automatic reset type.

D. Field-Add Interlocks: It shall be possible to field-add one or more extra normally open (NO) or normally closed (NC) interlocks to all (except size 00) motor controllers without removing existing wiring or removing the controller from its enclosure.

E. Allen-Bradley Bulletin Numbers: Any full voltage magnetic motor controller to be furnished under this Division shall be equal to the following:

1. Individual three phase motor controller - AB Bul. 709.
2. Individual single phase motor controller - AB Bul. 709SP.
3. Combination three phase motor controller with circuit breaker - AB Bul. 713.

F. Enclosures: In general, motor controller enclosures shall be NEMA Type 1 general purpose unless exposed to the weather or otherwise indicated on the Drawings. Any motor controller exposed to the weather within parking garages or within loading docks shall have NEMA Type 3R watertight enclosure and NEMA 4X grounded stainless steel enclosure for locations exposed to weather which are on the exterior of the building, on the roof, or within architectural exhaust and intake plenums unless noted otherwise on the Drawings.

G. Holding Coils and Control Transformers: Generally, holding coils in full voltage magnetic motor controllers shall be suitable for use with 120 volt, AC control voltage. Each controller shall have a control transformer with primary and secondary fuses.

H. Two Speed Motors: Two speed motor controllers for two speed motors shall have "decelerating relay" between high and low speeds.

I. Additional Requirements: Each motor controller shall be equipped with a 120 volt red running pilot light, control transformer, auxiliary contacts, and a hand-off-automatic switch. Two speed motor controllers shall have "fast-slow" pilot lights as well as a fast-slow selector switch in

conjunction with the H-O-A. Controllers for sump pumps and sewage ejectors shall have "test" position in lieu of "hand" position. Pilot lights will be operated by an interlock on the motor controller not placed across the operating coil.

J. Manual Controllers: Manual motor controllers where indicated on the Drawings, required, and/or specified shall be similar and approved equal to Allen-Bradley Bul. 600 in NEMA Type 1 enclosure or as otherwise required for the location of the installation.

PART 3 - EXECUTION

3.1 MOUNTING

A. Electric motors and controllers furnished under this Division and Section shall be factory mounted unless noted otherwise.

END OF SECTION 23 05 13

SECTION 22 05 30 - DIVISION 22 DEMOLITION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements", "Conditions of the Contract" and "Supplementary Conditions". Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Furnish all labor, equipment and materials required to demolish, cap, patch, repair and dispose of the Division 22 systems no longer required as part of an active system of the Project as specified herein and as indicated on the Drawings.

1.3 SCOPE OF WORK

A. It shall be the responsibility of the Contractor to review the Architectural Documents in addition to the Division 21, 22, 23, 25, 26 and 28 Documents to determine the complete scope of work.

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Under this Division, the Contractor shall furnish all labor, equipment, appliances, and materials to perform all operations in connection with the demolition of the Division 23 systems as indicated on the Drawings and as specified herein and/or as required by the Local Building Code. The work shall include but not be limited to the following:

1. The contractor shall review the existing conditions, in addition to the Architectural documents, and the Division 21, 22, 23,25, 26 and 28 documents, to determine the complete scope of work.

2. Field verify the exact sizes and locations of all existing piping prior to demolition of any existing work. The demolition work shall be coordinated with the architectural scope of the Project to assure proper limits of demolition.

3. Remove all existing plumbing equipment, plumbing fixtures, and associated piping within the area to be demolished, whether specifically indicated or not, that is no longer required unless otherwise indicated.

4. Where existing equipment is to be relocated, extreme care shall be taken to prevent damage during the removal and reinstallation. Where damage occurs, the equipment shall be replaced or repaired to the satisfaction and approval of the Architect at no additional cost to the Owner.

5. Unless otherwise indicated, all existing plumbing fixtures and trim in the area to be demolished shall be removed and stored in a manner to prevent loss or damage in the construction staging area for reuse.

6. Unless otherwise indicated, all plumbing piping systems required to remain operational throughout and after construction; including waste, vent and water systems; serving plumbing fixtures to be removed shall be capped in a suitable manner.

7. Contractor shall be responsible for cutting, patching and restoration of existing finishes.

8. Contractor shall be responsible for capping existing piping to remain.

9. Contractor shall be responsible for removal, disposal and salvage of equipment from job site.

10. Contractor shall be responsible for cleaning, testing and repair of existing equipment and piping to be reused.

11. Contractor shall be responsible for sealing all existing floor slab penetrations in order to maintain the integrity of the fire rating of the existing floor slabs.

12. Refer to Architectural documents and the Division 21, 22, 23, 25, 26 and 28 documents for the area and extent of the demolition.

13. Contractor shall inspect and test all existing water valves throughout this facility which are to remain and be reused to ensure each valve operates properly and is not damaged. Existing valves shall be able to adequately hold the maximum working water pressure of the system in which they are installed. The Contractor shall notify the Owner and Engineer of all valves needing replacement. Alternate: The Contractor shall submit with their bid, as an alternate, a unit price for replacement of defective valves including all necessary labor and material costs.

B. Where a system to be demolished must remain operational throughout the construction process until a new system that will take its place is functional, the contractor shall provide all products, services, materials, labor and supervision to maintain the operational status of the affected system.

## 3.2 EXISTING CONDITIONS

A. Under the scope of work, the Contractor shall perform a comprehensive field inspection of the existing conditions to thoroughly familiarize himself with the Project and determine the complete scope of new and demolition work. The Contractor shall verify in the field exact sizes, capacities, and locations of all existing equipment, ductwork, and piping prior to installation of any proposed work. Proposed work shall be coordinated with existing conditions to assure proper installation. This field survey and determination of the complete scope of work shall be performed prior to the final bid. Specific attention is brought to physically larger and heavy pieces of equipment that are new to be installed or existing to be removed and the routing of such equipment with the available means and site conditions.

## 3.3 CONFLICTS AND DEPARTURES

A. Conflicts

1. The Contractor shall remove all existing plumbing fixtures, equipment, piping, and associated accessories whether specifically indicated or not, that are no longer required. Any equipment or piping which must remain as part of an active system and is in conflict with the proposed work shall be relocated at no additional expense. There shall be no additional expense to the Owner for this type of demolition work, unless, in the Architect's and Owner's opinion, it is beyond the scope and intent of the Contract Documents.

B. Departures

1. The Contract Drawings are diagrammatic and schematic in nature and do not indicate inferred details. Any departures from the Drawings that are necessary to comply with the intent of Documents, requirements of applicable Building Codes or the Authority having jurisdiction shall be detailed in the form of shop drawing submittal data and submitted for review by the Architect and Engineer. All demolition and any such revisions for departures shall be made in accordance with the reviewed shop drawings without increased expense to the Owner. Departures shall not be made without prior written approval by the Architect.

## 3.4 COOPERATION

A. The Contractor shall comply with all the requirements of the Local Code Authority and the building Owner.

B. The Contractor shall give advance notice to the Architect when work is to be performed.

C. The Contractor shall cooperate and coordinate demolition of Division 22 systems as required with all of other trades. The Contractor shall supervise and assist in the removing and replacing of existing materials for installation of mechanical items and items related to all other trades.

## 3.5 OCCUPIED AREAS

A. Normal functions of occupied areas must continue during the construction phases. Every effort shall be made to insure such functions are not disturbed. The Contractor shall be responsible for the installation and removal of temporary systems as required to maintain the functions of occupied areas. The Contractor shall schedule any work which may be required in occupied areas during unoccupied hours.

## 3.6 BARRIERS

A. Where required by applicable health and safety regulations, the Contractor shall furnish and maintain safety and dust barriers. If necessary, the Contractor shall furnish and maintain temporary fencing and traffic barriers in accordance with all applicable health and safety regulations and as deemed necessary.

## 3.7 RESTORATION

A. Demolition shall be carried out with caution to assure that existing conditions to remain will be undamaged. Existing conditions to remain that are damaged or defaced by work under this contract shall be restored or replaced equal to the conditions at the time of award of the Contract.

B. Cut, patch, and restore all existing surfaces not receiving new finishes that have been disturbed during execution of this Contract. Materials and finishes used shall be similar, in all respect, to adjacent surfaces.

C. Existing piping which are to remain as part of an existing or proposed active system that the Contractor determines to be defective (leaks, corrosion, cracks, etc.) shall be brought to the attention of the Architect.

D. Existing equipment and plumbing fixtures to be removed shall be cleaned, repaired, and reused at the discretion of the Architect wherever applicable.

3.8 CONNECTIONS AND CAPS

A. Connections to existing work shall be similar, in all respects, to the existing system and conditions unless otherwise indicated. Existing work shall be altered and/or temporarily removed and replaced as required for completion of requirements of the project.

B. Remove and cap existing piping, no longer required to remain as part of an active system, at the appropriate main. Existing piping below grade shall be abandoned and capped flush with finish floor unless otherwise indicated. No sanitary, soil, water, vent, or waste pipe shall extend greater than two feet (2'-0") to a dead-end.

3.9 SALVAGE

A. The Owner assumes no responsibility for loss or damage to materials or structures on site for the salvage value of equipment which the Contractor may have reflected in his bid.

B. All existing plumbing fixtures, equipment, and piping removed during construction no longer required as part of an active system and not to be reused shall be removed from the job site and properly disposed of by the Contractor.

C. All existing equipment and piping not being reused shall be removed during construction and shall remain the property of the owner and shall be stored, removed from the job site, or disposed of as directed by the owner's representative.

3.10 CLEANING

A. The Contractor shall clear away all debris and demolished material at frequent intervals. The Contractor shall not allow debris to accumulate to the extent that it will interfere with work, passage of the workmen, and the operation of the existing occupied areas.

B. It is the intent of this Specification that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment.

3.11 INVENTORY

A. Contractor shall provide a detailed inventory of all existing plumbing fixtures and plumbing equipment to be stored for later reuse. Inventory shall include type, size, capacities, and quantities of each item or piece of equipment.

3.12 TESTING

A. Piping

1. No existing piping shall remain part of the new active system without prior approval from the engineer/owner. The Contractor shall replace any defective piping and shall repair any piping leaks if deemed to be reused.

B. Equipment

1. Existing equipment, instruments, and accessories to remain shall be tested for defects and shall be adjusted, repaired or replaced where required. Conditions, qualifications, and procedures regarding adjustments, reparations, or replacement of existing equipment as deemed necessary by the Contractor shall be submitted to the Architect for approval. The Contractor shall not proceed with equipment replacement without the

written approval of the Architect. The Architect shall determine whether replacement is required and shall specify the replacement equipment to be used.

C. Additional Requirements

1. For piping, equipment, etc., refer to the appropriate Specification Section for testing requirements and additional requirements of existing equipment and systems to be a part of the new active systems.

3.13 QUALIFICATIONS

A. Only Contractors, Subcontractors, and workmen experienced and regularly engaged in the demolition of mechanical, plumbing, and electrical systems shall be permitted to perform the demolition of existing systems.

END OF SECTION 22 05 30

SECTION 22 05 76 - FACILITY DRAINAGE PIPING CLEANOUTS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions". Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide sanitary and storm drainage piping cleanouts as noted below and as shown on the Drawings.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data
B. Section 22 13 16 - Sanitary Waste and Vent Piping

C. Section 22 40 00 - Plumbing Fixtures

1.4 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Section 22 00 10 and 22 00 70.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:
1. Josam
2. Smith
3. Wade
4. Watts
5. Zurn

2.2 TYPES

A. Cleanouts shall be similar in all respects to the types specified herein and as indicated on the Drawings. The cleanout connection type shall match the piping connection type used for each system and/or location.

1. Unfinished Areas:

a. Zurn Z-1440-BP, Smith 4283, or Watts CO-380 cleanout with bronze plug.

2. Finished Areas:

a. Walls: Zurn ZN-1440-BP, Smith 4435, or Watts CO-380-S7 cleanout with bronze plug, cast duraplate frame and 7" x 7" nickel bronze cover secured with screws.

b. Floor: Zurn ZN-1400-BP "Level-Trol", Smith 4048, or Watts CO-200-S cleanout with bronze plug, square nickel bronze frame and cover. Provide clamping collar when installed in floors having waterproof membrane flashing (Zurn ZN-1400-BP-T-KC, Smith 4048-F-C, or Watts CO-200-S-C).

3. Outside Building:

a. Zurn ZN-1400-BP-VP, Smith 4023-U, or Watts CO-200-R-6 cleanout with bronze plug, nickel bronze cover, non-slip vandal proof cover with anchoring lugs. Where not located in paved areas, these cleanouts shall be set in concrete pad 6" thick, flush with finished grade, 24" square and concrete shall be broom finished on top. Where located in custom paved areas cleanouts shall be mounted as indicated on the Architectural Documents. For areas subject to vehicular traffic, the cleanout shall be a heavy duty style identified by the manufacturer as suitable for the expected wheel loads similar to Zurn ZN-1400-BP-VP-HD or Watts CO-200-RX-4-6.

PART 3 - EXECUTION

3.1 GENERAL

A. Location
1. On vertical piping, cleanouts shall be provided no more than 5'-0" above the base of every riser and stack and no more than 5'-0" above the base of every horizontal offset. On horizontal piping, cleanouts shall be provided at each change in direction greater than 45 degrees, in all horizontal runs more than 40 feet in developed length at a spacing of not more than required by the governing code, and at all other locations as indicated on the drawings. In addition, clean outs shall be provided at the end of each horizontal run of sanitary piping serving water closets and lavatories. These clean outs shall turn upward and terminate above the elevation of the horizontal sanitary piping to prevent water from discharging onto the floor upon removal of the cap. For horizontal piping, cleanouts shall be provided at each change in direction, in all horizontal straight runs more than 50 feet long as required by the governing Code but at a spacing of not more than 50 feet apart, and at all other locations as indicated on the drawings. Cleanouts located in public areas areas shall need to be reviewed and approved by the Architect; if it is determined by the Architect that additional cleanouts are required to be provided, these additional cleanouts shall be provided by the Contractors at no additional cost to the Owner, Architect, or Engineer.

B. Style
1. For areas subject to vehicular traffic, the cleanout shall be a heavy duty style identified by the manufacturer as suitable for the expected wheel loads. For finished areas, the cleanout shall have a polished bronze finish, a scoriated cover and the letters "c.o." cast in the cover. For areas to be covered with carpet, the cleanout shall have a smooth polished bronze finished with the letters "c.o." cast in the cover.

C. Access
1. For areas to be covered with carpet, a permanent identifying mark shall be placed in the carpet directly above the cleanout. For walls, an access panel shall be provided as part of the cleanout or as a separate item. The access panel shall be located to provide clearance for rodding. Coordinate the exact location of the access panels with the architect and Owner.

END OF SECTION 22 05 76

SECTION 22 05 80 - CONNECTIONS TO UTILITIES AND UNDERGROUND PIPE INSTALLATION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Work shall include the connection of all applicable building systems to the appropriate utility connection. This shall include but not be limited to domestic cold water, fire water, storm drainage, and sanitary sewer waste drainage. Building piping systems shall be extended from the building to the location designated by the applicable utility. Refer to the Civil Site Documents for additional information. All utility connections shall be in accordance with and governed by the rules and regulations of the local utilities.

B. The Contractor is required to verify the locations and depths of all utility lines on the site and to govern himself accordingly. The Contractor shall make suitable permanent connections to all the utilities.

C. Excavation and backfill.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 11 16 - Domestic Water Piping
B. Section 22 13 16 - Sanitary Waste and Vent Piping
C.

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 SANITARY SEWER CONNECTIONS

A. Provide sanitary sewer outlets from the building and connect to sanitary sewers as shown on the Drawings. Sanitary sewer materials shall be as specified in Section 22 13 16.

B. The Contractor shall obtain and pay all connection fees, tap charges, if any, assessed by Local Authorities. The Owner shall reimburse the Contractor for all connection fees, tap charges, if any, assessed by Local Authorities.

C. .

3.2 WATER SERVICE CONNECTIONS

A. The water service to the building shall be furnished and installed as shown on the Drawings.

B. Water service materials shall be as specified in Section 22 11 16.

C. The Contractor shall obtain and pay all connection fees, tap charges, if any, assessed by Local Authorities. The Owner shall reimburse the Contractor for all connection fees, tap charges, if

any, assessed by Local Authorities.

3.3 UNDERGROUND PIPE TRENCH EXCAVATION, INSTALLATION, AND BACKFILL

A. All underground pipes shall be laid open in trench. All field measurements, layouts, batterboard alignment, grade establishments, etc., shall be done by a licensed contractor in accordance with the Local Jurisdiction requirements. The Architect will provide a "Bench Mark" reference for use by the Contractor.

B. Pipes shall be laid and maintained at required lines and grades during the course of the work to comply with the Drawings. All joints shall be aligned and complete.

C. The trench shall be excavated to alignment and depth as required. Trench shall be properly braced and dewatered. The trenches shall be kept free of water at all times during installation, testing of pipe, and backfilling. The Contractor shall provide proper facilities for discharging water into natural drainage channels. No water shall be discharged onto the street or freeway without approval by the Architect.

D. The trench shall be at least 18" wider than the diameter of largest bell on the pipe. If the pipe has no bells, the trench shall be at least 18" wider than the maximum diameter of the pipe. Pipe shall be laid in the center of the trench. The trench shall be excavated to a depth sufficient to provide for pipe cushions as specified herein.

E. Sheet pipe and brace excavations wherever necessary to prevent cave in. Trench width may be increased as required and piling left in place until sufficient compacted backfill is in place. The Contractor shall properly sheet and brace all open trenches to render them secure and shall remove all such sheeting and bracing before completing the backfill. The Contractor shall comply with local regulations or, in the absence thereof, with the "Manual of Accident Prevention in Construction" of the Associated General Contractors of America, Inc. The quantity of excavation required to install sheeting and the installation and removal of sheetings and bracings will not be regarded as extra work and all cost incurred for this excavation and the installation of sheeting shall be included in the Contract price.

F. Trenching may be done with trench digging machinery unless hand digging is required to avoid damage to existing structures or apparatus both above and below grade.

G. Upon completion of excavation and prior to the laying of the pipe, the trench bottom shall be brought up to the required elevation with a pipe cushion, except where the cushion has been eliminated by the Architect. Pipe cushions shall be select material deposited in the trench, and shall be compacted, leveled off, and shaped to obtain a smooth compacted bed along the laying length of the pipe. Material for pipe cushion shall comply with Local Codes. In absence of Local Code requirements the cushion shall be bank sand or select backfill material approved by the Architect. Any material used shall pass a one inch screen.

H. Pipes shall not be laid in water or when trench condition is unsuitable.

I. As sections of pipe are installed, they shall be temporarily sealed until remaining sections of the piping have been installed to complete the system.

J. Trenches shall be backfilled only after piping has been inspected, tested, and approved by the Architect. All backfill material shall be placed in the trench either by hand or by approved mechanical methods. The compaction of backfill material shall be accompanied by tamping, with hand tools or approved pneumatic tampers, by using vibratory compactors, by puddling, or by any combination of the three. The method of compaction shall be approved and all

compaction shall be done to the satisfaction of the Architect. Backfill completely around pipe, including 18" above the pipe, with suitable bank sand, tamped in 4" layers under, around, and over pipe. Water down backfill as required. The remainder of the backfill for all pipes shall be select backfill material tamped at intervals of no more than 12" depths, to attain Proctor Compaction Density specified in Section 22 00 10. All materials to be used as selected material backfill shall be approved by the Architect. If in the opinion of the Architect, the excavated material does not meet the requirements of selected material, the Contractor shall be required to screen the material prior to its use as selected material backfill. Material used in the upper portion of the backfill or subgrade shall not contain stone, rock, or other material larger than six inches in its longest dimension. No wood, vegetable matter, or other material which, in the opinion of the Architect, is unsuitable shall be included in the backfill. The upper 24" of backfill may be water jetted, if desired. Backfill shall be brought up to finish grade identified on the Architectural Drawings, including additional backfill required to offset settlement during consolidation.

K. When removal of unsuitable excavated material creates a shortage of backfill material, the Contractor shall, at no cost to the Owner, furnish material as specified in this Section in the amount required to complete the backfill.

L. Existing streets, driveways, and sidewalks damaged during the excavation work shall be restored to acceptable condition, subject to approval by the Architect.

M. Provide all street and sidewalk excavations with approved barricades, warning lights, and cover plates as required by Local Ordinances.

N. Submit complete installation and test procedures of the excavation and backfill work for approval by Architect prior to commencing any work.

O. Refer to guidelines and published regulations of the local code authority for additional requirements.

END OF SECTION 22 05 80

SECTION 22 05 93 - TESTING, ADJUSTING, AND BALANCING FOR PLUMBING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 SCOPE OF WORK

A. Provide all testing, balancing, and adjusting of plumbing systems as indicated on the Drawings and as specified herein. Tests shall include but not be limited to the items specified in this Section.

B. Testing, balancing, and adjusting shall in no way relieve the Contractor of the warranty requirements.

C. The Contractor shall furnish all fuel, water, and electricity required in performing the testing, balancing, and adjusting of plumbing systems.

1.3 CONTRACTOR REQUIREMENTS

A. The Air and Water Balancing Subcontractor shall work directly for the General Contractor. The Air and Water Balancing Subcontractor shall be separate and independent from the Division 22 Plumbing Contractor.

B. The air and water balancing subcontracting company shall have a minimum of five years experience in the testing, balancing and adjusting of commercial air conditioning systems.

C. The Air and Water Balancing Subcontractor and the personnel used for this Project shall be certified testing agents of the National Environmental Balancing Bureau (NEBB) or the Associated Air Balance Council (AABC).

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Testing

1. During the progress of the work, tests shall be made as specified herein and as required by authorities having jurisdiction, including Local Authorities Inspection Department, Owner, Architect, or Engineer. Tests shall be conducted by the Plumbing Contractor as part of the work of this Division and shall include all qualified personnel, equipment apparatus, and services required to perform the tests.

B. Job Site Reviews

1. The Testing, Balancing and Adjusting Subcontractor shall visit the site at intervals appropriate to the stage of construction to familiarize himself with the progress and quality of work. This Subcontractor shall report directly to the General Contractor in

written reports indicating any deficiencies, potential problems areas and conditions that could adversely affect the building systems and the testing, balancing, adjusting procedures.

C. Testing Procedures and Recording Forms

1. The Contractor shall submit proposed test procedures, recording forms, and test equipment for review prior to the execution of testing.

D. Repairs

1. Leaks, damage, or defects discovered or resulting from tests shall be repaired, or replaced to a like new condition. Leaky pipe joints, etc., shall be removed and replaced with acceptable materials.

E. Equipment

1. All equipment and instruments required for tests as well as additional thermometer wells, gauge and instrument connections shall be installed at no additional cost to the Owner.

F. Instrument Calibration

1. All instruments used for testing and balancing must have been calibrated within a period of six months prior to balancing. Instrument calibration shall be certified as specified in Section 22 00 10.

G. Report

1. Submit six copies of each complete testing and balancing report to the Engineer for review.

H. Acoustic Performance

1. The final results of all testing, adjusting and balancing shall achieve the range of indoor design goals for sound control for all occupied spaces as indicated in section 23 00 15. Refer to section 22 00 10 “Equipment Noise and Vibration” for additional requirements.

I. Equipment Identification

1. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

## 3.2 TESTING, BALANCING, AND ADJUSTING OF DIVISION 22

A. Vibration and Alignment

1. Vibration and alignment field measurements shall be taken when directed by the Engineer on equipment where vibration appears excessive. Readings shall include shaft alignment, equipment vibration, bearing housing vibration, foundation vibration, building

structure vibration, and other tests as directed by the Engineer. Readings will be made using portable IRD (or approved equal) equipment capable of filtering out various unwanted frequencies and standard reporting forms. Maximum vibration at any point listed above or specified shall not exceed 2 mils on pumps unless otherwise specified.

B. Sound Levels

1. Sound level readings shall be taken at ten locations in the building as selected by the Engineer. The readings shall be taken on an octave band analyzer in a manner acceptable to the Engineer. The Contractor shall submit test equipment data and reporting forms to the Engineer for review. In order to reduce the ambient noise level, the readings shall be taken at night. All tests shall be performed in the presence of the Engineer, if Engineer so desires.

C. Air and Water Flow Measurements

1. Water flow measurements shall be made in conjunction with the water balancing procedures as specified in this Section. Water balance procedures shall be performed by National Environmental Balancing Bureau or Associated Air Balance Council certified, independent Testing and Balancing Contractor(s) in accordance with the latest edition of the NEBB "Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems" or the latest edition of the AABC National Standards.

D. Piping Leak Testing

1. Leak tests shall be conducted in accordance with ANSI B31.1 "Power Piping Code" and as specified herein in Section 22 11 16 and 22 13 16 before piping of various systems has been covered or furred in. Note: Equipment not rated at or above the specified test pressure plus all sensitive system controls and instrumentation, excluding control valves, shall be valved off or removed and the connections plugged prior to hydrostatic testing. Removed items shall be replaced after testing has been completed. Controls and instrumentation shall be calibrated and tested after reinstallation.

E. Pumps

1. Pumps shall be tested to check impeller trim and operating characteristics. The pump manufacturer or a certified technician as recommended by the pump manufacturer shall perform a laser or dial indicator alignment of all pumps mounted on inertia bases to ensure the pump and motor are properly aligned and submit certified alignment test reports to the engineer for review. In addition, the following data shall be recorded and submitted to the Engineer for review. See Section 22 00 10 for requirements.

    a. Flow at operating conditions where flow venturi or orifices are installed in the system.
    b. Shut off pressure required to check impeller trim.
    c. Discharge pressure at operating conditions.
    d. Suction pressure at operating conditions.
    e. Motor amperage and voltage on each phase at operating conditions.
    f. Demonstrate operating controls and safety devices on domestic water pumping systems.

F. Field Capacity and Performance Testing

1. The Contractor shall prove the capacity and performance of each piece of equipment by field tests as specified herein in various Sections or as required by the Engineer. All

equipment and instruments required for tests as well as additional thermowells or gauge connections shall be installed at no additional cost to the Owner. A qualified representative of the equipment manufacturer, with credentials submitted in advance, shall be present at the test. These test requirements specifically include the pumps. The Engineer may witness tests, if Engineer so desires. The Contractor shall notify the Engineer, in writing, at least two weeks prior to the day of the test. See Section 22 00 10 for additional requirements.

G. Water Balancing and Adjusting

1. After piping systems have been installed complete with all pumps, piping, valves, coils, and other items as herein specified, the Contractor shall make adjustments as required to deliver the water volumes at each piece of equipment to within 5% of design flow as shown on the Drawings or as indicated in Section 22 06 00. Adjustments in water volumes shall be made in a manner satisfactory to the Engineer. The Contractor shall submit detailed balancing procedure and recording forms for the Engineer's review prior to commencing any water balancing work.

2. Submit six copies of complete comprehensive bound reports of the performance of the systems to the Architect for review and comments. Identify all data completely. Identify the manufacturer, size, type, location, room number, and zone of each piece of equipment reported. Give design and actual water flow. Give complete nameplate data for each piece of equipment reported.

H. Final Water Balance Readings

1. After water flow is balanced, measure and record all data necessary to compile a complete report to demonstrate the acceptability of the various plumbing systems. Include at least the following data in the report:

   a. Temperatures at all thermometer well locations.

   b. Suction and discharge pressure and shut off head of each circulating water pump and domestic water pump.

2. The balanced position of each controlling valve shall be permanently marked on the pipe or insulation.

I. Temperature Control Adjusting

1. Automatic temperature controls shall be calibrated and all valves, etc., adjusted so that control system is in the proper operating condition, subject to review by the Engineer. See Section 25 00 90 for additional requirements.

J. Drainage Piping System Tests

1. Refer to section 22 13 16 and 22 14 13 for additional requirements.

END OF SECTION 22 05 93

SECTION 22 07 00 - PLUMBING INSULATION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the insulation as specified herein for the Division 22 systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data

B. Section 22 11 16 - Domestic Water Piping
C. Section 22 11 19 - Domestic Water Piping Specialties
D. Section 22 13 16 - Sanitary Waste and Vent Piping

1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1. 90.1 Energy Standard for Buildings

B. ASTM International (formerly American Society for Testing and Materials)
1. C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus
2. C335 Standard Test Method for Steady-State Heat Transfer Properties of Pipe Insulation
3. C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
4. C534 Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form
5. C547 Standard Specification for Mineral Fiber Pipe Insulation
6. C585 Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing
7. C921 Standard Practice for Determining the Properties of Jacketing Materials for Thermal Insulation
8. C1136 Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
9. C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings
10. C1393 Standard Specification for Perpendicularly Oriented Mineral Fiber Roll and Sheet Thermal Insulation for Pipes and Tanks
11. D1056 Standard Specification for Flexible Cellular Materials-Sponge or Expanded Rubber
12. E84 Standard Test Method for Surface Burning Characteristics of Building Materials
13. E96 Standard Test Methods for Water Vapor Transmission of Materials
14. E119 Standard Test Methods for Fire Tests of Building Construction and Materials

C. DOE Code of Federal Regulations (CFR)
1. 40 CFR 59 Subpart D - National Volatile Organic Compound Emission Standards for

Architectural Coatings

D. North American Insulation Manufacturers Association (NAIMA)
   1. CI228 Guide to Insulating Chilled Water Piping Systems with Mineral Fiber Pipe Insulation

E. South Coast Air Quality Management District (SCAQMD)
   1. Rule 1168 Adhesive and Sealant Applications

F. Underwriters Laboratories (UL) Standard for Safety
   1. 723 Test Method for Surface Burning Characteristics of Building Materials
   2. 2818 GREENGUARD Certification Program for Chemical Emissions for Building Materials, Finishes and Furnishings

1.5 SUBMITTALS

A. Product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content.

1.6 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Sections 22 00 10 and 22 00 70.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Glass Fiber Material: The insulation shall have the same thermal properties, density, fire rating, vapor barrier, etc., as the types specified herein, subject to review by the Engineer. If it complies with these Specifications, insulating materials manufactured by one of the following manufacturers will be acceptable:

   1. Johns Manville
   2. Owens Corning
   3. CertainTeed
   4. Knauf
   5. Manson

B. Closed Cell Elastomeric Material: If it complies with these Specifications, insulating materials manufactured by one of the following manufacturers will be acceptable:

   1. Armacell
   2. Rubatex/K-Flex
   3. Aeroflex

C. Adhesives and mastics shall be as manufactured by MEI Inc., Arabol, Childers, Foster, Armacell, 3M or Insulmastic, Inc., and shall have the same adhesive properties, fire rating, vapor seal, etc., as the types specified herein, subject to review by the Engineer.

2.2 GENERAL

A. All insulation required for the work shall be new, of first class quality and shall be furnished, delivered, erected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class standard article as approved by the Architect shall be provided.

B. Pressure Sensitive Tape and Release Paper Systems: Unless otherwise noted, as in the application of piping insulation systems, the use of release paper systems, pressure sensitive tape systems, and peel & stick systems will not be acceptable and shall not be used by the Contractor. The use of release paper systems, pressure sensitive tape systems, and peel & stick systems shall be limited to piping insulation systems. Refer to the specifications herein for additional requirements.

2.3 FIRE AND SMOKE HAZARD RATING

A. All insulations (composite insulation, jacket or facing, tape, coating and adhesive) installed within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723 as required by applicable mechanical code. Accessories such as coatings, tapes, and adhesives shall have the same component ratings.

B. All insulations (composite insulation, jacket or facing, tape, coating and adhesive) installed within non-plenums shall be noncombustible or shall be tested and rated as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723. Accessories such as coatings, tapes, and adhesives shall have the same component ratings.

2.4 CONCEALED

A. The word "Concealed" as used in this Specification refers to insulation in furred spaces, pipe and duct shafts, unheated spaces immediately below roof, unexcavated spaces, crawl spaces, and tunnels. The word "exposed" refers to insulation in all other areas.

2.5 CORROSION PROTECTION

A. All insulation materials, adhesives and attachment devices used in direct contact with steel, copper, aluminum, galvanized steel, and other materials required by this Contract shall not cause corrosive effects. Materials to be used on stainless steel surfaces shall comply with ASTM C795 requirements. This non-corrosive requirement applies to all insulation systems for all operational conditions as installed on this Project. These requirements apply throughout the entire operational temperature range and moisture condition of each system.

2.6 PIPING SYSTEMS

A. Piping shall be insulated with the insulation materials and thickness as indicated herein.

B. Domestic cold water piping systems:

1. Heavy duty glass fiber material with non-combustible vapor barrier all service jacket (ASJ) shall comply with ASTM C547, C921 and C585. Insulation shall be Johns Manville "Micro-Lok", Knauf “Earthwool 1000° Pipe Insulation”, Owens Corning "SSL II with ASJ Max” or approved equal. Thermal conductivity (K) value shall be no more than 0.23 (btu-

in)/(hr-sf-°F) at 75°F mean temperature when tested by ASTM C335 and water permeability of no more than 0.02 perm-inch when tested by ASTM E96 (Procedure A). The insulation shall be UL Greenguard Gold Certified and listed under UL 2818. The insulation thickness shall be as follows:

| Nominal Pipe Size | Insulation Thickness |
|---|---|
| All Pipe Sizes | 1/2" |

C. Domestic hot water piping systems:

1. Same as domestic cold water piping systems except as follows:

a. For domestic hot water systems 140°F and below, thermal conductivity (K) value shall be no more than 0.24 (btu-in)/(hr-sf-°F) at 100°F mean temperature. The insulation thickness shall be as follows:

| Nominal Pipe Size | Insulation Thickness |
|---|---|
| Up to 1-1/4" | 1.0" |
| 1-1/2" and Larger | 1.5" |

b. For domestic hot water systems above 140°F, thermal conductivity (K) value shall be no more than 0.25 (btu-in)/(hr-sf-°F) at 125°F mean temperature. The insulation thickness shall be as follows:

| Nominal Pipe Size | Insulation Thickness |
|---|---|
| Up to 1-1/4" | 1.5" |
| 1-1/2" and Larger | 2.0" |

2. .

D. Elastomeric closed cell pipe insulation option:

1. Elastomeric insulation will be acceptable instead of glass fiber insulation for the following systems:

a. Domestic cold water piping
b. Domestic hot water and recirculation water piping
c. Domestic refrigerated drinking water piping
d. Waste lines from drinking water fountains using refrigerated water
e. Storm water piping

2. Plenum Application: Flexible elastomeric closed cell pipe insulation shall comply with ASTM C534 and shall be Armacell Armaflex Ultra or approved equal for installation within air plenums in accordance with the fire hazard ratings specified herein. Insulation shall have a thermal conductivity (K) of no more than 0.27 (btu-in)/(hr-sf-°F) at 75°F mean temperature when tested by ASTM C177 or C518.

3. Non-Plenum Application: Flexible elastomeric closed cell pipe insulation shall comply with ASTM C534 and shall be Armacell Proflex or approved equal for installation within

non-plenums in accordance with the fire hazard ratings specified herein. Insulation shall have a thermal conductivity (K) of no more than 0.27 (btu-in)/(hr-sf-°F) at 75°F mean temperature when tested by ASTM C177 or C518.

4. Insulation shall have a water permeability of 0.05 perm-inch or less when tested by ASTM E96, Procedure A.

5. Insulation shall be rated for upper and lower temperature limits of 220°F and -40°F when tested by ASTM C534.

6. Insulation adhesive shall be Armaflex BLV 520 or approved equal. Adhesive shall be specifically formulated to bond elastomeric insulation.

2.7 EQUIPMENT

A. Domestic water heaters:

1. Water heaters shall be factory insulated.

2.8 JACKETS AND FACINGS

A. Vapor barrier jackets and facings shall comply with ASTM C921 and C1136. Jackets and facings shall have a permeance rating not to exceed 0.02 perm-inch when tested by ASTM E96 (Procedure A), puncture resistance not less than 50 units, and tensile strength not less than 35 pounds per inch width when tested by ASTM C921.

B. Jackets and facings on insulation in exposed areas shall have a white finish suitable for painting without priming.

2.9 ADHESIVE, MASTIC AND SEALANT VOC REQUIREMENTS

A. For indoor applications, use adhesives, mastics and sealants that have a maximum volatile organic compound (VOC) content as listed below when calculated in accordance with 40 CFR 59, Subpart D (EPA Method 24).

1. Adhesives:
   a. Cellular Glass Adhesive: 50 grams/liter
   b. Flexible Elastomeric Adhesive: 50 grams/liter
   c.
   d. ASJ and FSK Jacket Adhesive: 50 grams/liter
   e. PVC Jacket Adhesive: 50 grams/liter
2. Mastics: 80 grams/liter

3. Sealants: 250 grams/liter

PART 3 - EXECUTION

3.1 APPLICATION

A. General

1. The execution of the insulation work shall be in strict accordance with the best practices of the trade, the listed Reference Standards, applicable local codes.

2. Unless otherwise specified, the application of all insulation materials shall be in accordance with the manufacturer's published recommendations. Where these Specifications are in conflict with manufacturer's instructions, such conflicts shall be brought to the attention of the Engineer for his decision.

3. Insulation shall not be applied until all surfaces are clean and dry. The insulation on pipe fittings, valves and pipe joints shall not be installed before the piping is tested and reviewed by the Engineer.

4. The insulation shall be handled and applied in a manner that will not adversely affect its structural or insulating properties.

5. A complete moisture and vapor seal shall be provided over insulation on cold surfaces where vapor barrier jackets, facings, or coatings are required. Anchors, hangers and other projections shall be insulated, and vapor sealed to prevent condensation. All openings and punctures shall be sealed with vapor barrier compound. The use of staples will not be acceptable.

6. Insulation shall be continuous through hangers and wall and floor openings.

7. Date of expiration for product installation shall not be exceeded. Products using release paper systems and pressure sensitive tapes shall not be older than one year from date of manufacturing at time of installation.

8. Temperature limitations for the application of products and systems shall not be exceeded.

9. Insulation shall be applied to strainers in such a manner as to allow removal of the end cap and perforated screen without destroying the strainer casing insulation.

B. Piping

1. Domestic cold, hot and recirculating water piping systems:

a. All above ground pipe, valves, fittings, flanges, etc., handling domestic cold, hot and recirculating water piping systems shall be insulated with glass fiber pipe insulation.

b. Prefabricated sectional insulation for straight pipes shall fit respective type of pipe indicated on drawings. It shall be applied, neatly fitted around piping, and sealed with adhesive. Adhesive shall be applied to only one side of each joint and shall not be applied to pipe surface.

c. A protection shield will be provided at each hanger. High density molded blocking or approved equal shall be installed between the pipe and the protection shield to prevent crushing of the insulation. Insulation blocking shall be not less than the same length and circumference as the pipe protection shield. The Insulation Contractor shall thoroughly vapor seal blocking material and finish to match adjoining pipe insulation.

d. Fittings, flanges, valves, and specialties shall be insulated with preformed or shop fabricated glass fiber covers the same thickness as the adjoining pipe insulation. Insulation shall be thoroughly vapor sealed with Foster 30 80 fire resistant vapor barrier mastic coating. Vapor barrier mastic shall be applied reinforced with 10 x

10 glass fabric.

e. Unless otherwise noted, if approved by the Local Authorities having jurisdiction, plastic 25/50 insulated fitting covers as manufactured by Johns Manville or approved equal may be used on piping systems. Interior fill of fitting covers shall be in strict accordance with the manufacturer's recommendations. Engineer reserve the right to have the contractor remove and repair several of the insulation covers at random as selected by the Engineer to inspect insulation fill. The joint between insulated fitting cover and pipe insulation shall be secured with 3" wide butt strip of vapor barrier jacket adhered neatly in place with approved vapor barrier adhesive. Insulated fitting covers shall not be used within main chilled water central plant area within 8'-0" of the plant floor. Fabricated glass fiber covers as specified hereinbefore shall be used in the central plant area.

f. Joints shall be sealed with mastic. Where required, oversized pipe sections of board type insulation may be used to fabricate and install insulation around pipe specialties.

g. Where piping is interrupted by fittings, flanges, valves, or hangers; at intervals not exceeding 21 feet on continuous runs; and at each vertical pipe slab penetration, an isolating seal shall be formed between the vapor barrier jacket and the bare pipe by liberal applications of Foster 30-80 flexible vapor barrier joint sealant to the ends of the pipe insulation. Identification of piping insulation isolating seal shall be labeled in accordance with Specification Section 22 05 53.

h. Longitudinal laps and 3" wide butt strips of the vapor barrier jacket shall be adhered neatly in place with Johns Manville PSS or Foster 85 75 fire resistant vapor barrier adhesive.

i. Sectional insulation shall be secured with A. J. Gerrard & Company "Steel Binder" 0.02" thick by 1/2" wide aluminum bands or approved equal, or Thomas & Betts "TY RAP" nylon ties, or approved equal, on 24" centers.

j. Supply pipes serving lavatories from angle stop to fixture connection do not require insulation.

k. Domestic hot water system shall use Foster 30-36 adhesive mastic in lieu of Fosters 30-80 adhesive mastic.

2. Air handling unit condensate and cooling tower drain piping:

a. Air handling unit condensate drain lines located above a ceiling or within a wall cavity or chase shall be insulated in the same way as domestic cold, hot and recirculating water piping. Condensate drain lines from floor mounted units located in return air plenums do not require insulation unless required by Local Codes. Cooling tower drain lines from the tower to the isolation valve shall be insulated in the same way as domestic cold, hot and recirculating water piping.

3. Elastomeric closed cell pipe insulation option:

a. The pipe insulation seams and joints shall be sealed with Armacell adhesive. The entire Armaflex insulation installation shall be in accordance with application

procedures and recommendations as described in Armacell's latest Armaflex Application Manual.

C. Equipment

1. Domestic water heaters:

a. Water heaters shall be factory insulated.

3.2 CLEANING

A. All of the above items shall be cleaned and all dust and debris removed.

END OF SECTION 22 07 00

SECTION 22 11 16 - DOMESTIC WATER PIPING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the piping systems as shown on the Drawings and as specified herein for the various systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 10 - Wall and Floor Penetrations
B. Section 22 00 15 - Division 22 Design Conditions
C. Section 22 00 70 - Division 22 Schedule of Submittal Data

D. Section 22 05 53 - Identification for Plumbing Piping and Equipment
E. Section 22 05 93 - Testing, Adjusting, and Balancing for Plumbing
F. Section 22 07 00 - Plumbing Insulation

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
1. Boiler and Pressure Vessel Code Section IX - Welding and Brazing Qualifications
2. A1.20.1 Pipe Threads, General Purpose, Inch
3. B16.1 Cast Iron Pipe Flanges and Flanged Fittings
4. B16.3 Malleable Iron Threaded Fittings Classes 150 and 300
5. B16.5 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
6. B16.9 Factory-Made Wrought Buttwelding Fittings
7. B16.11 Forged Fittings, Socket-Welding and Threaded
8. B16.15 Cast Copper Alloy Threaded Fittings: Classes 125 and 250
9. B16.18 Cast Copper Alloy Solder Joint Pressure Fittings
10. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
11. B31.9 Building Services Piping
12. B36.10M Welded and Seamless Wrought Steel Pipe
13. B36.19M Stainless Steel Pipe

B. ASTM International (formerly American Society for Testing and Materials)
1. A53/A53M Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
2. A135 Standard Specification for Electric-Resistance-Welded Steel Pipe
3. A234 Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
4. B32 Standard Specification for Solder Metal
5. B88 Standard Specification for Seamless Copper Water Tube
6. B306 Standard Specification for Copper Drainage Tube
7. B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube

8. F1476 Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications

C. American Welding Society (AWS)
1. A5.8 Specifications for Filler Metals for Brazing and Braze Welding
2. B2.1 Specification for Welding Procedure and Performance Qualification

D. American Water Works Association (AWWA)
1. C104/A21.4 Cement-Mortar Lining for Ductile-Iron Pipe and Fittings
2. C111/A21.11 Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
3. C151/A21.51 Ductile-Iron Pipe, Centrifugally Cast

E. Copper Development Association (CDA)
1. A4015 Copper Tube Handbook

F. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
1. SP-58 Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation
2. SP-69 Pipe Hangers and Supports - Selection and Application
3. SP-89 Pipe Hangers and Supports - Fabrication and Installation Practices

G. NSF International (formerly National Sanitation Foundation)
1. 61 Drinking Water System Components - Health Effects
2. 372 Drinking Water System Components - Lead Content

1.5 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all specification requirements listed herein plus certification data and shop drawing requirements listed in Section 22 00 10 and 22 00 70.

B. The Contractor shall provide, where applicable, product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content. Submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each field applied piping adhesive and sealant used within the building interior locations.

PART 2 - PRODUCTS

2.1 PIPING

A. All materials and specialties required for the work shall be new, of first class quality and shall be furnished, delivered, erected, connected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class standard article as approved by the Architect shall be provided.

B. All materials and specialties required for this project shall be the products of United States made materials. International or foreign made materials and specialties shall not be acceptable unless pre-approved by the Owner prior to bidding. If the Owner pre-approves the use of international or foreign made materials and specialties prior to bidding, the Contractor shall certify in writing to the Engineer that all international or foreign made materials and specialties furnished for this project are suitable for the service intended and complies with all the requirements specified herein. This certification shall be included within the submittal being issued to the Engineer for review and approval. In addition, the Contractor shall submit a 12” long sample of each type of international or foreign manufactured pipe material to be installed on this project to a certified metallurgy laboratory for complete properties and strength analysis.

The samples shall be cut from pipe material stored on this projects site or in the Contractor's fabrication facilities. The pipe material shall be identified as to its intended use, ASTM designation, project name, Contractor's name, and date the sample was obtained. If the pipe is coated, such as with a galvanized material, the thickness and properties of the coating material shall be indicated. A certified report of the test data shall be provided to the Owner and the Engineer for review and approval prior to the final acceptance and installation of any international or foreign made material and specialty. The cost of the certified laboratory testing shall be at the Contractors expense and the certified laboratory shall be subject to approval by the Owner and Engineer prior to testing. Any international or foreign made material and specialty that does not comply with the material standards and all other requirements specified herein shall be removed. Prior to purchase of any materials manufactured outside the United States, the Contractor shall submit complete certifications and typical mill reports in English language for review. Any foreign pipe delivered to the jobsite shall bear mill heat markings and the Contractor shall make available the corresponding mill test reports.

C. Connections between dissimilar piping materials shall be made with suitable dielectric insulating pipe fittings and unions.

## 2.2 DRAIN PIPING

A. Unless otherwise noted on the Drawings, drain piping from each air handling unit drain pan, strainer, water chilling unit, heat exchanger, pump base drain, tank and vessel overflow, auxiliary drain pan, piping system drain, blowdown, and any other system or equipment that has a drain connection shall be Schedule 40 PVC piping and fittings where allowed by code, Type K or DWV copper or cast iron. At the contractor's option, copper press-connect fittings may be used for drain piping 4" and smaller as specified herein below for Domestic Hot and Cold Water Piping.

## 2.3 DOMESTIC HOT AND COLD WATER PIPING

A. Pipe, valves, fittings, and materials shall conform to the working pressure requirements indicated in Section 22 00 15. All piping, specialties, and accessories including but not limited to fittings, gaskets, sealants, adhesives, coatings, lubricants, etc… which are installed directly in contact with potable water shall be certified and labeled in accordance with NSF 61 and NSF 372 standards.

B. Copper pipe shall be Class 150 psig working pressure, ASTM B88, Type L, UNS C12200, H58 drawn temper seamless copper water tube. Copper tube shall be manufactured in the United States by Mueller Streamline, Cerro, Cambridge-Lee, Howell Metal, Wieland or approved equal.

C. All pipe underground or under the concrete slab inside the building shall be Mueller Streamline, Cerro, Cambridge-Lee, Howell Metal, Wieland or approved equal, ASTM B88 Type K, UNS C12200, O60 soft anneal temper copper coil without fittings.

D. Fitting material shall be as follows:

1. Copper tube fittings shall be Mueller Streamline, NIBCO, Elkhart, or approved equal, wrought copper joint fittings conforming to the latest edition of ANSI B16.22. Bent 90° and 45° elbows using tubing bender on 3/4" copper pipe and below will be acceptable in lieu of elbow fittings.

2. Mechanically formed tee connections

a. At the Contractor's option mechanically extracted collars for tee connections shall be formed in continuous operation consisting of drilling a pilot hole and drawing out

the tube surface to form a collar having a height of not less than three times the thickness of the tube wall. The collaring device shall be fully adjustable as to insure proper tolerance and complete uniformity of the joint. The branch shall be notched to conform with the inner curve of the run tube and dimpled to ensure penetration of the branch tube into the collar is of sufficient depth for brazing and that the branch tube does not obstruct the flow in the main line tube.

b. All joints shall be brazed or soldered in accordance with the Copper Development Association Copper Tube Handbook and in accordance with these specifications.

c. All mechanically formed branch collars and joint system shall comply with International Plumbing Code, International Mechanical Code, National Standard Plumbing Code, V.A., and NFPA 13. The tee joint system shall also meet the applicable published standards of ANSI, ASME and the American Welding Society.

d. If it complies with these Specifications mechanically formed tee connections shall be provided by factory trained representatives of T-Drill Pipe Fabrication Systems or approved equal. All tools used to form the tees, notches and dimples shall be provided by T-Drill Pipe Fabrication Systems.

3. Mechanically formed couplings and end closures
   a. At the Contractor's option, couplings and end closures may be mechanically formed using a tube end spinning machine, which feeds a rotating tool against a stationary tube to close, reduce, or expand the end of the tube. Mechanically formed couplings shall be formed by expanding one tube end to form a solder-joint fitting cup using a tool specifically for this purpose in accordance with the tool manufacturer’s standard written procedure. For solder-joint connections, the formed fitting cup shall meet or exceed the dimensional requirements of ASME B16.22. For brazed joint connection, fitting cup depth shall be at least three times the thickness of the tube. Solder filler and brazing filler metals shall conform to the requirements of Article 2.4 “Joining Materials” from the Copper Development Association’s Guide Specification for Copper and Copper Alloy Building Piping Systems (for brazing, BCup Series copper phosphorus alloys, BCuP-2 through 7).

   b. All joints shall be brazed or soldered in accordance with the Copper Development Association Copper Tube Handbook and in accordance with these specifications.

   c. If it complies with these Specifications mechanically formed couplings and end closures shall be provided by factory trained representatives of T-Drill Pipe Fabrication Systems or approved equal. All tools used to close, reduce, and expand tube ends shall be provided by T-Drill Pipe Fabrication Systems.

E. All pipe joint materials shall be as follows:

1. Soldered Copper Joints: All fittings shall be soldered with lead free ASTM B32 Grade 95T tin-antimony solder, tin-bismuth-copper-silver solder, or brazed with AWS classification BCup-3 brazing alloy as required by the system temperature and working pressure. All solder and brazing materials shall meet the requirements of the Safe Drinking Water Act.

2. Flanged Fittings: At each flanged joint, flanges shall have matching raised or flat faces. Gasket material shall be suitable for the service intended and shall be installed as recommended by the Manufacturer. Gaskets manufactured by Crane, Garlock, or Schuller will be acceptable if they comply with these Specifications.

3. Copper Press-Connect Fittings: At the contractor's option, copper press-connect fittings may be used which conform to the material and sizing requirements of ASME B16.18 or ASME B16.22. All lead free wrought copper press fittings shall be made from commercially pure copper mill products per ASTM B75 Alloy C12200. These fittings shall be third-party certified to NSF 61 Annex G and comply with Section 116875 of the California Health and Safety Code and Vermont Act 193. Lead Free cast dezincification-resistant (DZR) fittings shall be made from a high quality lead free performance bronze alloy per ASTM B584 Alloy C87850. The press fittings connections shall be compatible with seamless K, L or M copper tube made to ASTM B88. Fittings shall have a maximum non-shock working pressure of 200 PSI between the temperatures of -20°F and +250°F. O-rings for the copper press-connect fittings shall be EPDM with leak detection design. If it complies with these specifications, copper press-connect fittings shall be manufactured by NIBCO, ApolloPress, Viega or approved equal which have an inboard bead design. Copper press-connect fittings shall be installed in accordance with the manufacturer's recommendations. Installing contractor(s) shall be required to be trained, and shall receive proof of training (certification or credential) by the manufacturer's representative prior to installing the press-connect fittings. The piping joints shall be crimped using pressing equipment comprised of a pressing tool, jaws, crimp sling, and crimp ring actuator approved by the manufacturer's representative. The jaw set pressing equipment shall be maintained within the calibration specifications in accordance with the manufacturer's recommendations. Jaws and crimp sling shall be dry brushed frequently to prevent accumulation of copper shavings. The piping shall be marked at the full required insertion depth using the insertion depth gauge provided by the manufacturer to confirm that the pipe is fully inserted into the fitting prior to final crimping. The fitting alignment shall be checked against the mark on the piping to ensure the piping is fully inserted into the fitting prior to crimping. The Contractors shall confirm that the positioning and alignments are correct and that the manufacturer's recommended installation guidelines have been followed prior to crimping of any pipe fitting. For press-connect fittings 2" and smaller, the jaws shall be aligned such that the O-ring bead on the fitting is set into the groove of the jaws prior to using the pressing tool for crimping. For press-connect fittings 2-1/2" through 4", the crimping sling shall be set to rest in the corresponding area on the fitting indicated for alignment by the manufacturer prior to attachment of the crimp ring actuator fitting and use of the pressing tool. The press-to-connect joint shall be made with pressing tools and jaw sets recommended and authorized by the manufacturer. All fittings, valves and tools shall be provided by same manufacturer

F. Provide piping of proper size as indicated on the Drawings, and as required for hot and/or cold water to all sinks, lavatories, toilets, urinals, etc. Each fixture shall be provided with a shut off valve for each supply line. All exposed pipe work in toilet rooms shall be chromium plated. Provide vacuum breakers as required by Local Code.

## 2.4 PIPE HANGERS AND SUPPORTS

A. All pipes throughout the building shall be thoroughly and substantially supported with UL listed hangers and supports. In addition to the items specified herein, furnish and install any special hangers or supports that may be required by the building construction. The design, selection, spacing, and application of horizontal pipe hangers, supports, restraints, anchors, and guides shall be in accordance with the Standard Code for Pressure Piping ANSI B31.1, the latest edition of MSS SP 58, and the applicable mechanical or plumbing code, whichever is most stringent. In no case shall piping be installed that is not in compliance with the applicable code.

B. Unless otherwise noted all horizontal piping shall be supported on National Pipe Hanger Corporation (NPHC) Figure 200, Carpenter & Paterson Figure 100, or approved equal adjustable wrought clevis hangers.

C. All vertical pipes not specified to be supported on vibration isolation shall be supported as follows:

1. The location and spacing of the vertical pipe supports specified above shall be in accordance with the manufacturer's recommendations. If supporting vertical pipe risers at every other floor is not adequate as determined by the manufacturer, provide vertical pipe supports at every floor. If supporting vertical pipe risers at every floor is not adequate as determined by the manufacturer, the manufacturer shall be consulted to provide alternate vertical pipe supports as necessary to adequately support the vertical pipe risers.

D. All hangers, hardware, rods, clamps, channels, base plates, and other factory-built pipe support devices shall be galvanized. All shop fabricated and welded steel supports shall be hot dipped galvanized after fabrication.

E. Refer to Section 22 00 10 "Equipment Mounting/Support Devices" for additional requirements.

## 2.5 PIPE COVERING PROTECTION SADDLES AND SHIELDS

A. Protection shield length and gauge thickness for use with clevis hangers shall be as specified for Type 40 protection shields in the current edition of MSS SP 58. Protection shields shall be galvanized and shall be arranged to cover one half of the circumference of the insulations and shall be mounted on the outside of the insulation with insulation blocking between the pipe and saddle to prevent crushing of insulation. Blocking on piping shall be furnished and installed as specified in Section 22 07 00.

## 2.6 ACCESSORIES

A. Shock Arresters

1. Furnish and install shock arresters conforming to the requirements of the Plumbing and Drainage Institute. The shock arresters shall be located immediately downstream of the service valve on all domestic water risers, upstream of all quick opening or quick closing flow regulating valves and other places as indicated on the Drawings. Shock arresters shall be as manufactured by Precision Plumbing Products, Watts, or approved equal and shall have the Plumbing and Drainage Institute ratings as identified on the Drawings.

B. Gauges

1. Gauges shall be bourdon tube with minimum 4-1/2" dial and die cast aluminum case with black enamel finish. The movement shall be all stainless steel with Grade A phosphor bronze bourdon tube brazed at socket and tip. The accuracy of the gauge shall be within 1/2 percent of the scale range. The pointer shall be the micrometer adjustment type recalibrated from the front. Pressure, compound, and differential pressure gauges shall have suitable scale ranges.

2. Gauges shall have 1/4" IPS connections and shall be Moeller "Vantage" gauges with Case Style No. 2, or approved equal.

3. Gauges shall be furnished complete with gauge test cocks.

4. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable: Ashcroft, Marsh, Trerice, Moeller, Weksler, Taylor, Weiss, Miljoco, or Midwest.

C. Gauge Test Cocks

1. Gauge test cocks shall be Jenkins Model 901 T, NIBCO 585 70, or Apollo 77CLF-A ball valves.

D. Thermometers

1. Thermometers shall be adjustable angle blue reading organic or spirit filled type having scale length of not less than 9", and scale divisions of 2°F, or less similar and approved equal to Moeller Instrument Company, Inc., Style AJ. Range shall be as specified or as required for the duty.

2. All thermometer wells shall be constructed of brass or stainless steel and where installed in insulated piping shall have at least 2 1/2" lagging extension. All wells shall be complete with caps and chains.

3. Thermometers and thermometer wells shall have the following insertion lengths.

a. Located in Horizontal Piping and Vertical Piping Below 6'-0":
1) 4" and 5" Pipe: 2-1/2"
2) 6" and 8" Pipe: 5"
3) 10" and Larger Pipe: 8"

b. Located in Vertical Piping Above 6'-0":
1) 3", 4" and 6" Pipe: 5"
2) 8" and 10" Pipe: 8"
3) 12" and Larger: 14"

4. Thermometers shall have the temperature ranges listed herein and shall be installed where indicated on the Drawings or specified elsewhere.

5. Thermometers and wells must be of at least the quality and design specified. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable: Moeller, Trerice, Miljoco, or Weksler. Submit a list detailing design and construction of thermometers for review by the Engineer.

E. Pressure and Temperature Test Plugs

1. Test plugs shall be Pete's Plug, 1/2" NPT, brass with Nordel core, Model 710. In addition, the Contractor shall supply the Owner with four pressure gauge adapters with 1/8" O.D. probe and four 5" stem pocket testing thermometers: Two -25-120°F, and two 0-220°F. Equivalent products by Sisco P/T Plugs or Miljoco PTB will be acceptable.

F. Flexible Pipe Connectors

1. Refer to Section 22 05 48 for additional requirements.

G. Dielectric Pipe Fittings/Unions

1. Fittings and unions shall be Watts Regulator 3000 series, Victaulic Style 47 waterway fittings, Apollo/Shurjoint DE30GG/C307 or approved equal.

PART 3 - EXECUTION

3.1 PIPING

A. General

1. All piping shall be made up and installed in a manner that permits expansion and contraction caused by changes in temperature and pressure. This shall be accomplished by the use of supports that will permit the movement of the pipe without undue stress, and by any other precautions that might be deemed necessary by the Engineer.

2. Piping exposed in all rooms shall be installed as nearly as possible parallel with or at right angles to the building walls. Install all pipe straight and true. Springing or forcing pipe into place will not be permitted unless specifically called for. Install piping in such a manner as to prevent strain on equipment connections.

3. Below grade piping shall be installed in such a manner that it does not bear directly on rocks or debris in ground. Any location where pipe passes close to or through walls or footings, it shall be protected from contact with concrete or cinder block. Any pipe passing through building walls shall be protected by a cast iron sleeve large enough to permit free movement of pipe. Refer to Section 22 00 10 for additional sleeve requirements.

4. All piping shall be protected at all times from dirt and moisture. During storage on the job site of construction, the Contractor shall keep pipe ends plugged or capped to prevent dirt or moisture entering the pipe.

5. The Contractor shall submit for the Engineer's review, a list of the proposed manufacturers of pipe, valves, fittings, and specialties.

6. All piping shall be carefully graded so as to eliminate traps and pockets. Where air pockets or water traps cannot be avoided, provide means for drainage with valved hose connections for water traps and with automatic air vents for air pockets. Automatic air vents shall be valved.

7. Where horizontal pipes change size, eccentric fittings shall be used, except where branch pipes connect into mains and in domestic water piping systems.

8. Make all joints smooth and unobstructed inside. Ream all pipe ends to remove burrs. Remove all obstructions and debris inside the piping systems prior to installation.

9. Install unions or flanges at all equipment and strainer connections.

10. Make welded joints on the piping system with continuous welds, without backing rings, and with pipe ends beveled before welding. Gas cuts shall be true and free from burned metal. Before welding, welding surfaces shall be thoroughly cleaned. The piping shall be carefully aligned and no weld metal shall project inside the pipe.

11. Connections between dissimilar piping materials shall be made with insulating couplings or dielectric fittings/unions. Fittings/unions shall be installed in piping systems in accordance with the manufacturer's recommendations and installation guidelines as required to prevent "Galvanic" and "Stray current" corrosion. Bronze bodied valves and devices installed in steel piping systems do not require dielectric fittings when both connections are made with steel piping, but do require dielectric fittings when the bronze bodied valves or devices are installed with one connection to steel piping and the other connection to copper piping. Iron bodied valves and devices installed in copper systems

do require dielectric fittings.

12. All piping shall be of the sizes indicated and shall be routed as indicated on the drawings, or as required, to serve all equipment and systems. Should there be a discrepancy in pipe sizes between the riser or schematic diagrams, and the floor plans; the riser and schematic diagrams will govern.

13. All piping materials, fabrication, erection, flanging, welding, tests, etc., shall be in accordance with American National Standard Code for Pressure Piping, current edition. Wire type "Short Arc" welding machines will not be acceptable for field welds. Wire type short arc welding machines will be acceptable for shop fabricated piping. If in the opinion of the Engineer any "short-arc" weld is found unsatisfactory, the use of the wire type short welding machine shall be discontinued for all remaining welding and re-welding the defective welds. The Engineer shall have access to the pipe fabrication facilities.

14. Piping insulation shall not be installed on connections, joints, flanges, etc., until pressure testing has been successfully accomplished.

15. Piping systems shall not be filled or put into operation until the vibration isolators have been installed.

B. Drain Piping

1. Drain piping from each air handling unit drain pan, strainer, water chilling unit, heat exchanger, pump base drain, tank and vessel overflow, auxiliary drain pan, piping system drain, blowdown, and any other system or equipment that has a drain connection shall be extended to the nearest floor drain, hub drain, or condensate drainage system. Drains shall be of the sizes indicated but not less than the full size of the drain pan connection. A valved 3/4" drainage line shall be provided at all low points in the piping systems. Air handling unit drains shall have deep seal traps to permit unit pan drainage. A deep seal trap shall be installed for each blow through or draw through air handling unit to maintain the water seal. Provide unions at all equipment drain connections.

C. Pipe Hangers and Supports

1. Hanger rods, inserts, etc., shall be sized and installed as recommended by the hanger manufacturer for the service intended.

2. The Contractor shall submit pipe hanger and support details for each type of floor construction to the Structural Engineer for review and approval prior to submission to the Engineer.

D. Pipe Covering Protection Saddles and Shields

1. Furnish and install pipe covering protection shields and saddles for all insulated piping at the locations of all supports. The insulation may be applied directly over the pipe hanger without the use of pipe saddles for domestic hot water and storm drain piping.

E. Gauges and Gauge Test Cocks

1. Gauges and gauge test cocks shall be installed at locations as indicated on the Drawings.

F. Thermometers and Thermometer Wells

1. Thermometers and thermometer wells shall be installed at locations as indicated on the Drawings and as specified herein.

G. Pressure and Temperature Test Plugs

1. Test plugs shall be provided in the entering and leaving piping to each vessel, tank, water heater, pressure reducing valve, pressure regulating valve, pump, orifice flange, flow meter, and at other locations indicated on the Drawings.

H. Flexible Pipe Connectors (FPC)

1. Refer to Section 22 05 48 for additional requirements.

I. Hose Bibb Drains

1. Refer to Section 22 40 00 for hose bibb requirements. Provide hose bibb drains at the following locations: At the low point of each domestic water riser downstream of the riser isolation valve and as shown on the drawings.

## 3.2 SCREW JOINTS

A. Screw joints shall be made with standard pipe thread and an approved compound applied to the male thread only.

## 3.3 SOLDERED PIPING JOINTS

A. Ends of pipe and inside of fittings shall be cleaned and flux applied to entire area of pipe end to be soldered. On pipe 1 1/2 inches and larger, flux shall be applied to pipe and fittings. Joints and fittings shall be soldered with ASTM B32 Grade 95T tin antimony solder, tin-bismuth-copper-silver solder, or brazed with AWS classification Bcup-3 brazing alloy as required by the system temperature and working pressure. Solder or brazing compounds containing lead shall be prohibited. Contractor shall thoroughly flush all soldered piping to remove excess flux so as to prevent corrosion of piping.

## 3.4 PRESS-CONNECT FITTINGS

A. Sealing element shall be verified for the intended use. Pipe tube ends shall be cut on a right angle (square) to the tube. Tube ends shall be reamed and chamfered. All grease, oil, or dirt shall be removed from the tube end with a clean rag. Visually examine the fitting sealing element to ensure there is no damage and that it is properly seated into the fitting. Insert the tube fully into the fitting and mark the tube wall at the face of the fitting. Always examine the tube to ensure it is fully inserted into the fitting prior to pressing the joint. Press fittings up to 4" shall be joined using appropriately sized approved Press Tools. Press fittings shall be installed according to the most current edition of the Manufacturers installation guidelines. Contractor or installer must be trained by the manufacturer representative prior to commencement of product installation.

## 3.5 PIPING SUPPORTS (ABOVE GRADE)

A. All pipe shall be supported from the building structure in a neat and workmanlike manner and wherever possible, parallel runs of horizontal piping shall be grouped together on trapeze type hangers. The use of wire or preformed metal to support pipes will not be permitted. Hanging pipes from other pipes will not be permitted. Piping shall be carefully coordinated before installation with other systems and equipment in chases and congested areas.

B. Vertical and horizontal pipes subject to sway in either direction shall have a snubber to prevent lateral movement.

## 3.6 PIPING TESTS

A. Scope: All piping installed on the Project unless specifically shown otherwise shall be hydraulically tested as specified herein, in accordance with ANSI B31.1 “Power Piping Code”, and as required by the local code authorities. This Contractor shall provide all equipment required to make the tests specified herein. Calibrated pressure gauges accurate to within ±1/2% of full range shall be used for pressure testing. System pressure gauges shall not be substituted for calibrated test gauges.

B. Sectionalizing: Piping may be tested a section at a time in order to facilitate the construction.

C. Tests to be performed: Before the piping of various systems is insulated, furred-in, or otherwise covered up; the subcontractor shall fill the section of the pipe to be tested with water and bring the section up to pressure with a test pump. The Contractor shall notify the Architect or his representative, in writing, at least five (5) working days prior to each test. The Architect or his representative may elect to witness the tests. These tests shall be conducted before any insulation is installed and any insulation installed prior to these tests shall be removed. Gauges used in the tests shall have been recently calibrated with a dead weight tester.

D. Duration of Tests: All tests shall apply full test pressure to the piping for a minimum of twenty four hours.

E. Pressure of Tests: All tests shall be conducted at the water working pressure of the pipe installed as listed in Section 22 00 15.

F. Inability to Hold Pressure: When the test pressure has fallen over five percent during the twenty four hour test period, the point of leakage shall be found, repaired and the test repeated. This procedure shall be followed until the piping system has been proven absolutely tight.

G. Use of Compounds: The use of chemicals or stop leak compounds will not be permitted at any time.

H. Delicate Controls: When delicate control mechanisms are installed in the piping section, they shall be removed during the tests to prevent shock damage. This does not apply to control valves.

I. Leaks: Leaks developing subsequent to these tests shall not be repaired by mastic or other temporary means. All leaks shall be repaired by removal of the valve, fitting, joint, or section that is leaking and reinstalling new material with joints as specified hereinbefore.

## 3.7 PIPE CLEANING AND STERILIZATION

A. The Contractor shall furnish all required pipe cleaning chemicals, chemical feed equipment, materials, and labor necessary to clean and sterilize the pipe as herein specified. In addition, the Contractor shall permanently install necessary chemical injection fittings complete with stop valves.

B. After final testing for leaks, all new potable water lines shall be thoroughly flushed by the Plumbing Contractor to remove foreign material. Before placing the systems in service, the Contractor shall engage a qualified service organization, NALCO, GE Betz, or approved equal, to sterilize the new water lines in accordance with the following procedure:

C. (Through a 3/4" hose connection in the main entering the building). Pump in sufficient sodium hypochlorite to produce a free available chlorine residual of not less than 200 PPM.

D. Proceed down stream from the point of chlorine application opening all faucets and taps until chlorine is detected. Close faucets and taps when chlorine is evident.

E. When chlorinated water has been brought to every faucet and tap with a minimum concentration of 200 PPM chlorine, retain this water in the system for at least three hours.

F. At the end of the retention period, no less than 200 PPM of chlorine shall be present at the extreme end of the system.

G. Proceed to open all faucets and taps and thoroughly flush all new lines until the chlorine residual in the water is less than 1.00 PPM. Obtain representative water samples from the system for analysis by a recognized bacteriological laboratory.

H. If all samples tested for coliform organisms are negative, a letter and laboratory reports shall be submitted by the service organization to the Contractor, certifying successful completion of the sterilization.

I. If any samples tested indicate the presence of coliform organisms, the entire sterilization procedure shall be repeated.

J. Contractor shall provide plumbing connections and power for pumping chlorine into the system.

K. Under no circumstances shall the Contractor permit the use of any portion of the domestic water system until it has been properly sterilized and certified same by the Local Water Department and the Local Code Authority.

L. The above procedure represents the minimum requirements for sterilization. Sterilization procedures shall also conform to all applicable regulations in accordance with the Local Code Authority.

M. Upon completion of the procedures and successful compliance with all applicable standards, the chemical company's authorized representative shall provide a written certified report documenting the domestic water lines have been cleaned and sterilized in accordance with these Specifications.

N. Owner Witness: The Contractor shall advise the Owner of the time and date of the pipe cleaning and sterilization procedures a minimum of two weeks prior to the procedures. The Owner shall have the option of witnessing the complete procedures. Any cleaning and sterilization procedures performed prior to Owner notification, at the Owner’s discretion, shall be performed again in the presence of the Owner.

END OF SECTION 22 11 16

SECTION 22 11 19 - DOMESTIC WATER PIPING SPECIALTIES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the valves and devices as shown on the Drawings and as specified herein for the plumbing systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 15 - Division 22 Design Conditions
B. Section 22 00 70 - Division 22 Schedule of Submittal Data

C. Section 22 11 16 - Domestic Water Piping
D. Section 22 40 00 - Plumbing Fixtures

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
   1. A1.20.1 Pipe Threads, General Purpose, Inch
   2. B16.18 Cast Copper Alloy Solder Joint Pressure Fittings
   3. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
   4. B31.9 Building Services Piping

B. ASTM International (formerly American Society for Testing and Materials)
   1. A53 Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
   2. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
   3. B32 Standard Specification for Solder Metal
   4. B88 Standard Specification for Seamless Copper Water Tube
   5. B584 Standard Specification for Copper Alloy Sand Castings for General Applications
   6. B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube

C. American Welding Society (AWS)
   1. A5.8 Specifications for Filler Metals for Brazing and Braze Welding

D. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
   1. SP-58 Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation
   2. SP-69 Pipe Hangers and Supports - Selection and Application
   3. SP-80 Bronze Gate, Globe, Angle, and Check Valves
   4. SP-89 Pipe Hangers and Supports - Fabrication and Installation Practices
   5. SP-110 Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends
   6. SP-125 Check Valves: Gray Iron and Ductile Iron, In-Line, Spring-Loaded, Center-Guided

E. NSF International (formerly National Sanitation Foundation)
   1. 61 Drinking Water System Components - Health Effects
   2. 372 Drinking Water System Components - Lead Content

1.5 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all specification requirements listed herein plus certification data and shop drawing requirements listed in Section 22 00 10 and 22 00 70.

PART 2 - PRODUCTS

2.1 VALVES AND SPECIALTIES

A. All valves required for the work shall be new, of first class quality, and shall be furnished, delivered, erected, connected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class standard article as approved by the Architect shall be provided.

B. Manufacturers

   1. Acceptable manufacturers will be listed in association with each valve and specialty type.

C. Packing and gaskets shall not contain PCBs or asbestos.

D. Pressure Rating: All valves shall have a pressure and temperature rating equal to or greater than the maximum working pressure of the system in which they are installed.

E. Unless otherwise noted all valves and specialties shall be the same size as the pipe in which they are installed.

F. As installed on this project and under the maximum system operating pressure, all balancing, isolation and shut off valves shall provide complete and total bubble tight shut off without leakage. Refer to individual valve specifications for additional requirements regarding valve shut off requirements.

G. Bronze valves shall be made with dezincification-resistant silicon bronze alloy materials. Bronze valves made with copper alloy (brass) containing more than 15 percent zinc are not permitted. Tinning flux which complies with ASTM B 813 shall be required to be used with all silicon bronze alloy materials.

H. Drain valves at riser heels and low points shall be ball valves and be NIBCO No. T/S/PC-585-80-LF-HC, Apollo 70HC, or approved equal by Crane for lead free applications. Pressure class shall be suitable for system in which it is installed as indicated in Specification Section 22 00 15. Provide ¾” hose connection and ¾” cap with chain.

I. All valves, specialties, and accessories including but not limited to fittings, gaskets, sealants, adhesives, coatings, lubricants, etc… which are installed directly in contact with potable water shall be certified and labeled in accordance with NSF/ANSI 61 and NSF/ANSI 372 standards. The weighted average lead content shall be ≤0.25% per the Safe Drinking Water Act.

J. Valves used for insulated piping service shall have valve handle extension and be insulated as detailed on the drawings.

## 2.2 DOMESTIC HOT AND COLD WATER VALVES

A. All domestic hot and cold water valves shall be in accordance with the following schedule:

1. Gate Valves and Ball Valves (Stop or Isolation Valve Application)

a.

b. 2 1/2" copper piping up to and including --NIBCO No. S/PC-111-LF or approved equal by Apollo or Crane, lead free gate valve, 300# CWP, dezincification-resistant silicon bronze body with soldered or press ends, rising stem, solid wedge and screwed bonnet. At the Contractor's option for piping up to and including 2", NIBCO No. S/PC-585-80-LF or approved equal by Apollo or Crane, lead free, 600# CWP, soldered or press ends, dezincification-resistant silicon bronze ball valves will be acceptable. Refer to additional domestic water ball valve requirements indicated below.

c. Copper piping 3" and 4"-- NIBCO No. F-637-33 or approved equal by Apollo or Crane, lead free gate valve, 250# CWP, Class 150 ductile iron OS&Y valve with Class 150 ANSI B16.1 flat or raised faced flanged ends and stainless steel trim: stems, disc faces, seat faces, seat rings, and bonnet bushings constructed of stainless steel. At the Contractors option for piping 3” through 12”, NIBCO No. F-607-RW or approved equal by Apollo or Crane, lead free gate valve, epoxy coated interior/exterior, 250# CWP, ductile iron OS&Y valve, resilient wedge, and flanged connections will be acceptable.

2. Ball Valves

a. Copper piping up to and including 2"-- NIBCO No. S/PC-585-80-LF or approved equal by Apollo or Crane, lead free, 600# CWP, soldered or press ends, dezincification-resistant silicon bronze ball valves. Refer to additional domestic water ball valve requirements indicated below.

B. If it complies with these Specifications and is approved as equal by the Engineer, gate, ball, and butterfly valves manufactured by one of the following manufacturers will be acceptable: Watts, Apollo, Victaulic, Milwaukee, Walworth, NIBCO, Crane, Fairbanks, and Lunkenheimer.

## 2.3 DOMESTIC HOT AND COLD WATER BALL VALVES

A. All domestic water valves up to and including 2” shall be ball valves. All domestic hot and cold water ball valves and butterfly valves shall be in accordance with the following:

1. General

a. Ball valves shall be constructed to permit repacking without removal of the valve body from the line and stem shall not blow out under pressure. Valves used for

insulated piping service shall have valve handle extension and be insulated as detailed on the Drawings.

b. Valves with special seats to meet temperature and throttling requirements shall have special markings clearly identified on the shop drawings. Resilient seated butterfly valves shall have EPDM or Viton seat material.

## 2.4 CHECK VALVES

A. Check valves 2-1/2" and smaller located in domestic water pipes shall be similar to NIBCO No. T/S/PC-413-Y-LF or approved equal, lead free, Class 125, 200# CWP, bronze body, swing type, teflon disc and screw in bonnet cap. Check valve non shock pressure temperature rating shall be same as specified for piping in which it is installed. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable: Hammond, NIBCO, Apollo, Walworth, or Crane will be acceptable.

## 2.5 RELIEF VALVES

A. Furnish and install relief valves where required and as indicated on the Drawings. Relief valve capacity shall be as required by the applicable Code, as required by the relief valve manufacturer, and as recommended by the associated equipment manufacturer served by the relief valve. Unless otherwise recommended by the manufacturers or as required by the applicable codes, the relief valve size shall not be less than the line size in which it is to be installed.

B. Inlet and outlet piping from the relief valve shall be at least the size of the valve connections unless otherwise indicated on the drawings. Provide union at piping connection to relief valve.

C. Relief valves shall be suitable for the working pressure and temperature of the system in which they are installed. Refer to Section 22 00 15.

D. Liquid relief valves shall be direct spring loaded type with a cast iron, bronze or steel body and stainless steel or bronze trim, lifting lever, field adjustable setting and ASME Unfired Pressure Vessel Code stamp. Valves shall be selected for a maximum overpressure of 10%. If they comply with these Specifications, relief valves manufactured by Kunkel, Consolidated, or Apollo will be acceptable.

## 2.6 REDUCED PRESSURE BACKFLOW PREVENTION ASSEMBLY (BFP)

A. Reduced pressure backflow prevention assembly located in the domestic water system shall be approved for use by the Local Water Authority and State Environmental Protection Agency.

B. Reduced pressure backflow prevention assembly shall be suitable for a working pressure of 150 psig. The pressure loss over the entire assembly shall not exceed 10 psi at building design flow. Size of the assembly shall not be smaller than the line size in which it is installed.

C. Assembly shall be complete with strainer, drain lines, inlet and outlet shut-off valves and integral air gap. Drain connections shall be routed to the nearest floor drain.

D. Reduced pressure backflow prevention assembly shall be ZURN 375AST Series. If it complies with the Specifications equipment manufactured by Hersey Products, Apollo (Series 4ALF-200 or 4ANLF-200), or Febco will be acceptable provided it is approved for use by the local code and water authority and complies with ASSE1013.

## 2.7 DOMESTIC COLD WATER PRESSURE REDUCING VALVE ASSEMBLIES

A. Domestic water pressure reducing assemblies shall be provided where indicated on the

Drawings. The pressure reducing assemblies shall have the capacities indicated on the drawings and shall have a total fall off pressure of 10 psi with the varying inlet pressure indicated over the minimum to maximum flow range listed on the drawings. The pressure reducing valves shall be selected to provide stable flow conditions without cavitation or valve chatter over the entire flow range specified.

B.

C. Furnish and install relief valves suitable for a working pressure of 300 psig where required and as indicated on the Drawings. Relief valve size shall be as required by the applicable Code, and as indicated on the Drawings. Domestic cold water pressure reducing assembly relief valve shall have an adjustable range of 20 to 200 psig and shall be Clayton Series 50 with X-105 limit switches, Watts Model No. ACV-116-52, Apollo Model No. A108-2, or approved equal.

D. Provide isolation stop valves upstream and downstream of each pressure reducing valve. Provide a separate bypass with a gate valve full line size to permit full flow to the zone in the event that the pressure reducing valves are taken off-line for service. High flow valve, low flow valve, and bypass assembly shall be piped in parallel with each other.

## 2.8 HYDRANTS

A. Wall Hydrants (WH-1)

1. Wall hydrants shall be the non freeze type with a stainless steel cover plate and either a modular or stainless steel box. All exposed parts of the hydrant viewable from the outside of the building shall be stainless steel. All interior parts shall be either bronze or cast brass. The hydrant seat shall be located in the heated area of the building. The hydrant shall be loose key operated and the washer shall be replaceable from the exterior face. The wall hydrant shall have an approved vacuum breaker. Hydrants shall be complete with all necessary accessories and options required for the application. Wall hydrants shall be Zurn Z 1300-SS-WC, Smith 5519-WC, Woodford MB67, or approved equal.

# PART 3 - EXECUTION

## 3.1 GENERAL

A. The Contractor shall install valves in all locations specified and as indicated on the Drawings.

B. Site Storage: All valves shall be protected at all times from dirt and moisture. During storage on the job site the Contractor shall keep valve ends plugged or capped to prevent dirt or moisture from entering the valves.

C. Installation: Valves shall be installed in the pipe lines without springing or forcing. Install all valves in a manner that allows future removal and service of the valve.

D. Temperature regulating valves, pressure reducing valves, and all equipment on the Project shall have pipe unions and stop valves installed on the system piping for removal and service.

E. Check valves on pump discharges shall be installed with the valve shaft perpendicular to the pump shaft. The check valves should be located at a minimum length of five times the pipe diameter away from the pump.

F. Provide each strainer with a drain line extending from the strainer to the nearest floor drain or hub drain. Drain line shall be of the size indicated but not less than the full size of the drain connection. Provide a stop valve and union immediately down stream of the strainer drain connection.

G. All hardware, including nuts, bolts, washers, and other miscellaneous hardware and accessories, used on piping, flanges for valves, and on other piping accessories located on the roof which are exposed to weather conditions shall be stainless steel or zinc coated.

H. General requirements for soldering joints of lead free piping and valves: Installer of lead free products shall be trained and certified prior to any installation by the local Sales Representative of selected manufacturer. Training shall consist of proper joint prep for solder cup and tube, the different and necessary soldering techniques, flame placement, the use of proper solder and flux selections, higher melt solders and tinning flux, sufficient joint heating, and a peal test by each mechanic The Contractor shall be responsible for documenting certification for each mechanic as provided by selected manufacturer and provide written certified documentation to Engineer.

END OF SECTION 22 11 19

SECTION 22 11 23 - DOMESTIC WATER PUMPS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the pumps and water pressure booster set as shown on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data
B. Section 22 05 13 - Common Motor Requirements for Plumbing Equipment
C. Section 22 05 48 - Vibration and Seismic Controls for Plumbing Piping and Equipment
D. Section 22 11 16 - Domestic Water Piping
E. Section 22 11 19 - Domestic Water Piping Specialties

1.4 REFERENCE STANDARDS

A. All Applicable Hydraulic Institute (HI) Standards

B. American Society of Mechanical Engineers (ASME)
1. Boiler and Pressure Vessel Code Section VIII Div 1 - Rules for Construction of Pressure Vessels
2. Boiler and Pressure Vessel Code Section IX - Welding and Brazing Qualifications
3. B16.5 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard

C. National Electrical Manufacturers Association (NEMA)
1. 250 Enclosures for Electrical Equipment (1,000 Volts Maximum)

D. NSF International (formerly National Sanitation Foundation)
1. 61 Drinking Water System Components - Health Effects
2. 372 Drinking Water System Components - Lead Content

E. Underwriters Laboratories (UL) Standard for Safety
1. 508 Industrial Control Equipment
2. 508A Industrial Control Panels
3. 778 Motor-Operated Water Pumps

1.5 SHOP DRAWINGS

A. Shop drawings shall be submitted for review as specified in Section 22 00 10. These shop drawings shall indicate specifically the exact construction, materials, etc., of the pumps to be furnished under these Specifications. The submittal, including capacity data, complete electrical control wiring diagrams, pumping assembly piping schematic, operating sequences, detailed dimensional physical layout drawings and certified factory test data, and shall be certified by the manufacturer as specified in Section 22 00 10.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. If they comply with these Specifications, the following manufacturers will be acceptable:

1. In-Line Circulating Pumps

a. Bell and Gossett
b. Grunfos
c. Taco
d. Paco
e. Patterson
f. Armstrong

2.2 INLINE CIRCULATING PUMP

A. In-line circulating pumps shall be bronze fitted with a dynamically balanced brass enclosed type impeller and with mechanical seal. Pump capacities shall be as indicated in the Schedule.

B. Pumps, casings, flanges, and seals shall be suitable for operation with the working pressures and temperatures indicated in Section 22 00 15. The scheduled working pressure applies to the entire pump assembly.

C. Refer to pump schedule on the drawings for pump selection design is based upon.

PART 3 - EXECUTION

3.1 FACTORY TESTING

A. Each factory assembled pumping system shall have a complete electric operating and sequence test, capacity performance test, and hydrostatic test prior to shipment. The test shall include a system operating flow test at the design flow rate with the design suction and net discharge pressure conditions scheduled.

B. The capacity performance test data shall include capacity, head, horsepower and efficiency at flow rates from zero to 100% of design flow with an X-Y graph test report.

C. The pump assembly shall be hydrostatically tested at least 50 psig pressure above the "Pump Working Pressure".

D. The factory test may be witnessed by the Engineer and Owner, if they so desire. The Contractor shall notify the Engineer in writing at least three weeks prior to the factory performance test.

E. Before the pumping assembly is shipped from the factory, the manufacturer shall transmit three copies of a certified letter stating that the pumps have been tested as specified hereinbefore. The certified factory test data and performance test curves shall accompany the certified letter. The pumping system shall not be installed before the test data has been reviewed by the Engineer.

3.2 PAINTING

A. The entire assembly shall be painted with two coats of enamel after shop testing.

3.3 START UP

A. The pump manufacturer shall assume unit responsibility and shall provide supervision at the time of installation and start up to insure proper operation of the pumping unit. In accepting this Contract, the domestic water pumping system manufacturer shall guarantee the system to give capacities and pressures not less than those specified with the pumping units installed as indicated on the Drawings.

END OF SECTION 22 11 23

SECTION 22 13 16 - SANITARY WASTE AND VENT PIPING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the sanitary waste and vent piping systems as shown on the Drawings, as specified herein, and/or as required by the local plumbing inspection department for the Division 22 systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 10 - Wall and Floor Penetrations
B. Section 22 00 70 - Division 22 Schedule of Submittal Data

C. Section 22 05 76 - Facility Drainage Piping Cleanouts
D. Section 22 05 80 - Connections to Utilities and Underground Pipe Installation

E. Section 22 13 19 - Sanitary Waste Piping Specialties
F. Section 22 40 00 - Plumbing Fixtures

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
1. A1.20.1 Pipe Threads, General Purpose, Inch
2. B16.1 Cast Iron Pipe Flanges and Flanged Fittings
3. B16.3 Malleable Iron Threaded Fittings Classes 150 and 300
4. B16.5 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
5. B16.9 Factory-Made Wrought Buttwelding Fittings
6. B16.11 Forged Fittings, Socket-Welding and Threaded
7. B16.15 Cast Copper Alloy Threaded Fittings: Classes 125 and 250
8. B16.18 Cast Copper Alloy Solder Joint Pressure Fittings
9. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
10. B31.1 Power Piping
11. B31.9 Building Services Piping
12. B36.10M Welded and Seamless Wrought Steel Pipe
13. B36.19M Stainless Steel Pipe

B. ASTM International (formerly American Society for Testing and Materials)
1. A53/A53M Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
2. A74 Standard Specification for Cast Iron Soil Pipe and Fittings
3. A135 Standard Specification for Electric-Resistance-Welded Steel Pipe
4. B32 Standard Specification for Solder Metal
5. B88 Standard Specification for Seamless Copper Water Tube
6. B306 Standard Specification for Copper Drainage Tube
7. B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube

8. C564 Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
9. D1785 Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120
10. D2461 Standard Test Method for Pressure Rating Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40 and 80 Socket-Type
11. D2467 Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
12. D2665 Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings
13. D2949 Standard Specification for 3.25-in. Outside Diameter Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings
14. D3311 Standard Specification for Drain, Waste, and Vent (DWV) Plastic Fittings Patterns
15. F891 Standard Specification for Coextruded Poly(Vinyl Chloride) (PVC) Plastic Pipe With a Cellular Core
16. F1476 Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications

C. American Welding Society (AWS)
1. A5.8 Specifications for Filler Metals for Brazing and Braze Welding
2. B2.1 Specification for Welding Procedure and Performance Qualification

D. Cast Iron Soil Pipe Institute (CISPI)
1. 301 Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
2. 310 Specification for Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications

E. Copper Development Association (CDA)
1. A4015 Copper Tube Handbook

F. CSA Group (formerly Canadian Standards Association)
1. B1800 Thermoplastic Non-Pressure Piping Compendium

G. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
1. SP-58 Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation
2. SP-69 Pipe Hangers and Supports - Selection and Application
3. SP-89 Pipe Hangers and Supports - Fabrication and Installation Practices

H. NSF International (formerly National Sanitation Foundation)
1. 14 Plastics Piping System Components and Related Materials

1.5 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Sections 22 00 10 and 22 00 70. Testing and installation guidelines shall be included with the shop drawing submittal data.

PART 2 - PRODUCTS

2.1 PIPING AND JOINTS

A. All materials and installation methods shall meet the requirements of applicable national and local codes. Cast iron pipe and fittings shall be marked with the collective trademark of the Cast

Iron Soil Pipe Institute and be listed by NSF International. Fabricated PVC DWV fittings shall be manufactured in the United States and shall conform to NSF International Standard 14.

B. General: All materials and specialties required for this project shall be the products of United States made materials. International or foreign made materials and specialties shall not be acceptable unless pre-approved by the Owner prior to bidding. If the Owner pre-approves the use of international or foreign made materials and specialties prior to bidding, the Contractor shall certify in writing to the Engineer that all international or foreign made materials and specialties furnished for this project are suitable for the service intended and complies with all the requirements specified herein. This certification shall be included within the submittal being issued to the Engineer for review and approval. In addition, the Contractor shall submit a 12" long sample of each type of international or foreign manufactured pipe material to be installed on this project to a certified metallurgy laboratory for complete properties and strength analysis. The samples shall be cut from pipe material stored on this projects site or in the Contractor's fabrication facilities. The pipe material shall be identified as to its intended use, ASTM designation, project name, Contractor's name, and date the sample was obtained. If the pipe is coated, such as with a galvanized material, the thickness and properties of the coating material shall be indicated. A certified report of the test data shall be provided to the Owner and the Engineer for review and approval prior to the final acceptance and installation of any international or foreign made material and specialty. The cost of the certified laboratory testing shall be at the Contractors expense and the certified laboratory shall be subject to approval by the Owner and Engineer prior to testing. Any international or foreign made material and specialty that does not comply with the material standards and all other requirements specified herein shall be removed. Prior to purchase of any materials manufactured outside the United States, the Contractor shall submit complete certifications and typical mill reports in English language for review. Any foreign pipe delivered to the jobsite shall bear mill heat markings and the Contractor shall make available the corresponding mill test reports.

C. Sanitary, Soil, and Waste – Under Slab Below Grade

1. Service weight cast iron, bell and spigot, pipe and fittings, ASTM A74 and NSF certified as approved by the local code authority, manufactured by Tyler Pipe Industries, Charlotte Pipe and Foundry, or approved equal.

2. Tyler "Ty-Seal", "Charlotte Seal Gasket", or approved equal, neoprene positive seal elastomeric compression type gasket shall be used at joints complying with ASTM C564.

3. Gaskets, joints, connectors, specialties and pipe shall be manufactured and provided by the same manufacturer.

4. At the Contractors option, for material for pipe 15 inches inside diameter and smaller, and where acceptable to the local plumbing code authority, sanitary, soil and waste piping located under slab below ground may be Type DWV PVC pipe and fittings complying with all applicable codes, rules and regulations. Type DWV PVC pipe shall conform to the following applicable standards: ASTM D2665, ASTM D2949 and ASTM F891 All piping that extends from a below slab condition up through a concrete slab to an above slab condition shall be cast iron. PVC fittings shall conform to the following applicable standard: ASTM D3311. The PVC pipe joining compound shall comply with the PVC pipe manufacturer's recommendations. The piping installation shall comply with all requirements of the local authority and the manufacturer's recommendations. Installation shall also comply with applicable requirements specified elsewhere. PVC pipe shall be manufactured by Charlotte Pipe and Foundry or approved equal.

5. Contractor shall provide camera inspection of existing building sanitary drain prior to

work. Pictures from camera inspection shall be included in O&M Manuals.

D. Sanitary, Soil, and Waste – Within Building Above Ground

1. Service weight cast iron, bell and spigot, pipe and fittings, ASTM A74 and NSF International certified as approved by the local code authority, manufactured by Tyler Pipe Industries, Charlotte Pipe and Foundry, or approved equal. Sanitary pipe from the urinals to the riser or to the waste branch piping that connects to other plumbing fixtures shall be DWV PVC pipe and fittings if acceptable and approved by the Local Code Authorities. DWV PVC pipe and fittings shall not be used from the urinals to the riser or to the waste branch piping that connects to other plumbing fixtures if prohibited by the Local Code Authorities. Copper piping shall not be acceptable for use in any sanitary, soil, and waste piping within the building above ground including, but not limited to, the toilet room waste battery piping assembly.

2. Tyler "Ty-Seal", “Charlotte Seal Gasket”, or approved equal, neoprene positive seal elastomeric compression type gasket shall be used at joints complying with ASTM C564.

3. Gaskets, joints, connectors, specialties and pipe shall be manufactured and provided by the same manufacturer.

4. Unless otherwise noted, hubless service weight cast iron soil pipe, fittings, and joints, CISPI-301, CISPI-310, and NSF International certified as approved by the local code authority with Tyler “no-hub” stainless steel connectors may be used on pipe 10” diameter and smaller. Equivalent systems by Charlotte Pipe and Foundry or approved equal will be acceptable. Hubless piping system shall not be used for the following applications.

   a. Horizontal and vertical piping 6” diameter and larger at or within 30 feet higher than the gravity main drainage pipe invert elevation.

   b. As noted on the drawings.

E. Vent Piping

1. Vent stacks shall be service weight cast iron as approved by the local code authority. Branch vents, revents, or circuit vents shall be service weight cast iron, DWV copper pipe and fittings or DWV plastic pipe and fittings as approved by the local code authority. Plastic vent pipe shall not be used if prohibited by Local Codes. Joints shall either be "No Hub" stainless connection or "Ty Seal" for cast iron, soldered for copper and solvent welded for plastic pipe.

2. Gaskets, joints, connectors, specialties and pipe shall be manufactured by the same manufacturer.

2.2 PIPE HANGERS AND SUPPORT SYSTEM

A. Horizontal Pipe

1. Clevis Hangers, supports, rods, structural inserts, etc., shall be as specified in Section 22 11 16.

B. Vertical Pipe

1. Riser clamps shall be as specified in Section 22 11 16.

C. Pipe Supports and Restraints

1. In general, all piping shall be supported and restrained in a manner recommended by the piping and joint manufacturer. In addition, for all piping above slab within building structure larger than 4” diameter, as well as dedicated grease waste piping, all vertical and horizontal offsets and changes in direction shall be restrained in an approved manner with socket clamps and all-thread rods.

D. Groups of Pipes

1. On large pipe or groups of pipes, use structural iron for support members. Concrete inserts for hanger rods attached to new structure shall be National Pipe Hanger Corporation Figure 606 with Figure 607 attached to new structure, or approved equal for pipe sizes up to 8". For 8" pipe and larger attached to the new structure reinforce the insert with steel anchor bars or suitable steel plates. Refer to Sections 22 11 16 and 22 00 10 for additional pipe hanger and support requirements.

E. Extension Bar Hangers

1. Perforated extension bar hangers will not be accepted on any of the work.

2.3 FIXTURE CONNECTIONS BETWEEN TRAPS AND SOIL PIPES

A. Water Closets

1. Plastic adjustable nipples.

B. Urinals

1. Plastic adjustable nipples. Copper nipples shall not be acceptable for use with urinals.

C. Lavatories and Sinks

1. Copper, cast iron, or malleable iron nipples with suitable adapters.

D. Service Sinks

1. Brass, cast iron, or malleable iron nipples with suitable adapters.

E. Drinking Fountains

1. Copper, cast iron, or malleable iron nipples with suitable adapters.

2.4 DEEP SEAL P-TRAPS

A. Provide deep seal P-traps for all drains shown with a P-trap only where noted on the Drawings.

PART 3 - EXECUTION

3.1 SEWAGE AND DRAINAGE PIPING

A. Neoprene Compression Type Gasket Joints

1. Compression type gasket joints shall be assembled in accordance with the manufacturer's written recommendations. Adhesive lubricant shall be recommended and approved in writing by the piping system manufacturer. Written recommendations and approvals shall be clearly indicated in the shop drawing data.

2. Horizontal joints shall be supported directly on each side of the joint with an approved hanger assembly.

3. For pipes larger than 4” diameter, vertical and horizontal offsets and changes in direction shall be restrained at each joint in an approved manner with socket clamps and all-thread rods.

B. "No hub" Joints

1. "No hub" type joints shall be assembled in accordance with the manufacturer's recommendations.

2. Horizontal joints shall be supported directly on each side of the joint with an approved hanger assembly.

3. For pipes larger than 4” diameter, vertical and horizontal offsets and changes in direction shall be restrained at each joint in an approved manner with socket clamps and all-thread rods.

4. For all piping systems located directly above ceilings in the computer areas, UPS system areas and battery room areas; each joint shall be reinforced and restrained in an approved manner with socket clamps and all-thread rods.

3.2 VENTING

A. General

1. Fixtures and piping shall be vented as indicated on the drawings and/or as required by Local Codes. The contractor shall be responsible to provide a code compliant venting system including vent piping as required by the local code authority. This requirement shall be reflected in the base bid.

B. Vent Pipe Flashing

1. All vent pipe flashing through the roof shall be flashed as required to provide a watertight and airtight seal to the roof covering. All flashing shall be done in cooperation with the Roofing Subcontractor. Refer to the Architectural Documents for additional requirements.

3.3 MINIMUM SLOPE REQUIREMENTS

A. All drainage lines shall have at least the minimum slope toward the main sewer as required by the Local Plumbing Code. Pipe must be so laid that the slope will be continuous. Permission shall be secured from the Architect before proceeding with any work where existing conditions prevent the installation at the minimum grade specified but in no case in direct violation of applicable codes and standards

3.4 CLEANOUTS

A. In each change of direction of soils and wastes, provide a cleanout plug connected to same with Y fittings and 45° ell made flush with floor or wall. In all horizontal straight runs more than fifty foot lengths provide cleanouts at a spacing indicated in Section 22 05 76. Cleanouts shall be brought up flush with floor or grade unless otherwise shown on the Drawings. The locations of all cleanouts shall be verified with the Architect. All cleanouts shall be of the same size as the pipe up to and including pipe four inches in diameter. Four inch cleanouts shall be used for all pipe larger than four inches unless otherwise noted on the Drawings.

3.5 PIPE HANGERS

A. All horizontal pipe throughout the building, shall be supported from the building construction by means of approved expansion ring hangers or clevis hangers. Pipe supports shall be prohibited from being supported from other building systems. Hangers shall be spaced no more than eight feet apart center to center on cast iron drainage and vent piping. Pipe support shall also be provided directly on each side of a joint. All pipe shall be straight and have uniform fall.

B. All vertical pipes shall be substantially supported at each floor line with approved steel or iron riser clamps.

C. Hanger rods, inserts, etc., shall be sized and installed as recommended by the hanger manufacturer for the service intended.

D. Refer to Sections 22 11 16 and 22 00 10 for additional requirements.

3.6 PIPING SUPPORTS (ABOVE GRADE)

A. All pipe shall be supported from the building structure in a neat and workmanlike manner and wherever possible, parallel runs of horizontal piping shall be grouped together on trapeze type hangers. The use of wire or preformed metal to support pipes will not be permitted. Hanging pipes from other pipes will not be permitted. Piping shall be carefully coordinated before installation with other systems and equipment in chases and congested areas.

B. Vertical and horizontal pipes subject to sway in either direction shall have a snubber to prevent lateral movement.

3.7 PIPE SUPPORTS (BELOW GRADE)

A. Earth shall be excavated to a minimum depth with an even surface to insure solid bearing of pipe for its entire length. Where pipe lines cross deeper excavations, these shall be filled and tamped to the proper level before pipe is installed.
   1. Interior: The water pipe shall be installed a minimum of 4 inches below the bottom of the slab and shall not be in direct contact with the concrete at any point.

   2. Exterior: The water pipe shall have a minimum of 3 feet of cover and shall comply with Local Codes.

B. The pipe shall have bearing along its entire length and, if installed on fill, shall be supported by concrete cradles to firm earth. All offsets, changes in grade, or changes in direction of run of pressure systems shall have poured in place concrete blocking from pipe to firm earth. Blocking shall be of adequate size to hold the pipe completely stationary regardless of water pressure. All exposed joints shall have anchor bolts or rods installed in such a manner that there is no chance of a joint being pushed apart.

C. Gravity systems shall have supports to undisturbed soil at no more than 5'-0" on center. Support shall be provided at the base of every stack. Backfill shall be accomplished in such a manner that alignment of piping is not disturbed.

3.8 RODDING SEWERS

A. All sewer lines, both inside and outside the building to the point of connection to the City sewer, shall be clear of obstructions. If the sewer lines are blocked, the Contractor shall rod out the sewers as required. This work shall be done in the presence of the Architect as part of the Contract, to make certain that all lines are clear. Any obstruction that may be discovered shall be removed immediately. Rodding shall be accomplished by utilizing a rotary cutter the full size of the pipe being cleaned.

3.9 EXCAVATION AND BACKFILL

A. Refer to Section 22 05 80 for excavation, installation and backfill requirements.

3.10 PIPE SIZES

A. Should there be a discrepancy in pipe size between the riser diagrams and floor plans, the riser diagrams shall govern.

3.11 TESTING AND INSPECTIONS FOR GRAVITY TYPE SANITARY PIPING

A. All testing and inspections shall be in accordance with the local code authorities and the requirements of the piping system manufacturer. Engineer shall retain the right to witness and inspect all piping systems. Contractor shall notify the local code authorities and the Engineer with adequate notice prior to testing and close-in of the systems so the code authorities and the Engineer have the option of inspection and witnessing test.

B. After soils and waste lines, etc., have been installed for at least the minimum amount of time recommended by the piping system manufacturer, all pipes in groups of two to three floors shall be temporarily plugged up and the pipe shall be filled with water full to the top and allowed to remain for a period of time not less than required by the local code authorities. However, in no case shall the time period of the test be less than two hours.

C. In addition to the water test, all pipes shall be tested by the peppermint or smoke test, if required by the Engineer or Local Authorities.

D. Sectionalizing: Piping may be tested a section at a time in order to facilitate the construction.

E. Tests to be performed: Before the piping of various systems is insulated, furred-in, or otherwise covered up; the subcontractor shall fill the section of the pipe to be tested with water and bring the section up to pressure with a test pump. The Contractor shall notify the Architect or his representative, in writing, at least five (5) working days prior to each test. The Architect or his representative may elect to witness the tests. These tests shall be conducted before any insulation is installed and any insulation installed prior to these tests shall be removed. Gauges used in the tests shall have been recently calibrated with a dead weight tester.

F. Duration of Tests: All tests shall apply full test pressure to the piping for a minimum of twenty four hours.

G. Pressure of Tests: All tests shall be conducted at the water working pressure of the installed domestic cold water system as listed in Section 22 00 15.

H. Inability to Hold Pressure: When the test pressure has fallen over five percent during the twenty four hour test period, the point of leakage shall be found, repaired and the test repeated. This procedure shall be followed until the piping system has been proven absolutely tight.

I. Use of Compounds: The use of chemicals or stop leak compounds will not be permitted at any time.

J. Delicate Controls: When delicate control mechanisms are installed in the piping section, they shall be removed during the tests to prevent shock damage. This does not apply to control valves.

K. Leaks: Leaks developing subsequent to these tests shall not be repaired by mastic or other temporary means. All leaks shall be repaired by removal of the valve, fitting, joint, or section that is leaking and reinstalling new material with joints as specified hereinbefore.

END OF SECTION 22 13 16

SECTION 22 13 19 - SANITARY WASTE PIPING SPECIALITIES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the devices as indicated on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data
B. Section 22 13 16 - Sanitary Waste and Vent Piping

1.4 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Sections 22 00 10 and 22 00 70.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. Josam
2. Smith
3. Wade
4. Zurn

2.2 TYPES

A. Drains and Backwater Valves shall be complete in every detail and shall be similar and approved equal in all respects to the model numbers specified herein.

2.3 DRAINS

A. General

1. Drains shall be provided complete with all connections. Drains shall be modified as required to be compatible with waterproof membranes, floor systems, topping slab systems, and roofing systems as shown on the Architectural Documents. The size shall be as shown on the Drawings. Roof drains shall be complete with flashing ring and deck clamps. Drains located in areas exposed to vehicular traffic shall have a secured top and be rated for traffic duty. All drains serving indirect drainage lines shall have a half grate or other approved equal opening to accommodate the drain line(s).

B. Drain Connection Type

1. The drain connection type shall match the piping connection type used for each system and/or location.

C. Floor Drains

1. Refer to the schedule on the drawings for all drain selections and appropriate accessories.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Drains and Flashings

1. Drains shall be installed at the low point of indicated slopes as shown on the Architectural Drawings. The location indicated on the MEP Drawings is diagrammatic and shown for informational purposes only. Installation shall be such that ponding will not occur around the perimeter of the drain.

2. All floor drains and roof drains installed in floors or roof having waterproof membrane flashing shall be flashed with lead not less than three pounds per square foot built 6" into the waterproof membrane. Include underdeck clamp and extension neck with drain if required by the type of roof or floor waterproofing furnished. Lead flashing may be omitted at roof if required by roofing and/or flooring contractor. Omission of lead roof drain flashing must be approved in writing by the Architect.

3.2 WORKMANSHIP

A. All equipment shall be installed in an approved manner. All finished surfaces shall be protected during construction. All items shall be set level and installed as recommended by the manufacturer.

3.3 CLEANING

A. All of the above items shall be cleaned and all construction dirt removed at the completion of the Project.

END OF SECTION 22 13 19

# SECTION 22 33 00 - ELECTRIC DOMESTIC WATER HEATERS

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

### 1.2 WORK INCLUDED

A. Provide the heater(s) as shown on the Drawings and as specified herein.

### 1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data
B. Section 22 11 16 - Domestic Water Piping
C. Section 22 11 19 - Domestic Water Piping Specialties

### 1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1. 90.1 Energy Standard for Buildings

B. American Society of Mechanical Engineers (ASME)

C. NSF International (formerly National Sanitation Foundation)
1. 5 Water Heaters, Hot Water Supply Boilers, and Heat Recovery Equipment
2. 61 Drinking Water System Components - Health Effects
3. 372 Drinking Water System Components - Lead Content

### 1.5 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data and shop drawing requirements listed in Sections 22 00 10 and 22 00 70 plus capacity and special requirements indicated in the Schedule.

## PART 2 - PRODUCTS

### 2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. A.O. Smith
2. Ruud
3. Rheem
4. State
5. Hubbell

### 2.2 ELECTRIC STORAGE WATER HEATER

A. Type

1. Each electric storage commercial grade water heater shall be a complete package, factory assembled and ready to be connected to service wiring and water piping. The unit shall be insulated with a high density insulating material covered with anodic zinc coated steel and finished with synthetic enamel. The heater shall comply with ASHRAE Standard 90.1, latest edition, and all local code authority requirements. The complete heater shall be listed by Underwriters Laboratories, Inc., and shall bear the UL Label. Each heater and accessories including but not limited to piping, valves, fittings, expansion tank, heat traps, gaskets, sealants, adhesives, coatings, lubricants, etc… which are installed directly in contact with potable water shall be certified and labeled in accordance with NSF 61 standards.

B. Capacity

1. The capacity and recovery shall be as indicated in the Schedules.

C. Tank

1. The tank shall have all welded seams. All tappings shall be welded in the tank prior to application of the lining. Tappings shall be the size of the piping shown on the Drawings or the Contractor shall provide brass reducer couplings and pipe nipples. Unless otherwise indicated, the tank and lining shall be designed and tested for 150 psig working pressure. All heaters located downstream of pressure regulating valves or located within domestic water systems without main house pumps, shall be suitable for the actual operating pressure of the related domestic water system and shall not require the 150 psig rating. Refer to the downstream set point of pressure reducing valve assemblies for maximum operating pressures. For systems using a circulating pump provide a tapping for the circulating line independent of the cold water tap.

D. Lining

1. Interior tank lining, unless otherwise stated, shall be a high temperature glass. Provide an unconditionally guaranteed five year non-prorated tank warranty against all defects in workmanship and materials.

E. Control Panel and Wiring

1. The electric control panel shall be mounted integrally with the heater. The panel housing shall serve as a junction box for the wiring and shall contain the thermostat. The thermostat shall be adjustable from 100°F to 180°F. Water heater setpoint shall be 120 degrees f unless otherwise required by the Owner. The heater shall be wired in compliance with the National Electrical Code and all Local Codes.

F. Heating Elements

1. Maximum size of heating elements shall be in accordance with the following table. Water heaters shall have multiple elements (not to exceed maximum limit) to meet the required capacity.

2. The immersion elements shall be screw in, individually removable direct contact immersion type sealed in a zinc or tin plated copper sheath or 80% nickel and 20% chromium sheath.

3. The heating elements shall have a maximum watt density of 75 watts per square inch.

4. The heaters shall be controlled with a thermostat through contactors if current requirements exceed thermostat capacity. Heaters shall have approved manual reset high limit thermostat to break all ungrounded conductors.

G. Expansion Tank

1. Provide an expansion tank for each water heater. Expansion tanks shall be labeled in accordance with the requirements of the Local Code Authority and shall be certified and labeled in accordance with NSF 61 standards. Expansion tanks shall be similar to Amtrol “Therm-X-Trol” Series “ST”or approved equal. The expansion tanks shall have a minimum total tank volume as detailed on the drawings.

H. Accessories

1. Accessories to include an ASME temperature and pressure relief valve, temperature/pressure plug inlet and at the tank outlet and an anode rod for corrosion protection. The discharge from the temperature and pressure relief valve shall be piped to the nearest floor drain independent of all other drain piping, auxiliary drain pan lines, etc. or as indicated on the Drawings.

I. Additional Requirements

1. Water heaters, trim, accessories, and piping shall be provided and installed in accordance with Local Code Authority Requirements. Tanks shall be labeled in accordance with the requirements of the Local Code Authority.

## PART 3 - EXECUTION

### 3.1 MOUNTING

A. Refer to electric water heater details on the drawings for mounting. The electric water heater shall be set dead level in both directions. Maintain access to any item which must be observed, serviced, or replaced. When the heater installation includes a circulating pump maintain clear access to the pump and service valves. Provide an auxiliary drain pan under heater with drain line extended to drain. Drain pan shall be at least 2" high and constructed of 16 gauge galvanized steel with all joints brazed. Pans shall be watertight and have hemmed edges. Drain pan shall have drain pipe connections of at least 1" IPS, or as shown on the Drawings. Drain connection shall be located at the lowest point of the drain pan.

### 3.2 PIPING CONNECTIONS

A. The electric water heater shall have piping connections as shown on the Drawings. Provide a service valve on the inlet and outlet piping.

B. Install expansion tank for each water heater as indicated on the drawings in accordance with the manufacturer’s recommendations and local code authority requirements.

C. Install heat traps at the cold water inlet and hot water outlet of each water heater in accordance

with the manufacturer's recommendations and as required by the local code authority. At the Contractor's option, heat traps may be integral to the water heater.

3.3 PROTECTION

A. The electric water heater shall be protected during construction and all solder or solder flux removed from the top.

3.4 CLEANING

A. The electric water heater and drain pan shall be cleaned and all construction dirt removed at the completion of the Project.

END OF SECTION 22 33 00

SECTION 22 40 00 - PLUMBING FIXTURES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

B. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 22 General Requirements" in Section 22 00 10.

1.2 WORK INCLUDED

A. Provide the plumbing fixtures as indicated on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 22 00 70 - Division 22 Schedule of Submittal Data
B. Section 22 11 16 - Domestic Water Piping
C. Section 22 13 16 - Sanitary Waste and Vent Piping

1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 18 Method of Testing for Rating Drinking-Water Coolers with Self-contained Mechanical Refrigeration

B. American Society of Mechanical Engineers (ASME)
   1. 112.19.2 Ceramic Plumbing Fixtures
   2. 112.19.3 Stainless Steel Plumbing Fixtures

C. NSF International (formerly National Sanitation Foundation)
   1. 61 Drinking Water System Components - Health Effects
   2. 372 Drinking Water System Components - Lead Content

D. Underwriters Laboratories (UL) Standard for Safety
   1. 399 Drinking Water Coolers

1.5 SHOP DRAWINGS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Section 22 00 10 and 22 00 70. Fixtures that will be used for handicapped or disabled person application shall clearly indicate compliance with the ADA and applicable ANSI standards. Submittal data shall list specific ADA and ANSI standards by section and paragraph.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Fixtures
   1. Acceptable manufacturers shall be listed under "Plumbing Fixtures" herein.

B. Fixture Trim
   1. Acceptable manufacturers shall be listed with the associated plumbing fixture.

C. Fixture Supports and Carriers

1. Josam
2. J. R. Smith
3. Wade
4. Watts
5. Zurn

## 2.2 FIXTURE SUPPORTS AND CARRIERS

A. Supports and carriers shall be by a company regularly engaged in the manufacturing of such devices. Material shall be steel or cast iron. Style shall be as recommended by the fixture manufacturer. Supports and carriers shall have the necessary adjustments to allow for irregularities of building construction.

## 2.3 FLUSH VALVES, FAUCETS, SUPPLIES AND TRAPS

A. Exposed fixture trim shall be chrome plated. All trim of a similar function shall be by the same manufacturer. Vacuum breakers shall be included where there is a possibility of the fixture inlet being submerged. Provide chrome plated escutcheon plates where pipes penetrate walls. Provide fixtures with individual stop valves for servicing the fixture. Fixture trim shall be as indicated hereinafter.

.

## 2.4 PLUMBING FIXTURES GENERAL

A. Fixtures of similar material shall be by the same manufacturer.

B. Fixtures are not scheduled on Drawings but are indicated by symbols. Carefully check drawings and furnish everything indicated and/or required for a complete installation. Fixtures shall be complete in every detail. The Contractor shall provide new fresh unused batteries for all battery operated plumbing fixtures immediately prior to Substantial Completion.

C. In addition to applicable National and Local Codes, plumbing fixtures provided for use by persons with disabilities or handicap persons shall meet the requirements of the "Americans with Disabilities Act" (ADA).

D. All plumbing fixtures including but not limited to faucets, drinking fountains, piping, valves, fittings, filters, sealants, gaskets, adhesives, coatings, lubricants, etc. which are installed directly in contact with potable water shall be certified and labeled in accordance with NSF 61 standards.

E. Fixtures and trim shall be as scheduled on the drawings

# PART 3 - EXECUTION

## 3.1 MOUNTING

A. Contractor shall obtain the necessary templates and dimensions from the Manufacturer prior to rough in. For masonry construction, provide backing plates or rods in the wall with 1/4" threaded rods anchored to the backing and extending thru the hanger supplied by the fixture manufacturer. Hanger shall be secured by nuts and washers. For construction other than

masonry the fixture shall be mounted on a carrier securely bolted to the building structure with thru bolts or preset inserts. Shop drawings shall include information on fixture mounting. Floor mounted fixtures shall have supports, blocking, or grout setting bed as necessary to prevent movement or flexing. Countertop fixtures shall be sealed watertight in the countertop. All fixtures shall be mounted level and/or plumb and properly located in the building. All methods of securing the fixture shall present a finished appearance. Refer to the Architectural documents for additional mounting height and dimensional location information.

3.2 WATER AND WASTE CONNECTION

A. During project rough in water and waste stub outs shall be properly located to prevent unsightly gaps between fixtures and walls and to allow exposed pipe to be installed straight and plumb from the stub out to the fixture. Piping shall be anchored to the wall construction, the fixture carrier, or the building structure to prevent movement after the fixture is set. Water and waste connections shall be made with trim identical in all respects to those listed in the Specifications.

3.3 FIXTURE PROTECTION

A. Fixtures shall be protected during construction from dirt and physical damage. Dirty fixtures shall be cleaned and damaged parts of fixtures replaced.

3.4 TESTS

A. Fixtures shall be given an in service normal and handicapped user test. Make the necessary adjustments and demonstrate the proper operation of the fixture to the satisfaction of the Architect.

END OF SECTION 22 40 00

# SECTION 22 4600 - SECURITY PLUMBING FIXTURES

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Stainless steel combination units.

B. Related Requirements:

1. Section 22 4000 "Plumbing Fixtures."

### 1.2 DEFINITIONS

A. Accessible Service Space: Service area in secure space behind wall-mounted fixtures.

B. Back-Access Fixture: Security plumbing fixture designed to mount on wall sleeve built into wall or on wall, so installation and removal of fixture, piping, and other components are accessible only from service space behind wall.

### 1.3 ACTION SUBMITTALS

A. Product Data:

1. Construction details, material descriptions, dimensions of individual components and profiles, and finishes for security plumbing fixtures.
2. Rated capacities, operating characteristics, and furnished specialties and accessories.

### 1.4 CLOSEOUT SUBMITTALS

A. Maintenance Data: For security plumbing fixtures and components.

### 1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials to Owner that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Flushometer-Valve Repair Kits: Equal to 10 percent of quantity of each type installed, but no fewer than one of each type.

## PART 2 - PRODUCTS

### 2.1 SOURCE LIMITATIONS

A. Obtain each product type from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Comply with ASME A112.19.3/CSA B45.4 for stainless steel plumbing fixtures.

B. Comply with ASSE 1037/ASME A112.1037/CSA B125.37 for flush valves.

C. Comply with ASME A112.18.1/CSA B125.1 for plumbing supply fittings.

D. Comply with ASME A112.18.2/CSA B125.2 for plumbing waste fittings.

E. Comply with ASME A112.6.1M for plumbing fixture supports.

F. Comply with ICC A117.1 for ADA-compliant, accessible plumbing fixtures and installation.

G. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

H. Faucets and bubblers intended to convey or dispense water for human consumption are to comply with the U.S. Safe Drinking Water Act (SDWA), with requirements of the Authority Having Jurisdiction (AHJ), and with NSF 61/NSF 372; or are certified in compliance with NSF 61/NSF 372 by an ANSI-accredited third-party certification body, that the weighted average lead content at wetted surfaces is less than or equal to 0.25 percent.

2.3 STAINLESS STEEL COMBINATION UNITS

A. Combination Units - Stainless Steel, Back Access, On or Off Floor, Cabinet, with Water Closet and Lavatory (CB-1)

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following or approved equivalent.:

    a. I-Con Systems, Inc. – 100689 ADA Combination Fixture.
    b. Metcraft Industries, Inc. – 3137 90R&L-LR.
    c. Willoughby Industries, Inc. – ECF-3696-FA Series

2. Material: 14-gauge, Type 304 stainless steel, seamless welded construction, with fire-resistant sound-deadening material in cabinet interior. Toilet to be able to withstand a 5000 lb (22 kN) load without deflection and/or damage.
3. Finish: Satin polished finish on exposed surfaces.
4. Configuration: On floor.
5. Cabinet: Rectangular apron, with backsplash.

    a. Water-Closet Bowl Location: Right of center on front of apron.
    b. Toilet-Paper Holder: Recessed; stainless steel located in left side of apron.

6. Optional Features: ADA compliant and ligature resistant.
7. Mounting: Bolts through wall sleeve into accessible service space.
8. Water Closet:

    a. Bowl:

        1) Type: Elongated, with back inlet, integral trap, and blowout design with back outlet and integral contoured seat.
        2) Seat Surface: Satin polished finish.

3) Outlet Connection: NPS 3 (DN 80), horizontal with cleanout and slip joint.

9. Flushometer Valve
10. Lavatory:

    a. Location: In top of cabinet.
    b. Receptor: rectangular bowl with integral soap depression.
    c. Hot- and Cold-Water and Bubbler Supply Valves: Pneumatic, push-button, hot and cold, metering type with individual check stops. Non-hold-open valve.
    d. Filler Spout: Backsplash or deck mounted.
    e. Drain: Integral punched grid with NPS 1-1/4 (DN 32) minimum waste and trap.

11. Wall Sleeve: Galvanized-steel frame of dimensions required to match fixture. Include steel bars or other design to prevent escape if fixture is removed.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Examine roughing-in for water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before fixture installation.

B. Examine walls and floors for suitable conditions where fixtures will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION OF SECURITY PLUMBING FIXTURES

A. Install security plumbing fixtures level and plumb.

B. Install back-access, stainless steel fixtures as follows:

1. Install wall sleeve in wall if indicated.
2. Install fixture on wall sleeve or wall, as indicated, with access from accessible service space.
3. Extend supply piping from service space to fixture.
4. Install soil and waste piping from fixture and extend into service space.
5. Install fixture trap in service space instead of below fixture drain.

C. Install fixture outlets with gasket seals.

D. Install fixtures designated "accessible" in accordance with ICC A117.1 for heights, dimensions, and clearances.

E. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible fixtures. Comply with requirements in Section 22 0700 "Plumbing Insulation."

F. Seal joints between fixtures, floors, and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 07 9219 "Security Joint Sealants."

G. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings. Comply with escutcheon requirements specified in Division 22 "Plumbing."

## 3.3 PIPING CONNECTIONS

A. Connect fixtures with water supplies, stops, and risers and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.

B. Comply with requirements for water piping specified in Section 22 1116 "Domestic Water Piping."

C. Comply with requirements for soil and waste drainage piping specified in Section 22 1316 "Sanitary Waste and Vent Piping."

## 3.4 ELECTRICAL CONNECTIONS

A. Connect wiring in accordance with Section 26 0519 "Low-Voltage Electrical Power Conductors and Cables."

B. Ground equipment in accordance with Section 26 0526 "Grounding and Bonding for Electrical Systems."

C. Install electrical devices furnished by manufacturer, but not factory mounted in accordance with NFPA 70.

D. Install nameplate for each electrical connection, indicating electrical equipment designation and circuit number feeding connection as specified in Section 26 0553 "Identification for Electrical Systems."

## 3.5 ADJUSTING

A. Operate and adjust flushometer valves and flow-control valves on fixtures. Replace damaged and malfunctioning fixtures, fittings, and controls.

B. Adjust water pressure at fixtures to produce proper flow.

C. Install new batteries in battery-powered, electronic-sensor mechanisms.

## 3.6 CLEANING AND PROTECTION

A. After installing fixtures, inspect and replace damaged finishes.

B. Clean fixtures, faucets, and other fittings with manufacturers' recommended cleaning methods and materials.

C. Provide protective covering for installed fixtures and fittings.

D. Do not allow use of fixtures for temporary facilities unless approved in writing by Owner.

END OF SECTION 22 4600

# SECTION 23 00 00 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

B. It is the intent of these Specifications for the Contractor to provide heating, ventilation and air conditioning systems complete, fully operational, fully adjusted, and ready for use.

C. All contract requirements indicated in the Division 21, 22, 23, 25, 26, and 28 documents, including Drawings and Specifications, shall be considered supplemental to and in addition to the contract requirements indicated in the documents of other trades and the Division 01 requirements.

### 1.2 REQUIRED WORK SPECIFIED ELSEWHERE

A. Section 23 00 10 - Division 23 General Requirements
B. Section 23 00 15 - Division 23 Design Conditions
C. Section 23 00 70 - Division 23 Schedule of Submittal Data
D. Section 23 05 13 - Common Motor Requirements for HVAC Equipment
E. Section 23 05 30 - Division 23 Demolition
F. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment
G. Section 23 05 53 - Identification for HVAC Piping and Equipment
H. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
I. Section 23 07 00 - HVAC Insulation
J. Section 23 21 13 - Hydronic Piping
K. Section 23 23 00 - Refrigerant Piping
L. Section 23 30 01 - Ductwork and Sheet Metal
M. Section 23 34 00 - HVAC Fans
N. Section 23 37 00 - Air Outlets and Inlets
O. Section 23 40 00 - HVAC Air Cleaning Devices
P. Section 23 81 26 - Split-System Air-Conditioners
Q. Section 23 82 01 - Electric Heating Coils
R. Section 23 82 33 - Convectors
S. Section 23 82 39 - Unit Heaters

### 1.3 PARTIAL LIST OF WORK NOT INCLUDED IN DIVISION 23

A. Electrical connections for motors
B. Individual motor controllers except motor controllers furnished as integral parts of pieces of equipment
C. Equipment pads
D. Metal louvers
E. Painting (except as otherwise specified herein)
F. Installing access doors (except as otherwise specified herein)

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. General: All equipment shall be of the capacity and type as specified herein, as shown in the Schedules and as indicated on the Drawings. Equipment shall be provided by a specified manufacturer and in accordance with the space limitations of the Project.

B. Single Source: To maximize ease of maintenance and part replacement, equipment of a similar nature shall be provided by a single manufacturer.

C. Approved Equal: Equipment and materials selected by the Contractor within the context of "equal as approved by the Engineer", "approved equal", "equivalent as determined by the Engineer" and similar terminology shall be submitted to the Engineer for review, approval and inclusion into the Contract Documents prior to the finalization of the contract between the Owner and the Contractor, and prior to the shop drawing submittal phase of the Project. All equipment and materials submitted to the Engineer under the terms of "approved equal" during the shop drawing phase of the Project without prior review and approval shall be returned to the Contractor without review under the status of "No Action".

2.2 MATERIAL

A. All material required for a complete and proper installation shall be as specified and as selected by the Contractor subject to the approval of the Architect.

2.3 CONTRACTOR APPROVALS

A. The contractor shall submit in writing and obtain written approval from the Owner, Architect, and/or Engineer for all equipment substitutions, installation deviations from that shown on the Contract Documents, and all other miscellaneous approvals required from the Owner, Architect, and/or Engineer as referenced throughout these Specifications.

B. A request for a substitution or deviation from the Contract Document constitutes that the Contractor:

1. Has investigated the proposed product and determined that it is equal to or superior in all respects to that specified.

2. Will provide the same warranties for the substitution as for the product specified.

3. Will coordinate the installation of the substitution into the work, and make such other changes as may be required to make the work complete in all respects.

4. Waives all claims for additional costs, under his responsibility, which may subsequently become apparent.

5. Will absorb all costs incurred by the substitution when affecting other trades including but not limited to electrical, structural, architectural, etc.

6. Will absorb any cost incurred by the Owner's Representative in review of the substituted product if the acceptance of the substituted item creates the need for system modification and/or redesign, or if the substituting Contractor exhibits negligence in his substituting procedure thus submitting inferior, misapplied or mis-sized equipment.

PART 3 - EXECUTION

3.1 CONDITIONS

A. Inspection: Prior to proceeding with the work of this Division, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation may properly commence. Verify that the work of this Division may be completed in strict accordance with all pertinent codes and regulations, the reviewed shop drawings, and the manufacturers' recommendations.

B. Discrepancies: In the event of discrepancy, immediately notify the Owner's Representative. Do not proceed in areas of discrepancy until all such discrepancies have been resolved. If there is a discrepancy between the Drawings and the Specifications, the Specifications shall typically govern. However, any discrepancy of this type shall be immediately brought to the attention of the Owner's Representative for formal interpretation prior to proceeding with the work.

C. Interpretation of Documents: Any and all contractual requirements may be indicated solely on the Drawings, solely in the Specifications, in both the Specifications and on the Drawings, in reference standards indicated in the Specifications and/or in the Owner's and Contractor's Contract. If Contract requirements are indicated in both the Specifications and the Drawings, the Contractor shall comply with both requirements unless the requirements are mutually exclusive of each other. If Contract requirements are indicated in both a reference standard and the Specifications, the more stringent requirement shall apply. Any and all contractual requirements shall be interpreted within the overall context of the complete scope of work. All materials, equipment, systems and installation methods shall be suitable for the intended service, coordinated with other trades and be complete, fully operational, adjusted, tested and ready for use by the Owner.

3.2 INSTALLATION OF EQUIPMENT

A. Locations: Install all equipment in the locations shown on the reviewed shop drawings, except where specifically otherwise approved on the job by the Architect and/or Owner's Representative.

B. Interferences: Avoid interference with structure and with work of other trades, preserving adequate headroom and clearing all doors and passageways to the approval of the Architect and/or Owner's Representative.

C. Inspection: Check each piece of equipment in the system for defects. Verify that all parts are properly furnished and installed, function properly, and that all adjustments have been made.

3.3 CONNECTIONS TO EQUIPMENT

A. The Contractor shall make final connections to all equipment for a complete and operational system.

3.4 CLOSING-IN OF UNINSPECTED WORK

A. General: Do not allow or cause any of the work of this Division to be covered up or enclosed until it has been inspected, tested, and approved by the Architect, Owner's Representative, and by all other authorities having jurisdiction.

B. Uncovering: Should any of the work of this Division be covered up or enclosed before it has been completely inspected, tested and approved, the Contractor shall provide all services,

labor, materials and equipment necessary to uncover such work without added expense to the Owner. After the work has been completely inspected, tested, and approved, the Contractor shall provide all services, labor, materials and equipment to make all repairs necessary to restore the work to its original and proper condition at no additional cost to the Owner.

3.5 COOPERATION WITH OTHER TRADES

A. Coordinate and cooperate with other trades in order that all systems in the scope of the Contract may be installed in the best arrangement. Coordinate and cooperate as required with all other trades which share space in common areas and maximize the access to each system and their respective components.

3.6 CLEANING

A. It is the intent of these Specifications that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment. Level of cleanliness shall be defined as "broom" clean unless noted otherwise. All exterior surfaces of Division 23 equipment shall be wiped down and cleaned of all dust and dirt. All interior surfaces of equipment shall be wiped down and vacuum cleaned so as to be delivered to the Owner in factory new condition. Surfaces to be painted shall be cleaned and prepared in accordance with architectural division of the contract and as noted in other sections herein.

3.7 COMPLETENESS

A. It is the intent of these Specifications to provide complete systems. Completeness shall mean that all materials, equipment, and systems as installed and operating on this project have been installed properly with the best practices of the trade; are suitable for the intended purpose, location, and environment; properly fit within the physical space limitations for the project; are in conformance with applicable codes and reference standards; have been started-up, tested, adjusted, and commissioned for the intended use; have maintained applicable UL listings; are in compliance with manufacturer's recommendations and warranty requirements; ready for the Owner’s use, and in the opinion of the Architect, performing as designed.

3.8 ADJUSTMENT OF CONTROLS

A. The Contractor shall provide the personnel and equipment to completely adjust the controls to the satisfaction of the Architect. At the completion of the project, the Architect will arrange a meeting at the job site to allow the Contractor to demonstrate the proper operation of the electrical controls.

3.9 NOISE

A. It is the intent of these Specifications to provide a system free from objectionable audible noise and vibration. Any equipment that is generating objectionable noise or vibration, in the opinion of the Architect, shall be corrected and dampened as required to eliminate the objectionable level.

3.10 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. The contractor shall be required to assist, coordinate, and comply with all owner commissioning requirements.

END OF SECTION 23 00 00

SECTION 23 00 10 - DIVISION 23 GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. General: The work under this Section shall conform to the requirements of the "General Conditions" and "Supplementary Conditions" of these Specifications.

B. Work Included: The work included in this Section consists of the general requirements for the work more specifically described in Division 21, 22, 23, 25, 26 and 28 where referenced.

C. Definitions: The Contractor shall provide all supervision, labor, material, equipment, machinery, and any and all other items necessary to complete the mechanical and electrical systems. All items of equipment are specified in the singular; however, the Contractor shall provide the number of items of equipment as indicated on the Drawings, and as required for complete systems.

D. Intent

1. It is the intention of these Specifications and Drawings to call for finished work that has been completed, tested, adjusted, commissioned, fully operational, approved by the local code authorities, and ready for the Owner’s use. Wherever the word "provide" is used, it shall mean "furnish and install complete and ready to use."

2. It is the intention of these specifications and drawings that wherever the words "weather" is used throughout the various Division 21, 22, 23, 25, 26 and 28 specification sections, it refers to "locations exposed to weather". "Locations exposed to weather" shall mean areas exposed to outside environmental temperature, humidity, or moisture conditions. These locations include but are not limited to areas on the exterior of the building, on the roof, within architectural exhaust and intake plenums, within loading docks, within above grade parking garages, within below grade parking structures, and other areas as indicated on the drawings or in the specifications.

3. All labor, materials, apparatus, and appliances essential to the complete functioning of the systems described and indicated herein, or which may be reasonably implied as essential whether mentioned in the Drawings and Specifications or not, shall be furnished and installed by the Contractor.

4. In addition to the requirements indicated throughout the Drawings and Specifications, as completely installed and operating on this project; equipment and systems provided for this project shall meet the following minimum requirements without additional cost to the Project.

    a. Be suitable for the intended service, location, and environment.
    b. Properly fit within the physical space limitations for the project.
    c. Meet the minimum requirements of applicable codes and reference standards as determined and approved by the local code authorities.
    d. Maintain applicable UL listings.
    e. Be coordinated between Divisions 21, 22, 23, 25, 26 and 28 in a timely manner.
    f. Be coordinated with the work of other trades in a timely manner.
    g. Not be purchased or installed prior to the resolution of any conflicts or coordination that could result in additional cost to the Owner.
    h. Be in accordance with the manufacturer's recommendations and warranty

requirements.

i. Be ready for use by the Owner.

5. In cases of doubt as to the Work intended, or in the event of need for explanation thereof, the Contractor shall request supplementary instructions from the Architect.

E. Codes, Rules, Regulations, Permits and Fees

1. Refer to the architectural specification Divisions for additional requirements. The architectural specification Divisions shall govern these requirements. The following information shall be in addition to the architectural specification Divisions.

2. The Contractor shall give all necessary notices, obtain all permits and pay all government sales taxes, fees, and other costs, including utility connections or extensions, in connection with the Contractor's work; file all necessary plans, prepare all documents and obtain all necessary approvals of all governmental departments having jurisdiction; obtain all required certificates of inspection of the Contractor's work and deliver same to the Architect before request for acceptance and final payment for work.

3. The Contractor shall comply with the most recently revised versions of all applicable laws, rules, regulations, and ordinances of Federal, State, and Local Authorities. Modifications required by the above said Authorities shall be made without additional charge to the Owner.

4. The Contractor shall comply with applicable building codes as adopted and modified by the Local Code Authorities. Modifications required by the above said building codes shall be made without additional charge to the Owner.

5. All equipment shall comply with all applicable requirements of laws, codes, ordinances, legislation, etc., of all Federal, State, and Local Authorities, whether indicated on the Contract Documents or not.

6. Where Contract Drawings and Specifications requirements are in excess of Code requirements and are permitted under the Code, the Contract Drawings and Specifications shall govern.

7. All rules and regulations of the State Fire Insurance Regulatory Body, Underwriters Laboratories, Americans with Disabilities Act (ADA), and National Electrical Code shall be complied with whether indicated in the Contract Drawings and Specifications or not.

8. All materials, equipment, and systems provided on this project shall be in compliance with the applicable energy codes and the energy efficiency ASHRAE Standard 90.1 as adopted by the applicable building codes.

9. Project shall meet or exceed the requirements of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality Guideline for Occupied Buildings Under Construction. In addition, all stored on-site or installed materials shall be protected at all times from moisture damage. Provide photographs to the owner, architect, or engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

10. Where alterations to or deviations from the Contract Drawings and Specifications are required by the above Authorities report the requirements to the Architect and secure

Architect's approval before starting the alterations.

11. Third Party Certification: All packaged equipment shall be independently Third Party labeled as a system for its intended use by a Nationally Recognized Testing Laboratory (NRTL), such as UL or ETL, in accordance with OSHA Federal Regulations 29 CFR 1910.303 & 1910.399, as well as NFPA 70 National Electric Code (NEC), Article 90-7.

1.2 QUALITY ASSURANCE

A. Surveys and Measurements

1. The Contractor shall base all measurements, both horizontal and vertical, from established benchmarks. All work shall agree with these established lines and levels. Verify all measurements at site and check the correctness of same as related to the work.

2. Should the Contractor discover any discrepancy between actual measurements and those indicated, which prevents following good practice or the intent of the Drawings and Specifications, the Contractor shall notify the Architect, and shall not proceed with the work until instruction has been received from the Architect.

B. Drawings

1. Drawings are diagrammatic and indicate the general arrangement of systems and work included in the contract. Drawings are not to be scaled. The Architectural Drawings and Details shall be examined for exact location of fixtures and equipment. Where they are not definitely located, this information shall be obtained from the Owner's Representative.

2. The Contractor shall follow Drawings in laying out work and check Drawings of other trades to verify spaces in which work will be installed. Maintain maximum headroom and space conditions at all points. Where headroom or space conditions appear inadequate, the Owner's Representative shall be notified before proceeding with installation.

3. If directed by the Owner's Representative, the Contractor shall, without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of the work, if such directions are made prior to the performance of the affected work.

C. Cooperation with Other Trades

1. The Subcontractors shall give full cooperation to other trades and shall furnish in writing to the General Contractor any information necessary to permit the work of all trades to be installed satisfactorily and with the least possible interference or delay.

2. Where Division 21, 22, 23, 25, 26 and 28 work will be installed in close proximity to, or will interfere with, work of other trades, composite shop drawings will be prepared to accomplish coordination among trades in working out space conditions. All necessary adjustments to work out space conditions shall be reflected on the composite shop drawings. The Contractor shall prepare composite shop drawings and sections at suitable scale not less than 1/4" = 1'-0", clearly showing how work of all affected trades is to be installed. If work is installed before coordinating with other trades, or so as to cause any interference with work of other trades, then the necessary changes in the work shall be made to correct the condition without additional expense to the Owner.

3. The Contractor shall obtain and transmit between affected trades, all necessary templates, patterns, setting plans, and shop details for the proper installation of work and for the purpose of coordinating adjacent work.

D. Protection

1. The Contractor shall protect all work and material from damage by the Contractor's work or workmen, subcontractors' work or workmen, and shall be liable for all damage thus caused.

2. The Contractor shall be responsible for work and equipment until "Final Review"; the Contractor shall protect work against theft, injury or damage; and shall carefully store material and equipment received on site which are not immediately installed. The Contractor shall close open ends of work with temporary covers or plugs during storage and construction to prevent entry of obstructing material.

3. All stored on-site or installed materials shall be protected at all times from moisture damage.

E. Material and Workmanship

1. Work shall be executed in strict accordance with the best practice of the trades in a thorough, substantial, workmanlike manner by competent workmen. The Contractor shall furnish the services of a full-time, experienced superintendent, who shall be constantly in charge of the installation of the work, together with all skilled workmen, fitters, metal workers, welders, helpers, and laborers required to unload, transfer, erect, connect, adjust, start, operate, and test each system.

F. Manufacturers' Recommendations

1. In addition to the requirements indicated in the drawings and specifications; manufactured products, articles, materials, and equipment shall be applied, installed, connected, erected, used, cleaned, conditioned, inspected, started-up, tested, operated, and commissioned in accordance with the manufacturer's current printed recommendations. Keep copies of such printed recommendations at job site and make them available as required. When conditions on this project are not covered by the manufacturer's printed recommendations, at the discretion of and as requested by the owner's representative, the manufacturer's authorized representative shall review the conditions and provide written supplemental recommendations to address the special situation. If the manufacturer’s recommendations are in conflict with the requirements of the drawings and specifications, the Contractor shall advise the Owner’s representative of the conflict prior to purchase and installation.

G. Space Limitations

1. MEP systems and equipment shall be provided and installed to accommodate the structural systems, architectural systems, MEP systems, other trades, and the physical space limitations of the project.

2. Equipment shall be provided which will fit into the physical spaces provided and indicated, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearance in accordance with the Code requirements, the requirements of the local inspection department, and the equipment manufacturers’ recommendations.

3. In the preparation of Drawings, a reasonable effort to accommodate equipment manufacturers' space requirements has been made. However, since space requirements, equipment arrangement, access and service clearances, conduit, piping, ductwork and other connections, manufacturers' recommendations, and other variables vary according to each manufacturer; due to the multiple manufacturers listed for each product and the possible number of different equipment combinations; the responsibility for access, proper fit, coordination, and accommodating the space limitations of the project rests with the Contractor. Refer to shop drawing submittal requirements herein for additional requirements.

4. MEP systems and equipment shall be installed to maximize headroom clearance below the bottom of material. In areas with ceilings or future ceilings provided by tenants; MEP systems and equipment shall be installed to maximize vertical clearances between the bottom of material and top of specified ceiling construction. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, lighting, and other MEP systems shall be prepared to indicate compliance with the Space Limitations and Maximum Headroom Clearance requirements. Conflicts with vertical and horizontal clearances, parking spaces, drive aisles, ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data, clouded, and clearly marked so as to bring the conflict to the attention of the Architect in a timely manner prior to installation. For the purpose of definition, the term MEP refers to all systems, material, equipment, services, means and methods described the Division 21, 22, 23, 25, 26 and 28 documents including the Division 21, 22, 23, 25, 26 and 28 specifications and drawings. Refer to the specifications for additional requirements.

5. Refer to the drawings for additional requirements.

6. Physical dimensions and arrangements of equipment to be installed shall be subject to the Architects' review.

H. Coordination with Installation Conditions: The requirements of the Division 21, 22, 23, 25, 26 and 28 systems shall be coordinated with the installation conditions. The Contractor shall visit the site, survey the installation conditions, and verify exact measurements. Means and methods of building access, rigging, and modifications to installation conditions to accommodate the installation of new Division 21, 22, 23, 25, 26 and 28 work shall be the responsibility of the Contractor. Disassembly, reassembly, and certification testing of equipment as required to accommodate the installation conditions shall be included in the base bid. Prior to the ordering, purchasing, and fabricating of Division 21, 22, 23, 25, 26 and 28 systems, coordination with installation conditions shall be completed by the Contractor.

I. Coordination Between Divisions 21, 22, 23, 25, 26 and 28: The Division 23 Contractors shall cooperate with the General Contractor and the Division 21, 22, 25, 26 and 28 Contractors to provide coordination between the Division 21, 22, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 23, it shall be the responsibility of the Division 23 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

J. Commissioning:

1. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience, and expertise at the general contractor and sub-contractor levels to properly install, check, verify, supervise, start-up, trouble-shoot, de-bug, pre-test, commission, and make the necessary corrections to all MEP systems in accordance with the Contract Documents. The Contractor shall be responsible for the operational demonstration of the completed MEP systems to the Owner's Representative, Architect and Engineer. The Contractor shall complete the MEP systems in accordance with the contract documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems prior to requesting that the Owner's Representative, Architect, and Engineer attend the operational demonstrations of the MEP systems. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner. Refer to the various sections of the specifications for additional commissioning requirements.

K. Additional Information:

1. Information provided by the Engineer to the Contractors during construction must be reviewed and approved by the Architect and Owner's Representative. The Contractor shall not proceed without prior review and authorization from the Architect and Owner's Representative. The Contractor shall not proceed with work that will increase the cost of the project or impact the project schedule without prior review and authorization from the Architect and Owner's Representative. If information provided by the Engineer to the Contractor will increase the cost of the project or impact the project schedule, the Contractor shall advise the Architect and Owner's Representative in a reasonably timely manner so as to minimize the cost and impact to the project. Refer to the General Conditions of the Contract and the Architectural documents for additional information.

1.3 SHOP DRAWINGS, PRODUCT DATA AND SAMPLES

A. After the Contract is awarded but prior to proceeding with the Work requiring final reviewed shop drawings, product data and samples, the Contractor shall provide the Architect and Engineer with a submittal log schedule indicating when each submittal is to be sent to the Architect and Engineer for review in order for the Architect and Engineer to schedule personnel and plan accordingly. Unless otherwise indicated, a minimum period of two (2) weeks, exclusive of transmittal time, will be required in the Engineer's office each time a shop drawing, product data and/or samples are submitted or resubmitted for review. This time period will commence once the shop drawing, product data and/or samples are received in the Engineer's office contingent upon the shop drawing, product data and/or samples being received in the Engineer's office before 3:00 PM. If the shop drawing, product data and/or samples are received in the Engineer's office after 3:00 PM, the time period will not start until the next business day. This time period shall be considered by the Contractor when scheduling the Work. The submittal log schedule should be updated and reissued by the Contractor on a regular basis to avoid submittal review delays. If a submittal arrives before or after the scheduled time indicated within the Contractor's submittal log schedule, or if a number of submittals arrive at the same time without scheduled notification, the Architect and Engineer may not be able to review all of the submittals within the minimum two (2) week time period. The Contractor shall obtain complete shop drawings, product data and samples from the manufacturers, suppliers, vendors, subcontractors, and sub-subcontractors for all materials, systems and equipment specified herein; and submit data and details of such materials, systems and equipment for review by the Architect and Engineer. Prior to submission of the shop drawings, product data and samples to the Architect and Engineer, the Contractor shall review and certify that the shop drawings, product data and samples are in compliance with the

Contract Documents. The Contractor shall check all materials and equipment after their arrival on the job site and verify their compliance with the Contract Documents. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience and expertise at the general contractor and sub-contractor levels to properly check, verify, and correct if necessary shop drawing submittal data to ensure compliance with the Contract Documents prior to delivery to the Architect and Engineer. The Division 23 shop drawing submittal requirements, including but not limited to the Division 23 certification requirements, shall be in addition to the requirements indicated in the Architectural Divisions of the Contract Documents.

B. The Architect's approval and Engineer's review of shop drawings, product data, and samples shall not relieve the Contractor of the responsibility for dimensions or errors that may be contained therein, or for deviations from requirements in the Contract Documents. It shall be clearly understood that the Engineer's noting some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents shall govern the Work and are neither waived nor superseded in any way by the review of shop drawings, product data and samples.

C. Before submitting shop drawings, product data and samples the Contractor shall certify that data and details set forth on each shop drawing, product data, or samples for each item of equipment and materials complies with the Contract Documents for this Project. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the Contractor, or by a person duly authorized to sign binding agreements for the Contractor. Unless certified, shop drawings, product data and samples will not be reviewed, and will be returned unchecked to the Contractor. Certifications shall be in the form of rubber stamp impressions or a digitally certified stamp which state:

> I hereby certify that this shop drawing, product data, and/or sample has been checked prior to submittal and that it has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents and physical space limitations for the Project.
>
> (Name of the Mechanical Subcontractor)
>
> Signed:_____
>
> Position:_____
>
> Date:_____

D. The Contractor shall submit shop drawings, product data and/or samples, equipment and other items listed in Section 23 00 70 and for any other equipment for which such submittals may be required in various Sections of the Specifications. Prior to submitting shop drawings, product data and samples for all items of equipment listed in Section 23 00 70 and shown with an asterisk (*), each submittal package including subsequent submittal packages shall be certified by the manufacturer as to performance and compliance with Contract Documents. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the manufacturer, or be a person duly authorized to sign for the Owner, Partner, or corporation. Unless certified, shop drawings, product data and/or samples will not be reviewed and will be returned unchecked to the Contractor. The certification shall be as follows:

> I hereby certify that the equipment shown on this

shop drawing, product data and/or sample, has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents for this Project. I further certify that all data shown herein as to performance dimensions, construction, materials and other pertinent items is true and correct.

(Name of the Manufacturer)

Signed:_____

Position:_____

Date:_____

E. Inadequate or incomplete shop drawings, product data and/or samples will not be reviewed by the Architect or the Engineer and shall be returned to the Contractor for resubmittal.

F. The Contractor shall observe the following rules when submitting shop drawings, product data and samples:

1. Identification: Each shop drawing shall indicate in the lower right hand corner, and each product data brochure shall indicate on the front cover the following: title of the sheet or brochure; name and location of the Project, names of the Architect and Engineer, Contractor, Subcontractor, manufacturer, supplier, vendor; the date of submittal; and the date of each correction and revision. So far as is practical, each shop drawing, product data and/or sample shall bear a cross-reference note to the page number or numbers of sheets of the Drawings and/or Specifications showing the Work. Unless the above information is included, the submittal will be returned for resubmittal.

2. Quality Assurance: Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does Not Comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark "C" or "NC" by each specification paragraph. A handwritten marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

3. Shop drawings shall be done in an easily legible scale and shall contain sufficient plans, elevations, sections and schematics to clearly describe the apparatus. Drawings shall be prepared by an engineer-draftsman skilled in this type of work. Unless otherwise indicated, all sheet metal, piping and similar shop drawings shall be drawn to at least 1/4" = 1'-0" scale. Typical floor sheet metal ductwork shop drawings may be drawn at 1/8" = 1'-0" scale. The Contractor shall submit shop drawings as described below. Shop drawings which do not comply with these requirements will be returned for resubmittal. Refer to "Cooperation with Other Trades" in this section, for additional composite shop drawing requirements. Unless otherwise indicated in the Contract Documents, the

submittal shall consist of three prints of the shop drawings. The Architect and Engineer will each retain one print and the remaining copies will be returned to the Contractor.

4. Product data to be submitted shall be published by the manufacturers and shall contain complete and detailed engineering and dimensional information. The Contractor shall submit product data as described below. Product data which does not comply with these requirements will be returned for resubmittal. Unless otherwise indicated in the Contract Documents, the Contractor shall submit eight (8) copies of each product data brochure. The Architect and Engineer will each retain one copy of the submittal and the remaining copies will be returned to the Contractor. Product data submitted shall contain only information relevant to the particular equipment or material to be furnished. The Contractor shall not submit catalogs which describe several different items in addition to those items to be used, unless all irrelevant information is marked out, or unless relevant information is clearly marked. The submittal shall consist of product data from each manufacturer. Product data from each manufacturer shall be identified and submitted separately.

5. Coordination between Division 21, 22, 23, 25, 26 and 28: Each Division 23 shop drawing and product data submitted for review, which has Division 23 equipment requiring electrical service, shall contain a statement indicating that the Division 23 Contractor has provided the Division 26 Contractor with the necessary electrical characteristics of the actual Division 23 equipment to be provided.

6. Shop Drawing and Product Data - Submittal Review Status:

    a. Shop drawing and product data submittals will be returned marked with one of the following status:

        1) "REVIEWED NO EXCEPTIONS"
        2) "REVIEWED EXCEPTIONS NOTED"
        3) "INFORMATION ONLY"
        4) "REVISE AND RESUBMIT"
        5) "REJECTED"

    b. If stamped "REVISE AND RESUBMIT" or "REJECTED", the shop drawing or product data shall be revised in accordance with the Contract Documents. If marked "REVIEWED, EXCEPTIONS NOTED" documentation that addresses only the noted exceptions in the form of a letter, supplemental information, or complete resubmittal shall be forwarded to the Owner, Architect and Engineer for record purposes only. If marked "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" no additional submittal will be required.

    c. If the copy stamped "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" is altered for any reason after it has been stamped, the "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" shall automatically be voided and the submittal status revised to "REVIEWED EXCEPTIONS NOTED", "REVISE AND RESUBMIT", or "REJECTED" depending on the alterations.

    d. All work shall be done in accordance with shop drawings stamped "REVIEWED NO EXCEPTIONS", "REVIEWED EXCEPTIONS NOTED", or "INFORMATION ONLY" insofar as these are in agreement with the Contract Documents. Wherever differences occur between the shop drawings and the Contract Documents, the Contract Documents shall govern the Work.

G. Record Shop Drawings

1. After receiving the Engineer's and Architect's final approval for each shop drawing, the Contractor shall provide the Engineer with AutoCAD files of each final approved shop drawing. All of the previous review comments by the Engineer and Architect associated with each shop drawing shall be incorporated into the AutoCAD files of each shop drawing prior to sending the AutoCAD files to the Engineer. The AutoCAD files shall be in AutoCAD 2015 version software or later version as approved by the Engineer. The Contractor shall be responsible for contacting the Engineer to determine the approved AutoCAD software to use prior to preparing the AutoCAD files.

H. Maximum Ceiling Clearance and Headroom Requirements

1. All material for systems and equipment shall be installed to accommodate the structural systems, architectural ceiling construction, base building and future tenant lighting systems, and other physical space limitations of the project. All systems and equipment shall be installed to maximize clearances between the bottom of material and top of ceiling construction. All systems and equipment shall be installed to maximize headroom clearance below the bottom of material. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, and other systems that will be located where ceiling plenum clearances and headroom clearances are limited; shall be prepared to address these issues. Conflicts with ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data and clearly marked so as to bring the situation to the attention of the Engineer.

I. Equipment and Layout Deviations

1. Layout, connections, supports, access & maintenance clearances, related architectural enclosures, and structural supports have been based on selected products and manufacturers. In some cases, the manufacturer's model number has been provided. Other products by specified manufacturers may be used if they comply with requirements of the contract documents including but not limited to capacity ratings. However, where the Contractor proposes to use an item of equipment that has alternative but equivalent minimum capacity ratings or that requires any redesign of the structure, partitions, foundations, piping, wiring or any other part of the mechanical, electrical, structural, civil, utility, or architectural systems; all such redesign, and all new drawings and detailing required therefore, shall be prepared by the Contractor at the Contractor's own expense as approved by the Owner's Representative. Where such approved deviation requires a different quantity, capacity, and arrangement of ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, and starters from that specified or indicated on the drawings, the Contractor shall furnish and install any such ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, starters and any other additional equipment required by the system, at no additional cost to the Owner.

1.4 RESPONSE TO SHOP DRAWING, SUBMITTAL, AND FIELD REPORT REVIEW COMMENTS

A. Contractor shall respond in written form to each shop drawing, submittal, or field report review comment from the Engineer within 10 business days from the date indicated on the Engineer's review form. Contractor's written response to each review comment should indicate the date and specifically what corrective action was or will be taken and include the name of the person and associated company responding to each review comment. Refer to the Architectural documents for additional requirements.

1.5 MOCK-UP REQUIREMENTS

A. Contractor shall provide the following mock-up requirements specified herein. Mock-up requirements shall be completed after the applicable shop drawing review but prior to the fabrication and installation of the multiple systems, equipment, and accessories throughout the facility which are to be included as part of the mock-up requirements. Mock-up arrangements shall be located on site within the facility at an actual useable location.

B. The Contractor shall advise the Engineer and Owner's Representative after the mock-up is complete and ready for review and inspection. The Contractor shall arrange a time mutually agreeable to these parties so they can meet at the project site, review the mock-up installation, and determine any changes that need to be made for the installation to be acceptable to the Owner's Representative and Engineer. The Contractor shall provide advance notice to the appropriate parties of the various mock-up inspections a minimum of seven working days prior to the meeting.

C. The Contractor shall make adjustments to the mock-up arrangement as requested by the Owner's Representative and Engineer prior to the fabrication and installation of the remaining systems, equipment, and accessories throughout the facility which are included as part of the mock-up arrangement. The Contractor shall account for this requirement in the schedule of construction, so this procedure does not delay the construction progress. If multiple systems, equipment, and accessories which are included as part of the mock-up arrangement are installed in multiple locations throughout the facility prior to the mock-up approval, the Contractor shall be responsible for the remedial work required to comply with the approved mock-up condition at no additional cost to the Owner.

D. Mock-up arrangements shall be located on site within the facility at an actual useable location. The Contractor shall provide advance notice to the appropriate parties of the various mock-up inspections a minimum of seven working days prior to the meeting. If additional or follow-up field inspections of the mock-up modifications are required, the Contractor shall provide these modifications and additional follow-up field inspections as required without additional cost to the Owner.

1. Primary Supply Air Ductwork Insulation Mock-Up Requirements: Refer to Section 23 30 01 for additional information.

2. Perimeter Slot Diffuser Mock-Up Requirements: Refer to Section 23 37 00 for additional information.

1.6 WARRANTY, SERVICE, AND REPLACEMENT PARTS

A. All equipment supplied for this Contract shall be free from defects in material, workmanship and title; shall be of the kind and quality described herein. If it appears within one (1) year from the date of the building final acceptance by the Owner that the equipment does not meet the warranties specified above or the equipment is recalled by the equipment manufacturer, the Contractor shall correct any defect or replace any recalled equipment, including nonconformance with these Specifications, at no additional cost to the Owner.

B. The manufacturers for each of the various selected Division 23 pieces of equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner's acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

1.7 CONFERENCE PRIOR TO START OF WORK

A. Immediately upon the award of this Contract, but prior to commencing any Work, the Contractor together with designated major Subcontractors shall confer with the Owner's Representative, the Architect and Engineer concerning the Work under this Contract.

B. Contractors and Subcontractors attending this meeting shall include but not be limited to the following: General; Mechanical; Plumbing; Fire Protection; Automatic Temperature Controls and Building Management; Sheet Metal; Insulation; Electrical; Fire Alarm; and Testing, Balancing and Adjusting. Contractor and Subcontractor representatives shall be familiar with the Drawings, Specifications and shop drawing submittal requirements and be prepared to intelligently discuss the project requirements.

C. The conference will be at a mutually agreed place and acceptable time.

1.8 ELECTRONIC FILE TRANSFER AGREEMENT

A. Release of the Dewberry CAD, BIM, and other electronic files will require the User of the files to sign the Dewberry Electronic File Transfer Agreement.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials and equipment shall be new and in good condition. The commercially standard items of equipment and the specific names mentioned herein are intended to identify standards of quality and performance necessary for the proper functioning of the Work.

B. Since manufacturing methods vary, reasonable minor variations are expected; however, performance and material requirements specified are the minimum standards acceptable. The Engineer retains the right to judge the quality of equipment that deviate from the Contract Documents.

2.2 PRODUCTS, COMPONENTS AND MATERIALS IN PLENUMS

A. All products, components, materials and composite materials within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723 as required by applicable mechanical code.

2.3 EQUIPMENT GUARDS

A. Provide equipment guards for the safe operation of equipment as recommended by the equipment manufacturer and as required by applicable codes. Provide easily removable expanded metal guards for all belts, couplings, exposed fan inlets and outlets, and other moving parts of machinery. Provide tachometer openings in the guards at least 2" in diameter, for all belt-driven or variable speed machinery.

2.4 EQUIPMENT MOUNTING/SUPPORT DEVICES

A. Unless otherwise required to accommodate the particular requirements and unique conditions of this project, all floor, wall or ceiling mounting/support devices shall be factory fabricated

devices as provided by the equipment manufacturer for the intended application. In addition to the factory-fabricated devices, the Contractor shall provide additional support members, framework, mounting hardware, brackets, steel angles, anchors, restraints, support stands, attachment devices, and other miscellaneous mounting and support devices as required for adequate equipment support.

B. The Engineer retains the right to judge the quality of the design, application, and installation of all field and shop fabricated mounting/support devices using steel angle, steel channels, Uni-Strut, brackets, all-thread rods, etc. Any mounting or support devices that will rust, corrode, or in the opinion of the Engineer become unsightly or a maintenance problem will not be acceptable.

C. Support devices not factory finished shall be hot dipped galvanized or properly primed and painted with a minimum of two coats of rust inhibiting paint. Refer to the Architectural Painting Systems for additional requirements.

D. All-thread rods shall be galvanized, or cadmium plated.

E. All plywood used as mounting boards in association with Division 23, unless specified otherwise by the Architect, shall be APA C-D plugged interior 5/8" to 3/4". All plywood used shall be painted (front, back and edges) with a minimum of one prime coat and one finish coat prior to equipment installation. Refer to the Architectural Specifications for additional requirements. Color shall be determined by the Architect.

2.5 EQUIPMENT PADS AND ANCHOR BOLTS

A. Concrete pads for various pieces of equipment will be furnished under another Division. Pads shall be provided in main machine rooms, all areas with floors below grade, upper level pump and storage tank rooms, penthouse equipment rooms, and in any other location as shown on the drawings. This shall include floor mounted equipment, equipment mounted on legs and pipe support stands. Unless otherwise noted, equipment pads shall generally conform to the shape of the piece of equipment it serves with a minimum 3" margin around the equipment and supports. Pads shall be a minimum of 4" high and made of a minimum 28 day, 2500 psi concrete reinforced with 6"x6" 6/6 gauge welded wire mesh. Top and sides of pad shall have smooth finishes, equal to those of the floors, with all external corners either bull-nosed to a 3/4" radius or provided with a 3/4" chamfer. Shop drawings stamped "FINAL REVIEW" or "REVIEWED EXCEPTIONS NOTED" shall be used for dimensional guidance in sizing pads. Pads for each water chilling unit in equipment rooms shall extend 6" beyond the footprint of each water chilling unit.

B. Unless noted otherwise furnish and install galvanized anchor bolts for all equipment placed on concrete equipment pads or on concrete slabs. Minimum dimension between anchor bolts and edge of slab condition shall not be less than 3". Bolts shall be of the size, type and number recommended by the manufacturer of the equipment and shall be located by means of suitable templates provided by the equipment manufacturer. When equipment is placed on vibration isolators, the equipment shall be secured to the isolator and the isolator secured to the floor, pad, or support as recommended by the vibration isolation manufacturer.

2.6 ACCESS DOORS

A. Wherever access is required through walls, ceilings or fire-rated enclosures to valves, fire dampers, heating coils, damper actuators, drain line cleanouts, or other concealed equipment installed under this Division, unless otherwise noted in the architectural documents, the Contractor shall furnish a hinged access door and frame as follows:

1. Drywall Construction - Milcor Style DW.

2. Finished Acoustical Tile Ceiling - Milcor Style AT in applications where there is not adequate clearance space to properly push an acoustical ceiling tile up for removal and replacement.

3. Finished Plaster Ceilings - Milcor Style AP.

4. Finished Plaster Walls or Ceramic Tile - Milcor Style M.

5. Plaster or Masonry Walls and Ceilings outside offices and in other finished areas exposed to view - Milcor Style K or M.

6. Access doors shall be "B" label fire construction where required.

7. Access doors shall be installed under another Division.

8. No access door shall be installed until location and type have been approved by the Architect.

2.7 ESCUTCHEONS

A. Furnish and install heavy chrome-plated or nickel-plated steel plates of approved pattern on all pipes passing through walls and ceiling in finished areas. Escutcheons shall be B & C No 10 or approved equal with concealed hinges. Pattern shall be approved by the Architect.

2.8 OVERCURRENT PROTECTION

A. Fused and circuit breaker type overcurrent protection devices required for any equipment furnished under Division 23 shall be in accordance with Division 26 Specification Section 26 28 00.

B. Unless specifically noted otherwise on the Drawings or in these Specifications, all Division 23 equipment shall be rated for use with both fused and circuit breaker overcurrent protection. This requirement shall be clearly indicated on the manufacturers' nameplate.

2.9 SEALANTS AND ADHESIVES

A. All sealants and adhesives shall comply with the current Volatile Organic Compound (VOC) content limits of the latest edition of the South Coast Air Quality Management District (SCAQMD) Rule 1168 titled "Adhesive and Sealant Applications". Piping sealants shall have no more than 250 grams/liter VOC content for all sealants and adhesives to be used within the building interior locations.

B. The Contractor shall submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each adhesive and sealant used within the building interior locations.

PART 3 - EXECUTION

3.1 SCAFFOLDING, RIGGING, HOISTING

A. The Contractor shall furnish all scaffolding, rigging, hoisting, and services necessary for erection and delivery of any equipment and apparatus furnished to the job site. Remove same from job

site when no longer required.

3.2 SLEEVES, CUTTING AND PATCHING

A. The Contractor shall be responsible for the timely placing of openings and sleeves for all piping passing through walls, partitions, beams, floors, louvers, and roofs while the same are under construction. A pipe sleeve shall be round and one size larger than the size of pipe it serves except where "Thunderline/Link Seal" casing seals are used in penetrations through walls below grade. If the pipe is insulated, its pipe sleeve shall be one size larger than the outside diameter of the insulation around the pipe. All sleeves in floors shall extend two inches above the finished floor. All pipes passing through concrete or masonry walls above grade shall have 18 gauge galvanized steel sleeves or plastic "hole-outs". All sleeves set in concrete floor construction shall be standard weight galvanized steel. Sleeves set in concrete floor construction directly supporting riser piping 4" and larger shall be standard weight galvanized steel with three 6" long, 3/8" diameter steel reinforcing rods welded at 120 degree spacing to the sleeve and shall be installed with the rods embedded in the concrete slab. Where the pipe passes through a sleeve, no point of the pipe or its insulation shall touch the sleeve. Caulk around such pipe with sufficient quantity of fire safing insulation to equal fire rating of construction penetrated by the sleeve. In addition, seal the fire safing material with a non-hardening mastic.

B. Seal all pipe penetrations and unused sleeves located in fire rated construction with factory built devices or with manufactured fill, void, or cavity materials "Classified" by Underwriters Laboratories for use as a Through-Penetration Firestop. All firestop devices and systems shall be approved for such use by the authority having jurisdiction. The fire safing system used shall maintain the fire resistance rating of the building component that is penetrated. Fire safing systems and devices shall comply with ASTM E814-13a and UL 1479. Submittal data for fire safing systems shall include the UL System Numbers listed in the UL Building Materials Directory under which the material was tested in accordance with ASTM E814-13a and UL 1479 for use in a Through-Penetration Firestop System. Excessive shrinkage of the fire stopping materials which would permit the transmission of smoke or water prior to exposure to a Fire condition is unacceptable. Where a mastic coating is used to seal the surface of the firestop, the mastic shall be non-hardening. The fire safing system used shall accommodate expansion and contraction of the floating mechanical piping systems without damaging the firestop or reducing its effectiveness as a smoke barrier or water seal. See paragraph titled "Vibration Isolation for Piping" in Section 23 05 48.

C. Sleeves penetrating walls below grade shall be "WS Wall Sleeves" by GPT (formerly Pipeline Seal & Insulator) with 2" water seal stop. The water seal stop shall be located in the middle of the wall. Seal off annular opening between pipe and sleeve with "Thunderline/Link-Seal" casing seal as manufactured by GPT. The pipe sleeve shall be sized to accommodate the "Thunderline/Link-Seal" casing seal. Casing seals shall be Model "C" and shall be sized per the manufacturer’s recommendations.

D. If holes and/or sleeves are not properly installed and cutting and patching becomes necessary, it shall be done at no expense to the Owner. The Contractor shall undertake no cutting or patching without first securing the Architect's written approval.

E. Sleeves in floor slabs with waterproof membranes shall be modified as required to accept a Zurn Z-197 riser sleeve and also meet all other sleeve requirements as specified herein before. The Z-197 riser sleeve shall be constructed of dura-coated cast iron body, double threaded pipe connector with combination membrane flashing flange.

F. The location and size of all penetrations and openings through existing walls, slabs, beams, roofs and other existing building construction shall be provided by the contractor to

accommodate the new work. Penetrations and openings through existing building construction shall be reviewed by the structural engineer and architect prior to cutting the penetration. Unless otherwise required by the architectural and structural divisions, metal sleeve inserts will not be required in pipe penetrations except as required to meet UL fire rating requirements. The contractor shall perform a structural survey, including x-rays or graphs of all floor slabs and supporting walls to locate existing structural bars, cables, in slab conduits and any other interferences that may be in conflict with proposed sleeve/penetration locations. The x-ray or graph shall be performed prior to any cutting or chipping of structural supporting walls. In the event of a conflict, the contractor shall notify the architect prior to proceeding with the work.

G. Provide sleeves for future use as indicated on the drawings. Provide a minimum of two 6" sleeves in each AHU room for the installation of future piping risers.

3.3 EXCAVATION AND BACKFILL

A. The Contractor shall make all necessary excavations, cutting of paving, concrete, etc., and do all backfilling and paving repairs necessary for the proper execution of the Work. Remove all dirt and debris out of and away from the building as directed. Backfill shall be mechanically compacted to a density of 95% of the maximum dry density at optimum moisture content as determined by the Standard Proctor Compaction Test.

B. Backfill shall be compacted and repairs to paving or concrete shall be accomplished to the satisfaction of the Architect and the Local Authorities.

C. See the various Sections of the Division 22 for additional excavation and backfill requirements.

3.4 PAINTING

A. Painting, except as specified or indicated otherwise, shall be done under another Division. This Division shall cooperate with the other Division to determine the size of equipment, sizes and lengths of pipes, etc., to be painted.

B. Equipment furnished under this Division shall be factory finished. If the factory finish is damaged during shipment, installation, etc., it shall be repainted subject to the Architect's approval.

C. Refer to Section 23 05 53 for additional requirements.

3.5 DRAYAGE AND HAULING

A. Include all drayage, hauling, hoisting, and placement in the building of equipment specified herein. The Contractor shall be responsible for the timely introduction of equipment to the Project. If any item of equipment is received prior to the time it is required, the Contractor shall be responsible for its proper storage and protection until such time as it may be required. The Contractor shall pay for all costs of demurrage or storage.

3.6 IDENTIFICATION OF EQUIPMENT

A. Equipment and piping shall be identified in accordance with the requirements indicated in specification Section 23 05 53.

B. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings;

wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; software, programming, and graphics.

## 3.7 EQUIPMENT NOISE AND VIBRATION

A. It is the intention to specify and for the Contractor to provide equipment and systems that, as defined herein, shall be quiet and free of apparent vibration in operation. To this end, both the balancing of rotating machinery and the installation of vibration isolation at various locations are required.

B. It shall be the responsibility of the Contractor to obtain equipment that is quiet in operation as compared to other available equipment of its size, capacity, and type; to install equipment so that a minimum amount of noise and/or vibration is transmitted to the building; and to fabricate the duct system so that air noises generated in the system are held to a minimum.

C. Any additional precautions deemed necessary to provide a quiet installation shall be done as part of the Work of the Contractor, subject to review by the Engineer, without additional cost to the Owner. After the system is in operation, it shall be the responsibility of the Contractor to make any changes to equipment or Work installed that may be required to provide a system which is quiet in operation as defined herein.

D. Air distribution equipment shall be sound tested as specified in Section 23 37 00. The test data shall be expressed in sound power levels with decibels referenced to ten watts to the negative twelfth power and in octave bands as set forth in Acoustical Society of America (ASA) Standard S1.6. The sound testing shall be performed in a reverberant room. The room effect defined as the sound power level (dB re $10^{-12}$ watts) minus sound pressure level (dB re 0.002 microbar) shall be based on "average" room surfaces and room construction as set forth in the ASHRAE Handbook Fundamentals and shall in no case be more than 10 dB. The additive allowance for multiple air terminals (air outlet): for the purpose of the sound test shall be considered as 5 dB in determining the sound power level per air terminal. The final sound pressure level test results shall be based on the above requirements and shall not exceed the noise criteria curve limitations specified herein.

E. The system noise level shall be equal to or less than those indicated in Section 23 00 15. Sound levels within the occupied spaces must meet the criteria described above and with all building, wall partition, floor and ceiling construction in place as they exist for the individual spaces. The attenuation through boundary construction of equipment rooms must be considered in selecting equipment for acceptable noise level.

## 3.8 CONTROL COORDINATION

A. It shall be the responsibility of any equipment manufacturer who furnishes equipment for this Project to coordinate and cooperate with the requirements of Division 25. Equipment automatically and remotely controlled and/or interlocked with some other piece of equipment as specified shall be complete with all necessary contact points and interface devices to accomplish the control strategies and sequences described. It shall be the responsibility of all applicable equipment manufacturers to fully participate in the coordination procedure. By accepting this Contract; the Contractor, all Subcontractors, and all applicable equipment manufacturers expressly agree to cooperate and expedite the obtaining of all necessary information.

3.9 LUBRICATION

A. Provide means for lubricating all bearings and other machine parts. If a part requiring lubrication is concealed or inaccessible, extend a lubrication tube with suitable fitting to an accessible location and suitably identify it.

B. After installation, properly lubricate all parts requiring lubrication and keep them adequately lubricated until final acceptance by the Owner.

3.10 SPARE PARTS

A. General: The Contractor shall provide all necessary spare parts for the equipment supplied in this Project as indicated hereafter and in various other Sections of the Specifications. Spare parts shall be obtained from the original equipment manufacturer.

B. Air Handling Unit and Fan Belts: Provide a matched set of spare belts for each fan/motor drive combination supplied for this project. Each spare belt set shall be bound together and clearly labeled with the tag number of their respective fan or AHU. Provide a summary sheet to the owner with equipment tag, belt, and filter sizes.

3.11 LOUVERS

A. Louvers shall be provided under the Architectural Division of this Contract. The exact size, location and type of louver shall be as indicated on the Architectural Documents. Unless otherwise indicated on the drawings, all louvers shown on the Division 23 Drawings are based on a minimum free area of 50% and a maximum static pressure loss of 0.16" wg at an air velocity through the free area of 1,000 fpm and maximum static pressure loss of 0.04" wg at an air velocity through the free area of 500 feet per minute. The louver static pressure loss characteristics shall be proportionally lower at lower velocities and proportionally higher at higher velocities. The Mechanical and General Contractors shall be responsible to verify these minimum performance levels are clearly met and indicated by the manufacturer in the louver submittal data. Provide a bird screen mounted on the inside of each louver. At locations where ductwork is connected to a louver for either intake or exhaust purposes, ductwork shall be installed, sloped, and connected to the louver so that all water entering the ductwork system positively drains back to and out of the louver. Refer to Section 23 30 01 for additional requirements.

B. Provide a sheet metal plenum connected to the inside portion of the louver to match the louver size indicated on the mechanical documents. This plenum shall be 36" deep unless otherwise indicated. Where the actual louver size indicated on the architectural documents exceeds the dimensions of the louver size indicated on the mechanical documents, the Contractor shall have the option of either increasing the size of the plenum to match the installed louver or provide insulated blank off panels at the unused portions of the louver. The insulated blank off panels shall meet the insulation values of the wall adjacent to the louver. Unless otherwise indicated on the drawings, plenums located within conditioned spaces shall be insulated with fiberglass flexible blanket type insulation as specified for rectangular sheet metal ductwork in Section 23 30 01. Unless otherwise indicated on the drawings or in Section 23 30 01, ductwork connected to plenums, up to and including the portion containing the associated automatic damper, shall also be insulated with fiberglass flexible blanket type insulation as specified for rectangular sheet metal ductwork in Section 23 30 01. Refer to the Architectural documents for additional information.

3.12 DOCUMENTATION

A. For all testing, training and review procedures the Contractor shall maintain adequate records in written form that clearly documents the event. The written documentation for each testing, training and review procedure shall include but not be limited to the following: Name and company of attending personnel, general description of procedure(s), date(s), time of day, deficiencies found, corrective action taken, and nature of final results. Documentation records shall be kept in an orderly, legible and up to date format. Contractor's documentation records shall be accessible for review by the Owner Representative upon request. Each testing, training and review procedure document shall bear the name and signature of the authorized representative of the General Contractor, Mechanical Contractor, and manufacturer or Subcontractor if applicable.

3.13 COMPLETION, COMMISSIONING AND FINAL REVIEW OF DIVISION 23 SYSTEMS

A. Start-up: After the Contractor and equipment manufacturer representative determine that the equipment and related systems have been properly installed and are ready for start-up; the equipment manufacturer representative shall perform the manufacturer’s start-up and check-out procedures to ensure proper operation of the equipment in accordance with the contract documents and the manufacturer’s recommendations. Start-up and check-out of each piece of equipment shall be in accordance with the equipment manufacturer’s recommendations and warranty requirements. Start-up services shall be performed by factory-trained personnel of the equipment manufacturer or factory-trained personnel authorized by the manufacturer and their official representatives. Start-up personnel shall be certified by the equipment manufacturer as properly trained in the manufacturer’s start-up and check-out procedures. Provide documentation of the start-up and check-out procedures in accordance with the contract documents and the manufacturer’s recommendations. Documentation of the start-up and check-out procedures shall be included within and submitted with the Operational and Maintenance Manuals.

B. Trouble-shooting, Debugging, and Pre-testing: After Start-Up but prior to the Operational Demonstrations; the contractor shall perform operational pre-testing, trouble-shooting, and de-bugging of all equipment and systems. At the completion of the Trouble-shooting, Debugging, and Pre-testing process, all equipment and systems shall be complete and operating in accordance with the requirements of the contract documents. Trouble-shooting, Debugging, and Pre-testing of equipment and systems shall be performed to the extent necessary for the Contractor to determine that all systems are complete and operating in accordance with the Contract Documents. Prior to final Commissioning and Operational Demonstrations all equipment and systems shall be certified by the Contractor as complete and operating in accordance with the Contract Documents.

C. Operational Demonstrations and Final Commissioning: Prior to Substantial Completion and within a schedule mutually agreed to by the contractor and the Owner’s Representative, the Contractor shall demonstrate the operation of all equipment and systems as necessary to demonstrate compliance with the Contract Documents. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner.

D. Final Review and Owner Acceptance:

1. After completion of the Operational Demonstrations, Final Commissioning, and Owner’s Close-out Package requirements; at a time designated and mutually agreed by the Contractor and Owner’s Representative; a final walk-through review of the project will be performed by the Contractor and Owner’s Representative. The review will be of a general

nature with the intent of addressing outstanding issues prior to the Owner's final acceptance of the project.

2. The entire system shall be clean and operating properly with all systems balanced and all controls adjusted. All labels shall be removed from the plumbing fixtures, and the fixtures shall be clean and in operating condition.

3. Certificates, Documents, and Close-out Package required herein shall be complete, in order and presented to the Architect at least two weeks prior to the review.

4. After the review, any changes or corrections noted as necessary for the Work to comply with these Specifications and the Drawings shall be accomplished without delay in order to secure final acceptance of the Work.

E. Schedule: The Contractor shall complete the MEP systems in accordance with the Contract Documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems a minimum period of time prior to the date of "Substantial Completion" as established by the Commissioning Agent. This period of time prior to the date of "Substantial Completion" shall be reserved by the Contractor for the Owner's Representative, Commissioning Agent, Architect, and Engineer to witness the operational demonstrations of the completed fully commissioned MEP systems. If more than this established time period is deemed necessary by the Contractor for the operational demonstrations, the Contractor shall notify the Owner's representative and Commissioning Agent in written form and determine an operational demonstration schedule mutually agreeable to the Owner's Representative, Commissioning Agent, and Contractor that completes the operational demonstrations prior to "Substantial Completion". Additional costs incurred by the Owner related to reviewing, witnessing, testing, inspecting, and commissioning of the mechanical systems by the Commissioning Agent, Architect and Engineer after the date of "Substantial Completion" shall be reimbursed by the Contractor to the Owner. As part of the shop drawing submittal process, the Contractor shall provide a detailed schedule of completion indicating when each MEP system will be complete and ready for owner acceptance. The schedule shall indicate time periods and milestone dates of Start-up; Trouble-shooting, Debugging, and Pre-testing; Operational Demonstrations and Final Commissioning; Final Review and Owner Acceptance.

F. Additional Requirements:

1. Owner's Close-out Package: Refer to the Owner's Close-out Package requirements within Section 23 00 10 for additional requirements.

2. Refer to the various sections of the specifications for additional requirements.

3.14 OPERATING INSTRUCTION

A. The Contractor shall provide the services of a general HVAC equipment operations specialist to supervise the operation of all equipment specified herein and to instruct the Owner's operators for a five-day, eight hours per day, operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. See individual Sections for additional instructions by the manufacturer's trained specialists.

B. Training schedule, duration, and subject matter shall also be in accordance with a training syllabus provided by owner's operating staff. This syllabus will be provided prior to the final review.

C. The Owner shall be notified in writing at least five days before each operating instruction period begins. The Contractor shall not commence instruction period until the Owner has issued a written acceptance of the starting time.

3.15 OPERATING AND MAINTENANCE BOOKS

A. The Contractor shall provide operating instruction and maintenance data books for all equipment and materials furnished under this Division.

B. Submit three (3) hard copies and three (3) electronic copies of the operating and maintenance data books for review at least eight weeks before final review of the Project. Assemble all data in a completely indexed volume or volumes and identify the size, model, and features indicated for each item. The Engineer’s review period for the operating and maintenance data books shall be a minimum of 4 weeks, exclusive of transmittal time. The Contractor’s schedule shall plan to accommodate this review period requirement.

C. Maintenance instruction manuals shall include complete oiling, cleaning, and servicing data compiled in clearly and easily understandable form. Data shall show all model numbers of each piece of equipment, complete lists of replacement parts, motor ratings, and actual loads.

D. Include the following information where applicable:

1. Identifying name and mark number as indicated on the Drawings and in the Specification.

2. Locations (where several similar items are used, provide a list).

3. Complete nameplate data.

4. Certified record drawings and "Final Reviewed" or completely corrected shop drawings and/or submittals.

5. Parts lists.

6. Performance curves and data including air and water design flow rate set points.

7. Wiring diagrams.

8. Lubrication charts.

9. Start-up and Check-out procedures and check-lists.

10. Manufacturers' recommended operating and maintenance instructions with all non-applicable information deleted. This shall include but not be limited to the manufacturer's service and repair manual.

11. Additional information as recommended by the equipment manufacturer or system provider for the proper operation, maintenance, and warranty of the equipment and systems.

3.16 RECORD DRAWINGS

A. The Contractor shall maintain on a daily basis at the project site a complete set of "Record Drawings". The "Record Drawings" shall consist of a set of black line prints and AutoCAD files of the Contract Drawings for this Division. The black line prints shall be marked and the

AutoCAD files shall be electronically updated regularly by the Contractor during the construction period to reflect an accurate dimensional record of all buried or concealed work. In addition, the set of black line prints shall be marked and the set of AutoCAD files shall be electronically updated regularly to show the precise location of concealed work and equipment including buried, concealed or embedded piping, conduit and valves and all changes and deviations in the mechanical work from that shown on the Contract Documents. This requirement shall not be construed as authorization for the Contractor to make changes in the layout or work without definite instructions from the Architect. The set of black line prints and AutoCAD files of the Contract Drawings for this Division shall remove or blacken out the Engineer's seal and Engineer's firm name.

B. Prior to commencing work, the Contractor shall need to purchase from the Architect or Engineer a set of black line prints to be used for the "Record Drawings" and obtain from the Engineer a set of AutoCAD files of the Contract Drawings for this Division in preparing "Records Drawings" for the project based on the Contractor's work. The AutoCAD files shall be in AutoCAD 2015 version software or later version as provided by the Engineer. In order to obtain a set of AutoCAD files of the Contract Drawings for this Division from the Engineer, the Contractor shall need to contact the Engineer, Architect or Client and obtain a written release and indemnification statement from the Engineer. This release and indemnification statement shall be signed by an owner, partner, corporate officer or other person duly authorized to agree to and sign legally binding documents for the Contractor's respective company. Upon the Engineer receiving the signed unedited release and indemnification statement back from the Contractor, the Engineer will forward the AutoCAD files of the Contract Drawings for this Division to the Architect or Client who in turn will forward the AutoCAD files of the Contract Drawings for this Division to the Contractor for his use in preparing "Records Drawings" for the project based on the Contractor's work. If the Contractor desires to have architectural and structural information in preparing "Records Drawings" for the project, the Contractor shall need to contact the Architect and Structural Engineer to obtain the pertinent architectural and structural information directly from the Architect and the Structural Engineer.

C. Record dimensions shall clearly and accurately delineate the Work as installed; locations shall be suitably identified by at least two dimensions to permanent structures.

D. The owner's representative shall retain the right to periodically review the "Record Drawings" as the project progresses. Final acceptance of as-built "Record Drawings" shall be conditional upon the Contractor clearly "red-marking" all changes from the contract documents, including "request for information" changes and other supplemental instruction changes, directly on the black line set of prints and clouding all of the changes in the electronic AutoCAD files. All of the clouded changes on each drawing in the electronic AutoCAD files shall be on a separate drawing layer which can be turned off without affecting the appearance of each drawing.

E. The Contractor shall mark all "Record Drawings" on the front lower right hand corner with a rubber stamp impression or an AutoCAD image that states the following:

"RECORD DRAWINGS (3/8" high letters)
To be used for recording Field Deviations and
Dimensional Data only" (5/16" high letters)

F. Upon completion of work, the Contractor shall certify all "Record Drawings" on the front lower right hand corner adjacent to the above marking with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWING CERTIFIED CORRECT (3/8" high letters)

(Name of Mechanical Subcontractor)

By:_____

Date:_____

G. For all black line prints to be used for the "Record Drawings", the Contractor must sign the above certification. For all AutoCAD files to be used for the "Record Drawings", the Contractor must fill in the name of the person responsible for certifying the above certification. Both the black line prints and the AutoCAD files shall be certified by the same person.

H. Prior to final acceptance of the Work of this Division, the Contractor shall submit properly certified "Record Drawings" to the Architect for review and shall make changes, corrections, or additions as the Architect may require to the "Record Drawings". After all changes have been made to the "Record Drawings", the Contractor shall deliver to the Owner the final set of "Record Drawings" consisting of the black line prints and the AutoCAD files. The AutoCAD files to be delivered by the Contractor to the Owner shall be in AutoCAD 2015 version software or later version as originally provided to the Contractor by the Engineer.

## 3.17 HVAC USE DURING CONSTRUCTION

A. The permanent HVAC systems serving the building may be used to circulate both supply air and return air throughout the building during the construction period only if the following conditions are adhered to:

1. A complete air filtration system with a MERV 8 rating based on ASHRAE Standard 52.2 is installed and properly maintained at the return air inlet of each air handling unit being used to circulate air.

2. No permanent air diffusers are used to circulate air.

3. No plenum type return air system is to be used.

4. The HVAC use during construction shall be in accordance with the standards set forth by the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality (IAQ) Guideline for Occupied Buildings Under Construction.

5. The complete HVAC ductwork system being used is properly sealed to prevent the spread of airborne particulate and other contaminants.

6. Following the final air flush-out of the entire building as described below, all ductwork systems used to circulate air are internally vacuumed with portable High-Efficiency Particulate Air (HEPA) vacuums and are documented clean in accordance with the National Air Duct Cleaners Association (NADCA) specifications.

## 3.18 FINAL AIR FLUSH-OUT

A. At the completion of all construction within the building and prior to occupancy of any interior/tenant space within the building, the Contractor shall operate all of the HVAC systems serving the building at maximum design capacity for a period of not less than two weeks to provide a final air flush-out of the entire building in accordance with ASHRAE Standard 62.1. During this final air flush-out period, the outside air system shall be operating at 100% of its design capacity. Prior to performing this final air flush-out, the Contractor shall need to obtain

approval from and coordinate the time period for this final air flush-out with the Owner's representative and the Engineer.

3.19 OWNER'S CLOSE-OUT PACKAGE

A. Contractor shall provide a master project close-out log and corresponding schedule for submission to the Owner and Engineer for review and approval.

B. Contractor shall provide the Owner with two (2) copies of all warranties bound in a notebook with a corresponding warranty log. Warranty log shall clearly indicate the subcontractor, address, telephone number, and contact person.

C. Contractor shall provide the Owner with one (1) complete set of approved submittals and a copy of the final submittal log.

D. Contractor shall provide the Owner with a complete list of all subcontractors and major suppliers with address, telephone number, and contact persons.

E. Contractor shall provide the Owner with a complete inventory of attic stock materials and spare parts as applicable, and a copy of the transmittal signed by the Owner's representative acknowledging receipt of the items.

F. Contractor shall provide the Owner and Engineer with documentation of compliance for all specified requirements associated with equipment start-up, commissioning, demonstrations, training and warranties. Documentation shall include, but not be limited to, equipment start-up sheets as completed by the manufacturer's representative. The Owner reserves the right to conduct performance tests on one or more HVAC units selected at random after the equipment start-up is complete. If the units fail to perform as specified for any reason, the Contractor shall be required to test all of the HVAC units and perform the necessary corrective work as required in order to have the HVAC units comply with the Contract Documents.

G. Contractor shall provide the Owner with approved air and water balancing reports including one (1) set of reproducible air balance drawings, three (3) copies of air and water balance reports, and two (2) sets of air and water balance prints.

H. Contractor shall provide the Owner and Engineer with a certified duct leakage test report for all ductwork.

I. Contractor shall provide the Owner with a comprehensive valve tag chart.

J. Contractor shall provide the Owner and Engineer with certification that the HVAC water piping has been properly cleaned and flushed. The Contractor shall advise the Owner of the time and date of the pipe cleaning, flushing, and sterilization procedures a minimum of two weeks prior to the procedures. The Owner shall have the option of witnessing the complete procedures. Any cleaning, flushing, and sterilization procedures performed prior to Owner notification, at the Owner's discretion, shall be performed again in the presence of the Owner.

K. Contractor shall provide the Owner and Engineer with a fire damper matrix in accordance with the requirements of NFPA 90A. The fire damper matrix shall include the location and size of the fire damper, the date tested, the person performing the test, and the signature of the Owner's representative who witnessed the test of each fire damper.

L. When the Contractor considers the work to be ready for final acceptance, the Contractor shall provide the Owner and Engineer with written certification stating the following:

1. All work has been completed in accordance with the Contract Documents.

2. All field review items and other deficiencies identified by the Certificate of Substantial Completion have been properly corrected.

3. All work has been inspected for compliance with the Contract Documents.

4. All mechanical systems have been tested in the presence of the Owner's representative and are fully operational in compliance with the Contract Documents.

5. As-Built documents have been reviewed, are complete and are accurate.

M. The Contractor shall provide the Owner with a letter from the Engineer of Record stating that the equipment installed fully complies with the Contract Documents and that the HVAC testing and balancing results have been reviewed and are in full compliance with the Contract Documents.

3.20 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. The contractor shall be required to assist, coordinate, and comply with all owner commissioning requirements.

END OF SECTION 23 00 10

SECTION 23 00 15 - DIVISION 23 DESIGN CONDITIONS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 DESIGN CONDITIONS

A. Outside Design Conditions:

| Description | Dry Bulb | Wet Bulb |
|---|---|---|
| Summer Outside Air Temperature | 95°F | 78°F |
| Winter Outside Air Temperature | 10°F | - |
| Coincident Summer Outside Air Condition for Conditioning Ventilation Outside Air | 93°F | 75°F |

B. Cooling Inside Design Conditions:

| Description | Dry Bulb | Wet Bulb |
|---|---|---|
| Offices and Lobbies | 73°F | 58.5°F * |
| Conditioned Equipment Rooms | 80°F | 68°F * |
| Other Interior Cooled Spaces | 73°F | 58.5°F * |

* Unless otherwise noted in the Contract Documents, humidification and humidity control are not provided.

C. Heating Inside Design Conditions:

| Description | Dry Bulb | Wet Bulb |
|---|---|---|
| Offices and Lobbies | 70°F | 55°F * |
| Heated Equipment Rooms and Soffit Areas | 65°F | * |
| Other Interior Heated Spaces | 70°F | 55°F * |

* Unless otherwise noted in the Contract Documents, humidification and humidity control are not provided.

D. Noise Criteria (NC): Range of indoor design goals for HVAC sound control for all occupied spaces shall have an NC curve range not to exceed the values indicated herein.

1. All office areas, and public spaces: NC 35 to 40.

2. All areas in close proximity of the rooftop units or AHUs: NC 40 to 45.

3. Design NC sound values indicated for tenant occupied spaces are based on the tenant spaces having an acoustical suspended ceiling installed and a ceiling cavity depth determined by the building standard ceiling height. The NC values are based on tenant spaces with light fixtures, grilles, diffusers, and other devices that affect the acoustic performance of the ceiling assembly positioned to minimize the transfer of sound energy to the occupied space. Design NC values are also based on the tenant space construction and the fan powered terminal devices installed and operating in accordance with the terminal device manufacturer's recommendations and acoustic installation guidelines.

E. Refer to the Drawings, specification section Automatic Temperature Control System, and Schedules for additional design conditions.

END OF SECTION 23 00 15

SECTION 23 00 70 - DIVISION 23 SCHEDULE OF SUBMITTAL DATA

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Furnish the shop drawing submittal data for the Architect's and Engineer's review as indicated herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Sections - Division 21, 22, 23, 25, 26 and 28

PART 2 - SHOP DRAWINGS AND SUBMITTALS

2.1 GENERAL

A. The submittal data to be furnished for this Project shall comply with the Specifications hereinbefore. The referenced Sections herein scheduled are for general information purposes only and shall not be construed to be limiting to the submittal data scheduled and/or required within the individual Sections of these Specifications.

2.2 CONTRACT REQUIREMENTS

A. Shop drawing submittal data shall clearly and concisely address all Contract Document requirements. Standard manufacturer data that does not indicate these requirements shall be supplemented with additional information provided by the Contractor or equipment manufacturer. Refer to sections 23 00 10 for additional requirements, including submittal log schedule requirements.

2.3 DEVIATIONS FROM THE CONTRACT DOCUMENTS AND PERTINENT DATA

A. All deviations from the contract documents shall be noted in the form of a cover letter included with the shop drawing submittal package. All pertinent data such as material options, equipment arrangements, control options, diagrams, etc. that applies to this Project shall be clearly marked. All data and diagrams that do not apply to this Project shall be omitted from the package or clearly marked out.

2.4 QUALITY ASSURANCE

A. Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does Not Comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark "C" or "NC" by each specification paragraph. A handwritten marked-up specification section will be acceptable. Along with this

specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

2.5 SPECIAL REQUIREMENTS

A. The shop drawing submittals for the Operation and Maintenance (O&M) Manuals shall require an Owner/Consultant review period of four (4) weeks. The Contractor shall plan for and arrange the shop drawing submittal review schedule to accommodate this requirement.

2.6 SPARE PARTS

A. The submittal data for each piece of equipment shall include a detailed itemized list of recommended spare parts. Spare parts shall be clearly identified by name and part number.

2.7 ORGANIZATION OF PACKAGES

A. Unless otherwise noted in the specifications and in the following schedule of shop drawing, the Contractor shall provide a single shop drawing submittal package for each of the listed specification sections. Unless specifically noted otherwise, multiple submittal packages per section will not be acceptable under the terms of the Contract.

2.8 CONTROL COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

2.9 COORDINATION BETWEEN DIVISIONS 21, 22, 23, 25, 26 AND 28

A. The Division 23 Contractors shall cooperate with the General Contractor and the Division 21, 22, 25, 26 and 28 Contractors to provide coordination between the Division 21, 22, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 23, it shall be the responsibility of the Division 23 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

2.10 MANUFACTURER'S CERTIFICATION

A. All items or equipment listed herein with asterisks (*) shall be certified by the manufacturer. Refer to Section 23 00 10 for certification requirements.

2.11 SHOP DRAWING SUBMITTAL SCHEDULE

B. Provide a comprehensive list of Divisions 21, 22, 23, 25, 26 and 28 shop drawing submittal packages and a schedule indicating the date when each package will be submitted to the contractor and the date when each package will be submitted to the design team. The intent of this list and schedule is to allow the contractor and design team to plan ahead and schedule the necessary review and coordination resources in advance of the arrival of shop drawing

submittal packages. Shop drawing submittal schedule shall be submitted to the Architect not less than four weeks prior to the submission of the first Division 21, 22, 23, 25, 26 and 28 shop drawing submittal packages. Shop drawing submittal schedule shall be periodically updated by the contractor and issued to the Architect to reflect current status of the project. Shop drawing submittal packages shall be listed on the schedule not less than four weeks prior to submission to the contractor and not less than six prior to the submission to the design team. Packages not listed on this schedule will required up to two additional weeks of review time.

2.12 SHOP DRAWING SUBMITTAL DATA

A. The submittal data to be furnished for this Project shall include but not be limited to the following:

1. Section 23 00 10 - Overcurrent Protection: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

2. Section 23 00 70 - Shop drawing submittal schedule

3. Section 23 05 13 - AC Electric Motors: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

4. Section 23 05 13 - Motor Controllers: Data shall be submitted with associated equipment. Separate submittal package not required for this section.

5. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment *

6. Section 23 05 93 - Air Balancing Procedures and Recording Forms

7. Section 23 05 93 - Testing, Adjusting, and Balancing HVAC Report for entire project

8. Section 23 07 00 - HVAC Insulation

9. Section 23 21 13 - Hydronic Piping

10. Section 23 30 01 - Ductwork and Sheet Metal

11. Section 23 30 01 - Dampers

12. Section 23 34 00 - HVAC Fans *

13. Section 23 37 00 - Air Outlets and Inlets *

14. Section 23 40 00 - HVAC Air Cleaning Devices *

15. Section 23 81 26 - Split-System Air-Conditioners *

16. Section 23 82 01 - Electric Heating Coils *

17. Section 23 82 33 - Convectors

18. Section 23 82 39 - Unit Heaters

19. Drawing Packages:
    a. Sheet metal ductwork drawings shall be limited to a maximum of two (2) packages.

b. Mechanical drawings shall be limited to maximum of two (2) packages.

2.13 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. The contractor shall be required to assist, coordinate, and comply with all owner commissioning requirements.

END OF SECTION 23 00 70

SECTION 23 05 13 - COMMON MOTOR REQUIREMENTS FOR HVAC EQUIPMENT

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

B. Furnish and install all electric motors required for equipment furnished under this Division. Electric motors shall be factory-mounted on equipment wherever possible and shall be constructed as specified in this Section. In general, all motor controllers will be furnished and installed under Division 26 unless the motor controller is an integral part of a piece of equipment and/or specifically required to be furnished under this Division.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Sections pertaining to equipment with electric motors and controllers.

1.4 REFERENCE STANDARDS

A. DOE Code of Federal Regulations (CFR)
1. 10 CFR 431 Subpart B - Electric Motors
2. 10 CFR 431 Subpart X – Small Electric Motors

B. Institute of Electrical and Electronics Engineers (IEEE)
1. 112 Standard Test Procedure for Polyphase Induction Motors and Generators

C. National Electrical Manufacturers Association (NEMA)
1. MG 1 Motors and Generators

D. National Fire Protection Association (NFPA)
1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems
2. 70 National Electrical Code (NEC)

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawings for all equipment requiring motors and controllers shall be submitted for review. Drawings shall state the motor manufacturer, motor horsepower, voltage, speed, efficiency, insulation class, motor winding material, special torque requirements, and other special requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. If they comply with these Specifications, the following manufacturers will be acceptable.

B. Motors

a. Baldor-Reliance/ABB
b. Siemens

c. Century
d. General Electric
e. Louis Allis
f. Toshiba
g. Weg

2.2 MOTORS

A. Voltage and Motor Horsepower: In general, motor voltages shall be as follows, unless specified or indicated otherwise.

1. 3/4 hp and larger: 480 volt, 3 phase, 60 hertz

2. Smaller than 3/4 hp: 120 volt, 1 phase, 60 hertz

3. All fan powered terminal device motors shall be single phase AC or DC ECM.

B. Motor Efficiency

1. Motors shall meet or exceed the NEMA "Premium Efficiency" standards and nominal full load efficiency values as required by applicable codes, currently adopted Table 12-12 of NEMA MG 1, and as defined in accordance with the currently adopted NEMA standards. Motor efficiency shall be determined and rated in accordance with IEEE 112 Test Method B.

2. Motors shall comply with the currently adopted DOE Energy Policy Act (EPACT), the Energy Independence and Security Act (EISA) and Code of Federal Regulations (CFR).

C. Motor Full Load Current:

1. Motor nameplate full load current shall not exceed the full load current listed in the National Electrical Code regardless of number of poles, motor speed or power factor.

D. Construction:

1. All motors will be started across-the-line, unless specified otherwise. Motors shall be selected with low starting current and shall be designed for continuous duty to provide the running torque and pull-in torque required to suit the load. Unless otherwise specified all motors shall be single speed, 1750 rpm. All motors shall have standard open drip proof (ODP) type enclosure unless otherwise specified. All motors exposed to the weather shall have totally enclosed fan cooled (TEFC), totally enclosed air over (TEAO) or totally enclosed non-ventilated (TENV) type enclosures. All motors, except TEFC, TEAO and TENV types, shall have a service factor of at least 1.15.

a. All motors shall have standard open drip proof (ODP) type enclosure unless otherwise specified. Single phase motors shall have cast aluminum end bells with steel frames. 3 phase motors 10 hp and smaller shall have cast aluminum end bells with steel frames. 3 phase motors 15 hp and larger shall have cast iron housings.

b. All motors exposed to the weather or as specified shall have totally enclosed fan cooled (TEFC), totally enclosed air over (TEAO) or totally enclosed non-ventilated (TENV) type enclosures. Motors shall have cast iron housings, Class F insulation, epoxy varnish coated for extra protection, cadmium plated hardware, stator and rotor to be epoxy enameled for extra protection, and corrosion resistant fan (TEFC only).

E. Single Phase Motors

1. Unless otherwise noted, single phase motors shall be capacitor start type with starting, pull-in, and running characteristics to suit the load.

2. 1/8 horsepower and smaller motors on direct drive fan applications may be of the shaded pole type.

3. Motors larger than 1/8 horsepower up to and including 1/3 horsepower on direct drive fan applications may be of the permanent split capacitor type.

F. Variable Speed Motors

1. Motors used for variable speed operation shall be specifically designed and UL listed for variable speed applications.

2. Motors shall be inverter duty type and comply with the requirements of NEMA Standard MG 1 Part 31.

3. Motors shall be selected for compatibility with specified variable speed drive.

4. Motors shall be designed and selected for the intended load and application.

5. Motors shall be selected for quiet operation throughout the entire operational range of the equipment. Quiet operation shall be defined as follows:

   a. Noise from a motor when operating at any point in the variable speed mode shall not increase more than 3 dB in any noise criteria (NC) curve frequency band when compared to the same motor operating in the full bypass non-variable speed mode.

   b. Noise from a motor, as installed as required by the Contract Documents, shall not cause the sound in areas outside of the air handling unit room to exceed the established NC curve range. This requirement applies to variable speed, non-variable speed, start-up and shut-down modes of operation.

6. Motors shall be designed and selected to operate at any continuous duty point from 10% to 100% of rated motor speed without motor damage and motor overheating.

7. Motors used in conjunction with variable speed drives shall be provided with a shaft grounding kit/bearing protection ring to prevent variable speed drive induced electrical current damage to the motor.

8. It shall be the joint responsibility of the motor manufacturer, equipment manufacturer, and variable speed drive manufacturer to coordinate and cooperate in meeting these requirements. The equipment manufacturer and drive manufacturer shall provide the motor manufacturer with all information necessary, such as fan and pump curves, to insure proper motor selection.

G. Electronically Commutated Motor (ECM):

1. ECM shall be DC brushless type with permanent magnet rotor and copper winding stator. Internal motor electronics shall convert AC power to DC power to operate the Motor.

2. ECM shall be complete with and operated by an integrated microprocessor based controller that sequentially energizes and de-energizes each winding of the stator with

pulsed voltage to spin the rotor and vary the speed.

3. ECM shall be permanently lubricated with heavy-duty ball bearings to match the load and pre-wired to the specific voltage and phase with overload protection.

4. ECM shall be constant flow type unless noted otherwise. Motor shall automatically adjust speed to maintain constant flow regardless of changes in pressure. Constant torque type shall not be provided.

5. Motor speed shall be controllable down to 20% of full speed (80% turndown). Load shall be controlled by a potentiometer dial mounted on the motor.

6. Motor efficiency shall not be less than 80% at all speeds.

7. Nameplate: A motor nameplate shall be securely affixed to each motor and shall clearly indicate the class of insulation, winding material, efficiency, power factor and service factor in addition to the usual electrical data.

8. Special Requirements: Refer to various Sections of this Division for special requirements for specific items of equipment requiring motors.

H. Nameplate: A motor nameplate shall be securely affixed to each motor and shall clearly indicate the class of insulation, winding material and the service factor in addition to the usual electrical data.

I. Special Requirements: Refer to various Sections of this Division for special requirements for specific items of equipment requiring motors.

PART 3 - EXECUTION

3.1 MOUNTING

A. Electric motors furnished under Division 23 shall be factory mounted unless noted otherwise.

END OF SECTION 23 05 13

SECTION 23 05 30 - DIVISION 23 DEMOLITION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements", "Conditions of the Contract" and "Supplementary Conditions". Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Furnish all labor, equipment and materials required to demolish, cap, patch, repair and dispose of the Division 23 systems no longer required as part of an active system of the Project as specified herein and as indicated on the Drawings.

B. It shall be the responsibility of the Contractor to review the Architectural Documents in addition to the Division 21, 22, 23, 25, 26 and 28 Documents to determine the complete scope of work.

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Under this Division, the Contractor shall furnish all labor, equipment, appliances, and materials to perform all operations in connection with the demolition of the Division 23 systems as indicated on the Drawings and as specified herein and/or as required by the Local Building Code. The work shall include but not be limited to the following:

1. The contractor shall review the existing conditions, in addition to the Architectural documents, and the Division 21, 22, 23, 25, 26 and 28 documents, to determine the complete scope of work.

2. Field verify the exact sizes and locations of all existing ductwork and piping prior to demolition of any existing work. The demolition work shall be coordinated with the architectural scope of the Project to assure proper limits of demolition.

3. Remove all existing mechanical equipment, ductwork, and associated piping within the area to be demolished, whether specifically indicated or not, that is no longer required unless otherwise indicated.

4. Openings remaining in ductwork as a result of demolition shall be sealed with an airtight sheet metal cap, unless otherwise indicated.

5. Where existing equipment is to be relocated, extreme care shall be taken to prevent damage during the removal and reinstallation. Where damage occurs, the equipment shall be replaced or repaired to the satisfaction and approval of the Architect at no additional cost to the Owner.

6. Unless otherwise indicated, all existing air distribution devices in the area to be demolished shall be removed and stored in a manner to prevent loss or damage in the construction staging area for reuse.

7. Unless otherwise indicated, all existing thermostats located throughout the area to be demolished shall be removed and stored in a manner to prevent loss or damage in the construction staging area for reuse. Note, if an existing thermostat serves a piece of HVAC equipment that is to remain, the existing thermostat shall need to remain and be protected or be stored in a manner to prevent loss or damage for reuse.

8. Unless otherwise indicated, all existing perimeter slot diffusers, mechanical system sheet metal ductwork, and terminal devices within the area to be demolished shall be retained in place for reuse.

9. Contractor shall be responsible for cutting, patching and restoration of existing finishes.

10. Contractor shall be responsible for capping existing piping and/or ductwork to remain.

11. Contractor shall be responsible for removal, disposal and salvage of equipment from job site.

12. Contractor shall be responsible for cleaning, testing and repair of existing equipment, ductwork and piping to be reused.

13. Contractor shall be responsible for sealing all existing floor slab penetrations in order to maintain the integrity of the fire rating of the existing floor slabs.

14. Refer to Architectural documents and the Division 21, 22, 23, 25, 26 and 28 documents for the area and extent of the demolition.

B. Where a system to be demolished must remain operational throughout the construction process until a new system that will take its place is functional, the contractor shall provide all products, services, materials, labor and supervision to maintain the operational status of the affected system.

## 3.2 EXISTING CONDITIONS

A. Under the scope of work, the Contractor shall perform a comprehensive field inspection of the existing conditions to thoroughly familiarize himself with the Project and determine the complete scope of new and demolition work. The Contractor shall verify in the field exact sizes, capacities, and locations of all existing equipment, ductwork, and piping prior to installation of any proposed work. Proposed work shall be coordinated with existing conditions to assure proper installation. This field survey and determination of the complete scope of work shall be performed prior to the final bid. Specific attention is brought to physically larger and heavy pieces of equipment that are new to be installed or existing to be removed and the routing of such equipment with the available means and site conditions.

## 3.3 CONFLICTS AND DEPARTURES

A. Conflicts: The Contractor shall remove all existing equipment, piping, ductwork, and associated accessories whether specifically indicated or not, that are no longer required. Any equipment, ductwork, or piping which must remain as part of an active system and is in conflict with the proposed work shall be relocated at no additional expense. There shall be no additional expense to the Owner for this type of demolition work, unless, in the Architect's and Owner's opinion, it is beyond the scope and intent of the Contract Documents.

B. Departures: The Contract Drawings are diagrammatic and schematic in nature and do not indicate inferred details. Any departures from the Drawings that are necessary to comply with the intent of Documents, requirements of applicable Building Codes or the Authority having

jurisdiction shall be detailed in the form of shop drawing submittal data and submitted for review by the Architect and Engineer. All demolition and any such revisions for departures shall be made in accordance with the reviewed shop drawings without increased expense to the Owner. Departures shall not be made without prior written approval by the Architect.

3.4 COOPERATION

A. The Contractor shall comply with all the requirements of the Local Code Authority and the building Owner.

B. The Contractor shall give advance notice to the Architect when work is to be performed.

C. The Contractor shall cooperate and coordinate demolition of Division 23 systems as required with all of other trades. The Contractor shall supervise and assist in the removing and replacing of existing materials for installation of mechanical items and items related to all other trades.

3.5 OCCUPIED AREAS

A. Normal functions of occupied areas must continue during the construction phases. Every effort shall be made to ensure such functions are not disturbed. The Contractor shall be responsible for the installation and removal of temporary systems as required to maintain the functions of occupied areas. The Contractor shall schedule any work which may be required in occupied areas during unoccupied hours.

3.6 BARRIERS

A. Where required by applicable health and safety regulations, the Contractor shall furnish and maintain safety and dust barriers. If necessary, the Contractor shall furnish and maintain temporary fencing and traffic barriers in accordance with all applicable health and safety regulations and as deemed necessary.

3.7 RESTORATION

A. Demolition shall be carried out with caution to assure that existing conditions to remain will be undamaged. Existing conditions to remain that are damaged or defaced by work under this contract shall be restored or replaced equal to the conditions at the time of award of the Contract.

B. Cut, patch, and restore all existing surfaces not receiving new finishes that have been disturbed during execution of this Contract. Materials and finishes used shall be similar, in all respect, to adjacent surfaces.

C. Existing piping and ductwork which are to remain as part of an existing or proposed active system that the Contractor determines to be defective (leaks, corrosion, cracks, etc.) shall be brought to the attention of the Architect.

D. Existing equipment to be removed shall be cleaned, repaired, and reused at the discretion of the Architect wherever applicable.

3.8 CONNECTIONS AND CAPS

A. Connections to existing work shall be similar, in all respects, to the existing system and conditions unless otherwise indicated. Existing work shall be altered and/or temporarily removed and replaced as required for completion of requirements of the project.

B. Remove and cap existing branch ducts and piping, no longer required to remain as part of an active system, at the appropriate main. Existing piping below grade shall be abandoned and capped flush with finish floor unless otherwise indicated.

3.9 SALVAGE

A. The Owner assumes no responsibility for loss or damage to materials or structures on site for the salvage value of equipment which the Contractor may have reflected in his bid.

B. All existing equipment, ductwork and piping removed during construction no longer required as part of an active system and not to be reused shall be removed from the job site and properly disposed of by the Contractor.

C. All existing diffusers, equipment and piping not being reused shall be removed during construction and shall remain the property of the owner and shall be stored, removed from the job site, or disposed of as directed by the owner's representative.

3.10 CLEANING

A. The Contractor shall clear away all debris and demolished material at frequent intervals. The Contractor shall not allow debris to accumulate to the extent that it will interfere with work, passage of the workmen, and the operation of the existing occupied areas.

B. It is the intent of this Specification that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment.

3.11 INVENTORY

A. Contractor shall provide a detailed inventory of all existing mechanical equipment including air distribution devices, terminal devices, exhaust fans, grilles, thermostats, etc. to be stored for later reuse. Inventory shall include type, size, capacities, and quantities of each item or piece of equipment.

3.12 TESTING

A. Piping: Existing piping which is to remain as part of an active system shall be completely cleaned, sanitized, flushed, and pressure tested similar to the proposed new piping. The Contractor shall replace any defective piping and shall repair any piping leaks.

B. Ductwork: Existing ductwork which is to remain as part of an active system shall be tested for leaks and replaced or repaired similar to the proposed new systems.

C. Equipment: Existing equipment, instruments, and accessories to remain shall be tested for defects and shall be adjusted, repaired or replaced where required. Conditions, qualifications, and procedures regarding adjustments, reparations, or replacement of existing equipment as deemed necessary by the Contractor shall be submitted to the Architect for approval. The Contractor shall not proceed with equipment replacement without the written approval of the Architect. The Architect shall determine whether replacement is required and shall specify the replacement equipment to be used.

D. Additional Requirements: For piping, ductwork, equipment, etc., refer to the appropriate Specification Section for testing requirements and additional requirements of existing equipment and systems to be a part of the new active systems.

3.13 QUALIFICATIONS

A. Only Contractors, Subcontractors, and workmen experienced and regularly engaged in the demolition of mechanical, plumbing, and electrical systems shall be permitted to perform the demolition of existing systems.

END OF SECTION 23 05 30

SECTION 23 05 48 - VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Furnish and install complete, fully adjusted, vibration and noise isolation systems and associated equipment as specified and as indicated on the contract documents.

B. It is the objective of this Specification to provide the necessary design requirements for the control of noise and vibration in the buildings due to the operation of machinery or equipment, and/or due to interconnected piping, ductwork and conduit.

C. The work of this section includes, but is not limited to the following:

1. Vibration isolation elements for piping and equipment
2. Equipment isolation bases
3. Wind restraints for outdoor equipment

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 70 - Division 23 Schedule of Submittal Data
B. Section 23 21 13 - Hydronic Piping
C. Section 23 23 00 - Refrigerant Piping
D. Section 23 30 01 - Ductwork and Sheet Metal
E. Section 23 34 00 - HVAC Fans
F. Section 23 81 26 - Split System Air Conditioners

1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1. HVAC Handbook Applications, Chapter 48

B. American Society of Civil Engineers / Structural Engineering Institute (ASCE/SEI)
1. 7 Minimum Design Loads for Buildings and Other Structures

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data listed in Sections 23 00 10 and 23 00 70 plus capacity and accessory data listed in the Schedules on the drawings.

B. In addition to these requirements, the submittal material shall include descriptive data for all products and materials including, but not limited to, the following:

1. Descriptive Data:

a. Catalog cuts and data sheets on specific vibration isolators to be utilized showing compliance with the specifications.

b. An itemized list showing the items of equipment or piping to be isolated, the isolator type and model number selected, isolator loading, deflection, material, strength, fastening provision, and finish for each type and size of vibration isolation

2. Shop Drawings:

a. Drawings showing equipment base constructions for each machine, including dimensions, structural member sizes and support point locations.

b. Drawings showing methods of suspension, support guides for piping and ductwork.

c. Drawings showing methods for isolation of pipes and ductwork piercing walls and slabs.

d. Concrete and steel details for bases, including anchor bolt locations.

e. Provide piping diagrams of vertical pipe risers prepared by the vibration isolation manufacturer showing anticipated expansion and contraction at each support point, initial and final loads on the building structure, and spring deflection changes. Submittal data shall include certification by the manufacturer that the piping system has been examined for excessive stresses and that none will exist in the design proposed.

## 1.6 MANUFACTURER RESPONSIBILITIES

A. Manufacturer of vibration isolation equipment shall have the following responsibilities:

1. Determine vibration isolation sizes and locations.

2. Provide piping and equipment isolation systems as scheduled or specified.

3. Guarantee specified isolation system deflection.

4. Provide wind restraint systems for outdoor equipment as scheduled or specified.

5. Provide installation instructions, drawings and field supervision to assure proper installation and performance.

6. Additional responsibilities as specified herein and as indicated on the Drawings.

# PART 2 - PRODUCTS

## 2.1 MANUFACTURERS

A. All vibration isolation devices shall be the product of a single manufacturer. Products of other manufacturers are acceptable provided that their systems comply with the design intent for system performance, static deflection and structural design of the base manufacturer; and the performance of these other products are certified by the single manufacturer taking responsibility for the project.

B. If it complies with these Specifications, equipment and systems manufactured by one of the following manufacturers will be acceptable:

1. Mason Industries
2. Kinetics Noise Control
3. VMC Group-Amber/Booth
4. Vibration Eliminator Co.
5. Vibro-Acoustics

C. Vibration isolation manufacturer and manufacturer's representatives providing systems and products for this project shall have a minimum 10 years of experience designing and installing vibration isolation systems.

D. The installation and final adjustment of all vibration isolation units, hangers and bases, shall be under the direct supervision of the vibration isolation manufacturer's representative.

2.2 GENERAL

A. Deflection Heights and Calibration: All vibration isolators shall have either known undeflected heights or calibration markings so that, after adjustment, when carrying their load, the deflection under load can be verified, thus determining that the load is within the proper range of the device and that the correct degree of vibration isolation is being provided according to the design.

B. Linear Characteristics: All isolators shall operate in the linear portion of their load versus deflection curve. Load versus deflection curves shall be furnished by the manufacturer, and must be linear over a deflection range of not less than 50% above the design deflection.

C. Spring Isolation: Where spring isolation systems are described in the following specifications, the mounting assemblies shall utilize bare springs with the spring diameter not less than 0.8 of the loaded operating height of the spring. Each spring isolator shall be designed and installed so that the ends of the springs remain parallel. The minimum deflection from loaded operating height to spring solid height shall be 50% of the rated static deflection of the spring.

D. Neoprene Isolation: Where neoprene in shear isolation systems are described in the following specifications, the mounting assemblies shall utilize bare neoprene elements with unit type design molded in oil resistant neoprene. The neoprene shall be compounded to meet the following:

1. Shore hardness of 35 to 65$\pm$5, after minimum aging of 20 days or corresponding oven aging.
2. Minimum tensile strength of 2000 psi.
3. Minimum elongation of 300%.
4. Maximum compression at 25% of original deflection.

E. Stiffness: The isolator ratio of lateral to vertical stiffness shall not be less than 0.9 nor more than 1.5.

F. Natural Frequency: The theoretical vertical natural frequency for each support point, based upon load per isolator and isolator stiffness, shall not differ from the design objectives for the equipment as a whole by more than ±10%.

G. Corrosion Protection: All mounting systems exposed to weather and other corrosive environments shall be provided with factory applied corrosion protection. All metal parts of

mountings (except springs and hardware) shall be hot dip galvanized. All vibration isolators shall be furnished with zinc electroplated or stainless steel hardware to prevent corrosion and bolt freeze up and to maintain attractive appearance. Isolators exposed to weather shall have the spring cadmium plated and neoprene or powdered coated. Nuts and bolts shall be cadmium plated.

H. Wind Loading: Contractor shall provide isolators for equipment subject to wind loading with uplift restraints which comply with all minimum code requirements.

I. Variable Speed Applications: Isolator natural frequency shall be 40% or less than the speed of the equipment per ASHRAE recommendations.

2.3 VIBRATION ISOLATOR SCHEDULE

| Equipment | Base Type | Isolator Type | Minimum Static Deflection (Inches) Under Load |
|---|---|---|---|
| Air Handling Units | | | |
| > 1hp - floor | - | A | 2” |
| < 1hp - floor | - | D | 3/8” |
| Fans | | | |
| > 1hp - hung | - | F | 1-1/2” |
| < 1hp - hung | - | E | 3/8” |
| > 1hp - floor | B-1 | A | 2” |
| < 1hp - floor | B-1 | D | 3/8” |

A. Equipment and systems shall be isolated in accordance with the schedule, as specified herein, and as indicated on the Drawings.

B. Cabinet fans shall be supported in the same manner as air handling units.

C. HVAC pumps include condenser water system pumps, chilled water system pumps, heating hot water system pumps, and all other HVAC system pumps.

2.4 VIBRATION ISOLATORS

A. Type A: Bare spring isolators shall incorporate the following:

1. Minimum 1/4" (6 mm) thick neoprene acoustical base pad or cup on underside, unless designated otherwise.

2. Non resonant with equipment forcing frequencies or support structure natural frequencies.

3. VMC Group-Amber/Booth Type SW, Mason Type SLF, Vibro-Acoustics Type FS, or as approved.

B. Type D: Elastomer Mounting Types/Elastomer Isolators, shall incorporate the following:

1. Bolt holes for bolting to equipment base.

2. Bottom steel plates for bolting or welding to sub-base as required.

3. Unit type design molded in oil resistant neoprene.

4. VMC Group-Amber/Booth Type RVD, Mason Type ND, Vibro-Acoustics Type RD, or as approved.

C. Type E: Elastomer hanger rod isolators shall incorporate the following:

1. Molded unit type neoprene element with projecting bushing lining rod clearance hole.

2. Neoprene element to be minimum 1-3/4" (45 mm) thick.

3. Steel retainer box encasing neoprene mounting.

4. Clearance between mounting hanger rod and neoprene bushing shall be minimum 1/8" (3 mm).

5. VMC Group-Amber/Booth Type HRD, Mason Type HD, Vibro-Acoustics Type NH, or as approved.

D. Type F: Combination spring/elastomer hanger rod isolators to incorporate the following:

1. Spring and neoprene isolator elements in a steel box retainer.

2. Other characteristics of steel box retainer and hanger rod swing as described for Type C isolators.

3. VMC Group-Amber/Booth Type BSRA, Mason Type 30N, Vibro-Acoustics Type SHRB, or as approved.

2.5 EQUIPMENT BASES

A. Integral Structural Steel Base, Type B-1:

1. Reinforced, as required, to prevent base flexure at start up and misalignment of drive and driven units. Centrifugal fan bases complete with motor slide rails. Drilled for drive and driven unit mounting template.

2. VMC Group-Amber/Booth Type SFB, Mason Type M, WF, Vibro-Acoustics Type IFB, or as approved.

PART 3 - EXECUTION

3.1 GENERAL VIBRATION ISOLATION REQUIREMENTS

A. The installation of all vibration isolation devices and systems shall be in accordance with the recommendations, instructions, and procedures of the vibration isolator manufacturer. Vibration isolators shall not cause any change of position of equipment or piping resulting in piping stresses or misalignment.

B. Unless otherwise noted, all mechanical equipment shall be isolated from the building structure by means of noise and vibration isolators as scheduled on the drawings and within these specifications. Air handling units and other air conditioning units provided with internal spring isolation systems shall not require external spring isolation; however, Type G pad type elastomer mountings as specified herein shall be required to be provided under such

equipment.

C. No rigid connections between equipment and building structure shall be made that degrades the noise and vibration isolation systems herein specified.

D. Provide temporary anchors and supports as required for the installation of the isolation systems. All temporary supports and anchors shall be removed prior to final review of the systems.

E. The contractor shall not install any equipment or piping which makes rigid contact with the building unless permitted in this Specification. Building includes, but is not limited to, slabs, beams, columns, studs and walls.

F. Isolation mounting deflection shall be at least the minimum specified.

G. Coordinate work with other trades to avoid rigid contact with the building. Inform other trades following work, such as plastering, drywall, masonry, electrical, etc. to avoid any contact which would reduce the vibration isolation.

H. Bring to the Architect's attention, prior to installation, any conflicts with other trades which will result in unavoidable rigid contact with equipment or piping as described herein, due to inadequate space or other unforeseen conditions. Corrective work necessitated by conflicts after installation shall be at the responsible contractor's expense.

I. Bring to the Architect's attention any discrepancies between the specifications and field conditions or changes required due to specific equipment selection, prior to installation. Corrective work necessitated by discrepancies after installation shall be at the contractor's expense.

J. Request inspection of any installation to be covered or enclosed, prior to such closure.

K. Manufacturer's Stock: The isolation manufacturer's representative shall maintain an adequate stock of springs and isolators of the type used so that any changes required during construction and checking can be accomplished promptly.

L. Coordination: The contractor and isolator manufacturer's representative shall coordinate the vibration isolation supports with the manufacturers of the equipment to be isolated. Any changes from the documents deemed necessary to provide adequate noise and vibration isolation in accordance with the requirements of the ASHRAE HVAC Applications, Chapter 48, shall clearly indicated in the submittal shop drawing data. Prior to submitting detailed shop drawings to the Engineer for review, the equipment manufacturer shall approve the shop drawings in writing. Refer to Section 23 00 10 for additional requirements.

M. Piping systems shall be filled prior to the installation of the vibration isolators as recommended by the vibration isolator manufacturer. However, the piping systems shall not be put into operation until the vibration isolators have been installed.

3.2 EQUIPMENT ISOLATION

A. Each fan and motor assembly shall be supported on a single structural steel frame. Flexible duct connections shall be provided at inlet and discharge of fan as specified in section 23 30 01.

B. Floor mounted machines to be isolated shall be supported by a structural steel frame or concrete inertia base.

C. Brackets, mounting plates, and miscellaneous metal angles shall be provided as required to accommodate the interface of the isolator and the respective isolated equipment. The position and size of the support devices shall be specified and provided by the isolator manufacturer. It shall be the responsibility of the isolator manufacturer to review the actual equipment being supplied to this project under other specification sections and coordinate the mounting arrangement of the isolators.

D. The minimum operating clearance between the equipment frame or rigid steel base frame and the housekeeping pad or floor shall be 1" (25 mm). Minimum operating clearance between concrete inertia base and housekeeping pad or floor shall be 2" (50 mm).

E. After the entire installation is complete and under full operational load, the isolators shall be adjusted so that the load from the isolated equipment is fully transferred to the isolators.

F. Prior to start up, clean out all foreign matter between bases and equipment. Verify that there are no isolation short circuits in the base or isolators.

G. Verify that all installed isolator and mounting systems permit equipment motion in all directions. Adjust or provide additional resilient restraints to flexibly limit start up equipment lateral motion to 1/4" (6 mm).

H. Diagonal thrust restraint shall be as described for Type C hanger with the same deflection as specified for the spring mountings. The spring element shall be designed so it can be preset for thrust and adjusted to allow for a maximum of 1/4" (6 mm) movement at start and stop. Diagonal restraints shall be attached at the centerline of thrust. Restraint shall be Mason Type WB, or as approved.

I. All piping and ductwork shall freely pass through walls and floors without rigid connections. Penetration points shall be sleeved or otherwise formed to allow passage of piping or ductwork, and maintain 3/4" to 1-1/4" (20 to 32 mm) clearance around the outside surfaces. This clearance space shall be tightly packed with fiberglass, and caulked airtight after installation of piping and ductwork.

J. Isolators shall be installed with the isolator hanger box attached to, or hung as close as possible to, the structure. Hanger rods shall be aligned to clear the hanger box.

K. Isolators shall be suspended from substantial structural members where possible. Suspending isolators from slab diaphragm should be avoided when possible.

L. The isolation system shall not be resonant with the structural steel support resonant frequencies or the driving frequencies of the supported equipment.

## 3.3 INSPECTION

A. After the completion of the vibration isolation systems in accordance with the contract documents, the local representative of the isolation materials manufacturer shall inspect the complete system and report in writing any installation errors, improperly selected isolation devices, and other faults that could affect the performance of the system. This report shall be submitted to the Contractor for corrective action. After the corrective action has been implemented, the local representative of the isolation materials manufacturer shall visit the site and verify the corrective action has been completed.

B. After the corrective action has been completed, the local representative of the isolation materials manufacturer shall provide a submittal to architect via the contractor that certifies in

writing that the vibration isolation systems have been installed in accordance with the manufacturer's recommendations and the requirements of the contract documents. All deviations from the contract document shall be noted in this final report.

END OF SECTION 23 05 48

SECTION 23 05 53 - IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. All ductwork systems, piping systems, valves, controls and equipment on the Project shall be identified as specified herein. All marks of identification shall be easily visible from the floor or usual point of vision.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Equipment Sections
B. Section 23 05 23 - General Duty Valves for HVAC Piping
C. Section 23 07 00 - HVAC Insulation
D. Section 23 11 00 - Facility Fuel Piping
E. Section 23 21 13 - Hydronic Piping
F. Section 23 30 01 - Ductwork and Sheet Metal

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
1. A13.1 Scheme for the Identification of Piping Systems

B. ASTM International (formerly American Society for Testing and Materials)
1. D6386 Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting

PART 2 - PRODUCTS

2.1 MANUFACTURER

A. Pipe markers, name plates, valve tags, "S" hooks, chains, ties, tapes and adhesives shall be manufactured by Seton or approved equal.

2.2 GENERAL

A. All work and products in this Section shall be in accordance with ASME Standard A13.1 and the Local Code Authority.

2.3 PAINTING

A. General: Refer to the Architectural Documents for central plant and other equipment room painting and finishing requirements. The Division 21, 22, 23, and 25 Contractors are not responsible for painting of these piping systems or equipment. Division 21, 22, 23, and 25 shall cooperate with the Architectural Division to properly identify the piping systems to be painted by the Division 09 Subcontractor under the requirements of the Division 09 specification section. The following requirements are in addition to the Architectural Division requirements.

B. Pipe/Equipment Paint Schedule and Color Codes:

1. The following piping and equipment shall be painted in accordance with the following Sherwin Williams (SW) Urethane Alkyd Enamel color codes or approved equal. However, the Contractor shall obtain approval from the Owner's representative prior to proceeding with any of the following colors:

2. Piping Systems

| Piping System | Color | SW Color Code |
|---|---|---|
| Condensate Drain | Black | SW 6258 Tricorn Black |
| Miscellaneous Drain | Color to match applicable system | |
| Natural Gas (inside building) | Yellow/Brown | SW ANSI/OSHA Yellow with Brown Band |
| Heating Hot Water Supply | Yellow | SW ANSI/OSHA Yellow |
| Heating Hot Water Return | Yellow | SW ANSI/OSHA Yellow |

3. Equipment

| Equipment | Color |
|---|---|
| Concrete Equipment Pads | Gray |
| Air Handling Units | Factory finish |
| Fans | Factory finish |
| Ductwork | Standard galvanized finish |
| Heating Hot Water Pumps | Yellow to match piping |
| Heating Hot Water Tanks and Pot Feeder | Yellow to match piping |

4. Additional Requirements:

a. Areas of pumps, tanks, vessels, pot feeders and any other items insulated with Armstrong Armaflex insulation shall be painted with Armstrong Armaflex paint. The Armstrong factory color is typically white. The Contractor shall add a compatible pigment to obtain the color that matches the appropriate piping system. The Contractor shall receive from Armstrong written approval of the pigment to be used before proceeding to work.

b. All items shall be painted with a minimum of two coats of the specified paint.

c. The Paint Schedule and Color Code specifications apply to the following areas: central plant, air handling unit rooms, electrical rooms, telephone rooms, stairwells, pump rooms, storage and janitor's closets, penthouse, generator room, garage, corridors, office space, computer rooms, truck dock, accessible chases, plus any other areas where piping and equipment are not concealed and are visible. This Specification does not apply to piping and equipment located above ceilings, above future tenant area ceilings, within inaccessible chases or below grade.

d. Provide Engineer and Architect with color samples for each system and application listed prior to beginning work. Refer to the Architectural Painting Systems for additional requirements.

2.4 NAME PLATES

A. Equipment nameplates shall be black surface, white core laminated bakelite with engraved letters. Plates shall be a minimum of 3" long by 1" wide with white letters 3/8" high.

2.5 VALVE TAGS

A. Valve tags shall be of 1-1/2" diameter brass.

2.6 PIPE SERVICE AND DIRECTION OF FLOW MARKERS

A. Pipe markers shall indicate the service, direction of flow and heat trace. Letters shall be a minimum of one inch high. Pipe, heat trace and flow markers shall be Seton "Opti-Code", "Setmark" or approved equal.

PART 3 - EXECUTION

3.1 DUCTWORK

A. All exposed ductwork shall be identified as to the service of the duct and the direction of flow. The letters shall be at least 2" high and the flow arrow shall be at least 6" long. The letters and flow arrow shall be made by precut stencils and black oil base aerosol paint. Concealed ducts need not be identified. Ductwork flow markers and identification shall be located at the connection to fans, air handling units, and other equipment; on each side of wall penetrations, and on ductwork visible from access doors. Subsequent flow markers shall have a spacing not to exceed 20 feet.

3.2 PIPING

A. All piping shall have pipe markers indicating the service and direction of flow. This requirement does not apply to piping located within non-accessible chases or below grade. Piping in the equipment rooms and other areas where piping is exposed shall be color coded as to type of use, service and direction of flow. Locate pipe markers at each valve, on each side of the wall where the pipe passes through, on piping and equipment visible from access doors, and at 20 foot centers on straight runs of pipe. Provide a flow arrow at each identification marker by wrapping ends of service description marker with banding showing flow direction. At each piping system provided with heat trace, provide a clearly visible marker at 10 foot centers on straight runs of pipe indicating "Heat Traced Piping". Refer to Section 23 05 34 for additional requirements. At each location where a pipe insulation isolating seal is provided, provide a clearly visible marker indicating "IS" for isolation seal. Refer to Section 23 07 00 for additional requirements.

3.3 VALVES

A. Each valve including control valves shall be tagged. The disc shall contain a number. A valve list shall be provided for the Operating and Maintenance Manual indicating the location of the valve, the service of the valve and any pertinent remarks regarding the operation of the valve. Securely fasten the discs to the valves with brass "S" hooks or chains. Valve tag designations shall correspond to the identification on the "Record Drawings".

3.4 EQUIPMENT AND CONTROL PANEL NAMEPLATES

A. Equipment and control panels shall be identified by means of nameplates permanently attached to the equipment. Name plate designations shall correspond to the identification on the "Record

Drawings". Controls and instruments shall be identified by name plates mounted under the control or instrument.

3.5 TIME OF APPLICATION

A. Equipment and system identification shall not be initiated until all painting required under the Architectural Section of these Specifications has been accomplished.

3.6 EQUIPMENT IDENTIFICATION

A. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; software, programming, and graphics.

END OF SECTION 23 05 53

SECTION 23 05 93 - TESTING, ADJUSTING, AND BALANCING FOR HVAC

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide all testing, balancing, and adjusting of mechanical systems as indicated on the Drawings and as specified herein. Tests shall include but not be limited to the items specified in this Section.

B. Testing, balancing, and adjusting shall in no way relieve the Contractor of the warranty requirements.

C. The Contractor shall furnish all fuel, and electricity required in performing the testing, balancing, and adjusting of mechanical systems.

1.3 REFERENCE STANDARDS

A. Associated Air Balance Council (AABC)
1. National Standards for Total System Balance

B. American Society of Mechanical Engineers (ASME)
1. B31.1 Power Piping

C. National Environmental Balancing Bureau (NEBB)
1. Procedural Standard for Testing, Adjusting and Balancing of Environmental Systems

1.4 CONTRACTOR REQUIREMENTS

A. The Air Balancing Subcontractor shall work directly for the General Contractor and shall be separate and independent from the Division 23 Mechanical Contractor.

B. The Air Balancing Subcontractor and the personnel shall have a minimum of five years of experience in the testing, balancing and adjusting of commercial air conditioning systems.

C. The Air Balancing Subcontractor and the personnel used for this Project shall be certified testing agents of the NEBB or the AABC.

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Testing: During the progress of the work, tests shall be made as specified herein and as required by authorities having jurisdiction, including Local Authorities Inspection Department, Owner, Architect, or Engineer. Tests shall be conducted by the Mechanical Contractor as part of

the work of this Division and shall include all qualified personnel, equipment apparatus, and services required to perform the tests.

B. Job Site Reviews: The Testing, Balancing and Adjusting Subcontractor shall visit the site at intervals appropriate to the stage of construction to familiarize himself with the progress and quality of work. This Subcontractor shall report directly to the General Contractor in written reports indicating any deficiencies, potential problems areas and conditions that could adversely affect the building systems and the testing, balancing, adjusting procedures.

C. Testing Procedures and Recording Forms: The Contractor shall submit proposed test procedures, recording forms, and test equipment for review prior to the execution of testing.

D. Repairs: Leaks, damage, or defects discovered or resulting from tests shall be repaired or replaced to a like new condition. Leaky pipe joints, ductwork, etc., shall be removed and replaced with acceptable materials.

E. Equipment: All equipment and instruments required for tests as well as additional thermometer wells, gauge and instrument connections shall be installed at no additional cost to the Owner.

F. Instrument Calibration: All instruments used for testing and balancing must have been calibrated within a period of six months prior to balancing. Instrument calibration shall be certified as specified in Section 23 00 10.

G. Report: Submit six copies of each complete testing and balancing report to the Engineer for review and comment.

H. Acoustic Performance: The final results of all testing, adjusting and balancing shall achieve the range of indoor design goals for sound control for all occupied spaces as indicated in section 23 00 15. Refer to section 23 00 10 “Equipment Noise and Vibration” for additional requirements.

I. Equipment Identification: The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

3.2 TESTING, BALANCING, AND ADJUSTING OF DIVISION 23

A. Vibration and Alignment: Vibration and alignment field measurements shall be taken when directed by the Engineer on equipment where vibration appears excessive. Readings shall include shaft alignment, equipment vibration, bearing housing vibration, foundation vibration, building structure vibration, and other tests as directed by the Engineer. Readings will be made using portable IRD (or approved equal) equipment capable of filtering out various unwanted frequencies and standard reporting forms. Maximum vibration at any point listed above or specified shall not exceed 2 mils on fans and 2 mils on pumps unless otherwise specified.

B. Sound Levels: Sound level readings shall be taken at 10 locations in the building as selected by the Engineer. The readings shall be taken on an octave band analyzer in a manner acceptable to the Engineer. The Contractor shall submit test equipment data and reporting forms to the Engineer for review. In order to reduce the ambient noise level, the readings shall be taken at night. All tests shall be performed in the presence of the Engineer, if Engineer so desires.

C. Air Flow Measurements: Air balance procedures shall be performed by National Environmental Balancing Bureau or Associated Air Balance Council certified, independent Testing and Balancing Contractor(s) in accordance with the latest edition of the NEBB "Procedural Standard for Testing, Adjusting and Balancing of Environmental Systems" or the latest edition of the AABC National Standards for Total System Balancing.

D. Piping Pressure/Leak Testing: Leak tests shall be conducted in accordance with ASME B31.1 and as specified herein in Section 23 21 13 and 23 11 00 before piping of various systems has been covered or furred in. Note: Equipment not rated at or above the specified test pressure plus all sensitive system controls and instrumentation, excluding control valves, shall be valved off or removed and the connections plugged prior to hydrostatic testing. Removed items shall be replaced after testing has been completed. Controls and instrumentation shall be calibrated and tested after reinstallation.

E. Field Capacity and Performance Testing: The Contractor shall prove the capacity and performance of each piece of equipment by field tests as specified herein in various Sections or as required by the Engineer. All equipment and instruments required for tests as well as additional thermowells or gauge connections shall be installed at no additional cost to the Owner. A qualified representative of the equipment manufacturer, with credentials submitted in advance, shall be present at the test. The Engineer may witness tests, if Engineer so desires. The Contractor shall notify the Engineer, in writing, at least two weeks prior to the day of the test. See Section 23 00 10 for additional requirements.

## 3.3 AIR BALANCE AND ADJUSTING

A. Supply air systems installed in finished areas of the building shall be balanced and adjusted as follows:

1. After duct systems have been installed complete with all VAV boxes, terminal devices, grilles, dampers, ducts, coils, and other items hereinafter specified the Contractor shall make adjustments, as required, to deliver the volume of air at each air outlet within 10% of design flow as shown on the Drawings. After the finished area is occupied, the air volumes shall be readjusted, if required, to properly balance the cooling and heating loads throughout the conditioned areas.

2. Air outlets shall be balanced with air pattern as shown on the Drawings.

B. Supply air systems installed in unfinished tenant areas of the building shall be balanced as follows:

1. After duct systems have been installed complete with all terminal units, air outlets and inlets, dampers, ducts, coils, and other items hereinafter specified, except the duct taps serving the air diffusers in the interior zones, the Contractor shall adjust, as required to deliver the air volumes within 10% of air handling unit design flow as shown on the Drawings or as indicated in the Schedules. Sufficient temporary openings are to be provided in the interior zone duct systems in order that the interior zone air volumes can be adjusted.

2. The final balanced position of manual balancing dampers such as in each air handling unit zone duct shall be permanently marked on the ductwork.

C. After all supply and exhaust air systems have been installed complete with all duct, grilles, dampers, fans, and other items as hereinafter specified, the Contractor shall make adjustments, as required to deliver the volumes of air at each inlet or outlet within 10% of design flow.

D. In Building Pressure: In addition to the TAB requirements; the overall pressurization of the project and the overall pressurization of individual areas of the project shall be adjusted to maintain the difference between sources of outdoor air supply into the building and exhaust air relieved from the building as indicated on the Drawings and in the Specifications. Pressurization may be of a positive pressure or negative pressure. Contractor shall make adjustments, as required to deliver the set point difference between supply air quantities and exhaust air quantities of related systems within 10% of design flow.

1. Unless otherwise indicated on the Drawings, the total setpoint difference between the outdoor air supply quantities and the exhaust air quantities on the typical floor of the building shall be 400 cfm.

2. Unless otherwise indicated on the Drawings, the total set point difference between the outside air supply quantities and the exhaust air quantities on the ground level floor of the building shall be 1500 cfm.

E. After all miscellaneous ventilation systems have been installed complete with all ducts, terminal units, air outlets and inlets, louvers, dampers, fans, and other items as hereinafter specified, the Contractor shall make adjustments as required to deliver the air volumes, or differential static pressures in the case of the pressurization fans, at each air inlet and/or outlet within 10% of design airflow. Differential pressure within the stairwell at any location shall not exceed 0.35” wg with all doors closed. The pressurization system shall provide a minimum pressure differential of 0.1" wg between the stairwell and the adjacent occupied areas with all doors closed.

F. Additional Requirements for Code Required Systems and Life Safety Systems: For all Code Required Systems and Life Safety Systems, the quantities indicated in the contract documents shall be the minimum values required. It shall not be acceptable to balance each of these systems below its minimum value. Air balance tolerance for each of these systems shall be between the minimum value up to 110% of the minimum value. In addition to this tolerance, systems that require a negative or positive pressurization shall be balanced to provide the necessary level of positive or negative pressurization by maintaining the relative difference in supply and exhaust air quantities. Code Required Systems and Life Safety Systems shall include but not be limited to smoke exhaust systems, smoke control systems, stairwell pressurization systems, parking area ventilation systems, refrigerant exhaust systems, elevator pressurization systems, and kitchen exhaust systems. These systems shall include the supply, make-up, exhaust, and relief air systems related to the applicable Code Required System and Life Safety System. All Code Required Systems and Life Safety Systems shall be installed, and performance tested in accordance with the applicable codes and the requirements of the Local Code Authority. Adjustments to these systems as required by the Local Code Authority during the performance TAB shall be provided by the Contractor without additional cost to the Owner.

G. All balancing must be accomplished with diffuser air patterns as indicated on the Drawings.

H. Adjust or replace the fan belt drives, motor sheaves, and belts on air handling units and fans as required to achieve the specified air volumes indicated on the contract documents. Air handling unit manufacturers and the fan manufacturers shall provide the fan belt drives, motor sheaves, and belts necessary to achieve the specified air volumes as indicated on the contract documents for their respective equipment. Refer to specification sections 23 34 00 and 23 73 00 for additional requirements. Do not make fan speed adjustments that result in motor overload. Consult with the equipment manufacturers concerning fan speed safety factors. Modulate

dampers and operate air handling units and fans through all modes of operation while measuring fan motor amperage to ensure no motor overload will occur.

I. The Contractor shall submit detailed balancing procedures and recording forms for the Engineer's review, prior to commencing any air balance work.

J. Submit complete comprehensive bound reports of the system performance for review and comments. Identify all data completely. Identify the manufacturer, size, type, location, room number, and zone of each coil and piece of equipment reported. Provide design and actual airflow. Provide complete nameplate data for each piece of equipment reported.

## 3.4 FINAL AIR BALANCE READINGS

A. After air flow are balanced, and with the temperature controls set to produce design cooling, measure and record all data necessary to compile a complete report to demonstrate the acceptability of the various mechanical systems. Include at least the following data in the report:

1. Outside dry bulb and wet bulb temperature.

2. Inside dry bulb and wet bulb temperatures in six selected areas of the conditioned space, room or area selected by the Engineer.

3. Dry bulb temperatures of air entering and leaving all coils.

4. Record control voltage range and set points for all adjustable fans typical to exhaust and supply fans, etc.

5. With controls set for design heating, repeat applicable procedures specified in the immediately preceding subparagraph. Data concerning dry bulb temperatures of air entering and leaving coils applies to all duct mounted electric heating coils, and air handling unit electric heating coils.

6. The balanced position of each controlling valve and damper shall be permanently marked on the pipe, insulation, or ductwork.

B. Temperature Control Adjusting: Temperature controls shall be calibrated and all thermostats, dampers, etc., adjusted in the proper operating condition, subject to review by the Engineer. Refer to Sequence of Operation on drawing M-502 for additional requirements.

C. Ductwork Leakage Testing: Refer to section 23 30 01 for additional information.

## 3.5 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. The contractor shall be required to assist, coordinate, and comply with all owner commissioning requirements.

END OF SECTION 23 05 93

SECTION 23 07 00 - HVAC INSULATION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the insulation as specified herein for the Division 23 systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. 23 00 70 - Division 23 Schedule of Submittal Data
B. 23 05 23 - General Duty Valves for HVAC Piping
C. 23 21 13 - Hydronic Piping
D. 23 23 00 - Refrigerant Piping
E. 23 30 01 - Ductwork and Sheet Metal

1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings

B. ASTM International (formerly American Society for Testing and Materials)
   1. C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus
   2. C335 Standard Test Method for Steady-State Heat Transfer Properties of Pipe Insulation
   3. C411 Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation
   4. C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
   5. C533 Standard Specification for Calcium Silicate Block and Pipe Thermal Insulation
   6. C534 Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form
   7. C547 Standard Specification for Mineral Fiber Pipe Insulation
   8. C553 Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
   9. C585 Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing
   10. C612 Standard Specification for Mineral Fiber Block and Board Thermal Insulation
   11. C916 Standard Specification for Adhesives for Duct Thermal Insulation
   12. C921 Standard Practice for Determining the Properties of Jacketing Materials for Thermal Insulation
   13. C1136 Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
   14. C1290 Standard Specification for Flexible Fibrous Glass Blanket Insulation Used to Externally Insulate HVAC Ducts
   15. C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings

16. C1393 Standard Specification for Perpendicularly Oriented Mineral Fiber Roll and Sheet Thermal Insulation for Pipes and Tanks
17. C1729 Specification for Aluminum Jacketing for Insulation
18. C1767 Specification for Stainless Steel Jacketing for Insulation
19. D1056 Standard Specification for Flexible Cellular Materials-Sponge or Expanded Rubber
20. E84 Standard Test Method for Surface Burning Characteristics of Building Materials
21. E96 Standard Test Methods for Water Vapor Transmission of Materials
22. E119 Standard Test Methods for Fire Tests of Building Construction and Materials
23. G21 Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

C. DOE Code of Federal Regulations (CFR)
1. 40 CFR 59 Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings

D. North American Insulation Manufacturers Association (NAIMA)
1. CI228 Guide to Insulating Chilled Water Piping Systems with Mineral Fiber Pipe Insulation
2. AH121 A Guide to Insulated HVAC Duct Systems

E. National Fire Protection Association (NFPA)
1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

F. South Coast Air Quality Management District (SCAQMD)
1. Rule 1168 Adhesive and Sealant Applications

G. Underwriters Laboratories (UL) Standard for Safety
1. 181 Factory-Made Air Ducts and Air Connectors
2. 723 Test for Surface Burning Characteristics of Building Materials
3. 2818 GREENGUARD Certification Program for Chemical Emissions for Building Materials, Finishes and Furnishings

1.5 SUBMITTALS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Sections 23 00 10 and 23 00 70.

B. Product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content.

1.6 CONCEALED

A. The word "Concealed" as used in this Specification refers to insulation in furred spaces, pipe and duct shafts, unheated spaces immediately below roof, unexcavated spaces, crawl spaces, and tunnels. The word "exposed" refers to insulation in all other are

PART 2 - PRODUCTS

2.1 GENERAL

A. All insulation required for the work shall be new, of first class quality and shall be furnished, delivered, erected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class

standard article as approved by the Architect shall be provided.

B. Pressure Sensitive Tape and Release Paper Systems: The use of release paper systems, pressure sensitive tape systems, and peel & stick systems shall be limited to piping insulation systems as specified herein. Unless otherwise noted, the use of release paper systems, pressure sensitive tape systems, and peel & stick systems will not be acceptable and shall not be used by the Contractor for other than piping insulation systems. Refer to the specifications herein for additional requirements.

2.2 FIRE AND SMOKE HAZARD RATING

A. All insulations (composite insulation, jacket or facing, tape, coating and adhesive) installed within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723 as required by applicable mechanical code. Accessories such as coatings, tapes, and adhesives shall have the same component ratings.

B. All insulations (composite insulation, jacket or facing, tape, coating and adhesive) installed within non-plenums shall be noncombustible or shall be tested and rated as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723. Accessories such as coatings, tapes, and adhesives shall have the same component ratings.

2.3 CORROSION PROTECTION

A. All insulation materials, adhesives and attachment devices used in direct contact with steel, copper, aluminum, galvanized steel, and other materials required by this Contract shall not cause corrosive effects. Materials to be used on stainless steel surfaces shall comply with ASTM C795 requirements. This non-corrosive requirement applies to all insulation systems for all operational conditions as installed on this Project. These requirements apply throughout the entire operational temperature range and moisture condition of each system.

2.4 PIPING SYSTEMS

A. Refrigerant Piping (Split System)

1. If it complies with these Specifications, one of the following manufacturers will be acceptable:

    a. Armacell
    b. Rubatex/K-Flex
    c. Aeroflex

2. Plenum Application: Flexible closed cell elastomeric pipe insulation shall comply with ASTM C534 (Grade 1) and shall be Armacell Armaflex Ultra or approved equal for installation within air plenums in accordance with the flame/smoke hazard ratings specified hereinbefore

3. Non-Plenum Application: Flexible closed cell elastomeric pipe insulation shall comply with ASTM C534 and shall be Armacell AP Armaflex or approved equal for installation within non-plenums in accordance with the flame/smoke hazard ratings specified hereinbefore.

4. Each refrigerant piping shall be insulated with 1" thick refrigerant piping insulation.

5. Insulation Wall Thickness - 1" and smaller:

   a. Insulation shall be NBR/PVC material and shall have a thermal conductivity (K value) of no more than 0.25 Btu-in/h-sf-°F at 75°F mean temperature per ASTM C177 or C518.

   b. Insulation shall have a water permeability of 0.05 perms or less when tested by ASTM E96, Procedure A.

   c. Insulation shall be rated for upper and lower temperature limits of 220°F and -40°F when tested by ASTM C534.

   d. Insulation shall withstand 250°F for up to 96 hour period without affecting thermal performance and water vapor permeability when tested by ASTM C411.

6. Insulation Wall Thickness - 1-1/2" and larger:

   a. Insulation shall be EPDM material and have a thermal conductivity (K value) of no more than 0.28 Btu-in/h-sf-°F at 75°F mean temperature per ASTM C177 or C518.

   b. Insulation shall have a water permeability of 0.08 perms or less when tested by ASTM E96, Procedure A.

   c. Insulation shall be rated for upper and lower temperature limits of 300°F and -40°F when tested by ASTM C534.

7. Insulation adhesive shall be Armaflex BLV 520 or approved equal. Adhesive shall be specifically formulated to bond elastomeric insulation.

8. For outdoor applications, provide Armaflex WB Finish water based enamel coating to protect insulation against ultraviolet ultraviolet damage. Aluminum jackets as specified hereinafter may be provided in lieu of UV protection coating.

B. AHU Condensate Piping:

1. Insulation shall be provided with factory applied vapor barrier all service jacket (ASJ) in accordance with ASTM C921. ASJ shall be composed of polymer film exterior surface on glass scrim reinforced kraft paper bonded to inner aluminum foil. ASJ water vapor permeance shall be less than 0.01 perms per ASTM E96 (Procedure A). At Contactor's option, AHU condensate drain piping may be insulated with 0.5" thick minimum refrigerant piping insulation specified hereinbefore.

C. Piping Systems Subject to Freezing:

1. All piping systems exposed to freezing conditions with or without heat tracing shall be insulated as specified herein for each type of piping. Refer to Section 23 05 34 for additional requirements.

D. Jackets and Facings

1. Vapor barrier jackets and facings shall comply with ASTM C921 and C1136. Jackets and facings shall have a permeance rating not to exceed 0.02 perm-inch when tested by ASTM E96 (Procedure A), puncture resistance not less than 50 units, and tensile strength not less than 35 pounds per inch width when tested by ASTM C921.

2. Jackets and facings on insulation in exposed areas shall have a white finish suitable for painting without priming.

3. Insulation exposed to outside environment or as noted herein shall be covered with a continuous 0.016" thick aluminum jacket which shall be secured in place by a continuous longitudinal friction Type "Z" joint to provide a waterproof seal along the entire length of the aluminum jacket. Aluminum jacket shall comply with ASTM C1729. Furnish end closures, fitting covers and caps to cover completely all valves, fittings, etc. Plastic covers for fittings, elbows, and pipe couplings will not be acceptable.

4. As an acceptable option to providing the aluminum jacket cover indicated above, insulation exposed to outside environments or as noted herein may be covered with Alumaguard 60 manufactured by Polyguard Products or approved equal. Alumaguard 60 is ultraviolet resistant having a laminated waterproofing and vapor barrier membrane composed of rubberized bitumen compound on a cross laminated high strength polyethylene film with an aluminum foil surface. The complete installation of Alumaguard 60 shall comply with the manufacturer's published written recommendations.

5. In addition to the vapor barrier jacket provided, provide either woven canvas covering or 10x10 open weave fiberglass mesh imbedded between coats of mastic for pipe insulation installed less than eight (8) feet above the floor within the main central plant area and penthouse area. If the open weave mesh with mastic option is provided, mastic shall be applied at a rate sufficient to completely cover and hide the open weave mesh. Mastic shall be rated as a vapor barrier for less than ambient piping systems or be rated as a breather type for above ambient piping systems. This additional requirement does not apply to piping insulation enclosed in a metal jacket

2.5 DUCTWORK SYSTEMS

A. Supply Air Ductwork: External ductwork insulation installation shall be as follows, unless indicated otherwise. All air conditioning supply ductwork that is not internally lined shall be externally insulated with fiberglass flexible blanket type insulation in lieu of duct liner insulation. Supply ductwork shall be defined as any ductwork, including but not limited to sheet metal ductwork, sheet metal plenums and flexible connections, that contains air conditioned air, contains air at a temperature less than 66°F, outside air system ductwork, or any other ductwork connected downstream of an air conditioning unit.

1. If it complies with these Specifications, one of the following manufacturers will be acceptable:

   a. Johns Manville
   b. Owens Corning
   c. CertainTeed
   d. Knauf
   e. Manson

2. Rectangular Ductwork:

   a. Flexible blanket type fiberglass external ductwork insulation shall be Knauf Performance+ Duct Wrap", Johns Manville "Microlite Duct Wrap", Owens Corning "SoftR Duct Wrap" or approved equal in accordance with ASTM C553, C1290 and NFPA 90A.

   b. Insulation shall be provided with laminated foil scrim kraft (FSK) vapor barrier facing and shall comply with ASTM C1136. Vapor barrier water permeance shall be no greater than 0.02 perms per ASTM E96 (Procedure A). Vapor barrier facing may be polypropylene scrim kraft (PSK) for open ceiling application to be painted as directed by the Architect.

   c. Insulation shall be 1.5" thick and 1.0 pound (per ft3) density.

      Installed thermal resistance (R value) shall be no less than 4.5 h-sf-°F/Btu and thermal conductivity (K value) shall be no more than 0.25 Btu-in/h-sf-°F at 75°F mean temperature with 25% compression in accordance with ASTM C177 or C518. Insulation shall be UL Greenguard Gold certified and listed under UL 2818.

3. Round and Oval Ductwork:

   a. Insulation material and construction shall be the same as rectangular ductwork except 1.5" thick and 0.75 pound (per ft3) density.

   b. Installed thermal resistance (R value) shall be no less than 4.2 h-sf-°F/Btu and thermal conductivity (K value) shall be no more than 0.27 Btu-in/h-sf-°F at 75°F mean temperature with 25% compression in accordance with ASTM C177 or C518. Insulation shall be UL Greenguard Gold certified and listed under UL 2818.

4. Flexible Ductwork: All flexible supply air ductwork not factory insulated shall be externally insulated. Insulation shall be the same as round and oval ductwork systems.

B. Refer to Section 23 30 01 for duct liner or internal duct insulation information.

## 2.6 EQUIPMENT

A. Air Handling Units: Factory-built air handling units shall be factory insulated.

## 2.7 ADHESIVE, MASTIC AND SEALANT VOC REQUIREMENTS

A. Adhesives and mastics shall be as manufactured by MEI, Childers, Foster, Armacell, and 3M shall have the same adhesive properties, fire rating, vapor seal, etc., as the types specified herein, subject to review by the Engineer.

B. For indoor applications, use adhesives, mastics and sealants that have a maximum volatile organic compound (VOC) content as listed below when calculated in accordance with 40 CFR 59, Subpart D (EPA Method 24).

1. Adhesives:

| | |
|---|---|
| a. Calcium Silicate Adhesive: | 80 grams/liter |
| b. Cellular Glass Adhesive: | 50 grams/liter |
| c. Flexible Elastomeric Adhesive: | 50 grams/liter |
| d. Mineral-Fiber Adhesive: | 80 grams/liter |

| | | |
|---|---|---|
| e. | ASJ and FSK Jacket Adhesive: | 50 grams/liter |
| f. | PVC Jacket Adhesive: | 50 grams/liter |
| 2. | Mastics: | 80 grams/liter |
| 3. | Lagging Adhesives: | 80 grams/liter |
| 4. | Sealants: | 250 grams/liter |

PART 3 - EXECUTION

3.1 APPLICATION - GENERAL

A. The execution of the insulation work shall be in strict accordance with the best practices of the trade, the listed Reference Standards, applicable local codes.

B. Unless otherwise specified, the application of all insulation materials shall be in accordance with the manufacturer's published recommendations. Where these Specifications are in conflict with manufacturer's instructions, such conflicts shall be brought to the attention of the Engineer for his decision.

C. Insulation shall not be applied until all surfaces are clean and dry. The insulation on pipe fittings, valves and pipe joints shall not be installed before the piping is tested and reviewed by the Engineer.

D. The insulation shall be handled and applied in a manner that will not adversely affect its structural or insulating properties.

E. A complete moisture and vapor seal shall be provided over insulation on cold surfaces where vapor barrier jackets, facings, or coatings are required. Anchors, hangers and other projections shall be insulated and vapor sealed to prevent condensation. All openings and punctures shall be sealed with vapor barrier compound. The use of staples will not be acceptable.

F. Insulation shall be continuous through hangers and wall and floor openings except at fire dampers.

G. Date of expiration for product installation shall not be exceeded. Products using release paper systems and pressure sensitive tapes shall not be older than one year from date of manufacturing at time of installation.

H. Temperature limitations for the application of products and systems shall not be exceeded.

I. Insulation shall be applied to strainers in such a manner as to allow removal of the end cap and perforated screen without destroying the strainer casing insulation.

3.2 PIPING SYSTEMS INSTALLATION

A. DX Split-System Air Conditioner Refrigerant Piping:

1. All suction piping shall be insulated. Liquid piping shall be insulated only if required by the split-system manufacturer.

2. The pipe insulation seams and joints shall be sealed with Armacell adhesive. The entire Armacell insulation installation shall be in accordance with application procedures and recommendations as described in Armacell’s latest Armaflex Application Manual.

B. AHU Condensate Drain Piping: Air handling unit condensate drain lines located above a ceiling

or within a wall cavity or chase shall be insulated in the same way as chilled water piping. Condensate drain lines from floor mounted units located in return air plenums do not require insulation unless required by Local Codes.

C. Air handling unit condensate elastomeric closed cell pipe insulation option: The pipe insulation seams and joints shall be sealed with Armacell adhesive. The entire Armacell insulation installation shall be in accordance with application procedures and recommendations as described in Armacell's latest Armaflex Installation Manual.

D. Piping systems subject to freezing: These systems shall be insulated as indicated in this section for each type of piping. Piping systems subject to freezing conditions, not indicated in this section to be insulated, shall be insulated in the same way specified for chilled water piping.

## 3.3 DUCTWORK SYSTEMS INSTALLATION

A. Rectangular Duct Insulation:

1. Insulation refers to the complete insulation system; including but not limited to insulation, facing, tapes, adhesives, mastics, fasteners, attachment devices, and other materials required for a complete installation. Insulation shall be installed in accordance with the insulation manufacturer's recommendations. Insulation as installed on this project shall form a continuous thermal and vapor barrier to minimize heat transfer and prevent condensation and moisture sweating from forming on the interior and exterior surfaces of the ductwork and ductwork insulation. Refer to the Schedule of Capacities for HVAC system air temperatures. Insulation damaged by condensation or moisture shall be replaced at no additional cost to the owner.

2. Duct wrap insulation shall be applied on clean, dry surfaces only after duct leak tests and inspections have been completed.

3. All duct wrap insulation shall be continuous through all wall and slab penetrations except at fire dampers. Insulation shall be applied to all sides of ductwork.

4. The insulation shall be precut by the manufacturer or field cut to provide a minimum 2" facing overlap at butt joints. Transverse and longitudinal edges shall be tightly butted. Overlap facing shall be secured to the adjacent insulation with an approved adhesive/mastic. In lieu of securing the lap joint material with adhesive/sealant, if recommended by the insulation manufacturer, the use of outward clinched staples to secure insulation seams and overlaps will be acceptable. Staples shall be spaced as recommended by the insulation manufacturer to achieve the specified performance, but not less than approximately 6" on center. All joints and overlap shall be covered with an approved vapor retarder mastic reinforced with 6" wide open-weave glass fabric. The mastic shall extend a minimum of 1" beyond the edge of the glass fabric in each direction. The mastic can be applied by either brush or spray at the Contractor's option. In lieu of mastic and glass fabric, 6" wide pressure sensitive tape centered over the overlap joint will be acceptable. Pressure sensitive tape shall be applied with moving pressure using a squeegee or other appropriate sealing tool as recommend by the insulation manufacturer to prevent creating an air pocket beneath the tape.

5. Duct wrap insulation shall be secured to the bottom of the duct with approved mechanical fasteners such as stick pins or weld pins on 18" centers along a line 3" from the leading edge of the insulation. Contractor's option for ductwork less than 24" in width: In lieu of mechanical fasteners, the duct wrap insulation shall be secured to the ducts with a Benjamin Foster No 85-60 adhesive applied in 6" wide strips on 12" centers.

6. Insulation shall be continuous over hangers. In order to maintain the performance integrity of the insulation vapor barrier, penetrations and damage to the duct wrap insulation facing shall be sealed or otherwise repaired using pressure sensitive tape matching the duct wrap insulation facing or mastic prior to system startup. The approved adhesive/sealant shall be continuous and extend a minimum of 1" beyond the edge of the pressure sensitive tape in each direction. Pressure sensitive tape shall be a minimum 3" wide and shall be applied with moving pressure using a squeegee or other appropriate sealing tool as recommend by the insulation manufacturer to prevent creating an air pocket beneath the tape.

7. Avoid over-compression of the insulation during installation and avoid damage to the duct wrap insulation. Insulation shall not be compressed beyond 25% of its nominal thickness.

B. Round and Flexible Ductwork: The insulation shall be secured to the ducts with a Foster No 85-60 adhesive applied in 6" wide strips on approximately 12" centers. Fiberglass tying cord shall be used for securing insulation until the adhesive sets. Insulation seams and joints shall be sealed in accordance with the requirements of the rectangular duct wrap insulation as specified in Section 23 30 01.

C. Painting:

1. Duct insulation with FSK or PSK facings shall be painted as directed by the Architect. Water based acrylic paint is recommended to be used. Exact type of paint shall be approved from the insulation manufacturer. All paints shall be field tested for adhesion and flexibility before applying to the installed ductwork.

3.4 EQUIPMENT INSULATION INSTALLATION

A. Air Handling Units: Factory-built air handling units shall be factory insulated.

3.5 CLEANING

A. All of the above items shall be cleaned, and all dust and debris removed.

END OF SECTION 23 07 00

SECTION 23 21 13 - HYDRONIC PIPING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the HVAC piping systems as shown on the Drawings and as specified herein for the various systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 10 - Wall and Floor Penetrations
B. Section 23 00 15 - Division 23 Design Conditions
C. Section 23 00 70 - Division 23 Schedule of Submittal Data
D. Section 23 05 23 - General Duty Valves for HVAC Piping
E. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment
F. Section 23 05 53 - Identification for HVAC Piping and Equipment
G. Section 23 05 93 - Testing, Adjusting and Balancing for HVAC
H. Section 23 07 00 - HVAC Insulation
I. Section 23 30 01 - Ductwork and Sheet Metal

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
1. Boiler and Pressure Vessel Code Section IX - Welding and Brazing Qualifications
2. A1.20.1 Pipe Threads, General Purpose, Inch
3. B16.1 Cast Iron Pipe Flanges and Flanged Fittings
4. B16.3 Malleable Iron Threaded Fittings Classes 150 and 300
5. B16.5 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
6. B16.9 Factory-Made Wrought Buttwelding Fittings
7. B16.11 Forged Fittings, Socket-Welding and Threaded
8. B16.15 Cast Copper Alloy Threaded Fittings: Classes 125 and 250
9. B16.18 Cast Copper Alloy Solder Joint Pressure Fittings
10. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
11. B31.1 Power Piping
12. B31.9 Building Services Piping
13. B36.10M Welded and Seamless Wrought Steel Pipe
14. B36.19M Stainless Steel Pipe

B. ASTM International (formerly American Society for Testing and Materials)
1. A53/A53M Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
2. A135 Standard Specification for Electric-Resistance-Welded Steel Pipe
3. A234 Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
4. B32 Standard Specification for Solder Metal
5. B88 Standard Specification for Seamless Copper Water Tube
6. B306 Standard Specification for Copper Drainage Tube
7. B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and

Copper Alloy Tube
8. F1476 Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications

C. American Welding Society (AWS)
1. A5.8 Specifications for Filler Metals for Brazing and Braze Welding
2. B2.1 Specification for Welding Procedure and Performance Qualification

D. Copper Development Association (CDA)
1. A4015 Copper Tube Handbook

E. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
1. SP-58 Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation
2. SP-69 Pipe Hangers and Supports - Selection and Application
3. SP-89 Pipe Hangers and Supports - Fabrication and Installation Practices

F. National Fire Protection Association (NFPA)
1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems
2. 211 Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all specification requirements listed herein plus certification data and shop drawing requirements listed in Section 23 00 10 and 23 00 70.

B. The Contractor shall provide, where applicable, product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content. Submit Material Safety Data Sheets (MSDS) clearly highlighting the VOC content for each field applied piping adhesive and sealant used within the building interior locations.

PART 2 - PRODUCTS

2.1 GENERAL

A. All materials and specialties required for the work shall be new, of first class quality and shall be furnished, delivered, erected, connected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class standard article as approved by the Architect shall be provided.

B. All materials and specialties required for this project shall be the products of United States made materials. International or foreign made materials and specialties shall not be acceptable unless pre-approved by the Owner prior to bidding. If the Owner pre-approves the use of international or foreign made materials and specialties prior to bidding, the Contractor shall certify in writing to the Engineer that all international or foreign made materials and specialties furnished for this project are suitable for the service intended and complies with all the requirements specified herein. This certification shall be included within the submittal being issued to the Engineer for review and approval. In addition, the Contractor shall submit a 12" long sample of each type of international or foreign manufactured pipe material to be installed on this project to a certified metallurgy laboratory for complete properties and strength analysis. The samples shall be cut from pipe material stored on this projects site or in the Contractor’s fabrication facilities. The pipe material shall be identified as to its intended use, ASTM designation, project name, Contractor’s name, and date the sample was obtained. If the pipe is coated, such as with a galvanized material, the thickness and properties of the coating material

shall be indicated. A certified report of the test data shall be provided to the Owner and the Engineer for review and approval prior to the final acceptance and installation of any international or foreign made material and specialty. The cost of the certified laboratory testing shall be at the Contractors expense and the certified laboratory shall be subject to approval by the Owner and Engineer prior to testing. Any international or foreign made material and specialty that does not comply with the material standards and all other requirements specified herein shall be removed. Prior to purchase of any materials manufactured outside the United States, the Contractor shall submit complete certifications and typical mill reports in English language for review. Any foreign pipe delivered to the jobsite shall bear mill heat markings and the Contractor shall make available the corresponding mill test reports.

C. Connections between dissimilar piping materials shall be made with suitable dielectric insulating pipe fittings and unions.

D. HVAC water piping shall be defined as chilled water, condenser water, heating hot water, and glycol system piping.

## 2.2 DRAIN PIPING

A. Unless otherwise noted on the Drawings, drain piping from each air handling unit drain pan, strainer, water chilling unit, heat exchanger, pump base drain, tank and vessel overflow, auxiliary drain pan, piping system drain, blowdown, and any other system or equipment that has a drain connection shall be Schedule 40 galvanized steel pipe with galvanized malleable iron fittings, Type K or DWV copper piping. At the contractor’s option, copper press-connect fittings may be used for drain piping 4" and smaller as specified in Section 22 11 16 for Domestic Water Piping.

## 2.3 PIPE HANGERS AND SUPPORTS

A. All pipes throughout the building shall be thoroughly and substantially supported with UL listed hangers and supports. In addition to the items specified herein, furnish and install any special hangers or supports that may be required by the building construction. The design, selection, spacing, and application of horizontal pipe hangers, supports, restraints, anchors, and guides shall be in accordance with ASME B31, MSS SP-58, and the applicable mechanical or plumbing code, whichever is most stringent. In no case shall piping be installed that is not in compliance with the applicable code.

B. Unless otherwise noted all horizontal piping shall be supported on National Pipe Hanger Corporation (NPHC) Figure 200, Carpenter & Paterson Figure 100, or approved equal adjustable wrought clevis hangers.

C. All vertical pipes not specified to be supported on vibration isolation shall be supported as follows:

1. Vertical pipes 8" in diameter and smaller shall be supported at least at every other floor with NPHC Figure 420, Carpenter & Paterson Figure 126, or approved equal riser clamps.

2. Vertical pipes 10" and larger in diameter shall be supported at least at every other floor, with four or six bolt riser clamps as recommended by the pipe hanger manufacturer, NPHC Figure 460 with four (4) bolts and extended ears, Carpenter & Paterson Figure 124, or approved equal.

3. The location and spacing of the vertical pipe supports specified above shall be in

accordance with the manufacturer's recommendations. If supporting vertical pipe risers at every other floor is not adequate as determined by the manufacturer, provide vertical pipe supports at every floor. If supporting vertical pipe risers at every floor is not adequate as determined by the manufacturer, the manufacturer shall be consulted to provide alternate vertical pipe supports as necessary to adequately support the vertical pipe risers.

D. Unless otherwise noted on the drawings, all horizontal water piping mounted on floor or roof deck shall be supported as indicated on the Drawings or by an adjustable pipe roller with fabricated steel angle assembly Anvil Figure 171, Carpenter & Paterson Figure 142, or approved equal. Pipe roller with fabricated steel angle assembly shall be suitable for mounting on top of vibration isolators and arranged to minimize overall height of pipe and support assembly. Refer to the Drawings and Section 23 05 48 for additional requirements.

E. Concrete inserts for hanger rods attached to new structure shall be NPHC Figure 600 with Figure 605, Anvil Figure 282, Anvil Figure 152, Carpenter & Paterson Figure 108, or approved equal, for pipe sizes up to 8". For over 8" pipe, an insert reinforced with steel anchor bars or suitable steel plate embedded in the concrete may be used.

F. Metal deck concrete insert applicable to new structure shall be galvanized fabricated steel metal deck ceiling bolt, Phillips Red Head or approved equal.

G. All hangers, hardware, rods, clamps, channels, base plates, and other factory-built pipe support devices shall be galvanized. All shop fabricated and welded steel supports shall be hot dipped galvanized after fabrication.

H. Refer to Section 23 00 10 "Equipment Mounting/Support Devices" for additional requirements.

2.4 PIPE COVERING PROTECTION SADDLES AND SHIELDS

A. Protection shield length and gauge thickness for use with clevis hangers shall be as specified for Type 40 protection shields in MSS SP-58. Protection shields shall be galvanized and shall be arranged to cover one half of the circumference of the insulations and shall be mounted on the outside of the insulation with insulation blocking between the pipe and saddle to prevent crushing of insulation. Blocking on piping shall be furnished and installed as specified in Section 23 07 00.

2.5 ACCESSORIES

A. Dielectric Pipe Fittings/Unions:

1. Fittings and unions shall be Watts 3000 series, Victaulic Style 47 waterway fittings, Shurjoint DE30GG or approved equal.

B. Piping Adhesives: All piping adhesives shall comply with the current Volatile Organic Compound (VOC) content limits of the latest edition of the South Coast Air Quality Management District (SCAQM) Rule 1168 entitled "Adhesive and Sealant Applications". Piping sealants shall have no more than 250 grams/liter VOC content for all field applied piping adhesives and sealants to be used within the building interior locations.

PART 3 - EXECUTION

3.1 GENERAL

A. All piping shall be made up and installed in a manner that permits expansion and contraction

caused by changes in temperature and pressure. This shall be accomplished by the use of supports that will permit the movement of the pipe without undue stress, and by any other precautions that might be deemed necessary by the Engineer.

B. Piping exposed in all rooms shall be installed as nearly as possible parallel with or at right angles to the building walls. Install all pipe straight and true. Springing or forcing pipe into place will not be permitted unless specifically called for. Install piping in such a manner as to prevent strain on equipment connections.

C. Below grade piping shall be installed in such a manner that it does not bear directly on rocks or debris in ground. Any location where pipe passes close to or through walls or footings, it shall be protected from contact with concrete or cinder block. Any pipe passing through building walls shall be protected by a cast iron sleeve large enough to permit free movement of pipe. Refer to Section 23 00 10 for additional sleeve requirements.

D. All piping shall be protected at all times from dirt and moisture. During storage on the job site of construction, the Contractor shall keep pipe ends plugged or capped to prevent dirt or moisture entering the pipe.

E. The Contractor shall submit for the Engineer's review, a list of the proposed manufacturers of pipe, valves, fittings, and specialties.

F. All piping shall be carefully graded so as to eliminate traps and pockets. Where air pockets or water traps cannot be avoided, provide means for drainage with valved hose connections for water traps and with automatic air vents for air pockets. Automatic air vents shall be valved.

G. Where horizontal pipes change size, eccentric fittings shall be used, except where branch pipes connect into mains.

H. Make all joints smooth and unobstructed inside. Ream all pipe ends to remove burrs. Remove all obstructions and debris inside the piping systems prior to installation.

I. Make welded joints on the piping system with continuous welds, without backing rings, and with pipe ends beveled before welding. Gas cuts shall be true and free from burned metal. Before welding, welding surfaces shall be thoroughly cleaned. The piping shall be carefully aligned, and no weld metal shall project inside the pipe.

J. Connections between dissimilar piping materials shall be made with insulating couplings or dielectric fittings/unions. Fittings/unions shall be installed in piping systems in accordance with the manufacturer's recommendations and installation guidelines as required to prevent "Galvanic" and "Stray current" corrosion. Bronze bodied valves and devices installed in steel piping systems do not require dielectric fittings when both connections are made with steel piping, but do require dielectric fittings when the bronze bodied valves or devices are installed with one connection to steel piping and the other connection to copper piping. Iron bodied valves and devices installed in copper systems do require dielectric fittings.

K. All hardware, including nuts, bolts, washers, and other miscellaneous hardware and accessories, used to connect the piping to the cooling towers shall be stainless steel. All hardware, including nuts, bolts, washers, and other miscellaneous hardware and accessories, used on piping, flanges for valves, and on other piping accessories located on the roof and outside of the enclosed Penthouse walls which are exposed to weather conditions shall be stainless steel or zinc coated.

L. All piping shall be of the sizes indicated and shall be routed as indicated on the drawings, or as

required, to serve all equipment and systems. Should there be a discrepancy in pipe sizes between the riser or schematic diagrams, and the floor plans; the riser and schematic diagrams will govern.

M. All piping materials, fabrication, erection, flanging, welding, tests, etc., shall be in accordance with American National Standard Code for Pressure Piping, current edition. Wire type "Short Arc" welding machines will not be acceptable for field welds. Wire type short arc welding machines will be acceptable for shop fabricated piping. If in the opinion of the Engineer any "short-arc" weld is found unsatisfactory, the use of the wire type short welding machine shall be discontinued for all remaining welding and re-welding the defective welds. The Engineer shall have access to the pipe fabrication facilities.

N. All welders shall be certified in accordance with ASME B31 or with ASME Boiler and Pressure Vessel Code Section IX. If required by Engineer, the Contractor shall furnish welder performance qualification test certificates for positions 2G, 5G, 6G made in strict compliance with the above codes. Welders shall be certified for the type of pipe material specified. All costs incident to procedures and welder's qualification tests shall be assumed by the Contractor. Copies of the qualification test report and certification with welder's identification number, letter, etc., shall be available from the Contractor at the request of the Engineer. Each weld shall bear the welder's identification mark permanently indented in the weld. Welding procedures shall also be in accordance with the requirements of the American Welding Society, current edition, where applicable.

O. Piping insulation shall not be installed on connections, joints, flanges, etc., until pressure testing has been successfully accomplished.

P. Piping systems shall not be filled or put into operation until the vibration isolators have been installed.

3.2 DRAIN PIPING

A. Drain piping from each air handling unit drain pan, strainer, water chilling unit, heat exchanger, pump base drain, tank and vessel overflow, auxiliary drain pan, piping system drain, blowdown, and any other system or equipment that has a drain connection shall be extended to the nearest floor drain, hub drain, or condensate drainage system.

B. Drains shall be of the sizes indicated, but not less than the drain pan connection size.

C. A valved 3/4" drainage line shall be provided at all low points in the piping systems.

D. Air handling unit drains shall have deep seal traps to permit unit pan drainage. A deep seal trap shall be installed for each blow through or draw through air handling unit to maintain the water seal.

E. Unless noted otherwise, drain piping shall be installed with a minimum of 1/8” per foot of slope.

3.3 PIPE HANGERS AND SUPPORTS

A. Hanger rods, inserts, etc., shall be sized and installed as recommended by the hanger manufacturer for the service intended.

B. The Contractor shall submit pipe hanger and support details for each type of floor construction to the Structural Engineer for review and approval prior to submission to the Engineer.

3.4 PIPE COVERING PROTECTION SADDLES AND SHIELDS

A. Furnish and install pipe covering protection shields and saddles for all insulated piping at the locations of all supports.

3.5 ACCESSORIES

A. Dielectric Pipe Fittings/Unions: Unions or flanges shall be provided at all connections of equipment, strainer, equipment drain, control valve, and other locations indicated on the Drawings.

3.6 SCREW JOINTS

A. Screw joints shall be made with standard pipe thread and an approved compound applied to the male thread only.

3.7 SOLDERED/BRAZING PIPING JOINTS

A. Ends of pipe and inside of fittings shall be cleaned and flux applied to entire area of pipe end to be soldered. On pipes 1-1/2" and larger, flux shall be applied to pipe and fittings. Joints and fittings shall be soldered with ASTM B32 Grade 95T tin antimony solder, tin-bismuth-copper-silver solder, or brazed with AWS classification Bcup 3 brazing alloy as required by the system temperature and working pressure. Solder or brazing compounds containing lead shall be prohibited. Contractor shall thoroughly flush all soldered piping to remove excess flux so as to prevent corrosion of piping.

3.8 GROOVED PIPE JOINNING SYSTEM

A. In addition to the piping execution and installation requirements specified herein, Victaulic piping systems, Anvil International Gruvlok, Apollo/Shurjoint, and other manufactured grooved joint coupling systems shall be installed in accordance with the manufacturer’s guidelines and recommendations. Grooves shall be made by rolling a groove around the end of the pipe. The depth and side to side dimensions of the groove and the distance from the end of the pipe shall be as recommended by the coupling manufacturer. Groove rolling machinery shall be as recommended and approved by the system manufacturer. The Contractor shall protect the gasket from damage during assembly of the joint. Groove rolling and installation of gaskets and couplings shall be in accordance with the manufacturers written recommendations. All grooved joint couplings and fittings must be the products of the same manufacturer. No cross mixing of one manufacturer’s couplings and fittings with another manufacturer’s couplings and fittings will be allowed.

B. All hardware, including nuts, bolts, washers, and other miscellaneous hardware and accessories, used on piping, flanges for valves, and on other piping accessories located on the roof and outside of the enclosed Penthouse walls which are exposed to weather conditions shall be stainless steel or zinc coated.

3.9 PIPING SUPPORTS (ABOVE GRADE)

A. All pipe shall be supported from the building structure in a neat and workmanlike manner and wherever possible, parallel runs of horizontal piping shall be grouped together on trapeze type hangers. The use of wire or preformed metal to support pipes will not be permitted. Hanging pipes from other pipes will not be permitted. Piping shall be carefully coordinated before installation with other systems and equipment in chases and congested areas.

B. Vertical and horizontal pipes subject to sway in either direction shall have a snubber to prevent lateral movement.

3.10 PIPING TESTS

A. Scope: All piping installed on the Project unless specifically shown otherwise shall be hydraulically tested as specified herein, in accordance with ASME B31.1, and as required by the local code authorities. Refrigeration piping, if used on the Project, shall be tested as specified under refrigeration piping. This Contractor shall provide all equipment required to make the tests specified herein. Calibrated pressure gauges accurate to within ±1/2% of full range shall be used for pressure testing. System pressure gauges shall not be substituted for calibrated test gauges.

B. Sectionalizing: Piping may be tested a section at a time in order to facilitate the construction.

C. Tests Procedure: Before the piping of various systems is insulated, furred-in, or otherwise covered up; the subcontractor shall fill the section of the pipe to be tested with water and bring the section up to pressure with a test pump. The Contractor shall notify the Architect or his representative, in writing, at least five (5) working days prior to each test. The Architect or his representative may elect to witness the tests. These tests shall be conducted before any insulation is installed. Any insulation installed prior to these tests shall be removed.

D. Pressure of Tests: All tests shall be conducted at 150% of the maximum system design pressure of the pipe installed as listed in Section 23 00 15, but not less than 100 psig.

E. Duration of Tests: All tests shall apply test pressure to the piping for a minimum of 6 hours.

F. Inability to Hold Pressure: When the test pressure has fallen over 5% during the 24-hour test period, the point of leakage shall be found, repaired and the test repeated. This procedure shall be followed until the piping system has been proven absolutely tight.

G. Use of Compounds: The use of chemicals or stop leak compounds will not be permitted at any time.

H. Delicate Controls: When delicate control mechanisms are installed in the piping section, they shall be removed during the tests to prevent shock damage. This does not apply to control valves.

I. Leaks: Leaks developing subsequent to these tests shall not be repaired by mastic or other temporary means. All leaks shall be repaired by removal of the valve, fitting, joint, or section that is leaking and reinstalling new material with joints as specified hereinbefore.

END OF SECTION 23 21 13

SECTION 23 23 00 - REFRIGERANT PIPING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the HVAC refrigerant piping systems as shown on the Drawings and as specified herein for the various systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 10 - Wall and Floor Penetrations
B. Section 23 00 70 - Division 23 Schedule of Submittal Data
C. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment
D. Section 23 05 53 - Identification for HVAC Piping and Equipment
E. Section 23 05 93 - Testing, Adjusting and Balancing for HVAC
F. Section 23 07 00 - HVAC Insulation
G. Section 23 81 29 - Variable Refrigerant Flow HVAC Systems

1.4 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)
1. A1.20.1 Pipe Threads, General Purpose, Inch
2. B16.15 Cast Copper Alloy Threaded Fittings: Classes 125 and 250
3. B16.22 Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
4. B31 Code for Pressure Piping
5. B31.1 Power Piping
6. B31.9 Building Services Piping

B. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1. 15 Safety Code for Mechanical Refrigeration Systems
2. 34 Designation and Safety Classification of Refrigerants

C. ASTM International (formerly American Society for Testing and Materials)
1. B75/B75M Standard Specification for Seamless Copper Tube
2. B88 Standard Specification for Seamless Copper Water Tube
3. B280 Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service
4. B306 Standard Specification for Copper Drainage Tube

D. American Welding Society (AWS)
1. A5.8 Specifications for Filler Metals for Brazing and Braze Welding
2. B2.1 Specification for Welding Procedure and Performance Qualification

E. Copper Development Association (CDA)
1. A4015 Copper Tube Handbook: Industry Standard Guide for the Design and Installation of Copper Piping Systems

F. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
   1. SP-58 Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation
   2. SP-69 Pipe Hangers and Supports - Selection and Application
   3. SP-89 Pipe Hangers and Supports - Fabrication and Installation Practices

G. National Fire Protection Association (NFPA)
   1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

H. Underwriters Laboratories (UL) Standard for Safety
   1. 109 Tube Fittings for Flammable and Combustible Fluids, Refrigeration Service, and Marine Use
   2. 207 Refrigerant-Containing Components and Accessories, Nonelectrical

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all specification requirements listed herein plus certification data and shop drawing requirements listed in Section 23 00 10 and 23 00 70.

B. The Contractor shall provide, where applicable, product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content. Submit Material Safety Data Sheets (MSDS) clearly highlighting the VOC content for each field applied piping adhesive and sealant used within the building interior locations.

PART 2 - PRODUCTS

2.1 GENERAL

A. All materials and specialties required for the work shall be new, of first class quality and shall be furnished, delivered, erected, connected and finished in every detail, and shall be so selected and arranged as to fit properly into the building spaces. Where no specific kind or quality is specified, a first class standard article as approved by the Architect shall be provided.

2.2 REFRIGERANT PIPING

A. Piping: Piping shall be ACR Type L hard copper.

B. Fittings: Fittings shall be Mueller Streamline wrought copper solder fittings with long radius bends or approved equal by Chase or Anaconda.

C. Insulation: Refer to Section 23 07 00 for requirements.

2.3 DRAIN PIPING

A. Unless otherwise noted on the Drawings, drain piping from each air handling unit drain pan, auxiliary drain pan, and any other system or equipment that has a drain connection shall be Schedule 40 galvanized steel pipe with galvanized malleable iron fittings, Type K or DWV copper or cast iron. At the contractor’s option, copper press-connect fittings may be used for drain piping 4" and smaller as specified in Section 22 11 16 for Domestic Hot and Cold Water Piping.

2.4 PIPE HANGERS AND SUPPORTS

A. All Pipe Hangers and Supports: Refrigerant piping shall be adequately supported to avoid pipe sagging during the life of the system in accordance with the manufacturer's recommendations. The split system manufacturer shall provide minimum requirements on support spacing, location of supports at indoor units, outdoor units and refrigerant distribution units, branch fittings, header fitting, take-offs, and changes in pipe direction.

B. All pipes throughout the building shall be thoroughly and substantially supported with UL listed hangers and supports. In addition to the items specified herein, furnish and install any special hangers or supports that may be required by the building construction. The design, selection, spacing, and application of horizontal pipe hangers, supports, restraints, anchors, and guides shall be in accordance with ASME B31, MSS SP-58, and the applicable mechanical or plumbing code, whichever is most stringent. In no case shall piping be installed that is not in compliance with the applicable code.

C. Unless otherwise noted on the drawings, all horizontal piping mounted on floor or roof deck shall be supported as indicated on the Drawings or by an adjustable pipe roller with fabricated steel angle assembly Anvil Figure 171, Carpenter & Paterson Figure 142, or approved equal. Pipe roller with fabricated steel angle assembly shall be suitable for mounting on top of vibration isolators and arranged to minimize overall height of pipe and support assembly. Refer to the Drawings and Section 23 05 48 for additional requirements.

D. Concrete inserts for hanger rods attached to new structure shall be NPHC Figure 600 with Figure 605, Anvil Figure 282, Anvil Figure 152, Carpenter & Paterson Figure 108, or approved equal, for pipe sizes up to 8". For over 8" pipe, an insert reinforced with steel anchor bars or suitable steel plate embedded in the concrete may be used.

E. All hangers, hardware, rods, clamps, channels, base plates, and other factory-built pipe support devices shall be galvanized. All shop fabricated and welded steel supports shall be hot dipped galvanized after fabrication.

F. Refer to Section 23 00 10 "Equipment Mounting/Support Devices" for additional requirements.

2.5 PIPE COVERING PROTECTION SADDLES AND SHIELDS

A. Protection shield length and gauge thickness for use with clevis hangers shall be as specified for Type 40 protection shields in MSS SP-58. Protection shields shall be galvanized and shall be arranged to cover one half of the circumference of the insulations and shall be mounted on the outside of the insulation with insulation blocking between the pipe and saddle to prevent crushing of insulation. Blocking on piping shall be furnished and installed as specified in Section 23 07 00.

PART 3 - EXECUTION

3.1 GENERAL

A. All piping shall be made up and installed in a manner that permits expansion and contraction caused by changes in temperature and pressure. This shall be accomplished by the use of supports that will permit the movement of the pipe without undue stress, and by any other precautions that might be deemed necessary by the Engineer.

B. Piping exposed in all rooms shall be installed as nearly as possible parallel with or at right

angles to the building walls. Install all pipe straight and true. Springing or forcing pipe into place will not be permitted unless specifically called for. Install piping in such a manner as to prevent strain on equipment connections.

C. Below grade piping shall be installed in such a manner that it does not bear directly on rocks or debris in ground. Any location where pipe passes close to or through walls or footings, it shall be protected from contact with concrete or cinder block. Any pipe passing through building walls shall be protected by a cast iron sleeve large enough to permit free movement of pipe. Refer to Section 23 00 10 for additional sleeve requirements.

D. All piping shall be protected at all times from dirt and moisture. During storage on the job site of construction, the Contractor shall keep pipe ends plugged or capped to prevent dirt or moisture entering the pipe.

E. Piping insulation shall not be installed on connections, joints, flanges, etc., until pressure testing has been successfully accomplished.

## 3.2 REFRIGERANT PIPING

A. Piping Install all refrigerant piping, fittings, connections, and insulation in accordance with the equipment manufacturer’s recommendations. Any refrigerant piping installation that does not meet the manufacturer’s recommendations and/or guidelines shall be replaced at no additional cost to the Owner.

B. It is imperative to keep the refrigerant piping system free of contaminants and debris during piping installation. Contaminant in refrigerant piping is detrimental to the split system operation.

C. Refrigerant piping shall be installed to accommodate the piping thermal contraction and expansion expected for the system operation in accordance with the split system manufacturer. split systems are intended to operate at a wide range of refrigerant temperatures and pressures that cause piping thermal expansion and contraction. Movement of piping due to expansion or contraction shall not cause leakage or equipment failure.

D. Pipe and fittings shall be cleaned by drawing through the tubing by means of a wire, a clean, lintless cloth. Next, a clean, lintless cloth saturated with a cleaning solvent approved by the split equipment manufacturer shall be pulled through the pipe as many times as required, until the cloth is not discolored by dirt. A lintless cloth saturated in compressor oil shall then be drawn through the pipe and the pipe visually inspected for dirt and lint.

E. All piping shall be protected at all times from dirt and moisture. During storage on the job site of construction, the Contractor shall keep pipe ends plugged or capped to prevent dirt or moisture entering the pipe

F. Tubing shall be cut so ends are perfectly square, using a pipe cutter, so that pipe will "bottom" in the fitting.

G. After the pipe is thoroughly cleaned before pipes and fittings are joined.

## 3.3 DRAIN PIPING

A. Drain piping from each air handling unit drain pan, strainer, water chilling unit, heat exchanger, pump base drain, tank and vessel overflow, auxiliary drain pan, piping system drain, blowdown, and any other system or equipment that has a drain connection shall be extended to the nearest

floor drain, hub drain, or condensate drainage system. Drains shall be of the sizes indicated but not less than the full size of the drain pan connection. A valved 3/4" drainage line shall be provided at all low points in the piping systems. Air handling unit drains shall have deep seal traps to permit unit pan drainage and maintain the water seal. Provide unions at all equipment drain connections.

## 3.4 PIPE HANGERS AND SUPPORTS

A. Hangers and Support: Piping shall be supported in such a manner so as to avoid the pipes from sagging during the life of the split system. Sagging pipes can become oil traps and can lead to equipment failures. Provide additional supports where potential sagging could occur. All horizontal piping shall be installed with a pipe hanging system that is adjustable in height. Provide rigid high compressive strength foam insulating pipe support at all support points in accordance with pipe insulation manufacturer's recommendations. Piping shall be attached to hangers directly. Maximum pipe support spacing and locations shall be determined by local codes or the manufacturer's recommendations, whichever is more stringent. Piping shall also be supported to avoid the transmission of machinery noise and/or vibration to the building structure. The split manufacturer shall provide minimum requirements on support spacing, location of support at indoor units, outdoor units and refrigerant distribution units, branch fittings, header fittings, take-offs, and changes in pipe direction.

B. Hanger rods, inserts, etc., shall be sized and installed as recommended by the hanger manufacturer for the service intended.

C. The Contractor shall submit pipe hanger and support details for each type of floor construction to the Structural Engineer for review and approval prior to submission to the Engineer.

## 3.5 PIPE COVERING PROTECTION SADDLES AND SHIELDS

A. Furnish and install pipe covering protection shields and saddles for all insulated piping at the locations of all supports.

## 3.6 SOLDERED/BRAZING PIPING JOINTS

A. Ends of pipe and inside of fittings shall be cleaned and flux applied to entire area of pipe end to be soldered. On pipes 1-1/2" and larger, flux shall be applied to pipe and fittings. Joints and fittings shall be soldered with ASTM B32 Grade 95T tin antimony solder, tin-bismuth-copper-silver solder, or brazed with AWS classification Bcup 3 brazing alloy as required by the system temperature and working pressure. Solder or brazing compounds containing lead shall be prohibited. Contractor shall thoroughly flush all soldered piping to remove excess flux so as to prevent corrosion of piping.

## 3.7 PIPING SUPPORTS (ABOVE GRADE)

A. All pipe shall be supported from the building structure in a neat and workmanlike manner and wherever possible, parallel runs of horizontal piping shall be grouped together on trapeze type hangers. The use of wire or preformed metal to support pipes will not be permitted. Hanging pipes from other pipes will not be permitted. Piping shall be carefully coordinated before installation with other systems and equipment in chases and congested areas.

B. Vertical and horizontal pipes subject to sway in either direction shall have a snubber to prevent lateral movement.

C. Outdoor piping below the cooling tower shall be supported from the cooling tower support

structure to minimize support from the roof.

## 3.8 PIPE SUPPORTS (BELOW GRADE)

A. Earth shall be excavated to a minimum depth with an even surface to insure solid bearing of pipe for its entire length. Where pipe lines cross deeper excavations, these shall be filled and tamped to the proper level before pipe is installed.

B. Interior: The water pipe shall be installed a minimum of 4" below the bottom of the slab and shall not be in direct contact with the concrete at any point.

C. Exterior: The water pipe shall have a minimum of 3 feet of cover and shall comply with Local Codes.

D. The pipe shall have bearing along its entire length and, if installed on fill, shall be supported by concrete cradles to firm earth. All offsets, changes in grade, or changes in direction of run of pressure systems shall have poured in place concrete blocking from pipe to firm earth. Blocking shall be of adequate size to hold the pipe completely stationary regardless of water pressure. Where a water service enters a building, the last joint of the mechanical joint pipe shall have anchor bolts to the building structure. All exposed joints shall have anchor bolts or rods installed in such a manner that there is no chance of a joint being pushed apart.

## 3.9 PIPING TESTS

A. Scope: All piping installed on the Project unless specifically shown otherwise shall be hydraulically tested as specified herein, in accordance with ASME B31.1, and as required by the local code authorities. Refrigeration piping, if used on the Project, shall be tested as specified under refrigeration piping. This Contractor shall provide all equipment required to make the tests specified herein. Calibrated pressure gauges accurate to within ±1/2% of full range shall be used for pressure testing. System pressure gauges shall not be substituted for calibrated test gauges.

B. Tests to be performed: Before the piping of various systems is insulated, furred-in, or otherwise covered up; the subcontractor shall fill the section of the pipe to be tested with water and bring the section up to pressure with a test pump. The Contractor shall notify the Architect or his representative, in writing, at least five (5) working days prior to each test. The Architect or his representative may elect to witness the tests. These tests shall be conducted before any insulation is installed and any insulation installed prior to these tests shall be removed. Gauges used in the tests shall have been recently calibrated with a dead weight tester.

C. Duration of Tests: All tests shall apply full test pressure to the piping for a minimum of twenty four hours.

D. Inability to Hold Pressure: When the test pressure has fallen over 5% during the 24-hour test period, the point of leakage shall be found, repaired and the test repeated. This procedure shall be followed until the piping system has been proven absolutely tight.

E. Leaks: Leaks developing subsequent to these tests shall not be repaired by mastic or other temporary means. All leaks shall be repaired by removal of the valve, fitting, joint, or section that is leaking and reinstalling new material with joints as specified hereinbefore.

3.10 PIPING EVACUATION

A. After piping pressure test has been completed, the piping system shall be evacuated when the surrounding ambient air is not less than 60°F. If the temperature is less, auxiliary heat must be provided to ensure proper evacuating conditions. A minimum vacuum of 500 microns of Hg shall be pulled on the piping system and maintained for 12 hours. The vacuum pump displacement shall be not less than 2 cfm for up to 15 tons. The system shall be charged as recommended by the equipment manufacturer. The test results shall be provided to the manufacturer.

END OF SECTION 23 23 00

# SECTION 23 30 01 - DUCTWORK AND SHEET METAL

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

### 1.2 WORK INCLUDED

A. Provide the ductwork and accessories as shown on the Drawings and as specified herein.

### 1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Fan and Air Handling Unit Sections
B. Section 23 00 70 - Division 23 Schedule of Submittal Data
C. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
D. Section 23 07 00 - HVAC Insulation
E. Section 23 37 00 - Air Outlets and Inlets
F. Drawing M-502 - Sequence of Operation

### 1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings
   2. Handbooks Fundamentals, Applications, Systems and Equipment

B. Air Movement and Control Association (AMCA)
   1. 500-D Laboratory Methods of Testing Dampers for Rating

C. ASTM International (formerly American Society for Testing and Materials)
   1. A90 Test Method for Weight (Mass) of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings
   2. A123 Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
   3. A653 Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
   4. A924 Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
   5. C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus
   6. C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
   7. C916 Standard Specification for Adhesives for Duct Thermal Insulation
   8. C1071 Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)
   9. C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings
   10. E84 Standard Test Method for Surface Burning Characteristics of Building Materials
   11. E96 Standard Test Methods for Water Vapor Transmission of Materials

12. E119 Standard Test Methods for Fire Tests of Building Construction and Materials
13. E477 Standard Test Method for Measuring Acoustical and Airflow Performance of Duct Liner Materials and Prefabricated Silencers
14. G21 Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

D. DOE Code of Federal Regulations (CFR)
1. 40 CFR 59 Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings

E. National Fire Protection Association (NFPA)
1. 54 National Fuel Gas Code
2. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems
3. 701 Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

F. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA)
1. 006 HVAC Duct Construction Standards - Metal and Flexible
2. 008 IAQ Guidelines for Occupied Buildings Under Construction

G. South Coast Air Quality Management District (SCAQMD)
1. Rule 1168 Adhesive and Sealant Applications

H. Underwriters Laboratories (UL) Standard for Safety
1. 181 Factory-Made Air Ducts and Air Connectors
2. 181A Closure Systems for Use with Rigid Air Ducts
3. 181B Closure Systems for Use with Flexible Air Ducts and Air Connectors
4. 555 Fire Dampers
5. 555S Smoke Dampers
6. 723 Test Method for Surface Burning Characteristics of Building Materials

1.5 SHOP DRAWINGS AND SUBMITTALS

A. General: Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein plus certification data and shop drawing requirements listed in Sections 23 00 10 and 23 00 70.

B. Requirements: Sheet metal ductwork shop drawings shall be made after actual job measurements are obtained. Sheet metal ductwork drawings shall indicate the coordination of the Contractor with sprinkler piping and other mechanical and electrical services and equipment installed under Division 21, 22, 23 ,25, 26 and 28. These "Shop Drawings" shall be submitted for review and inclusion with "Record Drawings".

C. Electrical Junction Boxes: Sheet metal ductwork drawings shall indicate the size and location of electrical junction boxes.

D. Devices: Sheet metal shop drawings shall show all applicable devices including but not limited to air flow monitoring devices and automatic dampers.

E. Special Requirements: Ductwork joint, connection, sealant, insulation, reinforcing, elbow hardware, liner, adhesives, attachment details, mounting details, regulators, access doors, dampers, diffuser blank-off end plate configurations, plenum/supply diffuser mounting arrangements, accessories, etc. shall be submitted with first ductwork shop drawing submittal. Details shall be indexed and index number shall appear on ductwork shop drawing at its point of use. Structural attachment details shall be submitted to the Structural Engineer for review.

F. Submission Requirements: Submit detailed shop drawings of all damper types for review by the Engineer.

G. Submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each field applied ductwork sealant and adhesives used within the building interior locations.

PART 2 - PRODUCTS

2.1 GENERAL

A. SMACNA References: Unless otherwise indicated or specified all sheet metal ductwork shall be constructed in accordance with the HVAC DUCT CONSTRUCTION STANDARDS-METAL AND FLEXIBLE, published by SMACNA, and herein referenced as the "SMACNA Manual." The SMACNA Manual edition utilized shall be the SMACNA edition referenced within the codes as adopted by the local code authorities. SMACNA Manual includes SMACNA supplements and addenda applicable to the referenced SMACNA edition. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

B. Project shall meet or exceed the requirements of the SMACNA Indoor Air Quality Guideline for Occupied Buildings Under Construction.

C. Various page numbers, section numbers, table numbers, plate numbers, detail numbers, and figure numbers herein cited refer to the SMACNA Manual. Various chapter titles, table titles, and figure titles herein cited refer to the SMACNA Manual. All ductwork shall be installed in accordance with the arrangements and sizes shown on the Drawings and specified herein.

D. Pressure Classes: The pressure class for each ductwork system shall be as indicated herein and as indicated on the Drawings.

E. Execution: The execution of the work shall be in strict accordance with the best practices of the trade and with these Specifications. In addition, the Contractors shall ensure that all stored on-site or installed materials are protected at all times from moisture damage.

F. Leakage: Air distribution system leakage in excess of 5% will not be acceptable. The total leakage for a particular system shall include air leakage at ductwork, spin-collar connections, branch taps, ductwork/diffuser interface connections, equipment connections, plenum/diffuser interface connections, and other areas in the air distribution system where leakage may occur. Refer to sealing requirements herein. Leakage requirements and limits shall apply to all ductwork pressure/seal classifications.

G. Construction Materials: Ductwork shall be constructed of "lock forming quality" galvanized steel of the gauge thickness and tolerances listed in the Appendix of the SMACNA Manual. All steel sheets shall comply with ASTM A924 and ASTM A653 with a hot dipped galvanized coating weight that complies with the G90 section of ASTM A653 and ASTM A90. Aluminum ductwork and galvanized G60 ductwork will not be acceptable. All sheet metal delivered to the site shall bare the "LFQ" and "G90" marks.

H. Lined Ductwork: Duct sizes indicated for internally lined ducts are the net duct dimensions and shall be increased in both dimensions by twice the thickness of the liner. Duct liner material and thickness shall be as specified herein.

I. Interior Surfaces: The interior surfaces of all ductwork shall be smooth. No sheet metal parts, tabs, angles, screws or anything else may project into the air ducts for any reason unless specified to do so. All seams and joints shall be external.

J. Prefabricated Joint Systems: At the Contractor's option, ductwork may be joined with prefabricated galvanized Ductmate, Exanno (Nexus), Elgen Manufacturing, or TDC sections. Prefabricated joint systems shall be manufactured and installed in strict accordance with the manufacturer's printed literature unless otherwise noted. The joint packing material, joint construction details and joint reinforcing details using this method shall be submitted to the Engineer for review. The SMACNA Manual and these Specifications will be used by the Engineer as the basis for determination of equivalent joint construction and reinforcing. If prefabricated joint systems are used at fire damper connections, the system shall be compatible with the fire damper and sleeve assembly provided and installed in strict accordance with the manufacturer's recommendations to maintain the UL listed fire rating of the architectural assemblies, fire damper, and fire damper sleeve assembly.

K. Division 28 Coordination: The requirements for the control wiring systems and/or power wiring systems for dampers and other equipment provided under this section shall be coordinated and compatible with the fire alarm system to the extent required for complete and operational systems. The Division 23 contractor providing equipment consisting of, but not limited to, dampers shall coordinate and request the requirements for the control wiring systems and/or power wiring systems for each piece of equipment. In addition, for dampers and other pieces of equipment controlled by the Division 28 fire alarm system, the Division 23 contractor shall coordinate with the Division 28 fire alarm system contractor and request the requirements for the control wiring systems and/or power wiring systems for each piece of equipment. Refer to Sequence of Operation on drawing M-502 and Specification Section 28 31 00 for additional requirements.

2.2 FIRE AND SMOKE HAZARD RATING

A. All duct components installed within supply, return, exhaust, relief and ventilation air plenums; all internal duct liners (composite liner, facing, and adhesive); all flexible ductwork shall be non-combustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723 as required by applicable mechanical code.

B. All duct components installed within supply, return, exhaust, relief and ventilation air plenums; all internal duct liners (composite liner, facing, and adhesive); all flexible ductwork shall be non-combustible or shall be tested and rated as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723.

C. Accessories, such as adhesives, mastics, sealant and cements, shall have the same component ratings listed above and be listed by the manufacturer to be "non-flammable" in the wet and dry states.

2.3 DUCTWORK PRESSURE CLASS

A. The rectangular sheet metal ductwork installed shall be of the following pressure classes:

1. 0.5" water gauge static pressure maximum positive or negative, indicated as 0.5" wg on the Drawings or hereinafter.

2. 1" water gauge static pressure maximum positive or negative, indicated as 1" wg on the Drawings or hereinafter.

3. 2" water gauge static pressure maximum positive or negative, indicated as 2" wg on the Drawings or hereinafter.

4. 3" water gauge static pressure maximum positive or negative, indicated as 3" wg on the Drawings or hereinafter.

5. 4" water gauge static pressure maximum positive or negative, indicated as 4" wg on the Drawings or hereinafter.

6. Refer to the Tables in the SMACNA Manual Titled "Rectangular Duct Reinforcement" for additional information.

B. Unless otherwise indicated on the Drawings or otherwise specified herein, the ductwork systems shall be constructed in accordance with the following pressure classes:

1. 2" wg - Outdoor air system ductwork

2. 2" wg - Toilet exhaust / relief air system ductwork downstream of exhaust / relief air terminal units

3. 1" wg - Toilet exhaust / relief air system ductwork upstream of exhaust / relief air terminal units

4. 2" wg - Supply ductwork downstream of single zone air handling units

5. 2" wg - Return air system ductwork

6. 1" wg - Return air elbows at air handling unit room

7. 1" wg - Miscellaneous single room exhaust system ductwork

8. 0.5" wg - Transfer air and return air duct elbows (except at air handling unit room).

9. Where no specific duct pressure designations are indicated in the Specifications or on the Drawings, the 2" wg pressure class shall be the basis of the Contract.

2.4 RECTANGULAR SHEET METAL DUCTWORK

A. Cross Breaking and Mechanical Beading: The rectangular ductwork shall have cross breaking or mechanical transverse beading for ductwork sizes 12" and wider. Duct beading used to stiffen ducts shall be at least 1/16" deep at the center of the bead and a maximum of 3/8" wide at the base of the bead.

B. Square Throat Elbows:

1. Square throat elbows and turning vanes shall be constructed in accordance with Type RE-2 as indicated in the Figures in the SMACNA Manual titled "Rectangular Elbows", "Vanes and Vane Runners" and "Vane Supports in Elbows" and the following specifications. Provide turning vanes in all 90° elbows. Shop fabricated turning vanes will not be acceptable unless they are machine shaped, punched, and assembled. Size transition in duct width up to 25% of the largest width dimension will be acceptable.

2. Ducts up to 18" wide: Turning vanes shall be minimum 24 gauge single vane type and 2" inside radius spaced no more than 1-1/2" apart.

3. Ducts up to and including 36" wide: Turning vanes shall be minimum 22 gauge single vane type with 2" inner radius spaced no more than 2-1/8" apart.

4. Ducts over 36" wide: Turning vanes shall be minimum 22 gauge single vane type with 4-1/2" inner radius spaced no more than 3-1/4" apart.

5. Acoustic Turning Vanes: Acoustic turning vane assembly shall be Airsan "Acoustiturn" airfoil shaped turning vanes for noise and directional air control. Manufacturer shall be responsible for sizing and constructing vane assemblies to optimize noise reduction for the particular application. Vanes shall be constructed of 14 gauge aluminum extrusion acoustically treated with glass fiber media that is protected by an open metal facing. Mounting rails shall be 24 gauge galvanized steel.

C. Radius Elbows:

1. Elbows and turning vanes shall be constructed in accordance with Types RE-1, RE-3, and RE-5 as indicated in the Figures in the SMACNA Manual titled "Rectangular Elbows", "Vanes and Vane Runners" and "Vane Supports in Elbows" and the following specifications. Radius elbows shall have an inner radius equal to the width of the elbow. Size transitions in duct depth and width up to 25% of the elbow's dimensions will be acceptable. Provide turning vanes within radius elbows only if shown on the drawings.

2. Radius elbows shall be used where indicated on the drawings. At the Contractor's option, radius elbows may be used in lieu of square throat elbows up to 24" in width and smaller.

3. Unless otherwise specified herein, all 45 degree elbows shall be radius without turning vanes. Mitered 45 degree elbows shall not be acceptable.

D. Joints and Seams:

1. Longitudinal Joints and Seams: Unless otherwise noted all longitudinal duct seams and joints shall be "Pittsburgh Lock" (L-1) or "Button Punch Snap Lock" (L-2) at the corners and "Grooved Seam" (L-3) or "Automatic Seam Weld" in sides between corners. Refer to the Figure in the SMACNA Manual titled "Rectangular Duct - Longitudinal Seams" for additional information. "Button Punch Snap Lock" longitudinal seams and joints will not be acceptable above 2" wg static pressure class.

2. Transverse Joints and Reinforcement: All transverse duct joints and reinforcements shall be in accordance with the SMACNA Manual and the following:
   a. Dimension of longest side of duct less than 55”: Minimum rigidity class and minimum gauge duct shall be based on a maximum reinforcement spacing of 5'-0".
   b. Dimension of longest side of duct 55” or greater: Minimum rigidity class and minimum gauge duct shall be based on a maximum reinforcement spacing of 2'-6".

3. Rigidity class options, dimensions of transverse joints and dimensions of intermediate reinforcement shall be in accordance with the physical space limitations of the Project and within the limitations indicated on the Drawings. Angle bracing shall not exceed 1-5/8" in depth. In lieu of angle bracing, prefabricated joint flanges, intermediate reinforcement, and any other ductwork construction that adds to the overall external dimensions of the ductwork; if required to meet the physical space limitations of the project; the contractor has the option to use internal tie-rod bracing and slip and drive joint construction for ductwork 36" or less in width and double "S" slip joint construction for ductwork above 36" in width in accordance with SMACNA.

4. Reinforcement for internally lined ductwork: In lieu of external reinforcing, reinforcing angles shall be mounted internally on lined ductwork as required to fit the ductwork within the physical limitations of the ceiling plenum. Duct liner thickness shall be modified to match or exceed the reinforcing thickness so as to present a smooth unobstructed internal surface for proper airflow.

E. Extractors: At all places where rectangular ducts take off a main trunk duct at right angles, and where side wall outlets tap into the ducts, Titus AG-45, or approved equal, shall be installed. Branch taps shall be equipped with Titus AG-45 with No 2 operator and regulators as specified for splitter dampers. Side outlet taps shall be equipped with Titus AG-45 with No 3 operator and no regulators.

2.5 ROUND SHEET METAL DUCTWORK

A. Duct gauges, girth reinforcing, girth joints, longitudinal seams, etc., shall be in accordance with the Chapter in the SMACNA Manual titled "Round, Oval and Flexible Duct". Draw band joint connection shall not be acceptable. Snaplock seam construction shall not be acceptable for systems in excess of 1" wg static pressure positive or negative.

2.6 DUCT ACCESS DOORS

A. Hinged Access Doors: Hinged access doors shall be furnished and installed in ductwork for required access and for fire dampers as specified herein. Access doors shall be insulated, gasketed, and at least 15"x15" when duct dimensions are large enough, complete with necessary hardware and be Amerloc 10 or Ventlock #100 latches and Ventlock Series #100 hinges, or Elgen Manufacturing Type "A or H" access door. Refer to the Figures in the SMACNA Manual titled "Duct Access Doors and Panels" and "Access Doors - Round Duct" for additional information.

B. Round Access Doors: Round spin-in type access doors, 18" and 24" diameter, will be acceptable in lieu of comparable size square or rectangular access doors specified herein. Each galvanized steel round access door shall be complete with insulation, spin-in frame, inner door, attachment cable, gaskets, 3 latches, and pull ring. Round spin-in type access door shall be "The Inspector Series" as manufactured by Flexmaster USA or Elgen Manufacturing round access door.

C. Hinged Walk-in Type: Hinged walk in type casing and plenum access doors shall be installed where required and indicated on the Drawings. Casing and plenum access doors shall be 57" high x 24" wide where possible and be complete with hardware, hinges, seals and latch handles. Refer to the Figures in the SMACNA Manual titled "Casing Access Doors - 2" wg" and "Casing Access Doors - 3-10" wg" and the Table in the SMACNA Manual Titled "Plenum and Casing Access Doors - 2" wg" for additional information. Latch handles shall be Ventlock Series #260 or approved equal. Hinges shall be Ventlock Series #200, #300 or approved equal.

D. Access doors for fire, smoke or fire/smoke dampers shall be permanently identified on the exterior of the access door by a label having letters not less than 0.5” in height reading: "FIRE/SMOKE DAMPER", "SMOKE DAMPER" or "FIRE DAMPER".

2.7 DAMPER REGULATORS

A. All damper control regulators serving dampers within externally insulated ductwork shall be mounted on 16 gauge elevated platform at least 1/8" higher than the thickness of the insulation. Damper shaft shall have Ventlok No 607 bearing mounted on ductwork within elevated platform.

B. If the ductwork is enclosed in a furring or in an inaccessible ceiling, provide the Bowden Remote Cable Control System as manufactured by Young Regulator or United Enertech in accordance with the following requirements:

1. For ceiling and wall applications, the Bowden Remote Cable Control System Model 270-896B rack and pinion controller/operator with 7/8" diameter flush mounted factory finished painted primer coated threaded metallic cap shall be used. For wall applications, this option will be acceptable only if there is enough wall cavity space to accommodate the concealed controller/operator.

2. The threaded metallic cap shall be painted in the field by the contractor to match the surrounding surfaces with the final color and finish of the cap selected and approved by the Architect. The location of all caps located on the face of ceilings and walls shall be approved by the Architect.

3. Each remote cable control system shall be furnished complete with the necessary length of flexible control cable and wire as required for each location and with all necessary hardware to ensure damper and damper shaft are coordinated and compatible with the Bowden Remote Cable Control System specified herein. Coordination shall be performed by the contractors prior to purchase of dampers and cable control system.

4. All damper control system regulators serving dampers within externally insulated ductwork shall be mounted on a 16 gauge elevated platform at least 1/8" higher than the thickness of the insulation. Damper shaft shall be extended as required to allow the remote cable control system to control the damper from the elevated platform.

5. Provide the Owner with two (2) Young Regulator Model 030-12 wrenches used to adjust and control the dampers through the Bowden Remote Cable Control System. Similar wrenches manufactured by United Enertech Corporation used to adjust and control the dampers through the Bowden Remote Cable Control System may also be provided.

C. As an option to using the Bowden Remote Cable Control System for ductwork enclosed in a furring or in an inaccessible ceiling, the contractors may use the United Enertech Power/Balance Remote Cable Control System in accordance with the following requirements:

1. For ceiling and wall applications, the United Enertech Power/Balance Remote Cable Control System Model No Hi-i-3 controller/operator with Model No J50 3.25" diameter flush mounted factory finished painted primer coated metallic cover plate shall be used. Cover plates shall have an RJ-11 connection. For wall applications, this option will be acceptable only if there is enough wall cavity space to accommodate the concealed controller/operator. For ceiling slot diffuser applications, if it is acceptable to the architect, the contractor may also conceal the cable inside the plenum slot diffuser in lieu of using a cover plate RJ-11 connector. If the cables are mounted inside of the slot diffusers plenum the contractor shall also furnish the appropriate adapter connection with each of the remote battery packs provided in paragraph 5 below.

2. The metallic cover plate shall be painted in the field by the contractor to match the surrounding surfaces with the final color and finish of the cover plates selected and approved by the Architect. The location of all cover plates located on the face of ceilings and walls shall be approved by the Architect.

3. Each remote cable control system shall be furnished complete with the necessary length of RJ-11 plenum rated control cable as required for each location and with all necessary hardware to ensure damper and damper shaft are coordinated and compatible with the United Enertech Power/Balance Remote Cable Control System specified herein.

Coordination shall be performed by the contractors prior to purchase of dampers and cable control system.

4. All damper control system regulators serving dampers within externally insulated ductwork shall be mounted on a 16 gauge elevated platform at least 1/8" higher than the thickness of the insulation. Damper shaft shall be extended as required to allow the remote cable control system to control the damper from the elevated platform.

5. Provide the Owner with two handheld power packs used to adjust and control the dampers through the United Enertech Power/Balance Remote Cable Control System.

2.8 FLEXIBLE DUCT CONNECTION

A. Provide flexible connections as specified and as shown on the Drawings.

B. Materials for Indoor Application: Flexible connection shall be rated up to 200°F with a minimum 30 oz/yd2 weight, heavy glass fabric with double-coated Neoprene, and shall be Ventfabrics, "Metaledge Ventglas", Elgen Manufacturing "Notch Lock (ZLN)" or approved equal.

C. Materials for Outdoor Application: Flexible connection shall be rated up to 275°F with a minimum 24 oz/yd2 weight, heavy glass fabric with double-coated Hypalon (resistance to long term degradation due to weather and sunlight), and shall be Ventfabrics, "Metaledge Ventlon", Elgen Manufacturing "Notch Lock (ZLH)" or approved equal.

D. Flexible connection shall comply with NFPA 701 as non-combustible for the fabric, and fire retardant for the coating. Ends of fabric must be overlapped 2" and glued with R-H Products Company "XL-8" contact glue. Sewing or stapling may be used in conjunction with gluing. At least, 1" slack shall be allowed in all flexible connection installations to ensure that no vibration is transmitted.

2.9 DUCTWORK HANGERS AND SUPPORTS

A. General: All ductwork shall be properly suspended or supported from the building structure. The duct hanging system is composed of three elements; the upper attachment to the building, the hanger itself, and the lower attachment to the duct. The attachments, hangers and supports for all ductwork shall be in accordance with the Chapter in the SMACNA Manual titled "Hangars and Supports" and the following.

B. Corrosion Resistance: All hangers, rods, straps, angles, inserts and other attachment/support devices shall be made of galvanized steel.

C. Strap Hangers: Each strap hanger shall be attached to the duct by a minimum of three screws: one on the bottom and two on the side of the duct.

D. Wire Hangers: As an acceptable option, the UL listed Gripple cable system utilizing stranded galvanized rope which has been tested and approved by SMACNA as an acceptable alternate to the SMACNA approved duct hanger systems may be used. The Gripple cable system shall be installed in accordance with the manufacturers written guidelines and requirements. All other types of wire hanger systems will not be acceptable.

E. Spacing: Hanger spacing shall not exceed 10'-0" for ducts up to 60" in the largest dimension. Hanger spacing shall not exceed 8'-0" for ducts over 61" in the largest dimension.

F. Upper Attachments:

1. New Structural Concrete:
   a. Drive pin fastener (No 4 as indicated in the Figure in the SMACNA Manual titled "Upper Attachment Devices - Typical") and expansion nail anchor (No 9 as indicated in the Figure in the SMACNA Manual titled "Upper Attachment Devices - Typical") may be used for ducts up to 18" maximum dimension.
   b. Threaded stud fastener (No 5 as indicated in the Figure in the SMACNA Manual titled "Upper Attachment Devices - Typical") may be used for ducts up to 36" maximum dimension.
   c. Expansion shield anchors (No 7 and 8 as indicated in the Figure in the SMACNA Manual titled "Upper Attachment Devices - Typical") shall be used for all other ductwork.
   d. All concrete attachments shall be made of steel. Powder actuated fasteners shall not be used in slab sections less than 4" thick.

2. Refer to Section 23 00 10 "Equipment Mounting/Support Devices" for additional requirements. Note, governing requirements for the interface between new equipment supports and existing structural conditions.

3. The Contractor shall be responsible for coordinating the installation and location of the attachments with structural elements. All structural attachment devices shall be submitted to and reviewed by the Structural Engineer prior to use on this project.

G. Outside Locations: All support devices including but not limited to steel angles used for support and bracing, base plates, rods, hangers, straps, screws bolts, etc. shall be hot dipped galvanized after fabrication or 304 stainless steel.

2.10 FLEXIBLE DUCTWORK

A. General: Flexible duct shall be provided in ductwork connected to various air outlets, inlets and equipment as shown on the Drawings.

B. Construction:

1. Flexible duct shall be a factory fiberglass insulated assembly with a minimum thermal resistance (R) value of 6 h-sf-°F/Btu. The inner core shall be constructed of polyethylene or trilaminate fabric mechanically locked into a spiral galvanized helix suitable for a positive working pressure of at least 3" wg. The inner core shall be wrapped with fiberglass insulation and sheathed in a metalized reinforced outer jacket. The flexible duct assembly shall be listed as a UL 181, Class 1 air duct and shall comply with NFPA 90A.

2. Alternate and Base Bid Requirements for Flexible Ductwork:
   a. Base Bid: Flexible ductwork shall be Flexmaster USA Type 1M, 3M, or 5M insulated flexible ductwork.
   b. Alternate Bid: Approved equal flexible ductwork may be provided as an alternate to the base bid by one of the listed manufacturers, provided they meet the requirements of the specifications. Alternate manufacturers: Buckleys Associates, Thermaflex.

C. Construction

1. Type 1 flexible duct shall be a factory fiberglass insulated assembly with a minimum thermal resistance (R) value of 6 h-sf-°F/Btu. Flexible duct shall be constructed of polyethylene fabric and mechanically locked into a spiral galvanized helix suitable for a positive working pressure of at least 3" wg. The inner core shall be wrapped with fiberglass insulation and sheathed in a metalized reinforced outer jacket. The flexible duct

assembly shall be listed as a UL 181, Class 1 air duct and shall comply with NFPA 90A. Type 1 flexible duct shall be used in all areas and applications except where Type 2 flexible duct is specified to be used.

2. Type 2 flexible duct shall be a factory fiberglass insulated assembly with a minimum thermal resistance R value of 6 h-sf-°F/Btu. Flexible duct shall be constructed of trilaminate film and mechanically locked into spiral helix without the use of adhesives suitable for a positive working pressure of at least 12" wg. The inner core shall be wrapped with fiberglass insulation and sheathed in a metalized reinforced outer jacket. It shall require no more than two 90° elbows as installed. The flexible duct assembly shall be listed as a UL 181, Class 1 air duct and shall comply with NFPA 90A. Type 2 flexible duct shall be used between the primary air trunk ductwork and the inlet of all fan powered terminal units.

3. Alternate and Base Bid Requirements for Type 1 and Type 2 Flexible Ductwork: Base Bid Type 1 flexible ductwork shall be Flexmaster USA Type 1M insulated flexible ductwork. Base bid Type 2 flexible ductwork shall be Flexmaster USA Type 3M or 5M insulated flexible ductwork. Approved equal flexible ductwork may be provided as an alternate to the base bid by one of the listed manufacturers, provided they meet the requirements of the specifications. Alternate manufacturers: Buckleys Associates, Thermaflex.

D. Flame/Smoke Rating: Refer to Section 2.2 - Fire and Smoke Developed Rating hereinbefore for UL 723 and ASTM E84 flame spread and smoke developed index requirements.

E. Connections to Diffusers: Oval ends on spiral aluminum helix flexible ductwork which may be required to connect to various air distribution devices shall be factory fabricated. At the Contractor's option oval ends may be field fabricated on special mandrels subject to the review of the Engineer.

F. Duct Clamps: Flexible duct clamps shall be stainless steel with swivel action screw or 100% nylon self-locking clamp manufactured by Panduit or approved equal.

G. Support: Flexible duct supports shall be in accordance with the Chapter in the SMACNA Manual titled "Hangars and Supports" and the Figures in the SMACNA Manual titled "Flexible Duct Supports". Refer to Section 23 00 10 "Equipment Mounting/Support Devices" for additional requirements concerning interface with existing structure.

2.11 SPIN-IN COLLAR FITTINGS

A. If it complies with these Specifications, one of the following manufacturers shall be acceptable:

1. Flexmaster USA
2. Thermaflex

B. General: Spin-in collar branch takeoff fittings shall be constructed of galvanized steel with plain or conical bellmouth shaped inlets. Fittings shall be provided with or without single blade volume dampers as specified herein and as indicated on the Drawings. Flanged and gasketed type collars attached to the ductwork with screws will not be acceptable.

C. Application:

1. Plain end fittings with volume dampers shall be used at all round branch taps serving air distribution devices.

2. Where depth of trunk duct permitting, conical bellmouth end fittings without volume damper shall be used at all round branch taps serving terminal units.

3. Where the depth of the ductwork does not permit the installation of conical bellmouth end fittings, plain end fittings without volume dampers shall be used at all round branch taps serving terminal units.

4. Plain end fittings with volume dampers shall be provided at any other round branch tap as required for the balancing of the supply, return and exhaust systems.

5. As indicated on the Drawings.

2.12 DUCT LINER

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Johns Manville
2. CertainTeed
3. Owens Corning
4. Knauf
5. Manson

B. Rectangular Duct Liner Thermal and Construction Characteristics:

1. Rectangular duct liner shall be constructed of fiberglass material with fire resistant coating on the surface facing the air stream and shall comply with ASTM C1071 (Type I - Flexible) and NFPA 90A. The fire resistant coating shall bind the surface fibers tightly to prevent any from becoming loose. The liner surface shall be guaranteed to remain free of surface erosion in ducts with air velocities up to 4,000 fpm.

2. The composite liners, facing, coating and adhesive shall comply with NFPA 90A, ASTM E84, and UL 723 for fire resistivity.

3. The liner shall be tested and shown to prevent the growth of mold, fungus and bacteria in accordance with ASTM C1338, ASTM G21, and UL 2824.

4. Duct liner shall be 1.5 pound per $ft^3$ density with various thicknesses as indicated hereinafter with a maximum thermal conductivity (K value) of 0.24 Btu-in/h-sf-°F at 75°F mean temperature when tested by ASTM C177 or C518.

5. Duct liner shall be UL Greenguard Gold Certified and listed under UL 2818.

6. Duct liner shall be Knauf "Performance+ Duct Liner", Johns Manville "Linacoustic RC", Owens Corning "QuietR Rotary Duct Liner" or approved equal.

C. Round Duct Liner Thermal and Construction Characteristics:

1. Round duct liner shall be constructed of fiberglass material with fire resistant coating on the cleanable surface facing the air stream, specifically designed to provide acoustical and thermal performance in round ducts and shall comply with ASTM C1071 (Type II, Rigid Board) and NFPA 90A. The fire resistant coating shall bind the surface fibers tightly to prevent any from becoming loose. The liner surface shall be guaranteed to remain free of surface erosion in ducts with air velocities up to 5,000 fpm.

2. The composite liners, facing, coating and adhesive shall comply with NFPA 90A, ASTM E84, and UL 723 for fire resistivity.

3. The liner shall be tested and shown to prevent the growth of mold, fungus and bacteria in accordance with ASTM C1338, ASTM G21, and UL 2824.

4. Duct liner shall be 4 pound per $ft^3$ density with various thickness as indicated hereinafter with a maximum thermal conductivity (K value) of 0.23 Btu-in/h-sf-°F at 75°F mean temperature when tested by ASTM C177 or C518.

5. The liner shall be UL Greenguard Certified and listed under UL 2818.

6. Flexible, low density liner material is prohibited on round ducts.

7. Duct liner shall be Johns Manville "Spiracoustic Plus", CertainTeed "Toughgard Ultra Round", Owens Corning "QuietR Spiral" or approved equal.

D. Liner adhesive shall be 3M, King, or approved equal. Adhesive shall meet the requirements of ASTM C916. For indoor applications, the adhesive shall have a VOC content of 80 g/L or less when calculated in accordance with 40 CFR, Subpart D (EPA Method 24). For the cotton batt type liner, the VOC content shall be less than 50 g/L.

E. Flame/Smoke Rating: Refer to Section 2.2 - Fire and Smoke Developed Rating hereinbefore for UL 723 and ASTM E84 flame spread and smoke developed index requirements.

F. Unless noted otherwise, all air conditioning supply ductwork that is not internally lined shall be externally insulated. Supply, return and exhaust air ductwork shall be lined as indicated hereinafter. Supply ductwork shall be defined as any ductwork, including but not limited to sheet metal ductwork, sheet metal plenums and flexible connections that contains any of the following:

1. Conditioned air (cooled and heated)
2. Air at a temperature less than 66°F
3. 100% outdoor air
4. Any other ductwork connected downstream of an air conditioning unit

G. All supply ductwork as defined above with internal fiberglass liner shall also be provided with 24 gauge perforated metal liner to separate the fiberglass liner from the air stream.

H. Duct Dimensions: Ducts with internal liner shall be increased in both dimensions from the sizes indicated on the Drawings, depending upon the thickness of the liner. The duct sizes indicated on the Drawings are inside clear dimensions.

I. Duct liner thickness shall be modified as required to match thickness of internally mounted reinforcing angle bracing as specified hereinbefore.

J. Duct Liner Location:

1. Supply air ductwork as noted herein and as indicated on Drawings, including plenums downstream of each air handling unit except spin in device, flexible connections, round ductwork and flexible branch ductwork to diffusers:

a. 2" Thick: Duct liner from the air handling unit discharge to air handling unit room walls or floors in all directions, except for flexible connections.

b. 2" Thick: Duct liner from the air handling unit room walls or floor up to and including the first transition or set of elbow fittings outside of the air handling unit room. Minimum total length of lined ductwork extending outside the air handling unit room shall not be less than 20 feet in all directions.

2. Outdoor Air Ductwork (outdoor air handling unit and outdoor air fan):

a. 2" Thick: Duct liner downstream of the air handling unit or fan. Minimum total length of lined ductwork downstream of the air handling unit or fan discharge shall not be less than 20 feet in all directions. The outdoor air ductwork upstream of the air handling unit or fan shall be externally insulated as specified in Section 23 07 00.

3. Exhaust Air and Relief Air Ductwork:

a. 2" Thick: Duct liner for exhaust system ductwork downstream main toilet exhaust/relief air fans and upstream of main toilet exhaust/relief air fans a Minimum total length of lined ductwork upstream of the fan shall not be less than 20 feet, except for flexible connections.

b. 1" Thick: Duct liner for ductwork which is within 10 feet upstream and downstream of room exhaust fans, except for flexible connections.

4. Return Air and Transfer Air Ductwork:
5.

a. 2" Thick: Duct liner for return air ductwork at air handling unit rooms.

b. 1" Thick: Duct liner for all open ended air transfer and return air ductwork.

6. All ductwork and return air paths as indicated on the Drawings.

2.13 DUCT SEALANT

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Ductwork located within thermal envelope:
   a. Design Polymerics - DP 1010

2. Ductwork exposed to outdoor locations:
   a. Design Polymerics - DP 1090
   b. Ductmate - SOLVseal

B. Duct sealant for indoor application shall be water based and shall be listed under UL 181. Duct sealant for outdoor application shall be solvent based.

C. Fire and Smoke Hazard Rating: Refer to Section 2.2 hereinbefore for UL 723 and ASTM E84 requirements.

D. VOC Content: Duct sealant shall have no more than 250 grams/liter volatile organic compound (VOC) content for all field applied duct sealant to be used within the building interior locations when calculated in accordance with 40 CFR 59, Subpart D (EPA Method 24).

2.14 FIRE DAMPERS (FD)

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Ruskin
2. Nailor
3. Greenheck
4. Pottorff
5. Air Balance
6. AWV

B. General: Furnish and install fire dampers at all locations as indicated, required by applicable codes, and/or herein specified.

C. Fire dampers shall be installed at duct penetrations or air transfer openings of fire rated assemblies. The minimum hourly rating of damper shall be suitable for the type of penetration of particular architectural or structural assembly in which it is installed. The entire fire damper assembly as installed on this project shall be in accordance with the most current edition of the UL 555, manufacturer's recommendations, and Local Code Authority requirements. All fire dampers to be provided shall be dynamic type fire dampers as defined by the most current edition of the UL 555. All fire dampers shall be tested at a velocity rating level and pressure rating level not less than the ductwork system velocity and pressure class in which the damper is to be installed. Fire dampers shall be constructed, tested, and installed to meet the most current edition of the UL 555 and NFPA 90A. Unless otherwise noted on the drawings or specified herein, the maximum static pressure loss through a fully open fire damper assembly under maximum design flow rate conditions shall not exceed 0.05" wg.

D. Submittal data shall indicate the design flow rate in cfm, corresponding face area velocity, and the associated pressure loss in inches of water column. A specific type of fire damper shall be provided at various locations as indicated on the drawings or as required to accommodate the project architectural systems, structural systems, and other physical space limitations of the project. Fire dampers shall be installed in sleeves in accordance with the most current edition of the UL 555. Sleeves shall be securely fastened to the wall or floor.

E. Fire dampers shall be ceiling radiation dampers for ceiling opening protection and shall conform to NFPA 90A and shall be UL listed.

F. Fire dampers in the electrical vault shall comply with local power company specifications.

G. Fusible Links: Dampers shall be actuated by UL approved fusible link. Upon operation of the link, the dampers shall close automatically and remain tightly closed. Fusible links shall operate approximately 50°F above the maximum temperature the duct system will encounter under normal operation, but not less than 50°F.

H. Access Doors: Provide access door for inspection and servicing of dampers and actuators. Where ducts containing dampers are concealed, access doors shall be provided in the ceiling or wall for access to the duct access door and actuator. Refer to Section 23 00 10 for additional requirements.

I. Ductwork Coordination: If prefabricated joint systems are used at fire damper connections, the fire damper and sleeve assembly shall be compatible with the type of prefabricated joint system provided and installed in strict accordance with the manufacturer's recommendations to maintain the UL listed fire rating of the architectural assemblies, fire damper, and fire damper sleeve assembly.

J. Inspection and Acceptance: Dampers shall be of types acceptable to the Local Code Authority Inspection Department and the Engineer. Submit detailed shop drawings to the Engineer for review.

K. and Acceptance: Dampers shall be of types acceptable to the Local Code Authority Inspection Department and the Engineer. Submit detailed shop drawings to the Engineer for review.

2.15 MANUAL DAMPERS (MD)

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Ruskin
2. Nailor
3. Greenheck
4. Pottorff
5. Air Balance
6. AWV

B. General: Furnish and install manual dampers where indicated on the Drawings. Each damper shall be sized as indicated on the Drawings.

C. Manual balancing dampers shall be installed at all rectangular branch taps to individual diffusers. Damper shall be installed at take-off from main duct. This requirement to provide manual balancing dampers at all rectangular branch taps to individual diffusers shall not apply to individual diffusers having a spin collar fitting with volume damper or to an individual diffuser where a Titus AG-45, or approved equal, extractor is provided.

D. Frame and Axles: Manual dampers shall have a minimum 16 gauge galvanized steel channel frame with mounting flanges and axles set in oil impregnated bronze bearings. Axles shall be minimum of 1/2" in diameter and shall be interlocked with blades to positively eliminate slippage.

E. Blades: Blades shall be minimum of 16 gauge galvanized steel. Manual dampers shall have a minimum of four breaks running the entire length of the damper to assure rigidity and tight close off. Blades shall be on centers no greater than 9" and no blade shall exceed 48" in length. All blades shall be interconnected to act in unison for opposed blade operation.

F. Locking Quadrants: Dampers shall be provided with Ventline 560, or approved equal, locking quadrants of sufficient size and design to securely hold the damper sections and blade linkage hardware.

G. Special Requirements: Manual dampers located within sections of stainless steel ductwork shall be constructed of stainless steel. 100% of damper construction shall be stainless steel.

2.16 AUTOMATIC DAMPERS (AD)

A. Automatic dampers shall be provided as indicated on drawing M-502. Automatic damper requirements shall be coordinated between the Division 23 Contractor. Automatic damper sizes, ductwork coordination, access requirements, and details shall be reviewed and approved by the Division 23 Contractor.

2.17 DRAIN PANS

A. Construction: Drain pans shall be constructed of 16 gauge galvanized steel with all joints brazed. Drain pans shall be watertight and shall have hemmed edges. Drain pans shall have drain pipe connections of at least 1" or as shown on the Drawings. Drain connection shall be located at the lowest point of the drain pan.

B. Dimensions: Drain pan shall be at least 6" larger in both dimensions than the equipment or piping it is serving and shall be at least 2" high.

2.18 AUXILIARY DRAIN PANS

A. Provide an auxiliary or secondary drain pan under the equipment as shown on the Drawings or where damage to building components could occur as a result of overflow from the equipment.

B. Construction and dimensions shall be identical to "Drain Pans" specified hereinbefore.

C. Drain Pan High Water Level Alarm: Refer to Sequence of Operation on drawing M-502 for the drain pan high water level alarm requirements. Unit manufacturer shall cooperate and for the location and installation of the high water level detection sensor and high water level alarms.

2.19 FIRE RATED ENCLOSURES FOR SHEET METAL SYSTEMS

A. UL Listing: All fire rated enclosures shall be rated assemblies classified by UL.

B. Coordination Requirements: Fire rated enclosures are shown on the Division 23 Drawings for location purposes only. The enclosure shall be furnished and installed by the General Contractor under the Architectural Division. Division 23 shall cooperate with the Architectural Divisions to accommodate the enclosures.

2.20 PREFABRICATED ROOF CURB (PRC)

A. Prefabricated roof curbs, flashing and counter flashing shall be constructed of minimum 20 gauge G90 galvanized steel. Roof curb shall be internally insulated with a minimum of 1-1/2" thick 1.0 pound density glass fiber insulation. Insulation shall be installed, and all seams sealed so as to prevent condensation. Welded or tabbed and riveted construction will be acceptable. Both types of construction shall be manufactured and sealed as required to be watertight and weatherproof. The entire roof curb assembly shall meet the 25/50 flame/smoke criteria of UL 723 and ASTM E84. Lower section of roof curb that will be integrated with the roofing system shall be constructed to accommodate the roofing system provided. Top surface of curb shall have a rubber weather seal pad. Provide wooden nailer sections as required for the installation. Sheet metal counter flashing shall be provided to accommodate rectangular or round ductwork. All sheet metal screws and rivets shall be stainless steel or coated with a corrosion resistant material. Height of roof curb shall be min. 12" unless otherwise indicated herein or on the Drawings. Length and width of roof curb shall be sized by the Contractor for the particular application. If they are constructed in accordance with these Specifications, prefabricated roof curbs manufactured by ACME, Cook, Greenheck or approved equal will be acceptable.

PART 3 - EXECUTION

3.1 DUCTWORK STORAGE AND PROTECTION

A. Protection: Ductwork shall be protected at all times from dirt, water, and debris. During storage on the jobsite the Contractor shall keep the ends of the ductwork covered to prevent foreign objects and water from entering the ductwork.

B. Damaged Ductwork: Prior to final system acceptance the Contractor shall take all necessary precautions to eliminate ductwork damage due to internal overpressure or negative pressure conditions. Ductwork with split seams, blown out joints, or collapsed sections shall be removed from the job site at the discretion of the Engineer. Application of additional ductwork sealant to split seams and blown out joints will not be an acceptable means of reparation.

C. Cleaning: The interior of all ductwork shall be thoroughly cleaned. All debris and foreign matter shall be removed from the ducts prior to fan operation.

D. Cutting of Seams: Where possible, all ductwork shall be fabricated in such a manner that seams and/or joints will not be cut for the installation of grilles, registers, or ceiling outlets. If cutting of seams or joints is unavoidable, the cut portion shall be properly reinforced to original strength.

E. The Contractors shall ensure that all stored on-site or installed materials are protected at all times from moisture damage. Contractors shall be required to provide photographs to the Owner, Architect, or Engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

3.2 INSTALLATION

A. Ductwork:

1. Ductwork and sheet metal products shall be installed in accordance with the SMACNA Manual unless specified otherwise.

2. All ductwork shall be installed to maximize clearance between the bottom of ductwork and the top of ceiling construction. Ductwork shall be configured, positioned, and installed to permit the installation of light fixtures as indicated on the Drawings. Provide all necessary rises, drops, offsets, and other fittings as required to accommodate this ceiling plenum criteria. In general, all ductwork shall be installed as close to the bottom of the slab as possible. Any ductwork which must transition and drop below piping or other ductwork shall be transitioned back to the bottom of the slab immediately adjacent to the obstruction.

3. Duct taps shall not be made in any elbows or fittings unless shown on the Drawings.

4. In addition, all ductwork systems shall be fabricated and installed in accordance with the requirements of the Local Code Authority.

B. Flexible Connection:

1. Flexible connection shall be provided in ductwork connected to the inlets and/or outlets of all air handling units, air conditioning units, fans, equipment creating vibration, etc.

2. Flexible connections shall be treated as part of the ductwork system in which installed. Flexible connection shall be sealed airtight.

C. Dampers:

1. All dampers shall be installed square and level. Damper blades and hardware shall operate freely without obstruction. Damper blades and hardware that bind within the frame or obstructed by adjacent construction will not be acceptable.

2. When installed, all damper frames shall be gasketed or caulked to eliminate leakage between duct and damper frames. Head and sill shall have stops. Damper shafts shall be marked after final balancing and adjusting to indicate blade position.

3. All dampers shall be suitable for installation in the mounting arrangement shown.

D. Fire Dampers:

1. Fire dampers shall be provided as required by code at all locations where ductwork and return air openings penetrate fire-rated assemblies including but not limited to fire-rated walls, partitions, ceilings, and floors. Refer to the Architectural Drawings for fire rated wall, partition, ceiling, and floor locations.

2. The entire fire damper assembly as installed on this project shall be in accordance with the most current edition of the UL No 555, manufacturer's recommendations, and Local Code Authority requirements.

3. Provide all necessary ductwork transitions upstream and downstream of fire dampers, fire safing materials, and miscellaneous metal framing members as required to accommodate the specific requirements of this project. If prefabricated joint systems are used at fire damper connections, the fire damper and sleeve assembly shall be compatible with the type of prefabricated joint system provided and installed in strict accordance with the manufacturer's recommendations to maintain the UL listed fire rating of the architectural assemblies, fire damper, and fire damper sleeve assembly.

4. Fire damper operation shall be tested after the installation is completed. The Contractor shall provide a matrix for testing to the Owner and Engineer for approval at least 2 weeks in advance of the scheduled testing. A written certificate shall be issued by the Contractor stating the identification of damper as to location, date tested, and names and signatures of testing personnel. The Owner shall be notified in writing at least 5 days before any testing and review procedure begins for fire damper operation. The Owner and Engineer shall retain the right to witness some or all of the fire damper tests.

E. Backdraft Dampers: Backdraft dampers mounted in roof curb shall be secured to galvanized steel support angles fastened to the curb.

F. Flexible Ductwork: Flexible ductwork shall be installed in accordance with the SMACNA manual, applicable Mechanical Codes, and the manufacturer's recommendations. Flexible ductwork shall not pass through any drywall/gypboard or other architectural assembly. Where a flexible duct must pass through drywall/gypboard or other architectural assembly, the flexible duct shall be replaced with a section of rigid metal ductwork that extends a minimum of 6" upstream and downstream of the architectural assembly. Maximum length of flexible ductwork shall not exceed 6'-0". Flexible ductwork shall be installed with no more than two 90° turns or bends.

G. Auxiliary Drain Pans: Under any equipment for which a drain pan is shown on the Drawings and under all horizontal air handling units located above any ceilings, piping located over electrical equipment, and piping located in ceiling space directly above computer facility areas; furnish and install auxiliary drain pans. Drain lines shall be routed to nearest floor or hub drain independent of any other drain. The drain pan shall not be attached to or supported from the equipment it serves.

H. Duct Liner:

1. The duct liner shall be applied to the inside of the duct with the coated/facing side to the air stream and shall be secured to the duct with adhesive completely coating the clean sheet metal.

2. All transverse joints in the liner shall be firmly butted and tightly sealed with adhesive. Longitudinal liner joints shall be firmly butted and edges sealed with adhesive to form a tightly sealed joint.

3. For rectangular ducts, the liner shall further be secured with Graham, Nelson, or DuraDyne welding pins and washers, "Grip Nails", or approved equal, located on not more than 18" centers along a line 3" from the leading transverse edge of the liner. The liner shall not be compressed more than 10% of its nominal thickness by the retaining pins.

4. Liner shall be accurately cut and ends thoroughly coated with adhesive after the liner has been attached to the duct so that when the duct section is installed, the liner shall make a firmly butted and tightly sealed joint. All rips and tears in the liner on the air stream surface shall be repaired by applying the damaged areas with the approved coating or be replaced as determined by the Engineer. Duct liner as installed on this project shall form a continuous thermal and vapor barrier to minimize heat transfer and prevent condensation and moisture sweating from forming on the interior and exterior surfaces of the ductwork and ductwork liner.

I. Painting of Ductwork: Ductwork not constructed of stainless steel exposed to the weather shall be cleaned and prepared for painting, painted with a primer suitable for use on galvanized steel and/or black steel, then painted with a topcoat of epoxy based exterior grade enamel. Paint shall be suitable for the intended service and applied in accordance with the manufacturer's recommendations. As an acceptable option to painting ductwork exposed to the weather as indicated above, ductwork exposed to the weather may be covered with Alumaguard 60 manufactured by Polyguard Products. Alumaguard 60 is ultraviolet resistant having a laminated waterproofing and vapor barrier membrane composed of rubberized bitumen compound on a cross laminated high strength polyethylene film with an aluminum foil surface. Alumaguard 60 shall be installed to prevent water from ponding on the top surface. The complete installation of Alumaguard 60 shall comply with the manufacturer's published written recommendations.

J. Prefabricated Roof Curb: Prefabricated roof curbs shall be provided for all ductwork roof penetrations and curb mounted roof fans. The roof curb installation, including flashing and counter flashing, shall provide a watertight weatherproof enclosure. Counter flashing shall be attached to the ductwork via rubber gasketed sheet metal screws. The space between counter flashing and ductwork shall be filled with a silicon-based sealant. Sealant shall also be applied at all sheet metal screw locations. It shall be the responsibility of this Contractor to coordinate and cooperate with the Roofing System Contractor to achieve a waterproof watertight installation. All changes to this prefabricated roof curb specification deemed necessary by the Roofing Contractor shall be documented in writing and submitted to the Architect for review.

K. Interface Between Ductwork and Louvers:

1. At all other locations where ductwork is connected to a louver for either intake or exhaust purposes, the ductwork shall be installed, sloped, and connected to the louver so that all water entering the ductwork system positively drains back to and out of the louver.

2. Refer to Section 23 00 10 for additional requirements.

L. Acoustic Turning Vanes: Acoustic turning vanes shall be installed in all ductwork system elbows as indicated herein:

1. Toilet Exhaust System: Between fan inlet and first riser tap.

2. Outdoor Air System: Between fan discharge and first riser tap.

3. Typical Floor Air Handling System: Between the air handling unit discharge through and including the first elbow in each duct run extending from the air handling unit room on all levels.

3.3 DUCTWORK LEAKAGE TESTING

A. General: The Contractor shall perform ductwork leakage tests for all ductwork systems as described herein. Tests shall be conducted on complete ductwork systems. Testing of partial installations or limited sections of ductwork will not be acceptable.

B. Tests for the primary supply air ductwork on the office building floors shall be based on representative samples of the installed ductwork. As the primary supply air ductwork for the office building floors are completed and ready for testing in groups of up to five floors, the Engineer shall select at random one floor for testing at the time of the scheduled test date. No less than 20% of all the office building floors will require the primary supply air ductwork to be tested. If any of the tests fail to perform in accordance with the specified leakage limits, the Engineer retains the right to witness additional testing of floors performed by the Contractor at no additional cost. Floors selected for additional testing shall be at the discretion of the Engineer.

C. Tests for each system type of ductwork risers shall be based on representative samples of installed ductwork. As the various ductwork risers for each system type are completed and ready for testing, the Engineer shall select at random one or more ductwork risers for each system type for testing at the time of the scheduled test date. No less than 20% of each system type of ductwork risers will be required to be tested. If any of the tests fail to perform in accordance with the specified leakage limits, the Engineer retains the right to witness additional testing of ductwork risers performed by the Contractor at no additional cost. Ductwork risers selected for additional testing shall be at the discretion of the Engineer.

D. The above leak testing requirements do not prohibit the Contractor from testing all primary supply air ductwork on all floors of the building and testing all ductwork risers of all system types in order to meet the specified leakage limits.

E. During start-up, testing, adjusting, and balancing; if it is determined that the ductwork systems are leaking in excess of the specified limits or if there is noticeable audible noise caused by ductwork air leakage, the Contractor shall seal leaks and take other corrective actions as necessary to meet the contract requirements. Under such circumstances, the Engineer retains the right to witness additional ductwork tests, performed by the Contractor at no additional cost.

F. Leakage Test Results: All ductwork leakage test procedures and results shall be submitted to the Engineer for review. Any remedial actions recommended based on the test results shall be applied to the systems tested as well as the remaining systems.

G. Witnessing: The Engineer shall retain the right to witness some or all of the ductwork leakage testing procedures. The Contractor shall notify the Engineer in writing at least five working days prior to ductwork testing.

H. Contractor Requirements: The Contractor shall be responsible for providing all necessary test fans and calibrated measuring devices to accomplish the ductwork leakage test and demonstrate that ductwork systems leakage rate is less than the maximum rate specified. Refer

to specification Section 23 05 93 for additional requirements including but not limited to the calibration of instruments used for testing and balancing.

3.4 SEALING OF AIR DISTRIBUTION SYSTEMS

A. General: All longitudinal and transverse joints, corners at transverse joints, spin-collar connections, branch taps, ductwork/diffuser interface connections, equipment connections, plenum/diffuser interface connections, and other areas in the air distribution system where leakage may occur shall be sealed with an approved sealant as specified herein. All prefabricated ductwork joint systems such as Lockformer TDC shall use flexible resilient gasket seals specifically manufactured and approved for use by the system manufacturer. In addition to gaskets, all corners of prefabricated ductwork joint systems such as Lockformer TDC shall be sealed with an approved sealant. Any deviation from a pre-fabricated duct system shall be approved in writing by the system manufacturer and engineer. Ductwork of all pressure/seal classifications shall comply with these specifications.

B. All ductwork must be sealed in accordance with the SMACNA Seal Class "A" requirements.

C. Air Outlet and Inlet Plenum Installations:

1. The interface point between a sheet metal plenum and its associated supply or return diffuser shall be sealed to prevent air leakage. Blank-offs plates shall be installed at both ends of a diffuser plenum to minimize leakage and to ensure the air is properly directed through the diffuser. The Sheet Metal Contractor shall work with the diffuser supplier to determine the physical configuration for the blank-off end plates required. End plates shall be painted flat black on all visible surfaces.

2. Plenum sizes shown on the Drawings shall be adjusted as required to accommodate the actual diffusers provided and the architectural conditions of the Project.

3. All interior surfaces of unlined supply or return plenums shall be painted flat black.

3.5 CLEANING

A. Ductwork shall be cleaned of all rust, dust, and debris both internally and externally before placing in operation.

3.6 RETURN AIR PATHS

A. Within each room or area of the space, provide return air paths throughout the space to achieve room design temperature, humidity, and air flow design set points at or below project noise criteria (NC) levels in accordance with the following requirements:

1. Return air paths shall be defined as the pathway where air travels from the occupied space, up through the ceiling via grilles, slots, light fixtures, or other openings into the ceiling plenum, through the return air plenum, in some cases through return air ductwork systems, and back to the respective air handling unit, air conditioning unit, fan, or HVAC unit. Unless specifically noted otherwise, return air paths shall be located above the ceiling. Areas below the ceiling shall not be used for return air path purposes. Code required egress or exit paths shall not be used for return air path purposes. Areas prohibited by applicable codes shall not be used for return air paths. Return air performance shall be met with all doors closed. For rooms with partitions or wall

construction that extends above the ceiling to the bottom of structure or that block the return air path, openings shall be cut in the wall above the ceiling to allow for the proper air flow of return air.

2. Square foot area of return air openings shall be sized in accordance with the following maximum allowable air velocity (fpm) schedule, except where return air sizes are specifically indicated on the Drawings:

   a. Return air opening serving one room or one continuously open area: 300 fpm.
   b. Return air opening serving two or more areas, total floor area not exceeding 5,000 sf: 400 fpm.
   c. Return air opening at typical demising corridor: 500 fpm.
   d. Return air opening at air conditioning unit or HVAC room wall: 700 fpm.
   e. Return air opening at all other areas: 500 fpm unless otherwise approved by Engineer.

3. Piping, ductwork, light fixtures, cable trays, conduit, cables, wiring, raceways, junction boxes, insulation, and other items located within the return air ceiling plenum shall be coordinated, arranged and installed to not obstruct or otherwise adversely affect the flow of return air from the occupied space, through the return air path, back to the respective HVAC unit. In all areas of the return air ceiling plenum, including congested areas of the ceiling plenum, the return air velocity shall not exceed 500 feet per minute. Refer to the mechanical drawings for additional return air path requirements.

4. Fire rated partitions shall have a fire damper within the return air opening. Smoke rated partitions shall have a smoke damper within the return air opening. Fire and smoke rated partitions shall have a fire/smoke damper within the return air opening. Refer to specifications herein for additional requirements for fire dampers, smoke dampers, and fire/smoke dampers.

5. Acoustically rated or STC rated walls and partitions shall have acoustically treated return air openings in the form of internally acoustically lined return air ductwork. Unless otherwise shown on the drawings, the inside clear area of the lined return air ductwork shall be sized to meet the maximum return air velocity limitations specified herein. Refer to the specifications herein for additional requirements for return air ductwork liner.

6. The contractor shall provide properly sized return air openings in partitions whether shown on the drawings or not at no additional cost to the Owner. Prior to cutting return air openings not shown on the drawings, the contractor shall obtain approval from the Architect and Engineer. Prior to the installation of the ceiling and prior to final air balancing, the Contractor shall perform a return air review of the spaces to verify that an adequate return air path exists for each room or area. Contractor shall be responsible for notifying both the Architect and Engineer, as early in the construction process as possible and prior to any ceiling close-ins, on any observed areas above ceilings in corridors or other areas used for return air that may be blocked by ductwork, piping, equipment, or other construction materials. However, this notification shall not be less than seven calendar days prior to ceiling installation.

3.7 COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

END OF SECTION 23 30 01

SECTION 23 34 00 - HVAC FANS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

B. Furnish and install the exhaust and ventilating fans as indicated on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 70 - Division 23 Schedule of Submittal Data
B. Section 23 05 13 - Common Motor Requirements for HVAC Equipment
C. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment
D. Section 23 30 01 - Ductwork and Sheet Metal
E. Drawing M-502 - Sequence of Operation

1.4 REFERENCE STANDARDS

A. Air Movement and Control Association (AMCA)
   1. 99 Publication - Standards Handbook
   2. 211 Publication - Certified Ratings Program - Product Rating Manual for Fan Air Performance
   3. 311 Publication - Certified Ratings Program - Product Rating Manual for Fan Sound Performance
   4. 204 Standard - Balance Quality and Vibration Levels for Fans
   5. 210 Standard - Laboratory Methods of Testing Fans for Certified Aerodynamic Performance Rating
   6. 300 Standard - Reverberant Room Method for Sound Testing of Fans
   7. 320 Standard - Laboratory Method of Sound Testing of Fans Using Sound Intensity
   8. 500-D Standard - Laboratory Methods of Testing Dampers for Rating

B. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings

C. National Electrical Manufacturers Association (NEMA)
   1. MG 1 Motors and Generators

D. National Fire Protection Association (NFPA)
   1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

E. Underwriters Laboratories (UL) Standard for Safety
   1. 507 Electric Fans
   2. 705 Power Ventilators

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data listed in Sections 23 00 10 and 23 00 70 plus capacity and accessory information indicated in the Schedule.

PART 2 - PRODUCTS

2.1 GENERAL

A. Fan Type and Capacity: Fans shall be of the types indicated on the Drawings and shall have the capacities listed in the Schedules. Speed and motor horsepower shall not exceed the values indicated in the Schedules.

B. Fan Drive and Accessories: Each fan shall be direct or belt drive as indicated in the Schedule. Each fan shall have the accessories as indicated in the Schedule and on the Drawings.

C. AMCA Certification: Manufacturer shall comply with AMCA Certified Ratings Program. Each fan shall be tested and rated in accordance with the latest version of AMCA Publication 211 and/or Publication 311 and shall bear an AMCA Seal for "Air Performance" and/or for "Sound and Air Performance" as applicable.

D. Balancing Requirements: Each fan shall be statically and dynamically balanced in accordance with AMCA Standard 204 at the factory and shall be so certified in writing to the Engineer. Certification shall be submitted prior to shipment of fans.

E. Special Requirements: The construction and installation of each fan shall be suitable for the intended service. The following Specifications outline the minimum requirements for fan construction. Refer to the Schedules and the Drawings for additional requirements including capacity, fan, motor, drive type, inlet condition, discharge arrangement, corrosive resistant coatings, spark resistant, explosion proof motors, weatherproof construction, limited access arrangements, acoustical linings, high temperature service, smoke control service and grease laden air service, accessories, etc.

F. Fan Drives

1. Motors: Each motor for a belt driven fan shall be mounted on an adjustable base rigidly supported on the fan and shall have extended shaft to accommodate the adjustable pitch sheave.

2. Fan drives shall be selected with a minimum belt horsepower capacity of 120% of the motor nameplate horsepower. All drives shall conform to industry standard tolerances in accordance with the "Engineering Standards for Multiple V-Belt Drives" as adopted by the Mechanical Power Transmission Association and the Rubber Manufacturers Association. The fan drive components shall be as follows:

a. Motor Sheaves

1) The motor sheaves for all fans having a motor smaller than 3/4 hp shall be Browning Type LVP or MVP adjustable type with double locking feature. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable: TB Woods or Browning.

2) The motor sheaves for all fans having a 3/4 hp and larger motor shall be nonadjustable type selected for the rated fan rpm as determined by the fan requirements scheduled. The Contractor shall furnish variable sheaves as required to determine correct fan rpm as determined by the fan requirements scheduled. After correct speed has been determined, the Contractor shall furnish additional fixed sheaves as required. Fixed sheaves shall be as manufactured by one of the manufacturers listed for the fan sheaves. All unused fixed sheaves shall become the property of the Owner.

b. Fan Sheaves: The fan sheaves shall be of the non-adjustable type with removable machined bushings. Fan sheaves shall be machined on all contact surfaces. Fan sheaves with over three grooves shall be dynamically balanced. Fan sheaves with three grooves or less shall be statically balanced. If weights are required for balancing, they shall be welded to the sheave. If it complies with these Specifications, fan sheaves manufactured by one of the following manufactures will be acceptable: Browning, Eaton, Yale and Towne, Dodge Manufacturing Company, Fort Worth Steel and Machinery Company, or TB Woods.

c. Belts: The belts shall be standard "V-groove" type suitable for the service intended with the capacities specified herein. The belts shall be closely matched and tagged prior to delivery to the job site. If the belts are not properly matched, they shall be replaced with another closely matched set of belts. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable: Gates, Durkee Atwood, Goodyear, Browning or Uniroyal.

G. Shafts and Bearings: Fan shafts shall have SKF, Sealmaster, Dodge Manufacturing Company, Timken or Fafnir externally or internally mounted grease lubricated self-aligning ball bearings. Bearings shall have zerk grease fittings.

H. Critical Speed: Fan shafts shall not pass through their first critical speed as the unit comes up to rated rpm.

I. Motors: Unless noted otherwise in the Schedule or specified herein motor controllers will be furnished under Division 26.

J. Motor Controllers: Unless noted otherwise in the Schedule or specified herein motor controllers will be furnished under Division 26.

K. Prefabricated Roof Curbs: Refer to Section 23 30 01 for additional requirements.

## 2.2 CENTRIFUGAL ROOF EXHAUST FANS (CREF)

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Greenheck
2. Cook
3. ACME
4. Twin City Fan
5. Aerovent
6. Carnes
7. Jenn Air

B. General: Fan hood, housing and base shall be constructed of heavy gauge aluminum. All external screws, bolts, fasteners, etc. shall be stainless steel. All other fasteners shall be stainless steel or zinc plated chromate treated. Unless otherwise noted provide each fan with a factory fabricated 12" high roof curb for fan mounting. The fan and roof curb assembly shall provide a weatherproof enclosure for the roof penetration.

C. Fan Wheels: Centrifugal fan wheels shall be aluminum and shall be backward inclined or airfoil non-overloading type.

D. Fan Arrangement: Upblast (CREF-U) or Downblast (CREF-D) airflow arrangement shall be as indicated on the Drawings.

E. Fan Drives: Belt driven fan shall have a variable pitch motor sheave as specified herein which will not overload the motor when set to its maximum pitch diameter.

F. Motor Mount: Each motor shall be mounted in a separate compartment outside the air stream and shall have a cooling system to continuously draw outside air through the motor and motor compartment.

G. Electrical Connections: Each fan shall have a wiring post from the motor compartment to the curb base. Each belt driven fan shall be factory wired from the motor to the wiring post with flexible metal conduit in accordance with the National Electrical Code using UL approved materials only. Each unit shall be furnished with a non fused disconnect switch factory mounted inside the drive housing.

2.3 SQUARE CENTRIFUGAL FANS (SCF)

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Greenheck
2. Cook
3. ACME
4. Twin City Fan
5. Aerovent

B. General: Inline centrifugal fans shall be non-overloading single width, single inlet and shall have backward inclined or airfoil fan blades. Fans shall be designed to be mounted and operated in any position.

C. Wheel Construction: Wheels shall be of riveted or spot-welded aluminum construction with venturi inlet.

D. Fan Housing:

1. Square shaped galvanized steel housing shall provide inlet and outlet flanges suitable for the connection of sheet metal ductwork and flexible ductwork connections as indicated on the Drawings.

2. Steel housing including all internal housing components shall be galvanized or painted with an enamel primer and an enamel topcoat. All housing surfaces on fans used for outdoor air service shall be painted with an epoxy primer and an epoxy topcoat. All fasteners shall be stainless steel or zinc plated chromate treated steel.

E. Housing access panels shall be provided for the inspection, service and removal of all internal fan

components without disturbing inlet and outlet ductwork. Access panels shall be located on the fan housing to accommodate the position and location of the fan as indicated on the Drawings.

F. Housing shall be constructed to allow inline discharge or side discharge arrangement as indicated on the Drawings.

G. Fan Shafts and Drives:

H. Belt Drive: Motor and drive components shall be isolated from the air stream. Unless otherwise noted in the Schedule, fan drive shall be adjustable V-belts.

I. Direct Drive: Direct drive fans shall have solid state manual speed controller wired and mounted on fan housing. Speed adjustment shall be continuously variable from 50% to 100% of maximum motor speed.

J. Bearings shall be of the permanently lubricated type with a minimum life of 200,000 hours.

K. Motors: Belt drive fan motors shall be mounted outside the air stream on an adjustable base with fan arrangement and location as shown on the Drawings. Motor and drive assembly shall be accessible for service and removal without disturbing duct connections and fan wheel.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Connect all ductwork, flexible connections, in line thrust resisters, electric devices, controls, vibration isolators, etc. as specified herein and as indicated on the Drawings.

3.2 VIBRATION ISOLATION

A. Refer to Section 23 05 48 for additional requirements.

3.3 CLEANING

A. Clean all dirt and debris from the inside and outside of the fans prior to start up.

END OF SECTION 23 34 00

# SECTION 23 37 00 - AIR OUTLETS AND INLETS

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

### 1.2 WORK INCLUDED

A. Furnish and install all air devices, grilles, registers, diffusers, etc. as indicated on the Drawings and as required for proper distribution of conditioned air within the conditioned space and for return of conditioned air from the conditioned space to the various air conditioning systems.

### 1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 70 - Division 23 Schedule of Submittal Data
B. Section 23 30 01 - Ductwork and Sheet Metal

### 1.4 REFERENCE STANDARDS

A. Air-Conditioning, Heating, and Refrigeration Institute (AHRI)
   1. 885 Procedure for Estimating Occupied Space Sound Levels in the Application of Air Terminals and Air Outlets

B. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 70 Method of Testing the Performance of Air Outlets and Air Inlets

C. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA)
   1. 006 HVAC Duct Construction Standards - Metal and Flexible

### 1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing data shall clearly and concisely indicate all requirements listed herein and on the Drawings, plus certification and performance requirements of Sections 23 00 10 and 23 00 70.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

A. The air distribution devices described herein are Titus products unless otherwise noted. If an alternate manufacturer is provided, the air distribution products submitted shall be of equal quality and performance to make and model numbers specified. Products of lower quality and lower performance will not be acceptable regardless of manufacturer. Please note limitations regarding diffuser manufacturers on selected air distribution devices specified herein.

B. Unless noted otherwise for a particular device, if they comply with these Specifications, equipment manufactured by the following manufacturers will be acceptable:

   1. Titus

2. Nailor
3. Price
4. Anemostat
5. Carnes
6. Greenheck/Metalaire
7. Krueger
8. Raymon Donco
9. Air Factors

C. Please note limitations regarding diffuser manufacturers on selected air distribution devices specified herein.

2.2 AIR DISTRIBUTION GENERAL

A. All air distribution devices shall be similar and approved equal to the types indicated on the Drawings and specified herein.

B. Grilles and registers shall have adjustable front and rear deflection bars with a bar space of 0.5" unless otherwise specified herein. Curved frame borders shall be maximum 0.75" wide unless specified otherwise.

C. All air distribution devices shall be suitable for the service intended and for installation on the ceiling, wall or floor as shown on the Architectural Drawings. Air devices shall be coordinated with the architectural material and dimensional limitations of the Project.

2.3 SPECIAL FINISH AND COORDINATION REQUIREMENTS

A. Special Finish Requirements: Where exposed to view, all air delivery or return devices with factory applied baked enamel or powder coated finish must conform to the color sample supplied by the Architect. Full size finished samples of typical device of each type and color shall be delivered to the Architect and Engineer for written approval. No devices shall be painted until samples are approved.

B. Ceiling, Wall and Floor Type Coordination: All ceiling mounted, wall mounted and floor mounted air distribution devices shall be suitable for installation in the scheduled ceiling, wall and floor construction. Where required, the air distribution device shall be provided with the appropriate manufacturer’s mounting frame, border type, and other accessories and hardware for installing the device in the scheduled ceiling, wall and floor construction.

C. Location: The Division 23 Drawings indicate the general location and relative position of the air distribution devices with respect to the architecture. The exact location and mounting details of all air distribution devices located and mounted in conjunction with walls, ceilings, floors and other architectural features shall be as indicated on the Architectural Drawings. It shall be the responsibility of the air distribution device supplier and the Contractor to thoroughly review the Architectural Documents in addition to the Mechanical Documents to determine the complete scope of work.

2.4 SQUARE CEILING DIFFUSERS

A. Supply Square Ceiling Diffuser - SCD-A:

1. Titus Model PAS perforated flush face supply diffuser or approved equal with removable core and aluminum or steel frame suitable for installation in the ceiling type as shown on the Architectural Documents (Titus Model PDS perforated drop face supply diffuser with removable core and aluminum or steel frame suitable for installation in dropped tile type Tegular or "Fineline" ceiling grids as shown on Architectural Documents.) Neck size of diffuser shall be the same as the flexible duct unless otherwise indicated. Face area shall be approximately 24"x24" or 12"x12" as indicated on the Drawings. Air pattern shall be 4-way adjustable unless otherwise indicated. Air volume shall be as indicated on the Drawings. Provide with baked enamel or powder coated finish.

B. Return/Exhaust Square Ceiling Diffuser - SCD-B:

1. Titus Model PAR perforated flush face return air grille or approved equal with aluminum or steel frame suitable for installation in the ceiling type as shown on the Architectural Drawings (Titus PDR perforated drop face return air grille suitable for installation in dropped tile type Tegular or "Fineline" ceilings as shown on Architectural Documents.) Face area shall be approximately 24"x24" or 12"x12" as indicated on the Drawings. Provide with baked enamel or powder coated finish.
   a. For unducted application, do not provide volume damper.
   b. For square ducted application, provide AG-95 opposed blade volume control damper and square neck for ducted application. Neck size shall match associated ductwork size as indicated on the Drawings.
   c. For round ducted application, provide volume damper at the rigid round duct. Neck size shall be the same as the flexible duct unless otherwise indicated on the Drawings.

2.5 REGISTERS AND GRILLES

A. Exposed Duct Supply Registers - SR-A:

1. Titus Model 272FS, Type A external fastening, extruded aluminum register or approved equal with short front vertical and long rear horizontal adjustable airfoil blades at 3/4" blade spacing and a screwdriver or key operated opposed blade damper for volume control. Provide AG-45 extractor and baked enamel or powder coated finish. Refer to "special finish and coordination requirements" hereinafter. Register shall be installed on a reverse knuckle joint in accordance with SMACNA Manual.

B. Sidewall/Ceiling Supply Registers - SR-B:

1. Titus Model 272FS, Type C concealed fastening, extruded aluminum register or approved equal with short front vertical and long rear horizontal adjustable airfoil blades at 3/4" blade spacing and a screwdriver or key operated opposed blade damper for volume control. Register shall be secured to frame with hidden recessed screws. Exposed mounting screws will not be acceptable. Provide AG-45 extractor and baked enamel or powder coated finish.

C. Sidewall/Ceiling Return-Exhaust Registers - RR-B:

1. Titus Model 4FL, Type C concealed fastening, extruded aluminum register or approved equal with one set of long horizontal 45° fixed down deflection flat blades at 1/2" blade spacing and a screwdriver or key operated opposed blade damper for volume control. Register shall be secured to frame with hidden recessed screws. Exposed mounting screws will not be acceptable. Provide with baked enamel or powder coated finish.

D. Sidewall/Ceiling Return-Exhaust Grilles RG-B:

1. Same as "RR-B", except without opposed blade damper.

2.6 LINEAR DIFFUSERS

A. Supply Linear Diffuser - LD-A:

1. Titus Model ML extruded aluminum linear diffuser or approved equal with continuous slots, air pattern controller, border, surface concealed mounting frame without holes, and baked enamel or powder coated finish. Mounting frame shall be suitable for installation in the ceiling type indicated on the Architectural Drawings. The diffuser length, number of slots, slot width, throw, and air volume shall be as indicated on the Drawings. Air shall discharge horizontally along the ceiling unless otherwise indicated. Each and every blade edge, etc., of linear diffusers, must be in perfect alignment and flush with adjacent ceiling material. Diffusers shall be provided with alignment strips for insertion in extruded alignment slot on each blade. Provide factory fabricated mitered corner accessories as required. Provide sheet metal blank-off plates painted flat black or perforated light shield hood in all inactive sections of supply diffusers.

B. Return Linear Diffuser - LD-B:

1. Titus Model MLR or approved equal, same as linear diffusers "LD-A" except without air pattern controller. Provide perforated light shield hood.

C. Unless otherwise noted on the Drawings, all sections of linear supply diffusers that are not covered with an active supply plenum shall not have volume controls and flow pattern control devices.

D. All items such as pattern controllers, support structure alignment strips, etc. visible through diffuser slots shall be painted flat black.

E. For all linear diffusers installed in a location which do not allow access for a flow measuring hood to be used for proper air balancing, provide a factory installed flow sensor at the inlet collar with an accuracy of ±5%. The accuracy of the flow sensor shall be verified by an independent testing laboratory at the flow rates specified in the schedule and the test data submitted to the Engineer for review. Plenum rated tubing shall be factory installed in a concealed location to allow for testing and air balancing from below the ceiling.

F. Refer to the Architectural Documents for mounting arrangements and border configurations.

2.7 PERIMETER SLOT DIFFUSERS

A. Perimeter Ceiling Combination Supply/Return Slot Diffusers - SD:

1. Combination supply/return slot plenum diffuser shall be Nailor 59N or 59BS Series with side round or oval inlet uninsulated supply plenum and integral 2" wide top outlet return plenum on the rear side of the diffuser (i.e., the side away from the direction of supply airflow). Supply air plenum and slot for each SD type shall be as specified hereinafter. Supply side inlet connection shall be at least 2-1/2" long neck with a 1/8" high raised bead located 1" from the inlet edge for flexible duct connection. The integral return air section shall be constructed as indicated on the Drawings. The plenum shall be constructed of at least 24 gauge galvanized steel and shall be substantially supported and reinforced as required. No spot welding visible from the face when installed.

2. The supply air volume, diffuser type and diffuser length shall be as indicated on the Drawings.

3. The supply diffuser shall be installed above the ceiling between the ceiling tee bars on above soffit in an architectural soffit slot as indicated on the Architectural Drawings.

4. The diffuser shall be designed, tested, and constructed in a manner so as to comply with the performance and sound level requirements specified hereinafter.

5. The diffuser manufacturer shall coordinate the attachment, support, etc., of the plenum with the ceiling subcontractor. The entire assembly shall be tested as a unit at the diffuser manufacturer's laboratory in accordance with ASHRAE 70 test requirements and ASHRAE 70 tolerances and may be witnessed by the Engineer. Six certified copies of the ASHRAE 70 test results shall be delivered to the Engineer for review. The test data shall include AK factors for an Alnor Velometer, the microprocessor programming data for the flow hood kit meter, sound data, diffuser static pressure drop, horizontal air throw and drop for the air supply rates per lineal foot of diffusers scheduled hereinafter. The test data shall be based on a 51°F cooling air supply temperature and a 24°F temperature differential. The unit tests are not required if previous tests were done and acceptable test data is available for the Engineer's review. The plenums shall be painted flat black on interior surfaces and the exposed surfaces viewed from below the ceiling system shall be painted flat black.

6. After the air distribution device manufacturer has submitted certified test results to the Engineer, the Engineer may desire to witness additional tests to verify compliance with the specifications. See Section 23 00 10 for additional requirements.

7. Refer to the architectural perimeter wall and ceiling details for dimensional limitations and mounting requirements.

8. All perimeter slot diffusers shall be provided by the same manufacturer.

9. After submittal review and prior to diffuser fabrication and finalization of purchase, the contractor shall provide diffuser samples to verify all diffuser types provided for this project will fit within the space provided as detailed on the architectural Drawings. Shop drawing submittal review status shall be conditional upon diffuser approval by the Architect.

10. The Engineer may desire to witness additional tests to verify compliance with the Specifications. See Section 23 00 10 for additional requirements. If it complies with these Specifications, perimeter ceiling supply diffusers manufactured by Titus will be acceptable.

B. Type SD-A:

1. Nailor Model 59NDR, 11" tall plenum slot diffuser with an 8" or 6" round inlet as indicated in the performance tables below or as indicated on the Drawings. The supply plenum diffuser shall have an 3/4" wide internal fixed curved aerodynamic shaped extruded aluminum outlet slot and a 1-5/8" wide center vertical down-below outlet slot with an adjustable air pattern controller. The diffuser shall be designed to provide the maximum amount of induced secondary room air. Aerodynamic supply deflector shall be constructed of extruded aluminum and held in position by set screws at both ends of the diffuser. The supply air shall be discharged horizontally along the ceiling toward the interior and the center section discharged vertically downward along the building exterior wall with the supply inlet positioned as indicated on the Drawings. SD-A diffusers shall meet the following performance and sound level characteristics:

a. 60" Long Diffuser with 8" Round Inlet and 15" Long Center Down-Blow Slot

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 30 | 0.030 | 13 ft | <20 | 0.011 |
| 40 | 0.060 | 16 ft | <20 | 0.018 |
| 50 | 0.090 | 18 ft | <23 | 0.027 |
| 60 | 0.130 | 20 ft | 29 | 0.038 |
| 70 | 0.180 | 21 ft | 33 | 0.050 |

b. 48" Long Diffuser with 8" Round Inlet and 12" Long Center Down-Blow Slot

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 30 | 0.030 | 12 ft | <20 | 0.011 |
| 40 | 0.050 | 15 ft | <20 | 0.018 |
| 50 | 0.070 | 16 ft | <20 | 0.027 |
| 60 | 0.110 | 18 ft | 24 | 0.038 |
| 70 | 0.160 | 19 ft | 29 | 0.050 |

c. 36" Long Diffuser with 8" Round Inlet and 12" Center Down-Blow Slot

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 30 | 0.020 | 10 ft | <20 | 0.011 |
| 40 | 0.040 | 12 ft | <20 | 0.018 |
| 50 | 0.050 | 14 ft | <20 | 0.027 |
| 60 | 0.080 | 17 ft | 20 | 0.038 |
| 70 | 0.110 | 18 ft | 25 | 0.050 |

d. 24" Long Diffuser with 6" Round Inlet and 8" Long Center Down-Blow Slot

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 30 | 0.010 | 10 ft | <20 | 0.011 |
| 40 | 0.030 | 12 ft | <20 | 0.018 |
| 50 | 0.050 | 15 ft | <20 | 0.027 |
| 60 | 0.080 | 16 ft | <23 | 0.038 |
| 70 | 0.120 | 18 ft | 28 | 0.050 |

2. Equivalent products by Raymon Donco or Price will be acceptable provided they meet the specification requirements and are equal in construction quality and performance to the products specified. The equivalent Raymon Donco slot shall be Model SOD with ice tong air deflector, internally lined plenum, and no factory cross-notch. The slot diffuser should not have notches for the ceiling grid to run through.

C. Type SD-B:

1. Nailor Model 59BSR, 11" tall plenum slot diffuser with an 8" or 6" round inlet as indicated in the performance tables below or as indicated on the Drawings. The supply plenum diffuser shall have 1-3/4" wide supply slot with a fully adjustable integral air pattern controller to direct airflow a full 180 degrees from horizontal left hand to vertical to horizontal right hand. On units 48" and longer, the air pattern controller shall be divided into equal lengths to allow variable air pattern control. The diffuser shall be designed to provide the maximum amount of induced secondary room air. Unless otherwise indicated on the Drawings, the initial air pattern shall be vertically downward along the building exterior wall. This pattern shall be adjusted in the field to avoid interference with discharge air from adjacent and nearby air distribution devices. SD-B diffusers shall meet the following performance and sound level characteristics:

a. 60" Long Diffuser with 8" Round Inlet

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 40 | 0.031 | 10 ft | <20 | 0.018 |
| 50 | 0.048 | 11 ft | <20 | 0.027 |
| 60 | 0.069 | 12 ft | <20 | 0.038 |
| 70 | 0.094 | 16 ft | 25 | 0.050 |

b. 48" Long Diffuser with 8" Round Inlet

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 40 | 0.029 | 10 ft | <20 | 0.018 |
| 50 | 0.045 | 15 ft | <20 | 0.027 |
| 60 | 0.065 | 17 ft | <20 | 0.038 |
| 70 | 0.088 | 20 ft | 20 | 0.050 |

c. 36" Long Diffuser with 8" Round Inlet

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 40 | 0.020 | 10 ft | <20 | 0.018 |
| 50 | 0.032 | 16 ft | <20 | 0.027 |
| 60 | 0.046 | 16 ft | <20 | 0.038 |
| 70 | 0.062 | 19 ft | <20 | 0.050 |

d. 24" Long Diffuser with 6" Round Inlet

| Diffuser (cfm/ft) | Max Diffuser SP Loss (in wg) | Horizontal Throw @ Vt = 50 fpm | Maximum NC Level | Max Return SP Loss (in wg) |
|---|---|---|---|---|
| 40 | 0.053 | 10 ft | <20 | 0.018 |
| 50 | 0.081 | 16 ft | <20 | 0.027 |
| 60 | 0.122 | 18 ft | <20 | 0.038 |
| 70 | 0.161 | 21 ft | <20 | 0.050 |

2. Equivalent products by Raymon Donco or Price will be acceptable provided they meet the specification requirements and are equal in construction quality and performance to the products specified. The slot diffuser should not have notches for the ceiling grid to run through.

D. After the air distribution device manufacturer has submitted six certified copies of the test results to the Engineer, the Engineer may desire to witness additional tests to verify compliance with the Specifications. Refer to Section 23 00 10 for additional requirements.

2.8 EXTRACTORS AND DAMPERS

A. Extractors shall be Titus AG-45 and opposed blade volume dampers shall be Titus AG-95. Similar devices as manufactured by an acceptable manufacturer will be acceptable, if the equipment furnished complies with these Specifications.

PART 3 - EXECUTION

3.1 AIR DISTRIBUTION EQUIPMENT

A. The air distribution outlets shall be sized as shown on the Drawings to accommodate the air volume and throw indicated so as to maintain a terminal velocity of 50 ft/min in the occupied area. The overall noise level produced by all of the supply air outlets and return air inlets in various rooms shall not exceed specified limits. In Section 23 00 10, see "Equipment Noise and Vibration" for requirements. The conditioned air shall be distributed in such a manner that the space temperature shall not vary more than 2°F over the entire conditioned area, which shall be defined as the area 2'-0" above the floor to 7'-0" above the floor, inclusive. It will be the responsibility of the bidder to size the air outlets to suit the volume and throw indicated on the Drawings so as to maintain a maximum terminal velocity specified herein in the occupied area and to fit the physical limitations imposed by the structure and surrounding interferences. If the bidder cannot comply with the above requirements by following the arrangement shown on the Drawings, he shall notify the Engineer eight days prior to the bid date, in writing, setting forth requested modifications.

3.2 TESTING

A. At the discretion of the Engineer, air outlets may be smoke tested to determine their compliance with these Specifications. See Section 23 05 93 for testing requirements. Any revisions required for compliance with terminal velocity requirements, noise level requirements, or maximum temperature variation requirements shall be made at no cost to the Owner.

3.3 DIFFUSER/PLENUM INTERFACE COORDINATION

A. All plenum sizes shown shall be considered approximate. Plenum sizes shall be coordinated and adjusted to fit the actual diffusers provided on this project. Plenums (factory fabricated) shall be provided by the air distribution device manufacturer.

B. At the Contractor's option, plenums may be shop fabricated and provided by the sheet metal contractor. The sheet metal contractor shall coordinate with the air distribution device manufacturer in developing sheet metal plenum details and arrangements that will indicate the attachment of the plenum to the actual diffusers provided. After the shop fabricated sheet metal plenum details are completed and prior to submitting to the Owner, the air distribution device manufacturer shall review the plenum details and provide written review comments and recommendations to the sheet metal contractor describing any revisions deemed necessary for proper diffuser and plenum installation. Shop fabricated plenums shall not create more air leakage when compared to the factory fabricated plenums.

C. End plates shall be installed at both ends of a diffuser plenum to minimize leakage and to ensure the air is properly directed through the diffuser. Factory fabricated end plates shall be provided by the diffuser manufacturer. The sheet metal contractor shall work with the air distribution device supplier to determine the physical configuration for the end plates required. End plates shall be painted flat black on all visible surfaces.

D. Refer to Section 23 30 01 for additional requirements.

3.4 BLANK-OFF PLATES

A. General: Blank-off plates shall be constructed of sheet metal painted flat black that have been fabricated for the particular application. The intent of the blank-off plate is to eliminate the line of sight into the ceiling plenum, wall cavities, and other areas unsuitable for direct view of the occupants.

B. Locations: Blank-off plates shall be provided at the following locations:

1. In between perimeter slot diffusers.

2. As indicated on the Drawings.

3.5 PERIMETER SLOT DIFFUSER MOCK-UP REQUIREMENTS

A. After shop drawing review but prior to the installation of perimeter slot diffusers, the Contractor shall install the perimeter slot diffusers within one typical window bay as indicated on the Drawings as a mock-up condition generally representing the typical ceiling plenum installation. The mock-up condition shall be complete with architectural ceiling system, architectural perimeter slot diffuser track if applicable, architectural return air opening track if applicable, and blank-off plates. Mock-up arrangements shall be located on site within the facility at an actual useable location.

B. The Contractor shall advise the Engineer and Owner after the mock-up is complete and ready for review and inspection. The Contractor shall arrange a time mutually agreeable to these parties so they can meet at the project site, review the mock-up installation, and determine any changes that need to be made for the installation to be acceptable to the Owner and Engineer. The Contractor shall provide advance notice to the appropriate parties of the mock-up inspection a minimum of seven working days prior to the meeting.

C. The Contractor shall make adjustments to the layout of the perimeter slot diffusers as requested by the Owner and Engineer prior to the installation of additional perimeter slot diffusers. The Contractor shall account for this requirement in the schedule of construction, so this procedure does not delay the construction progress. If additional perimeter slot diffusers are installed prior to the mock-up approval, the Contractor shall be responsible for the remedial work required to comply with the approved mock-up condition at no additional cost to the Owner.

D. If additional or follow-up field inspections of the mock-up modifications are required, the Contractor shall provide these modifications and additional follow-up field inspections as required without additional cost to the Owner.

END OF SECTION 23 37 00

SECTION 23 40 00 - HVAC AIR CLEANING DEVICES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide air cleaning devices as shown on the Drawings and as specified herein for the various air handling systems.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. All Sections pertaining to equipment containing cooling and/or heating coils.
B. Section 23 00 70 - Division 23 Schedule of Submittal Data
C. Section 23 05 93 - Testing, Adjusting, and Balancing for HVAC
D. Section 23 81 26 - Split-System Air-Conditioners

1.4 REFERENCE STANDARDS

A. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1. 52.2 Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

B. Air-Conditioning, Heating, and Refrigeration Institute (AHRI)
1. 850 Performance Rating of Commercial and Industrial Air Filter Equipment

C. ASTM International (formerly American Society for Testing and Materials)
1. E84 Standard Test Method for Surface Burning Characteristics of Building Materials

D. Code of Federal Regulations (CFR)
1. 40 CFR Part 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings

E. National Fire Protection Association (NFPA)
1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

F. Underwriters Laboratories (UL) Standard for Safety
1. 723 Test Method for Surface Burning Characteristics of Building Materials
2. 900 Air Filter Units

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely indicate all Specification requirements. Refer to Section 23 00 10 and Section 23 00 70 for additional requirements.

B. In addition to the above, the following submittal data shall be furnished according to the Conditions of the Contract, Division 01 Specifications, and Section 23 00 10 and shall include but not be limited to:

1. Dimensional data for all media and filter racks.
2. Maximum face and media velocities for each filter assembly at design air cfm.

3. Initial and final pressure drop data at design face velocities for each filter assembly.
4. Efficiency and MERV data for all media types.
5. ASHRAE Standard 52.2 Test Report.
6. All construction materials and details of construction.
7. Accessibility arrangement for each filter assembly: left, right, front, side, etc.
8. Installation, operating, and maintenance procedures.
9. Manufacturer's recommendations for a complete and fully operable installation.
10. Filter manufacturer's approval of air handling unit filter racks and frames.
11. Statement of coordination regarding requirements of air handling units.
12. Manufacturer's Certification Statement.
13. Materials Safety Data Sheet (MSDS) equivalent to OSHA Form 20 for any material deemed hazardous as determined by the manufacturer.
14. Filter Inventory Schedule: The size, type, and quantity of filters shall be listed for each air handling equipment by equipment tag designation. Refer to the Schedules and Drawings for equipment tag designations.

C. The Contractor shall provide product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content.

## PART 2 - PRODUCTS

### 2.1 GENERAL

A. MERV rating shall be determined based on the testing results in accordance with the entire procedure prescribed by the ASHRAE Standard 52.2.

B. Air Bypass: Construct filters, racks, and frames so as to prevent passage of unfiltered air. Provide gaskets and other permanent materials as required to prevent air bypass between filter frames and supporting devices.

C. Glass Fiber Filter Media Materials: Filter media shall not contain glass fibers, fiberglass or mineral wool materials.

D. Housing Material: If provided by the air handling unit manufacturer, refer to the air handling unit sections for requirements. If provided by the filter media supplier, filter racks and supporting devices shall be galvanized steel unless otherwise noted.

E. Identification: Provide each filter with an identification device mounted in a location where it will be visible after installation. Show model number and all other data necessary for ordering renewal filter media.

F. Supports: Provide supports such as frames, screens, grids, etc. as required for filters to adequately position the media for uniform air distribution and optimum performance without tearing, and to prevent excessive motion and deflection under all operating conditions. These criteria also apply to variable air volume applications.

G. VOC Content: Adhesive shall have a VOC content of 80 g/L or less when calculated in accordance with 40 CFR 59, Subpart D (EPA Method 24).

H. Air handling units with open return air inlets or ducted return air inlets shall have filter racks and replaceable media filters installed on the air handling unit as specified herein, as indicated on the Drawings, as indicated in the Schedules and as called for in the various Air Handling Units Sections. Filter media and racks shall be installed at the air handling unit manufacturer's factory or field installed at the project site at the Contractor's option.

I. Air handling units with ducted return shall have filter racks and replaceable media installed as specified herein, as indicated on the Drawings, as indicated in the Schedules and as called for in

the various Air Handling Units Sections. Filter media and racks shall be installed at the air handling unit manufacturer's factory or field installed at the project site at the Contractor's option.

J. Filter racks may be provided by the air handling unit manufacturer or the filtering media supplier at the Contractor's option provided the filter media and the filter racks are properly coordinated between manufacturers and coordinated with site conditions. Accessibility arrangement of filter racks and filter media shall be front or side access as appropriate for the particular job site conditions. Air handling unit manufacturer, filter manufacturer and Contractor shall coordinate job site specific requirements for this project. All filter racks; including but not limited to configurations, sizes, capacities, accessibility arrangements, etc.; shall be reviewed and approved in written form by the filter media supplier.

K. AHU and Job Site Coordination: In addition to the filter media and rack specifications, filter size, arrangement and access requirements shall be compatible and coordinated with the air handling units provided for this project. Refer to various air handling unit specification sections, Schedules and Drawings for additional requirements.

## 2.2 PARTICULATE AIR FILTERS

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Camfil
2. AAF Flanders
3. Parker/Airguard
4. Purolator
5. 3M

B. Type A - MERV 13 Filters: High capacity factory assembled filter media shall be Flanders RigidAir, Camfil Riga-Flo, Airguard Vari-Pak or another acceptable equal manufacturer. Filter media shall have a MERV of 13 as determined by the ASHRAE Standard 52.2. Filter media shall be 12" thick. Maximum face velocity shall be 500 fpm at the maximum scheduled unit cfm. Filters shall be UL 900 classified/listed. Filter media shall be installed in minimum 16 gauge galvanized steel holding frames, complete with gaskets and either spring fasteners or spring filter clips. The frame shall be coordinated with the air handling unit manufacturer and be selected to maximize filter area in the air handling unit plenum. The different filter sizes shall be limited to three (3) total for all units on the Project. The built up filter banks and frames shall be reinforced by welding the frame together or with galvanized angles or channels by the Contractor or unit manufacturer in accordance with the filter manufacturer's recommendations, so the maximum deflection at any point shall not exceed one quarter inch with the bank operating at a final resistance of 1.5" wg. Each filter bank shall be fully gasketed to prevent air bypass around the filter sections or between the filter modules. The Contractor shall caulk and seal between the plenum supplied by the Air Handling Unit Manufacturer and the holding frame as required. Filter bank shall be front accessible for units without ducted return. Provide side access slide-in type with gasketed hinged door for ducted air handling units as indicated on the drawings and as indicated in the Schedules. Provide galvanized sheet metal blank-offs to seal the filter holding frames in the plenum provided with the air handling units. Install caulk or gaskets to seal all joints of these blank-offs. Maximum initial air resistance through filter shall be 0.5" wg and recommended final air resistance shall be 1.5" wg.

C. Type B - MERV 8 Filters: High capacity factory assembled filter media shall be Camfil Farr 30/30, AAF Flanders PREpleat HC M8, or another acceptable equal manufacturer. Filter media shall have a MERV of 8 as determined by the ASHRAE Standard 52.2. Filter media shall be 2" thick and shall match the filter frame in the air handling unit or fan. Maximum face velocity shall be 500 fpm at the maximum scheduled unit cfm. Filters shall be UL 900 classified/listed. Filter frames and installation including the gaskets and either spring fasteners or spring filter clips shall comply with the requirements as indicated above for Filter Type “A”. Maximum initial air resistance through filter shall be 0.31” wg and recommended final air resistance shall be 1.0” wg.

D. Type E - MERV 13 Filters: High capacity factory assembled filter media shall be Camfil AP-Thirteen, AAF Flanders PREpleat M13, Airguard DP-Green13, or another acceptable manufacturer. Filter media shall have a MERV of 13 as determined by the ASHRAE Standard 52.2. Filter media shall be 4" thick. Maximum face velocity shall be 500 fpm at the maximum scheduled unit cfm. Filters shall be UL 900 classified/listed. Maximum initial air resistance through filter shall be 0.36” wg and recommended final air resistance shall be 1.0” wg.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General: All air filtering media, racks, and supporting equipment shall be installed in accordance with the latest industry standards, per the filter and air handling unit manufacturer’s recommendations, as indicated in the Schedules and as indicated on the Drawings.

B. Construction Phase Filters: Furnish and install one (1) complete set of the specified filters for each air handling and air conditioning unit to be used during the construction phase. Temporary filter media used during the construction shall have a MERV of 8 as determined by the ASHRAE Standard 52.2. Additional filters shall be provided as required during construction to protect the air handling equipment. Air moving equipment shall not be operated prior to the installation of the specified filter media.

C. Final Filters for Owners Use: In addition to the construction phase filters indicated above, furnish two (2) additional complete sets of the filters specified for each air handling and air conditioning unit in the Schedules on the Contract Documents. One (1) set of specified filters for each air handling unit shall be installed immediately prior to the air balance and testing required in the section titled "Testing, Adjusting and Balancing for HVAC". The second set of specified filters for each air handling unit shall be installed immediately prior to tenant occupancy as directed by the Owner’s representative.

3.2 FACTORY TESTING

A. All air filtering products shall be tested in accordance with the latest applicable industry standards and as specified herein.

3.3 FIELD TESTING

A. The Owner and Engineer reserve the right to have the Contractor field test and demonstrate performance of any or all filter media. Testing shall be in accordance with reference standards.

END OF SECTION 23 40 00

# SECTION 23 81 26 - SPLIT-SYSTEM AIR-CONDITIONERS

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements", "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

### 1.2 WORK INCLUDED

A. Furnish and install factory-built air cooled condensing units as specified herein and as indicated on the Drawings.

B. Furnish and install all necessary piping, fittings, valves, gauges, accessories, etc. for complete and operating refrigerant systems as specified herein and as indicated on the Drawings.

C. These Specifications set forth the minimum requirements for factory-built air cooled condensing units and refrigerant piping systems.

### 1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 23 00 70 - Division 23 Schedule of Submittal Data
B. Section 23 07 00 - HVAC Insulation
C. Drawing M-502 - Sequence of Operation

### 1.4 REFERENCE STANDARDS

A. Air-Conditioning, Heating, and Refrigeration Institute (AHRI)
   1. 210/240 Performance Rating of Unitary Air-conditioning & Air-source Heat Pump Equipment
   2. 340/360 Performance Rating of Commercial and Industrial Unitary Air-conditioning and Heat Pump Equipment
   3. 410 Forced Circulation Air-Cooling and Air-Heating Coils

B. Air Movement and Control Association (AMCA)
   1. 204 Balance Quality and Vibration Levels for Fans
   2. 210 Laboratory Methods of Testing Fans for Certified Aerodynamic Performance Rating
   3. 211 Certified Ratings Program - Product Rating Manual for Fan Air Performance
   4. 300 Reverberant Room Method for Sound Testing of Fans
   5. 301 Methods for Calculating Fan Sound Ratings from Laboratory Test Data

C. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
   1. 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings

D. ASTM International (formerly American Society for Testing and Materials)
   1. A653 Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
   2. C1071 Standard Specification for Fibrous Glass Duct Lining Insulation
   3. C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings
   4. G21 Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

E. Code of Federal Regulations (CFR)
   1. 40 CFR 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings

F. National Electrical Manufacturers Association (NEMA)
   1. 250 Enclosures for Electrical Equipment (1,000 Volts Maximum)

G. National Fire Protection Association (NFPA)
   1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

H. Underwriters Laboratories (UL) Standard for Safety
   1. 723 Test for Surface Burning Characteristics of Building Materials
   2. 1995 Heating and Cooling Equipment

I. Trane Reciprocating Refrigeration Manual

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein including but not limited to all certification requirements of Sections 23 00 10 and 23 00 70 plus capacity requirements indicated in the Schedules.

B. Air cooled condensing unit data shall be submitted concurrently with the direct expansion air conditioning unit submittal data.

C. Submittal data shall include capacity balance curves with direct expansion coil. Submit control wiring diagrams for the refrigeration machinery and its associated equipment such as time delay relays, refrigeration safety controls, etc. Interlock diagrams shall be prepared by the condensing unit manufacturer in arrangement as required to provide the sequence of system operation specified.

D. Submittal data shall include the refrigerant type, capacity/tonnage, and refrigerant charge for all equipment submitted on this project containing more than 0.5 lbs of refrigerant.

E. The Contractor shall provide product data for adhesives and sealants, including printed statement of Volatile Organic Compound (VOC) content.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Carrier
2. Trane
3. York/JCI
4. Daikin
5. Technical Systems (TSI)
6. EMI
7. United Cool Air

B. Air handling units shall be provided by the same manufacturer as the air cooled condensing unit it serves unless TSI or EMI equipment is utilized to serve Enviro-Tec air handling units or United Cool Air equipment is utilized to serve Daikin air handling units.

2.2 AIR COOLED CONDENSING UNITS

A. General: Air cooled condensing units shall be assembled and tested at the factory. They shall be shipped with refrigerant holding charges. The minimum cooling capacity of the unit shall be as indicated in the Schedules at a 100°F entering air temperature. Units shall be AHRI certified and UL listed. Air cooled condensing units shall be matched to the air handling unit that it serves with respect to capacity requirements.

B. Condenser: Condenser shall be sized to provide a minimum of 5°F subcooling when operating at design capacity. The saturated condensing temperature shall be limited to a maximum of 140°F at design conditions. Condenser coil casings shall house coils of non-ferrous construction with aluminum plate fins mechanically bonded to seamless copper or aluminum tubes.

C. Fans and Motors: Unit shall be furnished with direct-drive propeller type condenser fans arranged for vertical discharge. Fan motors shall have inherent thermal overload protection, be of the permanently lubricated type and be suitable for the intended outdoor service. Motors shall be resiliently mounted. Fans shall have safety guards.

D. Compressors: Compressors shall be of hermetic design with suction and discharge service valves. Each compressor shall be designed for operation with refrigerant R-410A. The unit shall be capable of operating satisfactorily with the electrical characteristics indicated on the electrical drawings. Compressors shall be mounted on factory installed internal spring vibration isolators. Provide hot gas bypass controls for partial load operation.

E. Controls: Safety devices shall consist of at least high and low pressure switch, crankcase heater, and compressor and condenser fan motor overload devices. Unit wiring shall incorporate a positive acting timer to prevent short cycling of the compressor if power is interrupted. The timer shall prevent the compressor from

restarting for a five-minute period. Units shall include a transformer for a maximum of 120 volt control circuit voltage.

F. Casing: Casings shall be fully weatherproofed for outdoor installation. Casings shall be of galvanized steel bonderized and finished with baked enamel. Openings shall be provided for power and refrigerant connections. Compressors and control box shall be located in a compartment other than that which contains the condenser coil.

G. Low Ambient Temperature Requirements: Factory installed low ambient head pressure control shall be provided for each unit. Low ambient control shall provide continuous operation of unit at full and/or partial load conditions in 0°F ambient. Low ambient control shall utilize solid state condenser fan speed control. The low ambient controls shall also use baffles and/or dampers as required. The Contractor shall charge the system with refrigerant in accordance with the manufacturer's requirements for the ambient conditions specified. Refer to Sequence of Operation on drawing M-502 for additional requirements.

H. Refrigerant Circuit: The initial refrigerant and oil charges shall be furnished by the Contractor in sufficient quantities to fully charge the system and leave it in satisfactory operating condition. All or any part of the charge that may be lost during the guarantee period shall be replaced by the Contractor without additional cost to the Owner.

I. Energy Efficiency: Energy efficiency rating (EER) and coefficient of performance (COP) shall be in accordance with all applicable National and Local Codes.

J. Special Considerations:

1. Fan Speed Options and Controls: Condensers serving evaporator units with variable speed or multi-speed fan motors shall be provided with multiple staged compressors or hot gas bypass and the necessary controls to accommodate the reduction in load at lower evaporator air flow rates.

2. Refrigerant Piping Systems: The refrigerant piping system provided shall accommodate the type of evaporator/condenser combination supplied by the manufacturer. For example, dual compressor condensers with independent refrigerant circuits shall require a dual system of refrigeration piping. The contractor shall be responsible for providing a refrigerant piping system in accordance with the evaporator/condenser manufacturer's requirements.

## 2.3 AIR HANDLING UNITS / FAN COIL UNITS

A. Type: Split system factory built air handling units with DX coils shall be either horizontal or vertical type with requirements, cooling-only or heat pump duty, and capacities as listed in the Schedules. Fan discharge condition and piping connections shall be as indicated on the drawings. Units shall be rated and certified in accordance with AHRI standards.

B. General: Unless otherwise noted, each unit shall be complete with casings, controls, centrifugal fans, insulation, drain pans, coils, filters, and direct drive fan motors. The units shall comply with the following minimum requirements. Units shall meet or

exceed the minimum requirements of the latest edition of the International Energy Conservation Code for operating efficiencies for this project's climate zone that eliminate the requirement that an economizer system be included in the building HVAC system.

C. All water connections and electrical knock-outs must be in the compressor compartment corner post so not to interfere with the serviceability of the unit.

D. Unit Casings:

1. Casing: Each unit shall have an airtight sectionalized single wall insulated casing constructed of galvanized steel with a minimum insulation thickness of 1/2". Casing of unit shall be internally lined with foil faced insulation on surfaces exposed to the air stream. Casing panels shall be finished with powder coated paint. Casing shall be factory assembled including coil, condensate drain pan, fan, motor, filters, and controls.

2. Drain Pan: The drain pan shall extend under the complete coil section. Drain pan shall be sloped and provide positive drainage to the drain connection so as to prevent any standing water within the pan. Drain pan shall be insulated to prevent condensation. A trapped condensate drain line shall be factory-installed and extend from the drain pan connection to outside the unit casing suitable for field piping connection.

3. Insulation: Unit casings shall be internally insulated with 1/2" thick (minimum) 1.5 pound density (minimum) to prevent condensation and be manufactured by Manville "Lina Coustic", "Exact 0 Mat", "Tuf Skin", or "Aeroflex" with foil faced cleanable surface on surfaces exposed to the air stream. All exposed parts such as angles, braces, etc., in contact with exterior surfaces shall be covered with insulation in such a manner to prevent condensation on the exterior casing. Insulation shall be installed in such a manner as not to be disturbed if panels are removed. All exposed unfinished edges of insulation shall be coated with adhesive mastic. Insulations shall be secured to the casing surfaces and framework with adhesive over entire surface.

4. Flame and Smoke Ratings: Composite flame and smoke hazard ratings of insulation shall comply with Section 23 07 00.

5. Refer to Condensation and Moisture Control requirements herein for additional requirements.

E. Centrifugal fan shall be forward curved. Fan motor shall be mounted on factory installed vibration isolators. Fan motor shall be permanently lubricated permeant split capacitor type with internal thermal overload protection and shall be direct drive with a minimum of 3 speeds.

F. Direct Expansion (DX) Cooling Coil and Refrigerant System:

1. Evaporator Coil: Coils shall be seamless copper tubing mechanically bonded to aluminum plate or spiral fins having a maximum of 12 fins per inch. Coils shall be minimum two (2) rows deep.

2. Testing: Coils shall be tested by air pressure under water. Coils shall be tested at one and one-half (1-1/2) times the pressure classification, but the test pressure shall in no case be less than 250 psig.

3. Coil Rating and Capacity: Cooling coil ratings shall be in accordance with AHRI certified data. Cooling coil capacity shall be in accordance with requirements listed in the schedule. Cooling coil face velocities shall be limited to the maximum face velocity indicated in the schedule.

4. Certification: Coil calculations and selection data shall be submitted with shop drawings. Prior to submission, all data shall be certified as specified in Section 23 00 10.

G. Filter Frames: Filter frame sections shall be provided by the fan coil unit manufacturer to accommodate two (2) inch media filters. Refer to the Schedules for HVAC and Section 23 40 00 for additional information.

H. Disconnect: Units shall be supplied with a factory mounted and wired non-fused disconnect switch.

I. Acoustics: Units shall have acoustic performance as indicated in the Schedules.

J. Condensation and Moisture Control: With the exception of the cooling coil surfaces and water collection surfaces of condensate drainage pan, fan coil unit construction shall prevent the formation of condensation on the interior and exterior surfaces of the fan coil unit as installed and operating on this project. If condensation forms on exterior or interior surfaces of a fan coil unit, the fan coil unit manufacturer will be responsible for corrective actions and expenses associated with modifying the units to eliminate condensation. Refer to specification Section 23 00 15 for design conditions.

## 2.4 CONTROLS

A. Controls shall be a factory installed solid state control system mounted on a microprocessor which is designed to protect against building electrical system noise contamination, EMI, and RFI interference.

B. The control system shall include anti-short cycle time delay on compressor operation, random start on power up mode, low and high voltage protection, and unit shutdown on high or low refrigerant pressures or low water temperature. The low-pressure switch shall not be monitored for the first 120 seconds of compressor operation to prevent nuisance safety trips. Controls shall include electronic condensate overflow protection, water coil and air coil low temperature sensing and the option to reset the unit at either the thermostat or the disconnect. Provide a 24 volt output to cycle the motorized water valve with the compressor contactor.

C. Provide factory mounted LED on the circuit board to indicate high or low pressure, high or low voltage, low water or air temperature cut-out, condensate overflow and control

voltage status. Unit shall automatically reset the unit 5 minutes after trip, if the fault has cleared. If a fault occurs 3 times sequentially without thermostat meeting temperature, then lockout requiring manual reset will occur. Provide the ability to defeat time delays for servicing.

D. Refer to Sequence of Operation on drawing M-502 for additional requirements.

PART 3 - EXECUTION

3.1 GENERAL

A. Installation shall be in accordance with the manufacturer's recommendations and, unless specifically noted to be otherwise by these recommendations, shall also be in accordance with the applicable sections of the Trane Reciprocating Refrigeration Manual. Pipe sizes shall be in strict accordance with the air conditioning system manufacturer's recommendations regardless of sizes shown on drawings.

B. Field verify the air cooled condensing units will fit within the space allotted. Verify air intake and discharge arrangements are clear and unobstructed. Verify access and operational clearances are in accordance with the manufacturer's recommendations. Verify vibration isolation mountings for fans and compressors are correctly adjusted. Repair any damage to unit.

3.2 CONTROL COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

3.3 MOUNTING AND ALIGNMENT

A. Mount unit’s level on equipment pads as indicated on the Drawings and as specified herein.

3.4 CONNECTING

A. Connect refrigerant piping to units as shown on the Drawings. All piping shall be supported from the building structure. Pipe shall not be supported by the unit.

B. Coordinate control requirements with Automatic Temperature Control Contractor.

C. Coordinate the electrical connections with Division 26.

3.5 CLEANING

A. All debris and dirt shall be removed from the unit.

3.6 START-UP AND CHECK-OUT

A. General: The air cooled condensing units and refrigerant piping systems shall be started-up and checked-out after all connections have been made.

B. Check:

1. Compressor operation and refrigerant flow
2. Control interlock connections and sequence of operation
3. Vibration and noise
4. Electrical connections
5. Rotation of fan(s)
6. Correct all functional deficiencies and place in operation

END OF SECTION 23 81 26

SECTION 23 82 01 - ELECTRIC HEATING COILS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the electric heating/reheat coils in ductwork as shown on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Drawing M-502 - Sequence of Operation
B. Section 23 00 70 - Division 23 Schedule of Submittal Data
C. Section 23 30 01 - Ductwork and Sheet Metal

1.4 REFERENCE STANDARD

A. National Electrical Manufacturers Association (NEMA)
   1. 250 Enclosures for Electrical Equipment (1,000 Volts Maximum)

B. National Fire Protection Association (NFPA)
   1. 70 National Electrical Code

C. Underwriters Laboratories (UL) Standard for Safety
   1. 1995 Heating and Cooling Equipment
   2. 1996 Electric Duct Heaters

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data and shop drawing requirements listed in Sections 23 00 10 and 23 00 70 plus capacity and special requirements in the Sequence of Operation on drawing M-502 and Schedules.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

B. If it complies with these Specifications, products manufactured by the following manufacturers will be acceptable:

   1. Indeeco
   2. Brasch
   3. Redd-i
   4. Tuttle
   5. Markel
   6. Warren

## 2.2 ELECTRIC HEATING COILS

A. General: Unless otherwise noted, electric heater coils shall consist of open coils of highest grade 80% nickel and 20% chromium resistance wire insulated with ceramic or lava insulators. All stages of all heater coils shall extend across the entire duct cross section regardless of the number of separate coils in the cross section. Duct mounted heater coils shall be constructed to slip into a duct through a rectangular opening in one side. The electric heater coil at its scheduled capacity and size shall run "black," with surface temperature below condition which would cause heater elements to glow. Electric heating coil manufacturer shall be responsible for coordinating heating coil size and other requirements to accommodate duct liner and insulation.

B. Outdoor Air Application: Heating coils used for pre-heating of outdoor air shall be finned tubular construction. Heating elements shall consist of a Grade A 80% nickel and 20% chromium, centered in a stainless steel tube which is filled with granular magnesium oxide. The entire assembly is compacted to maximize both the heat transfer and dielectric properties of the magnesium oxide. Stainless steel fins are helically wound onto the stainless steel tube. Particular attention is called to the control sequences and additional requirements that address the pre-heating of outdoor air. Coils used for outdoor air shall also be rated for variable flow conditions from the minimum operational condition to 120% of the maximum scheduled airflow conditions.

C. Capacity: Heating coils shall have the heating capacities as indicated in the Schedule.

D. Casing and Trim: Heater coils shall be supported in aluminized or galvanized steel brackets, supported in not less than 22 gauge aluminized or galvanized steel frames. Terminal bolts, nuts and washers shall be of corrosion resistant materials. Heater coils exposed to outdoor weather conditions shall be provided with watertight construction suitable for the intended service.

E. Controls: Heaters shall be controlled by the following control types:

1. SCR Modulating: Fully modulating solid state silicon controlled rectifier (SCR) controller with 0 to 100% proportional control. Modulation achieved by time base proportional firing which is varied in response to the load demands. Proportioning is achieved by varying the number of on cycles with regard to the off cycles.

F. Safety Protection:

1. Primary over temperature protection device shall be disc type or capillary type automatic reset thermal cutout in the control circuit.

2. Secondary over temperature protection device shall be heat limiters in power circuits.

3. Both primary and secondary over temperature devices shall be serviceable through terminal box without removing the heating element from the duct.

4. Provide static pressure airflow switch for each heater to be field installed and wired.

5. Overcurrent protection devices shall be in accordance with Division 26 Specification Section 26 28 00.

G. Terminal Box:

1. Terminal box shall have a dust tight cover and be thermally insulated from the air stream.

2. The location and orientation of the terminal box for duct mounted heating coils shall be as indicated on the Drawings. If the terminal box is not shown mounted in the vertical position on the side of the duct, it shall be mounted in the horizontal position on the bottom surface of the duct

3. Terminal boxes for heating coils located within inaccessible space shall be remotely mounted as show on the Drawings. This requirement applies but is not limited to heating coils located within fire rated enclosures

4. Terminal boxes for heating coils exposed to outdoor weather conditions shall be NEMA Type 4 watertight enclosure

5. All wiring of built in devices shall be brought to clearly marked terminal strips. A complete wiring diagram shall be permanently attached to terminal box cover.

H. Testing: Coils shall be given a 2,000 volt dielectric test by the manufacturer.

I. NEC and UL/ETL Requirements: Heater coils shall be constructed so that installation may be accomplished in accordance with the Local Authority and provisions of the National Electric Code for zero clearance. The complete unit shall be listed by Underwriters Laboratories or Edison Testing Laboratories and bear the appropriate UL or ETL label.

J. Downstream Temperature Profile: The arrangement, spacing, and configuration of individual coil elements and individual coil stages shall provide heating performance and downstream air temperature profile over the entire cross section of the duct to provide an air stream of which the temperature shall not vary more than 1.0°F over the duct cross section 24" downstream of the coil for each 20°F temperature difference between the temperatures of the entering and leaving air. The variation of temperature shall be proportionally less at smaller temperature differences.

K. Coordination: Electric heater coils shall be designed for operation with a room thermostat as specified. Refer to Sequence of Operation on drawing M-502 for additional requirements.

L. Electrical Coordination: Electric heater assemblies shall be designed for a single connection point of electrical service unless otherwise indicated on the electrical documents. All wiring, busing, and distribution of power downstream of the single point of electrical service shall be the responsibility of the heater manufacturer. This responsibility shall also include wiring between heating coils and controllers remotely mounted from the heating coils. The heater provided shall be coordinated with and be compatible with the electrical service indicated on the electrical documents. This electrical coordination shall take place prior to the purchase of the heater.

## PART 3 - EXECUTION

### 3.1 GENERAL

A. The installation shall be in accordance with the manufacturer’s recommendations. Particular attention is called to the manufacturer's restrictions concerning duct fittings, elbows, transitions, fan discharge conditions, etc. that will affect the air flow through the coil. Provide NEC code clearances and access for the terminal box. It shall be the responsibility of the Contractor to adjust the exact position of the duct heater based on actual field conditions to meet these requirements.

3.2 CONTROL COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

3.3 CONNECTIONS

A. Connect controls and thermostat.

3.4 CLEAN

A. Clean duct heater inside and outside and place in operation.

END OF SECTION 23 82 01

SECTION 23 82 33 - CONVECTORS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the electric baseboard heating convectors as indicated on the Drawings and as specified herein in cooperation with the Architectural Division.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Drawing M-502 - Sequence of Operation
B. All Division 26 Sections - Electrical System
C. Architectural Divisions

1.4 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA)
   1. 70 National Electrical Code
   2. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

B. Underwriters Laboratories (UL) Standard for Safety
   1. 1995 Heating and Cooling Equipment

1.5 SHOP DRAWINGS AND SUBMITTALS

A. The shop drawing submittal package shall clearly and concisely address all Specification requirements herein plus certification data listed in Sections 23 00 10 and 23 00 70.

PART 2 - PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

   1. Brasch
   2. Vulcan
   3. Markel
   4. Price
   5. Redd-i
   6. Stelpro

2.2 ELECTRIC BASEBOARD HEATING CONVECTORS (EBH) - PEDESTAL TYPE

A. Contractor shall furnish and install sections of UL or ETL listed and labeled electric baseboard heating convectors where indicated on the Drawings. Convector covers and bottom inlet and

top grilles shall be constructed of all extruded aluminum, anodized, or factory finished in accordance with the convector UL or ETL listing. The Architect shall select the convector custom color and finish from the available catalog selections. The Contractor shall provide the Architect with a custom color sample of the painted cover for approval prior to ordering product. The Contractor shall confirm color and finish selection with the Architect prior to ordering product. Convector covers shall be continuous and furnished with splices, ends, and shall be mounted as indicated on the Architectural Drawings.

B. Convectors shall be Vulcan Radiator Model SB, Markel Model DBF/DBT, or Engineer approved equal, and shall be of the rating and lengths as indicated on the Drawings. Convector units shall be a maximum be 6" high and 3.5" in depth front to back. Furnish units complete with thermostat, integral disconnect, and controls as specified hereinafter. The exact length and location of each unit shall be as indicated on the Architectural documents and field verified. Convectors shall be mounted a minimum of 2” above the finished floor.

C. Convectors shall be equipped with thermostats, disconnects, and limit controls. Wall mounted thermostat cover shall be similar in appearance to the typical space thermostat used on this Project. Thermostats mounted in the control section of each convector shall have an adjustable range of 0 to 100ºF with capillary bulb mounted at the bottom under the cover on a bracket. Access doors shall be provided in the cover for the thermostat and disconnect adjustment. Refer to section Sequence of Operation on drawing M-502 for additional information regarding controls and sequences of operation. Refer to the Drawings for additional information.

D. Continuous runs of heating convectors shall fill the space between columns as indicated on the Mechanical and Architectural Drawings. Provide concealed splice plates at each joint including blank sections and provide end caps at each end for a finished appearance. Provide a 1/4" mesh screen located beneath the outlet grille to deter insertion of foreign objects through grille.

E. Contractor shall rough-in conduits for power and control to access each convector section through the floor slab to the ceiling plenum below or down the columns from the ceiling above. Conduit connections shall be hidden from view.

F. Manufacturer shall provide all necessary accessories including but not limited to end caps, splice plates, inside and outside corner sections, disconnect switches, relays, blank sections, filler pieces, wall trim, mounting brackets, etc., for a complete installation.

G. Convectors shall be suitable for the intended service, location, environment, and mounting arrangement. UL or ETL listings and warranties shall apply to the convectors as installed on this project. This includes convectors installed within architectural enclosures. Convectors installed within architectural enclosures shall be provided with adequate clearances as indicated in the manufacturer’s product literature and as recommended by the manufacturer or manufacturer’s authorized representative. Due to the nature of custom architectural enclosures and variations in dimensions and clearances between convector equipment manufacturers, manufacturers providing convectors located within architectural enclosures shall indicate compliance with these enclosure requirements with their bids and shop drawing submittal data.

## PART 3 - EXECUTION

### 3.1 COMPLETE PROJECT REQUIREMENTS

A. Review Architectural and MEP Documents to ascertain complete Project requirements. Contractor shall verify dimensional data, finish requirements, mounting arrangements, controls, power connections, architectural enclosure requirements, service and access requirements, and suitability for the intended service prior to ordering product.

3.2 MOUNTING AND ALIGNMENT

A. Mount unit to fit the Project dimensions.

B. Mount unit dead level in both directions.

C. Align all enclosures.

3.3 CLEANING

A. All debris and dirt shall be removed from the electric baseboard heating units.

3.4 START UP AND CHECK OUT

A. Correct all functional deficiencies and place in operation.

END OF SECTION 23 82 33

SECTION 23 82 39 - UNIT HEATERS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 23 General Requirements" in Section 23 00 10.

1.2 WORK INCLUDED

A. Provide the electric unit heaters as shown on the Drawings and as specified herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Drawing M-502 - Sequence of Operation
B. Section 23 40 00 - HVAC Air Cleaning Devices
C. Section 26 00 00 - Division 26 Electrical Systems

1.4 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA)
1. 90A Standard for the Installation of Air-Conditioning and Ventilating Systems

B. Underwriters Laboratories (UL) Standard for Safety
1. 1995 Heating and Cooling Equipment

1.5 SHOP DRAWINGS AND SUBMITTALS

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data listed in Section 23 00 10 and 23 00 70 plus capacity and accessory information indicated in the Schedules.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. If it complies with these Specifications, one of the following manufacturers will be acceptable:

1. Q-Mark
2. Chromalox
3. Berko
4. Trane
5. Markel
6. Redd-i
7. Reznor

2.2 CABINET TYPE UNIT HEATER

A. Type: The unit heater shall be of the blow thru cabinet type with electric resistance heating coil. This type of unit heater is designated "CUH" on the Drawings and in the Schedules.

B. Capacity: The capacity of the unit heater shall be as indicated in the Schedule.

C. Arrangement: The arrangement and cabinet type shall be as indicated on the Drawings and in the Schedules.

D. Cabinet: The cabinet shall be constructed of 16 gauge furniture grade steel. All exposed corners and edges shall be rounded. The front panel shall be fastened by quick acting allen head cam locks for easy access. Metal surfaces shall be cleaned, phosphatized, prime coated, and all exposed surfaces shall be finished in a baked enamel, color to be selected by the Architect.

E. Fans: Fans shall be all steel, multiblade, forward curved, centrifugal type fan which has been statically and dynamically balanced to eliminate vibration. Fans shall be assembled to a fan board and mounted below the coil in the unit casing. Fan shafts shall be mounted on phosphor bronze bearings with graphite inserts. The bearing shall have large oil reservoirs allowing 2,000 hours of operation on each oiling. The fans shall be direct drive.

F. Electric Heating Coil: The electric heating coil shall be constructed of high quality, nickel chrome electric resistance coils embedded in a refractory material and enclosed within a steel sheath. The sheath shall be provided with specially wound extended fins. The coils shall be suitable for current characteristics shown on the Drawings and in the Schedules. All wiring in the heating element circuit shall be of the heat resistant type enclosed within metal boxes and conduits.

G. Filters: Provide one each throw away filter located in a frame in the return air stream. The filters shall be easily removable by removing the front access panel. Refer to Section 23 40 00 for additional requirements.

H. Control: The unit shall have an integral on off switch, 24 volt thermostat and 24 volt transformer. The unit shall have an automatically resetting thermal switch wired in series with each element and located in the discharge air stream to protect against overheating in the event the air stream becomes blocked. The unit shall have a thermally activated fan switch to keep the fan operating until all residual heat is dissipated.

I. UL/ETL Listing: The unit heater shall be UL or ETL listed and have a UL or ETL label attached to the unit.

## PART 3 - EXECUTION

### 3.1 INSTALLATION

A. Install as shown on the Drawings. Protect the entire unit with a cover during construction. Manufacturer's recommendations should be observed before installation.

### 3.2 CLEANING

A. Before putting in operation, the entire unit shall be cleaned, including coils, and any damage repaired with a satisfactory finish. The entire unit shall be vacuum cleaned after general construction is completed.

### 3.3 START UP

A. Start-up and check operation of thermostat and unit heater.

END OF SECTION 23 82 39

SECTION 25 00 00 - INTEGRATED AUTOMATION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 25 General Requirements" in Section 25 00 10.

B. It is the intent of these Specifications for the Contractor to provide an integrated automation system complete, fully operational, fully adjusted, and ready for use.

C. All contract requirements indicated in the Division 21, 22, 23, 25, 26, and 28 documents, including drawings and specifications, shall be considered supplemental to and in addition to the contract requirements indicated in the documents of other trades and the Division 01 requirements.

1.2 REQUIRED WORK SPECIFIED ELSEWHERE

A. Section 25 00 10 - Division 25 General Requirements
B. Section 25 00 70 - Division 25 Schedule of Submittal Data
C. Section 25 00 90 - Automatic Temperature Control System

1.3 ALTERNATE AND UNIT PRICING

A. Alternate Pricing, Unit Pricing, and Budgetary Cost Estimates: **Bid alternates and unit prices shall not be required with the Contractor's base bid.** These requirements shall apply to only those bidders contacted after the Owner's and Architect's review of the bids. If contacted by the Owner or Architect after the bid date, the Contractors shall provide alternate prices, unit prices, and budgetary cost estimates as indicated in the Architectural Documents, as specified herein, and as indicated on the Drawings.

B. If contacted by the Owner or Architect after the bid date, the contractor(s) shall provide complete alternate pricing information for the owner's evaluation of all alternate priced items specified herein. Alternates may be additive or deductive. An additive alternate, whether shown on the drawings or specified herein, shall be defined as an alternate that is not currently part of the base building scope of work with a price that would be required to add the alternate to the base building scope of work. A deductive alternate shall be defined as an alternate that is currently part of the base building scope of work with a price that would be required to delete the alternate from the base building scope of work. The owner reserves the right to accept or reject any alternate priced item.

C. Refer to the architectural specification section for Architect's and Owner's instructions to bidders for additional alternate pricing requirements.

D. Pricing information provided for unit costs, separate line items, alternates, and value engineering items shall be all inclusive pricing that accounts for the impact and ripple effects on adjacent or related systems affected by the alternate product, material, or system. Acceptance of an alternate product, material or system shall not result in additional cost to the project beyond the price indicated for the alternate product, material or system.

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. General: All equipment shall be of the capacity and type as specified herein, as shown in the Schedules and as indicated on the Drawings. Equipment shall be provided by a specified manufacturer and in accordance with the space limitations of the Project.

B. Single Source: To maximize ease of maintenance and part replacement, equipment of a similar nature shall be provided by a single manufacturer.

C. Approved Equal: Equipment and materials selected by the Contractor within the context of "equal as approved by the Engineer", "approved equal", "equivalent as determined by the Engineer" and similar terminology shall be submitted to the Engineer for review, approval and inclusion into the Contract Documents prior to the finalization of the contract between the Owner and the Contractor, and prior to the shop drawing submittal phase of the Project. All equipment and materials submitted to the Engineer under the terms of "approved equal" during the shop drawing phase of the Project without prior review and approval shall be returned to the Contractor without review under the status of "No Action".

2.2 MATERIAL

A. All material required for a complete and proper installation shall be as specified and as selected by the Contractor subject to the approval of the Architect.

2.3 CONTRACTOR APPROVALS

A. The contractor shall submit in writing and obtain written approval from the Owner, Architect, and/or Engineer for all equipment substitutions, installation deviations from that shown on the Contract Documents, and all other miscellaneous approvals required from the Owner, Architect, and/or Engineer as referenced throughout these specifications.

B. A request for a substitution or deviation from the Contract Document constitutes that the Contractor:

1. Has investigated the proposed product and determined that it is equal to or superior in all respects to that specified.

2. Will provide the same warranties for the substitution as for the product specified.

3. Will coordinate the installation of the substitution into the work, and make such other changes as may be required to make the work complete in all respects.

4. Waives all claims for additional costs, under his responsibility, which may subsequently become apparent.

5. Will absorb all costs incurred by the substitution when affecting other trades including but not limited to electrical, structural, architectural, etc.

6. Will absorb any cost incurred by the Owner's Representative in review of the substituted product if the acceptance of the substituted item creates the need for system modification and/or redesign, or if the substituting Contractor exhibits negligence in his substituting procedure thus submitting inferior, misapplied or mis-sized equipment.

PART 3 - EXECUTION

3.1 CONDITIONS

A. Inspection: Prior to proceeding with the work of this Division, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation may properly commence. Verify that the work of this Division may be completed in strict accordance with all pertinent codes and regulations, the reviewed shop drawings, and the manufacturers' recommendations.

B. Discrepancies: In the event of discrepancy, immediately notify the Owner's Representative. Do not proceed in areas of discrepancy until all such discrepancies have been resolved. If there is a discrepancy between the Drawings and the Specifications, the Specifications shall typically govern. However, any discrepancy of this type shall be immediately brought to the attention of the Owner's Representative for formal interpretation prior to proceeding with the work.

C. Interpretation of Documents: Any and all contractual requirements may be indicated solely on the Drawings, solely in the Specifications, in both the Specifications and on the Drawings, in reference standards indicated in the Specifications and/or in the Owner's and Contractor's Contract. If Contract requirements are indicated in both the Specifications and the Drawings, the Contractor shall comply with both requirements unless the requirements are mutually exclusive of each other. If Contract requirements are indicated in both a reference standard and the Specifications, the more stringent requirement shall apply. Any and all contractual requirements shall be interpreted within the overall context of the complete scope of work. All materials, equipment, systems and installation methods shall be suitable for the intended service, coordinated with other trades and be complete, fully operational, adjusted, tested and ready for use by the Owner.

3.2 INSTALLATION OF EQUIPMENT

A. Locations: Install all equipment in the locations shown on the reviewed shop drawings, except where specifically otherwise approved on the job by the Architect and/or Owner's Representative.

B. Interferences: Avoid interference with structure and with work of other trades, preserving adequate headroom and clearing all doors and passageways to the approval of the Architect and/or Owner's Representative.

C. Inspection: Check each piece of equipment in the system for defects. Verify that all parts are properly furnished and installed, function properly, and that all adjustments have been made.

3.3 CONNECTIONS TO EQUIPMENT

A. The Contractor shall make final connections to all equipment for a complete and operational system.

3.4 CLOSING-IN OF UNINSPECTED WORK

A. General: Do not allow or cause any of the work of this Division to be covered up or enclosed until it has been inspected, tested, and approved by the Architect, Owner's Representative, and by all other authorities having jurisdiction.

B. Uncovering: Should any of the work of this Division be covered up or enclosed before it has been completely inspected, tested and approved, the Contractor shall provide all services, labor, materials and equipment necessary to uncover such work without added expense to the Owner. After the work has been completely inspected, tested, and approved, the Contractor

shall provide all services, labor, materials and equipment to make all repairs necessary to restore the work to its original and proper condition at no additional cost to the Owner.

3.5 COOPERATION WITH OTHER TRADES

A. Coordinate and cooperate with other trades in order that all systems in the scope of the Contract may be installed in the best arrangement. Coordinate and cooperate as required with all other trades which share space in common areas and maximize the access to each system and their respective components.

3.6 CLEANING

A. It is the intent of these Specifications that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment. Level of cleanliness shall be defined as "broom" clean unless noted otherwise. All exterior surfaces of Division 25 equipment shall be wiped down and cleaned of all dust and dirt. All interior surfaces of equipment shall be wiped down and vacuum cleaned so as to be delivered to the Owner in factory new condition. Surfaces to be painted shall be cleaned and prepared in accordance with architectural division of the contract and as noted in other sections herein.

3.7 COMPLETENESS

A. It is the intent of these Specifications to provide complete systems. Completeness shall mean that all materials, equipment, and systems as installed and operating on this project have been installed properly with the best practices of the trade; are suitable for the intended purpose, location, and environment; properly fit within the physical space limitations for the project; are in conformance with applicable codes and reference standards; have been started-up, tested, adjusted, and commissioned for the intended use; have maintained applicable UL listings; are in compliance with manufacturer's recommendations and warranty requirements; ready for the Owner’s use, and in the opinion of the Architect, performing as designed.

3.8 ADJUSTMENT OF CONTROLS

A. The Contractor shall provide the personnel and equipment to completely adjust the controls to the satisfaction of the Architect. At the completion of the project, the Architect will arrange a meeting at the job site to allow the Contractor to demonstrate the proper operation of the electrical controls.

3.9 NOISE

A. It is the intent of these Specifications to provide a system free from objectionable audible noise and vibration. Any equipment that is generating objectionable noise or vibration, in the opinion of the Architect, shall be corrected and dampened as required to eliminate the objectionable level.

3.10 LEED FUNDAMENTAL BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 23 08 02 and the alternate pricing requirements indicated in Section 23 00 00 for additional fundamental building commissioning general requirements for LEED.

END OF SECTION 25 00 00

SECTION 25 00 10 - DIVISION 25 GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. General: The work under this Section shall conform to the requirements of the "General Conditions" and "Supplementary Conditions" of these Specifications.

B. Work Included: The work included in this Section consists of the general requirements for the work more specifically described in Division 21, 22, 23, 25, 26 and 28 where referenced.

C. Definitions: The Contractor shall provide all supervision, labor, material, equipment, machinery, and any and all other items necessary to complete the mechanical and electrical systems. All items of equipment are specified in the singular; however, the Contractor shall provide the number of items of equipment as indicated on the Drawings, and as required for complete systems.

D. Intent

1. It is the intention of these Specifications and Drawings to call for finished work that has been completed, tested, adjusted, commissioned, fully operational, approved by the local code authorities, and ready for the Owner’s use. Wherever the word "provide" is used, it shall mean "furnish and install complete and ready to use."

2. It is the intention of these specifications and drawings that wherever the word "weather" is used throughout the various Division 21, 22, 23, 25, 26 and 28 specification sections it refers to "locations exposed to weather". "Locations exposed to weather" shall mean areas exposed to outside environmental temperature, humidity, or moisture conditions. These locations include but are not limited to areas on the exterior of the building, on the roof, within architectural exhaust and intake plenums, within loading docks, within above grade parking garages, within below grade parking structures, and other areas as indicated on the drawings or in the specifications.

3. All labor, materials, apparatus, and appliances essential to the complete functioning of the systems described and indicated herein, or which may be reasonably implied as essential whether mentioned in the Drawings and Specifications or not, shall be furnished and installed by the Contractor.

4. In addition to the requirements indicated throughout the Drawings and Specifications, as completely installed and operating on this project; equipment and systems provided for this project shall meet the following minimum requirements without additional cost to the Project:

a. Be suitable for the intended service, location, and environment.

b. Properly fit within the physical space limitations for the project.

c. Meet the minimum requirements of applicable codes and reference standards as determined and approved by the local code authorities.

d. Maintain applicable UL listings.

e. Be coordinated between Divisions 21, 22, 23, 25, 26 and 28 in a timely manner.

f. Be coordinated with the work of other trades in a timely manner;

g. Not be purchased or installed prior to the resolution of any conflicts or coordination that could result in additional cost to the Owner;

h. Be in accordance with the manufacturer's recommendations and warranty requirements.

i. Be ready for use by the Owner.

5. In cases of doubt as to the Work intended, or in the event of need for explanation thereof, the Contractor shall request supplementary instructions from the Architect.

E. Codes, Rules, Regulations, Permits and Fees

1. Refer to the architectural specification Divisions for additional requirements. The architectural specification Divisions shall govern these requirements. The following information shall be in addition to the architectural specification Divisions.

2. The Contractor shall give all necessary notices, obtain all permits and pay all government sales taxes, fees, and other costs, including utility connections or extensions, in connection with the Contractor's work; file all necessary plans, prepare all documents and obtain all necessary approvals of all governmental departments having jurisdiction; obtain all required certificates of inspection of the Contractor's work and deliver same to the Architect before request for acceptance and final payment for work.

3. The Contractor shall comply with the most recently revised versions of all applicable laws, rules, regulations, and ordinances of Federal, State, and Local Authorities. Modifications required by the above said Authorities shall be made without additional charge to the Owner.

4. The Contractor shall comply with applicable building codes as adopted and modified by the Local Code Authorities. Modifications required by the above said building codes shall be made without additional charge to the Owner.

5. All equipment shall comply with all applicable requirements of laws, codes, ordinances, legislation, etc., of all Federal, State, and Local Authorities, whether indicated on the Contract Documents or not.

6. Where Contract Drawings and Specifications requirements are in excess of Code requirements and are permitted under the Code, the Contract Drawings and Specifications shall govern.

7. All rules and regulations of the State Fire Insurance Regulatory Body, Underwriters Laboratories, Americans with Disabilities Act (ADA), and National Electrical Code, shall be complied with whether indicated in the Contract Drawings and Specifications or not.

8. All materials, equipment, and systems provided on this project shall be in compliance with the applicable energy codes and the energy efficiency ASHRAE Standard 90.1 as adopted by the applicable building codes.

9. Project shall meet or exceed the requirements of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality Guideline for Occupied Buildings Under Construction. In addition, all stored on-site or installed materials shall be

protected at all times from moisture damage. Provide photographs to the owner, architect, or engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

10. Where alterations to or deviations from the Contract Drawings and Specifications are required by the above Authorities report the requirements to the Architect and secure Architect's approval before starting the alterations.

11. Third Party Certification: All packaged equipment shall be independently Third Party labeled as a system for it's intended use by a Nationally Recognized Testing Laboratory (NRTL), such as UL or ETL, in accordance with OSHA Federal Regulations 29 CFR 1910.303 & 1910.399, as well as NFPA 70, National Electric Code (NEC), Article 90-7.

1.2 QUALITY ASSURANCE

A. Surveys and Measurements

1. The Contractor shall base all measurements, both horizontal and vertical, from established benchmarks. All work shall agree with these established lines and levels. Verify all measurements at site and check the correctness of same as related to the work.

2. Should the Contractor discover any discrepancy between actual measurements and those indicated, which prevents following good practice or the intent of the Drawings and Specifications, the Contractor shall notify the Architect, and shall not proceed with the work until instruction has been received from the Architect.

B. Drawings

1. Drawings are diagrammatic and indicate the general arrangement of systems and work included in the contract. Drawings are not to be scaled. The Architectural Drawings and Details shall be examined for exact location of fixtures and equipment. Where they are not definitely located, this information shall be obtained from the Owner's Representative.

2. The Contractor shall follow Drawings in laying out work and check Drawings of other trades to verify spaces in which work will be installed. Maintain maximum headroom and space conditions at all points. Where headroom or space conditions appear inadequate, the Owner's Representative shall be notified before proceeding with installation.

3. If directed by the Owner's Representative, the Contractor shall, without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of the work, if such directions are made prior to the performance of the affected work.

C. Cooperation with Other Trades

1. The Subcontractors shall give full cooperation to other trades and shall furnish in writing to the General Contractor any information necessary to permit the work of all trades to be installed satisfactorily and with the least possible interference or delay.

2. Where Division 21, 22, 23, 25, 26 and 28 work will be installed in close proximity to, or will interfere with, work of other trades, composite shop drawings will be prepared to accomplish coordination among trades in working out space conditions. All necessary adjustments to work out space conditions shall be reflected on the composite shop

drawings. The Contractor shall prepare composite shop drawings and sections at suitable scale not less than 1/4" = 1'-0", clearly showing how work of all affected trades is to be installed. If work is installed before coordinating with other trades, or so as to cause any interference with work of other trades, then the necessary changes in the work shall be made to correct the condition without additional expense to the Owner.

3. The Contractor shall obtain and transmit between affected trades, all necessary templates, patterns, setting plans, and shop details for the proper installation of work and for the purpose of coordinating adjacent work.

D. Protection

1. The Contractor shall protect all work and material from damage by the Contractor's work or workmen, subcontractors' work or workmen, and shall be liable for all damage thus caused.

2. The Contractor shall be responsible for work and equipment until "Final Review"; the Contractor shall protect work against theft, injury or damage; and shall carefully store material and equipment received on site which are not immediately installed. The Contractor shall close open ends of work with temporary covers or plugs during storage and construction to prevent entry of obstructing material.

3. All stored on-site or installed materials shall be protected at all times from moisture damage.

E. Material and Workmanship

1. Work shall be executed in strict accordance with the best practice of the trades in a thorough, substantial, workmanlike manner by competent workmen. The Contractor shall furnish the services of a full-time, experienced superintendent, who shall be constantly in charge of the installation of the work, together with all skilled workmen, fitters, metal workers, welders, helpers, and laborers required to unload, transfer, erect, connect, adjust, start, operate, and test each system.

F. Manufacturers' Recommendations: In addition to the requirements indicated in the drawings and specifications; manufactured products, articles, materials, and equipment shall be applied, installed, connected, erected, used, cleaned, conditioned, inspected, started-up, tested, operated, and commissioned in accordance with the manufacturer's current printed recommendations. Keep copies of such printed recommendations at job site and make them available as required. When conditions on this project are not covered by the manufacturer's printed recommendations, at the discretion of and as requested by the owner's representative, the manufacturer's authorized representative shall review the conditions and provide written supplemental recommendations to address the special situation. If the manufacturer’s recommendations are in conflict with the requirements of the drawings and specifications, the Contractor shall advise the Owner’s representative of the conflict prior to purchase and installation.

G. Space Limitations

1. MEP systems and equipment shall be provided and installed to accommodate the structural systems, architectural systems, MEP systems, other trades, and the physical space limitations of the project.

2. Equipment shall be provided which will fit into the physical spaces provided and

indicated, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearance in accordance with the Code requirements, the requirements of the local inspection department, and the equipment manufacturers' recommendations.

3. In the preparation of Drawings, a reasonable effort to accommodate equipment manufacturers' space requirements has been made. However, since space requirements, equipment arrangement, access and service clearances, conduit, piping, ductwork and other connections, manufacturers' recommendations, and other variables vary according to each manufacturer; due to the multiple manufacturers listed for each product and the possible number of different equipment combinations; the responsibility for access, proper fit, coordination, and accommodating the space limitations of the project rests with the Contractor. Refer to shop drawing submittal requirements herein for additional requirements.

4. MEP systems and equipment shall be installed to maximize headroom clearance below the bottom of material. In areas with ceilings or future ceilings provided by tenants; MEP systems and equipment shall be installed to maximize vertical clearances between the bottom of material and top of specified ceiling construction. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, lighting, and other MEP systems shall be prepared to indicate compliance with the Space Limitations and Maximum Headroom Clearance requirements. Conflicts with vertical and horizontal clearances, parking spaces, drive aisles, ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data, clouded, and clearly marked so as to bring the conflict to the attention of the Architect in a timely manner prior to installation. For the purpose of definition, the term MEP refers to all systems, material, equipment, services, means and methods described the Division 21, 22, 23, 25, 26 and 28 documents including the Division 21, 22, 23, 25, 26 and 28 specifications and drawings. Refer to the specifications for additional requirements.

5. Refer to the drawings for additional requirements.

6. Not Used.

7. Physical dimensions and arrangements of equipment to be installed shall be subject to the Architects' review.

H. Coordination with Installation Conditions: The requirements of the Division 21, 22, 23, 25, 26 and 28 systems shall be coordinated with the installation conditions. The Contractor shall visit the site, survey the installation conditions and verify exact measurements. Means and methods of building access, rigging, and modifications to installation conditions to accommodate the installation of new Division 21, 22, 23, 26 and 28 work shall be the responsibility of the Contractor. Disassembly, reassembly, and certification testing of equipment as required to accommodate the installation conditions shall be included in the base bid. Prior to the ordering, purchasing, and fabricating of Division 21, 22, 23, 26 and 28 systems, coordination with installation conditions shall be completed by the Contractor.

I. Coordination Between Divisions 21, 22, 23, 25, 26 and 28: The Division 25 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 26 and 28 Contractors to provide coordination between the Division 21, 22, 23, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 25, it shall be the responsibility of the Division 25 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative

prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

J. Commissioning: The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience, and expertise at the general contractor and sub-contractor levels to properly install, check, verify, supervise, start-up, trouble-shoot, de-bug, pre-test, commission, and make the necessary corrections to all MEP systems in accordance with the Contract Documents. The Contractor shall be responsible for the operational demonstration of the completed MEP systems to the Owner's Representative, Architect and Engineer. The Contractor shall complete the MEP systems in accordance with the contract documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems prior to requesting that the Owner's Representative, Architect, and Engineer attend the operational demonstrations of the MEP systems. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner. Refer to the various sections of the specifications for additional commissioning requirements.

K. Additional Information: Information provided by the Engineer to the Contractors during construction must be reviewed and approved by the Architect and Owner's Representative. The Contractor shall not proceed without prior review and authorization from the Architect and Owner's Representative. The Contractor shall not proceed with work that will increase the cost of the project or impact the project schedule without prior review and authorization from the Architect and Owner's Representative. If information provided by the Engineer to the Contractor will increase the cost of the project or impact the project schedule, the Contractor shall advise the Architect and Owner's Representative in a reasonably timely manner so as to minimize the cost and impact to the project. Refer to the General Conditions of the Contract and the Architectural documents for additional information.

1.3 SHOP DRAWINGS, PRODUCT DATA AND SAMPLES

A. After the Contract is awarded but prior to proceeding with the Work requiring final reviewed shop drawings, product data and samples, the Contractor shall provide the Architect and Engineer with a submittal log schedule indicating when each submittal is to be sent to the Architect and Engineer for review in order for the Architect and Engineer to schedule personnel and plan accordingly. Unless otherwise indicated, a minimum period of two (2) weeks, exclusive of transmittal time, will be required in the Engineer's office each time a shop drawing, product data and/or samples are submitted or resubmitted for review. This time period will commence once the shop drawing, product data and/or samples are received in the Engineer's office contingent upon the shop drawing, product data and/or samples being received in the Engineer's office before 3:00 PM. If the shop drawing, product data and/or samples are received in the Engineer's office after 3:00 PM, the time period will not start until the next business day. This time period shall be considered by the Contractor when scheduling the Work. The submittal log schedule should be updated and reissued by the Contractor on a regular basis to avoid submittal review delays. If a submittal arrives before or after the scheduled time indicated within the Contractor's submittal log schedule, or if a number of submittals arrive at the same time without scheduled notification, the Architect and Engineer may not be able to review all of the submittals within the minimum two (2) week time period. The Contractor shall obtain complete shop drawings, product data and samples from the manufacturers, suppliers, vendors, subcontractors, and sub-subcontractors for all materials, systems and equipment specified herein; and submit data and details of such materials, systems and equipment for review by the Architect and Engineer. Prior to submission of the shop drawings, product data and samples to the Architect and Engineer, the Contractor shall review and certify that the shop drawings, product data and samples are in compliance with the Contract Documents. The Contractor shall check all materials and equipment after their arrival

on the job site and verify their compliance with the Contract Documents. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience and expertise at the general contractor and sub-contractor levels to properly check, verify, and correct if necessary shop drawing submittal data to ensure compliance with the Contract Documents prior to delivery to the Architect and Engineer. The Division 25 shop drawing submittal requirements, including but not limited to the Division 25 certification requirements, shall be in addition to the requirements indicated in the Architectural Divisions of the Contract Documents.

B. The Architect's approval and Engineer's review of shop drawings, product data, and samples shall not relieve the Contractor of the responsibility for dimensions or errors that may be contained therein, or for deviations from requirements in the Contract Documents. It shall be clearly understood that the Engineer's noting some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents shall govern the Work and are neither waived nor superseded in any way by the review of shop drawings, product data and samples.

C. Before submitting shop drawings, product data and samples the Contractor shall certify that data and details set forth on each shop drawing, product data, or samples for each item of equipment and materials complies with the Contract Documents for this Project. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the Contractor, or by a person duly authorized to sign binding agreements for the Contractor. Unless certified, shop drawings, product data and samples will not be reviewed, and will be returned unchecked to the Contractor. Certifications shall be in the form of rubber stamp impressions or a digitally certified stamp which state:

> I hereby certify that this shop drawing, product data, and/or sample has been checked prior to submittal and that it has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents and physical space limitations for the Project.
>
> (Name of the Automatic Temperature Control Subcontractor)
>
> Signed:_____
>
> Position:_____
>
> Date:_____

D. The Contractor shall submit shop drawings, product data and/or samples, equipment and other items listed in Section 25 00 70 and for any other equipment for which such submittals may be required in various Sections of the Specifications. Prior to submitting shop drawings, product data and samples for all items of equipment listed in Section 25 00 70 and shown with an asterisk (*), each submittal package including subsequent submittal packages shall be certified by the manufacturer as to performance and compliance with Contract Documents. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the manufacturer, or be a person duly authorized to sign for the Owner, Partner, or corporation. Unless certified, shop drawings, product data and/or samples will not be reviewed and will be returned unchecked to the Contractor. The certification shall be as follows:

I hereby certify that the equipment shown on this shop drawing, product data and/or sample, has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents for this Project. I further certify that all data shown herein as to performance dimensions, construction, materials and other pertinent items is true and correct.

(Name of the Manufacturer)

Signed:_____

Position:_____

Date:_____

E. Inadequate or incomplete shop drawings, product data and/or samples will not be reviewed by the Architect or the Engineer and shall be returned to the Contractor for resubmittal.

F. The Contractor shall observe the following rules when submitting shop drawings, product data and samples:

1. Identification: Each shop drawing shall indicate in the lower right hand corner, and each product data brochure shall indicate on the front cover the following: title of the sheet or brochure; name and location of the Project, names of the Architect and Engineer, Contractor, Subcontractor, manufacturer, supplier, vendor; the date of submittal; and the date of each correction and revision. So far as is practical, each shop drawing, product data and/or sample shall bear a cross-reference note to the page number or numbers of sheets of the Drawings and/or Specifications showing the Work. Unless the above information is included, the submittal will be returned for resubmittal.

2. Quality Assurance: Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark "C" or "NC" by each specification paragraph. A handwritten marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

3. Shop drawings shall be done in an easily legible scale and shall contain sufficient plans, elevations, sections and schematics to clearly describe the apparatus. Drawings shall be prepared by an engineer-draftsman skilled in this type of work. Unless otherwise indicated, all sheet metal, piping and similar shop drawings shall be drawn to at least 1/4" = 1'-0" scale. Typical floor sheet metal ductwork shop drawings may be drawn at 1/8" = 1'-0" scale. The Contractor shall submit shop drawings as described below. Shop drawings which do not comply with these requirements will be returned for resubmittal. Refer to "Cooperation with Other Trades", this section, for additional composite shop

drawing requirements. Unless otherwise indicated in the Contract Documents, the submittal shall consist of three prints of the shop drawings. The Architect and Engineer will each retain one print and the remaining copies will be returned to the Contractor.

4. Product data to be submitted shall be published by the manufacturers and shall contain complete and detailed engineering and dimensional information. The Contractor shall submit product data as described below. Product data which does not comply with these requirements will be returned for resubmittal. Unless otherwise indicated in the Contract Documents, the Contractor shall submit eight (8) copies of each product data brochure. The Architect and Engineer will each retain one copy of the submittal and the remaining copies will be returned to the Contractor. Product data submitted shall contain only information relevant to the particular equipment or material to be furnished. The Contractor shall not submit catalogs which describe several different items in addition to those items to be used, unless all irrelevant information is marked out, or unless relevant information is clearly marked. The submittal shall consist of product data from each manufacturer. Product data from each manufacturer shall be identified and submitted separately.

5. Coordination between Division 21, 22, 23, 25, 26 and 28: Each Division 25 shop drawing and product data submitted for review, which has Division 25 equipment requiring electrical service, shall contain a statement indicating that the Division 25 Contractor has provided the Division 26 Contractor with the necessary electrical characteristics of the actual Division 25 equipment to be provided.

6. Shop Drawing and Product Data - Submittal Review Status:

    a. Shop drawing and product data submittals will be returned marked with one of the following status:

        1) "REVIEWED NO EXCEPTIONS"
        2) "REVIEWED EXCEPTIONS NOTED"
        3) "INFORMATION ONLY"
        4) "REVISE AND RESUBMIT"
        5) "REJECTED"

    b. If stamped "REVISE AND RESUBMIT" or "REJECTED", the shop drawing or product data shall be revised in accordance with the Contract Documents. If marked "REVIEWED, EXCEPTIONS NOTED" documentation that addresses only the noted exceptions in the form of a letter, supplemental information, or complete resubmittal shall be forwarded to the Owner, Architect and Engineer for record purposes only. If marked "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" no additional submittal will be required.

    c. If the copy stamped "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" is altered for any reason after it has been stamped, the "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" shall automatically be voided and the submittal status revised to "REVIEWED EXCEPTIONS NOTED", "REVISE AND RESUBMIT", or "REJECTED" depending on the alterations.

    d. All work shall be done in accordance with shop drawings stamped "REVIEWED NO EXCEPTIONS", "REVIEWED EXCEPTIONS NOTED", or "INFORMATION ONLY" insofar as these are in agreement with the Contract Documents. Wherever differences occur between the shop drawings and the Contract Documents, the Contract Documents shall govern the Work.

G. Record Shop Drawings: After receiving the Engineer's and Architect's final approval for each shop drawing, the Contractor shall provide the Engineer with AutoCAD files of each final approved shop drawing. All of the previous review comments by the Engineer and Architect associated with each shop drawing shall be incorporated into the AutoCAD files of each shop drawing prior to sending the AutoCAD files to the Engineer. The AutoCAD files shall be in AutoCAD 2018 version software or later version as approved by the Engineer. The Contractor shall be responsible for contacting the Engineer to determine the approved AutoCAD software to use prior to preparing the AutoCAD files.

H. Maximum Ceiling Clearance and Headroom Requirements: All material for systems and equipment shall be installed to accommodate the structural systems, architectural ceiling construction, base building and future tenant lighting systems, and other physical space limitations of the project. All systems and equipment shall be installed to maximize clearances between the bottom of material and top of ceiling construction. All systems and equipment shall be installed to maximize headroom clearance below the bottom of material. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, and other systems that will be located where ceiling plenum clearances and headroom clearances are limited; shall be prepared to address these issues. Conflicts with ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data and clearly marked so as to bring the situation to the attention of the Engineer.

I. Equipment and Layout Deviations: Layout, connections, supports, access & maintenance clearances, related architectural enclosures, and structural supports have been based on selected products and manufacturers. In some cases, the manufacturer's model number has been provided. Other products by specified manufacturers may be used if they comply with requirements of the contract documents including but not limited to capacity ratings. However, where the Contractor proposes to use an item of equipment that has alternative but equivalent minimum capacity ratings or that requires any redesign of the structure, partitions, foundations, piping, wiring or any other part of the mechanical, electrical, structural, civil, utility, or architectural systems; all such redesign, and all new drawings and detailing required therefore, shall be prepared by the Contractor at the Contractor's own expense as approved by the Owner's Representative. Where such approved deviation requires a different quantity, capacity, and arrangement of ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, and starters from that specified or indicated on the drawings, the Contractor shall furnish and install any such ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, starters and any other additional equipment required by the system, at no additional cost to the Owner.

1.4 RESPONSE TO SHOP DRAWING, SUBMITTAL, AND FIELD REPORT REVIEW COMMENTS

A. Contractor shall respond in written form to each shop drawing, submittal, or field report review comment from the Engineer within 10 business days from the date indicated on the Engineer's review form. Contractor's written response to each review comment should indicate the date and specifically what corrective action was or will be taken and include the name of the person and associated company responding to each review comment. Refer to the Architectural documents for additional requirements.

1.5 WARRANTY, SERVICE, AND REPLACEMENT PARTS

A. All equipment supplied for this Contract shall be free from defects in material, workmanship, and title, and shall be of the kind and quality described herein. If it appears within one (1) year from the date of the building final acceptance by the Owner that the equipment does not meet the warranties specified above or the equipment is recalled by the equipment manufacturer, the Contractor shall correct any defect or replace any recalled equipment, including

nonconformance with these Specifications, at no additional cost to the Owner.

B. The manufacturers for each of the various selected Division 25 pieces of equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner's acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

1.6 CONFERENCE PRIOR TO START OF WORK

A. Immediately upon the award of this Contract, but prior to commencing any Work, the Contractor together with designated major Subcontractors shall confer with the Owner's Representative, the Architect and Engineer concerning the Work under this Contract.

B. Contractors and Subcontractors attending this meeting shall include but not be limited to the following: General; Mechanical; Plumbing; Fire Protection; Automatic Temperature Controls and Building Management; Sheet Metal; Insulation; Electrical; Fire Alarm; and Testing, Balancing and Adjusting. Contractor and Subcontractor representatives shall be familiar with the Drawings, Specifications and shop drawing submittal requirements and be prepared to intelligently discuss the project requirements.

C. The conference will be at a mutually agreed place and acceptable time.

1.7 ELECTRONIC FILE TRANSFER AGREEMENT

A. Release of the Dewberry CAD, BIM, and other electronic files will require the User of the files to sign the Dewberry Electronic File Transfer Agreement.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials and equipment shall be new and in good condition. The commercially standard items of equipment and the specific names mentioned herein are intended to identify standards of quality and performance necessary for the proper functioning of the Work.

B. Since manufacturing methods vary, reasonable minor variations are expected; however, performance and material requirements specified are the minimum standards acceptable. The Engineer retains the right to judge the quality of equipment that deviate from the Contract Documents.

2.2 PRODUCTS, COMPONENTS AND MATERIALS IN PLENUMS

A. All products, components, materials and composite materials within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723 as required by applicable mechanical code.

2.3 EQUIPMENT GUARDS

A. Provide equipment guards for the safe operation of equipment as recommended by the equipment manufacturer and as required by applicable codes. Provide easily removable

expanded metal guards for all belts, couplings, exposed fan inlets and outlets, and other moving parts of machinery. Provide tachometer openings in the guards at least 2" in diameter, for all belt-driven or variable speed machinery.

## 2.4 EQUIPMENT MOUNTING/SUPPORT DEVICES

A. Unless otherwise required to accommodate the particular requirements and unique conditions of this project, all floor, wall or ceiling mounting/support devices shall be factory fabricated devices as provided by the equipment manufacturer for the intended application. In addition to the factory-fabricated devices, the Contractor shall provide additional support members, framework, mounting hardware, brackets, steel angles, anchors, restraints, support stands, attachment devices, and other miscellaneous mounting and support devices as required for adequate equipment support.

B. The Engineer retains the right to judge the quality of the design, application, and installation of all field and shop fabricated mounting/support devices using steel angle, steel channels, Uni-Strut, brackets, all-thread rods, etc. Any mounting or support devices that will rust, corrode, or in the opinion of the Engineer become unsightly or a maintenance problem will not be acceptable.

C. Support devices not factory finished shall be hot dipped galvanized or properly primed and painted with a minimum of two coats of rust inhibiting paint. Refer to the Architectural Painting Systems for additional requirements.

D. All-thread rods shall be galvanized or cadmium plated.

E. All plywood used as mounting boards in association with Division 25, unless specified otherwise by the Architect, shall be APA C-D plugged interior 5/8" to 3/4". All plywood used shall be painted (front, back and edges) with a minimum of one prime coat and one finish coat prior to equipment installation. Refer to the Architectural Specifications for additional requirements. Color shall be determined by the Architect.

## 2.5 EQUIPMENT PADS AND ANCHOR BOLTS

A. Concrete pads for various pieces of equipment will be furnished under another Division. Pads shall be provided in main machine rooms, all areas with floors below grade, upper level pump and storage tank rooms, penthouse equipment rooms, and in any other location as shown on the drawings. This shall include floor mounted equipment, equipment mounted on legs and pipe support stands. Unless otherwise noted, equipment pads shall generally conform to the shape of the piece of equipment it serves with a minimum 3" margin around the equipment and supports. Pads shall be a minimum of 4" high and made of a minimum 28 day, 2500 psi concrete reinforced with 6"x6" 6/6 gauge welded wire mesh. Top and sides of pad shall be troweled to smooth finishes, equal to those of the floors, with all external corners bull-nosed to a 3/4" radius. Shop drawings stamped "FINAL REVIEW" or "REVIEWED EXCEPTIONS NOTED" shall be used for dimensional guidance in sizing pads. Pads for each water chilling unit in equipment rooms shall extend 6" beyond the footprint of each water chilling unit.

B. Unless noted otherwise furnish and install galvanized anchor bolts for all equipment placed on concrete equipment pads or on concrete slabs. Minimum dimension between anchor bolts and edge of slab condition shall not be less than 3". Bolts shall be of the size, type and number recommended by the manufacturer of the equipment and shall be located by means of suitable templates provided by the equipment manufacturer. When equipment is placed on vibration isolators, the equipment shall be secured to the isolator and the isolator secured to the floor, pad, or support as recommended by the vibration isolation manufacturer.

2.6 ACCESS DOORS

A. Wherever access is required through walls, ceilings or fire-rated enclosures to valves, fire dampers, heating coils, damper actuators, drain line cleanouts, or other concealed equipment installed under this Division, unless otherwise noted in the architectural documents, the Contractor shall furnish a hinged access door and frame as follows:

1. Drywall construction--Milcor Style DW.

2. Finished acoustical tile ceiling--Milcor Style AT in applications where there is not adequate clearance space to properly push an acoustical ceiling tile up for removal and replacement.

3. Finished plaster ceiling--Milcor Style AP.

4. Finished plaster walls or ceramic tile--Milcor Style M.

5. Plaster or masonry walls and ceilings outside offices and in other finished areas exposed to view--Milcor Style K or M.

6. Access doors shall be "B" label fire construction where required.

7. Access doors shall be installed under another Division.

8. No access door shall be installed until location and type have been approved by the Architect.

2.7 ESCUTCHEONS

A. Furnish and install heavy chrome-plated or nickel-plated steel plates of approved pattern on all pipes passing through walls and ceiling in finished areas. Escutcheons shall be B & C No. 10 or approved equal with concealed hinges. Pattern shall be approved by the Architect.

2.8 OVERCURRENT PROTECTION

A. Fused and circuit breaker type overcurrent protection devices required for any equipment furnished under Division 25 shall be in accordance with Division 26 Specification Section 26 28 00.

B. Unless specifically noted otherwise on the Drawings or in these Specifications, all Division 25 equipment shall be rated for use with both fused and circuit breaker overcurrent protection. This requirement shall be clearly indicated on the manufacturers' nameplate.

2.9 SEALANTS AND ADHESIVES

A. All sealants and adhesives shall comply with the current Volatile Organic Compound (VOC) content limits of the latest edition of the South Coast Air Quality Management District (SCAQM) Rule 1168 entitled "Adhesive and Sealant Applications". Piping sealants shall have no more than 250 grams/liter VOC content for all sealants and adhesives to be used within the building interior locations.

B. The Contractor shall submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each adhesive and sealant used within the building interior locations.

PART 3 - EXECUTION

3.1 SCAFFOLDING, RIGGING, HOISTING

A. The Contractor shall furnish all scaffolding, rigging, hoisting, and services necessary for erection and delivery of any equipment and apparatus furnished to the job site. Remove same from job site when no longer required.

3.2 SLEEVES, CUTTING AND PATCHING

A. The Contractor shall be responsible for the timely placing of openings and sleeves for all piping passing through walls, partitions, beams, floors, louvers, and roofs while the same are under construction. A pipe sleeve shall be round and one size larger than the size of pipe it serves except where "Thunderline/Link Seal" casing seals are used in penetrations through walls below grade. If the pipe is insulated, its pipe sleeve shall be one size larger than the outside diameter of the insulation around the pipe. All sleeves in floors shall extend two inches above the finished floor. All pipes passing through concrete or masonry walls above grade shall have 18 gauge galvanized steel sleeves or plastic "hole-outs". All sleeves set in concrete floor construction shall be standard weight galvanized steel. Sleeves set in concrete floor construction directly supporting riser piping 4" and larger shall be standard weight galvanized steel with three 6" long, 3/8" diameter steel reinforcing rods welded at 120o spacing to the sleeve and shall be installed with the rods embedded in the concrete slab. Where the pipe passes through a sleeve, no point of the pipe or its insulation shall touch the sleeve. Caulk around such pipe with sufficient quantity of fire safing insulation to equal fire rating of construction penetrated by the sleeve. In addition, seal the fire safing material with a non-hardening mastic.

B. Seal all pipe penetrations and unused sleeves located in fire rated construction with factory built devices or with manufactured fill, void, or cavity materials "Classified" by Underwriters Laboratories, Inc. for use as a Through-Penetration Firestop. All firestop devices and systems shall be approved for such use by the authority having jurisdiction. The fire safing system used shall maintain the fire resistance rating of the building component that is penetrated. Fire safing systems and devices shall comply with ASTM E 814 (UL 1479). Submittal data for fire safing systems shall include the UL System Numbers listed in the UL Building Materials Directory under which the material was tested in accordance with ASTM E 814 (UL 1479) for use in a Through-Penetration Firestop System. Excessive shrinkage of the fire stopping materials which would permit the transmission of smoke or water prior to exposure to a Fire condition is unacceptable. Where a mastic coating is used to seal the surface of the firestop, the mastic shall be non-hardening. The fire safing system used shall accommodate expansion and contraction of the floating mechanical piping systems without damaging the firestop or reducing its effectiveness as a smoke barrier or water seal.

C. Sleeves penetrating walls below grade shall be "WS Wall Sleeves" by Pipeline Seal and Insulator, Inc. Houston, Texas, with 2" water seal stop. The water seal stop shall be located in the middle of the wall. Seal off annular opening between pipe and sleeve with "Thunderline/Link-Seal" casing seal as manufactured by Pipeline Seal & Insulator, Inc., Houston, Texas. The pipe sleeve shall be sized to accommodate the "Thunderline/Link-Seal" casing seal. Casing seals shall be Model "C" and shall be sized per the manufacturer’s recommendations.

D. If holes and/or sleeves are not properly installed and cutting and patching becomes necessary, it shall be done at no expense to the Owner. The Contractor shall undertake no cutting or patching without first securing the Architect's written approval.

E. Sleeves in floor slabs with waterproof membranes shall be modified as required to accept a Zurn Z-197 riser sleeve and also meet all other sleeve requirements as specified herein before. The Z-197 riser sleeve shall be constructed of dura-coated cast iron body, double threaded pipe connector with combination membrane flashing flange.

F. The location and size of all penetrations and openings through existing walls, slabs, beams, roofs and other existing building construction shall be provided by the contractor to accommodate the new work. Penetrations and openings through existing building construction shall be reviewed by the structural engineer and architect prior to cutting the penetration. Unless otherwise required by the architectural and structural divisions, metal sleeve inserts will not be required in pipe penetrations except as required to meet UL fire rating requirements. The contractor shall perform a structural survey, including x-rays or graphs of all floor slabs and supporting walls to locate existing structural bars, cables, in slab conduits and any other interferences that may be in conflict with proposed sleeve/penetration locations. The x-ray or graph shall be performed prior to any cutting or chipping of structural supporting walls. In the event of a conflict, the contractor shall notify the architect prior to proceeding with the work.

G. Provide sleeves for future use as indicated on the drawings. Provide a minimum of two 6" sleeves in each AHU room for the installation of future piping risers.

## 3.3 PAINTING

A. Painting, except as specified or indicated otherwise, shall be done under another Division. This Division shall cooperate with the other Division to determine the size of equipment, sizes and lengths of pipes, etc., to be painted.

B. Equipment furnished under this Division shall be factory-finished. If the factory finish is damaged during shipment, installation, etc., it shall be repainted subject to the Architect's approval.

C. Refer to Sections 21 05 53, 22 05 53 and 23 05 53 for additional requirements.

## 3.4 DRAYAGE AND HAULING

A. Include all drayage, hauling, hoisting, and placement in the building of equipment specified herein. The Contractor shall be responsible for the timely introduction of equipment to the Project. If any item of equipment is received prior to the time it is required, the Contractor shall be responsible for its proper storage and protection until such time as it may be required. The Contractor shall pay for all costs of demurrage or storage.

## 3.5 IDENTIFICATION OF EQUIPMENT

A. Equipment and piping shall be identified in accordance with the requirements indicated in specification Sections 21 05 53, 22 05 53 and 23 05 53.

B. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; software, programming, and graphics.

3.6 EQUIPMENT NOISE AND VIBRATION

A. It is the intention to specify and for the Contractor to provide equipment and systems that, as defined herein, shall be quiet and free of apparent vibration in operation. To this end, both the balancing of rotating machinery and the installation of vibration isolation at various locations are required.

B. It shall be the responsibility of the Contractor to obtain equipment that is quiet in operation as compared to other available equipment of its size, capacity, and type; to install equipment so that a minimum amount of noise and/or vibration is transmitted to the building; and to fabricate the duct system so that air noises generated in the system are held to a minimum.

C. Any additional precautions deemed necessary to provide a quiet installation shall be done as part of the Work of the Contractor, subject to review by the Engineer, without additional cost to the Owner. After the system is in operation, it shall be the responsibility of the Contractor to make any changes to equipment or Work installed that may be required to provide a system which is quiet in operation as defined herein.

D. The system noise level shall be equal to or less than those indicated in Section 23 00 15. Sound levels within the occupied spaces must meet the criteria described above and with all building, wall partition, floor and ceiling construction in place as they exist for the individual spaces. The attenuation through boundary construction of equipment rooms must be considered in selecting equipment for acceptable noise level.

3.7 CONTROL COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

3.8 LUBRICATION

A. Provide means for lubricating all bearings and other machine parts. If a part requiring lubrication is concealed or inaccessible, extend a lubrication tube with suitable fitting to an accessible location and suitably identify it.

B. After installation, properly lubricate all parts requiring lubrication and keep them adequately lubricated until final acceptance by the Owner.

3.9 SPARE PARTS

A. General: The Contractor shall provide all necessary spare parts for the equipment supplied in this Project as indicated hereafter and in various other Sections of the Specifications. Spare parts shall be obtained from the original equipment manufacturer.

3.10 DOCUMENTATION

A. For all testing, training and review procedures the Contractor shall maintain adequate records in written form that clearly documents the event. The written documentation for each testing, training and review procedure shall include but not be limited to the following: Name and company of attending personnel, general description of procedure(s), date(s), time of day, deficiencies found, corrective action taken, and nature of final results. Documentation records shall be kept in an orderly, legible and up to date format. Contractor's documentation records shall be accessible for review by the Owner Representative upon request. Each testing, training and review procedure document shall bear the name and signature of the authorized

representative of the General Contractor, Mechanical Contractor, and manufacturer or Subcontractor if applicable.

## 3.11 COMPLETION, COMMISSIONING AND FINAL REVIEW OF THE DIVISION 25 SYSTEMS

A. Start-up: After the Contractor and equipment manufacturer representative determine that the equipment and related systems have been properly installed and are ready for start-up; the equipment manufacturer representative shall perform the manufacturer's start-up and check-out procedures to ensure proper operation of the equipment in accordance with the contract documents and the manufacturer's recommendations. Start-up and check-out of each piece of equipment shall be in accordance with the equipment manufacturer's recommendations and warranty requirements. Start-up services shall be performed by factory-trained personnel of the equipment manufacturer or factory-trained personnel authorized by the manufacturer and their official representatives. Start-up personnel shall be certified by the equipment manufacturer as properly trained in the manufacturer's start-up and check-out procedures. Provide documentation of the start-up and check-out procedures in accordance with the contract documents and the manufacturer's recommendations. Documentation of the start-up and check-out procedures shall be included within and submitted with the Operational and Maintenance Manuals.

B. Trouble-shooting, Debugging, and Pre-testing: After Start-Up but prior to the Operational Demonstrations; the contractor shall perform operational pre-testing, trouble-shooting, and de-bugging of all equipment and systems. At the completion of the Trouble-shooting, Debugging, and Pre-testing process, all equipment and systems shall be complete and operating in accordance with the requirements of the contract documents. Trouble-shooting, Debugging, and Pre-testing of equipment and systems shall be performed to the extent necessary for the Contractor to determine that all systems are complete and operating in accordance with the Contract Documents. Prior to final Commissioning and Operational Demonstrations all equipment and systems shall be certified by the Contractor as complete and operating in accordance with the Contract Documents.

C. Operational Demonstrations and Final Commissioning: Prior to Substantial Completion and within a schedule mutually agreed to by the contractor and the Owner's Representative, the Contractor shall demonstrate the operation of all equipment and systems as necessary to demonstrate compliance with the Contract Documents. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner.

D. Final Review and Owner Acceptance:

1. After completion of the Operational Demonstrations, Final Commissioning, and Owner's Close-out Package requirements; at a time designated and mutually agreed by the Contractor and Owner's Representative; a final walk-through review of the project will be performed by the Contractor and Owner's Representative. The review will be of a general nature with the intent of addressing outstanding issues prior to the Owner's final acceptance of the project.

2. The entire system shall be clean and operating properly with all systems balanced and all controls adjusted. All labels shall be removed from the plumbing fixtures, and the fixtures shall be clean and in operating condition.

3. Certificates, Documents, and Close-out Package required herein shall be complete, in order and presented to the Architect at least two weeks prior to the review.

4. After the review, any changes or corrections noted as necessary for the Work to comply with these Specifications and the Drawings shall be accomplished without delay in order to secure final acceptance of the Work.

E. Schedule: The Contractor shall complete the MEP systems in accordance with the Contract Documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems a minimum period of time prior to the date of "Substantial Completion" as established by the Commissioning Agent. This period of time prior to the date of "Substantial Completion" shall be reserved by the Contractor for the Owner's Representative, Commissioning Agent, Architect, and Engineer to witness the operational demonstrations of the completed fully commissioned MEP systems. If more than this established time period is deemed necessary by the Contractor for the operational demonstrations, the Contractor shall notify the Owner's representative and Commissioning Agent in written form and determine an operational demonstration schedule mutually agreeable to the Owner's Representative, Commissioning Agent, and Contractor that completes the operational demonstrations prior to "Substantial Completion". Additional costs incurred by the Owner related to reviewing, witnessing, testing, inspecting, and commissioning of the mechanical systems by the Commissioning Agent, Architect and Engineer after the date of "Substantial Completion" shall be reimbursed by the Contractor to the Owner. As part of the shop drawing submittal process, the Contractor shall provide a detailed schedule of completion indicating when each MEP system will be complete and ready for owner acceptance. The schedule shall indicate time periods and milestone dates of Start-up; Trouble-shooting, Debugging, and Pre-testing; Operational Demonstrations and Final Commissioning; Final Review and Owner Acceptance.

F. Additional Requirements:

1. Owner's Close-out Package: Refer to the Owner's Close-out Package requirements within Section 25 00 10 for additional requirements.

2. Refer to the various sections of the specifications for additional requirements.

## 3.12 OPERATING INSTRUCTION

A. The Contractor shall provide the services of a factory trained specialist to supervise the operation of all equipment specified herein and to instruct the Owner's operators for a five-day, eight hours per day, operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. See individual Sections for additional instructions by the manufacturer's trained specialists.

B. In addition, the manufacturer of the automatic temperature controls, water chilling units, cooling towers, pumps, chemical water treatment, and variable frequency drives shall furnish the services of factory trained specialists to instruct the Owner's operators as set forth in various Sections. The operating instruction periods shall be as defined as follows:

C. Controls: Refer to Section Sequence of Operation on Drawing M-502

D. Training schedule, duration, and subject matter shall also be in accordance with a training syllabus provided by owner's operating staff. This syllabus will be provided prior to the final review.

E. The Owner shall be notified in writing at least five days before each operating instruction period begins. The Contractor shall not commence instruction period until the Owner has issued a written acceptance of the starting time.

3.13 OPERATING AND MAINTENANCE BOOKS

A. The Contractor shall provide operating instruction and maintenance data books for all equipment and materials furnished under this Division.

B. Submit three (3) hard copies and three (3) electronic copies of the operating and maintenance data books for review at least eight weeks before final review of the Project. Assemble all data in a completely indexed volume or volumes and identify the size, model, and features indicated for each item. The Engineer's review period for the operating and maintenance data books shall be a minimum of 4 weeks, exclusive of transmittal time. The Contractor's schedule shall plan to accommodate this review period requirement.

C. Maintenance instruction manuals shall include complete oiling, cleaning, and servicing data compiled in clearly and easily understandable form. Data shall show all model numbers of each piece of equipment, complete lists of replacement parts, motor ratings, and actual loads.

D. Include the following information where applicable:

1. Identifying name and mark number as indicated on the Drawings and in the Specification.

2. Locations (where several similar items are used, provide a list).

3. Complete nameplate data.

4. Certified record drawings and "Final Reviewed" or completely corrected shop drawings and/or submittals.

5. Parts lists.

6. Performance curves and data including air and water design flow rate set points.

7. Wiring diagrams.

8. Lubrication charts.

9. Start-up and Check-out procedures and check-lists.

10. Manufacturers' recommended operating and maintenance instructions with all non-applicable information deleted. This shall include but not be limited to the manufacturer's service and repair manual.

11. Additional information as recommended by the equipment manufacturer or system provider for the proper operation, maintenance, and warranty of the equipment and systems.

3.14 RECORD DRAWINGS

A. The Contractor shall maintain on a daily basis at the project site a complete set of "Record Drawings". The "Record Drawings" shall consist of a set of black line prints and AutoCAD files of the Contract Drawings for this Division. The black line prints shall be marked and the AutoCAD files shall be electronically updated regularly by the Contractor during the construction period to reflect an accurate dimensional record of all buried or concealed work. In addition, the set of black line prints shall be marked and the set of AutoCAD files shall be electronically updated regularly to show the precise location of concealed work and equipment including

buried, concealed or embedded piping, conduit and valves and all changes and deviations in the mechanical work from that shown on the Contract Documents. This requirement shall not be construed as authorization for the Contractor to make changes in the layout or work without definite instructions from the Architect. The set of black line prints and AutoCAD files of the Contract Drawings for this Division shall remove or blacken out the Engineer's seal and Engineer's firm name.

B. Prior to commencing work, the Contractor shall need to purchase from the Architect or Engineer a set of black line prints to be used for the "Record Drawings" and obtain from the Engineer a set of AutoCAD files of the Contract Drawings for this Division in preparing "Records Drawings" for the project based on the Contractor's work. The AutoCAD files shall be in AutoCad 2015 version software or later version as provided by the Engineer. In order to obtain a set of AutoCAD files of the Contract Drawings for this Division from the Engineer, the Contractor shall need to contact the Engineer, Architect or Client and obtain a written release and indemnification statement from the Engineer. This release and indemnification statement shall be signed by an owner, partner, corporate officer or other person duly authorized to agree to and sign legally binding documents for the Contractor's respective company. Upon the Engineer receiving the signed unedited release and indemnification statement back from the Contractor, the Engineer will forward the AutoCAD files of the Contract Drawings for this Division to the Architect or Client who in turn will forward the AutoCAD files of the Contract Drawings for this Division to the Contractor for his use in preparing "Records Drawings" for the project based on the Contractor's work. If the Contractor desires to have architectural and structural information in preparing "Records Drawings" for the project, the Contractor shall need to contact the Architect and Structural Engineer to obtain the pertinent architectural and structural information directly from the Architect and the Structural Engineer.

C. Record dimensions shall clearly and accurately delineate the Work as installed; locations shall be suitably identified by at least two dimensions to permanent structures.

D. The owner's representative shall retain the right to periodically review the "Record Drawings" as the project progresses. Final acceptance of as-built "Record Drawings" shall be conditional upon the Contractor clearly "red-marking" all changes from the contract documents, including "request for information" changes and other supplemental instruction changes, directly on the black line set of prints and clouding all of the changes in the electronic AutoCAD files. All of the clouded changes on each drawing in the electronic AutoCAD files shall be on a separate drawing layer which can be turned off without affecting the appearance of each drawing.

E. The Contractor shall mark all "Record Drawings" on the front lower right hand corner with a rubber stamp impression or an AutoCAD image that states the following:

"RECORD DRAWINGS (3/8" high letters)
To be used for recording Field Deviations and
Dimensional Data Only". (5/16" high letters)

F. Upon completion of work, the Contractor shall certify all "Record Drawings" on the front lower right hand corner adjacent to the above marking with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWING CERTIFIED CORRECT (3/8" high letters)
(Name of Mechanical Subcontractor)

By: _____

Date:_____

For all black line prints to be used for the "Record Drawings", the Contractor must sign

the above certification. For all AutoCAD files to be used for the "Record Drawings", the Contractor must fill in the name of the person responsible for certifying the above certification. Both the black line prints and the AutoCAD files shall be certified by the same person.

G. Prior to final acceptance of the Work of this Division, the Contractor shall submit properly certified "Record Drawings" to the Architect for review and shall make changes, corrections, or additions as the Architect may require to the "Record Drawings". After all changes have been made to the "Record Drawings", the Contractor shall deliver to the Owner the final set of "Record Drawings" consisting of the black line prints and the AutoCAD files. The AutoCAD files to be delivered by the Contractor to the Owner shall be in AutoCAD 2015 version software or later version as originally provided to the Contractor by the Engineer.

3.15 OWNER'S CLOSE-OUT PACKAGE

A. Contractor shall provide a master project close-out log and corresponding schedule for submission to the Owner and Engineer for review and approval.

B. Contractor shall provide the Owner with two (2) copies of all warranties bound in a notebook with a corresponding warranty log. Warranty log shall clearly indicate the subcontractor, address, telephone number, and contact person.

C. Contractor shall provide the Owner with one (1) complete set of approved submittals and a copy of the final submittal log.

D. Contractor shall provide the Owner with a complete list of all subcontractors and major suppliers with address, telephone number, and contact persons.

E. Contractor shall provide the Owner with a complete inventory of attic stock materials and spare parts as applicable and a copy of the transmittal signed by the Owner's representative acknowledging receipt of the items.

F. Contractor shall provide the Owner and Engineer with documentation of compliance for all specified requirements associated with equipment start-up, commissioning, demonstrations, training and warranties. Documentation shall include, but not be limited to, equipment start-up sheets as completed by the manufacturer's representative. The Owner reserves the right to conduct performance tests on one or more HVAC units selected at random after the equipment start-up is complete. If the units fail to perform as specified for any reason, the Contractor shall be required to test all of the HVAC units and perform the necessary corrective work as required in order to have the HVAC units comply with the Contract Documents.

G. Contractor shall provide the Owner with a comprehensive valve tag chart.

H. When the Contractor considers the work to be ready for final acceptance, the Contractor shall provide the Owner and Engineer with written certification stating the following:

1. All work has been completed in accordance with the Contract Documents.

2. All field review items and other deficiencies identified by the Certificate of Substantial Completion have been properly corrected.

3. All work has been inspected for compliance with the Contract Documents.

4. All mechanical systems have been tested in the presence of the Owner's representative and are fully operational in compliance with the Contract Documents.

5. As-built documents have been reviewed, are complete and are accurate.

I. The Contractor shall provide the Owner with a letter from the Engineer of Record stating that the equipment installed fully complies with the Contract Documents.

END OF SECTION 25 00 10

SECTION 25 00 70 - DIVISION 25 SCHEDULE OF SUBMITTAL DATA

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of “Division 01, General Requirements,” “Conditions of the Contract” and “Supplementary Conditions.” Specific attention is called to the “Division 25 General Requirements” in Section 25 00 10.

1.2 WORK INCLUDED

A. Furnish the shop drawing submittal data for the Architect’s and Engineer’s review as indicated herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Division 21, 22, 23, 25, 26 and 28: All Sections

PART 2 - SHOP DRAWING SUBMITTAL DATA

2.1 GENERAL

A. The submittal data to be furnished for this Project shall comply with the Specifications hereinbefore. The referenced Sections herein scheduled are for general information purposes only and shall not be construed to be limiting to the submittal data scheduled and/or required within the individual Sections of these Specifications.

2.2 CONTRACT REQUIREMENTS

A. Shop drawing submittal data shall clearly and concisely address all Contract Document requirements. Standard manufacturer data that does not indicate these requirements shall be supplemented with additional information provided by the Contractor or equipment manufacturer. Refer to sections 25 00 10 for additional requirements, including submittal log schedule requirements.

2.3 DEVIATIONS FROM THE CONTRACT DOCUMENTS AND PERTINENT DATA

A. All deviations from the contract documents shall be noted in the form of a cover letter included with the shop drawing submittal package. All pertinent data such as material options, equipment arrangements, control options, diagrams, etc. that applies to this Project shall be clearly marked. All data and diagrams that do not apply to this Project shall be omitted from the package or clearly marked out.

2.4 QUALITY ASSURANCE

A. Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does Not Comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification paragraph. A handwritten marked-up specification section will be acceptable. Along with this

specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

2.5 SPECIAL REQUIREMENTS

A. The shop drawing submittals for the EMCS/ATC system and the Operation and Maintenance (O&M) Manuals shall require an Owner/Consultant review period of four (4) weeks. The Contractor shall plan for and arrange the shop drawing submittal review schedule to accommodate this requirement.

2.6 SPARE PARTS

A. The submittal data for each piece of equipment shall include a detailed itemized list of recommended spare parts. Spare parts shall be clearly identified by name and part number.

2.7 ORGANIZATION OF PACKAGES

A. Unless otherwise noted in the specifications and in the following schedule of shop drawing, the Contractor shall provide a single shop drawing submittal package for each of the listed specification sections. Unless specifically noted otherwise, multiple submittal packages per section will not be acceptable under the terms of the Contract.

2.8 CONTROL COORDINATION

A. By accepting this Contract, any equipment manufacturer who furnishes equipment for this Project shall coordinate and cooperate with the requirements of Division 25.

2.9 COORDINATION BETWEEN DIVISIONS 21, 22, 23, 25, 26 AND 28

A. The Division 25 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 26 and 28 Contractors to provide coordination between the Division 21, 22, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 25, it shall be the responsibility of the Division 25 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

2.10 MANUFACTURER'S CERTIFICATION

A. All items or equipment listed herein with asterisks (*) shall be certified by the manufacturer. Refer to Section 25 00 10 for certification requirements.

2.11 SHOP DRAWING SUBMITTAL SCHEDULE

A. Provide a comprehensive list of Divisions 21, 22, 23, 25, 26 and 28 shop drawing submittal packages and a schedule indicating the date when each package will be submitted to the contractor and the date when each package will be submitted to the design team. The intent of this list and schedule is to allow the contractor and design team to plan ahead and schedule the

necessary review and coordination resources in advance of the arrival of shop drawing submittal packages. Shop drawing submittal schedule shall be submitted to the Architect not less than four weeks prior to the submission of the first Division 21, 22, 23, 25, 26 and 28 shop drawing submittal packages. Shop drawing submittal schedule shall be periodically updated by the contractor and issued to the Architect to reflect current status of the project. Shop drawing submittal packages shall be listed on the schedule not less than four weeks prior to submission to the contractor and not less than six prior to the submission to the design team. Packages not listed on this schedule will required up to two additional weeks of review time.

2.12 SHOP DRAWING SUBMITTAL DATA

A. The submittal data to be furnished for this Project shall include but not be limited to the following:

1. Drawing M-502 - Sequence of Operation

2. Section 25 00 70 - Shop Drawing Submittal Schedule

3. Section 23 08 02 - Commissioning Plans and Documentation

END OF SECTION 25 00 70

SECTION 26 00 00 - ELECTRICAL

PART 1 - GENERAL

1.1 DESCRIPTION

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

B. It is the intent of these Specifications for the Contractor to provide an electrical system complete, fully operational, fully adjusted, and ready for use.

C. All contract requirements indicated in the Division 21, 22, 23, 25, 26 and 28 documents, including drawings and specifications, shall be considered supplemental to and in addition to the contract requirements indicated in the documents of other trades and the Division 01 requirements.

1.2 REQUIRED WORK DESCRIBED IN OTHER SECTIONS

A. Section 26 00 10 - Division 26 General Requirements
B. Section 26 00 70 - Division 26 Schedule of Submittal Data
C. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
D. Section 26 05 23 - Control-Voltage Electrical Power Cables
E. Section 26 05 26 - Grounding and Bonding for Electrical Systems
F. Section 26 05 30 - Division 26 Demolition
G. Section 26 05 33 - Raceway and Boxes for Electrical Systems
H. Section 26 05 48 - Vibration and Seismic Controls for Electrical Systems
I. Section 26 05 53 - Identification for Electrical Systems
J. Section 26 05 73 - Electrical Power System Study
K. Section 26 05 80 - Electrical Service
L. Section 26 05 81 - Public Telephone System
M. Section 26 05 93 - Testing
N. Section 26 08 00 - Commissioning of Electrical Systems
O. Section 26 09 23 - Automatic Control Devices
P. Section 26 24 16 - Panelboards
Q. Section 26 27 26 - Wiring Devices
R. Section 26 28 00 - Low-Voltage Circuit Protective Devices
S. Section 26 29 00 - Low-Voltage Controllers
T. Section 26 32 00 - Packaged Generator Assemblies
U. Section 26 36 23 - Automatic Transfer Switches
V. Section 26 41 00 - Facility Lightning Protection
W. Section 26 43 13 - Surge Protective Devices for Low-Voltage Electrical Power Circuits
X. Section 26 50 00 - Lighting

1.3 PARTIAL LIST OF WORK NOT INCLUDED IN DIVISION 26

A. Equipment pads:

B. Installing access doors:

C. Painting (except as otherwise specified herein):

D. Furnishing of motors: Division 21, 22, 23, and 25.

E. Furnishing of individual motor controllers that are factory-mounted and integral parts of pieces

of equipment: Division 21, 22, 23, and 25.

F. Furnishing, installing, and connecting telephone wiring, cables, and equipment unless otherwise indicated:

G. Compaction testing:

H. Division 21, 22, 23, and 25 shall furnish and install control wiring and conduit for the following:

1. Motor control interlock wiring for automatic temperature control of pumps, AHUs, and fans including chilled, heating hot water, and condenser water system controls, flow switches, pressure differential controls, compression tank alarms, water treatment system, etc.

2. Water system controls and alarms including but not limited to: House pumps, etc.

3. Interlocking of pumps, AHUs, and fans to building management system relays.

4. Any control wiring for thermostats, direct digital controls, control valves, dampers, ATC cabinets, etc. for automatic temperature control. Control wiring shall be considered both "line 120V." and "low" voltage wiring.

5. All other control wiring indicated in Section 25 00 90.

I. Any control wiring not listed above shall be furnished and installed under Division 26, including but not limited to conduit, other raceway components, and control wiring between water meters and remotely mounted water meters, and all necessary wiring to serve the electrically operated plumbing fixtures as specified under Division 22.

1.4 ALTERNATE AND UNIT PRICING

A. Alternate Pricing, Unit Pricing, and Budgetary Cost Estimates: **Bid alternates and unit prices shall not be required with the Contractor's base bid**. These requirements shall apply to only those bidders contacted after the Owner's and Architect's review of the bids. If contacted by the Owner or Architect after the bid date, the Contractors shall provide alternate prices, unit prices, and budgetary cost estimates as indicated in the Architectural Documents, as specified herein, and as indicated on the Drawings.

B. If contacted by the Owner or Architect after the bid date, the contractor(s) shall provide complete alternate pricing information for the owner's evaluation of all alternate priced items specified herein. Alternates may be additive or deductive. An additive alternate, whether shown on the drawings or specified herein, shall be defined as an alternate that is not currently part of the base building scope of work with a price that would be required to add the alternate to the base building scope of work. A deductive alternate shall be defined as an alternate that is currently part of the base building scope of work with a price that would be required to delete the alternate from the base building scope of work. The owner reserves the right to accept or reject any alternate priced item.

1. Alternate 26-1 - Extended Warranty and Service Contract Alternate for the Emergency Generator: Provide an additive alternate bid price to provide extended service contract for the stand-by emergency generator system. Refer to Section 26 32 00 for additional information.

2. Alternate 26-2 - Thermographic Testing Alternate: Provide a deductive alternate bid price to delete the thermographic testing requirement specified in Section 26 05 93.

C. Refer to the architectural specification section for Architect's and Owner's instructions to bidders

for additional alternate pricing requirements.

D. Pricing information provided for unit costs, separate line items, alternates, and value engineering items shall be all inclusive pricing that accounts for the impact and ripple effects on adjacent or related systems affected by the alternate product, material, or system. Acceptance of an alternate product, material or system shall not result in additional cost to the project beyond the price indicated for the alternate product, material or system.

1.5 ADDITIONAL REQUIREMENTS

A. It shall be the responsibility of the Electrical Contractor to review the complete set of Division 21, 22, 23, 25, and 28 drawings and specifications including but not limited to the following specification sections to determine the complete scope of the Division 26 work:
1. Section 23 05 14 - Variable Frequency Speed Controllers
2. Section 23 11 00 - Facility Fuel Piping
3. Section 25 00 90 - Automatic Temperature Control System

B. The Division 26 Contractor shall be responsible for providing the conduit system including, but not limited to, pull boxes, power circuits, outlet boxes, junction boxes and other electrical work as indicated on the various specialty consultant's design documents. The Division 26 Contractor shall be responsible to review the design documents of the various specialty consultants which include, but are not limited to, the Security Consultant, the Information Technology Consultant, Telephone and Communications Consultant, and Audio/Visual Consultant to determine the complete scope of work required to be provided by Division 26.

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. General: All equipment shall be new, of the capacity and type specified herein, and as shown on the Drawings. Equipment shall be of a listed manufacturer and model number and shall be in accordance with the space limitations of the project.

B. Single Source: To maximize ease of maintenance and part replacement, equipment of a similar nature shall be provided by a single manufacturer.

C. Approved Equal: Equipment and materials selected by the Contractor within the context of "equal as approved by the Engineer", "approved equal", "equivalent as determined by the Engineer" and similar terminology shall be submitted to the Engineer for review, approval and inclusion into the Contract Documents prior to the finalization of the contract between the Owner and the Contractor, and prior to the shop drawing submittal phase of the Project. All equipment and materials submitted to the Engineer under the terms of "approved equal" during the shop drawing phase of the Project without prior review and approval shall be returned to the Contractor without review under the status of "No Action".

2.2 MATERIAL

A. All material required for a complete and proper installation shall be as specified and as selected by the Contractor subject to the approval of the Architect.

B. Material shall be new, listed and approved by UL, and bear the inspection label if subject to such approval.

2.3 CONTRACTOR APPROVALS

A. The contractor shall submit in writing and obtain written approval from the Owner, Architect,

and/or Engineer for all equipment substitutions, installation deviations from that shown on the Contract Documents, and all other miscellaneous approvals required from the Owner, Architect, and/or Engineer as referenced throughout these specifications.

B. A request for a substitution or deviation from the Contract Document constitutes that the Contractor:

1. Has investigated the proposed product and determined that it is equal to or superior in all respects to that specified.

2. Will provide the same warranties for the substitution as for the product specified.

3. Will coordinate the installation of the substitution into the work and make such other changes as may be required to make the work complete in all respects.

4. Waives all claims for additional costs, under his responsibility, which may subsequently become apparent.

5. Will absorb all costs incurred by the substitution when affecting other trades including but not limited to electrical, structural, architectural, etc.

6. Will absorb any cost incurred by the Owner's Representative in review of the substituted product if the acceptance of the substituted item creates the need for system modification and/or redesign, or if the substituting Contractor exhibits negligence in his substituting procedure thus submitting inferior, misapplied or mis-sized equipment.

PART 3 - EXECUTION

3.1 CONDITIONS

A. Inspection: Prior to proceeding with the work of this Division, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation may properly commence. Verify that the work of this Division may be completed in strict accordance with all pertinent codes and regulations, the reviewed shop drawings, and the manufacturers' recommendations.

B. Discrepancies: In the event of discrepancy, immediately notify the Owner's Representative. Do not proceed in areas of discrepancy until all such discrepancies have been resolved. If there is a discrepancy between the Drawings and the Specifications, the Specifications shall typically govern. However, any discrepancy of this type shall be immediately brought to the attention of the Owner's Representative for formal interpretation prior to proceeding with the work.

C. Interpretation of Documents: Any and all contractual requirements may be indicated solely on the Drawings, solely in the Specifications, in both the Specifications and on the Drawings, in reference standards indicated in the Specifications and/or in the Owner's and Contractor's Contract. If Contract requirements are indicated in both the Specifications and the Drawings, the Contractor shall comply with both requirements unless the requirements are mutually exclusive of each other. If Contract requirements are indicated in both a reference standard and the Specifications, the more stringent requirement shall apply. Any and all contractual requirements shall be interpreted within the overall context of the complete scope of work. All materials, equipment, systems and installation methods shall be suitable for the intended service, coordinated with other trades and be complete, fully operational, adjusted, tested and ready for use by the Owner.

3.2 INSTALLATION OF EQUIPMENT

A. Locations: Install all equipment in the locations shown on the approved shop drawings, except where specifically approved otherwise on the job by the Architect and/or Owner's Representative.

B. Interferences: Avoid interference with structure and with work of other trades, while preserving adequate headroom and clearing all doors and passageways to the approval of the Architect and/or Owner's Representative.

C. Inspection: Check each piece of equipment in the system for defects. Verify that all parts are properly furnished and installed, function properly, and that all adjustments have been made.

3.3 CONNECTIONS TO EQUIPMENT

A. Mechanical Equipment: The Contractor shall make final electrical connections to all items of mechanical equipment, including all motors and unit heaters for a complete and operational system.

3.4 CLOSING-IN OF UNINSPECTED WORK

A. General: Do not allow or cause any of the work of this Division to be covered up or enclosed until it has been inspected, tested, and approved by the Architect and/or Owner's Representative and by the authorities having jurisdiction.

B. Uncovering: Should any of the work of this Division be covered up or enclosed before it has been completely inspected, tested and approved, the Contractor shall provide all services, labor, materials and equipment necessary to uncover such work without additional cost to the Owner. After the work has been completely inspected, tested, and approved, the Contractor shall provide all services, labor, materials and equipment to make all repairs necessary to restore the work to its original and proper condition at no additional cost to the Owner.

3.5 COOPERATION WITH OTHER TRADES

A. Coordinate and cooperate with other trades in order that all systems in the scope of the Contract may be installed in the best arrangement. Coordinate and cooperate as required with all other trades which share space in common areas and maximize the access to each system and their respective components.

3.6 CLEANING

A. It is the intent of these Specifications that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment. Level of cleanliness shall be defined as "broom" clean unless noted otherwise. All exterior surfaces of Division 26 equipment shall be wiped down and cleaned of all dust and dirt. All interior surfaces of electrical equipment including but not limited to switchboards, motor controllers, and panelboards shall be wiped down and vacuum cleaned so as to be delivered to the Owner in factory new condition. Surfaces to be painted shall be cleaned and prepared in accordance with architectural division of the contract and as noted in other sections herein.

3.7 COMPLETENESS

A. It is the intent of these Specifications to provide complete systems. Completeness shall mean that all materials, equipment, and systems as installed and operating on this project have been installed properly with the best practices of the trade; are suitable for the intended purpose, location, and environment; properly fit within the physical space limitations for the project; are in conformance with applicable codes and reference standards; have been started-up, tested, adjusted, and commissioned for the intended use; have maintained applicable UL Listings; are

in compliance with manufacturer's recommendations and warranty requirements; ready for the Owner's use, and in the opinion of the Architect, performing as designed.

3.8 ADJUSTMENT OF CONTROLS

A. The Contractor shall provide the personnel and equipment to completely adjust the controls to the satisfaction of the Architect. At the completion of the project, the Architect will arrange a meeting at the job site to allow the Contractor to demonstrate the proper operation of the electrical controls.

3.9 NOISE

A. It is the intent of these Specifications to provide a system free from objectionable audible noise and vibration. Any equipment that is generating objectionable noise or vibration, in the opinion of the Architect, shall be corrected and dampened as required to eliminate the objectionable level.

END OF SECTION 26 00 00

SECTION 26 00 10 - GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 DESCRIPTION

A. General

1. The work under this Section shall conform to the requirements of the "General Conditions" and "Supplementary Conditions" of these Specifications.

B. Work Included

1. The work included in this Section consists of the general requirements for the work more specifically described in Division 21, 22, 23 , 25, 26 and 28 where referenced.

C. Definitions

1. The Contractor shall provide all supervision, labor, material, equipment, machinery, and any and all other items necessary to complete the mechanical and electrical systems. All items of equipment are specified in the singular; however, the Contractor shall provide the number of items of equipment as indicated on the Drawings, and as required for complete systems.

D. Intent

1. It is the intention of these Specifications and Drawings to call for finished work that has been completed, tested, adjusted, commissioned, fully operational, approved by the local code authorities, and ready for the Owner’s use. Wherever the word "provide" is used, it shall mean "furnish and install complete and ready to use."

2. It is the intention of these specifications and drawings that wherever the word “weather” is used throughout the various Division 21, 22, 23, 25, 26 and 28 specification sections it refers to “locations exposed to weather”. “Locations exposed to weather” shall mean areas exposed to outside environmental temperature, humidity, or moisture conditions. These locations include but are not limited to areas on the exterior of the building, on the roof, within architectural exhaust and intake plenums, within loading docks, within above grade parking garages, within below grade parking structures, and other areas as indicated on the drawings or in the specifications.

3. All labor, materials, apparatus, and appliances essential to the complete functioning of the systems described and indicated herein, or which may be reasonably implied as essential whether mentioned in the Drawings and Specifications or not, shall be furnished and installed by the Contractor.

4. In addition to the requirements indicated throughout the Drawings and Specifications, as completely installed and operating on this project; equipment and systems provided for this project shall meet the following minimum requirements without additional cost to the Project:

a. Be suitable for the intended service, location, and environment.

b. Properly fit within the physical space limitations for the project.

c. Meet the minimum requirements of applicable codes and reference standards as determined and approved by the local code authorities.

d. Maintain applicable UL listings.

e. Be coordinated between Division 21, 22, 23, 25, 26 and 28 in a timely manner.

f. Be coordinated with the work of other trades in a timely manner.

g. Not be purchased or installed prior to the resolution of any conflicts or coordination that could result in additional cost to the Owner.

h. Be in accordance with the manufacturer's recommendations and warranty requirements.

i. Be ready for use by the Owner.

5. In cases of doubt as to the Work intended, or in the event of need for explanation thereof, the Contractor shall request supplementary instructions from the Architect.

E. Codes, Rules, Regulations, Permits and Fees

1. Refer to the architectural specification Divisions for additional requirements. The architectural specification Divisions shall govern these requirements. The following information shall be in addition to the architectural specification Divisions.

2. The Contractor shall give all necessary notices, obtain all permits and pay all government sales taxes, fees, and other costs, including utility connections or extensions, in connection with the Contractor’s work; file all necessary plans, prepare all documents and obtain all necessary approvals of all governmental departments having jurisdiction; obtain all required certificates of inspection of the Contractor’s work and deliver same to the Architect before request for acceptance and final payment for work.

3. The Contractor shall comply with the most recently revised versions of all applicable laws, rules, regulations, and ordinances of Federal, State, and Local Authorities. Modifications required by the above said Authorities shall be made without additional charge to the Owner.

4. The Contractor shall comply with applicable building codes as adopted and modified by the Local Code Authorities. Modifications required by the above said building codes shall be made without additional charge to the Owner.

5. All equipment shall comply with all applicable requirements of laws, codes, ordinances, legislation, etc., of all Federal, State and Local Authorities, whether indicated on the Contract Documents or not.

6. Where Contract Drawings and Specifications requirements are in excess of Code requirements and are permitted under the Code, the Contract Drawings and Specifications shall govern.

7. All rules and regulations of the State Fire Insurance Regulatory Body, Underwriters Laboratories, Americans with Disabilities Act (ADA), and National Electrical Code (NEC), shall be complied with whether indicated in the Contract Drawings and Specifications or not.

8. All materials, equipment, and systems provided on this project shall be in compliance with the applicable energy codes and the energy efficiency ASHRAE Standard 90.1 as adopted by the applicable building codes.

9. Project shall meet or exceed the requirements of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality Guideline for Occupied Buildings Under Construction. In addition, all stored on-site or installed materials shall be protected at all times from moisture damage. Provide photographs to the owner, architect, or engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

10. Where alterations to or deviations from the Contract Drawings and Specifications are required by the above authorities report the requirements to the Architect and secure Architect's approval before starting the alterations.

11. Third Party Certification: All packaged equipment shall be independently Third Party labeled as a system for its intended use by a Nationally Recognized Testing Laboratory (NRTL), such as UL or ETL, in accordance with OSHA Federal Regulations 29 CFR 1910.303 & 1910.399, as well as NFPA 70, National Electric Code (NEC), Article 90-7.

1.2 QUALITY ASSURANCE

A. Surveys and Measurements

1. The Contractor shall base all measurements, both horizontal and vertical, from established benchmarks. All work shall agree with these established lines and levels. Verify all measurements at site and check the correctness of same as related to the work.

2. Should the Contractor discover any discrepancy between actual measurements and those indicated, which prevents following good practice or the intent of the Drawings and Specifications, the Contractor shall notify the Architect and shall not proceed with the work until instruction has been received from the Architect.

B. Drawings

1. Drawings are diagrammatic and indicate the general arrangement of systems and work included in the Contract. Drawings are not to be scaled. The Architectural Drawings and Details shall be examined for exact location of fixtures and equipment. Where they are not definitely located, this information shall be obtained from the Owner's Representative.

2. The Contractor shall follow Drawings in laying out work and check Drawings of other trades to verify spaces in which work will be installed. Maintain maximum headroom and space conditions at all points. Where headroom or space conditions appear inadequate, the Owner's Representative shall be notified before proceeding with installation.

3. If directed by the Owner's Representative, the Contractor shall, without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of the work, if such directions are made prior to the performance of the affected work.

C. Cooperation with Other Trades

1. The Subcontractors shall give full cooperation to other trades and shall furnish in writing

to the General Contractor any information necessary to permit the work of all trades to be installed satisfactorily and with the least possible interference or delay.

2. Where Division 21, 22, 23, 25, 26 and 28 work will be installed in close proximity to, or will interfere with work of other trades, coordination among trades in working out space conditions to make a satisfactory adjustment shall be accomplished. The Contractor shall prepare composite shop drawings and sections at a suitable scale not less than 1/4" = 1'-0", clearly showing how work of all affected trades is to be installed. If work is installed before coordinating with other trades, or so as to cause any interference with work of other trades, then the necessary changes in work shall be made to correct the condition without extra charge.

3. The Contractor shall obtain and transmit between affected trades, all necessary templates, patterns, setting plans, and shop details for the proper installation of work and for the purpose of coordinating adjacent work.

D. Protection

1. The Contractor shall protect all work and material from damage by the Contractors' work or workmen, Subcontractors' work or workmen, and shall be liable for all damage thus caused.

2. The Contractor shall be responsible for work and equipment until work is final inspected, tested, and accepted; the Contractor shall protect work against theft, injury or damage; and shall carefully store material and equipment received on site which are not immediately installed. The Contractor shall close open ends of work with temporary covers or plugs during storage and construction to prevent entry of obstructing material.

3. All stored on-site or installed absorptive materials are protected at all times from moisture damage.

E. Material and Workmanship

1. Work shall be executed in strict accordance with the best practice of the trades in a thorough, substantial, workmanlike manner by competent workmen. The Contractor shall furnish the services of a full-time, experienced superintendent, who shall be constantly in charge of the installation of the work, together with all skilled workmen, fitters, metal workers, welders, helpers, and laborers required to unload, transfer, erect, connect, adjust, start, operate, and test each system.

F. Manufacturer’s Recommendations

1. In addition to the requirements indicated in the drawings and specifications; manufactured products, articles, materials, and equipment shall be applied, installed, connected, erected, used, cleaned, conditioned, inspected, started-up, tested, operated, and commissioned in accordance with the manufacturer's current printed recommendations. Keep copies of such printed recommendations at job site and make them available as required. When conditions on this project are not covered by the manufacturer's printed recommendations, at the discretion of and as requested by the owner's representative, the manufacturer's authorized representative shall review the conditions and provide written supplemental recommendations to address the special situation. If the manufacturer’s recommendations are in conflict with the requirements of the drawings and specifications, the Contractor shall advise the Owner’s representative of the conflict prior to purchase and installation.

G. Space Limitations

1. MEP systems and equipment shall be provided and installed to accommodate the structural systems, architectural systems, MEP systems, other trades, and the physical space limitations of the project.

2. Equipment shall be provided which will fit into the physical spaces provided and indicated, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearance in accordance with the Code requirements, the requirements of the local inspection department, and the equipment manufacturers' recommendations.

3. In the preparation of Drawings, a reasonable effort to accommodate equipment manufacturers' space requirements has been made. However, since space requirements, equipment arrangement, access and service clearances, conduit, piping, ductwork and other connections, manufacturers' recommendations, and other variables vary according to each manufacturer; due to the multiple manufacturers listed for each product and the possible number of different equipment combinations; the responsibility for access, proper fit, coordination, and accommodating the space limitations of the project rests with the Contractor. Refer to shop drawing submittal requirements herein for additional requirements.

4. MEP systems and equipment shall be installed to maximize headroom clearance below the bottom of material. In areas with ceilings or future ceilings provided by tenants; MEP systems and equipment shall be installed to maximize vertical clearances between the bottom of material and top of specified ceiling construction. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, lighting, and other MEP systems shall be prepared to indicate compliance with the Space Limitations and Maximum Headroom Clearance requirements. Conflicts with vertical and horizontal clearances, parking spaces, drive aisles, ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data, clouded, and clearly marked so as to bring the conflict to the attention of the Architect in a timely manner prior to installation. For the purpose of definition, the term MEP refers to all systems, material, equipment, services, means and methods described the Division 21, 22, 23, 25, 26 and 28 documents including the Division 21, 22, 23, 25, 26 and 28 specifications and drawings. Refer to the specifications for additional requirements.

5. Refer to the drawings for additional requirements.

6. Physical dimensions and arrangements of equipment to be installed shall be subject to the Architects' review.

7. Within any electrical room containing panelboards a minimum headroom of 6'-6" shall be maintained throughout the room measured from the finished floor to the lowest ceiling mounted element or component. Where the electrical equipment exceeds 6'-6" in height, the minimum headroom shall not be less than the height of the equipment. No ceiling mounted element including, but not limited to, light fixtures, conduit, ductwork, or piping shall be below the 6'-6" elevation or elevation established by the height of the electrical equipment. This is intended to ensure NEC required electrical equipment working space is maintained.

H. Coordination with Installation Conditions

1. The requirements of the Division 21, 22, 23, 25, 26 and 28 systems shall be coordinated

with the installation conditions. The Contractor shall visit the site, survey the installation conditions, and verify exact measurements. Means and methods of building access, rigging, and modifications to installation conditions to accommodate the installation of new Division 21, 22, 23, 25, 26 and 28 work shall be the responsibility of the Contractor. Disassembly, reassembly, and certification testing of equipment as required to accommodate the installation conditions shall be included in the base bid. Prior to the ordering, purchasing, and fabricating of Division 21, 22, 23, 25, 26 and 28 systems, coordination with installation conditions shall be completed by the Contractor.

I. Coordination Between Divisions 21, 22, 23, 25, 26 and 28:

1. The Division 26 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 25, and 28 Contractors to provide coordination between the Division 21, 22, 23, 25, 26, and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 21, 22, 23, 25, and 28, it shall be the responsibility of the Division 26 Contractors to acquire from the Division 21, 22, 23, 25, and 28 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

J. Commissioning:

1. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience, and expertise at the general contractor and sub-contractor levels to properly install, check, verify, supervise, start-up, trouble-shoot, de-bug, pre-test, commission, and make the necessary corrections to all MEP systems in accordance with the Contract Documents. The Contractor shall be responsible for the operational demonstration of the completed MEP systems to the Owner’s Representative, Architect and Engineer. The Contractor shall complete the MEP systems in accordance with the contract documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems prior to requesting that the Owner’s Representative, Architect, and Engineer attend the operational demonstrations of the MEP systems. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner. Refer to the various sections of the specifications for additional commissioning requirements.

K. Additional Information

1. Information provided by the Engineer to the Contractors during construction must be reviewed and approved by the Architect and Owner's Representative. The Contractor shall not proceed without prior review and authorization from the Architect and Owner's Representative. The Contractor shall not proceed with work that will increase the cost of the project or impact the project schedule without prior review and authorization from the Architect and Owner's Representative. If information provided by the Engineer to the Contractor will increase the cost of the project or impact the project schedule, the Contractor shall advise the Architect and Owner's Representative in a reasonably timely manner so as to minimize the cost and impact to the project. Refer to the General Conditions of the Contract and the Architectural documents for additional information.

1.3 SHOP DRAWINGS, PRODUCT DATA AND SAMPLES

A. After the Contract is awarded but prior to proceeding with the Work requiring final reviewed shop drawings, product data and samples, the Contractor shall provide the Architect and Engineer with a submittal log schedule indicating when each submittal is to be sent to the Architect and Engineer for review in order for the Architect and Engineer to schedule personnel and plan accordingly. Unless otherwise indicated, a minimum period of two (2) weeks, exclusive of transmittal time, will be required in the Engineer's office each time a shop drawing, product data and/or samples are submitted or resubmitted for review. This time period will commence once the shop drawing, product data and/or samples are received in the Engineer's office contingent upon the shop drawing, product data and/or samples being received in the Engineer's office before 3:00 PM. If the shop drawing, product data and/or samples are received in the Engineer's office after 3:00 PM, the time period will not start until the next business day. This time period shall be considered by the Contractor when scheduling the Work. The submittal log schedule should be updated and reissued by the Contractor on a regular basis to avoid submittal review delays. If a submittal arrives before or after the scheduled time indicated within the Contractor's submittal log schedule, or if a number of submittals arrive at the same time without scheduled notification, the Architect and Engineer may not be able to review all of the submittals within the minimum two (2) week time period. The Contractor shall obtain complete shop drawings, product data and samples from the manufacturers, suppliers, vendors, subcontractors, and sub-subcontractors for all materials, systems and equipment specified herein; and submit data and details of such materials, systems and equipment for review by the Architect and Engineer. Prior to submission of the shop drawings, product data and samples to the Architect and Engineer, the Contractor shall review and certify that the shop drawings, product data and samples are in compliance with the Contract Documents. The Contractor shall check all materials and equipment after their arrival on the job site and verify their compliance with the Contract Documents. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience and expertise at the general contractor and sub-contractor levels to properly check, verify, and correct if necessary shop drawing submittal data to ensure compliance with the Contract Documents prior to delivery to the Architect and Engineer. The Division 26 shop drawing submittal requirements, including but not limited to the Division 26 certification requirements, shall be in addition to the requirements indicated in the Architectural Divisions of the Contract Documents.

B. The Architect's approval and Engineer's review of shop drawings, product data, and samples shall not relieve the Contractor of the responsibility for dimensions or errors that may be contained therein, or for deviations from requirements in the Contract Documents. It shall be clearly understood that the Engineer's noting some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents shall govern the Work and are neither waived nor superseded in any way by the review of shop drawings, product data and samples.

C. Before submitting shop drawings, product data and samples the Contractor shall certify that data and details set forth on each shop drawing, product data, or samples for each item of equipment and materials complies with the Contract Documents for this Project. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the Contractor, or by a person duly authorized to sign binding agreements for the Contractor. Unless certified, shop drawings, product data and samples will not be reviewed, and will be returned unchecked to the Contractor. Certifications shall be in the form of rubber stamp impressions or a digitally certified stamp which state:

> I hereby certify that this shop drawing, product data, and/or sample has been checked prior to submittal and that it has been coordinated with existing conditions and complies in all respects

with the requirements of the Contract Documents and physical space limitations for the Project.

(Name of the Electrical Subcontractor)

Signed:_____

Position:_____

Date:_____

D. The Contractor shall submit shop drawings, product data and/or samples, equipment and other items listed in Section 26 00 70 and for any other equipment for which such submittals may be required in various Sections of the Specifications. Prior to submitting shop drawings, product data and samples for all items of equipment listed in Section 26 00 70 and shown with an asterisk (*), each submittal package including subsequent submittal packages shall be certified by the manufacturer as to performance and compliance with Contract Documents. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the manufacturer, or be a person duly authorized to sign for the Owner, Partner, or corporation.

1. Unless certified, shop drawings, product data and/or samples will not be reviewed and will be returned unchecked to the Contractor. The certification shall be as follows:

   I hereby certify that the equipment shown on this shop drawing, product data and/or sample, has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents for this Project. I further certify that all data shown herein as to performance dimensions, construction, materials and other pertinent items is true and correct.

   (Name of the Manufacturer)

   Signed:_____

   Position:_____

   Date:_____

E. Inadequate or incomplete shop drawings, product data and/or samples will not be reviewed by the Architect or the Engineer and shall be returned to the Contractor for resubmittal.

F. The Contractor shall observe the following rules when submitting shop drawings, product data and samples:

1. Identification: Each shop drawing shall indicate in the lower right hand corner, and each product data brochure shall indicate on the front cover the following: title of the sheet or brochure; name and location of the Project, names of the Architect and Engineer, Contractor, Subcontractor, manufacturer, supplier, vendor; the date of submittal; and the date of each correction and revision. So far as is practical, each shop drawing, product data and/or sample shall bear a cross-reference note to the page number or numbers of sheets of the Drawings and/or Specifications showing the Work. Unless the above information is included, the submittal will be returned for resubmittal.

2. Quality Assurance: Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor's option it will also be acceptable to make a copy of the specifications and mark "C" or "NC" by each specification paragraph. A hand written marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

3. Shop drawings shall be done in an easily legible scale and shall contain sufficient plans, elevations, sections and schematics to clearly describe the apparatus. Drawings shall be prepared by an engineer-draftsman skilled in this type of work. Unless otherwise indicated, all sheet metal, piping and similar shop drawings shall be drawn to at least 1/4" = 1'-0" scale. Typical floor sheet metal ductwork shop drawings may be drawn at 1/8"=1'-0" scale. The Contractor shall submit shop drawings as described below. Shop drawings which do not comply with these requirements will be returned for resubmittal. Refer to "Cooperation with Other Trades", this section, for additional composite shop drawing requirements. Unless otherwise indicated in the Contract Documents, the submittal shall consist of three prints of the shop drawings. The Architect and Engineer will each retain one print and the remaining copies will be returned to the Contractor.

4. Product data to be submitted shall be published by the manufacturers and shall contain complete and detailed engineering and dimensional information. The Contractor shall submit product data as described below. Product data which does not comply with these requirements will be returned for resubmittal. Unless otherwise indicated in the Contract Documents, the Contractor shall submit eight (8) copies of each product data brochure. The Architect and Engineer will each retain one copy of the submittal and the remaining copies will be returned to the Contractor. Product data submitted shall contain only information relevant to the particular equipment or material to be furnished. The Contractor shall not submit catalogs which describe several different items in addition to those items to be used, unless all irrelevant information is marked out, or unless relevant information is clearly marked. The submittal shall consist of product data from each manufacturer. Product data from each manufacturer shall be identified and submitted separately.

5. Coordination between Division 21, 22, 23, 25, 26 and 28: Each Division 26 shop drawing and product data submitted for review, which indicates electrical service to be provided to Division 21, 22, 23, 25, and 28 equipment, shall contain a statement indicating that the Division 26 Contractor has acquired from the Division 21, 22, 23, 25, and 28 Contractor the necessary electrical characteristics of the actual Division 21, 22, 23, 25, and 28 equipment to be provided.

6. Shop Drawing and Product Data - Submittal Review Status:

    a. Shop drawing and product data submittals will be returned marked with one of the following status:

        1) "REVIEWED NO EXCEPTIONS"
        2) "REVIEWED EXCEPTIONS NOTED"

3) "INFORMATION ONLY"
4) "REVISE AND RESUBMIT"
5) "REJECTED"

b. If stamped “REVISE AND RESUBMIT" or “REJECTED”, the shop drawing or product data shall be revised in accordance with the Contract Documents. If marked "REVIEWED, EXCEPTIONS NOTED" documentation that addresses only the noted exceptions in the form of a letter, supplemental information, or complete resubmittal shall be forwarded to the Owner, Architect and Engineer for record purposes only. If marked "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” no additional submittal will be required.

c. If the copy stamped "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” is altered for any reason after it has been stamped, the "REVIEWED NO EXCEPTIONS" or “INFORMATION ONLY” shall automatically be voided and the submittal status revised to "REVIEWED EXCEPTIONS NOTED", "REVISE AND RESUBMIT", or “REJECTED” depending on the alterations.

d. All work shall be done in accordance with shop drawings stamped "REVIEWED NO EXCEPTIONS", "REVIEWED EXCEPTIONS NOTED", or “INFORMATION ONLY” insofar as these are in agreement with the Contract Documents. Wherever differences occur between the shop drawings and the Contract Documents, the Contract Documents shall govern the Work.

G. Record Shop Drawings

1. After receiving the Engineer’s and Architect’s final approval for each shop drawing, the Contractor shall provide the Engineer with AutoCAD files of each final approved shop drawing. All of the previous review comments by the Engineer and Architect associated with each shop drawing shall be incorporated into the AutoCAD files of each shop drawing prior to sending the AutoCAD files to the Engineer. The AutoCAD files shall be in AutoCAD 2015 version software or later version as approved by the Engineer. The Contractor shall be responsible for contacting the Engineer to determine the approved AutoCAD software to use prior to preparing the AutoCAD files.

H. Maximum Ceiling Clearance and Headroom Requirements

1. All material for systems and equipment shall be installed to accommodate the structural systems, architectural ceiling construction, base building and future tenant lighting systems, and other physical space limitations of the project. All systems and equipment shall be installed to maximize clearances between the bottom of material and top of ceiling construction. All systems and equipment shall be installed to maximize headroom clearance below the bottom of material. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, and other systems that will be located where ceiling plenum clearances and headroom clearances are limited; shall be prepared to address these issues. Conflicts with ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data and clearly marked so as to bring the situation to the attention of the Engineer.

I. Equipment and Layout Deviations

1. Layout, connections, supports, access & maintenance clearances, related architectural enclosures, and structural supports have been based on selected products and manufacturers. In some cases, the manufacturer’s model number has been provided.

Other products by specified manufacturers may be used if they comply with requirements of the contract documents including but not limited to capacity ratings. However, where the Contractor proposes to use an item of equipment that has alternative but equivalent minimum capacity ratings or that requires any redesign of the structure, partitions, foundations, piping, wiring or any other part of the mechanical, electrical, structural, civil, utility, or architectural systems; all such redesign, and all new drawings and detailing required therefore, shall be prepared by the Contractor at the Contractor's own expense as approved by the Owner's Representative. Where such approved deviation requires a different quantity, capacity, and arrangement of ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, and starters from that specified or indicated on the drawings, the Contractor shall furnish and install any such ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, starters and any other additional equipment required by the system, at no additional cost to the Owner.

1.4 RESPONSE TO SHOP DRAWING, SUBMITTAL, AND FIELD REPORT REVIEW COMMENTS

A. Contractor shall respond in written form to each shop drawing, submittal, or field report review comment from the Engineer within 10 business days from the date indicated on the Engineer's review form. Contractor's written response to each review comment should indicate the date and specifically what corrective action was or will be taken and include the name of the person and associated company responding to each review comment. Refer to the Architectural documents for additional requirements.

1.5 WARRANTY, SERVICE, AND REPLACEMENT PARTS

A. All equipment supplied for this Contract shall be free from defects in material, workmanship, and title, and shall be of the kind and quality described herein. If it appears within one (1) year from the date of the building final acceptance by the Owner that the equipment does not meet the warranties specified above or the equipment is recalled by the equipment manufacturer, the Contractor shall correct any defect or replace any recalled equipment, including nonconformance with these Specifications, at no additional cost to the Owner.

B. The manufacturers for each of the various selected Division 26 pieces of equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner's acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

1.6 CONFERENCE PRIOR TO START OF WORK

A. Immediately upon the award of this Contract, but prior to commencing any Work, the Contractor together with designated major Subcontractors shall confer with the Owner's Representative, the Architect and Engineer concerning the Work under this Contract.

B. Contractors and Subcontractors attending this meeting shall include but not be limited to the following: General; Mechanical; Plumbing; Fire Protection; Automatic Temperature Controls and Building Management; Sheet Metal; Insulation; Electrical; Fire Alarm; and Testing, Balancing and Adjusting. Contractor and Subcontractor representatives shall be familiar with the Drawings, Specifications and shop drawing submittal requirements and be prepared to intelligently discuss the project requirements.

C. The conference will be at a mutually agreed place and acceptable time.

1.7 ELECTRONIC FILE TRANSFER AGREEMENT

A. Release of the Dewberry CAD, BIM, and other electronic files will require the User of the files to sign the Dewberry Electronic File Transfer Agreement.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials and equipment shall be new and in good condition. The commercially standard items of equipment and the specific names mentioned herein are intended to identify standards of quality and performance necessary for the proper functioning of the Work.

B. Since manufacturing methods vary, reasonable minor variations are expected; however, performance and material requirements specified are the minimum standards acceptable. The Engineer retains the right to judge equality of equipment that deviates from the Contract Drawings and Specifications.

2.2 PRODUCTS, COMPONENTS AND MATERIALS IN PLENUMS

A. All products, components, materials and composite materials within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723 as required by applicable mechanical code.

2.3 EQUIPMENT GUARDS

A. Provide equipment guards for the safe operation of equipment as recommended by the equipment manufacturer and as required by applicable codes.

2.4 EQUIPMENT MOUNTING/SUPPORT DEVICES

A. Unless otherwise required to accommodate the particular requirements and unique conditions of this project, all floor, wall or ceiling mounting/support devices shall be factory fabricated devices as provided by the equipment manufacturer for the intended application. In addition to the factory-fabricated devices, the Contractor shall provide additional support members, framework, mounting hardware, brackets, steel angles, anchors, restraints, support stands, attachment devices, and other miscellaneous mounting and support devices as required for adequate equipment support.

B. The Engineer retains the right to judge the quality of the design, application, and installation of all field and shop fabricated mounting/support devices using steel angle, steel channels, Uni-Strut, brackets, all-thread rods, etc. Any mounting or support devices that will rust, corrode, or in the opinion of the Engineer become unsightly or a maintenance problem will not be acceptable.

C. Support devices not factory finished shall be hot dipped galvanized or properly primed and painted with a minimum of two coats of rust inhibiting paint. Refer to the Architectural Painting Systems for additional requirements.

D. All-thread rods shall be galvanized or cadmium plated.

E. All plywood used as mounting boards in association with Division 26, unless specified otherwise by the Architect, shall be APA C-D plugged interior 5/8" to 3/4". All plywood used shall be painted (front, back and edges) with a minimum of one prime coat and one finish coat prior to equipment installation. Refer to the Architectural Specifications for additional requirements. Color shall be determined by the Architect.

## 2.5 EQUIPMENT PADS AND ANCHOR BOLTS

A. Concrete pads for various pieces of equipment will be provided under another Division. Pads shall be provided in main switchboard rooms, all areas with floors below grade, penthouse equipment rooms, electrical closets, and in any other location as shown on the drawings. This shall include floor-mounted equipment, equipment mounted on legs and pipe support stands. Unless otherwise noted, equipment pads shall generally conform to the shape of the piece of equipment it serves with a minimum 3" margin around the equipment and supports. Pads shall be a minimum of 4" high and made of a minimum 28 day, 2500 psi concrete reinforced with 6" x 6" 6/6 gauge welded wire mesh. Top and sides of pad shall have smooth finishes, equal to those of the floors, with all external corners either bull-nosed to a 3/4" radius or provided with a 3/4" chamfer. Shop drawings stamped "FINAL REVIEW" or "REVIEWED EXCEPTIONS NOTED" shall be used for dimensional guidance in sizing pads.

B. Unless otherwise noted furnish and install galvanized anchor bolts for all equipment placed on concrete equipment pads or on concrete slabs. Minimum dimension between anchor bolts and edge of slab shall not be less than 3". Bolts shall be of the size, type and number recommended by the manufacturer of the equipment and shall be located by means of suitable templates provided by the equipment manufacturer. When equipment is placed on vibration isolators, the equipment shall be secured to the isolator and the isolator secured to the floor, pad, or support as recommended by the vibration isolation manufacturer.

C. Provide two parallel structural galvanized steel I-beams or channel sills poured integrally within the 4” concrete equipment pad for each switchboard, switchgear, UPS, or other electrical distribution equipment over six feet in width with shipping splits and internal bussed connections. Bolt the equipment frame to the I-beams or channel sills.

## 2.6 ACCESS DOORS

A. Wherever access is required through walls, ceilings or fire-rated enclosures to valves, fire dampers, heating coils, damper actuators, drain line cleanouts, or other concealed equipment installed under this Division, unless otherwise noted in the architectural documents, the Contractor shall furnish a hinged access door and frame as follows:

1. Drywall construction--Milcor Style DW.

2. Finished acoustical tile ceiling--Milcor Style AT in applications where there is not adequate clearance space to properly push an acoustical ceiling tile up for removal and replacement.

3. Finished plaster ceiling--Milcor Style AP.

4. Finished plaster walls or ceramic tile--Milcor Style M.

5. Plaster or masonry walls and ceilings outside offices and in other finished areas exposed to view--Milcor Style K or M.

6. Access doors shall be "B" label fire construction where required.

7. Access doors shall be installed under another Division.

8. No access door shall be installed until location and type have been approved by the Architect.

2.7 ESCUTCHEONS

A. Furnish and install heavy chrome plated or nickel plated steel plates of approved pattern on all conduit passing through walls and ceilings in finished areas. Escutcheons shall be B & C No. 10 or approved equal with concealed hinges. Pattern shall be approved by the Architect.

2.8 SEALANTS AND ADHESIVES

A. All sealants and adhesives shall comply with the current Volatile Organic Compound (VOC) content limits of the latest edition of the South Coast Air Quality Management District (SCAQM) Rule 1168 entitled “Adhesive and Sealant Applications”. Piping sealants shall have no more than 250 grams/liter VOC content for all sealants and adhesives to be used within the building interior locations..

B. The Contractor shall submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each adhesive and sealant used within the building interior locations.

PART 3 - EXECUTION

3.1 SCAFFOLDING, RIGGING, HOISTING

A. The Contractor shall furnish all scaffolding, rigging, hoisting, and services necessary for erection and delivery of any equipment and apparatus furnished to the job site. Remove same from job site when no longer required.

3.2 SLEEVES, CUTTING, AND PATCHING

A. The Contractor shall be responsible for the timely placing of openings and sleeves for all conduit passing through walls, partitions, beams, floors, and roof while the same are under construction. A conduit sleeve shall be round and one size larger than the size of conduit it serves, except where "Thunderline/Link Seal" casing seals are used in sleeves through walls below grade. Sleeves set in concrete floor construction shall be 18 gauge galvanized steel. Sleeves shall extend two inches above the finished floor. All conduit passing through concrete or masonry walls shall have standard weight galvanized steel sleeves or plastic “hole-outs”. Block out(s) and core drill(s) shall have a 2" curb made of 2" x 2" x 1/4" angle iron welded at corners set in mastic or 2" wide by 2" high concrete curb extending above finished floor. Sleeves shall be set flush with finished wall. Caulk around conduit penetrating floors, roof, and fire-rated wall construction with sufficient layers of fire safing insulation and further seal off opening between conduit and sleeve with a fire rated non-hardening mastic. Fill all unused sleeves and block-out space with fire safing material. All fire safing materials shall be UL listed and comply with ASTM E 814 (UL 1479). The fire safing system used shall maintain the fire resistance rating of the building component that is penetrated.

B. Sleeves penetrating walls below grade shall be “WS Wall Sleeves” by Pipeline Seal and Insulator, Inc. Houston, Texas, with 2” water seal stop. The water seal stop shall be located in the middle of the wall. Seal off annular opening between pipe and sleeve with "Thunderline/Link-Seal" casing seal as manufactured by Pipeline Seal & Insulator, Inc., Houston, Texas. The pipe sleeve shall be sized to accommodate the “Thunderline/Link-Seal”

casing seal. Casing seals shall be Model “C” and shall be sized per the manufacturer’s recommendations.

C. If holes and/or sleeves are not properly installed and cutting and patching becomes necessary, it shall be done at no expense to the Owner. The Contractor shall undertake no cutting or patching without first securing the Architect's written approval.

D. The location and size of all penetrations and openings through existing walls, slabs, beams, roofs and other existing building construction shall be provided by the contractor to accommodate the new work. Penetrations and openings through existing building construction shall be reviewed by the structural engineer and architect prior to cutting the penetration. Unless otherwise required by the architectural and structural divisions, metal sleeve inserts will not be required in conduit penetrations except as required to meet UL fire rating requirements. The contractor shall perform a structural survey, including x-rays or graphs of all floor slabs and supporting walls to locate existing structural bars, cables, in slab conduits and any other interferences that may be in conflict with proposed sleeve/penetration locations. The x-ray or graph shall be performed prior to any cutting or chipping of structural supporting walls. In the event of a conflict, the contractor shall notify the architect prior to proceeding with the work.

## 3.3 EXCAVATION AND BACKFILL

A. The Contractor shall make all necessary excavations, cutting of paving, concrete, etc., and do all backfilling and paving repairs necessary for the proper execution of the Work. Remove all dirt and debris out of and away from the building as directed. Backfill shall be mechanically compacted to a density of 95% of the maximum dry density at optimum moisture content as determined by the Standard Proctor Compaction Test.

B. Backfill shall be compacted and repairs to paving or concrete shall be accomplished to the satisfaction of the Architect and the Local Authorities.

C. See the various Sections of the Division for additional excavation and backfill requirements.

## 3.4 PAINTING

A. Painting, except as specified or indicated otherwise, shall be done under another Division. This Division shall cooperate with the other Division to determine the size of equipment, sizes and lengths of pipes, etc., to be painted.

B. Equipment furnished under this Division shall be factory-finished. If the factory finish is damaged during shipment, installation, etc., it shall be repainted subject to the Architect's approval.

## 3.5 DRAYAGE AND HAULING

A. Include all drayage, hauling, hoisting, and placement in the building of equipment specified herein. The Contractor shall be responsible for the timely introduction of equipment to the Project. If any item of equipment is received prior to the time it is required, the Contractor shall be responsible for its proper storage and protection until such time as it may be required. The Contractor shall pay for all cost of demurrage or storage.

## 3.6 IDENTIFICATION OF CIRCUITS AND EQUIPMENT

A. Electrical equipment shall be identified by means of nameplates permanently attached to the equipment. Refer to Section 26 05 53 for additional requirements.

B. Cardholders and directory cards shall be provided for circuit identification in panelboards. Cardholder shall be located on inside of panel door. Directory cards shall be typewritten. Circuit descriptions shall include locations and names of items of equipment served.

C. Circuits and pull wires in empty conduit shall have tags attached to wiring at points where runs are interrupted at junction boxes or terminated in panels, boxes, etc. Feeder or branch circuit numbers shall be indicated.

D. Tag symbols shall correspond to the identifications on the record drawings.

E. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor's shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

## 3.7 EQUIPMENT NOISE AND VIBRATION

A. It is the intention to specify and for the Contractor to provide equipment and systems that, as defined herein, shall be quiet and free of apparent noise and vibration in operation.

B. It shall be the responsibility of the Contractor to obtain equipment that is quiet in operation as compared to the other available equipment of its size, capacity, and type, and to install equipment so that a minimum amount of noise and/or vibration is transmitted to the building.

C. Any additional precautions deemed necessary to provide a quiet installation shall be done as part of the Work of the Contractor, subject to review by the Engineer, without additional cost to the Owner. After the system is in operation, it shall be the responsibility of the Contractor to make any changes to equipment or Work installed as may be required to provide a system which is quiet in operation as defined herein.

## 3.8 SPARE PARTS

A. The Contractor shall provide all necessary spare parts for the equipment supplied in this Project as indicated hereafter and in various other Sections of the Specifications. Spare parts shall be obtained from the original equipment manufacturer.

## 3.9 DOCUMENTATION

A. For all testing, training and review procedures the Contractor shall maintain adequate records in written form that clearly documents the event. The written documentation for each testing, training and review procedure shall include but not be limited to the following: Name and company of attending personnel, general description of procedure(s), date(s), time of day, deficiencies found, corrective action taken, and nature of final results. Documentation records shall be kept in an orderly, legible and up to date format. Contractor's documentation records shall be accessible for review by the Owner Representative upon request. Each testing, training and review procedure document shall bear the name and signature of the authorized representative of the General Contractor, Mechanical Contractor, and manufacturer or Subcontractor if applicable.

## 3.10 COMPLETION, COMMISSIONING AND FINAL REVIEW OF THE DIVISION 26 SYSTEMS

A. Start-up: After the Contractor and equipment manufacturer representative determine that the equipment and related systems have been properly installed and are ready for start-up; the equipment manufacturer representative shall perform the manufacturer's start-up and check-out procedures to ensure proper operation of the equipment in accordance with the contract documents and the manufacturer's recommendations. Start-up and check-out of each piece of equipment shall be in accordance with the equipment manufacturer's recommendations and warranty requirements. Start-up services shall be performed by factory-trained personnel of the equipment manufacturer or factory-trained personnel authorized by the manufacturer and their official representatives. Start-up personnel shall be certified by the equipment manufacturer as properly trained in the manufacturer's start-up and check-out procedures. Provide documentation of the start-up and check-out procedures in accordance with the contract documents and the manufacturer's recommendations. Documentation of the start-up and check-out procedures shall be included within and submitted with the Operational and Maintenance Manuals.

B. Trouble-shooting, Debugging, and Pre-testing: After Start-Up but prior to the Operational Demonstrations; the contractor shall perform operational pre-testing, trouble-shooting, and de-bugging of all equipment and systems. At the completion of the Trouble-shooting, Debugging, and Pre-testing process, all equipment and systems shall be complete and operating in accordance with the requirements of the contract documents. Trouble-shooting, Debugging, and Pre-testing of equipment and systems shall be performed to the extent necessary for the Contractor to determine that all systems are complete and operating in accordance with the Contract Documents. Prior to final Commissioning and Operational Demonstrations all equipment and systems shall be certified by the Contractor as complete and operating in accordance with the Contract Documents.

C. Operational Demonstrations and Final Commissioning: Prior to Substantial Completion and within a schedule mutually agreed to by the contractor and the Owner's Representative, the Contractor shall demonstrate the operation of all equipment and systems as necessary to demonstrate compliance with the Contract Documents. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner.

D. Final Review and Owner Acceptance:

1. After completion of the Operational Demonstrations, Final Commissioning, and Owner's Close-out Package requirements; at a time designated and mutually agreed by the Contractor and Owner's Representative; a final walk-through review of the project will be performed by the Contractor and Owner's Representative. The review will be of a general nature with the intent of addressing outstanding issues prior to the Owner's final acceptance of the project.

2. The entire system shall be clean and operating properly with all systems balanced and all controls adjusted. All prohibited markings shall be removed from all switchboards, panelboards, motor control centers, switches, etc., and the equipment shall be clean and in operating condition.

3. Certificates, Documents, and Close-out Package required herein shall be complete, in order and presented to the Architect at least two weeks prior to the review.

4. After the review, any changes or corrections noted as necessary for the Work to comply with these Specifications and the Drawings shall be accomplished without delay in order to secure final acceptance of the Work.

E. Schedule: The Contractor shall complete the MEP systems in accordance with the Contract Documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems a minimum period of time prior to the date of “Substantial Completion” as established by the Commissioning Agent. This period of time prior to the date of “Substantial Completion” shall be reserved by the Contractor for the Owner’s Representative, Commissioning Agent, Architect, and Engineer to witness the operational demonstrations of the completed fully commissioned MEP systems. If more than this established time period is deemed necessary by the Contractor for the operational demonstrations, the Contractor shall notify the Owner’s representative and Commissioning Agent in written form and determine an operational demonstration schedule mutually agreeable to the Owner’s Representative, Commissioning Agent, and Contractor that completes the operational demonstrations prior to “Substantial Completion”. Additional costs incurred by the Owner related to reviewing, witnessing, testing, inspecting, and commissioning of the MEP systems by the Commissioning Agent, Architect and Engineer after the date of “Substantial Completion” shall be reimbursed by the Contractor to the Owner. As part of the shop drawing submittal process, the Contractor shall provide a detailed schedule of completion indicating when each MEP system will be complete and ready for owner acceptance. The schedule shall indicate time periods and milestone dates of Start-up; Trouble-shooting, Debugging, and Pre-testing; Operational Demonstrations and Final Commissioning; Final Review and Owner Acceptance.

F. Additional Requirements:

1. Commissioning Requirements: Refer to specification Section 26 08 00 for additional building commissioning general requirements.

2. Owner’s Close-out Package: Refer to the Owner’s Close-out Package requirements within Section 26 00 10 for additional requirements.

3. Refer to the various sections of the specifications for additional requirements.

## 3.11 OPERATING INSTRUCTION

A. The Contractor shall provide the services of a factory trained specialist to supervise the operation of all equipment specified herein and to instruct the Owner's operators for a five day, eight hours per day, operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. See individual Sections for additional instructions by manufacturer's trained specialists. In addition, a factory trained specialist shall be provided for an 8-hour operating instruction period for the standby generator and an 8 hour operating instruction period for the main switchboard.

B. The Owner shall be notified in writing at least five days before each operating instruction period begins. The Contractor shall not commence instruction period until the Owner has issued a written acceptance of the starting time.

C. Training schedule, duration, and subject matter shall also be in accordance with a training syllabus provided by owner's operating staff. This syllabus will be provided prior to the final review.

## 3.12 OPERATING AND MAINTENANCE BOOKS

A. The Contractor shall provide operating instruction and maintenance data books for all equipment and materials furnished under this Division.

B. Submit three (3) hard copies and three (3) electronic copies of the operating and maintenance data books for review at least eight weeks before final review of the Project. Assemble all data in a completely indexed volume or volumes and identify the size, model, and features indicated for each item. The Engineer's review period for the operating and maintenance data books shall be a minimum of 4 weeks, exclusive of transmittal time. The Contractor's schedule shall plan to accommodate this review period requirement.

C. Maintenance instruction manuals shall include complete oiling, cleaning, and servicing data compiled in clearly and easily understandable form. Data shall show all model numbers of each piece of equipment, complete lists of replacement parts, motor ratings, and actual loads.

D. Include the following information where applicable:

1. Identifying name and mark number as indicated on the Drawings and in the Specifications.

2. Locations (where several similar items are used, provide a list).

3. Complete nameplate data.

4. Certified record drawings and "Final Reviewed" or completely corrected shop drawings and/or submittals.

5. Parts lists.

6. Performance curves and data.

7. Wiring Diagrams.

8. Start-up and Check-out procedures and checklists.

9. Manufacturers' recommended operating and maintenance instructions for each piece of equipment requiring maintenance, with all non-applicable information deleted. Required routine maintenance actions shall be clearly identified. This shall include but not be limited to the manufacturer's service and repair manual.

10. Operation and maintenance manuals for each piece of lighting equipment and lighting controls with routine maintenance clearly identified including, at a minimum, a recommended relamping program and a schedule for inspecting and recalibrating all lighting controls.

11. Submittal data stating equipment rating and selected options for each piece of equipment requiring maintenance.

12. Submittal data indicating all selected options for each piece of lighting equipment and lighting controls.

13. Names and addresses of at least one qualified service agency.

14. A complete narrative of how each system is intended to operate.

15. A complete narrative of how each lighting control system is intended to operate including recommended settings.

16. Additional information as recommended by the equipment manufacturer or system provider for the proper operation, maintenance, and warranty of the equipment and systems.

3.13 RECORD DRAWINGS

A. The Contractor shall maintain on a daily basis at the project site a complete set of "Record Drawings". The “Record Drawings” shall consist of a set of black line prints and AutoCAD files of the Contract Drawings for this Division. The black line prints shall be marked and the AutoCAD files shall be electronically updated regularly by the Contractor during the construction period to reflect an accurate dimensional record of all buried or concealed work. In addition, the set of black line prints shall be marked and the set of AutoCAD files shall be electronically updated regularly to show the precise location of concealed work and equipment including buried, concealed or embedded piping, conduit and valves and all changes and deviations in the mechanical work from that shown on the Contract Documents. This requirement shall not be construed as authorization for the Contractor to make changes in the layout or work without definite instructions from the Architect. The set of black line prints and AutoCAD files of the Contract Drawings for this Division shall remove or blacken out the Engineer's seal and Engineer's firm name.

B. Prior to commencing work, the Contractor shall need to purchase from the Architect or Engineer a set of black line prints to be used for the "Record Drawings" and obtain from the Engineer a set of AutoCAD files of the Contract Drawings for this Division in preparing “Records Drawings” for the project based on the Contractor’s work. The AutoCAD files shall be in AutoCad 2015 version software or later version as provided by the Engineer. In order to obtain a set of AutoCAD files of the Contract Drawings for this Division from the Engineer, the Contractor shall need to contact the Engineer, Architect or Client and obtain a written release and indemnification statement from the Engineer. This release and indemnification statement shall be signed by an owner, partner, corporate officer or other person duly authorized to agree to and sign legally binding documents for the Contractor’s respective company. Upon the Engineer receiving the signed unedited release and indemnification statement back from the Contractor, the Engineer will forward the AutoCAD files of the Contract Drawings for this Division to the Architect or Client who in turn will forward the AutoCAD files of the Contract Drawings for this Division to the Contractor for his use in preparing “Records Drawings” for the project based on the Contractor’s work. If the Contractor desires to have architectural and structural information in preparing “Records Drawings” for the project, the Contractor shall need to contact the Architect and Structural Engineer to obtain the pertinent architectural and structural information directly from the Architect and the Structural Engineer.

C. Record dimensions shall clearly and accurately delineate the Work as installed; locations shall be suitably identified by at least two dimensions to permanent structures.

D. The owner's representative shall retain the right to periodically review the “Record Drawings” as the project progresses. Final acceptance of as-built “Record Drawings” shall be conditional upon the Contractor clearly "red-marking" all changes from the contract documents, including "request for information" changes and other supplemental instruction changes, directly on the black line set of prints and clouding all of the changes in the electronic AutoCAD files. All of the clouded changes on each drawing in the electronic AutoCAD files shall be on a separate drawing layer which can be turned off without affecting the appearance of each drawing.

E. The Contractor shall mark all "Record Drawings" on the front lower right hand corner with a rubber stamp impression or an AutoCAD image that states the following:

"RECORD DRAWINGS (3/8" high letters)
To be used for recording Field

Deviations and Dimensional Data
Only". (5/16" high letters)

F. Upon completion of work, the Contractor shall certify all "Record Drawings" on the front lower right hand corner adjacent to the above marking with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWING CERTIFIED CORRECT (3/8" high letters)

(Name of Electrical Subcontractor)

By: _____

Date:_____

For all black line prints to be used for the "Record Drawings", the Contractor must sign the above certification. For all AutoCAD files to be used for the "Record Drawings", the Contractor must fill in the name of the person responsible for certifying the above certification. Both the black line prints and the AutoCAD files shall be certified by the same person.

G. Prior to final acceptance of the Work of this Division, the Contractor shall submit properly certified "Record Drawings" to the Architect for review and shall make changes, corrections, or additions as the Architect may require to the "Record Drawings". After all changes have been made to the "Record Drawings", the Contractor shall deliver to the Owner the final set of "Record Drawings" consisting of the black line prints and the AutoCAD files. The AutoCAD files to be delivered by the Contractor to the Owner shall be in AutoCAD 2015 version software or later version as originally provided to the Contractor by the Engineer.

H. Within 90 days after the date of system acceptance, record drawings of the actual installation of the location, luminaire identifier, control and circuiting for each piece of lighting equipment shall be provided to the Building Owner.

I. Within 30 days after the date of system acceptance, record drawings of the actual installation of the single line diagram and building electrical distribution system, including floor plans indicating location and area served for all distribution, shall be provided to the Building Owner.

3.14 OWNER'S CLOSE-OUT PACKAGE

A. Contractor shall provide a master project close-out log and corresponding schedule for submission to the Owner and Engineer for review and approval.

B. Contractor shall provide the Owner with two (2) copies of all warranties bound in a notebook with a corresponding warranty log. Warranty log shall clearly indicate the subcontractor, address, telephone number, and contact person.

C. Contractor shall provide the Owner with one (1) complete set of approved submittals and a copy of the final submittal log.

D. Contractor shall provide the Owner with a complete list of all subcontractors and major suppliers with address, telephone number, and contact persons.

E. Contractor shall provide the Owner with a complete inventory of attic stock materials and spare parts as applicable and a copy of the transmittal signed by the Owner's representative acknowledging receipt of the items.

F. Contractor shall provide the Owner and Engineer with documentation of compliance for all specified requirements associated with equipment start-up, commissioning, demonstrations, training and warranties. Documentation shall include, but not be limited to, equipment start-up sheets as completed by the manufacturer's representative.

G. Contractor shall provide the Owner and Engineer with documentation of acceptable performance of the electrical service systems. The Owner reserves the right to contract with an independent consultant to perform additional electrical service testing. If it is determined that the electrical service systems are not operating properly in accordance with the contract documents, all costs associated with additional tests and time required to correct and uncover the deficiencies by the independent consultant will be the responsibility of the Contractor unless the deficiencies are not design related.

H. When the Contractor considers the work to be ready for final acceptance, the Contractor shall provide the Owner and Engineer with written certification stating the following:

1. All work has been completed in accordance with the Contract Documents.

2. All field review items and other deficiencies identified by the Certificate of Substantial Completion have been properly corrected.

3. All work has been inspected for compliance with the Contract Documents.

4. All electrical systems have been tested in the presence of the Owner's representative and are fully operational in compliance with the Contract Documents.

5. As-built documents have been reviewed, are complete and are accurate.

I. The Contractor shall provide the Owner with a letter from the Engineer of Record stating that the equipment installed fully complies with the Contract Documents.

3.15 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 26 08 00 and the alternate pricing requirements indicated in Section 26 00 00 for additional building commissioning general requirements.

END OF SECTION 26 00 10

SECTION 26 00 70 - DIVISION 26 SCHEDULE OF SUBMITTAL DATA

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED

A. Furnish the shop drawing submittal data for the Architect's and Engineer's review as indicated herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Division 21, 22, 23, 25, 26 and 28: All Sections

PART 2 - PART 2 - SHOP DRAWING SUBMITTAL DATA

2.1 GENERAL

A. The submittal data to be furnished for this Project shall comply with the Specifications hereinbefore. The referenced Sections herein scheduled are for general information purposes only and shall not be construed to be limiting to the submittal data scheduled and/or required within the individual Sections of these Specifications.

2.2 CONTRACT REQUIREMENTS

A. Shop drawing submittal data shall clearly and concisely address all Contract Document requirements. Standard manufacturer data that does not indicate these requirements shall be supplemented with additional information provided by the Contractor or equipment manufacturer. Refer to Section 26 00 10 for additional requirements including submittal log schedule requirements.

2.3 DEVIATIONS FROM THE CONTRACT DOCUMENTS AND PERTINENT DATA

A. All deviations from the contract documents shall be noted in the form of a cover letter included with the shop drawing submittal package. All pertinent data such as material options, equipment arrangements, control options, diagrams, etc. that applies to this Project shall be clearly marked. All data and diagrams that do not apply to this Project shall be omitted from the package or clearly marked out.

2.4 QUALITY ASSURANCE

A. Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification paragraph. A hand written marked-up specification section will be acceptable. Along with this

specification copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

2.5 SPECIAL REQUIREMENTS

A. The shop drawing submittal package containing the Short Circuit and Protective Device Coordination Study, Arc Flash Study, and the Operation and Maintenance (O&M) Manuals shall require an Owner/Consultant review period of four (4) weeks. The Contractor shall plan for and arrange the shop drawing submittal review schedule to accommodate this requirement.

2.6 SPARE PARTS

A. The submittal data for each piece of equipment shall include a detailed itemized list of recommended spare parts. Spare parts shall be clearly identified by name and part number.

2.7 ORGANIZATION OF PACKAGES

A. Unless otherwise noted in the specifications and in the following schedule of shop drawing, the Contractor shall provide a single shop drawing submittal package for each of the listed specification sections. Unless specifically noted otherwise, multiple submittal packages per section will not be acceptable under the terms of the Contract.

2.8 COORDINATION BETWEEN DIVISIONS 21, 22, 23, 25, 26 AND 28

A. The Division 26 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 25 and 28 Contractors to provide coordination between the Division 21, 22, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 21, 22, 23, 25, and 28, it shall be the responsibility of the Division 26 Contractors to acquire from the Division 21, 22, 23, 25 and 28 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

2.9 MANUFACTURER'S CERTIFICATION

A. All items or equipment listed herein with asterisks (*) shall be certified by the manufacturer. Refer to Section 26 00 10 for certification requirements.

2.10 SHOP DRAWING SUBMITTAL SCHEDULE

A. Provide a comprehensive list of Divisions 21, 22, 23, 25, 26 and 28 shop drawing submittal packages and a schedule indicating the date when each package will be submitted to the contractor and the date when each package will be submitted to the design team. The intent of this list and schedule is to allow the contractor and design team to plan ahead and schedule the necessary review and coordination resources in advance of the arrival of shop drawing submittal packages. Shop drawing submittal schedule shall be submitted to the Architect not less than four weeks prior to the submission of the first Division 21, 22, 23, 25, 26 and 28 shop drawing submittal packages. Shop drawing submittal schedule shall be periodically updated by the contractor and issued to the Architect to reflect current status of the project. Shop drawing

submittal packages shall be listed on the schedule not less than four weeks prior to submission to the contractor and not less than six prior to the submission to the design team. Packages not listed on this schedule will required up to two additional weeks of review time.

2.11 SHOP DRAWING SUBMITTAL DATA

A. The submittal data to be furnished for this Project shall include but not be limited to the following:

1. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
2. Section 26 05 26 - Grounding and Bonding for Electrical Systems
3. Section 26 05 33 - Raceway and Boxes for Electrical Systems
4. Section 26 05 48 - Vibration and Seismic Controls for Electrical Systems *
5. Section 26 05 53 - Identification for Electrical Systems
6. Section 26 05 73 - Electrical Power System Study
7. Section 26 05 80 - Electrical Service
8. Section 26 05 93 - Testing
9. Section 26 09 23 - Automatic Control Devices
10. Section 26 24 16 - Panelboards
11. Section 26 27 26 - Wiring Devices
12. Section 26 28 00 - Low-Voltage Circuit Protective Devices
13. Section 26 29 00 - Low-Voltage Controllers
14. Section 26 32 00 - Packaged Generator Assemblies *
15. Section 26 36 23 - Automatic Transfer Switches *
16. Section 26 41 00 - Facility Lightning Protection
17. Section 26 43 13 - Surge Protective Devices for Low-Voltage Electrical Power Circuits
18. Section 26 50 00 - Lighting
19. Section 26 00 70 - Shop drawing submittal schedule.
20. Section 26 08 00 - Commissioning plans and documentation.
21. Drawing Packages:
    a. Drawings shall be limited to a maximum of three (3) packages.

2.12 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 26 08 00 and the alternate pricing requirements indicated in Section 26 00 00 for additional building commissioning general requirements.

END OF SECTION 26 00 70

SECTION 26 05 19 – LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific Attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the wires and cables as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 33 - Raceway and Boxes for Electrical Systems
B. Section 26 05 53 - Identification for Electrical Systems
C. Section 26 05 93 - Testing

1.4 REFERENCE STANDARDS

A. All wire shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 20, 70, 72, 101, 110

2. ASTM B-3

3. UL 44, 83, 1569, 2196

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, specifications, engineering data, conductor supports, and insulation resistance test results for the wiring specific to this project.

PART 2 - PRODUCTS

2.1 GENERAL

A. Except as otherwise indicated, provide UL listed wire, cable and connectors of manufacturer's standard materials, as indicated by published product information and designed and constructed in accordance with applicable industry standards. All wire shall bear the stamped seal of approval of the National Board of Fire Underwriters.

2.2 WIRING

A. Material

1. All wire and cable shall be of soft drawn annealed copper wire having a conductivity of not less than 98% of that of pure copper. Each wire and cable shall be continuous without weld, splice, or joint throughout its length between panels, junction boxes, pull boxes, etc. Each wire and cable shall be uniform in cross section.

2. All wire size #8 AWG and larger shall be stranded and sizes #10 AWG, #12 AWG and #14 AWG shall be solid.

3. Where permitted, "AC" armor clad cable shall be three wire solid copper conductor with two #12 AWG THW insulated phase conductors and one #12 AWG insulated copper ground conductor. Contractor shall verify use of "AC" cable with local authority.

4. Where required, "MC" armor clad cable shall be three wire solid copper conductor with two #12 AWG THW insulated phase conductors and one #12 AWG insulated copper ground conductor. Contractor shall verify required use of "MC" cable with local authority. Four and five wire cable shall be similar to three wire with additional copper insulated phase conductors.

B. Insulation

1. All conductors shall be standard code type "THW", "THHN", "THWN" or "XHHW" insulated wire except as specified below, indicated on the contract documents, or as required per the NEC.

2. All conductors in buried conduit, conduit run in wet locations, conduit encased in concrete in contact with the earth and unprotected locations exposed to weather shall be type "THW", "THWN" or "XHHW".

3. All wire within lighting fixtures shall be temperature rated as required by the National Electrical Code, latest edition. Branch circuit wiring within 3 inches of a ballast shall be type "THHN" or "XHHW".

4. All wire to electric heaters shall be "THHN" or "XHHW".

C. Size

1. All wire shall be sized as indicated on the contract documents and as required by the adopted local codes for the load. However, the minimum wire size shall not be less than the following regardless of what wire size is indicated on the contract drawings.

a. Minimum size for power and lighting circuits 12 AWG.

b. Minimum size for 120V circuits over 60 feet to the first outlet, fixture, or device 10 AWG.

c. Minimum size for control circuit 14 AWG.

d. Feeder conductors shall be sized for a maximum voltage drop of 2% at design load.

e. Branch circuit conductors shall be sized for a maximum voltage drop of 3% at design load.

D. Marking

1. All conductors and cables shall be durably marked on the surface to indicate the following information:

a. The maximum rated voltage for which the conductor was listed.

b. The proper type letter or letters for the type of wire or cable.

c. The manufacturer's name, trademark, or other distinctive marking by which the organization responsible for the product can be readily identified.

2.3 COLOR CODING OF WIRE

A. General

1. Color code the entire power wiring system in the building.

a. 120/208 volt system:

1) Phase A Black, Phase B Red, Phase C Blue

2) Neutral White, Ground Green, Isolated Ground Green with Yellow Stripes

b. Extent

1) The color coding shall be continuous and shall extend throughout the entire building power wiring system to the last utilization device.

2) All branch and feeder conductors #6 and larger shall have manufacturer applied phasing color for full length. Use of phase colored tape is not acceptable.

2.4 CONDUCTOR FITTING

A. General

1. Provide all required fitting accessories.

B. Terminal Blocks

1. Terminal blocks shall be rated for 600 volts and shall be of the screw type.

C. Splices

1. Splices shall be made only in outlet boxes, junction boxes or splice boxes.

2. All #8 AWG and smaller conductors shall be spliced with preinsulated spring connectors. No other type of mechanical connector may be used for sizes #8 AWG and smaller wire. Connectors shall be Scotchlok, Buchanan B Cap, or approved equal.

3. All #6 AWG and larger copper conductors terminated on lugs of switchboards and panelboards with copper bus shall be terminated with copper UL listed compression connectors such as Thomas & Betts #54100 or #54200 series connectors.

4. All #6 AWG and larger copper conductor two way splices shall be made with barrel connectors requiring compression on each end. Connectors shall be UL listed for the type of conductors to be connected.

5. All #6 AWG and larger copper conductor tapping and pigtailing shall be made using "C" type compression taps such as Thomas & Betts #54700 series connectors. Connectors

shall be UL listed for type of conductors to be tapped.

6. Conductor termination provisions shall be listed and rated for use with conductor ampacities rated at 75°C minimum.

7. The manufacturer's recommended installing tools with required number of compressions shall be used for all terminations.

8. Stub splice connections for exterior lighting and similar applications shall be provided with a water-resistant cover kit with connector, TE Connectivity Raychem GelCap SL or approved equal.

## 2.5 TYPE A.C. CABLE AND M.C. CABLE

A. General

1. Type A.C. cable shall conform to Armored Cable: Type AC Article of the National Electrical Code.

2. Type M.C. cable shall conform to Metal-Clad Cable: Type MC Article of the National Electrical Code.

B. Use

1. Type A.C. cable may be used in the following locations and applications only:

a. For final connections to receptacles, switches and light fixtures installed in walls, concealed spaces and hung ceilings.

b. Type AC/MC cable shall not be used in visible public or back of house areas and shall not be terminated at panelboards. Type AC/MC cable shall terminate at lighting and receptacle junction boxes within spaces being served which in turn shall be connected with EMT or rigid conduit back to wiring gutters or panelboards.

2. Type M.C. cable shall be substituted for type A.C. cable for areas classified as "places of assembly" with occupancies over 100 as required by the N.E.C. Article 518. Type M.C. cable shall only be used for branch circuits as specified for A.C. cable.

## 2.6 CONDUCTOR SUPPORTS

A. Conductors in vertical raceways shall be supported by cleats or bushings at the top of each vertical raceway or as close to the top as practical, plus a support for each additional interval of spacing as required in NEC Article 300.

## 2.7 ALUMINUM CONDUCTOR OPTION

A. At the Contractor’s option, aluminum conductors may be used in lieu of copper conductors for all circuits protected by 100 amps and greater over current protection devices except for motor loads including but not limited to pumps, fans, and air handling units or as indicated on the drawings. All circuits protected by over current protection devices less than 100 amps are required to be copper. The use of aluminum for circuits protected by over current protection devices less than 100 amps will not be acceptable. If this option is selected by the Contractor, the Contractor shall be responsible for coordinating the additional work and space required for the larger or additional parallel sets of conduit and conductors required to meet the copper

conductor basis of design current carrying capacity. In addition, the Contractor shall provide all the necessary additional hangers, core drills, sleeves, fireproofing, etc. required to accommodate the optional aluminum conductors and bussing as indicated on the optional electrical distribution diagram system utilizing aluminum conductors. All aluminum conductors shall be terminated with aluminum UL listed compression connectors. The smallest allowable aluminum conductor for the optional aluminum system shall be #1/0 AWG.

PART 3 - EXECUTION

3.1 GENERAL

A. Install all wiring in accordance with the manufacturers' recommendations and the National Electrical Code.

3.2 INSTALLATION

A. All main power feeders shall be run their entire lengths in continuous pieces without joints or splices from points of origin to points of termination.

B. All feeder and branch circuit conductors shall be continuous without welds, splices or joints throughout their lengths between panels, junction boxes, pull boxes etc.

C. No wire shall be drawn into a conduit until all work of a nature which may cause injury is completed.

D. All circuits shall be identified per Section 26 05 53.

E. Conductor pulling tensions shall not exceed manufacturer's recommended values. Where necessary, UL listed lubricants, compatible with the type of insulation involved, may be used to facilitate cable pulling. Compounds similar to the following are acceptable:

1. 3M – Type “WL”

2. American Polywater Corporation - Type "J"

3. Ideal Industries - "Wire Lube"

F. At least 6 inches of free conductor shall be left at each outlet, junction, and switch point for splices or the connection of fixtures or devices.

G. For wire-binding screw posts, wrap the freshly stripped end of the conductor two-thirds to three-quarters of the distance around the post. The loop shall be made so that rotation of the screw in tightening will tend to wrap the wire around the post rather than unwrap it.

H. Cables and conduits shall be neatly routed parallel and perpendicular to building lines. Cables that are bundled together for support shall not exceed ten (10) in quantity in any one location. Multiple bundles neatly installed and separated by physical space shall be used where cable quantities exceed ten (10).

I. If conductors are deemed damaged (including but not limited to breaks/tears in the insulation or visible oxidation on the conductors) by the Engineer or Owner's Representative, the Contractor shall be responsible for replacing the conductors with damage-free conductors.

J. Sharing of neutrals is not allowed. Each branch circuit shall be provided with its own neutral

wire.

3.3 TESTING

A. Insulation resistance of conductors #6 AWG and larger shall be tested as indicated below. Each conductor shall have its insulation resistance tested after the installation is completed and all splices, taps, and connections are made except connection to or into its source and point(s) of termination.

B. Insulation resistance of conductors which are to operate at 600 volts or less shall be tested by using Biddle (or approved equal) Megger of not less than 1000 volts D.C. Resistance shall be measured by connecting one terminal of Megger to conductor and other terminal to conduit in which conductor is installed. Reading shall be observed after 15 seconds of operation of the Megger at slip speed. Insulation resistance of conductors rated at 600 volts shall be not less than the following for listed wire sizes:

1. #6,4, and 2 100,000 ohms

2. #1, 1/0, 2/0, 3/0, and 4/0 50,000 ohms

3. All sizes larger than 4/0 25,000 ohms

C. Wire sizes #12, #10, and #8 shall be tested for continuity with standard ohm meter indicating at least 250,000 ohms to ground resistance.

D. Conductors that do not exceed insulation resistance values listed above shall be removed and replaced and test repeated. The Contractor shall furnish all instruments and personnel required for tests, shall tabulate readings observed, and shall submit copies of test readings to the Engineer for review.

E. These test reports shall identify each main feeder conductor tested, date and time of test, weather conditions, test equipment, calibration date, and relative humidity. Each test shall be signed by party making test and person witnessing it. Any conductor or splice which is found defective shall be promptly removed and replaced, and additional tests shall be performed.

F. The above testing and report requirements shall apply to all main feeders. Branch circuits, control circuits, and signal circuits shall be checked in accordance with the National Electrical Code, latest edition.

END OF SECTION 26 05 19

SECTION 26 05 23 - CONTROL-VOLTAGE ELECTRICAL POWER CABLES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Furnish and install conduit and wiring necessary for complete and fully operational systems for all controls, pilot devices, safety devices, interlocks and interconnections as specified herein and shall include both `line (120V.)' and `low' voltage wiring/circuits as required, including but not limited to the following:

1. Division 28 shall furnish and install a complete independent and operational fire alarm system including but not limited to: Zoned fire alarm signal wiring, initiation devices, indicating devices, fire communications systems, sprinkler alarms, wiring for motor control/status at fire alarm panel, building management system interface, security station and their interconnecting wiring to fire alarm panel, interlocks between fire alarm system and fans, AHU's, door locks and dampers.

2. Interconnecting of controls and alarms between standby generator and fire alarm panel, security station, and building management system.

3. All interlock and control conduit and wiring between automatic transfer switches and standby generator.

4. Contactors, interconnection of data cable, indicated lighting circuits, telecommunication cable, printers and power circuits for a complete and operational lighting automation system.

5. Power connection to AHU's shall be supplied under Division 26. Automatic temperature control interlocks and building management system controls to AHU's shall be supplied under Division 25.

6. Connection of all individually mounted motor controllers and their associated auxiliary relays, pilot lights and H.O.A. switches, etc., as described under Sections 26 29 00 and 26 24 19. Interlock wiring connected to control circuit for automatic temperature control shall be supplied under Division 25.

7. All interlocks, controls and safety devices for a complete and operational standby generating system.

8. All control, interlock and safety interconnections for all systems listed under Division 26 and 28.

9. All underground conduit necessary for EMCS and ATC control functions shall be provided by Division 26 and 28 contractors as directed and coordinated with the Division 21, 22, 23, and 25 contractors. Control wiring within underground conduit shall be provided under Section 25 00 90.

10. Equipment furnished under Division 26 shall be provided with all status and alarm contacts which requires status monitoring and alarm initiation interface with EMCS. Refer to Section 25 00 90 for additional requirements.

11. Control, monitoring, and power wiring to HVAC systems shall be a listed 2-hour electrical circuit protective system.

B. It shall be the responsibility of any equipment manufacturer who furnishes equipment for this project to coordinate and cooperate with the requirements of Section 25 00 90. Equipment automatically and remotely controlled and/or monitored via the EMCS and/or interlocked with some other piece of equipment as specified in Section 25 00 90 shall be complete with all necessary contact points and interface devices to accomplish the control strategies and sequences described in Section 25 00 90 under "Composite Wiring Diagram". By accepting this contract; the contractor, all subcontractors, and all applicable equipment manufacturers expressly agree to cooperate with the EMCS contractor to fully comply with the "Composite Wiring Diagram" requirements and expedite the obtaining of all necessary information.

C. A piece of flexible conduit not less than 18" in length shall be connected between the conduit and the control device to permit adjusting and maintenance removal of the control device. All flexible conduit shall have grounding jumpers as specified.

D. The Contractors under other Divisions will furnish and set in place the necessary foundations or supports for all motors. Under this Division, the motors shall be set in place, belt drives aligned, couplings attached, etc., as required. The final alignment, belt tension, etc., shall be done in cooperation with the party that furnished the equipment. The Contractor is advised that alignment and vibration tests are required for each motor as part of Division 21, 22, and 23 and that the work performed in these Divisions must satisfy those requirements.

E. All motor controllers shall be furnished and installed under this Division unless the motor controller is an integral part of a piece of mechanical equipment furnished under Division 21, 22 or 23. See Division 21, 22 and 23. The Contractor shall install all motor controllers in suitable locations, subject to approval by the Architect. Under this Division, the Contractor shall furnish and install an externally operated disconnect device, either fused or non-fused as shown on the Drawings and/or as required by the motor nameplate and/or National Electrical Code. When the disconnect device is not near to and in sight of the motor controller, the Contractor shall furnish and install an externally operated disconnect device that will be near to and in sight of such motor and shall be approved for such use by the National Electrical Code.

F. Where possible, all conduits shall terminate in conduit boxes at motors. A piece of flexible conduit not less than 18" in length shall be connected between the conduit and the motor terminal to permit the maximum movement of the sliding base in each direction and to reduce vibration transmitted to conduit. Where motors are not provided with conduit boxes, terminate the conduit in suitable conduit and make motor connection. All flexible conduits shall have grounding jumpers as specified.

## 1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 25 00 90 - Automatic Temperature Control System
B. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
C. Section 26 05 33 - Raceway and Boxes for Electrical Systems
D. Section 26 05 93 - Testing
E. Section 26 29 00 - Low-Voltage Controllers
F. Section 26 32 00 - Packaged Generator Assemblies
G. Section 26 36 23 - Automatic Transfer Switches
H. Section 28 31 00 - Fire Detection and Alarm System

1.4 REFERENCE STANDARDS

A. NFPA 70 National Electrical Code

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 TESTING

A. All controls, interlocks, safety devices and alarms shall be tested by contractor after final installation prior to final review by engineer.

3.2 MATERIALS

A. All control wiring shall be copper conductor as specified installed in conduit.

END OF SECTION 26 05 23

SECTION 26 05 26 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the grounding as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceways and Boxes for Electrical Systems
C. Section 26 41 00 - Facility Lightning Protection

1.4 REFERENCE STANDARDS

A. The grounding system shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code

1.5 SUBMITTALS

A. Submit ground rods, ground clamps and connectors, expansion joint connectors, and graphed results of Fall of Potential Test with weather conditions stated, equipment used, and equipment calibration date.

PART 2 - PRODUCTS

2.1 GROUNDING EQUIPMENT

A. The equipment grounding system shall be designed so all metallic structures, enclosures, raceways, junction boxes, outlet boxes, cabinets, machine frames, portable equipment and other conductive items in close proximity with electrical circuits operate continuously at ground potential and provide a low impedance path for possible ground fault currents in conjunction with an equipment copper ground conductor. The system shall comply with the National Electrical Code, and as hereinafter specified. All branch circuits and feeders shall have green insulated equipment ground conductors sized in accordance with the NEC.

B. The building equipment grounding system shall consist of the electrically continuous metallic conduit system together with equipment copper ground wires extended from the main service switchboard as indicated. Every item of equipment served by the electrical system shall be bonded to the building equipment ground. Portions of metallic piping and duct systems which are isolated by flexible connections, insulating couplings, etc., shall be bonded to the equipment ground system with a flexible bonding jumper.

C. All expansion joints, isolated couplings, points of electrical discontinuity, or connections in conduit where firm mechanical bond is not possible shall be bonded with OZ Type "BJ" or

approved equal bonding jumper.

D. Mechanical equipment shall be bonded electrically to the building equipment grounding system. This includes but is not limited to flexible duct connections, fans, pumps, etc.

E. Grounding rods shall be 5/8 inch in diameter by 10 feet in length, made of copper-clad steel. Copper thickness shall be no less than 10 mils on the steel core of the rod.

F. A green insulated ground conductor is required in all raceways. Conduit or cable is not to be relied upon as the ground path.

## PART 3 - EXECUTION

### 3.1 GENERAL

A. The system neutral shall be grounded at the service entrance. The system neutral on interior wiring shall be kept isolated from other grounding systems throughout the building and shall be grounded only to the grounding electrode system.

B. Each system of electrically continuous metallic piping and ductwork shall be electrically grounded in accordance with the requirements of the Code for "Grounding Electrodes" and for "Grounding Conductors" as they apply to the grounding of "conductor enclosures and equipment only". Metallic piping and duct systems which enter the building shall be grounded at the point of entry with a grounding bond wire to the electric service ground. Isolated metallic piping and duct systems shall be bonded to the building equipment grounding system.

C. Bonding and grounding wires shall be sized, shall be run in conduit, and shall be clamped to various services in accordance with the requirements of the Local Authorities and the National Electrical Code. Where parallel conductors are used to comprise a single feeder, the equipment ground conductor in each conduit shall be fully rated sized in accordance with NEC Article 250.122.

D. Grounding shall be done in accordance with the requirements of, and subject to the approval of the Architect, Engineer, and local inspection authorities. Approved materials, devices, and workmanship shall be utilized.

E. A green insulated ground conductor is required in all raceways. Conduit, cable tray, or cable jackets are not to be relied upon as the ground path.

### 3.2 GROUNDING ELECTRODE

A. The grounding electrode conductor shall be bonded to the incoming underground metal cold water piping system to the building, structural steel, and supplemental made electrodes, as indicated on the contract documents. This bonding shall serve as the grounding electrode conductor in accordance with the requirements of the Code. The Contractor shall not connect any grounding bond wires to water piping systems throughout the building except at one point.

B. The grounding electrode conductor shall be installed in a separate conduit from the point of connection at the water main to the system neutral in the main switchboards. The grounding electrode conductor shall be insulated stranded copper, sized as required and/or as indicated on the contract documents and shall be without joints or splices over its entire length.

C. When a standby/emergency generator source is grounded to the building (normal) service

grounding electrode, a permanent laminated engraved nameplate shall be chained at the normal power source equipment stating the following:

WARNING
SHOCK HAZARD EXISTS IF GROUNDING ELECTRODE
CONDUCTOR OR BONDING JUMPER CONNECTION IN
THIS EQUIPMENT IS REMOVED WHILE ALTERNATE
SOURCE(S) IS ENERGIZED

D. Provide UL listed U-bolt ground clamp for connection of grounding electrode conductor to metal underground water pipe suitable for direct soil burial and compatible with water pipe material.

E. The following shall not be used as ground electrodes:

1. Metal underground gas piping systems
2. Aluminum

3.3 LIGHTNING PROTECTION SYSTEM CONNECTION

A. The lightning protection system shall be bonded at one location to the electric service grounding electrode system.

3.4 TESTS

A. The completed grounding electrode system shall be subjected to a metered test at switchboard ground bar to ensure that the ground resistance, without chemical treatment or other artificial means, does not exceed ten (10) ohms.

B. Testing shall be done using Fall of Potential method. Results shall be plotted on a graph with resistance in ohms on the Y axis and distance from the point being tested to potential test probe on the X axis. Current test probe shall be placed 1000' from point being tested. The potential probe placement shall be graphed at 30, 40, 50, 62, 75, 80, and 90% of the distance to the current probe. If the measured resistance does not level off at any point then repeat tests with the current test probe further away until measured resistance levels off at typically 62% the distance. Contractor shall confirm that the grounding system is isolated prior to testing (i.e. disconnected from neutral).

3.5 CLOSEOUT

A. Provide as-built drawings for the building grounding system including rod sizes, locations and configuration and connection details.

END OF SECTION 26 05 26

SECTION 26 05 30 - DIVISION 26 DEMOLITION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements", "Conditions of the Contract" and "Supplementary Conditions". Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Furnish all labor, equipment and materials required to demolish, cap, patch, repair and dispose of the Division 26 systems no longer required as part of an active system of the Project as specified herein and as indicated on the Drawings.

1.3 SCOPE OF WORK

A. It shall be the responsibility of the Contractor to review the Architectural Documents in addition to the Division 21, 22, 23, 25, 26 and 28 Documents to determine the complete scope of work

1.4 REFERENCE STANDARDS

A. NFPA 70 National Electrical Code

B. Underwriters' Laboratories, Inc. (UL)

C. National Fire Protection Association (NFPA)

D. National Electrical Manufacturers' Association (NEMA)

E. American National Standards Institute, Inc. (ANSI)

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Under this Division, the Contractor shall furnish all labor, equipment, appliances, and materials to perform all operations in connection with the demolition of the Division 26 systems as indicated on the Drawings and as specified herein and/or as required by the Local Building Code. The work shall include but not be limited to the following:

1. Field verify the exact sizes and locations of all existing electrical systems prior to demolition of any existing work. The demolition work shall be coordinated with the architectural scope of the Project to assure proper limits of demolition.

2. Remove all existing electrical equipment, conduit, raceways, light fixtures, wiring and associated equipment within the area to be demolished, whether specifically indicated or not, that is no longer required unless otherwise indicated. Refer to the electrical existing conditions, demolition, and new work single line diagrams for scope associated with distribution equipment.

3. Openings remaining in enclosures as a result of demolition shall be sealed with sheet metal cap or other approved appropriate means, unless otherwise indicated.

4. Where existing equipment is to be relocated, extreme care shall be taken to prevent damage during the removal and reinstallation. Where damage occurs, the equipment shall be replaced or repaired to the satisfaction and approval of the Architect at no additional cost to the Owner.

5. Contractor shall be responsible for cutting, patching and restoration of existing finishes.

6. Contractor shall be responsible for removal, disposal and salvage of equipment from job site.

7. Contractor shall be responsible for cleaning, testing and repair of existing equipment to be reused.

8. Contractor shall be responsible for sealing all existing floor slab penetrations in order to maintain the integrity of the fire rating of the existing floor slabs.

9. Refer to Architectural Documents and the Division 21, 22, 23, 25, 26 and 28 documents for area and extent of demolition.

B. Where a system to be demolished must remain operational throughout the construction process until a new system that will take its place is functional, the contractor shall provide all products, services, materials, labor and supervision to maintain the operational status of the affected system.

## 3.2 EXISTING CONDITIONS

A. Under the scope of work, the Contractor shall perform a comprehensive field inspection of the existing conditions to thoroughly familiarize himself with the Project and determine the complete scope of new and demolition work. The Contractor shall verify in the field exact sizes, capacities, and locations of all existing equipment prior to installation of any proposed work. Proposed work shall be coordinated with existing conditions to assure proper installation. This field survey and determination of the complete scope of work shall be performed prior to the final bid. Specific attention is brought to physically larger and heavy pieces of equipment that are new to be installed or existing to be removed and the routing of such equipment with the available means and site conditions.

## 3.3 CONFLICTS AND DEPARTURES

A. Conflicts

1. The Contractor shall remove all existing electrical equipment and associated accessories whether specifically indicated or not, that are no longer required. Any equipment which must remain as part of an active system and is in conflict with the proposed work shall be relocated at no additional expense. There shall be no additional expense to the Owner for this type of demolition work, unless, in the Architect's and Owner's opinion, it is beyond the scope and intent of the Contract Documents.

B. Departures

1. The Contract Drawings are diagrammatic and schematic in nature and do not indicate inferred details. Any departures from the Drawings that are necessary to comply with the

intent of Documents, requirements of applicable Building Codes or the Authority having jurisdiction shall be detailed in the form of shop drawing submittal data and submitted for review by the Architect and Engineer. All demolition and any such revisions for departures shall be made in accordance with the reviewed shop drawings without increased expense to the Owner. Departures shall not be made without prior written approval by the Architect.

3.4 COOPERATION

A. The Contractor shall comply with all the requirements of the Local Authorities having jurisdiction and the building Owner.

B. The Contractor shall give advance notice to the Architect when work is to be performed.

C. The Contractor shall cooperate and coordinate demolition of Division 26 systems as required with all of other trades. The Contractor shall supervise and assist in the removing and replacing of existing materials for installation of electrical items and items related to all other trades.

3.5 BARRIERS

A. Where required by applicable health and safety regulations, the Contractor shall furnish and maintain safety and dust barriers. If necessary, the Contractor shall furnish and maintain temporary fencing and traffic barriers in accordance with all applicable health and safety regulations and as deemed necessary.

3.6 RESTORATION

A. Demolition shall be carried out with caution to assure that existing conditions to remain will be undamaged. Existing conditions to remain that are damaged or defaced by work under this contract shall be restored or replaced equal to the conditions at the time of award of the Contract.

B. Cut, patch, and restore all existing surfaces not receiving new finishes that have been disturbed during execution of this Contract. Materials and finishes used shall be similar, in all respect, to adjacent surfaces.

C. Existing conduit and wiring which are to remain as part of an existing or proposed active system that the Contractor determines to be defective shall be brought to the attention of the Architect.

D. Existing equipment to be removed shall be cleaned, repaired, and reused at the discretion of the Architect wherever applicable.

3.7 CONNECTIONS

A. Connections to existing work shall be similar, in all respects, to the existing system and conditions unless otherwise indicated. Existing work shall be altered and/or temporarily removed and replaced as required for completion of requirements of the project.

3.8 SALVAGE

A. The Owner assumes no responsibility for loss or damage to materials or structures on site for the salvage value of equipment which the Contractor may have reflected in his bid.

B. All existing electrical equipment, conduit and wiring removed during construction no longer

required as part of an active system and not to be reused shall be removed from the job site and properly disposed of by the Contractor.

C. All existing electrical equipment not being reused shall be removed during construction and shall remain the property of the owner and shall be stored, removed from the job site, or disposed of as directed by the owner's representative.

3.9 CLEANING

A. The Contractor shall clear away all debris and demolished material at frequent intervals. The Contractor shall not allow debris to accumulate to the extent that it will interfere with work, passage of the workmen, and the operation of the existing occupied areas.

B. It is the intent of this Specification that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment.

3.10 QUALIFICATIONS

A. Only Contractors, Subcontractors, and workmen experienced and regularly engaged in the demolition of mechanical, plumbing, and electrical systems shall be permitted to perform the demolition of existing systems.

END OF SECTION 26 05 30

SECTION 26 05 33 - RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the raceway systems as shown on the contract documents and as specified herein. Provide raceway systems for electrical, telephone, fire protection and signal systems. All electric conductors shall be installed in conduit, except where armor clad cable is allowed. No metallic conduit smaller than 1/2" shall be installed.

B. Furnish junction and pull boxes where required by the Code or where indicated or required to facilitate pulling wires, either feeders or branch circuits, regardless of whether shown on contract documents or not.

C. The approval of the Architect shall be obtained for any outlet location about which there may be any question. Outlets previously placed without specific approval in locations not suitable to the furnished room or area shall be removed and relocated without cost to the Owner when so directed by the Architect. This shall include the cutting and patching of work of others as may be necessary.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 27 26 - Wiring Devices
C. Section 26 05 53 - Identification for Electrical Systems

1.4 REFERENCE STANDARDS

A. All raceways shall be designed, manufactured, tested and installed in compliance with the following standards:

1. ANSI C80.1: Rigid steel conduits

2. NFPA 70 National Electrical Code

3. ACI 318 Building Code

4. BOCA Codes as adopted by the local authority

5. ANSI/TIA-569-C

B. All outlet boxes shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code

2. Underwriters' Laboratories

1.5 CONDUIT DIAGRAMS

A. Provide shop drawing submittal data indicating layout of all conduits and junction boxes located within the concrete slabs. Shop drawings shall require review and approval by the structural engineer prior to conduit installation.

1.6 GUARANTEE

A. Contractor shall provide a one-year guarantee for items furnished that shall cover:

1. Faulty or inadequate design
2. Improper assembly or erection
3. Defective workmanship and materials

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, outlet boxes manufactured by one of the following manufacturers will be acceptable:

1. Crouse Hinds
2. Appleton
3. Steel City
4. Red Dot
5. RACO

2.2 OUTLET BOXES

A. Boxes shall be steel, hot dipped galvanized after fabrication, and shall have industry standard knockouts necessary to accommodate the conduits at point of installation. Sectionalized boxes shall be used wherever possible to group adjacent devices under a single plate. All boxes shall have lugs or ears inside to secure covers.

B. Ceiling outlet boxes shall be not less than four inches square to provide a seat for the fixture canopy and listed for such use.

C. Outlet boxes for wall fixtures where conduit is concealed shall be deep type, four inch square, and shall have device covers with center openings as required per fixture.

D. Switch and receptacle outlet boxes:

1. Outlet boxes for switches and receptacles in finished walls shall be of suitable size for the device to be mounted and the partitions in which they are installed (boxes shall be deep type, 4" square minimum when containing more than one circuit). The boxes shall have covers with rectangular openings of proper size and shape. Provide covers with raised openings on all outlets in masonry walls with plaster or tile finishes. Wall switch outlets shall be set flush in walls. When located near doors, they shall be close to the trim on lock side of door.

2. Unless otherwise shown, specified, or directed, outlet boxes, etc., shall be located with their center lines at the following elevations above the finished floor line. Boxes shall be mounted vertically unless indicated otherwise.

a. Wall switches (toggle type) vertical 4' 0" (above floor).
b. Wall receptacles (duplex type) vertical 1' 6" (above floor).
c. Wall telephone outlets (desk type) vertical 1' 6" (above floor).
d. Wall telephone outlets (pay station) 4' 9" (above floor).

Other special outlets shall be installed as shown or specified in the symbol list on the contract documents. Confirm all mounting heights with Architect.

E. Outlet, device, pull and junction boxes in walls or ceilings of concrete, tile, or other noncombustible material, shall be so installed that the front edge of the box or fitting will not set back of the finished surface more than 1/4 inch. In walls and ceilings constructed of wood or other combustible material, outlet boxes and fittings shall be flush with the finished surface or project there from. Plaster, drywall or plasterboard surfaces that are broken or incomplete shall be repaired so there will be no gaps or open spaces greater than 1/8 inch at the edge of the box or fitting.

F. Outlet boxes for exposed switches, receptacles, and pull boxes shall be of the cast aluminum "condulet" type, Crouse-Hinds, Stonco or equal. Exposed shall be defined as located in view of a person within or outside occupiable space not concealed within plenum or wall spaces. Exposed shall include both interior and exterior locations where surface mounting or conduit supported stand alone boxes are required. Outlet boxes in outdoor or wet locations shall also have gasketed weatherproof aluminum cast-metal covers with individual gasketed spring-latched hinged outlet covers.

G. Unused openings in boxes, raceways, auxiliary gutters, cabinets, equipment cases or housings shall be effectively closed to afford protection substantially equivalent to the wall of the equipment.

H. Outlet boxes for 20 ampere 120 volt circuits shall have a green insulated #12 AWG solid copper conductor grounding pigtail with ground screw.

I. The highest operable part of electrical and communication outlets shall be placed within the reach ranges as required in The American with Disabilities Act (ADA). These requirements do not apply where the use of special equipment dictates otherwise or where electrical and communication outlets are not normally indicated for use by building occupants.

J. Junction boxes shall be sized per NEC Article 314 to accommodate proper conductor fill without the use of extension rings where possible.

K. Exterior junction and pull boxes and their associated fittings, connectors, covers, etc. that are mounted below grade and where susceptible to immersion by water shall be of the watertight type.

2.3 CONDUIT

A. General:

1. Provide all new and full length materials. All rigid, EMT, PVC, ENT, and flexible conduit shall be UL listed. All conduit shall be suitable for the intended service and location. Any materials located within environmental air plenums shall be UL listed for the application or approved in writing by all applicable local code authorities.

2. Electrical Metallic Tubing (EMT):

a. EMT shall be aluminum or steel with a hot-dipped galvanized coating.

b. EMT shall be formed to permit the use of compression connectors.

c. EMT shall be in accordance with UL 797 and ANSI C80.3.

3. Rigid Conduit:

a. Rigid conduit shall be heavy wall, rigid steel conduit and shall have hot-dipped galvanized coating inside and out with overcoating of zinc-chromate, or transparent enamel or lacquer.

b. Running thread shall not be used. Where the use is required, use T&B “Erickson” type union or OZ type SSP split coupling.

4. PVC Conduit:

a. The PVC conduit shall consist of UL approved Schedule 40 extruded Type II high impact virgin polyvinyl chloride conduit, similar and approved equal to Carlon PVC conduit.

b. All types of conduit joints shall be made up using plastic couplings in accordance with the manufacturer’s recommendations. The tapered ends of joints shall be swabbed with bituminous or joint-sealing compound before the coupling is applied.

c. All below grade and concrete encased conduits shall include metallic tracer tape for above grade detection.

5. Flexible metal conduit:

a. Flexible metallic conduit shall be galvanized steel.

b. Continuity of the equipment ground across flexible conduit connections shall be maintained for all systems that are over 150 volts to ground. The continuity shall be maintained by installing a grounding conductor sized in accordance with Article 250 of the National Electrical Code. The grounding conductor shall be inside the flexible conduit and shall be connected to both ends of the flexible conduit by a suitable binding post. All grounding conductors shall be solid copper conductors.

c. In conduit sizes 1-1/4” and smaller and lengths of 6 feet or less, watertight flexible conduit with watertight fittings which is UL rated as a grounding conductor may be used in lieu of a flexible metallic conduit and separate grounding conductor described above.

6. Electrical Nonmetallic Tubing (ENT).

a. The contractor shall have the option to use ENT in lieu of conduit provided the following conditions are met:

1) ENT shall only be used as permitted by the NEC within the concrete slabs provided fittings identified for these purposes are used for connections.

2) The conduit proposed meets all applicable local and national codes.

3) The material and conduit system provided are UL or ETL listed specifically for the intended service.

4) The material and conduit system provided are recommended by the manufacturer for the intended service.

5) ENT shall not be smaller than 1/2 in. electrical trade size or larger than 2-in. electrical trade size.

6) Product data shall be submitted to the Owner's representative and the engineer for review prior to final acceptance by the Owner.

B. Fittings:

1. Unions, couplings and fittings for rigid conduit shall be galvanized steel of conventional dimensions and shall be internally threaded at each end to fit the tapered thread standard for the corresponding size conduit.

2. Couplings and fittings for EMT shall be of steel, cast pot metal is not acceptable. Couplings shall be compression type.

3. All conduit shall be terminated into panels, cabinets, boxes and outlets with insulated throat connectors for ½" through 1" and for greater than 1" with insulated bushings and locknuts. All conduits not terminated into an enclosure shall have insulated bushings to protect the exposed rough edges of the conduit; bushing material shall be listed for the location and environment installed within.

4. EMT conduit shall be terminated with liquid tight compression connectors where exposed to weather, moisture or in concrete.

5. Liquid-tight flexible metal conduit into panels, cabinets, boxes and outlets shall be secured with insulated throat clamp. Where exposed to weather or moisture or where indicated on the drawings, flexible metal conduit shall have polyvinyl liquid-tight jacket.

6. Flexible metal conduit shall be terminated into cabinets, boxes and outlets and shall be secured using steel or malleable iron squeeze connectors.

7. Any conductors pulled through non-insulated connectors shall be removed and replaced with new after insulated connector is installed.

2.4 WIREWAY

A. Wireway shall be used where indicated and for mounting groups of switches and/or starters. Wireways shall be the standard manufactured product of a company regularly producing wireways and shall not be a local shop assembled unit. Wireway shall be UL listed, and of sizes indicated or as required by NEC if not indicated. The opening, including connectors, shall be completely open without interference. Finish shall be ASA #49 medium light gray enamel over rust inhibitor. Wireway shall be as manufactured by G.E., Wiremold, or Airey Thompson.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General:

1. Install all conduit and outlet boxes as shown on the contract documents and as required for a complete system. Each metallic conduit system shall be installed complete before conductors are drawn in. All wiring in exposed locations in visible public areas and back of house locations including mechanical rooms, electrical rooms, telephone rooms, pump rooms, and similar spaces shall be installed in conduit.

B. Runs:

1. Install all exposed conduit grouped together insofar as possible. Install all conduits parallel or perpendicular to the building surfaces. All conduit shall be concealed except where indicated on the drawings.

2. Metallic conduit shall be continuous from outlet to outlet and from outlet to cabinets, junction or pull boxes and shall enter and be secured at all boxes in such a manner that each system shall be electrically continuous throughout. The equipment grounding conductors shall be bonded to all raceway terminations, i.e. junction boxes, outlet boxes, etc. Where #8 or larger equipment grounding conductor is used, the bonding shall be achieved through the use of bonding bushings. The grounding conductor shall be run continuously with the insulation peeled back for connection to bonding bushing.

3. Wherever possible, conduit shall be typically routed tight to or near the bottom of slab. The exact position of all conduit shall be adjusted to accommodate the site conditions and coordination requirements of other trades. Unless otherwise approved by the owner or specifically shown on the drawings, conduit shall be routed as nearly as possible parallel with or at right angles to the building walls. Conduit shall not block equipment service or removal access spaces. Prior to the installation of any conduit, its location shall be coordinated with the requirements of other trades.

C. Bends:

1. All bends shall be made with standard ells or field bends in which the conduit is bent to a radius not less than allowed in applicable National Electrical Code Article for conduit type. All bends shall be free from dents or flattening.

a. Conduit systems used for 120 volt or higher applications shall not have more than the equivalent of four quarter bends used in any run between terminals and cabinets, or between outlets and junction or pull boxes.

b. Refer to Section 26 05 81 for additional limitations on conduits for telephone, audiovisual, data, and other low-voltage applications.

D. Damaged Conduit:

1. Damaged conduit shall not be used and shall be removed.

E. Location: The location and application of raceway materials shall be in accordance with applicable codes, suitable for the intended application, and the following.

1. Rigid Steel Conduit:

a. Unless otherwise specified herein, rigid steel conduit shall be used where conduit is underground outside the footprint of the building, exposed to the weather, in damp locations, where subject to striking after installation, and where conduit is greater than 4" in diameter.

b. Where exposed to weather, conduit shall have a manufacturer applied plastic coating such as Robroy Industries "Plasti-bond" conduit. Complete with "Plasti-bond" fittings.

c. Rigid steel conduit installed in the ground shall have watertight joints and shall be painted the entire length with two coats of protective finish. The first shall be a 5 mils thick coat of PPG "Coal Cat Epoxy Coating". All coating shall be applied in accordance with the manufacturer's recommendations. The entire length of metallic conduit, including fittings, in contact with the ground, to a point 6" above the ground (or concrete slab), shall be completely coated, subject to the Architect's approval.

2. EMT Conduit:

a. For conduit 4" in diameter and smaller, conduit shall be EMT, except where rigid conduit is required.

b. Aluminum EMT may be used when exposed to the weather with liquid tight fittings and not subject to striking or physical damage after installation. Aluminum EMT shall not be encased in concrete.

c. Steel EMT is to be used when conduit is encased in concrete with "liquid tight" connectors.

d. Unless otherwise indicated herein or structurally prohibited due to the size of the conduit, whenever and wherever possible, conduits shall be located within the concrete slabs. With the exception of the electrical distribution downstream of the in-slab junction boxes, routing conduit exposed in tenant area ceiling plenums will not be acceptable. Conduits embedded in structural slabs shall be installed in a manner approved by the structural engineer. Maintain a minimum concrete coverage of 1” except where penetration is made.

3. Flexible Metal Conduit:

a. Flexible metal conduit shall be used for connections to the following:

1) Motors
2) Transformers
3) Control equipment and devices
4) Lighting fixtures not connected by rigid conduit
5) Appliances
6) Equipment and devices requiring adjustment or removal for maintenance

b. Plastic jacketed liquid tight flexible conduit UL listed as a grounding means shall be used when exposed to weather or moisture in addition to a grounding conductor.

c. Refer to Section 26 05 26 for grounding requirements.

d. When flexible metallic conduit is used for vibration isolation, it shall be 18" minimum length and 36" maximum length with no support. When used to connect light fixtures, it shall be 6'-0" long with no support.

4. PVC Conduit:

a. Provide as indicated on the drawings and as specified herein.

b. PVC conduit shall be used, if approved by the local code authority, for exterior electric utility service entrance, telephone service, conduit below the lowest level concrete slab within the perimeters of the building and parking lot, site lighting applications, and as indicated on the drawings. PVC conduit system shall be provided with pull boxes of approved sizes as required after two right angle bends and at intervals as required by the NEC and as indicated on the contract documents. Boxes shall be in accessible locations. Utility telephone, utility power and other conduits as indicated on the contract documents shall be encased in concrete with a minimum 3" of concrete cover. Minimum concrete strength and maximum aggregate size shall be per the Division 3 specifications and the local utility company requirements where applicable.

c. At the Contractor's option, PVC conduit may extend from a below grade under slab condition up through the concrete slab unless otherwise specified herein.

d. PVC conduit material shall not extend above the surface of concrete slabs within central plant areas, domestic water pump rooms, HVAC pump rooms, and as indicated on the drawings. Exception: If the PVC conduit terminates within an enclosed piece of equipment, such as a motor control center, it will be acceptable for the PVC conduit to extend above the concrete slab.

e. Unless otherwise indicated herein or structurally prohibited due to the size of the conduit, PVC conduit may be used within concrete slabs if acceptable to the local code authority. PVC conduit shall not be exposed to return air plenums. Conduits embedded in structural slabs shall be installed in a manner approved by the structural engineer. Maintain a minimum concrete coverage of 2" except where penetration is made. Refer to the additional requirements for conduits installed within concrete slabs specified herein.

f. PVC conduit shall not be located within environmental air plenums and other areas prohibited by applicable building codes.

F. Outlet box mounting:

1. Wall outlet boxes shall be recess mount except for the following locations which shall be surface mount:

a. Fire rated walls where fur-out is not approved by the Architect.

b. Structural columns or CMU walls.

3.2 COORDINATION

A. Coordinate all conduit runs with other sections of these Specifications requiring electrical connections.

3.3 CONNECTION

A. Connect conduits and wires as shown on the contract documents as required for a complete system.

3.4 CONDUIT SUPPORT

A. General:

1. All horizontal conduits throughout the building shall be thoroughly and substantially supported with straps, hangers, or trapeze arrangements at the highest possible elevation to building structure and not to other trades equipment. Perforated extension bar hanger will not be accepted in any part of the work. All vertical conduits shall be substantially supported to carry the weight of the conduit and cable. All support items shall be galvanized.

B. Grouping:

1. Group conduits together and support on Unistrut or Kindorf channel supports. Hanger rod sizes shall be as recommended by the hanger manufacturer for the service intended.

C. Support Spacing:

1. Vertical – Maximum spacing of supports = 10’-0”.

2. Horizontal – Maximum spacing of supports = 8’-0”.

3. Where conduits are concrete encased and where conduits are direct buried below grade, provide snap lock type duct spacers every five feet along the entire length of run.

D. Upper Attachment to Existing Structure:

1. Refer to Section 26 00 10 “Equipment Mounting/Support Devices” for additional requirements.

## 3.5 BOX SUPPORT

A. All boxes shall be rigidly supported to the building structure with bolts or anchors.

## 3.6 CLEANING

A. All conduit and boxes stored on the job shall have end caps and shall be kept clean inside and outside during the duration of the construction.

B. After installation, keep ends plugged until boxes are closed.

C. Clean entire raceway system prior to installation of conductors and/or pull cords.

D. Contractor shall pull a short mandrel or plug which approximates the diameter of the conduit in all building conduit systems 3" and larger. Contractor shall remove any burrs or obstructions prior to pulling conductors. This shall be performed before and after pouring of concrete in systems requiring concrete cover.

E. Vacuum clean all dirt and debris from the outlet boxes before connections are made.

## 3.7 PULL WIRE

A. A 1/8" braided polypropylene rope or #14 galvanized iron fish wire shall be left in all empty conduit systems. At least 12" of properly secured rope or wire shall be tied off at each end of the empty conduits.

3.8 COLOR CODED IDENTIFICATION

A. All conduit and boxes shall be labeled and clearly marked as indicated in Section 26 05 53.

3.9 SITE CONDUIT DOCUMENTATION

A. The Contractor shall provide as-builts of all exterior conduit installed, including conduit serving site lighting, car chargers, fuel stations, etc.

3.10 EXPANSION JOINTS

A. Raceways shall be provided with expansion fittings at all building expansion joints, parking structure expansion joints, and other areas where necessary to compensate for thermal expansion and contraction. Furnish and install O-Z/Gedney type "DX" or "AXDX" expansion fittings with bonding jumper where conduits cross building expansion joints and parking structure expansion joints. Install per manufacturer's recommendations. Expansion fittings shall be provided which allow for deflection and expansion in any direction.

END OF SECTION 26 05 33

SECTION 26 05 48 - VIBRATION AND SEISMIC CONTROLS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of Division 01, “General Requirements,” “Conditions of the Contract" and “Supplementary Conditions.” Specific attention is called to "Division 26 General Requirements" in Section 26 00 10.

1.2 DESCRIPTION OF WORK

A. Furnish and install complete, fully adjusted, vibration and noise isolation systems and associated equipment as specified and as indicated on the contract documents.

B. It is the objective of this Specification to provide the necessary design requirements for the control of noise and vibration in the buildings due to the operation of machinery or equipment, and/or due to interconnected piping, ductwork and conduit.

C. The work of this section includes, but is not limited to the following:

1. Vibration isolation elements for equipment.

2. Equipment isolation bases.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Section 26 00 70 - Division 26 Schedule of Submittal Data:
B. Section 26 32 00 - Packaged Generator Assemblies

1.4 SHOP DRAWING AND SUBMITTAL DATA

A. Shop drawing submittal data shall clearly and concisely address all Specification requirements listed herein, certification data listed in Sections 26 00 10 and 26 00 70.

B. In addition to these requirements, the submittal material shall include descriptive data for all products and materials including, but not limited to, the following:

1. Descriptive Data:

a. Catalog cuts and data sheets on specific vibration isolators to be utilized showing compliance with the specifications.

b. An itemized list showing the items of equipment to be isolated, the isolator type and model number selected, isolator loading and deflection.

2. Shop Drawings:

a. Drawings showing equipment base constructions for each machine, including dimensions, structural member sizes and support point locations.

3. Concrete and steel details for bases, including anchor bolt locations.

1.5 CODE AND STANDARD REQUIREMENTS

A. 2011 ASHRAE HVAC Applications, Chapter 48

1.6 MANUFACTURER RESPONSIBILITIES

A. Manufacturer of vibration isolation equipment shall have the following responsibilities:

1. Determine vibration isolation sizes and locations.

2. Provide equipment isolation systems as scheduled or specified.

3. Guarantee specified isolation system deflection.

4. Provide installation instructions, drawings and field supervision to assure proper installation and performance.

5. Additional responsibilities as specified herein and as indicated on the drawings.

1.7 MANUFACTURERS AND MANUFACTURER'S REPRESENTATIVES

A. All vibration isolation devices shall be the product of a single manufacturer. Products of other manufacturer's are acceptable provided that their systems comply with the design intent for system performance, static deflection and structural design of the base manufacturer; and the performance of these other products are certified by the single manufacturer taking responsibility for the project.

B. Unless otherwise noted the vibration isolation devices described herein are products of the VMC Group-Amber/Booth and Mason Industries.

C. If it complies with these Specifications, equipment and systems manufactured by one of the following manufacturers will be acceptable:

1. Mason Industries, Inc. Hauppauge, NY
2. Kinetics Noise Control, Inc. Dublin, OH
3. VMC Group-Amber/Booth Houston, TX
4. Vibration Eliminator Co. Copiague, NY
5. Vibro-Acoustics Markham, ON

D. Vibration isolation manufacturer and manufacturer's representatives providing systems and products for this project shall have a minimum ten years experience designing and installing vibration isolation systems.

E. The installation and final adjustment of all vibration isolation units, hangers and bases, shall be under the direct supervision of the vibration isolation manufacturer's representative.

PART 2 - PRODUCTS

2.1 GENERAL

A. Deflection Heights and Calibration: All vibration isolators shall have either known undeflected heights or calibration markings so that, after adjustment, when carrying their load, the deflection under load can be verified, thus determining that the load is within the proper range of the device and that the correct degree of vibration isolation is being provided according to the design.

B. Linear Characteristics: All isolators shall operate in the linear portion of their load versus deflection curve. Load versus deflection curves shall be furnished by the manufacturer, and must be linear over a deflection range of not less than 50% above the design deflection.

C. Spring Isolation: Where spring isolation systems are described in the following specifications, the mounting assemblies shall utilize bare springs with the spring diameter not less than 0.8 of the loaded operating height of the spring. Each spring isolator shall be designed and installed so that the ends of the springs remain parallel. The minimum deflection from loaded operating height to spring solid height shall be 50% of the rated static deflection of the spring.

D. Neoprene Isolation: Where neoprene in shear isolation systems are described in the following specifications, the mounting assemblies shall utilize bare neoprene elements with unit type design molded in oil resistant neoprene. The neoprene shall be compounded to meet the following:

1. Shore hardness of 35 to 65+_ 5, after minimum aging of 20 days or corresponding oven aging.

2. Minimum tensile strength of 2000 PSI.

3. Minimum elongation of 300 %.

4. Maximum compression at 25 % of original deflection.

E. Stiffness: The isolator ratio of lateral to vertical stiffness shall not be less than 0.9 nor more than 1.5.

F. Natural Frequency: The theoretical vertical natural frequency for each support point, based upon load per isolator and isolator stiffness, shall not differ from the design objectives for the equipment as a whole by more than ±10%.

G. Corrosion Protection: All mounting systems exposed to weather and other corrosive environments shall be provided with factory applied corrosion protection. All metal parts of mountings (except springs and hardware) shall be hot dip galvanized. All vibration isolators shall be furnished with zinc electroplated or stainless steel hardware to prevent corrosion and bolt freeze up and to maintain attractive appearance. Isolators exposed to weather shall have the spring cadmium plated and neoprene or powdered coated. Nuts and bolts shall be cadmium plated.

H. Wind Loading: Contractor shall provide isolators for equipment subject to wind loading with uplift restraints which comply with all minimum code requirements.

## 2.2 VIBRATION ISOLATOR SCHEDULE

| Equipment | Base Type | Isolator Type | Minimum Static Deflection (Inches) Under Load |
|---|---|---|---|
| Generators: | -- | B | 2” |

Equipment and systems shall be isolated in accordance with the schedule, as specified herein, and as indicated on the drawings.

## 2.3 VIBRATION ISOLATORS

A. Type A: Bare spring isolators shall incorporate the following:

1. Minimum 1/4" (6 mm) thick neoprene acoustical base pad on underside, unless designated otherwise.

2. Non resonant with equipment forcing frequencies or support structure natural frequencies.

3. VMC Group-Amber/Booth Type SW, Mason Type SLF, Vibro-Acoustics Type FS, or as approved.

B. Type B: Spring isolators shall be same as Type A, except:

1. Provide built in vertical limit stops with minimum 1/4"(6 mm) clearance under normal operation.

2. Tapped holes or through holes in top plate for bolting to equipment.

3. Capable of supporting equipment at a fixed elevation during equipment erection. Installed and operating heights shall be identical.

4. Spring package shall be of such design that it can be installed after equipment is erected and removed in the future without disturbing the installation.

5. VMC Group-Amber/Booth Type CT, Mason Type SLR, Vibro-Acoustics Type CSR, or as approved.

C. Type C: Not Used.

D. Type D: Elastomer Mounting Types/Elastomer Isolators, shall incorporate the following:

1. Bolt holes for bolting to equipment base.

2. Bottom steel plates for bolting or welding to sub base as required.

3. Unit type design molded in oil resistant neoprene.

4. VMC Group-Amber/Booth Type RVD, Mason Type ND, Vibro-Acoustics Type RD, or as approved.

E. Type E: Elastomer hanger rod isolators shall incorporate the following:

1. Molded unit type neoprene element with projecting bushing lining rod clearance hole.

2. Neoprene element to be minimum 1 3/4" (45 mm) thick.

3. Steel retainer box encasing neoprene mounting.

4. Clearance between mounting hanger rod and neoprene bushing shall be minimum 1/8" (3 mm).

5. VMC Group-Amber/Booth Type HRD, Mason Type HD, Vibro-Acoustics Type NH, or as approved.

F. Type F: Not Used.

G. Type G: Pad type elastomer mountings to incorporate the following:

1. 0.750" (19 mm) minimum thickness.

2. 50 psi (345 KN/m2) maximum loading.

3. Ribbed or waffled design.

4. 0.10" (2.5 mm) deflection per pad thickness.

5. 1/16" (1.6 mm) galvanized steel plate between multiple layers of pad thickness.

6. Suitable bearing plate to distribute load.

7. VMC Group-Amber/Booth Type NR, Mason Type Super W, Vibro-Acoustics Type Neo+, or as approved.

2.4 EQUIPMENT BASES

A. Type B 1: Not used.

B. Type B 2: Not Used.

C. Isolated Rail Base, Type B 3:

1. Rails shall be constructed from structural steel angles, as required, to prevent flexure and misalignment under load.

2. Each rail shall be the full length of the supported equipment and be welded to a series of vertically restrained spring isolators as Type B described above.

3. Angles shall have bolt together ties at the ends and center to form one rigid base platform.

4. VMC Group-Amber/Booth Type SR with Type CT Isolators, Mason Type TRSLR, or as approved.

PART 3 - EXECUTION

3.1 GENERAL VIBRATION ISOLATION REQUIREMENTS

A. The installation of all vibration isolation devices and systems shall be in accordance with the recommendations, instructions, and procedures of the vibration isolator manufacturer. Vibration isolators shall not cause any change of position of equipment or conduit resulting in conduit stresses or mis-alignment.

B. Unless otherwise noted, all electrical equipment shall be isolated from the building structure by means of noise and vibration isolators as scheduled on the drawings and within these specifications.

C. No rigid connections between equipment and building structure shall be made that degrades the noise and vibration isolation systems herein specified.

D. Provide temporary anchors and supports as required for the installation of the isolation systems. All temporary supports and anchors shall be removed prior to final review of the systems.

E. The contractor shall not install any equipment or conduit which makes rigid contact with the building unless permitted in this Specification. Building includes, but is not limited to, slabs, beams, columns, studs and walls.

F. Isolation mounting deflection shall be at least the minimum specified.

G. Coordinate work with other trades to avoid rigid contact with the building. Inform other trades following work, such as plastering, drywall, masonry, electrical, etc. to avoid any contact which would reduce the vibration isolation.

H. Bring to the Architect's attention, prior to installation, any conflicts with other trades which will result in unavoidable rigid contact with equipment or conduit as described herein, due to inadequate space or other unforeseen conditions. Corrective work necessitated by conflicts after installation shall be at the responsible contractor's expense.

I. Bring to the Architect's attention any discrepancies between the specifications and field conditions or changes required due to specific equipment selection, prior to installation. Corrective work necessitated by discrepancies after installation shall be at the contractor's expense.

J. Request inspection of any installation to be covered or enclosed, prior to such closure.

K. Manufacturer's Stock: The isolation manufacturer's representative shall maintain an adequate stock of springs and isolators of the type used so that any changes required during construction and checking can be accomplished promptly.

L. Coordination: The contractor and isolator manufacturer's representative shall coordinate the vibration isolation supports with the manufacturers of the equipment to be isolated. Any changes from the documents deemed necessary to provide adequate noise and vibration isolation in accordance with the requirements of the 2011 ASHRAE HVAC Applications, Chapter 48, shall clearly indicated in the submittal shop drawing data. Prior to submitting detailed shop drawings to the Engineer for review, the equipment manufacturer shall approve the shop drawings in writing. Refer to Section 26 00 10 for additional requirements.

3.2 EQUIPMENT ISOLATION

A. Floor mounted machines to be isolated shall be supported by a structural steel frame or concrete inertia base.

B. Brackets, mounting plates, and miscellaneous metal angles shall be provided as required to accommodate the interface of the isolator and the respective isolated equipment. The position and size of the support devices shall be specified and provided by the isolator manufacturer. It shall be the responsibility of the isolator manufacturer to review the actual equipment being supplied to this project under other specification sections and coordinate the mounting arrangement of the isolators.

C. The minimum operating clearance between the equipment frame or rigid steel base frame and the housekeeping pad or floor shall be 1" (25 mm). Minimum operating clearance between concrete inertia base and housekeeping pad or floor shall be 2" (50 mm).

D. After the entire installation is complete and under full operational load, the isolators shall be

adjusted so that the load from the isolated equipment is fully transferred to the isolators.

E. Prior to start up, clean out all foreign matter between bases and equipment. Verify that there are no isolation short circuits in the base or isolators.

F. Verify that all installed isolator and mounting systems permit equipment motion in all directions. Adjust or provide additional resilient restraints to flexibly limit start up equipment lateral motion to 1/4" (6 mm).

G. All conduit shall freely pass through walls and floors without rigid connections. Penetration points shall be sleeved or otherwise formed to allow passage of conduit and maintain 3/4" to 1 1/4" (20 to 32 mm) clearance around the outside surfaces. This clearance space shall be tightly packed with fiberglass, and caulked airtight after installation of conduit.

H. Isolators shall be installed with the isolator hanger box attached to, or hung as close as possible to, the structure. Hanger rods shall be aligned to clear the hanger box.

I. Isolators shall be suspended from substantial structural members where possible. Suspending isolators from slab diaphragm should be avoided when possible.

J. The isolation system shall not be resonant with the structural steel support resonant frequencies or the driving frequencies of the supported equipment.

3.3 INSPECTION

A. After the completion of the vibration isolation systems in accordance with the contract documents, the local representative of the isolation materials manufacturer shall inspect the complete system and report in writing any installation errors, improperly selected isolation devices, and other faults that could affect the performance of the system. This report shall be submitted to the Contractor for corrective action. After the corrective action has been implemented, the local representative of the isolation materials manufacturer shall visit the site and verify the corrective action has been completed.

B. After the corrective action has been completed, the local representative of the isolation materials manufacturer shall provide a submittal to architect via the contractor that certifies in writing that the vibration isolation systems have been installed in accordance with the manufacturer's recommendations and the requirements of the contract documents. All deviations from the contract document shall be noted in this final report.

END OF SECTION 26 05 48

SECTION 26 05 53 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this section shall conform to the requirements of Division 01, “General Requirements,” “Conditions of the Contract” and “Supplementary Conditions.” Specific attention is called to the “Division 26 General Requirements” in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Types of electrical identification work specified in this section include the following:

1. Electrical power conductors
2. Electrical Equipment
3. Raceways including all conduit and junction boxes
4. Operational instructions and warnings
5. Danger signs
6. Equipment/system identification signs

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 26 - Grounding and Bonding for Electrical Systems
C. Section 26 05 33 - Raceway and Boxes for Electrical Systems
D. Section 26 24 16 - Panelboards
E. Section 26 28 00 - Low-Voltage Circuit Protective Devices
F. Section 26 29 00 - Low-Voltage Controllers
G. Section 26 36 23 - Automatic Transfer Switches
H. Section 26 43 13 - Surge Protective Devices for Low-Voltage Electrical. Power Circuits

1.4 REFERENCE STANDARDS

A. All signs, labels, colors, lettering, styles and sizes of electrical identification shall be in accordance with the applicable requirements of the American National Standards Institute (ANSI).

1.5 SUBMITTALS

A. Submit a list of equipment requiring a nameplate as specified herein identifying nomenclature and color field.

PART 2 - PRODUCTS

2.1 ELECTRICAL IDENTIFICATION MATERIALS

A. General

1. Except as otherwise indicated, provide manufacturer's standard identification products of categories and types required for each application. Where more than one single identification type is specified for an application, selection is Installer's option. However, identification product shall be consistent for each application.

B. Color-Coded Plastic Tape

1. Wiring:

a. Not allowed.

2. Raceways:

a. General: Provide manufacturer's standard self-adhesive vinyl tape not less than 1-1/2” wide and not more than 2-1/2” wide for use on conduit.

b. Colors: Conforming to color scheme indicated for “circuit and conduit identification”.

C. Circuit and Conduit Identification

1. Mark with an indelible marker the painted covers of all junction boxes to indicate the circuits contained therein.

2. The covers of junction boxes shall be painted as follows:

| SYSTEM | COLOR |
|---|---|
| Communications/Telephone/Data | Yellow |
| EMCS/Control | Blue |
| Power (Normal) | Black |
| Power (Emergency) | Orange |
| Power (UPS) | Brown |
| Security | Green |

3. The color-code scheme of 2.1-C-2 above shall also apply to all conduit systems as follows:

a. All conduit shall be clearly marked, at intervals not to exceed 20 feet in length of conduit, with color-coded bands. Bands shall be 1-1/2 inches to 2-1/2 inches in width.

b. All conduit shall also be clearly marked with color coded bands as it passes through any barrier or obstruction such as a wall or floor. The bands shall be installed within 12 inches of the obstruction and on both sides of the obstruction.

4. Provide Engineer and Architect with color samples for each system and application listed prior to beginning work. Refer to the Architectural Painting Systems for additional requirements.

D. Self-Adhesive Plastic Signs

1. General: Provide manufacturer's standard, self-adhesive or pressure-sensitive, pre-printed, flexible vinyl signs for operational instructions or warnings; of sizes suitable for application areas and adequate for visibility, with proper wording for each application.

2. Colors: Unless otherwise indicated or required by governing regulations, provide black signs with white lettering for all normal power systems. Provide red signs with white lettering for all emergency power systems.

E. Engraved Plastic Laminate Signs

1. General: Provide engraving stock melamine plastic laminate with engraver's standard letter style of sizes and wording indicated, with white lettering in a field to match color coding identified in 2.1-C-2 except as otherwise indicated, punched for mechanical fastening except where adhesive mounting is necessary because of substrate.

2. Thickness: 1/16”

3. Fasteners: Self-tapping stainless steel screws, except contact-type permanent adhesive where screws cannot or should not penetrate substrate.

F. Baked Enamel Danger Signs

1. General: Provide manufacturer's standard “DANGER” signs of baked enamel finish on 20 gage steel; of standard red, black and white graphics; 14” x 10” size except where 10” x 7” is the largest size which can be applied where needed, and except where larger size is needed for adequate vision; with recognized standard explanation wording e.g., HIGH VOLTAGE, KEEP AWAY.

2.2 LETTERING

A. General: Coordinate names, abbreviations and other designations used in electrical identification work, with corresponding designations shown on construction documents, specified or scheduled. Provide numbers, lettering and wording as indicated in the examples below.

2.3 EXAMPLES

A. Lighting control panel; circuit breakers within switchboards, distribution panelboards, or generators:

**AHU-LL-1**

1. Minimum Dimensions: 1.5” x 3”
2. Minimum Text Height: 3/4”

B. Enclosed circuit breaker, disconnect switch, or variable frequency device:

SERVES
AHU-LL-1
FED FROM HL1A-1,3,5

1. Minimum Dimensions: 2” x 4”
2. Minimum Text Height: 1/8” (line 1) and 1/4” (lines 2 and 3)

C. Panelboard:

**2A**
208Y/120V, 3Ø, 4W +G
NORMAL PANELBOARD
150A, MLO, 22 KAIC RATING
FED FROM PANELBOARD 1A

1. Minimum Dimensions: 3” x 5”
2. Minimum Text Height: 1/2” (line 1) and 1/8” (lines 2 - 5)
3. Required information:
   a. Line 1: Equipment Designation
   b. Line 2: Electrical Characteristics (voltage, phase, number of wires + ground)
   c. Line 3: Power supply system (e.g., Normal OR Emergency) and Equipment Type (e.g., Panelboard)
   d. Line 4: Equipment Ratings (Main Rating; MLO or MCB for panelboards or switchboards, fused or non-fused for switches; minimum AIC rating)
   e. Line 5: Source (equipment name and feeder number where applicable)

D. Automatic Transfer Switches or Equipment with two sources:

**ATS**
208Y/120V, 3Ø, 4W +G
OPTIONAL STANDBY ATS
260A, 22 KAIC RATING
NORMAL FED FROM PANELBOARD 1A
STANDBY FED FROM GENERATOR

1. Minimum Dimensions: 3” x 5”
2. Minimum Text Height: 1/2” (line 1) and 1/8” (lines 2 – 5)
3. Required information:
   a. Line 1: Equipment Designation
   b. Line 2: Electrical Characteristics (voltage, phase, number of wires + ground)
   c. Line 3: Power supply system (e.g., Emergency, Legally Required Standby, Optional Standby) and Equipment Type (i.e., ATS)
   d. Line 4: Equipment Ratings (Main Rating; minimum AIC rating)
   e. Line 5: Normal Source (equipment name and feeder number where applicable)
   f. Line 6: Emergency/Standby Source (equipment name and feeder number where applicable)

## PART 3 - EXECUTION

### 3.1 APPLICATION AND INSTALLATION

A. General Installation Requirements

1. Install electrical identification products as indicated, in accordance with manufacturer's

written instructions, and requirements of NEC.

2. Coordination: Where identification is to be applied to surfaces, which require finish, install identification after completion of painting.

B. Operational Identification and Warnings

1. General: In addition to the requirements indicated herein, wherever reasonably required to ensure safe efficient operation and maintenance of electrical systems, electrically connected mechanical systems, general systems, and equipment, including prevention of misuse of electrical facilities by unauthorized personnel; install self-adhesive plastic signs or similar equivalent identification, instruction or warnings on switches, outlets and other controls, devices, and covers of electrical enclosures. Where detailed instructions or explanations are needed, provide plasticized tags with clearly written messages adequate for intended purposes.

C. Warning Signs

1. Entrances to rooms and other guarded locations that contain exposed live parts, including but not limited to all main electrical service entrance rooms, electrical vaults, and as otherwise indicated on the contract documents, shall be marked with conspicuous warning signs forbidding unqualified persons to enter. Refer to architectural specifications for additional requirements.

D. Equipment/System Identification

1. General: Install engraved plastic laminate sign on each major unit of electrical equipment in the building; including central or master unit of each electrical system, unless unit is specified with its own self-explanatory identification or signal system. Except as otherwise indicated, provide single line of text, 1/2” high lettering on 1-1/2” high sign (2” high where 2 lines are required), white lettering in a field to match color coding identified in 2.1-C-2 Provide text matching terminology and numbering of the contract documents and shop drawings. Provide signs for each unit of the following categories of electrical work.

   a. Panelboards, electrical cabinets, pull boxes, and enclosures (individual sub-components for distribution panelboards).
   b. Motor starters.
   c. Disconnect switches.
   d. Automatic transfer switches.
   e. Transient voltage surge suppressors.
   f. Other equipment as indicated on the drawings and as specified herein.

2. Install engraved plastic laminate sign on each switchboard and panelboard supplied by a feeder to indicate the device or equipment where the power supply originates.

3. Install signs on the equipment at location for best convenience of viewing without interference with operation and maintenance of equipment. For recessed panels in finished areas, provide the sign on the inside of the panel door. Secure to substrate with fasteners, except use adhesive where fasteners should not or cannot penetrate the substrate.

4. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match

and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor's shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

5. Provide computed NEC demand load on all panels and sub-panels utilizing NEC demand factors. In addition, show the load corresponding to the "feeder ampacity" and the "connected load" to determine the adequacy of the panel per NEC.

E. Available Fault Current

1. Service equipment in other than dwelling units shall be legibly marked in the field with the maximum available fault current. The field markings shall include the date the fault current calculations were performed and be of sufficient durability to withstand the environment.

END OF SECTION 26 05 53

# SECTION 26 05 73 - ELECTRICAL POWER SYSTEM STUDY

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of “Division 01, General Requirements,” “Conditions of the Contract” and “Supplementary Conditions.” Specific attention is called to the “Division 26 General Requirements” in Section 26 00 10.

### 1.2 WORK INCLUDED IN THIS SECTION

A. Furnish an Electrical Power System Study for the electrical distribution system as defined herein.

B. The study shall begin at the point of electrical service for the facility and include all downstream distribution and branch panelboards, and significant motor locations (50 horsepower and larger). In addition, all equipment that is required to be rated for the available fault current shall be evaluated in the study including but not limited to transformers, enclosed switches, individual motor controllers, contactors, variable speed drives, and enclosed circuit breakers. The project shall include all new generators and all associated emergency power distribution equipment, including automatic transfer switches.

C. Where any part of the electrical system is served by multiple power sources the system configuration that delivers the highest level of fault current shall be used for the evaluation.

### 1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 80 - Electrical Service
C. Section 26 24 16 - Panelboards
D. Section 26 28 00 - Low-Voltage Circuit Protective Devices
E. Section 26 29 00 - Low-Voltage Controllers
F. Section 26 32 00 - Packaged Generator Assemblies
G. Section 26 36 23 - Automatic Transfer Switches
H. Section 26 43 13 - Surge Protective Devices for Low-Voltage Electrical Power Circuits

### 1.4 REFERENCED STANDARDS

A. All studies shall be performed in accordance with the latest applicable industry standards including the following:

1. IEEE/ANSI Standard 242 – Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems

2. IEEE 1584 – Guide for Performing Arc-Flash Hazard Calculations

3. NEMA AB 3 – AB 3 Molded Case Circuit Breakers and their Application

4. NFPA 70 – National Electrical Code

5. NFPA 70E – Standard for Electrical Safety in the Workplace

### 1.5 SUBMITTALS

A. The following submittal data shall be furnished according to the General Conditions and Section 26 00 10 and shall include, but not be limited to:

1. The final report shall be indexed and contain individual, tabbed sections. The tabbed section shall contain the information as outlined in Part 2 of this document including the following:

a. Firm Name, Address, Phone Number and Professional Engineer's signature and seal of the registered professional Engineer that performed the study,
b. Method used to perform the study and analysis,
c. Short-circuit analysis with protective device evaluation,
d. Protective device coordination study,
e. Arc Flash Analysis,
f. Input Data,
g. One-line diagram,
h. Electronic file used in creating the study including all non-standard curves created to produce the study. The software version including revision number shall be clearly indicated.

1.6 QUALITY ASSURANCE

A. Short-Circuit Analysis, Arc Flash Analysis and Coordination Study shall be performed by a Professional Engineer registered in the Commonwealth of Virginia. Study shall be signed and sealed by the Engineer.

PART 2 - PRODUCTS

2.1 MANUFACTURER

A. If it complies with the specifications, the engineering study specified herein shall be prepared by the following manufacturer providing the electrical switchboards for this project as specified in Section 26 24 13 or manufacturer's authorized engineer:

1. Eaton/Cutler Hammer
2. Siemens
3. Square D
4. General Electric

2.2 SHORT-CIRCUIT ANALYSIS WITH PROTECTIVE DEVICE EVALUATION

A. Systematically calculate fault currents based on the available fault current at the facility service entrance. Study preparer shall obtain the maximum available fault current and power factor or X/R ratio of the fault current at the service entrance from the local utility.

B. Fully rated circuit breakers shall be used. Series connected rated circuit breakers are not to be used.

C. Evaluate the distribution device and equipment ratings compared to the calculated fault current and make recommendations where equipment is overdutied. Analyze the short-circuit currents by preparing a tabulation comparing the fault levels to the device interrupting ratings.

D. When the power factor or X/R ratio of the maximum symmetrical fault calculated at the device location is determined to be more inductive than that used to establish the device interrupting rating adjust the available short circuit current in accordance with ANSI/IEEE standard 242.

E. Calculations shall be presented in tabular form and shall include:

1. Location identification
2. Voltage
3. Manufacturer and type of equipment
4. Equipment short-circuit current rating
5. Calculated short-circuit current (and adjustments due to high X/R ratio where required)
6. Calculated X/R ratio of the fault
7. Recommendations where equipment is calculated to be overdutied.

F. This study and the results of this study shall be provided to and approved by the Engineer and Authority Having Jurisdiction prior to purchasing and installing all new electrical equipment on this project. All necessary modifications to the new electrical equipment in order to maintain the UL listing of the equipment as a result of this study shall be provided by the Contractor and the costs associated with these modifications shall be included in the Contractor's bid.

## 2.3 PROTECTIVE DEVICE COORDINATION STUDY

A. Prepare coordination time-current characteristic curves to determine the required settings of the protective devices to achieve selectivity. The utility upstream protective device feeding the facility shall be maintained as the upper limit for coordination. These settings shall be obtained by the preparer, along with any other protective device setting requirements. The coordination curves shall be prepared on log-log paper and illustrate adequate clearing times between series devices. The curves provided shall reflect actual protective devices (manufacturer and model number) to be installed. Adequate time-current curves shall be generated to depict coordination.

B. The coordination study shall include ground fault protection coordination and recommended device settings for the devices provided with ground fault protection.

C. Protective device characteristics shall be plotted to reflect calculated short-circuit levels at the location.

D. A narrative analysis shall accompany each coordination curve sheet and describe the coordination and protection. All curve sheets shall be multi-color or use hatching for improved clarity. Areas lacking complete coordination shall be highlighted and reasons provided for allowing condition to remain or provide recommendations to improve coordination.

E. The following information shall be provided on all curve sheets:

1. Device identification and associated settings/size
2. Voltage at which curves are plotted
3. Current multiplier
4. ANSI frequent fault damage curve
5. Cable insulation damage curves
6. Transformer inrush point
7. Single-line for the portion of the system
8. Motor starting profiles (where applicable)

F. The recommended device settings shall be provided in tabular form and shall include:

1. Location Identification
2. Voltage
3. Device Manufacturer

4. Device catalog number/series
5. Adjustable long time pickup and delay
6. Adjustable short time pickup and delay
7. Adjustable ground fault pickup and delay
8. Adjustable instantaneous pickup

## 2.4 ARC FLASH ANALYSIS

A. An arc flash analysis shall be performed based on the short circuit values and device settings in conjunction with a short circuit and coordination study. The results from the short circuit and coordination study shall be used to determine arc energy levels at each power distribution location in the facility where work could be performed on energized parts.

B. Where any part of the electrical system is served by multiple power sources the system configuration that delivers the highest level of arc flash incident energy shall be used for the evaluation. Include significant motor contribution (all motors 50 horsepower or greater) in the calculations.

C. For each location with a main device, the line and load side fault contributions shall be included in the calculations.

D. Arc Flash Calculation results shall be presented in tabular form and shall include:

1. Location identification
2. Voltage
3. Arcing fault magnitude
4. Protective device clearing time
5. Duration of arc
6. Arc flash boundary
7. Working distance
8. Incident energy
9. Hazard Risk Category

E. Equipment Labeling shall be provided in accordance with NFPA 70E. Arc Flash Warning Labels shall be provided as part of the report. Labels shall be provided in accordance with ANSI Z535.4. The labels shall be approximately 4 inch x 4 inch thermal transfer type labels of high adhesion polyester for each work location analyzed. Labels shall be machine printed with no field markings. The labels shall include the following information:

1. Location designation
2. Nominal voltage
3. Arc flash boundary
4. At least one of the following:
    a. Available incident energy and the corresponding working distance, or the arc flash PPE category for the equipment, but not both
    b. Minimum arc rating of clothing
    c. Site-specific level of PPE

## 2.5 INPUT DATA

A. The study shall be conducted based on the equipment and conductors being installed. Input data for the report shall be compiled from the contractor, manufacturers, and codes and standards as required providing the studies and conducting a proper evaluation.

B. Conductor lengths, sizes, material, and raceway information shall be provided by the Electrical Subcontractor to the engineer performing the study so that the study is performed based on the final installation.

C. Input data used for the studies shall be provided in the final report. The input data for the report shall include supporting data from the manufacturer used for the evaluation, lengths of all feeders provided by the contractor, transformer data, motor data, utility company and motor generator data and any other supporting data to the report.

2.6 SINGLE-LINE DIAGRAM

D. The final report shall include a single-line diagram of the electrical distribution system within the scope of the project. The single-line shall include:

1. Transformer rating, voltage ratio, impedance, and winding connection.

2. Feeder cables per phase, neutral and ground sizes, length of cable, conductor material, and conduit size and type.

3. Panelboards, fuses, circuit breakers, automatic transfer switches, and enclosed switches.

4. Protective relays with appropriate device numbers and CTs and PTs with associated ratios.

PART 3 - EXECUTION

3.1 INSTALLATION

A. The electrical equipment and protective devices shall not be installed prior to confirmation of adequate equipment fault ratings as specified herein.

3.2 FIELD SETTINGS

A. Adjustments shall be made to the protective devices as required for placing the equipment in final operating condition. The settings shall be in accordance with the approved short circuit study and protective device evaluation / coordination study.

B. Arc Flash Labels shall be affixed to the equipment after the study is approved by the Engineer.

1. One label shall be affixed at a height and location that is clearly visible on the front of panelboards, switchboards, switchgear, motor control centers and other electrical equipment that is less than 8 feet in length unless otherwise noted.

2. Labels shall be affixed at similar locations on each type of equipment. For example, multiple panelboards of similar size shall have labels located at the same location on the front of each panelboard cover.

3. Where multiple labels are affixed to equipment, the labels shall be identical and identify the worst case information for that equipment.

4. Where equipment is rear connected affix labels on both the front and rear of the equipment. Locate the labels on the rear at a similar height and location as on the front.

C. Device settings and adjustments and affixing of Arc Flash Hazard Labels shall be by one of the

following companies: Cutler Hammer Engineering Services, Siemens Industrial Services, Square D Technical Services, General Electric, or the equipment manufacturer's authorized service and testing organization.

END OF SECTION 26 05 73

SECTION 26 05 80 - ELECTRICAL SERVICE

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. The Contractor shall obtain from the Utility Company's project representative the most recently revision of the utility company's drawings and specifications for electrical service. The Contractor shall familiarize itself with the Utility Company's requirements for electrical service and installation. Service shall be in accordance with all applicable utility company's requirements.

B. As part of the work under this Division, the Contractor shall furnish and install the electric service laterals from the main panelboard to the utility transformer(s) as required by the electrical utility company.

C. Furnish and install a metering cabinet as required by the electrical utility company.

D. Prior to order, Contractor shall submit service disconnect, metering cabinet, and panelboard drawings to electrical utility for approval. Copies of approved switchboard drawings shall be included in Engineers' submittals.

E. Service shall be in accordance with all applicable utility company's requirements.

F. Where a service raceway enters from an underground distribution system, it shall be sealed. Spare or unused raceways shall also be sealed. Sealants shall be identified for use with the cable insulation, shield or other components.

G. Service-entrance conductors shall not be spliced.

H. Service equipment shall be suitable for the short-circuit current available at its supply terminals.

I. Project specific details shall be in accordance with the local utility company's recommendations and published guidelines acquired directly from the utility company by the contractor. Requirements shall be processed through the submittal process which shall include the review and approval by the local utility company. Utility company approved shop drawings shall be included within engineers' submittal.

J. Contractor shall obtain all necessary permits directly from the local electrical utility as required for performing any work related to or governed by the electrical utility.

K. Contractor shall be responsible for obtaining all utility company inspections, tests, and approvals directly from the local utility prior to building occupancy.

L. Contractor shall be responsible for providing all necessary temporary power for the construction process. Refer to the General Conditions for additional requirements.

M. Contractor shall install all utility accessories, including but not limited to vault access ladders,

padlocks, etc. provided by electrical utility company.

N. Contractor shall provide all compression terminations for primary service lateral feeders.

O. In addition to items shown on the contract documents, contractor shall provide all additional pull boxes and cable pulling structures modifications as required to connect the electrical service from the utility transformer(s) to the building panelboard.

P. Contractor shall provide all service feeder terminations not provided by the electrical utility.

Q. The Contractor shall obtain and pay all connection fees, tap charges, if any, assessed by Local Authorities. The Owner shall reimburse the Contractor for all connection fees, tap charges, if any, assessed by Local Authorities.

1.3 SYSTEM VOLTAGE

A. Refer to the drawings for the building system voltage requirements.

END OF SECTION 26 05 80

SECTION 26 05 81 - PUBLIC TELEPHONE SYSTEM

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Contractor shall furnish and install an empty conduit system for the telephone service throughout the project as shown on the contract documents to be utilized by the telephone company for installation of service cables.

B. Contractor shall furnish and install telephone entrance conduits as required by the telephone company. See Section 26 00 10 for sleeve requirements.

C. Unless otherwise required by the telephone service utility provider, Contractor shall install flame retardant 3/4" thick 4' x 8' plywood telephone boards in the location determined by the utility provider. Extent, arrangement, and type of plywood mounted within these rooms and remote closets shall be in accordance with the requirements of the telephone service utility provider.

D. Any conduit run inside the building must be metallic and must be grounded.

E. Telecommunications pathways should be coordinated with the electrical distribution system in order to maintain a minimum 12" separation between parallel runs of telecommunications and electrical cabling. Where electrical and telecommunications cable paths intersect, they must cross at right angles only.

F. A maximum of two bends not to exceed 90 degrees each, or 100 feet of linear run, may be included in the backbone conduits between pull boxes or access points. A third 90 degree bend can be utilized in a conduit route if the third bend is within 12" of a pulling access point.

G. Refer to Section 26 05 53 for conduit and junction box identification requirements. The identification system shall be in accordance with the requirements and recommendations of ANSI/TIA-606. All horizontal and backbone conduits shall be assigned a unique alphanumeric designation for identification purposes, and permanent, non-removable labels affixed at both ends and in line as required.

H. Telephone and data outlets located in insulated walls shall have a 1” conduit with pull string, cord, or tape extending from wall mounted J-box up to ceiling space for each gang. Grommets shall be provided at the end of the conduit in the ceiling space. Telephone and data outlets located in non-insulated walls will only require pull string, cord, or tape extending from wall mounted J-box up to ceiling space for each gang. Each pull string, cord, or tape shall have a minimum pulling strength of 600 lbs.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables

B. Section 26 05 33 - Raceway and Boxes for Electrical Systems

C. Section 26 05 53 - Identification for Electrical Systems

1.4 REFERENCE STANDARDS

A. All raceways and telephone backbone systems shall be designed, manufactured, tested and installed in compliance with the following standards:

1. ANSI/TIA-568-C

2. ANSI/TIA-569-C

3. ANSI/TIA-607-B

4. ANSI/TIA-606-B

END OF SECTION 26 05 81

SECTION 26 05 93 - TESTING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN OTHER SECTIONS

B. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
C. Section 26 05 23 - Control-Voltage Electrical Power Cables
D. Section 26 05 26 - Grounding and Bonding for Electrical Systems
E. Section 26 09 23 - Automatic Control Devices
F. Section 26 24 16 - Panelboards
G. Section 26 28 00 - Low-Voltage Circuit Protective Devices
H. Section 26 29 00 - Low-Voltage Controllers
I. Section 26 32 00 - Packaged Generator Assemblies
J. Section 26 36 23 - Automatic Transfer Switches
K. Section 26 41 00 - Facility Lightning Protection
L. Section 26 43 13 - Surge Protective Devices for Low-Voltage Electrical Power Circuits
M. Section 26 50 00 - Lighting
N. Section 28 31 00 - Fire Detection and Alarm System

1.3 WORK INCLUDED IN THIS SECTION

A. During the progress of the work and upon completion, tests shall be made as specified herein and as required by authorities having jurisdiction including inspectors, Owner, Architect, or Engineer. Tests shall be conducted by the Contractor as part of the work of this Division and shall include the services of qualified personnel as well as all equipment, apparatus, and services required. Each wiring system with devices connected must test free from short circuits and from grounds and must have an insulation resistance between conductors and ground, based on maximum load, not less than the requirements of the latest edition of the National Electrical Code.

B. Prior to the execution of testing, the Contractor shall notify the Engineer of proposed test procedures and forms. The procedures and documentation shall include the items listed herein, applicable code requirements, and manufacturer’s recommended installation and acceptance test procedures.

C. Tests shall include, but not be limited to, the following:

1. Wire and cable insulation, refer to Sections 26 05 19 and 26 05 16.

2. Grounding system continuity, refer to Section 26 05 26.

3. Running of mechanical and plumbing equipment with demonstration of controls and interlocks.

4. Operation of electrical equipment and appliances whether provided under this Division or not including any portions of electrical power system as required by Engineer, Architect or Owner.

5. Demonstration and operating test of the fire alarm and detection system including all indicating devices, initiation devices, communication system, on/off controls, door locks, annunciation, fire protection systems, and any other devices or systems. Any tests required by the City, County or State Authorities will be performed. Engineer shall be notified two weeks prior to test for witnessing test. Refer to Section 28 31 00.

6. Operation of standby power system while the system is fully functional to assure proper operation of the emergency generator system, transfer switches, etc. Provide load bank connected on the load side of each automatic transfer switch to allow testing of generator, automatic transfer switches and distribution system under loaded conditions. Engineer shall be notified two weeks prior to test for witnessing test. Refer to Sections 26 32 00, 26 13 20, and 26 36 23. Test shall include the following plus any other required manufacturer's tests.

    a. Follow NFPA 110 Installation Acceptance Testing.
    b. Witness test all of the generator system safety, trouble, alarm features.
    c. Automatic starting through loss of utility power.
    d. Witness test all automatic transfer switch interfaces and operations.
    e. Confirm interfaces with EMCS, fire alarm system, remote annunciator, etc.
    f. Confirm proper functioning of associated equipment (i.e. fuel system, dampers, ventilation, exhaust, etc.).

7. After the electrical distribution system including distribution panels, lighting panels, disconnects, automatic transfer switches, motor starters, and equipment feeder terminations has been final checked, adjusted, and calibrated, and is under load, it shall be subjected to a thermographic test using an infrared temperature scanning unit. Siemens, General Electric, Eaton Cutler-Hammer or Square D instrumentation division shall perform this test. The required quantity of copies of the test report shall be submitted to the Engineer upon completion of test. Connections indicated having higher temperatures than acceptable shall be tightened or corrected as required. After corrections have been made the connections shall be rechecked to confirm that the problem has been corrected. Non-loaded typical floor tenant panels shall be tested under the tenant contract drawing package. Submit list of equipment to be tested for confirmation prior to testing.

8. Test the operation and program the lighting schemes for the dimming system(s); confirm proper settings and programming with Owner's Representative. Confirm interfaces and operation with the EMCS.

9. Test the operation and program the timing schemes for the lighting relay, time clock, and contactor control systems. Confirm proper settings and programming with Owner's Representative. Confirm interfaces and operation with the EMCS.

10. Provide the lightning protection grounding system tests per Section 26 41 00.

11. Test the operation of all automatic transfer switches. Loss of power on the line side of each transfer switch shall confirm proper standby power sequence of operation. All transfer switch timer settings shall be confirmed/adjusted and written in report. The contractor and manufacturer shall provide the necessary equipment to ensure that the required that the Standby Systems are energized and operational within sixty seconds. These systems shall be as defined by the local, state, and federal codes which shall include but not be limited to lighting, exit signs, fire alarm system, etc.

12. Test the operation of all the SPD's with verification of unit status, indicators, diagnostics, common alarm to EMCS, etc.

13. After work of project is complete and during Final Review, the contractor shall open (remove and reinstall covers upon inspection) all electrical equipment that was affected by work of this project for visual inspection to include but not limited to visually confirm terminations, cleanliness, color coding, circuit identification, grounding, and wiring. Coordinate inspection with Engineer.

14. Lighting control devices and control systems shall be tested to ensure that control hardware and software are calibrated, adjusted, programmed, and in proper working condition in accordance with the construction documents and manufacturer's installation instructions. For all occupancy sensors, time switches, programmable schedule controls, and photocells:

    a. Confirm that the placement, sensitivity, and time-out adjustments for occupancy sensors yield acceptable performance, lights turn off only after the space is vacated and do not turn on unless the space is occupied.

    b. Confirm that the time switches and programmable schedule controls are programmed to turn the lights off.

    c. Confirm that the photosensor controls reduce electric light levels based on the amount of usable daylight in the space as specified.

    d. The responsible party for this testing shall not be directly involved in either the design or construction of the project, and shall provide documentation certifying that the installed lighting controls meet or exceed all documented performance criteria. Certifications shall be specific enough to verify compliance.

D. All tests to be witnessed by Engineer shall be completely pre-tested by Contractor and all systems shall be operational prior to Engineer's review. Should a second trip be necessary to re review a test for an inoperable system, Contractor shall be responsible for compensating Engineer for his time based on Engineer's hourly rates and reimburse any travel expenses.

E. All tests must be performed and test results received by Engineer including witnessing of tests prior to final review by Engineer.

F. Contractor shall submit the required quantity of copies of each complete test report specified herein to the Engineer. Each test report shall include at a minimum all model numbers of test equipment with last calibration date, test personnel, date, time, weather conditions, witnesses, passing requirements, test procedures, test results, equipment settings, etc.

G. The foregoing shall in no way relieve the Contractor of the warranty requirements.

1.4 BUILDING COMMISSIONING GENERAL REQUIREMENTS

A. Refer to specification Section 26 08 00 and the alternate pricing requirements indicated in Section 26 00 00 for additional building commissioning general requirements.

END OF SECTION 26 05 93

SECTION 26 09 23 - AUTOMATIC CONTROL DEVICES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the control devices as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 93 - Testing
D. Section 26 27 26 - Wiring Devices
E. Section 26 50 00 - Lighting

1.4 REFERENCE STANDARDS

A. All wiring devices shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code

2. Underwriters Laboratories

3. Americans with Disabilities Act (ADA)

4. ASHRAE/IESNA 90.1

5. International Energy Conservation Code (IECC)

6. NEMA Guide Publication WD 7-2011

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, specifications, and engineering data for each type and size of device with the appropriate cover plate.

B. Submit occupancy sensor illustration pattern tested in accordance with NEMA Guide Publication WD 7-2011. The illustration shall include grid patterns with a maximum 3’x3’ cell size, clearly differentiating between major and minor motion detections. In addition, the illustration shall include the following:

1. Mounting height
2. Maximum horizontal coverage angle
3. Dimensions of coverage area
4. Dimensions of room (length, width, height)
5. Floor surface material

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:
1. Hubbell
2. Lutron Vive
3. Legrand – Pass & Seymour
4. Or approved equal

B. The entire room control system shall be provided by one manufacturer.

2.2 WALL MOUNTED OCCUPANCY SENSORS

A. Furnish and install occupancy sensors where required per adopted codes and as indicated on the Drawings. Occupancy sensors shall be provided in quantities to adequately cover the intended areas. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron MA-AS-XX series and shall include the following features:

a. LED status indicator
b. Time delay adjustable between 1 to 30 minutes in 1 minute increments
1) Default setting shall be set to 15 minutes.
c. Default manual on/occupancy off (vacancy sensor)
d. Two on/off buttons for bi-level switching (unless noted otherwise)

2. Device color shall be confirmed with Architect

2.3 CEILING MOUNTED OCCUPANCY SENSORS

A. Furnish and install occupancy sensors where required per adopted codes and as indicated on the Drawings. Occupancy sensors shall be provided in quantities to adequately cover the intended areas. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron LRF2-OCR2B series and shall include the following features:

a. Time delay adjustable between 1 to 30 minutes in 1 minute increments.

1) Default setting shall be set to 15 minutes.

b. Detection settings of Ultrasonic/PIR/Both/Either with 'Both' as default.

c. Retrigger settings of Ultrasonic/PIR/Both/Either with 'Either' as default.

2. Device color shall be confirmed with Architect.

2.4 DAYLIGHT SENSORS

A. Furnish and install daylight sensors where required per adopted codes and as indicated on the

Drawings. Daylight sensors shall be provided in quantities to adequately cover the intended areas. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Single zone switching devices shall be similar to Lutron LRF2-DCRB series and shall include the following features:

    a. Capable of automatically controlling the lighting zone to output steps of 100%, between 50% to 70%, and below 35%, at a minimum.

    b. Capable of being installed in a ceiling up to 1.25" in depth.

2. Multi zone switching devices shall be similar to Lutron LRF2-DCRB series and shall include the following features:

    a. Capable of automatically controlling the lighting zone to output steps of 100%, between 50% to 70%, and below 35%, at a minimum.

    b. Capable of being installed in a ceiling up to 1.25" in depth.

3. Device color shall be confirmed with Architect.

## 2.5 PLUG LOAD CONTROLLERS

A. Furnish and install plug load controllers where required per adopted codes and as indicated on the Drawings. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron RMJS-20R series and shall include the following features:

    a. Capable of controlling a load of 20A at 120V.

## 2.6 DIMMER SWITCHES

A. Furnish and install dimmer switches where required per adopted codes and as indicated on the Drawings. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron PJ2-2BRL-XXX series and shall include the following features:

    a. Capable of multiple on/off buttons in a single gang.

    b. Option for custom engraving of each on/off button.

2. Device color shall be confirmed with Architect.

## 2.7 OVERRIDE SWITCHES

A. Furnish and install override switches where required per adopted codes and as indicated on the Drawings. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron PJ2-2B-XXX series and shall include the following features:

    a. Capable of multiple on/off buttons in a single gang.

    b. Option for custom engraving of each on/off button.

2. Device color shall be confirmed with Architect.

2.8 ROOM CONTROLLERS

A. Furnish and install room controllers where required per adopted codes and as indicated on the Drawings. All devices shall be located per the Manufacturer's recommendations with approval of the Architect.

1. Devices shall be similar to Lutron HJS-0-FM series and shall include the following features:

    a. Capable of controlling a load of 20A at 120V.

    b. Interface with occupancy sensors, daylight sensors, plug load controllers, override switches, and all other room controlling devices via Class 2 low voltage wiring.

PART 3 - EXECUTION

3.1 INSTALLATION

A. All rooms with ceiling mounted occupancy sensors shall be provided with an override switch. The quantity of on/off buttons on the override switch shall meet or exceed the combined quantity of lighting and plug load control zones in each room/area.

B. All occupancy sensors shall be set by default to manual on/automatic off for lighting. The following locations shall be set to automatic on/automatic off unless noted otherwise:

1. Restrooms
2. Public Lobbies
3. Public Corridors
4. Stairwells
5. Other areas specifically noted on the Drawings

C. All occupancy sensors shall be set by default to automatic on/automatic off for receptacle control.

D. Occupancy sensor time-outs shall be set to 20 minutes.

E. Provide all accessories to include but not limited to power packs, control and power wiring/conduit, relays, cover plates, junction boxes, backboxes, etc. for a complete and operational automatic control system. Equipment located within plenums shall be plenum rated for intended space.

F. All duplex 120V 15 and 20A receptacles located in private offices, open offices and computer classrooms shall have the top receptacle(s) controlled by the plug load controllers unless otherwise noted. All quadruplex 120V 15 and 20A receptacles located in private offices, open offices, and computer classrooms shall have one duplex fully controlled. The Contractor shall

install an additional 120V switch leg from each room controller to each receptacle for control.

3.2 SWITCH MOUNTING

A. Switches shall be mounted vertically. Notify Architect of discrepancies before roughing in outlet and obtain a new location as necessary. Mounting height shall be in accordance with Section 26 05 33 unless otherwise noted on the contract documents.

3.3 LOW VOLTAGE CONTROLS

A. Where low voltage controls are incorporated for local switching, a layout of relays shall be permanently placed in the facility's main electrical room and panels and circuits shall be clearly identified.

3.4 COMMISSIONING

A. Contractor shall commission, set, and confirm all adjustable settings and sensor placement per Manufacturer's recommendations to ensure a trouble-free occupancy based lighting control system. Refer to Sections 26 05 93 and 26 08 00 for additional requirements.

END OF SECTION 26 09 23

SECTION 26 24 16 - PANELBOARDS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide dead front panelboards sized, rated, and complete with the quantity and size of circuit breakers as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 53 - Identification for Electrical Systems
D. Section 26 05 93 - Testing
E. Section 26 28 00 - Low-Voltage Circuit Protective Devices

1.4 REFERENCE STANDARDS

A. All panelboards shall be designed, manufactured, tested and installed in compliance with the following standards:

B. National Electrical Manufacturers Association (NEMA)
   1. PB 1 Panelboards

C. National Fire Protection Association (NFPA)
   1. 70 National Electrical Code

D. Underwriters Laboratories (UL)
   1. 50 Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
   2. 67 Standard for Safety Panelboards

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, engineering data, specifications, directory card sample and tabulation of panelboards and overcurrent devices including panel rating, dimensions, lugs, covers, breaker arrangement, breaker type, frame, trip and interrupting rating, ground bus and provision for future.

B. Submit ¼” = 1’-0” scale layouts of all electrical and data/communications rooms showing dimensioned switchboards, panelboards, disconnects, troughs, and any other electrical equipment or other building system components taking up substantial space.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURES

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. Siemens
2. Eaton Cutler-Hammer
3. Square D
4. General Electric

## 2.2 GENERAL

A. Panelboards shall be standard catalog items complying with NEC, UL, and NEMA standards and bear the label of Underwriters Laboratories.

B. Panelboards shall be identified by laminated plastic nameplates indicating panelboard designation. Refer to Section 26 05 53 for identification requirements.

## 2.3 PANELBOARDS

A. Provide copper full size phase and neutral busses which have been rated in accordance with UL 67 heat rise tests. Bus bar connections shall be column consecutive phase sequence type. 120/208 volt panelboards shall have bus bars drilled and equipped with all hardware for bolt-in breakers.

B. Provide bolted on grounding bus(es) with main lug(s) unless otherwise noted.

C. Panelboard short circuit bracing and breaker interrupting capacity shall be as indicated on the manufacturer's short circuit and fault current coordination study. Levels shall not be less than 10,000 amperes rms symmetrical for 120/208 volt service.

D. Provide manufacturers' standard #16 gage (minimum) galvanized sheet steel cabinets with enamel hinged front cover for door in door construction to box, master keyed door locks, multiple knockouts, wiring gutters, and typed circuit directory depicting each pole position in a two-column fashion.

E. Busbars shall be sized to limit the temperature rise within the panelboard to 50°C with a 40°C ambient temperature. Busbars shall be round edge copper with bolted joint connections. Bolted joint connection locations shall be readily accessible for maintenance. This applies to phase, neutral, and ground busbars.

F. Panelboard main lugs connected to #6 AWG or larger conductors shall be furnished to accommodate compression connectors. Adequate wiring space shall be provided to accommodate the compression connectors. Refer to Section 26 05 19 for compression connector requirements.

G. All two-section panelboards shall have equivalent short circuit bracing and be connected with copper cable equal to or greater than the main bus amperage capacity.

H. Panelboard neutral bars shall be sized to accommodate the neutral feeder sizes indicated on the contract documents. This shall include but not be limited to the up-sizing of neutral bars as much as 200% above the NEC minimum requirements.

I. Where indicated on the contract documents, provide insulated/isolated ground bus bars.

J. Provide pad locking hardware on circuit breakers for small pieces of equipment which are

hardwired without a local disconnecting means (i.e. electric water cooler, dishwasher, ice maker, etc.).

K. Contractor shall provide branch circuit wiring, conduit, and overcurrent protection meeting submitted and approved equipment's respective nameplate data (maximum overcurrent protection (MOP) and minimum circuit ampacity (MCA)).

2.4 ALUMINUM OPTION

A. It would be an acceptable option for the Contractors to use aluminum bus within panelboards in lieu of copper bus within panelboards. The Contractors shall be responsible for coordinating the additional space required due to the aluminum equipment.

PART 3 - EXECUTION

3.1 SURFACE-MOUNTED PANELS

A. All surface mounted panels shall be mounted on 12 gauge formed steel channel having a cross section dimension of at least 1 1/2 inches by 1 1/2 inches. The channel and fittings shall have Galv-Krom or hot dipped galvanized finish. Channels shall be installed vertically and/or horizontally.

3.2 RECESSED/FLUSH-MOUNTED PANELS

A. All recessed- or flush-mounted panels shall be coordinate by the Contractor to ensure the associated wall is of sufficient inside clear depth to accommodate the panelboard cabinet. Coordinate with other Divisions to ensure work of other trades such as wall framing, piping, ductwork, structural members, etc. will not be in conflict with the panel. Install suitable panel cover flush with finished wall surface without gaps.

3.3 COORDINATION

A. Coordinate conduit rough in for panelboards.

3.4 CONNECTIONS

A. Connect conduit, wireways, and wiring to panelboards.

3.5 CLEANING

A. Vacuum clean all dirt and debris from the inside and outside of the panelboards.

3.6 CIRCUIT DIRECTORY CARDS

A. Type-in the panelboard's circuit directory card clearly identifying the use of each circuit and place behind clear plastic on the inside of the panelboard door depicting each pole position in a two-column fashion. The card shall be removable at the completion of the project.

B. Contractor shall provide updated typed panelboard's circuit directory cards reflecting all updates and revisions which have occurred during the course of construction. Hand-written notes on the panelboard's circuit directory cards will not be accepted.

C. The directory card shall indicate "Fed from ------," indicating the upstream electrical source. This "Fed from -----" is in addition to the panel labeling required in section 260553.

3.7 CLEARANCE

A. Contractor shall install panelboards with proper NEC clearances. No piping, ducts, or equipment foreign to the electrical equipment or architectural appurtenances shall be permitted to be installed, enter or pass through such required clearance spaces.

3.8 SPARE PARTS

A. Provide one stock circuit breaker for each type installed for each panelboard. These spares will not be indicated on the drawings and should be provided installed in addition to what is shown on the drawings.

END OF SECTION 26 24 16

SECTION 26 27 26 - WIRING DEVICES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of “Division 01, General Requirements,” “Conditions of the Contract” and “Supplementary Conditions.” Specific attention is called to the “Division 26 General Requirements” in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the wiring devices as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 09 23 - Automatic Control Devices

1.4 REFERENCE STANDARDS

A. All wiring devices shall be designed, manufactured, tested and installed in compliance with the following standards:
   1. NFPA 70 National Electrical Code (NEC)
   2. Underwriters Laboratories
   3. Americans with Disabilities Act (ADA)

1.5 SUBMITTALS

A. Submit manufacturers’ literature, illustrations, specifications, and engineering data for each type and size of device with the appropriate cover plate.

PART 2 - PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

   1. Hubbell
   2. Pass & Seymour
   3. General Electric
   4. Leviton
   5. Lutron

2.2 LINE VOLTAGE FLUSH SWITCHES

A. Furnish and install commercial specification grade line voltage flush toggle switches where indicated on the Drawings. Switches shall have oversized silver alloy contacts, brass terminal screws for # 10 or # 12 copper wire, molded phenolic or urea housing and shall be UL listed. Switches shall be as follows or Engineer approved equivalent:

   1. Load of 20 amps or less, single pole, 120 volts: Pass & Seymour # CS20AC1-GRY or as

specified by the Architect. Contractor shall confirm selection with the Architect.

2. Load of 20 amps or less, located outside, 120 volts: Switch as noted above with Hubbell # HBL1795 cover plate or as specified by the Architect. Contractor shall confirm selection with the Architect.

B. Where more than one switch occurs in the same location, they shall be set in gangs under one cover plate. Switches shall be installed in multiple gang boxes in a uniform position so that the same direction will open and close the circuits throughout the project.

C. Switch color shall be gray unless otherwise indicated on the drawings or as specified by Architect. Contractor shall confirm selection with the Architect.

2.3 RECEPTACLES

A. Furnish and install commercial specification grade plug in receptacles of the type and location as indicated on the Drawings. Receptacles shall have large brass grounding contacts, triple wipe bronze hot and neutral contacts, back wiring screws capable of accepting #10 and #12 copper wire, high impact lexan or urea body and shall be UL listed. Receptacles shall be as follows or Engineer approved equivalent.

1. 20 amps, 120 volts, duplex: Pass & Seymour # CR20-GRY or as specified by the Architect. Contractor shall confirm selection with the Architect.

2. 20 amps, 120 volts, duplex, weatherproof: Pass & Seymour # WR5362 GRY with weatherproof “While in Use” cover plate or as specified by the Architect. Contractor shall confirm selection with the Architect.

3. 20 amps, 120 volts, duplex with GFCI: Pass & Seymour # 2097 GRY or as specified by the Architect. Contractor shall confirm selection with the Architect.

4. 20 amps, 120 volts, duplex, tamper-resistant: Pass & Seymour # TR20GRY or as specified by the Architect. Contractor shall confirm selection with the Architect.

5. Special outlet NEMA numbers shall be as noted on the Drawings.

6. 20 amps, 120 volts, duplex, controlled: Pass & Seymour #5362CDW or as specified by the Architect. Receptacle shall be clearly identified with “controlled” marking and symbol printed on the face. Contractor shall confirm selection with the Architect.

7. 20 amps, 120 volts, duplex, top-half controlled: Pass & Seymour #5362CH or as specified by the Architect. Receptacle shall be clearly identified with “controlled” marking and symbol printed on the top outlet face. Contractor shall confirm selection with the Architect.

B. Where more than one receptacle occurs in the same location, they shall be set in gangs under one cover plate installed in multiple gang boxes in a uniform position.

C. Receptacle color shall be gray for back of house areas or white in finished spaces unless otherwise indicated on the drawings or as specified by Architect. Contractor shall confirm selection with the Architect.

2.4 COVER PLATES

A. Cover plates shall be as follows for both receptacles and switches:

1. Plates shall be Type 302 brushed stainless steel for back of house areas or white nylon for finished spaces unless otherwise indicated on the drawings or as directed by Architect to match device it covers. Contractor shall confirm selection with the Architect.

2. Provide multiple gang cover plate whenever more than one device is indicated together on a wall.

3. Plates for exposed or surface mounted boxes shall be of the galvanized type for interior use and cast metal type with gaskets for moist locations and exterior use.

4. Telephone cover plates shall be supplied and installed by telephone installer.

5. Back of house receptacle outlets connected with emergency power shall have red nylon cover plates.

## 2.5 DIMMERS

A. Refer to specification section 260923.

## 2.6 DECORATOR STYLE DEVICES

A. Furnish and install commercial specification grade decorator style devices where indicated on the Drawings. In addition to the receptacle and switch construction requirements above, devices shall be as follows or Engineer approved equivalent:

1. 20 amps, 120 volts, duplex: Pass & Seymour # 26352-W or as specified by the Architect. Contractor shall confirm selection with the Architect.

2. 15 amps, 120 volts, duplex, weather-resistant: Pass & Seymour # 885TRWRW W with weatherproof “While in Use” cover plate or as specified by the Architect. Contractor shall confirm selection with the Architect.

3. 20 amps, 120 volts, duplex, tamper-resistant: Pass & Seymour # TR26342W or as specified by the Architect. Contractor shall confirm selection with the Architect.

4. 20 amps, 120 volts, duplex with USB A charger: Pass & Seymour #TR5362USBW or as specified by the Architect. Contractor shall confirm selection with the Architect.

5. 20 amps, 120 volts, duplex with USB C charger: Pass & Seymour #TR20USBPDW or as specified by the Architect. Contractor shall confirm selection with the Architect.

6. 20 amps, 120 volts, duplex, controlled: Pass & Seymour #TR26362CDW or as specified by the Architect. Receptacle shall be clearly identified with “controlled” marking and symbol printed on the face. Contractor shall confirm selection with the Architect.

7. 20 amps, 120 volts, duplex, controlled: Pass & Seymour #TR26362HW or as specified by the Architect. Receptacle shall be clearly identified with “controlled” marking and symbol printed on the face. Contractor shall confirm selection with the Architect

B. Where more than one switch or receptacle occurs in the same location, they shall be set in gangs under one cover plate. Devices shall be installed in multiple gang boxes in a uniform position so that the same direction will open and close the circuits throughout the project.

C. Device color shall be gray for back of house areas or white in finished spaces unless otherwise indicated on the drawings or as specified by Architect. Contractor shall confirm selection with the Architect.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Both switch and receptacle screw terminals shall be wrapped with at least one wrap of electrical tape before device is installed in outlet box. Vacuum clean all switch and receptacle outlets prior to and after installation.

B. The highest operable part of electrical and communication outlets shall be placed within the reach ranges as required in The Americans with Disabilities Act (ADA). These requirements do not apply where the use of special equipment dictates otherwise or where electrical and communication outlets are not normally intended for use by building occupants.

C. Receptacles and switches shall be grounded through the use of grounding screws. Grounding through the use of self-grounding receptacles and switches is not acceptable.

D. All receptacle cover plates in back of house spaces shall be labeled indicating panelboard and circuit number serving the associated receptacle(s). Labels shall be clear adhesive tape with black lettering. Provide similar labeling in finished spaces if required by the Owner.

E. GFCI and Arc fault receptacle circuits shall not share neutral wires with normal circuits or multiple receptacle circuits.

3.2 SWITCH MOUNTING

A. Switches shall be mounted vertically. Notify Architect of discrepancies before roughing in outlet and obtain a new location as necessary. Mounting height shall be in accordance with Section 26 05 33 unless otherwise noted on the contract documents.

3.3 RECEPTACLE MOUNTING

A. Receptacles shall be mounted vertically, unless noted otherwise, and properly grounded per the NEC. Notify the Owner’s Representative of any discrepancies before roughing in outlet and obtain a new location as necessary. Mounting height shall be in accordance with Section 26 05 33 unless otherwise noted on the contract documents. Receptacles shall be wired using screw terminals. Quick connect or spring loaded terminals shall not be acceptable.

B. Receptacles shall be mounted with the grounding pole on top. Horizontal receptacles shall be mounted with the grounding pole to the right.

END OF SECTION 26 27 26

SECTION 26 28 00 - LOW-VOLTAGE CIRCUIT PROTECTIVE DEVICES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide manufacturer's standard, UL listed, molded case circuit breakers as shown on the drawings and as specified herein.

B. Provide UL listed fuses as shown on the drawings and as specified herein.

C. Provide main switchboard overcurrent protective devices.

D. Provide disconnect switches as shown on the drawings and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 53 - Identification for Electrical Systems
D. Section 26 05 73 - Electrical Power System Study
E. Section 26 05 93 - Testing
F. Section 26 24 16 - Panelboards

1.4 REFERENCED STANDARDS

A. All protective devices shall be designed, manufactured, tested and installed in compliance with the following standards:

1. National Electrical Manufacturers Association (NEMA)
    a. AB 1 Molded Case Circuit Breakers
    b. KS 1 Heavy Duty Enclosed and Dead-Front Switches (600 Volts Maximum)

2. National Fire Protection Association (NFPA)
    a. 70 National Electrical Code
    b. 70E Standard for Electrical Safety in the Workplace

3. Underwriters Laboratories (UL)
    a. 98 Standard for Safety Enclosed and Dead-Front Switches
    b. 198D Standard for Safety Class K Fuses
    c. 489 Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures
    d. 943 Ground-Fault Circuit-Interrupters
    e. 977 Fused Power-Circuit Devices

4. ANSI C97.1 Low Voltage Cartridge Fuses.

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, engineering data, and specifications for each type and size of device specific to this project. Include a listing of all fuse locations cross referenced with fuse characteristics (type, size, manufacturer, etc.).

B. Submit manufacturer's literature, illustrations, engineering data and specifications for each type and size of disconnect switch specific to this project. A comprehensive list of equipment with each piece of equipment identified by contract document designation shall be submitted with cross reference to type of disconnect switch being supplied for it. Include enclosure NEMA type, fused, non-fused, rejection fuse clips, number of poles, amperage, etc.

PART 2 - PRODUCTS

2.1 MOLDED CASE CIRCUIT BREAKERS IN PANELBOARDS

A. The breakers shall be operated by a toggle type handle and shall have a quick make, quick break, over center switching mechanism that is mechanically trip free from the handle so that the contacts cannot be held closed against short circuits and abnormal currents. Tripping due to overload or short circuit shall be clearly indicated by the handle automatically assuming a position midway between the manual ON and OFF positions. All latch surfaces shall be ground and polished. Poles shall be so constructed that they can open, close and trip simultaneously.

B. Breakers shall be completely enclosed in a molded case with the ampere ratings clearly visible. Contacts shall be non-welding silver alloy. Breakers shall be UL listed for use with 60°, 75°, and 90°C insulated wire.

C. Circuit breaker ratings, modifications, etc., shall be as indicated on the drawings. Molded case circuit breakers shall be as follows:

1. Thermal magnetic type that provides inverse time delay overload and instantaneous short circuit protection by means of a thermal magnetic element.

2. Ambient compensated standard that provides inverse time delay overload and instantaneous short circuit protection by means of a thermal magnetic element. Compensation shall allow the breaker to carry rated current between 25°C and 50°C with tripping characteristics that are approximately the same throughout this temperature range.

3. Multi pole breakers shall be of the common trip type having a single operating handle. Multi-pole 240 volt breakers smaller than fifty amperes may consist of factory connected single pole breakers with a common trip handle.

4. All circuit breakers shall be full size and bolt-on.

5. The minimum interrupting rating of breakers shall be 10,000 amperes RMS symmetrical. Refer to manufacturer's short circuit and fault current coordination study for exact interrupting rating.

D. Provide one stock circuit breaker for each type installed per panelboard.

2.2 FUSES

A. Fuses shall be listed and meet UL and/or NEMA standards for Class K5, J, R, and L fuses.

B. Fuses shall be dual element, high interrupting capacity with current limiting effect, 200,000

ampere RMS symmetrical at rated voltage where possible to coordinate with the series rating of the electrical equipment. Single element devices will only be used where series rating conflicts arise due to UL tested combinations. Bussmann LPS-RK1 and Littelfuse LLS-RK1 are preferred where proper coordination is possible.

C. Unless otherwise indicated on drawings, Class R or J time-delay fuses shall be used for individual motor circuit protection. Switches shall be provided with proper mounting clips

D. Fuse voltage rating shall be 250 volt for 120/208 volt.

E. Provide one complete set of fuses for all switches requiring fuses, including coordination panels. Provide one set of three fuses for each type of fuse installed on the job to be stored in the original boxes in a cabinet in the main electrical room. Cabinet shall be furnished and installed by Contractor. Contractor shall replenish any stored fuses used during construction. For all fuses, one complete spare set of fuses with respective pullers shall be provided to the Owner at completion of project.

F. If it complies with these Specifications, fuses shall be acceptable as manufactured by Bussmann, Mersen/Ferraz Shawmut, or Littelfuse. Refer to specifications for specific manufacturers for certain applications.

## 2.3 DISCONNECT SWITCHES

A. If it complies with these Specifications, disconnect switches manufactured by one of the following manufacturers will be acceptable:

1. Siemens
2. Eaton Cutler-Hammer
3. Square D
4. General Electric

B. Disconnect switches shall be direct drive, quick make, quick break type switches with spring reinforced wire grips and self-aligning switch contacts. Switches shall be enclosed in a heavy sheet metal enclosure with hinged interlocking cover which shall prevent the cover from being opened when switch is "on". Provide rejection type fuse clips for cartridge fuses in switches as indicated and accommodation for R fuses. Switches shall be NEMA type HD. Provide NEMA 3R enclosure for locations exposed to weather which are on the exterior of the building or on.

C. Fused disconnect switches shall be substituted for non-fused switches where required by manufacturer or local inspecting authority and/or fuse protection is indicated on motor nameplate. Refer to fuse requirements specified herein for additional requirements. Contractor shall provide fuses meeting equipment nameplate requirements.

D. Disconnect switches for 208-volt circuits shall be similar except that 250 volt NEMA type HD with rejection type fuse clips for R fuses shall be acceptable.

E. All switches shall have provision for padlocking.

F. Provide extra large legible on/off label.

G. Contractor shall provide readily accessible disconnect switches within sight of all Division 21, 22, and 23 equipment where an integral disconnecting means from the manufacturer or an external disconnecting means via a controller or similar device within sight of the equipment is not present whether indicated on the drawings on not. In lieu of a disconnect switch, a snap

switch shall be permissible for 120V, 1Ø equipment with maximum overcurrent protection requirement of less than 30 amperes. Disconnect switch and snap switch sizing shall be based upon the stated maximum overcurrent protection requirements from the submitted equipment manufacturers' product data.

PART 3 - EXECUTION

3.1 GENERAL

A. Install circuit breakers securely in panels as indicated on the contract drawings. Breakers with adjustable magnetic trip shall be set at the highest set point per the manufacturer's recommendation in coordination with the fault current coordination study except where indicated otherwise on the drawings.

B. Circuit breakers requiring current limiters shall be provided complete with removable current limiters.

C. Circuit breakers shall be fully rated. Series connected rated circuit breakers shall not be used.

D. Provide fuses in all required devices requiring fuses as required by the drawings, local inspecting authorities, and the National Electrical Code.

E. Mount all switches 60A and larger on 12 gauge formed steel channels having a cross section dimension of at least 1 1/2 inches x 1 1/2 inches. The channel and fittings shall have galvcrom or hot dipped galvanized finish to resist rust formation. Install channels vertically. Channels shall be Kindorf or equal.

F. Vacuum clean all dirt and debris from the inside and outside of disconnect switches.

G. All circuit protective devices shall be readily accessible and shall be installed so that the center of the grip of the operating handle of the switch or circuit breaker, when in its highest position, is not more than six foot seven inches above the floor except as otherwise permitted by NFPA 70.

3.2 TESTING

A. The switchboards' protective devices shall be factory tested at the manufacturer's factory prior to shipment as specified herein.

1. Additional auxiliary, pilot, control relays, switches etc., shall have the proper voltages applied to their circuits and satisfactory operation demonstrated.

2. Four (4) copies of all test results certified by the switchboard manufacturer shall be submitted by the Contractor to the Engineer for review. See Section 26 00 10 for requirements.

3. Upon completion of the factory ground fault protection system tests the current and time adjustments on each relay will be set on their minimum values.

3.3 SHORT CIRCUIT AND PROTECTIVE DEVICE COORDINATION STUDY

A. Refer to specification Section 26 05 73 for the short circuit and protective device coordination study requirements.

3.4 ARC FLASH STUDY

A. Refer to specification Section 26 05 73 for the arc flash study requirements.

3.5 EQUIPMENT NAMEPLATE DATA

A. Contractor shall provide overcurrent protection for equipment that does not exceed respective nameplate maximum overcurrent protection (MOP). Contractor shall provide branch circuit wiring and conduit for equipment that will be sufficient to meet the respective nameplate minimum circuit ampacity (MCA).

END OF SECTION 26 28 00

SECTION 26 29 00 - LOW-VOLTAGE CONTROLLERS

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the motor starters as shown on the contract documents and as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 53 - Identification for Electrical Systems
D. Section 26 05 93 - Testing

1.4 REFERENCE STANDARDS

A. All starters shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code

2. IC Standards

3. IEC 947

4. UL 508

1.5 SUBMITTALS

A. Submit manufacturer's literature, illustrations, engineering data and specifications for each type and size of motor starter specific to this project. A comprehensive list of equipment with each piece of equipment identified by contract document designation shall be submitted with cross reference to type of motor starter being supplied for it. Submit schematic wiring diagrams indicating all accessories for each type of motor starter specific to this project. All accessories shall be integrated within overall wiring diagram and the automatic temperature control and fire alarm interface point(s) shall be indicated. Catalog numbers shall contain suffix and/or prefix numbers/letters representing accessory modifications. Suffix and/or prefix numbering definitions shall be included in submittal.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:
1. Siemens
2. Eaton Cutler-Hammer

3. Square D
4. General Electric

## 2.2 MOTOR STARTERS

A. General: All motor starters, required for a particular application, indicated by details or on the Drawings shall be of one manufacturer. Each starter and its component and related parts shall be properly designed and coordinated to suit the characteristics of the motor it controls and the driven equipment. Starters provided with automatic controls shall be capable of making as many frequent starts as the control devices may demand. The horsepower rating of each starter shall be not less than the rating of the motor it controls. Starters shall be in accordance with NEMA standards.

B. Three phase and single phase magnetic motor starters:

1. All controllers shall be molded case three pole (two pole for single phase) magnetic only circuit breakers or disconnect switches in combination with motor starters as shown on the Drawings and specified herein. Interrupting rating of combinations shall be a minimum as indicated in the manufacturer's short circuit and fault current coordination study. Rating shall be UL listed for breaker or disconnect in combination with starter. Fused switch combination starters shall be supplied for motors where required by manufacturer, inspecting authority, and/or fuse protection is indicated on motor nameplate. Where indicated on the drawings for magnetic motor starters only, requirements specified herein shall apply, however, without circuit breakers or disconnect.

2. The motor controllers shall be steel mounted and front wired with all terminals accessible for wiring directly from the front. No slate or ebony asbestos shall be permitted on any size controller from Size 00 through Size 8. All contacts shall be solid silver cadmium oxide alloy which will not require any filing, dressing, or cleaning throughout the life of the control equipment. Bare copper or silver flashed type shall not be permitted. Operating coils shall be pressure molded and so designed that if accidentally connected to excessive voltage they will not expand, bubble, or melt. When a coil fails from over voltage conditions, the motor controller shall definitely drop out and not freeze the contacts in the "ON" position.

3. Provide solid state overload relay. Trip current rating will be established by selection of overload relay and shall be adjustable (minimum 2 to 1 current range). The overload shall contain a built in normally closed contact, be self-powered and ambient insensitive, provide phase loss protection down to 60% of the dial setting and have +/- 2% repeat trip accuracy. In addition, the unit should have a mechanical test function, conformally coated circuit board and provisions to accept a tamper guard. Overload shall also contain a manual reset, conform to NEMA Trip class 10 and be designed for the addition of either normally open or normally closed auxiliary contacts.

4. It shall be possible to field add one or more extra normally open or normally closed interlocks to all motor controllers except Size 00 without removing existing wiring or removing the controller from its enclosure.

5. Each motor controller circuit breaker/disconnect shall be capable of being padlocked in the "OFF" position with up to three padlocks.

6. In general, motor controller enclosures shall be NEMA 1, general purpose unless exposed to the weather or unless otherwise specified on the Drawings. Motor controllers

exposed to the weather shall have NEMA Type 3R watertight.

7. Generally, holding coils in full voltage magnetic motor controllers shall be suitable for use on 120 volt, AC control voltage. Each controller supplying loads above 120 volts nominal shall have a control power transformer with primary fuses and one secondary fuse in the ungrounded phase. The control power transformer shall be properly sized to meet the requirements of Section 25 00 90.

8. All three phase full voltage magnetic motor controllers shall have overload protection in all three phases. All single phase full voltage magnetic motor controllers shall have overload protection in ungrounded phases. All two speed full voltage magnetic motor controllers shall have overload protection in all six legs of the controller.

9. Motor controller shall be provided to properly coordinate with the motors to be furnished by the Mechanical Contractor and by others as set forth in Section 26 00 00. See Divisions 21, 22, 23, and 25 for motor types, horsepower, etc. Electrical Contractor shall coordinate starter size and overload protection with Mechanical Contractor prior to ordering.

10. Two speed motor controllers for two speed motors shall have "decelerating relay" between high and low speed. On dual motor arrangements, motor starters shall be mechanically and electrically interlocked so that both motors cannot operate simultaneously.

11. Motor controllers shall be provided with accessories such as HOA’s, selector switches, pilot lights, terminals, etc., and all internal wiring shall be copper.

12. Each motor controller shall be equipped with a 120 volt green running pilot light, 120 volt red off pilot light, and a hand off automatic switch. All requirements of the Mechanical Contractor and automatic operational control sequences of Section 25 00 90 shall be met. Two speed motor controllers will have "fast slow" pilot lights as well as a fast slow selector switch in conjunction with the hand-off-automatic switch. Starters for cooling tower motors shall have a reverse rotation selector switch that will allow the operator to run the cooling tower fans in a reverse flow mode. Reversing motor controllers shall have normal-reverse pilot lights as well as a normal-reverse selector switch in conjunction with the hand-off automatic switch. Pilot lights will be operated by an interlock on the motor controller not placed across the operating coil. In addition to the "holding interlock and pilot light interlocks", each controller shall have two extra interlock contacts which shall be field convertible to normally open or normally closed.

13. Terminal blocks shall be provided for remote control functions as required by the fire alarm subcontractor Section 28 31 00 and automatic temperature control subcontractor Section 25 00 90. These terminal blocks shall be numbered for incorporation into wiring diagrams.

14. Provide phase loss and undervoltage relay protection for all three phase magnetic starters on motors 3 HP and larger. Provide all the associated accessories for a complete and operational system.

C. Manual Starters shall be provided for fractional horsepower motors where indicated on the Drawings. Each starter shall be equipped with a manually operated trip free switch. 120 volt manual starters shall have a single thermal overcurrent device. Manual starters shall be from Allen Bradley.

PART 3 - EXECUTION

3.1 LOCATION

A. Starters shall be located with proper NEC clearance where indicated on the Drawings.

3.2 MOUNTING

A. Mount all individual starters on Kindorf or Unistrut Channels.

3.3 CONNECTION

A. Connect conduit and wiring to starters as shown on the Drawings and as required for a complete system.

3.4 CLEANING

A. Vacuum clean all dirt and debris from the inside and outside of all starters.

END OF SECTION 26 29 00

SECTION 26 32 00 - PACKAGED GENERATOR ASSEMBLIES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Furnish and install standby diesel generator system of the latest commercial type and design as shown on the drawings and as specified herein.

1. These specifications cover requirements for the design, fabrication, testing and furnishing of a complete and operable standby diesel generator system.

2. Provide all material and parts for the system.

3. The maximum installed generator height including the silencer, exhaust pipe, concrete pad, fuel storage tank, spring isolators, railings, etc. shall not exceed the available space where the generator is located. The generator provided with all of its accessories including the manufacturers recommended clearances shall fit within the physical space limitations of the project and not exceed the available space where the generator is located. Refer to the drawings for the generator location.

4. The maximum installed generator height including the outdoor acoustical weatherproof housing with sound attenuating discharge and intake plenums, silencer, exhaust pipe, flapper cap, concrete pad, fuel storage tank, spring isolators, railings, etc. shall not extend above the top elevation of the architectural screen wall enclosure. The generator provided with all of its accessories including the manufacturers recommended clearances shall fit within the physical space limitations of the project and not exceed the available space where the generator is located. Refer to the drawings for the generator location.

5. The drawings and specification indicate minimum requirements. In addition to these requirements, the generator system as installed on this project shall meet the requirements of the local code authorities without additional cost to the Owner.

B. The installation of the standby diesel generator system shall include the following:

1. Diesel engine driven electric set to provide standby power.

2. Engine start stop control system mounted on the electric set. Control system shall be able to manually and automatically start or stop the set.

3. Unit mounted radiator.

4. Accessories as specified, and/or standard with the specified equipment, including integral frame mounted double wall fuel oil storage tank with interstitial fuel leak detection, flapper cap, oil heaters, immersion type jacket water heater, fuel pumping system, outdoor

acoustical weatherproof housing with discharge and intake plenums, and any other accessory necessary for an unprotected outdoor installation.

5. Energy storage batteries, battery cables, battery tray, and charger.

6. Engine silencer with mounting and isolation brackets.

7. Output circuit breaker and enclosure. Enclosure shall be sized and constructed such that the primary generator output feeders and conduits may exit out of the top, side, or bottom of the generator. Arrangement of enclosure and circuit breaker within shall be field adjustable and shall contain additional knockouts to facilitate top, side, or bottom exit.

8. Necessary relays for remote start stop control from fire alarm panel. In addition, provide a remote recessed flush mounted alarm annunciator panel located in the building engineer's office, as indicated on the drawings, or in a location as directed by the Owner's representative. Coordinate the exact location of this remote annunciator with the Owner's representative.

9. Necessary contacts and open communication protocol for monitoring/control as defined in Section 25 00 90.

10. Necessary contacts for monitoring generator circuit breaker by the fire alarm system.

C. System Responsibility:

1. The manufacturer of the system shall be regularly engaged in the production of diesel engine driven sets, and associated control products for the last ten years, and shall be able to provide maintenance service on a 24 hour basis. The manufacturer shall have local representation including installation and service personnel that are familiar with local codes and the requirements of the local code authorities.

D. Environmental Conditions:

1. The brake horsepower of the engine and kW rating of the alternator with all specified accessories and efficiency losses accounted for shall be capable of delivering full rated kW output as indicated on the drawings at a minimum 0.8 power factor under the environmental conditions as follows:
   a. Altitude: 600 ft above sea level
   b. Maximum ambient temperature: 100ºF
   c. Minimum ambient temperature: 10ºF

2. The generator manufacturer shall review the installation location and conditions as shown of the contract documents, including the MEP and architectural documents, for the potential of generator overheating and capacity de-rating due to localized air and exhaust recirculation. The generator system installed on this project shall account for these issues and deliver the specified capacity under the actual operational conditions for this project.

E. Exceptions:

1. Major exceptions to these specifications will be considered sufficient cause for rejection of bids.

1.3 WORK INCLUDED IN OTHER SECTIONS:

A. Division 23 - Heating, Ventilating and Air Conditioning (HVAC)
B. Section 25 00 90 - Automatic Temperature Control System
C. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
D. Section 26 05 26 - Grounding and Bonding for Electrical Systems
E. Section 26 05 33 - Raceway and Boxes for Electrical Systems
F. Section 26 05 48 - Vibration and Seismic Controls for Electrical Systems
G. Section 26 05 93 - Testing
H. Section 26 36 23 - Automatic Transfer Switches
I. Section 28 46 00 - Fire Detection and Alarm

1.4 REFERENCE STANDARDS

A. American National Standards Institute, Inc. (ANSI)

B. Department of Environmental Quality (DEQ)

C. Environmental Protection Agency (EPA)

D. International Organization for Standardization (ISO 3046, ISO 8528)

E. National Electrical Code (NFPA 70) (NEC)

F. National Electrical Manufacturers' Association (NEMA MG1)

G. National Fire Protection Association (NFPA)

H. State Corporate Commission

I. Underwriters' Laboratories, Inc. (UL 2200)

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, dimensional data, engineering data, specifications, and other information as required to demonstrate compliance with the contract documents and these Specifications.

B. In addition to the submittal requirements of Section 26 00 10 provide:

1. A list of all exceptions and deviations to the specifications.

2. Battery size calculations.

3. Air flow requirements for cooling and combustion.

4. Heat rejection of engine and alternator when operating at full rated load.

5. State guaranteed fuel consumption at 100%, 75%, and 50% load. Also state guaranteed lube oil consumption when submitting test report.

6. Dimensional scale drawings of generator systems in plan view and elevations including but not limited to engine generator, enclosures, fuel systems, and exhaust system including method of mounting exhaust silencer. Drawings shall also indicate manufacturer’s recommended operational, service and maintenance clearances including those clearances required for proper intake and exhaust of ventilation and combustion air

flow rates. Drawings shall also indicated type, size, and location of connection points for other trades including but not limited to fuel piping, controls, exhaust piping, and electrical feeders and conduit. Drawings shall demonstrate the Contractor's intent with regards to primary feeder conduit exit (top, side, or bottom exit).

7. Maximum motor starting capacity at 20% voltage drop.

8. A list of interface, monitoring, and control points as required by Section 25 00 90.

9. Manufacturer's guidelines and recommendations for installation, start-up, check-out and troubleshooting, operational testing, operation, service, and maintenance as installed on this project.

C. Prior to final inspection, provide installation test procedures and operation and maintenance manuals in accordance with Section 26 00 10.

## PART 2 - PRODUCTS

### 2.1 ACCEPTABLE MANUFACTURERS

A. A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. Caterpillar
2. Cummins/Onan
3. Rehlko
4. MTU Onsite Power

B. Generator engine manufacturer shall be Caterpillar, Cummins, Kohler, or MTU.

C. Integral circuit breaker manufacturer shall be Eaton Cutler-Hammer, General Electric, Siemens, Square D.

### 2.2 DIESEL ENGINE DRIVEN ELECTRIC SET

A. Rating:

1. The output rating of the diesel generator shall be based on operation of the set with all necessary accessories operating. The generator shall be capable of carrying an average power output of 85% during a 24 hour period of operation. The electric set shall be rated and capable of producing the capacity shown on the drawings plus reserve factor at 0.8 power factor for standby emergency power service under the environmental conditions specified.

2. Generator output voltage shall be 120/208 volt, 60 hertz, 3 phase, 4 wire.

B. The manufacturer shall certify that this electric set series has been developed through rigorous, comprehensive tests including:

1. Maximum power level.

2. Maximum motor starting capacity.

3. Endurance tests.

4. Structural soundness.

5. Fuel consumption.

6. Engine alternator cooling air flow.

7. Transient response and steady state governing.

8. Alternator temperature rise per NEMA MG1 22.40 definition.

9. Single step load pickup.

10. Three phase short circuit test for mechanical and electrical strength.

11. EPA and Department of Environmental Quality (DEQ) certified emissions standards. Standby diesel generator system shall implement Best Available Control Technology (BACT) as defined by the DEQ with regards to emission levels of nitrogen oxides (NO2), carbon monoxides (CO), sulfur dioxides (SO2) and other products of combustion and visible emissions. The Contractor shall contact the DEQ to confirm current BACT requirements. Include written documentation with submittal package demonstrating compliance with DEQ BACT requirements. The installed generator system shall meet the local, state, and federal codes, EPA, and Department of Environment Quality (DEQ) certified emissions standards.

2.3 ENGINE SPECIFICATIONS

A. Engine Type: Stationary, liquid cooled, full diesel, compression ignition, four cycle, multi cylinder of either vertical in line or V type. Arrange engine for direct connection to alternating current generator. Fuel injection and valves shall not require adjustment while in service. The engine may be naturally aspirated or turbo charged.

B. Horsepower: Manufacturer shall certify engine as capable of developing sufficient brake horsepower to drive an alternator yielding kW capacity as shown on the drawings on a continuous standby basis under the environmental conditions specified in Paragraph 1.2 D hereinbefore.

C. Speed of engine and alternator: Not to exceed 1800 RPM at full rated load.

D. Fuel: Engine shall give satisfactory performance on ASME classified commercial grade of petroleum fuels Designation D 975 Grades 1 D and 2 D. Diesel engines requiring premium fuel will not be considered. Engine shall be capable of operating at light loads for extended periods of time and shall provide for precombustion of the fuel or a similar means for prevention of carbonization.

E. Governor: Equip engine with high performance governor to maintain frequency within limits, as specified below, by controlling engine and alternator speed.

1. Stability: +/- 0.25% maximum variation at any constant load from no load to full load.

2. Regulation: +/- 0.25% maximum frequency deviation between no load steady state and full load steady state.

3. Frequency deviation on transients on applying or removing load shall not exceed 1 second.

F. Fuel system:

1. Include primary and secondary fuel filters with replaceable elements and engine driven mechanical, positive displacement fuel pump, all mounted on the engine. Generator set shall have an automatic fuel primer and a hand fuel primer.

2. A frame mounted double wall fuel oil storage tank (belly tank) with fuel gauge and interstitial fuel leak detection integral with the generator package shall be provided by the generator manufacturer. Double wall tank with leak containment construction shall comply with the requirements of these specifications, have the approval of the local Fire Marshal, be UL 142 listed unless Local Code Authorities requires UL 2085 listed, and complies with all applicable local code requirements. Fuel oil tank construction and fire rating shall be in accordance with all applicable local codes for the particular tank location and application.

3. Double wall fuel oil storage tank shall have adequate capacity for a minimum of forty eight (48) hours operation at full load operation conditions including the minimum fuel circulation rate required to sufficiently cool the generator as recommended by the generator manufacturer. However, under no circumstances shall the total amount of fuel exceed the minimum and maximum limits imposed by the local code authorities.

4. Fuel oil storage tank shall have double wall leak containment construction with fuel gauge and interstitial fuel leak detection. Double wall tank shall comply with all applicable codes and environmental regulations for fuel leak containment to avoid the need for a containment curb system and additional secondary fuel containment devices. Refer to specification Section 23 13 23 for additional information including the required control and alarm monitoring points for the fuel oil storage tank.

5. The footprint of the fuel oil storage tank shall not extend beyond the footprint of the generator.

G. Lubrication system:

1. Include positive displacement mechanical lubrication oil pump to supply oil under pressure to main bearings, crankpin bearing, pistons, piston pins, timing gears, camshaft bearings and valve rocker mechanism.

2. Full flow lubrication oil filters, conveniently located for servicing shall be provided. The filters shall be equipped with a bypass valve to insure oil circulation if filters are clogged.

3. The oil system shall include lubricating oil filter and dipstick oil level indicator.

H. Air Cleaner:

1. Provide a replaceable oil bath or dry element air cleaner of size recommended by engine manufacturer.

I. Starting:

1. Equip engine with electric starting motor, 12 volt or 24 volt system, of sufficient capacity to crank the engine at a speed which will allow full diesel starting of the engine. Starter pinion shall disengage automatically when engine starts. Provide a relay, connected to the alternator, to disconnect the starting circuit when the alternator develops voltage.

2. Provide batteries which will crank the engine as specified in Paragraph 2.5- E when the battery temperature is 32°F and oil temperature is 50°F without discharging below 58%. The batteries shall have sufficient capacity to start the diesel engine a minimum of four times. Battery voltage shall match starter motor voltage. Furnish all necessary intercell connecting and battery cables.

3. Provide an automatic float, voltage regulated battery charger which will recharge the batteries from 58% to 90% within 24 hours and will maintain batteries at full capacity during standby conditions. An ammeter shall indicate the charge rate. Input and output circuits shall be protected by either fuses or circuit breakers. Design charger so that it will not be damaged during engine cranking. Charger shall operate on 120 volt AC, single phase power. Provide auxiliary dry trouble contact for trouble panel alarm. Alarms shall include charger malfunction and high current output.

4. Equip engine with engine driven, battery charging alternator, 30 amp DC output, 12 volt or 24 volts as required to recharge batteries to normal starting requirement after starting of electric set.

5. Provide engine mounted, thermostatically controlled water jacket heater as a starting aid to insure a minimum coolant temperature of 80°F. Heater shall be suitable for voltage, phase, and kW rating as indicated on drawings.

6. The signal for starting and stopping the engine is to be sensed through an auxiliary contact located in the automatic transfer switch(es). Division 26 shall provide interconnecting conductors and conduit.

J. Cooling:

1. Furnish engine with a cooling system having sufficient capacity for cooling engine when diesel engine driven electric set is delivering full rated load in the ambient temperature stated in Paragraph 1.2-D hereinbefore

2. Provide radiator cooled engine with pusher fan of type and capacity as recommended by engine manufacturer. Cooling system shall contain necessary piping, thermostat temperature control, engine mounted intercooler and intercooler system where required. Provide 50% ethylene glycol in cooling system with protection to 0ºF.

3. Factory installed cooling water piping system shall contain inlet solenoid valve, strainer, self-contained thermostat flow single seat control valve, isolating gate valves and valved bypass.

K. Exhaust System:

1. Furnish critical grade insulated exhaust silencer enclosed within the outdoor acoustical weatherproof enclosure. Silencer shall be provided with tailpipe and flapper rain cap terminating upward through the generator acoustical enclosure. Silencer shall be sized to assure proper operation without excessive back pressure when installed in the exhaust system.

2. The exhaust system shall include necessary equipment for testing emissions. This shall include Pitot tube for testing stack gas velocity and volumetric flow rates as required by EPA and Department of Environmental Quality (DEQ).

L. Mounting:

1. Provide diesel generator set with suitable base for mounting on suitable level surface. Mounting shall be sufficiently rigid to maintain alignment of alternator and to minimize engine and alternator stresses.

2. Provide 6” tall concrete equipment pad below each generator. Concrete equipment pad shall be reinforced with No. 6 concrete reinforcing bars at 12" on center both ways top and bottom or in accordance with the Structural Engineers requirements. Refer to Section 26 00 10 for additional requirements.

3. Equip diesel generator set with spring type vibration isolators. Spring type vibration isolators shall be installed between the generator set and base or separate isolators of suitable number installed between diesel generator base and the surface as recommended by the manufacturer. Springs shall be restrained type with minimum 2" deflection equal to amber booth type CT, Mason Industries type SLR or approved equal. Housed or caged springs will not be acceptable. Refer to Section 26 05 48 for additional requirements.

M. Outdoor Acoustical Weatherproof Housing:

1. The complete generator set shall be enclosed in an outdoor weatherproof acoustical housing supplied by generator manufacturer. The enclosure shall be internally insulated with acoustical insulation and be provided with an intake sound attenuating plenum and discharge sound attenuating plenum. The intake plenum and discharge plenum shall be located on the ends of the generator as indicated on the drawings. The intake plenum shall be provided with louvered side air inlets to allow air to enter the generator. The discharge plenum shall be open on the top to direct the air upward. In addition, the discharge plenum shall be provided with adequate drainage to prevent rainwater from ponding and entering the generator enclosure. The roof of the housing shall be pitched so that no ponding of water will occur. Side panels shall be easily removable allowing access to all engine/alternator components with latches that may be locked preventing unauthorized entry. Supply and exhaust ventilation louvers shall be built in allowing adequate ventilation of unit when operating under full load. Louvers shall prevent rain from entering the generator enclosure. Ventilation openings shall contain a wire mesh bird/varmint screen to prevent entry by rodents, etc. Batteries with rack shall be contained within enclosure designed for ease of access. Enclosure shall be designed to allow exhaust silencer to be mounted inside the weatherproof housing. Access to controller shall be accessible and visible from a door while standing outside the enclosure. The output circuit breaker operation device(s) shall not be higher than 6’-6” at its highest point. The output circuit breaker(s) shall also be accessible and visible from a door while standing outside the enclosure while maintaining a minimum clearance of 4 feet in front of the circuit breaker(s). Housing shall be aluminum or steel coated with a rust inhibitive primer and painted with two coats of enamel, color to be determined by the Owner and the Architect.

2. The generator acoustical housing with discharge and intake sound attenuating plenums shall provide sound pressure levels not to exceed a maximum of 75 dBA measured at 23 feet from all sides of the enclosure under all operational conditions expected for this project. Any deviation from these values will require review and approval by the acoustical consultant, engineer, and the owner.

3. The generator to be used on this project shall be completely tested in a testing laboratory. The laboratory must have a suitable ARI rated reverberant chamber. The unit manufacturer shall submit complete details, brochures, instrumentation information, etc., for review. The engineer shall have the option to witness the test. The test may be waived at the owner's discretion provided the submitted generator for this project has

already been laboratory tested and has documented sound pressure data complying with the specified values indicated above that have been approved by the acoustical consultant.

4. If the generator is tested and found to exceed these noise limits indicated above, the generator shall be modified, at no additional cost to the owner, to bring the generator to within the required noise limits. Contractor shall bear the cost of all subsequent testing required to verify the results of the generator modifications.

2.4 ALTERNATOR

A. Type:

1. Engine driven, single bearing, synchronous type. Directly connect to engine by means of flexible disc coupling for permanent self-alignment. Revolving field, 4 pole, brushless type employing rotor amortisseur windings to improve the AC wave form and reduce field heating. Capacity shall be as shown on the drawings.

B. Exciter shall respond adequately to accommodate motor starting.

1. Salient pole, wound field, brushless exciter with series boost system or

2. Permanent magnet generator exciter system.

C. Class H insulation, as defined by NEMA MG1.

D. Temperature Rise: Rated load and power factor, NEMA Class F, 130°C rise and 40°C ambient, as defined by NEMA MG1 22.40.

E. Cooling: Direct centrifugal blower cooled and furnished with heavy duty ball bearing with pre lubricated cartridge.

F. Instantaneous Voltage Dip:

1. Generator (engine and alternator) shall be sized such that the instantaneous voltage dip will be less than 20% of rated voltage for the standby ATS services when respective full three phase locked rotor (LRkVA) load at rated power factor as indicated on the drawings is applied to alternator.

G. Voltage Regulation:

1. Equip alternator with solid state voltage regulator to maintain voltage within limits specified below:
   a. Stability: 1% of its mean value at any constant load from no load to full load.
   b. Regulation: +/- 2% maximum voltage deviation between no load steady state and full load steady state.

H. Frequency Stability and Regulation:

1. In accordance with that defined by engine governor performance.

2.5 DIESEL GENERATOR CONTROL CONSOLE

A. Set shall have unit mounted NFPA 110 Level 1 control console wired, tested and shock mounted by manufacturer of diesel generator.

B. Control console shall contain generator main circuit breaker with two sets of monitoring contacts.

C. Control console shall contain auto start input which starts engine on closing contact and stops engine on opening contact. Provide an emergency stop EPO.

D. Include manual selector switch providing three control positions: AUTO-OFF-RUN. RUN position shall permit engine to be started locally at the set and run unloaded; OFF position shall prevent local starting or automatic starting of engine; AUTO position shall ready the system for automatic start or stop on demand of automatic transfer switch(es) or programmed exerciser.

E. Console shall include adjustable solid state thermal protection cranking control circuit providing an adjustable cranking period from 5 to 15 seconds followed by a minimum of 5 seconds rest time. A sensing device shall automatically disconnect the starting circuit when engine has started. If engine has not started after three attempts, engine starting control shall be locked out and overcrank alarm initiated. Lock out shall be manually reset.

F. Provide a minimum of five (5) auxiliary contacts which close when generator runs for fire alarm panel indication, input to building management system, input to building security console, and to achieve the EMCS monitoring and control functions specified in Section 25 00 90. Console shall contain alarm lights and contacts energized by engine protection devices as specified. Provide relays to signal when generator is set in the "stop" position to signal security console, fire alarm system and building management system. Refer to Section 25 00 90 for monitoring and control system interface requirements. It shall be the responsibility of the generator manufacturer to cooperate and coordinate with the requirements of specification Section 25 00 90.

G. Alternator Instruments:

1. Console shall contain the following controls and instruments for proper alternator surveillance:
    a. AC voltmeter 2% accuracy.
    b. AC ammeter.
    c. Voltmeter ammeter phase selector switch with "OFF" position.
    d. Frequency meter.
    e. Automatic starting controls.
    f. Voltage level adjustment potentiometer.
    g. Dry contacts for remote alarms wired to terminal strips.

2. Provide a digital display for the following engine protection devices. Each device shall be provided with an alarm contact.
    a. High engine temperature safety shutdown. Safety shall be pre alarm type to indicate danger is being approached.
    b. Low oil pressure safety shutdown. Safety shall be pre alarm type to indicate danger is being approached. A time delay shall be incorporated in the low oil pressure safety circuit to prevent premature shutdown of the engine before reaching operating speed.
    c. Engine overspeed safety shutdown.
    d. Engine failure to start (overcrank).
    e. Low engine temperature.
    f. Provide a relay wired in common with all safety devices to indicated generator "trouble" at fire command station, and EMCS.

g. Provide an indicator on alternator console which signals when main circuit serving generator is in open position or trips off.
h. Low fuel.
i. Additional features and devices as required by the local code authorities.

3. Provide a common alarm relay or contact for all of the above alarms for the fire alarm annunciator panel.

H. Engine Instruments: Unit mounted console shall contain the following gauges and alarms for proper engine surveillance and maintenance:

1. Gauges and Meters
   a. Water outlet temperature gauge
   b. Oil pressure gauge
   c. Engine running time meter
   d. Battery charging ammeter. Omit on engine instruments if supplied on battery charger.
   e. Fuel level (may be on fuel tank or remote mounted on wall).

2. Lights for indicating the following alarms:
   a. High outlet water temperature
   b. Engine overspeed safety
   c. Low oil pressure
   d. Failure to start (overcrank)
   e. Low outlet water temperature

I. Remote Communications:

1. Each generator shall be provided with remote communications for monitoring by the EMCS. The communications connection and protocol shall be coordinated with the EMCS Contractor and shall provide all monitoring and alarm functions available to the operator at the system control panel. The communications protocol shall be one of the open protocols listed in specification Section 25 00 90, or an open published proprietary protocol acceptable to the EMCS Contractor. The Standby Diesel Generator System manufacturer shall be responsible for coordinating this requirement with the EMCS contractor to ensure proper interface between the generator system communications interface module and the EMCS. The manufacturer shall provide all required on-site technical support to demonstrate the proper command and transfer of information between the standby generators and the EMCS. Refer to Section 25 00 90 for additional requirements.

J. Remote Alarm Annunciator Panel:

1. Provide a manufacturer’s remote alarm annunciator panel with LED or digital display and audible alarm annunciation in the building engineer’s office, as indicated on the drawings, or in a location as directed by the Owner’s representative. Coordinate the exact location of this remote annunciator with the Owner’s representative. Remote alarm annunciator panel shall be recessed flush mounted with all necessary dry contact modules, controllers, conduits, and conductors for proper operation. The panel shall include but not be limited to the following status parameters: Overspeed, Overcrank, Low Fuel, Emergency Stop, High Engine Temperature, Auxiliary Fault, High Fuel, Prealarm Low Oil Pressure, Prealarm High Engine Temperature, Low Battery Voltage, Battery Charger Fault, Line Power, Generator Power, Generator Not in Auto, and System Ready.

2.6 PORTABLE GENERATOR / LOAD BANK DOCKING STATION

A. Provide a docking station for load bank or portable generator connection to the generator distribution equipment as indicated on the electrical drawings.

1. Docking stations shall consist of insulated case circuit breakers, male cam-style inlet connectors, female cam-style outlet connectors, distribution lugs and bus bars, and grounding terminals, all housed within a single enclosure.

2. Docking station enclosures shall be NEMA 3R. The main access shall be through an interlocked, hinged door that extends the full height of the enclosure. Access for both portable generator cables with female cam-style plugs and for load bank cables with male cam-style plugs shall be via hinged lower door. A hinged flap door shall be provided to cover the cable openings when cables are not connected; the hinged flap door shall allow cable entry only after the main access door has been opened. Docking station enclosures shall be painted to match the associated generator enclosure or as directed by the Owner and the Architect.

3. Cam-style male connectors (inlets) and cam-style female connectors (outlets) shall be UL Listed single-pole separable type and rated 400 amps at 600VAC. All cam-style connectors shall be color coded. Cam-style connectors shall be provided for each phase and for ground, and shall also be provided for neutral. The ground cam-style male connectors shall be bonded to the enclosure, and a ground lug shall be provided for connection of the facility ground conductor. Access to cam-style connections for portable generator shall be kirk key interlocked with the respective permanent generator output circuit breaker.

4. A power distribution block shall be provided for load-side field wiring. The power distribution block shall be factory wired to the molded case circuit breakers.

5. Molded case circuit breakers shall be UL Listed 3-pole and the short circuit interrupt rating shall be a minimum of 50 kAIC at 208V. Trip ratings of the molded case circuit breakers shall be as shown on the drawings. The circuit breakers shall be provided with shunt trip capability such that in the event of a utility power outage during load bank testing, the docking station shall automatically open the closed circuit breaker allowing the connected generator to pick up the building loads.

6. Docking station shall be Trystar Dual-Purpose Generator Docking Station or approved equal.

## PART 3 - EXECUTION

### 3.1 PRIOR TO SHIPMENT

A. Prior to shipment, the complete engine generator unit shall be tested under actual load conditions for performance and proper functioning of component parts. A copy of manufacturer’s test report shall be included in the electrical O&M manuals.

B. The design and construction of the unit shall be such that no rotating or hazardous components are exposed, and adjustments, repairs and normal maintenance can be conveniently performed without the use of special tools.

C. All exposed metal parts shall be primed with a rust inhibitor and finished in durable machinery enamel.

### 3.2 INSTALLATION

A. Install standby diesel generator system including all connections, as indicated on drawings and wiring diagrams as specified, and in accordance with approved shop drawings and manufacturer's written instructions.

B. Connect the generator control cabinet to the fire alarm system; ATS's, remote annunciator, etc., with #12 minimum wires in conduit. Conductor sizes shall be increased when distance dictates per manufacturer's recommendations.

3.3 INITIAL START UP, SYSTEM CHECKOUT AND TEST

A. Complete installation shall be initially started and checked out for operational compliance by representatives of the manufacturer. Engine lubrication oil and antifreeze, as recommended by manufacturer for operation under environmental conditions specified, shall be provided by the supplier of the electric set.

B. Upon completion of initial startup and system checkout after all systems have been pre-tested, the supplier of the diesel generator set shall perform a field test, witnessed by the Owner's representative, to demonstrate full specified power, stability, voltage, and frequency. Provide a load bank of sufficient capacity to complement the existing load. Field test shall include running the diesel generator under loads as specified below:

1. Follow NFPA 110 Installation Acceptance Testing.

2. Witness test all of the generator system safety, trouble, alarm features.

3. Automatic starting through loss of utility power.

4. Witness test all automatic transfer switch interfaces and operations.

5. Confirm interfaces with BMCS, fire alarm system, remote annunciator, etc.

6. Confirm proper functioning of associated equipment (i.e. fuel system, dampers, ventilation, exhaust, etc.).

7. DEQ Air Quality testing as required.

C. Voltage current and frequency readings shall be recorded at 15 minute intervals throughout the test. There shall be a 30 minute unloaded run at the conclusion of the test to allow engine to cool before shutdown. Three copies of the field test data shall be furnished to the engineer.

D. Contractor shall make all necessary hookups to facilitate field test and shall furnish all fuel necessary for field test.

E. All fuel fill requirements, including but not limited to the fuel required for testing and the final fuel tank fill prior to Owner acceptance, shall be provided by Division 26. At the time the building is turned over to the Owner, the Division 26 contractor shall provide complete fill of diesel fuel oil in all storage tanks.

F. A factory trained specialist shall provide training of owner's representative(s) on basic familiarization and proper operation of standby generator for an 8 hour operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. Training shall be separate and independent from the start-up commissioning.

3.4 CAPACITY GUARANTEE

A. In accepting this contract, the generator manufacturer shall guarantee to give capacities not less than those specified in the contract documents, with the generator installed as shown on the contract drawings and will be required to take whatever steps are necessary to meet the guarantee, at no additional cost to the Owner, regardless of the extent of the revisions required.

3.5 EXTENDED WARRANTY AND SERVICE

A. Standby Generator Extended Service Contract Alternate: The manufacturer shall submit with their bid the cost, as an alternate, to provide four (4) additional years of service beyond the minimum one (1) year warranty period. Provide a separate price for each of the four (4) additional years. This additional service shall commence one (1) year following the date of substantial completion. The first year of service after substantial completion shall be included in the base bid price.

END OF SECTION 26 32 00

SECTION 26 36 23 - AUTOMATIC TRANSFER SWITCHES

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of “Division 01, General Requirements,” “Conditions of the Contract,” and “Supplementary Conditions.” Specific attention is called to the “Division 26 General Requirements” in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide automatic transfer switch(es) as noted on the contract documents and as specified herein. Automatic transfer switches shall be bypass-isolation type.

B. Test the transfer switch and all operational features.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 53 - Identification for Electrical Systems
B. Section 26 05 93 - Testing
C. Section 26 32 00 - Packaged Generator Assemblies

1.4 REFERENCE STANDARDS

A. Automatic transfer switches shall be designed, manufactured, tested and installed in compliance with the following standards:

B. Institute of Electrical and Electronics Engineers (IEEE)

C. National Electrical Manufacturers Association (NEMA)

D. National Fire Protection Association (NFPA)
1. 70 National Electrical Code
2. 20 Standard for the Installation of Stationary Pumps for Fire Protection
3. 110 Standard for Emergency and Standby Power Systems

E. Underwriters Laboratories (UL)
1. 1008 Standard for Safety Transfer Switch Equipment

1.5 SUBMITTALS

A. Submit manufacturers’ literature, illustrations, specifications, operation/maintenance manuals, timer set points, and engineering data for each ATS specific to this project. Contractor shall confirm and program the transfer switches to the set points provided.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. ASCO

2. Cummins
3. Zenith

B. Provided they comply with these Specifications, the Contractor shall have the option to submit automatic transfer switches manufactured by the acceptable generator manufacturers listed in Specification Section 26 32 00.

2.2 GENERAL

A. Provide UL listed 3 pole, solid neutral, 208 volt, 60 hertz automatic transfer switch. Continuous duty current rating as noted on the contract drawing. The switches shall be electrically operated and mechanically held in each direction by a single solenoid mechanism momentarily energized from the source to which the load is to be transferred. The operating mechanism shall mechanically lock the switches in each position without the use of hooks, latches, springs, or semi-permanent magnets. Disarrangement of any part or failure of any coil shall neither permit a neutral position nor shall it cause both sides to be closed at the same time. The switches must be inherently double throw with both sets of main contacts moving simultaneously. The operating mechanism shall not be dependent upon critical mechanical and/or electrical adjustments or utilize miniature type limit switches or similar non-industrial type components. A handle or knob for manual operation of the main poles shall be provided. Transfer time for switch(es) shall be less than 170 milliseconds.

B. Transfer switch(es) shall be suitable for use with engine driven alternators. The switches shall be specifically designed for automatic transfer switch service and must be capable of handling all classes of loads on a make, carry and break basis. Switches utilizing adapted devices such as molded case circuit breakers, circuit interrupters, disconnect switches, etc., which were not originally intended to repeatedly open and close load current to 600% of rated current at 0.4 to 0.5 power factor in accordance with test procedures of UL 508, 10th Edition (Industrial Control Equipment), are not acceptable.

C. The main contacts shall be silver alloy protected by arc barriers and arc quenchers. Switches shall have separate arcing contacts. Switch shall have brush type main contacts. Switching shall be accomplished without any evidence of source to source arc over.

D. All contacts, coils, springs, and control elements shall be conveniently removable from the front of the transfer switch without major disassembly or disconnection of power conductors.

E. Provide control module with protective cover, solid state sensing and control logic on keyed plug in printed circuit boards, and enclosed industrial control grade plug in interface relays.

F. Control module shall sense normal and emergency sources and provide positive automatic operation as follows:

1. Provide close differential voltage sensing and monitor the three phase to phase voltages of the normal supply.

2. Provide adjustable 85 to 100% pick up and 75 98% dropout voltage relays. Initiate transfer to emergency upon reduction of normal source to 85 percent of normal voltage, with emergency power source available.

3. Initiate re transfer to normal upon restoration of normal source to 95% of normal voltage and frequency.

G. Automatic transfer switches shall conform to requirements of NEMA Standards ICS 10 2005,

Part 2 and Underwriters' Laboratories UL 1008 and shall be UL listed as follows:

1. For use in emergency systems in accordance with Articles 517, 700, 701, and 702 of the National Electrical Code.

2. Rated in amperes for total system transfer including control of motors, electric discharge lamps, electric heating and tungsten filament lamp loads as referred to in Paragraph 30.9 of UL 1008.

3. Transfer switches shall be suitable for 100 percent tungsten filament lamp load.

H. Automatic transfer switch shall have a momentary withstand rating sufficient to meet incoming fault current indicated at the main switchboards and as indicated on the manufacturer's short circuit and fault current coordination study. Withstand time shall be 1/2 cycle with current limiting fuses as the overcurrent device, 3 cycles when molded case circuit breakers are the overcurrent device, and 5 cycles when insulated case or metal clad breakers are the overcurrent device.

1. X/R ratio during withstand shall be as per UL 1008.

2. Contacts shall not weld or burn during withstand interval or switch operation.

3. After withstand, transfer switch shall be completely operable and pass a dielectric test of 1960 volts RMS.

I. Provide the following features:

1. Transfer switches shall be provided with solid state timers to signal the generating set to start after an adjustable time delay of 0.5 to 15 seconds. Provide a lockout relay to prevent transfer until the generating set has reached 90% of rated voltage and frequency. Provide an adjustable time delay of 2 seconds for transfer switches to override momentary dips in the normal source. The load must remain connected to the normal source during this period.

2. Time delay on retransfer to normal. This time delay shall be automatically bypassed if the emergency source fails and normal source is available. Shall be field adjustable from 0 to 30 minutes and factory set at 10 minutes.

3. An unloaded running time delay for emergency generator cool down. Time delay shall be field adjustable from 0 to 60 minutes and factory set at 5 minutes.

4. Provide independent single phase voltage and frequency sensing of emergency source. Pickup voltage shall be adjustable from 85 percent to 100 percent of nominal. Pickup frequency shall be adjustable from 90 percent to 100 percent of nominal. Transfer to emergency upon normal source failure when emergency source voltage is 90 percent or more of nominal and frequency is 95 percent or more of nominal.

5. Switch position and source availability indicator lights shall be provided on all transfer switches.

6. One auxiliary contact which closes when the load is connected to normal and one auxiliary contact which closes when the load is connected to emergency. Rated 10 amps at 250 volts AC.

7. Each transfer switch shall include a manually operated momentary test switch to simulate a normal source failure and to test the complete system.

8. Automatic exerciser to start the generator, transfer load, run for one half hour, retransfer load back and shutdown automatically. Exerciser shall include a programmable cycle timer to allow automatic testing to occur at a selectable day and time and be repeated at 2 week or 4 week intervals.

9. NEMA 1 wall mounted enclosure with sufficient space for cable bending and compression lugs.

10. Each transfer switch shall be equipped with an in phase monitor. In Phase Monitor shall be a factory installed and adjusted solid state device which inhibits switch operation until normal and emergency sources are in phase. Transfer shall be completed within+_ 60° phase angle between sources.

11. Provide each automatic transfer switch with a common alarm contactor which shall send an alarm signal to the EMCS should the automatic transfer switch fail to operate upon loss of normal power for any reason including but not limited to a problem with the logic board, the controller issuing a transfer command but the transfer does not occur, and the transfer switch being set in a manual position in lieu of being set in an automatic position. Refer to Section 25 00 90 for additional control system interface requirements.

2.3 GENERATOR ENGINE CONTROL PANEL INTERFACE

A. Provide a separate 1-1/4” conduit with 16-cat5 control wires from the ATS to the generator engine control panel.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Mount transfer switches as noted on the contract drawings. Provide at least three (3) feet front clearance. Floor mounted units shall be mounted on a 4” high concrete pad.

3.2 CONNECTION

A. Completely wire and test all operations of the transfer switches prior to final inspection. Provide wiring (for control, monitoring, etc.) and conduit between the transfer switches, generator controller, fire alarm system, and the building management system.

B. Control conductors between automatic transfer switches and the standby generator shall be protected to resist potential damage by fire or structural failure. Control conductors shall include but not be limited to start signaling circuit installed between automatic transfer switches and the standby generator system. Similar control conductors shall be provided from the automatic transfer switches to the mobile generator connection box. These control conductors/conduits shall be encased in 2” minimum of concrete, be circuit protective systems with a minimum of 2 hour fire resistance rating or installed in spaces or areas that are fully protected by an approved automatic fire suppression system. Conductor sizes shall be sized per generator manufacturer recommendations incorporating voltage drop and distance considerations.

3.3 CLEANING

A. Vacuum clean all dirt and debris from the internal and external surfaces of the automatic transfer switches.

3.4 FIELD TESTING

A. The contractor and manufacturers shall provide the necessary equipment to ensure that the required Emergency Systems are energized and operational within ten seconds and that the Legally Required Standby Systems are energized and operational within sixty seconds. These systems shall be as defined by the local, state, and federal codes which shall include but not be limited to lighting, exit signs, ventilation fans, fire alarm system, smoke removal systems, pressurization fans, etc.

3.5 EXTENDED WARRANTY AND SERVICE

A. Automatic Transfer Switch Extended Service Contract Alternate: The manufacturer shall submit with their bid the cost, as an alternate, to provide four (4) additional years of service beyond the minimum one (1) year warranty period. Provide a separate price for each of the four (4) additional years. This additional service shall commence one (1) year following the date of substantial completion. The first year of service after substantial completion shall be included in the base bid price.

END OF SECTION 26 36 23

SECTION 26 41 00 - FACILITY LIGHTNING PROTECTION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. The work covered by this section of the specifications consists in furnishing all labor, materials, equipment and services, and in performing all operations in connection with the concealed lightning protection system installation, complete for the project, in strict accordance with this section of the specifications and the applicable drawings, and subject to the terms and conditions of the contract. This project is to include the buildings, exterior structures, and all appurtenances attached.

B. The lightning protection system shall be installed in strict compliance with this section of the specifications. If any departures from these specifications are deemed necessary by the Contractor, details of such departures and the reasons therefore shall be submitted as soon as practicable to the Owner's Representative for approval. No such departures shall be made without the prior written approval of the Owner's Representative.

C. The system shall be designed to comply with requirements of the National Fire Protection Association, Lightning Protection Institute, and the Underwriters' Laboratories, Inc. The system shall be designed to qualify for the maximum insurance rate reduction allowed for lightning protection for this type of construction. A UL master label shall be obtained for this project.

D. Provide a concealed parallel path lightning protection system complete with air terminals, conductors, ground terminals, interconnecting conductors, and other connectors or fittings required for a complete system as specified herein.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 26 - Grounding and Bonding for Electrical Systems
C. Section 26 05 33 - Raceway and Boxes for Electrical Systems
D. Section 26 05 93 - Testing

1.4 REFERENCE STANDARDS

A. The lighting protection system shall be designed, manufactured, tested and installed in compliance with the following standards:

1. Lightning Protection Institute (LPI) 175 – Standard for the Design - Installation - Inspection of Lightning Protection Systems
2. National Fire Protection Association (NFPA) 70 – National Electrical Code
3. National Fire Protection Association (NFPA) 780 – Lightning Protection Code
4. Underwriters Laboratories (UL) 96A – Standard for Safety Installation Requirements for Lightning Protection Systems

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, maintenance/operating manuals, engineering data, specifications, mounting details and installation procedures for the lightning protection system specific to this project.

B. Provide 1/8"=1'-0" shop drawings (AutoCAD 2015 Version software or later version as approved by the engineer) indicating the layout of the lightning protection system. The shop drawings shall indicate horizontal and vertical dimensional data showing the position of the cables, air terminals, architecture, structure and connections.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. If they comply with these Specifications, installer and equipment manufactured by one of the following manufacturers will be acceptable:

1. Bonded Lightning Protection System
2. Thompson Lightning Protection Inc.
3. Lightning Prot. Co. Inc.
4. Advanced Lightning Systems
5. Warren Lightning Rod Company
6. National Lightning Protection Corporation
7. Dillon Lightning Protection System, Inc.

2.2 STANDARD PRODUCTS

A. Standard Products - The system to be furnished under this specification shall be the standard product of a manufacturer regularly engaged in the production of lightning protection systems and shall be the manufacturer's latest approved design. All materials specified for this work shall be Underwriters' Laboratories, Inc., approved and shall be manufactured by Bonded Lightning Protection, Inc., Rockville, Maryland, Thompson Lightning Protection, Inc., St. Paul, Minnesota, Independent Protection Co., Goshen, Indiana, Harger, Grayslake, IL, Robbins Lightning Protection Co., Maryville, Missouri, or East Coast Lightning Equipment, Inc.

B. Shop Drawings - Complete shop drawings shall be submitted to the Owner's Representative for approval prior to start of work. Shop drawings shall include complete details with description of all air terminals, air terminal bases, conductors, conductor fasteners, splicers, bonding clamps, ground rods, etc.

C. Class - Lightning conductors, air terminals, ground terminals, and associated fittings required shall be Class I materials for structures not exceeding 75 feet in height and Class II materials for structures exceeding 75 feet in height.

D. Material Requirements

1. Class I Material Requirements:

| Type of Conductor | | Copper | Aluminum |
|---|---|---|---|
| Air Terminal, Solid | Minimum Diameter | 3/8 inch | 1/2 inch |
| | Minimum Size each Strand | 17 AWG | 14 AWG |

| | | | |
|---|---|---|---|
| Main Conductor, Cable | Weight per Length | 187 lbs/1000 feet | 95 lbs/1000 feet |
| | Cross Sectional Area | 57,400 CM | 98,600 CM |
| Main Conductor, Solid Strip | Thickness | 0.051 inch | 0.064 inch |
| | Width | 1 inch | 1 inch |
| Bonding Conductor, Cable (Solid or Stranded) | Minimum Size each Strand | 17 AWG | 14 AWG |
| | Cross Sectional Area | 26,240 CM | 41,100 CM |
| Bonding Conductor, Solid Strip | Thickness | .051 inch | .064 inch |
| | Width | 1/2 inch | 1/2 inch |

2. Class II Material Requirements:

| Type of Conductor | | Copper Stranded | Aluminum Stranded |
|---|---|---|---|
| Air Terminal, Solid | Minimum Diameter | 1/2 inch | 5/8 inch |
| Main Conductor, Cable | Minimum Size each Strand | 15 AWG | 13 AWG |
| | Weight per Length | 375 lbs/1000 feet | 190 lbs/1000 feet |
| | Cross Sectional Area | 115,000 CM | 192,000 CM |
| Bonding Conductor, Cable (Solid or Stranded) | Minimum Size each Strand | 17 AWG | 14 AWG |
| | Cross Sectional Area | 26,240 CM | 41,100 CM |
| Bonding Conductor, Solid Strip | Thickness | 0.051 inch | 0.064 inch |
| | Width | 1/2 inch | 1/2 inch |

E. The materials of which protection systems are made shall be resistant to corrosion or shall be acceptably protected against corrosion. No combination of materials shall be used that forms an electrolytic couple of such nature that in the presence of moisture corrosion is accelerated. One or more of the following materials shall be used:

1. Copper: Where copper is used it shall be of the grade ordinarily required for commercial electrical work, generally designated as being of 95 percent conductivity when annealed.

2. Copper Alloys: Where alloys of copper are used they shall be substantially as resistant to corrosion as copper under similar conditions.

3. Aluminum: Where aluminum is used, care shall be taken not to use it in contact with the earth or elsewhere where it will rapidly deteriorate. Conductors shall be of electrical grade aluminum.

4. Copper lightning protection materials shall not be installed on aluminum roofing, siding, or other aluminum surfaces.

5. Aluminum lightning protection materials shall not be installed on copper surfaces.

## 2.3 CONDUCTORS

A. Conductors shall interconnect all air terminals and shall form a two-way path from each air

terminal horizontally or downward to connections with ground terminals.

B. Conductors shall maintain a horizontal or downward coursing, free from "U" or "V" (down and up) pockets.

C. No bend of a conductor shall form an included angle of less than 90 degrees; nor shall it have a radius of bend less than 8 in.

D. Roof conductors shall be around the perimeter of flat roofs; behind or on top of parapets; and across flat or gently sloping roof areas as required to interconnect all air terminals. Conductors shall be coursed through or around obstructions in a horizontal plane with the main conductor.

E. Cross-run conductors (main conductors) are required to interconnect the air terminals on flat or gently sloping roofs that exceed 50 ft in width. For example, roofs from 50 ft to 100 ft in width require one cross run; roofs 100 ft to 150 ft in width require two cross runs; etc. Cross-run conductors shall be connected to the main perimeter cable at intervals not exceeding 150 ft. These cables shall be flat braided type meeting Class I or Class II requirements and installed beneath or in between pavers on roof.

F. Down conductors shall be as widely separated as practicable. The use of building steel as down conductors may be acceptable if present and approved for use in writing by the Owner. Conduits shall be roughed-in for down conductors; however, it is the responsibility of the lightning protection system contractor to verify the quantity, conduit size and type, and locations of all down conductors required early in the project so that they may be roughed-in by the electrical contractor.

G. At least two down conductors shall be provided on any kind of structure. Structures exceeding 250 ft in perimeter shall have a down conductor for every 100 ft of perimeter or fraction thereof. The total number of down conductors on structures having flat or gently sloping roofs shall be such that the average distance between all down conductors does not exceed 100 ft.

H. Down conductors coursed on or in reinforced concrete columns shall be connected to the reinforcing steel at its upper and lower extremities. Roof conductors or other concealed horizontal conductor runs shall be connected to the reinforcing steel at intervals not exceeding 100 ft.

I. Conductors shall be sized as stated in the Class I or Class II material requirements.

J. Conductors used for the bonding of grounded metal bodies, or isolated metal bodies, requiring connection to the lightning protection system shall be sized in accordance with bonding conductors as stated in the Class I or Class II material requirements. Bonding is required to roof mounted grounded metal bodies to prevent side flash.

K. Each down conductor shall terminate at a ground terminal consisting of a ground rod.

L. All ground terminals shall be in direct contact with earth at a depth of not less than 2-1/2 feet and at least 2 feet away from foundation walls. The conductor/ground terminals shall be completely independent of other grounding systems except for the one connection to the electrical service (power) grounding electrode system by connecting to the supplementary grounding system.

## 2.4 AIR TERMINALS

A. Air terminals shall be provided for all parts of a structure that are likely to be damaged by direct

lightning flashes. Metal parts of a structure that are exposed to direct lightning flashes and that have a metal thickness of 3/16 in. or greater only require connection to the lightning protection system. Such connections shall provide a two-way path to ground as is required for air terminals.

B. Air terminals shall be placed on ridges of pitched roofs and around the perimeter of flat or gently sloping roofs at intervals not exceeding 20 ft. except that air terminals 24 in. or higher may be placed at intervals not exceeding 25 ft. Air terminals shall be placed at or within 2 ft. of the ends of ridges or edges and corners of roofs.

C. Flat or gently sloping roofs that exceed 50 ft. in width or length shall have additional air terminals located at intervals not to exceed 50 ft. on the flat or gently sloping areas.

D. Air terminals shall be secured against overturning either by attachment to the object to be protected or by means of braces which shall be permanently and rigidly attached to the building. An air terminal exceeding 24 in. in height shall be supported at a point not less than one-half its height. If additional supports are required, they shall be integrated into the lightning protection grounding system.

E. Air terminals shall be solid copper, bronze, or aluminum with blunt personnel safety points. Air terminals shall extend not less than 10" above the object they are to protect. Diameter shall be as stated in the Class I or Class II material requirements.

F. Air terminals shall be spring loaded swivel type. Provide personnel protective tips for air terminals.

2.5 FASTENERS, CLAMPS AND CONNECTORS

A. Conductors shall be securely fastened to the structure upon which they are placed, at intervals not exceeding 3 ft. The fasteners shall not be subject to breakage and shall be of the same material as the conductor or of a material equally resistant to corrosion as that of the conductor. No combination of materials shall be used that forms an electrolytic couple of such nature that, in the presence of moisture, corrosion will be accelerated.

B. Connector fittings shall be used at all "end-to-end", "tee", or "Y" splices of lightning conductors. They shall be attached to withstand a pull test of 200 lb.

C. Special bimetal clamps which are designed and manufactured for the use in splicing copper to aluminum conductors shall be used at each ground location and shall be located not less than 18 inches above finished grade and/or finished floor.

D. All below grade copper to copper connections, connections to ground rods, and connections to steel shall be made with CADWELD connections.

E. All above grade copper to copper and copper to steel primary connections shall be made with CADWELD connections.

F. All above grade aluminum to aluminum and aluminum to copper transitions shall be made with approved mechanical connections and installed by trained personnel.

G. All secondary connections shall use approved bolted type mechanical connections.

2.6 GROUND RODS

A. Ground rods shall not be less than 1/2 inch in diameter by 8 feet in length, made of copper clad steel. The portion of copper-on-copper clad rods shall be approximately 27% of the weight of the rod. Ground rods shall be driven to a depth of 24" below finished grade and/or finished floor, minimum. All ground rods shall have a target resistance of 10 ohms. Augmentation of grounding rod system with approved ancillary earth electrode subsystem is permitted with the Owner approval.

PART 3 - EXECUTION

3.1 INSTALLATION

A. The lightning protection system shall be installed by a firm regularly engaged and experienced in installing UL Master Labeled lightning protection systems and shall be listed with the Underwriters' Laboratories, Inc.

B. All equipment shall be installed in a neat, workmanlike manner. The system shall consist of a complete cable network on the roof including all air terminals, connections, and bonds with down leads routed down the building and terminating into a proper ground terminal.

C. The lightning protection installer shall work with other trades to insure a correct, neat and unobtrusive installation.

D. It shall be the responsibility of the lightning protection installer to assure a sound bond equal to the main size lightning conductor to the electrical service grounding electrode system as indicated on the contract documents.

E. The Contractor shall coordinate the exact locations of the lightning protection system air terminals with the Architect and Owner. Conductor fasteners shall be attached to the building using a method approved by the Architect and Owner. The Contractor shall provide a sample of an attached fastener to the Architect for approval prior to installation.

3.2 TESTING

A. Perform tests to verify continuity of all system components especially those concealed.

B. Conduct ground resistance tests of the ground termination system and its individual ground electrodes in accordance with NFPA 780 Appendix J.

C. Provide the owner's representative with the written tests described above.

D. Owner shall witness all tests at his discretion.

E. A UL master label shall be obtained for this project and provided to owner.

F. Prior to covering up/concealing work, coordinate inspection by engineer of down conductor ground rod installations and single point connection to power ground.

G. Refer to Section 26 05 93 for additional requirements.

END OF SECTION 26 41 00

# SECTION 26 43 13 - SURGE PROTECTIVE DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

### 1.2 WORK INCLUDED IN THIS SECTION

A. Provide the surge protective devices (SPD) as shown on the contract documents and as specified herein.

### 1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 53 - Identification for Electrical Systems
D. Section 26 05 93 - Testing
E. Section 26 28 00 - Low-Voltage Circuit Protective Devices

### 1.4 REFERENCE STANDARDS

A. All surge protective devices shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code
2. UL 1449 Third Edition
3. UL 1283
4. ANSI/IEEE C62.41 and C62.45
5. UL 96A

### 1.5 SUBMITTALS

A. Submit manufacturer's literature, illustrations, engineering data and specifications for each type and size of surge protective device specific to this project.

B. Submit physical wire lead lengths from system bus protected to surge protective device.

### 1.6 WARRANTY

A. Manufacturer warrants equipment to be free from defects in materials and workmanship for minimum 5 years from date of purchase.

## PART 2 - PRODUCTS

### 2.1 ACCEPTABLE MANUFACTURERS

A. If it complies with these Specifications, equipment manufactured by one of the following manufacturers will be acceptable:

1. Siemens
2. Eaton Cutler-Hammer
3. Innovative Technology
4. Square D
5. APT
6. Current Technology

2.2 ELECTRICAL REQUIREMENTS

A. The unit shall provide effective high energy transient voltage surge suppression, surge current diversion, and high frequency noise attenuation in all electrical modes for equipment connected downstream. The unit shall be connected in parallel with the facility's wiring system.

B. The maximum continuous operating voltage (MCOV) of all suppression components utilized in the unit shall not be less than 125% for 120/208V systems at the point of attachment within power distribution system.

C. The unit shall include an integral safety interlocked disconnect located in the unit enclosure with an externally mounted manual operator. The switch shall be rated for 600 VAC. Where fuses are included, the fusing shall be capable of conducting a transient equal to the nameplate transient rating of the surge protective device without failure. The fuses' minimum interrupting rating shall be 200 KAIC.

D. The maximum single impulse surge current rating shall be 120 kA per mode or higher and the maximum surge current capacity shall be 240 kA per phase or higher.

E. The nominal discharge current rating (In) shall be 20 kA.

F. Surge protective devices shall provide protection in each of the following modes: L-N, L-G, N-G, and L-L.

G. Voltage protection ratings (VPR) shall be at a minimum: L-L 1200V, L-N 700V, L-G 700V, and N-G 700V.

H. Provide Type 1 surge protective devices when installed on the supply side of the main service disconnect. Provide Type 2 surge protective devices when installed on the load side of the main service disconnect.

I. Provide a surge protective device at each of the following locations whether indicated on the drawings or not:

1. At each service entrance main switchboard or disconnect switch.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Surge protective devices for new switchboards shall be internally factory mounted. Surge protective devices for all other equipment shall be mounted externally. External installations shall be mounted on 12 gauge formed steel channels having a cross section dimension of at least 1-1/2 inches x 1-1/2 inches. The channel and fittings shall have galvcrom or hot dipped

galvanized finish to resist rust formation. Install channels vertically. Channels shall be Kindorf or equal.

B. The unit shall be mounted as close as possible to point of attachment with the shortest leads/conductors possible. Connection conductor and overcurrent protective device sizes and types shall meet manufacturer's recommendations.

3.2 CLEANING

A. Vacuum clean all dirt and debris from the inside and outside of surge protective devices.

3.3 INITIAL START UP, SYSTEM CHECKOUT, AND TEST

A. Each unit shall be factory tested before shipment. Testing shall include, but not be limited to production-line tests, quality assurance checks, MCOV, and benchmark clamping voltage tests. A copy of the benchmark clamping tests for each individual surge protective device shall be included with each unit.

END OF SECTION 26 43 13

SECTION 26 50 00 - LIGHTING

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 26 00 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the lighting and accessories as shown on the contract documents and as specified herein. The specifications in this section shall apply to the light fixtures specified by the engineer and the fixtures specified by the lighting consultant. Refer to the drawings for additional information.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables
B. Section 26 05 33 - Raceway and Boxes for Electrical Systems
C. Section 26 05 93 - Testing
D. Section 26 09 23 - Automatic Control Devices
E. Section 25 00 90 - Automatic Temperature Control System

1.4 REFERENCE STANDARDS

A. Lighting fixtures shall be designed, manufactured, tested and installed in compliance with the following standards:

1. NFPA 70 National Electrical Code
2. Underwriters' Laboratories (UL)
3. Illuminating Engineering Society (IES)
4. ANSI C62 and C82
5. FCC Part 18, Subpart C
6. ASHRAE/IESNA 90.1
7. FDA Food Code
8. International Energy Conservation Code (IECC)
9. NEMA SSL-1

1.5 SUBMITTALS

A. Submit manufacturers' literature, illustrations, voltage, ballast specifications, and lamp characteristics for each light fixture type specific to this project.

B. Substitute fixtures shall be submitted with current photometric certified test data from an independent testing laboratory of the substitute fixture and the specified fixture.

C. For exterior surface parking lighting, manufacturer shall submit point to point foot candle calculations, based on maintained factor. All calculations shall conform to IES recommended values and design criteria.

D. Submit a schedule that indicates each fixture type with specified input wattage, submitted input wattage, specified line voltage, and submitted line voltage.

PART 2 - PRODUCTS

2.1 FIXTURES AND LAMPS

A. Fixtures shall be provided as specified herein and as indicated on the drawings. Fixtures shall be complete with all required lamps, sockets, wiring, glassware, reflectors, hangers, fittings, plaster frames, etc., necessary for a complete installation.

1. All fixtures shall be cleaned and have all lamps at the time of final acceptance of the building. All burned-out lamps shall be replaced at the time of final acceptance of building.

2. All fixtures specified herein scheduled and/or detailed on the drawings, shall conform to the standards and bear the label of the Underwriters Laboratories, Inc.

3. Prior to the application of any finish, all metal parts of all fixtures shall be protected by a rust inhibiting process approved by the Architect. The rust inhibiting process shall be chemical. No type of sprayed, painted, or dipped primer may be used as the basic rust inhibitor. Any fixtures and/or parts of any fixtures which shall have begun to show signs of rusting or corroding at the time of completion of the job shall be removed and replaced by properly protected metal parts, subject to the approval of the Architect, and this shall be done before a final certificate of acceptance will be issued.

4. Where required, fixtures shall be furnished complete with internal wiring and drivers. All joints and splices within the fixture housing shall be made as specified in Section 26 05 19. Substitute material will not be acceptable.

5. All fixtures provided for this project shall be designed and rated by the fixture manufacturer for the intended application and for the location installed. All fixtures provided for this project shall be UL listed for the intended application and for the location installed. All fixtures exposed to outdoor weather conditions shall be suitable for outdoor weather conditions. All fixtures exposed to outside ambient temperatures shall be designed and rated for operation throughout the entire range of temperatures between the minimum and maximum outside ambient temperatures for this project. All fixtures exposed to damp locations, including but not limited to parking structure areas, shall be of damp rated construction. All fixtures shall be in accordance with the requirements of the Local Code Authority and the National Electrical Code. The requirements of this paragraph shall be in addition to all the other requirements of this specification. The requirements of the specification shall be in addition to the requirements indicated in the lighting fixture schedule. All fixtures shall meet these requirements whether specifically indicated in the lighting fixture schedule or not. It shall be the responsibility of the fixture manufacturer to provide all necessary accessories and modifications to the fixtures specified to meet these requirements. Catalog numbers of fixtures in the fixture schedule are intended to establish manufacturer, type, quality, aesthetic appearance, and lighting characteristics of the fixtures and do not necessarily indicate all the requirements of the specifications.

B. Provide the proper fixture type for the type of ceiling or wall construction in which the fixture is to be installed. Regardless of fixture numbers given in the fixture schedule, the fixtures supplied shall have the proper trim, frames, mounting devices and configuration and accessories necessary to be properly installed in the building construction. Catalog numbers of fixtures in the fixture schedule are to establish a type of fixture and not to determine a method of mounting.

C. Light fixtures used for supply air are not allowed.

D. Supply two (2) loose exit signs to owner at completion of project. Contractor shall use these where fire marshal instructs to add exit signs during inspection at no additional cost to owner.

E. All lighting fixtures in areas where there is exposed food, clean equipment, utensils, linens, or unwrapped single service and single use articles shall be lensed or provided with shatter resistant lamps in compliance with the most current FDA Food Code and other adopted codes.

F. Provide LED lighting fixtures as indicated. Provide LED lighting fixtures complete with lamps of number, type, and wattage indicated. Details, shapes, and dimensions are indicative of the general type desired, but are not intended to restrict selection to LED lighting fixtures of a particular manufacturer. LED lighting fixtures of similar designs and of equal finish and quality will be acceptable as approved.

1. Lamp maximum wattage shall be as indicated on the Drawings. LED arrays shall have an average rated life of 50,000 hours (minimum) as confirmed by standard industry testing life span data which complies with IESNA LM-80. LED arrays shall have a color temperature of 4000K and a minimum color rending index (CRI) of 80.

    a. Confirm color temperature with the Architect, Interior Designer, or Lighting Consultant. Where a discrepancy exists for color temperature, the engineering specifications shall take the least precedence.

2. Drivers for LED lighting fixtures shall be high efficiency, full range dimming from 0-10V control signal (or as otherwise specified), voltage as indicated. Drivers shall be individually addressable and shall include a lumen management system to ensure constant light output over the life of the fixture.

## 2.2 LIGHTING CONTROLS

A. Dimmers and Dimming Systems

1. Refer to the electrical drawings for dimming system requirements.

2. Refer to specification Section 26 27 26 and the electrical drawings for wall box dimmer requirements.

B. Lighting Contactors

1. Lighting contactors shall be similar or equal to ASCO #917 Series, NEMA 1 enclosure.

2. Refer to specification Section 25 00 90 for additional requirements.

# PART 3 - EXECUTION

## 3.1 INSTALLATION

A. Recessed lay in type lighting fixtures shall be installed in the lay in type ceiling in such a manner that the lens or louver housing may be easily opened and so that the fixtures may be removed and relocated without forcing the fixtures or changing the grid system tie wires. The electrical subcontractor shall coordinate with the ceiling subcontractor before the ceiling grid is installed to assure a mutually satisfactory installation of ceiling and light fixtures, which will permit the fixtures to be relocated at a future date.

3.2 SUPPORTS

A. Recessed downlight lighting fixtures installed in lay in type ceilings shall be supported by means of hanger bars extending across the main ceiling support members supported by wires at all four corners. Lighting fixtures (1'x1', 1'x4', 2'x2', 2'x4', etc.) shall be supported independently of the ceiling grid by wires on all four corners. The four wires shall be independent to the structure above. All other type fixtures in lay in type ceilings shall have the outlet box rigidly supported from the building structural system. In addition, each fixture shall be supported independently of the ceiling grid and all other MEP equipment by support wires connected to the building structure. Provide removable clips to securely fasten light fixtures in place to the ceiling construction; however, support shall be via tie wires. In addition to above support methods, all Local Code Authority requirements shall be adhered to. Support means and methods shall be in accordance with manufacturer's recommendations and the Local Code Authority seismic requirements and wind loading.

3.3 CLEANING

A. All fixtures shall be cleaned of dirt and debris inside and outside and left in a clean condition at the end of the construction work.

3.4 SAMPLES

A. Provide samples of each fixture type as requested by the architect.

3.5 COORDINATION

A. The Contractors shall coordinate the control of all light fixtures with the lighting controls indicated in the Automatic Temperature Control System specification section.

3.6 LIGHT POLES

A. Light poles shall have a hand hole near the bottom of the pole with a secure grommeted cover for access. A ground lug shall be provided with access from hand hole. Provide concrete bases for pole-mounted light fixtures. Concrete bases shall meet the respective region's wind loading and seismic requirements. Refer to the structural engineer's documents for additional requirements.

B. All pole mounted parking lot light fixtures shall be full cut-off type light fixtures.

3.7 UNDERGROUND AND GRADE MOUNTED LIGHT FIXTURES

A. For exterior mounted light fixtures, provide watertight fittings, seals, connectors, junction boxes, terminations, concrete collars, etc for a complete and operational system. Provide all accessories required per manufacturer's recommendations for each light fixture installation/system. Provide appropriate backfill material for proper water drainage away from the light fixtures; refer to the contract drawings and manufacturer's recommendations for additional requirements and details.

END OF SECTION 26 50 00

SECTION 27 0500 – COMMON WORK RESULTS FOR TELECOMMUNICATIONS

# PART 1 - GENERAL

## 1.1 DESCRIPTION OF WORK

A. Demolition

1. Refer to 'TD' plans for details defining the scope and extent of the demolition of the existing Information Communications Technology cabling system.
2. Prior to the start of demolition, contactor is required to coordinate the disconnection and removal of all active network hardware. All network hardware is to be turned over the Owner/User. This is inclusive but not limited to switches, routers, servers, and storage devices.
3. Wireless access points and CCTV Cameras are to be removed, stored, and protected from theft and damage. Contractor will be responsible for reinstalling wireless access points and returning existing cameras to the owner.
4. The existing horizontal cabling system is to be removed in its entirety to source. The owner's communications group gets first right of refusal, it will be the contractor's responsibility to dispose of unwanted faceplates, jacks, cable, empty patch panels, and racks.

B. Provide all labor, equipment, supplies and materials for the complete installation of fully functional Information Communication Technology systems in accordance with the Contract Documents. Information Communication Technology systems include but are not limited to:

1. Telecommunications Grounding Network
2. Cable Tray Network
3. Horizontal cabling system
4. CATV cabling system

C. The work includes the following as well as work not listed below but described elsewhere as it applies to all Information Communication Technology systems:

1. Raceway and cable trays
2. Underground raceways
3. Horizontal cabling and terminations
4. Coaxial cable and terminations
5. Telecommunications outlets
6. Equipment racks and cabinets
7. Patch Panels
8. Telecommunications grounding network
9. Patch Cords and Telecommunications Outlet Cords
10. Surge Protection
11. Power Distribution
12. Fire Stopping
13. System testing
14. Documentation and submissions
15. Maintenance and Service

D. Information Communication Technology Cabling Systems are to be bid as follows-

1. All aspects of the Information Communication Technology Cabling Systems are to be included in the Base Bid.

E. Interpretation of Contract Documents

1. The technologies that are specified by this document include telephone, data, and CATV cabling. See section 27 4100 for general provisions related to Audio Visual Systems. See section 28 0500 for general provisions related to Security Electronics systems.
2. It is the division 27 contractor's responsibility to coordinate the pathway and raceway systems with the division 26 contractor and all low voltage systems included on the responsibilities matrix provided on the drawings.
3. Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, appurtenances and devices incidental to or necessary for a sound, secure and complete installation.
4. Mention in these specifications or indications and/or reasonable implications whereby articles, materials, operation or methods related to execution of the work are noted, specified, drawn or described, thereby requires execution of each such item of work and provision of all labor, materials, equipment and accessories required for execution thereof.
5. No exclusions from, or limitations in, the language used in the specifications shall be interpreted as meaning that the accessories necessary to complete any required system or item of equipment are to be omitted.
6. The use of words in the singular shall not be considered as limiting where other indications denote that more than one item is referred to.
7. Furnish and install all materials for systems, resulting upon completion, in functioning systems in compliance with performance requirements specified. The omission of express reference to any parts necessary for, or reasonably incidental to, a complete installation shall not be construed as a release from furnishing such parts.
8. Drawings are diagrammatic and indicate general arrangement of system and equipment, except when specifically dimensioned or detailed. They are to show size, capacity, approximate location, direction and general relationship of one work phase to another, but not exact detail or arrangement.
9. Refer to dimensioned architectural/structural drawings for exact, locations of building elements.
10. Field verification of measurements takes priority over dimensioned drawings.
11. Dimensions indicated anywhere, are limiting dimensions.
12. The Owner/User reserves the right to make any reasonable change in location of devices and equipment prior to rough installation without involving additional expense. All changes from the drawings as are necessary to make the work of the Contractor conform to the building as constructed shall be included and installed without extra cost.

1.2 SCOPE AND RESPONSIBILITY

A. Provide full time on-site field representation for Division 27 Information Communication Technology Cabling Systems scope of work for duration of installation and prior to turn over.

B. Include detailed scheduling information for Information Communication Technology Cabling Systems installation and testing in the construction schedule. Provide detailed GANTT chart construction schedule showing all tasks referenced in the project phasing plans. Include:

1. Engineering
2. Shop drawing preparation
3. Shop Fabrication
4. Equipment installation
5. Electrical Work
6. Testing, commissioning, and training.

C. This schedule must be submitted fourteen (14) calendar days after receipt of contract. The duration of this schedule must also comply with the completion dates of the construction schedule contained

in the contract documents.

D. Provide coordination to ensure Telecommunication spaces are completed, cleaned and have conditioned air as early as possible to facilitate completion of Information Communication Technology Cabling Systems wiring and terminations. Space shall be free of air-borne particles prior to installation of any Information Communication Technology Cabling Systems Equipment. The Architect shall inspect and approve the condition of these rooms prior to the installation of any active or passive equipment.

E. Conduct periodic coordination meetings between Contractors to make everyone aware of critical areas of construction. Distribute the meeting minutes and attendance to the Owner/User's Representative, the Architect and the Owner/User in a timely fashion.

F. Contractor shall provide coordination of mechanical and electrical installation requirement with the General Contractor and the Electrical Contractor.

G. Provide coordination of the Information Communication Technology Cabling Systems installation.

H. Provide coordination as required to complete the inspection described in paragraph 3.1 INSPECTION.

I. Furnish and install a complete conduit raceway system including back boxes, junction boxes, mortar boxes, and cable tray for all Information Communication Technology Cabling Systems. The conduit size shall allow for a maximum conductor fill of 40% in accordance with NEC guidelines.

J. Inspect conduit raceway system including back boxes, junction boxes, and mortar boxes for all Information Communication Technology Cabling Systems furnished by others. Notify the Architect of any discrepancies immediately.

K. Conduit from telecommunications outlets shall continue from box to location above accessible ceiling. J-hook shall be used to support cabling from conduit stub to cable tray. Maximum spacing of J-hooks shall not exceed 5'.

L. Furnish and install all nonstandard back boxes.

M. Furnish and install all equipment racks and cabinets.

N. Furnish and install Telecommunications Grounding System as shown in the Project Documents. Provide connection of the Telecommunications Grounding System to the electrical ground system per NEC, and as shown in the Project Documents.

O. Furnish all 120 volt AC wiring and connections for power panels and/or terminal strips in electrical panels, cabinets, enclosures, and or consoles, as indicated in the contract documents and approved shop drawings.

P. Furnish and install all devices, equipment, and appurtenances resulting in complete, functional, and fully operational systems as specified herein, indicated on the drawings and listed below;

1. Section 270526 – Grounding and Bonding for Telecommunications
2. Section 270536 – Cable Trays for Telecommunications
3. Section 271116 – Cabinets and Enclosures for Telecommunications
4. Section 271120 – CATV Systems
5. Section 271500 – Horizontal Cabling for Telecommunications

Q. Provide a detailed installation, which is prepared and certified by a Registered Communications Distribution Designer (RCDD).

R. Furnish and install the miscellaneous systems equipment and materials as required for a complete and fully functional Information Communication Technology Cabling Systems as specified and or indicated on the drawings.

S. Provide termination of all cabling for the Information Communication Technology Cabling Systems.

T. Prior to fabrication, coordinate exact location and installation of devices with other trades.

U. Provide coordination to complete the inspection described in Paragraph 3.1 INSPECTION.

V. Provide complete system test, Manufacturer's minimum 20-year performance certification and Owner/User training by the personnel responsible for the Information Communication Technology Cabling Systems installation.

W. Provide complete factory certified training to the Owner/User.

X. Coordinate the work of this Section with that of other Sections to ensure that the entire work of this project will be carried out in an orderly, complete and coordinated fashion.

1.3 RELATED DOCUMENTS

A. General

1. Drawings, specifications and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification sections, apply to work of this section. The Contractor and all subcontractors are responsible for locating information pertaining to required items of work specified or indicated elsewhere in the Contract Documents.

B. Related Work Specified Elsewhere

1. Division 00 - BIDDING REQUIREMENTS
2. Division 01 - GENERAL REQUIREMENTS
3. Division 26 - ELECTRICAL

C. Reference Specifications, Materials, and/or Codes

1. Submit all items necessary to obtain all required permits to the appropriate Regulatory Agencies obtain all required permits and pay all required fees.
2. All work shall conform to the National Electrical Code (NEC) and to applicable National Fire Protection Association (NFPA) codes.
3. All work shall conform to all Federal, State and local ordinances.
4. Where applicable, all fixtures, equipment and materials shall be as approved or listed by the following:

    a. Factory Mutual Laboratories (FM).
    b. Underwriters Laboratories, Inc. (UL).
    c. National Electrical Manufacturers Association (NEMA).

5. References to the National Electrical Code and National Fire Protection Association (NFPA) are a minimum installation requirement standard. Design drawings and specification

sections shall govern in those instances where requirements are greater than those specified in the NEC and NFPA.

6. All material and equipment shall be listed, labeled or certified by Underwriters' Laboratories, Inc. where such standards have been established. Equipment and material, which are not covered by UL Standard, will be accepted provided equipment and material are listed, labeled, certified or otherwise determined to meet safety requirements of a nationally recognized testing laboratory. Equipment of a class, which no nationally recognized testing laboratory accepts, certifies, lists, labels, or determines to be safe will be considered, if inspected or tested in accordance with national industrial standards such as NEMA, ICEA or ANSI. Evidence of compliance shall include certified test reports and definitive shop drawings. NOTE: It is not required that the final installed system be UL listed as a single product.

D. All work shall meet or exceed the standards and procedures of the following:

1. National Fire Protection Association (NFPA): NFPA 70, NFPA 72, NFPA 90A
2. American National Standards Institute (ANSI)
3. Telecommunications Industry Association (TIA)

    a. 568-C.0-2 Generic Telecommunications Cabling for Customer Premise
    b. 568-C.1-1 Commercial Building Telecommunications Cabling Standard
    c. 568-C.2 Balanced Twisted Pair Telecommunications Cabling and Component Standard
    d. 568-C.3 Optical Fiber Cabling Components Standard.
    e. 569-C-1 Commercial Building Standard for Telecommunications Pathways and Spaces
    f. 598-B Commercial Building Standard for Telecommunications Optical Fiber Cable Color Coding.
    g. 606-B Administration Standard for the Telecommunications Infrastructure of Commercial Buildings.
    h. 607 Commercial Buildings Grounding (Earthing) and Bonding Requirements for Telecommunications
    i. 758-B Customer Owned Outside Plant Telecommunications Cabling Standard.
    j. TSB-67 Performance Specifications for Field Testing of Unshielded Twisted Pair Cabling Systems.

4. National Electrical Manufacturers Association (NEMA)
5. American Society of Testing Materials (ASTM)
6. Institute of Electronic & Electrical Engineers (IEEE)
7. Underwriters Laboratory (UL)
8. Americans With Disabilities Act (ADA)
9. Building Industry Consulting Service International (BICSI)

E. Include all items of labor and material required to comply with such standards and codes. Where quantity, sizes or other requirements indicated on the drawings or herein specified are in excess of the standard or code requirements, the specifications or drawings, respectively, shall govern.

F. Installation shall be performed in accordance with the applicable standards, requirements and recommendations of the current local codes and any additional authorities having jurisdiction.

1.4 QUALITY ASSURANCE

A. General

1. Furnish and install only new equipment and materials required (less than 1 year from manufacture), unused without blemish or defect.
2. Each major component of equipment shall have the manufacturer's name, address, model number, and rating on a plate securely affixed in a conspicuous place. The nameplate of a distributing agent will not be acceptable. NEMA Code Ratings, UL label, or other data, which is die-stamped into the surface of the equipment, shall be stamped in a location easily visible. Performance as delineated in schedules and in the specifications shall be interpreted as minimum performance. In many cases, equipment is oversized to allow for pickup loads which cannot be delineated under the minimum performance.
3. All equipment of the same type shall be the product of one manufacturer.
4. The original factory condition of manufactured equipment shall not be modified without the written approval of the Architect.

B. Qualifications

1. The Contractor shall have been in existence for a minimum of 5 years.
2. The Contractor shall specialize in the installation of Information Communication Technology Cabling Systems of equal scope, quality, type, and complexity to that required herein.
3. The Contractor shall be a member in good standing in BICSI for at least 5 years prior to bid time. Provide verification of membership.
4. The Contractor will own, at time of bid, all required testing equipment called for in the specifications. Technicians responsible for operating testing equipment will have successfully completed all manufacturers approved training courses for the successful operation of the testing equipment.
5. The Contractor shall be an authorized dealer for the Structure Cabling solution for at least one year prior to bid time. This is to include fiber optic cable, copper unshielded twisted pair cable, and jacks and patch panels.
6. The Contractor shall be factory trained on the installation and termination of the Information Communication Technology Cabling Systems solution.
7. The principal members and key personnel to be assigned to the project shall each have a minimum of 10 years experience in completing projects of equal scope, quality, type, and complexity to that required herein.
8. Minimum of five Information Communication Technology Cabling Systems, of similar size and scope, installed and operational for a minimum of one year.
9. Minimum of one BICSI Registered Communications Distribution Designer (RCDD) employed or contracted by the Contractor.
10. Project Lead Technician

    a. Trained and certified by the Information Communication Technology Cabling Systems solution manufacturer.
    b. Minimum 5 years experience as Lead Technician on Information Communication Technology Cabling Systems projects of similar size and scope.
    c. BICSI Installer Level 2 certification.

11. Project Technicians

    a. Trained and certified by the Information Communication Technology Cabling Systems solution manufacturer.
    b. Minimum 2 years experience on Information Communication Technology Cabling Systems projects of similar size and scope.
    c. BICSI level 1 or level 2 installer certification.

12. Local services facility within 100-mile radius of project location.

C. Qualification Proposal Requirements

1. THE CONTRACTOR IS RESPONSIBLE FOR PROVIDING DOCUMENTED PROOF THAT ALL QUALIFICATIONS HAVE BEEN MET. THE FOLLOWING DOCUMENTATION IS REQUIRED WITHIN 48 HOURS OF THE BID OPENING FOR THE APPARENT LOW BIDDER AND THE RUNNER UP. QUALIFICATION PROPOSAL SHALL INCLUDE THE FOLLOWING-

    a. A history of the Contractor that reflects the length of time they have performed services similar to those required for this project.
    b. Contractor Qualification Statement AIA-305A
    c. Audited financial statement for the previous year.
    d. Name, address, and telephone number of organization's current Bonding company and level of bonding capacity.
    e. Resume's of the principal members and key personnel of the organization indicating a minimum of 10 years experience each in successfully completing projects of equal scope, quality, type and complexity to that required herein. Include resume(s) of personnel in the employ of the organization that have at least 5 years experience in the design, fabrication, and installation of comparable systems. Include a delineation of each individual's responsibilities on this project. Resumes must include information about the individual experience and systems integration capabilities.
    f. Provide a description of installation capabilities and indicate:

        1) Name, registration number and resume of certifying BICSI RCDD.
        2) Name, Information Communication Technology Cabling Systems solution manufacturer certification number and resume of Lead Technician.
        3) Name, Information Communication Technology Cabling Systems solution manufacturer certification number and resume of Technician(s).
        4) Name and resume of the project manager that will be assigned the project.

    g. List of at least 5 comparable, completed projects (of similar size and scope, on which work has been performed, that have been operational for minimum of 1 year. Include for each facility the following:

        1) Name and location of project.
        2) Date of occupancy.
        3) Name, address, and telephone number of Owner/User.
        4) Name, address, and telephone number of Architect.
        5) Name, address, and telephone number of organization's Bonding Company for the individual project.
        6) A description of the systems involved, the construction dates, and the contract amount for the equipment and services provided.

    h. List of all projects, within the last 5 years, in which the Contractor has been involved in litigation with a City, County, State or Federal government agency. Include summary of final decisions and status of pending litigation.
    i. The Contractor shall be an authorized dealer for at least 75% of the equipment specified for at least one year prior to bid time. Provide line list of all manufactures represented and period of time represented.
    j. Complete list of all testing equipment and date of purchase. Include serial numbers for each piece of equipment.
    k. The technical proposal shall also indicate an understanding of the scope, quality, and technical aspects of the work and shall include the following:

1) A technical description of the Contractors approach to implementing each of the major systems included in the work. Interfaces with other contractor's work, if any, must be specifically addressed.
2) For each major system, a delineation of the tasks to be performed by the Contractor's staff and those tasks, if any, to be performed by subcontractors.
3) A list of major systems components, including the equipment manufacturer's product numbers, to be used on this project. Detailed technical specifications and catalog cut sheets must accompany any proposed substitutions for the specified equipment in accordance with Division 1 requirements.
4) A description of the overall system integration, which shall include a functional block diagram of the integrated system. For each functional unit shown on the diagram, the Contractor shall provide a description of the functional capabilities and characteristics of the unit, including the hardware and software systems associated with the functional unit.
5) A functional description of the software to be furnished. Software that is currently available and software that is to be developed by the Contractor shall be identified.

l. Copy of manufacturer's certification indicating the capability to provide a minimum 20-year guaranteed system.
m. THE ARCHITECT WILL EVALUATE ALL QUALIFICATION PROPOSALS. THE CONTRACTOR WILL BE REQUIRED TO MEET ALL QUALIFICATIONS LISTED PRIOR TO THE SUBMISSION OF THE SYSTEM SHOP DRAWING SUBMITTALS.

1.5 SUBMITTALS

A. Refer to Section 013300 - Submittal Procedures

B. Duplicate copies of the bidding documents are not acceptable and will be rejected.

C. All submittals provided must be certified by an RCDD employee.

D. The formal submittal shall be transmitted 30 days after award of contract.

E. Submit electronic submittals via email as PDF electronic files.

F. Product Submittal

1. Submittal must consist of a complete package including Bill of Material, Product Data for each Section of the Division 27 Information Communication Technology Cabling Systems, and Shop Drawings as applicable. PARTIAL OR INCOMPLETE SUBMITTALS ARE NOT ACCEPTABLE. The Submittal shall include the following:

a. A Title Page complete with the following required information:

1) Project name.
2) Date.
3) Name and address of the Architect.
4) Name of Construction Manager
5) Name and address of the Electrical Contractor
6) Name and address of the General Contractor
7) Name and address of any Subcontractors.

b. An Index Page complete with the following required information:

1) Name of the Supplier.
2) Name of the Manufacturer.
3) Title, section and paragraph of the Specification Sections. (Example section 271500, paragraph 2.4)
4) Products in order as specified in PART II of the related specification.

c. Bill of Materials

1) Provide complete bill of materials for all major components, accessories, and hardware to be provided in order to assemble a complete functioning system.
2) Bill of materials shall include-

i) Manufacturer Name
ii) Model
iii) Version
iv) Quantity

d. Each Specification section shall be separated, collated in order, and complete with the following information:

1) Title sheet.
2) Descriptive purpose of the system, stating how each product is to function.

e. Each Data Sheet shall have the specific reference to the Specification it is to be used for, noting the section and paragraph.
f. Product Data showing multiple products, models or options shall be clearly marked identifying the specific product, model and options, which are submitted for review. Unmarked submittals or facsimile copies shall not be acceptable.
g. Submit product data for all equipment showing:

1) Original Data Sheets Only.
2) Product performance, mechanical and electrical specifications.
3) Manufacturer's installation instructions.
4) Certification from the submitted manufacturers that the Contractor’s designated personnel are trained on the installation of the system. Include Contractor installer's name, experience and responsibility.
5) Product test compliance certificates if required.

G. Shop Drawings

1. Contract documents are diagrammatic in nature and intended to define the general scope and complexity of the systems. They do not reflect the detail necessary to construct the specified system. Assembly and submission of detailed shop drawings are required.
2. Submittals consisting of reproduced copies of the original bidding documents will be rejected. The contractor is required to develop a complete set of drawings specific to the final configuration of the system based on the manufacture and models of all components included. Shop drawings are to include all changes noted in addenda, as well as any changes included in architects special instructions or change orders issued prior to the submittal of the shop drawings.
3. Shop drawings shall be submitted with product data.
4. All drawings shall be created using an industry recognized computer aided design program. Recognized programs include AutoDesk Revit, AutoCAD, and Microstation.
5. Submit shop drawings for all equipment showing:

a. Location and type of all field equipment on floor plans. Include all device revisions form addenda.

   1) Telecommunications Outlets Of All Types
   2) Wall Phone Outlets
   3) Telecommunication Rooms
   4) Equipment Rooms
   5) Conduit Routing
   6) Cable Tray Routing
   7) Cabinets and Equipment Racks
   8) Wall Penetrations for Conduit, Tray, and Sleeves

b. All telecommunications outlets shall be shown with the respective label.
c. Large scale (minimum 1/4"=1') floor plan and elevation view drawings of all telecommunications closets depicting all racks, consoles, cabinets, equipment, outlets, backboard, cable ladders, cable tray, backbone conduits, sleeves, penetrations, etc.
d. Size and spacing of all anchors, wall penetrations, joinery construction, etc., required for complete system installation.
e. Electrical riser diagrams identifying all signal, power, and ground circuitry. Identify circuit numbers corresponding with electrical drawings.
f. Elevations of all equipment racks showing equipment mounting locations (front and rear if any equipment is rear mounted). Rack identification/labeling shall be shown. Elevations are to scale and representative of specific manufacturer and model of products provided. All patch panels, horizontal/vertical cable management, power distribution, grounding hardware, and un-interruptible power supplies. Identify space allocation required for owner furnished and installed hardware and security electronics hardware furnished and installed as part of Division 28. Quantities of devices shall reflect final configuration of system based on total number of outlets and devices served from rack.
g. Wall elevations showing mounting of cabinets, and wall mount hardware and electronics. Rack identification/labeling shall be shown. Elevations are to scale and representative of specific manufacturer and model of products provide. All patch panels, horizontal/vertical cable management, power distribution, grounding hardware, and un-interruptible power supplies. Identify space allocation required for owner furnished and installed hardware and security electronics hardware furnished and installed as part of Division 28. Quantities of devices shall reflect final configuration of system based on total number of outlets and devices served from rack.
h. Wire management details for the installation of cable harnesses inside racks, equipment cabinets and other areas of exposed cable.
i. Block diagrams depicting wire connectivity to all data closets. Identify specific cable manufacture/model numbers for cables (horizontal and backbone) and patch panels.
j. Wiring diagrams for all equipment and devices (active and passive).

   1) Wiring diagrams are required to show point to point connection of all terminations for all devices.
   2) Indicate wire type used for connection.
   3) Indicated type of termination to be made.

k. Telecommunication Backboard layouts including wall mounted equipment, equipment ladders, ground bus bar mounting, wall penetrations, cable routing, and cable support methods.
l. Detail typical copper patch panel. Scaled drawings shall include specific manufacturer and model of device provided and associated labeling.

m. Detail fiber termination devices. Scaled drawings shall include specific manufacturer and model of devices provided, layout including all inserts/cards, and associated labeling.
n. Telecommunication Grounding system block diagram. Identify all bounding/grounding conductors, size of conductors, size and placement of bus bars.

H. Samples

1. Provide samples as requested for review and approval of substitutions or as specified in Division 27 Information Communication Technology Cabling Systems specifications.

I. Test Plan and Documentation

1. Submit a complete testing plan for all systems for approval within thirty (30) days of receipt of approved shop drawing/product data submittal.
2. Plan submitted must include shop and field testing of each and every field device and control function. Refer to standards for more detailed testing requirements for structured UTP and fiber optic cabling.
3. Plan submitted must include examples of testing documentation. Testing documentation must be submitted when requesting final Demonstration/Inspection (described below).
4. Detailed testing agenda and testing documentation forms for all systems. Detailed agenda outlining the "hands-on" training sessions to be provided to the Owner/User. The operation, programming/debugging, troubleshooting, repair and maintenance of all systems shall be covered.

J. Training Plan and Documentation

1. Submit a complete training plan for all systems for approval.
2. As a minimum, the plan submitted must include individualized training paths for the following personnel on all aspects of the Information Communication Technology Cabling Systems.

    a. Administrators
    b. Supervisors
    c. Operators
    d. Maintenance Personnel

3. As a minimum, the plan submitted must include the following:

    a. Proposed classes or sessions.
    b. Recommended attendees for each class or session.
    c. Proposed class or session objectives.
    d. Proposed class or session agenda and duration.
    e. Proposed supporting materials for each class or session.
    f. Proposed testing procedure for evaluation purposes.
    g. Proposed documentation of testing and evaluation.

4. Do not commence training until the plan has been approved.
5. Plan submittal must include names of technical instructors. Contractor must provide engineer responsible for project as instructor. Contractor instructor shall submit evidence of factory training on system provided.

K. Operating and Maintenance Data

1. Provide operating and maintenance manual for each Division 27 Information Communication

Technology Cabling Systems.
2. Manuals shall be submitted in both paper and electronic format.
3. Provide one (1) copy of operating and maintenance manual for each Division 27 Information Communication Technology Cabling Systems. Manual shall be bound in "D-ring" binder with a detailed table of contents.
4. Provide electronic submittal in PDF format of each Division 27 Information Communication Technology Cabling Systems.
5. The O & M manuals shall be cross-referenced to the Record Documents and contain the following information for all systems:

    a. Product catalog cut sheets and specifications of all equipment.
    b. "Hands-on" operational description of all equipment and performance features in each system using clear and understandable terminology.
    c. Detailed programming instructions for all systems and all software programs.
    d. Copy of all software programs stored on thumb drive or CD. Provide one copy for each O & M Manual, and one copy that are to be left in the rear of the equipment rack.
    e. Troubleshooting procedures to diagnose malfunctions in each system.
    f. Procedures for addressing repairs and replacement of defective equipment.
    g. Preventative maintenance procedures for all equipment.
    h. Table listing the model numbers for all equipment in each system including the names and phone numbers of the manufacturer and their representative directly responsible for this project.

L. Record Documents

1. Provide Record Documents for each Division 27 Information Communication Technology Cabling Systems.
2. Records shall be submitted in both paper and electronic format.
3. Provide one (1) black line prints.
4. Provide electronic submittal in PDF format.
5. Record documents shall detail "As-built" condition of all systems including:

    a. A set of updated shop drawings showing all Contract changes.
    b. A set of updated product data showing all Contract changes.
    c. Hands on operational description of all equipment and performance features in each system using clear and understandable terminology.
    d. Floor plan showing conduit raceway routing including all equipment rack, cabinet and pull box locations, and conduit sizes.
    e. Complete point-to-point wiring diagrams showing ALL equipment, devices, wire and cable (Signal, power and ground). This document shall also include all terminal block designations, abbreviations and color-coding.
    f. Electronic copy shall use the most recent version of the software platform used for creating shop drawings.

6. Provide report showing results of all tests required and outlined in individual 270500 series specifications.
7. Two (2) copies of training session DVD.
8. Transmittal letter listing delivery of complete spare parts inventory.
9. Copy of warranty statement.
10. Procedures for addressing warranty/repair issues.

1.6 ADDITIONAL CONSTRUCTION ADMINSTRATION SERVICES BY THE ARCHITECT

A. The Contractor shall be obligated to reimburse the Owner/User for reasonable costs for the

following additional professional services for which the Owner/User will be required to compensate the Architect. Reimbursement shall be on an hourly basis at the Architects standard rates, plus reasonable reimbursable expenses billed at a rate of 1.1 times the actual cost to the Architect. A deductive change order to the Contractor's sum will be processed periodically for the cost of such services incurred by the Owner/User.

1. More than two reviews of the same submittal.
2. Responding to multiple "Requests for Information" (RFI) for which the answer is readily obtainable in the Contract Documents.
3. Preparation and follow up review of more than one "Punch List" for the same items of work.
4. Construction administration services that extend more than 60 days past the initial date of Substantial Completion. These services include, but are not limited to, all work associated with the following: attending pay/progress meetings, on-site observation, and processing the Application and Certification for Payment.

B. The Architect will endeavor to notify the Owner/User and the Contractor prior to performing any work that will result in the aforementioned additional services; and to review any proposed modifications to the contract. Contract modification shall not be contingent, however, upon issuance of any such notice or prior review.

## PART 2 - PRODUCTS

### 2.1 GENERAL REQUIREMENTS

A. Refer to Section 016000 - Product Requirements

B. Products that have been in manufacture for at least one year, unless otherwise specified by performance requirements.

### 2.2 SUBSTITUTIONS

A. Submit substitutions in accordance with Section 012500 – Substitutions Procedures. No other method of substitution is acceptable.

B. Deviations from specifications

1. Any deviations from the specifications must be approved 10 calendar days prior to the bid date. This includes changes to the scope of work, equipment substitutions, and changes to the general provision.
2. Changes to the scope of work in the bid proposal are not acceptable. Any proposed change is to be submitted to the Architect for review. Any approved changes will be listed by addendum prior to the bid opening.
3. Any proposed equipment substitution must be submitted 10 calendar days prior to the bid date. Accompany the request, the Contractor must provide manufactures product specifications for the exact model be substituted. This literature must clearly state all specifications called for in the bidding documents, as well as performance characteristics not specified but inherent to the product listed in the specifications. Any items approved as a substitute will be listed by addendum prior to the bid opening. Substitutions after the award of bid will only be allowed in case of discontinued equipment, or if an item of equal or better quality is available and will not affect the contract cost of the system.
4. Changes to the general provisions are not acceptable. Any proposed change is to be submitted to the Architect for review. Any approved changes will be listed by addendum prior to the bid opening.

5. When a specified item is found to be discontinued or obsolete by the manufacturer, the contractor is required to substitute the manufacturer recommended equivalent for that product. If an equivalent is not available, the contractor is instructed to notify the Architect in writing prior to bid time.

C. Where specific products are sole source specified, no substitutions will be allowed.

2.3 MANUFACTURED PRODUCTS

A. Materials and equipment furnished shall be of current production by manufacturers regularly engaged in the manufacture of such items, for which replacement parts are readily available.

B. Equipment Assemblies and Components

1. Excluding cable solutions, all components of an assembled unit need not be products of the same manufacturer; however, all components must be acceptable to the Architect.
2. Manufacturers of equipment assemblies, which include components made by others, shall assume complete responsibility for the final assembled unit.
3. When more than one unit of the same class of equipment or material is required, such units shall be the products of a single manufacturer.
4. Components shall be compatible with each other and with the total assembly for the intended service.
5. Constituent parts which are similar shall be the product of a single manufacturer.

C. All factory wiring shall be identified on or provided with the equipment being furnished and on all wiring diagrams and included with O & M manuals.

D. When factory testing is specified

1. The Architect shall have the option of witnessing factory tests. The Contractor shall notify the Architect a minimum of 15 working days prior to the manufacturers making the factory tests.
2. Four copies of certified test reports containing all test data shall be furnished to the Architect prior to final inspection and not more than 90 days after completion of tests.
3. When equipment fails to meet factory test and reinspection is required, the Contractor shall be liable for all additional expenses.

2.4 EQUIPMENT IDENTIFICATION

A. In addition to the requirements of the National Electrical Code, install an identification sign which will clearly indicate information required for use and maintenance of items such as cabinets, control devices and other significant equipment.

B. Nameplates shall be laminated black phenolic resin with a white core and engraved lettering, a minimum of 1/4-inch high. Nameplates that are furnished by manufacturer, as a standard catalog item, or where other methods of identification are herein specified, are exceptions.

C. All inputs and outputs are to be clearly labeled. Inputs to include the source location and signal type. Outputs should indicate location signal is sent to.

D. All custom input plates labeling shall be engraved and paint filled or laser etched with a contrasting color as shown on the specification.

E. Any and all user serviceable devices shall be clearly labeled

## PART 3 - EXECUTION

### 3.1 INSPECTION

A. Before installing equipment, carefully inspect the installed Work of all other Trades. Verify that all such Work is complete to the point where the installation of electronic equipment may properly commence. Verify that the telecommunications closets are free of airborne contaminants prior to the installation of electronic equipment.

B. Verify that all equipment is installed in accordance with all pertinent codes and regulations, the original design, and the referenced standards.

C. In the event of discrepancy, immediately notify the Architect.

D. Do not proceed with installation in areas of discrepancy until such discrepancies have been fully resolved.

E. Return to original (preconstruction) condition any work disturbed during system installation.

### 3.2 INSTALLATION

A. Install all equipment in strict accordance with the manufacturer's recommendations, reviewed shop drawings, BICSI TDMM 14th edition, and TIA Standards for UTP and fiber optic cable.

B. Secure equipment with fasteners suitable for the use, materials, and loads encountered. If requested, submit evidence proving suitability. Do not attach electrical materials to roof decking, removable or knockout panels, or temporary walls and partitions, unless indicated otherwise.

C. National Electrical Code requirements are applicable to all work.

D. Working spaces shall be not less than specified in the National Electrical Code for all voltages specified.

E. Where the Architect determines that the Contractor has installed equipment not conveniently accessible for operation and maintenance, equipment shall be removed and reinstalled as directed at no additional cost to the Owner/User. "Conveniently accessible" is defined as being capable of being reached without the use of ladders, or without climbing or crawling under or over obstacles such as motors, pumps, belt guards, transformers, piping, and duct work.

F. Cabling must not be supported by building systems equipment, ductwork, mechanical system supports, ceiling grid system or supports. All cable support hardware shall be independently supported by hardware and installation methods as dictated on building or architectural drawings

G. Cabling must not be attached to any part of a Fire Sprinkler System, and must not obstruct sprinkler heads

H. Cabling must be supported at a maximum interval of 5 feet.

I. All new cable pathways using cable ladders, trays, J-hooks, or conduits must be designed and installed to be at 40% or less of maximum fill capacity.

J. Cabling must be grouped together in pathways where possible.

K. Latch and hook type straps shall be used when bundling cables. Nylon ZIP ties are prohibited.

L. Plywood backboards with the following specs will be used on the walls of the Telecommunication Room. Hardwood ¾", A/C, fire retardant, plywood free of defects (knots and voids shall be considered a defect) must be mounted on walls indicated on the plans. Plywood shall be mounted with the best side exposed to interior of room. Plywood shall be mounted so as to be capable of supporting 50 pounds per linear foot of wall space. The plywood shall be painted with two coats of fire-retardant white, low-gloss paint leaving the UL fire-rating symbol unpainted and visible.

3.3 WORK PERFORMANCE

A. Coordinate location of equipment with other trades to minimize interferences.

B. Holes through concrete and masonry in new and existing structures shall be cut with a diamond core drill or concrete saw. Pneumatic hammer, impact electric, hand or manual hammer type drills shall not be allowed, except where permitted by the Architect as required by limited working space.

C. Holes shall not affect structural sections such as ribs or beams.

D. Holes shall be laid out in advance. The Architect shall be advised prior to drilling through structural sections for determination of proper layout.

E. Any holes created in walls, floors, or ceilings by the Contractor are to be sealed with a fire rated method according to all national, state, and local codes.

F. Hangers and other supports shall support only equipment and materials. Provide not less than a safety factory of 5, which shall conform to any specific requirements in the Construction Documents.

G. The Contractor is responsible for repairing and or replacing any damage caused by their workforce at no additional cost to the Owner/User, or the Owner/User's representatives.

3.4 PROTECTION AND CLEANING OF SYSTEMS AND EQUIPMENT

A. Protect all materials and equipment from damage during storage at the Site and throughout the construction period. Equipment and materials shall be protected during shipment and storage against physical damage, dirt, moisture, cold and rain.

B. Prevent damage from rain, dirt, sun and ground water by storing equipment on elevated supports and covering all sides with securely fastened protective rigid or flexible waterproof coverings.

C. Protect piping by storing it on elevated supports and capping the ends with suitable closure material to prevent dirt accumulation in the piping.

D. During construction, cap the top of all conduits and raceway installed vertically.

E. During installation, protect equipment against entry of foreign matter on the inside, and vacuum clean both inside and outside before testing and operating.

F. Damaged equipment, as determined by the Architect, shall be replaced.

G. Protect painted surfaces with removable heavy kraft paper, sheet vinyl or equal, installed at the factory and removed prior to final inspection.

H. Repaint damaged “FINISH” paint on equipment and materials with painting equipment and finished with same quality of paint and workmanship as used by the original manufacturer so repaired areas are not obvious.

I. Conduit back boxes, floor boxes, and poke thru’s shall be vacuumed clean prior to the installation of cable.

3.5 IDENTIFICATION

A. Nameplates shall be laminated black phenolic resin with white core and engraved lettering, a minimum of ¼ inch high. Nameplates that are furnished by manufacturer, as a standard catalog item, or where other methods of identification are herein specified, are exceptions.

B. Uppercase letters of uniform height; centered on device, cover plate, or enclosure; engraved letters filled with a contrasting color; and all characters made clearly and distinctly.

C. Use abbreviations defined in the contract documents whenever possible. Use plan designations for labeling unless indicated otherwise.

3.6 LABELING

A. All cables shall be properly identified using an industrial quality thermal transfer labeling device. Labels shall be high contrast labels with white background and black lettering. Hand written labels are prohibited.

B. Cable

1. Provide typewritten labels on both ends of all system cabling. Locate label within 6” of cable termination. Cover label with clear heat shrink tubing.

C. Power Outlets

1. Power outlet labels are to be mechanically generated.
2. All power outlets designated for equipment shall be labeled on top.
3. Provide a second label on the bottom of the outlet cover plate indicating service panel number and circuit breaker number.
4. Text lettering to be 1/8” high.

D. Rack Mounted Electronic Components

1. Electronic component labels are to be mechanically generated.
2. All inputs are to be labeled identifying source location of signal.
3. All outputs are to be labeled identifying signal destination.
4. Power switches shall be clearly labeled indicating switch is to be left in the on position at all times.

3.7 CABLE TERMINATIONS AND DRESS

A. Installation of communication conductors shall adhere to the following:

1. Furnish and install cable supports as specified.
2. All cables and/or conductors shall be terminated with approved cable termination connectors compatible with the specific termination.
3. Label all cables on both ends and on all patch panel/termination points.

4. Cables shall be neatly bundled using latch and hook type straps.

3.8 CLEANING

A. Refer to Section 01 7419 – Construction Waste Management and Disposal

B. Daily during construction and prior to Owner/User acceptance of the building, remove from the premises and dispose of all packing material and debris caused by work performed under Division 27 Information Communication Technology Cabling Systems specifications.

C. Remove all dust and debris from interiors and exteriors of electrical equipment. Clean accessible current carrying elements prior to being energized.

D. Upon completion of the work, remove excess debris, materials, equipment, apparatus, tools and the like and leave the premises clean, neat and orderly.

E. All bright metal or plated work shall be thoroughly polished. All pasted labels, dirt and stains shall be removed from the devices.

3.9 TESTING EQUIPMENT

A. Copper Certification Tester

1. Cable Types

a. Shielded and unshielded twisted pair LAN cabling

1) TIA Category 3, 4, 5, 5e, 6, and 6A: 100 ohm
2) ISO/IEC Class C, D, E, EA, F, and FA: 100 ohm and 120 ohm
3) Permanent Link Adapters: shielded RJ45 plug type
4) Channel Adapters: Shielded RJ45 jack type
5) TIA Test Standards: Cat 3, 4, 5, 5E, 6, and 6A per TIA568-C.2
6) ISO/IEC Test Standards: Class C and D, E, Ea, F, FA certification per ISO/IEC 11801:2002 and amendments
7) Maximum Frequency: 1200 MHz
8) Display: 5.7” LCD display with protected capacitance touchscreen

2. Supported Test Parameters

a. Wire Map
b. Length
c. Propagation Delay
d. Delay Skew
e. DC Loop resistance
f. Insertion Loss (Attenuation)
g. Return Loss
h. NEXT
i. Attenuation to cross talk Ratio
j. ACR-F
k. Power Sum ACR-F
l. Power Sum NEXT
m. Power Sum ACR-N
n. Power Sum Alien Near End XTalk
o. Power Sum Alien Attenuation XTalk Ratio for End

B. Fiber Certification Tester

1. Wavelengths
   a. Multimode: 850 nm, 1300 nm
   b. Single Mode: 1310 nm, 1550 nm

2. Compatible Fiber Types:

   a. Multimode: 50/125 um, 62.5/125 um
   b. Single Mode: Single Mode

3. Event Dead Zone:

   a. Multi Mode: 850 nm: .5 m, 1300 nm: .7 m
   b. Single Mode: 1310: 3.6 m, 1550: 3.7 m

4. Attenuation Dead Zone:

   a. Multi Mode: 850 nm: 2.2 m, 1300 nm: 4.5 m
   b. Single Mode: 1310: 3.6 m, 1550: 3.7 m

5. Dynamic Range:

   a. Multi Mode: 850 nm: 28 dB, 1300 nm: 30 dB
   b. Single Mode: 1310: 32 dB, 1550: 30 dB

6. Max Distance Range Setting:

   a. Multi Mode: 40 km
   b. Single Mode: 130 km

7. Distance Measurement Range:

   a. Multi Mode: 850 nm: 9 km, 1300 nm: 35 km
   b. Single Mode: 1310: 80 km, 1550: 130 km

C. Acceptable Testing Devices

1. Copper

   a. Fluke DSX 5000 CableAnalyzer

2. Fiber

   a. Fluke OptiFIBER Pro OTDR

3.10 COMPLETION

A. Results Expected

1. All equipment and materials shall be in place and all systems shall be demonstrated to be operationally complete.
2. All testing, start-up and cleaning work shall be complete.
3. All documented testing results are submitted and approved by the Architect. THE

CONTRACTOR AT NO ADDITIONAL COST SHALL REPLACE ANY COMPONANT OF THE SYSTEM THAT FAILS TO MEET TEST PERFORMANCE STANDARDS.

4. The Contractor shall provide test equipment to the Architect to spot check test result documentation.
5. All O & M Manuals and Record Documents are reviewed and accepted.
6. Substantial Completion inspection is performed and granted. The Substantial Completion inspection punch list is completed by the Contractor.
7. All facility training shall be complete and DVD of the training sessions are reviewed.
8. All "minimum" twenty (20) year structured cable plant certifications and warranties are received.
9. Transmittal letter listing delivery of complete spare parts inventory.

3.11 DEMONSTRATION/INSPECTION

A. Upon completion of approved testing procedure and submittal of testing documentation as described above, the Contractor shall notify the Architect, who will visit the project for a demonstration of the systems and an inspection of the completed work in conformance with the Construction Documents. It is mandatory for a representative from the Contractor directly responsible for the project to be present during demonstration and inspection periods.

B. Items which do not comply with the Contract Documents, or which function incorrectly, will be listed and the list will be submitted by the Architect to the Contractor.

C. After all corrections have been made, the Contractor shall notify the Architect who will recheck the system for compliance of all items listed. IF THE ARCHITECT IS REQUIRED TO RETURN TO THE SITE FOR FURTHER INSPECTIONS, ALL LABOR AND TRAVEL EXPENSES SHALL BE REIMBURSED BY THE CONTRACTOR.

3.12 TRAINING

A. Provide training after all final tests and adjustments have been completed. Instruct the Owner/User's personnel according to the approved Training Plan specified above in "Training Plan and Documentation." Training shall cover all aspects of systems operation, maintenance and programming, and will be provided by the responsible Contractor technical representatives.

B. Provide written operating instructions specific to the basic operation of the system. Basic instructions should cover all operational aspects of the system required for basic operation including system powering, signal patching and routing, and level control. These instructions are to be laminated and located with the main system power control.

C. Schedule training sessions to meet the needs of facility personnel as determined by the Owner/User. Provide training during second and third shifts if the Owner/User determines it is necessary.

D. **Record all training sessions**. Submit an **edited** DVD copy (maximum of 2 hours) to the Architect for review and approval. Present all materials used in the training session on the video. Any DVD found to be inferior in recording or material content shall be reproduced at no cost until an acceptable tape is submitted.

E. Provide a minimum of 4 hours of training. All training is to be completed during the time frame established during scheduling with the Owner/User and training may not necessarily be in continuous periods.

F. Refer to individual sections for system specific training requirements.

3.13 MAINTENANCE MATERIALS

A. All special tools for proper operation and maintenance of the equipment provided under this Specification shall be delivered in conformance with Section 017700 Closeout Procedures.

B. Provide spare materials as indicated in Contract Documents and as required for proper maintenance of systems.

3.14 WARRANTY, WARRANTY SERVICE AND MAINTENANCE

A. Warranty

1. The Information Communication Technology Cabling Systems shall have a factory warranty for minimum of twenty (20) years beginning on the date of Owner/User acceptance.
2. The Division 27 Information Communication Technology Cabling Systems not covered under the Information Communication Technology Cabling Systems “minimum” twenty (20) year warranty shall be fully guaranteed by the Contractor for a period of two (2) years beginning on the date of Owner/User acceptance.
3. The guarantee shall include the entire Division 27 Information Communication Technology Cabling Systems scope of work including all equipment, devices, materials, cable/wire, software and installation.
4. Work shall be guaranteed to be free from defects. Any defective materials or workmanship, as well as damage to the Work of all other Trades resulting from same, shall be replaced or repaired as directed by the Owner/User for a period of two (2) years from the date of Owner/User final acceptance. The Contractor shall provide written warranties for all systems and all buildings to the Owner/User.
5. The guarantee shall exclude acts of God, vandalism, physical abuse or operator misuse.
6. Acceptance by a manufacturer of an order for equipment for this contract signifies acceptance of this warranty. During the warranty period there shall be no charge to the Owner/User for equipment, material, software, etc. for guarantee work.
7. During the warranty period, there shall be no charges to the Owner/User for service calls (mileage, labor, travel, expenses, etc.) for guarantee work.

B. Warranty Service
1. Warranty service shall be provided as part of this Contract by the Contractor during the warranty period.
2. Warranty service agreement must be submitted separately and must contain the name, voice phone number, facsimile phone number and pager number of the service provider capable of meeting the response time as defined by the Architect.
3. This service shall consist of the following:

a. 24-hour phone number.
b. Technician’s factory trained by the manufacturers of the system’s components.
c. Authorized representation of the manufacturer via an agreement of factory support.
d. Five years experience (minimum) servicing systems of the type included in this project.
e. Capability of making additions or changes to the software systems used in this project.
f. Capability of servicing the individual system components and the total Information Communication Technology Cabling Systems.
g. Equipment and knowledge to test all specified equipment and devices.
h. Current system documentation including but not limited to the following:

1) Wiring Diagrams

2) Operation and Maintenance Manuals
3) Software Programs
4) Other documentation as required to provide assistance to the Owner/User in the operation and maintenance of the systems.
5) All documents shall be made available to the Owner/user upon request.
6) Upon termination of maintenance agreement, all system documents shall be furnished to Owner/User for maintenance continuity.

4. The Owner/User and/or Architect shall notify the Contractor in writing, outlining operational malfunctions or defects in the Division 27 Information Communication Technology Cabling Systems. This report shall be forwarded to the service provider, which will establish the date and time or problem notification.
5. Upon successful completion of warranty service, the responsible technician shall return a copy of the original service request to the facility with a detailed description of the problem found, and corrective action taken including a list of equipment/parts/software repaired or replaced. The technician shall also sign the on-site service log at the facility.
6. Response Times:

   a. Telephone Call Response: Within 1 hour of request
   b. On-Site Response: 8:00 am to 5:00 pm – Within 24 hours of request

END OF SECTION 27 0500

SECTION 27 0526 – GROUNDING, BONDING, UPS FOR TELECOMMUNICATIONS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish all materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere.

C. General

1. Provide a complete Telecommunications Grounding and Bonding system as indicated in Project Documents in accordance with ANSI/TIA-607-E.
2. Provide all necessary materials and labor for the Backbone Cabling system and the Telecommunications Grounding and Bonding system in all Telecommunication Entrance Facilities, Telecommunications Rooms, Telecommunications Equipment Rooms, and roof top cable entrances (antennas), as indicated in Project Documents.
3. System shall include but not be limited to Primary Bonding Bus Bar (PBB), Secondary Bonding Bus Bar (SBB), Telecommunications Bonding Backbones (TBB), Grounding Equalizer (GE), and Bonding Conductor for Telecommunications (TBC) as indicated in Project Documents.
4. It shall be the responsibility of this contractor to ensure that the telecommunication grounding system for this facility is continuous, complete, and meets or exceeds all applicable codes and standards.
5. Security systems shall utilize the telecommunications grounding network outlined in 27 0526 and within the TT series drawings.
6. UPS Back Up Power
    a. All microprocessor-based equipment shall be powered by an un-interruptible power supply (UPS).
    b. All equipment located within the shell of the storm shelter/dispatch area, shall be served by the centralized UPS and generator separated from the rest of the facility.
    c. Each rack located within the IT- Main server room and second floor data room shall be equipped with a local online-double conversion rack mount UPS, equipped with a network card for central monitoring of UPS health and status.

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality. Installing contractor is to verify prior to installation that all equipment meet or exceed the performance and operational requirements noted by the manufacturer of the field device that is to be connected.

B. Substitutions of equal quality will be accepted, unless noted as otherwise. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the schedule bid time. Refer to Division 01

C. Provide materials listed by UL or ETL.

2.2 PRIMARY AND SECONDARY BONDING BUSBAR (PBB & SBB)

A. Specifications

1. Pre-drilled electro-tin plated copper
2. Hole patterns accommodate two-hole lugs
3. Bolt hole sizing and spacing: NEMA standard
4. Minimum thickness: 0.25"
5. Minimum height: 4" H
6. Mounting: 300 series stainless steel insulated standoff brackets
7. Listed for use, meet TIA/EIA 607, conform to BICSI recommendations, comply with NEMA standards.
8. The busbar shall be cleaned and an anti-oxidant applied prior to fastening connectors to the busbar.

B. Acceptable Manufacturers

1. PBB/SBB: Min 12” Length
   a. Chatsworth: 12” - 40153-012
   b. Harger: 12” - TGBIP14412TMGB
   c. Legrand: 12” - GB4X12TMGBKIT
   d. Erico: 12” – TMGBA12L15PT

2.3 TELECOMMUNICATIONS BONDING CONDUCTOR & TELECOMMUNICATIONS BONDING BACKBONE CONDUCTOR (TBC, TBB, SBC)

A. Specifications

1. Gage: #6 AWG through 750KCMIL copper conductor
2. Insulation: PVC 600V insulated, green or green with yellow-stripe
3. Where installed in ferrous metallic conduit, the conductor shall be bonded to each end of the conduit using a grounding bushing or using a minimum sized conductor of 6AWG tapped to the backbone at both ends of the conduit.

B. Acceptable Manufacturers

1. General Cable
2. Belden Cable
3. Alpha Cable
4. Harger
5. Southwire

2.4 TELECOM EQUIPMENT BONDING CONDUCTOR (TEBC)

1. Gage: Min. #6 AWG copper conductor.
2. Insulation: Green or green with yellow stripe, PVC 600V insulated THHN
3. Support at intervals not exceeding 3’ and terminate to RBB from TGB

B. Acceptable Manufacturers

1. General Cable

2. Belden Cable
3. Alpha Cable

2.5 TWO HOLE COMPRESSION LUGS (TBC, TBB, SBC, AND TEBC CONNECTIONS TO PBB/SBB)

A. Specifications

1. UL listed for use up to 35 KV
2. Temperature rated 90 degrees Celsius
3. CSA certified to 600V
4. Meets J-STD-607-A
5. Electroplated tinned copper
6. Two hole minimum

B. Acceptable Manufacturer

1. Chatsworth
2. Hubbell
3. Panduit
4. Square D
5. Storm Cooper Components
6. Burndy

2.6 RACK BUSBAR (RBB) KIT – VERTICAL MOUNTING

A. Specifications

1. Vertical Rack Bus bar Kit
2. 72” long x ¼” thick
3. Copper
4. Minimum cross-sectional area equal to a 6awg wire
5. The busbar shall be cleaned and an anti-oxidant applied prior to fastening connectors to the busbar.
6. Provide one along with all necessary mounting hardware in each four post rack.

B. Acceptable Manufacturer

1. Chatsworth
2. Hubbell
3. Panduit
4. Square D
5. Storm Cooper Components

2.7 RACK BUSBAR KIT (RBB) –HORIZONTAL MOUNTING

A. Specifications

1. Horizontal 19” Rack EIA-310-D Busbar Kit
2. 19” horizontal rack mount.
3. 3/4” wide minimum
4. Copper/ Tin-plated
5. UL 467
6. The busbar shall be cleaned and an anti-oxidant applied prior to fastening connectors to the busbar.

7. Provide one along with all necessary mounting hardware in each two post rack

B. Acceptable Manufacturer

1. Panduit RGRB19U
2. Chatsworth 10610-019
3. Hubbell HGRKTHT
4. Siemon VP-GRD

2.8 TWO HOLE GROUND TERMINAL BLOCK

A. Specifications

1. ¾” H x 1 ½” W
2. 5/8” Hole Spacing
3. #14 AWG to 3/0

B. Acceptable Manufacturers

1. Chatsworth
2. Hubbell
3. Panduit
4. Square D
5. Storm Cooper Components

2.9 IRREVERSIBLE CRIMP TAP

A. Specifications

1. Continuous conductor splicing or pigtail.
2. Range of conductor sizes and combinations as required.
3. Slotted design
4. Single or multiple taps
5. Color coded
6. Requires crimping tool.
7. UL Listed for application sup to 600V
8. Tin plated

B. Acceptable Manufacturer

1. Chatsworth
2. Harger
3. Panduit

2.10 CABLE TRAY BONDING CLAMP - SPLIT BOLT GROUNDING LUG

A. Specifications

1. Installation shall comply with the requirements of NEC 392.60 and tray manufacturer’s instructions.
2. Provide #6awg THHN (600v rated) green insulated copper bonding conductor (refer to item 2.3 – TEBC) continuous along outside of tray, bonding each section of tray with split bolt.
3. Copper sized

4. RoHS compliant electrical grade
5. UL Listed

B. Acceptable Manufacturer

1. Cablofil
2. Eaton
3. MonoSystems, Inc.
4. Chatsworth
5. Hoffman

2.11 COPPER BONDING TAPE

A. Specifications

1. Designed for use with ESD type floor tiles.
2. Copper
3. RoHS compliant electrical grade
4. Select and install in accordance with floor tile manufacturer's instructions.

B. Acceptable Manufacturer

1. 3M
2. Harger
3. Panduit
4. United Static Control Products

2.12 RACK MOUNT 2KVA UNINTERRUPTIBLE POWER SUPPLY

A. Specifications

| | | |
|---|---|---|
| 1. | Nominal Input Voltage: | 100-127 VAC |
| 2. | Rated Input Current: | 16A |
| 3. | UPS Input Connection Type: | 5-20P |
| 4. | Input Circuit Breakers: | 30 A |
| 5. | Recommended Electrical Service: | 20A 120V |
| 6. | Input Phase: | Single |
| 7. | Frequency: | 50/60 Hz |
| 8. | Output kVA Capacity: | 2.2 kVA |
| 9. | Output Watt Capacity: | 1800 Watts |
| 10. | Power Factor: | .82 |
| 11. | Crest Factor: | 3:1 |
| 12. | Output Voltage Regulation: | +/- 2% (Line Mode), +/- 3% (battery mode) |
| 13. | Output Voltage: | 100v, 110V, 115V, 120V, 127V |
| 14. | Output Receptacles: | (6) 5-15/20R, (1) L5-20R |
| 15. | Battery: | Valve Regulated Lead Acid |
| 16. | Voltage Regulation: | Online, double-conversion power conditioning maintains 2% output voltage regulation |
| 17. | Overvoltage Correction: | Corrects over voltages to 150V |
| 18. | Undervoltage Correction: | Corrects undervoltage to 100V at full load |
| 19. | UL 1778 listed | |
| 20. | Double conversion on-line | |

21. Network interface

B. Acceptable Manufacturer

1. Eaton Tripp-Lite SU2200RTXLCD2U with WEBCARDLX. Furnish one UPS with two extended run battery packs (BP48RT) in each cabinet and two post rack.
2. APC
3. Vertiv

PART 3 - EXECUTION

3.1 INSTALLATION

A. Ground all equipment as per manufacturer's recommendations, IEEE 1100, NEC and TIA/EIA guidelines.

B. Provide equipment grounding conductor from equipment grounding lugs to ground bar. Size conductor based on length; see drawings for sizing requirements.

C. Provide green insulated grounding conductors from ground bar to the following-

1. Telecom Equipment Racks and Cabinets
2. Building Steel
3. Local electrical panel
4. Conduits and conduit sleeves
5. Cable tray

D. Provide a green insulated grounding conductor from each end of the metallic sheath on the telephone backbone cable to the ground bar.

E. Remove paint and or finish on all equipment racks, cable trays, etc., exposing bare metal at all locations where grounding conductor is terminated.

3.2 LABELING

A. Use PBB for the Telecommunications Primary Bonding Busbar.

B. Use SBB for the Telecommunications Secondary Bonding Busbar.

C. Provide an identifier for each Telecommunications Grounding Busbar that incorporates the identifier of the Telecommunications Space the busbar serves.

D. Label both ends of all grounding conductors with both bus bars connected.

3.3 TESTING

A. Certify system is complete and functional.

B. Test all cabling and connections. Perform final functional tests in presence of the Architect.

C. Complete certified testing report.

END OF SECTION 27 0526

SECTION 27 0536 - CABLE TRAYS FOR TELECOMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Provide a complete cable support system comprised of cable tray, and J-Hooks as indicated in Project Documents.

C. Provide all necessary materials and labor for the cable support system including supporting/suspension hardware, and miscellaneous accessories resulting in a complete system.

D. Refer to division 26 for conduit specification.

E. Open, visible cable installation outside of telecom rooms is not permissible. All cabling is to be concealed from view.

1.3 SUMMARY

A. Section Includes:
1. Wire-Mesh cable tray.
2. Ladder cable tray.
3. J-Hooks to support distribution through- out building. J-hooks are only permissible above accessible finished ceilings, all other areas cabling is to be encased in conduit pathway.
4. Cable tray accessories.
5. Cable supports.
6. Warning signs.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of cable tray.

1. Include data indicating dimensions and finishes for each type of cable tray indicated.

B. Shop Drawings: For each type of cable tray.

1. Show fabrication and installation details of cable trays, including plans, elevations, and sections of components and attachments to other construction elements. Designate components and accessories, including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths, and fittings.
2. Cable tray layout, showing cable tray route to scale, with relationship between the tray

and adjacent structural, electrical, and mechanical elements. Include the following:

a. Vertical and horizontal offsets and transitions.
b. Clearances for access above and to sides of cable trays.
c. Vertical elevation of cable trays above the floor or bottom of ceiling structure.
d. Load calculations to show dead and live loads as not exceeding manufacturer's rating for tray and its support elements.

C. Delegated-Design Submittal: For seismic restraints.

1. Seismic-Restraint Details: Signed and sealed by a qualified professional engineer, licensed in the state where Project is located, who is responsible for their preparation.
2. Design Calculations: Calculate requirements for selecting seismic restraints.
3. Detail fabrication, including anchorages and attachments to structure and to supported cable trays.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Floor plans and sections, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Scaled cable tray layout and relationships between components and adjacent structural, electrical, and mechanical elements.
2. Vertical and horizontal offsets and transitions.
3. Clearances for access above and to side of cable trays.
4. Vertical elevation of cable trays above the floor or below bottom of ceiling structure.

B. Seismic Qualification Data: Certificates, for cable trays, accessories, and components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

C. Field quality-control reports.

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality.

B. Substitutions of equal quality will be accepted. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the schedule bid time. Refer to Division 01.

C. Provide materials listed by UL or ETL.

2.2 WIRE-MESH BASKET CABLE TRAY

A. Specifications
1. Materials: Electroplated Zinc Galvanized Finish for use in controlled interior.
2. Splice per ASTMB633
3. Configuration: Galvanized- steel wire mesh, complying with NEMA VE 1.
4. Width: 18 inches unless otherwise indicated on Drawings.
5. Minimum Usable Load Depth: 4 inches where horizontally mounted in ceiling space. 2 inches where vertically mounted to support vertical routing of cable.
6. Straight Section Lengths: 10 feet or 12 feet except where shorter lengths are required to facilitate tray assembly.
7. Structural Performance: Capable of supporting a maximum cable load, with a safety factor of 1.5, plus a 200-lb concentrated load, when tested according to NEMA VE 1.
8. Class Designation: Comply with NEMA VE 1.
9. Splicing Assemblies: Bolted splice bar kit.
10. Splice-Plate Capacity: Splices located within support span shall not diminish rated loading capacity of cable tray.

B. Acceptable Manufacturers

1. Eaton (B-line).
2. MonoSystems, Inc.
3. Chatsworth
4. Hoffman
5. Legrand - Cablofil

2.3 WIRE MESH BASKET TRAY ACCESSORIES

A. Specifications
1. Fittings: Tees, crosses, risers, elbows, and other fittings as indicated, of same materials and finishes as cable tray.
2. Barrier Strips: Same Materials and finishes as for cable tray.
3. Cable tray supports and connectors, including bonding jumpers, as recommended by cable tray manufacturer.

B. Acceptable Manufacturers

1. Eaton (B-line).
2. MonoSystems, Inc.
3. Chatsworth
4. Hoffman
5. Legrand - Cablofil

2.4 LADDER CABLE TRAY

A. Specifications

1. Materials: Alloy 6063-T6 according to ANSI H 35.1/H 35.1M for extruded components and Alloy 5052-H32 or Alloy 6061-T6 according to ANSI H 35.1/H 35.1M for fabricated parts.
2. Hardware: Chromium-zinc-plated steel, ASTM F 1136.
3. Configuration: Two longitudinal side rails with transverse rungs swaged or welded to side rails, complying with NEMA VE 1.
4. Width: 6 inches (150 mm) 9 inches (225 mm) 12 inches (300 mm) 18 inches (450 mm)

24 inches (600 mm) 30 inches (750 mm) 36 inches (900 mm). Refer to plans for dimensions.
5. Minimum Usable Load Depth: 4 inches (100 mm).
6. Straight Section Lengths: 10 feet (3.0 m) 12 feet (3.7 m) 20 feet (6.0 m) 24 feet (7.4 m), except where shorter lengths are required to facilitate tray assembly.
7. Rung Spacing: 6 inches (150 mm) o.c.
8. Radius-Fitting Rung Spacing: 9 inches (225 mm) at center of tray's width.
9. Minimum Cable-Bearing Surface for Rungs: 7/8-inch (22-mm) width with radius edges.
10. No portion of the rungs shall protrude below the bottom plane of side rails.
11. Structural Performance of Each Rung: Capable of supporting a maximum cable load, with a safety factor of 1.5, plus a 200-lb (90-kg) concentrated load, when tested according to NEMA VE 1.
12. Fitting Minimum Radius: 12 inches (300 mm).
13. Class Designation: Comply with NEMA VE 1.
14. Splicing Assemblies: Bolted type using serrated flange locknuts.
15. Splice-Plate Capacity: Splices located within support span shall not diminish rated loading capacity of cable tray.

B. Acceptable Manufacturers

1. Eaton (B-line).
2. MonoSystems, Inc.
3. Chatsworth
4. Hoffman
5. Legrand - Cablofil

2.5 LADDER CABLE TRAY ACCESSORIES

A. Specifications

1. Fittings: Tees, crosses, risers, elbows, and other fittings as indicated, of same materials and finishes as cable tray.
2. Barrier Strips: Same Materials and finishes as for cable tray.
3. Cable tray supports and connectors, including bonding jumpers, as recommended by cable tray manufacturer.

B. Acceptable Manufacturers

1. Eaton (B-line).
2. MonoSystems, Inc.
3. Chatsworth
4. Hoffman
5. Legrand - Cablofil

2.6 J-HOOKS

A. Specifications
1. UL Listed for support of Category 6 4-pair UTP cables.
2. Pre-galvanized steel finish
3. Fill shall not exceed 40%
4. Static load capacity minimum 30 lbs.

| | |
|---|---|
| ¾” Single Hook | Eaton (B-Line) BCH12 |
| 2” Single Hook | Eaton (B-Line) BCH32 |
| 4” Single Hook | Eaton (B-Line) BCH64 |

| | |
|---|---|
| 2” Multi-Tier (2 high) | Eaton (B-Line) BCH32-2S |
| 4” Multi-Tier (2 high) | Eaton (B-Line) BCH64-2S |
| 2” Multi-Tier (3 high) | Eaton (B-Line) BCH32-3S |
| 4” Multi-Tier (3 high) | Eaton (B-Line) BCH64-3S |
| Structural support hardware to be selected and rated for the application | Eaton (B-Line) |

B. Acceptable Manufacturers (see component list above)
1. Eaton (B-line)
2. Legrand - Cablofil
3. Panduit
4. Hubbell
5. Chatsworth

2.7 WARNING SIGNS

A. Specifications
1. Lettering: 1-1/2-inch high, black letters on yellow background with legend "Warning! Not To Be Used as Walkway, Ladder, or Support for Ladders or Personnel”
2. Machine generated label applied to label holder designed for surface mounting to cable tray.
3. Mount to external side of cable tray where clearly visible at intervals not greater than 25’. One per each horizontal mounted tray run and one per each vertical mount tray.

B. Acceptable Manufacturers
1. Eaton (B-line) Label Clip + Machine generated Label
2. Legrand - Cablofil
3. Panduit
4. Hubbell
5. Chatsworth

2.8 CABLE SUPPORTS – WALL MOUNTED IN TELECOMMUNICATIONS ROOMS

A. Specifications
1. UL Listed for support of Category 6 4-pair UTP cables.
2. Designed to prevent degradation of cable performance and pinch points that could damage cable.
   a. Support brackets with cable tie slots for fastening cable ties to brackets.
   b. Lacing bars, spools, J-hooks, and D-rings
   c. Straps and other devices.

B. Acceptable Manufacturers
1. Eaton (B-line).
2. MonoSystems, Inc.
3. Chatsworth
4. Hoffman
5. Legrand

2.9 PULL STRING

A. Specifications
1. Application: Light duty cable pulling
2. Special Features: Rot and mildew resistant

| | | |
|---|---|---|
| 3. | Tensile Strength: | 210 lb. |
| 4. | Material: | Polypropylene |
| 5. | Color: | White/Contrasting color |

B. Acceptable Manufacturer
1. Ideal Powr-Fish Pull Line. Provide pull in in all conduits.
2. Klein Tools 56110 poly pull line
3. Greenlee

2.10 PULL ROPE

A. Specifications

| | | |
|---|---|---|
| 1. | Applications: | Below grade duct bank |
| 2. | Construction: | ¼" minimum diameter braided |
| 3. | Special Features: | Rot and mildew resistant |
| 4. | Tensile Strength: | 1,125 lb. |
| 5. | Material: | Polypropylene |

B. Acceptable Manufacturer
1. Ideal Pro-Pull Polypropylene Pull Rope. Provide pull rope in all duct banks.
2. Klein Tools
3. Greenlee

PART 3 - EXECUTION

3.1 GENERAL REQUIREMENTS FOR CABLE TRAYS

A. Cable Trays and Accessories: Identified as defined in NFPA 70 and marked for intended location, application, and grounding.
1. Source Limitations: Obtain cable trays and components from single manufacturer.

B. Sizes and Configurations: See the plans for specific requirements for types, materials, sizes, and configurations.

C. Structural Performance: See articles for individual cable tray types for specific values for the following parameters:
1. Uniform Load Distribution: Capable of supporting a uniformly distributed load on the indicated support span when supported as a simple span and tested according to NEMA VE 1.
2. Concentrated Load: A load applied at midpoint of span and centerline of tray.
3. Load and Safety Factors: Applicable to both side rails and rung capacities.

3.2 SOURCE QUALITY CONTROL

A. Testing: Test and inspect cable trays according to NEMA VE 1.

3.3 CABLE TRAY INSTALLATION

A. Install cable trays according to NEMA VE 2.

B. Install cable trays as a complete system, including fasteners, hold-down clips, support systems, barrier strips, adjustable horizontal and vertical splice plates, elbows, reducers, tees, crosses, cable dropouts, adapters, covers, and bonding.

C. Install cable trays so that the tray is accessible for cable installation and all splices are accessible for inspection and adjustment.

D. Remove burrs and sharp edges from cable trays.

E. Join aluminum cable tray with splice plates; use four square neck-carriage bolts and locknuts.

F. Fasten cable tray supports to building structure.

G. Design fasteners and supports to carry cable tray, the cables, and a concentrated load of 200 lb.

H. Place supports so that spans do not exceed maximum spans on schedules and provide clearances shown on Drawings. Install intermediate supports when cable weight exceeds the load-carrying capacity of the tray rungs.

I. Construct supports from channel members, threaded rods, and other appurtenances furnished by cable tray manufacturer. Arrange supports in trapeze or wall-bracket form as required by application.

J. Support bus assembly to prevent twisting from eccentric loading.

K. Install center-hung supports for single-rail trays designed for 60 versus 40 percent eccentric loading condition, with a safety factor of 3.

L. Locate and install supports according to NEMA VE 2. Do not install more than one cable tray splice between supports.

M. Support wire-basket cable trays with trapeze hangers.

N. Support trapeze hangers for wire-basket trays with 3/8-inch diameter rods.

O. Make connections to equipment with flanged fittings fastened to cable trays and to equipment. Support cable trays independent of fittings. Do not carry weight of cable trays on equipment enclosure.

P. Install expansion connectors where cable trays cross building expansion joints and in cable tray runs that exceed dimensions recommended in NEMA VE 2. Space connectors and set gaps according to applicable standard.

Q. Make changes in direction and elevation using manufacturer's recommended fittings.

R. Make cable tray connections using manufacturer's recommended fittings.

S. Seal penetrations through fire and smoke barriers. Comply with requirements in Section 078413 "Penetration Firestopping."

T. Install capped metal sleeves for future cables through firestop-sealed cable tray penetrations of fire and smoke barriers.

U. Install cable trays with enough workspace to permit access for installing cables.

V. Install warning signs in visible locations on or near cable trays after cable tray installation.

3.4 CABLE TRAY GROUNDING

A. Ground cable trays according to NFPA 70 unless additional grounding is specified. Comply with requirements in Section 270526 "Grounding and Bonding for Telecommunications."

B. Cable trays shall be bonded together with splice plates listed for grounding purposes or with listed bonding jumpers.

C. Cable trays with single-conductor power conductors shall be bonded together with a grounding conductor run in the tray along with the power conductors and bonded to the tray at 72-inch intervals. The grounding conductor shall be sized according to NFPA 70, Article 250.122, "Size of Equipment Grounding Conductors," and Article 392, "Cable Trays."

D. When using epoxy- or powder-coat painted cable trays as a grounding conductor, completely remove coating at all splice contact points or ground connector attachment. After completing splice-to-grounding bolt attachment, repair the coated surfaces with coating materials recommended by cable tray manufacturer.

E. Bond cable trays to power source for cables contained within with bonding conductors sized according to NFPA 70, Article 250.122, "Size of Equipment Grounding Conductors."

3.5 CABLE INSTALLATION

A. Install cables only when each cable tray run has been completed and inspected.

B. Fasten cables on horizontal runs with cable clamps or cable ties according to NEMA VE 2. Tighten clamps only enough to secure the cable, without indenting the cable jacket. Install cable ties with a tool that includes an automatic pressure-limiting device.

C. Fasten cables on vertical runs to cable trays every 18 inches.

D. Fasten and support cables that pass from one cable tray to another or drop from cable trays to equipment enclosures. Fasten cables to the cable tray at the point of exit and support cables independent of the enclosure. The cable length between cable trays or between cable tray and enclosure shall be no more than 72 inches.

3.6 CONNECTIONS

A. Remove paint from all connection points before making connections. Repair paint after the connections are completed.

B. Connect pathways to cable trays according to requirements in NEMA VE 2.

3.7 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:
   1. After installing cable trays and after electrical circuitry has been energized, survey for compliance with requirements.
   2. Visually inspect cable insulation for damage. Correct sharp corners, protuberances in cable trays, vibrations, and thermal expansion and contraction conditions, which may cause or have caused damage.
   3. Verify that the number, size, and voltage of cables in cable trays do not exceed that permitted by NFPA 70. Verify that communications or data-processing circuits are separated from power circuits by barriers or are installed in separate cable trays.

4. Verify that there are no intruding items such as pipes, hangers, or other equipment in the cable tray.
5. Remove dust deposits, industrial process materials, trash of any description, and any blockage of tray ventilation.
6. Visually inspect each cable tray joint and each ground connection for mechanical continuity. Check bolted connections between sections for corrosion. Clean and retorque in suspect areas.
7. Check for improperly sized or installed bonding jumpers.
8. Check for missing, incorrect, or damaged bolts, bolt heads, or nuts. When found, replace with specified hardware.
9. Perform visual and mechanical checks for adequacy of cable tray grounding; verify that all takeoff raceways are bonded to cable trays. Test entire cable tray system for continuity. Maximum allowable resistance is 1 ohm.

B. Prepare test and inspection reports.

3.8 PROTECTION

A. Protect installed cable trays and cables.
1. Install temporary protection for cables in open trays to safeguard exposed cables against falling objects or debris during construction. Temporary protection for cables and cable tray can be constructed of wood or metal materials and shall remain in place until the risk of damage is over.
2. Repair damage to galvanized finishes with zinc-rich paint recommended by cable tray manufacturer.
3. Repair damage to paint finishes with matching touchup coating recommended by cable tray manufacturer.

END OF SECTION 270536

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SECTION 27 1116 – CABINETS AND ENCLOSURES FOR TELECOMMUNICATIONS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish and install all equipment, materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere:

1. Provide equipment racks to support backbone and horizontal system cabling.
2. Provide required horizontal and vertical cable management.
3. Coordinate placement and installation of telecommunications equipment racks provided by Information Communication Technology Cabling Systems contractor with the Security Electronics equipment racks and or equipment specified under Division 28 scope of work, and all other equipment within telecom spaces identified on the plans.
4. Provide rack mount power distribution units.
5. Telecommunications equipment racks shown in IT closets will be used for mounting of Owner/User furnished equipment. Coordinate work with Owner/User during shop drawing submittal.
6. Provide grounding of equipment racks.

C. Un-Interruptible Power Supply

1. Furnish and install un-interruptible power supplies as called for in Project Documents (Refer to section 27 0526).

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality.

B. Substitutions of equal quality will be accepted. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the schedule bid time. Refer to Division 01

C. Provide materials listed by UL or ETL.

2.2 FOUR POST FREE STANDING EQUIPMENT RACKS

A. Specifications
1. (4) post open frame equipment rack.
2. Provide rack for mounting standard 19-inch rack width equipment.
3. 84" mountable height. 45U available rack mounting space.
4. Minimum 36” deep
5. Independent adjustable front and rear mounting rails. Capable of being adjusted after rack is secured to the floor.
6. Printed rack space identification all equipment rails.
7. UL Listed for 2500 lbs load rating.

8. Mounting holes in top flanges for securing ladder rack.
9. Provide 6” x 6” vertical finger ducts, front and back, on both sides for the full height of the rack frame.
10. Provide double sided horizontal finger ducts between each patch panel, and space allocated for active equipment.
11. Bottom of rack with holes for bolting to floor.
12. Threaded equipment mounting rails.
13. Black painted finish.
14. Provide quantity as shown on plans.
15. Coordinate installation with Security Electronics racks and or equipment specified under Division 28 scope of work.
16. Coordinate installation, grounding, and placement with other racks.

B. Acceptable Manufacturers

1. Panduit R4P36 Four-Post Rack; PR2VD06 Vertical Manager X 2 per each rack.
2. Chatsworth Products
3. Middle Atlantic
4. Eaton
5. Legrand Mighty Mo

2.3 TWO POST FREE STANDING EQUIPMENT RACKS

A. Specifications

1. Provide two post open frame rack for mounting standard 19-inch rack width equipment.
2. 84" mountable height. 45U available rack mounting space.
3. Printed rack space identification on all equipment rails
4. UL Listed for 1500 lbs load rating.
5. Mounting holes in top flanges for securing ladder rack.
6. Provide dual sided 6” x 6” vertical finger ducts on both sides of rack, front and back, for the full height of the rack frame.
7. Provide covered double sided horizontal finger ducts between each patch panel, and space allocated for active equipment.
8. Bottom of rack with holes for bolting to floor.
9. 12-24 threaded equipment mounting rails.
10. Black finish.
11. Provide quantity as shown on plans.
12. Coordinate installation with Security Electronics racks and or equipment specified under Division 28 scope of work.
13. Coordinate installation, grounding, and placement with other racks.

B. Acceptable Manufacturers
1. Panduit R2P6S two-post rack. PR2VD06 Dual Sided Vertical cable manager.
2. Chatsworth Products
3. Middle Atlantic
4. Eaton
5. Siemon

2.4 HORIZONTAL WIRE MANAGEMENT

A. Specification

1. Compatible with standard 19” equipment racks.
2. Welded steel construction
3. Finished in durable powder coat.
4. 2 space construction
5. Provide horizontal management suited for cable load above and below all patch bays, routers, and hubs.

B. Acceptable Manufacturer

1. Siemon HCM-4-2U
2. Chatsworth 13930-702
3. Panduit PR2HF2
4. Legrand DHMC2RU

2.5 VERTICAL RACKMOUNT POWER DISTRIBUTION UNIT

A. Specifications

| | | |
|---|---|---|
| 1. | Input Voltage: | 120VAC |
| 2. | Output Voltage: | 120VAC |
| 3. | Output Capacity: | 2.03 kW/20 amp circuit |
| 4. | Maximum Input Current: | 20A |
| 5. | Form Factor: | Zero U vertical rack mount |
| 6. | Input Receptacles: | NEMA 5-20P |
| 7. | Output Receptacles: | 12xNEMA 5-20R |
| 8. | Front Panel LCD Display: | N/A |
| 9. | Communications: | NA |

10. Provide one strip for each data cabinet and two post equipment rack.
11. Mount vertically in rear of rack. Plug into UPS/Generator power.

B. Acceptable Manufacturer

1. Panduit P12E22M
2. Tripp Lite
3. APC
4. Eaton

2.6 QUAZITE HANDHOLE (Refer to TY site plan for required outside plant raceways and enclosures)

A. Specifications

| | | |
|---|---|---|
| 1. | Polymer Concrete | |
| 2. | Open bottom | |
| 3. | 36”W x 36”H x 36”L | |
| 4. | Design/ Test Load: | 22,500/33,750 lb |
| 5. | Tier Rating: | 22 |

6. 3/8” stainless steel hex head bolt/washer, typical of two
7. ½” x 4” pull slot, typical of two
8. .5 coefficient of friction skid resistant surface
9. 4”x4” mouse hole, typica two.
10. UL Listed

B. Acceptable Manufacturer

1. Quazite PG3636BB36 box with PG3636HH00 cover. Refer to plans for quantity and location.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install racks in location as indicated on the Project Documents.

B. Install vertical wire management ducts.

C. Install horizontal wire management system

D. Install racks as per manufacturer's recommendations.

E. Provide rack alignment kits for all racks installed side-by-side.

F. Secure top of racks to structure above per the manufacturer's instructions, and as detailed herein.

G. Bolt relay racks to floor using machine bolts and expansion anchors.

H. Ground racks using the manufacturer's ground lug kit for all bonding conductors.

I. Maintain proper bend radius for all cable transitioning into and out of the racks using drop out fittings.

3.2 LABELING

A. General
1. Provide labeling based on ANSI/TIA-606 administration concepts
2. Provide typewritten labels, hand labeling is not acceptable
3. Administration database shall utilize identifiers on labels

B. Telecommunication Spaces and Pathways

1. Use the identifier EF for the Telecommunications Entrance Facility
2. Use the identifier MDF for the Main Telecommunications Room for the networks identified on plans.
3. Use the identifier IDF for each Intermediate Telecommunications Room that indicates the floor and an ordinal designation (e.g. IDF2A is the first IDF telecommunications room on the second floor). Refer to the drawings for the networks serviced by the IDF.
4. Provide an identifier for each Relay Rack in each Telecommunications Space that incorporates the Telecommunications Space identifier (e.g. IDF2A-RR3 is the third relay rack in the first intermediate telecommunications room on the second floor).
5. Provide an identifier for each Telecommunications Backboard in each Telecommunications Space that incorporates the Telecommunications Space identifier (e.g. IDF2A-TBB3 is the third backboard in the first intermediate telecommunications room on the second floor).

END OF SECTION 27 1116

SECTION 271120 – CATV SYSTEMS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish and install all equipment, materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere:

1. The system will provide in house multi-channel RF distribution with its input being provided by the User. The provided signal will be amplified and distributed to the CATV jack locations indicated on the plans.
2. System will include cable plant, head end electronics, equipment mounting hardware, and outlet plates.

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality.

B. Substitutions of equal quality will be accepted. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the schedule bid time. Refer to Division 01.

C. Provide materials listed by UL or ETL.

2.2 AMP

A. Specifications

| | | |
|---|---|---|
| 1. | Frequency Range: | 49-1000 MHz |
| 2. | Gain: | 30 dB |
| 3. | Output Level: | 32/40 dBmV |
| 4. | Flatness: | +/- .75 dB |
| 5. | Hybrid Technology: | Push Pull |
| 6. | Cross Modulation: | -60 dB |
| 7. | Impedance: | 75 ohms |

B. Acceptable Manufacturers

1. Blonder Tongue BIDA 100A-30 distribution amplifier.
2. Pico Macom
3. RL Drake

2.3 SPLITTERS

A. Specifications

1. Frequency Range: 5-1000 MHz
2. 2, 3, 4, and 8 way models
3. RFI Shielding: -120 dB
4. Splitter Loss: 3.5-11.5 db, frequency and output dependent.
5. -60 dBmV Spurious Signals and 2nd Order Harmonics with a +55 dBmV Return Input Carrier.
6. Category A3 Standard 6kV, 100 KHz Ring Wave Surge Withstand Capability on all Ports.
7. Zinc Alloy Die-Cast Housing
8. Machine Threaded Flat F Ports.
9. Return Loss- Input: 25-28 dB, frequency and output dependent
10. Isolation: 25-40 dB, frequency and output dependent
11. Return Loss – Output: 25-30 dB, frequency and output dependent

B. Acceptable Manufacturers

1 Blonder Tongue DGS series splitters. Provide termination resistors as required. Splitters are to be wall mounted in the tele/data rooms as shown on the project documents.
2 Pico Macom
3 RL Drake
4 Toner

2.4 TAP-OFFS

A. Specifications

1. Frequency Range: 5-1000 MHz
2. RFI Shielding: 120 dB
3. Isolation Output to Tap: 27-40 dB, frequency and output dependent
4. Tap Values: 4-26 dB
5. -60 dBmV Spurious Signals and 2nd Order Harmonics with a +55 dBmV Return Input Carrier.
6. Category A3 Standard 6kV, 100 KHz Ring Wave Surge Withstand Capability on all Ports.
7. Zinc Alloy Die-Cast Housing
8. Machine Threaded Flat F Ports.
9. Return Loss- In/Out: 16-20 dB, frequency and output dependent
10. Isolation Tap-Out: 20-40 dB, frequency and output dependent
11. Return Loss – Tap: 18-20 dB, frequency and output dependent
12.

B. Acceptable Manufacturers

1. Blonder Tongue DGT series tap offs. Provide termination resistors as required. Tap-offs are to be wall mounted in the tele/data rooms as shown on the project documents.
2. Pico Macom
3. RL Drake

2.5 OUTLET PLATES

A. Specifications

1. Single gang plate
2. 4 ports capacity
3. UL 1863 compliant
4. CATV outlets shall share the same faceplate as data outlets.
5. Text to read “TV” above connector.
6. Flush mount isolated F feed thru connector.

B. Acceptable Manufacturers:
1. Refer to details and specification 27 1500. F feed thru connector to terminate to shared data port faceplate.

2.6 CATV HORIZONTAL FEEDER

A. Specifications

1. RG-6/U
2. 6.5 ohm/1000’
3. 18 AWG, solid bare copper conductor
4. 16.2 pf/ft nominal capacitance
5. 82% nom. vel of prop
6. 75 ohms nominal impedance
7. Quad shield consisting of aluminum foil with 100% coverage, aluminum braid with 65% coverage, aluminum foil with 100% coverage, and aluminum braid with 55% coverage
8. Gas injected foam polyethylene
9. PVC insulation (CMR/non-plenum applications)
10. Polymer alloy insulation (CMP/plenum applications)

B. Acceptable Manufacturer

1. West Penn Wire

   Q841, CMR/non-plenum applications
   25Q841, CMP/Plenum applications

2. Liberty

   RG6Q CCS 3 GHZ CM, CMR/Non-plenum applications
   RG6Q CC 3 GHZ CMP, CMP/Plenum applications
3. Gepco

   VB18Q, CMR/Non-plenum applications
   VB18CTS, CMP/Plenum applications

4. Windy City Wire

   RG6Q, CMR/Non-plenum applications
   RG6QP, CMP/Plenum applications

2.7 CONNECTORS, AS REQUIRED

A. In-line Cable Connectors. Provide quantity and type as needed

| | | |
|---|---|---|
| 1. | Phono (RCA) plug: | Neutrik NF2CB//2 |
| 2. | 75 Ohm BNC Crimp Plug: | Canare BCP-C4B |
| 3. | 75 Ohm F-Connectors: | Canare FP series |

B. Chassis Mount Connectors. All chassis mount connectors are to be isolated from the panel. Provide quantity and type as needed

| | | |
|---|---|---|
| 1. | BNC feed through: | Neutrik NBB75DFI |
| 2. | F feed through: | Canare FJ-JRU |
| 3. | RCA (audio) feed through: | Canare RJ-RU |
| 4. | RCA (video) feed through: | Canare RJ-BCJRU |

PART 3 - EXECUTION

3.1 INSTALLATION

A. The System shall be installed by qualified personnel in strict compliance with manufacturers instructions.

B. Wiring shall be color coded, uniform and in accordance with national electric codes and manufacturer's instructions.

C. Equipment shall be firmly secured, plumb and level.

D. All cable runs to the telecom room shall be tagged and identified.

E. The Contractor is responsible for selection tap-offs appropriate for establishing optimal signal level at the outlet based on as-built conditions. As part of the shop drawing submittal, provide calculated tap-off values for all drops as based on existing field conditions and cable lengths.

F. After cable routing is established between the head end equipment rack and the outlet location, Contractor shall determine required tap off setting based on attenuation of cable run and additional splitters that are inline between the outlet and the head end equipment rack.

G. All unused tap-off and splitter outputs are to be terminated with 75 ohm terminating resistors.

H. Coordinate all work with other trades and Contractors.

I. Grounding of cables, and peripheral equipment shall be installed per manufacturer's direction to eliminate noise induction and achieve optimum system performance.

J. As part of the project record documents, the Contractor shall provide as-built documentation showing all splitters, tap-offs, and cable routing for the CATV system. Documentation to include signal level measurements at all outlet locations.

3.2 SYSTEM INITIALIZING AND PROGRAMMING

A. The System shall be turned on and adjustment made to meet requirements of the specification and on-site conditions.

B. The System shall be programmed to function as specified.

C. Any special programming shall be documented and a written copy given to the User.

3.3 SYSTEM TEST PROCEDURES

A. Testing to be performed in the Contractors shop prior to delivery-

1. Test 120VAC power equipment and hardware. Test all conductors for shorts, opens, and polarity.

B. Testing to be performed at the job site prior to powering the system

1. Verify all signal cables are free of shorts and opens prior to termination of head-end electronics.
2. Test all 120VAC power sources for correct polarity and voltage. Test grounding system for continuity. Notify Electrical Contractor of any problems.

C. The System shall be completely tested to assure that all components are hooked up and in working order. The System shall be adjusted for optimum signal level at each CATV outlet location.

D. Inspect system for defects. Correct all causes of such defects. If the cause is outside of the scope of the Division 27 series scope of work, promptly notify the Architect in writing, indicating the cause of the defect and suggested corrective procedures.

E. Test all cabling for continuity. Verify all lines are free from shorts and opens. Testing shall include all passive components including tap-offs and splitters.

F. Test all cabling verifying that all cables are terminated accordingly.

G. All outlet signal level readings shall be +10dBmV, +/-5 db. The Contractor is responsible for providing tap-offs required for establishing consistent levels.

H. Provide riser diagram of system components indicating calculated visual carrier level at 55.25 MHz (channel 2) and 433.25 MHz (channel WW). Indicate levels at the inputs and outputs of active components and multi-taps, and at TV set outlets. Indicate physical location (e.g., room number, column line, etc.) of each component.

I. Provide written documentation showing all test results.

J. The System shall be final tested in the presence of the Architect.

END OF SECTION 27 1120

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SECTION 27 1500 – HORIZONTAL CABLING FOR TELECOMMUNICATIONS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish and install all equipment, materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere:

1. Provide a complete UTP Horizontal Cabling system that supports voice, and data as indicated in Project Documents.
2. Provide telecommunications room rack mounted termination equipment, telecommunications room cross connect and equipment patch cables, telecommunications outlets, telecommunications outlet equipment cables, terminations, labeling and testing for the Horizontal Cabling system as indicated in Project Documents.
3. Provide Telecommunications Grounding and Bonding of all Horizontal Cabling system components as indicated in Project Documents.
4. Provide Fire stopping Materials for Horizontal Cabling as indicated in Project Documents, provide all labor and materials to maintain Fire stopping at all penetrations at all times during construction and to ensure adequate Fire stopping at all penetrations at completion.
5. Provide all necessary materials and labor for the Horizontal Cabling system in all Telecommunication Entrance Facilities and Telecommunications Rooms as indicated in Project Documents.
6. Cable shall be plenum rated.

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality.
B. Substitutions of equal quality will be accepted. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the schedule bid time. Refer to Division 01.
C. Provide materials listed by UL or ETL.

2.2 CATEGORY 6 UTP HORIZONTAL CABLE

A. Specifications

1. All cable must be NEC type OFNP or NEC type CMP unless otherwise noted.
2. ANSI/TIA Category 6 compliant
3. Quantities and locations as indicated in Project Documents
4. 4 pair, 23 AWG, solid bare annealed copper
5. Sweep tested through 250 MHz
6. Longitudinal rip-cord
7. Operating temperature -20 to +75 degrees Celsius
8. 80V Maximum Operating Voltage
9. 7 ohms/100 m Maximum dc Resistance

10. 3% Maximum dc Resistance Unbalance
11. .6 Amps/conductor Ampacity

B. Acceptable manufacturers

1. CommScope Uniprise Category 6
2. PanGen General Cable GenSpeed 6 Category 6
3. Hubbell NEXTSPEED Category 6
4. Belden Mohawk 6 LAN
5. Superior Essex

2.3 CATEGORY 6 RJ-45 JACKS

A. Specifications

1. Category 6 module featuring 110-termination technology
2. Keystone compliant form factor.
3. Backwards compatible to Category 3
4. ANSI/TIA 568-B.2-1 Category 6 compliant.
5. Sweep tested through 250 MHz
6. UL 94VO Polycarbonate construction
7. Phosphor bronze 90/10 tin lead IDC contacts
8. Accepts 24 through 22 AWG solid copper conductors

B. Acceptable Manufacturers

1. CommScope Uniprise
2. PanGen NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

2.4 CATEGORY 6 PATCH PANELS (48 PORT)

A. Specifications

1. For mounting on 19” rack
2. Modular 48 port capacity (empty)
3. Angled
4. UL 94VO polycarbonate construction
5. UL 1863 compliant
6. Field designation strips.
7. Category 6 RJ-45 modular jack openings.
8. Furnish rear cable management bars

B. Acceptable Manufacturers

1. CommScope Uniprise
2. PanGen NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

2.5 CATEGORY 6 CROSS CONNECT AND PATCH CABLE ASSEMBLIES

A. Specifications

1. ANSI/TIA Category 6
2. 4 pair RJ45
3. Strain relief boot
4. Tested to 250 MHz
5. Patch cable and connector color based on cross-connect field identification color code, TDMM 14th edition.
6. All cables are to be factory assembled, no field fabrication or field termination allowed.
7. Furnish (2) per each horizontal permanent link, unless noted otherwise below, + 25 spare of each length. Length shall be standard size (5', 7', and 10') and sufficient for patching to associated active hardware. Contractor is to coordinate final lengths with owner and owner furnished equipment.

B. Acceptable Manufacturers

1. CommScope Uniprise
2. PanGen NetKey
3. Hubbell NextSpeed
4. Belden KeyConnect
5. Superior Essex/Ortronics

2.6 CATEGORY 6 TELECOMMUNICATIONS OUTLET

A. Specifications

1. Wall Faceplate

a. Single gang plate, stainless steel.
b. 4 ports capacity
c. UL 1863 compliant
d. Voice/data outlets shown adjacent CATV drops shall share the same faceplate. Coordinate mounting and termination of CATV cable and drop with Voice/Data faceplate.
e. Clear label cover.
f. All unused positions shall have blanking plates installed. Blanking plates shall match faceplate color.
g. All installed modules will be oriented to prevent dust and debris from settling on pins and for ease of use.
h. Back box/faceplate orientation and elevation shall match adjacent power receptacle

2. Jack

a. Refer to "Category 6 RJ-45 Jacks"

3. Shutter

a. Footprint equal to jack
b. Color to match jack
c. Icon included on shutter mount

B. Acceptable Manufacturers

1. CommScope Uniprise
2. Panduit NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

2.7 SURFACE MOUNT CATEGORY 6 TELECOMMUNICATIONS OUTLET (WIRELESS ACCESS POINT OUTLETS)

A. Specifications

1. Modular Boxes

a. 1 or 2 ports capacity
b. UL 1863 compliant
c. UL 94VO polycarbonate construction
d. Clear label cover.

2. Jack

a. Refer to “Category 6 RJ-45 Jacks”

3. Shutter

a. Footprint equal to jack
b. Color to match jack
c. Icon included on shutter mount

B. Acceptable Manufacturers

1. CommScope Uniprise
2. Panduit NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

2.8 CATEGORY 6 WALL PHONE OUTLET

A. Specifications

1. Faceplate

a. Single gang plate
b. Stainless Steel Construction
c. 1 ports capacity
d. Integrated telephone mounting posts
e. UL 1863 compliant

2. Jack

a. Category 6 RJ-45 Jacks
b. Category 6 module featuring 110-termination technology
c. Keystone compliant form factor.
d. Backwards compatible to Category 3

e. ANSI/TIA 568-B.2-1 Category 6 compliant.
f. Sweep tested through 250 MHz
g. UL 94VO Polycarbonate construction
h. Phosphor bronze 90/10 tin lead IDC contacts
i. Accepts 24 through 22 AWG solid copper conductors

B. Acceptable Manufacturers
1. CommScope
2. Panduit NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

2.9 FURNITURE MOUNTED CATEGORY 6 TELECOMMUNICATIONS OUTLET

A. Specifications

1. Faceplate

a. Coordinate faceplate furnished with cubical type office furniture furnished by owner, contractor is to ensure compatibility of face plate with furniture.
b. 4 ports capacity stainless steel construction, or to match furniture's finish.
c. UL 1863 compliant
d. Coordinate mounting and termination of cable drop with Voice/Data faceplate.
e. Clear label cover.
f. All unused positions shall have blanking plates installed. Blanking plates shall match faceplate color.
g. All installed modules will be oriented to prevent dust and debris from settling on pins and for ease of use.
h. Back box/faceplate orientation and elevation shall match adjacent power receptacle

2. Jack

a. Refer to "Category 6 RJ-45 Jacks"

3. Shutter

a. Footprint equal to jack
b. Color to match jack
c. Icon included on shutter mount

B. Acceptable Manufacturers

1. CommScope Uniprise
2. Panduit NetKey
3. Hubbell NEXTSPEED
4. Belden KeyConnect
5. Ortronics

PART 3 - EXECUTION

3.1 INSTALLATION

A. General

1. Follow all ANSI/TIA installation guidelines.
2. Follow all cabling solution provider installation guidelines.
3. Install all cabling and secure utilizing wire management and hook and loop (Velcro) straps. No plastic cable ties shall be used to secure any cabling.

B. Telecommunications Rooms

1. Install all Horizontal Cable System Telecommunications Room equipment as indicated in Project Documents.
2. Install cable management to support and train cables with spacing as required to meet bend radius and support requirements.

C. Telecommunications Outlet

1. Install all Horizontal Cabling System telecommunications outlet equipment and cabling as indicated in Project Documents.
2. Do not exceed recommended pulling tensions.
3. Route cabling from outlets to patch panel in the nearest telecommunication closet.
4. Do not exceed 90 meter horizontal length for Category 6 cables. Notify Architect immediately if any Category 6 route exceeds 90 meters.
5. Terminate data cabling per T568B wiring method.

    a. Maintain wire twists to within 0.5 inches of termination.
    b. Remove no more than 1.0 inches of cable jacket.

D. Elevator Telecommunications Outlet

1. Coordinate final placement of telecommunications outlet with elevator control panel for use with elevator cab communications. Refer to electrical drawings for placement of control panel.

E. Rescue Assistance Communications Telecommunications Outlet.

1. Coordinate final placement of telecommunications outlet with Rescue Assistance System Control panel. Refer to Electrical Drawings for placement of control panel.

3.2 LABELING

A. Telecommunications Room Patch Panel Labeling

1. Provide each Patch Panel with a laminated black phenolic resin label with a white core and engraved lettering, a minimum of 1/4-inch high.
2. Provide permanent, computer-generated labels at each Patch Panel port.

B. Horizontal Cable Labeling

1. Provide permanent, computer generated labels on both ends of every Horizontal Cable, Patch Cord and Work Area Equipment Cable.
2. Example of horizontal wiring label: 1A-A001 1A=Equipment Room number, A=the patch panel, and 001= patch panel port.

C. Telecommunications Outlet Labeling

1. Provide permanent, computer generated labels at the station faceplate.

D. General

1. Provide labeling based on ANSI/TIA-606 administration concepts.
2. Provide typewritten labels, hand labeling is not acceptable.

E. Telecommunication Space Terminating Hardware

1. Provide an identifier for each Patch Panel that incorporates the Relay Rack identifier (e.g. TR2A-RR3-PP1 is the first patch panel in the third relay rack in the first intermediate telecommunications room on the second floor).
2. Provide an identifier for each Patch Panel port that incorporates the Patch Panel identifier (e.g. TR2A-RR3-PP1-09 is the ninth port on the first patch panel in the third relay rack in the first intermediate telecommunications room on the second floor).

F. Telecommunications Outlet

1. Provide an identifier for each telecommunications outlet that indicates the room number and an ordinal designation (e.g. 142A is the first activation point in room 142).
2. Provide an identifier for each telecommunications outlet jack designation (e.g. 142A-2 is the second jack in the first TO in room 142).

G. Horizontal Cable

1. Provide the cable's Patch Panel Port identifier and Work Area Activation Point Jack identifier on both ends of all Horizontal cables.
2. Insure that label is intact after termination of cable.

3.3 TESTING

A. Copper cable testing

1. Test all Category 6 horizontal cabling using a digital cable analyzer and certify that the system is complete and functional.
2. A level IIe or better test unit is required and must be updated to include the requirements of ANSI/TIA-568-B.
3. The basic tests required are:

    a. Wire Map
    b. Length
    c. Attenuation
    d. NEXT (Near and crosstalk)
    e. Return Loss
    f. Propagation Delay
    g. Delay Skew
    h. PSNEXT (Power sum near-end crosstalk loss)
    i. PSELFET (Power sum equal level far-end crosstalk loss)

4. Submit all test data to the architect for review and approval.
5. Spot test cabling in the presence of the architect to verify the accuracy of the data submitted.
6. Spot tests will be reflective of the test data submitted or the entire system will be retested in the presence of the architect, and at the contractor's expense.

END OF SECTION 27 1500

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SECTION 274130 – AUDIO-VIDEO FOR AUDIO/VISUAL SYSTEMS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish and install all equipment, materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere:
   1. A/V Display - Roll Call 109
      a. Provide in wall blocking to support owner furnished and owner installed wall mount flat panel display and display mounting hardware.
      b. System will accommodate CATV, USB, and HDMI sources. Digital tuners to be provided by the Owner/User.
      c. System sources will interface with the display via wall mount media interface.
      d. HDMI & USB extenders utilizing shielded Category 6a cable are required for transmission of HDMI and USB signal over long distances.
      e. Provide complete video cable plant, including patch cables allowing connection between AV input panels and owner furnished sources.
      f. Refer to construction documents for details regarding the conduit and power system.
   2. Private Office – Chief of Police 106, Captain 107, Patrol Lt. Office 110, Patrol Sgts. 111, CID Lieut. 209
      a. Provide in wall blocking to support owner furnished and owner installed wall mount flat panel display and display mounting hardware.
      b. System will accommodate CATV, USB, and HDMI sources. Digital tuners to be provided by the Owner/User.
      c. System sources will interface with the display via wall mount media interface.
      d. Provide complete video cable plant, including patch cables allowing connection between AV input panels and owner furnished sources.
      e. Refer to construction documents for details regarding the conduit and power systems
   3. Digital Signage Display – Lobby 002
      a. Provide in wall blocking to support owner furnished and owner installed wall mount flat panel display and display mounting hardware.
      b. System will accommodate HDMI sources.
      c. System sources will interface with the display via wall mount media interface.
      d. Provide complete video cable plant, including patch cables allowing connection between AV input panel and owner furnished sources.
      e. Refer to construction documents for details regarding the conduit and power systems
   4. Security Video Monitor – Roll Call 109
      a. Provide in wall blocking to support security vendor (Div. 28) furnished wall mount flat panel display and display mounting hardware.
      b. System will accommodate HDMI sources.
      c. System sources will interface with the display via rack mount media interface.
      d. Provide complete video cable plant, including patch cables allowing connection between rack mount media extender and rack mounted security vendor furnished source.

e. Refer to construction documents for details regarding the conduit and power systems

5. A/V Display – CID 204
   a. Provide in wall blocking to support owner furnished and owner installed wall mount flat panel display and display mounting hardware.
   b. System will accommodate CATV and HDMI sources. Digital tuners to be provided by the Owner/User.
   c. System sources will interface with the display via wall mount media interface.
   d. Provide complete video cable plant, including patch cables allowing connection between AV input panels and owner furnished sources.
   e. Refer to construction documents for details regarding the conduit and power systems

C. The work includes the following, as well as work not listed below but described elsewhere as it applies to all audio, video and control systems:
   1. Raceway
   2. Audio, video, and control cabling
   3. Equipment Racks and Cabinets
   4. Source and head-end equipment
   5. Field devices
   6. Surge Protection
   7. Power Distribution and Control
   8. Fire Stopping
   9. Systems Testing
   10. Documentation and submissions
   11. Maintenance and Service

D. A/V systems are to be bid as follows-
   1. All aspects of the audio/visual systems are to be included in the Base Bid.

E. Interpretation of Contract Documents

   1. This section of the specifications describes general provisions applicable to Division 27 4100 series systems. See section 27 0500 for general provisions related to Structured Cabling systems. See section 28 0500 for general provisions related to Security Electronics systems.
   2. Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, appurtenances and devises incidental to or necessary for a sound, secure and complete installation.
   3. Mention in these specifications or indications and/or reasonable implications whereby articles, materials, operation or methods related to execution of the work are noted, specified, drawn or described, thereby requires execution of each such item of work and provision of all labor, materials, equipment and accessories required for execution thereof.
   4. No exclusions from, or limitations in, the language used in the specifications shall be interpreted as meaning that the accessories necessary to complete any required system or item of equipment are to be omitted.
   5. The use of words in the singular shall not be considered as limiting where other indications denote that more than one item is referred to.
   6. Furnish and install all materials for systems, resulting upon completion, in functioning systems in compliance with performance requirements specified. The omission of express reference to any parts necessary for, or reasonably incidental to, a complete installation shall not be construed as a release from furnishing such parts.

7. Drawings are diagrammatic and indicate general arrangement of system and equipment, except when specifically dimensioned or detailed. They are to show size, capacity, approximate location, direction and general relationship of one work phase to another, but not exact detail or arrangement.
8. Refer to dimensioned architectural/structural drawings for exact, locations of building elements.
9. Field verification of measurements takes priority over dimensioned drawings.
10. Dimensions indicated anywhere, are limiting dimensions.
11. The Owner/User reserves the right to make any reasonable change in location of devices and equipment prior to rough installation without involving additional expense. All changes from the drawings as are necessary to make the work of the Contractor conform to the building as constructed shall be included and installed without extra cost.

1.1 SCOPE AND RESPONSIBILITY

A. Provide full time on site field representation for Division 274100 series scope of work for duration of installation and prior to turn over.

B. Include detailed scheduling information for Audio/Visual System installation and testing in the construction schedule. Provide detailed GANTT chart construction schedule showing all tasks referenced in the project phasing plans. Include:

1. Engineering
2. Shop drawing preparation
3. Software Programming
4. Shop Fabrication
5. Equipment installation
6. Electrical Work
7. Testing, commissioning, and training.

C. This schedule must be submitted fourteen (14) calendar days after receipt of contract. The duration of this schedule must also comply with the completion dates of the construction schedule contained in the contract documents.

D. Provide coordination to ensure rooms housing Audio Visual equipment are completed, cleaned and have conditioned air as early as possible to facilitate completion of control wiring and terminations. Space shall be free of air-borne particles prior to installation of any Audio/Visual Equipment. The Architect shall inspect and approve the condition of these rooms prior to the installation of any active equipment.

E. Conduct periodic coordination meetings between Contractors to make everyone aware of critical areas of construction. Distribute the meeting minutes and attendance to the Owner/User's Representative, the Architect and the Owner/User in a timely fashion.

F. Contractor shall provide coordination of mechanical and electrical installation requirement with the General Contractor and the Electrical Contractor.

G. Provide coordination of the Audio/Visual system installation.

H. Provide coordination as required to complete the inspection described in paragraph 3.1 INSPECTION.

I. Furnish and install a complete conduit raceway system including back boxes, junction boxes, mortar boxes, and pull string for all audio/visual and paging systems as called for in the contract documents. The conduit size shall allow for a maximum conductor fill of 40% in accord-

ance with NEC Guidelines.

J. Inspect conduit raceway system including back boxes, junction boxes, and mortar boxes for all Audio/Visual systems furnished by others. Notify the Architect of any discrepancies immediately.

K. Provide all custom casework required for housing electronic equipment.

L. Furnish and install a complete Audio/Visual power cable plant including power, and ground wiring, per NEC.

M. Coordinate termination of telecommunications grounding network to UTP shielded patch panels.

N. Furnish and install all nonstandard back boxes.

O. Furnish and install all equipment racks and cabinets.

P. Furnish all 120 volt AC wiring and connections for power panels and/or terminal strips in electrical panels, cabinets, enclosures, and/or consoles, as indicated in the Contract Documents and approved shop drawings.

Q. Furnish and install all required surge protection field devices.

R. Provide all required circuit protection internal to the head end equipment racks.

S. Furnish and install all devices, equipment, and appurtenances resulting in complete, functional, and fully operational systems as specified herein, indicated on the drawings and listed below:

1. Section 274107 – Horizontal Cabling for Audio/Visual Systems

T. Furnish and install the audio/visual equipment.

U. Furnish and install a complete Audio/Visual signal cable plant including audio, video, signal, antenna, communication, and control cable, per NEC and as described in the contract documents.

V. Furnish and install the miscellaneous systems equipment and materials as required for a complete and fully functional audio/visual system as specified and/or indicated on the drawings.

W. Prior to fabrication, coordinate exact location and installation of audio/visual devices.

X. Provide coordination to complete the inspection described in Paragraph 3.1 INSPECTION.

Y. Provide complete system test, operational certification and Owner/User training as called for in the contract documents.

Z. Provide complete factory certified training to the Owner/User.

AA. Coordinate the work of this Section with that of other Sections to ensure that the entire work of this Project will be carried out in an orderly, complete and coordinated fashion.

1.2 RELATED DOCUMENTS

A. General

1. Drawings, specifications and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification sections, apply to work of this section. The Contractor and all subcontractors are responsible for locating information pertaining to required items of work specified or indicated elsewhere in the Contract Documents.

B. Related Work Specified Elsewhere

1. Division 00 – PROCUREMENT AND CONTRACTING REQUIREMENTS
2. Division 01 - GENERAL REQUIREMENTS
3. Division 26 – ELECTRICAL

C. Reference Specifications, Materials, and/or Codes

1. Submit all items necessary to obtain all required permits to the appropriate Regulatory Agencies, obtain all required permits and pay all required fees.
2. All work shall conform to the National Electrical Code (NEC) and to applicable National Fire Protection Association (NFPA) codes.
3. All work shall conform to all Federal, State and local ordinances.
4. Where applicable, all fixtures, equipment and materials shall be as approved or listed by the following:

    a. Factory Mutual Laboratories (FM).
    b. Underwriters Laboratories, Inc. (UL).
    c. National Electrical Manufacturers Association (NEMA).

5. References to the National Electrical Code and National Fire Protection Association (NFPA) are a minimum installation requirement standard. Design drawings and specification sections shall govern in those instances where requirements are greater than those specified in the NEC and NFPA.
6. All material and equipment shall be listed, labeled or certified by Underwriters' Laboratories, Inc. where such standards have been established. Equipment and material which are not covered by UL Standard will be accepted provided equipment and material are listed, labeled, certified or otherwise determined to meet safety requirements of a nationally recognized testing laboratory. Equipment of a class, which no nationally recognized testing laboratory accepts, certifies, lists, labels, or determines to be safe will be considered, if inspected or tested in accordance with national industrial standards such as NEMA, ICEA or ANSI. Evidence of compliance shall include certified test reports and definitive shop drawings. NOTE: It is not required that the final installed system be UL listed as a single product.
7. All work shall meet or exceed the standards and procedures of the following:

    a. National Fire Protection Association (NFPA)
    b. National Electrical Code (NEC)
    c. American National Standards Institute (ANSI/TIA)

        1) 568-C.0 Generic Telecommunications Cabling for Customer Premise
        2) 568-C.1 Commercial Building Telecommunications Cabling Standard
        3) 568-C.2 Balanced Twisted Pair Telecommunications Cabling and Component Standard

4) 568-C.3 Optical Fiber Cabling Components Standard.
5) 569-B Commercial Building Standard for Telecommunications Pathways and Spaces
6) 598-B Commercial Building Standard for Telecommunications Optical Fiber Cable Color Coding.
7) 606-A Administration Standard for the Telecommunications Infrastructure of Commercial Buildings.
8) 607 Commercial Buildings Grounding (Earthing) and Bonding Requirements for Telecommunications
9) 758-A Customer Owned Outside Plant Telecommunications Cabling Standard.
10) TSB-67 Performance Specifications for Field Testing of Unshielded Twisted Pair Cabling Systems.

d. National Electrical Manufacturers Association (NEMA)
e. American Society of Testing Materials (ASTM)
f. Institute of Electronic & Electrical Engineers (IEEE
g. Underwriters Laboratory (UL)
h. Americans With Disabilities Act (ADA)
i. Building Industry Consulting Service International (BICSI)
j. Audio/Visual & Integrated Experience Association (AVIXA)

8. Include all items of labor and material required to comply with such standards and codes. Where quantity, sizes or other requirements indicated on the drawings or herein specified are in excess of the standard or code requirements, the specifications or drawings, respectively, shall govern.
9. All systems shall be installed according to standards outlined in the following-

a. “Sound System Engineering” Third Edition, Davis and Patronis.
b. “Handbook for Sound Engineers” Fourth Edition, Glen M. Ballou.
c. “Audio and Visual Systems Design” Revised Edition, Wadsworth.

1.3 QUALITY ASSURANCE

A. General

1. Furnish and install only new equipment and materials required (less than 1 year from manufacture), unused without blemish or defect.
2. Each major component of equipment shall have the manufacturer's name, address, model number and rating on a plate securely affixed in a conspicuous place. The nameplate of a distributing agent will not be acceptable. NEMA Code Ratings, UL label, or other data which is die-stamped into the surface of the equipment shall be stamped in a location easily visible. Performance as delineated in schedules and in the specifications shall be interpreted as minimum performance. In many cases, equipment is oversized to allow for pickup loads which cannot be delineated under the minimum performance.
3. All equipment of the same type shall be the product of one manufacturer.
4. The original factory condition of manufactured equipment shall not be modified without the written approval of the Architect.

B. Qualifications

1. The Contractor shall have been in existence for a minimum of 5 years.
2. The contractor shall specialize in the installation of Audio/Visual systems of equal

scope, quality, type, and complexity to that required herein.

3. The Contractor shall be a member in good standing in an industry recognized trade organization such as the AVIXA or National Systems Contractors Association for at least 5 years prior to bid time. Provide verification of membership.
4. The Contractor will own, at time of bid, all required testing equipment called for in the specifications. Technicians responsible for operating testing equipment will have successfully completed all manufacturers approved training courses for the successful operation of the testing equipment.
5. The Contractor shall be an authorized dealer for at least 75% of the primary components specified for at least one year prior to bid time. This is to include speaker systems, audio digital signal processing, integrated control systems, amplification, and video processing equipment.
6. The Contractor shall be trained and certified on the programming, configuration, and setup of all micro processor based electronics including audio digital signal processors, integrated control systems, and video processing equipment.
7. The principal members and key personnel to be assigned to the project shall each have a minimum of 10 years experience in completing Audio/Visual projects of equal scope, quality, type, and complexity to that required herein.
8. Minimum of five Audio/Visual systems, of similar size and scope, installed and operational for a minimum of one year.
9. On-site personnel will have successfully completed all training requirement set forth by the primary Audio/Visual electronics product manufacturers. A copy of the training certificates will be available upon request.
10. Project Lead Technician

    a. Trained and certified by the primary Audio/Visual product manufacturers specified for the project including all audio digital signal processors, integrated control systems, and video processing equipment.
    b. Minimum 5 years experience as Lead Technician on Audio/Visual System projects of similar size and scope.
    c. AVIXA CTS-I certification.

11. Project Technicians

    a. Minimum 2 years experience on Audio/Visual System projects of similar size and scope.
    b. AVIXA CTS certification.

12. Local services facility within 100-mile radius of project location.

C. Qualification Proposal Requirements

1. THE CONTRACTOR IS RESPONSIBLE FOR PROVIDING DOCUMENTED PROOF THAT ALL QUALIFICATIONS HAVE BEEN MET. THE FOLLOWING DOCUMENTATION IS REQUIRED WITHIN 48 HOURS OF THE BID OPENING FOR THE APPARENT LOW BIDDER AND THE RUNNER UP. QUALIFICATION PROPOSAL SHALL INCLUDE THE FOLLOWING-

    a. A history of the Contractor that reflects the length of time they have performed services similar to those required for this project.
    b. Contractor Qualification Statement AIA-305A.
    c. Audited financial statement for the previous year.
    d. Name, address, and telephone number of organization's current Bonding company and level of bonding capacity.

e. Resume's of the principal members and key personnel of the organization indicating years of experience in successfully completing projects of equal scope, quality, type and complexity to that required herein. Include resume(s) of personnel in the employ of the organization responsible for the engineering, fabrication, and installation of comparable systems. Include a delineation of each individual's responsibilities on this project. Resumes must include information about the individual's audio/visual experience and systems integration capabilities.

f. List at least five comparable, completed projects, on which work has been performed, that have been operational for at least 1 year. Include for each facility the following:

   1) Name and location of project.
   2) Date of occupancy.
   3) Name, address, and telephone number of Owner/User.
   4) Name, address, and telephone number of Architect
   5) Name, address, and telephone number of General Contractor.
   6) Name, address, and telephone number of organization's Bonding Company for the individual project.
   7) A description of the systems involved, the construction dates, and the contract amount for the equipment and services for which the Contractor had responsibility.

g. List all projects, within the last 5 years, in which the organization has been involved in litigation with a City, County, State or Federal government agency. Include summary of final decisions and status of pending litigation.

h. Provide line list of all manufactures represented (established dealer) and period of time represented.

i. Complete list of all testing equipment and date of purchase. Include serial numbers for each piece of equipment. Provide documentation proving those responsible for operating the testing equipment have been factory certified.

j. Documentation verifying lead technicians are trained and certified for all audio digital signal processors, integrated control systems, and video processing equipment.

k. The technical proposal shall also indicate an understanding of the scope, quality, and technical aspects of the work and shall include the following:

   1) A technical description of the Contractors approach to implementing each of the major systems included in the work. Interfaces with other contractor's work, if any, must be specifically addressed.
   2) For each major system, a delineation of the tasks to be performed by the Contractor's staff and those tasks, if any, to be performed by subcontractors.
   3) A list of major systems components, including the equipment manufacturer's product numbers, to be used on this project. Detailed technical specifications and catalog cut sheets must accompany any proposed substitutions for the specified equipment in accordance with Division 01 requirements.
   4) A description of the overall system integration, which shall include a functional block diagram of the integrated system. For each functional unit shown on the diagram, the Contractor shall provide a description of the functional capabilities and characteristics of the unit, including the hardware and software systems associated with the functional unit.

5) A functional description of the software to be furnished. Software that is currently available and software that is to be developed by the Contractor shall be identified.

l. THE QUALIFICATION PROPOSAL WILL BE EVALUATED BY THE ARCHITECT. THE CONTRACTOR WILL BE REQUIRED TO MEET ALL QUALIFICATIONS LISTED PRIOR TO THE SUBMISSION OF THE SYSTEM SHOP DRAWINGS SUBMITTALS.

1.4 SUBMITTALS

A. Refer to Section 01 3300 - Submittal Procedures

B. Duplicate copies of the bidding documents are not acceptable and will be rejected.

C. The formal submittal shall be transmitted 30 days after Pre-Submittal Meeting.

D. Product Submittal

1. Submittal must consist of a complete package, including, Product Data for each Section of the 27 4100 series specifications, and Shop Drawings as applicable. PARTIAL OR INCOMPLETE SUBMITTALS ARE NOT ACCEPTABLE. The Submittal shall include the following:
2. A Title Page complete with the following required information:

a. Project name.
b. Date.
c. Name and address of the Architect.
d. Name and address of the Electrical Contractor
e. Name and address of the General Contractor
f. Name and address of any Subcontractors.

3. An Index Page complete with the following required information:

a. Name of the Supplier.
b. Name of the Manufacturer.
c. Title, section and paragraph of the Specification Sections. (Example section 27 4140, paragraph 2.4)
d. Products in order as specified in PART II of the related specification.

e. Bill of Materials

1) Provide complete bill of materials for all major components, accessories, and hardware to be provided in order to assemble a complete functioning system.
2) Bill of materials shall include-

a) Manufacturer Name
b) Model
c) Version
d) Quantity

4. Each Specification section shall be separated, collated in order, and complete with the following information:

    a. Title sheet.
    b. Descriptive purpose of the system, stating how each product is to function.

5. Each Data Sheet shall have the specific reference to the Specification it is to be used for, noting the section and paragraph.
6. Product Data showing multiple products, models or options shall be clearly marked identifying the specific product, model and options, which are submitted for review. Unmarked submittals or facsimile copies shall not be acceptable.
7. Submit product data for all equipment showing:

    a. Original Data Sheets Only. Fax copies are not acceptable.
    b. Product performance, mechanical and electrical specifications.
    c. Manufacturer's installation instructions.
    d. Certification from the submitted manufacturers that the Contractor’s designated personnel are trained on the installation of the system. Include Contractor installer's name, experience and responsibility.
    e. Product test compliance certificates if required.

E. Shop Drawings

1. Contract documents are diagrammatic in nature and intended to define the general scope and complexity of the systems. They not reflect the detail necessary to construct the specified system. Assembly and submission of detailed shop drawings are required.
2. Submittals consisting of reproduced copies of the original bidding documents will be rejected. The contractor is required to develop a complete set of drawings specific to the final configuration of the system based on the manufacture and models of all components included. Shop drawings are to include all changes noted in addenda, as well as any changes included in architects special instructions or change orders issued prior to the submittal of the shop drawings.
3. Shop drawings shall be submitted with product data.
4. All drawings shall be created using an industry recognized computer aided design program. Recognized programs include AutoDesk Revit, AutoCAD, and Microstation. All drawings are to be made using the latest software release available.
5. Submit shop drawings for all equipment showing:

    a. Location and layout of all field equipment on floor plans. Include all device revisions from addenda.
    b. All devices shall be shown with the respective label.
    c. Large scale (minimum 1/4"=1') floor plan and elevation view drawings of all Audio/Visual equipment in rooms depicting all racks, consoles, cabinets, equipment, outlets, etc.
    d. Size and spacing of all anchors, wall penetrations, joinery construction, etc., required for complete system installation.
    e. Sizes, shapes, thickness and finishes of all materials and equipment surfaces.
    f. Electrical riser diagrams identifying all signal, power and ground circuitry.
    g. Block diagram(s) depicting system integration details.
    h. Scaled elevations of all audio/visual equipment racks showing equipment

mounting locations (front and rear if any equipment is rear mounted).

i. Wall elevations showing mounting of cabinets and enclosures, conduit, routing, etc.
j. Wire management details for the installation of cable harnesses inside racks, equipment cabinets, consoles, control panels and other areas of exposed cable.
k. Wiring diagrams for all equipment and devices (active and passive).

    1) Wiring diagrams are required to show point-to-point connection of all terminations for all devices.
    2) Indicate wire type used for connection.
    3) Indicated type of termination to be made. (i.e. terminal strip, euro block, XLR, etc.)
    4) Indicate wire tag number.

l. Scaled layouts of all custom integrated control system graphic user interface panels.
m. Scaled layouts of all custom AVI input/output interface panels. Panels should show final descriptive labels and connector type.
n. Installation detail of all surface or flush mounted devices including but not limited to speakers, cameras, projectors, etc. Detail to show all required mounting and suspension hardware, as well as connection made to structure. Provide front, side, and top views.
o. Installation detail of all suspended devices including but not limited to speaker systems, projection systems, etc. Detail to show all required mounting and suspension hardware, as well as connection made to structure. Provide front, side, and top view.
p. Printouts of all digital signal processing programs showing signal routing, level control, and processing.

F. Samples

1. Provide samples as requested for review and approval of substitutions or as specified in Division 27 4100 series.

G. Test Plan and Documentation

1. Submit a complete testing plan for all systems for approval within thirty (30) days of receipt of approved shop drawing/product data submittal.
2. Plan submitted must include shop and field testing of each and every field device and control function.
3. Plan submitted must include examples of testing documentation. Shop testing documentation must be submitted for approval prior to control system shipment to job site. Field testing documentation must be submitted when requesting final Demonstration/Inspection (described below).
4. Detailed testing agenda and testing documentation forms for all systems. Detailed agenda outlining the "hands-on" training sessions to be provided to the Owner/User. The operation, programming/debugging, troubleshooting, repair and maintenance of all systems shall be covered.

H. Training Plan and Documentation

1. Submit a complete training plan for all systems for approval.
2. As a minimum, the plan submitted must include individualized training paths for the following personnel on all aspects of the Audio/Visual system.

a. Administrators
b. Supervisors
c. Operators
d. Maintenance Personnel

3. As a minimum, the plan submitted must include the following:

a. Proposed classes or sessions.
b. Recommended attendees for each class or session.
c. Proposed class or session objectives.
d. Proposed class or session agenda and duration.
e. Proposed supporting materials for each class or session.
f. Proposed testing procedure for evaluation purposes.
g. Proposed documentation of testing and evaluation.

4. Do not commence training until the plan has been approved.
5. Plan submittal must include names of technical instructors. Contractor must provide engineer responsible for project as instructor. Contractor instructor shall submit evidence of factory training on system provided.

I. Operating and Maintenance Data

1. Provide operating and maintenance manual for each Division 27 Audio/Visual System.
2. Manuals shall be submitted in both paper and electronic format.
3. Provide one (1) copy of operating and maintenance manual for each Division 27 Audio/Visual System. Manual shall be bound in “D-ring” binder with a detailed table of contents.
4. Provide electronic submittal in PDF format of each Division 27 Audio/Visual System.
5. The O & M manuals shall be cross-referenced to the Record Documents and contain the following information for all systems:

a. Printed hard copy "Hands-on" operational instructions. Operational instruction are to be specific to each individual system and should be written using clear and understandable terminology. Photos and illustrations should be used to the fullest extent to convey the successful operation of the system. Instructions shall define the operation of the system including but not limited to the following –
1) Manual shall include trouble shooting instructions for common issues and failures that can be expected of each system.
2) Product catalog cut sheets and specifications of all equipment.
3) Manufacturer operation manuals for all equipment
4) Detailed programming instructions for all systems and all software programs.
5) Printed copy of all equipment settings. This is to include the final settings for all source level control, analog and digital processing, and amplification.

6. Provide electronic copy of each O & M Manual, saved on a minimum of three USB thumb drives. Two thumb drives are to be turned over to the Owner/User. The third is to be left in the rear of the equipment rack.
7. Repair procedures for all equipment.
8. Preventative maintenance procedures for all equipment.

J. Record Documents

1. Provide Record Documents for each Division 27 Audio/Visual System.
2. Records shall be submitted in both paper and electronic format.
3. Provide one (1) black line print.
4. Provide electronic submittal in PDF format.
5. Record documents shall detail "As-built" condition of all systems including:

    a. A set of updated shop drawings showing all Contract changes.
    b. A set of updated product data showing all Contract changes.
    c. Floor plan showing all field devices, conduit raceway routing including all equipment rack, cabinet and pull box locations, and conduit sizes.
    d. Floor plans shall show labeling and identification for all field devices, cables, and conductors.
    e. Finalized rack elevations, front and rear. Rack mounted devices shall reflect labeling and identification.
    f. Complete point-to-point wiring diagrams showing ALL equipment, devices, wire and cable (Signal, power and ground). This document shall also include all terminal block designations, wire tag numbering, abbreviations and color-coding.
    g. Copies of all software programs required for any digital signal processing shall be provided. Copy shall include applied version of manufacturer's software, and all final programming/settings. Software shall be stored on USB thumb drive. All passwords shall be turned over.
    h. Table listing the model numbers for all equipment in each system including the names and phone numbers of the manufacturer and their representative directly responsible for this project.
    i. Electronic copy shall use the most recent version of the software platform used for creating shop drawings.

6. Provide report showing results of all tests required and outlined in individual 27 4100 series specifications.
7. Two (2) copies of training session on USB thumb drive.
8. Transmittal letter listing delivery of complete spare parts inventory.
9. Copy of warranty statement.
10. Procedures for addressing warranty/repair issues.

PART 2 - PRODUCTS

2.1 GENERAL

A. Equipment specified is intended as a reference standard for level of quality.

B. Submit substitutions in accordance with Document 00 26000 – Substitution Procedures. Substitution requests must be approved through the process outlined in Document 00 26000 for procedures during the procurement process and Section 012500 after contract award.

C. Provide materials listed by UL or ETL.

2.2 HDMI & USB TRANSMITTER & RECEIVER (AUDIO VISUTAL INTERFACE, TYPE 1)

A. Specifications
    1. Transmit Device:

a. Video, audio, power, and data transmission up to 100 meters 4K/UHD using Cat 6A cable.
b. USB 2.0 extension up to 100 meters.
c. HDCP 2.2 compliant
d. PoE powering from remote receiver device
e. RS232 control
f. Connector Input: (1) HDMI, (1) USB Type B Host, (1) USB Type A device
g. Connector Output: (1) RJ45 HDBaseT
h. Video: Max. Data Rate: 10.2 Gbps, Max. Resolution: 4K@60 Hz (4:2:0) 24bpp.
i. Extension Line: Up to 100 M 4K @ 60Hz (4:2:0)
j. Decora style interface plate
k. 3 year warranty

2. Receiver Device:
a. Video, audio, power, and data transmission up to 100 meters 4K/UHD using Cat 6A cable.
b. USB 2.0 extension up to 100 meters.
c. HDCP 2.2 compliant
d. Local powering supports device plus POE powering of transmit device – 48V DC, furnish external 120VAC-power supply.
e. RS232 control
f. Connector Input: (1)RJ45 HDBaseT IN, (1) RJ45 Ethernet
g. Output: (1) HDMI, (2) USB Type A, (1) RS-232, (1) IR (1) USB Type A device
h. Video: Max. Data Rate: 10.2 Gbps, Max. Resolution: 4K@60 Hz (4:2:0) 24bpp.
i. Extension Line: Up to 100 M at 4K @ 60Hz (4:2:0)
j. Furnish surface mounting hardware
k. 3 year warranty

B. Acceptable Manufacturer
1. Atlona AT-OME-EX-TX-WP and AT-OME-EX-RX
2. Legrand C2G
3. Extron
4. Crestron

2.3 HDMI PASS THROUGH WALL PLATE (AUDIO VISUTAL INTERFACE, TYPE 2)

A. Specifications
1. Connector 1: (1) HDMI female -female pass-through
2. Video resolution: 1080p, 4K
3. Single gang, brushed aluminum

B. Acceptable Manufacturer
1. C2G CG-39870
2. Extron
3. Crestron

2.4 HDMI TRANSMITTER & RECEIVER (AUDIO VISUTAL INTERFACE, TYPE 3)

A. Specifications
1. Transmit Device:
a. 4k/60 4:2:0 video plus embedded audio transmission up to 100 meters 4K/UHD using Cat 6A cable.
b. HDCP 2.2 compliant
c. PoE powering from remote receiver device
d. Connector Input: (1) HDMI

e. Connector Output: (1) RJ45 HDBaseT
f. Decora style interface plate
g. 3 year warranty

2. Receiver Device:
   a. 4k/60 4:2:0 video plus embedded audio transmission up to 100 meters 4K/UHD using Cat 6A cable.
   b. Local powering supports device plus POE powering of transmit device – 48V DC, furnish external 120VAC-power supply.
   c. Audio de-embedding
   d. Connector Input: (1)RJ45 HDBaseT IN, (1) RJ45 Ethernet
   e. Output: (1) HDMI, (1) RS-232, (1) IR
   f. Furnish surface mounting hardware
   g. 3 year warranty

B. Acceptable Manufacturer
   1. Atlona AT-OME-EX-TX11-WP and AT-OME-RX11
   2. Legrand C2G
   3. Extron
   4. Crestron

2.5 HDMI TRANSMITTER & RECEIVER (AUDIO VISUTAL INTERFACE, TYPE 4)

A. Specifications
   1. Transmit Device:
      a. 4k/60 4:2:0 video plus embedded audio transmission up to 100 meters 4K/UHD using Cat 6A cable.
      b. HDCP 2.2 compliant
      c. PoE powering from remote receiver device
      d. Connector Input: (1) HDMI
      e. Connector Output: (1) RJ45 HDBaseT
      f. RS232 control
      g. Furnish rack shelf, mounting kit
      h. 3 year warranty
   2. Receiver Device:
      a. 4k/60 4:2:0 video plus embedded audio transmission up to 100 meters 4K/UHD using Cat 6A cable.
      b. Local powering supports device plus POE powering of transmit device – 48V DC, furnish external 120VAC-power supply.
      c. Audio de-embedding
      d. RS232 control
      e. Connector Input: (1)RJ45 HDBaseT IN, (1) RJ45 Ethernet
      f. Output: (1) HDMI, (1) RS-232, (1) IR
      g. Furnish surface mounting hardware
      h. 3 year warranty

B. Acceptable Manufacturer
   1. Atlona AT-AVA-EX100CE-BP-TX and AT-AVA-EX100CE-BP-RX (AT-AVA-EX100CE-BP-KIT)
   2. Legrand C2G
   3. Extron
   4. Crestron

2.6 PRETERMINATED VIDEO CABLES

A. Specifications
1. Molded, pre-terminated connectivity
2. Refer to plans for connectivity requirements.
3. Typical of cables between AVI pass thru's and displays.
4. Typical of cables for connection to AV connectivity in flip top table top boxes. 3' length.
5. Plenum rated jacket
6. Active connectivity for HDMI cables exceeding 15'

B. Acceptable Manufacturers
1. Crestron
2. AMX
3. Extron
4. C2G
5. Atlona

2.7 FLAT PANEL DISPLAY WALL BOX

A. Specifications

1. 16 gauge galvanized steel back box
2. .062" aluminum painted front cover
3. UL listed AC back box
4. Mounts to wood and steel studs
5. (2) 1"x2" oblong knockouts
6. (3) single gang box knock outs
7. (2) 1'x1 ¼" conduit knock outs
8. Provisions for cable management
9. Knockout for IPS series modular connector inserts

B. Acceptable Manufacturer

1. FSR PWB-320-ESK wall box with trim ring and equipment shelf. Refer to installation details for quantity of conduits and methods of termination. Coordinate installation of 120VAC power with electrical contractor. Coordinate installation of telecom and CATV with structured cabling contractor.

2.5 FLAT PANEL DISPLAY SURGE PROTECTION

A. Specifications

1. Load Rating: 8 amps @120 volts
2. Power Requirements: 10 Watts
3. UL Adjunct Class. Test Results: 1000 surges, 6000 volts, 3000 amps, C1pulse, measured suppressed voltage 290 volts, no failures.
4. Federal Guidelines: Grade A, Class 1, Mode 1
5. EMI/RFI Filter, Normal Mode: 40 dB@ 100 kHz, 50 dB @ 300 kHz,
6. Max. Applied Surge Pulse Joule Rating: Unlimited, due to current limiting.
7. Maximum Applied Surge Pulse Voltage: 6000 volts (1.2 x 50 us pulse)
8. Overvoltage Shutdown: 145 volts

B. Acceptable Manufacturer

1. Ametek SurgeX SX-DS-154. Provide one for each flat panel display location.
2. Middle Atlantic
3. Furman

PART 3 - EXECUTION

3.1 INSTALLATION

A. All rack assembly shall be done at the contractor’s facility prior to installation on site.

B. All system programming shall be done at the contractor’s facility prior to installation on site.

C. The System shall be installed by qualified personnel in strict compliance with manufacturer’s instructions.

D. Wiring shall be color coded, uniform and in accordance with national electric codes and manufacturer's instructions.

E. Equipment shall be firmly secured, plumb, and level.

F. All cable runs shall be tagged and identified.

G. Coordinate all work with other trades and Contractors.

H. Grounding of cables and peripheral equipment shall be installed per manufacturer's direction to eliminate noise induction and achieve optimum system performance.

I. All unused tap-off and splitter outputs are to be terminated according to manufactures guidelines.

J. Install and program AV local area network as required for control and communication between system devices. When required, provide necessary coordination, termination, and programming associated with integrating AV local area network with facility network.

3.2 SYSTEM INITIALIZING AND PROGRAMMING

A. All programming shall occur in the Contractors shop prior to installation on site.

B. The System shall be turned on and adjustment made to meet requirements of the specification and on-site conditions.

C. The System shall be programmed to function as specified.

D. The System shall be programmed to optimize image resolution.

E. Any special programming shall be documented and a written copy given to the Owner/User.

F. Default software passwords shall be changed. Coordinate new passwords with owner/user representative.

3.3 SYSTEM TEST PROCEDURES

A. Testing to be performed in the Contractors shop prior to delivery-
   1. Test 120VAC power equipment and hardware internal to all equipment racks. Test all conductors for shorts, opens, and polarity.
   2. Test routing of AV signals between devices.
   3. Verify video signal resolution is optimized for intended display.
   4. Verify all switchers route inputs to intended outputs.

B. Testing to be performed at the job site prior to powering the system
   1. Verify all signal and control cables are free of shorts and opens prior to termination of head-end electronics.
   2. Test all 120VAC power sources for correct polarity and voltage. Test grounding system for continuity. Notify Electrical Contractor of any problems.

C. The System shall be completely tested to assure that all components are hooked up and in working order. Inspect system for defects. Correct all causes of such defects. If the cause is outside of the scope of the Division 27 4100 series scope of work, promptly notify the Architect in writing, indicating the cause of the defect and suggested corrective procedures.

D. The system shall be capable of operating free of interference, hum bars, and distortion under normal operating conditions.

E. The System shall be pre-tested by the Contractor and certified, in writing, to function in accordance with the plans and specification.

F. Test all cabling for continuity. Verify all lines are free from shorts and opens. Testing shall include all passive components including tap-offs and splitters.

G. Test all cabling verifying that all cables are terminated accordingly.

H. All outlet signal level readings shall be within +/-2 db of manufacturers recommended signal level for that component.

I. Verify display systems are operating free of banding, bending, blooming, chroma delay, convergence, ghosting, humbars/ground loops, jitter, luma display, smearing, and all other symptoms of defective equipment, incorrect termination, and incomplete system setup.

J. Verify all components are responding to all integrated control system commands.

K. Provide written documentation showing all test results.

L. The System shall be final tested in the presence of the Architect.

3.4 TRAINING

A. Contractor is responsible for providing operational and maintenance training applicable to the audio video extender system. Training is to include, but not be limited to the following-
   1. Review all O+M manuals with Owner/User’s representatives present for training.
   2. Perform a tour of the entire facility. During the tour the trainer shall point out all video equipment and provide a brief description of its purpose and use. This is to include but not be limited to all projectors, screens/projection surfaces, interface panels, equipment rack, field devices, and control locations.

3. Explain and demonstrate the powering sequence of the system. Note all locations where the system can be powered from. Note any devices that are required to remain on at all times.
4. Explain and demonstrate signal routing as it applies to all interface panels and patch panels. Note origination and destination for all signal cabling.
5. Note all control locations. Describe the basic operation and purpose of all components associated with each control location.
6. Explain and demonstrate the operation of all control equipment. This is to include all source selection and signal routing.
7. Explain and demonstrate the operation of all video source devices.
8. Explain and demonstrate the operation of all computer interfaces.

END OF SECTION 27 4130

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SECTION 27 4107 – HORIZONTAL CABLING FOR AUDIO/VISUAL SYSTEMS

PART 1 - GENERAL

1.1 SYSTEM DESCRIPTION

A. Drawings and General Provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

B. Furnish and install all equipment, materials and labor necessary to complete the installation of specific systems described herein and integration of all systems as indicated, specified herein or both. The work includes the following, as well as work not listed below but described elsewhere:

1. Provide a complete F/UTP Horizontal Cabling system that supports data as indicated in Project Documents.
2. Provide telecommunications room rack mounted termination equipment, cross connect and equipment patch cables, outlets, outlet equipment cables, terminations, labeling and testing for the Horizontal Cabling system as indicated in Project Documents.
3. Provide Grounding and Bonding of all Horizontal Cabling system components as indicated in Project Documents.
4. Provide Fire stopping Materials for Horizontal Cabling as indicated in Project Documents, provide all labor and materials to maintain Fire stopping at all penetrations at all times during construction and to insure adequate Fire stopping at all penetrations at completion.
5. Provide all necessary materials and labor for the Horizontal Cabling system in all Audio/Visual equipment cabinets as indicated in Project Documents.
6. The horizontal cabling system shall be comprised of a single end to end solution with all cable and components manufactured by the same manufacturer.
7. Cable shall be plenum rated.

PART 2 - PRODUCTS

2.1 GENERAL

A. Cable specified is intended as a reference standard for level of quality. Installing contractor is to verify prior to installation that all cables meet or exceed the performance and operational requirements noted by the manufacturer of the field device that is to be connected.

B. Substitutions of equal quality will be accepted. All substitutions are to be approved prior to bid and listed by addenda. Substitutions will not be accepted after the scheduled bid time. Refer to Division 01.

C. All cable must be NEC type OFNP or NEC type CMP unless otherwise noted

2.2 SHIELDED CATEGORY 6A F/UTP HORIZONTAL CABLE

A. Specifications

1. All cable must be NEC type OFNP or NEC type CMP unless otherwise noted.
2. ANSI/TIA Category 6a compliant
3. Quantities and locations as indicated on Project Documents
4. 4 pair, 23 AWG, solid bare copper
5. Sweep tested through 500 MHz

6. Longitudinal rip cord
7. Aluminum/Polyester Tape Outer shield, 100% coverage.
8. Drain wire TC-tinned solid copper
9. Operating temperature -20 to +75 degrees Celsius
10. 300 V RMS UL Voltage Rating
11. 82 ohm/km Maximum dc Resistance
12. 3% Maximum dc Resistance Unbalance
13. .5 Amps/conductor Ampacity
14. Color of cable to be Black

B. Acceptable Manufacturers

1. Commscope Systimax
2. Panduit
3. Hubbell
4. Belden
5. Leviton/Berk-Tek
6. Hitachi
7. OCC
8. Superior Essex

2.3 SHIELDED CATEGORY 6A PATCH PANELS

A. Specifications

1. For mounting on 19” rack
2. Modular 48 port capacity
3. Metal connector body
4. Phosphor bronze with 50u inch gold over nickel plating front connection
5. Phosphor bronze with tin platting over nickel rear connection
6. Sweep tested through 500 MHz
7. UL 1863 compliant
8. IDCs accept 26 through 22 AWG solid copper conductors
9. Color based on cross-connect field identification color code, TDMM 12th edition, chapter 7.
10. Field designation strips.
11. Category 6a RJ-45 modular jack
12. Patch panels to be fully populated.

B. Acceptable Manufacturer

1. Commscope Systimax
2. Panduit
3. Hubbell
4. Belden
5. Leviton/Berk-Tek
6. Hitachi
7. OCC
8. Ortronics

2.4 SHIELDED CATEGORY 6A RJ-45 JACKS

A. Specifications

1. Category 6a module featuring 110-termination technology
2. Keystone compliant form factor.
3. Backwards compatible to Category 3
4. ANSI/TIA 568-B.2-1 Category 6a compliant.
5. Sweep tested through 500 MHz
6. Metal connector body
7. Phosphor bronze with 50u inch gold over nickel plating front connection
8. Phosphor bronze with tin platting over nickel rear connection
9. Accepts 24 through 22 AWG solid copper conductors
10. Standard jack color is White.

B. Acceptable Manufacturers

1. Commscope Systimax
2. Panduit
3. Hubbell
4. Belden
5. Leviton/Berk-Tek
6. Hitachi
7. OCC
8. Ortronics

2.5 SHIELDED CATEGORY 6A CROSS CONNECT AND PATCH CABLE ASSEMBLIES

A. Specifications

1. ANSI/TIA Category 6a
2. 4 pair RJ45
3. Strain relief boot
4. Tested to 500 MHz
5. Laminate aluminum-polyester tape floating shielded pair
6. 30% coverage tinned copper braid floating shield
7. Patch cable and connector color based on cross-connect field identification color code, TDMM 10th edition, chapter 7.
8. All cables are to be factory assembled, no field fabrication or field termination allowed.
9. Patch cables are to be turned over to the Owner/User for use in installation of active equipment outside the scope of the project.
10. Furnish one patch cable for each active Category 6 port.

B. Acceptable Manufacturers

1. Commscope Systimax
2. Panduit
3. Hubbell
4. Belden
5. Leviton/Berk-Tek
6. Hitachi
7. OCC
8. Ortronics

2.6 SHIELDED CATEGORY 6A TELECOMMUNICATIONS OUTLET

A. Specifications

1. Wall Faceplate

a. Single gang plate
b. 4 ports capacity
c. UL 1863 Compliant
d. Typical faceplate shall be stainless steel construction.
e. Clear label cover.
f. All unused positions shall have blanking plates installed.
g. All installed modules will be oriented to prevent dust and debris from settling on pins and for ease of use.
h. Back box/faceplate orientation and elevation shall match adjacent power receptacle.

2. Jack

a. Category 6a module featuring 110-termination technology
b. Keystone compliant form factor.
c. Backwards compatible to Category 3
d. ANSI/TIA 568-B.2-1 Category 6a compliant.
e. Sweep tested through 500 MHz
f. Metal connector body
g. Phosphor bronze with 50u inch gold over nickel plating front connection
h. Phosphor bronze with tin platting over nickel rear connection
i. Accepts 24 through 22 AWG solid copper conductors
j. Standard jack color is White.

3. Shutter

a. Footprint equal to jack
b. Color to match jack
c. Icon included on shutter mount

B. Acceptable Manufacturers

1. Commscope Systimax
2. Panduit
3. Hubbell
4. Belden
5. Leviton/Berk-Tek
6. Hitachi
7. OCC
8. Ortronics

PART 3 - EXECUTION

3.1 INSTALLATION

A. General

1. Follow all ANSI/TIA installation guidelines.
2. Follow all cabling solution provider installation guidelines.

3. Install all cabling and secure utilizing wire management and hook and loop (Velcro) straps. No plastic cable ties shall be used to secure any cabling.

B. Telecommunications Rooms

1. Install all Horizontal Cable System Telecommunications Room equipment as indicated in Project Documents.
2. Install cable management to support and train cables with spacing as required to meet bend radius and support requirements.

C. Outlet

1. Install all Horizontal Cabling System telecommunications outlet equipment and cabling as indicated in Project Documents.
2. Do not exceed recommended pulling tensions.
3. Route cabling from outlets to patch panel in the nearest Audio/Visual cabinet.
4. Do not exceed 90 meter horizontal length for Category 6a cables. Notify Architect immediately if any Category 6a route exceeds 90 meters.
5. Terminate data cabling per T568B wiring method.

a. Maintain wire twists to within 0.5 inches of termination.
b. Remove no more than 1.0 inches of cable jacket.

3.2 LABELING

A. Patch Panel Labeling

1. Provide each Patch Panel with a laminated black phenolic resin label with a white core and engraved lettering, a minimum of 1/4-inch high.
2. Provide permanent, computer-generated labels at each Patch Panel port.

B. Horizontal Cable Labeling

1. Provide permanent, computer generated labels on both ends of every Horizontal Cable, Patch Cord and Work Area Equipment Cable.
2. Example of horizontal wiring label: 1A-A001 1A=Equipment Room number, A=the patch panel, and 001= patch panel port.

C. Outlet Labeling

1. Provide permanent, computer generated labels at the station faceplate.

D. General

1. Provide labeling based on ANSI/TIA-606 administration concepts.
2. Provide typewritten labels, hand labeling is not acceptable.

E. Terminating Hardware

1. Provide an identifier for each Patch Panel that incorporates the Audio/Visual cabinet identifier (e.g. TR2A-RR3-PP1 is the first patch panel in the third relay rack in the first intermediate telecommunications room on the second floor).

2. Provide an identifier for each Patch Panel port that incorporates the Patch Panel identifier (e.g. TR2A-RR3-PP1-09 is the ninth port on the first patch panel in the third relay rack in the first intermediate telecommunications room on the second floor).

F. Outlet

1. Provide an identifier for each telecommunications outlet that indicates the room number and an ordinal designation (e.g. 142A is the first activation point in room 142).
2. Provide an identifier for each outlet jack designation (e.g. 142A-2 is the second jack in the first outlet in room 142).

G. Horizontal Cable

1. Provide the cable's Patch Panel Port identifier and Work Area Activation Point Jack identifier on both ends of all Horizontal cables.
2. Insure that label is intact after termination of cable.

3.3 TESTING

A. Copper cable testing

1. Test all Category 6a horizontal cabling using a digital cable analyzer and certify that the system is complete and functional.
2. A level IIe or better test unit is required and must be updated to include the requirements of ANSI/TIA-568-B.
3. The basic tests required are:

    a. Wire Map
    b. Length
    c. Attenuation
    d. NEXT (Near and crosstalk)
    e. Return Loss
    f. Propagation Delay
    g. Delay Skew
    h. PSNEXT (Power sum near-end crosstalk loss)
    i. PSELFET (Power sum equal level far-end crosstalk loss)

4. Submit all test data to the architect for review and approval.
5. Spot test cabling in the presence of the architect to verify the accuracy of the data submitted.
6. Spot tests will be reflective of the test data submitted or the entire system will be retested in the presence of the architect, and at the contractor's expense.

END OF SECTION 27 4107

SECTION 28 0500 - COMMON WORK RESULTS FOR ELECTRONIC SECURITY

PART 1 GENERAL

1.1 DESCRIPTION OF WORK

A. Demolition

1. Refer to 'TD' plans for details defining the scope and extent of the demolition of the existing Security Electronics systems.
2. Prior to the start of demolition, contactor is required to coordinate the disconnection and removal of all Security Electronics Systems hardware and components. All systems and hardware is to be turned over the Owner/User. This is inclusive but not limited to system electronics, field devices, and cabinets.
3. The Security Electronics cabling system is to be removed in its entirety. Faceplates, jacks, cable, and patch panels are to be removed and disposed of by the Contractor.

B. Provide all labor, equipment, supplies and materials for a complete installation of raceways supporting owner vendor furnished and installed new electronic security systems including Access Control System (ACS) Video Management System (VMS, CCTV) and door intercom in accordance with the Contract Documents. Refer to responsibility matrix and plans.

C. Work includes the following, as well as work not listed below but described elsewhere as it applies to Security Systems:

1. All electrical work associated with the ACS, VMS and door intercom.
2. Four post rack supporting the security electronics head-end equipment installation and termination
3. Grounding system supporting security electronics.
4. Raceways supporting addition of owner vendor furnished field devices.
5. Documentation

D. Interfacing work furnished by the owner's security integrator:

1. Access Control System
   a. Utilize proximity style readers and combination proximity/keypad style readers for validation of credential and initiation of door open sequence.
   b. System will control locking devices for architectural style doors, as well as motorized vehicle gates.
   c. Provide programming that integrates control of devices indicated in the project documents.
   d. Furnish security workstations. Workstations will be configured to work as graphic user workstation for operation and monitoring of the access control alarm system. Custom graphics illustrating controlled doors, gates, overhead doors, and alarm points will be shown overlaying floor plans for the facility. The user interfaces will be used for unlocking, locking, and hold open functions of controlled doors, as well as alarm notification and acknowledgement.
   e. See plans for required security workstations.
   f. Access control system will include control of doors. Under normal operation conditions, staff will utilize proximity reader or keypad/proximity reader for

unlocking controlled door. The security workstations will also be used to remotely lock/unlock and hold open doors as needed.

g. Access control system will interface with ADA accessible doors. When the building is secured, a valid credential read unlocks the door, then activates the paddle activating the door opener.

h. Remote door release is required. Security workstations and Intercom Master Stations will allow momentarily unlocking of identified doors allowing entry.

2. Access control system will interface with motorized gates for Secure Parking.
   a. Plan West and East gates will act as entry to Secure Parking.
   b. Dual proximity readers are required on pedestal outside entry gates. Valid card read will initiate open sequence.
   c. Upon entering, detection loops will be used to verify the vehicle has entered the secure parking and cleared the entry gate. Additionally, safety devices integral to the gate system will be used to verify personnel have cleared the gate. Upon clearing the gate, close sequence will be initiated.
   d. Vehicles lacking valid credential requesting access to the Secure Parking will communicate with Security Workstation via pedestal mounted intercom stations. The dual intercoms will be mounted on the pedestal located outside the secure parking gate.
   e. Exit gate will require wireless remote or push button internal to facility to initiate the open sequence. Detection loops and devices integral to the gate system will be used to verify the vehicle and any personnel have cleared motorized gate before initiating the close sequence.
   f. Security workstations can be used to open, close, and hold open all motorized gates.

3. Access control system will interface with intercom system allowing for remote door release functionality.
4. Under no circumstances shall access control system headend components be installed at a door. Exception is credential readers, DPS and Locks. There is no power available at the door for modules, boards, etc.
5. Alarm System
   a. Furnish and install a complete and functional alarm system. Provide coordination to ensure that the system furnished includes integration of, or interfacing to all devices and systems
   b. The alarm system will be capable of monitoring alarm points throughout the building. Monitoring devices will include but not be limited to panic alarm push buttons and door position switches.
   c. Local alarm notification is required on Security Workstations as well as initiate a call out to Dispatch.
   d. Local acknowledgement of panic alarms is required by resetting alarm with a key.
   e. Alarm notification and acknowledgement will be presented on the Security Workstation. Visual and audible notification is required.
   f. During programming the contractor will coordinate with the Owner/User to define and schedule alarm points, arm/disarm schedule, and alarm notification.
   g. Alarm system interfaces with CCTV system. When an alarm is initiated, (panic, DPS, etc.) the CCTV system shall display image of the camera nearest the alarm point on Security Workstations.
   h. When a panic alarm is initiated, an audible announcement is to be generated throughout the public address system. This verbal announcement shall provide clear instruction on the location of the initiated panic alarm.

6. Intercom System:
   a. Network based SIP/VoIP compatible intercom system.

b. The purpose of the intercom system shall be to provide clear, two-way, remote reply intercommunication between intercom master stations and remote intercom substations.
c. The extent, size, locations, and layout of the various systems shall be shown on the plans.
d. The system is configured to allow for visual and audible notification to the operator of intercom call in and active intercom channels on the intercom master station.
e. When applicable, stations will be equipped with mega-pixel camera that is ONVIF profile. Camera stream to be shared with facilities CCTV System for viewing and recording purposes.
f. Master stations to be equipped with integral video monitor.
g. Intercom system will integrate with the access control system allowing remote door release.

7. Video Surveillance System:
   a. Video Surveillance throughout the building and site via IP communications network.
   b. Video monitoring of programmed cameras on wall mount monitors at locations identified on the plans, identified by Owner/User.
   c. Video monitoring of all cameras at work stations identified by the Owner/User.
   d. Viewing capabilities on two security workstations are required.
   e. In addition to viewing and control capability on security workstation, wall mount displays will be required. Each display will be capable of displaying multiple simultaneous images with the quantity of images selectable by the user and scrolling capability between all cameras.
   f. Review capabilities for the digitally recorded images via the facility LAN on Owner/User furnished workstations.
   g. Stored image transfer capabilities to allow archival functions and stored image transfer via removable media such as CD/DVD.
   h. System is comprised of POE IP high resolution fixed color cameras. Refer to plans for quantity and location of cameras.
   i. Fixed cameras will utilize Category 6 cable to transport video signal to the recording system.
   j. Camera system interfaces with Intercom system.
   k. Camera system interfaces with the access control system. When a panic alarm pushbutton or door position switch is activated, the access control system initiates call up of the camera nearest the duress or door. Image is then displayed on the CCTV monitor.
   l. All IP cameras, recording devices, viewing stations, and controllers will reside on the dedicated Security System LAN.
   m. Provide all software and licensing as required for servers, workstations, management systems, and devices.
   n. Furnish CCTV network storage. Configure based on H.265 compression, medium motion, and maximum resolution of each camera. A minimum of 30 days of storage is required. Minimum frame rate requirements as follows –
      1) 1 frame per second increasing to 15 frames per second when motion activated.

E. Interview Record System:

1. The interview record system will be furnished and installed by the Owner/User.

F. Interpretation of Contract Documents

1. This section of the specifications describes general provisions applicable to all Division 28 Security Systems.
2. Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, appurtenances and devices incidental to or necessary for a sound, secure and complete installation.
3. Mention in these specifications or indications and/or reasonable implications whereby articles, materials, operation or methods related to execution of the work are noted, specified, drawn or described, thereby requires execution of each such item of work and provision of all labor, materials, equipment and accessories required for execution thereof.
4. No exclusions from, or limitations in the language used in the specifications shall be interpreted as meaning that the accessories necessary to complete any required system or item of equipment are to be omitted.
5. The use of words in the singular shall not be considered as limiting where other indications denote that more than one item is referred to.
6. Drawings are diagrammatic and indicate general arrangement of system and equipment, except when specifically, dimensioned or detailed. They are to show size, capacity, approximate location, direction and general relationship of one work phase to another, but not exact detail or arrangement. The division 28 contractor is responsible for final verification of field device counts.
7. Field verification of measurements take priority over dimensioned drawings.
8. Dimensions indicated anywhere, are limiting dimensions.
9. The Owner reserves the right to make any reasonable change in location of devices and equipment prior to rough installation without involving additional expense. All changes from the drawings as are necessary to make the work of the SEC conform to the building as constructed shall be included and installed without extra cost.
10. All raceways must be coordinated with owner’s security vendor’s shop drawings prior to the commencement of work.

1.2 SCOPE AND RESPONSIBILITY

A. Provide full-time on-site field representation for Division 28 Security Control Systems series scope for work for the duration of installation and prior to turn over. This requirement shall commence upon the start of work inside the secure area of the facility.

B. Include detailed scheduling information for the expansion of the existing systems installation and testing in the construction schedule. Provide detailed GHANT chart construction schedule showing all tasks referenced in the project phasing plans. Include:

1. Shop Drawing Preparation
2. Equipment Installation
3. Electrical Work
4. This schedule must be submitted fourteen (14) calendar days after receipt of contract. The duration of this schedule must also comply with the completion dates of the construction schedule contained in the contract documents.
5. Work with the owner and engineer to schedule/conduct periodic coordination meetings (either weekly or bi-weekly) between trades to make everyone aware of critical areas of construction. Distribute the meeting minutes and attendance to the Owner Representative, the Engineer, and other trades in a timely fashion.
6. Provide coordination of the Security Electronics System installation.
7. Provide coordination as required to complete the inspection described in paragraph 3.1 INSPECTION.
8. Furnish and install the four post equipment rack for use by the owner’s security vendor, refer to section 27 1116 CABINETS AND ENCLOSURES.

9. Surge protection of field devices are to be furnished by the owner's security vendor, furnish and install the telecommunications grounding system to support grounding, refer to section 27 0526 GROUNDING BONDING & UPS FOR TELECOMMUNICATIONS.
10. Owner's security vendor shall provide all required circuit protection internal to the head end equipment racks.
11. Owner's security vendor will furnish and install all devices, equipment, and appurtenances resulting in complete, functional, and fully operational systems.

C. Contractor is to coordinate the work of this Section with the Engineer and that of other Sections to ensure that the entire work of this Project will be carried out in an orderly, complete and coordinate fashion.

1.3 RELATED DOCUMENTS

A. General

1. Drawings, specifications and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification sections, apply to work of this section. The SEC and all subcontractors are responsible for locating information pertaining to required items of work specified or indicated elsewhere in the Contract Documents.

B. Related Work Specified Elsewhere

1. Division 00 - BIDDING REQUIREMENTS
2. Division 01 - GENERAL REQUIREMENTS
3. Division 08 – DOOR AND DOOR HARDWARE
4. Division 32 – SECURITY GATES

C. Reference Specifications, Materials, and/or Codes

1. Submit all items necessary to obtain all required permits to the appropriate Regulatory Agencies, obtain all required permits and pay all required fees.
2. All work shall conform to the National Electrical Code (NEC) and to applicable National Fire Protection Association (NFPA) codes.
3. All work shall conform to all Federal, State and local ordinances.
4. Where applicable, all fixtures, equipment and materials shall be as approved or listed by the following:

    a. Factory Mutual Laboratories (FM).
    b. Underwriters Laboratories, Inc. (UL).
    c. National Electrical Manufacturers Association (NEMA).

5. References to the National Electrical Code and National Fire Protection Association (NFPA) are a minimum installation requirement standard. Design drawings and specification sections shall govern in those instances where requirements are greater than those specified in the NEC and NFPA.
6. All material and equipment shall be listed, labeled or certified by Underwriters' Laboratories, Inc. where such standards have been established. Equipment and material which are not covered by UL Standard will be accepted provided equipment and material are listed, labeled, certified or otherwise determined to meet safety requirements of a nationally recognized testing laboratory. Equipment of a class, which no nationally recognized testing laboratory accepts, certifies, lists, labels, or determines to be safe will

be considered, if inspected or tested in accordance with national industrial standards such as NEMA, ICEA or ANSI. Evidence of compliance shall include certified test reports and definitive shop drawings. NOTE: It is not required that the final installed system be UL listed as a single product.

7. All work shall meet or exceed the standards and procedures of the following:

   a. National Fire Protection Association (NFPA): NFPA 70, NFPA 72, NFPA 90A
   b. National Electrical Code (NEC)
   c. American National Standards Institute (ANSI)
   d. National Electrical Manufacturers Association (NEMA)
   e. American Society of Testing Materials (ASTM)
   f. Institute of Electronic & Electrical Engineers (IEEE)
   g. Underwriters Laboratory (UL)
   h. Americans With Disabilities Act (ADA)
   i. American Corrections Association (ACA)

8. Include all items of labor and material required to comply with such standards and codes. Where quantity, sizes or other requirements indicated on the drawings or herein specified are in excess of the standard or code requirements, the specifications or drawings, respectively, shall govern.

## 1.4 QUALITY ASSURANCE

A. General

1. Furnish and install only new equipment and materials required (less than 1 year from manufacture), unused without blemish or defect.
2. Each major component of equipment shall have the manufacturer's name, address, model number and rating on a plate securely affixed in a conspicuous place. The nameplate of a distributing agent is not permitted. NEMA Code Ratings, UL label, or other data which is die-stamped into the surface of the equipment shall be stamped in a location easily visible. Performance as delineated in schedules and in the specifications shall be interpreted as minimum performance. In many cases, equipment is oversized to allow for pickup loads which cannot be delineated under the minimum performance.
3. All equipment of the same type shall be the product of one manufacturer.
4. The original factory condition of manufactured equipment shall not be modified without the written approval of the Architect /Engineer.
5. Provide only non-proprietary software.

## 1.5 SUBMITTALS

A. Refer to Section 013300 – Submittal Procedures

B. Duplicate copies of the bidding documents are not acceptable and will be rejected.

C. Shop Drawings

1. Submittals consisting of reproduced copies of the original bidding documents will be rejected. The SEC is required to develop a complete set of drawings specific to the final configuration of the system based on the manufacture and models of all components included. Shop drawings are to include all changes noted in addenda, as well as any

changes included in architects' special instructions or change orders issued prior to the submittal of the shop drawings.

2. All drawings shall be created using an industry recognized computer aided design program. Recognized programs include AutoDesk Revit, or Auto Cad. All drawings are to be made using the latest software release available.
3. Submit shop drawings for all equipment showing:

    a. Location and layout of all field equipment on floor plans.
    b. Large scale (minimum ¼"=1') floor plan and elevation view drawings of all security electronics rooms and control rooms depicting all racks, consoles, cabinets, equipment, outlets, etc.
    c. Size and spacing of all anchors, wall penetrations, joinery construction, etc., required for complete system installation.
    d. Sizes, shapes, thickness and finishes of all materials and equipment surfaces.
    e. Electrical riser diagrams identifying all signal, power and ground circuitry.
    f. Installation detail of all surface or flush mounted devices including but not limited to speakers, cameras, intercom, card readers, etc. Detail to show all required mounting and suspension hardware, as well as connection made to structure. Provide front, side, and top views.
    g. Installation detail of all suspended devices including but not limited to speaker systems, cameras, etc. Detail to show all required mounting and suspension hardware, as well as connection made to structure. Provide front, side, and top views.
    h. Functional and operational written reports of all security systems.

D. Record Documents

1. Submit four (4) black line prints, and electronic copy of the "As-built" condition of all systems including:

    a. A set of updated shop drawings showing all Contract changes.
    b. Floor plan showing conduit raceway routing including all equipment rack, cabinet and pull box locations, and conduit sizes.

## PART 2 PRODUCTS

### 2.1 GENERAL REQUIREMENTS

A. All products shall be new (less than one year from manufacture and unused and without blemish or defect).

### 2.2 SUBSTITUTIONS

A. Submit substitutions in accordance with Section 013300 – Submittals and Substitutions. No other method of substitution is acceptable.

B. Deviations from specifications

1. Any deviations from the specifications must be approved 10 calendar days prior to the bid date. This includes changes to the scope of work, equipment substitutions, and changes to the general provision.

2. Changes to the scope of work in the bid proposal are not acceptable. Any proposed change is to be submitted to the Engineer for review. Any approved changes will be listed by addendum prior to the bid opening.
3. Any proposed equipment substitution must be submitted 14 calendar days prior to the bid date. Accompany the request, the contractor must provide manufactures product specifications for the exact model to be substituted. This literature must clearly state all specifications called for in the bidding documents, as well as performance characteristics not specified but inherent to the product listed in the specifications. Any items approved for substitution will be listed by addendum prior to the bid opening. Substitutions after the award of bid will only be allowed in case of discontinued equipment, or if an item of equal or better quality is available and will not affect the contract cost of the system.
4. Changes to the general provisions are not acceptable. Any proposed change is to be submitted to the Engineer for review. Any approved changes will be only listed by addendum prior to the bid opening.
5. When a specified item is found to be discontinued or obsolete by the manufacturer, the SEC is required to substitute the manufacturer recommended equivalent for that product. If an equivalent is not available, the contractor is instructed to notify the Engineer in writing prior to bid time.
6. Where specific products are sole source specified, no substitutions will be allowed.

## 2.3 MANUFACTURED PRODUCTS

A. Materials and equipment furnished shall be of current production by manufacturers regularly engaged in the manufacture of such items, for which replacement parts are available.

B. Equipment Assemblies and Components

1. All components of an assembled unit need not be products of the same manufacturer; however, all components must be acceptable to the Architect Engineer and the Owner.
2. Manufacturers of equipment assemblies, which include components made by others, shall assume complete responsibility for the final assembled unit.
3. When more than one unit of the same class of equipment or material is required, such units shall be the products of a single manufacturer.
4. Components shall be compatible with each other and with the total assembly for the intended service.
5. Constituent parts which are similar shall be the product of a single manufacturer.

## 2.4 EQUIPMENT IDENTIFICATION

A. In addition to the requirements of the National Electrical Code, install an identification sign that clearly indicates information required for use and maintenance of items such as cabinets and other significant equipment.

B. Nameplates shall be laminated black phenolic resin with a white core and engraved lettering, a minimum of 1/4-inch high. Nameplates that are furnished by manufacturer, as a standard catalog item, or where other methods of identification are herein specified, are exceptions.

C. All inputs and outputs are to be clearly labeled. Inputs to include the source location and signal type. Outputs should indicate location signal is sent to.

D. All custom input plates labeling shall be engraved and paint filled or laser etched with a contrasting color as shown on the specification.

E. Any and all user serviceable devices shall be clearly labeled

PART 3 EXECUTION

3.1 INSPECTION

A. Before the owner's vendor installs electronic equipment, carefully inspect the installed Work of all other Trades. Verify that all such Work is complete to the point where the installation of electronic equipment may properly commence. Verify that the security/telecommunications closets are free of airborne contaminants prior to the installation of electronic equipment.

B. Verify that all equipment is installed in accordance with all pertinent codes and regulations, the original design, and the referenced standards.

C. In the event of discrepancy, immediately notify the Architect.

D. Do not proceed with installation in areas of discrepancy until such discrepancies have been fully resolved.

3.2 INSTALLATION

A. Install all equipment in strict accordance with the manufacturer's recommendations, reviewed shop drawings and ANSI/EIA/TIA standards.

B. Do not attach electrical materials to roof decking, removable or knockout panels, or temporary walls and partitions, unless indicated otherwise.

C. Secure equipment with fasteners suitable for the use, materials and loads encountered. If requested, submit evidence proving suitability.

D. National Electrical Code requirements are applicable to all work.

E. Working spaces shall be not less than specified in the National Electrical Code for all voltages specified.

F. Where the Architect determines that the SEC has installed equipment not conveniently accessible for operation and maintenance, equipment should be removed and reinstalled as directed at no additional cost to the Owner. "Conveniently accessible" is defined as being capable of being reached without the use of ladders, or without climbing or crawling under or over obstacles such as motors, pumps, belt guards, transformers, piping, and duct work.

3.3 WORK PERFORMANCE

A. Coordinate location of equipment with other trades to minimize interferences.

B. Holes through concrete and masonry in new and existing structures shall be cut with a diamond core drill or concrete saw. Pneumatic hammer, impact electric, hand or manual hammer type drills shall not be allowed, except where permitted by the Engineer and the Owner as required by limited working space.

C. Holes shall be located so as not to affect structural sections such as ribs or beams.

D. Holes shall be laid out in advance. The Architect/Engineer shall be advised prior to drilling through structural sections for determination of proper layout.

E. Any holes created in walls, floors, or ceilings by the SEC are to be fire-stopped according to all national, state, and local codes.

F. Hangers and other supports shall support only equipment and materials. Provide not less than a safety factory of 5, which shall conform to any specific requirements in the Construction Documents.

G. The SEC is responsible for repairing and or replacing any damage caused by their workforce at no additional cost to the Owner.

3.4 PROTECTION AND CLEANING OF SYSTEMS AND EQUIPMENT

A. Protect all materials and equipment from damage during storage at the Site and throughout the construction period. Equipment and materials shall be protected during shipment and storage against physical damage, dirt, moisture, cold and rain.

B. Damage from rain, dirt, sun and ground water shall be prevented by storing the equipment on elevated supports and covering them on all sides with securely fastened protective rigid or flexible waterproof coverings.

C. Piping shall be protected by storing it on elevated supports and capping the ends with suitable closure material to prevent dirt accumulation in the piping.

D. During construction, cap the top of all conduits and raceway installed vertically.

E. During installation, protect equipment against entry of foreign matter on the inside, and vacuum clean both inside and outside before testing and operating.

F. Damaged equipment, as determined by the Engineer, shall be replaced.

G. Painted surfaces shall be protected with removable heavy craft paper, sheet vinyl or equal, installed at the factory and removed prior to final inspection.

H. Repaint damaged “FINISH” paint on equipment and materials with painting equipment and finished with same quality of paint and workmanship as used by the original manufacturer so repaired areas are not obvious.

I. Conduit back boxes, floor boxes, and poke through’s shall be vacuumed clean prior to the installation of cable.

3.5 IDENTIFICATION

A. Nameplates shall be laminated black phenolic resin with a white core and engraved lettering, a minimum of 1/4-inch high. Nameplates that are furnished by manufacturer, as a standard catalog item, or where other methods of identification are herein specified, are exceptions.

B. Uppercase letters of uniform height; centered on device, cover plate, or enclosure; engraved letters filled with a contrasting color; and all characters made clearly and distinctly.

C. Use abbreviations defined in the contract documents whenever possible. Use plan designations for labeling unless indicated otherwise.

3.6 LABELING

A. All cables shall be properly identified using a high-quality thermal transfer-labeling device such as the Kroy K4100 or equal.

B. Power Outlets

1. Power outlet labels are to be mechanically generated.
2. All power outlets designated for security equipment shall be labeled “SECURITY ELECTRONICS EQUIPMENT ONLY”. On top.
3. Provide a second label on the bottom of the outlet cover plate indicating service panel number and circuit breaker number.
4. Text lettering to be 1/8” high.

3.7 CLEANING

A. Refer to Section 017400 – Cleaning

B. Daily during construction and prior to User acceptance of the building, remove from the premises and dispose of all packing material and debris caused by work performed under Division 28 Security Systems.

C. Remove all dust and debris from interiors and exteriors of electrical equipment. Clean accessible current carrying elements prior to being energized.

D. Upon completion of the work, remove excess debris, materials, equipment, apparatus, tools and the like and leave the premises clean, neat and orderly.

E. All bright metal or plated work shall be thoroughly polished. All pasted labels, dirt and stains shall be removed from the devices.

3.8 COMPLETION

A. Results Expected

1. Substantial Completion inspection is performed and granted.

END OF SECTION 28 0500

SECTION 28 30 00 - ELECTRONIC DETECTION AND ALARM

PART 1 - GENERAL

1.1 DESCRIPTION

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 28 General Requirements" in Section 28 30 10.

B. It is the intent of these Specifications for the Contractor to provide an electronic detection and alarm system complete, fully operational, fully adjusted, and ready for use.

C. All contract requirements indicated in the Division 21, 22, 23, 25, 26 and 28 documents, including drawings and specifications, shall be considered supplemental to and in addition to the contract requirements indicated in the documents of other trades and the Division 01 requirements.

1.2 REQUIRED WORK DESCRIBED IN OTHER SECTIONS

A. Section 28 30 10 - Division 28 General Requirements
B. Section 28 30 53 - Identification for Fire Detection and Alarm Systems
C. Section 28 30 70 - Division 28 Schedule of Submittal Data
D. Section 28 31 00 - Fire Detection and Alarm

1.3 PARTIAL LIST OF WORK NOT INCLUDED IN DIVISION 28

A. Equipment pads:

B. Installing access doors:

C. Painting (except as otherwise specified herein):

D. Furnishing of motors: Division 21, 22, 23, and 25.

E. Furnishing of individual motor controllers that are factory-mounted and integral parts of pieces of equipment: Division 21, 22, 23, and 25.

F. Furnishing, installing, and connecting telephone wiring, cables, and equipment unless otherwise indicated:

1.4 ALTERNATE AND UNIT PRICING

A. Alternate Pricing, Unit Pricing, and Budgetary Cost Estimates: **Bid alternates and unit prices shall not be required with the Contractor's base bid.** These requirements shall apply to only those bidders contacted after the Owner's and Architect's review of the bids. If contacted by the Owner or Architect after the bid date, the Contractors shall provide alternate prices, unit prices, and budgetary cost estimates as indicated in the Architectural Documents, as specified herein, and as indicated on the Drawings.

B. If contacted by the Owner or Architect after the bid date, the contractor(s) shall provide complete alternate pricing information for the owner's evaluation of all alternate priced items specified herein. Alternates may be additive or deductive. An additive alternate, whether shown on the drawings or specified herein, shall be defined as an alternate that is not currently

part of the base building scope of work with a price that would be required to add the alternate to the base building scope of work. A deductive alternate shall be defined as an alternate that is currently part of the base building scope of work with a price that would be required to delete the alternate from the base building scope of work. The owner reserves the right to accept or reject any alternate priced item.

1. Alternate 28-1 - Extended Warranty and Service Contract Alternate for the Fire Alarm and Detection System: Provide an additive alternate bid price to provide extended service contract for the fire alarm and detection system. Refer to Section 28 31 00 for additional information.

C. Refer to the architectural specification section for Architect's and Owner's instructions to bidders for additional alternate pricing requirements.

D. Pricing information provided for unit costs, separate line items, alternates, and value engineering items shall be all inclusive pricing that accounts for the impact and ripple effects on adjacent or related systems affected by the alternate product, material, or system. Acceptance of an alternate product, material or system shall not result in additional cost to the project beyond the price indicated for the alternate product, material or system.

1.5 ADDITIONAL REQUIREMENTS

A. It shall be the responsibility of the Electronic Detection and Alarm System Contractor to review the complete set of Division 21, 22, 23, 25, 26, and 28 drawings and specifications including but not limited to the following specification sections to determine the complete scope of the Division 28 work:

| | |
|---|---|
| Section 21 10 00 | Water Based Fire Suppression Systems |
| Section 25 00 90 | Automatic Temperature Control System |

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. General: All equipment shall be new, of the capacity and type specified herein, and as shown on the Drawings. Equipment shall be of a listed manufacturer and model number and shall be in accordance with the space limitations of the project.

B. Single Source: To maximize ease of maintenance and part replacement, equipment of a similar nature shall be provided by a single manufacturer.

C. Approved Equal: Equipment and materials selected by the Contractor within the context of "equal as approved by the Engineer", "approved equal", "equivalent as determined by the Engineer" and similar terminology shall be submitted to the Engineer for review, approval and inclusion into the Contract Documents prior to the finalization of the contract between the Owner and the Contractor, and prior to the shop drawing submittal phase of the Project. All equipment and materials submitted to the Engineer under the terms of "approved equal" during the shop drawing phase of the Project without prior review and approval shall be returned to the Contractor without review under the status of "No Action".

2.2 MATERIAL

A. All material required for a complete and proper installation shall be as specified and as selected by the Contractor subject to the approval of the Architect.

B. Material shall be new, listed and approved by UL, and bear the inspection label if subject to such approval.

2.3 CONTRACTOR APPROVALS

A. The contractor shall submit in writing and obtain written approval from the Owner, Architect, and/or Engineer for all equipment substitutions, installation deviations from that shown on the Contract Documents, and all other miscellaneous approvals required from the Owner, Architect, and/or Engineer as referenced throughout these specifications.

B. A request for a substitution or deviation from the Contract Document constitutes that the Contractor:

1. Has investigated the proposed product and determined that it is equal to or superior in all respects to that specified.

2. Will provide the same warranties for the substitution as for the product specified.

3. Will coordinate the installation of the substitution into the work and make such other changes as may be required to make the work complete in all respects.

4. Waives all claims for additional costs, under his responsibility, which may subsequently become apparent.

5. Will absorb all costs incurred by the substitution when affecting other trades including but not limited to electrical, structural, architectural, etc.

6. Will absorb any cost incurred by the Owner's Representative in review of the substituted product if the acceptance of the substituted item creates the need for system modification and/or redesign, or if the substituting Contractor exhibits negligence in his substituting procedure thus submitting inferior, misapplied or mis-sized equipment.

PART 3 - EXECUTION

3.1 CONDITIONS

A. Inspection: Prior to proceeding with the work of this Division, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation may properly commence. Verify that the work of this Division may be completed in strict accordance with all pertinent codes and regulations, the reviewed shop drawings, and the manufacturers' recommendations.

B. Discrepancies: In the event of discrepancy, immediately notify the Owner's Representative. Do not proceed in areas of discrepancy until all such discrepancies have been resolved. If there is a discrepancy between the Drawings and the Specifications, the Specifications shall typically govern. However, any discrepancy of this type shall be immediately brought to the attention of the Owner's Representative for formal interpretation prior to proceeding with the work.

C. Interpretation of Documents: Any and all contractual requirements may be indicated solely on the Drawings, solely in the Specifications, in both the Specifications and on the Drawings, in reference standards indicated in the Specifications and/or in the Owner's and Contractor's Contract. If Contract requirements are indicated in both the Specifications and the Drawings, the Contractor shall comply with both requirements unless the requirements are mutually exclusive of each other. If Contract requirements are indicated in both a reference standard

and the Specifications, the more stringent requirement shall apply. Any and all contractual requirements shall be interpreted within the overall context of the complete scope of work. All materials, equipment, systems and installation methods shall be suitable for the intended service, coordinated with other trades and be complete, fully operational, adjusted, tested and ready for use by the Owner.

3.2 INSTALLATION OF EQUIPMENT

A. Locations: Install all equipment in the locations shown on the approved shop drawings, except where specifically approved otherwise on the job by the Architect and/or Owner's Representative.

B. Interferences: Avoid interference with structure and with work of other trades, while preserving adequate headroom and clearing all doors and passageways to the approval of the Architect and/or Owner's Representative. Where busway is installed on a job, Electrical Contractor shall coordinate location early with other trades. Horizontal runs of bus shall be run above all piping and ductwork so as to maximize clear headroom below busway and maintain manufacturers recommended access clearances to all sides of busway. All section joints shall be accessible.

C. Inspection: Check each piece of equipment in the system for defects. Verify that all parts are properly furnished and installed, function properly, and that all adjustments have been made.

3.3 CONNECTIONS TO EQUIPMENT

A. Mechanical Equipment: The Contractor shall make final electrical connections to all items of mechanical equipment, including all motors and unit heaters for a complete and operational system.

3.4 CLOSING-IN OF UNINSPECTED WORK

A. General: Do not allow or cause any of the work of this Division to be covered up or enclosed until it has been inspected, tested, and approved by the Architect and/or Owner's Representative and by the authorities having jurisdiction.

B. Uncovering: Should any of the work of this Division be covered up or enclosed before it has been completely inspected, tested and approved, the Contractor shall provide all services, labor, materials and equipment necessary to uncover such work without additional cost to the Owner. After the work has been completely inspected, tested, and approved, the Contractor shall provide all services, labor, materials and equipment to make all repairs necessary to restore the work to its original and proper condition at no additional cost to the Owner.

3.5 COOPERATION WITH OTHER TRADES

A. Coordinate and cooperate with other trades in order that all systems in the scope of the Contract may be installed in the best arrangement. Coordinate and cooperate as required with all other trades which share space in common areas and maximize the access to each system and their respective components.

3.6 CLEANING

A. It is the intent of these Specifications that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment. Level of cleanliness shall be defined as "broom" clean unless noted otherwise. All exterior surfaces of Division 28 equipment shall be wiped down and cleaned of all dust and dirt. All interior surfaces

of electrical equipment including but not limited to switchboards, motor controllers, and panelboards shall be wiped down and vacuum cleaned so as to be delivered to the Owner in factory new condition. Surfaces to be painted shall be cleaned and prepared in accordance with architectural division of the contract and as noted in other sections herein.

3.7 COMPLETENESS

A. It is the intent of these Specifications to provide complete systems. Completeness shall mean that all materials, equipment, and systems as installed and operating on this project have been installed properly with the best practices of the trade; are suitable for the intended purpose, location, and environment; properly fit within the physical space limitations for the project; are in conformance with applicable codes and reference standards; have been started-up, tested, adjusted, and commissioned for the intended use; have maintained applicable UL Listings; are in compliance with manufacturer's recommendations and warranty requirements; ready for the Owner’s use, and in the opinion of the Architect, performing as designed.

3.8 ADJUSTMENT OF CONTROLS

A. The Contractor shall provide the personnel and equipment to completely adjust the controls to the satisfaction of the Architect. At the completion of the project, the Architect will arrange a meeting at the job site to allow the Contractor to demonstrate the proper operation of the electrical controls.

3.9 NOISE

A. It is the intent of these Specifications to provide a system free from objectionable audible noise and vibration. Any equipment that is generating objectionable noise or vibration, in the opinion of the Architect, shall be corrected and dampened as required to eliminate the objectionable level.

END OF SECTION 28 30 00

SECTION 28 30 10 - DIVISION 28 GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 DESCRIPTION

A. General

1. The work under this Section shall conform to the requirements of the "General Conditions" and "Supplementary Conditions" of these Specifications.

B. Work Included

1. The work included in this Section consists of the general requirements for the work more specifically described in Division 21, 22, 23, 25, 26 and 28 where referenced.

C. Definitions

1. The Contractor shall provide all supervision, labor, material, equipment, machinery, and any and all other items necessary to complete the mechanical and electrical systems. All items of equipment are specified in the singular; however, the Contractor shall provide the number of items of equipment as indicated on the Drawings, and as required for complete systems.

D. Intent

1. It is the intention of these Specifications and Drawings to call for finished work that has been completed, tested, adjusted, commissioned, fully operational, approved by the local code authorities, and ready for the Owner’s use. Wherever the word "provide" is used, it shall mean "furnish and install complete and ready to use."

2. It is the intention of these specifications and drawings that wherever the word “weather” is used throughout the various Division 21, 22, 23, 25, 26 and 28 specification sections it refers to “locations exposed to weather”. “Locations exposed to weather” shall mean areas exposed to outside environmental temperature, humidity, or moisture conditions. These locations include but are not limited to areas on the exterior of the building, on the roof, within architectural exhaust and intake plenums, within loading docks, within above grade parking garages, within below grade parking structures, and other areas as indicated on the drawings or in the specifications.

3. All labor, materials, apparatus, and appliances essential to the complete functioning of the systems described and indicated herein, or which may be reasonably implied as essential whether mentioned in the Drawings and Specifications or not, shall be furnished and installed by the Contractor.

4. In addition to the requirements indicated throughout the Drawings and Specifications, as completely installed and operating on this project; equipment and systems provided for this project shall meet the following minimum requirements without additional cost to the Project:

a. Be suitable for the intended service, location, and environment

b. Properly fit within the physical space limitations for the project

c. Meet the minimum requirements of applicable codes and reference standards as determined and approved by the local code authorities

d. Maintain applicable UL listings

e. Be coordinated between Division 21, 22, 23, 25, 26 and 28 in a timely manner

f. Be coordinated with the work of other trades in a timely manner

g. Not be purchased or installed prior to the resolution of any conflicts or coordination that could result in additional cost to the Owner

h. Be in accordance with the manufacturer's recommendations and warranty requirements

i. Be ready for use by the Owner.

5. In cases of doubt as to the Work intended, or in the event of need for explanation thereof, the Contractor shall request supplementary instructions from the Architect.

E. Codes, Rules, Regulations, Permits and Fees

1. Refer to the architectural specification Divisions for additional requirements. The architectural specification Divisions shall govern these requirements. The following information shall be in addition to the architectural specification Divisions.

2. The Contractor shall give all necessary notices, obtain all permits and pay all government sales taxes, fees, and other costs, including utility connections or extensions, in connection with the Contractor’s work; file all necessary plans, prepare all documents and obtain all necessary approvals of all governmental departments having jurisdiction; obtain all required certificates of inspection of the Contractor’s work and deliver same to the Architect before request for acceptance and final payment for work.

3. The Contractor shall comply with the most recently revised versions of all applicable laws, rules, regulations, and ordinances of Federal, State, and Local Authorities. Modifications required by the above said Authorities shall be made without additional charge to the Owner.

4. The Contractor shall comply with applicable building codes as adopted and modified by the Local Code Authorities. Modifications required by the above said building codes shall be made without additional charge to the Owner.

5. All equipment shall comply with all applicable requirements of laws, codes, ordinances, legislation, etc., of all Federal, State and Local Authorities, whether indicated on the Contract Documents or not.

6. Where Contract Drawings and Specifications requirements are in excess of Code requirements and are permitted under the Code, the Contract Drawings and Specifications shall govern.

7. All rules and regulations of the State Fire Insurance Regulatory Body, Underwriters Laboratories, Americans with Disabilities Act (ADA), and National Electrical Code (NEC), shall be complied with whether indicated in the Contract Drawings and Specifications or not.

8. All materials, equipment, and systems provided on this project shall be in compliance with the applicable energy codes and the energy efficiency ASHRAE Standard 90.1 as adopted by the applicable building codes. All panelboard enclosures and bus plugs shall have sufficient space to permit check metering with metering CT's as required by ASHRAE 90.1.

9. Project shall meet or exceed the requirements of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) Indoor Air Quality Guideline for Occupied Buildings Under Construction. In addition, all stored on-site or installed materials shall be protected at all times from moisture damage. Provide photographs to the owner, architect, or engineer at regular intervals throughout the complete construction period of the project documenting and illustrating that all stored on-site or installed materials are protected at all times from moisture damage.

10. Where alterations to or deviations from the Contract Drawings and Specifications are required by the above authorities report the requirements to the Architect and secure Architect's approval before starting the alterations.

11. Third Party Certification: All packaged equipment shall be independently Third Party labeled as a system for its intended use by a Nationally Recognized Testing Laboratory (NRTL), such as UL or ETL, in accordance with OSHA Federal Regulations 29 CFR 1910.303 & 1910.399, as well as NFPA 70, National Electric Code (NEC), Article 90-7.

1.2 QUALITY ASSURANCE

A. Surveys and Measurements

1. The Contractor shall base all measurements, both horizontal and vertical, from established benchmarks. All work shall agree with these established lines and levels. Verify all measurements at site and check the correctness of same as related to the work.

2. Should the Contractor discover any discrepancy between actual measurements and those indicated, which prevents following good practice or the intent of the Drawings and Specifications, the Contractor shall notify the Architect, and shall not proceed with the work until instruction has been received from the Architect.

B. Drawings

1. Drawings are diagrammatic and indicate the general arrangement of systems and work included in the Contract. Drawings are not to be scaled. The Architectural Drawings and Details shall be examined for exact location of fixtures and equipment. Where they are not definitely located, this information shall be obtained from the Owner's Representative.

2. The Contractor shall follow Drawings in laying out work and check Drawings of other trades to verify spaces in which work will be installed. Maintain maximum headroom and space conditions at all points. Where headroom or space conditions appear inadequate, the Owner's Representative shall be notified before proceeding with installation.

3. If directed by the Owner's Representative, the Contractor shall, without extra charge, make reasonable modifications in the layout as needed to prevent conflict with work of other trades or for proper execution of the work, if such directions are made prior to the performance of the affected work.

C. Cooperation with Other Trades

1. The Subcontractors shall give full cooperation to other trades and shall furnish in writing to the General Contractor any information necessary to permit the work of all trades to be installed satisfactorily and with the least possible interference or delay.

2. Where Division 21, 22, 23, 25, 26 and 28 work will be installed in close proximity to, or will interfere with work of other trades, coordination among trades in working out space conditions to make a satisfactory adjustment shall be accomplished. The Contractor shall prepare composite shop drawings and sections at a suitable scale not less than 1/4" = 1'-0", clearly showing how work of all affected trades is to be installed. If work is installed before coordinating with other trades, or so as to cause any interference with work of other trades, then the necessary changes in work shall be made to correct the condition without extra charge.

3. The Contractor shall obtain and transmit between affected trades, all necessary templates, patterns, setting plans, and shop details for the proper installation of work and for the purpose of coordinating adjacent work.

D. Protection

1. The Contractor shall protect all work and material from damage by the Contractors' work or workmen, Subcontractors' work or workmen, and shall be liable for all damage thus caused.

2. The Contractor shall be responsible for work and equipment until work is final inspected, tested, and accepted; the Contractor shall protect work against theft, injury or damage; and shall carefully store material and equipment received on site which are not immediately installed. The Contractor shall close open ends of work with temporary covers or plugs during storage and construction to prevent entry of obstructing material.

3. All stored on-site or installed absorptive materials are protected at all times from moisture damage.

E. Material and Workmanship

1. Work shall be executed in strict accordance with the best practice of the trades in a thorough, substantial, workmanlike manner by competent workmen. The Contractor shall furnish the services of a full-time, experienced superintendent, who shall be constantly in charge of the installation of the work, together with all skilled workmen, fitters, metal workers, welders, helpers, and laborers required to unload, transfer, erect, connect, adjust, start, operate, and test each system.

F. Manufacturer’s Recommendations

1. In addition to the requirements indicated in the drawings and specifications; manufactured products, articles, materials, and equipment shall be applied, installed, connected, erected, used, cleaned, conditioned, inspected, started-up, tested, operated, and commissioned in accordance with the manufacturer's current printed recommendations. Keep copies of such printed recommendations at job site and make them available as required. When conditions on this project are not covered by the manufacturer's printed recommendations, at the discretion of and as requested by the owner's representative, the manufacturer's authorized representative shall review the conditions and provide written supplemental recommendations to address the special situation. If the manufacturer’s recommendations are in conflict with the requirements of the drawings and specifications, the Contractor shall advise the Owner’s representative

of the conflict prior to purchase and installation.

G. Space Limitations

1. MEP systems and equipment shall be provided and installed to accommodate the structural systems, architectural systems, MEP systems, other trades, and the physical space limitations of the project.

2. Equipment shall be provided which will fit into the physical spaces provided and indicated, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearance in accordance with the Code requirements, the requirements of the local inspection department, and the equipment manufacturers' recommendations.

3. In the preparation of Drawings, a reasonable effort to accommodate equipment manufacturers' space requirements has been made. However, since space requirements, equipment arrangement, access and service clearances, conduit, piping, ductwork and other connections, manufacturers' recommendations, and other variables vary according to each manufacturer; due to the multiple manufacturers listed for each product and the possible number of different equipment combinations; the responsibility for access, proper fit, coordination, and accommodating the space limitations of the project rests with the Contractor. Refer to shop drawing submittal requirements herein for additional requirements.

4. MEP systems and equipment shall be installed to maximize headroom clearance below the bottom of material. In areas with ceilings or future ceilings provided by tenants; MEP systems and equipment shall be installed to maximize vertical clearances between the bottom of material and top of specified ceiling construction. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, lighting, and other MEP systems shall be prepared to indicate compliance with the Space Limitations and Maximum Headroom Clearance requirements. Conflicts with vertical and horizontal clearances, parking spaces, drive aisles, ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data, clouded, and clearly marked so as to bring the conflict to the attention of the Architect in a timely manner prior to installation. For the purpose of definition, the term MEP refers to all systems, material, equipment, services, means and methods described the Division 21, 22, 23, 25, 26 and 28 documents including the Division 21, 22, 23, 25, 26 and 28 specifications and drawings. Refer to the specifications for additional requirements.

5. Refer to the drawings for additional requirements.

6. Physical dimensions and arrangements of equipment to be installed shall be subject to the Architects' review.

H. Coordination with Installation Conditions

I. The requirements of the Division 21, 22, 23, 25, 26 and 28 systems shall be coordinated with the installation conditions. The Contractor shall visit the site, survey the installation conditions, and verify exact measurements. Means and methods of building access, rigging, and modifications to installation conditions to accommodate the installation of new Division 21, 22, 23, 25, 26 and 28 work shall be the responsibility of the Contractor. Disassembly, reassembly, and certification testing of equipment as required to accommodate the installation conditions shall be included in the base bid. Prior to the ordering, purchasing, and fabricating of Division 21, 22, 23, 25, 26 and 28 systems, coordination with installation conditions shall be completed

by the Contractor.

J. Coordination Between Divisions 21, 22, 23, 25, 26 and 28:

K. The Division 28 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 25 and 26 Contractors to provide coordination between the Division 21, 22, 23, 25 and 26 trades in a timely manner. For all equipment requiring electrical service provided under Division 28, it shall be the responsibility of the Division 28 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

L. Commissioning:

1. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience, and expertise at the general contractor and sub-contractor levels to properly install, check, verify, supervise, start-up, trouble-shoot, de-bug, pre-test, commission, and make the necessary corrections to all MEP systems in accordance with the Contract Documents. The Contractor shall be responsible for the operational demonstration of the completed MEP systems to the Owner’s Representative, Architect and Engineer. The Contractor shall complete the MEP systems in accordance with the contract documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems prior to requesting that the Owner’s Representative, Architect, and Engineer attend the operational demonstrations of the MEP systems. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner. Refer to the various sections of the specifications for additional commissioning requirements.

M. Additional Information

1. Information provided by the Engineer to the Contractors during construction must be reviewed and approved by the Architect and Owner's Representative. The Contractor shall not proceed without prior review and authorization from the Architect and Owner's Representative. The Contractor shall not proceed with work that will increase the cost of the project or impact the project schedule without prior review and authorization from the Architect and Owner's Representative. If information provided by the Engineer to the Contractor will increase the cost of the project or impact the project schedule, the Contractor shall advise the Architect and Owner's Representative in a reasonably timely manner so as to minimize the cost and impact to the project. Refer to the General Conditions of the Contract and the Architectural documents for additional information.

1.3 SHOP DRAWINGS, PRODUCT DATA AND SAMPLES

A. After the Contract is awarded but prior to proceeding with the Work requiring final reviewed shop drawings, product data and samples, the Contractor shall provide the Architect and Engineer with a submittal log schedule indicating when each submittal is to be sent to the Architect and Engineer for review in order for the Architect and Engineer to schedule personnel and plan accordingly. Unless otherwise indicated, a minimum period of two (2) weeks, exclusive of transmittal time, will be required in the Engineer’s office each time a shop drawing, product data and/or samples are submitted or resubmitted for review. This time period will commence once the shop drawing, product data and/or samples are received in the Engineer’s

office contingent upon the shop drawing, product data and/or samples being received in the Engineer's office before 3:00 PM. If the shop drawing, product data and/or samples are received in the Engineer's office after 3:00 PM, the time period will not start until the next business day. This time period shall be considered by the Contractor when scheduling the Work. The submittal log schedule should be updated and reissued by the Contractor on a regular basis to avoid submittal review delays. If a submittal arrives before or after the scheduled time indicated within the Contractor's submittal log schedule, or if a number of submittals arrive at the same time without scheduled notification, the Architect and Engineer may not be able to review all of the submittals within the minimum two (2) week time period. The Contractor shall obtain complete shop drawings, product data and samples from the manufacturers, suppliers, vendors, subcontractors, and sub-subcontractors for all materials, systems and equipment specified herein; and submit data and details of such materials, systems and equipment for review by the Architect and Engineer. Prior to submission of the shop drawings, product data and samples to the Architect and Engineer, the Contractor shall review and certify that the shop drawings, product data and samples are in compliance with the Contract Documents. The Contractor shall check all materials and equipment after their arrival on the job site and verify their compliance with the Contract Documents. The Contractor shall provide adequate personnel with the appropriate level of specialized training, experience and expertise at the general contractor and sub-contractor levels to properly check, verify, and correct if necessary shop drawing submittal data to ensure compliance with the Contract Documents prior to delivery to the Architect and Engineer. The Division 28 shop drawing submittal requirements, including but not limited to the Division 28 certification requirements, shall be in addition to the requirements indicated in the Architectural Divisions of the Contract Documents.

B. The Architect's approval and Engineer's review of shop drawings, product data, and samples shall not relieve the Contractor of the responsibility for dimensions or errors that may be contained therein, or for deviations from requirements in the Contract Documents. It shall be clearly understood that the Engineer's noting some errors but overlooking others does not grant the Contractor permission to proceed in error. Regardless of any information contained in the shop drawings, product data and samples, the Contract Documents shall govern the Work and are neither waived nor superseded in any way by the review of shop drawings, product data and samples.

C. Before submitting shop drawings, product data and samples the Contractor shall certify that data and details set forth on each shop drawing, product data, or samples for each item of equipment and materials complies with the Contract Documents for this Project. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the Contractor, or by a person duly authorized to sign binding agreements for the Contractor. Unless certified, shop drawings, product data and samples will not be reviewed, and will be returned unchecked to the Contractor. Certifications shall be in the form of rubber stamp impressions or a digitally certified stamp which state:

> I hereby certify that this shop drawing, product data, and/or sample has been checked prior to submittal and that it has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents and physical space limitations for the Project.
>
> (Name of the Fire Detection and Alarm Subcontractor)
>
> Signed:_____
>
> Position:_____

Date:_____

D. The Contractor shall submit shop drawings, product data and/or samples, equipment and other items listed in Section 28 30 70 and for any other equipment for which such submittals may be required in various Sections of the Specifications. Prior to submitting shop drawings, product data and samples for all items of equipment listed in Section 28 30 70 and shown with an asterisk (*), each submittal package including subsequent submittal packages shall be certified by the manufacturer as to performance and compliance with Contract Documents. Such certification shall be made by the Owner, a Partner, a Corporate Officer of the manufacturer, or be a person duly authorized to sign for the Owner, Partner, or corporation. Unless certified, shop drawings, product data and/or samples will not be reviewed and will be returned unchecked to the Contractor. The certification shall be as follows:

> I hereby certify that the equipment shown on this shop drawing, product data and/or sample, has been coordinated with existing conditions and complies in all respects with the requirements of the Contract Documents for this Project. I further certify that all data shown herein as to performance dimensions, construction, materials and other pertinent items is true and correct.
>
> (Name of the Manufacturer)
>
> Signed:_____
>
> Position:_____
>
> Date:_____

E. Inadequate or incomplete shop drawings, product data and/or samples will not be reviewed by the Architect or the Engineer and shall be returned to the Contractor for resubmittal.

F. The Contractor shall observe the following rules when submitting shop drawings, product data and samples:

1. Identification: Each shop drawing shall indicate in the lower right hand corner, and each product data brochure shall indicate on the front cover the following: title of the sheet or brochure; name and location of the Project, names of the Architect and Engineer, Contractor, Subcontractor, manufacturer, supplier, vendor; the date of submittal; and the date of each correction and revision. So far as is practical, each shop drawing, product data and/or sample shall bear a cross-reference note to the page number or numbers of sheets of the Drawings and/or Specifications showing the Work. Unless the above information is included, the submittal will be returned for resubmittal.

2. Quality Assurance: Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification paragraph. A hand written marked-up specification section will be acceptable. Along with this specification

copy mark-up, provide an abbreviated version of the Statement of Compliance that addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

3. Shop drawings shall be done in an easily legible scale and shall contain sufficient plans, elevations, sections and schematics to clearly describe the apparatus. Drawings shall be prepared by an engineer-draftsman skilled in this type of work. Unless otherwise indicated, all sheet metal, piping and similar shop drawings shall be drawn to at least 1/4"=1'-0" scale. Typical floor sheet metal ductwork shop drawings may be drawn at 1/8"=1'-0" scale. The Contractor shall submit shop drawings as described below. Shop drawings which do not comply with these requirements will be returned for resubmittal. Refer to "Cooperation with Other Trades", this section, for additional composite shop drawing requirements. Unless otherwise indicated in the Contract Documents, the submittal shall consist of three prints of the shop drawings. The Architect and Engineer will each retain one print and the remaining copies will be returned to the Contractor.

4. Product data to be submitted shall be published by the manufacturers and shall contain complete and detailed engineering and dimensional information. The Contractor shall submit product data as described below. Product data which does not comply with these requirements will be returned for resubmittal. Unless otherwise indicated in the Contract Documents, the Contractor shall submit eight (8) copies of each product data brochure. The Architect and Engineer will each retain one copy of the submittal and the remaining copies will be returned to the Contractor. Product data submitted shall contain only information relevant to the particular equipment or material to be furnished. The Contractor shall not submit catalogs which describe several different items in addition to those items to be used, unless all irrelevant information is marked out, or unless relevant information is clearly marked. The submittal shall consist of product data from each manufacturer. Product data from each manufacturer shall be identified and submitted separately.

5. Coordination between Division 21, 22, 23, 25, 26 and 28: Each Division 28 shop drawing and product data submitted for review, which has Division 28 equipment requiring electrical service, shall contain a statement indicating that the Division 28 Contractor has provided the Division 26 Contractor with the necessary electrical characteristics of the actual Division 28 equipment to be provided.

6. Shop Drawing and Product Data - Submittal Review Status:

   a. Shop drawing and product data submittals will be returned marked with one of the following status:

      1) “REVIEWED NO EXCEPTIONS”
      2) "REVIEWED EXCEPTIONS NOTED"
      3) “INFORMATION ONLY”
      4) “REVISE AND RESUBMIT"
      5) “REJECTED”

   b. If stamped “REVISE AND RESUBMIT" or “REJECTED”, the shop drawing or product data shall be revised in accordance with the Contract Documents. If marked "REVIEWED, EXCEPTIONS NOTED" documentation that addresses only the noted exceptions in the form of a letter, supplemental information, or complete resubmittal shall be forwarded to the Owner, Architect and Engineer for record purposes only. If marked "REVIEWED NO EXCEPTIONS" or “INFORMATION

ONLY" no additional submittal will be required.

c. If the copy stamped "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" is altered for any reason after it has been stamped, the "REVIEWED NO EXCEPTIONS" or "INFORMATION ONLY" shall automatically be voided and the submittal status revised to "REVIEWED EXCEPTIONS NOTED", "REVISE AND RESUBMIT", or "REJECTED" depending on the alterations.

d. All work shall be done in accordance with shop drawings stamped "REVIEWED NO EXCEPTIONS", "REVIEWED EXCEPTIONS NOTED", or "INFORMATION ONLY" insofar as these are in agreement with the Contract Documents. Wherever differences occur between the shop drawings and the Contract Documents, the Contract Documents shall govern the Work.

G. Record Shop Drawings

1. After receiving the Engineer's and Architect's final approval for each shop drawing, the Contractor shall provide the Engineer with AutoCAD files of each final approved shop drawing. All of the previous review comments by the Engineer and Architect associated with each shop drawing shall be incorporated into the AutoCAD files of each shop drawing prior to sending the AutoCAD files to the Engineer. The AutoCAD files shall be in AutoCAD 2015 version software or later version as approved by the Engineer. The Contractor shall be responsible for contacting the Engineer to determine the approved AutoCAD software to use prior to preparing the AutoCAD files.

H. Maximum Ceiling Clearance and Headroom Requirements

1. All material for systems and equipment shall be installed to accommodate the structural systems, architectural ceiling construction, base building and future tenant lighting systems, and other physical space limitations of the project. All systems and equipment shall be installed to maximize clearances between the bottom of material and top of ceiling construction. All systems and equipment shall be installed to maximize headroom clearance below the bottom of material. Shop drawing submittal drawings for equipment, piping, ductwork, conduit, and other systems that will be located where ceiling plenum clearances and headroom clearances are limited; shall be prepared to address these issues. Conflicts with ceilings, room sizes, lighting, structure, and other physical space limitations of the projects shall be indicated on the shop drawing submittal data and clearly marked so as to bring the situation to the attention of the Engineer.

I. Equipment and Layout Deviations

1. Layout, connections, supports, access & maintenance clearances, related architectural enclosures, and structural supports have been based on selected products and manufacturers. In some cases, the manufacturer's model number has been provided. Other products by specified manufacturers may be used if they comply with requirements of the contract documents including but not limited to capacity ratings. However, where the Contractor proposes to use an item of equipment that has alternative but equivalent minimum capacity ratings or that requires any redesign of the structure, partitions, foundations, piping, wiring or any other part of the mechanical, electrical, structural, civil, utility, or architectural systems; all such redesign, and all new drawings and detailing required therefore, shall be prepared by the Contractor at the Contractor's own expense as approved by the Owner's Representative. Where such approved deviation requires a different quantity, capacity, and arrangement of ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, and starters from that specified or indicated on the

drawings, the Contractor shall furnish and install any such ductwork, piping, wiring, conduit, other electrical service components, equipment, utilities, structure, architectural features, insulation, controllers, motors, starters and any other additional equipment required by the system, at no additional cost to the Owner.

1.4 RESPONSE TO SHOP DRAWING, SUBMITTAL, AND FIELD REPORT REVIEW COMMENTS

A. Contractor shall respond in written form to each shop drawing, submittal, or field report review comment from the Engineer within 10 business days from the date indicated on the Engineer's review form. Contractor's written response to each review comment should indicate the date and specifically what corrective action was or will be taken and include the name of the person and associated company responding to each review comment. Refer to the Architectural documents for additional requirements.

1.5 WARRANTY, SERVICE, AND REPLACEMENT PARTS

A. All equipment supplied for this Contract shall be free from defects in material, workmanship, and title, and shall be of the kind and quality described herein. If it appears within one (1) year from the date of the building final acceptance by the Owner that the equipment does not meet the warranties specified above or the equipment is recalled by the equipment manufacturer, the Contractor shall correct any defect or replace any recalled equipment, including nonconformance with these Specifications, at no additional cost to the Owner.

B. The manufacturers for each of the various selected Division 28 pieces of equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner's acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

1.6 CONFERENCE PRIOR TO START OF WORK

A. Immediately upon the award of this Contract, but prior to commencing any Work, the Contractor together with designated major Subcontractors shall confer with the Owner's Representative, the Architect and Engineer concerning the Work under this Contract.

B. Contractors and Subcontractors attending this meeting shall include but not be limited to the following: General; Mechanical; Plumbing; Fire Protection; Automatic Temperature Controls and Building Management; Sheet Metal; Insulation; Electrical; Fire Alarm; and Testing, Balancing and Adjusting. Contractor and Subcontractor representatives shall be familiar with the Drawings, Specifications and shop drawing submittal requirements and be prepared to intelligently discuss the project requirements.

C. The conference will be at a mutually agreed place and acceptable time.

1.7 ELECTRONIC FILE TRANSFER AGREEMENT

A. Release of the Dewberry CAD, BIM, and other electronic files will require the User of the files to sign the Dewberry Electronic File Transfer Agreement.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Materials and equipment shall be new and in good condition. The commercially standard items of equipment and the specific names mentioned herein are intended to identify standards of quality and performance necessary for the proper functioning of the Work.

B. Since manufacturing methods vary, reasonable minor variations are expected; however, performance and material requirements specified are the minimum standards acceptable. The Engineer retains the right to judge equality of equipment that deviates from the Contract Drawings and Specifications.

2.2 PRODUCTS, COMPONENTS AND MATERIALS IN PLENUMS

A. All products, components, materials and composite materials within supply, return, exhaust, relief and ventilation air plenums shall be noncombustible or shall be listed/classified and labeled as having a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723 as required by applicable mechanical code.

2.3 EQUIPMENT GUARDS

A. Provide equipment guards for the safe operation of equipment as recommended by the equipment manufacturer and as required by applicable codes.

2.4 EQUIPMENT MOUNTING/SUPPORT DEVICES

A. Unless otherwise required to accommodate the particular requirements and unique conditions of this project, all floor, wall or ceiling mounting/support devices shall be factory fabricated devices as provided by the equipment manufacturer for the intended application. In addition to the factory-fabricated devices, the Contractor shall provide additional support members, framework, mounting hardware, brackets, steel angles, anchors, restraints, support stands, attachment devices, and other miscellaneous mounting and support devices as required for adequate equipment support.

B. The Engineer retains the right to judge the quality of the design, application, and installation of all field and shop fabricated mounting/support devices using steel angle, steel channels, Uni-Strut, brackets, all-thread rods, etc. Any mounting or support devices that will rust, corrode, or in the opinion of the Engineer become unsightly or a maintenance problem will not be acceptable.

C. Support devices not factory finished shall be hot dipped galvanized or properly primed and painted with a minimum of two coats of rust inhibiting paint. Refer to the Architectural Painting Systems for additional requirements.

D. All-thread rods shall be galvanized or cadmium plated.

E. All plywood used as mounting boards in association with Division 28, unless specified otherwise by the Architect, shall be APA C-D plugged interior 5/8" to 3/4". All plywood used shall be painted (front, back and edges) with a minimum of one prime coat and one finish coat prior to equipment installation. Refer to the Architectural Specifications for additional requirements. Color shall be determined by the Architect.

2.5 EQUIPMENT PADS AND ANCHOR BOLTS

A. Concrete pads for various pieces of equipment will be provided under another Division. Pads shall be provided in main switchboard rooms, all areas with floors below grade, penthouse

equipment rooms, electrical closets and in any other location as shown on the drawings. This shall include floor-mounted equipment, equipment mounted on legs and pipe support stands. Unless otherwise noted, equipment pads shall generally conform to the shape of the piece of equipment it serves with a minimum 3" margin around the equipment and supports. Pads shall be a minimum of 4" high and made of a minimum 28 day, 2500 psi concrete reinforced with 6" x 6" 6/6 gauge welded wire mesh. Top and sides of pad shall have smooth finishes, equal to those of the floors, with all external corners either bull-nosed to a 3/4" radius or provided with a 3/4" chamfer. Shop drawings stamped "FINAL REVIEW" or "REVIEWED EXCEPTIONS NOTED" shall be used for dimensional guidance in sizing pads.

B. Unless otherwise noted furnish and install galvanized anchor bolts for all equipment placed on concrete equipment pads or on concrete slabs. Minimum dimension between anchor bolts and edge of slab shall not be less than 3". Bolts shall be of the size, type and number recommended by the manufacturer of the equipment and shall be located by means of suitable templates provided by the equipment manufacturer. When equipment is placed on vibration isolators, the equipment shall be secured to the isolator and the isolator secured to the floor, pad, or support as recommended by the vibration isolation manufacturer.

## 2.6 ACCESS DOORS

A. Wherever access is required through walls, ceilings or fire-rated enclosures to valves, fire dampers, heating coils, damper actuators, drain line cleanouts, or other concealed equipment installed under this Division, unless otherwise noted in the architectural documents, the Contractor shall furnish a hinged access door and frame as follows:

1. Drywall construction--Milcor Style DW.

2. Finished acoustical tile ceiling--Milcor Style AT in applications where there is not adequate clearance space to properly push an acoustical ceiling tile up for removal and replacement.

3. Finished plaster ceiling--Milcor Style AP.

4. Finished plaster walls or ceramic tile--Milcor Style M.

5. Plaster or masonry walls and ceilings outside offices and in other finished areas exposed to view--Milcor Style K or M.

6. Access doors shall be "B" label fire construction where required.

7. Access doors shall be installed under another Division.

8. No access door shall be installed until location and type have been approved by the Architect.

## 2.7 ESCUTCHEONS

A. Furnish and install heavy chrome plated or nickel plated steel plates of approved pattern on all conduit passing through walls and ceilings in finished areas. Escutcheons shall be B & C No. 10 or approved equal with concealed hinges. Pattern shall be approved by the Architect.

## 2.8 SEALANTS AND ADHESIVES

A. All sealants and adhesives shall comply with the current Volatile Organic Compound (VOC)

content limits of the latest edition of the South Coast Air Quality Management District (SCAQM) Rule 1168 entitled “Adhesive and Sealant Applications”. Piping sealants shall have no more than 250 grams/liter VOC content for all sealants and adhesives to be used within the building interior locations..

B. The Contractor shall submit Material Safety Data Sheets (MSDS) clearly highlighting the Volatile Organic Compound (VOC) content for each adhesive and sealant used within the building interior locations.

PART 3 - EXECUTION

3.1 SCAFFOLDING, RIGGING, HOISTING

A. The Contractor shall furnish all scaffolding, rigging, hoisting, and services necessary for erection and delivery of any equipment and apparatus furnished to the job site. Remove same from job site when no longer required.

3.2 SLEEVES, CUTTING, AND PATCHING

A. The Contractor shall be responsible for the timely placing of openings and sleeves for all conduit passing through walls, partitions, beams, floors, and roof while the same are under construction. A conduit sleeve shall be round and one size larger than the size of conduit it serves, except where "Thunderline/Link Seal" casing seals are used in sleeves through walls below grade. Sleeves set in concrete floor construction shall be 18 gauge galvanized steel. Sleeves shall extend two inches above the finished floor. All conduit passing through concrete or masonry walls shall have standard weight galvanized steel sleeves or plastic “hole-outs”. Block out(s) and core drill(s) shall have a 2" curb made of 2" x 2" x 1/4" angle iron welded at corners set in mastic or 2" wide by 2" high concrete curb extending above finished floor. Sleeves shall be set flush with finished wall. Caulk around conduit penetrating floors, roof, and fire-rated wall construction with sufficient layers of fire safing insulation and further seal off opening between conduit and sleeve with a fire rated non hardening mastic. Fill all unused sleeves and block-out space with fire safing material. All fire safing materials shall be UL listed and comply with ASTM E 814 (UL 1479). The fire safing system used shall maintain the fire resistance rating of the building component that is penetrated.

B. Sleeves penetrating walls below grade shall be “WS Wall Sleeves” by Pipeline Seal and Insulator, Inc. Houston, Texas, with 2” water seal stop. The water seal stop shall be located in the middle of the wall. Seal off annular opening between pipe and sleeve with "Thunderline/Link-Seal" casing seal as manufactured by Pipeline Seal & Insulator, Inc., Houston, Texas. The pipe sleeve shall be sized to accommodate the “Thunderline/Link-Seal” casing seal. Casing seals shall be Model “C” and shall be sized per the manufacturer’s recommendations.

C. If holes and/or sleeves are not properly installed and cutting and patching becomes necessary, it shall be done at no expense to the Owner. The Contractor shall undertake no cutting or patching without first securing the Architect's written approval.

D. The location and size of all penetrations and openings through existing walls, slabs, beams, roofs and other existing building construction shall be provided by the contractor to accommodate the new work. Penetrations and openings through existing building construction shall be reviewed by the structural engineer and architect prior to cutting the penetration. Unless otherwise required by the architectural and structural divisions, metal sleeve inserts will not be required in conduit penetrations except as required to meet UL fire rating requirements. The contractor shall perform a structural survey, including x-rays or graphs of all floor slabs and

supporting walls to locate existing structural bars, cables, in slab conduits and any other interferences that may be in conflict with proposed sleeve/penetration locations. The x-ray or graph shall be performed prior to any cutting or chipping of structural supporting walls. In the event of a conflict, the contractor shall notify the architect prior to proceeding with the work.

3.3 EXCAVATION AND BACKFILL

A. The Contractor shall make all necessary excavations, cutting of paving, concrete, etc., and do all backfilling and paving repairs necessary for the proper execution of the Work. Remove all dirt and debris out of and away from the building as directed. Backfill shall be mechanically compacted to a density of 95% of the maximum dry density at optimum moisture content as determined by the Standard Proctor Compaction Test.

B. Backfill shall be compacted and repairs to paving or concrete shall be accomplished to the satisfaction of the Architect and the Local Authorities.

C. See the various Sections of the Division for additional excavation and backfill requirements.

3.4 PAINTING

A. Painting, except as specified or indicated otherwise, shall be done under another Division. This Division shall cooperate with the other Division to determine the size of equipment, sizes and lengths of pipes, etc., to be painted.

B. Equipment furnished under this Division shall be factory-finished. If the factory finish is damaged during shipment, installation, etc., it shall be repainted subject to the Architect's approval.

3.5 DRAYAGE AND HAULING

A. Include all drayage, hauling, hoisting, and placement in the building of equipment specified herein. The Contractor shall be responsible for the timely introduction of equipment to the Project. If any item of equipment is received prior to the time it is required, the Contractor shall be responsible for its proper storage and protection until such time as it may be required. The Contractor shall pay for all cost of demurrage or storage.

3.6 IDENTIFICATION OF CIRCUITS AND EQUIPMENT

A. Electrical equipment shall be identified by means of nameplates permanently attached to the equipment. Refer to Section 28 30 53 for additional requirements.

B. Cardholders and directory cards shall be provided for circuit identification in panelboards. Cardholder shall be located on inside of panel door. Directory cards shall be typewritten. Circuit descriptions shall include locations and names of items of equipment served.

C. Circuits and pull wires in empty conduit shall have tags attached to wiring at points where runs are interrupted at junction boxes or terminated in panels, boxes, etc. Feeder or branch circuit numbers shall be indicated.

D. Tag symbols shall correspond to the identifications on the record drawings.

E. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This

includes, but is not limited to, the Contractor's shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

3.7 EQUIPMENT NOISE AND VIBRATION

A. It is the intention to specify and for the Contractor to provide equipment and systems that, as defined herein, shall be quiet and free of apparent noise and vibration in operation.

B. It shall be the responsibility of the Contractor to obtain equipment that is quiet in operation as compared to the other available equipment of its size, capacity, and type, and to install equipment so that a minimum amount of noise and/or vibration is transmitted to the building.

C. Any additional precautions deemed necessary to provide a quiet installation shall be done as part of the Work of the Contractor, subject to review by the Engineer, without additional cost to the Owner. After the system is in operation, it shall be the responsibility of the Contractor to make any changes to equipment or Work installed as may be required to provide a system which is quiet in operation as defined herein.

3.8 SPARE PARTS

A. The Contractor shall provide all necessary spare parts for the equipment supplied in this Project as indicated hereafter and in various other Sections of the Specifications. Spare parts shall be obtained from the original equipment manufacturer.

3.9 DOCUMENTATION

A. For all testing, training and review procedures the Contractor shall maintain adequate records in written form that clearly documents the event. The written documentation for each testing, training and review procedure shall include but not be limited to the following: Name and company of attending personnel, general description of procedure(s), date(s), time of day, deficiencies found, corrective action taken, and nature of final results. Documentation records shall be kept in an orderly, legible and up to date format. Contractor's documentation records shall be accessible for review by the Owner Representative upon request. Each testing, training and review procedure document shall bear the name and signature of the authorized representative of the General Contractor, Mechanical Contractor, and manufacturer or Subcontractor if applicable.

3.10 COMPLETION, COMMISSIONING AND FINAL REVIEW OF THE DIVISION 28 SYSTEMS

A. Start-up: After the Contractor and equipment manufacturer representative determine that the equipment and related systems have been properly installed and are ready for start-up; the equipment manufacturer representative shall perform the manufacturer's start-up and check-out procedures to ensure proper operation of the equipment in accordance with the contract documents and the manufacturer's recommendations. Start-up and check-out of each piece of equipment shall be in accordance with the equipment manufacturer's recommendations and warranty requirements. Start-up services shall be performed by factory-trained personnel of the equipment manufacturer or factory-trained personnel authorized by the manufacturer and their official representatives. Start-up personnel shall be certified by the equipment manufacturer as properly trained in the manufacturer's start-up and check-out procedures. Provide documentation of the start-up and check-out procedures in accordance with the contract documents and the manufacturer's recommendations. Documentation of the start-up and check-out procedures shall be included within and submitted with the Operational and

Maintenance Manuals.

B. Trouble-shooting, Debugging, and Pre-testing: After Start-Up but prior to the Operational Demonstrations; the contractor shall perform operational pre-testing, trouble-shooting, and de-bugging of all equipment and systems. At the completion of the Trouble-shooting, Debugging, and Pre-testing process, all equipment and systems shall be complete and operating in accordance with the requirements of the contract documents. Trouble-shooting, Debugging, and Pre-testing of equipment and systems shall be performed to the extent necessary for the Contractor to determine that all systems are complete and operating in accordance with the Contract Documents. Prior to final Commissioning and Operational Demonstrations all equipment and systems shall be certified by the Contractor as complete and operating in accordance with the Contract Documents.

C. Operational Demonstrations and Final Commissioning: Prior to Substantial Completion and within a schedule mutually agreed to by the contractor and the Owner's Representative, the Contractor shall demonstrate the operation of all equipment and systems as necessary to demonstrate compliance with the Contract Documents. Additional costs incurred by the Owner related to re-reviewing, re-witnessing, re-testing, re-inspecting, and re-commissioning all or part of the MEP systems due to incomplete or non-compliant work shall be reimbursed by the Contractor to the Owner.

D. Final Review and Owner Acceptance:

1. After completion of the Operational Demonstrations, Final Commissioning, and Owner's Close-out Package requirements; at a time designated and mutually agreed by the Contractor and Owner's Representative; a final walk-through review of the project will be performed by the Contractor and Owner's Representative. The review will be of a general nature with the intent of addressing outstanding issues prior to the Owner's final acceptance of the project.

2. The entire system shall be clean and operating properly with all systems balanced and all controls adjusted. All prohibited markings shall be removed from all switchboards, panelboards, motor control centers, switches, etc., and the equipment shall be clean and in operating condition.

3. Certificates, Documents, and Close-out Package required herein shall be complete, in order and presented to the Architect at least two weeks prior to the review.

4. After the review, any changes or corrections noted as necessary for the Work to comply with these Specifications and the Drawings shall be accomplished without delay in order to secure final acceptance of the Work.

E. Schedule: The Contractor shall complete the MEP systems in accordance with the Contract Documents including start-up, trouble-shooting, de-bugging, pre-testing, and commissioning of the MEP systems a minimum period of time prior to the date of "Substantial Completion" as established by the Commissioning Agent. This period of time prior to the date of "Substantial Completion" shall be reserved by the Contractor for the Owner's Representative, Commissioning Agent, Architect, and Engineer to witness the operational demonstrations of the completed fully commissioned MEP systems. If more than this established time period is deemed necessary by the Contractor for the operational demonstrations, the Contractor shall notify the Owner's representative and Commissioning Agent in written form and determine an operational demonstration schedule mutually agreeable to the Owner's Representative, Commissioning Agent, and Contractor that completes the operational demonstrations prior to "Substantial Completion". Additional costs incurred by the Owner related to reviewing, witnessing, testing, inspecting, and commissioning of the MEP systems by the Commissioning

Agent, Architect and Engineer after the date of “Substantial Completion” shall be reimbursed by the Contractor to the Owner. As part of the shop drawing submittal process, the Contractor shall provide a detailed schedule of completion indicating when each MEP system will be complete and ready for owner acceptance. The schedule shall indicate time periods and milestone dates of Start-up; Trouble-shooting, Debugging, and Pre-testing; Operational Demonstrations and Final Commissioning; Final Review and Owner Acceptance.

F. Additional Requirements:

1. Owner’s Close-out Package: Refer to the Owner’s Close-out Package requirements within Section 28 30 10 for additional requirements.

2. Refer to the various sections of the specifications for additional requirements.

3.11 OPERATING INSTRUCTION

A. The Contractor shall provide the services of a factory trained specialist to supervise the operation of all equipment specified herein and to instruct the Owner's operators for a five day, eight hours per day, operating instruction period. The operating instruction period shall be defined as straight time working hours and shall not include nights, weekends, or travel time to and/or from the Project. See individual Sections for additional instructions by manufacturer's trained specialists.

B. The Owner shall be notified in writing at least five days before each operating instruction period begins. The Contractor shall not commence instruction period until the Owner has issued a written acceptance of the starting time.

C. Training schedule, duration, and subject matter shall also be in accordance with a training syllabus provided by owner's operating staff. This syllabus will be provided prior to the final review.

3.12 OPERATING AND MAINTENANCE BOOKS

A. The Contractor shall provide operating instruction and maintenance data books for all equipment and materials furnished under this Division.

B. Submit four (4) hard copies and four (4) electronic copies of the operating and maintenance data books for review at least eight weeks before final review of the Project. Assemble all data in a completely indexed volume or volumes and identify the size, model, and features indicated for each item. The Engineer’s review period for the operating and maintenance data books shall be a minimum of 4 weeks, exclusive of transmittal time. The Contractor’s schedule shall plan to accommodate this review period requirement.

C. Maintenance instruction manuals shall include complete oiling, cleaning, and servicing data compiled in clearly and easily understandable form. Data shall show all model numbers of each piece of equipment, complete lists of replacement parts, motor ratings, and actual loads.

D. Include the following information where applicable:

1. Identifying name and mark number as indicated on the Drawings and in the Specifications.

2. Locations (where several similar items are used, provide a list).

3. Complete nameplate data.

4. Certified record drawings and "Final Reviewed" or completely corrected shop drawings and/or submittals.

5. Parts lists.

6. Performance curves and data.

7. Wiring Diagrams.

8. Start-up and Check-out procedures and checklists.

9. Manufacturers' recommended operating and maintenance instructions with all non-applicable information deleted. This shall include but not be limited to the manufacturer's service and repair manual.

10. Additional information as recommended by the equipment manufacturer or system provider for the proper operation, maintenance, and warranty of the equipment and systems.

## 3.13 RECORD DRAWINGS

A. The Contractor shall maintain on a daily basis at the project site a complete set of "Record Drawings". The “Record Drawings” shall consist of a set of black line prints and AutoCAD files of the Contract Drawings for this Division. The black line prints shall be marked and the AutoCAD files shall be electronically updated regularly by the Contractor during the construction period to reflect an accurate dimensional record of all buried or concealed work. In addition, the set of black line prints shall be marked and the set of AutoCAD files shall be electronically updated regularly to show the precise location of concealed work and equipment including buried, concealed or embedded piping, conduit and valves and all changes and deviations in the mechanical work from that shown on the Contract Documents. This requirement shall not be construed as authorization for the Contractor to make changes in the layout or work without definite instructions from the Architect. The set of black line prints and AutoCAD files of the Contract Drawings for this Division shall remove or blacken out the Engineer's seal and Engineer's firm name.

B. Prior to commencing work, the Contractor shall need to purchase from the Architect or Engineer a set of black line prints to be used for the "Record Drawings" and obtain from the Engineer a set of AutoCAD files of the Contract Drawings for this Division in preparing “Records Drawings” for the project based on the Contractor’s work. The AutoCAD files shall be in AutoCad 2015 version software or later version as provided by the Engineer. In order to obtain a set of AutoCAD files of the Contract Drawings for this Division from the Engineer, the Contractor shall need to contact the Engineer, Architect or Client and obtain a written release and indemnification statement from the Engineer. This release and indemnification statement shall be signed by an owner, partner, corporate officer or other person duly authorized to agree to and sign legally binding documents for the Contractor’s respective company. Upon the Engineer receiving the signed unedited release and indemnification statement back from the Contractor, the Engineer will forward the AutoCAD files of the Contract Drawings for this Division to the Architect or Client who in turn will forward the AutoCAD files of the Contract Drawings for this Division to the Contractor for his use in preparing “Records Drawings” for the project based on the Contractor’s work. If the Contractor desires to have architectural and structural information in preparing “Records Drawings” for the project, the Contractor shall need to contact the Architect and Structural Engineer to obtain the pertinent architectural and

structural information directly from the Architect and the Structural Engineer.

C. Record dimensions shall clearly and accurately delineate the Work as installed; locations shall be suitably identified by at least two dimensions to permanent structures.

D. The owner's representative shall retain the right to periodically review the “Record Drawings” as the project progresses. Final acceptance of as-built “Record Drawings” shall be conditional upon the Contractor clearly "red-marking" all changes from the contract documents, including "request for information" changes and other supplemental instruction changes, directly on the black line set of prints and clouding all of the changes in the electronic AutoCAD files. All of the clouded changes on each drawing in the electronic AutoCAD files shall be on a separate drawing layer which can be turned off without affecting the appearance of each drawing.

E. The Contractor shall mark all "Record Drawings" on the front lower right hand corner with a rubber stamp impression or an AutoCAD image that states the following:

"RECORD DRAWINGS (3/8" high letters)
To be used for recording Field Deviations and
Dimensional Data Only". (5/16" high letters)

F. Upon completion of work, the Contractor shall certify all "Record Drawings" on the front lower right hand corner adjacent to the above marking with a rubber stamp impression or an AutoCAD image that states the following:

RECORD DRAWING CERTIFIED CORRECT (3/8" high letters)

(Name of Electronic Detection and Alarm Subcontractor)

By: _____

Date:_____

For all black line prints to be used for the "Record Drawings”, the Contractor must sign the above certification. For all AutoCAD files to be used for the "Record Drawings”, the Contractor must fill in the name of the person responsible for certifying the above certification. Both the black line prints and the AutoCAD files shall be certified by the same person.

G. Prior to final acceptance of the Work of this Division, the Contractor shall submit properly certified "Record Drawings" to the Architect for review and shall make changes, corrections, or additions as the Architect may require to the "Record Drawings". After all changes have been made to the “Record Drawings”, the Contractor shall deliver to the Owner the final set of “Record Drawings” consisting of the black line prints and the AutoCAD files. The AutoCAD files to be delivered by the Contractor to the Owner shall be in AutoCAD 2015 version software or later version as originally provided to the Contractor by the Engineer.

## 3.14 OWNER’S CLOSE-OUT PACKAGE

A. Contractor shall provide a master project close-out log and corresponding schedule for submission to the Owner and Engineer for review and approval.

B. Contractor shall provide the Owner with four (4) copies of all warranties bound in a notebook with a corresponding warranty log. Warranty log shall clearly indicate the subcontractor, address, telephone number, and contact person.

C. Contractor shall provide the Owner with four (4) complete set of approved submittals and a copy of the final submittal log.

D. Contractor shall provide the Owner with four (4) complete lists of all subcontractors and major suppliers with address, telephone number, and contact persons.

E. Contractor shall provide the Owner with a complete inventory of attic stock materials and spare parts as applicable and a copy of the transmittal signed by the Owner's representative acknowledging receipt of the items.

F. Contractor shall provide the Owner and Engineer with documentation of compliance for all specified requirements associated with equipment start-up, commissioning, demonstrations, training and warranties. Documentation shall include, but not be limited to, equipment start-up sheets as completed by the manufacturer's representative.

G. Contractor shall provide the Owner and Engineer with documentation of acceptable performance of the electrical service systems. The Owner reserves the right to contract with an independent consultant to perform additional electrical service testing. If it is determined that the electrical service systems are not operating properly in accordance with the contract documents, all costs associated with additional tests and time required to correct and uncover the deficiencies by the independent consultant will be the responsibility of the Contractor unless the deficiencies are not design related.

H. When the Contractor considers the work to be ready for final acceptance, the Contractor shall provide the Owner and Engineer with written certification stating the following:

1. All work has been completed in accordance with the Contract Documents.

2. All field review items and other deficiencies identified by the Certificate of Substantial Completion have been properly corrected.

3. All work has been inspected for compliance with the Contract Documents.

4. All electrical systems have been tested in the presence of the Owner's representative and are fully operational in compliance with the Contract Documents.

5. As-built documents have been reviewed, are complete and are accurate.

I. The Contractor shall provide the Owner with a letter from the Engineer of Record stating that the equipment installed fully complies with the Contract Documents.

J. Provide a locked document box adjacent to the main ire alarm panel that contains a full size set of as-built drawings, ½ size set of as-built drawings, a flash drive with PDFs of the as-built drawings, and all product data and calculations. The box shall be keyed to match the fire alarm panel.

END OF SECTION 28 30 10

SECTION 28 30 30 - DIVISION 28 DEMOLITION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements", "Conditions of the Contract" and "Supplementary Conditions". Specific attention is called to the "Division 28 General Requirements" in Section 28 30 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Furnish all labor, equipment and materials required to demolish, cap, patch, repair and dispose of the Division 28 systems no longer required as part of an active system of the Project as specified herein and as indicated on the Drawings.

1.3 SCOPE OF WORK

A. It shall be the responsibility of the Contractor to review the Architectural Documents in addition to the Division 21, 22, 23, 25, 26 and 28 Documents to determine the complete scope of work

1.4 REFERENCE STANDARDS

A. NFPA 70 National Electrical Code

B. Underwriters' Laboratories, Inc. (UL)

C. National Fire Protection Association (NFPA)

D. National Electrical Manufacturers' Association (NEMA)

E. American National Standards Institute, Inc. (ANSI)

PART 2 - NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Under this Division, the Contractor shall furnish all labor, equipment, appliances, and materials to perform all operations in connection with the demolition of the Division 28 systems as indicated on the Drawings and as specified herein and/or as required by the Local Building Code. The work shall include but not be limited to the following:

1. Unless noted otherwise in the Architectural Documents, all existing electrical equipment within the building shall be demolished. The contractor shall review the existing conditions, in addition to the Architectural documents, and the Division 21, 22, 23, 25, 26 and 28 documents, to determine the complete scope of work.

2. Field verify the exact sizes and locations of all existing fire alarm systems prior to demolition of any existing work. The demolition work shall be coordinated with the architectural scope of the Project to assure proper limits of demolition.

3. Remove all existing notification devices, initiating devices, conduit, raceways, fire alarm panels, wiring and associated equipment within the area to be demolished, whether specifically indicated or not, that is no longer required unless otherwise indicated.

4. Openings remaining in enclosures as a result of demolition shall be sealed with sheet metal cap or other approved appropriate means, unless otherwise indicated.

5. Where existing equipment is to be relocated, extreme care shall be taken to prevent damage during the removal and reinstallation. Where damage occurs, the equipment shall be replaced or repaired to the satisfaction and approval of the Architect at no additional cost to the Owner.

6. Contractor shall be responsible for cutting, patching and restoration of existing finishes.

7. Contractor shall be responsible for removal, disposal and salvage of equipment from job site.

8. Contractor shall be responsible for cleaning, testing and repair of existing equipment to be reused.

9. Contractor shall be responsible for sealing all existing floor slab penetrations in order to maintain the integrity of the fire rating of the existing floor slabs.

10. Refer to Architectural Documents and the Division 21, 22, 23, 25, 26 and 28 documents for area and extent of demolition.

B. Where a system to be demolished must remain operational throughout the construction process until a new system that will take its place is functional, the contractor shall provide all products, services, materials, labor and supervision to maintain the operational status of the affected system.

## 3.2 EXISTING CONDITIONS

A. Under the scope of work, the Contractor shall perform a comprehensive field inspection of the existing conditions to thoroughly familiarize himself with the Project and determine the complete scope of new and demolition work. The Contractor shall verify in the field exact sizes, capacities, and locations of all existing equipment prior to installation of any proposed work. Proposed work shall be coordinated with existing conditions to assure proper installation. This field survey and determination of the complete scope of work shall be performed prior to the final bid. Specific attention is brought to physically larger and heavy pieces of equipment that are new to be installed or existing to be removed and the routing of such equipment with the available means and site conditions.

## 3.3 CONFLICTS AND DEPARTURES

A. Conflicts

1. The Contractor shall remove all existing fire alarm equipment and associated accessories whether specifically indicated or not, that are no longer required. Any equipment which must remain as part of an active system and is in conflict with the proposed work shall be relocated at no additional expense. There shall be no additional expense to the Owner for this type of demolition work, unless, in the Architect's and Owner's opinion, it is beyond the scope and intent of the Contract Documents.

B. Departures

1. The Contract Drawings are diagrammatic and schematic in nature and do not indicate inferred details. Any departures from the Drawings that are necessary to comply with the intent of Documents, requirements of applicable Building Codes or the Authority having jurisdiction shall be detailed in the form of shop drawing submittal data and submitted for review by the Architect and Engineer. All demolition and any such revisions for departures shall be made in accordance with the reviewed shop drawings without increased expense to the Owner. Departures shall not be made without prior written approval by the Architect.

## 3.4 COOPERATION

A. The Contractor shall comply with all the requirements of the Local Authorities having jurisdiction and the building Owner.

B. The Contractor shall give advance notice to the Architect when work is to be performed.

C. The Contractor shall cooperate and coordinate demolition of Division 28 systems as required with all of other trades. The Contractor shall supervise and assist in the removing and replacing of existing materials for installation of electrical items and items related to all other trades.

## 3.5 BARRIERS

A. Where required by applicable health and safety regulations, the Contractor shall furnish and maintain safety and dust barriers. If necessary, the Contractor shall furnish and maintain temporary fencing and traffic barriers in accordance with all applicable health and safety regulations and as deemed necessary.

## 3.6 RESTORATION

A. Demolition shall be carried out with caution to assure that existing conditions to remain will be undamaged. Existing conditions to remain that are damaged or defaced by work under this contract shall be restored or replaced equal to the conditions at the time of award of the Contract.

B. Cut, patch, and restore all existing surfaces not receiving new finishes that have been disturbed during execution of this Contract. Materials and finishes used shall be similar, in all respect, to adjacent surfaces.

C. Existing conduit and wiring which are to remain as part of an existing or proposed active system that the Contractor determines to be defective shall be brought to the attention of the Architect.

D. Existing equipment to be removed shall be cleaned, repaired, and reused at the discretion of the Architect wherever applicable.

## 3.7 CONNECTIONS

A. Connections to existing work shall be similar, in all respects, to the existing system and conditions unless otherwise indicated. Existing work shall be altered and/or temporarily removed and replaced as required for completion of requirements of the project.

## 3.8 SALVAGE

A. The Owner assumes no responsibility for loss or damage to materials or structures on site for the salvage value of equipment which the Contractor may have reflected in his bid.

B. All existing fire alarm equipment, conduit and wiring removed during construction no longer required as part of an active system and not to be reused shall be removed from the job site and properly disposed of by the Contractor.

C. All existing fire alarm equipment not being reused shall be removed during construction and shall remain the property of the owner and shall be stored, removed from the job site, or disposed of as directed by the owner's representative.

3.9 CLEANING

A. The Contractor shall clear away all debris and demolished material at frequent intervals. The Contractor shall not allow debris to accumulate to the extent that it will interfere with work, passage of the workmen, and the operation of the existing occupied areas.

B. It is the intent of this Specification that all work, including the inside of equipment, be left in a clean condition. All construction dirt shall be removed from material and equipment.

3.10 INVENTORY

A. Contractor shall provide a detailed inventory of all existing fire alarm devices including notification devices, initiating devices, fire alarm panels, etc. to be stored for later reuse. Inventory shall include type, size, capacities, and quantities of each item or piece of equipment.

3.11 TESTING

A. Equipment

1. Existing equipment, instruments, and accessories to remain shall be tested for defects and shall be adjusted, repaired or replaced where required. Conditions, qualifications, and procedures regarding adjustments, reparations, or replacement of existing equipment as deemed necessary by the Contractor shall be submitted to the Architect for approval. The Contractor shall not proceed with equipment replacement without the written approval of the Architect. The Architect shall determine whether replacement is required and shall specify the replacement equipment to be used.

B. Additional Requirements

1. For fire alarm devices, etc., refer to the appropriate Specification Section and manufacturer’s recommendations for testing requirements and additional requirements of existing equipment and systems to be a part of the new active system.

3.12 QUALIFICATIONS

A. Only Contractors, Subcontractors, and workmen experienced and regularly engaged in the demolition of fire alarm systems shall be permitted to perform the demolition of existing systems.

END OF SECTION 28 30 30

# SECTION 28 30 53 - IDENTIFICATION FOR FIRE DETECTION AND ALARM SYSTEMS

## PART 1 - GENERAL

### 1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of Division 01, "General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 26 General Requirements" in Section 28 30 10.

### 1.2 WORK INCLUDED IN THIS SECTION

A. Types of electrical identification work specified in this section include the following:

1. Electrical power conductors.
2. Electrical Equipment.
3. Raceways including all conduit and junction boxes.
4. Operational instructions and warnings.
5. Danger signs.
6. Equipment/system identification signs.

### 1.3 WORK INCLUDED IN OTHER SECTIONS

A. Section 28 31 00 - Fire Detection and Alarm

### 1.4 REFERENCE STANDARDS

A. All signs, labels, colors, lettering, styles and sizes of electrical identification shall be in accordance with the applicable requirements of the American National Standards Institute (ANSI).

### 1.5 SUBMITTALS

A. Submit a list of equipment requiring a nameplate as specified herein identifying nomenclature and color field.

## PART 2 - PRODUCTS

### 2.1 ELECTRICAL IDENTIFICATION MATERIALS

A. General

1. Except as otherwise indicated, provide manufacturer's standard identification products of categories and types required for each application. Where more than one single identification type is specified for an application, selection is Installer's option. However, identification product shall be consistent for each application.

B. Color-Coded Plastic Tape

1. Wiring:

a. General: Provide manufacturer's standard self-adhesive vinyl tape not less than 3 mils thick by 3/4" wide for use on non-color-coded wires.

b. Colors: Conforming to Section 26 05 19, "Low-Voltage Electrical Power Conductors and Cables".

2. Raceways:

a. General: Provide manufacturer's standard self-adhesive vinyl tape not less than 1-1/2" wide and not more than 2-1/2" wide for use on conduit.

b. Colors: Conforming to color scheme indicated for "circuit and conduit identification".

C. Circuit and Conduit Identification

1. Mark with an indelible marker the painted covers of all junction boxes to indicate the circuits contained therein.

2. The covers of junction boxes shall be painted as follows:

| SYSTEM | COLOR |
|---|---|
| Fire Alarm | Red |

3. The color-code scheme of 2.1-C.2 above shall also apply to all conduit systems as follows:

a. All conduit shall be clearly marked, at intervals not to exceed 20 feet in length of conduit, with color-coded bands. Bands shall be 1-1/2 inches to 2-1/2 inches in width.

b. All conduit shall also be clearly marked with color coded bands as it passes through any barrier or obstruction such as a wall or floor. The bands shall be installed within 12 inches of the obstruction and on both sides of the obstruction.

4. Provide Engineer and Architect with color samples for each system and application listed prior to beginning work. Refer to the Architectural Painting Systems for additional requirements.

D. Self-Adhesive Plastic Signs

1. General: Provide manufacturer's standard, self-adhesive or pressure-sensitive, pre-printed, flexible vinyl signs for operational instructions or warnings; of sizes suitable for application areas and adequate for visibility, with proper wording for each application.

2. Colors: Unless otherwise indicated or required by governing regulations, provide black signs with white lettering.

E. Engraved Plastic Laminate Signs

a. General: Provide engraving stock melamine plastic laminate with engraver's standard letter style of sizes and wording indicated, with white lettering in a field to match color coding identified in 2.1-C. 2 except as otherwise indicated, punched for mechanical fastening except where adhesive mounting is necessary because of substrate.

b. Thickness: 1/16."

c. Fasteners: Self-tapping stainless-steel screws, except contact-type permanent adhesive where screws cannot or should not penetrate substrate.

F. Baked Enamel Danger Signs

1. General: Provide manufacturer's standard "DANGER" signs of baked enamel finish on 20 gage steel; of standard red, black and white graphics; 14" x 10" size except where 10" x 7" is the largest size which can be applied where needed, and except where larger size is needed for adequate vision; with recognized standard explanation wording e.g., HIGH VOLTAGE, KEEP AWAY.

## 2.2 LETTERING

A. General: Coordinate names, abbreviations and other designations used in electrical identification work, with corresponding designations shown on construction documents, specified or scheduled. Provide numbers, lettering and wording as indicated or, if not otherwise indicated, as recommended by manufacturers or as required for proper identification and operation/maintenance of electrical systems and equipment. Comply with ANSI A13.1 pertaining to minimum sizes for letters and numbers.

# PART 3 - EXECUTION

## 3.1 APPLICATION AND INSTALLATION

A. General Installation Requirements

1. Install electrical identification products as indicated, in accordance with manufacturer's written instructions, and requirements of NEC.

2. Coordination: Where identification is to be applied to surfaces, which require finish, install identification after completion of painting.

B. Operational Identification and Warnings

1. General: In addition to the requirements indicated herein, wherever reasonably required to ensure safe efficient operation and maintenance of electrical systems, electrically connected mechanical systems, general systems, and equipment, including prevention of misuse of electrical facilities by unauthorized personnel; install self-adhesive plastic signs or similar equivalent identification, instruction or warnings on switches, outlets and other controls, devices, and covers of electrical enclosures. Where detailed instructions or explanations are needed, provide plasticized tags with clearly written messages adequate for intended purposes.

C. Danger Signs

1. Locate on doors to all main electrical service entrance rooms, electrical vaults, as indicated in the contract documents. Refer to architectural specifications for additional requirements.

D. Equipment/System Identification

1. General: Install engraved plastic laminate sign on each major unit of electrical equipment in the building and parking structure; including central or master unit of each electrical system, unless unit is specified with its own self-explanatory identification or signal

system. Except as otherwise indicated, provide single line of text, 1/2" high lettering on 1-1/2" high sign (2" high where 2 lines are required), white lettering in a field to match color coding identified in 2.1-C.2. Provide text matching terminology and numbering of the contract documents and shop drawings. Provide signs for each unit of the following categories of electrical work.

a. Panelboards, electrical cabinets, pull boxes, and enclosures (individual sub-components for distribution panelboards).

b. Disconnect switches.

c. Automatic transfer switches.

d. Transient voltage surge suppressors.

e. Other equipment as indicated on the drawings and as specified herein.

2. Install signs on the equipment at location for best convenience of viewing without interference with operation and maintenance of equipment. For recessed panels in finished areas, provide the sign on the inside of the panel door. Secure to substrate with fasteners, except use adhesive where fasteners should not or cannot penetrate the substrate.

3. The Contractors shall coordinate the equipment tag designations and nameplates used to identify each piece of the Division 21, 22, 23, 25, 26 and 28 equipment so as to match and correspond to the equipment tag designations indicated on the contract documents. This includes, but is not limited to, the Contractor’s shop drawings; submittals; record drawings; wiring diagrams; testing and balancing reports; Operating and Maintenance (O&M) manuals; Energy Management and Control System (EMCS) documents, software, programming, and graphics.

END OF SECTION 28 30 53

SECTION 28 30 70 - DIVISION 28 SCHEDULE OF SUBMITTAL DATA

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 28 General Requirements" in Section 28 30 10.

1.2 WORK INCLUDED

A. Furnish the shop drawing submittal data for the Architect's and Engineer's review as indicated herein.

1.3 RELATED WORK SPECIFIED ELSEWHERE

A. Division 21, 22, 23, 25, 26 and 28: All Sections

PART 2 - SHOP DRAWING SUBMITTAL DATA

2.1 GENERAL

A. The submittal data to be furnished for this Project shall comply with the Specifications hereinbefore. The referenced Sections herein scheduled are for general information purposes only and shall not be construed to be limiting to the submittal data scheduled and/or required within the individual Sections of these Specifications.

2.2 CONTRACT REQUIREMENTS

A. Shop drawing submittal data shall clearly and concisely address all Contract Document requirements. Standard manufacturer data that does not indicate these requirements shall be supplemented with additional information provided by the Contractor or equipment manufacturer. Refer to Section 28 30 10 for additional requirements.

2.3 DEVIATIONS FROM THE CONTRACT DOCUMENTS AND PERTINENT DATA

A. All deviations from the contract documents shall be noted in the form of a cover letter included with the shop drawing submittal package. All pertinent data such as material options, equipment arrangements, control options, diagrams, etc. that applies to this Project shall be clearly marked. All data and diagrams that do not apply to this Project shall be omitted from the package or clearly marked out.

2.4 QUALITY ASSURANCE

A. Product submittal data shall contain a Statement of Compliance. The Statement of Compliance shall individually list each specification paragraph by its alphanumeric designation along with a statement of "Complies" (C) or a statement of "Does not comply" (NC) for each of these alphanumeric designations. If an NC statement is provided, supplemental information shall be provided in the Statement of Compliance that explains the non-compliance and what will be provided to meet the intent of the specification. At the Contractor’s option it will also be acceptable to make a copy of the specifications and mark “C” or “NC” by each specification paragraph. A handwritten marked-up specification section will be acceptable. Along with this specification copy mark-up, provide an abbreviated version of the Statement of Compliance that

addresses only the NC statements and deviations from the contract documents. If there is a conflict between the Statement of Compliance and the product literature, the Statement of Compliance will govern. Submittal data received without a Statement of Compliance will be returned without review.

2.5 SPECIAL REQUIREMENTS

A. The shop drawing submittal package containing the Fire Alarm composite wiring diagrams and the Operation and Maintenance (O&M) Manuals shall require an Owner/Consultant review period of four (4) weeks. The Contractor shall plan for and arrange the shop drawing submittal review schedule to accommodate this requirement.

2.6 SPARE PARTS

A. The submittal data for each piece of equipment shall include a detailed itemized list of recommended spare parts. Spare parts shall be clearly identified by name and part number.

2.7 ORGANIZATION OF PACKAGES

A. Unless otherwise noted in the specifications and in the following schedule of shop drawing, the Contractor shall provide a single shop drawing submittal package for each of the listed specification sections. Unless specifically noted otherwise, multiple submittal packages per section will not be acceptable under the terms of the Contract.

2.8 COORDINATION BETWEEN DIVISIONS 21, 22, 23, 25, 26 AND 28

A. The Division 28 Contractors shall cooperate with the General Contractor and the Division 21, 22, 23, 25 and 26 Contractors to provide coordination between the Division 21, 22, 23, 25, 26 and 28 trades in a timely manner. For all equipment requiring electrical service provided under Division 28, it shall be the responsibility of the Division 28 Contractors to provide to the Division 26 Contractors the electrical characteristics of the actual equipment to be provided. Should there be a discrepancy between the electrical service characteristics of the equipment to be provided and what is indicated on the documents, the contractor shall obtain written direction from the Owner's representative prior to proceeding. This coordination and transfer of information shall take place in a timely manner prior to the purchasing and installation of the electrical service.

2.9 MANUFACTURER'S CERTIFICATION

A. All items or equipment listed herein with asterisks (*) shall be certified by the manufacturer. Refer to Section 28 30 10 for certification requirements.

2.10 SHOP DRAWING SUBMITTAL SCHEDULE

A. Provide a comprehensive list of Divisions 21, 22, 23, 25, 26 and 28 shop drawing submittal packages and a schedule indicating the date when each package will be submitted to the contractor and the date when each package will be submitted to the design team. The intent of this list and schedule is to allow the contractor and design team to plan ahead and schedule the necessary review and coordination resources in advance of the arrival of shop drawing submittal packages. Shop drawing submittal schedule shall be submitted to the Architect not less than four weeks prior to the submission of the first Division 21, 22, 23, 25, 26 and 28 shop drawing submittal packages. Shop drawing submittal schedule shall be periodically updated by the contractor and issued to the Architect to reflect current status of the project. Shop drawing submittal packages shall be listed on the schedule not less than four weeks prior to submission

to the contractor and not less than six prior to the submission to the design team. Packages not listed on this schedule will required up to two additional weeks of review time.

2.11 SHOP DRAWING SUBMITTAL DATA

A. The submittal data to be furnished for this Project shall include but not be limited to the following:

1. Section 28 30 53 - Identification for Fire Detection and Alarm Systems

2. Section 28 31 00 - Fire Detection and Alarm System *

3. Section 28 30 70 - Shop drawing submittal schedule.

4. Drawing Packages:
   a. Drawings shall be limited to a maximum of three (3) packages.

END OF SECTION 28 30 70

SECTION 28 31 00 - FIRE DETECTION AND ALARM

PART 1 - GENERAL

1.1 GENERAL REQUIREMENTS

A. The work required under this Section shall conform to the requirements of "Division 01, General Requirements," "Conditions of the Contract" and "Supplementary Conditions." Specific attention is called to the "Division 28 General Requirements" in Section 28 30 10.

1.2 WORK INCLUDED IN THIS SECTION

A. Provide the Fire Detection and Alarm Systems as shown on the drawings and as specified herein. System shall be wired, connected, verified and left in first class operating condition.

B. The fire alarm system shall comply with the currently enforced editions of the following NFPA articles: 1) 72, Fire Alarm Code, 2) the installation with 70 National Electrical Code (NEC), and 3) 101, Life Safety Code. In addition, the fire alarm system shall comply with the requirements of the applicable building codes as adopted and modified by the Local Code Authorities.

C. The Fire Alarm system shall include the following integrated systems:

1. Supervised, electronic manual and automatic fire detection/initiation system.
2. Supervised, electronic public alarm signaling system.
3. Related Systems Annunciation and Programmed Control.

D. All conduit and wire (control wire) for a complete and operational system will be provided and shall include both `line (120V.)' and 'low' voltage wiring.

E. Fire alarm system control wiring and conduit from all fan's, AHU's, dampers and other mechanical equipment shall be furnished and installed by the fire alarm system subcontractor provided under Division 28. This fire alarm control wiring system shall be completely independent of the mechanical control wiring system. Dampers and associated control circuits not indicated on the Division 26 and Division 28 drawings shall be the responsibility of the Division 25 EMCS contractor. The Division 26 and Division 28 contractors shall be responsible to coordinate with the Division 23 and Division 25 contractors for the dampers to be provided and connected to their respective systems.

F. The Division 28 contractor with the assistance and cooperation of the Division 21, 22, 23, 25, and 26 contractor shall be responsible for operational verification of alarms, annunciation and initiation sequences.

G. All necessary fire alarm system wiring diagrams and control diagrams required by fire alarm contractor shall be supplied to general contractor from automatic temperature controls (EMCS) contractor. Final fire alarm system control wiring diagrams shall be provided to EMCS contractor from fire alarm contractor for EMCS contractor review and comment prior to installation.

H. These specifications outline the minimum system requirements. The Contractor shall provide all additional equipment, devices, programming, hardware, and software above and beyond the minimum requirements specified herein, necessary for a complete, fully commissioned, code compliant, and operational system.

1.3 WORK INCLUDED IN OTHER SECTIONS

A. Generator Section 26 32 00. The fire alarm system shall monitor, supervise, alarm, trouble, etc. the generator system and generator circuit breakers.

B. Automatic Transfer Switch Section 26 36 23.

C. Door hardware (prop sensors, electric strikes, power operators, etc) requiring power and respective low voltage transformers shall be provided by Division 8. Fire alarm control wiring and power circuit wiring to the individual door locations as identified by the Architect shall be provided by Divisions 26 and 28. Install 120V (1P-20A circuit breaker, 2# 12 1/2 C.) circuits to door hardware. Where required, pass circuiting through the auxiliary fire alarm 20A contacts to the various door locations. Upon a fire alarm, the circuit is disconnected and the door locks fail safe open. Divisions 26 and 28 shall rough in proper conduit/wire from door hardware up to ceiling plenum and connect to Division 8 supplied transformer with primary and secondary fuses. Refer to Architectural drawings and specifications for door hardware schedules and locations. The above requirements shall include prop sensors; however, the circuits shall not pass through the fire alarm contacts if the doors are not electrically locked. Rough in 3/4" conduit with pull string to nearest accessible tenant ceiling plenum.

D. Divisions 26 and 28 shall rough-in conduit from doors, door hardware, junction boxes, etc. to accessible area where required for various security devices to include but not limited to motion sensors, exit buttons, card readers, sounders, electric strikes, door contacts, cameras, etc. Refer to security system drawings and information.

E. Provide all fire alarm system provisions as required for remote monitoring of all fire alarm trouble and alarm conditions on the Division 25 EMCS system. Refer to Section 25 00 90 for additional information.

F. All dampers and associated control circuits not indicated on the Division 26 drawings that require control wiring systems and/or power wiring systems up to and including 120 volt, including but not limited to, automatic dampers, combination fire, smoke and automatic dampers (FDS), combination fire and automatic dampers (AFD), and other motorized dampers shall be provided by Division 23 and Division 25. Refer to the Division 23 and Division 25 documents for additional information. The Division 28 contractor shall be responsible to coordinate with the Division 23 and Division 25 contractors for the dampers to be provided and connected to their respective systems.

G. The Contractor shall be responsible for information referenced herein and requirements in the Architectural Documents including, but not limited to equipment finishes and device mounting locations, locations and provisions of fire rated partitions and doors for fire alarm rooms for system survivability, and project square footage.

1.4 SUBMITTALS

A. Provide fire alarm drawings and calculations prepared by or under direct supervision of an individual having a NICET Level III or IV certification in fire alarm systems or a Professional Engineer that passed the NCEES Fire Protection Engineering exam.

B. Provide manufacturer's literature and illustrations including cut sheets of all alarm initiation devices (i.e. smoke detectors, pull stations, flow and tamper switch interface monitors, etc.) and notification devices (i.e., horns, strobes, etc.).

C. Provide a description of the system operation which includes the method of operation and

supervision of each type of circuit (alarm initiation, signaling, control, annunciation, etc.), operation of manual controls, and sequence of automatic and manual operation. The system description shall be written specifically for this project. The manufacturer's standard descriptions which refer only to general operation are not acceptable.

D. In addition to the written operational description, the system manufacturer shall provide a comprehensive fire alarm system matrix. The matrix shall indicate all devices by type, floor and zone; and the associated fire alarm system response. The matrix shall also reflect the operational response of all equipment listed herein under "system operation". All Division 21, 22, 23, and 25 equipment shall be listed by tag number. The matrix shall reflect all specification, NFPA 72, and local Fire Marshal's office requirements. Matrix shall be sufficiently detailed to be used as the basis of the contractor's testing and owner demonstration procedures.

E. Provide wiring diagrams which show the method of wiring for each type of circuit for each function performed. These shall include the following:

1. Each type of initiation circuit.
2. Each type of notification circuit.
3. Trunk SLC loop.
4. Annunciation methods.
5. Control methods (separate diagrams shall be provided for each type of device controlled).

F. Each wiring diagram shall indicate:

1. Method of circuit protection.
2. Wiring type and size, method of ground or shielding if used, and size of conduit.
3. Terminal identification at control panel and remote device. Each components address.
4. End of line device, type, rating and location.

G. Submit tables indicating the following:

1. List of each horn circuit, with area served, number of horns connected, amperage draw, horn tap used (1/2 watt minimum), used capacity, spare capacity, anticipated sound level at 10' at tap used and at next higher tap.

2. List of each amplifier with rated output (intermittent and continuous output shall be indicated if they are different), list of horn circuits connected to each amplifier. Number of standby amplifiers provided.

3. List of each strobe circuit, with area served, number of strobes connected, amperage draw, used capacity, spare capacity, etc.

4. List of each initiation circuit, with area served, number of devices connected, used capacity, spare capacity, etc

H. Provide total system operating and supervisory power requirements including battery size calculations

I. Provide four complete sets of the Operation and Maintenance Manual:

1. General:

a. Provide the manual required in Section 28 30 10 as required to fully describe the operation and maintenance of the Fire Detection and Alarm Systems.

b. The manual shall include at least the data as given by the outline listed below.

2. Installation Instructions:

a. Installation
b. Adjustment
c. Checking
d. As-built drawings

J. Operation Instructions:

a. Startup
b. Routine and Normal Instructions
c. Regulation and control
d. Periodic testing

K. Record of Completion:

1. Provide a record of completion per NFPA 72.

2. Provide a written certified report attesting to the proper operation of the completed system.

L. Provide 1/8"=1'-0" scale drawings in blackline printed form produced utilizing AutoCAD 2015 version software or later version if approved by the Engineer of entire project identifying each fire alarm device, location, associated address, the fire control panel, terminal cabinets, remote transponder panels, notification appliance circuit extender panels, fire service annunciator panels, loops, conduit/wire, etc.

M. Drawings illustrating annunciator with circuit connections. Drawings shall be produced using AutoCAD 2015 version software or later version if approved by the Engineer.

N. The authority having jurisdiction shall be notified prior to installation of equipment or wiring. Complete information regarding the system or system alterations, including specifications, wiring diagrams, and floor plans shall be submitted for approval. Submit a copy of the local Fire Marshal approved drawings.

O. Shop drawing submittal packages shall also indicate all requirements to be code compliant within the local code jurisdiction.

PRODUCTS

1.5 ACCEPTABLE MANUFACTURERS AND CONTRACTORS

A. Fire Alarm and Detection System Contractor

1. The Contractor shall have a minimum of ten (10) years' experience in the design installation and maintenance of fully electronic addressable fire alarm and detection systems.

2. The Contractor shall have a minimum of five (5) years' experience as a manufacturer's authorized representative in the design, installation and maintenance of the

manufacturer's system and products.

3. The Contractor shall be thoroughly experienced and knowledgeable concerning the requirements of local code authority having jurisdiction including but not limited to the shop drawing submittal process, building permit process, inspection process, all necessary approval processes, local code supplements, and special requirements above and beyond the code. The contractor shall have a minimum of five (5) years' experience as an installer of fire detection and alarm systems within the local code jurisdiction.

4. The Contractor shall maintain a factory trained certified engineering and commissioning staff and shall have complete offsite training facilities. The Contractor shall have the necessary facilities to provide the Owner with a complete maintenance, periodic inspection and service contract. This service contract shall include 24-hour emergency service, seven days per week. Warranty service shall be provided on a 24-hour, 7 day per week basis during the warranty period. Response time of service personnel shall not exceed one hour from the time of the service call. Repairs under warranty shall be complete within 72 hours of the service call.

B. System Manufacturer

1. The work in this section shall be accomplished by one of the following if they comply with the specifications.
   a. Notifier
   b. Simplex Grinnell
   c. Edwards Systems Technology
   d. Siemens
   e. Gamewell-Fire Control Instruments (FCI)

1.6 GENERAL

A. The complete installation shall be listed by the Underwriters Laboratories Inc. and bear the UL Label. The complete installation shall conform to NFPA. 72 and 90, the local Fire Marshal's requirements for Low Rise Fire Alarm and Detection System and the National Electrical Code (NEC). In addition, the fire alarm system shall comply with the requirements of the applicable building codes as adopted and modified by the Local Code Authorities.

B. The equipment furnished under this section shall be the standard product of one manufacturer. Equipment descriptions are intended to indicate the type and quality of design and material as well as the operating features desired. All devices, combinations of devices, appliances, and equipment installed shall be listed for the protective signaling purpose for which they are used.

C. Provide fire alarm terminal cabinets on each floor for terminating circuits and housing control modules, end of line resistors, etc. Division 26 shall provide emergency power * (battery) (circuits (conduit/wiring/circuit breaker)) wherever particular vendors equipment requires power.

D. An addressable, intelligent device fire alarm system with alarm verification shall be provided. Upon initiation of any smoke detector the fire alarm system control module will delay and test the detector for up to 20 seconds prior to putting system into alarm. Should the detector indicate a fire condition after the 20 second period, a general fire alarm sequence will be initiated. Each initiating and supervisory device shall be individually addressable and identifiable from the main control panel alphanumeric display.

E. The system shall provide a hard copy written record of all alarms, troubles and system activity.

A built-in printer which is part of the control panel or a remote printer shall be provided. If a remote printer is provided, it shall be located within the Fire Command Room unless otherwise directed by the local code authority. Print out shall include type of alarm/trouble, address, zone, date, and time.

F. If required by the local authority, a means of manually notifying the local fire department of a fire alarm shall be provided adjacent to the central control panel. A coded fire alarm pull box or pull box which initiates an auto dial phone system with prerecorded message or other local authority approved system shall be provided. A separate phone with dedicated phone line shall also be furnished adjacent to the central control panel.

G. All indoor components of the fire alarm system shall be so designed as to perform all intended functions under the following conditions:

1. At 85% and 110% of nameplate input voltage.

2. At ambient temperatures of minimum 32°F ± 4°F and maximum 120°F ± 4°F for a minimum duration at each extreme of 3 hours.

3. At a relative humidity of 85% ± 5% and a ambient temperature of 90°F ± 4°F for a duration of 24 hours.

H. Exact location of all alarm and detection devices shall be coordinated with the owner's representative prior to installation. Exact locations of devices to be installed in the building main lobby shall be reviewed and approved by the Architect, Engineer and Owner prior to installation.

I. Space and locations for fire alarm equipment has been provided on the drawings; however, the exact location of all fire alarm system panels and cabinets shall be coordinated with the owner's representative prior to final design of the system by the contractor and/or vendor. The scope of work for the fire alarm system shall include location of system panels and cabinets at any location within the building main lobby or vestibule with no additional cost to the owner.

J. Provide system capacity, wiring, terminals, etc., in fire alarm riser and each floor's fire alarm terminal cabinets for future tenant fire alarm and detection system devices including horns and visual alarms based on the design capacities specified herein. In addition to the tenant requirements, provide 20% spare system capacity (amplification, inputs, etc.) in total and on each floor individually for future modification.

K. The fire alarm system and components shall include the following intelligent features: Auto detector test, sensitivity adjust from panel, and maintenance alert.

L. Any concealed initiation device shall have a remote readily visible red LED light with description label and be mounted so as to be fully accessible for maintenance and inspection. Provide access doors as required.

M. Provide one (1) remote LCD annunciator per annunciation node similar to Simplex #4603-9101 or Notifier LCD-160 in a location as directed by the Owner’s representative. Coordinate the exact location of the remote annunciator(s) with the owner’s representative. Connect the remote annunciator(s) to the appropriate Fire Alarm Control Panel (FCP).

N. Standard NFPA 72 System shall be modified and amended as required to comply with all local rules, regulations, and code supplements.

O. Two dedicated telephone outlets are required for each auto-dialer. A jack shall be provided for

each line to facilitate maintenance and testing.

1. Auto-dialer program and format codes shall be transmitted to the Owner at the substantial completion walk-through, and a copy shall be included in the operations manual for the fire alarm system.

2. Provide an auto-dialer connected to the County's contracted monitoring service as directed by the Owner.

3. The auto-dialer shall not serve any other equipment.

4. The auto-dialer shall be connected to support the FACP.

5. The auto-dialer shall have manual onsite reset capabilities and shall automatically reset once the trouble or phone line or power interruption has cleared.

6. The auto-dialer shall automatically restore and or reset upon power or phone line interruption, once the power or the phone line restores or stabilizes.

7. The auto-dialer test timer test shall be programmed to perform between 7:00 AM and 1:00 PM.

8. The Contractor shall be required to arrange to receive and respond to all trouble and alarms received by the County's monitoring service prior to building occupancy by the County.

9. The auto-dialers shall have external surge suppression to protect the system from lightning and voltage surges both from the line voltage and the phone lines. The surge suppression shall be in addition to the manufacturer's built-in suppression system. The surge suppression shall be installed and located adjacent to the FACP and the auto-dialer. A separate surge suppressor shall be installed for each of the FACPs and the auto-dialers.

## 1.7 SYSTEM OPERATION

A. The fire alarm system shall provide the following alarm indication, control functions, sequence of operation and system components:

1. Alarm System: Activation of any manual station, any automatic smoke, heat, sprinkler waterflow switch, or any initiation device shall cause the following:

   a. Initiate an audible and a visual indication alarm signal at the central fire alarm panel for each initiating and supervisory device. The appropriate initiation alarm shall light or flash and audible alarm shall sound until manually acknowledged.

   b. Initiate alarm tone(s) and activate all indicating devices within the building or as otherwise required by the local authority. Alarm tone(s) shall comply with currently enforced local regulations. The alarm signal shall sound until the sequence has been halted at the control panel or the alarm condition has been acknowledged.

   c. Duplicate the visual alarms and alphanumeric indication on the one (1) remote LCD annunciator located within the building engineer's office or in a location as directed by the Owner's representative. Initiate a fire alarm signal at the EMCS.

d. Initiate remote alarm, supervisory, and trouble signal transmission to the appropriate central monitoring station and/or city fire department as required by the local code authority.

e. Cause all required doors to automatically unlock. Doors shall fail in the unlocked position upon loss of power.

f. Audible alarm devices shall be provided on the exterior of the building in an approved location. Alarm shall be activated by the equivalent flow of a single sprinkler of the smallest orifice size installed in the system.

2. Automatic Sequence: Unless otherwise indicated, the air handling units, toilet exhaust fans, outside air handling units, and other miscellaneous fans shall automatically function upon activation of any automatic smoke, heat, sprinkler waterflow switch or initiation device (excluding manual station where required by the Authority Having Jurisdiction) as follows:

a. Exhaust and relief air fan(s) serving the area in which the alarm initiating device is activated shall automatically stop.

b. The air handling units shall shutdown and associated damper(s) shall operate appropriately when respective return air smoke detector is activated.

c. Unless otherwise noted, all other fans serving the floor or area in which the alarm initiating device is activated shall stop.

d. The outside air handling units shall stop. The associated dampers, louvers, etc. shall operate appropriately.

e. All automatic fire/smoke dampers (FDS) shall close.

f. Automatic sequences of operation for building systems shall also be in accordance with the requirements of the local code authority as required to acquire the building occupancy permit.

3. Alarm acknowledgment and reset:

a. Upon completion of alarm verification, audible alarm signals shall sound and the Control Panel alarm indication LEDs/LCD shall flash upon alarm initiation until the alarm has been manually acknowledged by means of a control panel switch at the main panel. Upon acknowledgment, audible alarm signals shall silence and alarm indication LEDs/LCD shall be steadily energized until the alarm condition has been reset.

b. Receipt of a subsequent alarm shall cause the appropriate audible signal circuits to energize and the alarm indication LEDs/LCD for the new alarm to flash.

4. System Power Supply and Distribution:

a. System power shall be supplied by building's emergency power system. Upon loss of building power, system shall automatically switch to battery power, indicate a trouble condition and sound a local alarm. Local alarm shall have a manual silence switch. Alarm shall resound upon new trouble or alarm conditions.

b. All circuits requiring system operating power shall be fused at the control panel

(e.g., audible alarm annunciation, visual signaling, door holder control circuits, etc.). Auxiliary fused circuits shall not exceed 2 amperes.

c. As per NFPA 72, under maximum quiescent load to operate system, provide battery backup for 24 hours and at the end of that period, shall be capable of operating all alarm notification appliances used for evacuation for five minutes. The power supply for emergency voice/alarm communications service shall be capable of operating the system under maximum quiescent load for 24 hours and then shall be capable of operating the system during a fire or other emergency condition for a period of 2 hours. Fifteen minutes of evacuation alarm operation at maximum connected load shall be considered the equivalent of 2 hours of emergency operation. Provide battery chargers. Submit battery calculations with fire alarm system submittals.

d. The battery charger shall provide integral means by which the battery voltage and charging current can be determined.

e. The battery shall be protected by overcurrent devices having a rating of not less than 150% and not more than 200% of the maximum operating load applied to the battery. After the fully charged battery is subjected to a single discharge cycle the charging current shall be such that the battery capacity after 48 hours is recharged. Upon attaining a fully charged condition, the trickle charge rate shall not be so excessive as to result in battery damage. The batteries shall be located that the signaling system equipment, including overcurrent devices, are not adversely affected by battery gases, and shall conform to the requirements of Article 480, NFPA 70, National Electrical Code.

f. Provide wire, conduit and connections to door holder power supplies.

g. Power conductors for the fire alarm system shall be on dedicated branch circuits. The circuit disconnecting means shall have a red marking, shall be accessible only to authorized personnel, and shall be identified as "FIRE ALARM CIRCUIT CONTROL".

5. System Supervision:

a. System alarm indicating, initiation and remote annunciation wiring shall be continuously supervised for open, shorted and grounded conditions. Refer to NFPA 72.

b. Supervised monitoring circuits shall monitor the status of all sprinkler systems, the generator, and the generator circuit breaker. Any abnormal condition shall activate the system's trouble circuits and annunciate the source by floor, type of device and specific device identification or address number. Initiation circuits shall be as specified under "Main Control Panel". Style D initiation circuits shall comply with NFPA 72 for supervisory requirements.

c. Manual bypass of an automatic alarm function or manual activation of an alarm controlled function shall cause the circuit to indicate trouble.

d. Any system trouble shall energize audible and visual circuits on the system's central control module. Trouble signal silencing switches shall be furnished each with its associated pilot lamp, so that faults on the alarm initiating circuits and the alarm signal circuits can initiate trouble signals and be silenced independently of

each other. When trouble signals are silenced after being sounded by faults on either the alarm initiating circuit or the alarm signaling circuit, the trouble signal circuit shall restore to normal automatically without manual restoration of the associated trouble signal silencing switch. A separate LED shall be furnished that shall light in the event a trouble signal is caused by a fault on any alarm signal circuit. Any relays used for sounding alarm signals or trouble signals shall have their coils electrically supervised and shall sound a trouble signal in the event of an open coil.

e. When signals from manual fire alarm boxes and other fire alarm initiating devices within a building are transmitted over the same signaling circuit, there shall be no interference with the fire alarm box signals when both types of signaling devices are operated at or near the same time. Provision of the shunt noninterfering method of operation is acceptable for this performance.

f. When both sprinkler supervisory signals and fire or water flow alarm signals are transmitted over the same signaling line circuit, provision shall be made to either obtain alarm signal precedence, sufficient repetition of the fire alarm signal or other approved method to prevent the loss of any alarm signal.

g. The system shall have dry contacts to transmit system alarm and trouble to the building management and control system (EMCS).

h. System shall be capable of operating in a "degraded" mode of operation. During any main system processor failure or loss of communication network, remote panel initiation device circuits shall be capable of receiving and reporting both alarm and trouble conditions and shall be capable of activating local alarm indicating circuits.

i. In the event of an amplifier, a preamplifier or a tone oscillator failure, the system shall automatically switch all functions performed by that failed unit to an assigned standby unit.

j. Trouble indicators shall be automatically de-energized during a system alarm condition and reappear when the alarm condition has been reset.

6. The system panel shall include a local speaker and switch for testing the prerecorded message.

1.8 MAIN CONTROL PANEL

A. Provide a modular designed, networked, fire alarm control system. Construction shall be modular, solid state; all visual indicators shall be LED or LCD type. All components shall be mounted in locked cabinets meeting UL 894.

B. Daughterboard/motherboard combinations shall be held in place by a tamper proof retainer. Modular circuit boards shall be constructed of protective coated fiberglass. Each component designation shall be silkscreened on the board. Composite disconnects shall be provided to allow testing of field wiring without disturbing panel connections. All functions shall be performed by the manufacturer's standard modules.

1. Master control module with system trouble audible device, system trouble LED/LCD, system reset, internal loop circuit monitor, alarm acknowledgment, and trouble silence features.

2. Module to provide constant supervision of the system's power 120V AC power supply. Module shall be provided with a trouble LED/LCD for incoming phase.

3. The data/phone/voice SLC communication riser loop between fire alarm distribution cabinets (i.e. transponders, etc.) shall be Class X. The SLC wiring shall be UL classified fire-resistive cables meeting NFPA 70 and 72 circuit survivability requirements, NEC Article 760 installation requirements, and manufacturer's recommendations.

4. Initiating circuit modules shall provide intelligent analogue double loop Class "X" SLC circuits to all initiation devices. Each SLC circuit shall support individually addressable devices.

5. Provide power limited supply modules to provide operating power for remote annunciation circuits.

6. A supervised means of manually controlling any door lock release circuits shall be provided.

7. Provide power limited and supervised circuits for power supplies. Auxiliary contacts and relays shall be activated upon device alarm even when other devices within the zone are in alarm.

8. Notification circuits shall be provided from a UL listed solid state signaling module. Circuits shall be capable of driving both audible and visual devices.

9. A power supply module shall provide and supervise operating power and 120V AC power distribution to the system. One back up power supply with automatic switchover from primary power supply will be installed.

10. Contractor shall determine number of initiation circuits required for devices as specified herein and as indicated on the drawings. Contractor shall furnish and install an additional 20% spare initiation control circuits and circuit capacity beyond the base requirements for building owners future use. Contractor shall not load required individual initiation circuits beyond circuit limitations as outlined in NFPA 72. At a minimum, each floor shall be comprised of two (2) initiation SLC circuit loops.

11. Contractor shall furnish and install the required number of main control panels necessary to support the building initiation and signaling audible/visual circuits.

12. NACs shall provide synchronized control of fire alarm flashing strobes. Pulse rate shall comply with latest ADA requirements.

C. Audible/Visual Alarm System

1. Solid state plug-in UL listed control module(s) shall be provided. Modules shall supply Class B (Style Y) type output circuits for control of listed audible devices.

2. Each output circuit shall be capable of being controlled either automatically or manually from the central control panel.

3. Each control module shall have two programmable solid state open collector type outputs in addition to audible applicable circuits.

4. Each module and output circuit shall be supervised for open and shorted circuits which

will indicate with an audible and visual trouble condition at the central control panel.

5. Contractor shall determine number of notification circuits required for devices as specified herein and indicated on the drawings for the base building. Contractor shall furnish and install an additional 20% spare notification circuits and circuit capacity above and beyond the base requirements for building owner's future use. Contractor shall not load required individual base notification circuits beyond the circuit limitations as outlined in NFPA 72. At a minimum, each floor shall be comprised of two (2) independent notification loops (one per zone).

6. Control module shall be provided, which contains a central oscillator motherboard, microphone inputs and dual channel mixer/pre amplifier circuits necessary for proper system operation. Provide two (2) in phase alarm oscillators and two (2) independent auxiliary oscillators. Provide Slow Whoop alarm oscillators which make a slowly ascending tone from 200 to 830Hz in 2.5 seconds.

7. Dual preamplifier module shall provide continuous supervision of the dual pre amplifier, dual alarm oscillators and microphone output levels. This module shall provide individual channel "trouble" LED's, automatic monitoring and switching for all system amplifiers.

8. A control switch module shall be furnished to provide manual access for authorized fire fighting personnel of the audio control module functions. Include "all circuits", "manual alarm", "auxiliary 1 tone", "auxiliary 2 tone", "oscillator stop", and "audio trouble reset" pushbuttons.

9. Audio power amplifiers shall be furnished with self-contained filtered 24v DC power supply and transformer. Each amplifier shall provide a minimum 50 watt RMS output at 25v RMS with a frequency of 400 to 4,000Hz.

10. Each horn circuit shall be supervised so that an opening in any circuit will result in audible and visual trouble indication.

D. Auxiliary Functions

1. Provide auxiliary relays as required to perform the door lock functions specified herein.

2. Relays with 20 amp contacts shall be provided to unlock electrically locked egress doors upon fire-alarm or loss of power. All electric locks on egress doors shall fail safe "open". At a minimum, provide two contacts for future security and door lock requirements.

E. Maintenance Testing

1. Provide a supervised maintenance system to test system components that is acceptable to the local code authorities. The intent is to be able to test any initiation device and to be able to individually or by group disable the automatic audible response, the automatic visual response, the automatic mechanical equipment shut down response, and the door unlock response. The fire alarm system shall indicate trouble/testing mode during the test until returned to fully operational mode. The testing parties shall have procedures to follow as approved by the local authority. These procedures shall be similar to a fire watch mode. The intent is to test the applicable device, confirm the FCP indication, and confirm the FSA and any remote FSA indications without disrupting the building occupants. Provide all the necessary hardware, software, devices, controllers, etc. for a complete and operational supervised maintenance test system.

1.9 MANUAL STATIONS

A. Manual stations shall be double action, individually addressable devices and constructed of Lexan polycarbonate or equal material designed for semi-flush mounting. Stations shall not be of the break-glass design. Stations must be reset with a key. It shall not be possible to close a station without first resetting it. Provide three keys for manual-station reset. Each manual station shall be securely mounted within five feet from the entrance to each exit with the operable part of the manually actuated alarm-initiating device located not less than 42” and not more than 48” above the finished floor. Additional boxes shall be provided so that travel distance to the nearest box will not be in excess of 200 feet horizontal distance on the same floor. Manual stations shall be provided at the following locations and as indicated on the drawings:

1. Each entry/exit to the building.
2. At other locations as required by the Local Code Authority.

1.10 DUCT SMOKE DETECTORS

A. Each duct smoke detector shall be individually addressable and mounted in a housing which will work in conjunction with air sampling tubes supplied to detect products of combustion in the ducts of air handling systems listed for the air velocity present. The housing shall be suitable for mounting directly on the duct or remotely, as indicated on the drawings, where direct mounting is not possible. The unit will be installed under the mechanical section of these specifications, but furnished, wired and tested under this electrical section. The detectors shall be suitable for operation from the power supplied from the central panel. The detector shall include a remote test station with "power on" light, an alarm light, and a test reset switch. Remote test station shall be located in closest mechanical room or back of house space to detector. Coordinate location of station with Engineer. Label test station as "Duct Smoke Detector _____" (insert fan description in blank). The detector shall meet NFPA 90A requirements. The detector's sampling tubes shall span the entire width of the duct. Detector location shall be at least six duct widths downstream from any duct openings, deflection plates, sharp bends, or branch connections. Where this is physically impossible, place as far as possible from the opening, bend, or deflection plate. Each detector base will be required to have one auxiliary dry contact for shutting down any associated supply air fan or Division 23 provided damper. Whether shown on the drawings or not, provide duct smoke detectors in the return air path downstream of air handling units and fans with an air quantity of 2,000 cfm and larger and as required by IMC section 606. Where required by the local AHJ, provide additional duct smoke detectors in the supply air path of air handling units and fans with an air quantity of 2,000 cfm and larger.

1.11 SMOKE DETECTORS

A. Ionization or photoelectric type smoke detectors shall be provided where indicated on the drawings and as indicated below. The type of detector provided at each location shall be whichever type is best suited for that particular application and is acceptable to the Local Code Authorities. Automatic ionization detectors shall be individually addressable, in accordance with NFPA 72, UL listed for 900 square feet, and operate on the ionization principal utilizing the dual chamber design. Automatic photoelectric detectors shall be individually addressable, in accordance with NFPA 72, UL listed for 900 square feet, and operate on the light scattering, photodiode principle. Devices shall be selected based on appropriate air velocity requirements for each application and shall be activated by the presence of products of combustion. An integral indicator light on the device shall provide visual indication of the initiation of an alarm. The base shall have terminals for making all connections and provisions for a remote relay output. One base mounted dry contact shall be furnished as required for local air handling unit, damper, fan, etc… shutdown. Power for detector operation and auxiliary devices shall be via a separate supervised pair of wires. Detectors located in air handling unit plenum spaces shall be

mounted on a conduit box located directly in the path of air to the unit. All smoke detector covers in finished areas shall be white. Ten percent (10%) spare of the installed photoelectric type smoke detectors shall be provided for the owners use (minimum of 1). Smoke detectors shall be provided as follows and as indicated on the drawings:

1. Main electrical room.
2. Each telephone closet
3. Smoke detectors shall be provided in the return air path for all supply, return, and exhaust air distribution systems with an air quantity of 2,000 cfm and larger.
4. Mechanical, plumbing, and air handling unit equipment rooms.
5. At other locations as required by the Local Code Authority.

1.12 VISUAL APPLIANCES

A. Visual appliances shall be a flashing strobe (Xenon flashtube) and shall have a rugged clear Lexan lens with the word "FIRE" lettered in red on the outside. Housings shall have a white or red finish as selected by Owner/Architect. The strobe shall produce minimum visible intensity levels as required by ADA/NFPA at approximately one flash per second and comply with American Disabilities Act (ADA) standards. Strobes shall comply with UL 1971 testing standard.

B. Provide appliances as follows, as indicated on the drawings, and as required per NFPA 72:

1. Each entry/exit from the building.
2. Each restroom.
3. At other locations as required by the Local Code Authority.
4. Provide five (5) additional visual appliances to the Owner. Under this contract, install these additional appliances where directed by the local Fire Marshal. All unused appliances shall remain the property of the Owner.

C. All appliances shall be UL listed as an approved visual device for fire alarm signaling. All appliances installed in the same room or non-partitioned space including corridors shall utilize flash synchronization. Appliances shall incorporate multiple field selectable candela intensity levels.

D. Any appliances located exterior to the building shall be UL cross listed for 1971 and 1638. Devices directly exposed to wet locations shall be installed in a weatherproof and vandal resistant enclosure.

E. Where visual alarms and audible devices are shown on the contract drawings in close proximity, then combination audible/visual devices may be provided. Combination devices shall be listed for the intended service.

F. Ten percent (10%) spare of the installed visual and audio-visual notification devices shall be provided for the owners use (minimum of 1 each).

1.13 FIRE ALARM HORNS

A. Alarm signal horns shall be UL listed wall/ceiling mounted similar to Simplex Model 4901-9820 or approved equal housed in approximately a 5" square circular white or red textured decorative grilles with finish as approved by the Owner/Architect. Provide one (1) alarm signal horn at each location as required by the Local Code Authority and as indicated on the Drawings.

B. Horns located outdoors and in high noise level mechanical spaces shall be water, humidity, and

vermin proof. This unit shall be adjusted to produce 10dB more volume than indoor horns. Outdoor horns shall be similar to Simplex Model No. 4901-9813 or approved equal with selectable volume control.

C. All horns shall be UL listed as an approved audio appliance for fire alarm signaling.

D. Provide 1 additional horn to the Owner. Under this contract, install this additional horn where directed by the local Fire Marshal. All unused horns shall remain the property of the Owner.

1.14 FIRE ALARM TERMINAL CABINET (FATC)

A. Where indicated on the drawings and as required by fire alarm manufacturer and local authority, provide red NEMA 1 metal enclosures to house terminal blocks for appropriate fire alarm sub-systems. Enclosure size shall meet fire alarm manufacturer's requirements and include hinged front access door with key locking provision. Provide five FATC keys to building owner if a FATC is provided. Terminal blocks shall be organized, numbered, and include 25% spare connections.

1.15 NOTIFICATION APPLIANCE CIRCUIT EXTENDER PANEL (NAC)

A. Where required by fire alarm manufacturer and local authority, provide addressable auxiliary power supply, notification appliance extender panels with vented, lockable, NEMA 1 enclosures. Provide five NAC keys to building owner. Each NAC shall include but not be limited to the following:

1. 6 amps of notification appliance power.

2. Advanced microprocessor Control.

3. Four 2 amp minimum power limited notification outputs.

4. Built in strobe synchronization.

5. Supervision of low AC power, low battery condition, earth ground fault, auxiliary output power limit condition, and EOL supervision trouble or power limited condition at an output. Interface these conditions to FCP.

6. Indicator lights for AC power on, battery trouble, ground fault, auxiliary trouble, and output troubles.

7. 120 volt AC input and appropriate output DC voltage.

8. Two inputs for appropriate Class A or Class B systems.

9. Appropriate amp-hours of sealed, lead acid, rechargeable batteries per applicable adopted codes.

1.16 FIRE ALARM TRANSPONDER PANEL (FTR)

A. Wherever a Fire Alarm Transponder Panel (FTR) is required by the manufacturer, provide a UL listed, modular, programmable, microprocessor-based network interface remote fire alarm control panel with the following features for a distributed networked fire alarm system. The FTR shall control all initiating devices and circuits, status inputs, control signals between the devices and the FCP, and all notification appliance circuit extender panels within the zone of

transponder service and/or auxiliary control outputs.

PART 2 - EXECUTION

2.1 FIRE ALARM WIRING

A. Furnish and install all wiring, conduit, and outlet boxes required for the erection of a complete and properly operating system in accordance with applicable National, State and local codes, the manufacturer's recommendations, and the plans and specifications. Specific attention is brought to NFPA 70, Article 760 and Article 770.

B. Fire alarm wiring protection in the designated NEC 708 areas shall comply with NEC 708.14.

C. All wiring shall be in a completely separate Class A conduit system except for a Class B conduit system for the typical floor notification appliance circuits. All system circuit cabling shall be sized and provided so as to allow an additional 25% distance. The minimum wiring requirements shall be provided as follows, unless otherwise required by the local authority:

1. Audible (horn) alarm circuits: twisted jacketed minimum #16 pairs.

2. Telephone circuits: twisted, shielded #18 pairs (Beldon 8760 or equal).

3. SLC & Network cables: Cables shall be protected to resist potential damage by fire or structural failure. Where installed in an Architectural 2 hour fire rated enclosure, the network circuits may be provided as type FPLR or similar as allowed by NEC Article 760. Horizontal offsets located outside Architectural 2 hour fire rated enclosures shall be encased in 2” minimum of concrete. Refer to paragraph 2.4-B-3 for SLC circuiting requirements.

4. Fan Control wiring and other wiring: THHN, size as required.

5. Visual (strobe) notification appliance alarm circuits: twisted jacketed minimum #14 AWG FPL.

D. All system circuiting shall be resistance tested and shall test free of grounds prior to system testing.

2.2 AUDIBLE/VISUAL SIGNALING ALARM CIRCUITING AND AMPLIFICATION

A. Separate audible/visual appliance circuits shall be provided for each floor of the building.

B. No two audible or visual notification appliance circuits located on the same floor shall be assigned to the same control module.

C. No limitation of component power supply or other consideration shall cause the sound output level to deteriorate from those specified in any mode of system operation.

D. All signaling appliance circuiting shall be monitored for tampering, open, short, and ground circuits in compliance with NFPA 72. Signaling J-boxes on the outside of the building, shall be supervised for tampering.

E. A maximum of ten (10) circuits shall be connected to any amplifier.

F. One (1) standby amplifier shall be provided for every system amplifier.

G. Sufficient horn amplification capacity shall be provided to operate all horns indicated on the plans, plus an additional 20% spare future capacity. Calculations shall accompany the contractor's submission.

H. Sufficient strobe device capacity shall be provided to operate all strobes indicated on the plans, plus an additional 20% spare future capacity. Calculations shall be based on the type of strobe. Calculations shall accompany the contractor’s submission.

2.3 SYSTEM TESTS

A. Prior to any fire alarm system testing by the local fire marshal's office or system demonstrations to the owner's representative, the electrical contractor and fire alarm system supplier shall have thoroughly tested, trouble-shot and de-bugged the entire fire alarm system in accordance with the testing requirements listed hereinafter. The Contractor shall provide a written certification statement for review by the Owner's representative indicating the system has been thoroughly tested, trouble-shot, and debugged prior to demonstrating the system to the local fire marshal and Engineer.

B. The manufacturer's authorized representative shall provide supervision of final connections to the system panel, perform a complete functional test of the system and submit a written report to the contractor attesting to the proper operation of the completed system. The written report shall be certified and signed by the electrical contractor and fire alarm system manufacturer in accordance with the certification requirements of Section 28 30 10. The report shall list measured sound pressure levels of 10% of the horns on each floor. Measurements shall be taken by the manufacturer's technical representative.

C. After system testing, it shall be the responsibility of the contractor to demonstrate the fire alarm system to the owner's representative and the local fire marshal's representative as required to obtain system approval and building occupancy permit. Owner and fire marshal demonstrations may be independent of each other at owner's discretion. A representative for the building owner and the manufacturer's authorized technical representative shall be present during all tests and inspections conducted by the local Fire Marshal's office.

D. A sound pressure level meter shall be provided by the manufacturer for the use of the fire marshal during testing. A Record of Completion in accordance with NFPA 72 shall be completed and accepted by the fire department prior to final acceptance by building owner.

E. Any required retests shall be made by the Contractor at no expense to the Owner including engineering fees to witness system retests. The Contractor shall provide at least three weeks written notice to the Owner, Engineer, Local Building Department and the local Fire Marshal's office prior to the date of testing.

F. Upon satisfactory completion of all system tests, the manufacturer's representative shall present for the owner's consideration a proposal to provide maintenance, inspections, and tests of the system. This proposal and the requirements shall conform to NFPA 72.

2.4 OPERATING INSTRUCTIONS AND DRAWINGS

A. Upon completion of installation, printed system instructions and as built wiring diagrams illustrating complete system wiring, central control station schematics and device addresses and locations shall be furnished to the Owner.

B. The manufacturer shall provide an authorized representative to instruct and train building fire department personnel in the operation and maintenance of the system. Manufacturer should

allow a minimum of forty (40) man-hours of instruction time.

2.5 LICENSED SOFTWARE

A. The fire alarm system vendor shall provide to the Owner prior to final review and acceptance a copy of fire alarm system data and program messages.

2.6 WARRANTY, SERVICE AND REPLACEMENT PARTS

A. Basic Warranty and Service: All equipment and wiring shall be guaranteed against defects in materials and workmanship for a one (1) year period from substantial completion. Warranty service for the equipment shall be provided by the manufacturer's factory trained service representative during normal working hours (Monday through Friday, 8:00 a.m. to 5:00 p.m. excluding holidays). Emergency service provided at times other than as stipulated above, shall be available on a 24 hour basis from the same source at additional cost to the Owner.

B. The manufacturer for each of the fire alarm system equipment provided on this project shall certify that authorized factory support and an adequate supply of replacement parts will be available to the Owner for a minimum of ten (10) years from the date of the Owner’s acceptance of the system. The requirement for replacement parts shall apply to all specialty items and proprietary components that are only available from the system manufacturer.

C. Fire Alarm System Extended Warranty and Service Contract Alternate: The manufacturer shall submit with their bid the cost, as an alternate, to provide four (4) additional years of service beyond the minimum one (1) year warranty period. Pricing shall be provided for each additional year of service. This additional service shall commence one (1) year following the date of substantial completion. The first year of service after substantial completion shall be included in the base bid price.

END OF SECTION 28 31 00

# SECTION 311000 - SITE CLEARING

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Protecting existing vegetation to remain.
2. Removing existing vegetation.
3. Clearing and grubbing.
4. Stripping and stockpiling topsoil.
5. Stripping and stockpiling rock.
6. Removing above- and below-grade site improvements.
7. Disconnecting, capping or sealing, and removing site utilities.
8. Temporary erosion and sedimentation control.

B. Related Requirements:

1. Section 015000 "Temporary Facilities and Controls" for temporary erosion- and sedimentation-control measures.

### 1.3 DEFINITIONS

A. Subsoil: Soil beneath the level of subgrade; soil beneath the topsoil layers of a naturally occurring soil profile, typified by less than 1 percent organic matter and few soil organisms.

B. Surface Soil: Soil that is present at the top layer of the existing soil profile. In undisturbed areas, surface soil is typically called "topsoil," but in disturbed areas such as urban environments, the surface soil can be subsoil.

C. Topsoil: Top layer of the soil profile consisting of existing native surface topsoil or existing in-place surface soil; the zone where plant roots grow.

D. Topsoil: Top layer of the soil profile consisting of existing native surface topsoil or existing in-place surface soil; the zone where plant roots grow. Its appearance is generally friable, pervious, and black or a darker shade of brown, gray, or red than underlying subsoil; reasonably free of subsoil, clay lumps, gravel, and other objects larger than 2 inches (50 mm) in diameter; and free of weeds, roots, toxic materials, or other nonsoil materials.

E. Plant-Protection Zone: Area surrounding individual trees, groups of trees, shrubs, or other vegetation to be protected during construction and indicated on Drawings.

F. Tree-Protection Zone: Area surrounding individual trees or groups of trees to be protected during construction and indicated on Drawings.

G. Vegetation: Trees, shrubs, groundcovers, grass, and other plants.

## 1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

## 1.5 MATERIAL OWNERSHIP

A. Except for materials indicated to be stockpiled or otherwise remain Owner's property, cleared materials shall become Contractor's property and shall be removed from Project site.

## 1.6 INFORMATIONAL SUBMITTALS

A. Existing Conditions: Documentation of existing trees and plantings, adjoining construction, and site improvements that establishes preconstruction conditions that might be misconstrued as damage caused by site clearing.

1. Use sufficiently detailed photographs or video recordings.
2. Include plans and notations to indicate specific wounds and damage conditions of each tree or other plant designated to remain.

B. Topsoil stripping and stockpiling program.

C. Record Drawings: Identifying and accurately showing locations of capped utilities and other subsurface structural, electrical, and mechanical conditions.

## 1.7 QUALITY ASSURANCE

A. Topsoil Stripping and Stockpiling Program: Prepare a written program to systematically demonstrate the ability of personnel to properly follow procedures and handle materials and equipment during the Work. Include dimensioned diagrams for placement and protection of stockpiles.

B. Rock Stockpiling Program: Prepare a written program to systematically demonstrate the ability of personnel to properly follow procedures and handle materials and equipment during the Work. Include dimensioned diagrams for placement and protection of stockpiles.

## 1.8 FIELD CONDITIONS

A. Traffic: Minimize interference with adjoining roads, streets, walks, and other adjacent occupied or used facilities during site-clearing operations.

1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction.
2. Provide alternate routes around closed or obstructed trafficways if required by Owner or authorities having jurisdiction.

B. Utility Locator Service: Notify Miss Utility for area where Project is located before site clearing.

C. Do not commence site clearing operations until temporary erosion- and sedimentation-control measures are in place.

D. Tree- and Plant-Protection Zones: Protect according to requirements in Section 015639 "Temporary Tree and Plant Protection."

E. Soil Stripping, Handling, and Stockpiling: Perform only when the soil is dry or slightly moist.

## PART 2 - PRODUCTS

### 2.1 MATERIALS

A. Satisfactory Soil Material: Requirements for satisfactory soil material are specified in Section 312000 "Earth Moving" when applicable.

1. Obtain approved borrow soil material off-site when satisfactory soil material is not available on-site.

## PART 3 - EXECUTION

### 3.1 PREPARATION

A. Protect and maintain benchmarks and survey control points from disturbance during construction.

B. Verify that trees, shrubs, and other vegetation to remain or to be relocated have been flagged and that protection zones have been identified and enclosed according to requirements in Section 015639 "Temporary Tree and Plant Protection."

C. Protect existing site improvements to remain from damage during construction.

1. Restore damaged improvements to their original condition, as acceptable to Owner.

### 3.2 TEMPORARY EROSION AND SEDIMENTATION CONTROL

A. Provide temporary erosion- and sedimentation-control measures to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways, according to erosion- and sedimentation-control Drawings and requirements of authorities having jurisdiction.

B. Verify that flows of water redirected from construction areas or generated by construction activity do not enter or cross protection zones.

C. Inspect, maintain, and repair erosion- and sedimentation-control measures during construction until permanent vegetation has been established.

D. Remove erosion and sedimentation controls and restore and stabilize areas disturbed during removal.

3.3 TREE AND PLANT PROTECTION

A. Protect trees and plants remaining on-site according to requirements in Section 015639 "Temporary Tree and Plant Protection."

B. Repair or replace trees, shrubs, and other vegetation indicated to remain or be relocated that are damaged by construction operations according to requirements in Section 015639 "Temporary Tree and Plant Protection."

3.4 EXISTING UTILITIES

A. Owner will arrange for disconnecting and sealing indicated utilities that serve existing structures before site clearing, when requested by Contractor.

1. Verify that utilities have been disconnected and capped before proceeding with site clearing.

B. Locate, identify, disconnect, and seal or cap utilities indicated to be removed or abandoned in place.

1. Arrange with utility companies to shut off indicated utilities.
2. Owner will arrange to shut off indicated utilities when requested by Contractor.

C. Locate, identify, and disconnect utilities indicated to be abandoned in place.

D. Interrupting Existing Utilities: Do not interrupt utilities serving facilities occupied by Owner or others, unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

1. Notify Architect not less than two days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without Architect's written permission.

E. Excavate for and remove underground utilities indicated to be removed.

3.5 CLEARING AND GRUBBING

A. Remove obstructions, trees, shrubs, and other vegetation to permit installation of new construction.

1. Do not remove trees, shrubs, and other vegetation indicated to remain or to be relocated.
2. Grind down stumps and remove roots larger than 2 inches in diameter, obstructions, and debris to a depth of 18 inches below exposed subgrade.
3. Use only hand methods or air spade for grubbing within protection zones.
4. Chip removed tree branches and dispose of off-site.

B. Fill depressions caused by clearing and grubbing operations with satisfactory soil material unless further excavation or earthwork is indicated.

1. Place fill material in horizontal layers not exceeding a loose depth of 8 inches and compact each layer to a density equal to adjacent original ground.

## 3.6 TOPSOIL STRIPPING

A. Remove sod and grass before stripping topsoil.

B. Strip topsoil to depth of 6 inches in a manner to prevent intermingling with underlying subsoil or other waste materials.

1. Remove subsoil and nonsoil materials from topsoil, including clay lumps, gravel, and other objects larger than 2 inches in diameter; trash, debris, weeds, roots, and other waste materials.

C. Stockpile topsoil away from edge of excavations without intermixing with subsoil or other materials. Grade and shape stockpiles to drain surface water. Cover to prevent windblown dust and erosion by water.

1. Limit height of topsoil stockpiles to 72 inches.
2. Do not stockpile topsoil within protection zones.
3. Dispose of surplus topsoil. Surplus topsoil is that which exceeds quantity indicated to be stockpiled or reused.

## 3.7 SITE IMPROVEMENTS

A. Remove existing above- and below-grade improvements as indicated and necessary to facilitate new construction.

B. Remove slabs, paving, curbs, gutters, and aggregate base as indicated.

1. Unless existing full-depth joints coincide with line of demolition, neatly saw-cut along line of existing pavement to remain before removing adjacent existing pavement. Saw-cut faces vertically.
2. Paint cut ends of steel reinforcement in concrete to remain with two coats of antirust coating, following coating manufacturer's written instructions. Keep paint off surfaces that will remain exposed.

## 3.8 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Remove surplus soil material, unsuitable topsoil, obstructions, demolished materials, and waste materials including trash and debris, and legally dispose of them off Owner's property.

B. Separate recyclable materials produced during site clearing from other nonrecyclable materials. Store or stockpile without intermixing with other materials and transport them to recycling facilities. Do not interfere with other Project work.

END OF SECTION 311000

# SECTION 311020 – EROSION AND SEDIMENT CONTROL

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section Includes:

1. Requirements for temporary erosion and sedimentation control.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site. The Contractor shall present a general overview of erosion and sediment control measures will be implemented of the project.

### 1.3 CONTRACTOR PERSONNEL

A. The Contractor shall assign a Certified Employee to the project site in the capacity of the Responsible land Disturber (RLD).

### 1.4 STANDARDS AND SPECIFICATIONS

A. Temporary erosion and sediment control measures shall be constructed in conformance with “Chapter 5 Erosion and Sediment Control and Stormwater Management Requirements” and “Chapter 7 Design Specifications for Construction BMPs” of the latest Virginia Stormwater Management Handbook, published by the Virginia Department of Environmental Quality (VA DEQ).

B. The installation and maintenance of temporary erosion and sediment control devices shall be in accordance with the handbook.

### 1.5 CONSTRUCTION OPERATION

A. Storage of construction vehicles, trucks, and equipment is limited to the area within the limits of the temporary 6 foot chain link safety fencing.

## PART 2 - PRODUCTS

### 2.1 STABILIZATION MATERIALS

A. Seed, mulch, fertilizer, soil conditioner and other materials for seeding and soil stabilization shall meet the requirements of the Virginia Stormwater Management Handbook, latest edition.

2.2 GEOTEXTILE (FILTER CLOTH)

A. Geotextiles shall conform to the standards and specifications of the Contract Drawings and shall meet the requirements of the Virginia Stormwater Management Handbook, latest edition.

2.3 STONE

A. Stone 2 to 3 inches in size shall conform to AASHTO M43, No. 2.

B. Stone 3/4 –inch to 12-inch shall conform to AASHTO M43, No.4.

C. VDOT No. 1 Coarse Aggregate, Section 203.

D. VDO No. 57 Coarse Aggregate, Section 203.

2.4 SILT FENCE

A. Silt fence shall be installed in the locations reflected on the Contract Drawings.

B. Virginia Stormwater Management Handbook Chapter 7, Figure C-PCM-04-4.

2.5 INLET PROTECTION

A. Silt fence shall be installed in the locations reflected on the Contract Drawings. The storm drain inlet protection shall be constructed at the drop inlets as indicated and maintained in accordance with Virginia Stormwater Management Handbook, to the satisfaction of the inspector until the site has been stabilized.

B. Virginia Stormwater Management Handbook Chapter 7, Figure C-SCM-04-7.

PART 3 - EXECUTION

3.1 PREPARATION

A. Protect and maintain benchmarks and survey control points from disturbance during construction.

B. Install the Phase 1 Erosion and Sediment Controls as shown on the Contract Drawings. All measures shall be inspected and approved by the owner and the local jurisdiction prior to beginning any land disturbance. The Contractor shall ensure all runoff from the disturbed area is directed to the erosion and sediment control measures.

C. Verify that flows of water redirected from construction areas or generated by construction activity do not enter or cross protection zones.

D. Inspect, maintain, and repair erosion- and sedimentation-control measures during construction until permanent vegetation has been established.

E. The Contractor shall not remove any erosion and sediment control measure without the approval of the local jurisdiction.

3.2 TEMPORARY EROSION AND SEDIMENTATION CONTROL SEQUENCE OF CONSTRUCTION

A. The Contractor shall implement the Erosion and Sediment Control Plan (E&S Plan) and sequence of construction reflected in the Contract Documents. Minor adjustments to the sediment control measures may be made in the field with the prior approval of the inspector.

B. Major revisions, deletions, or substitutions to the E&S Plan will require a formal review and approval by the Engineer and the inspector prior to implementing the change. Changes to the approved E&S Plan shall be submitted to the inspector in writing at least seven (7) days prior to implementing the change.

C. The Contractor shall apply for and obtain the Virginia Pollutant Discharge Elimination System (VPDES) Construction Permit.

3.3 RESPONSIBLE LAND DISTURBER

A. At least 10 days prior to applying to the permit, the name and credentials of the Responsible Land Disturber (RLD) shall be submitted to the Engineer for approval. Any substitutes for the RLD will be subject to the approval of the Engineer. The substitution shall be timed to ensure that a RLD is assigned to the project at all times.

B. The RLD shall be thoroughly experienced in all aspects of construction and have satisfactorily completed the Erosion and Sediment Control Training Program either conducted or authorized by the Commonwealth of Virginia pursuant to the appropriate legislation. The RLD shall have the primary responsibility and authority for the implementation of the approved erosion and sediment control plans, schedules and methods of operation for both onsite and offsite activities.

C. The RLD’s duties shall include”:

1. Inspect all erosion and sediment controls on a daily basis to ensure that all controls are in place at all times and to develop a list of activities and schedules to ensure conformance with the Contract Documents.

2. Maintain a daily log of these inspections, including actions taken and submit a written report to the Engineer at the end of the week.

3. Conduct after storm inspections with the Engineer both during and beyond normal working hours/days and submit a written report.

4. Be assigned the authority by the Contractor to mobilize crews to make immediate repairs to the controls during working and nonworking hours.

5. When requested, accompany the Engineer on Quality Assurance Inspections and inspections made by the regulating agencies.

6. Coordinate with the Engineer to ensure all the corrections are made immediately and that the project is in compliance with the approved plans.

3.4 MEETINGS

A. At least seven (7) working days prior to the start of work, the Contractor shall initiate and conduct an Erosion and Sediment Control Field Meeting.

3.5 INITIAL CONTROLS

A. All perimeter controls such as silt fence, earth dikes/swales, check dams, traps, basins, etc., shall be installed prior to the clearing and grubbing operation. Remove sod and grass before stripping topsoil.

B. If the inspector determines that the clearing area has been disturbed and a potential for sediment runoff or erosion exists, the inspector will direct the Contractor to install the controls at that time.

3.6 STABILIZATION REQUIREMENTS

A. Areas flatter than 3:1 and stockpile areas shall be permanently or temporarily stabilized as soon as possible, but not later than fourteen days after grubbing and grading activities have ceased in the area. Trap embankments and slopes, earth dikes, temporary swales, perimeter dike/swales, ditches, and slopes 3:1 or steeper shall be permanently or temporarily stabilized as soon as possible, but not later than seven days after grubbing and grading activities have ceased in the area. Slope 2:1 and steeper shall be stabilized with slope stabilization mat. The seven and fourteen day requirements mean that the stabilization operation is complete within the applicable seven or fourteen day time frame.

B. When permanently stabilized areas are disturbed by the Contractor's grading operation or other activities not specifically approved by the Engineer, the restabilization will be at no additional cost to the Town. Stabilization requirements may be reduced to less than seven days for sensitive areas and can be required at any point in time by the local jurisdiction. Maintenance shall be performed as necessary to ensure continued stabilization.

C. All slopes shall be tracked within five days of establishment with cleated type equipment operating perpendicular to the slope.

3.7 MAINTENANCE

A. All erosion and sediment control devices shall be maintained during the construction season, the winter months, and other times when the project is shut down. Access shall be maintained to all erosion and sediment controls until the controls are removed. Lack of maintenance by the Contractor will be considered as noncompliance with the E&S Plan.

B. Controls shall be inspected immediately following storm events. The Contractor shall repair controls when damaged and clean out controls as necessary as the first order of business after a storm event.

C. Any pumping activity, including dewatering sediment traps and basins, shall be directed through a dewatering device approved by the Engineer.

D. The Contractor shall remove accumulated sediment from sediment controls or other areas during routine maintenance of sediment controls, or as directed by the Engineer. Silt fence, super silt fence, stone outlet structures, stone check dams, sediment control logs, and straw bales shall have sediment removed when it reaches 50 percent of the height of the control device.

E. Sediment removed from control devices shall be placed in an approved waste site either on or off the project. Material stored on-site may be reused once it has dried and it conforms to project requirements for embankment.

3.8 WASTE AREAS

A. Waste areas on the site shall be in locations approved by the owner/inspector and shall be protected by erosion and sediment control measures and stabilized within the seven or fourteen day stabilization requirement. Debris and other demolition items that will ultimately be transported off site to the landfill are not to be temporarily stored off of the Project limits of disturbance.

3.9 INSPECTIONS

A. Field inspections relative to erosion and sediment control compliance may be conducted. If noncompliance with erosion and sediment control provisions are found, the Contractor shall conduct corrective action immediately. This corrective action may require a shutdown of construction activities until the noncompliance is satisfactorily corrected, and no claims against the project will be considered due to a shutdown of the grading operations or the entire project.

3.10 REMOVAL OF CONTROLS

A. No erosion and sediment control measures shall be removed until all previously disturbed areas are vegetated and the removal has been approved by the inspector and local jurisdictions. The sediment controls shall be backfilled, graded, and stabilized as specified in the Contract Documents.

B. All control devices shall be removed, except where an attempt to remove a particular control may severely disturb an area that has been stabilized. When a sediment trap or stone outlet structure is placed at the bottom of a fill greater than 8 feet, the controls may be left in place as determined by the Engineer. Sediment traps left in place shall be stabilized by placing erosion control matting over a permanent seed mix.

3.11 SILT FENCE

A. The geotextile shall be trenched a minimum of 8 inches into the ground and extend a minimum of 22 inches above ground. Silt fence shall be removed and reset when and as directed by the Engineer. All of the requirements for the original placement of the silt fence shall be strictly adhered to when the fence is reset.

3.12 STABILIZED CONSTRUCTION ENTRANCE

A. Stabilized construction entrances shall be located as specified in the Contract Documents or as directed by the Engineer. Rehabilitation of stabilized construction entrance shall consist of periodic top dressing with additional VDOT No. 1 aggregate, replacement of pipe, or other repairs to the entrance and sediment trapping devices as needed or as directed by the Engineer.

END OF SECTION 311020

SECTION 312000-EARTH MOVING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
1. Excavating and filling for rough grading the Site.
2. Preparing subgrades for slabs-on-grade, walks, pavements, turf and grasses and plants.
3. Excavating and backfilling for buildings and structures.
4. Drainage course for concrete slabs-on-grade.
5. Subbase course for concrete walks and pavements.
6. Subbase course and base course for asphalt paving.
7. Excavating and backfilling trenches for utilities and pits for buried utility structures.

B. Related Sections:
1. Division 01 Section “Construction Progress Documentation” for recording pre-excavation and earth moving progress.
2. Division 01 Section "Temporary Facilities, Utilities, Access and Construction Aids" for temporary controls, utilities, and support facilities; also for temporary site fencing if not in another Section.
3. Division 03 Section "Cast-in-Place Concrete" for granular course if placed over vapor retarder and beneath the slab-on-grade.
4. Divisions 21, 22, 23, 26, 27, 28, and 33 Sections for installing underground mechanical and electrical utilities and buried mechanical and electrical structures.
5. Division 31 Section "Site Clearing" for site stripping, grubbing, stripping and stockpiling topsoil, and removal of above- and below-grade improvements and utilities.
6. Division 31 Section "Dewatering" for lowering and disposing of ground water during construction.
7. Division 31 Section "Excavation Support and Protection" for shoring, bracing, and sheet piling of excavations.
8. Division 31 Section "Drilled Concrete Piers and Shafts" for excavation of shafts and disposal of surplus excavated material.
9. Division 33 Section "Subdrainage" for drainage of foundations, slabs-on-grade walls, and landscaped areas.

1.03 UNIT PRICES

Section left blank on purpose.

1.04 DEFINITIONS

A. Backfill: Soil material or controlled low-strength material used to fill an excavation.
1. Initial Backfill: Backfill placed beside and over pipe in a trench, including haunches to support sides of pipe.
2. Final Backfill: Backfill placed over initial backfill to fill a trench.

B. Base Course: Aggregate layer placed between the subbase course and hot-mix asphalt paving.

C. Bedding Course: Aggregate layer placed over the excavated subgrade in a trench before laying pipe.

D. Borrow Soil: Satisfactory soil imported from off-site for use as fill or backfill.

E. Drainage Course: Aggregate layer supporting the slab-on-grade that also minimizes upward capillary flow of pore water.

F. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.
   1. Authorized Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by the Engineer. Authorized additional excavation and replacement material will be paid for according to Contract provisions for changes in the Work.
   2. Bulk Excavation: Excavation more than 10 feet in width and more than 30 feet in length.
   3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction by the Engineer. Unauthorized excavation, as well as remedial work directed by the Engineer, shall be without additional compensation.

G. Fill: Soil materials used to raise existing grades.

H. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material that exceed 1 cu. yd. for bulk excavation or 3/4 cu. yd. for footing, trench, and pit excavation that cannot be removed by rock excavating equipment equivalent to the following in size and performance ratings, without systematic drilling, ram hammering, ripping, or blasting, when permitted:
   1. Excavation of Footings, Trenches, and Pits: Late-model, track-mounted hydraulic excavator; equipped with a 42-inch-wide, maximum, short-tip-radius rock bucket; rated at not less than 138-hp flywheel power with bucket-curling force of not less than 28,700 lbf and stick-crowd force of not less than 18,400 lbf with extra-long reach boom; measured according to SAE J-1179.
   2. Bulk Excavation: Late-model, track-mounted loader; rated at not less than 230-hp flywheel power and developing a minimum of 47,992-lbf breakout force with a general-purpose bare bucket; measured according to SAE J-732.

I. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material 3/4 cu. yd. or more in volume that exceed a standard penetration resistance of 100 blows/2 inches when tested by a geotechnical testing agency, according to ASTM D 1586.

J. Structures: Buildings, footings, foundations, retaining walls, slabs, tanks, curbs, mechanical and electrical appurtenances, or other man-made stationary features constructed above or below the ground surface.

K. Subbase Course: Aggregate layer placed between the subgrade and base course for hot-mix asphalt pavement, or aggregate layer placed between the subgrade and a cement concrete pavement or a cement concrete or hot-mix asphalt walk.

L. Subgrade: Uppermost surface of an excavation or the top surface of a fill or backfill immediately below subbase, drainage fill, drainage course, or topsoil materials.

M. Utilities: On-site underground pipes, conduits, ducts, and cables, as well as underground services within buildings.

1.05 SUBMITTALS

A. Product Data: For each type of the following manufactured products required:
   1. Geotextiles.
   2. Controlled low-strength material, including design mixture.
   3. Geofoam.
   4. Warning tapes.

B. Samples for Verification: For the following products, in sizes indicated below:
   1. Geotextile: 12 by 12 inches.
   2. Warning Tape: 12 inches long; of each color.

C. Qualification Data: For qualified testing agency.

D. Material Test Reports: For each on-site and borrow soil material proposed for fill and backfill as follows:
   1. Classification according to ASTM D 2487.
   2. Laboratory compaction curve according to ASTM D 698 and ASTM D 1557.

E. Blasting plan approved by authorities having jurisdiction.

F. Seismic survey report from seismic survey agency.

G. Pre-excavation Photographs or Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces that might be misconstrued as damage caused by earth moving operations. Submit before earth moving begins.

1.06 QUALITY ASSURANCE

A. Blasting: Comply with applicable requirements in NFPA 495, "Explosive Materials Code," and prepare a blasting plan reporting the following:
   1. Types of explosive and sizes of charge to be used in each area of rock removal, types of blasting mats, sequence of blasting operations, and procedures that will prevent damage to site improvements and structures on Project site and adjacent properties.
   2. Seismographic monitoring during blasting operations.
   3. It is the Contractors responsibility to ensure that the Blasting Plan meets the requirements of all applicable local jurisdictions/agencies (Town of Strasburg, Shenandoah County, etc.) and to get it approved by all applicable jurisdictions/agencies prior to any beginning blasting operations

B. Seismic Survey Agency: An independent testing agency, acceptable to authorities having jurisdiction, experienced in seismic surveys and blasting procedures to perform the following services:
   1. Report types of explosive and sizes of charge to be used in each area of rock removal, types of blasting mats, sequence of blasting operations, and procedures that will prevent damage to site improvements and structures on Project site and adjacent properties.
   2. Seismographic monitoring during blasting operations.

C. Geotechnical Testing Agency Qualifications: Qualified according to ASTM E 329 and ASTM D 3740 for testing indicated.

D. Pre-excavation Conference: Conduct conference at Project site in accordance with the inspection schedule associated with the Approved Grading Permit.

1.07 FIELD CONDITIONS

A. Traffic: Minimize interference with adjoining roads, streets, walks, and other adjacent occupied or used facilities during earth moving operations.
   1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction.
   2. Provide alternate routes around closed or obstructed traffic ways if required by Owner or authorities having jurisdiction.

B. Utility Locator Service: Notify utility locator service "Miss Utility" for area where Project is located before beginning earth moving operations.

C. Do not commence earth moving operations until temporary erosion- and sedimentation-control measures, specified in Division 31 Section "Site Clearing," are in place.

D. Do not commence earth moving operations until plant-protection measures specified in Division 01 Section "Temporary Tree and Plant Protection" are in place.

E. The following practices are prohibited within protection zones:
1. Storage of construction materials, debris, or excavated material.
2. Parking vehicles or equipment.
3. Foot traffic.
4. Erection of sheds or structures.
5. Impoundment of water.
6. Excavation or other digging unless otherwise indicated.
7. Attachment of signs to or wrapping materials around trees or plants unless otherwise indicated.

F. Do not direct vehicle or equipment exhaust towards protection zones.

G. Prohibit heat sources, flames, ignition sources, and smoking within or near protection zones.

PART 2 - PRODUCTS

2.01 SOIL MATERIALS

A. General: Provide borrow soil materials when sufficient satisfactory soil materials are not available from excavations.

B. Unsatisfactory Soils: Soil Classification Groups CL, ML, OL, CH, MH, OH, and PT according to ASTM D 2487, or a combination of these groups.
1. Unsatisfactory soils also include satisfactory soils not maintained within 2 percent of optimum moisture content at time of compaction.

C. Subbase Material: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940; with at least 90 percent passing a 1-1/2-inch sieve and not more than 12 percent passing a No. 200 sieve.

D. Base Course: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940; with at least 95 percent passing a 1-1/2-inch sieve and not more than 8 percent passing a No. 200 sieve.

E. Engineered Fill: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940/D 2940M; with at least 90 percent passing a 1-1/2-inch sieve and not more than 12 percent passing a No. 200 sieve. Fill materials shall have a minimum standard Proctor (ASTM D 698) maximum dry density of approximately 100 pounds per cubic feet for fill materials. A mixture of on-site soils and boulders/cobbles is not an acceptable fill material.

F. Bedding Course: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940; except with 100 percent passing a 1-inch sieve and not more than 8 percent passing a No. 200 sieve.

G. Drainage Course: Narrowly graded mixture of washed crushed stone or crushed or uncrushed gravel; ASTM D 448; coarse-aggregate grading Size 57; with 100 percent passing a 1-1/2-inch sieve and 0 to 5 percent passing a No. 8 sieve.

H. Filter Material: Narrowly graded mixture of natural or crushed gravel, or crushed stone and natural sand; ASTM D 448; coarse-aggregate grading Size 67; with 100 percent passing a 1-inch sieve and 0 to 5 percent passing a No. 4 sieve.

I. Sand: ASTM C 33; fine aggregate.

J. Impervious Fill: Clayey gravel and sand mixture capable of compacting to a dense state.

2.02 GEOTEXTILES

A. Subsurface Drainage Geotextile: Nonwoven needle-punched geotextile, manufactured for subsurface drainage applications, made from polyolefins or polyesters; with elongation greater than 50 percent; complying with AASHTO M 288 and the following, measured per test methods referenced:
1. Survivability: Class 2; AASHTO M 288.
2. Grab Tensile Strength: 157 lbf; ASTM D 4632.
3. Sewn Seam Strength: 142 lbf; ASTM D 4632.
4. Tear Strength: 56 lbf; ASTM D 4533.
5. Puncture Strength: 56 lbf; ASTM D 4833.
6. Apparent Opening Size: No. 40 sieve, maximum; ASTM D 4751.
7. Permittivity: 0.5 per second, minimum; ASTM D 4491.
8. UV Stability: 50 percent after 500 hours' exposure; ASTM D 4355.

B. Separation Geotextile: Woven geotextile fabric, manufactured for separation applications, made from polyolefins or polyesters; with elongation less than 50 percent; complying with AASHTO M 288 and the following, measured per test methods referenced:
1. Survivability: Class 2; AASHTO M 288.
2. Grab Tensile Strength: 247 lbf; ASTM D 4632.
3. Sewn Seam Strength: 222 lbf; ASTM D 4632.
4. Tear Strength: 90 lbf; ASTM D 4533.
5. Puncture Strength: 90 lbf; ASTM D 4833.
6. Apparent Opening Size: No. 60 sieve, maximum; ASTM D 4751.
7. Permittivity: 0.02 per second, minimum; ASTM D 4491.
8. UV Stability: 50 percent after 500 hours' exposure; ASTM D 4355.

2.03 CONTROLLED LOW-STRENGTH MATERIAL

A. Controlled Low-Strength Material: Self-compacting, low-density, flowable concrete material produced from the following:
1. Portland Cement: ASTM C 150, Type I, Type II, or Type III.
2. Fly Ash: ASTM C 618, Class C or F.
3. Normal-Weight Aggregate: ASTM C 33, 3/4-inch nominal maximum aggregate size.
4. Foaming Agent: ASTM C 869.
5. Water: ASTM C 94/C 94M.
6. Air-Entraining Admixture: ASTM C 260.

B. Produce low-density, controlled low-strength material with the following physical properties:
1. As-Cast Unit Weight: 30 to 36 lb/cu. ft. at point of placement, when tested according to ASTM C 138/C 138M.
2. Compressive Strength: 80 psi, when tested according to ASTM C 495.

C. Produce conventional-weight, controlled low-strength material with 80-psi compressive strength when tested according to ASTM C 495.

2.04 ACCESSORIES

A. Warning Tape: Acid- and alkali-resistant, polyethylene film warning tape manufactured for marking and identifying underground utilities, 6 inches wide and 4 mils thick, continuously inscribed with a description of the utility; colored as follows:
1. Red: Electric.
2. Yellow: Gas, oil, steam, and dangerous materials.
3. Orange: Telephone and other communications.
4. Blue: Water systems.
5. Green: Sewer systems.

B. Detectable Warning Tape: Acid- and alkali-resistant, polyethylene film warning tape manufactured for marking and identifying underground utilities, a minimum of 6 inches wide and 4 mils thick, continuously inscribed with a description of the utility, with metallic core encased in a protective jacket for corrosion protection, detectable by metal detector when tape is buried up to 30 inches deep; colored as follows:
1. Red: Electric.
2. Yellow: Gas, oil, steam, and dangerous materials.
3. Orange: Telephone and other communications.
4. Blue: Water systems.
5. Green: Sewer systems.

PART 3 - EXECUTION

3.01 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earth moving operations.

B. Protect and maintain erosion and sedimentation controls during earth moving operations.

C. Protect subgrades and foundation soils from freezing temperatures and frost. Remove temporary protection before placing subsequent materials.

3.02 DEWATERING

A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.
1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.

3.03 EXPLOSIVES

A. Explosives: Obtain written permission from the Owner and authorities having jurisdiction before bringing explosives to Project site or using explosives on Project site.
1. Blasting operations are to be in accordance with the approved Blasting Plan (approved by all applicable jurisdictions/agencies). It is the Contractor’s responsibility to ensure that this Blasting Plan is properly approved and that blasting operations are performed according to it. It is also the Contractor’s responsibility to ensure that all applicable regulations regarding blasting (including setbacks from existing utilities and structures) are adhered to.

2. Perform blasting without damaging adjacent structures, property, or site improvements.
3. Perform blasting without weakening the bearing capacity of rock subgrade and with the least practicable disturbance to rock to remain.

3.04 EXCAVATION, GENERAL

A. Unclassified Excavation: Excavate to subgrade elevations regardless of the character of surface and subsurface conditions encountered. Unclassified excavated materials may include rock, soil materials, and obstructions. No changes in the Contract Sum or the Contract Time will be authorized for rock excavation or removal of obstructions.
   1. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.
   2. Remove rock to lines and grades indicated to permit installation of permanent construction without exceeding the following dimensions:
      a. 24 inches outside of concrete forms other than at footings.
      b. 12 inches outside of concrete forms at footings.
      c. 6 inches outside of minimum required dimensions of concrete cast against grade.
      d. Outside dimensions of concrete walls indicated to be cast against rock without forms or exterior waterproofing treatments.
      e. 6 inches beneath bottom of concrete slabs-on-grade.
      f. 6 inches beneath pipe in trenches, and the greater of 24 inches wider than pipe or 42 inches wide.

3.05 EXCAVATION FOR STRUCTURES

A. Excavate to indicated elevations and dimensions within a tolerance of plus or minus 1 inch. If applicable, extend excavations a sufficient distance from structures for placing and removing concrete formwork, for installing services and other construction, and for inspections.
   1. Excavations for Footings and Foundations: Do not disturb bottom of excavation. Excavate by hand to final grade just before placing concrete reinforcement. Trim bottoms to required lines and grades to leave solid base to receive other work.
   2. All foundation subgrades shall be observed, evaluated, and verified for the design bearing pressure by the geotechnical engineer after excavation and prior to reinforcement steel placement. If low consistency soils are encountered during foundation construction, localized undercutting and/or in-place stabilization of foundation subgrades will be required. The actual need for and extent of undercutting shall be based on field observations made by the geotechnical engineer at the time of construction.
   3. Pile Foundations: Stop excavations 6 to 12 inches above bottom of pile cap before piles are placed. After piles have been driven, remove loose and displaced material. Excavate to final grade, leaving solid base to receive concrete pile caps.
   4. Excavation for Underground Tanks, Basins, and Mechanical or Electrical Utility Structures: Excavate to elevations and dimensions indicated within a tolerance of plus or minus 1 inch. Do not disturb bottom of excavations intended as bearing surfaces.

B. Excavations at Edges of Tree- and Plant-Protection Zones:
   1. Excavate by hand to indicated lines, cross sections, elevations, and subgrades. Use narrow tine spading forks to comb soil and expose roots. Do not break, tear, or chop exposed roots. Do not use mechanical equipment that rips, tears, or pulls roots.
   2. Cut and protect roots according to requirements in Division 01 Section "Temporary Tree and Plant Protection."

3.06 EXCAVATION FOR WALKS AND PAVEMENTS

A. Excavate surfaces under walks and pavements to indicated lines, cross sections, elevations, and subgrades.

3.07 EXCAVATION FOR UTILITY TRENCHES

A. Excavate trenches to indicated gradients, lines, depths, and elevations.

B. Excavate trenches to uniform widths to provide the following clearance on each side of pipe or conduit. Excavate trench walls vertically from trench bottom to 12 inches higher than top of pipe or conduit unless otherwise indicated.
   1. Clearance: 12 inches each side of pipe or conduit.

C. Trench Bottoms: Excavate and shape trench bottoms to provide uniform bearing and support of pipes and conduit. Shape subgrade to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits. Remove projecting stones and sharp objects along trench subgrade.
   1. For pipes and conduit less than 6 inches in nominal diameter, hand-excavate trench bottoms and support pipe and conduit on an undisturbed subgrade.
   2. For pipes and conduit 6 inches or larger in nominal diameter, shape bottom of trench to support bottom 90 degrees of pipe or conduit circumference. Fill depressions with tamped sand backfill.
   3. For flat-bottomed, multiple-duct conduit units, hand-excavate trench bottoms and support conduit on an undisturbed subgrade.
   4. Excavate trenches 6 inches deeper than elevation required in rock or other unyielding bearing material to allow for bedding course.

D. Trenches in Tree- and Plant-Protection Zones:
   1. Hand-excavate to indicated lines, cross sections, elevations, and subgrades. Use narrow tine spading forks to comb soil and expose roots. Do not break, tear, or chop exposed roots. Do not use mechanical equipment that rips, tears, or pulls roots.
   2. Do not cut main lateral roots or taproots; cut only smaller roots that interfere with installation of utilities.
   3. Cut and protect roots according to requirements in Division 01 Section "Temporary Tree and Plant Protection."

3.08 SUBGRADE INSPECTION

A. Notify Geotechnical Engineer when excavations have reached required subgrade.

B. If Geotechnical Engineer determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed.

C. Proof-roll subgrade below new fill, slabs, pavements and foundations with a pneumatic-tire 20- to 30-ton truck to identify soft pockets and areas of excess yielding. Do not proof-roll frozen, wet or saturated subgrades.
   1. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Geotechnical Engineer, and replace with compacted backfill or fill as directed.

D. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, determined by the Geotechnical Engineer and as directed by Architect, without additional compensation.

3.09 UNAUTHORIZED EXCAVATION

A. Fill unauthorized excavation under foundations or wall footings by extending bottom elevation of concrete foundation or footing to excavation bottom, without altering top elevation. Lean concrete fill, with 28-day compressive strength of 2500 psi, may be used when approved by Architect.

1. Fill unauthorized excavations under other construction, pipe, or conduit as directed by Architect.

3.10 STORAGE OF SOIL MATERIALS

A. Stockpile borrow soil materials and excavated satisfactory soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.

3.11 BACKFILL

A. Place and compact backfill in excavations promptly, but not before completing the following:
1. Construction below finish grade including, where applicable, subdrainage, damp proofing, waterproofing, and perimeter insulation.
2. Surveying locations of underground utilities for Record Documents.
3. Testing and inspecting underground utilities.
4. Removing concrete formwork.
5. Removing trash and debris.
6. Removing temporary shoring and bracing, and sheeting.
7. Installing permanent or temporary horizontal bracing on horizontally supported walls.

B. Place backfill on subgrades free of mud, frost, snow, or ice.

3.12 UTILITY TRENCH BACKFILL

A. Place backfill on subgrades free of mud, frost, snow, or ice.

B. Place and compact bedding course on trench bottoms and where indicated. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.

C. Trenches under Footings: Backfill trenches excavated under footings and within 18 inches of bottom of footings with satisfactory soil; fill with concrete to elevation of bottom of footings. Concrete is specified in Division 03 Section "Cast-in-Place Concrete”.

D. Trenches under Roadways: Provide 4-inch-thick, concrete-base slab support for piping or conduit less than 30 inches below surface of roadways. After installing and testing, completely encase piping or conduit in a minimum of 4 inches of concrete before backfilling or placing roadway subbase course. Concrete is specified in Division 03 Section "Cast-in-Place Concrete".

E. Backfill voids with satisfactory soil while removing shoring and bracing.

F. Initial Backfill: Place and compact initial backfill of subbase material, free of particles larger than 1 inch in any dimension, to a height of 12 inches over the pipe or conduit.
1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.

G. Controlled Low-Strength Material: Place initial backfill of controlled low-strength material to a height of 12 inches over the pipe or conduit. Coordinate backfilling with utilities testing.

H. Final Backfill: Place and compact final backfill of satisfactory soil to final subgrade elevation.

I. Controlled Low-Strength Material: Place final backfill of controlled low-strength material to final subgrade elevation.

J. Install warning tape directly above utilities, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.13 SOIL FILL

A. Plow, scarify, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material.

B. Place and compact fill material in layers to required elevations as follows:
1. Under grass and planted areas, use satisfactory soil material.
2. Under walks and pavements, use satisfactory soil material.
3. Under steps and ramps, use engineered fill.
4. Under building slabs, use engineered fill.
5. Under footings and foundations, use engineered fill.

C. Place soil fill on subgrades free of mud, frost, snow, or ice.

3.14 SOIL MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percent of optimum moisture content.
1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air dry, otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.15 COMPACTION OF SOIL BACKFILLS AND FILLS

A. General: Do not fill or backfill until all debris, water, unsatisfactory soil materials, obstructions, and deleterious materials have been removed from excavation. For fill and backfill, use excavated materials and borrow meeting the criteria specified by the Geotechnical Engineer. Do not use unsuitable excavated materials. Do not backfill until foundation walls have been completed above grade and adequately braced, waterproofing or dampproofing applied, foundation drainage, and pipes coming in contact with backfill have been installed and work inspected and approved by Geotechnical Engineer..

3.16 GRADING

A. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
1. Provide a smooth transition between adjacent existing grades and new grades.
2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.

B. Site Rough Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:
1. Turf or Unpaved Areas: Plus or minus 1 inch.
2. Walks: Plus or minus 1 inch.
3. Pavements: Plus or minus 1/2 inch.

C. Grading inside Building Lines: Finish subgrade to a tolerance of 1/2 inch when tested with a 10-foot straightedge.

3.17 SUBSURFACE DRAINAGE

A. Subdrainage Pipe: Specified in Division 33 Section "Subdrainage."

B. Subsurface Drain: Place subsurface drainage geotextile around perimeter of subdrainage trench. Place a 6-inch course of filter material on subsurface drainage geotextile to support subdrainage pipe. Encase subdrainage pipe in a minimum of 12 inches of filter material, placed in compacted layers 6 inches thick, and wrap in subsurface drainage geotextile, overlapping sides and ends at least 6 inches.
1. Compact each filter material layer with a minimum of two passes of a plate-type vibratory compactor.

C. Drainage Backfill: Place and compact filter material over subsurface drain, in width indicated, to within 12 inches of final subgrade, in compacted layers 6 inches thick. Overlay drainage backfill with one layer of subsurface drainage geotextile, overlapping sides and ends at least 6 inches.
1. Compact each filter material layer with a minimum of two passes of a plate-type vibratory compactor.
2. Place and compact impervious fill over drainage backfill in 6-inch-thick compacted layers to final subgrade.

3.18 SUBBASE AND BASE COURSES UNDER PAVEMENTS AND WALKS

A. Place subbase course and base course on subgrades free of mud, frost, snow, or ice.

B. On prepared subgrade, place subbase course and base course under pavements and walks as follows:
1. Install separation geotextile on prepared subgrade according to manufacturer's written instructions, overlapping sides and ends.
2. Place base course material over subbase course under hot-mix asphalt pavement.
3. Shape subbase course and base course to required crown elevations and cross-slope grades.
4. Place subbase course and base course 6 inches or less in compacted thickness in a single layer.
5. Place subbase course and base course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.
6. Compact subbase course and base course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than 95 percent of maximum dry unit weight according to ASTM D 698.

C. Pavement Shoulders: Place shoulders along edges of subbase course and base course to prevent lateral movement. Construct shoulders, at least 12 inches wide, of satisfactory soil materials and compact simultaneously with each subbase and base layer to not less than 95 percent of maximum dry unit weight according to ASTM D 1557.

3.19 DRAINAGE COURSE UNDER CONCRETE SLABS-ON-GRADE

A. Place drainage course on subgrades free of mud, frost, snow, or ice.

B. On prepared subgrade, place and compact drainage course under cast-in-place concrete slabs-on-grade as follows:
1. Install subdrainage geotextile on prepared subgrade according to manufacturer's written instructions, overlapping sides and ends.
2. Place drainage course 6 inches or less in compacted thickness in a single layer.
3. Place drainage course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.

4. Compact each layer of drainage course to required cross sections and thicknesses to not less than 95 percent of maximum dry unit weight according to ASTM D 698.

3.20 FIELD QUALITY CONTROL

A. Special Inspections: Owner will engage a qualified special inspector to perform special inspections.

B. Testing Agency: Contractor will engage a qualified geotechnical engineering testing agency to perform tests and inspections.

C. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earth moving only after test results for previously completed work comply with requirements.

D. Footing Subgrade: At footing subgrades, at least one test of each soil stratum will be performed to verify design bearing capacities. Subsequent verification and approval of other footing subgrades may be based on a visual comparison of subgrade with tested subgrade when approved by Architect.

E. Testing agency will test compaction of soils in place according to ASTM D 1556, ASTM D 2167, ASTM D 2922, and ASTM D 2937, as applicable and the requirements of Shenandoah County, VDOT and the Geotechnical Engineer. Field density tests to verify fill compaction should be performed for every 2,500 square feet (approximately 50 feet square) of fill area, with a minimum of two tests per lift. In confined areas, a greater frequency shall be required.

F. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil materials to depth required; recompact and retest until specified compaction is obtained.

3.21 PROTECTION

A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing, and erosion. Keep free of trash and debris.

B. Repair and reestablish grades to specified tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.

C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.
1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

3.22 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Remove waste materials, including unsatisfactory soil, trash, and debris, and legally dispose of them off Owner's property.

END OF SECTION 312000

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# SECTION 312319 - DEWATERING

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes construction dewatering.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.3 FIELD CONDITIONS

A. Survey Work: Engage a qualified land surveyor or professional engineer to survey adjacent existing buildings, structures, and site improvements; establish exact elevations at fixed points to act as benchmarks. Clearly identify benchmarks and record existing elevations.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Dewatering Performance: Design, furnish, install, test, operate, monitor, and maintain dewatering system of sufficient scope, size, and capacity to control hydrostatic pressures and to lower, control, remove, and dispose of ground water and permit excavation and construction to proceed on dry, stable subgrades.

## PART 3 - EXECUTION

### 3.1 PREPARATION

A. Provide temporary grading to facilitate dewatering and control of surface water.

B. Protect and maintain temporary erosion and sedimentation controls, which are specified during dewatering operations.

### 3.2 INSTALLATION

A. Install dewatering system utilizing wells, well points, or similar methods complete with pump equipment, standby power and pumps, filter material gradation, valves, appurtenances, water disposal, and surface-water controls.

1. Space well points or wells at intervals required to provide sufficient dewatering.
2. Use filters or other means to prevent pumping of fine sands or silts from the subsurface.

B. Place dewatering system into operation to lower water to specified levels before excavating below ground-water level.

C. Provide standby equipment on-site, installed and available for immediate operation, to maintain dewatering on continuous basis if any part of system becomes inadequate or fails.

## 3.3 OPERATION

A. Operate system continuously until drains, sewers, and structures have been constructed and fill materials have been placed or until dewatering is no longer required.

B. Operate system to lower and control ground water to permit excavation, construction of structures, and placement of fill materials on dry subgrades. Drain water-bearing strata above and below bottom of foundations, drains, sewers, and other excavations.

1. Do not permit open-sump pumping that leads to loss of fines, soil piping, subgrade softening, and slope instability.
2. Reduce hydrostatic head in water-bearing strata below subgrade elevations of foundations, drains, sewers, and other excavations.
3. Maintain piezometric water level a minimum of 24 inches below bottom of excavation.

C. Remove dewatering system from Project site on completion of dewatering. Plug or fill well holes with sand or cut off and cap wells a minimum of 36 inches below overlying construction.

## 3.4 FIELD QUALITY CONTROL

A. Survey-Work Benchmarks: Resurvey benchmarks regularly during dewatering and maintain an accurate log of surveyed elevations for comparison with original elevations. Promptly notify Architect if changes in elevations occur or if cracks, sags, or other damage is evident in adjacent construction.

END OF SECTION 312319

# SECTION 315000 - EXCAVATION SUPPORT AND PROTECTION

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes temporary excavation support and protection systems.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.3 ACTION SUBMITTALS

A. Shop Drawings: For excavation support and protection system, prepared by or under the supervision of a qualified professional engineer.

B. Delegated-Design Submittal: For excavation support and protection systems, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

### 1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For the following:

1. Professional Engineer: Experience with providing delegated-design engineering services of the type indicated, including documentation that engineer is licensed in the state in which Project is located.

B. Contractor Calculations: For excavation support and protection system. Include analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

C. Existing Conditions: Using photograph or video recordings, show existing conditions of adjacent construction and site improvements that might be misconstrued as damage caused by inadequate performance of excavation support and protection systems. Submit before Work begins.

### 1.5 CLOSEOUT SUBMITTALS

A. Record Drawings: Identify locations and depths of capped utilities, abandoned-in-place support and protection systems, and other subsurface structural, electrical, or mechanical conditions.

### 1.6 FIELD CONDITIONS

A. Interruption of Existing Utilities: Do not interrupt any utility-serving facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary utility according to requirements indicated:

1. Notify Architect no fewer than two days in advance of proposed interruption of utility.
2. Do not proceed with interruption of utility without Architect's written permission.

B. Survey Work: Engage a qualified land surveyor or professional engineer to survey adjacent existing buildings, structures, and site improvements; establish exact elevations at fixed points to act as benchmarks. Clearly identify benchmarks and record existing elevations.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design excavation support and protection systems to resist all lateral loading and surcharge, including but not limited to, retained soil, groundwater pressure, adjacent building loads, adjacent traffic loads, construction traffic loads, material stockpile loads, and seismic loads, based on the following:

1. Compliance with OSHA Standards and interpretations, 29 CFR 1926, Subpart P.
2. Compliance with AASHTO Standard Specification for Highway Bridges or AASHTO LRFD Bridge Design Specification, Customary U.S. Units.
3. Compliance with requirements of authorities having jurisdiction.
4. Compliance with utility company requirements.

### 2.2 MATERIALS

A. Provide materials that are either new or in serviceable condition.

B. Structural Steel: ASTM A 36/A 36M, ASTM A 690/A 690M, or ASTM A 992/A 992M.

C. Steel Sheet Piling: ASTM A 328/A 328M, ASTM A 572/A 572M, or ASTM A 690/A 690M; with continuous interlocks.

1. Corners: Site-fabricated mechanical interlock.

D. Wood Lagging: Lumber, mixed hardwood, nominal rough thickness of size and strength required for application.

E. Cast-in-Place Concrete: ACI 301, of compressive strength required for application.

F. Reinforcing Bars: ASTM A 615/A 615M, Grade 60, deformed.

G. Tiebacks: Steel bars, ASTM A 722/A 722M.

## PART 3 - EXECUTION

### 3.1 INSTALLATION - GENERAL

A. Locate excavation support and protection systems clear of permanent construction, so that construction and finishing of other work is not impeded.

B. Install excavation support and protection systems to ensure minimum interference with roads, streets, walks, and other adjacent occupied and used facilities.

1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction.
2. Provide alternate routes around closed or obstructed traffic ways if required by authorities having jurisdiction.

C. Install excavation support and protection systems without damaging existing buildings, structures, and site improvements adjacent to excavation.

## 3.2 SOLDIER PILES AND LAGGING

A. Install steel soldier piles before starting excavation.

1. Extend soldier piles below excavation grade level to depths adequate to prevent lateral movement.
2. Space soldier piles at regular intervals not to exceed allowable flexural strength of wood lagging.
3. Accurately align exposed faces of flanges to vary not more than 2 inches from a horizontal line and not more than 1:120 out of vertical alignment.

B. Install wood lagging within flanges of soldier piles as excavation proceeds.

1. Trim excavation as required to install lagging.
2. Fill voids behind lagging with soil, and compact.

C. Install wales horizontally at locations indicated on Drawings and secure to soldier piles.

## 3.3 SHEET PILING

A. Before starting excavation, install one-piece sheet piling lengths and tightly interlock vertical edges to form a continuous barrier.

B. Accurately place the piling using templates and guide frames unless otherwise recommended in writing by the sheet piling manufacturer.

1. Limit vertical offset of adjacent sheet piling to 60 inches.
2. Accurately align exposed faces of sheet piling to vary not more than 2 inches from a horizontal line and not more than 1:120 out of vertical alignment.

C. Cut tops of sheet piling to uniform elevation at top of excavation.

## 3.4 TIEBACKS

A. Drill, install, grout, and tension tiebacks.

B. Test load-carrying capacity of each tieback, and replace and retest deficient tiebacks.

1. Have test loading observed by a qualified professional engineer responsible for design of excavation support and protection system.

C. Maintain tiebacks in place until permanent construction is able to withstand lateral earth and hydrostatic pressures.

3.5 BRACING

A. Locate bracing to clear columns, floor framing construction, and other permanent work. If necessary to move brace, install new bracing before removing original brace.

1. Do not place bracing where it will be cast into or included in permanent concrete work unless otherwise approved by Architect.
2. Install internal bracing if required to prevent spreading or distortion of braced frames.
3. Maintain bracing until structural elements are supported by other bracing or until permanent construction is able to withstand lateral earth and hydrostatic pressures.

3.6 MAINTENANCE

A. Monitor and maintain excavation support and protection system.

B. Prevent surface water from entering excavations by grading, dikes, or other means.

C. Continuously monitor vibrations, settlements, and movements to ensure stability of excavations and constructed slopes and to ensure that damage to permanent structures is prevented.

3.7 FIELD QUALITY CONTROL

A. Survey-Work Benchmarks: Resurvey benchmarks regularly during installation of excavation support and protection systems, excavation progress, and for as long as excavation remains open.

1. Maintain an accurate log of surveyed elevations and positions for comparison with original elevations and positions.
2. Promptly notify Architect if changes in elevations or positions occur or if cracks, sags, or other damage is evident in adjacent construction.

B. Promptly correct detected bulges, breakage, or other evidence of movement to ensure that excavation support and protection system remains stable.

C. Promptly repair damages to adjacent facilities caused by installation or faulty performance of excavation support and protection systems.

3.8 REMOVAL AND REPAIRS

A. Remove excavation support and protection systems when construction has progressed sufficiently to support excavation and earth and hydrostatic pressures.

1. Remove in stages to avoid disturbing underlying soils and rock or damaging structures, pavements, facilities, and utilities.
2. Remove excavation support and protection systems to a minimum depth of 48 inches below overlying construction, and abandon remainder.

3. Fill voids immediately with approved backfill compacted to density specified in Section 312000 "Earth Moving."
4. Repair or replace, as approved by Architect, adjacent work damaged or displaced by removing excavation support and protection systems.

B. Leave excavation support and protection systems permanently in place.

END OF SECTION 315000

# SECTION 321216 - ASPHALT PAVING

## 1.1 SUMMARY

A. Section Includes:

1. Cold milling of existing asphalt pavement.
2. Hot-mix asphalt patching.
3. Hot-mix asphalt paving.
4. Hot-mix asphalt overlay.

B. Related Requirements:

1. Section 312000 "Earthwork" for subgrade preparation, fill material, separation geotextiles, unbound-aggregate subbase and base courses, and aggregate pavement shoulders.

## 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

## 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

## 1.4 INFORMATIONAL SUBMITTALS

A. Material Certificates: For each paving material. Include statement that mixes containing recycled materials will perform equal to mixes produced from all new materials.

## 1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: A paving-mix manufacturer registered with and approved by the Virginia Department of Transportation.

B. Testing Agency Qualifications: Qualified according to ASTM D 3666 for testing indicated.

C. Regulatory Requirements: Comply with materials, workmanship, and other applicable requirements of the Virginia Department of Transportation standards and specifications, latest editions for asphalt paving work.

1. Measurement and payment provisions and safety program submittals included in standard specifications do not apply to this Section.

1.6 FIELD CONDITIONS

A. Environmental Limitations: Do not apply asphalt materials if subgrade is wet or excessively damp, if rain is imminent or expected before time required for adequate cure, or if the following conditions are not met:

1. Prime Coat: Minimum surface temperature of 60 deg F.
2. Tack Coat: Minimum surface temperature of 60 deg F.
3. Slurry Coat: Comply with weather limitations in ASTM D 3910.
4. Asphalt Base Course: Minimum surface temperature of 40 deg F and rising at time of placement.
5. Asphalt Surface Course: Minimum surface temperature of 60 deg F at time of placement.

PART 2 - PRODUCTS

2.1 AGGREGATES

A. Aggregates: As specified in Section 211 of VDOT Specifications.

B. Mineral Filler: As specified in Section 201 of VDOT Specifications.

2.2 ASPHALT MATERIALS

A. Emulsified Asphalt Prime Coat: Liquid asphalt conforming to Sections 210 and 311 of VDOT Specifications.

B. Tack Coat: Emulsified asphalt conforming to Section 210 and 310 of VDOT Specifications.

C. Water: Potable.

2.3 AUXILIARY MATERIALS

A. Recycled Materials for Hot-Mix Asphalt Mixes: Reclaimed asphalt pavement; reclaimed, unbound-aggregate base material; and recycled tires, asphalt shingles, or glass from sources and gradations that have performed satisfactorily in previous installations, equal to performance of required hot-mix asphalt paving produced from all new materials.

B. Herbicide: Commercial chemical for weed control, registered by the EPA, and not classified as "restricted use" for locations and conditions of application. Provide in granular, liquid, or wettable powder form.

2.4 MIXES

A. Hot-Mix Asphalt: Dense-graded, hot-laid, hot-mix asphalt plant mixes per Section 211 of the VDOT Specifications. :

1. Provide mixes with a history of satisfactory performance in geographical area where Project is located.
2. Base Course: As specified on the Drawings.

3. Surface Course: As specified on the Drawings.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

A. Verify that subgrade is dry and in suitable condition to begin paving.

B. Proceed with paving only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

A. Protection: Provide protective materials, procedures, and worker training to prevent asphalt materials from spilling, coating, or building up on curbs, driveway aprons, manholes, and other surfaces adjacent to the Work.

B. Proof-roll subgrade below pavements with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.

1. Completely proof-roll subgrade in one direction, repeating proof-rolling in direction perpendicular to first direction. Limit vehicle speed to 3 mph.
2. Proof roll with a loaded 10-wheel, tandem-axle dump truck weighing not less than 15 tons.
3. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Architect, and replace with compacted backfill or fill as directed

### 3.3 COLD MILLING

A. Clean existing pavement surface of loose and deleterious material immediately before cold milling. Remove existing asphalt pavement by cold milling to grades and cross sections indicated.

1. Mill to a depth of 2 inches.
2. Mill to a uniform finished surface free of excessive gouges, grooves, and ridges.
3. Control rate of milling to prevent tearing of existing asphalt course.
4. Repair or replace curbs, driveway aprons, manholes, and other construction damaged during cold milling.
5. Excavate and trim unbound-aggregate base course, if encountered, and keep material separate from milled hot-mix asphalt.
6. Patch surface depressions deeper than 1 inch after milling, before wearing course is laid.
7. Handle milled asphalt material according to approved waste management plan."
8. Keep milled pavement surface free of loose material and dust.
9. Do not allow milled materials to accumulate on-site.

### 3.4 PATCHING

A. Asphalt Pavement: Saw cut perimeter of patch and excavate existing pavement section to sound base. Excavate rectangular or trapezoidal patches, extending 12 inches into perimeter of adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Remove

excavated material. Re-compact existing unbound-aggregate base course to form new subgrade.

B. Tack Coat: Before placing patch material, apply tack coat uniformly to vertical asphalt surfaces abutting the patch. Apply at a rate of 0.05 to 0.15 gal./sq. yd.

1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.

C. Placing Patch Material: Fill excavated pavement areas with hot-mix asphalt base mix for full thickness of patch and, while still hot, compact flush with adjacent surface.

## 3.5 REPAIRS

A. Leveling Course: Install and compact leveling course consisting of hot-mix asphalt surface course to level sags and fill depressions deeper than 1 inch in existing pavements.

1. Install leveling wedges in compacted lifts not exceeding 3 inches thick.

B. Crack and Joint Filling: Remove existing joint filler material from cracks or joints to a depth of 1/4 inch.

1. Clean cracks and joints in existing hot-mix asphalt pavement.
2. Use emulsified-asphalt slurry to seal cracks and joints less than 1/4 inch wide. Fill flush with surface of existing pavement and remove excess.
3. Use hot-applied joint sealant to seal cracks and joints more than 1/4 inch wide. Fill flush with surface of existing pavement and remove excess.

## 3.6 SURFACE PREPARATION

A. Immediately before placing asphalt materials, remove loose and deleterious material from substrate surfaces. Ensure that prepared subgrade is ready to receive paving.

B. Herbicide Treatment: Apply herbicide according to manufacturer's recommended rates and written application instructions. Apply to dry, prepared subgrade or surface of compacted-aggregate base before applying paving materials.

C. Tack Coat: Apply uniformly to surfaces of existing pavement at a rate of 0.05 to 0.15 gal./sq. yd.

1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.

## 3.7 PLACING HOT-MIX ASPHALT

A. Machine place hot-mix asphalt on prepared surface, spread uniformly, and strike off. Place asphalt mix by hand in areas inaccessible to equipment in a manner that prevents segregation of mix. Place each course to required grade, cross section, and thickness when compacted.

1. Spread mix at a minimum temperature of 250 deg F.

2. Regulate paver machine speed to obtain smooth, continuous surface free of pulls and tears in asphalt-paving mat.

B. Place paving in consecutive strips not less than 10 feet wide unless infill edge strips of a lesser width are required.

C. Promptly correct surface irregularities in paving course behind paver. Use suitable hand tools to remove excess material forming high spots. Fill depressions with hot-mix asphalt to prevent segregation of mix; use suitable hand tools to smooth surface.

## 3.8 JOINTS

A. Construct joints to ensure a continuous bond between adjoining paving sections. Construct joints free of depressions, with same texture and smoothness as other sections of hot-mix asphalt course.

1. Clean contact surfaces and apply tack coat to joints.
2. Offset longitudinal joints, in successive courses, a minimum of 6 inches.
3. Offset transverse joints, in successive courses, a minimum of 24 inches.
4. Construct transverse joints at each point where paver ends a day's work and resumes work at a subsequent time. Construct these joints using either "bulkhead" or "papered" method according to AI MS-22, for both "Ending a Lane" and "Resumption of Paving Operations."

## 3.9 COMPACTION

A. General: Begin compaction as soon as placed hot-mix paving will bear roller weight without excessive displacement. Compact hot-mix paving with hot, hand tampers or with vibratory-plate compactors in areas inaccessible to rollers.

1. Complete compaction before mix temperature cools to 185 deg F.

B. Breakdown Rolling: Complete breakdown or initial rolling immediately after rolling joints and outside edge. Examine surface immediately after breakdown rolling for indicated crown, grade, and smoothness. Correct laydown and rolling operations to comply with requirements.

C. Intermediate Rolling: Begin intermediate rolling immediately after breakdown rolling while hot-mix asphalt is still hot enough to achieve specified density. Continue rolling until hot-mix asphalt course has been uniformly compacted to the following density:

1. Average Density: 92 percent of reference maximum theoretical density according to ASTM D 2041/D 2041M, but not less than 90 percent or greater than 96 percent.

D. Finish Rolling: Finish roll paved surfaces to remove roller marks while hot-mix asphalt is still warm.

E. Edge Shaping: While surface is being compacted and finished, trim edges of pavement to proper alignment. Bevel edges while asphalt is still hot; compact thoroughly.

F. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened.

G. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

## 3.10 INSTALLATION TOLERANCES

A. Pavement Thickness: Compact each course to produce the thickness indicated within the following tolerances:

1. Base Course: Plus or minus 1/2 inch.
2. Surface Course: Plus 1/4 inch, no minus.

B. Pavement Surface Smoothness: Compact each course to produce a surface smoothness within the following tolerances as determined by using a 10-foot straightedge applied transversely or longitudinally to paved areas:

1. Base Course: 1/4 inch.
2. Surface Course: 1/8 inch.
3. Retain "Crowned Surfaces" Subparagraph below if required.
4. Crowned Surfaces: Test with crowned template centered and at right angle to crown. Maximum allowable variance from template is 1/4 inch.

## 3.11 FIELD QUALITY CONTROL

A. Testing Agency: Contractor will engage a qualified testing agency to perform tests and inspections.

B. Replace and compact hot-mix asphalt where core tests were taken.

C. Remove and replace or install additional hot-mix asphalt where test results or measurements indicate that it does not comply with specified requirements.

D. All test reports shall be submitted to the Owner.

## 3.12 WASTE HANDLING

A. General: Handle asphalt-paving waste according to approved waste management plan required in Division 1.

END OF SECTION 321216

SECTION 321313 - CONCRETE PAVING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. “Virginia Department of Transportation Road and Bridge Specifications”, current edition.

1.02 SUMMARY

A. Section Includes:
1. Curbs and gutters.
2. Walks.
3. Parking Pad.

B. Related Sections:
1. Division 03 Section "Cast-in-Place Concrete for general building applications of concrete.

1.03 DEFINITIONS

A. VDOT Specifications: Virginia Department of Transportation, Road and Bridge Specifications, current edition.

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Samples: For each type of product, ingredient, or admixture requiring color selection.

C. Design Mixtures: For each concrete paving mixture. Include alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

1.05 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer of ready-mix concrete manufacturer and testing agency.

B. Material Test Reports: From a qualified testing agency indicating and interpreting test results for compliance of the following:
1. Aggregates.

C. Material Certificates: For the following, from manufacturer:
1. Cementitious materials.
2. Steel reinforcement and reinforcement accessories.
3. Fiber reinforcement.
4. Admixtures.
5. Curing compounds.
6. Applied finish materials.
7. Bonding agent or epoxy adhesive.
8. Joint fillers.

D. Field quality-control reports.

1.06 QUALITY ASSURANCE

A. Ready-Mix-Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.
   1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities" (Quality Control Manual - Section 3, "Plant Certification Checklist").

B. Testing Agency Qualifications: Qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.
   1. Personnel conducting field tests shall be qualified as ACI Concrete Field-Testing Technician, Grade 1, according to ACI CP-1 or an equivalent certification program.

C. Concrete Testing Service: Engage a qualified testing agency to perform material evaluation tests and to design concrete mixtures.

D. ACI Publications: Comply with ACI 301 unless otherwise indicated.

E. Preinstallation Conference: Conduct conference at Project site.
   1. Review methods and procedures related to concrete paving, including but not limited to, the following:
      a. Concrete mixture design.
      b. Quality control of concrete materials and concrete paving construction practices.
   2. Require representatives of each entity directly concerned with concrete paving to attend, including the following:
      a. Contractor's superintendent.
      b. Independent testing agency responsible for concrete design mixtures.
      c. Ready-mix concrete manufacturer.
      d. Concrete paving subcontractor.
      e. Manufacturer's representative of stamped concrete paving system used for detectable warnings.

1.07 PROJECT CONDITIONS

A. Traffic Control: Maintain access for vehicular and pedestrian traffic as required for other construction activities.

B. Cold-Weather Concrete Placement: Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing, or low temperatures. Comply with ACI 306.1 and the following:
   1. When air temperature has fallen to or is expected to fall below 40 deg F, uniformly heat water and aggregates before mixing to obtain a concrete mixture temperature of not less than 50 deg F and not more than 80 deg F at point of placement.
   2. Do not use frozen materials or materials containing ice or snow.
   3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in design mixtures.
   4. Hot-Weather Concrete Placement: Comply with ACI 301 and as follows when hot-weather conditions exist:
   5. Cool ingredients before mixing to maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated in total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
   6. Cover steel reinforcement with water-soaked burlap, so steel temperature will not exceed ambient air temperature immediately before embedding in concrete.

7. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade moisture uniform without standing water, soft spots, or dry areas.

PART 2 - PRODUCTS

2.01 FORMS

A. Curb and Gutter: VDOT Section 502.

B. Sidewalks: VDOT Section 504.

C. Form Materials: Plywood, metal, metal-framed plywood, or other approved panel-type materials to provide full-depth, continuous, straight, and smooth exposed surfaces.
1. Use flexible or uniformly curved forms for curves with a radius of 100 feet or less. Do not use notched and bent forms.

D. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and that will not impair subsequent treatments of concrete surfaces.

2.02 CONCRETE, GENERAL

A. VDOT Section 217

2.03 STEEL REINFORCEMENT

A. Plain-Steel Welded Wire Reinforcement: ASTM A 185, fabricated from as-drawn steel wire into flat sheets.

B. Deformed-Steel Welded-Wire Reinforcement: ASTM A 1064/A 1064M, flat sheet.

C. Reinforcing Bars: ASTM A 615/A 615M, Grade 60; deformed.

D. Joint Dowel Bars: Plain steel bars, ASTM A 615/A 615M, Grade 60. Cut bars true to length with ends square and free of burrs.

E. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars, welded wire reinforcement, and dowels in place. Manufacture bar supports according to CRSI's "Manual of Standard Practice" from steel wire, plastic, or precast concrete of greater compressive strength than concrete specified, and as follows:
1. Equip wire bar supports with sand plates or horizontal runners where base material will not support chair legs.
2. For epoxy-coated reinforcement, use epoxy-coated or other dielectric-polymer-coated wire bar supports.

2.04 CONCRETE MATERIALS

A. Concrete Materials: VDOT Section 217.

2.05 CURING MATERIALS

A. Curing Materials: VDOT Section 220.

2.06 RELATED MATERIALS

A. Joint Fillers: ASTM D 1751, asphalt-saturated cellulosic fiber or ASTM D 1752, cork or self-expanding cork in preformed strips.

B. Slip-Resistive Aggregate Finish: Factory-graded, packaged, rustproof, nonglazing, abrasive aggregate of fused aluminum-oxide granules or crushed emery with emery aggregate containing not less than 50 percent aluminum oxide and not less than 20 percent ferric oxide; unaffected by freezing, moisture, and cleaning materials.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine exposed subgrades and subbase surfaces for compliance with requirements for dimensional, grading, and elevation tolerances.

B. Proof-roll prepared subbase surface below concrete paving to identify soft pockets and areas of excess yielding.
1. Completely proof-roll subbase in one direction and repeat in perpendicular direction. Limit vehicle speed to 3 mph.
2. Proof-roll with a pneumatic-tired and loaded, 10-wheel, tandem-axle dump truck weighing not less than 15 tons.
3. Correct subbase with soft spots and areas of pumping or rutting exceeding depth of 1/2 inch according to requirements in Division 31 Section "Earth Moving."

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Remove loose material from compacted subbase surface immediately before placing concrete.

3.03 EDGE FORMS AND SCREED CONSTRUCTION

A. Set, brace, and secure edge forms, bulkheads, and intermediate screed guides to required lines, grades, and elevations. Install forms to allow continuous progress of work and so forms can remain in place at least 24 hours after concrete placement.

B. Clean forms after each use and coat with form-release agent to ensure separation from concrete without damage.

3.04 STEEL REINFORCEMENT

A. General: Comply with CRSI's "Manual of Standard Practice" for fabricating, placing, and supporting reinforcement.

B. Clean reinforcement of loose rust and mill scale, earth, ice, or other bond-reducing materials.

C. Arrange, space, and securely tie bars and bar supports to hold reinforcement in position during concrete placement. Maintain minimum cover to reinforcement.

D. Install welded wire reinforcement in lengths as long as practicable. Lap adjoining pieces at least one full mesh, and lace splices with wire. Offset laps of adjoining widths to prevent continuous laps in either direction.

E. Install fabricated bar mats in lengths as long as practicable. Handle units to keep them flat and free of distortions. Straighten bends, kinks, and other irregularities, or replace units as required before placement. Set mats for a minimum 2-inch overlap of adjacent mats.

3.05 JOINTS

A. General: Form construction, isolation, and contraction joints and tool edges true to line, with faces perpendicular to surface plane of concrete. Construct transverse joints at right angles to centerline unless otherwise indicated.
   1. When joining existing paving, place transverse joints to align with previously placed joints unless otherwise indicated.

B. Construction Joints: Set construction joints at side and end terminations of paving and at locations where paving operations are stopped for more than one-half hour unless paving terminates at isolation joints.
   1. Continue steel reinforcement across construction joints unless otherwise indicated. Do not continue reinforcement through sides of paving strips unless otherwise indicated.
   2. Provide tie bars at sides of paving strips where indicated.
   3. Doweled Joints: Install dowel bars and support assemblies at joints where indicated. Lubricate or coat with asphalt one-half of dowel length to prevent concrete bonding to one side of joint.

C. Isolation Joints: Form isolation joints of preformed joint-filler strips abutting concrete curbs, catch basins, manholes, inlets, structures, other fixed objects, and where indicated.
   1. Locate expansion joints at intervals of 50 feet unless otherwise indicated.
   2. Extend joint fillers full width and depth of joint.
   3. Terminate joint filler not less than 1/2 inch or more than 1 inch below finished surface if joint sealant is indicated.
   4. Place top of joint filler flush with finished concrete surface if joint sealant is not indicated.
   5. Furnish joint fillers in one-piece lengths. Where more than one length is required, lace or clip joint-filler sections together.
   6. During concrete placement, protect top edge of joint filler with metal, plastic, or other temporary preformed cap. Remove protective cap after concrete has been placed on both sides of joint.

D. Contraction Joints: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of the concrete thickness, as follows:
   1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint with grooving tool to a 1/4-inch radius. Repeat grooving of contraction joints after applying surface finishes. Eliminate grooving-tool marks on concrete surfaces.
      a. Tolerance: Ensure that grooved joints are within 3 inches either way from centers of dowels.
   2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inchwide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before developing random contraction cracks.
      a. Tolerance: Ensure that sawed joints are within 3 inches either way from centers of dowels.
   3. Doweled Contraction Joints: Install dowel bars and support assemblies at joints where indicated. Lubricate or coat with asphalt one-half of dowel length to prevent concrete bonding to one side of joint.

E. Edging: After initial floating, tool edges of paving, gutters, curbs, and joints in concrete with an edging tool to a 1/4-inch radius. Repeat tooling of edges after applying surface finishes. Eliminate edging-tool marks on concrete surfaces.

3.06 CONSTRUCTION METHODS

A. Curbs, Gutters, and Combination Curb and Gutter: VDOT Section 502, “Road and Bridge Standards,” and as indicated on the Drawings.

B. Sidewalks: VDOT Section 504.

C. Equipment Pads: Refer to Section 03 3000 “Cast-in-Place Concrete” of these Specifications.

D. Curing: VDOT Section 31 6.04.

3.07 FINISHING

A. Handicap Ramps: Shall be constructed in accordance with the requirements and procedures of ADA and or VDOT Standards.

3.08 PAVEMENT MARKING

A. Do not apply pavement-marking paint until layout, colors, and placement have been verified with Architect.

B. Allow concrete paving to cure for a minimum of 10 days and be dry before starting pavement marking.

C. Sweep and clean surface to eliminate loose material and dust.

D. Apply paint with mechanical equipment to produce markings of dimensions indicated with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils.
   1. Apply graphic symbols and lettering with paint-resistant, die-cut stencils, firmly secured to concrete surface. Mask an extended area beyond edges of each stencil to prevent paint application beyond stencil. Apply paint so that it cannot run beneath stencil.
   2. Broadcast glass beads uniformly into wet markings at a rate of 6 lb/gal.

3.09 FIELD QUALITY CONTROL

A. Testing Agency: Contractor will engage a qualified testing agency to perform tests and inspections.

B. Testing Services: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:
   1. Testing Frequency: Obtain at least one composite sample for each 5000 sq. ft. or fraction thereof of each concrete mixture placed each day.
      a. When frequency of testing will provide fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.
   2. Slump: ASTM C 143/C 143M; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
   3. Air Content: ASTM C 231, pressure method; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
   4. Concrete Temperature: ASTM C 1064/C 1064M; one test hourly when air temperature is 40 deg F and below and when it is 80 deg F and above, and one test for each composite sample.
   5. Compression Test Specimens: ASTM C 31/C 31M; cast and laboratory cure one set of three standard cylinder specimens for each composite sample.

6. Compressive-Strength Tests: ASTM C 39/C 39M; test one specimen at seven days and two specimens at 28 days.
    a. A compressive-strength test shall be the average compressive strength from two specimens obtained from same composite sample and tested at 28 days.

C. Strength of each concrete mixture will be satisfactory if average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength, and no compressive-strength test value falls below specified compressive strength by more than 500 psi.

D. Test results shall be reported in writing to Architect, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.

E. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by Architect but will not be used as sole basis for approval or rejection of concrete.

F. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Architect.

G. Concrete paving will be considered defective if it does not pass tests and inspections.

H. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

I. Prepare test and inspection reports.

3.10 REPAIRS AND PROTECTION

A. Remove and replace concrete paving that is broken, damaged, or defective or that does not comply with requirements in this Section. Remove work in complete sections from joint to joint unless otherwise approved by Architect.

B. Drill test cores, where directed by Architect, when necessary to determine magnitude of cracks or defective areas. Fill drilled core holes in satisfactory paving areas with portland cement concrete bonded to paving with epoxy adhesive.

C. Protect concrete paving from damage. Exclude traffic from paving for at least 14 days after placement. When construction traffic is permitted, maintain paving as clean as possible by removing surface stains and spillage of materials as they occur.

D. Maintain concrete paving free of stains, discoloration, dirt, and other foreign material. Sweep paving not more than two days before date scheduled for Substantial Completion inspections.

END OF SECTION 321313

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# SECTION 321723 - PAVEMENT MARKINGS

## PART 1 - GENERAL

### 1.1 SUMMARY

A. Section includes painted markings applied to asphalt and concrete pavement.

### 1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples: For each exposed product and for each color and texture specified.

## PART 2 - PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: Comply with applicable provisions in the USDOJ's "2010 ADA Standards for Accessible Design" and ICC A117.1.

### 2.2 PAVEMENT-MARKING PAINT

A. Pavement-Marking Paint: MPI #32, solvent-borne traffic-marking paint.

1. Color: As indicated on Drawings.

## PART 3 - EXECUTION

### 3.1 PAVEMENT MARKING

A. Do not apply pavement-marking paint until layout, colors, and placement have been verified with Architect.

B. Allow paving to age for a minimum of 30 days before starting pavement marking.

C. Sweep and clean surface to eliminate loose material and dust.

D. Apply paint with mechanical equipment to produce pavement markings, of dimensions indicated, with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils.

1. Apply graphic symbols and lettering with paint-resistant, die-cut stencils. Apply paint so that it cannot run beneath stencil.

END OF SECTION 321723

# SECTION 32 3113 - HIGH SPEED ELECTRONIC SECURITY GATE

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section includes Vehicle bi-fold high speed electronic gate. The work in this section consists of furnishing all labor, materials, equipment and appliances necessary to complete all motorized folding gates.

### 1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings:

1. Include plans, elevations, sections, gate locations, post spacing, and mounting attachment details, and grounding details.
2. Contractor shall coordinate shop drawings with actual field conditions where gates are to be installed with final grading and adjacent conditions included.

### 1.4 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

B. Test reports:

1. Drive unit shall bear a label indicating that the gate controller/operator mechanism has been tested certified to UL 325 and CSA C22.2 No. 247 standards for all electrical components.

### 1.5 CLOSEOUT SUBMITTALS

A. Provide operation and maintenance data for gate for incorporation into manual.

B. Conduct comprehensive demonstration for maintenance staff on operation and care of gate.

### 1.6 QUALITY ASSURANCE

A. Manufacturer: A company specializing in the manufacture of electric gate systems.

B. Installer: A minimum of three years' experience installing similar equipment and approved by manufacturer. Capable of providing a four hour response time for warranty and service items.

1.7 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of high speed electronic security gate that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures including rupturing, cracking, or puncturing.
b. Deterioration of metals and other materials beyond normal weathering.
c. Software and firmware failures.

2. Warranty Period: Two years from date of Substantial Completion.

B. Special Warranty: Manufacturer's standard form in which manufacturer agrees to replace the gates hinges

1. Finish Warranty Period: Lifetime date of Substantial Completion.

PART 2 - PRODUCTS

2.1 AUTOMATIC ENTRANCE ASSEMBLIES

A. Source Limitations: Obtain folding swinging automatic speed gate from single source from single manufacturer.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.2 HIGH SPEED ACCESS GATE

A. Basis-of-Design Product: Subject to compliance with requirements, Provide Wallace Foldsmart Trackless Bi-folding Gate System, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

a. Wallace Perimeter Security

B. No Equal Products or Substitutions are allowed.

C. Configuration: Two pairs of folding doors (bi-folding) that fold to each side of door opening.

a. Traffic Pattern: One-way Entry and one-way Exit Layout

2.3 MATERIALS

A. Steel sheet: hot dipped galvanized to ASTM A653/A653M, A36 pre galvanized steel

B. Steel sections: to ASTM Grade 300W.

C. Welding materials: to ASWD1.1

D. Electrical components: Complete gate system to be UL325 listed and complying with local requirements.

E. Power Supply: 208V – 20A, single phase.

2.4 COMPONENTS

A. Gate Columns:

1. Formed steel columns, anchored to concrete foundation.
2. Columns to be 12” square with a wall thickness of .250”.

B. Gate:

1. Width: per Drawings Max 30 ft opening x 8 ft high.
2. Panels to be capable of fully opening within 8 seconds
3. Panels Infill: Rampart 280 Welded Wire or equivalent.
4. Panel Hinges: Double stacked ball bearing at each hinge point with hex bolt and lock nut. Column hinges: Alloy rod end bearing and graphite impregnated bronze bushings
5. End post mounting brackets for fence attachment.
6. Bottom skirt blank off panels to meet site requirements. Gate bottom distance to finished surface shall not exceed 6” maximum.

C. Safety Devices:

1. Provide reduced speed sensor, - Absolute encoder mounted directly to drive motor rotor to act as primary entrapment detection device
2. Photoelectric transmitter and receiver: Equip each interior column with 2 built-in photocells at 24” & 60 inches above the base plate. To be mounted within the columns. Control unit to be located within 20 ft. of the gate structure, 2-2” conduits for communication/power cabling to gate structure.
3. Underground Loop: Provide embedded underground loops, lead-in wires and 2-channel relay card for integration of embedded vehicle loop detectors with gate operation.

D. Drive Unit:

1. Provide variable frequency drive with programmable logic controller for controlling electro-mechanical drive system. Drive system to incorporate encoders with reduced speed sensing software as primary entrapment detection device.
2. All drive electrical components to be enclosed in weather-resistant housing.
3. Provide 24VDC drive with printed circuit board - integrated motor control circuitry for controlling electro-mechanical drive system. Drive system to incorporate current sensing resistors and adaptive monitoring software as inherent entrapment protection
4. All electrical drive components to be enclosed in weather-resistant housing.
5. Dual 0.5HP, 24 VDC motors, gear reduction box with synthetic lubricant.
6. Integrated battery backup.
7. All units and accessories shall be installed on the secure side of the gate.

E. Vehicle Loop Detector: System includes automatic closing timer with adjustable time delay, timer cutoff switch, and loop detector designed to open and close gate. System includes electronic detector with adjustable detection patterns, adjustable sensitivity and frequency

settings, and panel indicator light designed to detect presence or transit of a vehicle over an embedded loop of wire and to emit a signal activating the gate operator. System includes number of loops consisting of multiple strands of wire, number of turns, loop size, and method of placement, as recommended in writing by detection system manufacturer for function indicated, at locations indicated on Drawings.

F. Emergency Access:

1. Provide secured access panel for manual opening and closing in case of power failure/malfunction to be located on inside of property.
2. Provide Knox box for fire department access on the public side, one per entrance column to allow emergency access to the site. Inside column provide cable to disengage the motor to allow gates to be opened manually.

2.5 FINISHES

A. Powder coated finish 0.5 kg/m2 zinc coating to ASTM A653/A653M

PART 3 - EXECUTION

3.1 INSTALLATION

A. Provision of concrete foundations as determined by local engineer according to drawings provided.

B. Install high-speed security gate to manufacturer's written instructions, by contractor certified by manufacturer.

C. Submit certificate of installation to manufacturer upon completion of installation for warranty validation

3.2 CLEANING AND MAINTENANCE

A. Perform cleaning and maintenance procedures in strict accordance with manufacturer’s written instructions.

B. Maintain logbook of repairs and maintenance.

END OF SECTION 32 3116

# SECTION 32 3116 - WELDED WIRE SECURITY FENCES AND GATES

## PART 1 - GENERAL

### 1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

### 1.2 SUMMARY

A. Section Includes:

1. Metallic-coated-steel, welded-wire anti-climb security fences.
2. Swing gates.

B. Related Requirements:

1. Section 03 3000 "Cast-in-Place Concrete" for cast-in-place concrete post footings.
2. Section 31 2000 "Earth Moving" for site excavation, fill, and backfill where welded-wire fences and gates are located.

### 1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

### 1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For fencing and gates.

1. Include plans, elevations, sections, gate locations, post spacing, and mounting attachment details, and grounding details.

C. Samples: For each fence material and for each color specified.

1. Provide Samples 12 inches (300 mm) in length for linear materials.
2. Provide Samples 12 inches (300 mm) square for wire mesh.

### 1.5 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For gate operators to include in maintenance manuals.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: Fabricator of products.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for fabrication and installation.

1. Include 10-foot (3-m) length of fence complying with requirements.
2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Wind Loading:

1. Fence Height: 0 to 15 feet (0 to 4.57 m).
2. Wind Exposure Category: B.
3. Design Wind Speed: 124 mph (195 kph).
4. Design Wind Pressure: As indicated on Structural Drawing S-001.

B. Lightning-Protection System: Maximum grounding-resistance value of 25 ohms under normal dry conditions.

2.2 METALLIC-COATED-STEEL, WELDED-WIRE FENCES

A. Metallic-Coated-Steel, Welded-Wire Anti-Climb Fences:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Wallace Rampart 358 Anti-Climb Fence, or equivalent product from available manufacturers offering products that may be incorporated into the Work include:

a. Ameristar Perimeter Security; ASSA ABLOY. (Wire Works Anti-Climb)
b. Betafence; Praesidiad Group Limited. (Duty Guard Anti-Climb)
c. Wallace Perimeter Security. (Rampart 358 Anti-Climb)

B. Fence Fabric: Metallic-coated-steel wire.

1. Spacing of Vertical Wires: 3 inches (76.2 mm).
2. Vertical Wire Size: 0.187 inch (4.76 mm).
3. Spacing of Horizontal Wires: 0.5 inches (12.7 mm).
4. Horizontal Wire Size: 0.187 inch (4.76 mm).

C. Posts:

1. Line Posts: Square tubes or I-beam. 4 by 2-3/4 inches (101.6 by 69.8 mm) formed from 0.119-inch (3.03-mm) nominal-thickness, metallic-coated steel sheet.
2. End and Corner Posts: Square tubes or I-beam. 4 by 2-3/4 inches (101.6 by 69.8 mm) formed from 0.119-inch (3.03-mm) nominal-thickness, metallic-coated steel sheet.
3. Posts at Swing Gate Openings: Square tubes or I-beam. 4 by 2-3/4 inches (101.6 by 69.8 mm formed from 0.187-inch (4.74-mm) nominal-thickness, metallic-coated steel sheet.

D. Post Caps: Formed from steel sheet and hot-dip galvanized after forming.

E. Fence Top and Gate Rails: Rectangular tubes.

1. Size: 2 by 2 inches (50.8 by 50.8 mm).
2. Metal and Thickness: 0.104-inch (2.6-mm) nominal-thickness, metallic-coated steel sheet.

F. Fasteners: Manufacturer's standard tamperproof, corrosion-resistant, color-coated fasteners matching fence components with resilient polymer washers or clips.

G. Finish: Powder coating.

## 2.3 SWING GATES

A. Gate Configuration: As indicated.

B. Gate Frame Height: As indicated.

C. Gate Opening Width: As indicated.

D. Frame Corner Construction: Welded or assembled with corner fittings and 5/16-inch- (7.9-mm-) diameter, adjustable truss rods for panels 5 feet (1.52 m) wide or wider.

E. Additional Rails: Provide as indicated, complying with requirements for fence rails.

F. Infill: Welded-wire fence fabric matching adjacent fence.

G. Hardware: Latches permitting operation from both sides of gate, hinges, and keepers for each gate leaf more than 5 feet (1.52 m) wide. Provide center gate stops and cane bolts for pairs of gates.

1. Comply with Section 119814 "Detention Door Hardware" for detention door hardware and locks for swing gates.

H. Spring Hinges: BHMA A156.17, Grade 1, suitable for exterior use.

I. Mortise Locks: BHMA A156.13, Grade 1, suitable for exterior use.

J. Electric Strikes: BHMA A156.31, Grade 1, of configuration required for use with lock specified, fail-secure, and suitable for exterior use.

1. Mounting Plate: Configuration necessary for mounting electric strikes. Fabricate from 1/8-inch- (3.2-mm-) thick, steel plate; hot-dip galvanize after fabrication.
2. Mounting: Mortise into post.

K. Exit Hardware: BHMA A156.3, Grade 1, Type 1 (rim exit device), with push pad actuating bar, suitable for exterior use.

L. Cane Bolts: Provide for inactive leaf of pairs of gates. Fabricated from 1/2-inch- (12.7-mm-) diameter, round steel bars, hot-dip galvanized after fabrication. Finish to match gates. Provide galvanized-steel pipe strikes to receive cane bolts in closed position.

M. Finish exposed welds to comply with NOMMA Guideline 1, Finish #4 - good-quality, uniform undressed weld with minimal splatter.

N. Metallic-Coated-Steel Finish: High-performance coating.

## 2.4 FENCE AND GATE MATERIALS

A. Metallic-Coated-Steel Wire: Welded-wire fence fabric, hot-dip galvanized after fabrication. Weight of zinc coating shall be not less than 1.0 oz./sq. ft. (305 g/sq. m).

B. Plates, Shapes, and Bars: ASTM A36/A36M.

C. Tubing: ASTM A500/A500M, cold-formed steel tubing.

D. Metallic-Coated Steel Sheet: Galvanized-steel sheet or aluminum-zinc, alloy-coated steel sheet.

E. Galvanized-Steel Sheet: ASTM A653/A653M, structural quality, Grade 50 (Grade 340), with G60 (Z180) coating.

F. Aluminum-Zinc, Alloy-Coated Steel Sheet: ASTM A792/A792M, structural quality, Grade 50 (Grade 340), with AZ60 (AZM180) coating.

G. Iron Castings: Either gray or malleable iron unless otherwise indicated.

1. Gray Iron: ASTM A48/A48M, Class 30.
2. Malleable Iron: ASTM A47/A47M.

H. Aluminum Castings: ASTM B26/B26M, Alloy A356.0-T6.

I. Galvanizing: For components indicated to be galvanized and for which galvanized coating is unspecified, hot-dip galvanize to comply with ASTM A123/A123M. For hardware items, hot-dip galvanize to comply with ASTM A153/A153M.

## 2.5 MISCELLANEOUS MATERIALS

A. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.

B. Concrete: Normal-weight, air-entrained, ready-mix concrete complying with requirements in Section 033000 "Cast-in-Place Concrete" with a minimum 28-day compressive strength of 3000 psi (20 MPa), 3-inch (75-mm) slump, and 1-inch (25-mm) maximum aggregate size.

C. Nonshrink Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M and specifically recommended by manufacturer for exterior applications.

2.6 GROUNDING MATERIALS

A. Comply with requirements of Section 260526 "Grounding and Bonding for Electrical Systems."

B. Grounding Conductors: Size as indicated on Drawings. Bare, solid wire for No. 6 AWG and smaller; stranded wire for No. 4 AWG and larger.

1. Material above Finished Grade: Copper.
2. Material on or below Finished Grade: Copper.
3. Bonding Jumpers: Braided copper tape, 1-5/8 inch (41 mm) wide and 1/16 inch (1.6 mm) thick, woven of No. 30 AWG bare copper wire, terminated with copper ferrules.

C. Grounding Connectors and Grounding Rods: Comply with UL 467.

1. Connectors for Below-Grade Use: Exothermic-welded type.
2. Grounding Rods: Copper-clad steel.

a. Size: 5/8 by 96 inches (16 by 2440 mm).

2.7 METALLIC-COATED-STEEL FINISHES

A. Powder Coating: Immediately after cleaning and pretreating, apply manufacturer's standard two-coat finish consisting of zinc-rich epoxy prime coat and TGIC polyester topcoat to a minimum dry film thickness of 2 mils (0.05 mm). Comply with coating manufacturer's written instructions to achieve a minimum total dry film thickness of 4 mils (0.10 mm).

1. Color and Gloss: As selected by Architect from manufacturer's full range.
2. Comply with surface finish testing requirements in ASTM F2408.

B. High-Performance Coating: Apply epoxy primer, polyurethane intermediate coat, and polyurethane topcoat to prepared surfaces. Comply with coating manufacturer's written instructions and with requirements in SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting. Apply at spreading rates recommended by coating manufacturer.

1. Match approved Samples for color, texture, and coverage. Remove and refinish or recoat work that does not comply with specified requirements.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and conditions, with Installer present, for compliance with requirements for site clearing, earthwork, pavement work, construction layout, and other conditions affecting performance of the Work.

B. Do not begin installation before final grading is completed unless otherwise permitted by Architect.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Stake locations of fence lines, gates, and terminal posts. Do not exceed intervals of 500 feet (152.5 m) or line of sight between stakes. Indicate locations of utilities, lawn sprinkler system, underground structures, benchmarks, and property monuments.

1. Construction layout and field engineering are specified in Section 010000 "General requirements."

3.3 FENCE INSTALLATION

A. Install fences according to manufacturer's written instructions.

B. Post Excavation: Drill or hand-excavate holes for posts in firm, undisturbed soil. Excavate holes to a diameter of not less than 4 times post size and a depth of not less than 24 inches (600 mm) plus 3 inches (75 mm) for each foot (300 mm) or fraction of a foot (300 mm) that fence height exceeds 4 feet (1.2 m).

C. Post Setting: Set posts in concrete at indicated spacing into firm, undisturbed soil.

1. Verify that posts are set plumb, aligned, and at correct height and spacing, and hold in position during setting with concrete or mechanical devices.
2. Concrete Fill: Place concrete around posts and vibrate or tamp for consolidation. Protect aboveground portion of posts from concrete splatter.

a. Exposed Concrete: Extend 2 inches (50 mm) above grade. Finish and slope top surface to drain water away from post.
b. Concealed Concrete: Top 2 inches (50 mm) below grade to allow covering with surface material. Slope top surface of concrete to drain water away from post.

3. Posts Set in Concrete: Extend post to within 6 inches (150 mm) of specified excavation depth, but not closer than 3 inches (75 mm) to bottom of concrete.
4. Space posts uniformly at 8 feet (2.44 m) o.c.

3.4 GATE INSTALLATION

A. Install gates according to manufacturer's written instructions, level, plumb, and secure for full opening without interference. Attach hardware using tamper-resistant or concealed means. Install ground-set items in concrete for anchorage. Adjust hardware for smooth operation and lubricate where necessary.

3.5 GROUNDING AND BONDING

A. Comply with Section 260526 "Grounding and Bonding for Electrical Systems."

B. Fence Grounding: Install at maximum intervals of 1500 feet (450 m) except as follows:

1. Fences within 100 Feet (30 m) of Buildings, Structures, Walkways, and Roadways: Ground at maximum intervals of 750 feet (225 m).

a. Gates and Other Fence Openings: Ground fence on each side of opening.

1) Bond metal gates to gate posts.

C. Protection at Crossings of Overhead Electrical Power Lines: Ground fence at location of crossing and at a maximum distance of 150 feet (45 m) on each side of crossing.

D. Fences Enclosing Electrical Power Distribution Equipment: Ground as required by IEEE C2 unless otherwise indicated.

E. Grounding Method: At each grounding location, drive a grounding rod vertically until the top is 6 inches (150 mm) below finished grade. Connect rod to fence with No. 6 AWG conductor. Connect conductor to each fence component at grounding location.

F. Bonding Method for Gates: Connect bonding jumper between gate post and gate frame.

G. Connections: Make connections so possibility of galvanic action or electrolysis is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact are galvanically compatible.

1. Use electroplated or hot-tin-coated materials to ensure high conductivity and to make contact points closer in order of galvanic series.
2. Make connections with clean, bare metal at points of contact.
3. Make aluminum-to-steel connections with stainless-steel separators and mechanical clamps.
4. Make aluminum-to-galvanized-steel connections with tin-plated copper jumpers and mechanical clamps.
5. Coat and seal connections having dissimilar metals with inert material to prevent future penetration of moisture to contact surfaces.

H. Bonding to Lightning-Protection System: If fence terminates at lightning-protected building or structure, ground the fence and bond the fence grounding conductor to lightning-protection down conductor or lightning-protection grounding conductor, complying with NFPA 780.

## 3.6 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

1. Grounding-Resistance Tests: Subject completed grounding system to a megger test at each grounding location. Measure grounding resistance not less than two full days after last trace of precipitation, without soil having been moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural grounding resistance. Perform tests by two-point method according to IEEE 81.
2. Excessive Grounding Resistance: If resistance to grounding exceeds specified value, notify Architect promptly. Include recommendations for reducing grounding resistance and a proposal to accomplish recommended work.
3. Report: Prepare test reports of grounding resistance at each test location, certified by a testing agency. Include observations of weather and other phenomena that may affect test results.

3.7 ADJUSTING

A. Gates: Adjust gates to operate smoothly, easily, and quietly, free of binding, warp, excessive deflection, distortion, nonalignment, misplacement, disruption, or malfunction, throughout entire operational range. Confirm that latches and locks engage accurately and securely without forcing or binding.

B. Lubricate hardware and other moving parts.

3.8 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's personnel to adjust, operate, and maintain fences and gates.

END OF SECTION 32 3116